BC−760[B]/BC−760[R]/BC−780[R]

Service Manual

Version 17.0

P/N: 3128B-IS001

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1 Preface

1.1 Revision History

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<tr><td>17.0</td><td>2024–11–28</td><td>EJ523</td><td></td></tr>
<tr><td>16.0</td><td>2024–11–11</td><td>1. Replaced the Chinese pictures in the manual</td><td>2024–11–11</td></tr>
<tr><td>15.0</td><td>2024–10–12</td><td>1. Manual optimization and upgrade</td><td>2024–10–12</td></tr>
<tr><td>14.0</td><td>2024–08–16</td><td>EJ517</td><td>2024–08–16</td></tr>
<tr><td>13.0</td><td>2024–07–08</td><td>1. Changed 5x6 autoloader, and performed the material upgrade for 10x5 autoloader due to the change of its communication cable with the mian unit</td><td>2024–07–08</td></tr>
<tr><td>12.0</td><td>2024–05–28</td><td>1.Added the following FRU in the FRU Replacement section
Table 1–1<table><tr><th>FRU Code</th><th>Name</th></tr><tr><td>023-002657-00</td><td>ROUTER 4G USB</td></tr><tr><td>011-000022-00</td><td>Optical Sensor</td></tr></table>2. Updated the Product Knowledge- Optical System section.</td><td>2024–05–29</td></tr>
<tr><td>11.0</td><td>2024–04–17</td><td>1. Hydraulic diagram, updated FRU list</td><td>Not effective, voided</td></tr>
</table>

Version	Revision Date	Revision Description	Effective Date
10.0	2024–02–05	1. Reentered the content in the maintenance chapter. 2.Added code 115-102319-00	2024–02–05
9.0	2023–11–24	Modify FRU code	2023–11–23
8.0	2023–08–25	1. Added a fluidic diagram with search boxes 2. Added a list of components and parts 3. Added the section about FRU Replacement and modified some FRUs	2023–09–13
7.0	2023–05–08	Mechanical mechanism confirmation and newly added content of the commissioning topic	2023–05–08
6.0	2023–04–13	Mother Board, Main Control Board FRU Name, Code change.	2023–04–13
5.0	2023–02–06	Added the high pressure valve, ESR measurement assembly, and upgrade chapter; changed the blending assembly and aspiration module	2023–02–06
4.0	2022–12–29	Blending assembly and connecting line, rotary scanning and blending motor photocoupler line modification	Not effective, voided
3.0	2022–11–22	Diluent heating assembly (FRU) change	2022–11–22
2.0	2022–10–19	New FRU	2022–10–26
1.0	2022–6–19	Initial Release	2022–6–19

1.2 Audience

This manual is intended for service professionals who:

- Have comprehensive knowledge of circuitry and fluidics;

- Have comprehensive knowledge of reagents;
- Have comprehensive knowledge of quality control;
- Have comprehensive knowledge of troubleshooting;
- Get familiar with the operations of the system;
- Are able to use basic mechanical tools and understand the terminology;
- Are skilled users of the digital voltmeter and oscillograph;
- Are able to analyze the circuit diagrams and hydraulic diagram.

1.3 Common Operation Terms

Name	It means...
Click	to press the **××** lightly with your finger; or to left-CLICK **××** with the mouse.
Input	Click **××** edit box to move the cursor there and use the external keyboard or the pop-up keyboard to enter the desired characters or digits; or to scan the number by using the bar-code scanner.
Delete	Click the left button of the mouse, or directly tap on the touch screen or use ←→**HomeEnd** on an external keyboard to move the cursor to the character or digit to be deleted, and then delete the character after the cursor by pressing **Del**, or delete the character before the cursor by pressing **BackSpace** (← on the upper right part of the soft keyboard).
SELECT from ×× pull-down list (for pull-down list)	Click the down arrow button of the **××** box to display the pull-down list, (and drag scroll bar) to browse and then CLICK the desired item; or to press the keys (↑↓**PageUpPageDown** on an external keyboard) to browse the current list and press **Enter** to select the desired item.

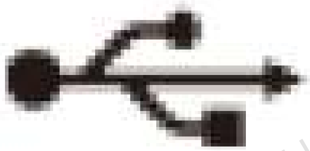
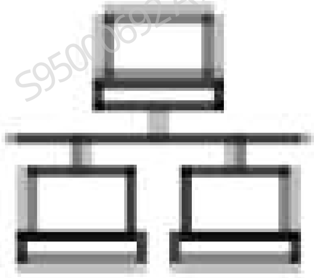
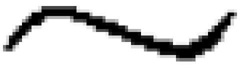
1.4 Safety Warning Messages

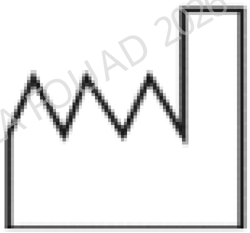
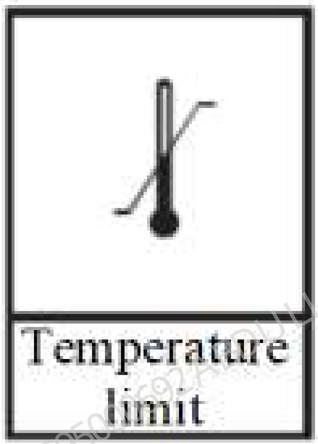
You may find the following symbols on package or the body of the instrument:

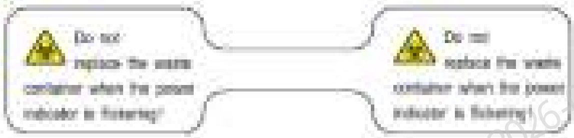
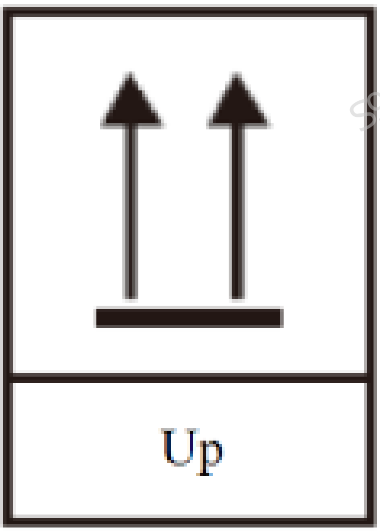
CAUTION

- **During the daily use of the instrument, especially the cleaning process, the operator shall ensure the intactness of the labels.**

When you see...	**Indication**
	Caution Note: Indicates the need for the user to consult the instruction for use for important cautionary information such as warnings and precautions that cannot, for a variety of reasons, be presented on the medical device itself.
	Biological risks
	WARNING, LASER BEAM
	Protective earth (ground)

When you see...	Indication
	Off (Power)
	On (Power)
	USB interface
	Computer network
	Alternating current
	SERIAL NUMBER
	FOR IN VITRO DIAGNOSTIC USE

When you see...	Indication
	DATE OF MANUFACTURE
	Temperature limit
	Humidity limitation
	Atmospheric pressure limitation

When you see...	Indication
	This electronic information product contains certain toxic and hazardous substances. The environmentally friendly use period is 20 years. You can use it safely within this period. After this period, it should enter the recycling system.
	Biological risks (on the tube of the waste container cap assembly) Do not replace the waste container when the power indicator is blinking!
	Fragile! Handle with caution!
	This way up

When you see...	Indication
	Keep dry
	Do not roll

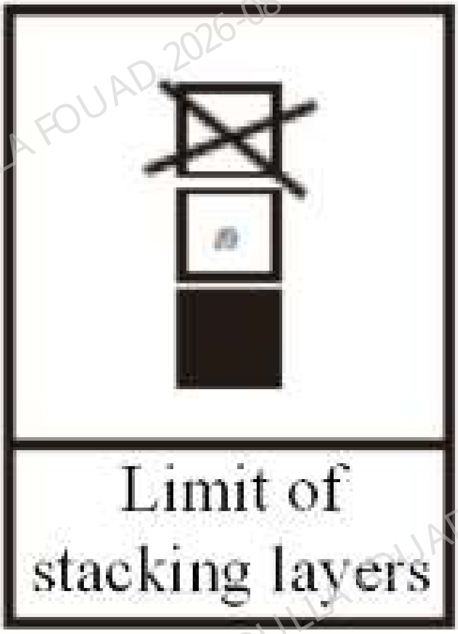
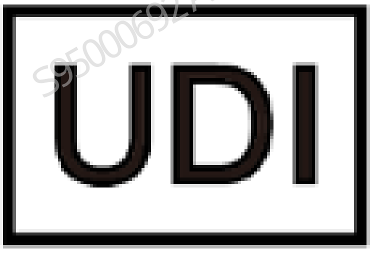
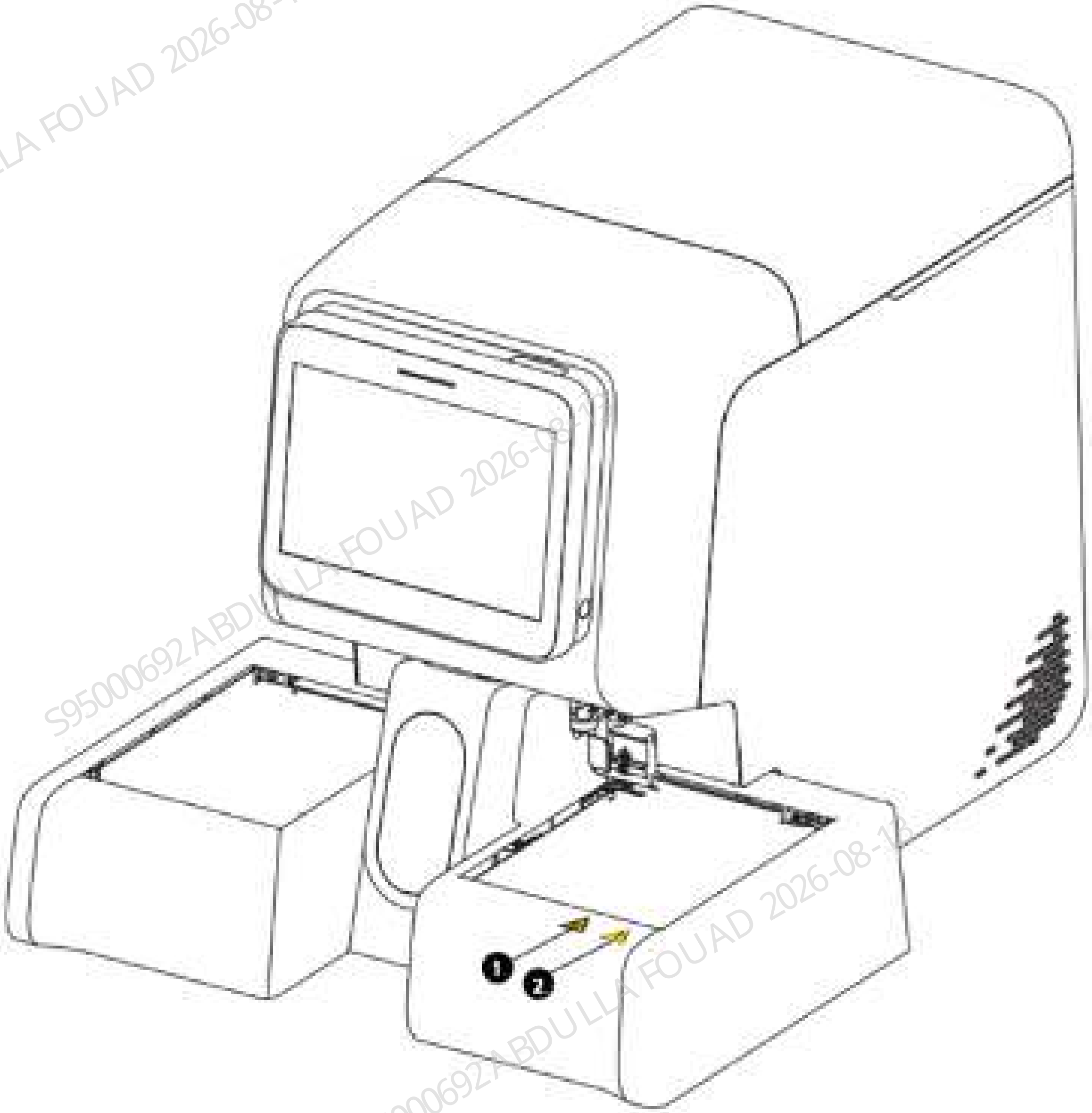
When you see...	Indication
	Max stacking mass
	Unique identification code of medical device

Figure1–1 Warnings on the Front Side of Analyzer

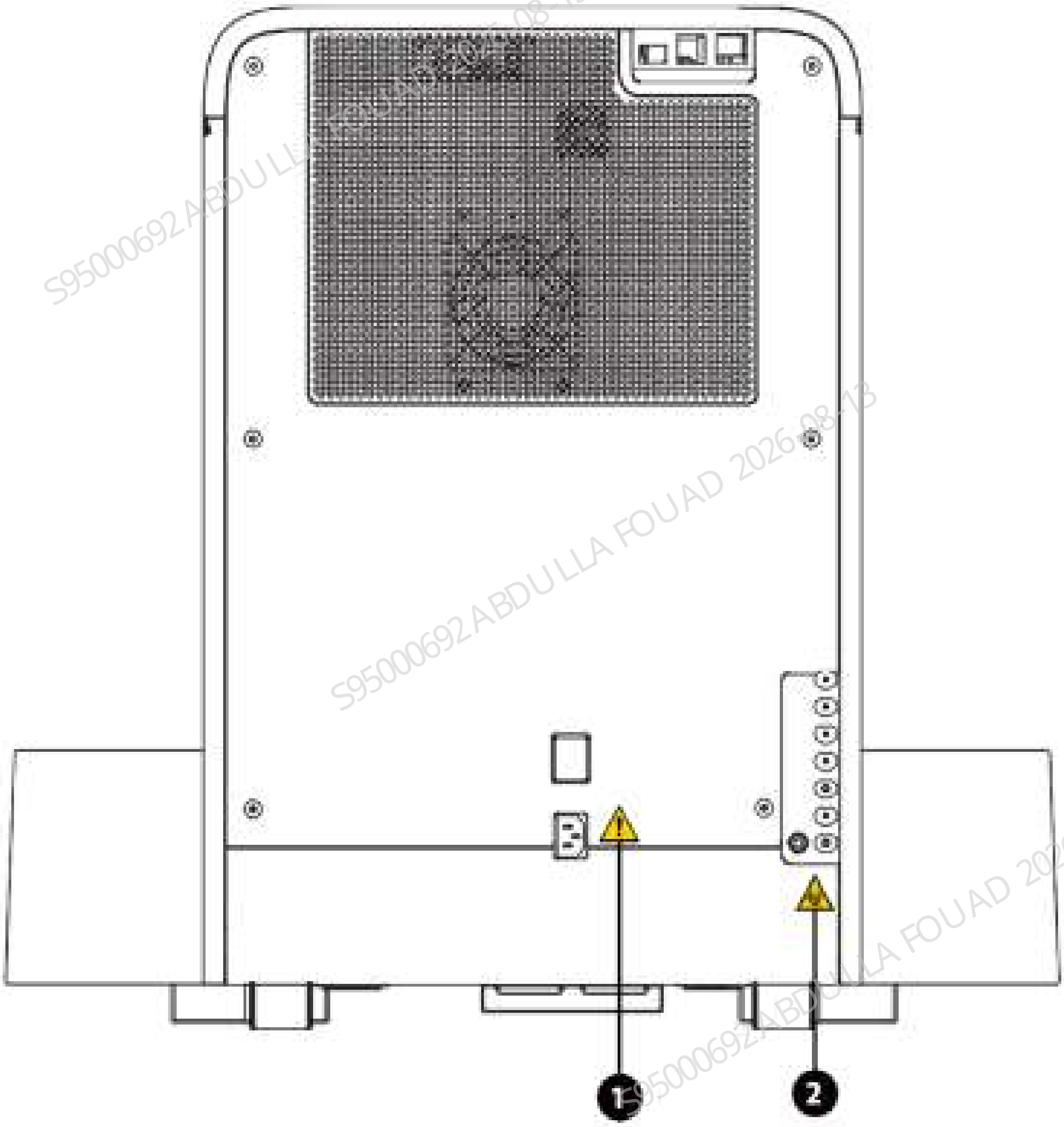


1 Biological risks
Biological risk

2 WARNING

1. Slide the tube into the tube rack. If it was not feasible, replace the tube bar code.
2. Make sure the tube cap is secured.
3. Make sure the unloading tray is not full and the detecting optocoupler is not covered, and then start analysis.

Figure1–2 Warnings on the Rear Side of Analyzer



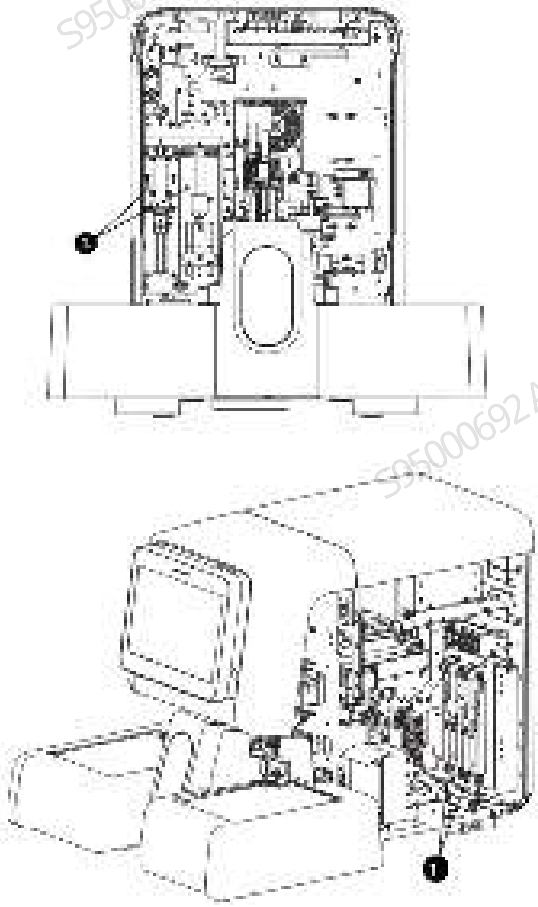
1 WARNING

1. Connect only to a properly earth grounded outlet.
2. To avoid electric shock, disconnect power cord prior to maintenance.

2 Biological risks

Biological risk

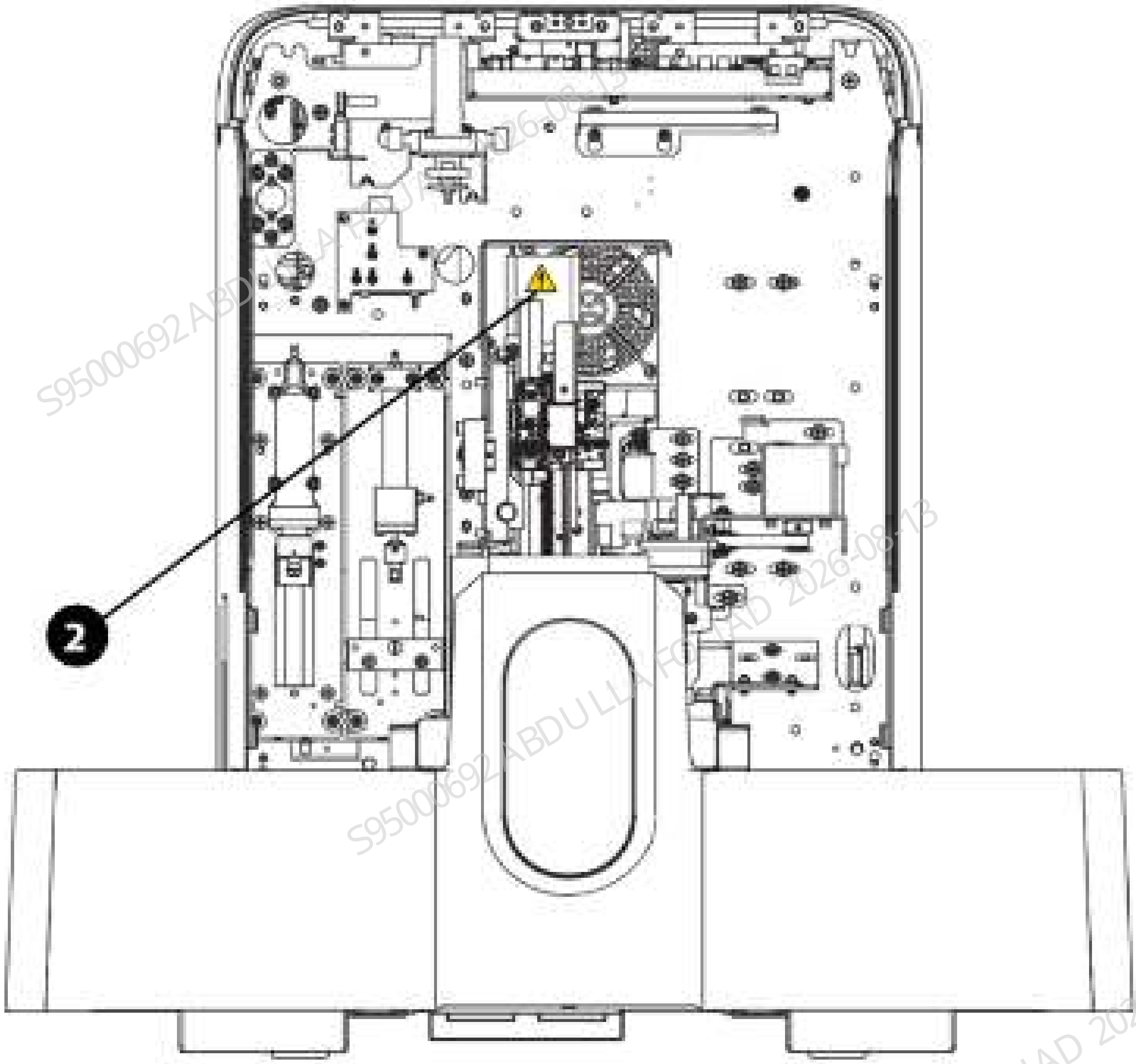
Figure1–3 Warning Symbol 1 of Moving Parts



1 WARNING

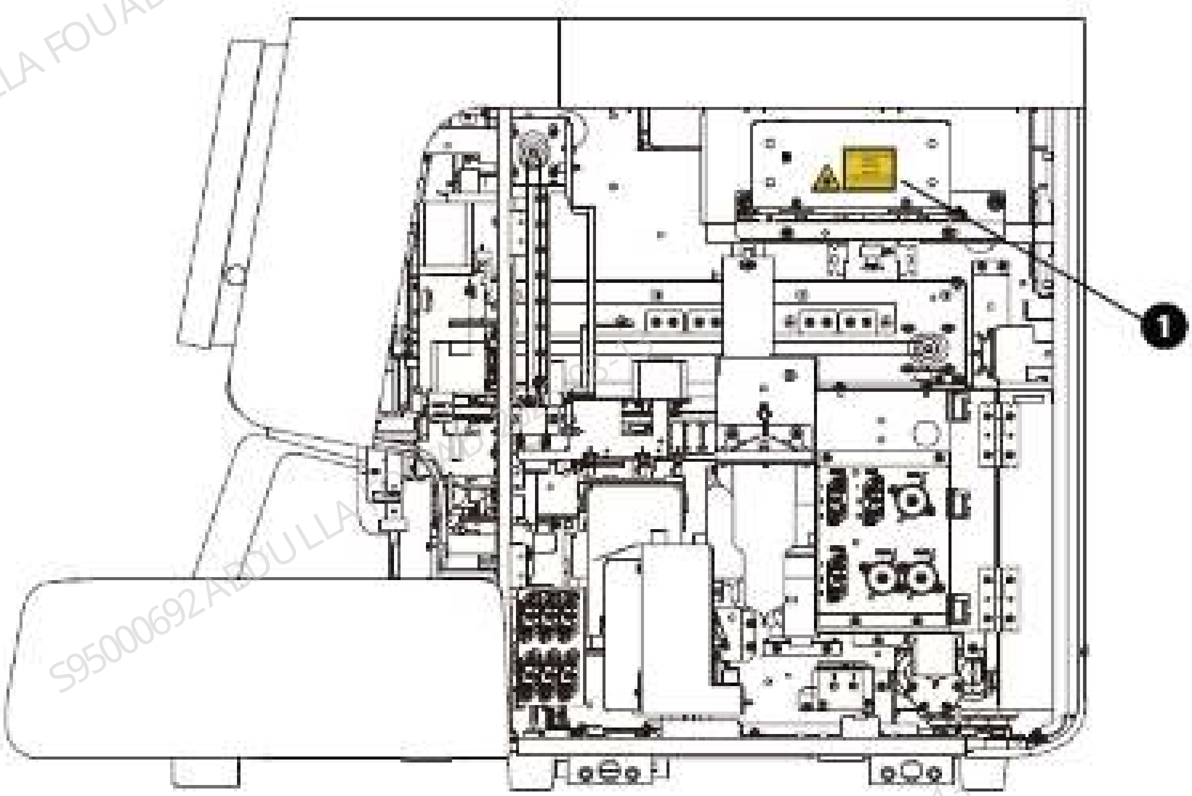
To avoid personal injury, do not put your hand under the syringe or inside the slot!

Figure1–4 Warning Symbol 2 of Moving Parts



2 WARNING
To avoid personal injury, do not put your hand under the sampling probe or guide rail!

Figure1–5 Warning Symbol 1 of Optical Assembly Laser



1 Precautions
Class 3B laser radiation when it is open and interlock fails
You will find the following symbols in this manual:

When you see...	Indication
	The statement is alerting you to the potential risk of biological infection.
	Read the statement below the symbol. The statement is alerting you to an operating hazard that can cause personnel injury.
	Read the statement below the symbol. The statement is alerting you to a possibility of minor personal injury, product damage or unreliable analysis results.
	Read the statement below the symbol. The statement is alerting you to information that requires your attention.

1.5 General Prompts

Biological Risk

- **All the samples, controls, calibrators, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective instrument (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.**
- **Be sure to dispose of reagents, waste, samples, consumables, etc. according to government regulations.**
- **The analyzer scrap must be disposed of according to local regulations.**

WARNING

- **Fuses used in this analyzer are not external replaceable fuses. For any problem, contact Mindray's Customer Service Department or your local agent.**

CAUTION

- **Use this analyzer strictly according to the Instructions for Use. Otherwise the protection for the analyzer may be destroyed.**
- **Be sure to use peripherals of the specified models only, which shall be protected from moisture.**
- **External analyzer connected to the analyzer and digital interfaces must have passed CCC (S&E) compulsory certification. Any persons who connects additional analyzer to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system works normally and complies within the safety and EMC requirements. If you have any questions, consult the technical service department of your local representative.**
- **Make sure all the safety measurements are adopted. Never disable any device or sensor for safety purposes.**
- **Clothes, hair, hands must be kept at a distance from the moving parts to prevent pinching by moving parts. Do not touch parts in motion.**
- **Be sure to operate the analyzer under the situation specified in this manual; Otherwise, the analyzer will not work normally and the analysis results will be unreliable, which would damage the analyzer components and cause personal injury.**
- **This system is only intended for use by test professionals, doctors and testers trained by Mindray or Mindray's agents.**
- **In case of sudden power failure, please turn off the power supply of the analyzer immediately, and immerse the sampling probe in the probe cleanser after the next startup.**
- **The analyzer must be within the calibration validity period. Otherwise, the measurement results may be inaccurate.**

1.6 Analyzer Transportation and Installation-related Safety Messages

WARNING

- **Before turning on the system, make sure the input voltage meets the requirements.**
- **Do not install the analyzer at a place where it is difficult to disconnect analyzer.**

CAUTION

- **Unpacking, handling or installation by personnel not authorized or trained by Mindray may cause personal injury or damage your analyzer. Do not unpack, handle or install your analyzer without the presence of the personnel authorized by Mindray.**
- **The installation, authorization, upgrade and modification of the system software must be performed by the personnel authorized by Mindray. Be sure to only install the software licensed by Mindray on the analyzer's computer.**
- **Using power strip may result in electrical interference and the unreliable analysis results. Please place the analyzer near the electrical outlet to avoid using the pinboard.**
- **Use the accompanying power cord provided by the manufacturer. Using other electrical wire may damage the system or lead to unreliable analysis results.**
- **When connecting the reagents, make sure the color of the reagent container cap assembly is the same as that of the reagent inlet to which it is connected.**
- **Before using the analyzer, check that the reagents are properly connected. Otherwise, the measurement results may be compromised.**

1.7 EMC Performance

EMC performance

- This IVD device meets the transmission and immunity requirements of GB/T 18268.1 and GB/T 18268.26.
- This analyzer is intended for the typical health care environment (hospitals, clinics, doctors' offices, etc.), and is designed and tested according to the standards for Group 1 Class A devices in GB 4824.
- For a household use, this product may cause radio interference and require protective measures.
- It is recommended to evaluate the electromagnetic environment before using the analyzer.
- Do not use this product near strong radiation sources (such as unshielded RF sources). Otherwise, it may interfere with the normal operation of the product.

Table 1–2 Electromagnetic Emission

Emission test	Conformance
GB 4824 conducted emission	Group 1 Class A
GB 4824 radiated emission	
GB 17625.1 harmonic emission	Not applicable
GB 17625.2 voltage fluctuation/scintillation emission	Not applicable

Table 1–3 Electromagnetic Immunity

Immunity test items	Basic standards	Test value	Conformance
Electrostatic discharge (ESD)	GB/T 17626.2	Contact discharge: ±2kV, ±4kV Air discharge: ±2kV, ±4kV, ±8kV	B
Radio frequency electromagnetic field	GB/T 17626.3	3V/m,80MHz~2.0GHz,80%AM	A
Pulse train	GB/T 17626.4	Power cord: ±1kV (5/50ns,5KHz)	B
Surge	GB/T 17626.5	Line to ground: ±2kV Line to line: ±1kV	B
Radio frequency transmission	GB/T 17626.6	Power cord: 3V, 150kHz~80MHz, 80%AM	A

Table 1–3 Electromagnetic Immunity(continued)

Immunity test items	Basic standards	Test value	Conformance
Power frequency magnetic field	GB/T 17626.8	3A/m, 50Hz and 60Hz	A
Voltage sags/ interruption	GB/T 17626.11	1 cycle 0% 5/6 cycles 40% 25/30 cycles 70% 250/300 cycles 5%	B C C C
Remarks: "5/6 cycles" means "5 cycles when testing at 50Hz" and "6 cycles when testing at 60Hz" Performance judgment basis: A. During the test, the performance is normal within the specifications. B. During the test, the function or performance is temporarily reduced or lost, but it can recover by itself. C. During the test, the function or performance is temporarily reduced or lost, and operator intervention or system reset is required.			

NOTICE

- **It is the responsibility of the manufacturer to provide the customer or user with electromagnetic compatibility information for the product.**
- **It is the responsibility of the user to ensure the product's EMC environment so that the product can work properly.**

1.8 Relevant Prompts of Reagents, Controls and Calibrators

WARNING

- **The reagents are irritating to eyes, skin and mucosa. Wear proper personal protective instrument (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory. If reagents accidentally spill on your skin or in your eyes, rinse the area with ample amount of clean water, and seek medical attention immediately.**
- **Use the reagents, controls and calibrators specified by the manufacturer only. Store and use the calibrators and reagents as instructed by their instructions for use. The use of other controls, calibrators and reagents will damage the analyzer and result in the wrong control, calibration and measurement results.**
- **Make sure that reagents, controls and calibrators are used before the expiration dates indicated in the Instructions for Use. Otherwise, the measurement results will be wrong due to the use of the expired reagents, controls or calibrators.**
- **Do not mix the new container of reagent with the residue in the replaced container to ensure accurate measurement. Follow the laboratory practice to protect the reagents from being contaminated.**

1.9 Maintenance-related Prompts

Biological Risk

- **Mindray assumes no responsibility for the effectiveness of the listed chemicals as a means of controlling infections. For methods of controlling infections, consult the hospital's infection prevention department or an epidemiologist.**
- **Remove the waste container cap and replace the waste container only when the power indicator is not blinking, in order not to make the waste overflow from the container and avoid any biological pollution so caused.**
- **When using the waste container, make sure that the catheter of the waste container cap assembly is located above the waste container and that the tubes are clear and unbent.**
- **After replacing the reagent container/bag, check the pipeline connected to the cap assembly to ensure that it is secure and not bent. Otherwise, operators are exposed to infection risks or the measurement results may be wrong.**

CAUTION

- **Improper maintenance may damage the analyzer. Make sure you service the instrument strictly as instructed by this manual. For problems not mentioned in this manual, contact Mindray's Customer Service Department for service advice.**
- **Only use the spare parts provided by Mindray for maintenance. Otherwise, the analyzer may be damaged.**
- **If samples and reagents are accidentally splashed on the analyzer surface, the clean and disinfect the surface. Recommended cleansers and disinfectants include water and 75% ethanol. Do not use 3% hydrogen peroxide and other materials that may be corrosive to metals. During cleaning and disinfection operations, it is required to observe laboratory safety practices and wear personal protective instrument (e.g., laboratory gowns, gloves, goggles).**
- **The analyzer cover shall be cleaned on a daily basis. Clean the analyzer with the materials specified in the Instruction for Use only. Mindray does not provide any guarantee for damage or accidents caused by using other materials.**
- **Disinfection can cause a certain degree of damage to the analyzer. It is advisable to disinfect the analyzer only if it is deemed necessary in your hospital maintenance plan. Disinfect the analyzer according the hospital's disinfection requirements. Clean the analyzer before disinfecting.**
- **Do not use cleaning agents that may cause danger due to chemical reaction with analyzer parts or materials contained in the analyzer. For any questions about the compatibility of cleaning agent with analyzer parts or materials contained in the analyzer, please contact Mindray's Customer Service Department or your local sales representative.**
- **Only use the accessories specified by Mindray to ensure the performance and safety of the analyzer. For more information, please contact Mindray's Customer Service Department or your local sales representative.**
- **If any of the pipes or fluidic components are worn out, stop using the analyzer and contact Mindray Customer Service Department immediately for inspection or replacement.**
- **If the indicator blinks frequently, the sampling probe is moving downward. Never place your hands under the sampling probe when it is moving downward. Otherwise, you will hurt your hand!**

1.10 Laser Warning

Class 1 laser products

NOTICE

- **Class 3B laser radiation if the shield is open or the interlock fails.**
- **Protect from beam.**

1.11 Cyber Security

CAUTION

- **This analyzer software adopts closed network, and its application system runs in exclusive mode without interference from other pages. Users can only operate the product-side software interface and cannot directly access the software operating system or install the software.**
- **If the analyzer is connected to an external computer, do not use the computer for any purposes other than the intended use. It is recommended to install antivirus software and periodically run and update it to prevent computer viruses.**
- **Data must be transmitted in a closed network or virtual network environment to ensure network isolation.**
- **Operators must protect the security of network authorization information (such as passwords and user information) and ensure that the authorization information is not accessible by the unauthorized.**

2 Product Knowledge

2.1 Product Specifications

2.1.1 Product Name and Intended Use

The auto hematology analyzer (BC-760[B]/BC-760[R]/BC-780[R]) is intended for blood cell count, white blood cell classification, hemoglobin concentration measurement, reticulocyte measurement, erythrocyte sedimentation rate measurement, and body fluid cytometry in clinical trials.

NOTICE

- **The types of body fluid samples currently supported by the analyzer include: cerebrospinal fluid, pleural fluid, ascites, and synovial fluid. You may not get accurate results if you analyze body fluid samples other than specified above.**
- **The analyzer identifies the normal patient, with all normal system-generated parameters for In Vitro Diagnostic Use. The product flags or identifies patient results that require additional studies.**

2.1.2 Product Configuration

Analyzer Configuration

Model	Difference in channel configurations of different modes			AL, CT
	#PLT-H\IPF RUO parameters	RET	ESR	
BC-760[B]	X	X	√	√
BC-760[R]	√	√	√	√
BC-780[R]	√	√	√	√

Analyzer Interface

NOTICE

- **The USB interface on the side of the analyzer is only used to connect the specified external equipment. For the supported equipment and model information, please refer to 2.2.3 External PC Configuration, 2.2.4 Keyboard, 2.2.5 Mouse, 2.2.6 External Barcode Scanner, 2.2.7 Printer and 2.2.8 USB Drive.**

One network port (10/100/1000M Ethernet compatible, conforming to 802.3u/802.3ab standards)
Four USB ports (specification: DC 5V; 500mA; USB 2.0 (3); USB 3.0 (1))

External PC Configuration

CPU: Equal or superior to Intel® 2.6GHz/AMD® 3.5GHz
Memory: 4 GB or above
Disk: 500 GB or above
NIC: 2 or more
Monitor resolution: 1920 × 1080 or higher
Operating system: Windows 10 Home 64bit Chinese

Keyboard

Keyboard with USB port (USB2.0 and above supported).

Mouse

Mouse with USB port (USB2.0 and above supported).

External Barcode Scanner

Hand-held barcode scanner with USB port (USB2.0 and above supported).

Printer

Printer with USB port (USB2.0 and above supported).

NOTICE

- **Users may purchase printers from Mindray or purchase the printers certified by Mindray.**
- **Printers certified by Mindray: Lenovo Lj2655dn, HP LaserJet Pro M203dn**

USB Drive

USB2.0 and above supported

2.1.3 Introduction and Principles of the Device

Overview of Overall Unit

Product Structure and Components

BC-760 [B]/BC-760 [R]/BC-780 [R] Automatic Hematology Analyzer consists of analysis unit, information management unit, result output unit, and accessories.

The analyzer supports three types of autoloaders: 10-Position and 10-Row, 10-Position and 5-Row, and 5-Position and 6-Row-.

Modules and Assemblies

Front View of Analyzer

Figure2–1 Front View of Analyzer (10-Position and 10-Row Autoloader)

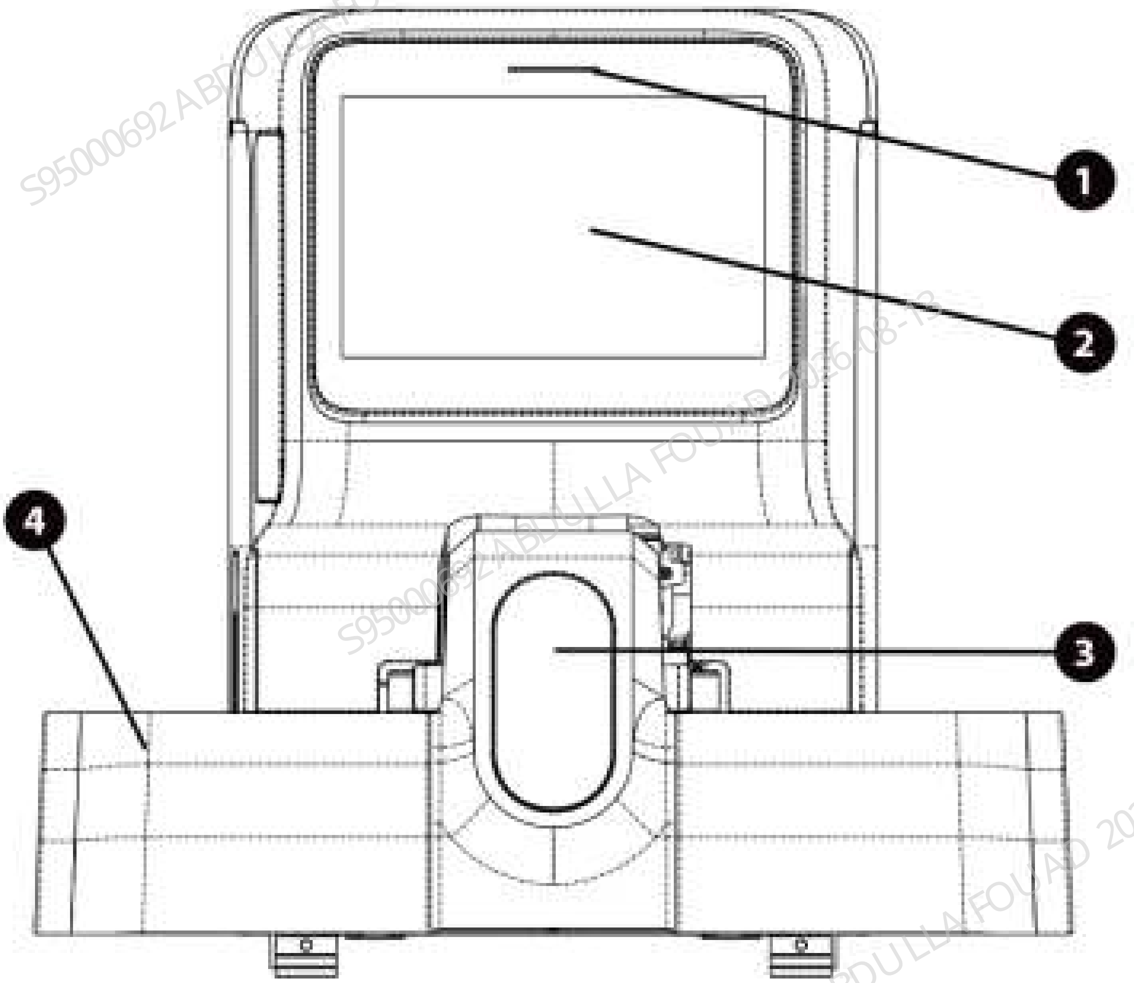


Figure2–2 Front View of Analyzer (10-Position and 5-Row Autoloader)

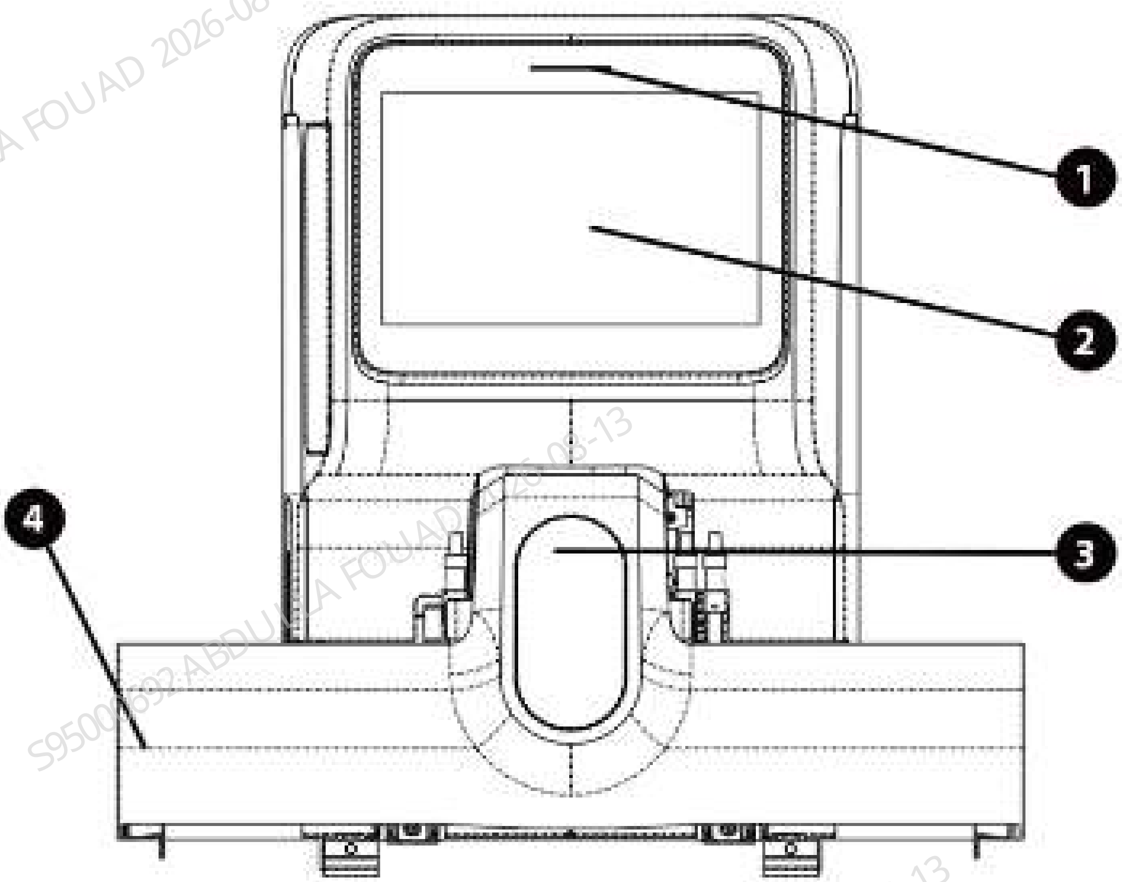
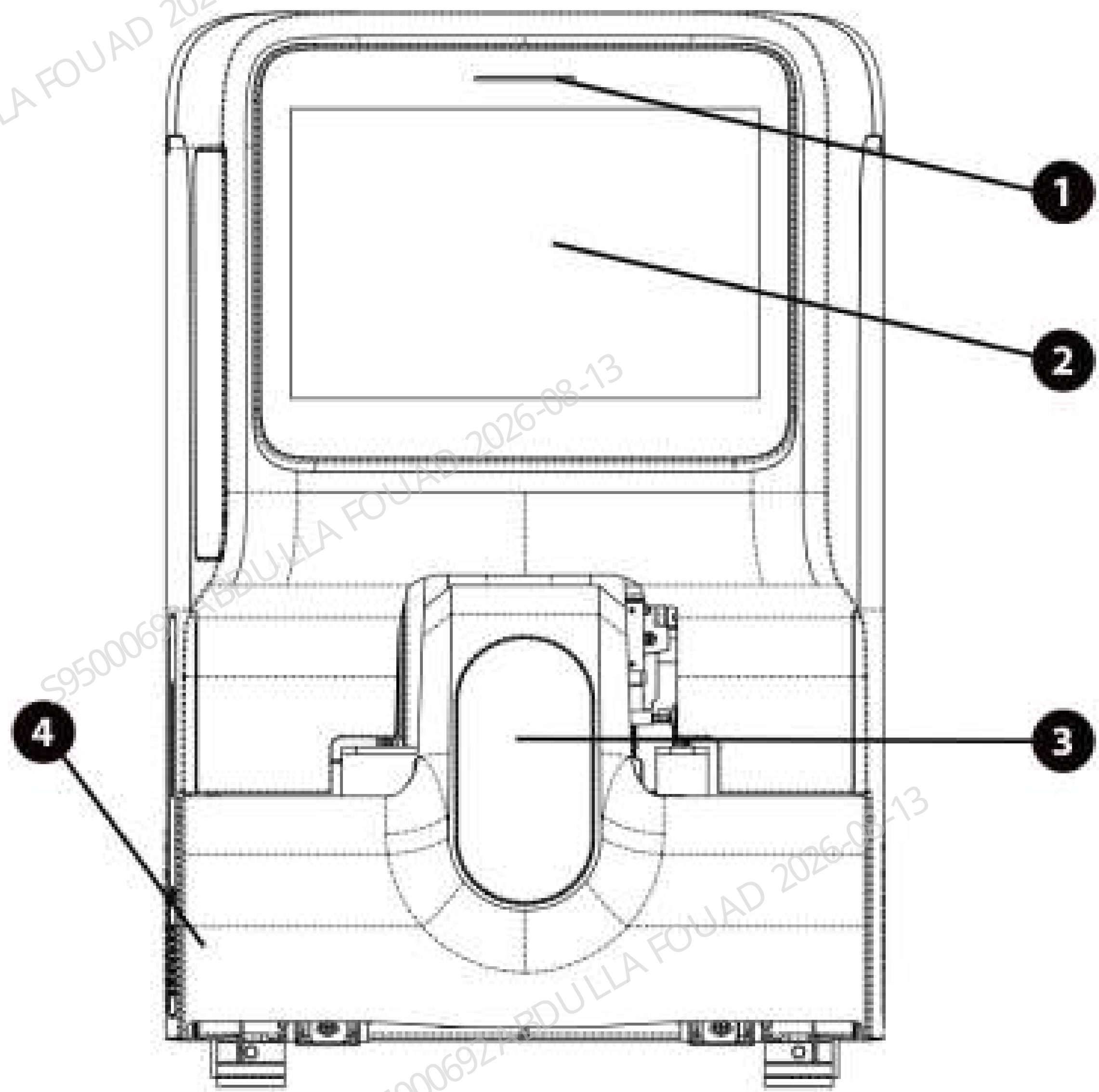


Figure2–3 Front View of Analyzer (5-Position and 6-Row Autoloader)



1 Status Indicator
2 Touch screen
3 Closed sample compartment
4 Autoloader

1	Status Indicator	The status indicator is located above the touch screen to tell you about the statues of the instrument, including ready, running, error, sleep and power-off, etc.	Ready: indicator stays in green
			Running: indicator flickers in green
			Sample probe: Flashing quickly
			Error: indicator stays in red

			Sleep: indicator stays in orange
			Off: indicator off
2	Touch screen	The touch screen is located on the front side of the analysis for executing screen operations and displaying information.	/
3	Closed sampling compartment	When the closed sampling mode is selected on the touch screen (2), the system will automatically open the closed sampling compartment. Place the applicable adapter and sample test tube, and perform a closed sampling test.	/
4	Autoloader	The autoloader is located in the front of the analyzer. You can use it to load sample automatically.	/

Back View of Analyzer

Figure2–4 Back View of Analyzer (10-Position and 10-Row & 10-Position and 5-Row Autoloader)

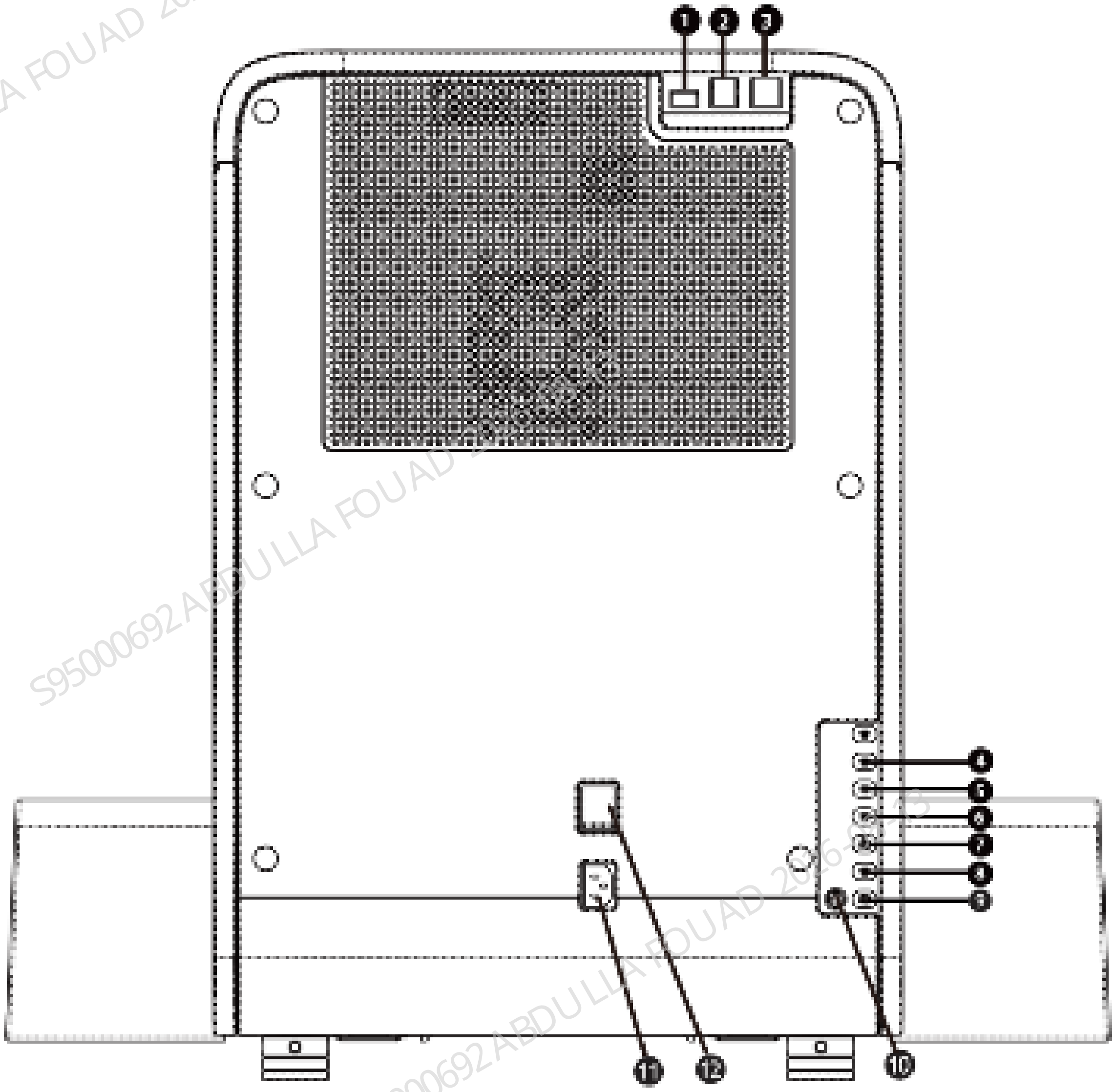
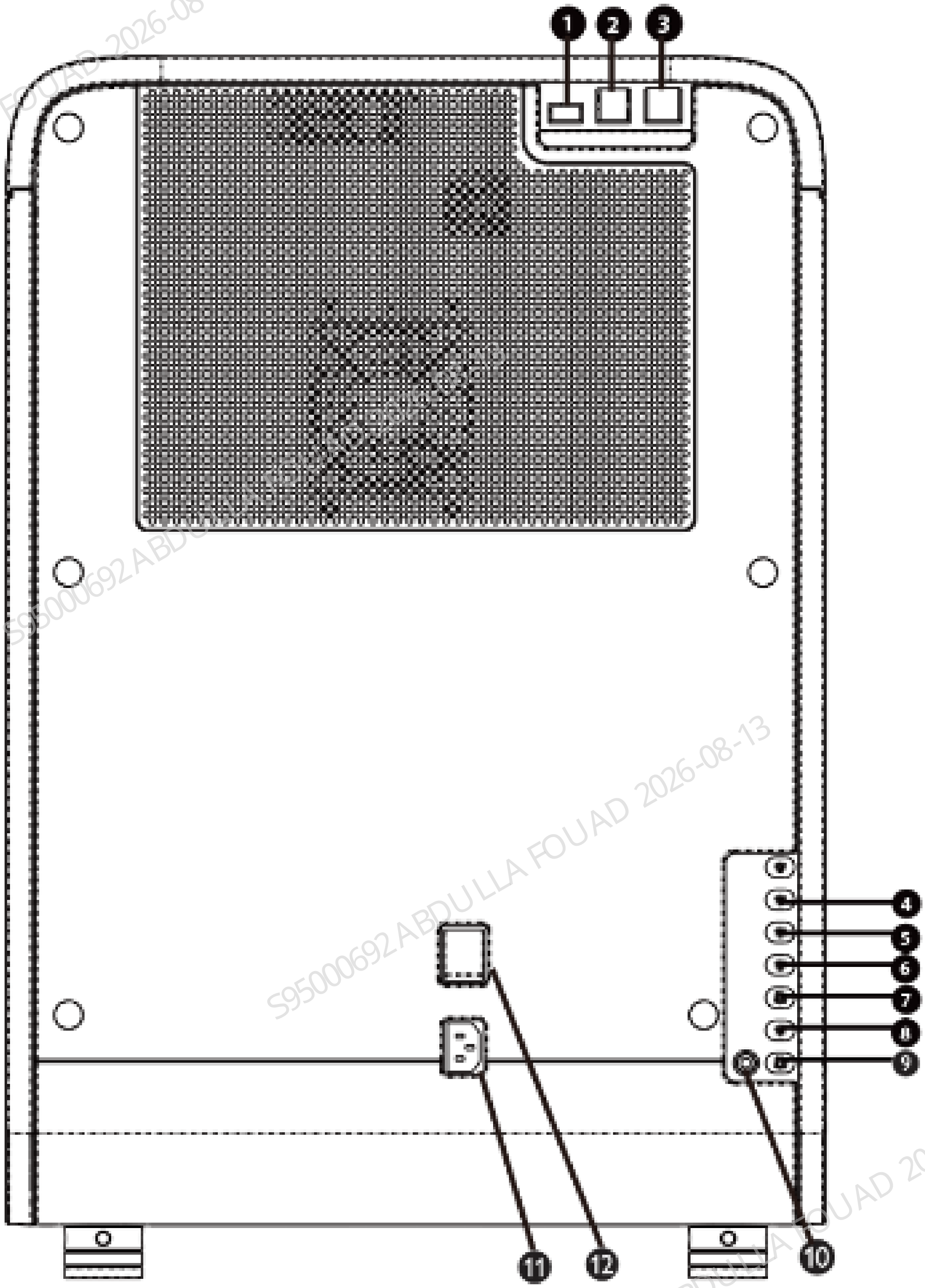


Figure2–5 Back View of Analyzer (5-Position and 6-Row Autoloader)



1 USB interface (protocol version 3.0)	2 USB interface (protocol version 2.0)
3 Network port	4 Interface for DR blood cell analysis diluent (applicable to BC-760[R]/BC-780[R] models)
5 Interface for LD hematology analysis lyse	6 Interface for LH hematology analysis lyse
7 Interface for wash solution (model: ESR)	8 Interface for DS hematology analysis diluent
9 Waste outlet	10 Waste sensor
11 Power Input Interface	12 Power switch

Right View of Analyzer

Figure2–6 Right View of Analyzer (10-Position and 10-Row Autoloader)

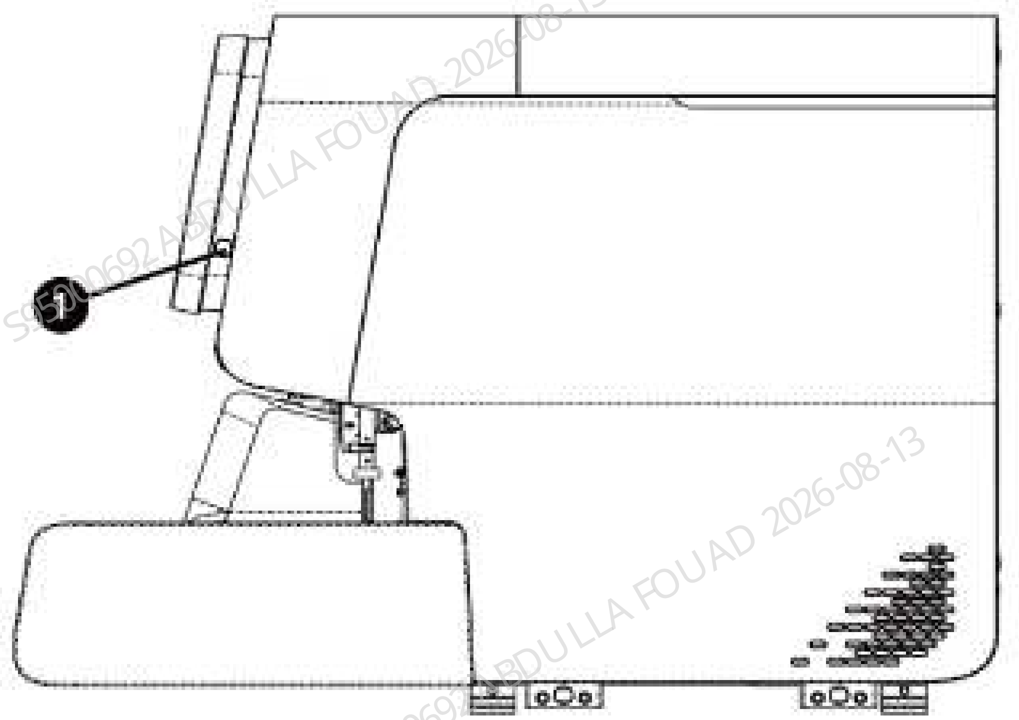
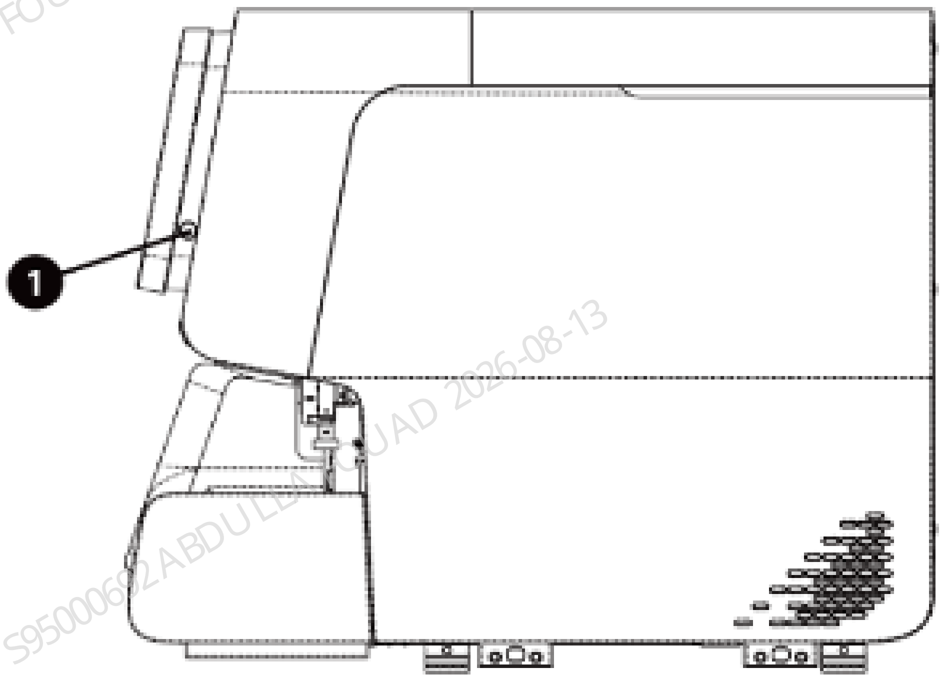


Figure2–7 Right View of Analyzer (10-Position and 5-Row & 5-Position and 6-Row Autoloader)



1 Standby key

Left View of Analyzer

Figure2–8 Left View of Analyzer (10-Position and 10-Row Autoloader)

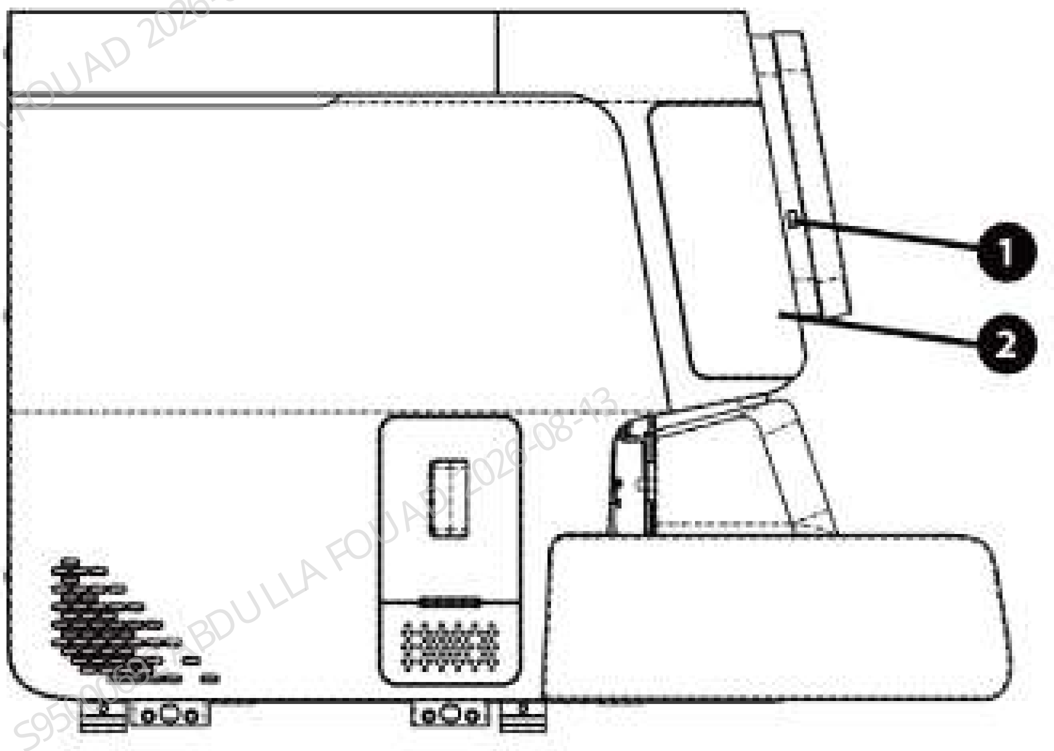
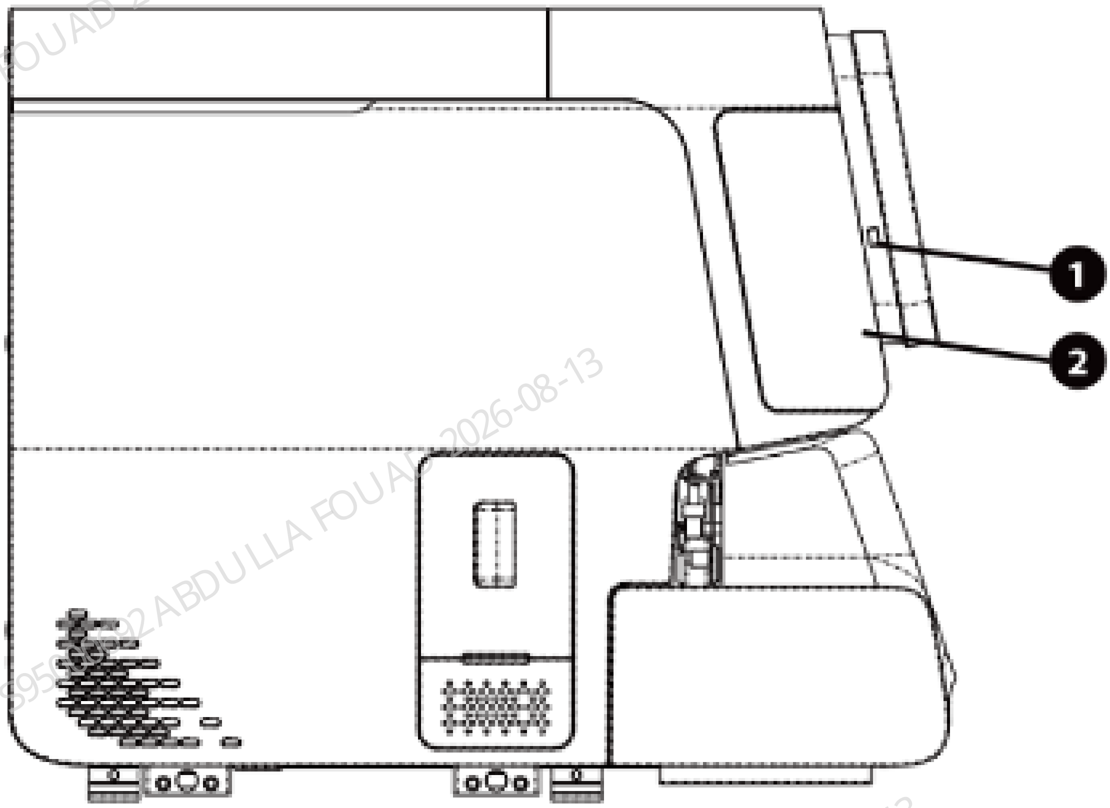


Figure2–9 Left View of Analyzer (10-Position and 5-Row & 5-Position and 6-Row Autoloader)



1 USB interface (protocol version 2.0)
2 Dye replacement access

List of Analyzer Accessories

The list of accessories and optional components of the analyzer is shown as bellow.

Table 2–1 List of Accessories and Optional Components

	Standard	Optional
Reagent cap assembly	√	
Waste tube assembly	√	
Power cord	√	
Autoloader	√	
Tube rack	√	
External barcode scanner		√
Printer		√

NOTICE

- **For any questions about the configured/optional accessories, consult your sales representative.**
- **The actual accessories accompanying the product may vary depending on your product configuration.**

CAUTION

- **Only use the accessories and consumables manufactured or recommended by Mindray to achieve the promised system performance and safety. For more information, contact Mindray Customer Service Department or your local distributor.**
- **Use the original power cord provided by the manufacturer. Using other electrical wire may damage the system or lead to unreliable analysis results.**

System Principles

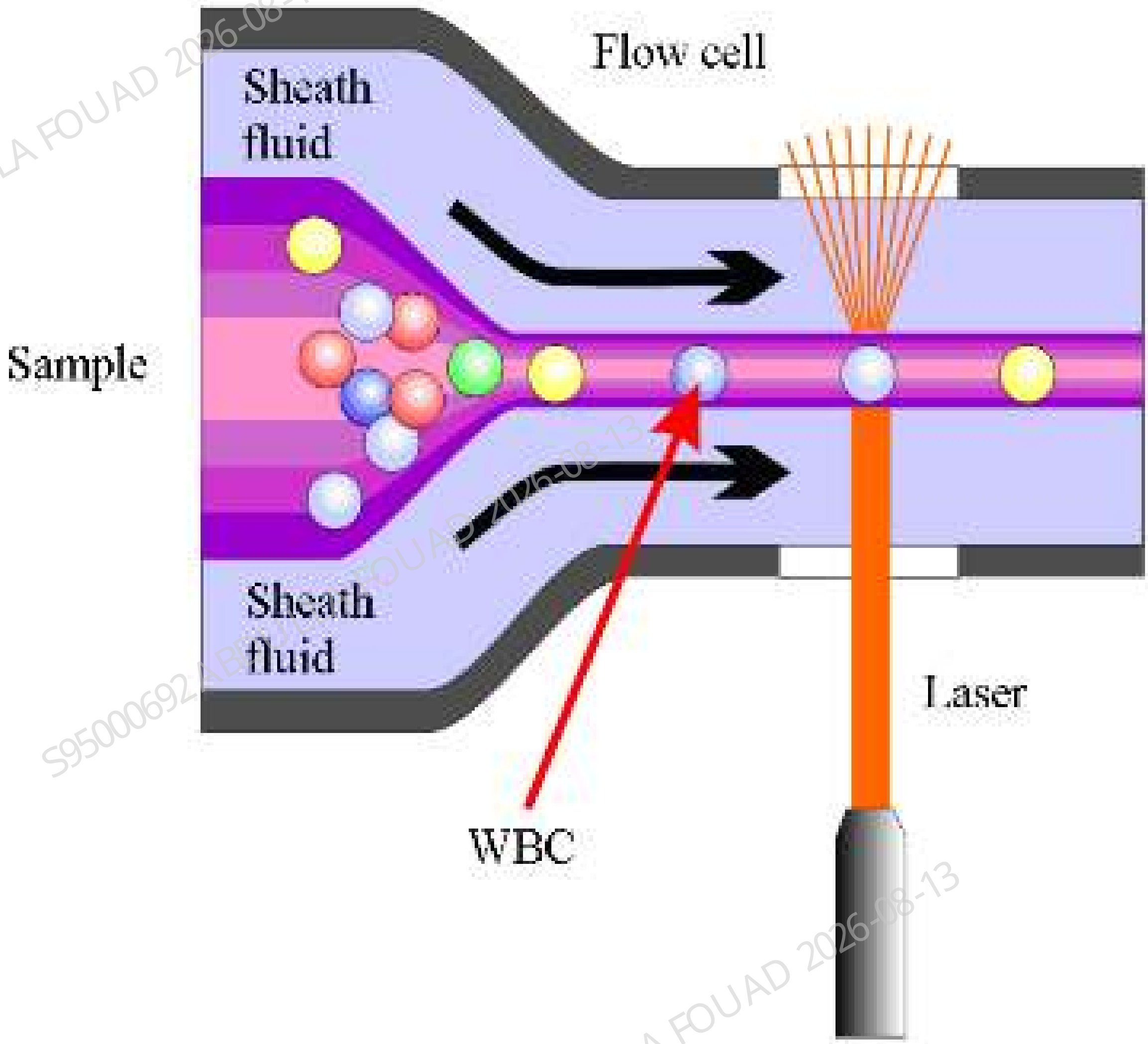
Overview

The analyzer adopts the sheath-low impedance method, laser scattering method and flow cytometry (FCM)+fluorescence staining technology for cell differentiation and counting. It also use a colorimetric method for hemoglobin measurement. Based on the above data, the analyzer calculates other parameters.

WBC Measurement

Laser FCM + fluorescence staining detection technology

Figure2–10 Diagram of FCM Technology

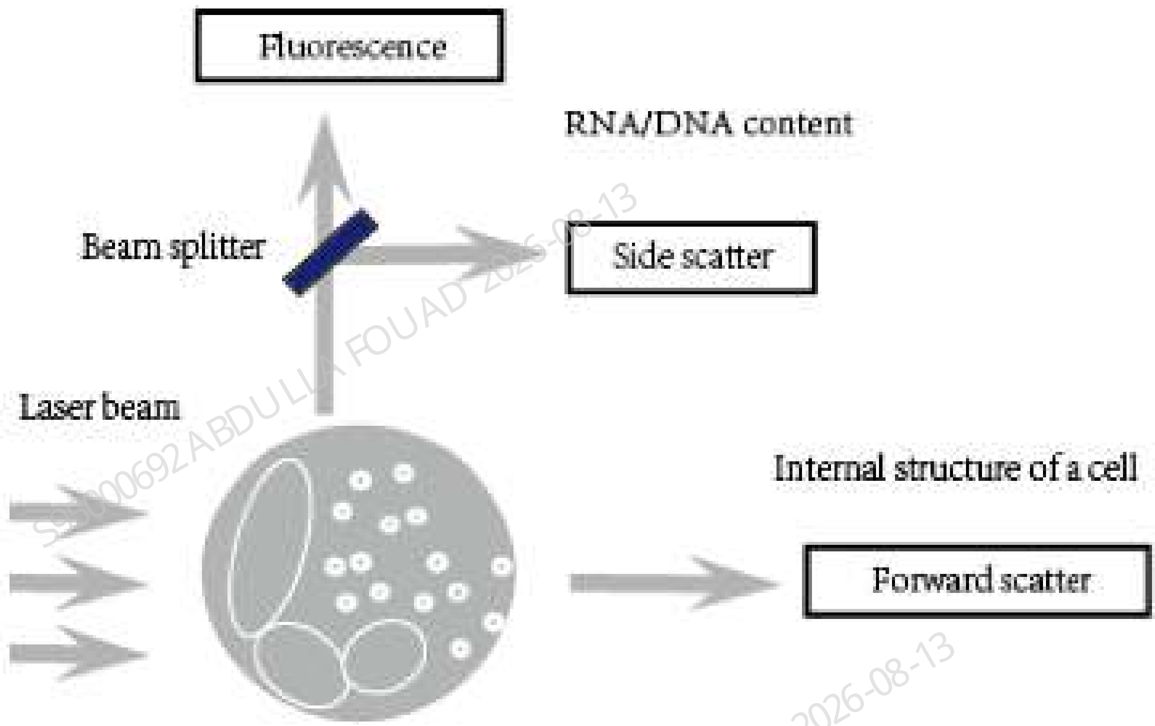


In normal peripheral blood, white blood cells can be classified into 5 categories: lymphocytes, monocytes, neutrophils, eosinophils and basophils. Analyzing all types of white blood cells will provide a great deal of useful information for the clinical diagnosis of diseases. Under the influence of certain diseases, the peripheral blood may contain various abnormal cells apart from the five subpopulations of normal cells, such as atypical lymphocytes, immature cells, etc. Most of these abnormal cells are different kinds of immature cells in the cell generation process. The cells contain a great deal of nucleic acid (DNA and RNA), the content of which decreases as the cell gets maturer. Therefore, normal cells and immature cells can be differentiated by detecting the content of nucleic acid in the cells.

Body fluid refers to the fluid in side body cavities except blood vessels. The common body fluids include cerebrospinal fluid, pleural fluid, peritoneal fluid and serous cavity fluid. Both cerebrospinal fluid and serous cavity fluid are colorless and transparent in normal case, but in abnormal cases, there could be increase of cells (including leukocytes and erythrocytes). Leukocytes in body fluid can be categorized into mononuclear cells (MN) and polymorphonuclear cells (PMN). The analysis of the cells in body fluid can provide useful information for clinical diagnosis.

The analyzer adopts the FCM +fluorescence staining technology to recognize and detect the immature cells in blood accurately on top of WBC 5-part differentiation. as well as identify the nucleated cells in body fluid.

Figure2–11 SF Cube Technology



The analyzer adopts the fluorescence staining technology in its DIFF channel. The RBCs are lysed and the WBC subpopulations are made different in size and complexity by the lyse; the nucleic acid substances in WBCs are marked by the new asymmetric cyanine fluorescent substance. Due to the different content of nucleic acid in different WBC subpopulations, maturity stages or abnormal development status, the volume of fluorescent dye staining the nucleic acid substances can be different; The low-angle light scatter reflects cell size, the high-angle light scatter reflects intracellular granularity, and the intensity of fluorescent signal reflects the degree that the cell is stained. By sensing the signal difference in three-dimensions of the cells treated with reagents, the DIFF channel differentiates major WBC subpopulations (lymphocytes, monocytes, neutrophil and eosinophils), as well as identifies and flags abnormal cells like immature granulocytes, atypical lymphocytes, and blast cells.

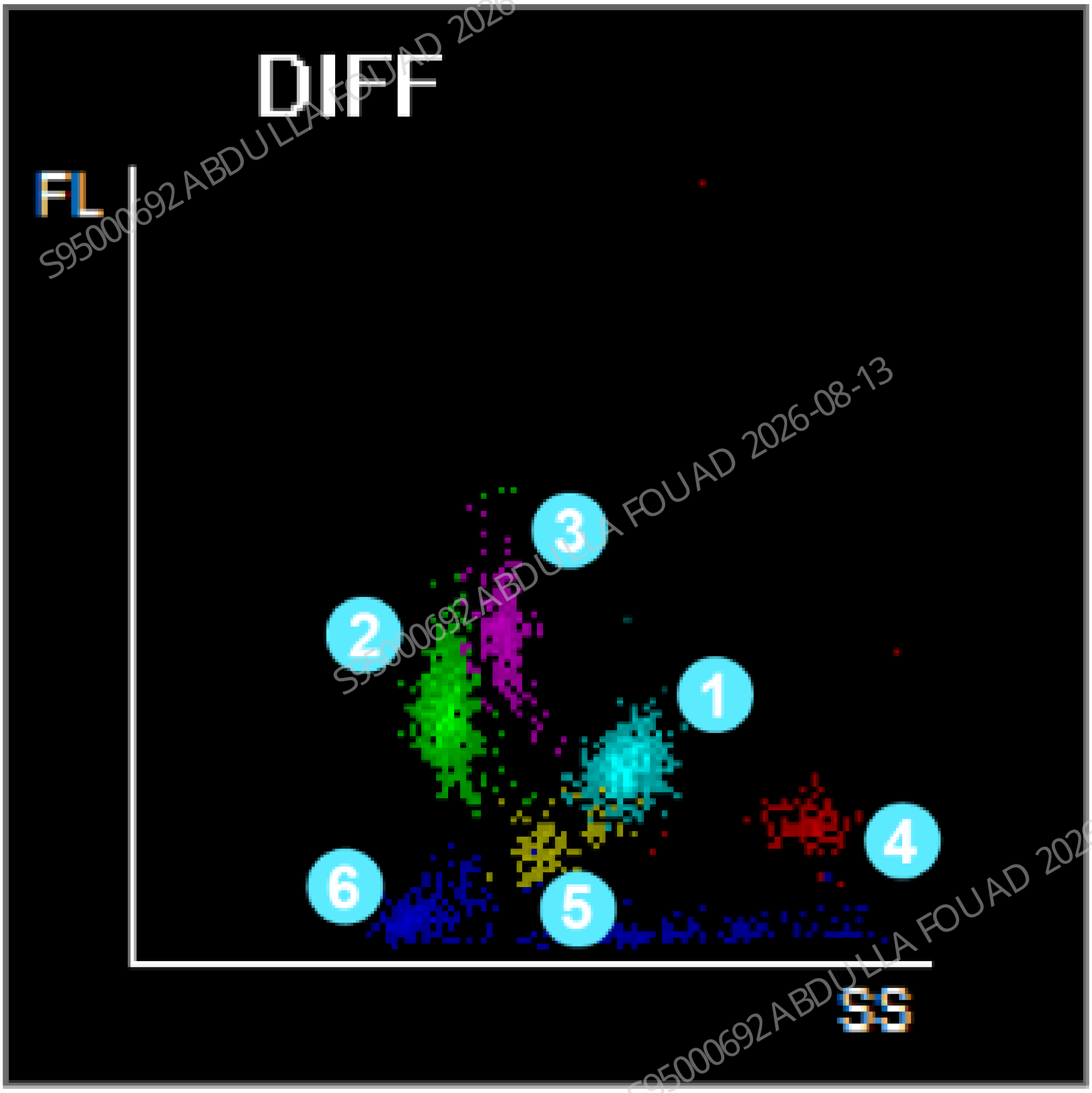
The lymphocytes are smaller in size with the nucleus taking most part of them. Lymphocytes have a high nucleus-to-cytoplasm ratio, but their nucleic acid content is low, therefore they are at a lower position in the direction of fluorescence and side scatter. The monocytes are larger in size, with high nucleus-to-cytoplasm ratio and high nucleic acid content, and less complex in structure, therefore they are at a higher position in the direction of fluorescence, and have stronger side scatter. The neutrophils and basophils are larger in size, and have medium nucleus-to-cytoplasm ratio and low nucleic acid content, therefore they are at a lower position in the direction of fluorescence, but they have stronger side scatter. The characteristics of the eosinophils are similar to those of the neutrophils, but they contain a lot of alkaline grains, so they have very strong side scatter. The blast cells, atypical lymphocytes and immature granulocytes have high nucleic acid content, so they are at a higher position in the direction of fluorescence on the scattergram.

In body fluid samples, the mononuclear cells (MN) are less complex in intracellular granularity, so the side scatter is weaker, while polymorphonuclear cells are more complex in intracellular granularity, so the side scatter is stronger.

WBC−Related Parameters

DIFF Scattergram

Figure2–12



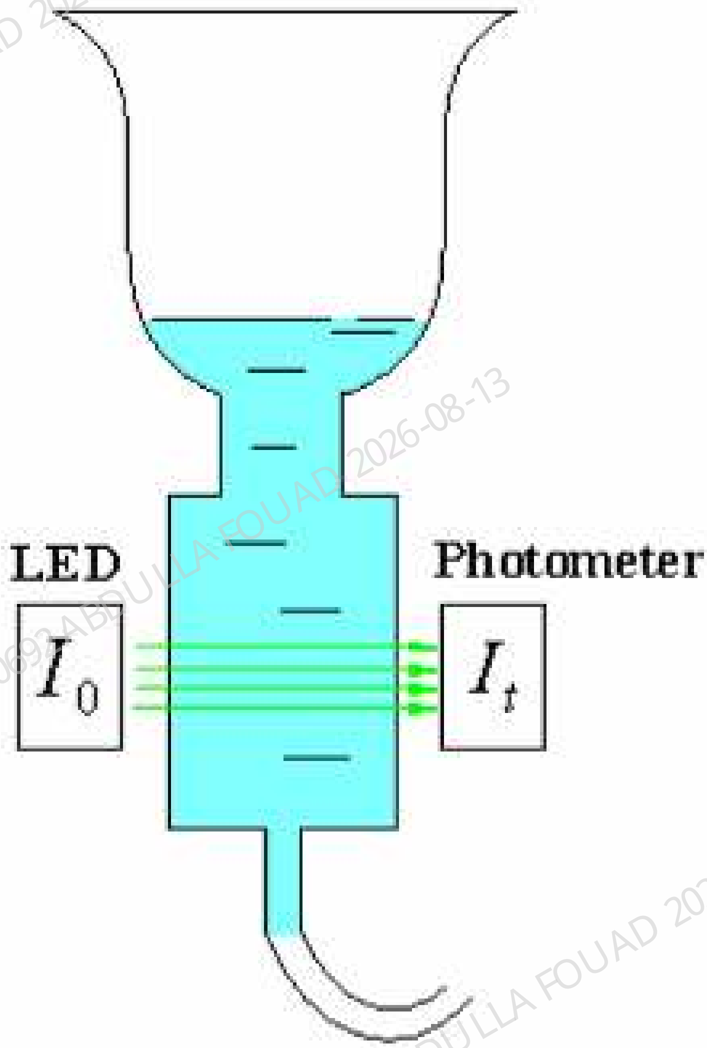
1. Neu region
2. Lym region
3. Mon region
4. Eos region
5. Baso region
6. Ghost region

Parameters	Name	Formula/Test Methods	Unit
WBC	White blood cell count	WBC = Sum of all particles in the WBC area in the DIFF channel	10^9/L
Bas#	Basophils number	Bas# = WBC × Bas%	10^9/L
Bas%	Basophils percentage	[illegible]	%
Neu#	Neutrophils number	Neu# = WBC × Neu%	10^9/L
Neu%	Neutrophils percentage	[illegible]	%
Eos#	Eosinophils number	Eos# = WBC × Eos%	10^9/L
Eos%	Eosinophils percentage	[illegible]	%
Lym#	Lymphocytes number	Lym# = WBC × Lym%	10^9/L
Lym%	Lymphocytes percentage	[illegible]	%
Mon#	Monocytes number	Mon# = WBC × Mon%	10^9/L
Mon%	Monocytes percentage	[illegible]	%
IMG#	Immature Granulocyte number	IMG# = WBC × IMG%	10^9/L
IMG%	Immature Granulocyte percentage	[illegible]	%

Hemoglobin Concentration Measurement

Colorimetric method

Figure2–13 Colorimetric Method



$$A = \lg \frac{I_o}{I_t} = k \times C \times L$$

According to the Lambert-Beer Principle, when a beam of monochromatic light passes through a well-proportioned non-scattering light-absorbing solution, the absorbance A is proportional to the product of the thickness L and the concentration C. The HGB acts as the light absorbing substance after being treated by reagent, therefore the HGB concentration can be measured by measuring the absorbance.

HGB Measurement

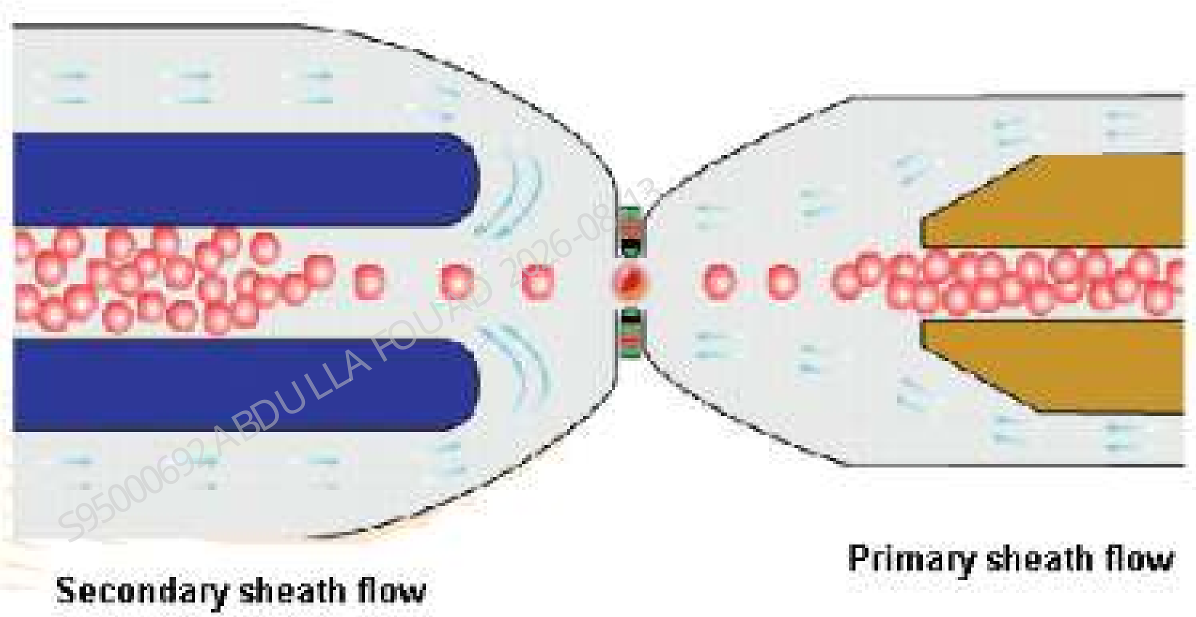
Hemoglobin concentration (HGB) is calculated using the following equation and expressed in g/L.

Parameters	Name	Formula/Test Methods	Unit
HGB	Hemoglobin Concentration	$HGB = Constant \times Ln\left(\frac{Blank\ Photocurrent}{Sample\ Photocurrent}\right)$	g/L

RBC/PLT Test

Sheath fluid impedance method

Figure2–14 Structural sketch of sheath fluid impedance method



The analyzer adopts sheath flow impedance method to count RBCs/PLTs. When passing through the aperture under the focusing effect in the fluid, RBC and PLT generate corresponding pulses according to Coulter's principle. The back-end processing device amplifies these collected pulses and compares them to the channel voltage threshold corresponding to the normal RBC/PLT volume range and then calculates the number of electrical pulses with amplitudes falling within the RBC or PLT channels. All the collected pulses are thus classified based on the voltage thresholds of different channels, and the number of the pulses in the RBC/PLT channel indicates the number of the RBC/PLT particles. The number of cells in each channel range divided according to the pulse voltage amplitude determines the cell volume distribution. The 2-dimensional distribution diagram (scattergram), with the X-axis representing cell volume, and Y-axis representing relative cell numbers, shows the distribution of cell population.

Compared with the common simple impedance method, the sheath fluid impedance method has higher test efficiency, less reagent consumption, better signal quality, and more accurate test results.

DIFF Channel PLT−H Measurement Principle

Conventional impedance methods detect impedance signals that are positively correlated with cell volume and count PLT by identifying differences in these impedance signals. However, when small red blood cells or fragments are present in the blood, their volumes are similar to the volume of large platelets, resulting the inaccurate result of the impedance-based PLT measurement. The analyzer addresses this issue by lysing red blood cells in the DIFF channel, thereby allowing for unaffected detection of large platelets. By combining the results of large platelets obtained from the DIFF channel with the unaffected small platelet results from the impedance channel, an accurate PLT count can be obtained.

SF CUBE Cell Analysis Technology

The RET channel operates similarly to the DIFF channel, with the key difference being that RBCs are spherized by RET diluent in RET channel rather than being lysed and then the nucleic acids of RBC and PLT are stained by fluorescent dyes.

NOTICE

RET channel is only available for BC-760[R]/BC-780[R] model.

RBC−Related Parameters

Parameters	Name	Formula/Test Methods	Unit
RBC	RBC count	The analyzer provides the number of red blood cells (RBC) directly by counting the red blood cells passing through the aperture.	10^{12}/L
MCV	Mean Corpuscular Volume	Calculated based on the red blood cell histogram	fL
HCT	Hematocrit	$HCT = \frac{RBC \times MCV}{10}$	□
MCH	Mean Corpuscular Hemoglobin	$MCH = \frac{HGB}{RBC}$	pg
MCHC	Mean Corpuscular Hemoglobin Concentration	$MCHC = \frac{HGB}{HCT} \times 100$	g/L
RDW-CV	Red Blood Cells Distribution Width - Coefficient of Variation	RBC Histogram	%
RDW-SD	Red Blood Cells Distribution Width - Standard Deviation	Standard deviation of RBC distribution	fL

Parameters	Name	Formula/Test Methods	Unit
NRBC%1	Nucleated red blood cell percent-age	Calculated based on the information derived from LYM region 1 and Ghost region 2 of DIFF scattergram	/100WBC
NRBC#	Nucleated red blood cell count	NRBC#=WBC x NRBC%	10^9/L

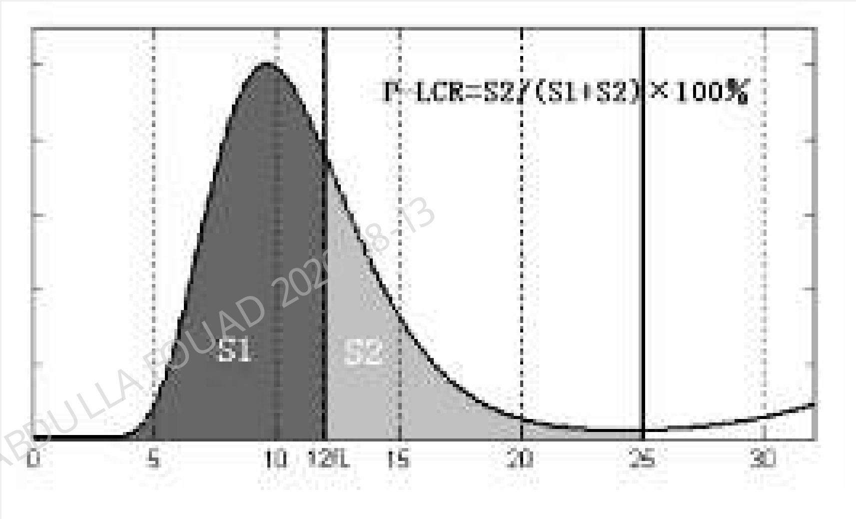
PLT−Related Parameters

Parameters	Name	Formula/Test Methods	Unit
PLT	Platelet count	Method 1: The PLT count is obtained by directly measuring the number of pulses corresponding to PLT. Method 2: The PLT count is obtained by correcting the PLT result from the impedance channel in combination with the large PLT result from the DIFF channel.	10^9/L
MPV	Mean Platelet Volume	Mean Platelet Volume (MPV) is calculated based on the PLT histogram.	fL
PDW	Platelet Distribution Width	Platelet distribution width is derived from the platelet histogram, and is reported as 10 geometric standard deviation (10 GSD)	/
PCT	Plateletcrit	$PCT = \frac{PLT \times MPV}{10000}$	□
P-LCR[1]	Platelet-large cell ratio	Derived from the PLT histogram, it represents the ratio of the number of platelets with a size over 12fL to the total platelets number.	%
P-LCC	Platelet-large cell count	P-LCC = PLT × P-LCR	10^9/L
IPF[2]	Immature Platelet Fraction*	For BC-760[R]/BC-780[R], , the parameter is derived based on PLT-O scattergram:	%

Parameters	Name	Formula/Test Methods	Unit
		[illegible] For BC-760[B], this is an optional parameter and is derived based on PLT-H scattergram. [illegible]	
PLT-I	Platelet count-Impedance	Platelet number (PLT) is measured directly by counting the platelets passing through the aperture.	109/L
PLT-H	Platelet count hybrid	PLT results derived from impedance channel, which are corrected by the DIFF channel large size platelet results.	109/L
PLT-O	Optical platelet count	PLT results derived from the RET channel	109/L

NOTE

1. **Platelet-Large Cell Ratio (P-LCR) is derived from the platelet histogram; it represents the ratio of the number of platelets with a size over 12fL to the total platelets number. The ratio is represented in %. In the following figure, S2 represents the number of larger platelet cells, and S1+S2 represents the total PLT count.**



2. **IPF parameters are applicable to the BC-760[R]/BC-780[R]model only. For BC-760[B], this is an optional parameter. For more information, contact Mindray Customer Service Department.**
3. **PLT-O is applicable to BC-760[R]/BC-780[R]model only.**
4. **PLT-H is applicable to BC-760 [R]/BC-780 [R] model only. For BC-760 [B], this is an optional parameter. For more information, contact Mindray Customer Service Department.**

RET Parameters *

Parameters	Name	Formula/Test Methods	Unit
*RET%	Reticulocyte percentage	[illegible]	□
*RET#	Reticulocyte number	RET# = RBC × RET%	10^{12}/L
*HFR	High Fluorescent Reticulocyte Ratio	$HFR = \frac{\text{Number of cells in HFR region}}{\text{Number of cells in RET region}} \times 100\%$	□
*MFR	Middle Fluorescent Ratio	$MFR = \frac{\text{Number of cells in MFR region}}{\text{Number of cells in RET region}} \times 100\%$	□
*LFR	Low Fluorescent Ratio	$LFR = \frac{\text{Number of cells in LFR region}}{\text{Number of cells in RET region}} \times 100\%$	□

Parameters	Name	Formula/Test Methods	Unit
		NOTE **RET count = Particle count in LFR area + Particle count in MFR area + Particle count in HFR area**	
*IRF	Immature Reticulocyte Fraction	IRF = MFR + HFR	%
RHE	Reticulocyte Hemoglobin Expression	Calculated based on RET scattered light information	pg

*Note: The reticulocyte parameters are only available on the BC-760[R]/BC-780[R] model.

ESR Measurement

The analyzer employs a photometric method to measure the degree and rate of erythrocyte aggregation within a specified time, thus estimating the sedimentation rate. Erythrocytes in whole blood samples form rouleaux structures, and then aggregate and begin to sediment, with the sedimentation rate primarily depending on the degree and rate of aggregation. Changes in erythrocyte aggregation state affect light scattering of the measurement beam, specifically leading to increased transmittance as aggregation increases. By measuring the change in transmittance over time, the degree and rate of erythrocyte aggregation can be determined, thus further obtaining the ESR.

Body Fluid Parameters

Body fluid refers to the fluid in side body cavities except blood vessels. The common body fluids include cerebrospinal fluid, pleural fluid, peritoneal fluid and serous cavity fluid. Both cerebrospinal fluid and serous cavity fluid are colorless and transparent in normal case, but in abnormal cases, there could be increase of cells (including leukocytes and erythrocytes). Leukocytes in body fluid can be categorized into mononuclear cells (MN) and polymorphonuclear cells (PMN). The analysis of the cells in body fluid can provide useful information for clinical diagnosis.

Parameters	Name	Formula/Test Methods	Unit
WBC-BF	White Blood Cell count-body fluid	WBC-BF = Sum of all particles in the DIFF channel other than the ghost and high fluorescence regions	$10^9/L$
TC-BF#	Total nucleated cell counts-body fluid	TC-BF #= Sum of all particles in the DIFF channel other than the ghost and fluorescence regions	$10^9/L$
MN%	Mononuclear cell percentage	$MN\% = \frac{\text{[illegible]}}{WBC-BF} \times 100\%$	□
PMN%	Polymorphonu clear cell percentage	$PMN\% = \frac{\text{[illegible]}}{WBC-BF} \times 100\%$	□
MN#	Mononuclear cell number	MN#=WBC-BF ×MN%	$10^9/L$
PMN#	Polymorphonu clear cell number	PMN#=WBC-BF×PMN%	$10^9/L$
RBC-BF	Red Blood Cell count-body fluid	The analyzer provides the number of red blood cells in body fluid (RBC-BF) directly by counting the red blood cells passing through the aperture.	$10^{12}/L$

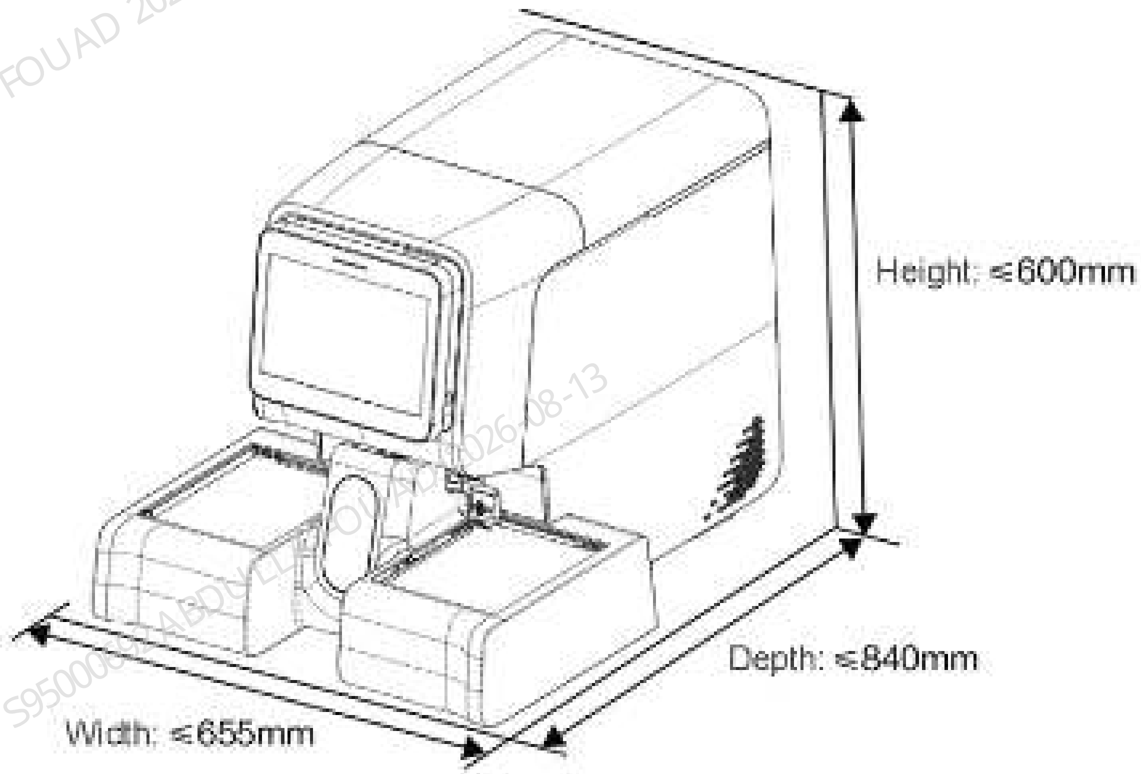
Flush

After each analysis cycle, all elements of the analyzer that the sample runs through are washed to ensure no residue is left.

2.1.4 System Specification

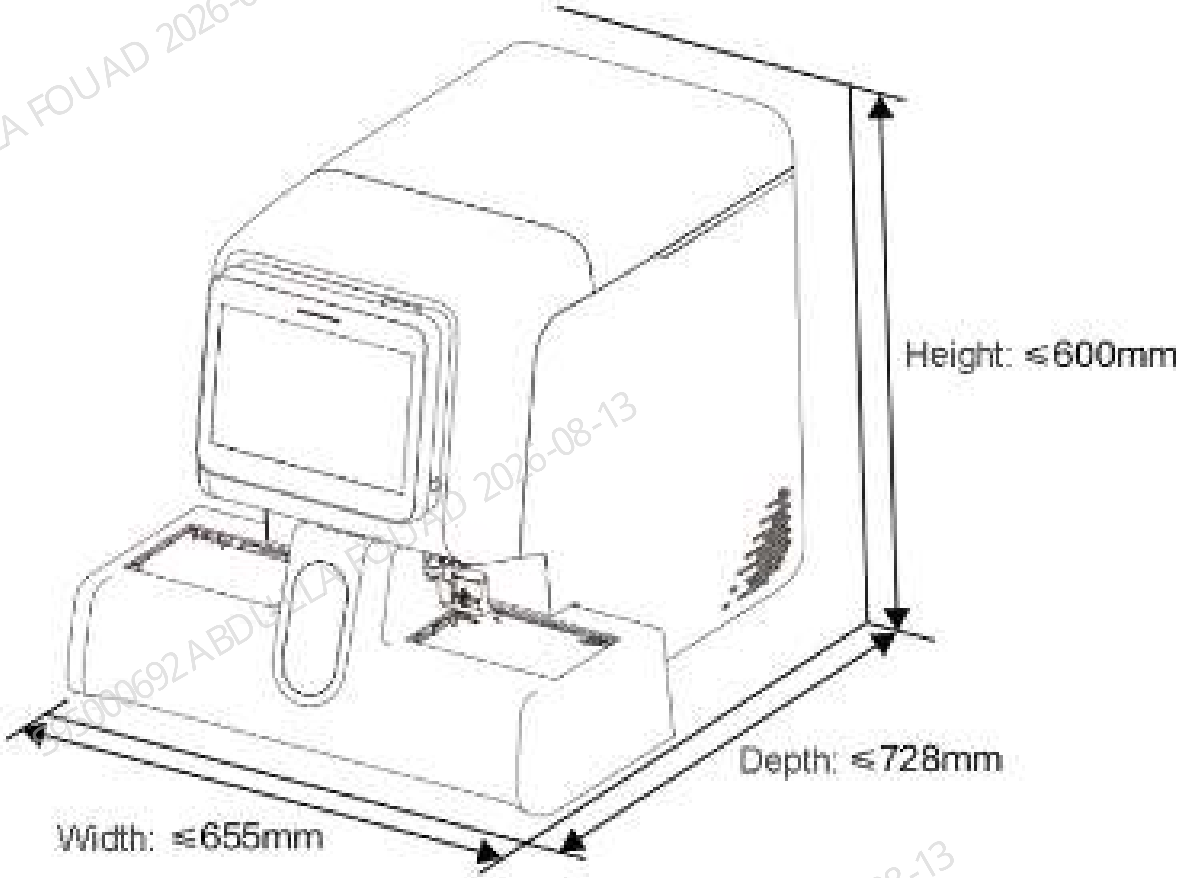
Dimensions and Weight

Figure2–15 Dimension of Main Unit (10-Position and 10-Row Autoloader)



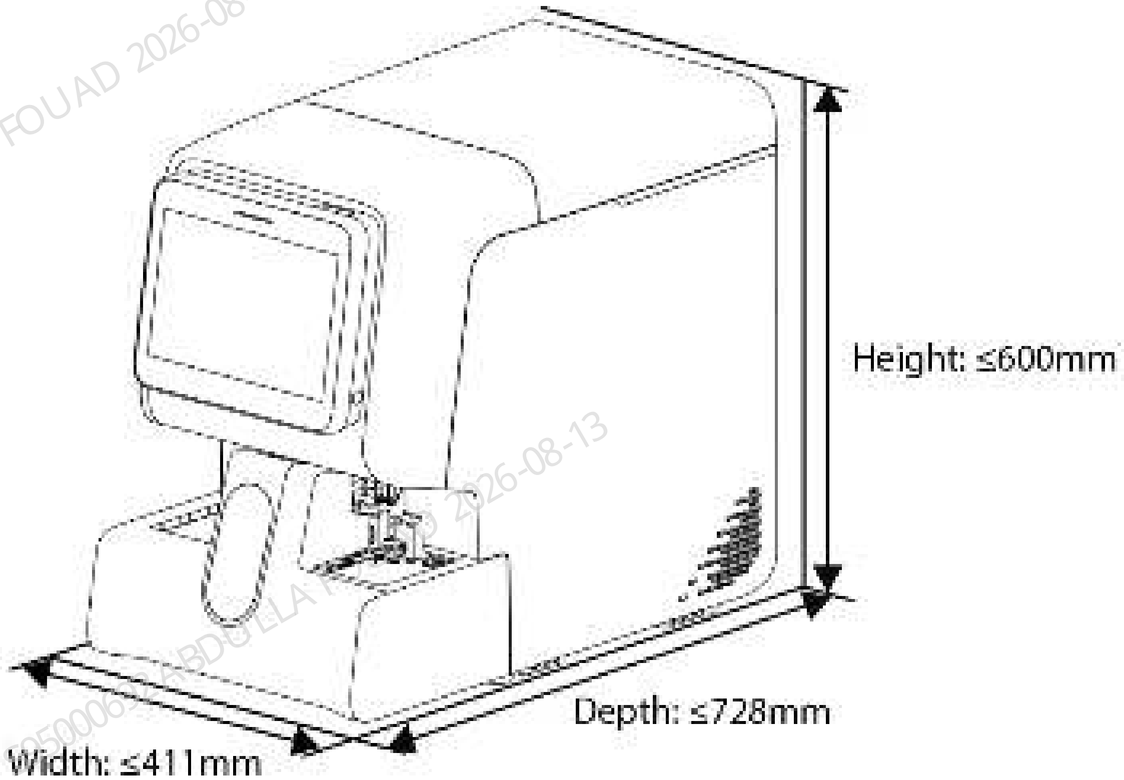
Dimensions and Weight of the Main Unit (10-Position and 10-Row Autoloader)	Value
Width (including the autoloader)	≤ 655mm
Height (including the autoloader)	≤ 600mm
Depth (including the autoloader)	≤ 840mm
Weight (including the autoloader)	≤ 82 kg

Figure2–16 Dimension of Main Unit (10-Position and 5-Row Autoloader)



Dimensions and Weight of the Main Unit (10-Position and 5-Row Autoloader)	Value
Width (including the autoloader)	≤ 655mm
Height (including the autoloader)	≤ 600mm
Depth (including the autoloader)	≤ 728mm
Weight (including the autoloader)	≤ 78 kg

Figure2–17 Dimension and Weight of the Main Unit (5-Position and 6-Row Autoloader)



Dimensions and Weight of the Main Unit (5-Position and 6-Row Autoloader)	Value
Width (including the autoloader)	≤ 411mm
Height (including the autoloader)	≤ 600mm
Depth (including the autoloader)	≤ 728mm
Weight (including the autoloader)	≤ 73 kg

Electrical Specification

Power supply

	Voltage	Input power	Frequency
Main unit	100V to 240V□ (allowance ±10%)	600 VA	50Hz/60Hz (allowance ±1 Hz)

Fuse

WARNING

Fuses used in this analyzer are not external replaceable fuses. For any problem, contact Mindray's Customer Service Department or your local agent.

Environmental Specification

Normal Working Environment

- Normal operating temperature: 10 ℃ to 35 ℃
- Normal operating humidity: 30 % to 85 %
- Normal operating atmospheric pressure: 70.0 kPa to 106.0 kPa

Storage and Transportation Environment

- Ambient temperature: -10℃ to 40℃
- Relative humidity: 10% to 90%;
- Atmospheric pressure: 50.0 kPa to 106.0 kPa

Operating Environment

- Ambient temperature: 5℃ to 40℃
- Relative humidity: 10% to 90%;
- Atmospheric pressure: 70.0 kPa to 106.0 kPa

Acoustic Noise Level

Sound Pressure

Maximum sound pressure: 80dBA

NOTICE

Be sure to store and operate the analyzer under the situation specified in this manual.

Accessories

The configured and optional accessory list of the analyzer is as shown in the table below.

Table 2–2 List of Accessories and Optional Components

	Standard	Optional
Reagent cap assembly	√	
Waste tube assembly	√	
Power cord	√	
Autoloader	√	
Tube rack	√	
External barcode scanner		√
Printer		√
PC		√
Display		√

NOTICE

- **For any questions about the standard/optional accessories, consult your sales representative.**
- **The actual accessories accompanying the product may vary depending on your product configuration.**

CAUTION

- **Only use the accessories and consumables manufactured or recommended by Mindray to achieve the promised system performance and safety. For more information, contact Mindray Customer Service Department or your local distributor.**
- **Use the original power cord provided by the manufacturer. Using other electrical wire may damage the system or lead to unreliable analysis results.**

Product Specifications

Throughput

Table 2–3 Throughput

Sample mode	Analysis mode	Throughput (tests/h)	Applicable model
WB	Routine analysis mode, CBC-4D, ESR	Not less than 80	BC-760[B]/BC-760[R]/BC-780[R]
	RET analysis mode	Not less than 45	BC-760[R] BC-780[R]
	CBC+ESR, CD+ESR	Not less than 40	BC-760[B]/BC-760[R]/BC-780[R]
	CDR+ESR	Not less than 30	BC-760[R] BC-780[R]
Low WBC	CD/WBC-3X	Not less than 60	BC-760[B]/BC-760[R]/BC-780[R]
	CDR/WBC-3X	Not less than 40	BC-760[R] BC-780[R]
Low PLT	CR/PLT-5X, CDR/PLT-5X	Not less than 35	BC-760[R] BC-780[R]
PD	Routine analysis mode	Not less than 50	BC-760[B]/BC-760[R]/BC-780[R]
	RET analysis mode	Not less than 33	BC-760[R] BC-780[R]
BF	CD	Not less than 50	BC-760[B]/BC-760[R]/BC-780[R]

NOTICE

Routine analysis includes CBC and CD modes; RET analysis includes CDR, CR and RET modes.

Sample Mode, Analysis Mode and Applicable Models

Table 2–4 Table of Sample Mode, Analysis Mode and Applicable Models

Sample mode		Analysis mode	Applicable model
Whole blood samples	CT-WB/AL-WB	CBC-4D, CBC, CD, Ret, CR, CDR ESR, CBC+ESR, CD+ESR, CDR+ESR	BC-760[R] BC-780[R]
		CBC-4D, CBC, CD ESR, CBC+ESR, CD+ESR	BC-760[B]
	CT-Low WBC/AL-Low WBC	CD/WBC-3X	BC-760[B]
		CD/WBC-3X, CDR/ WBC-3X	BC-760[R] BC-780[R]
	CT-Low PLT/AL-Low PLT	CR/PLT-5X, CDR/ PLT-5X	BC-760[R] BC-780[R]
CT-PD		CBC, CD, Ret, CR, CDR	BC-760[R] BC-780[R]
		CBC, CD	BC-760[B]
CT-BF		CD	Universal

Blood Sample Requirements

Sample Volume Required for Each Analysis

Table 2–5 Sample Volume Required for Each Analysis

Sample mode	Analysis mode	Sample Volume required for each analysis (μL)	Applicable model
WB	CBC-4D, CD	25±2	BC-760[B]/BC-760[R]/BC-780[R]
	CDR	33	BC-760[R] BC-780[R]
	ESR	140	BC-760[B]/BC-760[R]/BC-780[R]

Table 2–5 Sample Volume Required for Each Analysis(continued)

Sample mode	Analysis mode	Sample Volume required for each analysis (μL)	Applicable model
	CD+ESR	160	BC-760[B]/BC-760[R]/BC-780[R]
Low WBC	CD/WBC-3X	37	BC-760[B]/BC-760[R]/BC-780[R]
	CDR/WBC-3X	42	BC-760[R] BC-780[R]
Low PLT	CR/PLT-5X, CDR/PLT-5X	33	BC-760[R] BC-780[R]
PD		20	BC-760[B]/BC-760[R]/BC-780[R]
BF		85	BC-760[B]/BC-760[R]/BC-780[R]

Preparing Sample to be Analyzed

Preapring Whole Blood Samples

To obtain accurate analysis results, make sure the sample volume meets the following requirements:

Table 2–6 Preparing Whole Blood Sample

Load & Sample mode	Sample setting positions	Cap open?
CT-WB	Common test tube position/ micro-test tube position	Either
AL-WB	Tube rack	No

1. Collect venous blood samples using vacuum collection tubes, or collect peripheral blood samples using centrifuge tubes.

NOTICE

Use tubes compatible with the adapter; for details, see the supported tubes, tube racks, and adapters. Consult your local sales representative or the Mindray after-sales service department for information on unsupported tubes.

2. Rapidly mix the blood sample in the tube with the EDTA K2/EDTA K3 anticoagulant.

CAUTION

- **Use disposable supplies specified by the manufacturer, such as tubes for blood collection. Do not reuse disposable supplies as this may lead to inaccurate results.**
- **To obtain accurate analysis results, make sure the sample volume under various analysis modes meets the following requirements; otherwise, inaccurate results may occur.**
- **Prolonged uncapped time may cause sample volatilization and inaccurate results.**
- **Samples stored in refrigerated condition (2°C to 8°C) should be placed at room temperature for at least 15 minutes before analysis, as failure to do so may lead to inaccurate results.**
- **Samples that have been placed for a period of time should be re-mixed before analysis, as failure to do so may result in inaccurate measurement results.**
- **Misleading results may occur if the body fluid sample has clots or floccules. Follow your laboratory protocol to deal with such samples.**
- **If samples for ESR analysis are stored in refrigerated condition (2°C to 8°C), perform the ESR analysis within 24 hours of collection.**
- **For the samples to be tested for WBC differential or PLT count, store them at the room temperature and run them within 8 hours after collection, as failure to do so may result in inaccurate measurement results.**
- **The samples not to be tested for PLT, MCV or WBC differential can be stored in a refrigerator at the temperature of 2℃-8℃ for 24 hours. Samples stored in refrigerated conditions should be placed at room temperature for at least 30 minutes before analysis, as failure to do so may lead to inaccurate results.**

NOTICE

- **Capillary blood sample can not be used for ESR-related mode analysis.**
- **Gentle force can be applied when collecting peripheral blood. However, excessive force may introduce body fluids into the blood, reducing the reliability of the test results.**
- **After collecting the peripheral blood, please wait at least 5 minutes before performing the analysis. It is recommended to complete the testing within 1 hour.**

Preparing Predilute Samples

Prepare the pre-diluted samples at the ratio of 20:100 (venous blood/peripheral blood: diluent).

To obtain accurate analytical results, please ensure that the volume of the pre-diluted samples meets the following requirements:

Table 2–7 Volume of Pre-diluted Sample Prepared

Load & Sample mode	Sample setting positions	Cap open?
CT-PD	Micro-test tube position	Yes

1. Collect venous blood or peripheral blood using centrifuge tubes.

NOTICE

Gentle force can be applied when collecting peripheral blood. However, excessive force may introduce body fluids into the blood, reducing the reliability of the test results.

2. Tap  **button** on the **Sample Analysis** screen.
   √ Expand the suspension analysis window.
3. Switch to **Closed-Sampling** and select **PD**.
   √ The analyzer switches to PD mode.
   √ Under the PD mode, the software status bar lights in orange.
4. Click the **Add Diluent** button in the Suspension Analysis window.
5. Make sure to turn the tube holder position switch to the general tube position side. After installing the appropriate adapter, place the centrifuge tube with sample into the micro-test tube position of sample holder.
6. Click **OK**.
   √ The analyzer starts to add diluent.

NOTICE

- **Press the OK button to dispense the diluent multiple times, depending on your specific needs. The number in the Cancel button indicates how many times diluent has been dispensed.** 

In this example, the diluent has been added twice.
- **The analyzer adds 100 μL of diluent each time.**

7. After adding the liquid, tap the **Cancel** button.

CAUTION

- **After mixing the peripheral blood with the diluent, allow it to stand at least for 3 minutes, and mix the sample again before running the test to ensure accurate measurement results.**
- **Ensure a thorough reaction between the blood sample and the diluent. Make sure to run the samples within 30 minutes after the dilution; otherwise the results may be unreliable.**
- **Samples that have been placed for a period of time should be re-mixed before analysis, as failure to do so may result in inaccurate measurement results.**
- **Prolonged uncapped time may cause sample volatilization and inaccurate results.**
- **If there is much prediluted sample adhering to the tube cap or the tube wall, the analyzer may not be able to aspirate sufficient volume of the sample.**
- **When preparing pre-diluted samples, slightly tap the bottom of the test tube with a finger. After mixing, there should be not less than 100 uL of pre-diluted sample settled at the tube bottom.**

NOTICE

- **You can also add diluent by pipette to prepare the pre-diluted sample.**
- **Be sure to evaluate predilute stability based on your laboratory's sample population and sample collection techniques or methods.**

Preparing Body Fluid Samples

The types of body fluid samples currently supported by the analyzer include: cerebrospinal fluid, pleural fluid, ascites, and synovial fluid.

To attain accurate analysis results, make sure the sample volume meets the following requirements:

Table 2–8 Volume of Body Fluid Sample Prepared

Load & Sample mode	Sample setting positions	Cap open?
CT-BF	Common test tube position/ micro-test tube position	Yes

1. Collect blood using vacuum collection tubes and collect body fluid using centrifuge tubes.

CAUTION

- **To ensure sample stability, process serous cavity fluid (pleural fluid and ascitic fluid) samples and synovial fluid samples with EDTA anticoagulant. Do not process cerebrospinal fluid samples with anticoagulant.**
- **Use disposable supplies specified by the manufacturer, such as tubes for blood collection. Do not reuse disposable supplies as this may lead to inaccurate results.**
- **To obtain accurate analysis results, make sure the sample volume under various analysis modes meets the following requirements; otherwise, inaccurate results may occur.**
- **Samples stored in refrigerated condition (2°C to 8°C) should be placed at room temperature for at least 15 minutes before analysis, as failure to do so may lead to inaccurate results.**
- **Prolonged uncapped time may cause sample volatilization and inaccurate results.**
- **To obtain accurate analysis results, test body fluid samples within 1 hour as soon as possible after collection.**
- **Misleading results may occur if the body fluid sample has clots or floccules. Follow your laboratory protocol to deal with such samples.**

2. Mix the sample according to your laboratory's protocol.

NOTICE

Slightly mix the body fluid samples.

Placing Barcode Labels

To ensure good readability of the barcode, you must place the label right on the region. When using the Autoloader function, you must paste the barcode right on the region as shown in Figure **Figure2–18**, and place the barcode correctly as shown in Figure **Figure2–19**.

Figure2–18 Where to paste the barcode label

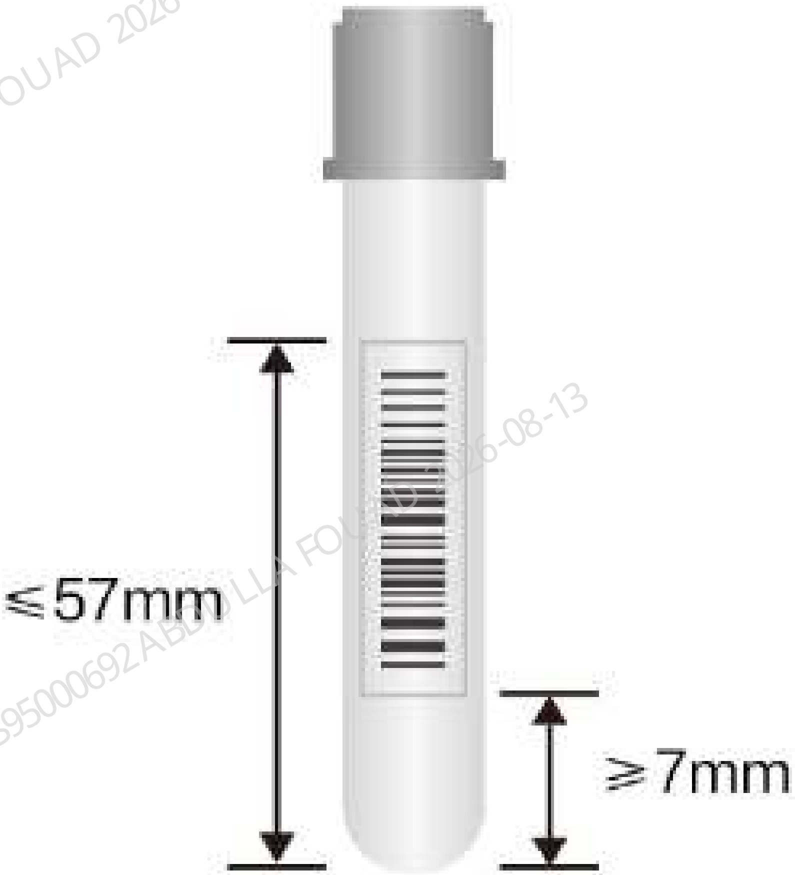


Figure2–19 How to paste the barcode label

Test Parameters

The analyzer is used for quantitative determination of the following report parameters and Research Use Only (RUO) parameters of blood sample tests and body fluid sample tests, and provides histograms and scattergrams.

Blood Sample Test Parameters, Histograms, and Scattergrams

Table 2–9 Blood sample test report parameters

Group	Name	Abbreviation	Applicable Model
WBC group	White blood cell count	WBC	General
	Basophil count	Bas#	General
	Basophil percentage	Bas%	General
	Neutrophil count	Neu#	General
	Neutrophil percentage	Neu%	General
	Eosinophil count	Eos#	General

Table 2–9 Blood sample test report parameters(continued)

Group	Name	Abbreviation	Applicable Model
	Eosinophil percentage	Eos%	General
	Lymphocyte count	Lym#	General
	Lymphocyte percentage	Lym%	General
	Monocyte count	Mon#	General
	Monocyte percentage	Mon%	General
	Immature granulocyte count	IMG#	General
	Immature granulocyte percentage	IMG%	General
RET group	*Reticulocyte percentage	RET%	BC-760[R] BC-780[R]
	*Reticulocyte count	RET#	
	*Reticulocyte hemoglobin expression	RHE	
	*Immature reticulocyte fraction	IRF	
	*Low fluorescent ratio	LFR	
	*Middle fluorescent ratio	MFR	
	*High fluorescent ratio	HFR	
RBC group	Red blood cell count	RBC	General
	Hemoglobin concentration	HGB	General
	Mean corpuscular volume	MCV	General
	Mean corpuscular hemoglobin	MCH	General
	Mean corpuscular hemoglobin concentration	MCHC	General
	Red blood cell distribution width coefficient of variation	RDW-CV	General

Table 2–9 Blood sample test report parameters(continued)

Group	Name	Abbreviation	Applicable Model
	Red blood cell distribution width standard deviation	RDW-SD	General
	Hematocrit	HCT	General
	Nucleated red blood cell count	NRBC#	General
	Nucleated red blood cell percentage	NRBC%	General
Platelet group	Platelet count	PLT	General
	Mean platelet volume	MPV	General
	Platelet distribution width	PDW	General
	Plateletcrit	PCT	General
	Platelet-large cell ratio	P-LCR	General
	Platelet-large cell count	P-LCC	General
	*Immature platelet fraction	IPF	BC-760[R] BC-780[R] this parameter is optional, for more information, consult Mindray Customer Service Department For BC-760[B], this parameter is optional, for more information, consult Mindray Customer Service Department
	Platelet count-impedance	PLT-I	General (find the parameter result on the "Other Para." - "Analysis Para."
	*Platelet count hybrid	PLT-H	BC-760[R] BC-780[R] (find the parameter result on the "Other

Table 2–9 Blood sample test report parameters(continued)

Group	Name	Abbreviation	Applicable Model
			Para." - "Analysis Para.") For BC-760[B], this parameter is optional. For more information, consult Mindray Customer Service Department
	*Optical platelet count	PLT-O	BC-760[R] BC-780[R] (find the parameter result on the "Other Para." - "Analysis Para.")
ESR	Erythrocyte sedimentation rate	ESR	General

NOTICE

The parameters marked with * are parameters for specified models.

Table 2–10 Blood sample test RUO parameters

Name	Abbreviation	Applicable Model
High fluorescent cell count	HFC#	General
High fluorescent cell percentage	HFC%	General
White blood cell count -DIFF	WBC-D	General
Total nucleated cell count-DIFF	TNC-D	General
Immature eosinophil percentage	IME%	General
Immature eosinophil count	IME#	General
Neutrophil-to-lymphocyte ratio	NLR	General
Infected red blood cell count	InR#	General
Infected red blood cell permillage	InR‰	General
NEU# Minus IMG#	Neu#&	General

Table 2–10 Blood sample test RUO parameters(continued)

Name	Abbreviation	Applicable Model
NEU% Minus IMG%	Neu%&	General
LYM# Minus HFC#	Lym#&	General
LYM% Minus HFC%	Lym%&	General
Microcyte count	Micro#	General
Microcyte percentage	Micro%	General
Macrocyte count	Macro#	General
Macrocyte percentage	Macro%	General
*Optical red blood cell count	RBC-O	BC-760[R] BC-780[R]
*Optical white blood cell count	WBC-O	
*RBC fragment count	FRC#	
*RBC fragment percentage	FRC%	
*Reticulocyte production index	RPI	
*Mean reticulocyte volume	MRV	
*RET scattergram, mean reticulocyte distribution-forward scatter intensity	RET-Y	
*RET scattergram, mean reticulocyte distribution-side fluorescent intensity	RET-X	
*RET scattergram, mean immature reticulocyte fraction distribution-forward scatter intensity	IRF-Y	
*RET scattergram, mean immature reticulocyte fraction distribution-side fluorescent intensity	IRF-X	
*RET scattergram, mean red blood cell distribution-forward scatter intensity	RET-RBC-Y	
*RET scattergram, mean red blood cell distribution-side fluorescent intensity	RET-RBC-X	

Table 2–10 Blood sample test RUO parameters(continued)

Name	Abbreviation	Applicable Model
***High fluorescent immature platelet fraction**	H-IPF	
***Immature platelet count**	IPF#	
Platelet count- impedance	PLT-I	General
Platelet-to-lymphocyte ratio	PLR	General
Platelet distribution width standard deviation	PDW-SD	General
Dimorphic population, smaller distribution RBC count	SRBC	General
Dimorphic population, larger distribution RBC count	LRBC	General
Dimorphic population, smaller distribution mean corpuscular volume	SMCV	General
Dimorphic population, larger distribution mean corpuscular volume	LMCV	General
DIFF scattergram, mean neutrophil distribution-side scatter intensity	Neu-X	General
DIFF scattergram, mean neutrophil distribution-side fluorescent light intensity	Neu-Y	General
DIFF scattergram, mean neutrophil distribution-forward scatter intensity	Neu-Z	General
DIFF scattergram, mean lymophocyte distribution-side scatter intensity	Lym-X	General
DIFF scattergram, mean lymophocyte distribution-side fluorescent intensity	Lym-Y	General
DIFF scattergram, mean lymophocyte distribution-forward scatter intensity	Lym-Z	General

Table 2–10 Blood sample test RUO parameters(continued)

Name	Abbreviation	Applicable Model
DIFF scattergram, mean monocyte distribution-side scatter intensity	Mon-X	General
DIFF scattergram, mean monocyte distribution-side fluorescent light intensity	Mon-Y	General
DIFF scattergram, mean monocyte distribution-forward scatter intensity	Mon-Z	General
DIFF scattergram, neutrophil side scatter distribution width	Neu-XW	General
DIFF scattergram, neutrophil side fluorescent light distribution width	Neu-YW	General
DIFF scattergram, neutrophil forward scatter distribution width	Neu-ZW	General
DIFF scattergram, lymophocyte side scatter distribution width	Lym-XW	General
DIFF scattergram, lymophocyte side fluorescent light distribution width	Lym-YW	General
DIFF scattergram, lymophocyte forward scatter distribution width	Lym-ZW	General
DIFF scattergram, monocyte side scatter distribution width	Mon-XW	General
DIFF scattergram, monocyte side fluorescent light distribution width	Mon-YW	General
DIFF scattergram, monocyte forward scatter distribution width	Mon-ZW	General
*Immature platelet fraction-DIFF	IPF-D	These parameters are available on BC-760[R]/BC-780[R] (when the user-
*Reticulocyte percentage-DIFF	RET%-D	

Table 2–10 Blood sample test RUO parameters(continued)

Name	Abbreviation	Applicable Model
*Reticulocyte count- DIFF	RET#-D	defined test panel includes DIFF tests, but not Ret tests.) For BC-760[B], this parameter is optional. For more information, consult Mindray Customer Service Department
*Immature reticulocyte fraction- DIFF	IRF-D	
*Low fluorescent ratio- DIFF	LFR-D	
*Middle fluorescent ratio- DIFF	MFR-D	
*High fluorescent ratio- DIFF	HFR-D	
Corrected erythrocyte sedimentation rate	ESR-Corr.	General
Surface area	SA	General
Amplitude	AMP	General
Aggregation index	AI	General
Minimum	MIN	General
Aggregation half time	T1/2	General

NOTICE

- **RUO parameters are for research purpose only. They cannot be used for diagnosis purpose.**
- **The parameters marked with * are parameters for specified models.**
- **To print the RUO parameter results, tap Setup>System Setup>Print Setup>Printing Content, and check Print RUO Parameters.**

Table 2–11 Blood sample test histograms

Name	Abbreviation	Applicable Model
Red blood cell histogram	RBC histogram	General
Platelet histogram	PLT histogram	General
* Platelet hybrid histogram	PLT-H histogram	BC-760[R] BC-780[R]For BC-760[B], this histogram is optional. For more information, consult Mindray Customer Service Department.
White blood cell –FSC histogram	WBC-FSC histogram	General

NOTICE

The parameters marked with * are parameters for specified models.

Table 2–12 Blood sample test scattergrams

Name	Abbreviation	3D Scattergram	Applicable Model
Differential(FS/FL) scattergram	DIFF(FS/FL) scattergram	Available	General
Differential(FS/SS) scattergram	DIFF(FS/SS) scattergram		General
Differential(FL/SS) scattergram	DIFF(FL/SS) scattergram		General
*Reticulocyte scattergram	RET scattergram	Available	BC-760[R] BC-780[R]
*Optical platelet scattergram	PLT-O scattergram	Unavailable	
*Reticulocyte - extension scattergram	RET-EXT scattergram	Unavailable	
*Platelet hybrid scattergram	PLT-H scattergram	Available	BC-760[R] BC-780[R] For BC-760[B], this scattergram is optional. For more information, consult Mindray Customer Service Department.
Differential(FS/FL) scattergram	DIFF(FS/FL) scattergram	Available	General
Differential(FS/SS) scattergram	DIFF(FS/SS) scattergram		General
Differential(FL/SS) scattergram	DIFF(FL/SS) scattergram		General
*Reticulocyte scattergram	RET scattergram	Available	BC-760[R] BC-780[R]
*Optical platelet scattergram	PLT-O scattergram	Unavailable	

Table 2–12 Blood sample test scattergrams(continued)

Name	Abbreviation	3D Scattergram	Applicable Model
*Reticulocyte - extension scattergram	RET-EXT scattergram	Unavailable	
*Platelet hybrid scattergram	PLT-H scattergram	Available	BC-760[R] BC-780[R] For BC-760[B], this scattergram is optional. For more information, consult Mindray Customer Service Department.

NOTICE

The parameters marked with * are parameters for specified models.

Body Fluid Sample Test Parameters, Histograms, and Scattergrams

Table 2–13 Body fluid sample test report parameters

Series	English Name	Abbreviation	Applicable Model
WBC group	White blood cell count-body fluid	WBC-BF	General
	Total nucleated cell counts-body fluid	TC-BF#	General
	Mononuclear cell count	MN#	General
	Mononuclear cell percentage	MN%	General
	Polymorphonuclear cell count	PMN#	General
	Polymorphonuclear cell percentage	PMN%	General
RBC group	Red blood cell count-body fluid	RBC-BF	General

Table 2–14 Body fluid sample test RUO parameters

Name	Abbreviation	Applicable Model
Eosinophil count- body fluid	Eos-BF#	General
Eosinophil percentage- body fluid	Eos-BF%	General
Neutrophil count- body fluid	Neu-BF#	General
Neutrophil percentage- body fluid	Neu-BF%	General
Lymphocyte count- body fluid	Lym-BF#	General
Lymphocyte percentage- body fluid	Lym-BF%	General
Monocyte count- body fluid	Mon-BF#	General
Monocyte percentage- body fluid	Mon-BF%	General
High fluorescent cell count - body fluid	HFC-BF#	General
High fluorescent cell percentage- body fluid	HFC-BF%	General
Red blood cell count-body fluid	RBC-BF(R)	General

NOTICE

- **RUO parameters are for research purpose only. They cannot be used for diagnosis purpose.**
- **To print the RUO parameter results, tap Setup>System Setup>Print Setup>Printing Content, and check Print RUO Parameters.**

Table 2–15 Body fluid sample test histograms

Name	Abbreviation	Applicable Model
Red blood cell histogram	RBC histogram	General
White blood cell–FSC histogram	WBC-FSC histogram	General

Table 2–16 Body fluid sample test scattergrams

Name	Abbreviation	3D Scattergram	Applicable Model
Differential scattergram	DIFF scattergram	Available	General
Differential-extension scattergram	DIFF-EXT scattergram	Unavailable	General

2.1.5 Performance Specifications

Background/Blank Count Requirements

Table 2–17 Background/blank count requirements for blood samples

Parameters	Acceptable Range
WBC	$\leq 0.10 \times 10^{9}$ / L
RBC	$\leq 0.02 \times 10^{12}$/ L
RBC-O	$\leq 0.02 \times 10^{12}$/ L
HGB	≤ 1 g/L
PLT	$\leq 3 \times 10^{9}$/ L

Table 2–18 Background/blank count requirements for body fluid samples

Parameters	Acceptable Range
WBC-BF/ TC-BF#	$\leq 0.001 \times 10^{9}$ / L
RBC-BF	$\leq 0.003 \times 10^{12}$/ L

Repeatability

Table 2–19 Repeatability requirements for blood samples

Parameters	Range	Whole Blood (CV/ Absolute Deviation d*/SD)	Predilute (CV/ Absolute Deviation d*)
WBC	$(3.50 \sim 4.50) \times 10^{9}$/L	≤ 3.0%	≤ 4.0%
	$\geq 4.51 \times 10^{9}$/L	≤ 2.5%	≤ 3.5%
RBC	$\geq 3.50 \times 10^{12}$/L	≤ 1.5%	≤ 2.0%
HGB	(110 ~ 180)g/L	≤ 1.0%	≤ 2.0%
MCV	(80.0 ~ 100.0) fL	≤ 1.0%	≤ 3.0%
HCT	(30.0 ~ 50.0)%	≤ 1.5%	≤ 3.0%
MCH	/	≤ 1.5%	/

Table 2–19 Repeatability requirements for blood samples(continued)

Parameters	Range	Whole Blood (CV/ Absolute Deviation d*/SD)	Predilute (CV/ Absolute Deviation d*)
MCHC	/	≤ 1.5%	/
RDW-SD	/	≤ 2.0%	/
RDW-CV	/	≤ 2.0%	/
PLT	≥100×10^9/L	≤ 4.0%	≤ 8.0%
		≤ 2.5% Applicable to the CDR/PLT-5X and CR/ PLT-5X test panels	/
	(20~100)×10^9/L	≤ 5.0% Applicable to the CDR/PLT-5X and CR/ PLT-5X test panels	/
	≤20×10^9/L	≤ 1.5(SD) Applicable to the CDR/PLT-5X and CR/ PLT-5X test panels	/
PDW	PLT≥100×10^9/L	≤ 10.0%	/
MPV	PLT≥100×10^9/L	≤ 3.0%	/
P-LCR	PLT≥100×10^9/L	≤ 15.0%	/
P-LCC	PLT≥100×10^9/L	≤ 15.0%	/
PCT	PLT≥100×10^9/L	≤ 5.0%	/
Neu%	Neu%≥30.0% WBC≥ 4.00 × 10^9/L	≤6.0%	≤12.0%
Lym%	Lym %≥15.0% WBC≥ 4.00 × 10^9/L	≤6.0%	≤12.0%
Mon%	Mon %≥5.0% WBC≥ 4.00 × 10^9/L	≤16.0%	≤32.0%
Eos%	WBC≥ 4.00 × 10^9/L	≤20.0% or ±1.5%(d)	≤40.0% or ±3.0%(d)
Bas%	WBC≥ 4.00 × 10^9/L	≤30.0% or ±1.0%(d)	≤60.0% or ±2.0%(d)
Neu#	≥1.20×10^9/L	≤6.0%	≤12.0%
Lym#	≥0.60×10^9/L	≤6.0%	≤12.0%
Mon#	≥0.20×10^9/L	≤16.0%	≤32.0%
Eos#	WBC≥ 4.00 × 10^9/L	≤20.0% or ±0.12 × 10^9/L(d)	≤40.0% or ±0.24 × 10^9/L(d)

Table 2–19 Repeatability requirements for blood samples(continued)

Parameters	Range	Whole Blood (CV/ Absolute Deviation d*/SD)	Predilute (CV/ Absolute Deviation d*)
Bas#	WBC≥4.00 × 10^9/L	≤30.0% or ±0.06×10^9/L(d)	≤60.0% or ±0.12×10^9/L(d)
Gran%	Gran%≥30.0% WBC≥ 4.00 × 10^9/L	≤6.0%	/
Gran#	≥1.20×10^9/L	≤6.0%	/
IMG%	WBC≥4.00 × 10^9/L IMG%≥2.0%	≤25.0% or ±1.5%(d)	/
IMG#	≥0.10×10^9/L	≤25.0% or ±0.12 × 109/L(d)	/
RET%	RBC≥ 3.00 × 10^{12}/L RET%: 1.00% ~ 4.00%	≤ 15%	≤ 30%
RET%	RBC≥ 3.00 × 10^{12}/L RET%: 1.00% ~ 4.00%	≤ 15%	≤ 30%
RHE	RET#≥ 0.0200 × 10^{12}/L	≤ 5%	/
LFR	RBC≥ 3.00 × 10^{12}/L RET%: 1.00% ~ 4.00% LFR≥20%	≤ 30%	/
MFR	RBC≥ 3.00 × 10^{12}/L RET%: 1.00% ~ 4.00% MFR≥20%	≤ 50%	/
HFR	RBC≥ 3.00 × 10^{12}/L RET%: 1.00% ~ 4.00%	≤ 100% or ±2.0%(d)	/
IRF	RBC≥ 3.00 × 10^{12}/L RET%: 1.00% ~ 4.00% IRF≥20%	≤ 30%	/
IPF	PLT ≥ 50 × 10^9/L IPF≥3.0%	≤ 25%	/
ESR	0 ~ 20mm/h	≤1.8(SD)	/
	> 20mm/h	≤ 9%	/

NOTE

1. *Absolute deviation d = MAX|Measured value - measured mean|
2. **Range = Maximum measurement value – Minimum measurement value

Table 2–20 Repeatability Requirements for Body Fluid Samples

Parameters	Measurement Range	CV or Range
WBC-BF/TC-BF#	(0.015-0.100) × 10^9/L	≤30%
RBC-BF	(0.003-0.050) × 10^{12}/L	≤40% or range ≤7000/μL

Carryover

Table 2–21 Carryover requirements for blood samples

Parameters	Carryover
WBC	≤1.0%
RBC	≤1.0%
HGB	≤1.0%
HCT	≤1.0%
PLT	≤1.0%
ESR	≤1.0%

Table 2–22 Carryover requirements for body fluid samples

Parameters	Carryover
WBC-BF / TC-BF#	≤0.3% or ≤0.001×10^9/L
RBC-BF	≤0.3% or ≤0.003×10^{12}/L

Linearity Ranges

Table 2–23 Linearity requirements for blood samples

Parameters	Linearity Range	Acceptable Deviation Range (WB)	Acceptable Deviation Range (PD)	Correlation Coefficient
WBC	(0 ~ 100.00)×10^9/L	±0.20×10^9/L±2%	±0.50×10^9/L±5%	≥ 0.990
	(100.01 ~ 350.00)×10^9/L	±6%	±6%	≥ 0.990

Table 2–23 Linearity requirements for blood samples(continued)

Parameters	Linearity Range	Acceptable Deviation Range (WB)	Acceptable Deviation Range (PD)	Correlation Coefficient
	(350.01 ~ 500.00)×10^9/L	±11%	±11%	≥ 0.990
RBC	(0 ~ 8.60)×10^{12}/L	±0.03×10^{12}/L±2%	±0.05×10^{12}/L ±5%	≥ 0.990
HGB	(0 ~ 260) g/L	±2g/L±2%	±2g/L or ±3%	≥ 0.990
HCT	(0.0 ~ 75.0)%	±1.0% (HCT) or ±2% (percentage error)	±2.0% (HCT) or ±4% (percentage error)	/
PLT	(0 ~ 1000)×10^9/L	±10×10^9/L±5%	±10×10^9/L±10%	≥ 0.990
	(1001 ~ 5000)×10^9/L	±6%	±10%	≥ 0.990
RET%	(0.00 ~ 30.00)%	±0.30% (RET% value) or ±20% (percentage error)	/	/
RET%	(0.0000~0.8000)×10^{12}/L	±0.0150×10^{12}/L±20%	/	/

Table 2–24 Linearity requirements for body fluid samples

Parameters	Linearity Range	Deviation range
WBC-BF/TC-BF#	(0~0.050)×10^9/L	±0.010×10^9/L
	(0.051~1.000)×10^9/L	±20%
	(1.001~10.000)×10^9/L	±20%
RBC-BF	(0.000~0.100)×10^{12}/L	±0.010×10^{12}/L or ±5%
	(0.101~5.000)×10^{12}/L	±0.030×10^{12}/L or ±2%

2.1.6 Display Ranges for Major Parameters

Table 2–25 Display Range of Blood Test (CBC-4D Analysis Mode Excluded) Report Parameters

Parameters	Display range
WBC	0.00~999.99×10^9/L
Neu%/ Lym%/ Mon%/ Eos%/ Bas%/ IMG%	0~100.0%
Neu#	0.00~999.99×10^9/L
Lym#	0.00~999.99×10^9/L
Mon#	0.00~999.99×10^9/L
Eos#	0.00~999.99×10^9/L
Bas#	0.00~999.99×10^9/L
IMG#	0.00~999.99×10^9/L
RBC	0.00~99.99×10^{12}/L
HGB	0~999g/L
MCV	0.0~999.9 fL
MCH	0.0~999.9pg
MCHC	0~9999g/L
RDW-CV	0.0~100.0%
RDW-SD	0.0~999.9fL
HCT	0.0~100.0%
PLT	0~9999×10^9/L
MPV	0.0~99.9fL
PDW	0.0~99.9
PCT	0.000~9.999%
P-LCR	0.0~100.0%
P-LCC	0~9999×10^9/L
*IPF	0.0~100.0%
*RET%	0~100 %
*RET□	0.0000~9.9999×10^{12}/L
*IRF	0.0~100.0%
*LFR	0.0~100.0%
*MFR	0.0~100.0%
*HFR	0.0~100.0%
*RHE	0.0~999.9pg
ESR	0.00~999.00mm/L

NOTICE

- **Items with * are applicable to for BC-760[R] and BC-780[R] only.**

Table 2–26 Display Range of Blood Test (CBC-4D Analysis Mode) Report Parameters

Parameters	Display range
WBC	0.00~999.99×10^9/L
Gran#	0.00~999.99×109/L
Lym%/ Mon%/ Eos%/ Gran%	0~100.0%
Lym#	0.00~999.99×10^9/L
Mon#	0.00~999.99×10^9/L
Eos#	0.00~999.99×10^9/L
RBC	0.00~99.99×10^{12}/L
HGB	0~999g/L
MCV	0.0~999.9 fL
MCH	0.0~999.9pg
MCHC	0~9999g/L
RDW-CV	0.0~100.0%
RDW-SD	0.0~999.9fL
HCT	0.0~100.0%
PLT	0~9999×10^9/L
MPV	0.0~99.9fL
PDW	0.0~99.9
PCT	0.000~9.999%
P-LCR	0.0~100.0%
P-LCC	0~9999×10^9/L

Table 2–27 Display Range of Body Fluid Test Report Parameters

Parameters	Display range
WBC-BF/TC-BF#	0.000~999.999×10^9/L
RBC-BF	0.000 ~ 99.999×10^{12}/L
MN#	0.000~999.999×10^9/L
MN%	0~100 %
PMN#	0.000~999.999×10^9/L
PMN%	0.00~100.00%

2.1.7 Reagents, Controls and Calibrators

As the analyzer, reagents, controls and calibrators are components of a system, performance of the system depends on the combined integrity of all components. You must only use the Mindray-specified reagents, controls, and calibrators which are formulated specifically for the fluidic system of your analyzer in order to provide optimal system performance. Do not use the analyzer with reagents, controls, and calibrators from multiple suppliers. In such case, the analyzer may not meet the performance specified in this manual and may provide unreliable results.

All the reagents, controls, and calibrators mentioned in this manual refer to the reagents, controls, and calibrators specifically formulated for this analyzer. You must buy those reagents, controls, and calibrators from Mindray or Mindray-authorized distributors. When you need to buy reagents and consumables, please call Mindray Customer Service Department.

Reagent

All reagents used by this analyzer are specifically matched with Mindray equipment. Use for any other purpose is prohibited.

Correctly use ans store reagents according to their Instructions for Use.

NOTICE

- **For the reagent model, intended use, test principle, main components, storage conditions, expiration date, applicable models, and other information, please refer to the reagent IFU.**
- **For any questions related to reagents, controls, and calibrators, please consult your sales representatives.**

Table 2–28 Reagents

Applicable Channel	BC-760[B]	BC-760[R]/BC-780[R]
HGB Channel	LH Lyse	LH Lyse
DIFF Channel	LD Lyse	LD Lyse
	FD Dye	FD Dye
RET Channel	/	*DR Diluent
	/	*FR Dye
/	DS Diluent	DS Diluent
	Probe Cleanser	Probe Cleanser
ESR channel	ESR Solution Reagent	ESR Solution Reagent

Controls and Calibrators

The controls and calibrators are used to verify accurate operation and calibration of the analyzer.

The controls are suspension of simulated human blood, specifically manufactured to monitor and evaluate the analysis precision of the analyzer. The controls are prepared with three levels, namely low, normal and high. Daily use of all levels verifies the operation of the analyzer and ensures reliable results are obtained. The calibrators are commercially prepared whole-blood products used to calibrate some parameters (WBC, RBC, HGB, MCV and PLT etc.) of analyzer to build the metrological traceability of analysis results. For the use and storage of controls and calibrators, please refer to the Instruction for Use of each product.
All references related to controls in this manual refer to the "controls" and "calibrators" specifically formulated for this analyzer by Mindray. You must buy those controls and calibrators from Mindray or Mindray-authorized distributors.
The following models of controls are used with the analyzer:

Table 2–29 Controls for complete blood count tests

Name	Model	Level	Applicable Model	*QC Parameters
Hematology Control	BR60	High, Low, Normal	BC-760[B]/BC-760[R]/BC-780[R]	All blood sample test report parameters except ESR
Hematology Control	BC-6D	High, Low, Normal	BC-760[B]/BC-760[R]/BC-780[R]	All blood sample test report parameters except RET series
Hematology Control	BC-RET	High, Low, Normal	BC-760[R]/BC-780[R]	RET group report parameters

Table 2–30 Controls for body fluid tests

Name	Model	Level	Applicable Model	QC Parameters
Hematology Control	BC-BF	High, Normal, Low	BC-760[B]/BC-760[R]/BC-780[R]	Body fluid sample test report parameters

Table 2–31 Controls for ESR tests

Name	Model	Level	Applicable Model	QC Parameters
Hematology Control	BC-6D	Normal, Low	BC-760[B]/BC-760[R]/BC-780[R]	ESR

The following models of calibrators are used with the analyzer:

Table 2–32 Calibrator for complete blood count tests

Name	Model	Applicable Model
Hematology Calibrator	SC-CAL PLUS	BC-760[B]/BC-760[R]/BC-780[R]

NOTICE

For the specific test parameters and reference values of parameters, see the reference value sheets of controls and calibrators.

Table 2–33 Calibrator for ESR tests

Name	Model	Applicable Model
Hematology Calibrator	SC-CAL PLUS	BC-760[B]/BC-760[R]/BC-780[R]

2.2 Mechanical System

2.2.1 Introduction to Mechanical Structure

2.2.1.1 Introduction to Mechanical Structure

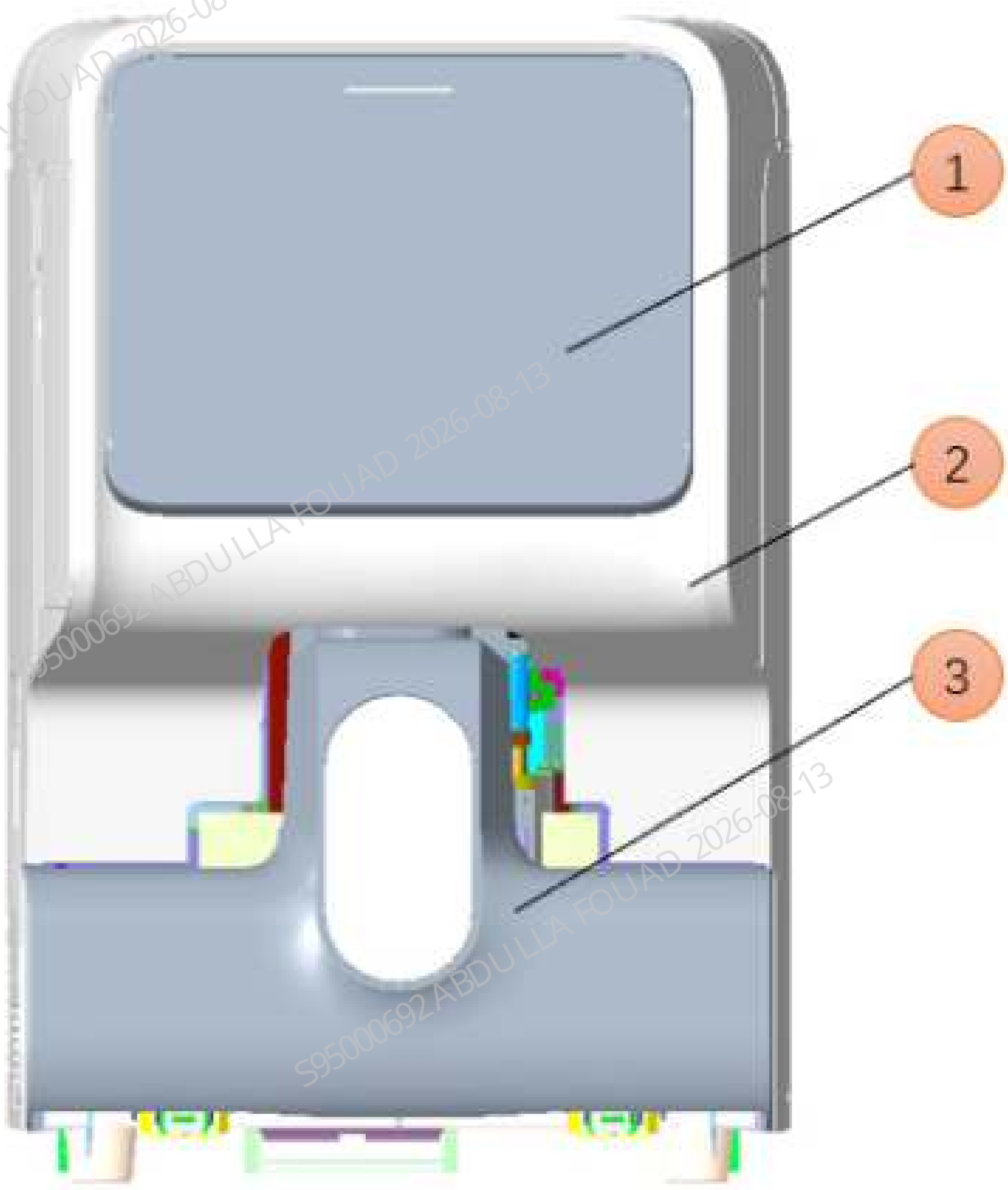
The analyzer mainly consists of the following parts: frame, cover, autoloader, sampling assembly, reaction bath, sheath fluid impedance, optical assembly, syringe, dosing pump, valve assembly, various baths, power supply, hardware board, wire and tube.

2.2.2 Appearance

2.2.2.1 Appearance

The front of the analyzer is shown as below.

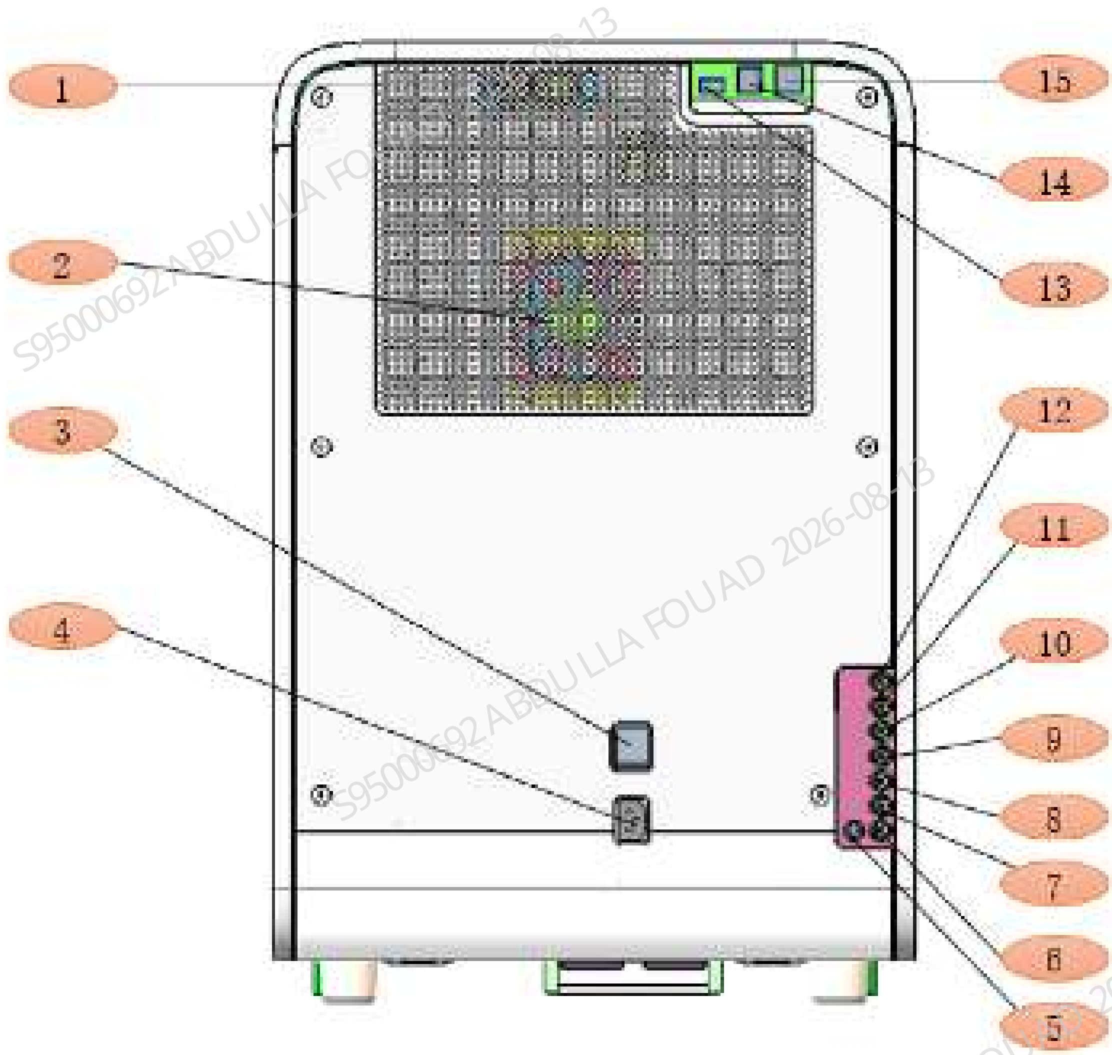
Figure2–20 Front View of the Analyzer



No.	Name	Function Description	No.	Name	Function Description
1	Display	Analyzer operation and man-machine interaction screen	3	Autoloader	Autoloading of tube rack and closed-sampling of tubes
2	Cover	Appearance			

The rear side of the analyzer is shown as below .

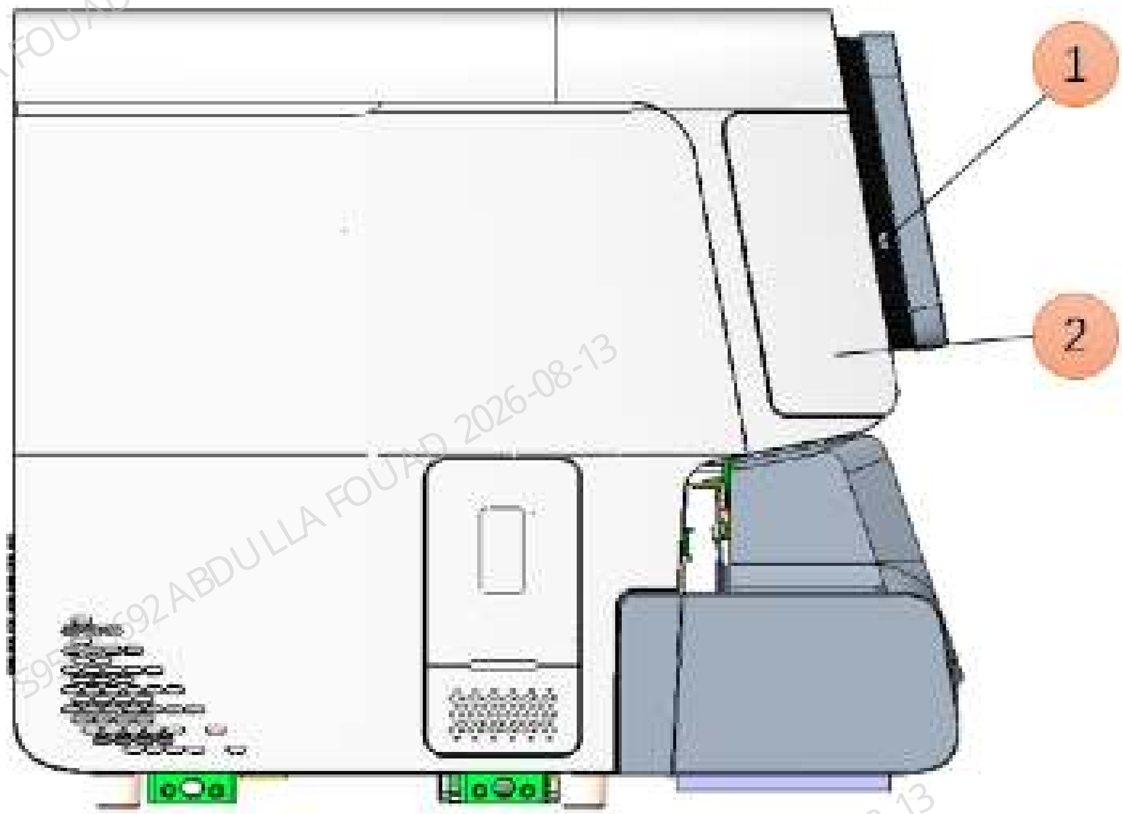
Figure2–21 Rear View of the Analyzer



No.	Name	Function Description	No.	Name	Function Description
1	Board fan	Cool the top board area	9	LH lyse connector	LH lyse inlet
2	Analyzer fan	Cool the analyzer	10	LD lyse connector	LD lyse inlet
3	Power switch	Power switch	11	DR diluent connector	DR diluent inlet
4	Power input connector	AC input	12	LS lyse connector	LS lyse inlet
5	BNC socket	Test the waste	13	USB port	Connect to peripherals (e.g. USB drive, mouse, keyboard, printer and scanner)
6	Waste connector	Waste discharge	14	USB port	Connect to peripherals (e.g. USB drive, mouse, keyboard, printer and scanner)
7	DS diluent connector	DS Diluent inlet	15	Network port	Connect to the network port on the PC
8	ESR cleaner connector	DS lyse inlet			

The left side of the analyzer is shown as below .

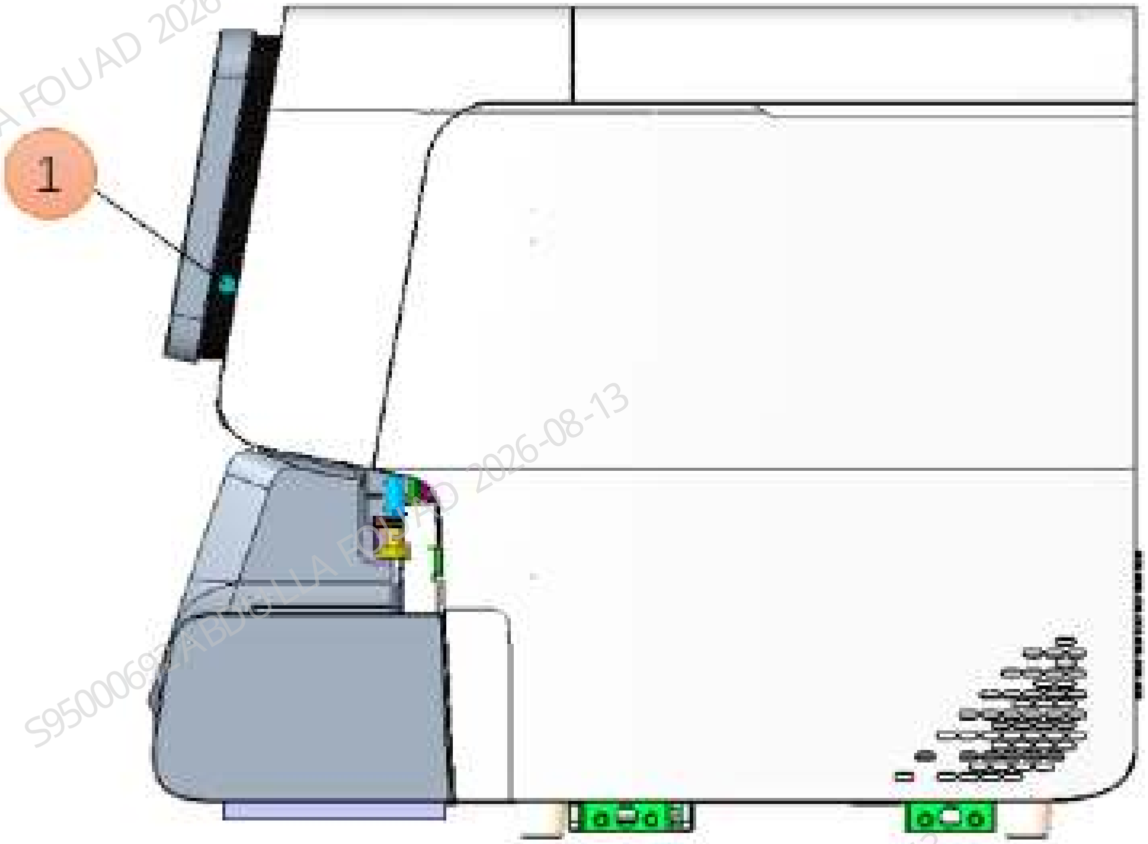
Figure2–22 Left Side of the Analyzer



No.	Name	Function Description	No.	Name	Function Description
1	USB port	Connect to peripherals (e.g. USB drive, mouse, keyboard, printer and scanner)	2	Dye replacement access	Access to replace dye bag

The right side of the analyzer is shown as below .

Figure2–23 Right Side of the Analyzer



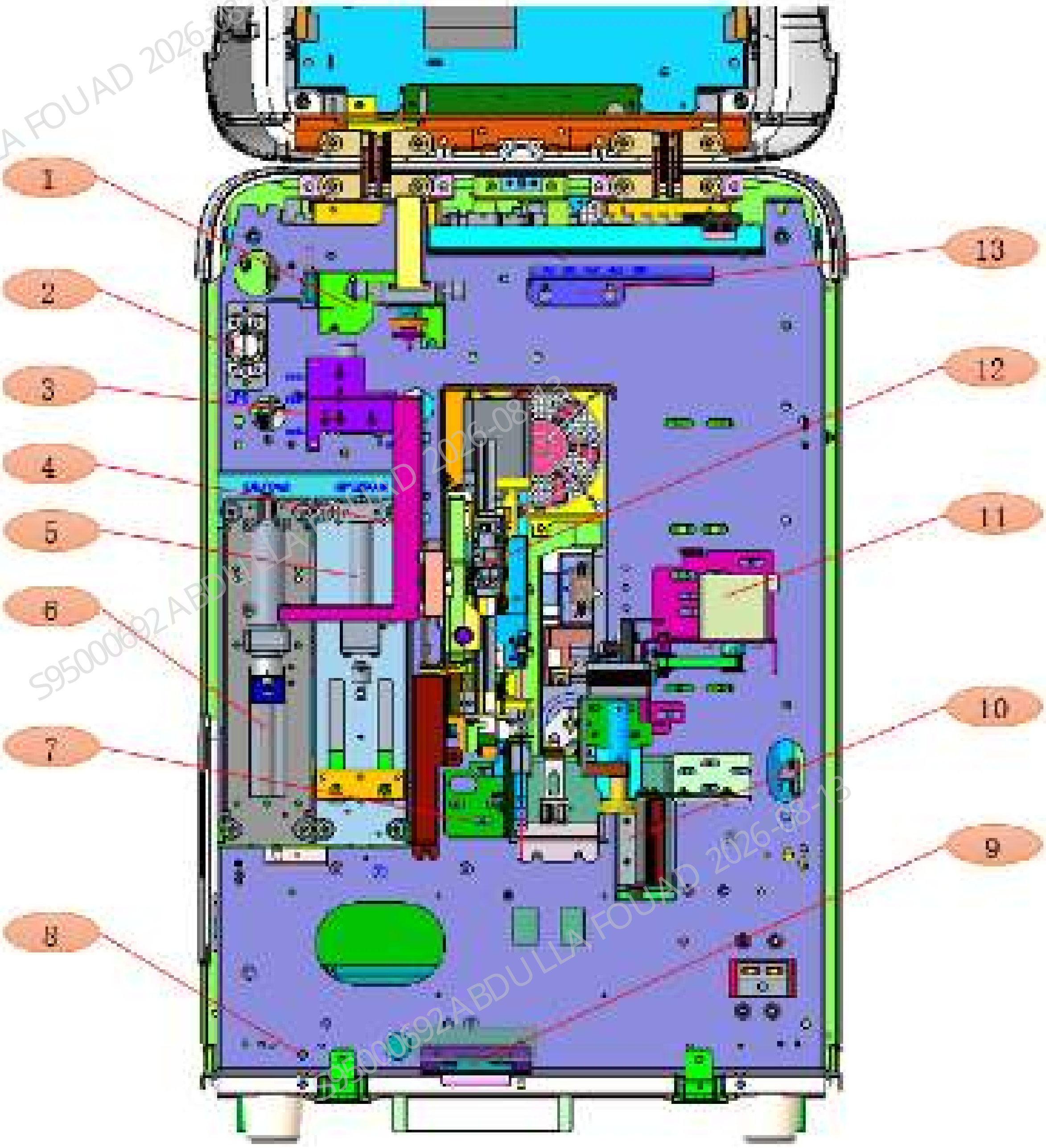
No.	Name	Function Description	No.	Name	Function Description
1	Startup/ shutdown keys	Provide the soft switch			

2.2.3 Structure

2.2.3.1 Structure

The components located in the front side of the analyzer after removal of autoloader are shown as below.

Figure2–24 Front View of the Analyzer (Front Cover Open)

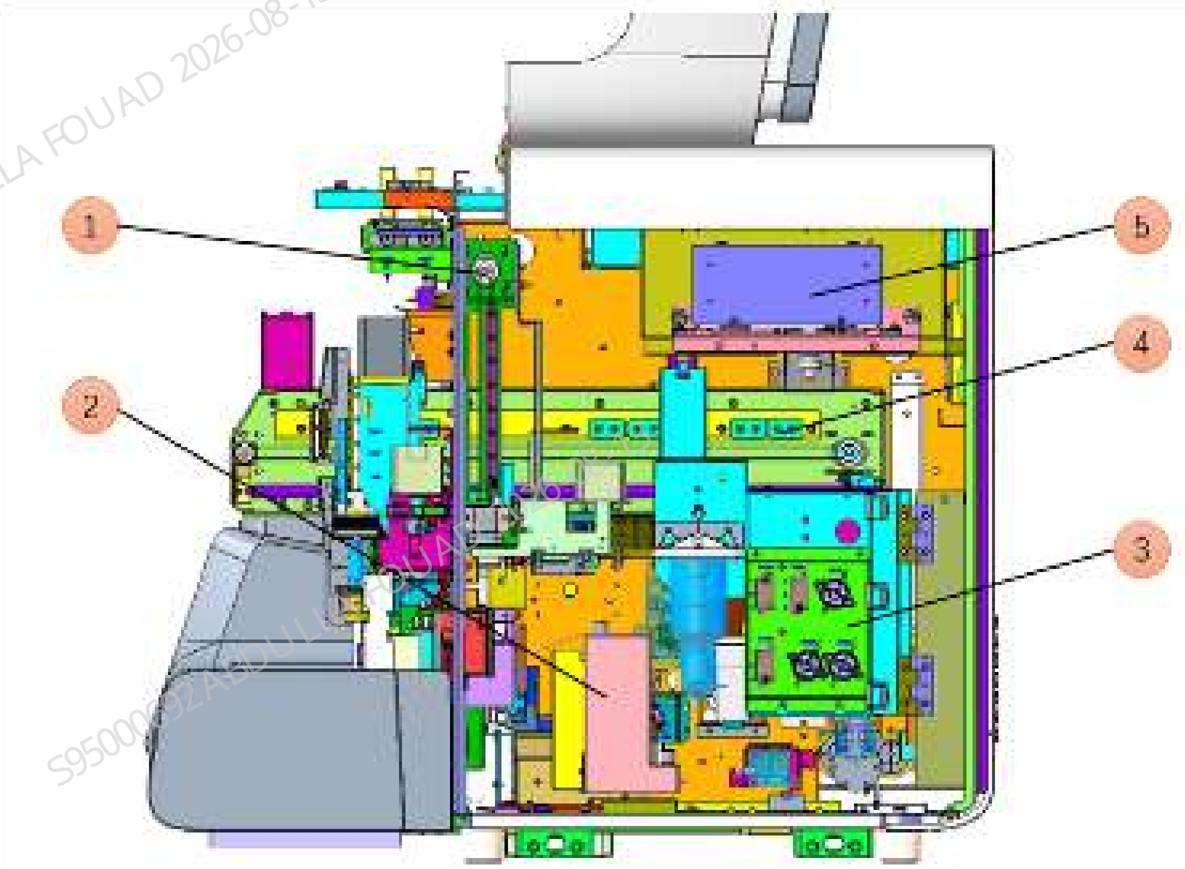


No.	Name	Function Description	No.	Name	Function Description
1	Dye pump assembly	Absorb the dye reagent and detect the reagent	8	Ambient tempera ture sensor	Detect the temperature of the surrounding environment where the analyzer is located
2	Fluid pressure detection assembly	Detect the fluid pressure at the outlet of the 250 μL syringe.	9	Trumpet Assembly	Provide alarm tones and beeps.
3	Sampling manifold assembly	Switch between sampling probe and 10 mL syringe assembly/250 μL syringe, and switch between 250 μL syringe and sheath fluid sample booster/optical sample booster.	10	Barcode scanner	Scan the tube barcode information
4	Dye bag assembly	Place and replace the dye bags	11	Rotary Compress ing Assembly	In the autoloading, it is used to clamp and rotate the test tube, so that the scanner can scan the barcode information on the test tube
5	250 μL syringe assembly (43F4J)	Quantitatively absorb and distribute the samples; serve as the power source for optical sample flow formation, RBC sample flow formation, ESR measurement and diluent function.	12	ESR Measure ment Assembly	Measure the erythrocyte sedimentation rate of whole blood samples.

No.	Name	Function Description	No.	Name	Function Description
6	10 mL lead screw driving syringe assembly	Provide diluent for sample dilution in HGB bath, pipeline cleaning, SCI bath filling and sheath liquid supply in flow chamber; serve as the power source for optical sample transfer, RBC sample transfer and quantitative absorption and distribution of probe solution	13	Air pressure detection board	It is used to detect the pressure of each air pipe in this analyzer
7	Hold Assembly	The stabilizing assembly is used to eliminate the random oscillation of the tube in the tube rack during pinching and make the tube oscillate in a fixed direction, which can prevent the sample probe from sticking on the tube cap, then the probability of bending the probe can be reduced.			

The components located in the right side in the analyzer are shown as below.

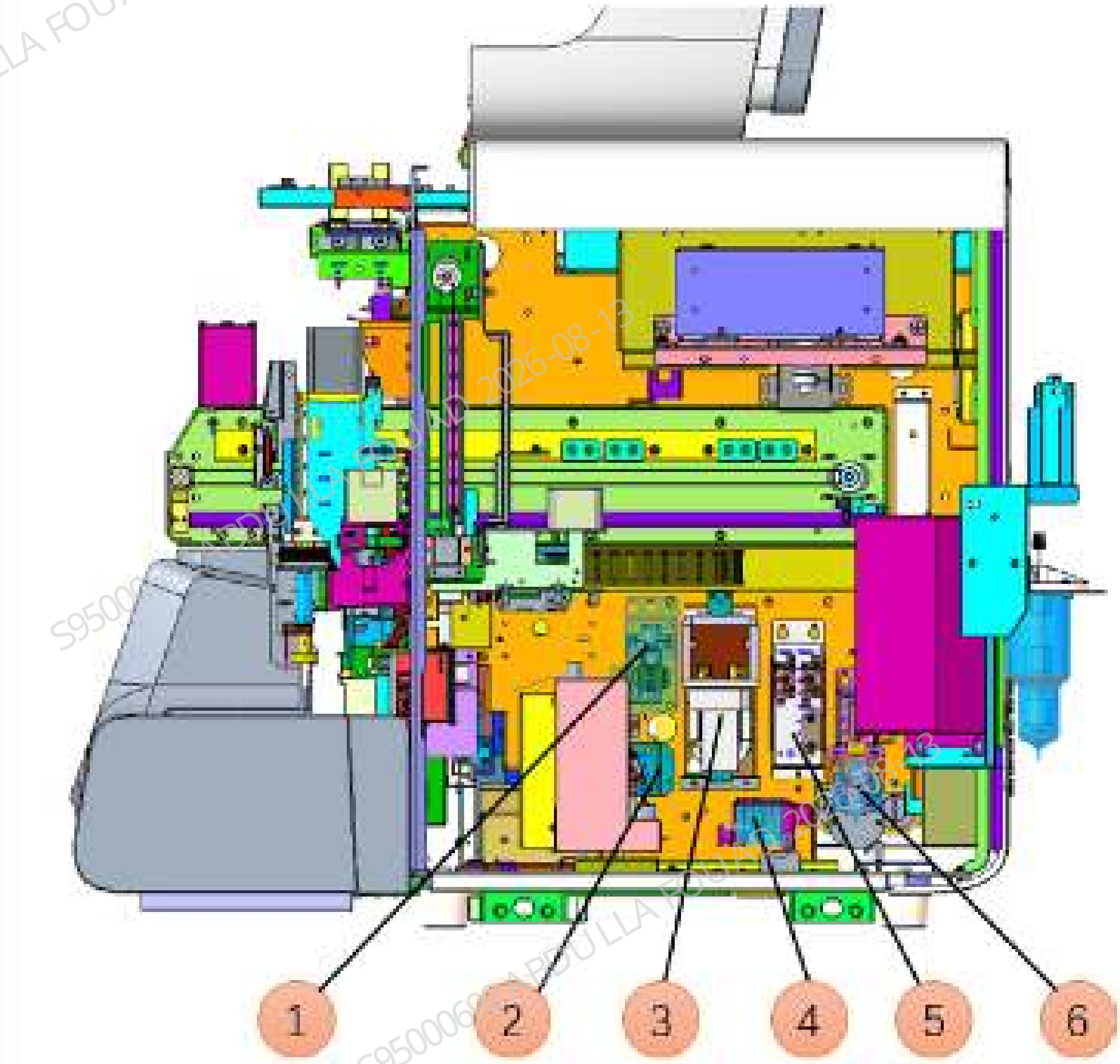
Figure2–25 Right Side of the Analyzer (Right Cover Open)



No.	Name	Function Description	No.	Name	Function Description
1	Blending Assembly	Mix the blood samples, making the serum and blood cells mixed evenly	4	Sampling assembly	Aspirate the blood samples in the closed sampling position and automatic sampling position, and dispense the collected sample into each reaction bath
2	Sheath fluid impedance assembly	Provide a test platform for the detection of RBC/ PLT and its related parameters	5	Optical system	Detect cell morphology, etc.
3	Right bath assembly	Collect and drain reagents			

The components located in the right side in the analyzer (open the assembly panel of the right bath) are shown as below.

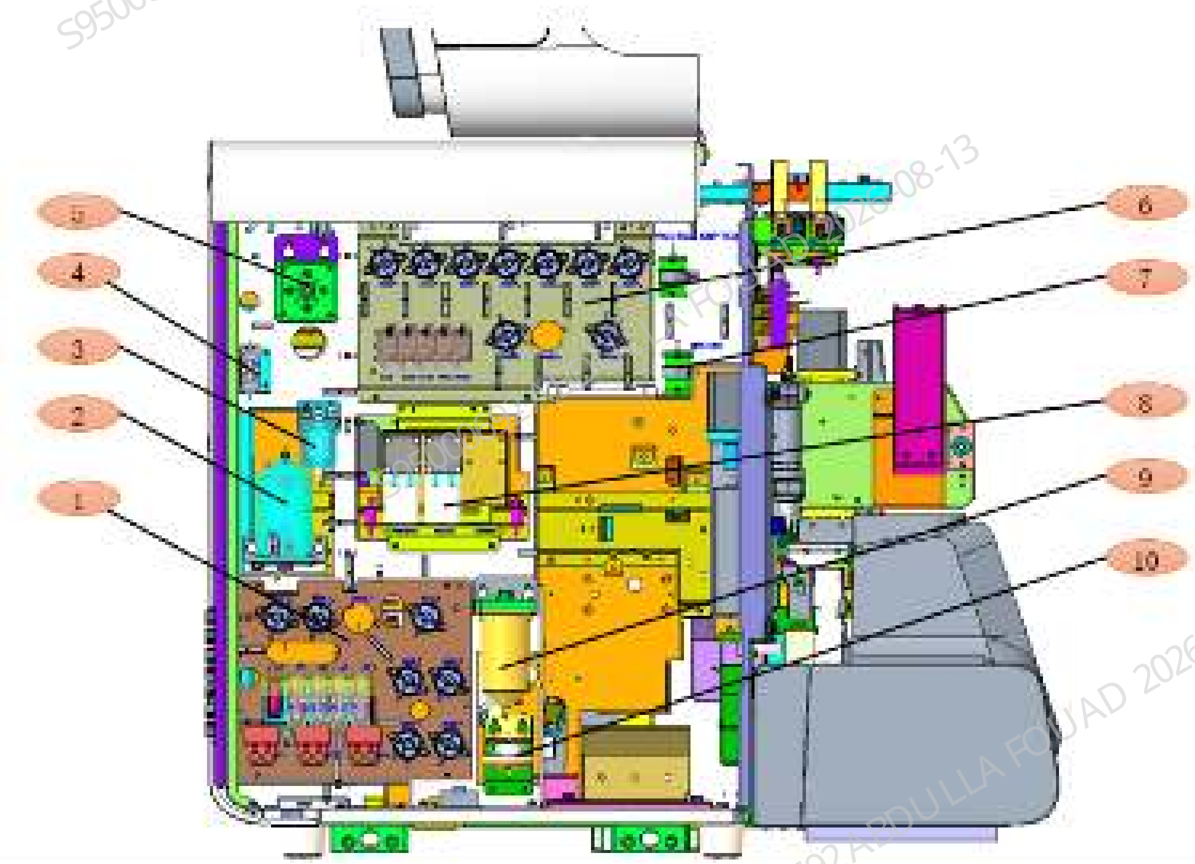
Figure2–26 Components in the Right Side in the Analyzer (Assembly Panel Open)



No.	Name	Function Description	No.	Name	Function Description
1	HGB assembly	Measure hemoglobin in blood	4	Sheath impedance confluence board assembly	Control the connection of RBC sample preparation tube
2	Sheath fluid impedance sample preparation valve assembly	Control sheath fluid sample preparation channel	5	Optical sample prepara tion valve assembly	Control optical sample preparation channel
3	Reaction bath assembly	Incubate samples at constant temperature	6	ESR cleaning assembly	It is used to transport ESR cleaner

The components located in the left side in the analyzer are shown as below.

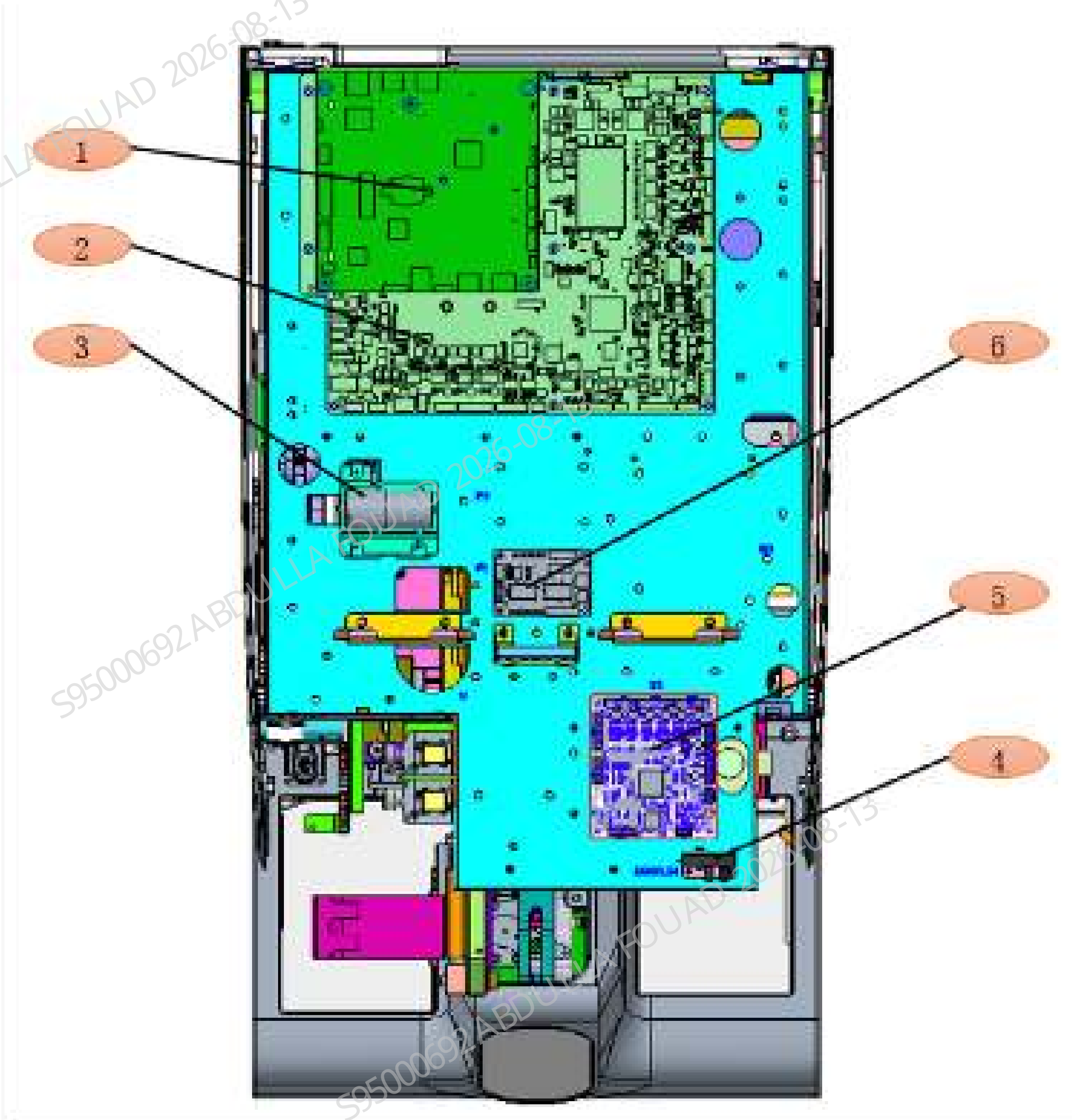
Figure2–27 Left Side of the Analyzer (Left Cover Open)



No.	Name	Function Description	No.	Name	Function Description
1	Bottom left valve assembly	Install reagent detection board, dosing pump and valve	6	Top left valve assembly	Install the valve controlling the fluidics connection
2	Diluent heating assembly	Heat diluent	7	Isolation chamber	Prevent the liquid from entering the gas pressure detection board
3	Sheath fluid filter assembly (3201)	Eliminate air bubble in diluent	8	Waste pump assembly	Build negative for fluidics system and pump fluid.
4	Liquid Detection Board PCBA	Detect the existence of fluid in tube	9	WC2 Cistern	Collect waste
5	Air tank assembly	Store and release gas	10	Filter assembly	Filter the waste from the probe wipe cleaning.

The components located on the top of the analyzer are shown as below.

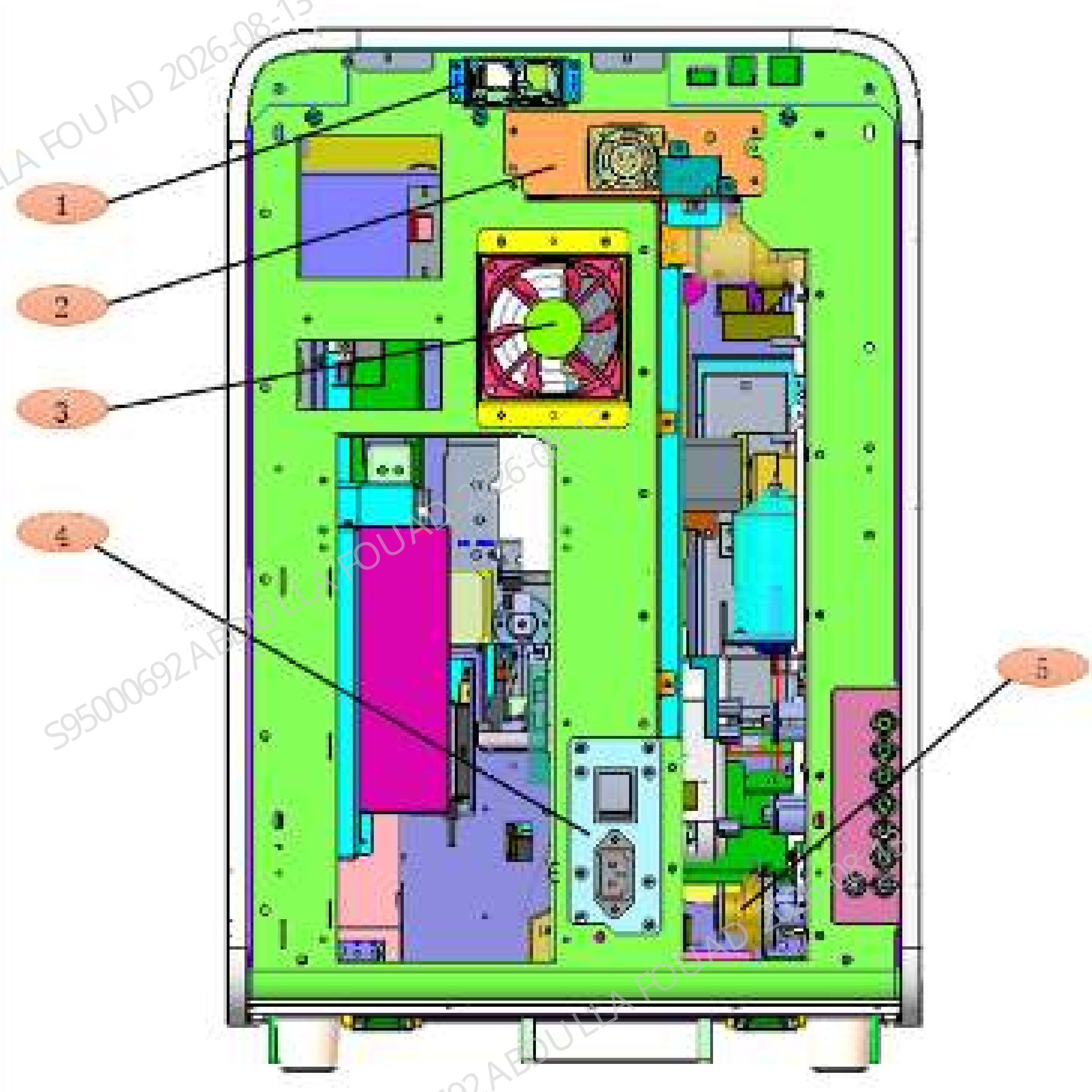
Figure2–28 Vertical View of the analyzer (Top Cover Open)



No.	Name	Function Description	No.	Name	Function Description
1	PCBA, Main Control Board, iMX8	Provide USB port, network port, display interface, touch screen interface and speaker interface, and provide data processing and calculation.	5	Motor drive broad	Drive and control the stepping motor inside the analyzer
2	PCBA, Motherboard	Energize all analyzer components, drive valves, pumps and motors	6	Valve driver board PCBA	Drive valves and pumps
3	Air pump 12VDC	Build gas pressure for the analyzer			
4	Sensor	Detect whether the front cover is open			

The components located on the rear side of the analyzer are shown as below.

Figure2–29 Rear View of the Analyzer (Rear Cover Open)



No.	Name	Function Description	No.	Name	Function Description
1	Board Fan Assembly	Cool the top board area	4	Power Switch Assembly	Control the power supply for the analyzer and its power on/off
2	Power Supply Assembly	Provide constant power for the analyzer	5	0.5mL diaphragm pump (same direction as EPDM connector)	Aspirate reagent of specified volume
3	Main unit fan assembly	Cool the analyzer			

2.3 Fluidics System

2.3.1 Fluidics System Overview

2.3.1.1 Sample Distributing Procedure and Measurement Principles

The analyzer uses multi-section sample distribution method. After being aspirated into the sampling probe, the sample is dispensed into each channel in sequence to complete measurement. The distributing procedure is as below (taking the CT-WB CDR +ESR mode for example):

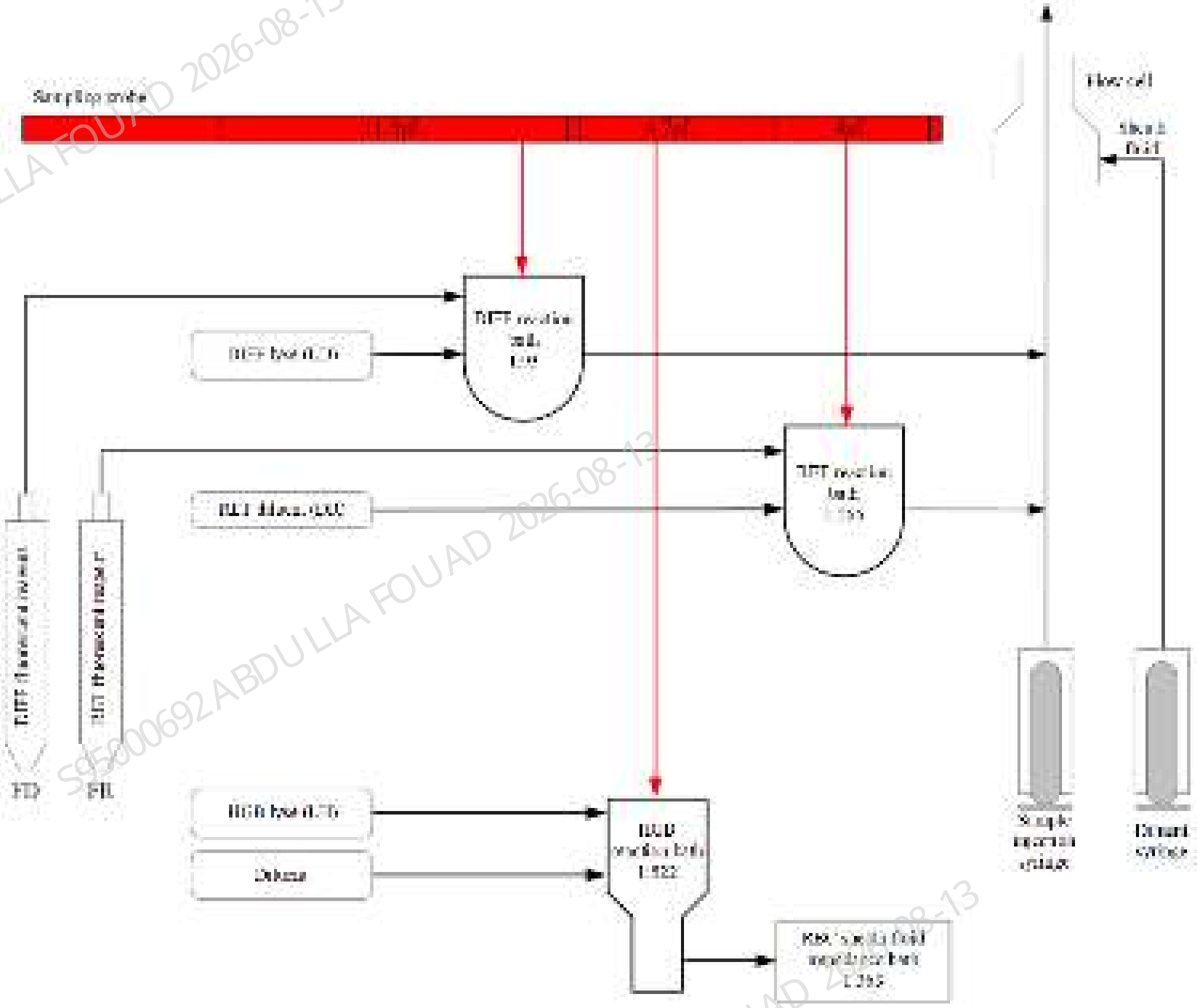
Sampling→RET Distributing→HGB Distributing→DIFF Distributing→ESR Measurement→Cleaning and Restore

ESR measurement requires no distributing and can be completed directly in the sampling tube.

All measurement modules included in other modes share the same distributing sequence with that of CDR+ESR mode.

The sample will be incubated and measured in each channel individually after being dispensed. Where, the HGB channel is measured by colorimetric method, the DIFF and RET channels are measured by laser scatter+sheath fluid method, and the RBC channel is measured by sheath fluid impedance method. The distributing and measurement procedure for each channel is as shown in the figure below:

Figure2–30 Measurement Procedure of Each Channel

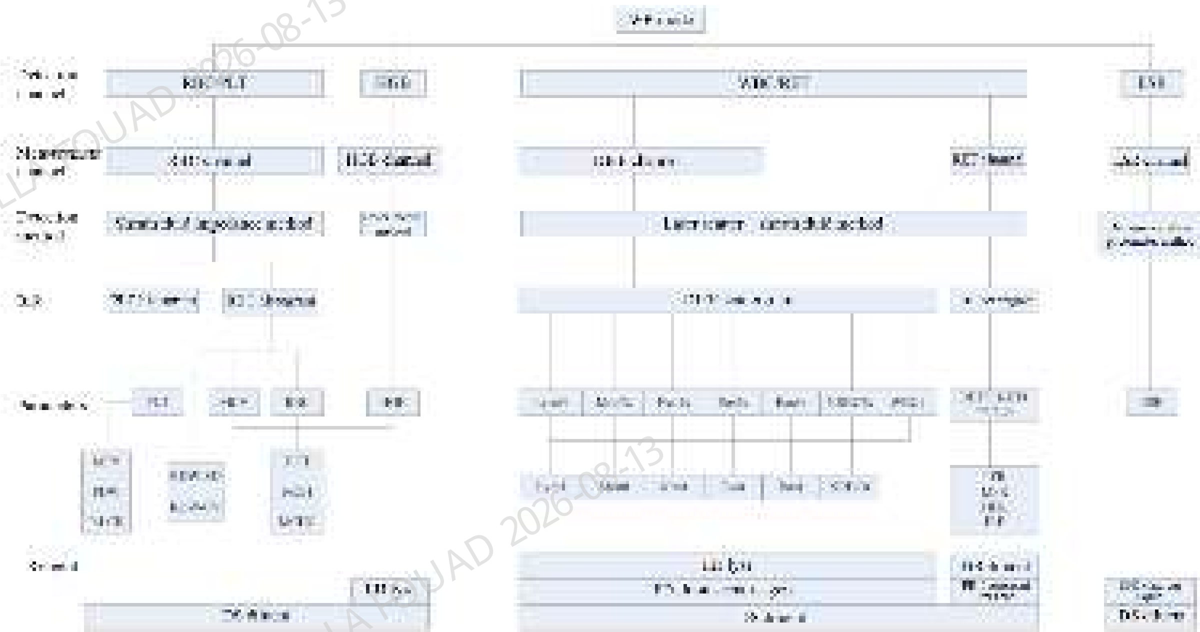


The measurement procedure and principle in predilution mode are the same as those in WB mode, with the only difference in distributing volume. Refer to for the details on the distributing volumes in different modes.

2.3.1.2 Analysis Flow

Taking the WB mode for example, the analysis procedure is as below:

Figure2–31 Analysis Flowchart



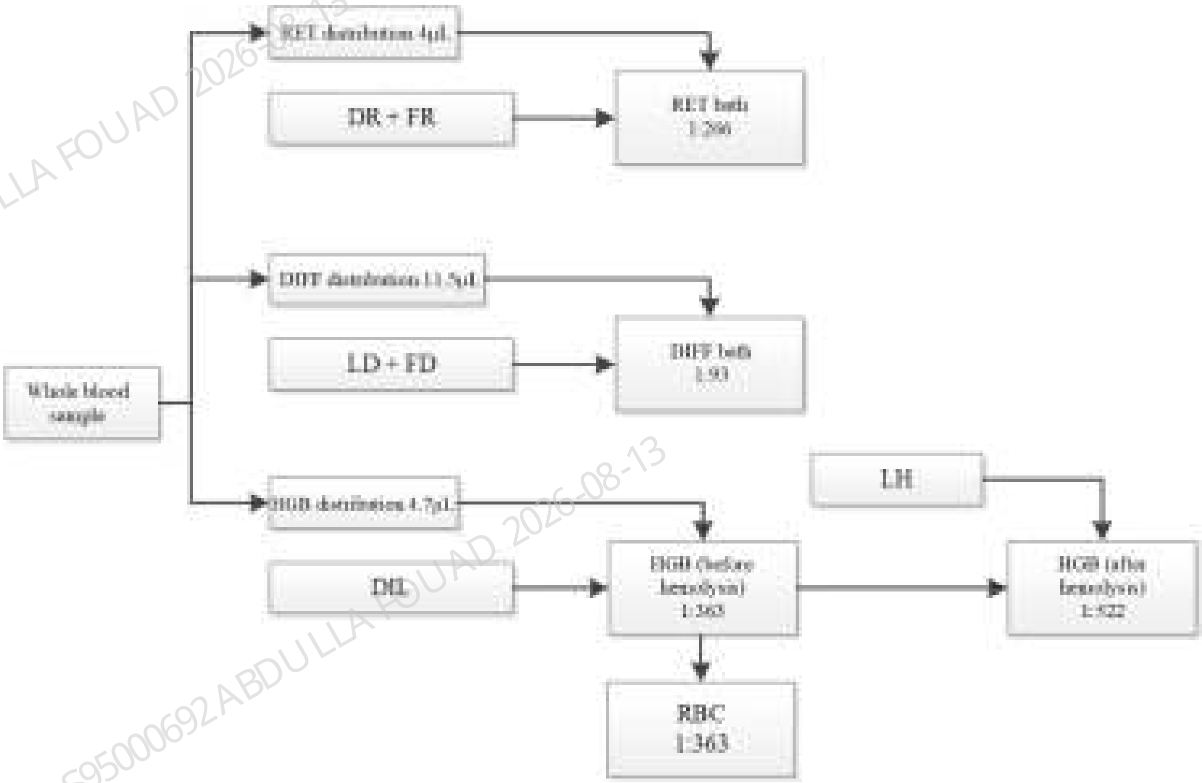
Predilution mode and WB mode share the same analysis procedure. No ESR detection is conducted in the predilution mode.

2.3.1.3 Sample Dilution Flowchart

Blood sample dilution procedures for two different modes (WB and Predilution):

WB Mode

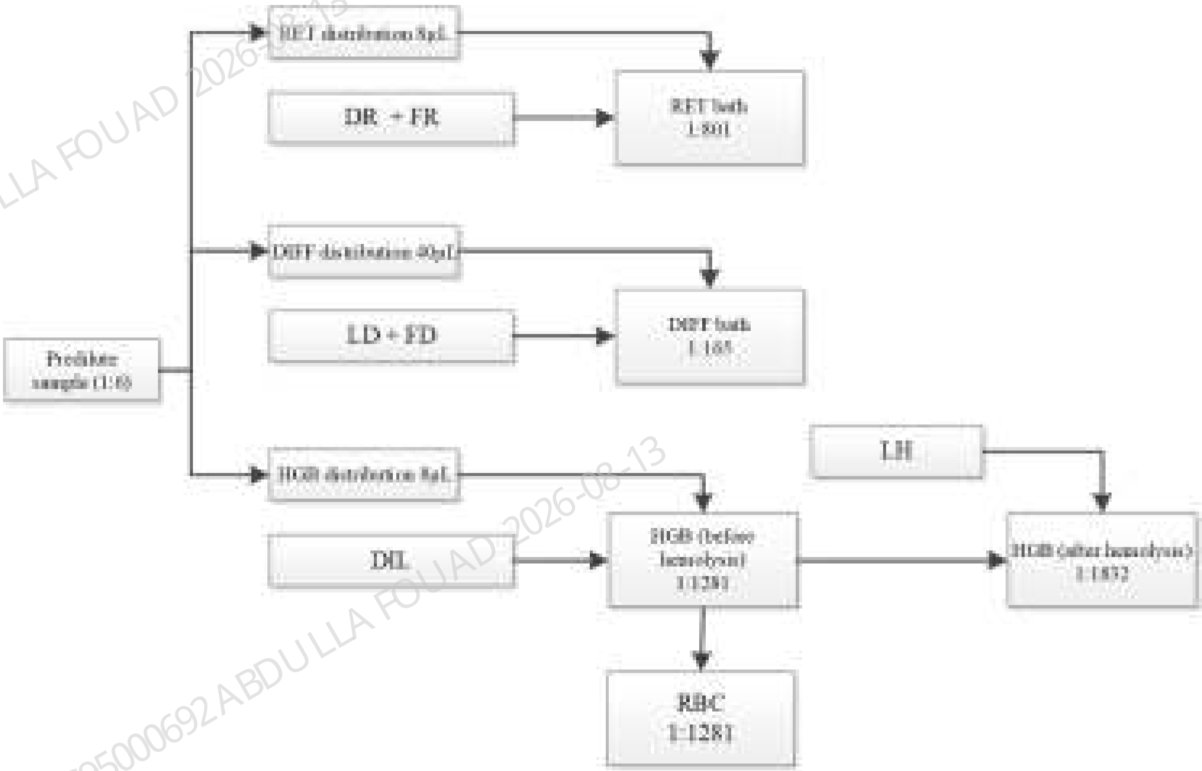
Figure2–32 WB Sample Dilution Flowchart



Predilution Mode

In the predilution mode, the blood sample is first diluted before analysis according to the ratio of 1:6 (20 μL blood sample + 100 μL diluent), and then the diluted sample is used for measurement. The process is as follows:

Figure2–33 Prediluted Sample Dilution Flowchart



2.3.1.4 Sample Volume

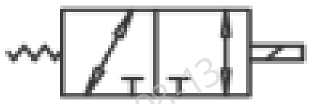
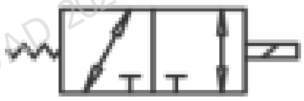
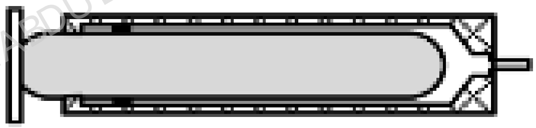
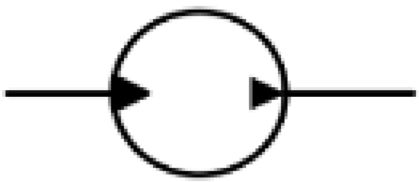
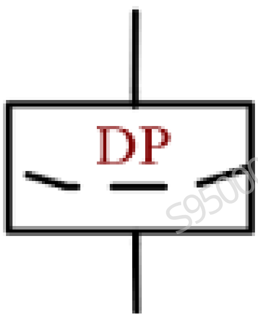
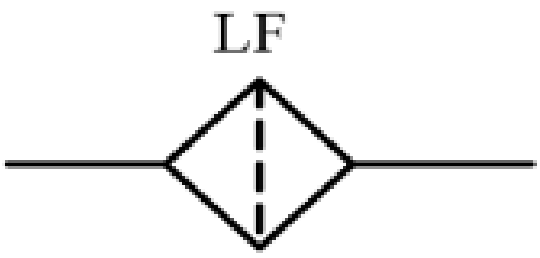
Blood consumption under different modes are as shown in the table below:

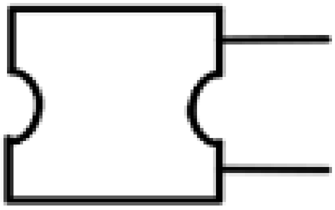
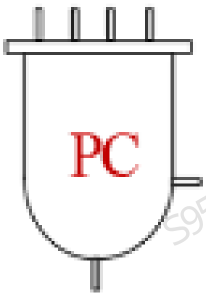
Table 2–34 Blood Consumption Under Different Modes

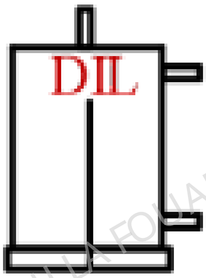
Items	WB Mode	Predilution Mode
RET sample	4μL	8μL
HGB sample	4.7μL	8μL
DIFF sample	11.5μL	40μL

2.3.2 Introduction to Fluidic Parts/Components Functions

2.3.2.1 Introduction to Fluidic Parts

Name	Fluidics symbol		Functions	Whether calibration is required after replace-ment
Valves	Solenoid valve (two-position ways valve)		Control the connection of tube	No
	Solenoid valve (three-position ways valve)		Control the connection of tube	No
Syringe			Aspirate/drain sample and reagent	Yes
Gas pump, waste pump			Build up pressure and drain waste	No
Dosing pump			Aspirate and drain reagent of specified volume	Yes
Filter			Filter impurities	No
Sampling probe			Sampling and distributing	No

Name	Fluidics symbol	Functions	Whether calibration is required after replace-ment
Probe wipe		Clean sampling probe	No
Pressure Detection Assembly		Monitor pressure	No
SCI bath		Provide sheath fluid for sheath fluid impedance measurement	No
PC tank		Build positive pressure	No

Name	Fluidics symbol	Functions	Whether calibration is required after replace-ment
WC2 cistern		Build negative pressure and collect waste	No
DIL preheating bath		Preheat diluent at the analyzer inlet	No

2.3.2.2 Valves

Solenoid Valve

Two-position ways valve: Build or cut a passage. The two-position ways valve for this analyzer is normally closed one. When de-electrified, the passage between the inlet and outlet of the valve is cut off; when electrified, the passage is built up.

Three-position ways valve: Switch among passages. When the three-position ways valve is not energized, the common end (COM) and the normally open end (NO) are connected. When the three-position ways valve is energized, the common end and the normally closed end (NC) are connected.

Figure2–34 Solenoid Valve

1	Valve nozzle of two-position ways valve
2	Valve nozzle of three-position ways valve

Pressure-proof Solenoid Valve

Function:

Compared with ordinary solenoid valves, pressure-proof valves are used for diluent channels, with a pressure range greater than and the working principles same as those of ordinary solenoid valves. The label for an ordinary valve is grayish, and the marked pressure-proof parameter is 0.2 MPa, while the label for a pressure-proof valve is whiter, and the marked pressure-proof parameter is 0.25 MPa. Pressure-proof solenoid valves are also divided into two-position ways pressure-proof ones and three-position ways pressure-proof ones.

Figure2–35 Pressure-proof Valve

1 Valve nozzle of two-position ways pressure-proof solenoid valve

2 Valve nozzle of three-position ways pressure-proof solenoid valve

SMC Valve

Function:

Same as solenoid valve. SMC valve is also divided into two-position ways SMC valve and three-position ways SMC valve. These two types are completely consistent in appearance but can be distinguished by the model information on the valve body.

The model LVM10R3-6A-2-Q is a two-way SMC valve, and the Nozzle ID "1" and "2" are available.

The model LVM10R6-6A-2-Q is a two-position ways SMC valve, and the Nozzle ID "1" and "3" are available.

The model LVM105R-6A-2-Q is a three-position ways SMC valve, and the Nozzle ID "1", "2" and "3" are available.

Figure2–36 SMC Valve

1 Model information

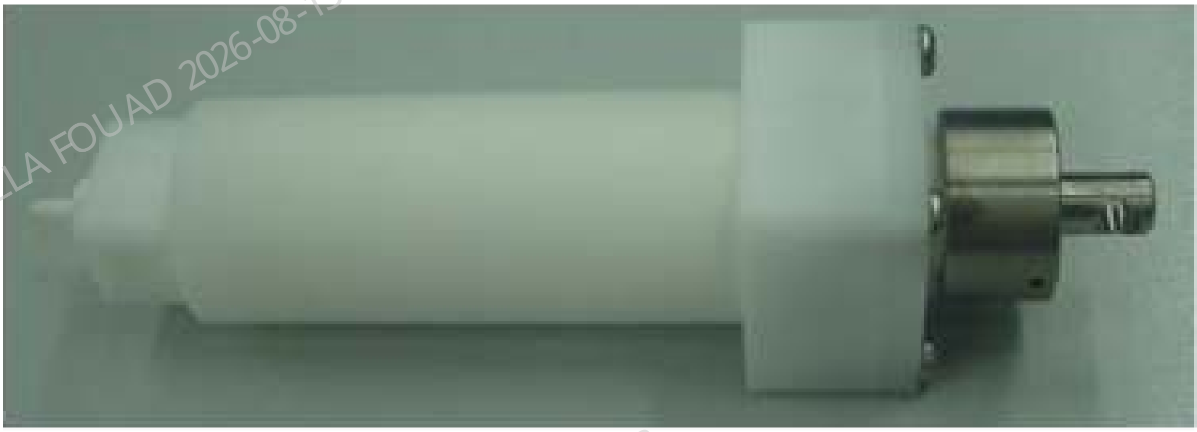
2 Valve nozzle ID

3 Manual control switch

2.3.2.3 Syringe

Appearance:

Figure2–37 Diluent syringe (upper) and sample injection syringe (lower)



Function:

There are 2 kinds pf syringes, and their specifications and functions of them are shown as the following table:

Name	Label	Specifications	Functions
Diluent syringe	DIL_SYRINGE	10mL	Clean the sampling probe, HGB bath and optical channel, prime the SCI bath, and supply the diluent to the reaction bath and the sheath fluid to the flow cell
Sample injection syringe	SP_SYRINGE	250μL	Aspirate and distribute the blood samples of specified volume, depolymerize the ESR samples, dispense the sample to the flow cell and measure in the sheath fluid impedance count bath.

2.3.2.4 Pumps

Gas Pump (GP)

Provide a pressure source for the pneumatic system, and use compressed air to drive the fluidics.

Waste Pump (PUMP_VAC)

Collect and drain the wipe waste, reaction bath waste and RBC sample preparation tube waste.

Waste Pump (PUMP_WASTE)

Empty the waste cisterns WC2.

Figure2–38 Gas Pump (Left) and Waste Pump (Right)

Dosing pump

Control the reagent dosing, there are 5 dosing pumps in the analyzer, whose functions are as follows:

- DP1: 0.52mL dosing pump, dispense the DR diluent of specified volume to the RET bath.
- DP2: 0.52mL dosing pump, dispense the LD lyse of specified volume to the DIFF bath.
- DP3: 0.52mL dosing pump, dispense the LH lyse of specified volume to the HGB bath.
- DP4: 20μL dosing pump, dye dosing pump, dispense the FR dye of specified volume to the RET bath.
- DP5: 20μL dosing pump, dye dosing pump, dispense the FD dye of specified volume to the DIFF bath.
- DP6: 0.52mL dosing pump, dispense the ESR dye of specified volume to the RET bath.

Figure2–39 Dosing pump

1 Liquid chamber connector

2 Gas chamber connector

Figure2–40 Dye dosing pump assembly

2.3.2.5 Filter

Diluent Filter LF1

Filter the impurities in the diluent, protect the optical channel, flow along the direction of letters.

Figure2–41 Diluent Filter LF1

Waste Filter LF2

Filter the impurities from the HGB bath, protect the solenoid valve SV09.

Figure2–42 Waste Filter LF2

1 Flow along the direction indicated by the arrow

Waste Filter LF3

Filter the impurities from the RET bath, protect the solenoid valve SV06. LF3 and LF2 share the same materials.

Waste Filter LF4

Filter the impurities from the DIFF bath, protect the solenoid valve SV07. LF4 and LF2 share the same materials.

Sheath Fluid Filter LF5

Intercept tiny bubbles in the optical sheath fluid channel.

Figure2–43 Sheath Fluid Filter LF5

Probe Wipe Filter LF6

Filter the impurities from the probe wipe, protect the solenoid valve SV08 and waste pump PUMP_VAC.

Figure2–44 Probe Wipe Filter

Waste Filter LF8

Filter the impurities from the sheath fluid impedance assembly, protect the waste pump PUMP_ VAC. LF8 and LF2 share the same materials.

2.3.2.6 Sampling probe

Appearance

Figure2–45 Sampling probe

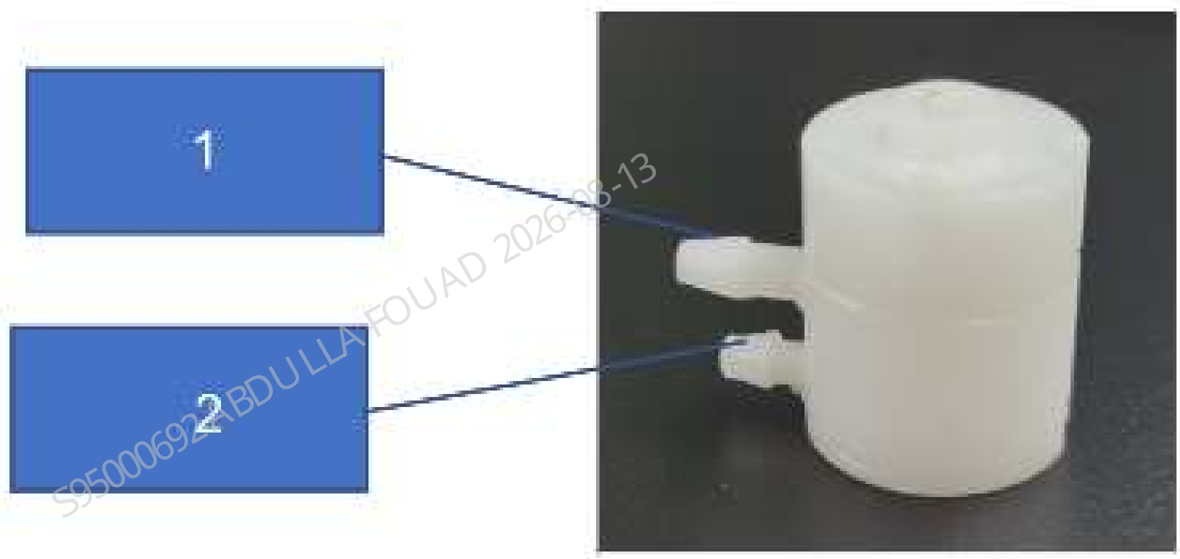
Functions

Aspirate and dispense samples. The sampling probe move between each incubation bath through horizontal and vertical motors to distribute samples.

2.3.2.7 Probe wipe

Appearance

Figure2–46 Probe wipe



1 Outlet

2 Inlet

Functions

Clean the exterior of sample probe, dry and recover the cleaning waste of sample probe interior and exterior.

2.3.2.8 Pressure Detection Assembly

Sampling Pressure Sensor LPS

Collect the liquid pressure data in the sampling tube, which are mainly used for pressure detection in the sampling process and flow chamber measurement process. Measurement readiness [50 kPa, 120 kPa], sensor detection range [0 kPa, 206 kPa]. Inlet and outlet are not distinguished for the sensor.

Figure2–47 Sampling Pressure Sensor

Pressure Sensor PS1~PS4

PS1: Monitor the pressure of air chamber: measurement readiness [45kPa, 60 kPa], sensor detection range [0kPa, 200kPa]
PS2: Monitor the pressure of SCI bath: measurement readiness [35kPa, 45kPa], sensor detection range [0kPa, 200kPa]
PS3: Monitor the pressure of waste channel: measurement readiness [-35kPa, 0kPa], sensor detection range [-100kPa, 0kPa]
PS4: Monitor the pressure of waste cisterns WC2: measurement readiness [-45kPa, 0kPa], sensor detection range [-100kPa, 0kPa]

2.3.3 Fluidic Structure Description

2.3.3.1 DIFF Channel

Reagents Used

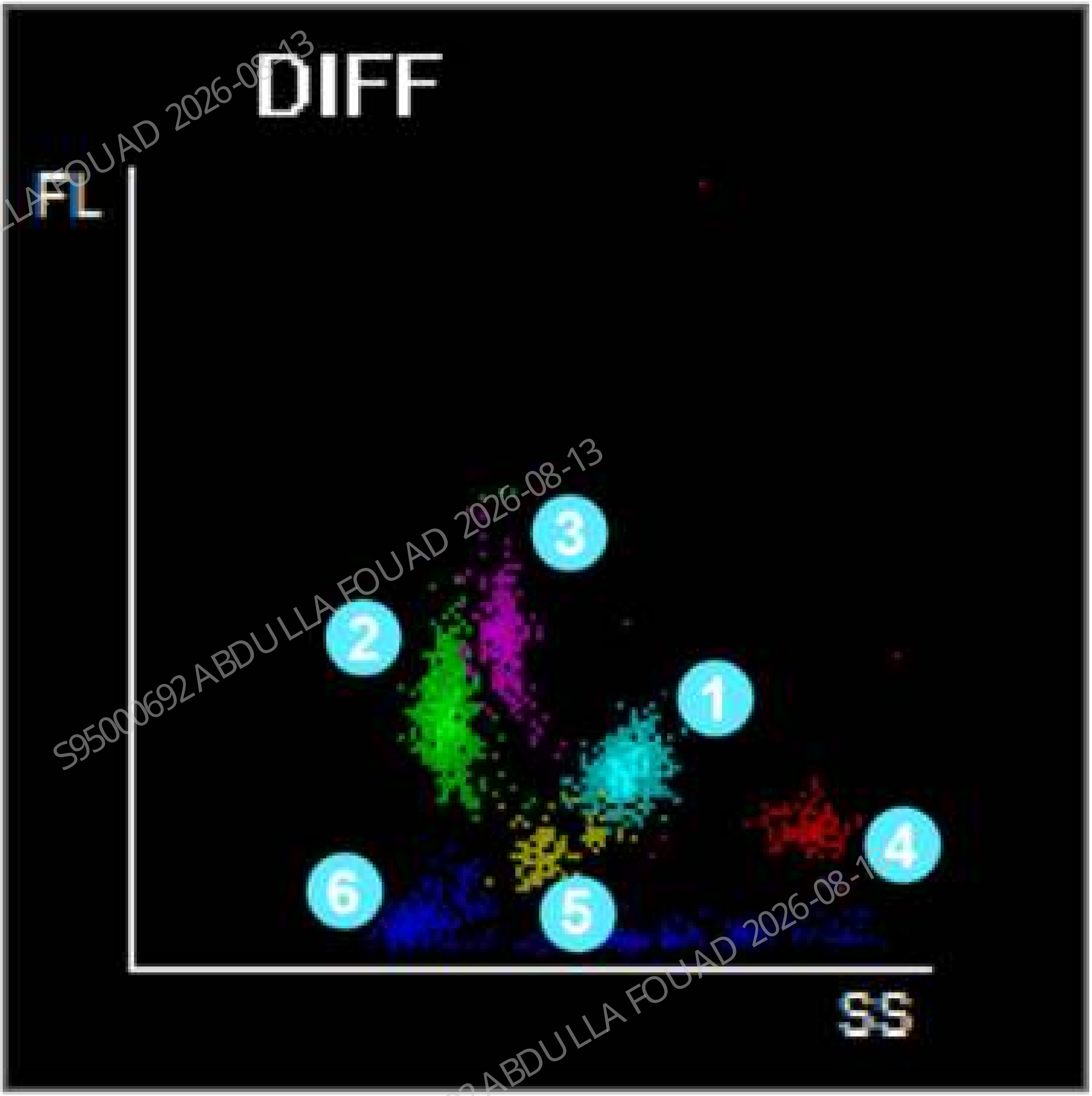
DS diluent, LD lyse, FD fluorescent dye

Detection Principle

Laser scatter + sheath fluid method

Chart Information

DIFF scattergram



1	Neu area
2	Lym area
3	Mon area
4	Eos area
5	Baso area
6	Achromatocyte

Detection Parameters

Lym%,Mon%,Neu%,Eos%,Bas%,NRBC%,Lym#,Mon#,Neu#,Eos#,,Bas#,NRBC#,WBC#

Dilution Ratio

1:93 (WB mode)

Process Description

1. Empty DIFF bath: Turn on SV07 to empty the DIFF bath (DIFF-T139-T140-T141-T142-T194-T193-T189-T187-T183-T182-PUMP_VAC).
2. Dispense initial volume: With the dosing pump DP2, the LD lyse is carried by the LD reagent and transferred to the DIFF bath through SV02 and preheat tube T107 (shown as the green path T7-T10-T13-T105-T106-T107-T108-T110-DIFF in the figure).
3. Dispense FD dye: Before distribution and with the dosing pump DP5, the FD dye is carried by the FD dye bag and transferred to the DIFF bath through SV26 and T114 tube to stain the distributed WBC (shown as the blue path T113-T114-DIFF in the figure).
4. DIFF distribution: Move the sample probe to the DIFF bath, and transfer the blood sample in the sample probe to the DIFF bath for distribution by using the sample injection syringe (SP_SYRINGE).
5. Dispense and swirl LD lyse for mixing: With the dosing pump DP2, the LD lyse is carried by the LD reagent and transferred to the DIFF bath through SV02 and preheat tube T107 (shown as the green path T7-T10-T13-T105-T106-T107-T108-T110-DIFF in the figure). The LD lyse dispensed to the DIFF bath will swirl with the sample in the bath, in which case, the sample solution is well mixed.
6. Transfer samples: Energize SV17, SV12 and SV37. After incubation, the sample is transferred by using diluent syringe (DIL_SYRINGE) to the sample preparation tube T131 through the DIFF bath and SV37, and then to the tube T132 downstream (shown as the red path T129-T130-T131-T132 in the figure).
7. Generate sample flow: Energize SV23. The sample injection syringe (SP_SYRINGE) transfers the sample in the sample preparation tube T131 to the flow cell (SP_SYRINGE-T38-T39-T40-T134-T131-FC) through SV23, SV24 and SV39. Meanwhile, turn on SV17 and SV11, inject the sheath solution through tube T121 to the flow cell by using diluent syringe (DIL_SYRINGE) to form the sheath fluid which wrap the sample to form the sample flow (shown as the light blue path DIL_SYRINGE-T25-T26-T115-T116-T119-T120-T121-FC in the figure). As shown below, turn on SV21 to release the WC bath to normal pressure. Before the operation of syringe, energize SV14. After testing, the waste may flow to the WC2 bath through SV14 (shown as the brown path FC-T122-T123-T124-T206-T204-T203-WC2 in the figure).
8. Start testing: The sample flow goes through the flow cell at a stable flow rate as driven by the sheath solution and sample injection syringe (SP_SYRINGE). At the same time, the optical system starts working, and the DIFF measurement starts.
9. Cleaning: Energize SV17, SV12 and SV37. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the sample preparation tube T129, T130, T131, T132 and DIFF bath (DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T1129-DIFF). Energize SV17, SV11, SV12 and SV14. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the flow cell (DIL_SYRINGE-T25-T26-T115-T116-T119-T120-T121-FC, DIL_SYRINGE-T25-T26-T27-T28-T133-T132-FC), the waste flows into the WC2 bath through SV14 (FC-T122-T126-T124-T206-

T204-T203-WC2). Energize SV12, SV17, SV25, SV23, SV37 and SV39. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the three-way connector C53 and C54. The waste flows into the DIFF bath (DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-C53-T130-T129-DIFF, IL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T37-T38-T39-T40-T128-T127-C54-T132-T131-T130-T129-DIFF). Then, turn on SV07 to empty the DIFF bath waste (DIFF-T139-T140-T141-T142-T194-T193-T189-T187-T183-T182-PUMP_VAC). Finally, energize SV17, SV12 and SV37. The diluent syringe (DIL_SYRINGE) dispenses the initial volume to the DIFF bath (DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T129-DIFF). The sample injection syringe (SP_SYRINGE) is restored to the original status.

Figure2–48 DIFF Measurement Channel

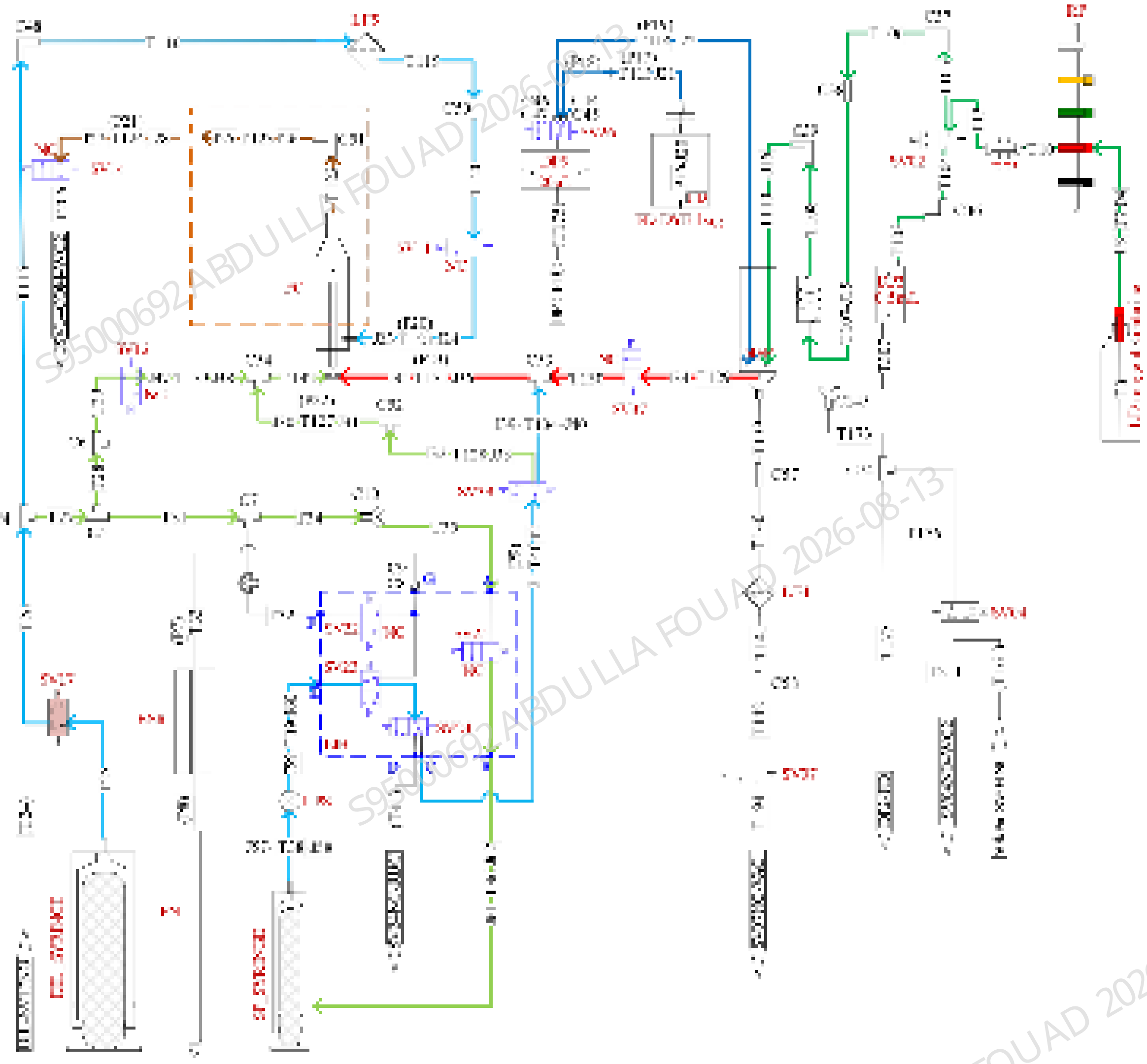
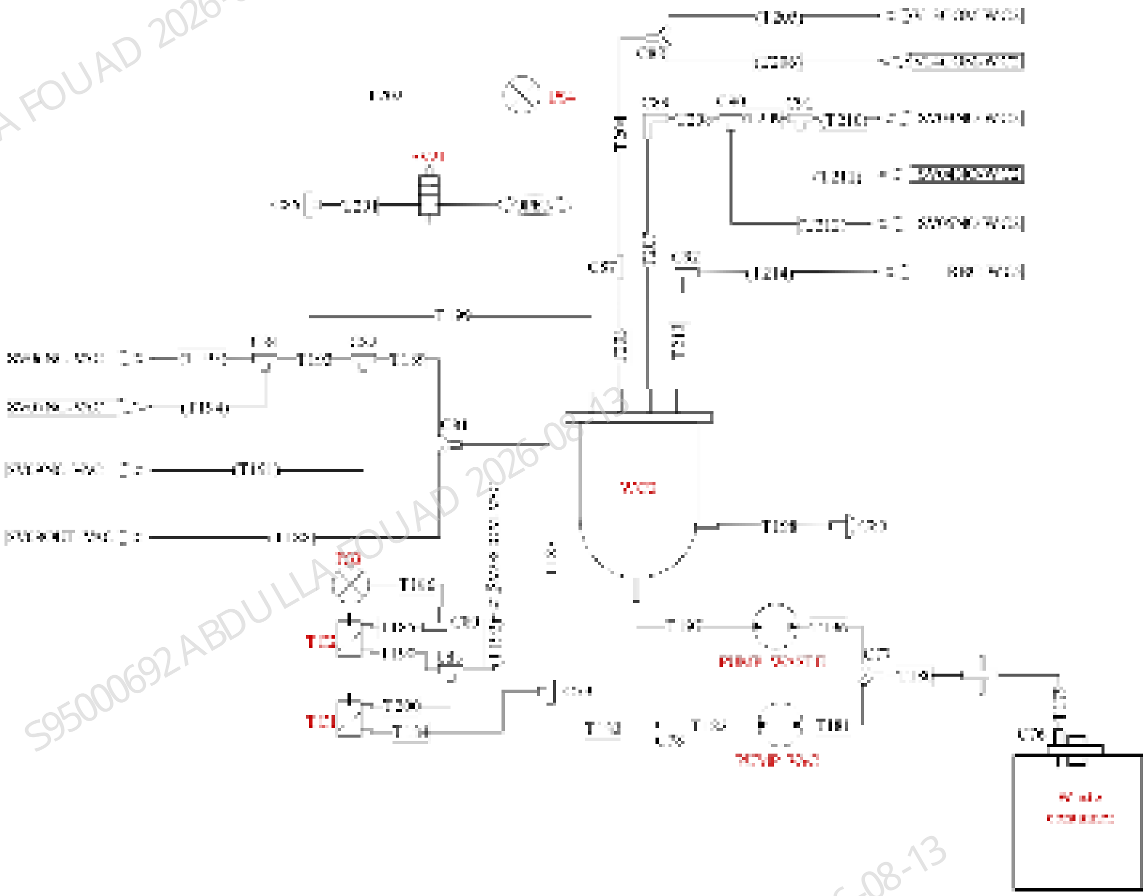


Figure2–49 DIFF Measurement Waste Channel



2.3.3.2 RET Channel

Reagents Used

DS diluent, DR diluent, FR fluorescent dye

Detection Principle

Laser scatter + sheath fluid method

Chart Information

RET scattergram

Detection Parameters

RET%,RET#,PLT-O

Dilution Ratio

1:266 (WB mode)

Process Description

1. Empty the RET bath: Turn on valve SV06 to drain the last measured diluent in the RET bath (RET-T135-T136-T137-T138-T195-T193-T189-T187-T183-T182-Pump _ VAC).
2. Dispense initial volume: With the dosing pump DP1, the DR diluent is carried by the DR reagent and transferred to the RET bath through SV01 and preheat tube T99 (as the green path T8-T11-T14-T97-T98-T99-T100-T102-RET in the figure).
3. Dispense FR dye: Before distribution and with the dosing pump DP4, the FR dye is carried by the FR dye bag and transferred to the RET bath through SV27 and T112 tube to bind to the nucleic acid of distributed reticulocyte (as the dark blue path T111-T112-RET in the figure).
4. RET distribution: Move the sample probe to the RET bath, and transfer the blood sample in the sample probe to the RET bath for distribution by using the sample injection syringe (SP_ SYRINGE).
5. Dispense and swirl DR diluent for mixing: With the dosing pump DP1, the DR diluent is carried by the DR reagent tank and transferred to the RET bath through SV01 and preheat tube T99 (as the green path T8-T11-T14-T97-T98-T99-T100-T308-T102-RET in the figure). The DR diluent dispensed to the RET bath will swirl with the sample in the bath, in which case, the sample solution is well mixed.
6. Transfer samples: Energize SV17, SV12 and SV38. After incubation, the sample is transferred by using diluent syringe (DIL_SYRINGE) to the sample preparation tube T127 through the RET bath and SV38, and then to the tube T133 (as the red path T125-T126-T127-T133 in the figure).
7. Generate sample flow: Energize SV23 and SV39. The sample injection syringe (SP_ SYRINGE) transfers the sample in the sample preparation tube T127 to the flow cell through SV23, SV24 and SV39 (SP_SYRINGE-T38-T39-T40-T128-T127-T132-FC). Meanwhile, turn on SV17 and SV11, inject the sheath solution through tube T121 to the flow cell by using diluent syringe (DIL_SYRINGE) to form the sheath fluid which wrap the sample to form the sample flow (shown as the light blue path DIL_SYRINGE-T25-T26-T115-T116-T119-T120-T121-FC in the figure). As shown below, turn on SV21 to release the WC bath to normal pressure. Before the operation of syringe, energize SV14. After testing, the waste may flow to the WC2 bath through SV14 (shown as the brown path FC-T122-T123-T124-T206-T204-T203-WC2 in the figure).
8. Start testing: The sample flow goes through the flow cell at a stable flow rate as driven by the sheath solution and sample injection syringe (SP_SYRINGE). At the same time, the optical system starts working, and the RET measurement starts.
9. Cleaning: Energize SV17, SV12 and SV38. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the sample preparation tube T125, T126, T127, T133 and RET bath (DIL_ SYRINGE-T25-T26-T27-T28-T29-T133-T127-T126-T125-RET). Energize SV17, SV11, SV12 and SV14. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the flow cell (DIL_ SYRINGE-T25-T26-T115-T116-T119-T120-T121-FC, DIL_SYRINGE-T25-T26-T27-T28-T133-

T132-FC), the waste flows into the WC2 bath through SV14 (FC-T122-T123-T124-T206-T204-T203-WC2). Energize SV12, SV17, SV25, SV23, SV38 and SV39. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the three-way connector C52 and C54 (DIL_SYRINGE-T25-T26-T27-T28-T29-T133-C54-T127-T126-T125-RET. DIL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T38-T39-T40-T128-C52-T126-T125-RET). The waste flows to the RET bath. Then, turn on SV06 to drain the waste in the RET bath (RET-T135-T136-T137-T138-T195-T193-T189-T187-T183-T182-PUMP_VAC). Finally, energize SV17, SV12 and SV38. The diluent syringe (DIL_SYRINGE) dispenses the diluent to the RET bath as initial volume (DIL_SYRINGE-T25-T26-T27-T28-T29-T133-C54-T127-T126-T125-RET), until the next measurement. The diluent syringe (DIL_SYRINGE) and sample injection syringe (SP_SYRINGE) are restored to the original status.

Figure2–50 RET Measurement Channel

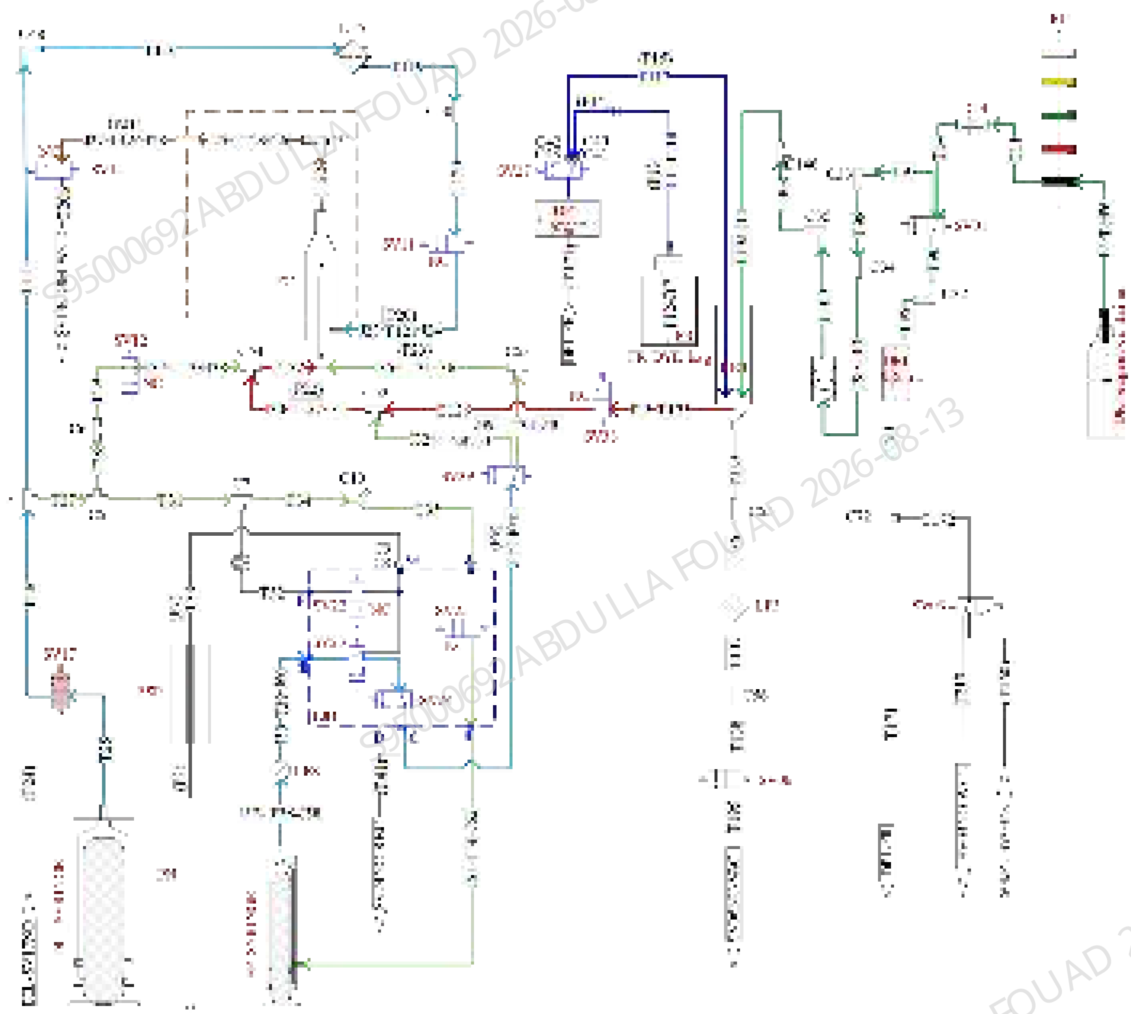
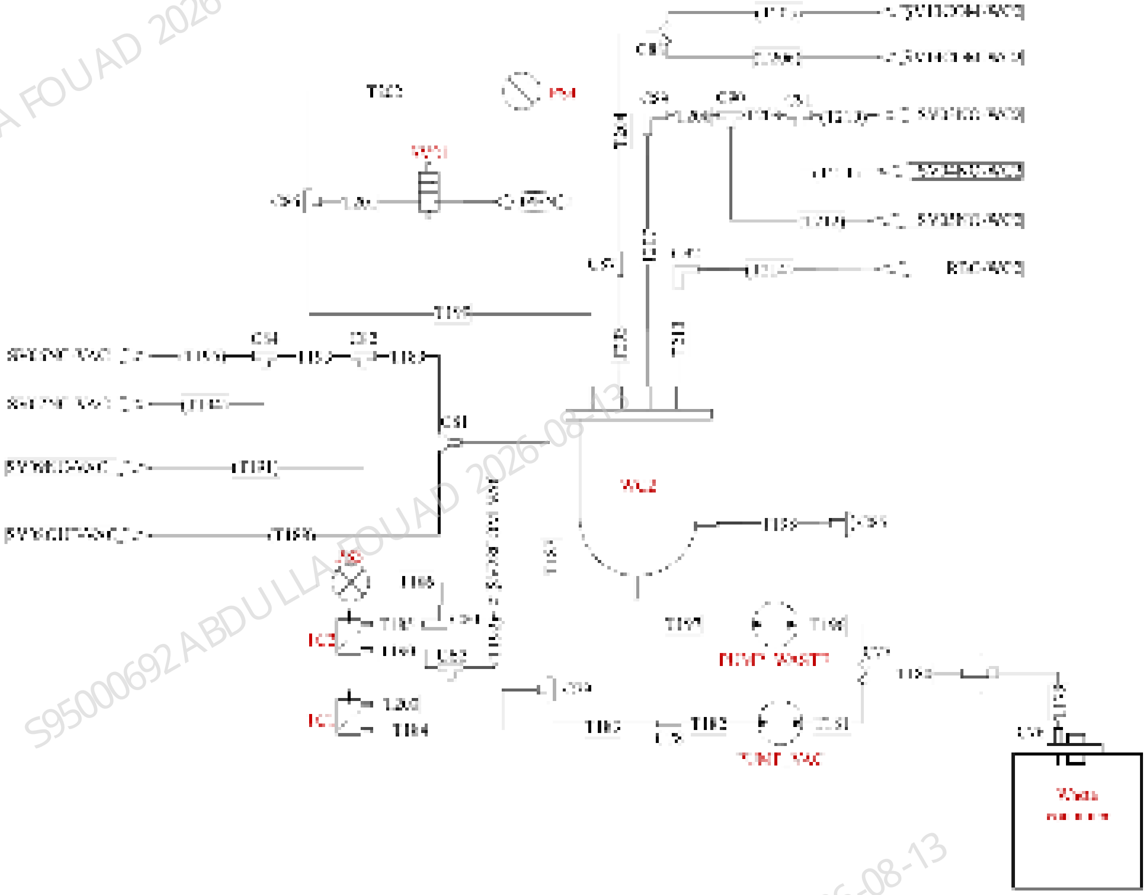


Figure2–51 RET Measurement Waste Channel



2.3.3.3 HGB Channel

Reagents Used

DS diluent, LH lyse

Detection Principle

Colorimetric method

Chart Information

N/A

Detection Parameters

HGB

Dilution Ratio

1:522 (WB mode)

Process Description

1. Empty the HGB bath: Turn on SV09 to empty the HGB bath (HGB-T88-T89-T90-T91-T191-T190-T189-T187-T183-T182-PUMP_VAC).
2. HGB background voltage measurement: Energize SV17 and SV16. The diluent syringe (DIL_SYRINGE) drives the diluent to the HGB bath (as the green path DIL_SYRINGE-T25-T26-T27-T28-T30-T84-T85-T86-HGB in the figure). The light emitted by the LED passes through the HGB bath to measure the HGB background voltage.
3. Initial volume dispensing: Turn on SV09 to empty the HGB bath (HGB-T88-T89-T90-T91-T191-T190-T189-T187-T183-T182-PUMP_VAC). Turn on SV17 and SV16. The diluent syringe (DIL_SYRINGE) drives the diluent to the HGB bath (as the green path DIL_SYRINGE-T25-T26-T27-T28-T30-T84-T85-T86-HGB in the figure).
4. HGB distribution: Move the sample probe to the HGB bath, and transfer the blood sample in the sample probe to the HGB bath for distribution by using the sample injection syringe (SP_SYRINGE).
5. Sample mixing by swirling with diluent: Energize SV17 and SV16. The diluent syringe (DIL_SYRINGE) drives the diluent to the HGB bath (as the green path DIL_SYRINGE-T25-T26-T27-T28-T30-T84-T85-T86-HGB in the figure). The sample is mixed by swirling with the diluent.
6. RBC sample preparation: Energize SV17, SV29 and SV36. The diluent syringe (DIL_SYRINGE) drives some diluted sample to the RBC sample preparation tube T43 through SV36 (as the red path HGB-T86-T87-T42-T43 in Figure 6-23, refer to 6.3.4 **2.3.3.4** RBC/PLT Channel for details).
7. HGB lyse: Energize SV05 and SV10. With the dosing pump DP3, the LH lyse flows through the LH reagent tank and SV10 to the HGB bath (as the blue path T2-T6-T9-T12-T94-T85-T86-HGB in the figure). The sample is mixed by swirling. The left red blood cells in the HGB bath are dissolved.
8. HGB measurement: Dwell the sample, the light emitted by the LED lamp passes through the HGB bath, and the HGB concentration in the sample is measured by colorimetry.
9. HGB bath cleaning: Energize SV17 and SV16. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the HGB bath (DIL_SYRINGE-T25-T26-T27-T28-T30-T84-T85-T86-HGB). After cleaning, energize SV09 to empty the HGB bath (HGB-T88-T89-T90-T91-T191-T190-T189-T187-T183-T182-PUMP_VAC). The diluent syringe (DIL_SYRINGE) dispense the diluent to the HGB bath through SV17 and SV16 as the initial volume (DIL_SYRINGE-T25-T26-T27-T28-T30-T84-T85-T86-HGB) to restore the status ready for measurement.

Figure2–52 HGB Measurement Channel

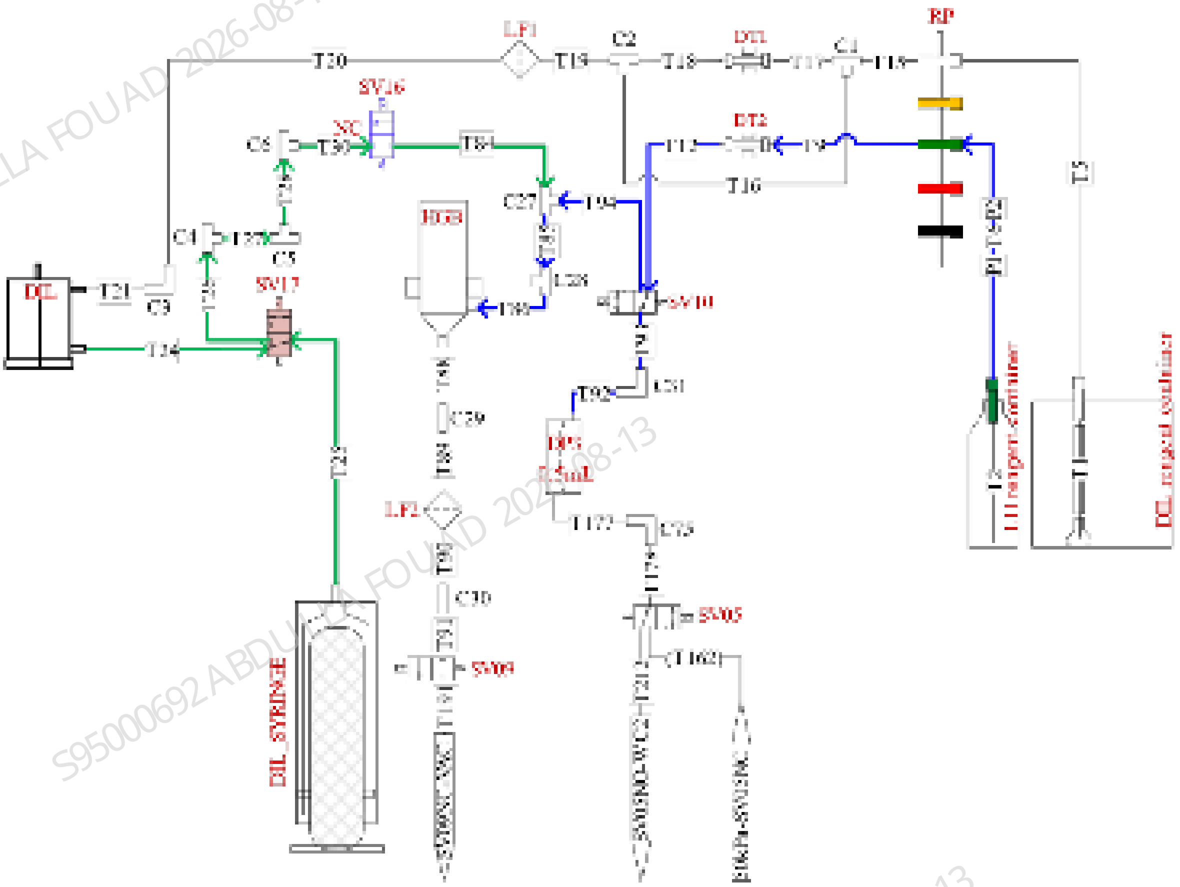


Figure2–53 RBC Measurement Channel

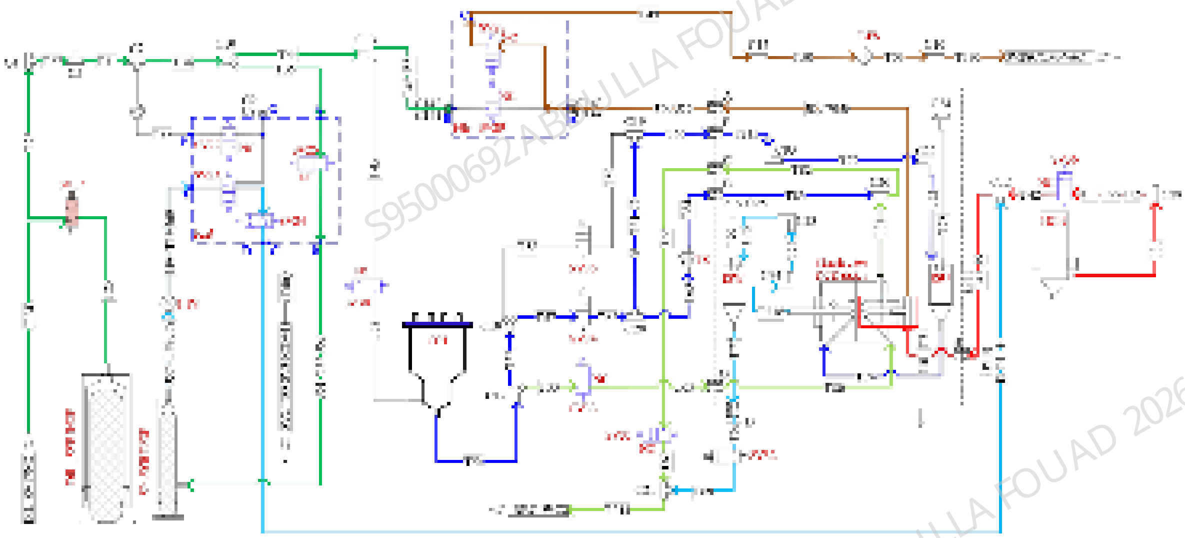
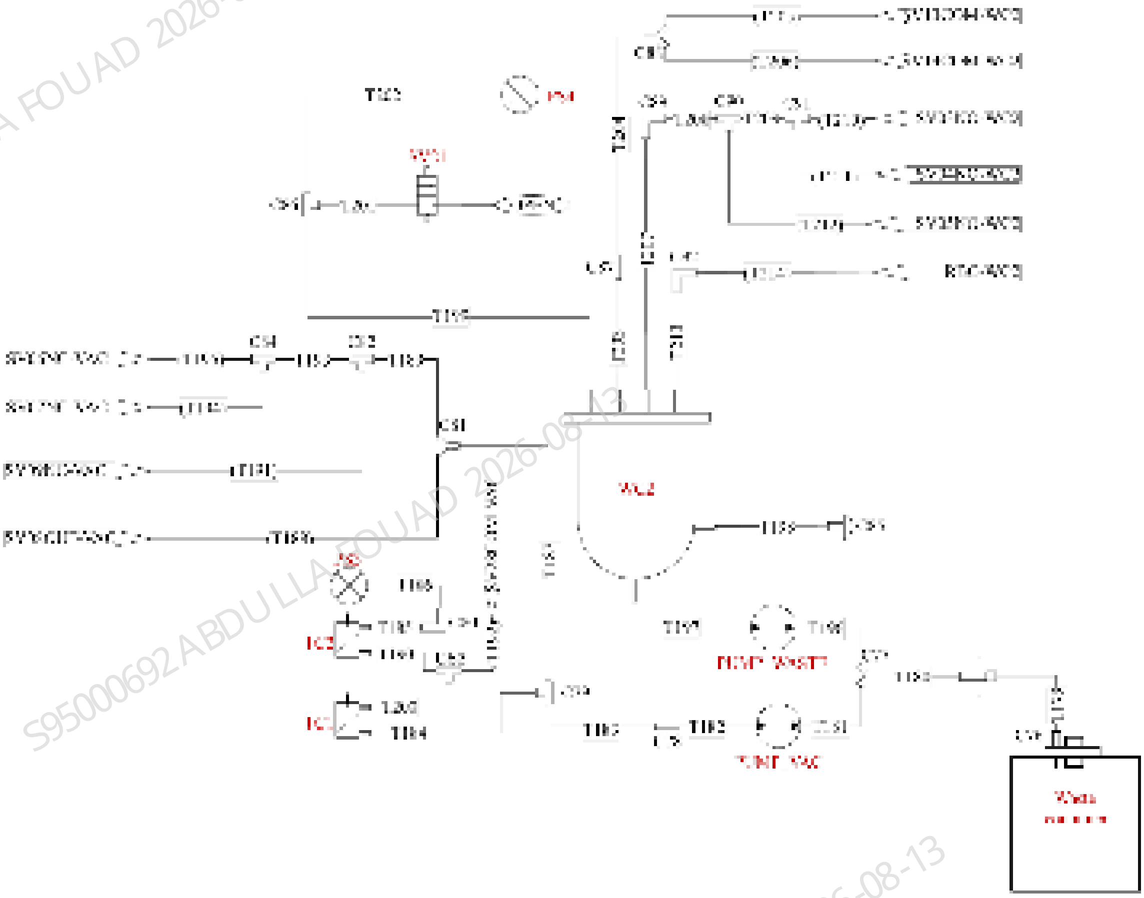


Figure2–54 HGB Measurement Waste Channel



2.3.3.4 RBC/PLT Channel

Reagents Used

DS diluent

Detection Principle

Sheath fluid impedance method

Chart Information

RBC histogram, PLT histogram

Detection Parameters

RBC,MCV,PLT,MPV

Dilution Ratio

1:363 (WB mode)

Process Description

1. RBC sample preparation: After distributing the sample to the HGB bath, energize SV17, SV29 and SV36. The diluent syringe (DIL_SYRINGE) drives some diluted sample to the RBC sample preparation tube T43 through SV36 (as the red path HGB-T86-T87-T42-T43 in the figure).
2. Front bath cleaning: Energize SV33 and SV31. With positive pressure, the diluent in the SCI bath is driven to flush the front bath of sheath fluid impedance bath, and finally flows into the WC2 bath (as the light green path SCI-T54-T55-T65-T80-T81-T82-T66-T67-T214-T213-WC2 in the figure).
3. Sample preparation three-way connector cleaning: Energize SV33, SV35 and SV28. With the negative pressure of waste pump (PUMP_VAC), the diluent in the front bath flushes the sample probe between J9 and J10 and the corresponding three-way connector (SCI-T54-T55-T65-T80-T70,SCI-T54-T56-T58-T59-T60-T215-T82-T73-T74-T70. Sample probe and three-way connector are integrated to the sheath fluid impedance count bath, as the red box in Figure 6-24).
4. RBC measurement: Energize SV23 and SV24. The sample injection syringe (SP_SYRINGE) drives the RBC sample in the sample preparation tube slowly to the sheath fluid impedance bath (as the blue path SP_SYRINGE-T38-T39-T41-T43-T71 in the figure). At the same time, energize SV34 and SV32. With positive pressure, the diluent in the SCI bath flows into the front and rear baths (as the light green path SCI-T54-T56-T57-T62-T63-T64-T83-T81-T67-T216-T75-T76-ISW. SCI-T54-T56-T57-T62-T61-T60-T215-T72-T73-ISU-T74-T216-T75-T76-ISW in the figure), to form the RBC sample wrapped by sheath fluid, which flows through the aperture for RBC counting. The waste finally flows through the ISW bath to the WC2 bath (ISW-T77-T68-T69-T214-T213-WC2).
5. Background measurement: After RBC counting, energize SV28 to flush the sample probe and three-way connector (same with Step 3). Turn on SV34 and SV32. With positive pressure, the diluent in the SCI bath flows into the front and rear baths (as the light green path in the figure). The sheath fluid formed flows through the aperture for background counting. The waste finally flows through the ISW bath to the WC2 bath.
6. Cleaning after measurement: Turn on SV17, SV29, SV28, SV36, SV30, SV25, SV23 and SV24 at the same time of background measurement. With negative pressure, the diluent driven by the diluent syringe (DIL_SYRINGE) and the diluent in the HGB reaction bath flush the sample preparation tube (HGB-T86-T87-T42-T43-T71-T70-T52-T49) and three-way connector C14 (DIL_SYRINGE-T25-T26-T27-T31-T34-T44-T46-T47-C14-T48), and valve SV28, SV 29 (DIL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T38-T39-T41-C12-T43-T71-T70-T52-T49). The waste is drained by the waste pump (PUMP_VAC) through SV28 (T49-T50-T51-T192-T190-T189-T187-T183-T182-PUMP_VAC).

Figure2–55 RBC Measurement Channel

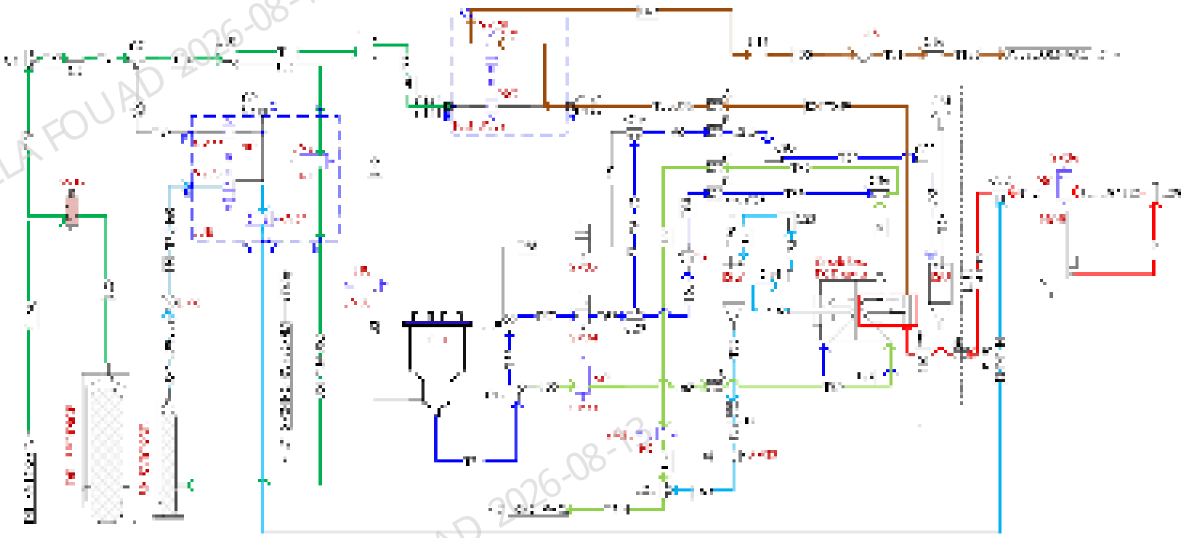
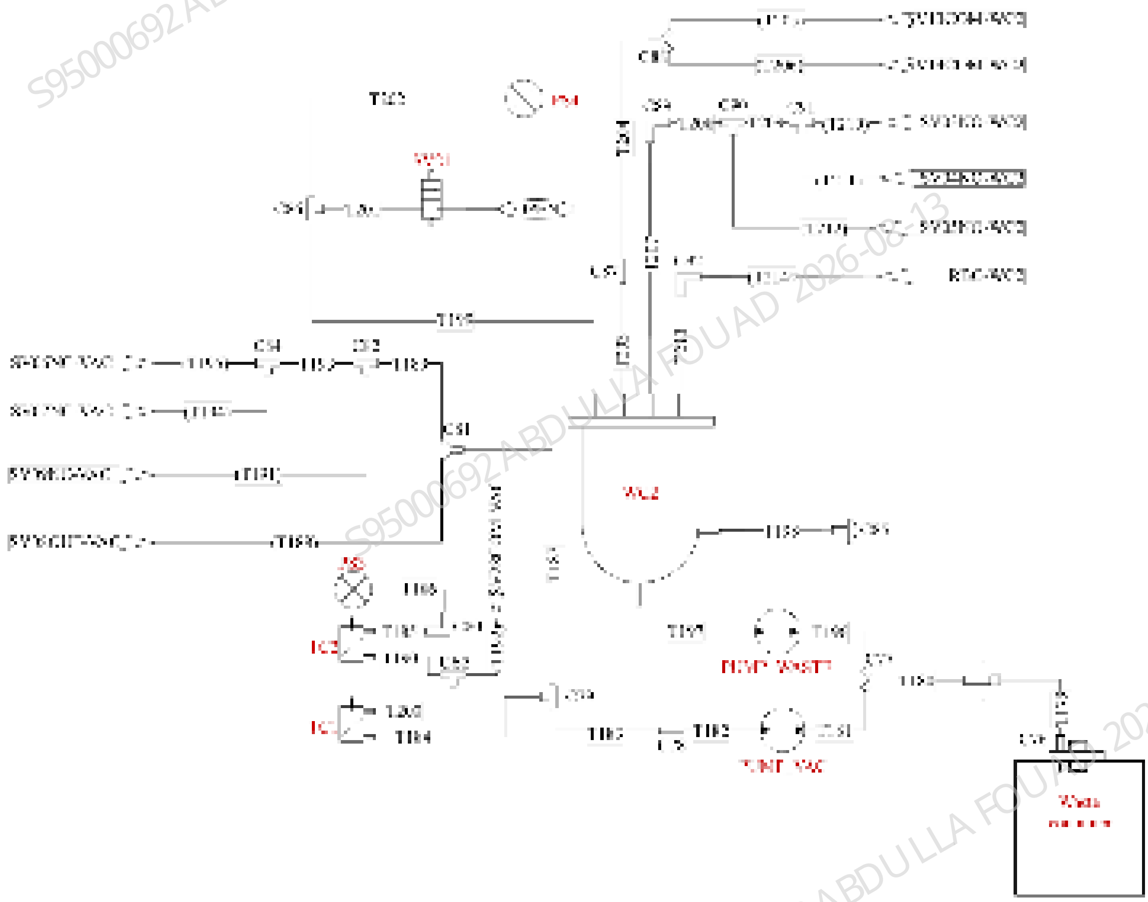


Figure2–56 RBC Measurement Waste Channel



2.3.3.5 ESR Channel

Reagents Used

DS diluent, ESR cleaner

Detection Principle

Dynamic capillary photometer method

Chart Information

N/A

Detection Parameters

ESR

Dilution Ratio

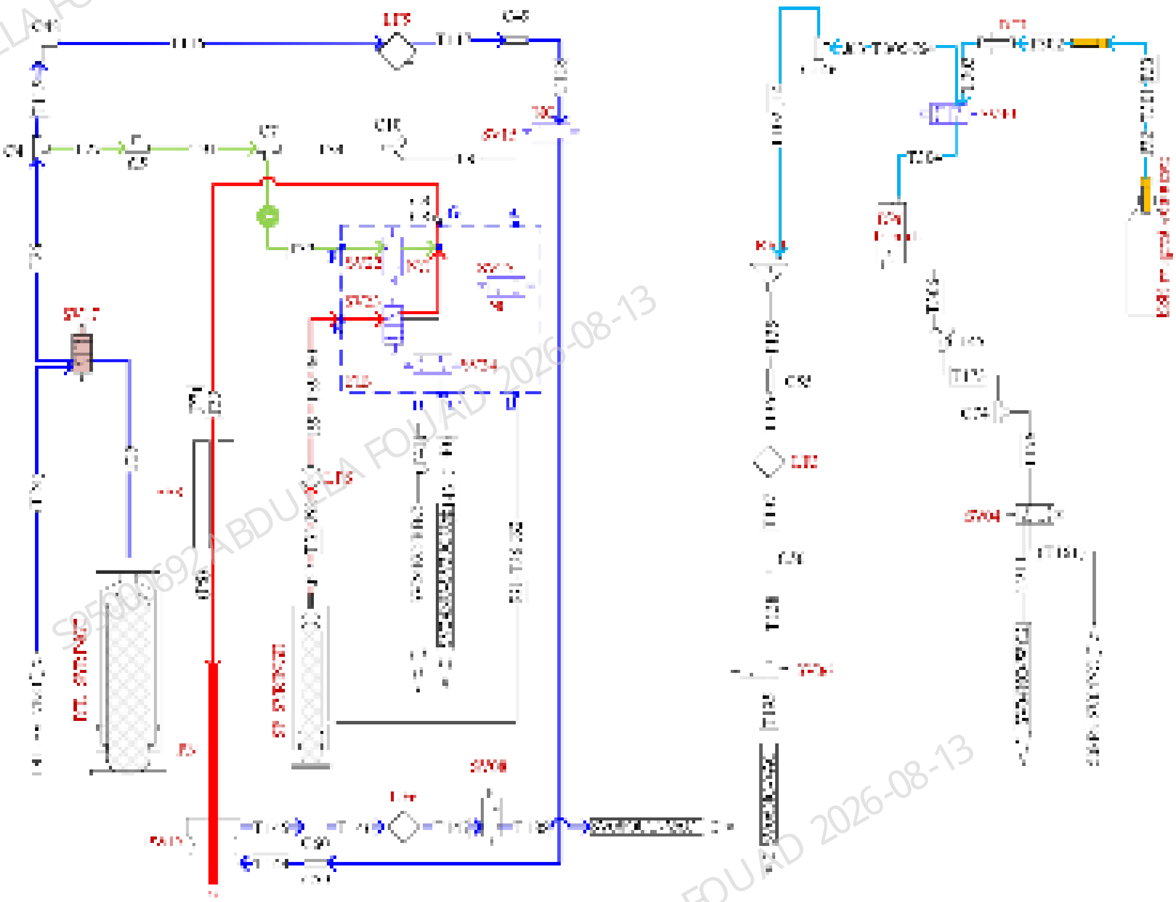
N/A

Process Description

1. Preparation before measurement: The sample injection syringe (SP_SYRINGE) drives the redundant sample out of the sample probe to the wipe PWD (as the red path SP_SYRINGE-T38- T39-SV23-T33-PN-PWD in the figure). Meanwhile, clean the exterior of sample probe and wipe. Energize SV17, SV15, SV08 and waste pump (PUMP_VAC). The diluent syringe (DIL_SYRINGE) drives the diluent for cleaning (as the blue path DIL_SYRINGE-T25-T26-T115-T117-PWD-T145-PUMP_VAC in the figure). Then, the sample injection syringe (SP_SYRINGE) pipettes the sample to be tested into the ESR module through this channel (the position where tube T33 coincides with ESR as shown in the figure).
2. ESR depolymerization and measurement: The sample injection syringe (SP_SYRINGE) repeatedly pipettes and discharges the sample to make it flow back and forth for depolymerization (as the red path in the figure), and then measure ESR.
3. Cleaning after measurement: Energize SV17, SV22, SV08 and waste pump (PUMP_VAC), and the diluent syringe (DIL_SYRINGE) drives the diluent to clean the interior of sample probe (as the light green path DIL_SYRINGE-T25-T26-T27-T31-T32-T33-PN-PWD-T145-PUMP_VAC in the figure). Energize valve SV40 and SV04, and dispense the ESR cleaner to the RET bath through the dosing pump DP6 (shown as the light blue path T301-T302-T303-T305-T102-RET in the figure). Move the sample probe to the RET bath, energize valve SV17 and SV22, and drive the ESR cleaner into T33 with use of DIL syringe. Energize SV17, SV22, SV08 and waste pump (PUMP_VAC), and the diluent syringe (DIL_SYRINGE) drives the diluent to clean the interior of sample probe (as the light green path DIL_SYRINGE-T25-T26-T27-T31-T32-T33-

PN-PWD-T145-PUMP_VAC in the figure). Energize SV17, SV15, SV08 and waste pump (PUMP_VAC). The diluent syringe (DIL_SYRINGE) drives the diluent to clean the exterior of sample probe (as the blue path in the figure).

Figure2–57 ESR Measurement Channel



2.3.4 Detailed Measurement Flow

2.3.4.1 Open-vial Sample Introduction WB Analysis Procedure

Details on Analysis Procedure

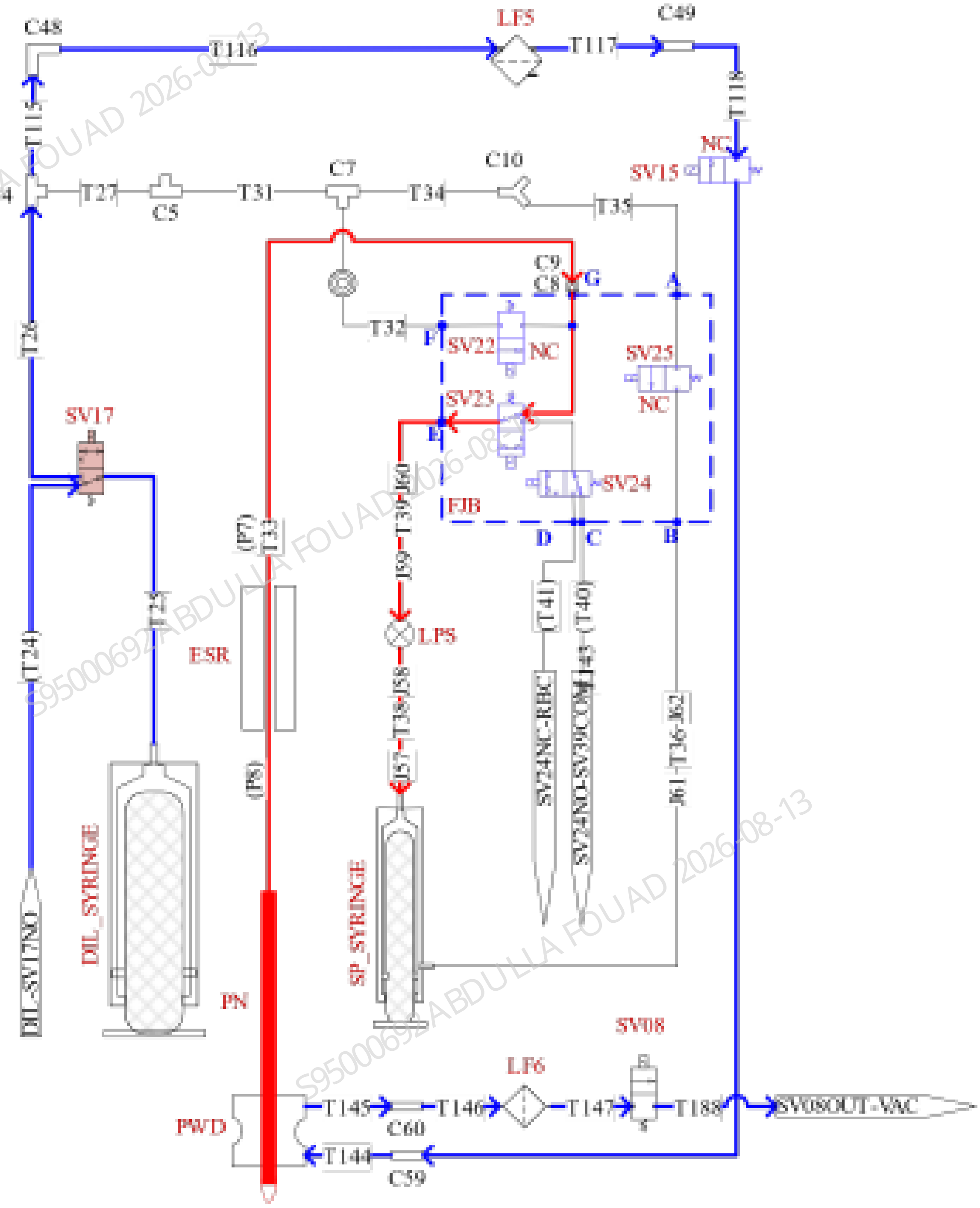
Taking CT-WB CDR+ESR mode for example, the sample probe successively travels to RET bath, HGB bath and DIFF bath for distribution after sampling. Afterwards, each analysis channel completes the processes of incubation, measurement and cleaning. Finally, analyze ESR and clean the sampling channel. The analysis procedure is as below:

Figure2–58 Analysis Procedure

The steps of each procedure are respectively detailed as below.

Sampling Process

1. Pipette the WB sample in a quantitative manner with the sample injection syringe (SP_ SYRINGE).
2. Withdraw the probe after pipetting, turn on SV17, SV15 and SV08. The diluent syringe (DIL_ SYRINGE) drives out the diluent and the waste pump (PUMP_VAC) pumps the cleanser to clean the exterior of sample probe.



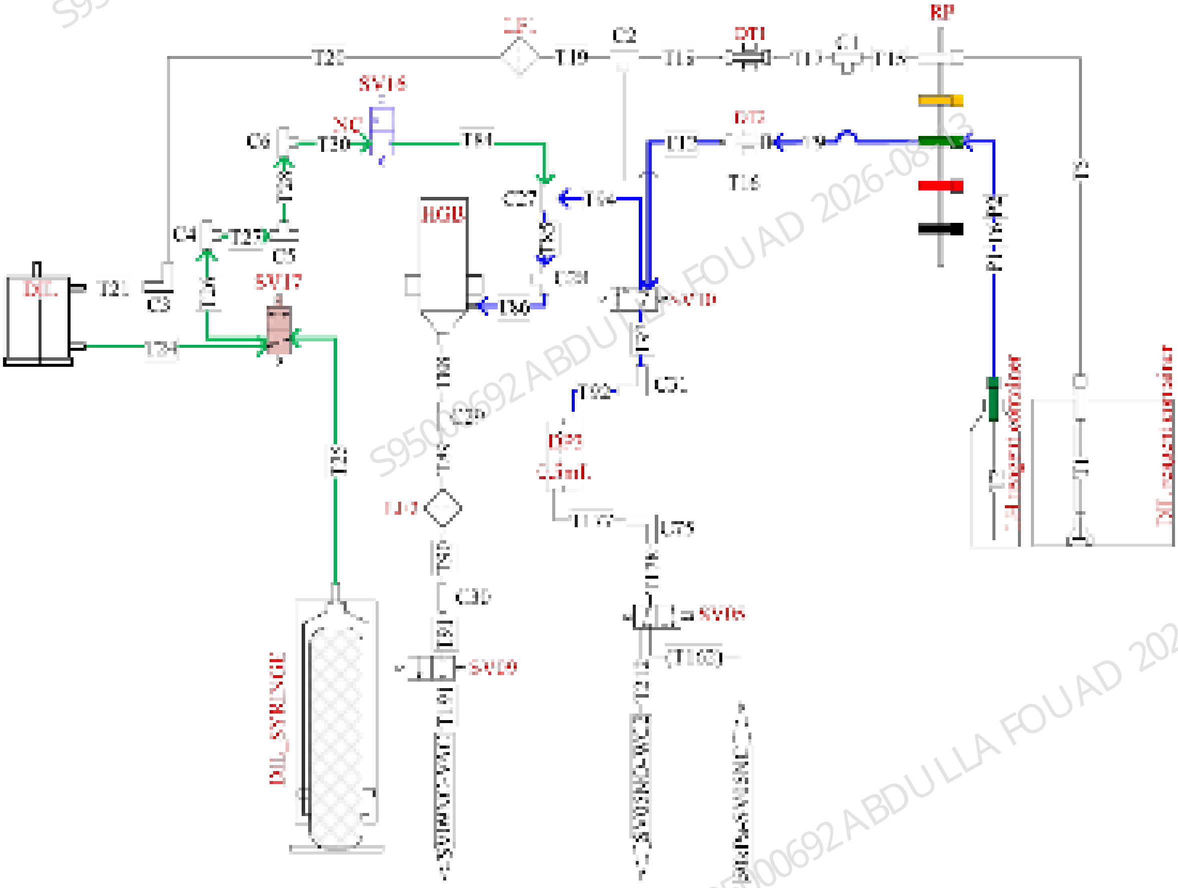
Distributing Procedure

1. The sample probe successively travels to RET bath, HGB bath and DIFF bath. Withdraw the probe after distributing, turn on SV17, SV15 and SV08 while withdrawing the probe. The diluent syringe (DIL_SYRINGE) drives out the diluent and the waste pump (PUMP_VAC) pumps the cleanser to clean the exterior of sample probe.

2. After distributing, initialize the sampling assembly and restore to the upper CT aspiration position (micro tube).

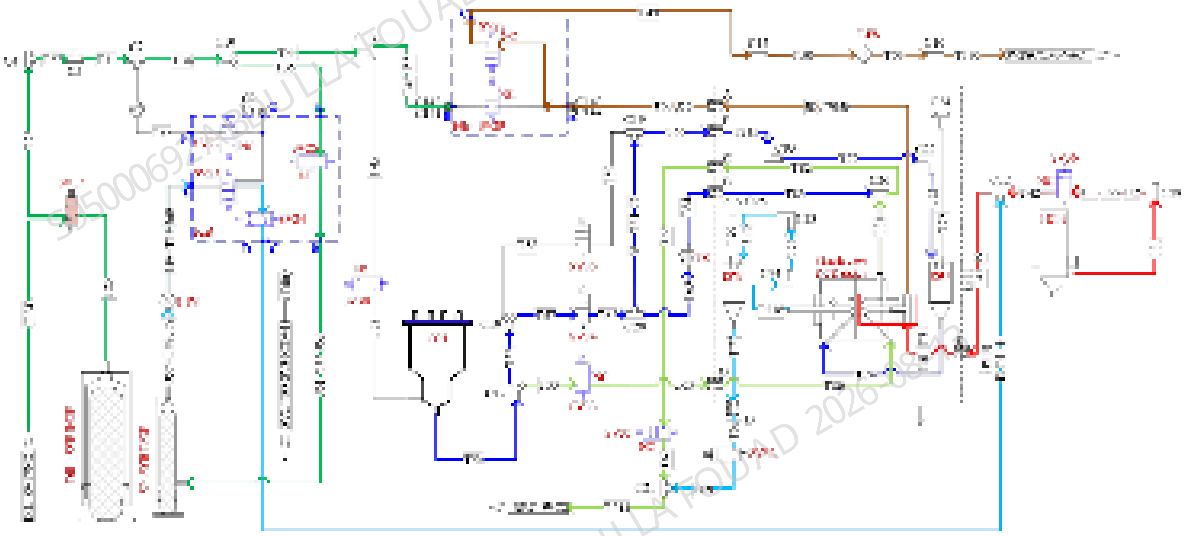
Incubation Analysis and Cleaning Restore Process 1 - HGB Channel

1. Drain the initial volume in the HGB bath. Dispense the diluent to the HGB bath with a diluent syringe (DIL_SYRINGE). Measure the background voltage and then empty the HGB bath. Dispense the diluent to the HGB bath with a diluent syringe (DIL_SYRINGE).
2. After distribution to HGB bath, dispense the diluent to the HGB bath with a diluent syringe (DIL_SYRINGE) for swirling and mixing the sample. After mixing, transfer the sample from the HGB bath to the RBC sample preparation tube with a diluent syringe (DIL_SYRINGE) (refer to the next section "RBC Channel" for details).
3. Dispense the LH lyse to the HGB bath through the dosing pump DP3. The sample reaction starts in the HGB bath.
4. Dwell the sample in the HGB bath and start analysis. After analysis, empty the HGB bath, and dispense the cleanser with a diluent syringe (DIL_SYRINGE) for cleaning and restoring to the status ready for analysis.
5. The HGB channel measurement has been finished (refer to Chapter **2.3.3.3** HGB Channel for the details on measurement process).



Incubation Analysis and Cleaning Restore Process 2 - RBC Channel

1. Transfer the sample from the HGB bath to the RBC sample preparation tube T71 and T43 with a diluent syringe (DIL_SYRINGE).
2. With positive pressure, the SCI bath drives the diluent to clean the front bath of sheath fluid impedance bath and the three-way connector of sample probe (between J10 and J9). It is now ready for analysis.
3. Drive the RBC sample in the sample preparation tube with a diluent syringe (SP_SYRINGE) to the sheath fluid impedance bath.
4. Energize valve 34, dispense sheath fluid to form the sample flow. Start RBC channel counting (measure the RBC and PLT of sheath fluid impedance channel).
5. After counting, clean the sheath fluid impedance bath.
6. The RBC channel measurement has been finished (refer to Chapter **2.3.3.4** RBC/PLT Channel for the details on measurement process).

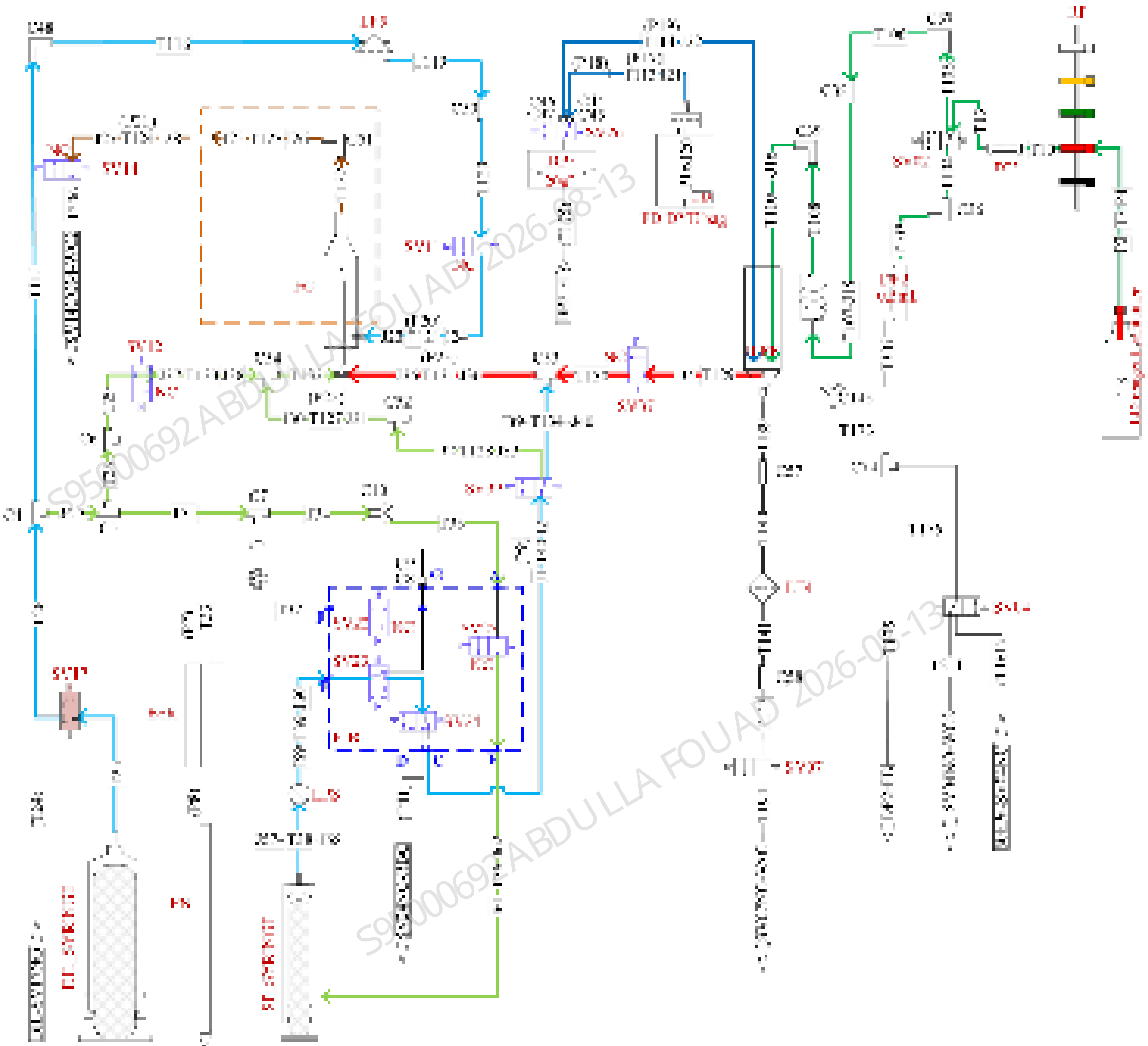


Incubation Analysis and Cleaning Restore Process 3 - DIFF Channel

1. Empty the DIFF bath, dispense the LD lyse with the DP2 dosing pump to the DIFF bath to clean the DIFF bath.
2. Empty the DIFF bath, dispense the LD lyse with the DP2 dosing pump to the DIFF bath as the initial volume. It is ready for analysis.
3. Dispense the FD fluorescent dye to the DIFF bath with the dosing pump DP5.
4. Distribute the sample to the DIFF bath with a sample probe.
5. Dispense the LD lyse to the DIFF bath with the dosing pump DP2 to mix the sample by swirling. The sample in the bath start incubation.
6. Transfer the sample from the DIFF bath to the sample preparation tube T131 with a diluent syringe (DIL_SYRINGE).
7. The diluent syringe (DIL_SYRINGE) dispense the sheath fluid to the flow cell, while the sample injection syringe (SP_SYRINGE) transfers the DIFF sample in the sample preparation tube to

the flow cell. The sample is wrapped by the sheath fluid and form the stable sample flow, which flows through the flow cell and goes through the optical detection

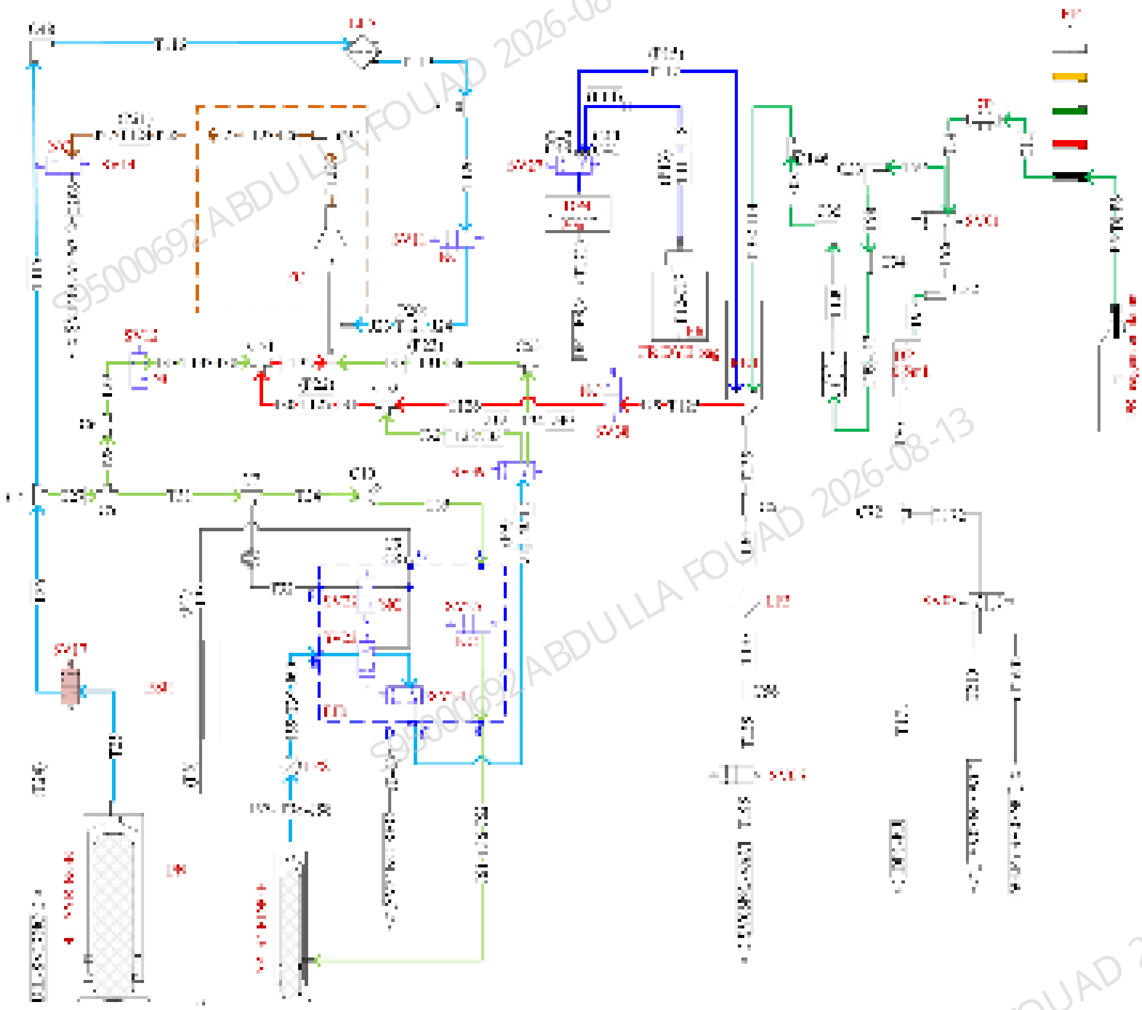
8. Empty the DIFF bath after detection, dispense the cleanser with a diluent syringe (DIL_ SYRINGE) to clean the reaction bath, sample preparation tube and flow cell (refer to Chapter **2.3.3.1** DIFF Channel for details).
9. The DIFF channel measurement has been finished (refer to Chapter **2.3.3.1** DIFF Channel for the details on measurement process).



Incubation Analysis and Cleaning Restore Process 4 - RET Channel

1. Empty the RET bath, dispense the DR diluent with the DP1 dosing pump to the RET bath to clean the RET bath.
2. Empty the RET bath, dispense the DR diluent with the DP1 dosing pump to the RET bath. It is ready for analysis.
3. Dispense the FR fluorescent dye to the RET bath with the dosing pump DP4.
4. Distribute the sample to the RET bath with a sample probe.

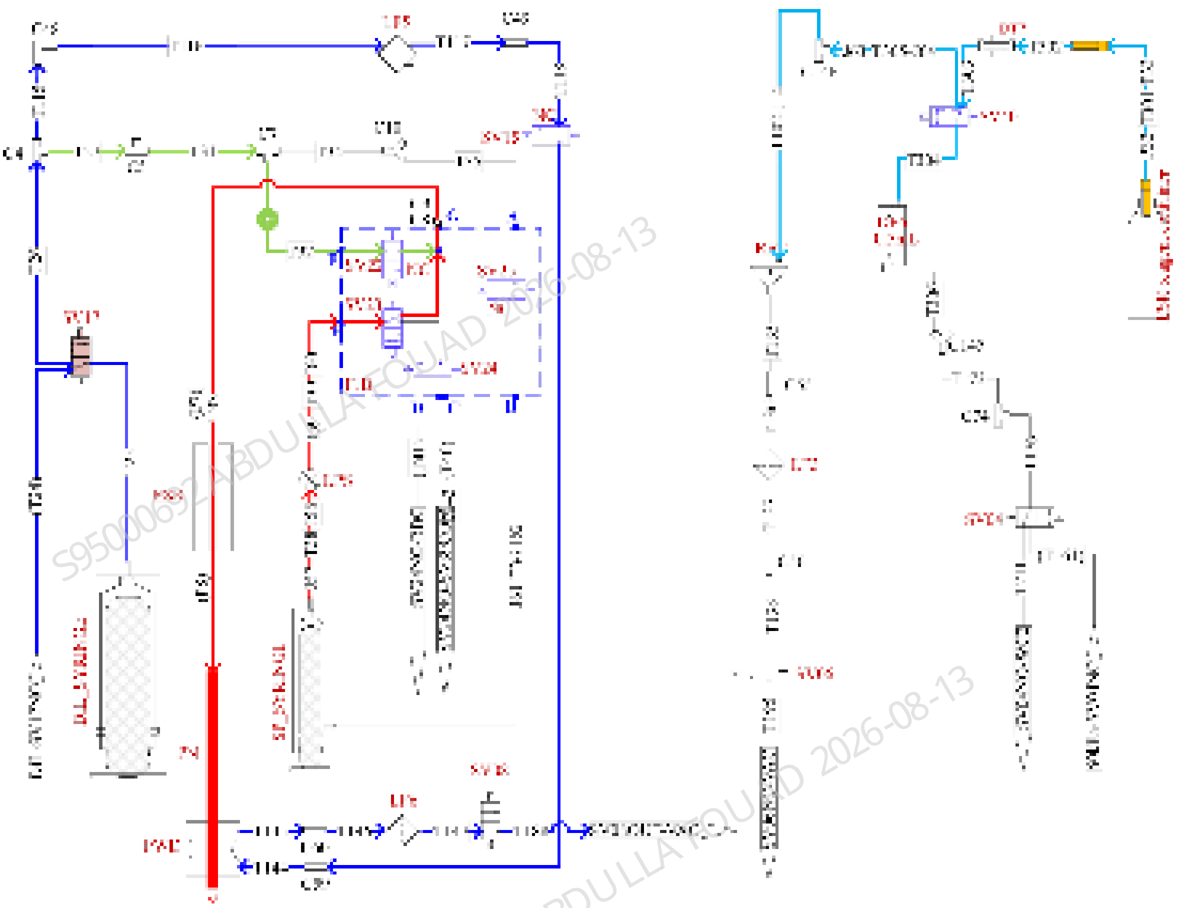
5. Dispense the DR diluent to the RET bath with the dosing pump DP1 to mix the sample by swirling. The sample in the bath start incubation.
6. Transfer the sample from the RET bath to the sample preparation tube with a diluent syringe (DIL_SYRINGE).
7. The diluent syringe (DIL_SYRINGE) dispense the sheath fluid to the flow cell, while the sample injection syringe (SP_SYRINGE) transfers the RET sample in the sample preparation tube to the flow cell. The sample is wrapped by the sheath fluid and form the stable sample flow, which flows through the flow cell and goes through the optical detection
8. Empty the RET bath after detection, dispense the cleanser with a diluent syringe to clean the reaction bath, sample preparation tube and flow cell.
9. The RET channel measurement has been finished (refer to Chapter **2.3.3.2** RET Channel for the details on measurement process).



ESR Measurement and Cleaning

1. Preparation before measurement: use the wipe to recover the redundant sample driven by the sample injection syringe (SP_SYRINGE).

2. ESR measurement and cleaning: measure the ESR background, measure ESR after sample depolymerization with a sample injection syringe (SP_SYRINGE). Use a diluent syringe (DIL_SYRINGE) to clean the interior and exterior of the sample probe. Restore the sampling assembly to the status ready for analysis.
3. The ESR channel measurement has been finished (refer to Chapter **2.3.3.5** ESR Channel for the details on measurement process).



2.3.4.2 Open-vial Sample Introduction Predilution Analysis Procedure

Predilution analysis sequences and WB measurement sequences share the similar main procedure.

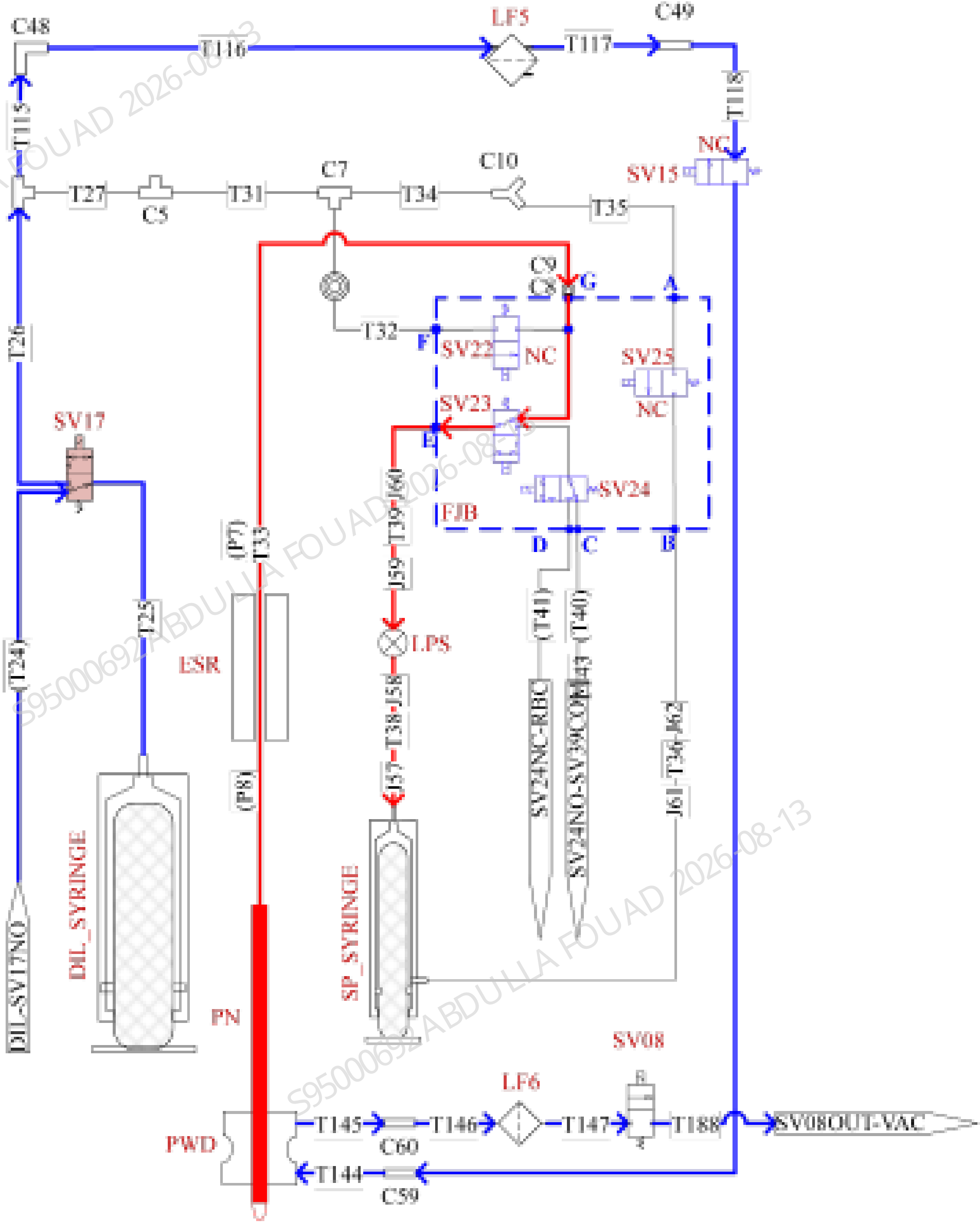
The differences in the two modes are:

- The dilution ratio and analysis duration are different.
- The sample for predilution mode is diluted at a specified ratio before analysis. The sample for WB mode is the venous blood not diluted.

Dispensing Diluent Procedure

The dispensing diluent procedure for predilution is as follows:

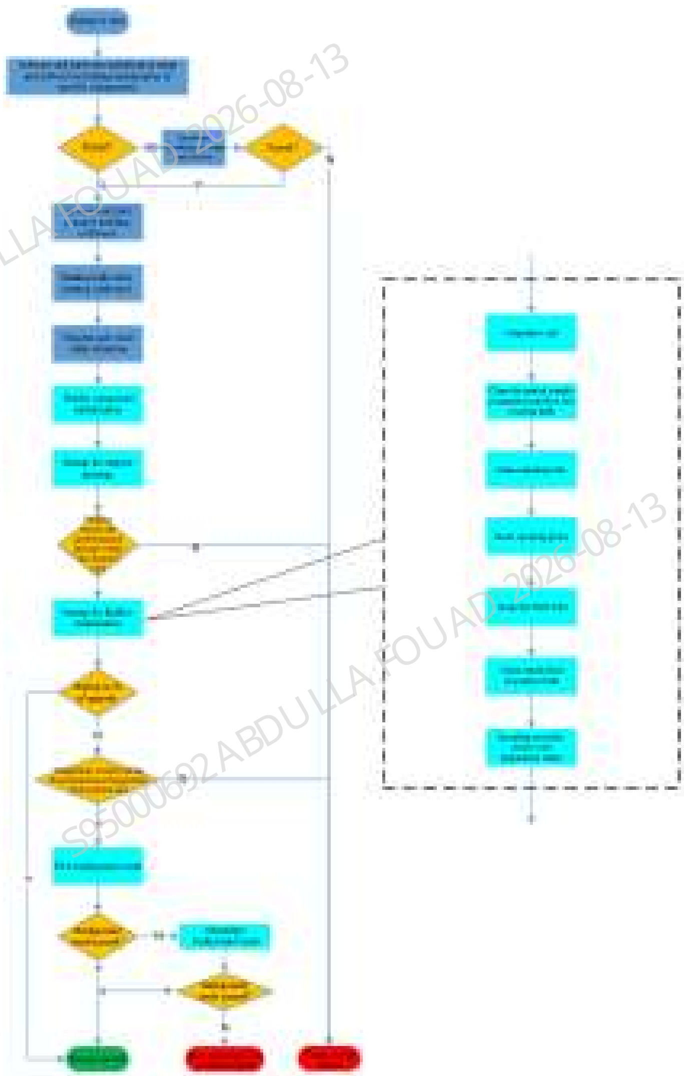
1. Get into the dispensing diluent state: The diluent syringe (DIL_SYRINGE) drives the diluent to clean the exterior of sampling probe (DIL_SYRINGE-T25-T26-T115-T116-T117-T118-T143-T144-PWD-T145-T146-T147-T188-PUMP_VAC), and initialize the sample injection syringe (SP_SYRINGE).
2. Dispensing diluent: Close the autoloader door (closed-sampling model), and have the sample probe in place. Energize SV23, SV25, SV17 and SV15, and the sample injection syringe (SP_SYRINGE) aspirates the liquid from T144 tube (SP_SYRINGE-T36-T35-T34-T31-T27-T115-T116-T117-T118-T143-T144). Power off SV23, SV25, SV17 and SV15, and the sample injection syringe (SP_SYRINGE) drives 100uL diluent through the sampling probe (SP_SYRINGE-T38-T39-T33-PN). Energize SV23, SV25, SV17 and SV15, and initialize the sample injection syringe (SP_SYRINGE). Then the diluent syringe (DIL_SYRINGE) drives the diluent to refill T143 with the diluent (DIL_SYRINGE-T25-T26-T115-T116-T117-T118-T143-T144-PWD-T145-T146-T147-T188-PUMP_VAC). One cycle of dispensing is completed. Then have the sample probe in place, and open the autoloader door (closed-sampling model).
3. Exit the dispensing diluent state: The diluent syringe (DIL_SYRINGE) drives the diluent to clean the exterior of sampling probe (DIL_SYRINGE-T25-T26-T115-T116-T117-T118-T143-T144-PWD-T145-T146-T147-T188-PUMP_VAC). Initialize the sample injection syringe (SP_SYRINGE) after air aspiration through the sampling probe (SP_SYRINGE-T38-T39-T33-PN), and initialize the diluent syringe (DIL_SYRINGE) after air aspiration through T144 (DIL_SYRINGE-T25-T26-T115-T116-T117-T118-T143-T144).



2.3.4.3 Startup/Shutdown Procedure

Startup Procedure

The startup flowchart is shown as below:

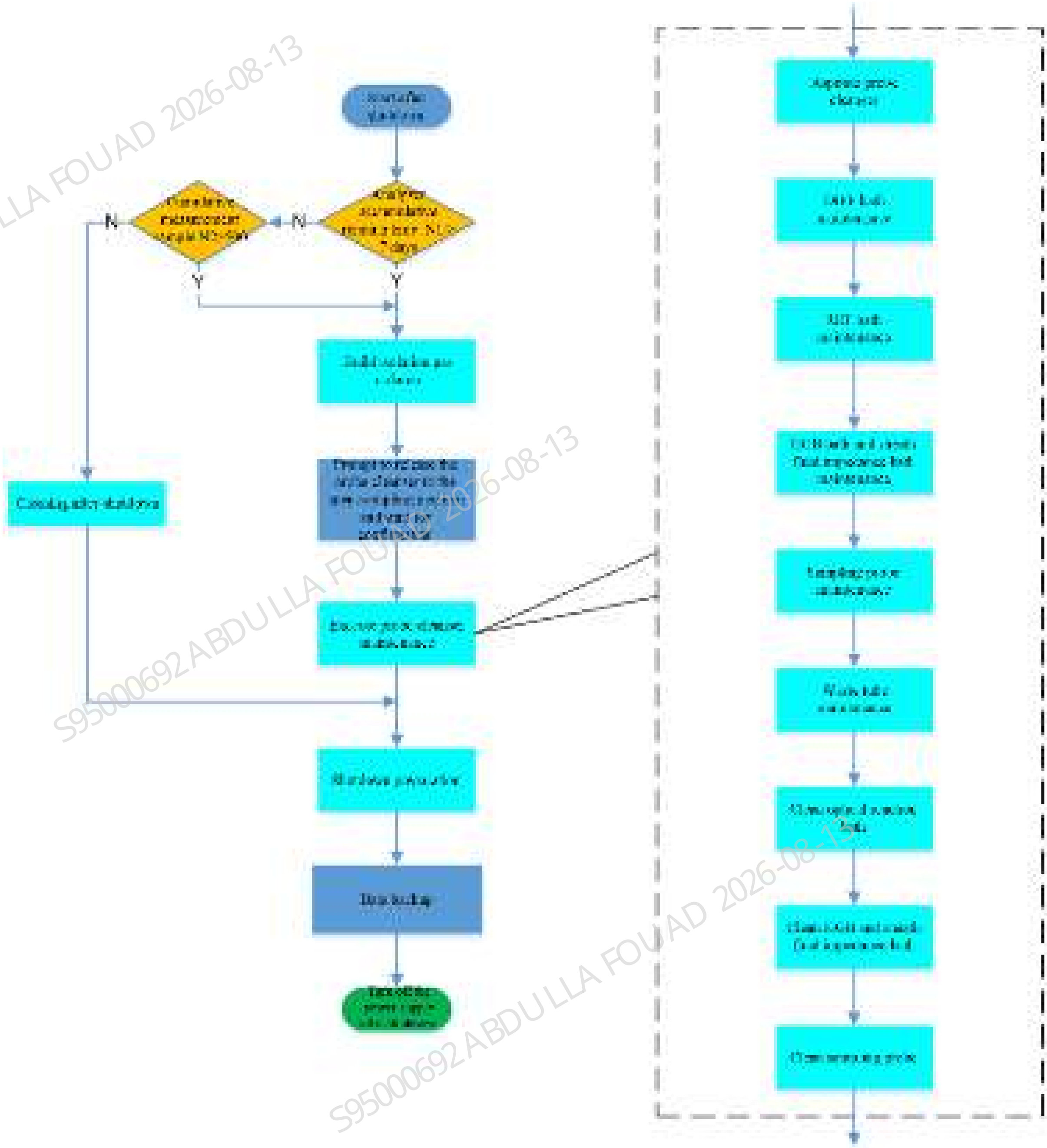


The startup process can be briefly introduced as below:

1. When the power key is pressed to turn on the analyzer, the hardware and software system will start initialization and self-test.
2. After the self-test is passed, the gas source starts to rise pressure, the preheating bath starts heating, and the fluidic components start self-test.
3. After pressure rise, start the fluidics maintenance, clean the analyzer (refer to **2.3.4.4** Analyzer Cleaning Procedure for the procedure), and clean the fluidic components.
4. Run the background test and the analyzer is successfully started after passing the test.

Shutdown Procedure

The shutdown flowchart is shown as below:



The shutdown procedure can be briefly introduced as below:

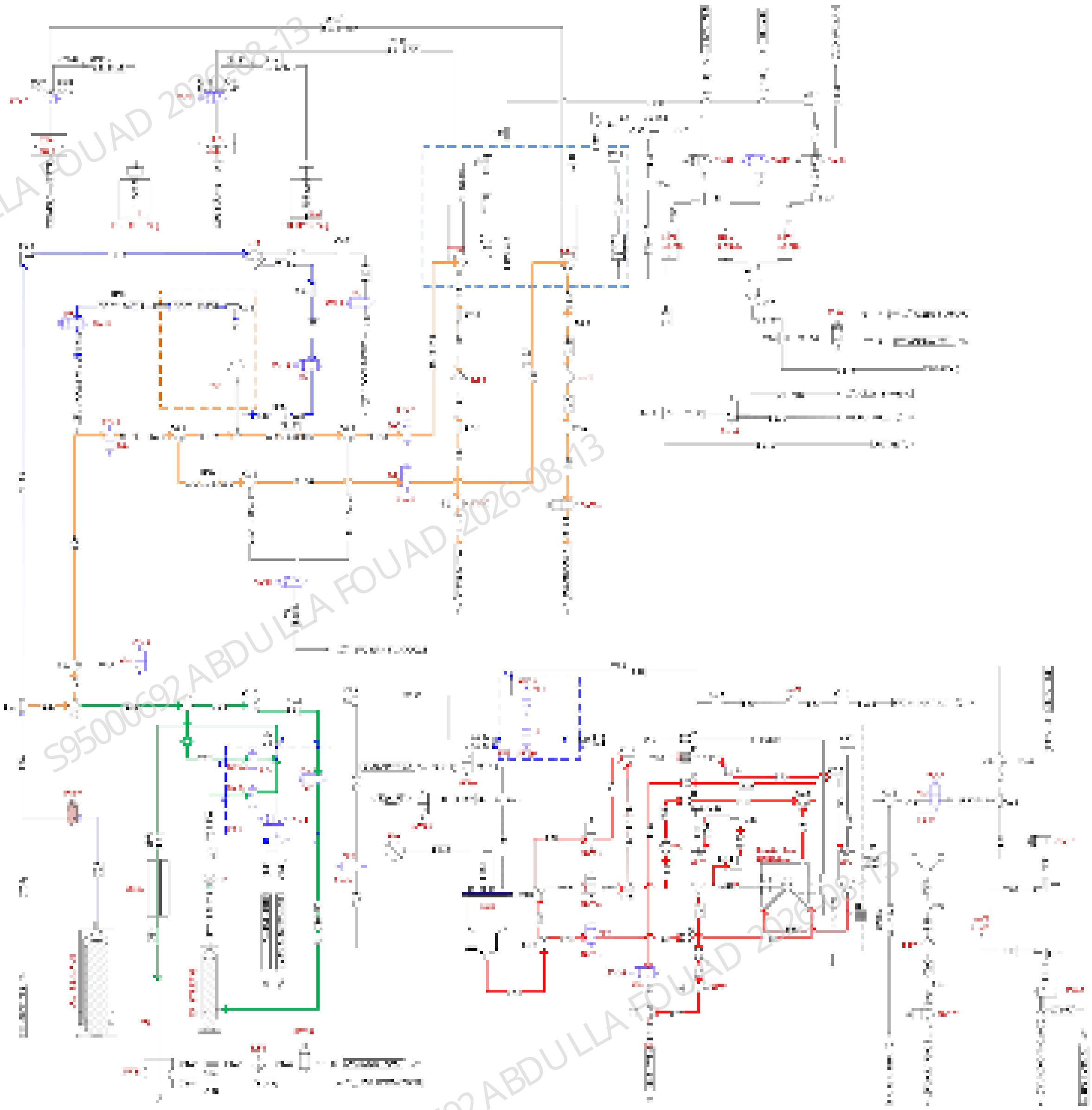
1. Click the shutdown button, and the analyzer can judge the necessity of probe cleanser maintenance.
2. Run shutdown cleaning if it is not qualified for probe cleanser maintenance.
3. Run probe cleanser maintenance if it is qualified for probe cleanser maintenance. The sample probe builds an isolation gas column, suggests to pipette probe cleanser. After preparation, press the sampling key to pipette the fluid and run the probe cleanser maintenance procedure (refer to **2.3.4.5** Probe Cleanser Maintenance Procedure for the procedure).
4. After shutdown cleaning or shutdown probe cleanser maintenance, prepare for shutdown and back up the data.

5. Power off and shut down the analyzer.

2.3.4.4 Analyzer Cleaning Procedure

The analyzer cleaning procedure is as follows (the procedure for exiting deep standby status is the same as the analyzer cleaning procedure):

1. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the flow cell (as the blue path DIL_SYRINGE-T25-T26-T115-T116-T119-T120-T121-FC-T122-T123-T124-T206 in the figure).
2. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the optical sample preparation channel and DIFF reaction bath (as the orange path DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T129-DIFF-T139-T140-T141-T142-T194 in the figure), RET reaction bath (as the orange path DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T127-T126-T125-RET-T135-T136-T137-T138-T195 in the figure).
3. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the probe cleanser storage tube (DIL_SYRINGE-T25-T26-T27-T31-T32-T33-PN), sample injection syringe (SP_SYRINGE) tube, manifold and sampling probe (as the dark green path DIL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T38-T39-T33-PN in the figure).
4. The diluent syringe (DIL_SYRINGE) drives the diluent to clean the HGB bath (as the purple path DIL_SYRINGE-T25-T26-T27-T28-T30-T84-T85-T86-HGB-T88-T89-T90-T91-T191 in the figure).
5. With positive pressure, SCI bath drives the diluent to clean the front bath of sheath fluid impedance bath (as the red path SCI-T54-T55-T65-T80-T81-T82-T66-T67-T214 in the figure SCI-T54-T56-T57-T62-T63-T64-T83-T82-T66-T67-T214.) rear bath of sheath fluid impedance bath (as the red path SCI-T54-T56-T58-T59-T60-T215-T72-ISU-T74-T216-T75-T76-T77-T68-T69-T214 in the figure)
6. Restore the sampling assembly to the status ready for analysis.
7. Analyzer cleaning completed.



2.3.4.5 Probe Cleanser Maintenance Procedure

The probe cleanser maintenance procedure is as below:

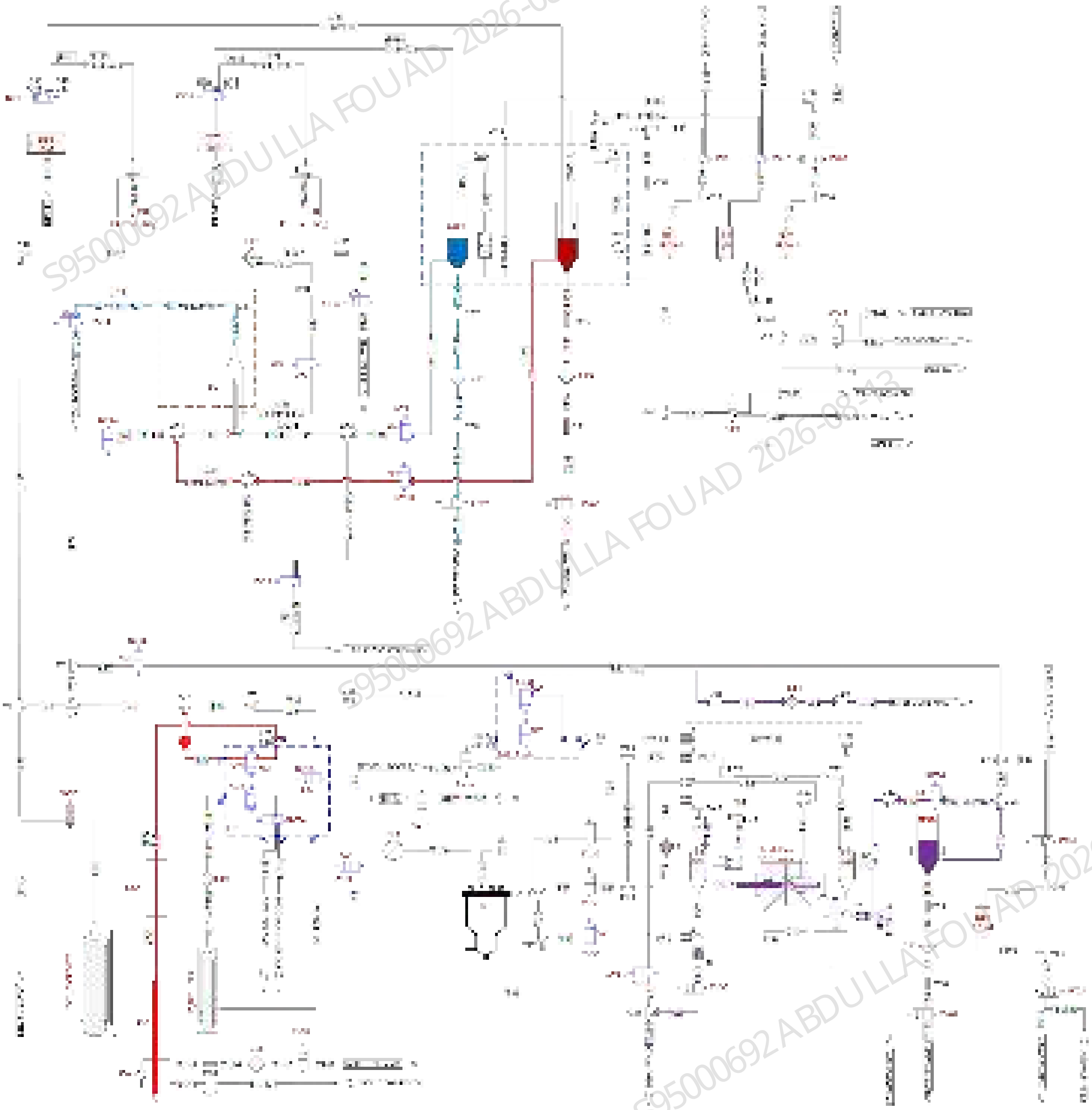
1. Probe cleanser pipetting: Pipette 2.2 mL probe cleanser (as the red path PN-T33-T32 in the figure) with diluent syringe (DIL_SYRINGE) and sample probe (PN), during which the valves SV17 and SV22 shall be energized. After pipetting the cleanser, clean the exterior of sample probe by using diluent syringe (DIL_SYRINGE) and the valves SV15 and SV17 (energized) to add the cleanser. Meanwhile, waste pump PUMP_VAC and SV08 are energized to pump the exterior cleaning waste.
2. DIFF bath maintenance: Move the sampling probe to DIFF bath, energize SV17 and SV22, and dispense the probe cleanser of certain volume in the sample probe to the reaction bath by using diluent syringe (DIL_SYRINGE) for the purpose of RET bath maintenance. Then transfer

some probe cleanser to the sample preparation tube T129-T130-T131 by using diluent syringe (DIL_SYRINGE), during which SV17, SV12 and SV37 are energized. Based on the negative pressure of WC2 bath, transfer some probe cleanser in sample preparation tube to flow cell, during which SV14 and SV37 are energized. The maintenance path is shown as the blue arrows T129-T130-T131-T122-T123-124 in the figure.

3. RET bath maintenance: RET bath maintenance is required when the instrument has conducted more than 200 tests under RET mode. Move the sampling probe to RET bath, energize SV17 and SV22, and dispense the probe cleanser of certain volume in the sample probe to the reaction bath by using diluent syringe (DIL_SYRINGE) for the purpose of RET bath maintenance. Then transfer some probe cleanser to the sample preparation tube T125-T126-T127 by using diluent syringe (DIL_SYRINGE), during which SV17, SV12 and SV38 are energized. The path is shown as the red arrows T125-T126-T127-T132-T123-124 in the figure.
4. HGB and sheath fluid impedance bath maintenance: Move the sampling probe to HGB position, energize SV17 and SV22, and dispense the probe cleanser of certain volume in the sample probe to the HGB bath by using diluent syringe (DIL_SYRINGE) for the purpose of HGB bath maintenance. Energize SV17, SV29 and SV36, and use the diluent syringe (DIL_SYRINGE) to dispense the probe cleanser of certain volume from HGB bath to RBC sample preparation tube by T87-T42-T43-T71-T70-T52. Then energize SV23, SV25, SV29 and SV30, and use SP syringe to drive the probe cleanser of certain volume from RBC sample preparation tube to sheath fluid impedance bath for the purpose of sheath fluid impedance bath maintenance. The path is shown as the purple arrows T87-T42-T43-T71-T70-T52-T49 and T87-T42-T43-T71-sample probe-T216-T75-T76-T77-T68-T230-T69 in the figure
5. Sample probe maintenance: Move the sample probe to the DIFF bath, and energize SV17 and SV22. Pipette a small volume of probe cleanser from the DIFF bath by using diluent syringe (DIL_SYRINGE). Immerse the interior of sample probe for maintenance.
6. Reaction bath waste tube maintenance: Energize SV17, SV29 and SV36, and transfer the probe cleanser from the flow cell to the flow cell waste tube's immersion waste tube by using diluent syringe (DIL_SYRINGE). Energize SV07 and PUMP_VAC to build the negative-pressure environment, and transfer the probe cleanser from the DIFF bath to the DIFF waste tube T139-T140-T141-T142, waste filter LF4 and waste valve SV07 for immersion. Energize SV09 and PUMP_VAC to build the negative-pressure environment, and transfer the probe cleanser from the HGB bath to the HGB waste tube T88-T89-T90-T91, waste filter LF2 and waste valve SV09 for immersion; Energize SV28, SV36 and PUMP_VAC to build the negative-pressure environment, and transfer the probe cleanser from the RBC sample preparation channel to SV28 and waste filter LF8 for immersion.
7. Optical reaction bath cleaning: Turn on the waste pump PUMP_VAC, energize SV07 and SV06 successively, and clear the probe cleanser in DIFF and RET optical reaction baths respectively. Then clean the DIFF bath with cleanser by using diluent syringe (DIL_SYRINGE) jointly with SV17, SV12 and SV37 (energized), and clean the RET bath with cleanser by using diluent syringe (DIL_SYRINGE) jointly with SV17, SV12 and SV38 (energized). And then, turn on PUMP_VAC, energize SV07 and SV06 successively, clear the waste in DIFF and RET baths, and dispense the initial volume to restore the test preparation status by using diluent syringe

(DIL_SYRINGE) jointly with SV17, SV12, SV37 and SV38. Clear the probe cleanser in the flow cell by using diluent syringe (DIL_SYRINGE) jointly with SV17, SV11, SV12 and SV14 (energized), and dispense diluent for flushing.

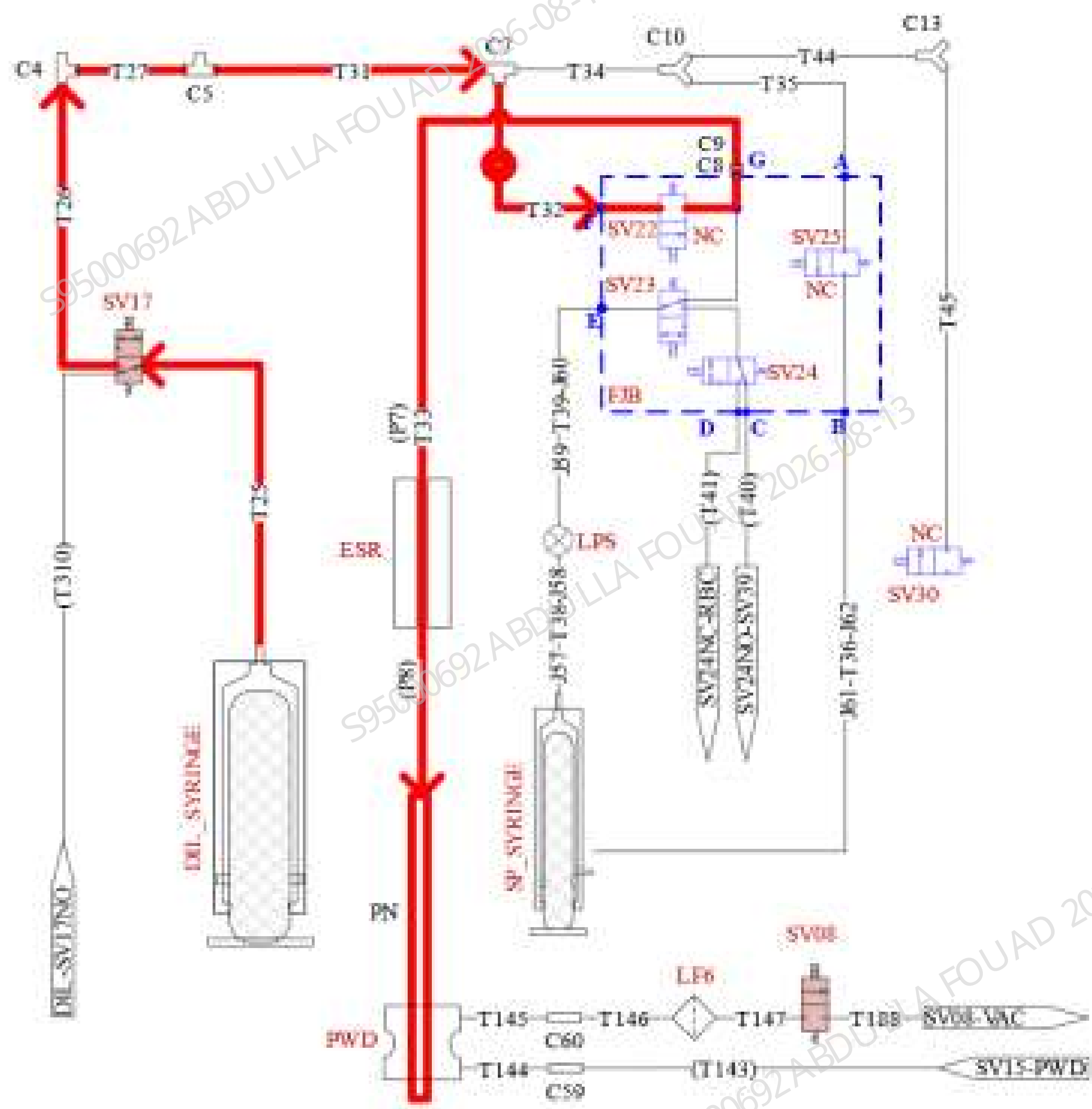
8. HGB and sheath fluid impedance bath cleaning: Turn on the waste pump PUMP_VAC, energize SV09, clear the probe cleanser in HGB bath, and clean the HGB bath by using diluent syringe (DIL_SYRINGE) jointly with SV17 and SV16. After cleaning, turn on the waste pump PUMP_VAC and SV09 for emptying. Turn on the waste pump PUMP_VAC, energize SV36, SV28 and SV32, and clear the probe cleanser in the sheath fluid impedance bath. Then clean the sheath fluid impedance bath by using SCI bath jointly with SV31, SV33, SV34 and SV35 (energized), diluent syringe (DIL_SYRINGE) jointly with SV17, SV25, SV23 and SV24 (energized). Finally, dispense the initial volume to the HGB bath by using diluent syringe (DIL_SYRINGE) jointly with SV17 and SV16 (energized) to restore the test preparation status.



2.3.4.6 Fluidic initialization process

1. Initializing pressure - PC (+50 kPa) - SCI (+40kpa) bath pressurization.
2. **Initializing syringe**: SP_SYRINGE/DIL_SYRINGE/LC_SYRINGE/CRP_SYRINGE
3. **Initializing sample probe assembly (clean the sample probe interior and exterior with DIL syringe)** (diluent syringe DIL_SYRINGE-T25-T26-T27-T31-T32-T33-PN).
4. Wipe establishes the separation bubble.
5. Empty the waste bath.
6. Restore the sampling assembly to the status ready for analysis.
7. Complete the initialization.

Figure2–59 Diagram of Initializing Sample Probe Assembly



2.3.4.7 Pack-up Startup Procedure

Pack-up Startup Procedure

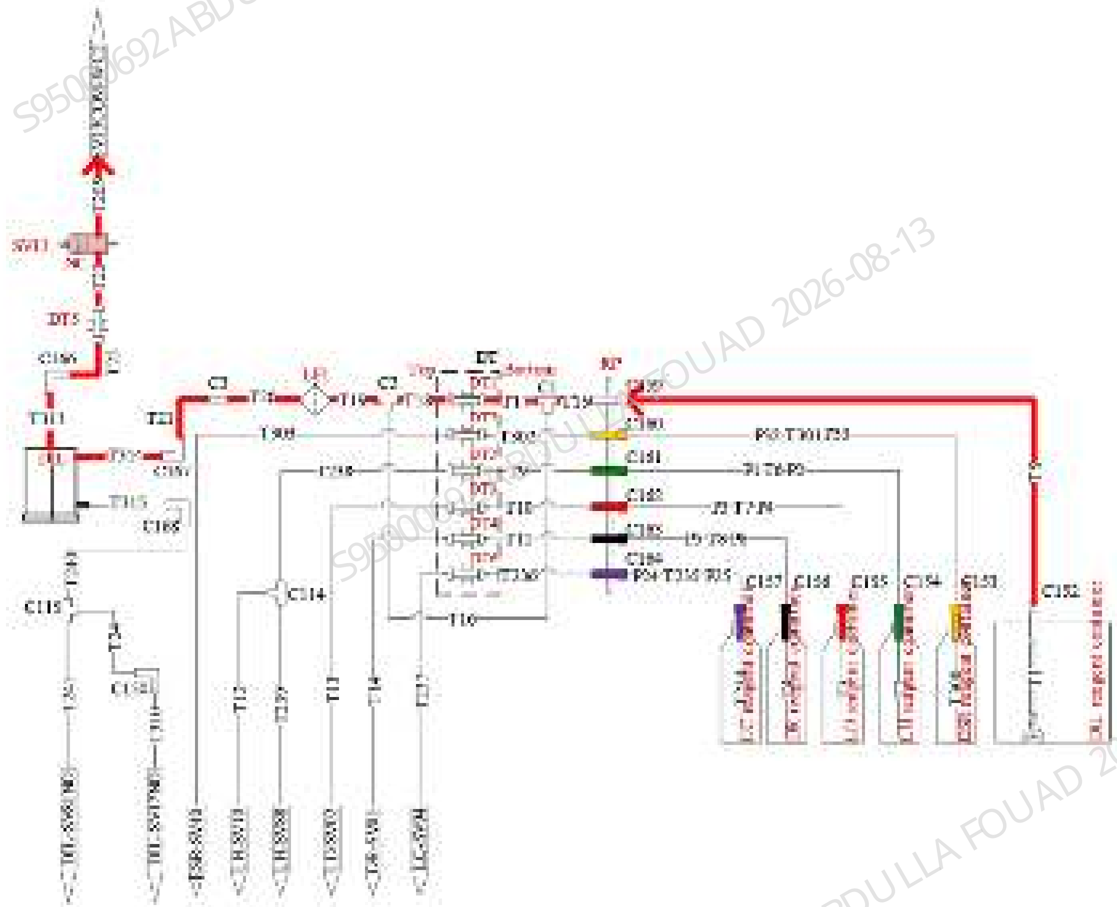
Initializing Pack−up Startup Fluidic Components

1. Initializing pressure - PC (+50 kPa) - SCI (+40kpa) bath pressurization
2. Initializing syringe: SP_SYRINGE/DIL_SYRINGE/LC_SYRINGE/CRP_SYRINGE
3. Initializing sample probe assembly
4. Wipe establishes the separation bubble.
5. Empty the waste bath.
6. Restore the sampling assembly to the status ready for analysis.
7. Complete the initialization.

DIL Bath

Priming DIL bath: DS reagent tank-T5-T15-T18/T16-T19-T20-T314-T313-T22-T23-T205-T204-T203 (as indicated by the red path in the figure)

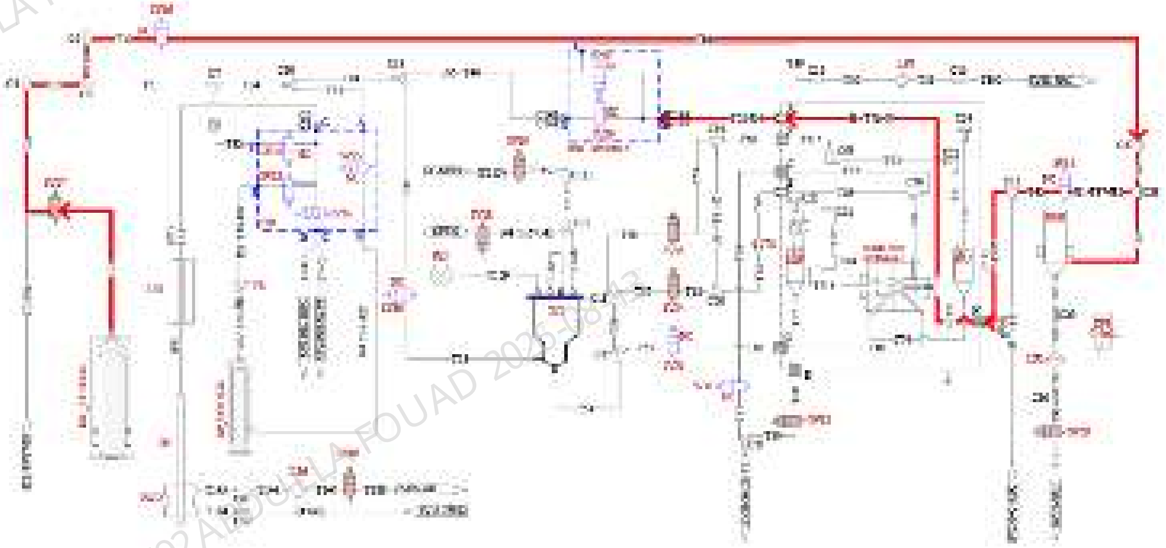
Figure2–60 Diagram of Priming DIL Bath



Diluent Priming and Cleaning

1. Priming DIL- HGB tubing: Diluent syringe (DIL_SYRINGE) pushes out the diluent to prime the HGB tubing: DIL_SYRINGE-T25-T26-T28-T30-T84-T85-T86-T87-T42-T43-T71-T70-T52 (as indicated by the red path in the figure).

Figure2–61 Diagram of Priming DIL-HGB Tubing



2. Priming RBC sample preparation tubing
   Prime the SCI (DIL_SYRINGE-T25-T26-T27-T31-T34-T44-T45-T53-SCI); the positive pressure pushes out the front bath of sheath fluid impedance bath (as indicated by the red path in the **Figure2–62**: SCI-T54-T55-T65-T80-T81-T82-T66-T67-T214) SCI-T54-T56-T57-T62-T63-T64-T83-T82-T66-T67-T214.) and back bath (as indicated by the red path in the **Figure2–62**: SCI-T54-T56-T58-T59-T60-T215-T72-ISU-T74-T216-T75-T76-T77-T68-T69-T214), while the syringe and negative pressure prime the RBC sample tubing (as indicated by the green path in **Figure2–63**: DIL_SYRINGE-T25-T26-T27-T31-T34-T44-T46-T49-T50-T51-T192 and by the purple path in **Figure2–63**: HGB-T87-T42-T43-T71-T70-T56- T49-T50-T51-T192)

Figure2–62 Diagram 1: Priming RBC Sample Preparation Tubing

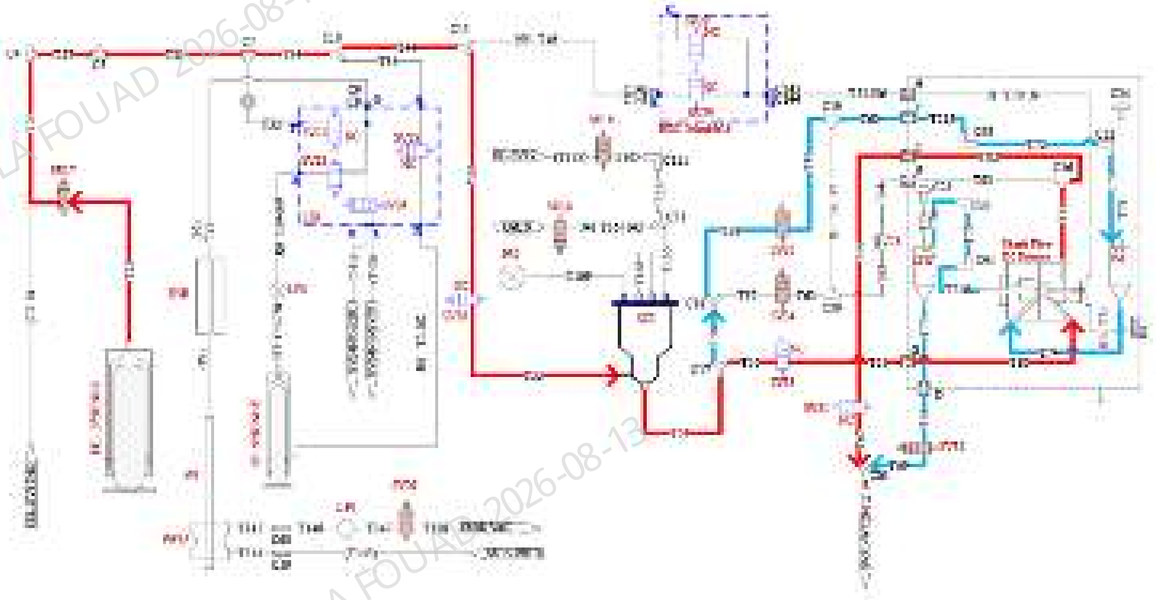
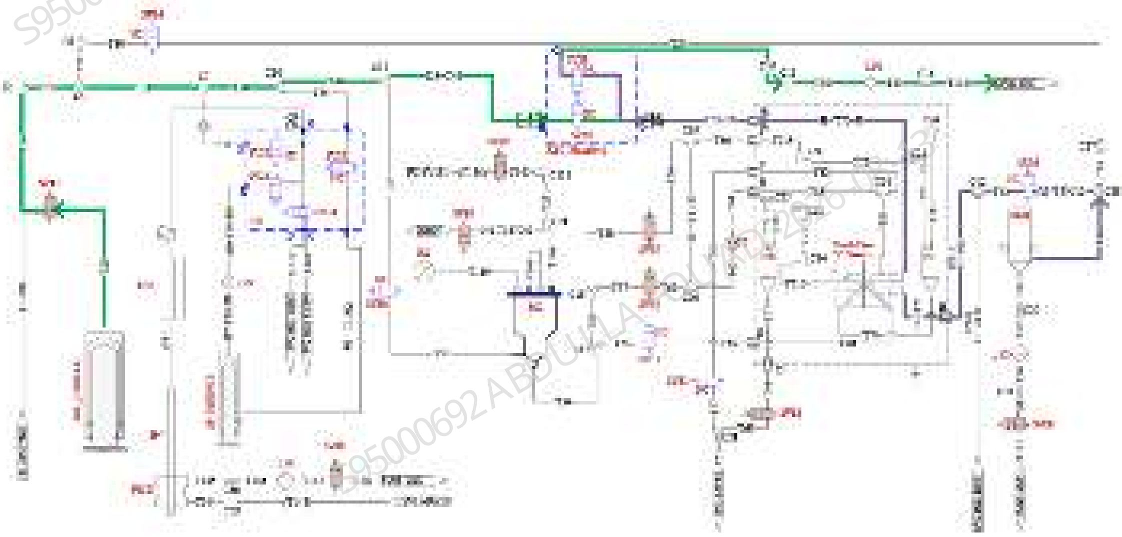


Figure2–63 Diagram 2: Priming RBC Sample Preparation Tubing



3. Priming flow cell

1) Priming sheath fluid-wipe channel: DIL_SYRINGE-T25-T26-T115-T116-T117-T118-T143-T144-T145-T146-T147-T188 (as indicated by the red path in **Figure2–64**)

2) Priming sheath fluid channel: DIL_SYRINGE-T25-T26-T115-T116-T119-T120-T121-flow cell-T122-T123-T124-206 (as indicated by the red-blue-green paths in **Figure2–64**)

3) Priming sample probe: DIL_SYRINGE-T25-T26-T115-T116-T119-T120-T121-flow cell-T131-T130-T129-DIFF bath-T139-T140-T141-T142-T194 (as indicated by the red-blue paths in **Figure2–64**)

4) Priming T-joint of flow cell: DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T129- DIFF bath-T139-T140-T141-T142-T194 (as indicated by the orange path in **Figure2–65**)

Figure2–64 Diagram 1 of Priming Flow Cell

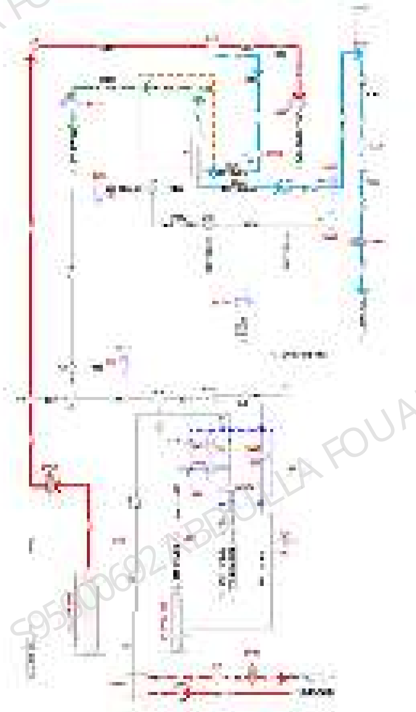
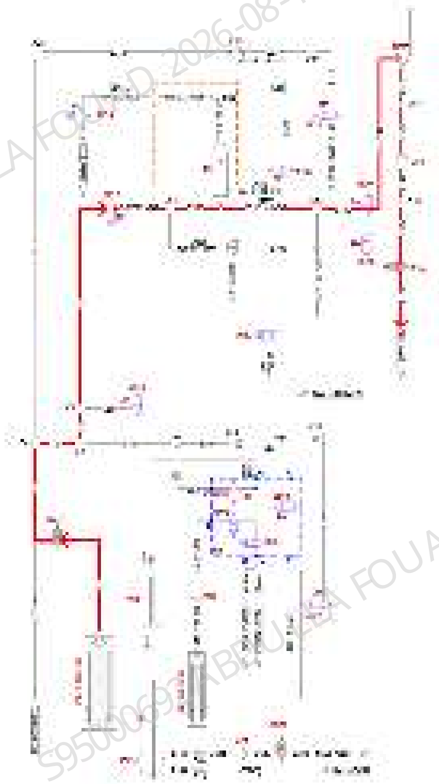


Figure2–65 Diagram 2 of Priming Flow Cell



4. DIFF Bath

1) Priming DIFF sample preparation tubing: DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T129- DIFF bath-T139-T140-T141-T142-T194 (as indicated by the red path in **Figure2–66**)

2) DIFF syringe tubing: DIL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T38-T39-T40-T134-T130-T129-DIFF bath-T139-T140-T141-T142-T194 (as indicated by the blue path in the **Figure2–67**)

Figure2–66 Diagram 1 of Priming DIFF

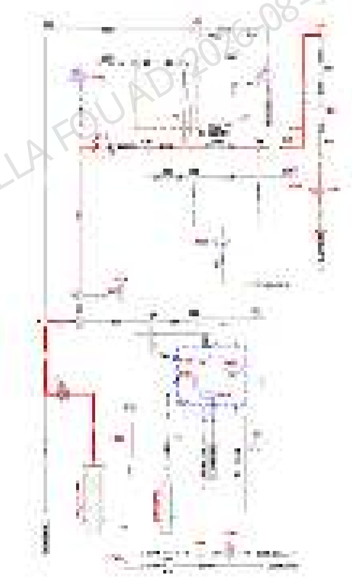
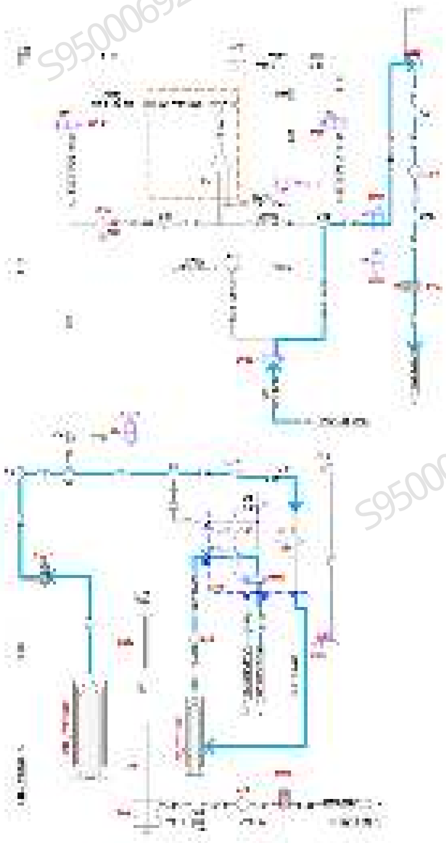


Figure2–67 Diagram 2 of Priming DIFF



5. RET Bath

1) Priming RET sample preparation tubing: DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T127-T126-T125- RET bath-T135-T136-T137-T138-T195 (as indicated by the red path in **Figure2–68**)

2) Priming T-joint of RET sample preparation tubing: DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T129-DIFF bath (as indicated by the red-blue paths in **Figure2–68**)

3) RET syringe tubing: DIL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T38-T39-T40-T128-T126-T125-RET bath-T135-T136-T137-T138-T195 (as indicated by the orange path in **Figure2–69**)

Figure2–68 Diagram 1 of Priming RET

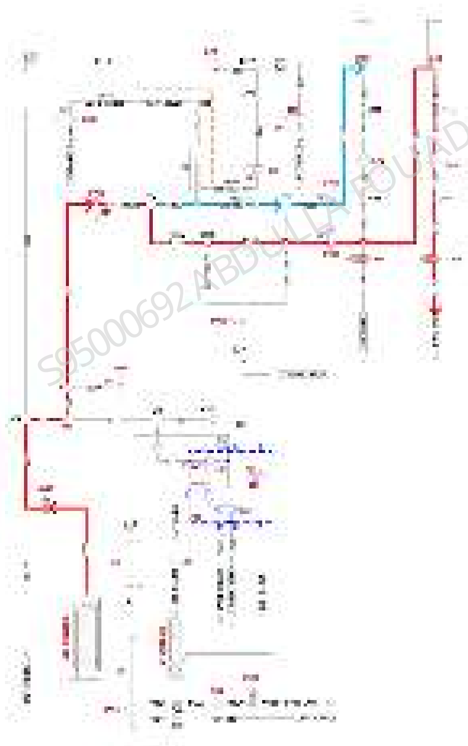
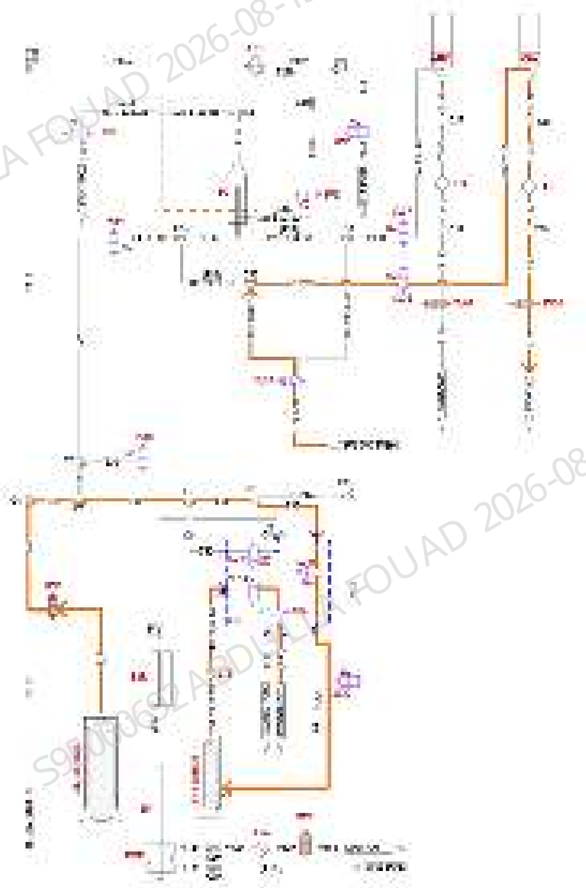
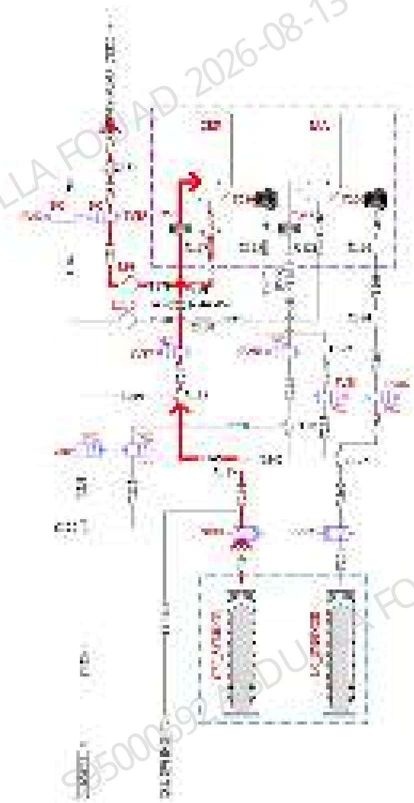


Figure2–69 Diagram 2 of Priming RET



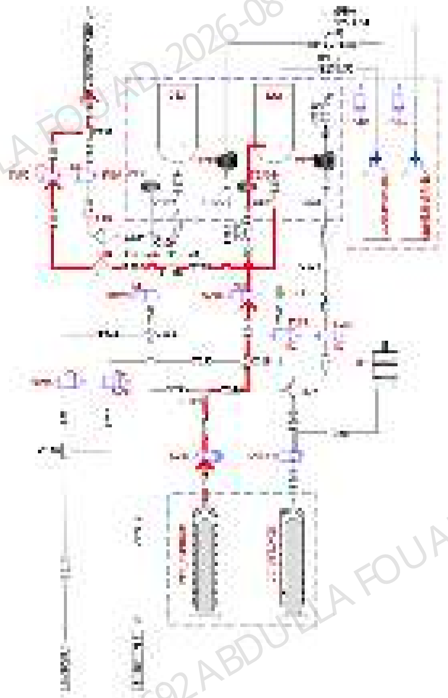
6. Priming CRP: CRP_SYRINGE-T242-T244-T245-T247-T253-CRP bath-T275-T277-T279-T281-T283-T285 (as indicated by the red path in the following figure)

Figure2–70 Diagram of Priming CRP



7. Priming SAA: CRP_SYRINGE-T242-T244-T246-T248-T254-SAA bath-T276-T278-T280-T282-T284-T285 (as indicated by the red path in the following figure)

Figure2–71 Diagram of Priming SAA



8. Priming RBC sample preparation tubing

1) RBC helps to push sample tubing for priming: Diluent syringe (DIL_SYRINGE) pushes out the diluent to prime RBC tubing (as indicated by the red path in **Figure2–72**: DIL_SYRINGE-T25-T26-T27-T31-T34-T35-T36-T38-T39-T41-T42-T87-T86)

2) Priming RBC sheath fluid impedance: Prime SCI bath (as indicated by the red-blue paths in **Figure2–72**: DIL_SYRINGE-T25-T26-T27-T31-T34-T44-T45-T53-SCI); the positive pressure of SCI bath pushes out the front bath of sheath fluid impedance bath (as indicated by the red-blue paths in the **Figure2–72**: SCI-T54-T55-T65-T80-T81-T82-T66-T67-T214. SCI-T54-T56-T57-T62-T63-T64-T83-T82-T66-T67-T214), and back bath (as indicated by the red-blue-green paths in the **Figure2–72**: SCI-T54-T56-T58-T59-T60-T215-T72-ISU-T74-T216-T75-T76-T77-T68-T69-T214), while the syringe and negative pressure prime the RBC sample preparation tubing (as indicated by the orange path in **Figure2–73**: DIL_SYRINGE-T25-T26-T27-T31-T34-T44-T46-T49-T50-T51-T192, HGB-T87-T42-T43-T71-T70-T56- T49-T50-T51-T192)

3) Sampling assembly restores test preparation status

Figure2–72 Diagram 1: Priming RBC Sample Preparation Tubing

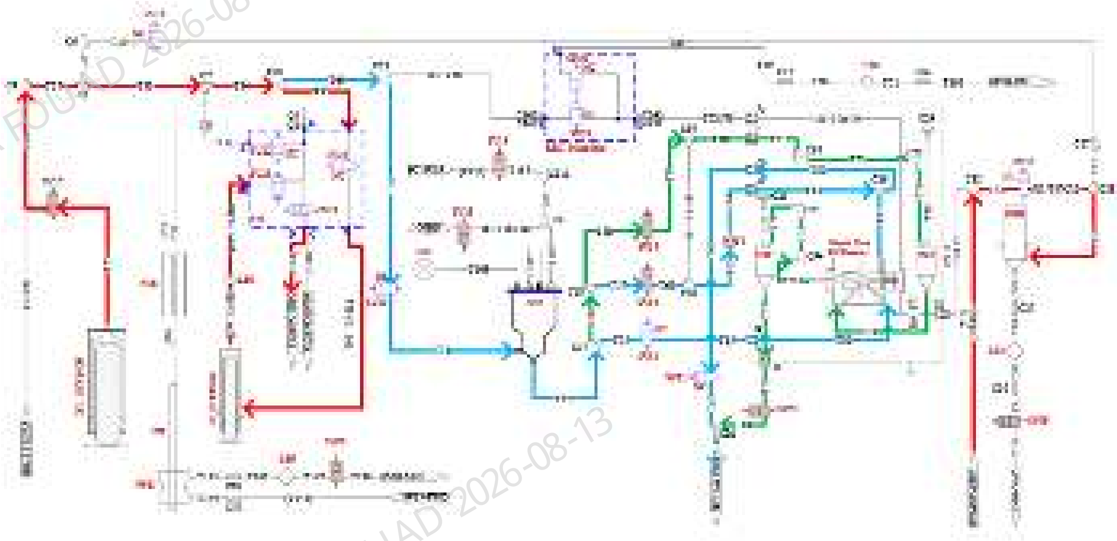
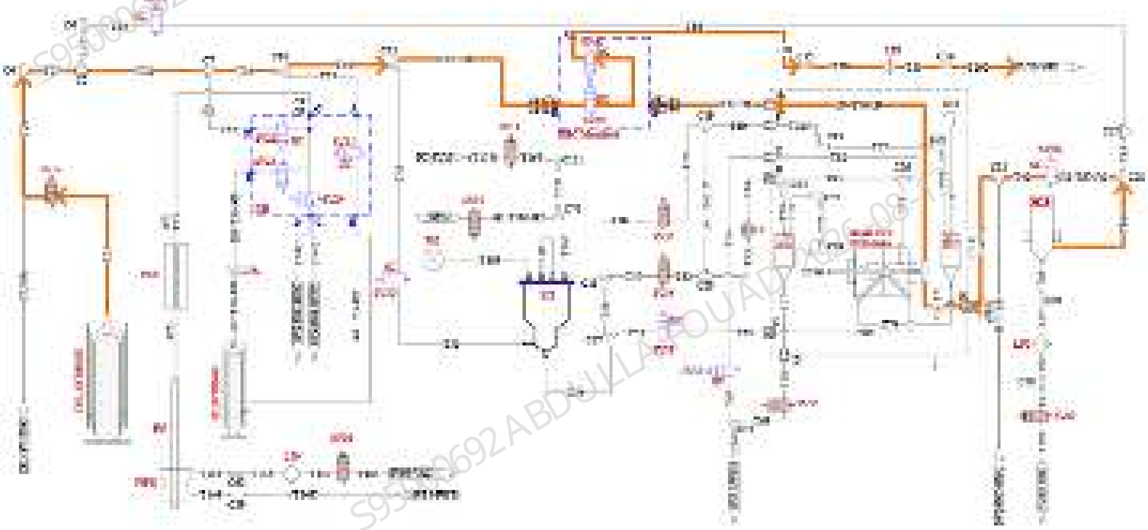


Figure2–73 Diagram 2: Priming RBC Sample Preparation Tubing



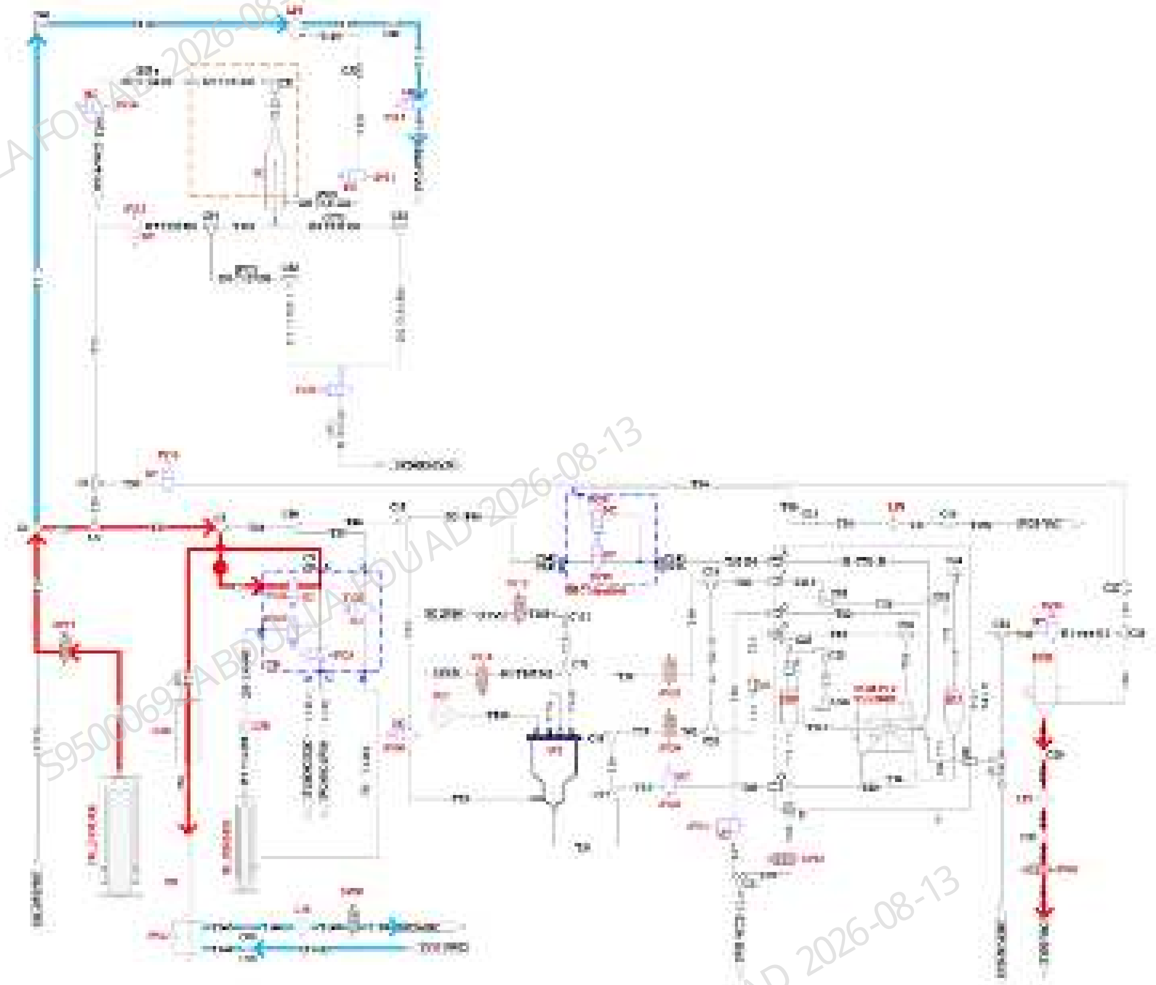
9. Priming probe cleanser reservoir tubing:

1) Priming probe cleanser reservoir tubing: Move sample probe to HGB bath: DIL_SYRINGE-T25-T26-T27-T31-T32-T33-PN-HGB bath-T88-T89-T90-T91-T191 (as indicated by the red path in the following figure)

2) Cleaning the tubing and sample probe exterior: DIL_SYRINGE-T25-T26-T27-T31-T34-T44-T46-T49-T150-T151-T192, and DIL_SYRINGE-T25-T26-T115-T116-T117-T118-T143-T146-T147-T188 (as indicated by the red-blue paths in the following figure)

3) Restore the sampling assembly to the status ready for analysis.

Figure2–74 Diagram of Priming Probe Cleanser Reservoir Tubing



10.Priming flow cell: Same as 3- 5.

Priming DIFF/RET/HGB−Related Reagents

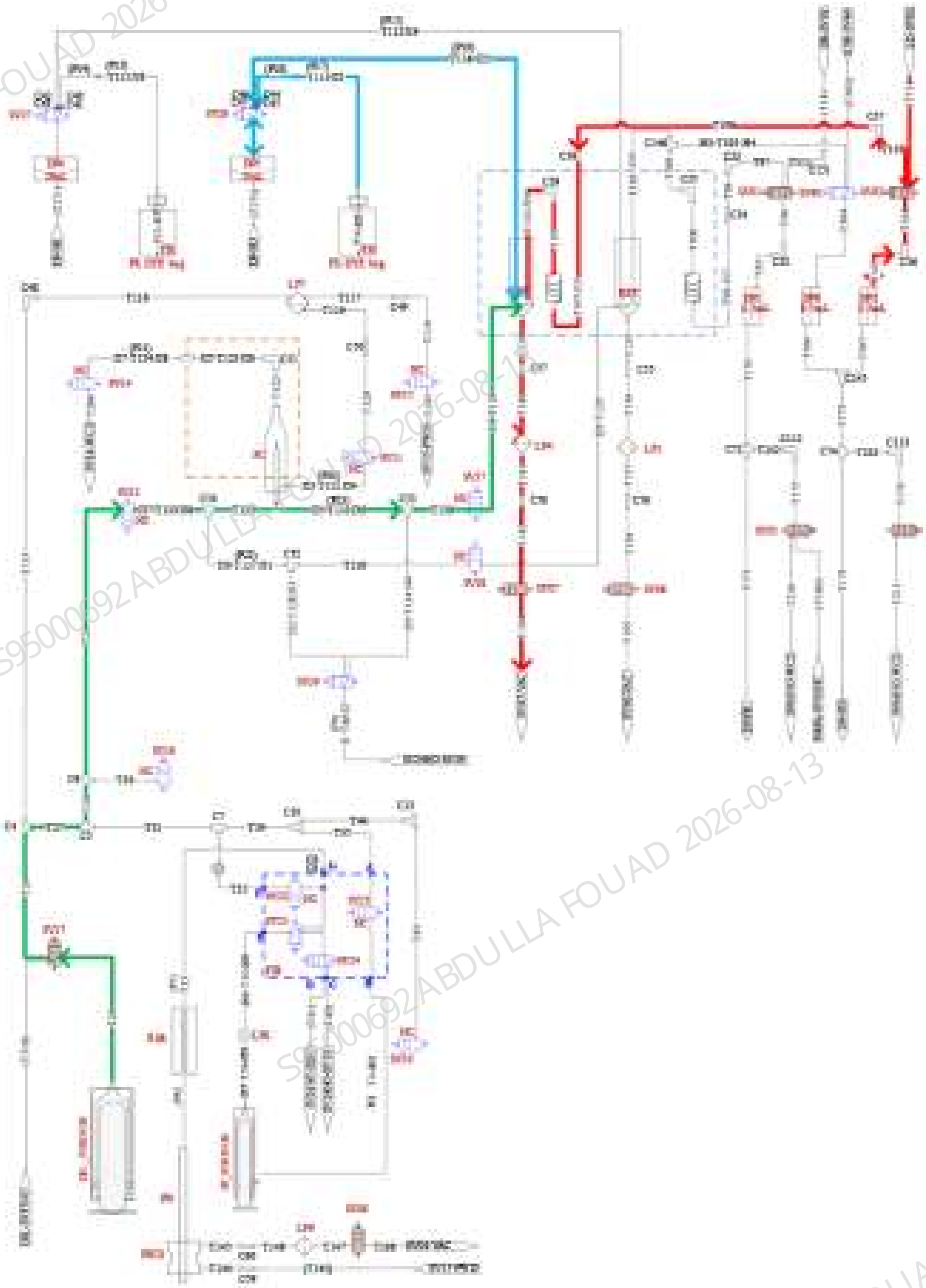
1. Priming DIFF-related reagents

1) Priming LD: Cap assemble of LD reagent bottle-T13-T104-T103-DP2-T103-T104-T105-T106-T107-T108-T110-DIFF bath-T139-T140-T141-T142-T194 (as indicated by the red path in the following figure).

2) Priming FD: Cap assemble of FD reagent bottle-T113-DP05-T114-DIFF bath-T139-T140-T141-T142-T194 (as indicated by the blue-red path in the following figure).

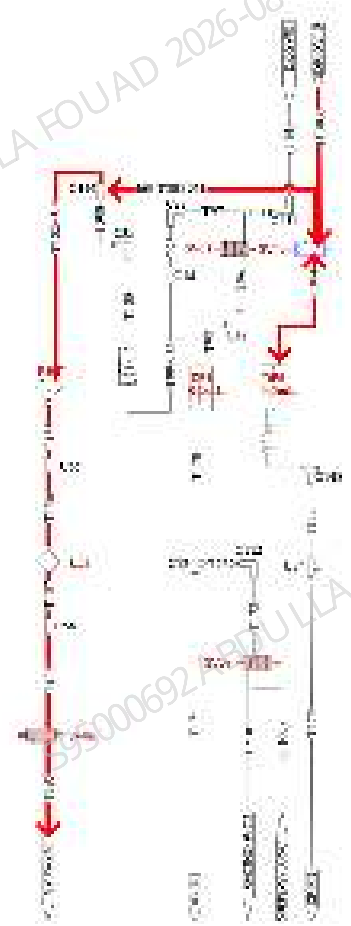
3) Priming DIFF channel: DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T132-T131-T130-T129-DIFF bath-T139-T140-T141-T142-T194 (as indicated by the green path in the following figure).

Figure2–75 Diagram of Priming DIFF-Related Reagents



2. Priming ESR cleaning solution
   Priming ESR: Cap assembly of ESR reagent bottle-T303-T304-DP06-T304-T305-T102-RET bath-T135-T136-T137-T138-T195 (as indicated by the red path in the following figure).

Figure2–76 Diagram of Priming ESR Reagent

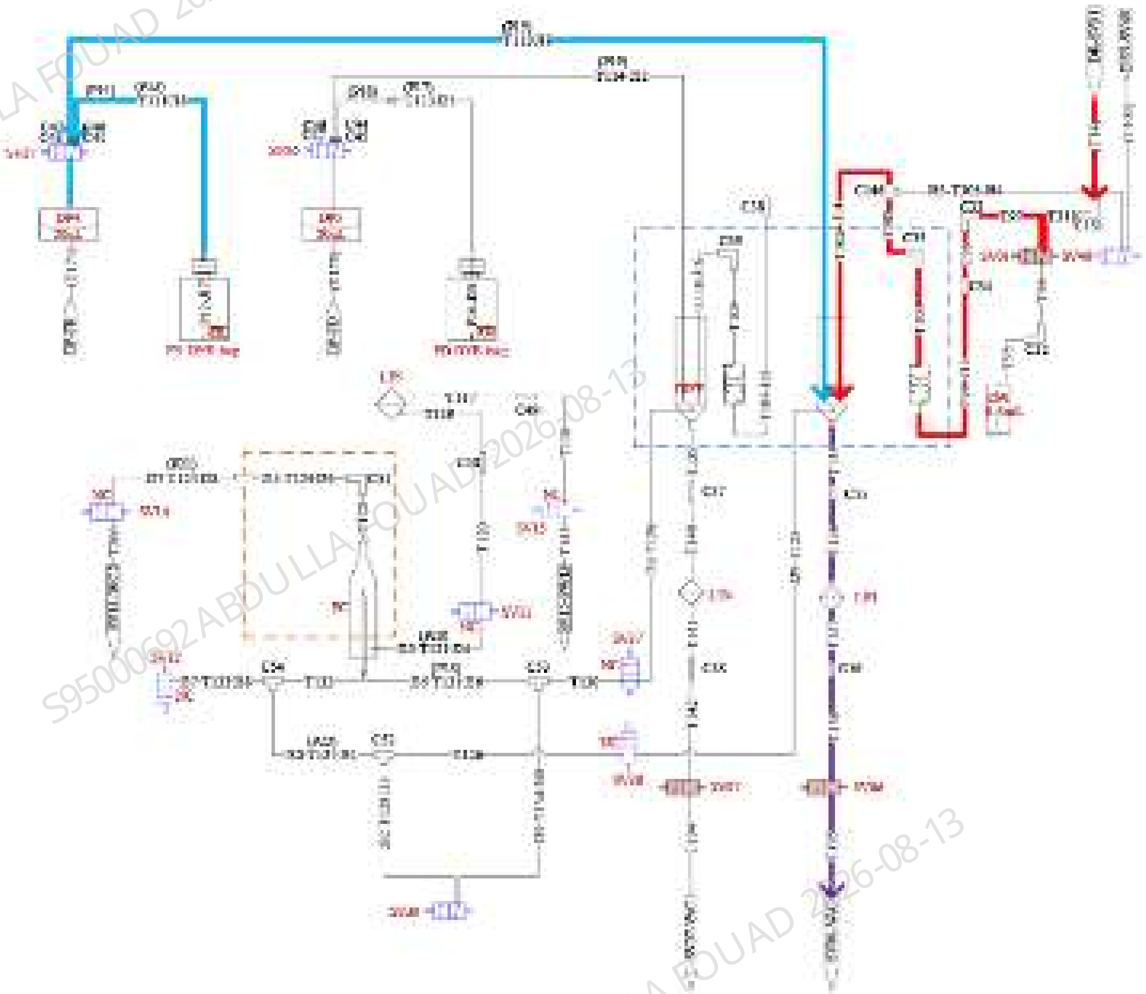


3. Priming RET-related reagents

1) Priming DR: Cap assembly of DR reagent bottle-T14-T311-T97-T98-T99-T100-T308-T102-RET bath-T135-T136-T137-T138-T195 (as indicated by the red path in the following figure).

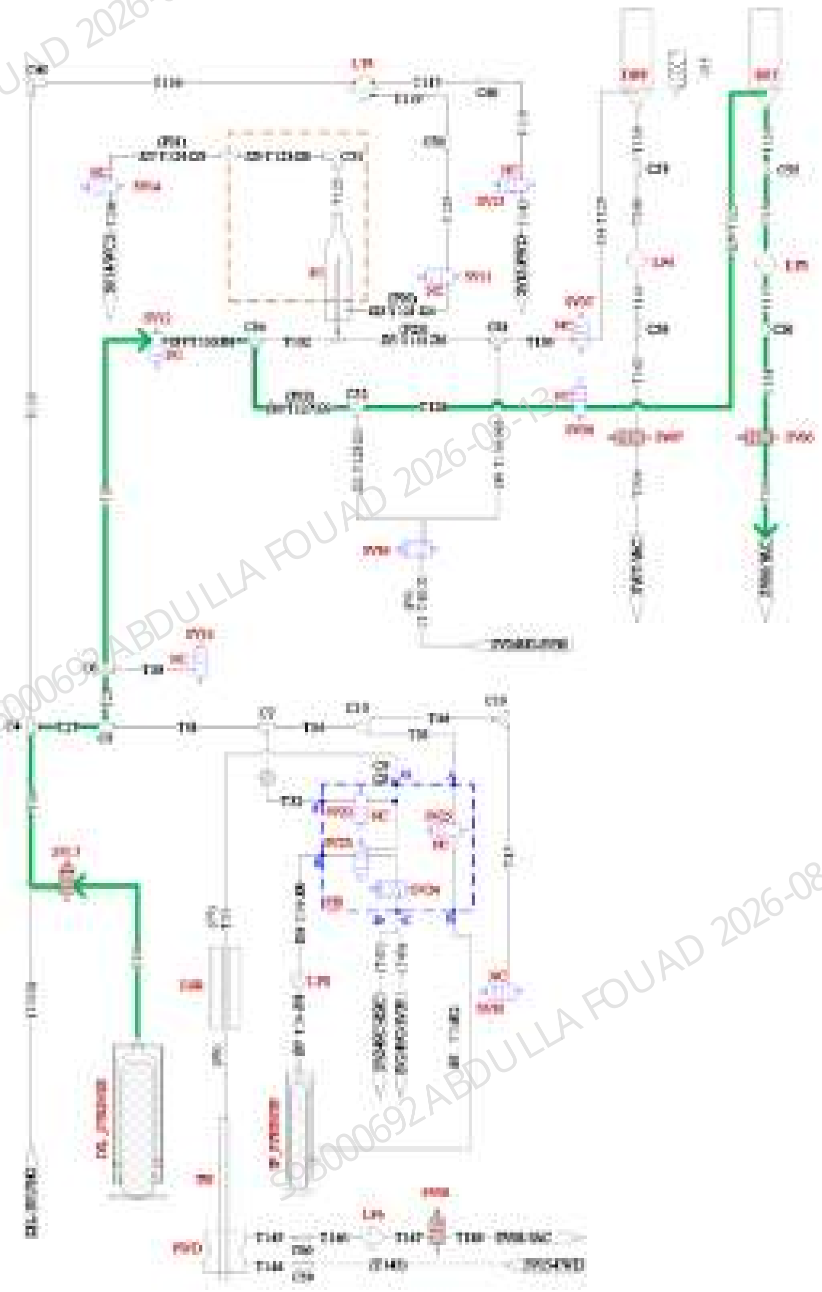
2) Priming FR: Cap assembly of FR reagent bottle-T111-DP04-T112-RET bath-T135-T136-T137-T138-T195 (as indicated by the blue path in the following figure).

Figure2–77 Diagram 1 of Priming RET-Related Reagents



3) Cleaning RET channel: DIL_SYRINGE-T25-T26-T27-T28-T29-T133-T127-T126-T125- RET bath-T135-T136-T137-T138-T195 (as indicated by the green path in the following path).

Figure2–78 Diagram 2 of Priming RET-Related Reagents

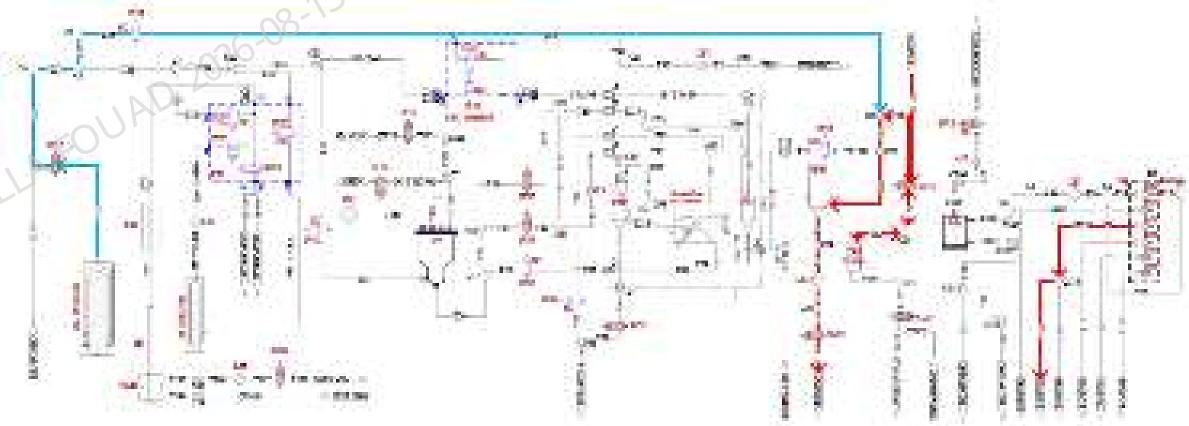


4. Priming HGB reagent

1) Priming LH: Cap assembly of LH reagent bottle-T12-T93-T92-DP03-T92-T93-T94-T85-T86-HGB bath-T88-T29-T90-T91-T191 (as indicated by the red path in the following figure).

2) Cleaning HGB channel: DIL_SYRINGE-T25-T26-T28-T30-T84-T85-T86-HGB bath-T88-T29-T90-T91-T191 (as indicated by the blue-red paths in the following figure).

Figure2–79 Diagram of Priming HGB Reagent



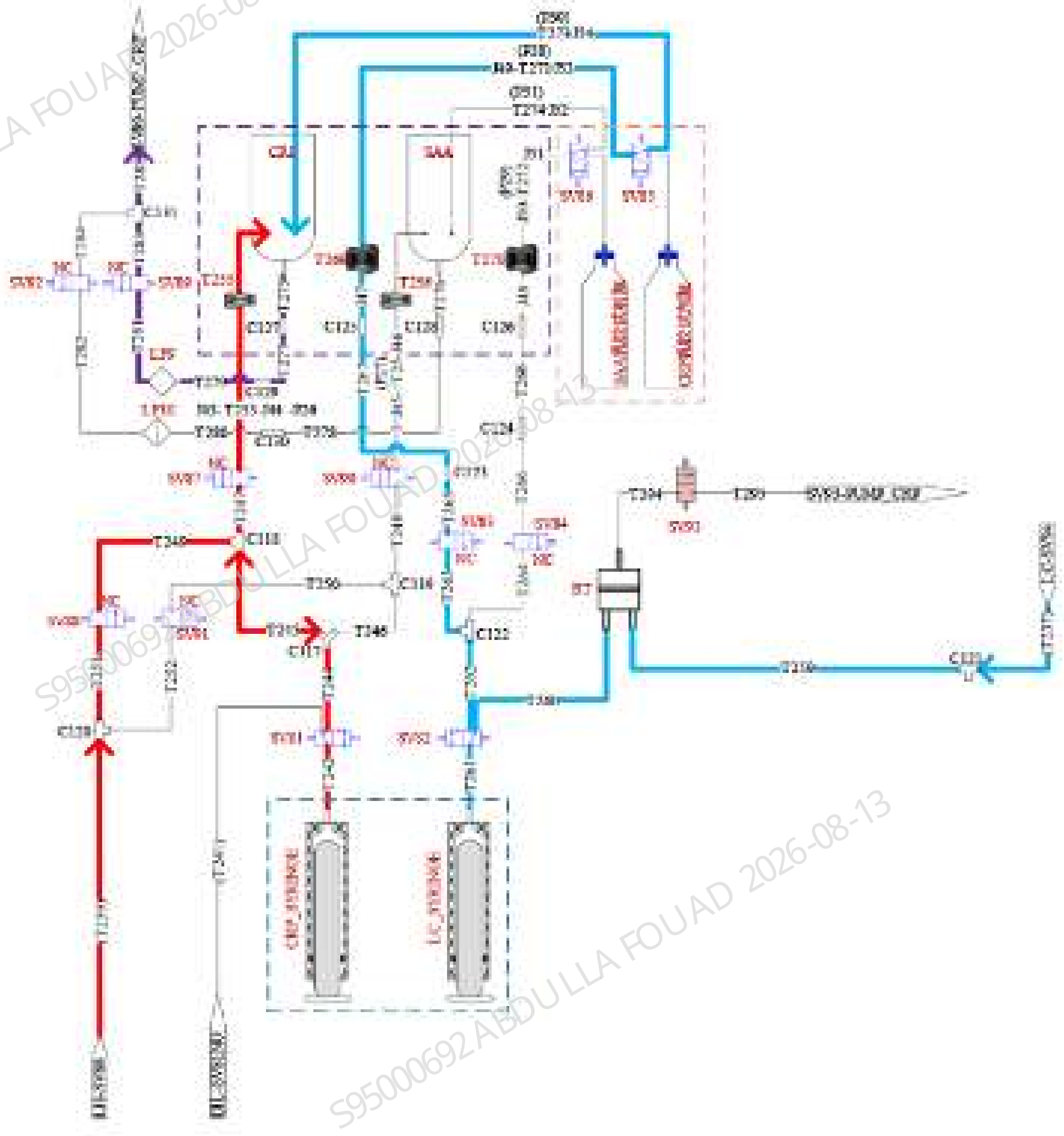
Priming CS−Related Reagents

1. Priming CRP reagent

1) Priming CRP-LH branch: T239-T251-T249-T245-T244-T242-CRP_SYRINGE-T242-T244-T245-T247-T253-CRP bath-T275-T277-T279-T281-T283-T285 (as indicated by the red path in the following figure).

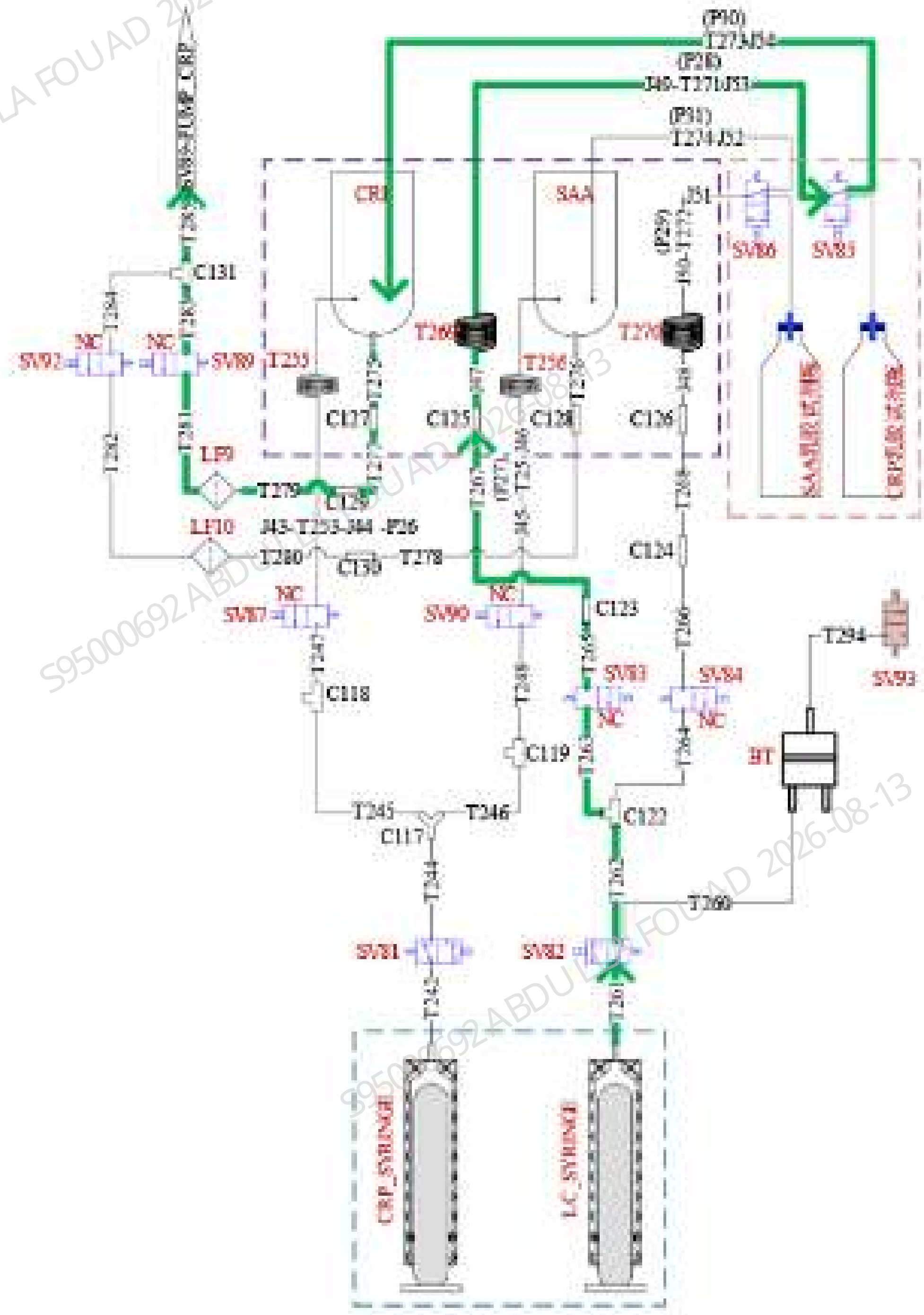
2) Priming CRP-LC branch: Cap assembly of LC bottle-T237-T259-T260-T261-LC_SYRINGE-T261-T262-T263-T265-T267-T271-T273-CRP bath-T275-T277-T279-T281-T283-T285 (as indicated by the blue path in the following figure).

Figure2–80 Diagram of Priming CRP Reagent



3) Priming CRP latex branch: LC_SYRINGE-T261-T262-T263-T265-T267-T271-T273-CRP bath-T275-T277-T279-T281-T283-T285 (as indicated by the green path in the following figure).

Figure2–81 Diagram of Priming CRP Latex Branch

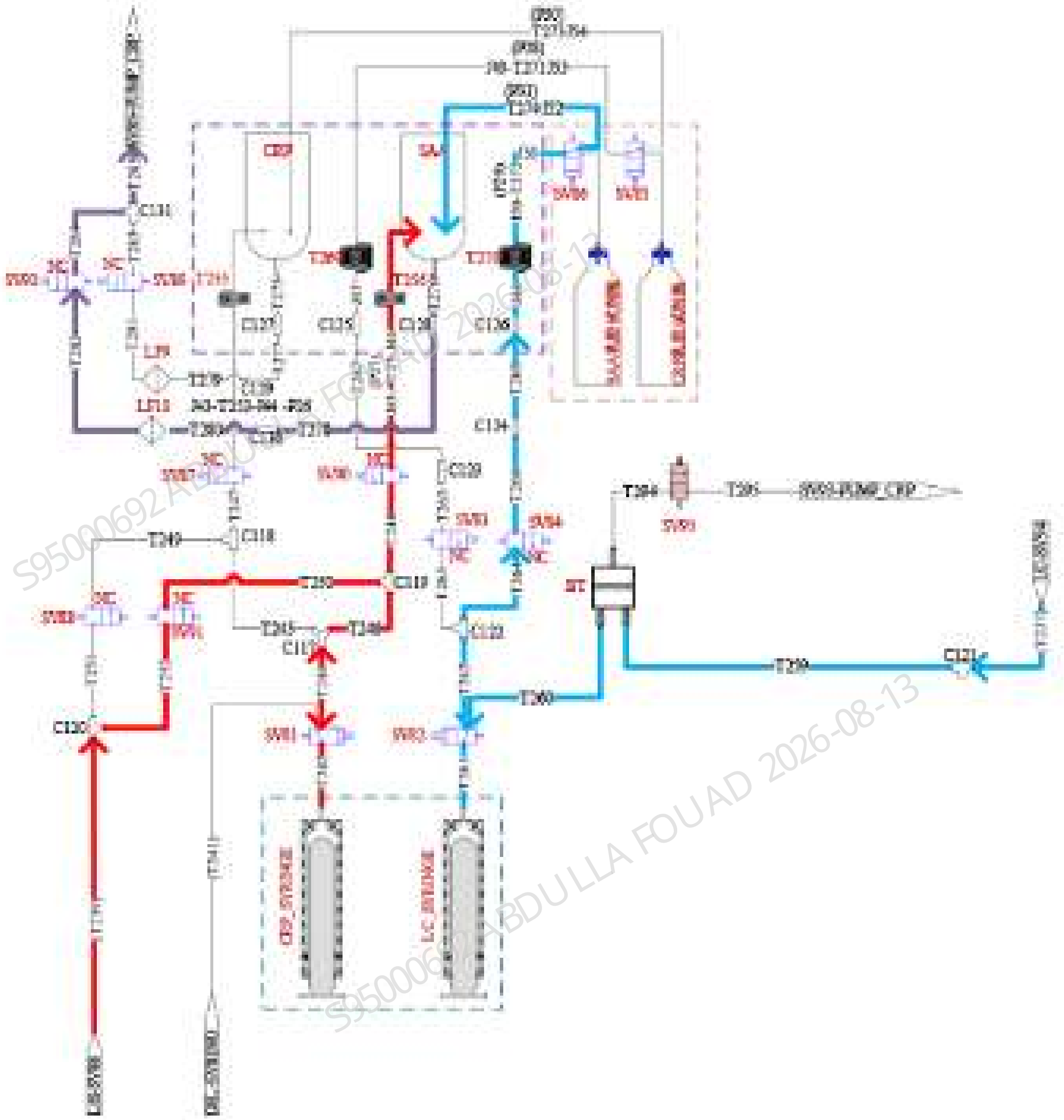


1. Priming SAA reagent

1) Priming SAA-LH branch: T239-T252-T250-T246-T244-T242-CRP_SYRINGE-T242-T244-T246-T248-T254-SAA bath-T276-T278-T280-T282-T284-T285 (as indicated by the red path in the following figure).

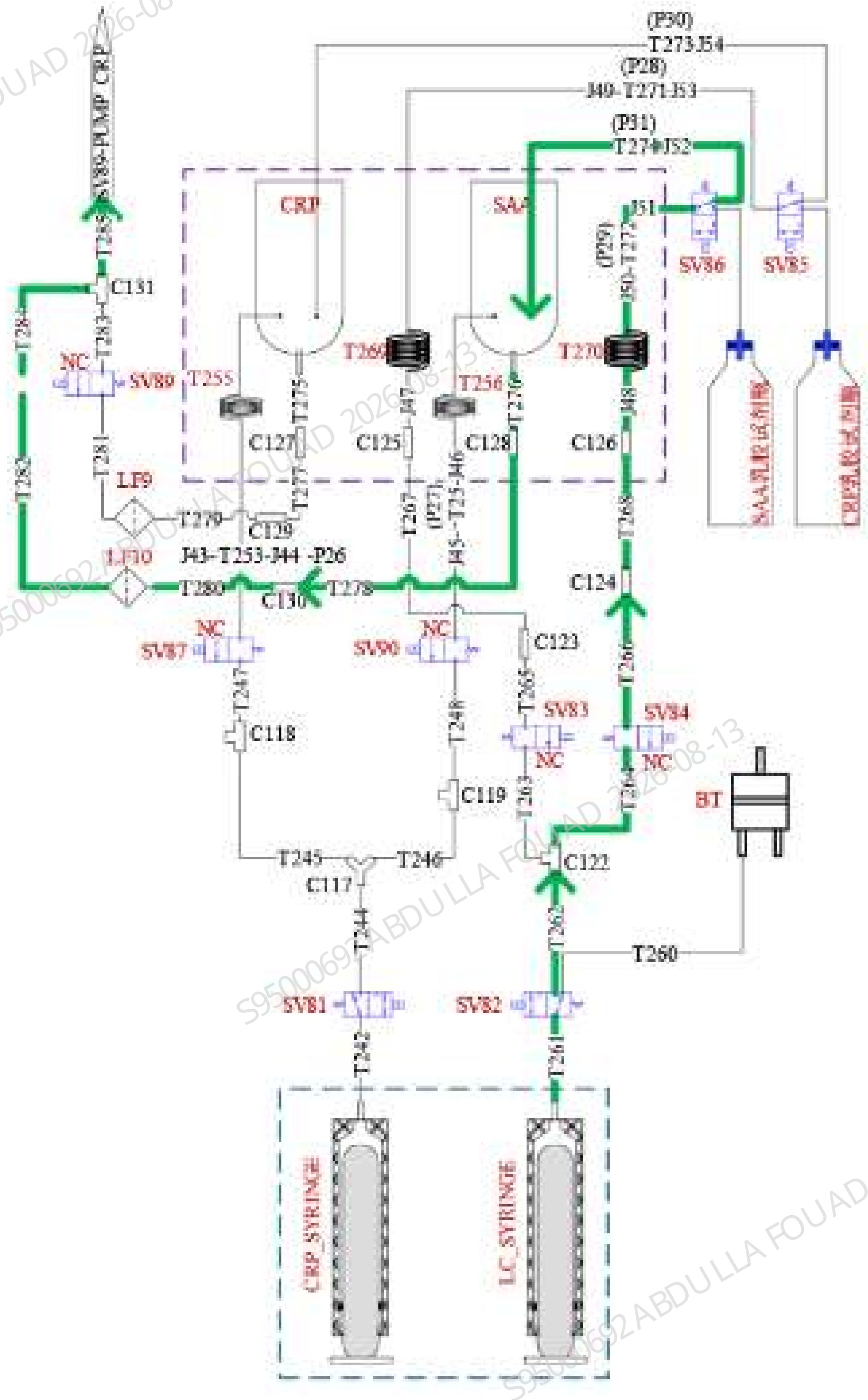
2) Priming SAA-LC latex branch: Cap assembly of LC bottle-T237-T259-T260-T261-LC_SYRINGE-T261-T262-T264-T266-T268-T272-T273- SAA bath-T276-T278-T280-T282-T284-T285 (as indicated by the blue path in the following figure).

Figure2–82 Diagram 1 of Priming SAA Reagent



3) Priming SAA latex branch: LC_SYRINGE-T261-T262-T264-T266-T268-T272-T273- SAA bath-T276-T278-T280-T282-T284-T285 (as indicated by the green path in the following figure).

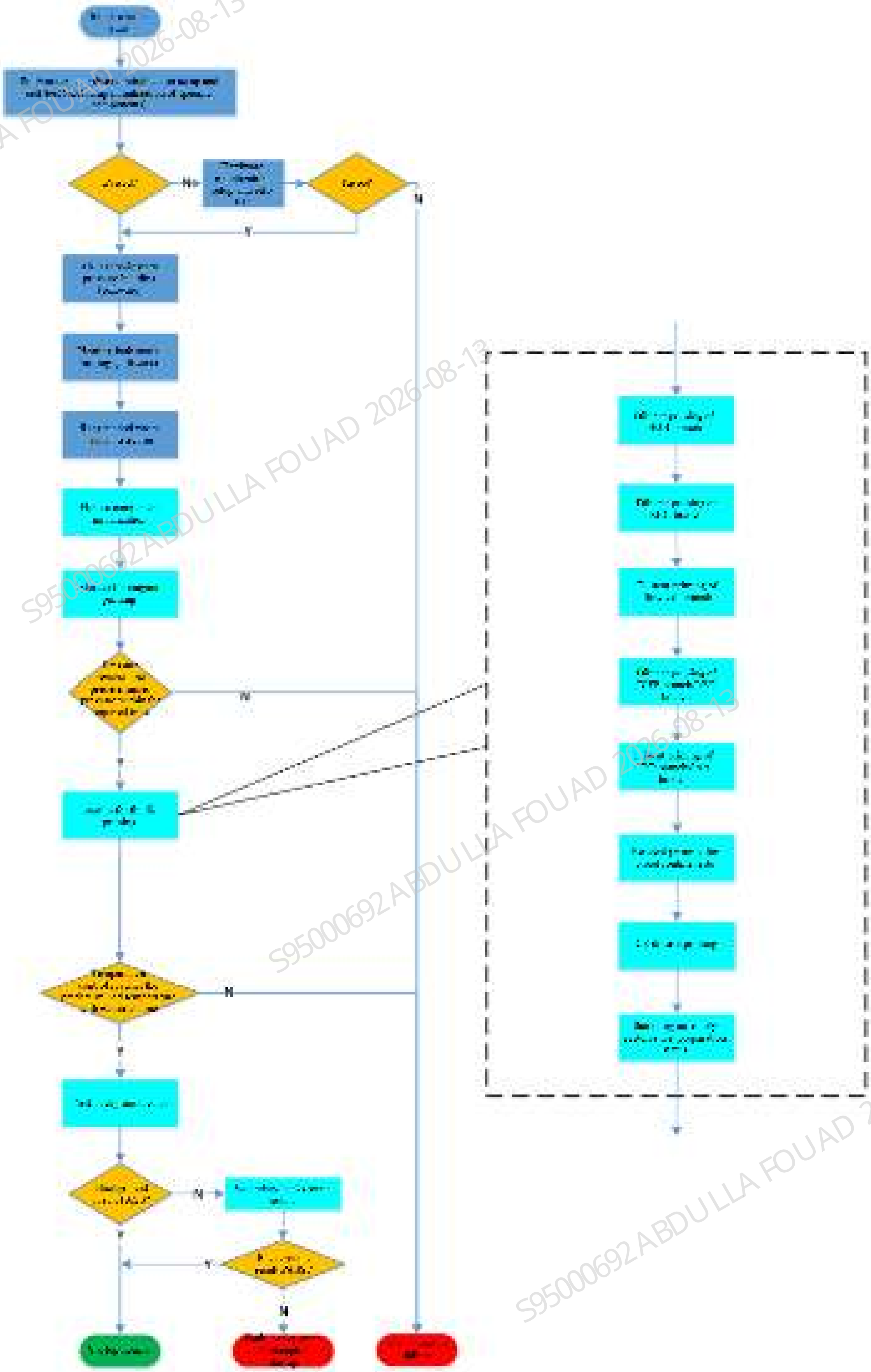
Figure2–83 Diagram 2 of Priming SAA Reagent



Restoring Analyzer Status

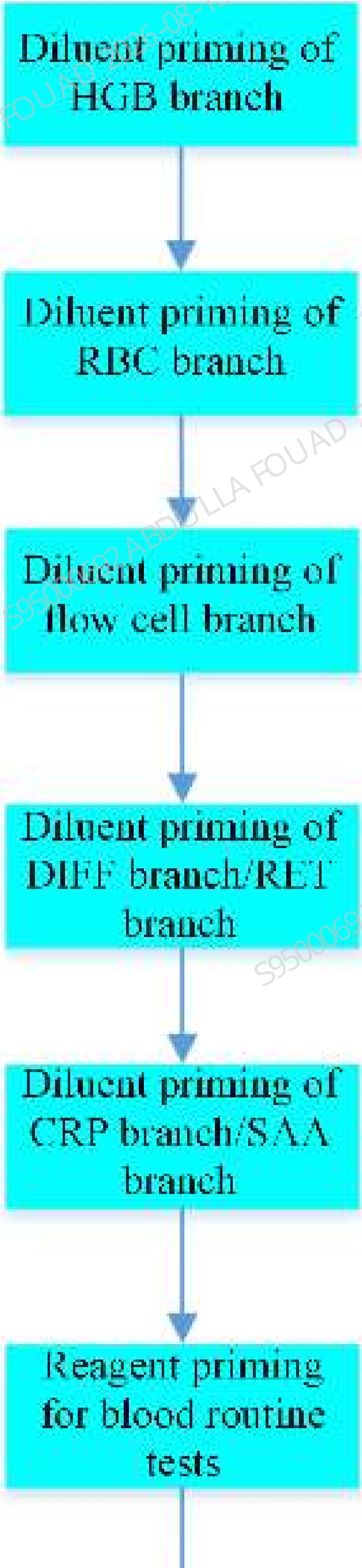
1. For measurement status parameters for reaction bath cleaning and recovery, refer to **4**- **7**.
2. For recovery of HGB/RBC tubing, refer to instrument cleaning **2** and **8**.
3. For cleaning of flow cell, refer to **3**.
4. Sampling assembly restores test preparation status.

Diagram of Pack-up Startup Procedure



2.3.4.8 Priming Flow of Whole Machine

Refer to the Priming Startup Fluidics in the **2.3.4.7** Pack-up Startup Procedure. For detailed operations, refer to pack-up startup priming.


Diluent priming of HGB branch
Diluent priming of RBC branch
Diluent priming of flow cell branch
Diluent priming of DIFF branch/RET branch
Diluent priming of CRP branch/SAA branch
Reagent priming for blood routine tests

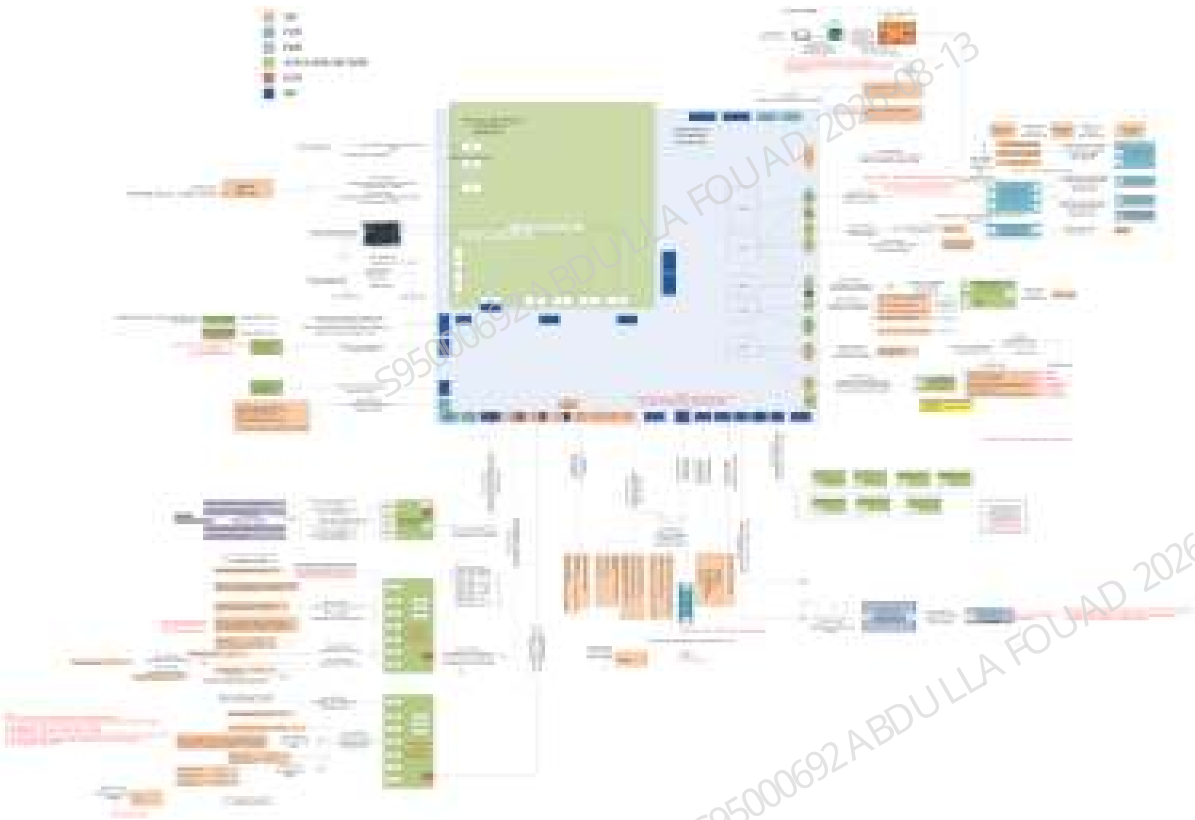
2.4 Hardware System

2.4.1 Hardware System Overview

The hardware system is a support system for the analyzer operation. It is also a parameter sensor drive and signal detection system, and mainly includes the following functions.

- The hardware system provides a main control platform for the user end software, fluidics sequence, algorithm operation, which is the main control of the analyzer man-machine interaction and the whole system, and communicate with the PC.
- The hardware system provides drive platforms and interfaces to drive the execution components such as motors, valves, pumps, to complete the test preparation of the sample collection, reaction, and delivery, and to monitor each state of the whole device during the execution process.
- The hardware system provides drive and signal detection for the impedance count bath, HGB sensor, ESR sensor, optical system and other parameter sensors, obtains parameter signals then offers them to the algorithm to calculate the clinical parameters.
- The hardware system provides the power supply for each module, board, and assembly.
- The hardware system provides the wires for interconnection between modules, boards, and assemblies.

Figure2–84 Function Block Diagram of Hardware System



2.4.2 Composition and Functions of Hardware System

The hardware system mainly includes the following assemblies, boards and modules (the wires for connection only are not demonstrated here):

Table 2–35 Composition of Hardware System

No.	FRU Code	FRU Name	Function Description	Remarks
1	115-074574-00	Power Supply Assembly	Supply power to the analyzer, mainly including the 24V/600W power supply and fan for power supply assembly.	
2	051-003910-00	IMX8 main control board	Provide various ports including USB port, network port, hard disk interface and screen assembly interface. Analog/digital conversion and digital processing of optical signals, impedance signals, HGB signals and analyzer status monitoring signals.	
3	051-004008-00	Motherboard	Convert the 24V output by the 600W power supply to P24V, P12V, D5V, analog +/-12V, and analog +/-5V, and provide the power for different boards. Analog/digital conversion and digital processing of optical signals, impedance signals, HGB signals and analyzer status monitoring signals. Drives motors, pumps, heatings, fans and other power components. Detect temperature sensors, hydraulic sensors, dye sensors, floaters and sensors, etc. Interfaces for level board, air pressure board, valve driver board and ESR assembly, etc.	

Table 2–35 Composition of Hardware System(continued)

No.	FRU Code	FRU Name	Function Description	Remarks
4	021-000591-00	One-glass-solution TFT 12.1" 1280*800	Display function and the touch function are provided, including display and capacitive touch screen. Connect display assembly to main control board.	It is impossible to replace the display assembly only, which shall be replaced with the cover.
5	051-001621-00	Liquid detection board	Detect diluent, lyse and the reagents of 6 channels. Output to motherboard.	
6	051-004205-00	Air pressure detection board PCBA	+50a, +40a, +90 kPa, -30 kPa and -40 kPa, output analog voltage signal (digital conversion on driver board). Built-in EEPROM, storage pressure calibration factor.	
7	051-004706-00	RFID Reader PCBA	With RFID antenna board, allow to read and write RFID label for closed reagent model. Serial communication with main control board.	
8	051-003263-00	Startup/ shutdown key board	Provide physical keys for user input operation, detect the on/off state of keys and control the on/off status of analyzer. The keys are equipped with indicators.	
9	051-001240-00	Indicator board PCBA	Provides the analyzer status indication, including the RGB tricolor indicator array and its driver.	
10	051-004012-00	PCBA, impedance signal board	The drive and impedance signal amplification of the counting bath sensor.	

Table 2–35 Composition of Hardware System(continued)

No.	FRU Code	FRU Name	Function Description	Remarks
11	051-002805-00	Valve driver board PCBA	Provide the drive of each valve in the fluidic system.	
12	051-004344-00	PCBA, Motor board	Two boards in total. The one installed on the host provides the drive of motor and sensor for rotary blending assembly; and the other one installed on the autoloader realizes the drive of motor and sensor for the autoloading mechanism.	

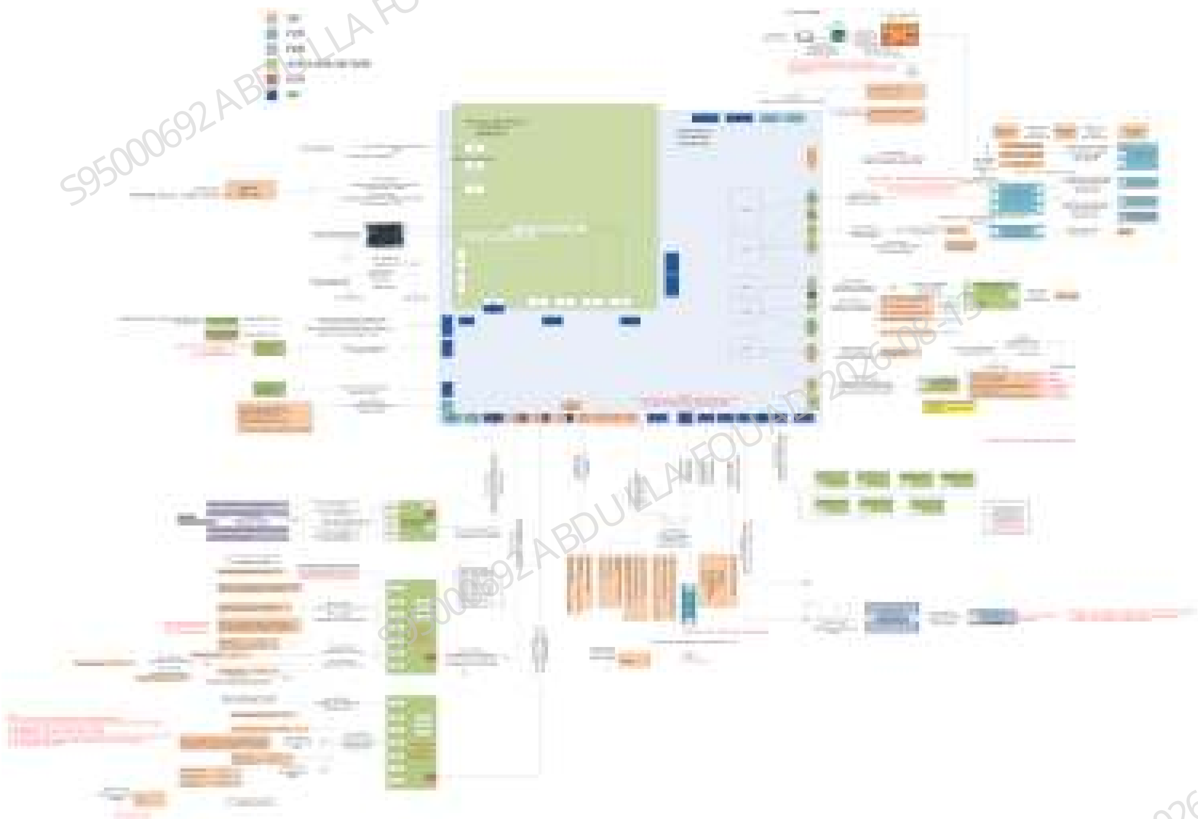
Table 2–36 Category and Purpose of Power Supply

Power supply symbol	Power supply description	Power supply range	Purpose
P24V	Power+24V	22.8~25.2V	Motor drive, temperature control heating
P12V	Power+12V	11.5~12.5V	Driver of valve/pump/board fan and optical system fan
D12V	Digital+12V	11.5~12.5V	Power supply for main control board, display
D5V	Digital+5V	4.75~5.25V	Power supply for digital circuits of motherboard, liquid detection board, indicator board, RFID reader board; sensor power supply
D3V3	Digital+3.3V	3.1~3.5V	Power supply for digital circuits of motherboard, air pressure detection board, touch screen control
A+12V	Analog+12V	11.5~12.5V	Power supply for analog circuits of impedance analog board, optical signal board, laser drive board, laser control board.
A-12V	Analog-12V	-11.5~-12.5V	Power supply for analog circuits of impedance analog board, optical signal board, laser drive board, laser control board.

Table 2–36 Category and Purpose of Power Supply(continued)

Power supply symbol	Power supply description	Power supply range	Purpose
A+5V	Analog+5V	4.75~5.25V	Power supply for analog circuits of motherboard, optical signal board and air pressure detection board
A-5V	Analog-5V	-4.75~-5.25V	Power supply for motherboard and optical signal board

Hardware block diagram



Cable Checklist

Cable material silkscreen	FRU Code	Name
C012952	009-012952-00	Paired wire, motherboard to motor board A, motherboard end
C012959	009-012959-00	Wire, sample type detection sensor
C012960	009-012960-00	Wire, motherboard to motor board B
C012970	009-012970-00	Wire, motherboard to scanner
C012981	009-012981-00	Wire, motherboard to fan, ext.
C012982	009-012982-00	Wire, HGB sensor, ext.
C012983	009-012983-00	Wire, fluid pressure sensor wire, ext.
C012984	009-012984-00	Wire, impedance detection module, ext.
C012985	009-012985-00	Wire, temperature detection wire, ext.
C012986	009-012986-00	Wire, dye detecting sensor, ext.
C012987	009-012987-00	Wire, motherboard to air pressure detection board, ext.
C012989	009-012989-00	Paired wire, motherboard to RFID board 1&2, motherboard end
C012990	009-012990-00	Wire, motherboard to floater, ext.
C012991	009-012991-00	Wire, ESR assembly
C012993	009-012993-00	Wire, motherboard to heater, ext.
C013000	009-013000-00	Paired wire, valve board 1 valve group 3, board end
C013002	009-013002-00	Wire, valve board 1 valve group 5
C013010	009-013010-00	Wire, motherboard to jog switch (automatic)
C013020	009-013020-00	Wire, rotary scanner and blending motor
C013021	009-013021-00	Wire, sampling asm&syringe motor sensor
C013072	009-013072-00	Wire, motherboard to valve board 1 Pow & Com, ext.
C013193	009-013193-00	Wire, valve board 1 valve group 2
C013235	009-013235-00	Wire, syringe grounding
C012607	009-012607-00	Wire, front cover sheet metal ground wire
C013186	009-013186-00	Wire, front cover
C013064	009-013064-00	Wire, RFID reader to antenna
C012969	009-012969-00	Paired wire, sensor vertical motor of sampling assembly and ESR

Cable material silkscreen	FRU Code	Name
C012962	009-012962-00	Paired wire, motor and sensor of latex probe assembly, latex probe assembly end
C013234	009-013234-00	Lead-out wire, latex reader
C008191	009-008191-00	Micro-switch connection cable
C005387	009-005387-00	Wire, semiconductor refrigeration sheet
C013019	009-013019-00	Wire, digital sensor (S), A.5 (Design)
C013018	009-013018-00	Wire, motor (S)
C012948	009-012948-00	Wire, pusher (S)
C012953	009-012953-00	Paired wire, motherboard to motor board A, motor board end
C49040	3100-20-49040	Micro-switch connection cable
C013022	009-013022-00	Wire, digital sensor (M)
C013023	009-013023-00	Wire, motor (M)
C013024	009-013024-00	Wire, pusher (M)

2.5 Optical system

2.5.1 Introduction to Optical System Principles

After being processed by reagents, the blood cells flow through the flow cell and radiated by the laser beam generated by the analyzer's optical system. Collect the forward scatter, lateral scatter and lateral fluorescence from each cell. The forward scatter reflects the volume information of the cell, the lateral scatter reflects the internal nucleus and particle information of the cell, and the lateral fluorescence reflects the internal nucleic acid information of the cell. The blood cells for the blood sample can be analyzed through analyzing and processing the above three optical signals.

2.5.2 Optical Path and Workflow of Optical System

Figure2–85 Workflow Diagram of Optical System

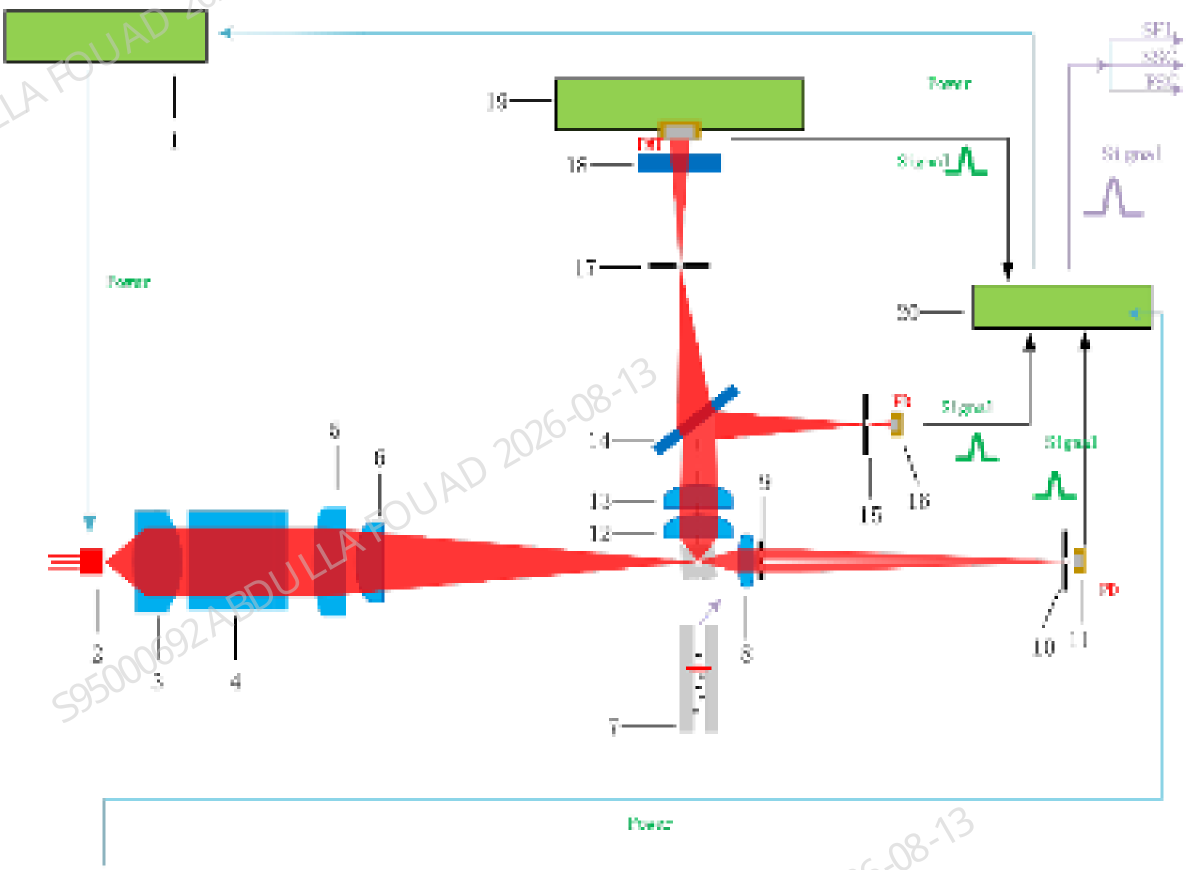
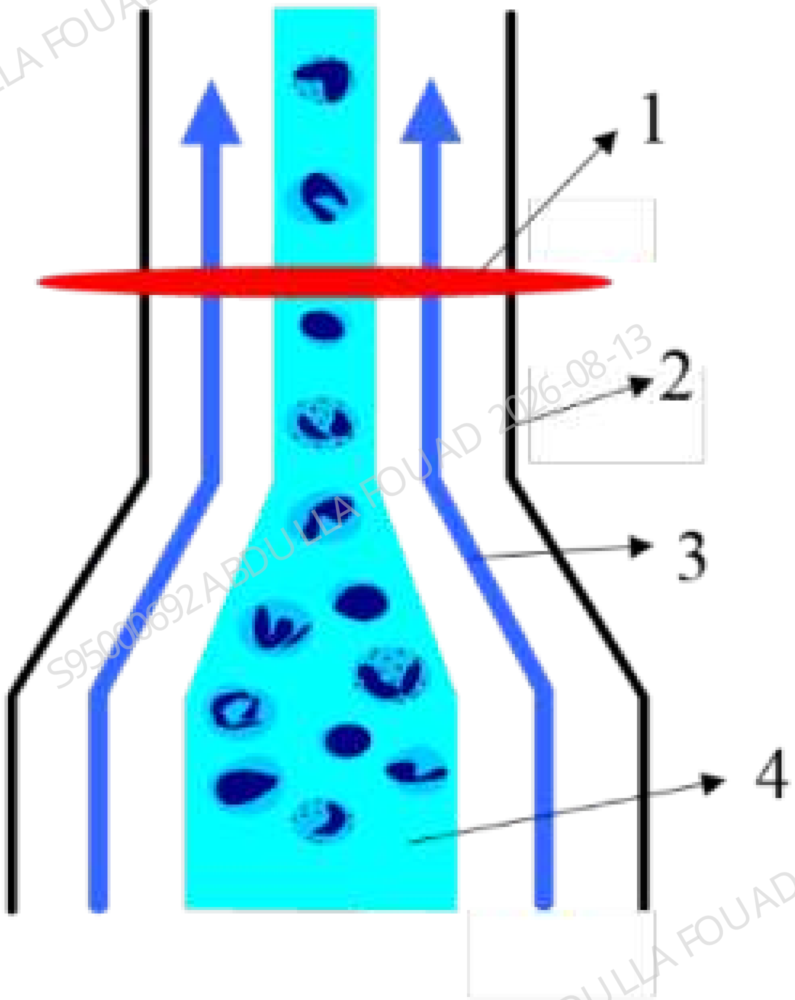


Table 2–37 Optical Components of Optical System

1	Laser driver board	11	Forward PD
2	Laser	12	Plano-convex lens A
3	Collimating lens	13	Plano-convex lens B
4	Opto-isolator	14	Dichroic lens
5	Cylindrical lens	15	Lateral aperture diaphragm
6	Spherical lens	16	Lateral PD
7	Flow cell	17	Fluorescent aperture diaphragm
8	Biconvex lens	18	Filter
9	Radiation-proof diaphragm	19	Fluorescence preamplifier board
10	Forward aperture diaphragm	20	Optical signal board

The analyzer optical system uses semiconductor laser 2 as the source of light. The laser driver board 1 provides the current for the laser to emit a red laser beam of 635nm. The laser beam passes through the collimating lens 3, opto-isolator 4, cylindrical lens 5 and spherical lens 6, to form a flat elliptic laser beam going through the sample flow in the center of the flow cell 7.

Figure2–86 Laser Facula of Flow Cell



1	Laser light	3	Sheath fluid
2	Flow cell	4	Sample solution

Blood cells treated with lyse and fluorescent reagents are wrapped by a high speed diluent flow (also known as sheath fluid) that forms a hydrodynamic focusing phenomenon at the flow cell, which in turn passes through the flow cell as shown in Figure2–86. The flat elliptic laser beam irradiates light on each blood cell passing through the flow cell, excites scattered light and fluorescence, and direct light that has not been scattered by the cell is blocked by the radiation-proof diaphragm 9.

1. The 1~10° forward scatter is collected by the biconvex lens 8, passes through the radiation-proof diaphragm 9 and then the forward aperture diaphragm 10. It converges on the forward PD 11 to form a forward scattered signal.

2. The lateral scatter is collected by the lens group of plano-convex lens A 12 and B 13, and is reflected by dichroic lens 14, passes through lateral aperture diaphragm 15. It converges on the lateral PD 16 to form the lateral scattered signal.
3. The lateral fluorescence is collected by the lens group of plano-convex lens A 12 and B 13, and is transmitted by dichroic lens 14, passes through fluorescent aperture diaphragm 17 and filter 18 to further filter the scatter. It converges on the fluorescence preamplifier board 19 to form the fluorescence signal at the Si-PMT site. The three channels of optical signals are further amplified and processed by the optical signal board 20, and then are uploaded to the analyzer signal processing system to form a scattergram.

The three channels of optical signals are further amplified and processed by the optical signal board 20, and then are uploaded to the analyzer signal processing system to form a scattergram.

2.5.3 Physical Components of Optical System

Remove the top cover of the insulation box for analyzer's optical system, and the components are shown in the figure below.

Figure2–87 Mechanical Structure of Optical System

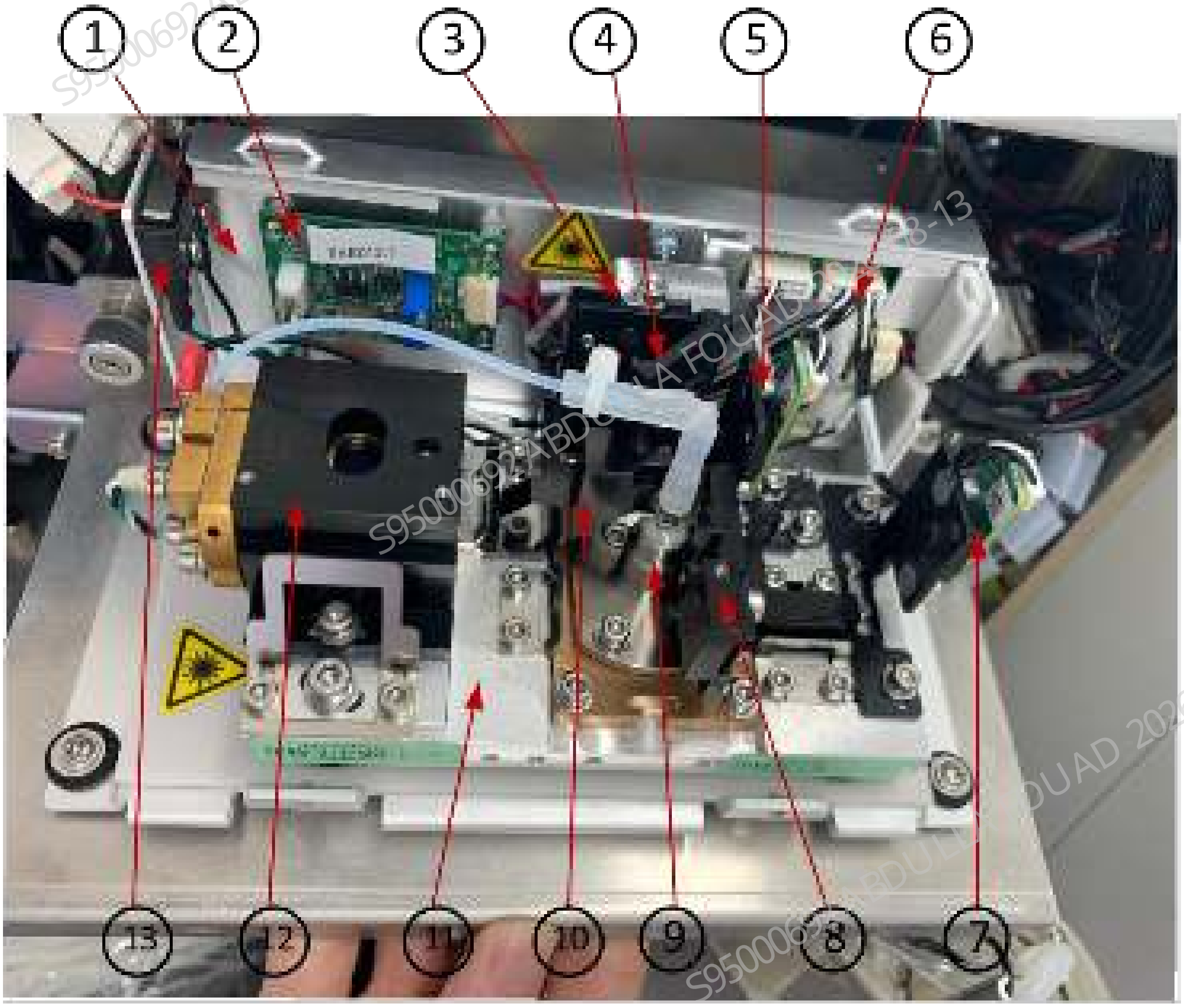


Table 2–38 Mechanical Components of Optical System

1	Optical lateral cover assembly	8	Rear collection assembly
2	Laser driver board	9	Flow cell assembly
3	Fluorescent preamplifier board assembly	10	Lateral collection assembly
4	Dichroic lens assembly	11	Optical base plate assembly
5	Lateral PD assembly	12	Front optical assembly
6	Optical signal board	13	Micro-switch
7	Forward PD assembly		

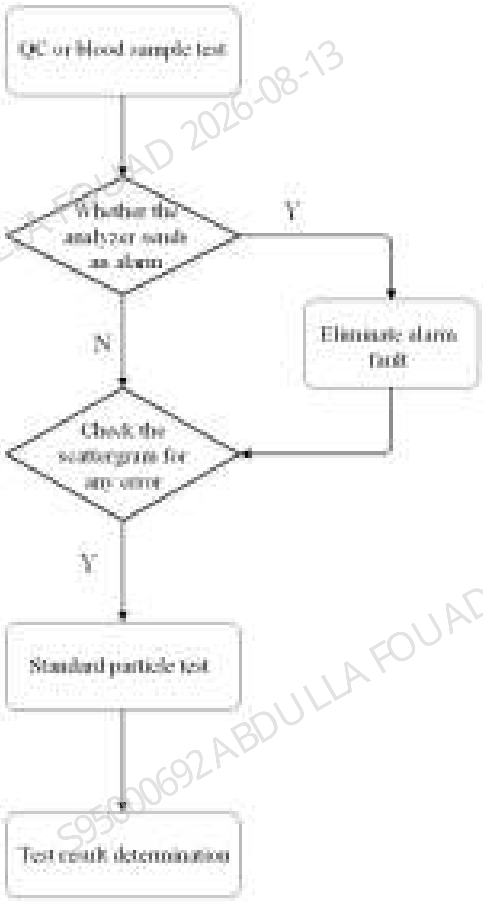
1. The front optical module 2 consists of the semiconductor laser (635 nm), a collimator lens, opto-isolator, cylindrical lens, and spherical lens. The laser drive board on the optical lateral cover assembly (1) provides a stable current input to the front optical assembly, and the front optical assembly is responsible for irradiating the shaped laser beam to the flow cell detection area.
2. The flow cell assembly 3 includes a flow cell and a rectifying assembly, and creates a fluid state in the flow cell where the stable sheath fluid wraps the sample flow. The reagent-treated cells in the laser irradiation flow cell excite scattered light and fluorescence.
3. The rear collection assembly 4 includes the biconvex lens and radiation-proof diaphragm. The 1~10° forward scattered light emitted by the cells converges on the PD photosensitive surface of the forward PD module 5.
4. The lateral collection assembly 6 has a lateral collection lens group to ensure the collection and convergence of lateral scattered light and fluorescence, which are then respectively reflected and transmitted by the dichroic lens assembly 7. The lateral scattered light converges to the photosensitive surface of the lateral PD assembly 8. The fluorescent signals transmitted through the dichroic lens converge at the diaphragm of the fluorescent aperture in the fluorescent preamplifier board assembly 9 and reach the photosensitive surface of Si-PMT after passing the filter.

The above three channels of signals are transmitted to the optical signal board 10 at the same time, so as to further amplify and process the blood cell scattering signal, and then the signal is uploaded to the analyzer signal processing system, and finally a scattergram is formed.

The optical assembly is installed on the base plate assembly 12. The entire optical system is provided with an insulation box to shield outside light, heat, vibration, electromagnetic field and dust.

2.5.4 Status Determination of Optical System

The optical system status may be determined according to the following procedure



Scattergram Based Determination

NOTICE

- **Laser radiation.**
- **ESD protection.**
- **It is not allowed to open the top cover of the insulation box during startup.**

1. Flag: After QC or blood sample testing, confirm whether there is any flag of the analyzer. Eliminate the error in case of any flag, and proceed to the next step in case of no flag.
2. Check the scattergram for any error

1) Check whether the scattergrams of both DIFF and RET are abnormal. The error of a single channel is rarely caused by the optical system. The fluidics, reagents or temperature control system of any failed channel shall be inspected.

2) If the scattergrams of both DIFF and RET are abnormal, the causes are likely to lie in the optical system, in which case, the standard particle test may be conducted.

Standard Particle Test

NOTICE

- **Laser radiation.**
- **ESD protection.**
- **It is not allowed to open the top cover of the insulation box during startup.**

1. Log in the analyzer with the service level account, and click **Menu**>**>Service >>Commissioning and Self-Test** >>**Optical Commissioning** to access the *Optical Commissioning* screen.
2. Gently mix the 4k-07 standard particles in a vial, and dispense 4 to 5 drops of 4k-07 standard particles to 1mL deionized water to prepare the standard particle samples.
3. Place the standard particle samples under the OV sampling probe and press the Aspirate key to enter the standard particle (Commissioning) mode. After counting, the scattergram of the standard particles will be automatically displayed. Note that the total number of particles shall be greater than 1000, otherwise, the particle statistics are insufficient, in which case, pipette standard particles into the centrifuge tube appropriately and run the counting process again.
4. The center of gravity and CV of the counting results FS of 4k-07 standard particles shall meet the requirements in Table 11-3.
5. Gently mix the A-7312 standard particle in a vial, and dispense 1 drop of standard particles to 1mL deionized water to prepare the standard particle samples.
6. Place the standard particle samples under the OV sampling probe and press the Aspirate key to enter the standard particle (Commissioning) mode. After counting, the scattergram of the standard particles will be automatically displayed. Note that the total number of particles shall be greater than 2000, otherwise, the particle statistics are insufficient, in which case, pipette standard particles into the centrifuge tube appropriately and run the counting process again.
7. The center of gravity and CV of the counting results FS of A-7312 standard particles shall meet the requirements in Table 11-4.

Table 2–39 Index Range of 4k-07 Standard Particles

Items		Reference
FS	Center of gravity	658.0~865.0
	CV	≤ target+0.7%
Total number of particles		1000~6000
Note: You can look up the target value of 760_DIFF FS CV in the target value sheet [4k07 (7μm) Standard Particle Single Bottle CV Standard Value] based on the batch number of standard particle.		

Table 2–40 Index Range of A-7312 Standard Particles

Items		Reference
SS	Center of gravity	455.0~695.0

Table 2–40 Index Range of A-7312 Standard Particles(continued)

Items		Reference
	CV	≤ 5.2%
FL	Center of gravity	1368.0~3080.0
	CV	≤ 3.3% or ≤ target+0.9%
Total number of particles		1000~6000

Determination of Standard Particle Test Results:

1. If the CV, background and center of gravity all meet the requirements in **Table 2–39**, but the blood sample scattergrams are still abnormal, the optical system is OK, in which case, check the fluidics, reagents or temperature control system.
2. If the CV and background, other than the center of gravity, meet the requirements in **Table 2–39**, perform the optical gain calibration according to the instructions in the Optical Gain Calibration Process.
3. If either the CV or the background does not meet the requirements in the range of **Table 2–39**, the causes may lie in the contaminated flow cell, in which case, perform the maintenance of flow cell.

Flow Cell Maintenance

1. Tap**Menu** >>**Service** >>**Maintenance** to enter the maintenance screen, tap **Flow Cell**in **Probe Cleanser Maint**, and then follow the software instruction to maintain the flow cell with probe cleanser (the purpose is to eliminate impurities on the internal wall of the flow cell, which always cause high background).
2. Afterward, click **Menu**>>**Maintenance**>>**Cleaning** to access the Cleaning screen, click **Flow Cell Flush** button, and perform flow cell flushing, and then test 4k-07 standard particles to confirm whether the scattergrams are improved. In case of no or poor improvement, repeat Step 9)~10) again.
3. If the test results of 4k-07 standard particles are inconsistent with the indexes in Table 11-3 after performing Step 2, replace the optical system.

2.5.5 Maintenance and Replacement of Optical System

NOTICE

- **Laser radiation.**
- **ESD protection.**
- **It is not allowed to open the top cover of the insulation box during startup.**
- **Make sure that the optical system wire and flow cell tube are connected correctly and securely.**

Optical System Maintenance

1. Analyzer cleaning: Click **Menu**>>**Service**>>**Maintenance**>>**Cleaning** to access the "Cleaning" screen, and click **Analyzer Cleaning** button to flush the analyzer.
2. Probe cleanser maintenance: Click **Menu**>>**Service**>>**Maintenance**>>**Probe Cleanser Maintenance**to access the screen and then click **Flow Cell Probe Cleanser Maintenance**button. Afterwards, conduct the flow cell probe cleanser maintenance as per the instructions on the screen.
3. Flow cell cleaning: Click **Menu**>>**Service**>>**Maintenance**>>**Sensor Maintenance** to access the screen and then click **Flow Cell Cleaning** button to clean the flow cell.

Optical System Replacement Steps

1. Disconnect the wires, optical system wire, flow cell connecting tube, and waste box connecting tube at the top of the analyzer that will affect the opening of board-fixing sheet metal (as shown in **Figure2–89** and **Figure2–90**).
2. As shown in **Figure2–88** and **Figure2–89**, attach a new optical system to the main unit with two captive screws and two M4X8 stainless steel cross-head screw assemblies. Insert the optical system connection cable into the motherboard J49 OPTIC (routing as shown in **Figure2–88** and **Figure2–89**, which shall be fixed with wire clips).
3. Well connect the flow cell tubes and waste container tube as per **Figure2–90**.
4. After assembling the optical system, start up the analyzer and successively click **Status**>>**Voltage & Current** to check whether the laser drive current and laser output power are within specification. If not, replace the optical assembly.

Figure2–88 Assembly Procedure for Replacing Optical System

1	Installation position of optical system	2	Wiring of optical system

Figure2–89 Wiring of Optical System

1	Optical system connection cable		

Figure2–90 Flow Cell - Waste Container Tube Connection

1	DIFF sample batch preparation tube	2	RET sample batch preparation tube
3	Optical sheath fluid	4	Optical waste

2.5.6 Optical Gain Commissioning

NOTICE

- **Generally, no manual modification is required for PMT gain. If the target gain of FL channel is out of specification, however, it is required to manually modify the PMT gain and restart counting, as shown in Figure2–92.**

The optical gain commissioning steps are as below:

1. Click **Menu>>Calibration>>Optical Gain Calibration** to access the Optical Gain Calibration screen (as shown in **Figure2–91**). Select **Calibrator** to enter the optical target of calibrator in the target column.

Figure2–91 Optical Gain Calibration Screen

1	Optical target of calibrator	2	Current gain deviation

2. Place a calibrator at the position 1# of the tube rack, and tap **Start Count**. After measurement, a prompt of saving current counts appears. In case of normal counting, click **Yes** and continue the following automatic counts.
3. After continuous measurement, if it is ≤2.0% in the deviation column. Click **Save** and **Export** to save the data and record the gain factor in the table.
4. Click **Menu>>Setup>>Gain Setup** to check whether the **Set Value** of optical gain has been updated to the one saved in Step 3.

Figure2–92 Manual Modification of Optical Gain

	DIFF channel			RET channel			Current
	Set	Range	Regulation	Set	Range	Regulation	
FS	110	[0,220]	100.0%	115	[1,255]	100.0%	2
SS	15	[0,220]	100.0%	140	[0,250]	100.0%	234
SSPMT	110	[100,130]	100.0%	85	[55,105]	100.0%	175
FL	100	[4,255]	100.0%	140	[46,255]	100.0%	70
PMT	124	[10,160]	100.0%	117	[10,160]	100.0%	213
WAS	15	[0,220]	100.0%	140	[0,251]	100.0%	149

2.6 Temperature Control System

2.6.1 Overview of Temperature Control System

Overview of Temperature Control System

The temperature control system is an assembly preheating or maintaining the temperature of the assemblies including diluent (DIL), blood routine and CRP reaction baths. The temperature control system supports the temperature control and reaction protocols for blood routine and CRP, and mainly consists of the diluent preheating bath assembly, blood routine reaction bath assembly and CRP measurement assembly (temperature control of ESR assembly is excluded here).

The temperature control system builds the reaction environment suitable for the analyzer's reaction system under the normal working temperature.

Table 2–41 Temperature Control Indexes

Temperature control system	Assembly	Object	Evaluation point	Target temperature (°C)	Required temperature (°C)	Ambient temperature (°C)
Blood routine	Diluent Preheating Assembly	Diluent flowing in the instrument	Preheating bath temperature sensor	Track ambient tempera ture	N/A	15~35
	Blood routine reaction bath assembly	DIFF Lyse	Sample for DIFF batch reaction	42.5	42~43.5	15~35
		RET diluent	Sample for RET batch reaction	42.5	42~43.5	15~35
CRP	CRP measure ment assembly	LC Lyse	Sample for CRP measurement	37	36~38	15~35

2.6.2 Diluent Preheating Assembly

Diluent Preheating Assembly

The preheating assembly heats the diluent to the ambient temperature at the diluent inlet of the instrument, so that the diluent stored at a low temperature can be used without a long time for heating.

2.6.3 Blood routine reaction bath assembly

Blood routine reaction bath assembly

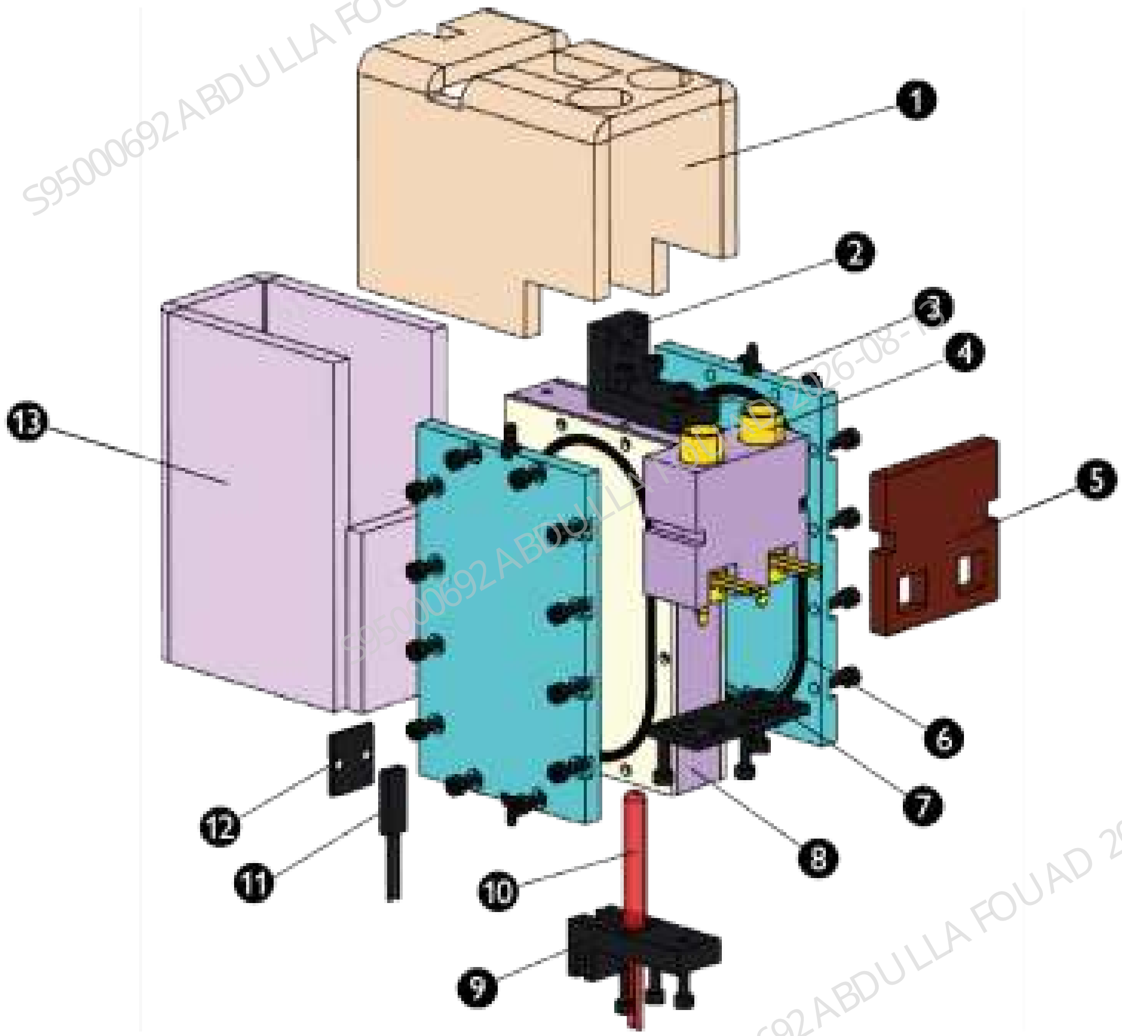
Blood routine reaction bath assembly supports the following reagent preheating functions:

- Lyse for DIFF hematology analysis.
- Diluent for RET hematology analysis.

The target incubation temperature for above reagents is 42.5℃ under the normal working environment (ambient temperature of 15~35℃). The heating assembly of incubation reaction batch supports two functions for temperature control:

- Lyse/diluent preheating.
- Reaction bath incubation temperature stabilization.

Figure2–93 Figure: Exploded View of Blood Routine Reaction Bath Assembly Structure



1	Heat insulation cotton	8	Membrane basal plate
2	L-shape bracket	9	L-shape bracket
3	Preheating bath body	10	Heating rod
4	Incubation bath assembly	11	Temperature Sensor
5	Heat insulation cotton	12	Fixing plate
6	Seal ring	13	Heat insulation cotton
7	Positioning plate		

2.6.4 Assembly Replacement

See Chapter 7.37 of the manual for maintenance and replacement of diluent preheating assembly
See Chapter 7.29 of the manual for maintenance and replacement of reaction bath assembly

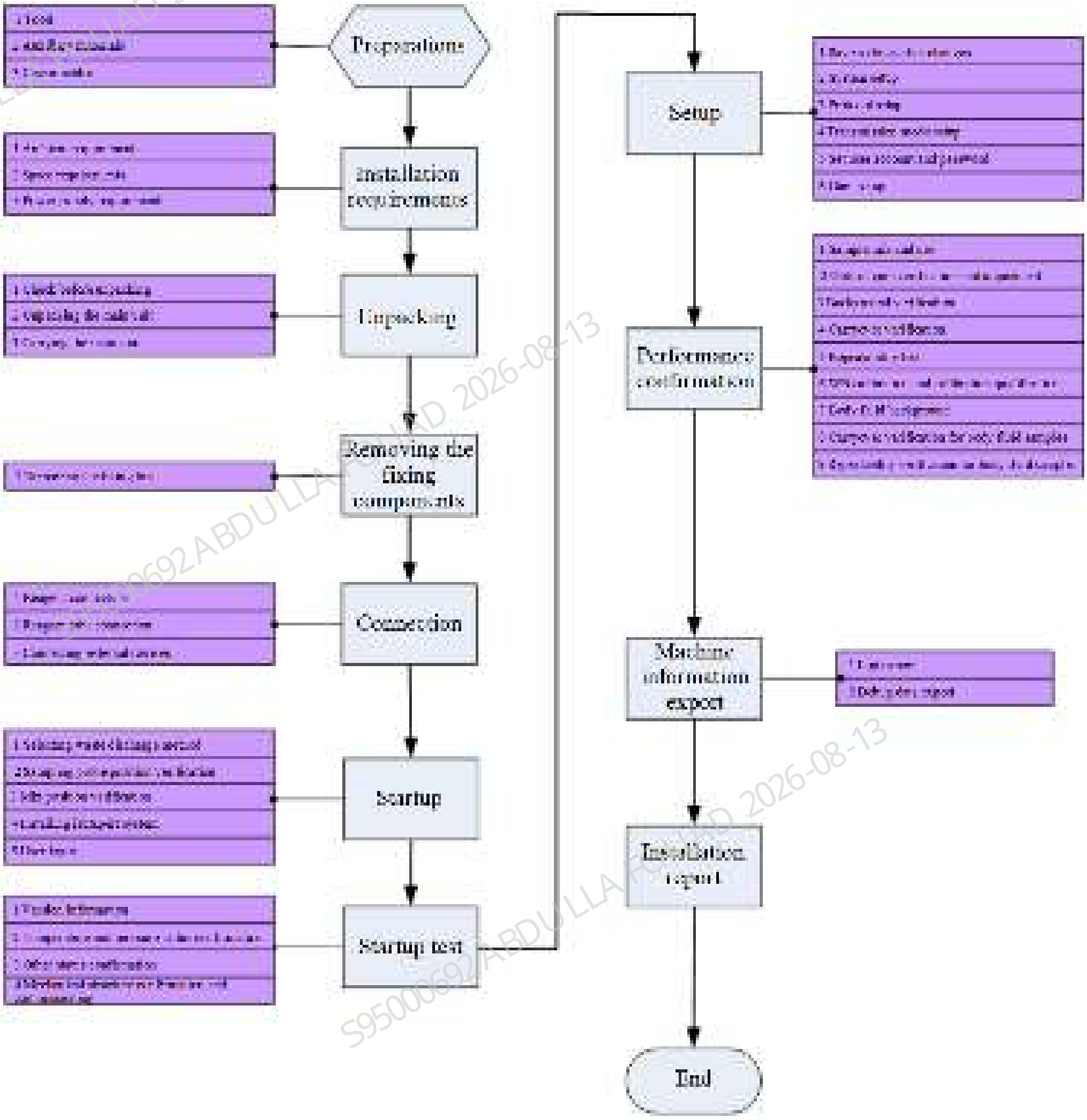
3 Installation

3.1 Overview

The installation guide is a normative document instructing the service engineers to install the product on user site. The whole installation process includes the following procedures: preparation before installation, installation, commissioning, validation and acceptance.

3.2 Installation Procedure

Figure3–1 Installation Flowchart



3.3 Site Survey

Dimensions and Weight of Outer Package

The gross weight of the product is about 150 kg. A forklift is needed to transport it. The instrument transport route must be investigated to ensure that the instrument can be transported to the destination.

Dimensions and Weight of Main Unit (Outer Package)

Length	Width	Height	Gross Weight
820mm	700mm	810mm	About 84 kg

Dimensions and Weight of Packaged Autoloader (10X5)

Length	Width	Height	Gross Weight
715mm	340mm	370mm	About 15 kg

Dimensions and Weight of Packaged Autoloader (10X5)

Length	Width	Height	Gross Weight
715mm	340mm	370mm	About 12 kg

Figure3–2 Outer Package

Dimensions and Weight of Main Unit and Autoloader

The following table lists net weight and size of the instrument. Use handles to carry the instrument. By default, the instrument requires three persons to carry. Make sure there is sufficient space for instrument transport.

Table 3–1 Dimensions and Weight of Main Unit (10X5 Autoloader)

Dimensions and Weight of Main Unit (10X5 Autoloader)	Value
Width (including autoloader)	About 655mm
Height (including autoloader)	About 600mm
Depth (including autoloader)	About 728mm
Weight (including autoloader)	About 78kg

Table 3–2 Dimensions and Weight of Main Unit (5X6 Autoloader)

Dimensions and Weight of Main Unit (5X6 Autoloader)	Value
Width (including autoloader)	About 411mm
Height (including autoloader)	About 600mm
Depth (including autoloader)	About 728mm
Weight (including autoloader)	About 73kg

Dimensions and Weight of Autoloader (10X5 Autoloader)	Value
Width (including autoloader)	About 655mm
Height (including autoloader)	About 600mm
Depth (including autoloader)	About 728mm
Weight (including autoloader)	About 78kg

3.4 Preparations

Tools

Table 3–3 Tool List

Type	Name	Quantity
Installation tools	Scissors or cutting pliers	1
	Inner hexagon spanner (M8)	1
	Cross-headed screwdriver (M3)	1
	Cross-headed screwdriver (M4)	1
	Flat-headed screwdriver (M4)	1
	Flat-headed screwdriver (M1)	1
	Pipette (200 µL)	1

Consumables

Table 3–4 Consumable list

Type	Name	Quantity
Frequently-used	Pipette	20
	WB tube rack	2

Table 3–4 Consumable list(continued)

Type	Name	Quantity
	Gloves	2
	Micro-WB tube rack	1
	Tissues	1 (roll)
	Empty waste container and carton	1
Mechanical mechanism confirmation and adjustment	Empty WB tubes	5 to 8
	Empty peripheral blood tube (depending on configuration)	5 to 8
Background verification	Empty tubes	3 to 5
Optical gain verification and adjustment	Calibrator	1

NOTICE

- **Use controls and fresh blood samples as alternatives.**
- **Either QC or fresh blood samples are prepared for repeatability test.**
- **Either calibrators or random fresh blood samples are prepared for calibration and calibration verification.**

3.5 Installation Requirements

The analyzer has been strictly tested before shipment. The analyzer has been well packaged before shipment to avoid any impact during transportation. When you receive your analyzer, carefully inspect the carton. In case of any signs of mishandling or damage, contact Customer Service department or your local distributor immediately.

Environment Requirements

	Normal working environment	Storage and Transportation Environment	Operation Environment
Ambient temperature	10 ℃□ 35 ℃	-10 ℃□ 40 ℃	5 ℃□ 40 ℃
Relative humidity	30% □ 85%	10% □ 90%	10% □ 90%
Atmospheric pressure	70.0kPa □ 106.0kPa (Note)	50.0kPa □ 106.0kPa	70.0kPa □ 106.0kPa

NOTE

The altitude requirement for normal operation is -400m~3,000m.

- Keep the instrument away from water.
- Keep the instrument away from dust, mechanical vibrations, loud noises and electrical interference.
- Do not use this instrument in close proximity to sources of strong electromagnetic radiation.
- Keep the instrument away from brush-type motors, blinking fluorescent lights, and electrical contacts which are frequently on/off.
- Protect the instrument against direct sunlight, heat source and wind source.
- Do not use this instrument in environment filled with conductive or flammable gases.
- Well ventilated environment is preferred.
- No inclined arrangement is preferred.
- Indoor use only.

Space Requirements

To guarantee the space for repair and maintenance, favorable heat radiation performance and sufficient room for the rear fluidic tubes of the analyzer (for the purpose of smooth reagent flow), the installation environment shall satisfy the following requirements:

- Proper placement height for analyzer.
- Minimum 500mm between analyzer side and wall.
- Minimum 600mm space above the analyzer.
- Minimum 250mm space behind the analyzer.
- The height of diluent relative to the analyzer shall be less than 1.0m. The lyse and the analyzer shall on the same level.
- The table where the analyzer is placed shall be able to withstand minimum 180kg weight.

CAUTION

- **When installing or using the analyzer, make sure at least the two inside supporters of the autoloader are on the supporting table of the analyzer.**
- **Do not install the equipment at a place where it is difficult for power supply switch operation.**

Power Requirements

WARNING

- **Make sure the system is properly grounded.**
- **Confirm that the analyzer switch is turned off before connecting it to a power supply.**
- **Use a multimeter to measure the zero-to-earth voltage, which is required to be less than 5V.**

CAUTION

- **Using power strip may result in electrical interference and the unreliable analysis results. Please place the analyzer near the electrical outlet to avoid using the pinboard.**
- **Please use the accompanying power cord. Using other power cord may damage the analyzer or lead to unreliable analysis results.**

	Voltage	**Frequency**	**Input power**
Main Unit (Analyzer)	100V-240V~ (allowance ±10%)	50Hz/60Hz (allowance ±1Hz)	600VA

3.6 Unpacking

CAUTION

- **Installation by personnel not authorized or trained by Mindray may damage your analyzer. Do not install your analyzer without the presence of Mindray-authorized personnel.**

Checking before Unpacking

Check for Missing Items

The packaging box of BC-760[B]/BC-760[R]/BC-780[R] series mainly consists of one package for analyzer and one package for autoloader, including one analyzer, one autoloader and one package for accessories (including manual, handle and capping assembly).

Package Inspection

1. Check the outer package for any damage. If the package is found damaged, please take photos for evidence, and contact your distributor and the logistic agency to check if there is any

accident during transport. Make sure you install the instrument after confirming that there is no damage to the instrument caused during transport.

2. Check the two tilt-watch labels on the outer package of the main unit. If it is red, the package was tilted over 80 degree during transportation. Record the issue in the installation log, and notify Mindray Customer Service Department or your local distributor immediately. As shown in the following figure.

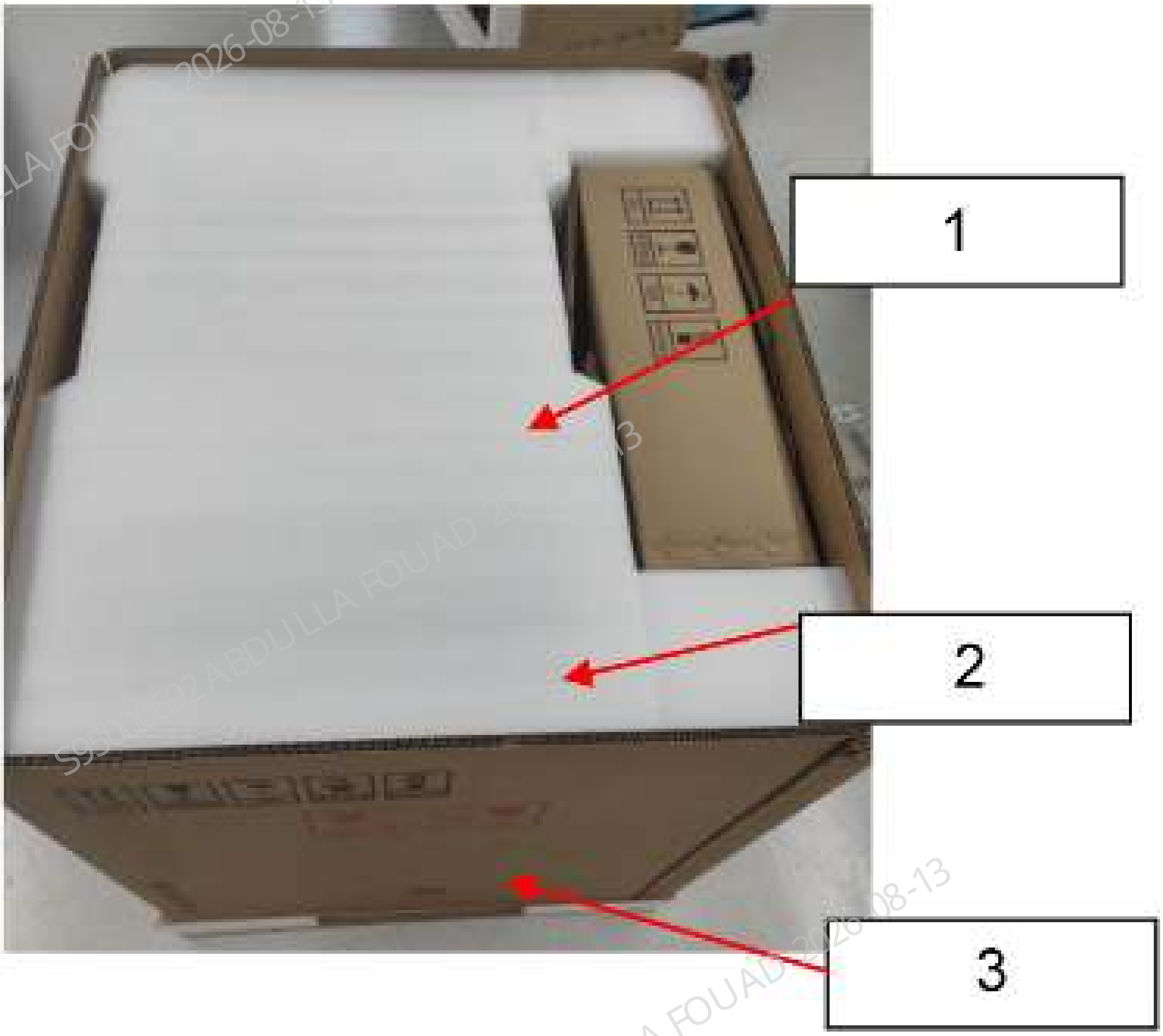
Figure3–3 Outer Package of Analyzer

1	Top cover of outer package

Unpacking the Main Unit

1. Cut off the binding straps with scissors or cutting pliers, and take the packing list. Then take down the top cover of the outer package to open the outer package.

Figure3–4 Taking Down the Top Cover of Outer Package



1	Accessory package
2	Limit position foam
3	Main unit (analyzer) carton

2. Take down the accessory package and limit position foam.

Figure3–5 Taking Down the Protective Foam

3. Take down the main unit (analyzer) carton.

Figure3–6 Taking Down the Carton

4. Use an Allen wrench to remove the L-shape sheet metal block connecting the main unit (both sides) and pallet.

Figure3–7 Removing the L-shape Sheet Metal Block

1	M8X30 inner hexagon bolt

Unpacking the Autoloader

There are two autoloader models: 5x6 and 10x5. They are same in packaging form and structure, but different in packaging size. Therefore, 10x5 is taken as an example here.

1. Use a blade to cut the tape of the package.

Figure3–8 Remove the Tape

:

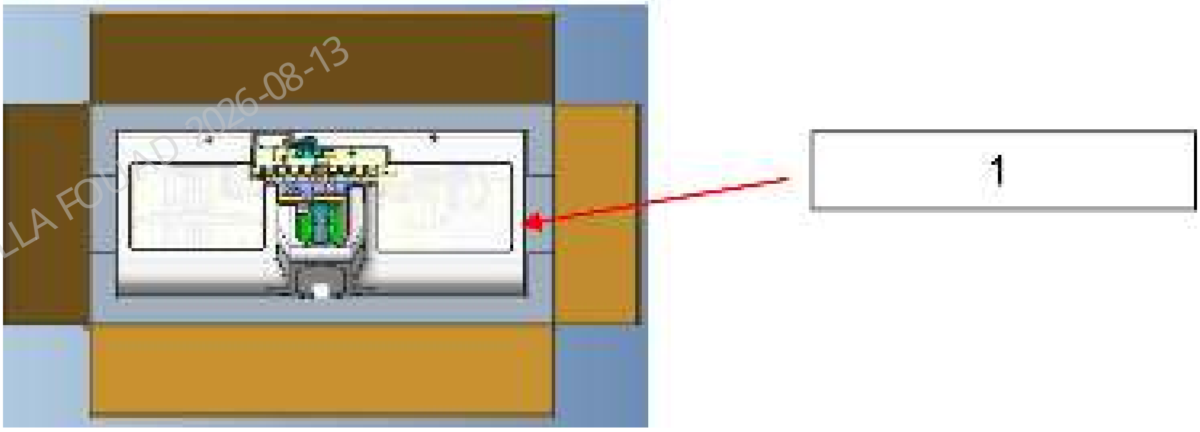
1	Tape

2. Open the package and take out the bridging beam of the autoloader and the upper autoloader foam.

Figure3–9 Taking Out the Bridging Beam of the Autoloader

1	Bridging beam of the autoloader
2	Upper autoloader foam

3. Take out the autoloader.



1 Autoloader

Verifying Contents

Check the delivered components against the packing list to see if everything is delivered.

NOTICE

- **Any inconformity between the delivered parts and items on the packing list, contact Mindray Customer Service department or your local distributor.**

Inspecting Appearance of Key Parts of Main Unit and Autoloader

1. Check the analyzer cover for deformation and damage.
2. Check the reagent ports on the rear side of the analyzer for breakage or bending.
3. In case of any issue, contact Mindray Customer Service Department or your local distributor immediately.
4. Check the cover of the autoloader for deformation and damage.

Moving the Main Unit

1. Insert the transportation handles into the 4 holes reserved for transportation purpose on the bottom plate. Note the handles should be in their horizontal position, and be inserted to the limit position. As shown in the following figure.

Figure3–10 Inserting the Handles

2. Rotate the handles to their vertical position, as shown the figure below. Try to pull the handles and make sure they cannot be pulled out. Do not do the transportation if the handles are not tightly inserted.

Figure3–11 Placing the Handles to Vertical Position

NOTICE

1. **During handling, two persons are required to use two handles respectively for handling at the same time, and one person shall hold the front of the main unit to prevent rollover and other accidents.**
2. **Reserve sufficient space to place the main unit. Ensure that the two inner legs of the autoloader can be placed on the desk of the main unit after the autoloader is installed.**
3. **Avoid main unit collision during transport, especially the back-panel connector position. After transport, check the connector.**

3.7 Removing the Fixing Components

NOTICE

- **Remove the red ties only. Do not cut any ties in other colors.**

Removal of Fixing Parts for Analyzer

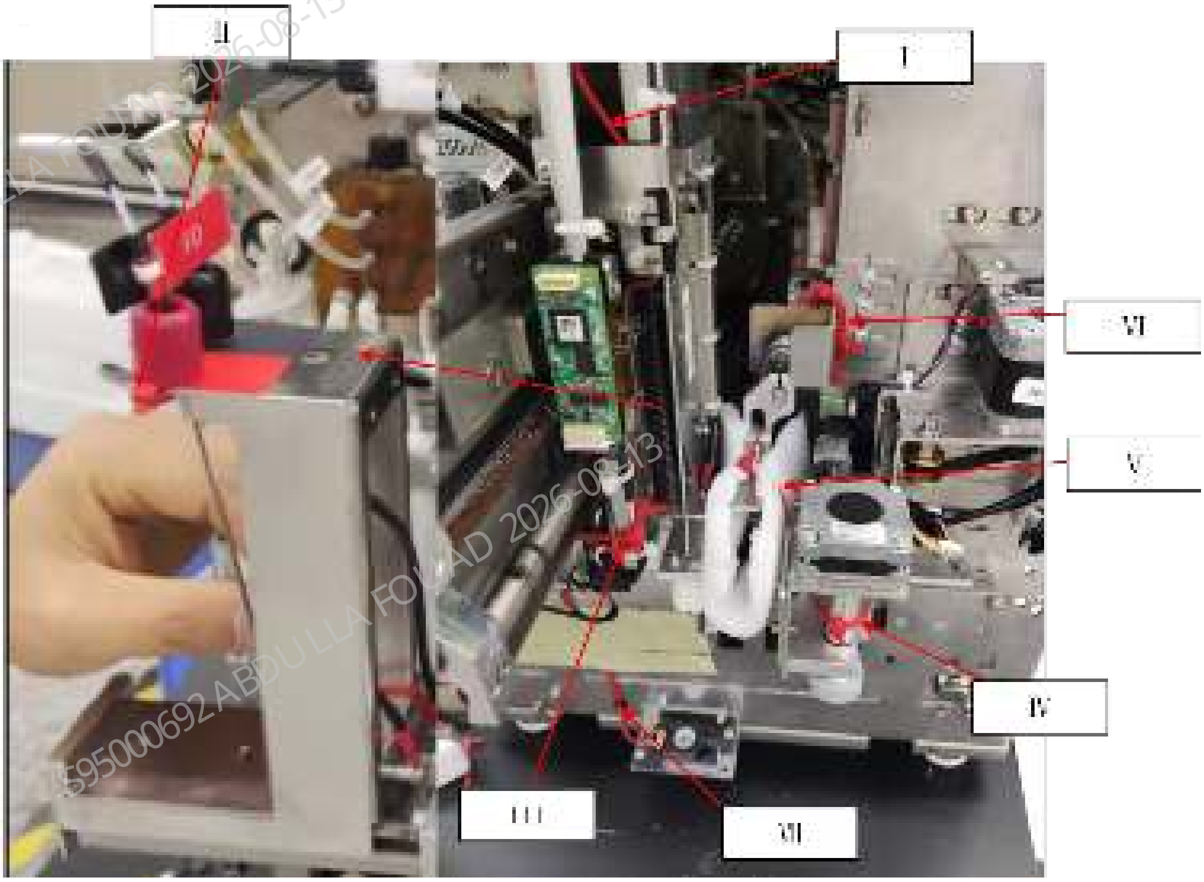
Remove the wrapping films from the analyzer.

1 Wrapping films

The wire ties for the main unit (analyzer) are distributed as follows:

Remove the ties at the corresponding positions in turn, including one tie at Position I, three ties at Position II, two ties at Position III, one tie at Position IV, one tie at Position V and two ties at Position VI.

Figure3–12 Layout of wire ties



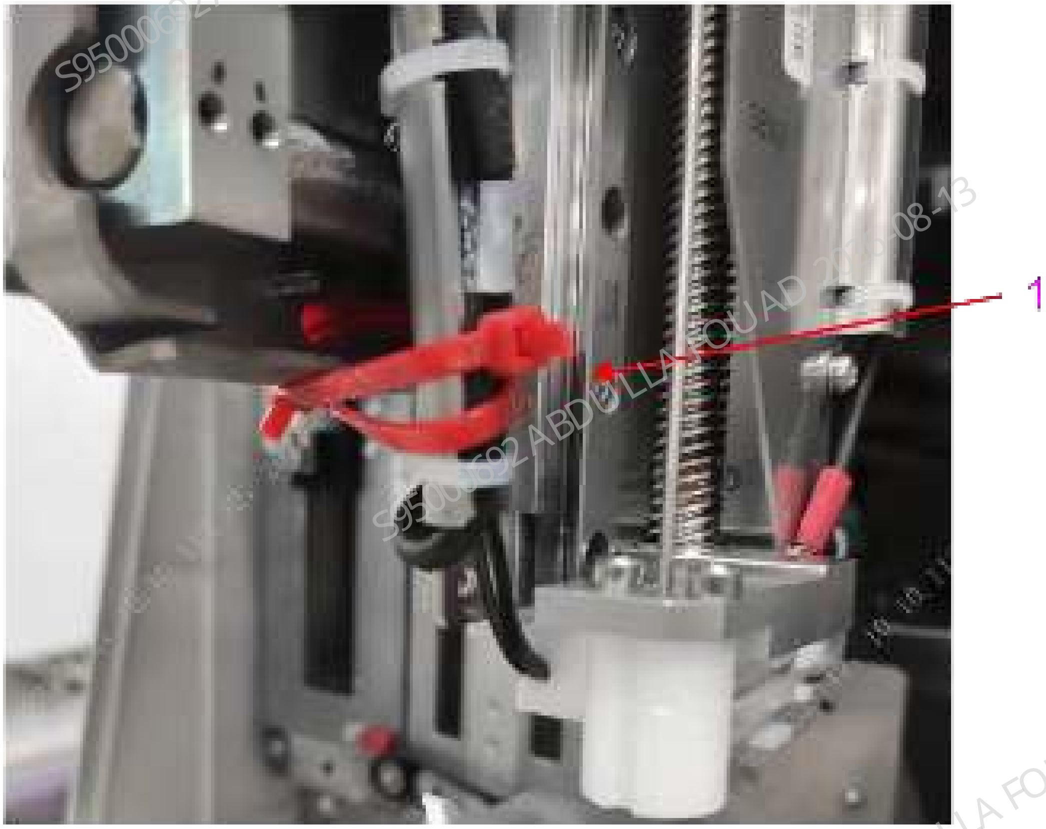
□	At sampling Z-axis assembly
□	At dye container cap assembly
III	At sampling tube rack
□	At rotary compressing assembly*
V	At tube clamp*
VI	At blending Z-axis assembly
VII	Serial port line outlet of autoloader analyzer*

(The places with * only involve the autoloader models, not the open models)

I: Position of sampling Z-axis assembly.

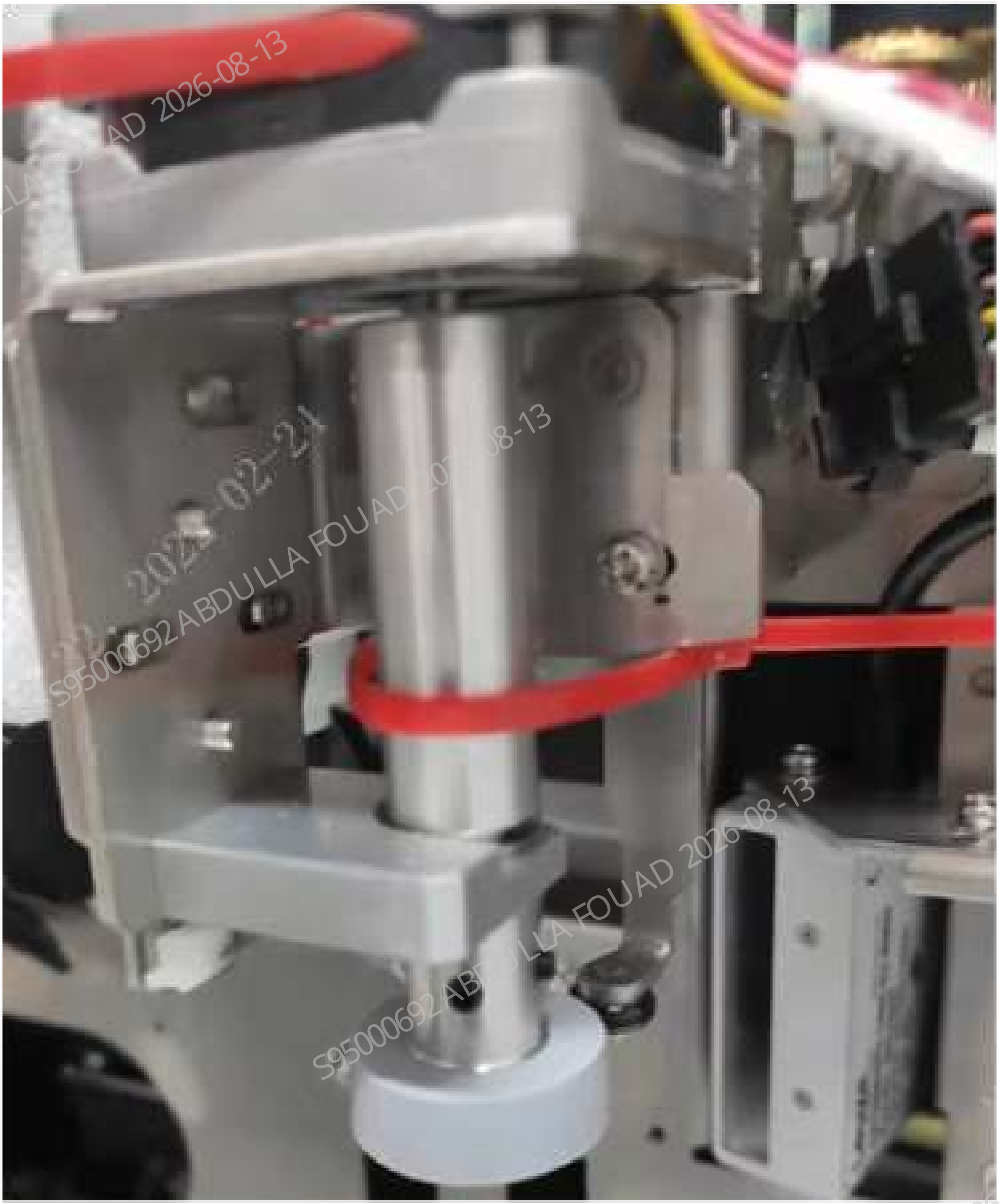
1 The compact bag covers the dye probe

□: Dye container cap assembly.



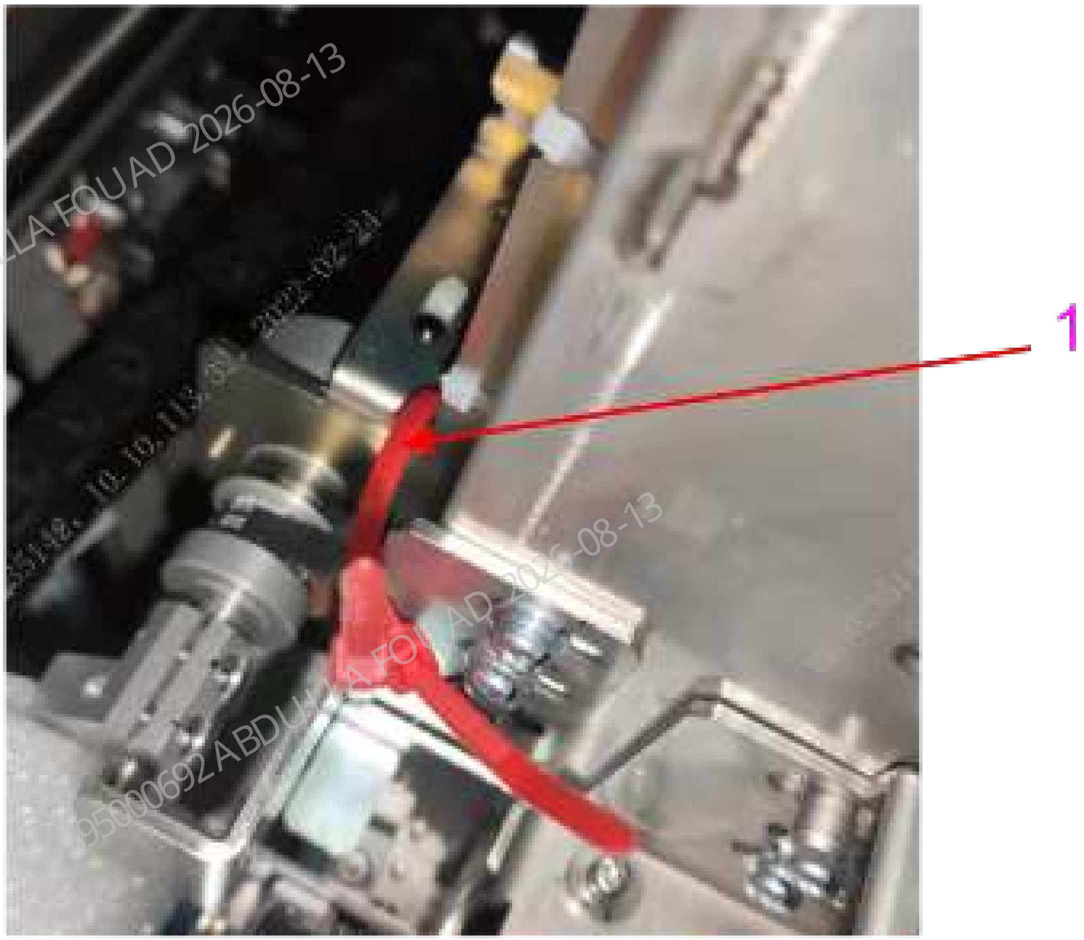
1 Two wire ties

III: Sampling tube rack.



IV. Rotary compressing assembly.

V. Tube clamp.

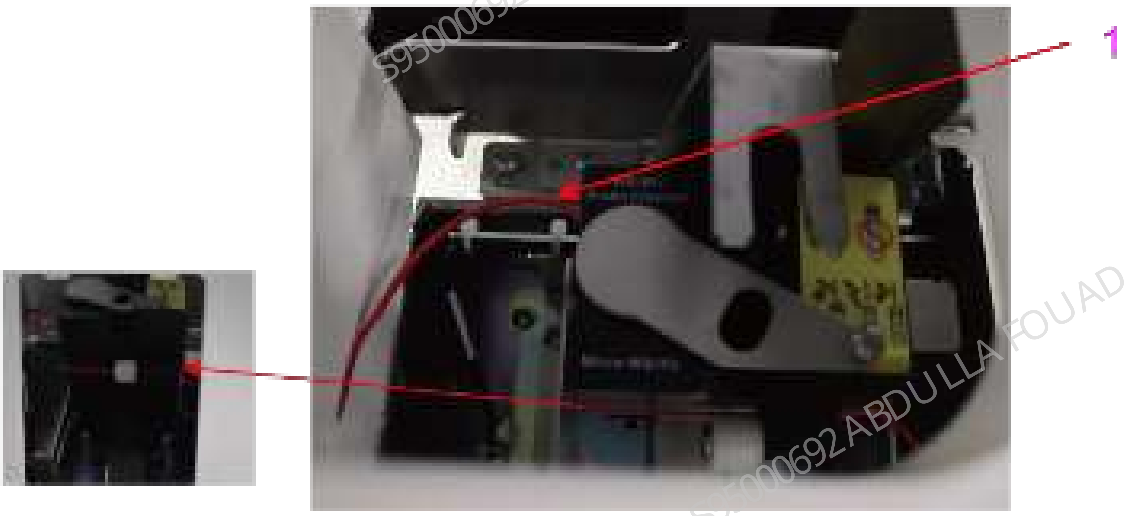


1	Two wire ties

VI. Blending Z-axis assembly

Removing the Wire Ties for Autoloader

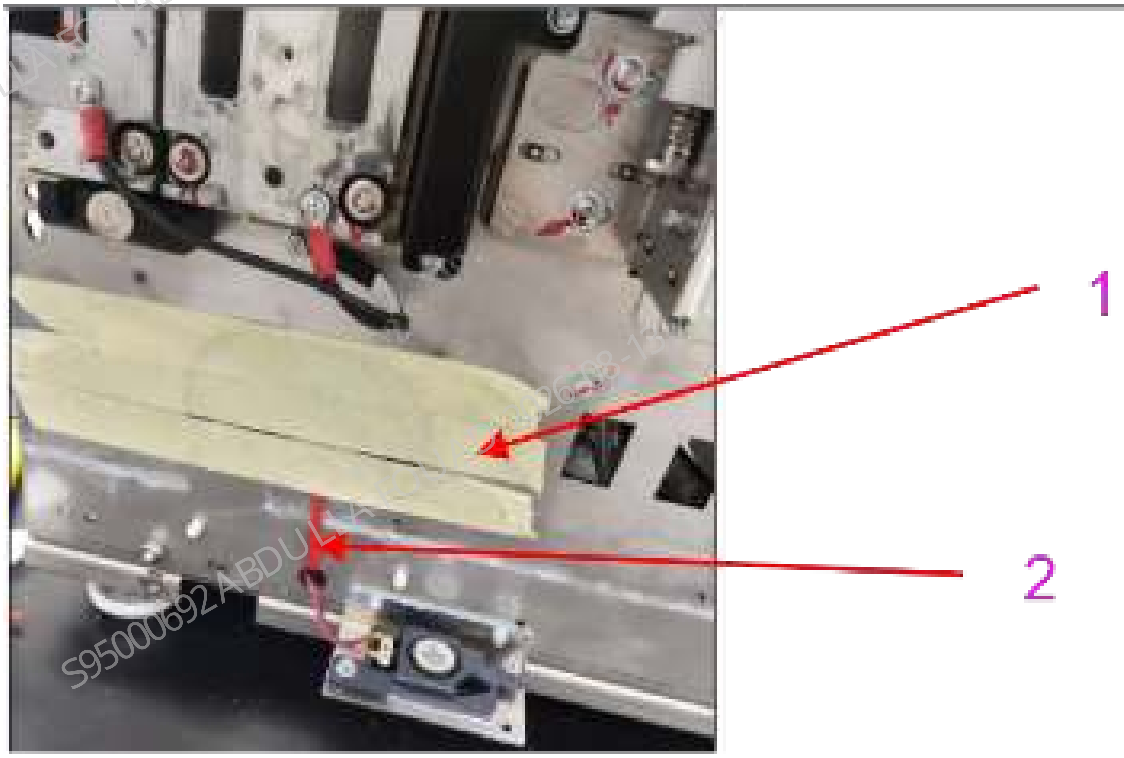
1. Remove the ties from sample compartment.



1	Wire ties

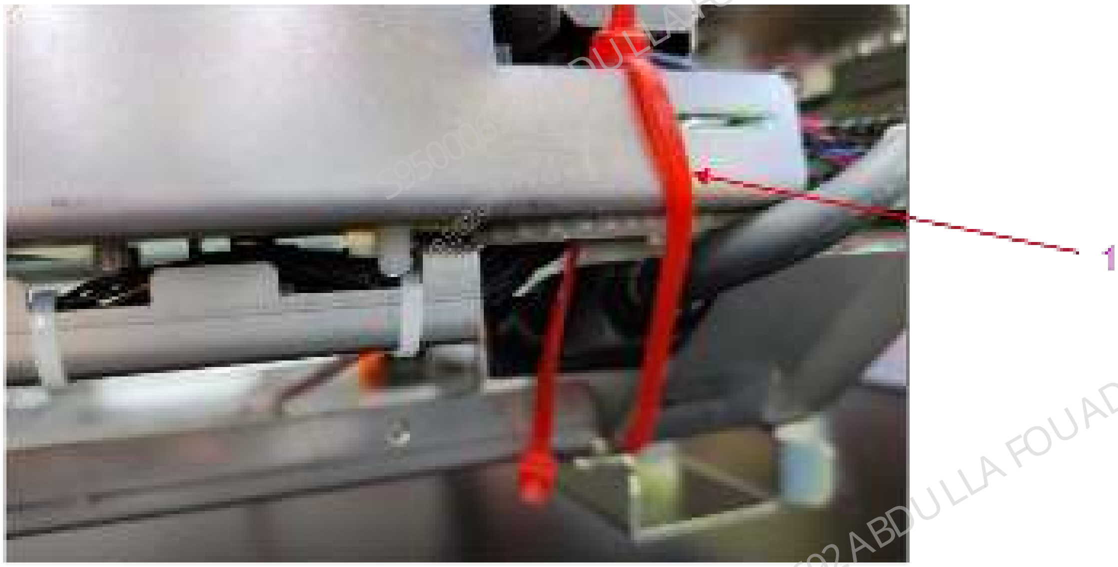
Remove the tape

1. Remove the high-temperature adhesive paper and wire ties from the connecting line of the autoloader.



1	High temperature adhesive paper
2	Wire ties

2. Cut off the ties securing the serial port line of the autoloader.



1	Wire ties

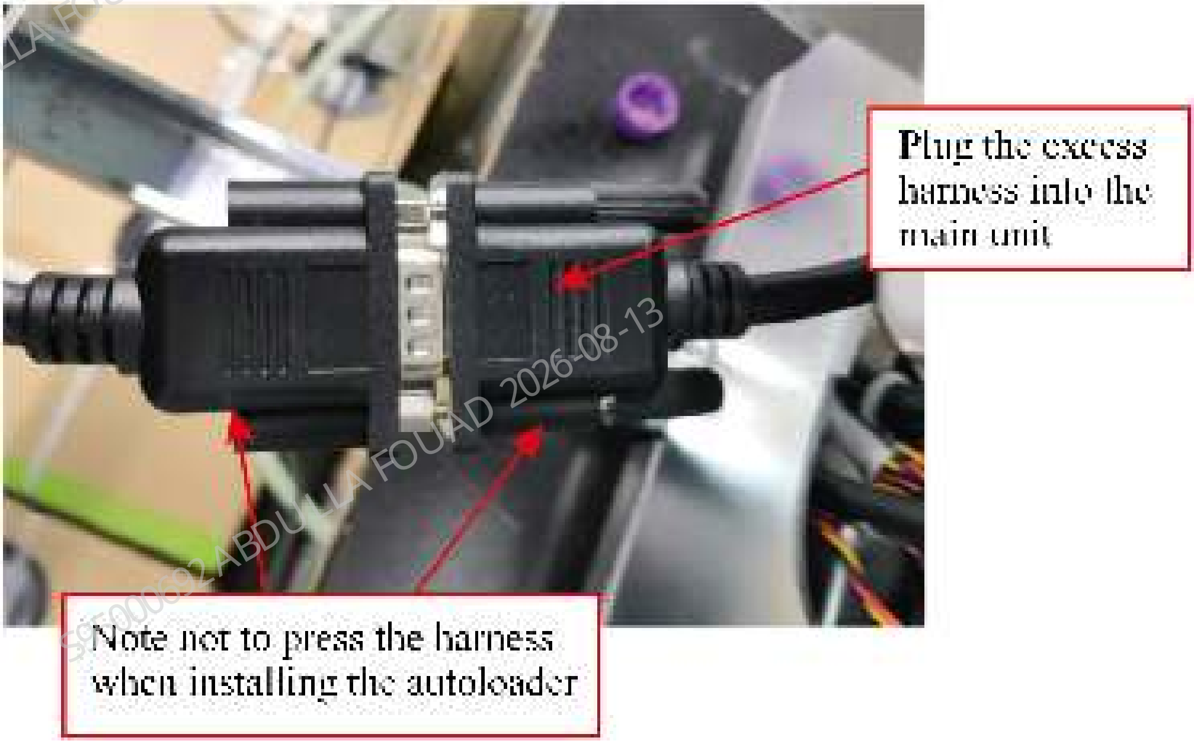
Remove the high-temperature adhesive paper from dye compartment door and latex comportment door (depending on configuration).

3.8 Autoloader Connection

1. Use M4X8 stainless steel cross pan head combination screws to fix the two sets of autoloader crossbeams to the main unit respectively.

2. Check whether the harness at the back of the autoloader protrudes from the sheet metal plane (as shown in the figure below). If yes, sort out the harness first.
3. Insert the supporting board of the autoloader crossbeam into the autoloader crossbeam, and push the autoloader to an appropriate position towards the main unit so that the autoloader can be stably maintained on the autoloader crossbeam when not support by two hands.

4. After connecting the wiring harness connector of the autoloader with the connector led out from the main unit, insert the connector and the excess harness into the hole in the lower left corner of the front side of the main unit. When pushing the autoloader towards the main unit, do not clamp the harness, otherwise the autoloader cannot fit with the main unit.



5. Push the autoloader to the bottom, make sure that the positioning pin of the autoloader is inserted into the corresponding pin hole of the main unit, and that the left and right sides of the autoloader back metal fit the baseplate of the main unit. If they do not fit the main unit baseplate, the wire is caught or the autoloader is not pushed into the correct position. Then use two M4X8 stainless steel cross pan head combination screws to lock the autoloader bottom and crossbeam.

3.9 Reagent Installation

Model Description

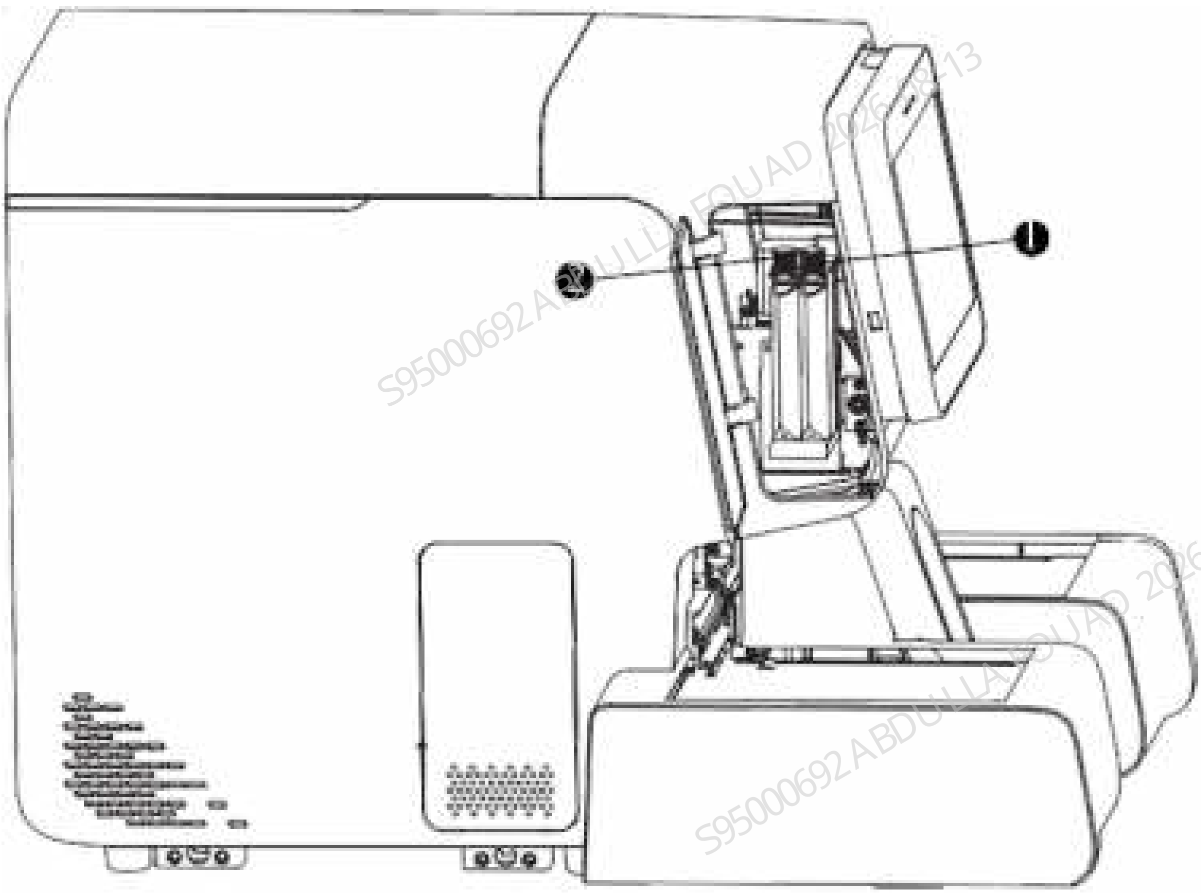
Analyzer configuration.

Series	Model	Differences in configuration		
		RET	ESR	Closed-sampling
760 series	760[B]		•	•
	760[R]	•	•	•
780 series	780[R]	•	•	•

3.10 Reagent Tube Connection

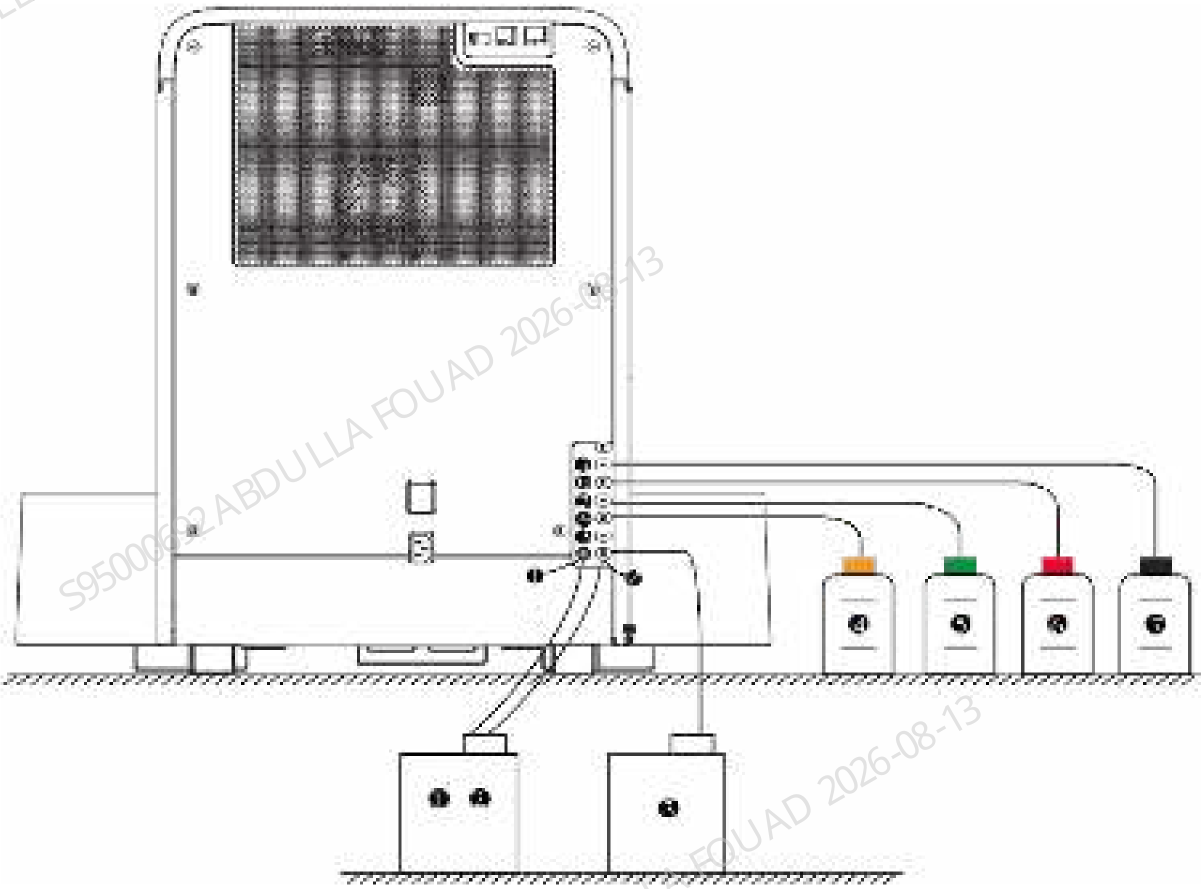
When connecting the reagents, make sure the color of the reagent container cap assembly is the same as that of the reagent inlet to which it is connected.
Connect the reagent tubes to the analyzer as shown in the figure below (10x5 autoloader model is taken as an example here):

Figure3–13 Reagent Connection - Dye



Port No.	Name	Description
1	Dye for FD hematology analysis	N/A
2	Dye for FR hematology analysis	Applicable to BC-760[R]/BC-780[R] model

Figure3–14 Reagent Connection - Lyse, Diluent, Waste

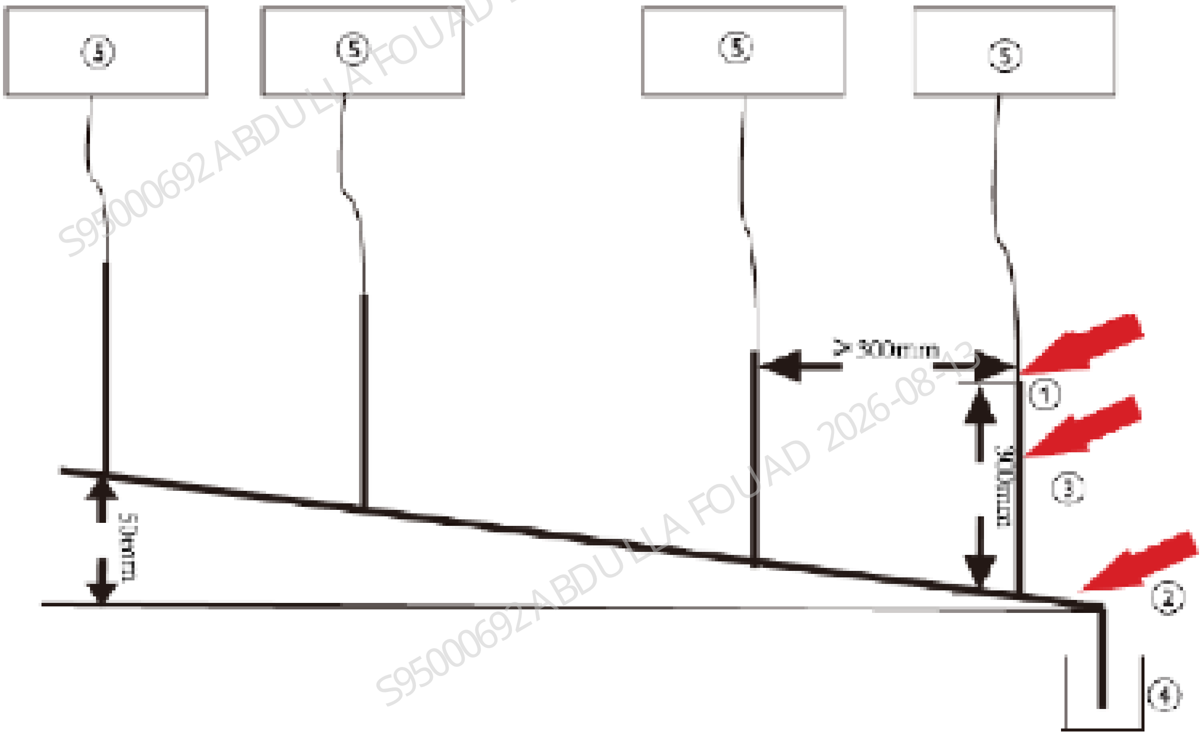


Port No.	Name	Description
1 2	Waste container	Port (2) is for waster container connection, while port (1) is for waste floater sensor connection (detect whether the waste container is full)
3	Diluent for DS hematology analysis	N/A
4	ESR cleaner	N/A
5	Lyse for LH hematology analysis	N/A
6	Lyse for LD hematology analysis	N/A
7	Diluent for DR hematology analysis	Applicable to BC-760[R]/BC-780[R] model

Installation Requirements for Waste Direct-Discharge

When you are using the waste direct-discharge method, follow below requirements:

1. The inner diameter of the common pipes ≥2".
2. The common pipes to the instruments must include both horizontal and vertical sections of pipes. The vertical pipe should be in the length of 300mm to avoid foams spilling over.
3. If multiple devices are installed in your laboratory, the distance between the vertical pipe to this analyzer and the vertical pipe to the neighboring instrument must be ≥300mm.
4. The outlet of the vertical pipe connected to this analyzer should be open.
5. Place the common pipes at an angle, with its upper-stream section 50mm higher than the lower stream section, so that the waste automatically flows down into the waterway by gravity, as shown in the figure below.
6. The direct-discharge pipe of the instruments must not be bent, and is properly fixed into the common pipe.

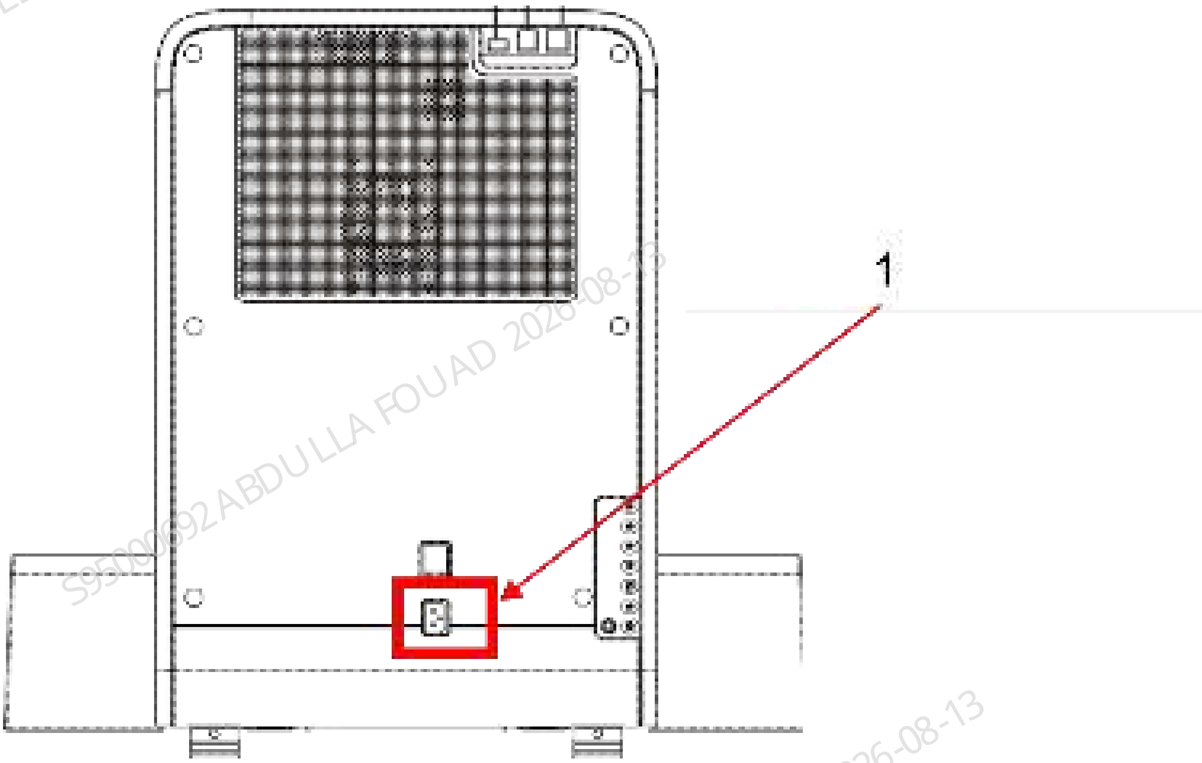


1	Vertical pipe connected to the analyzer
2	Common pipe
3	Vertical pipe
4	Sewer
5	Analyzer

3.11 Connect Peripherals

Plug the power cord to the outlets on the back panel of the analyzer. Make sure all the wires are tightly and properly connected. (10x5 autoloader model is taken as an example here).

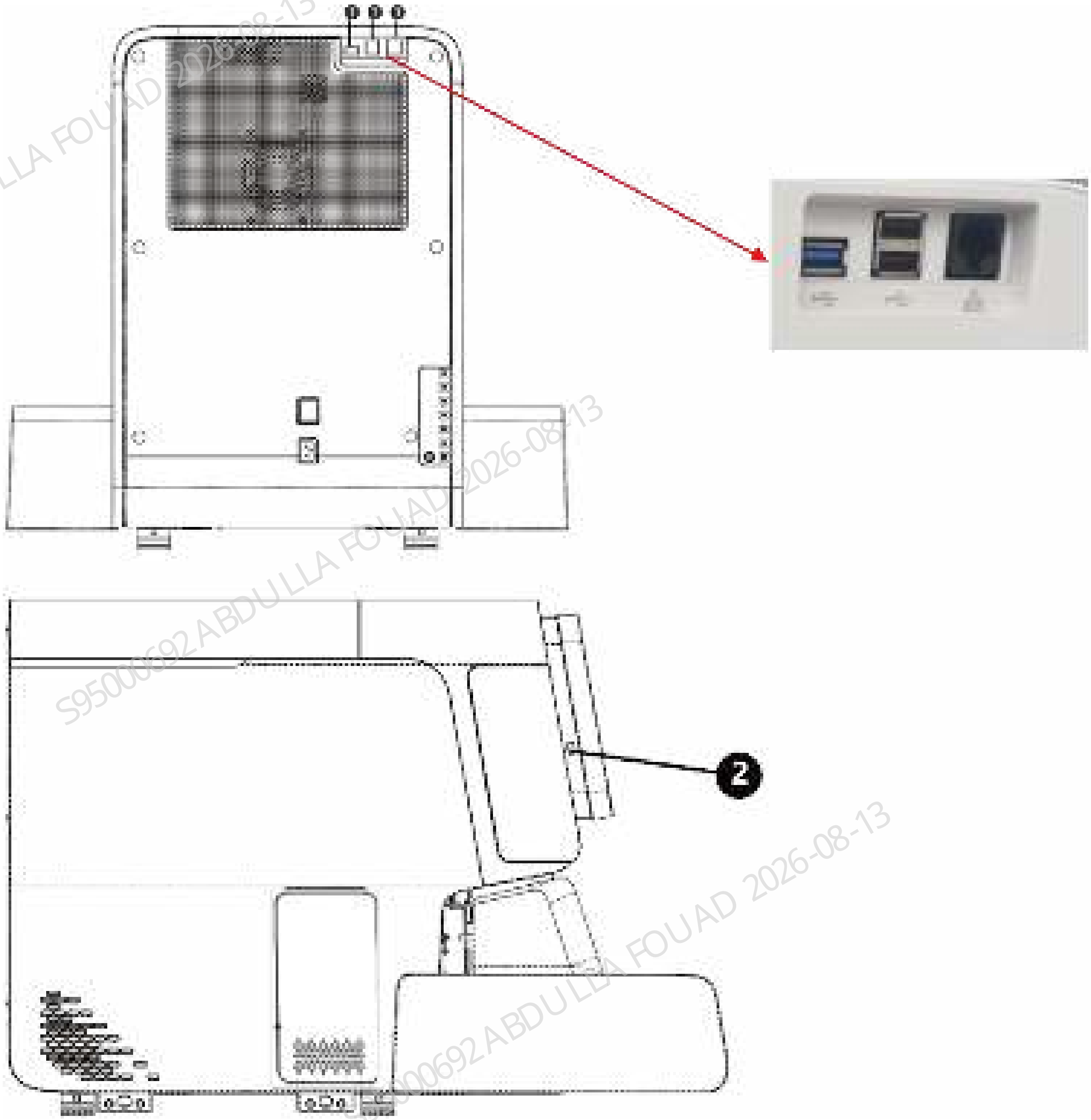
Figure3–15 Power Supply Connection



1	External power inlet

USB and network port connection

Figure3–16 Connect Peripherals



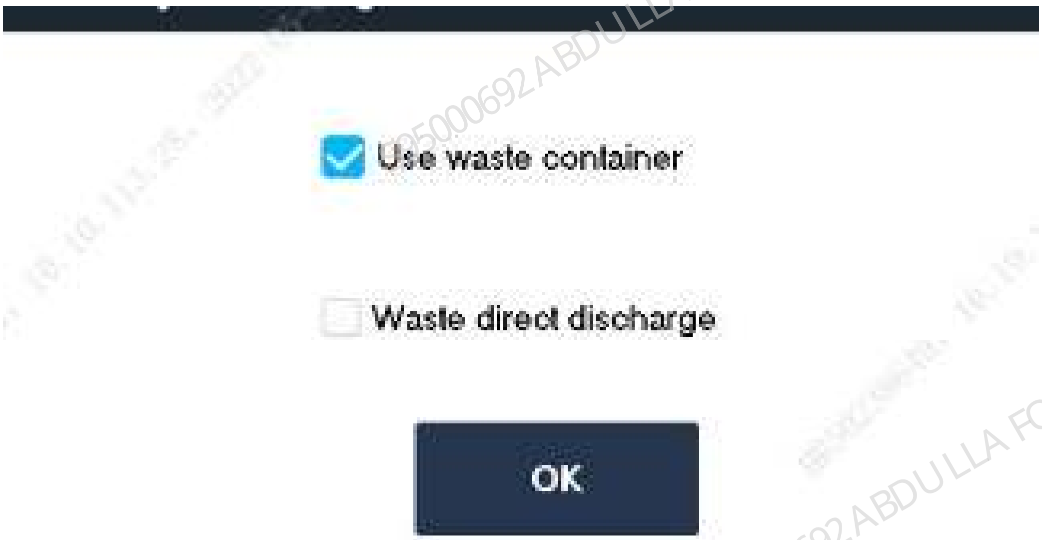
Port No.	Port	Connection Instructions
1	USB port (protocol 2.0)	Connect to the USB ports for peripherals (e.g. printers and scanners)
2	USB port (protocol 3.0)	
3	Network port	Connect to the network port on the PC

3.12 Initial Startup Flow

Startup Flow

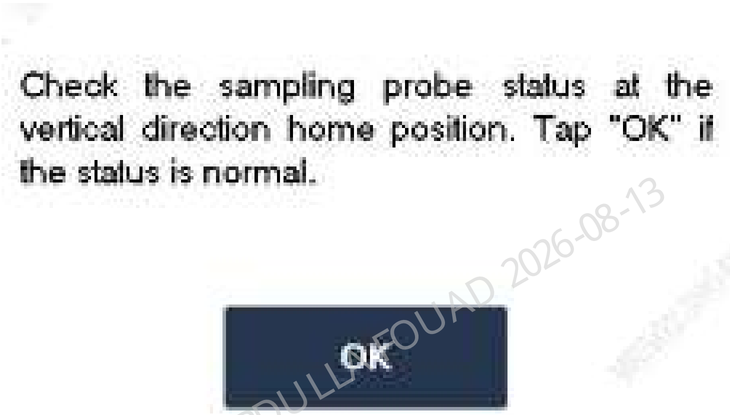
Waste Discharge Method Selection

Power on and start up the analyzer. The analyzer software displays the **Waste Discharge Method Selection** dialog box. Select **Discharge through Waste Container**, and click OK.



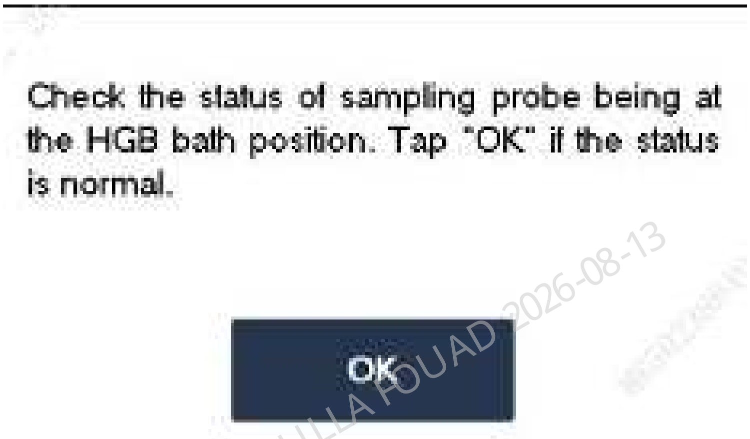
Confirming Sample Probe Position

1. After the waste discharge mode is selected and confirmed, the status confirmation interface of the vertical initial position of the sample probe will be displayed. Observe whether the sample probe meets the requirements (for the acceptance criteria, refer to the acceptance criteria for the observation position of the sampling assembly, as described in **8.13.3** Commissioning and Verification), and click **OK** if it meets the requirements. If not, after startup, perform commissioning according to the adjustment method of vertical initial position of the sampling assembly for commissioning, as described in **8.13.3** Commissioning and Verification.

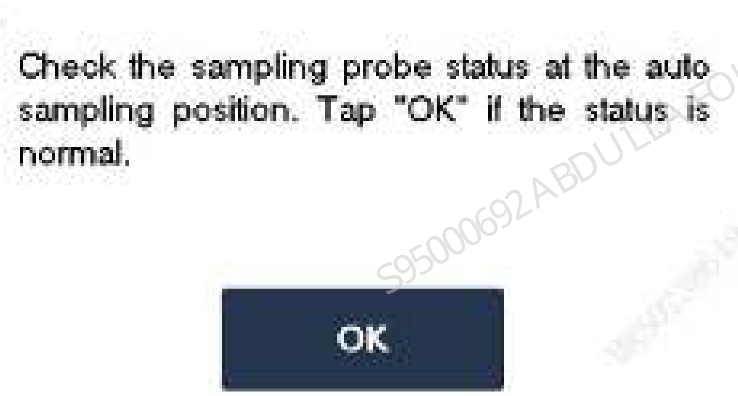


2. Enter the status confirmation interface of the sample probe reaction bath position, observe whether the sample probe meets the requirements (for the acceptance criteria, refer to the acceptance criteria for the reaction bath observation position of the sampling assembly, as described in **8.13.3** Commissioning and Verification), and click **OK** if it meets the requirements. If not, after startup, perform commissioning according to the adjustment method of reaction bath position of sampling assembly for commissioning, as described in **8.13.3** Commissioning and Verification.
3. Enter the status confirmation interface of sample probe DIFF bath position, observe whether the sample probe meets the requirements (for the acceptance criteria, refer to the acceptance criteria for the DIFF bath observation position of the sampling assembly, as described in **8.13.3** Commissioning and Verification), and click **OK** if it meets the requirements. If not, after startup, perform commissioning according to the adjustment method of DIFF bath position of sampling assembly for commissioning, as described in **8.13.3** Commissioning and Verification.

4. Enter the status confirmation interface of sample probe HGB bath position, observe whether the sample probe meets the requirements (for the acceptance criteria, refer to the acceptance criteria for the HGB bath observation position of sampling assembly, as described in **8.13.3** Commissioning and Verification), and click **OK** if it meets the requirements. If not, after startup, perform commissioning according to the adjustment method of HGB bath of sampling assembly for commissioning, as described in **8.13.3** Commissioning and Verification.



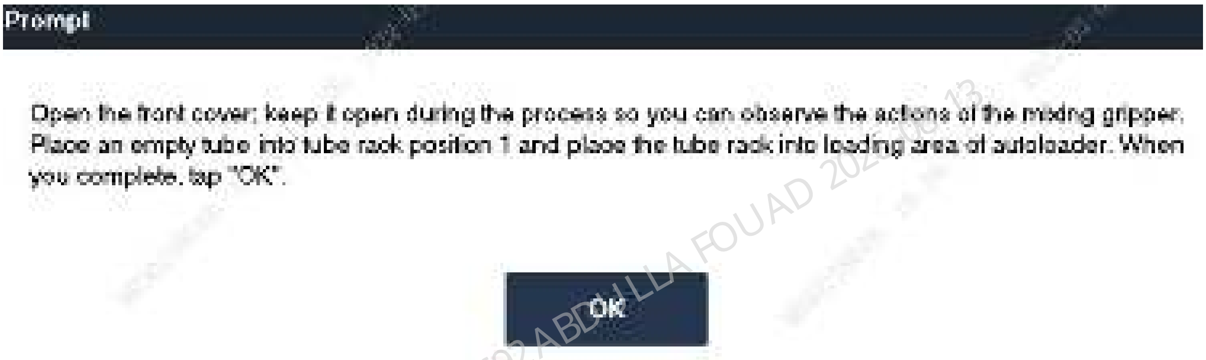
5. Enter the status confirmation interface of auto aspirating position, observe whether the sample probe meets the requirements (for the acceptance criteria, refer to the acceptance criteria for the auto aspirating position observation position of sampling assembly, as described in **8.13.3** Commissioning and Verification), and click **OK** if it meets the requirements. If not, after startup, perform commissioning according to the adjustment method of auto aspirating position of sampling assembly for commissioning, as described in **8.13.3** Commissioning and Verification.



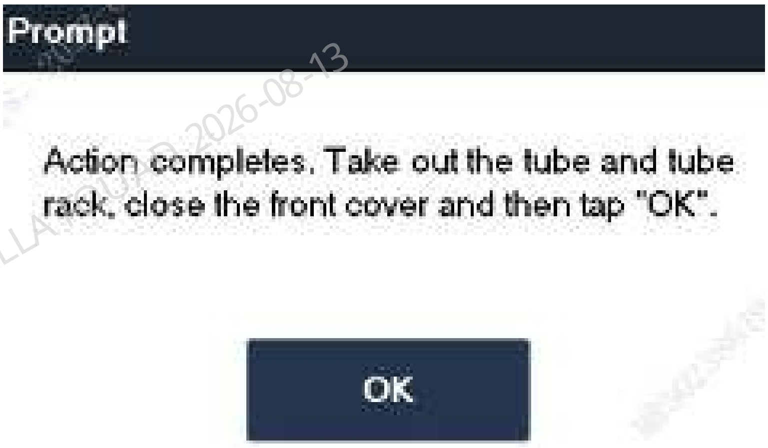
6. Enter the status confirmation interface of CT-WB position, observe whether the sample probe meets the requirements (for the acceptance criteria, refer to the acceptance criteria for the CT-WB position observation position of sampling assembly, as described in **8.13.3** Commissioning and Verification), and click **OK** if it meets the requirements. If not, after startup, perform commissioning according to the adjustment method of CT-WB position of sampling assembly for commissioning, as described in **8.13.3** Commissioning and Verification.

Confirming Mixing Position

After confirming the position of the sample probe, open the front cover as prompted, prepare one tube rack and one empty test tube, place the test tube in position 1, and place the test tube in the loading area. After confirming that it is placed well, click **OK** on the instrument interface, and the instrument will automatically load and feed the tube rack, and then grab the test tube for mixing. Observe whether the mixing operation is normal and whether there is interference. If it is normal, go to the next step. If it is abnormal, refer to section 15.5.5 for confirmation after starting up the analyzer. Confirm that the position is improper, and perform commissioning by referring to the commissioning content in the section about for the blending assembly RU.

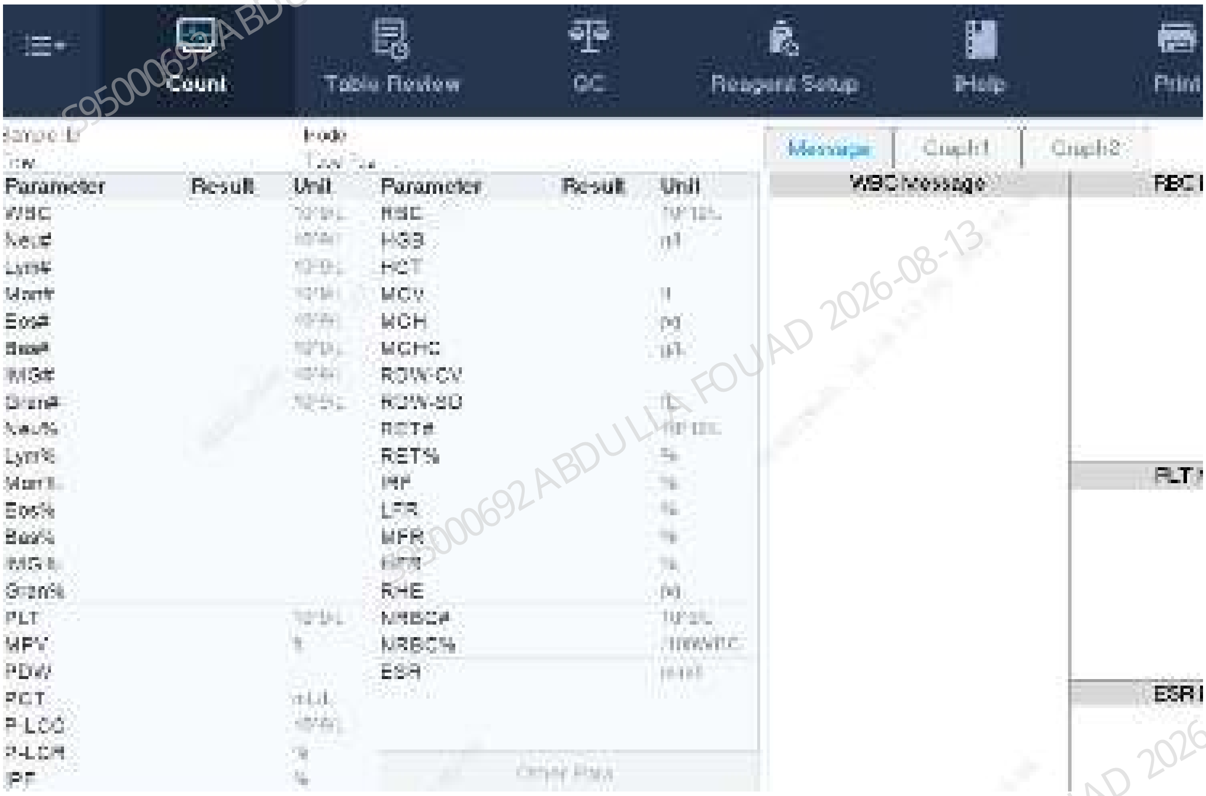


After the mixing operation is completed, the tube rack will be automatically pushed to the unloading area, take out the tube rack and test tube, close the front cover, and click **OK**.



Startup Initialization

After the above confirmation is completed, the startup procedure will run the analyzer initialization and fluidics initialization. The instrument is normally started after running the background process. After the initialization succeeds, the software operation interface is accessed.



Install LabXpert (optional)

1. Install Mindray labXpert on the PC client according to the installation procedure.
2. Click the icon on the desktop, or select from the *Start* menu **labXpert Client Folder>>labXpert Client**. Run the client to access the login screen as below:

Figure3–17 Login Screen

3. Select the current client type, and click **Next**. The following dialog box pops up.

Figure3–18 Set Server IP

4. If the client software and the server software are installed on the same PC, the default IP address 127.0.0.1 can be used. Otherwise, manually enter the IP address of the server PC before logging in again.
5. When you do not know the server IP address, check if the server software runs normally, and the network connection is proper. If so, use the IP Auto Detection function to acquire the server IP address. When the client software finds the server, it enters the server IP address to the IP address field.
6. Click **Next** after setting the server IP. The following prompt appears:

Figure3–19 Current System Configuration

7. Confirm the current client configuration, and click "**Next**".
8. The default account ID and password are both admin which has the administrator authority. The service account and password are consistent with other products.

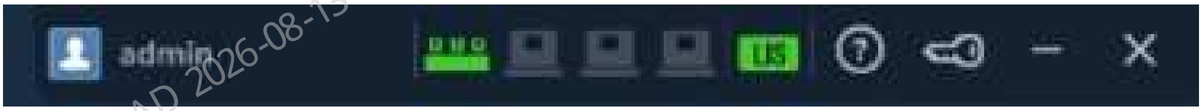
Figure3–20 System Login

9. After the instrument is started, click **System Setup**>>**Instrument Connection** in labXpert to access the "Instrument Connection" screen.

Figure3–21 Instrument Connection Screen

10. Alternatively, click **New Instrument** to add a new analyzer. Save the change and quit the screen.
11. In labXpert, click **System Setup**>>**Communication** to access the "Communication" screen; enter the IP address, port number, protocol type and protocol version of LIS.
12. After the connection, corresponding LIS indicator in the labXpert status bar is on.

Figure3–22 Connection Status



13. If there are more than two instruments, connect them using switches so that the instruments can be controlled through a single PC.

Login

After clicking **OK**, the startup procedure will run the analyzer initialization and fluidics initialization. The instrument is normally started after running the background process. De-register the original account and log in again using the service level account and password.
Service level account: Service
Password: Se s700

Figure3–23 Login with the Service Account

3.13 Inspection after Startup

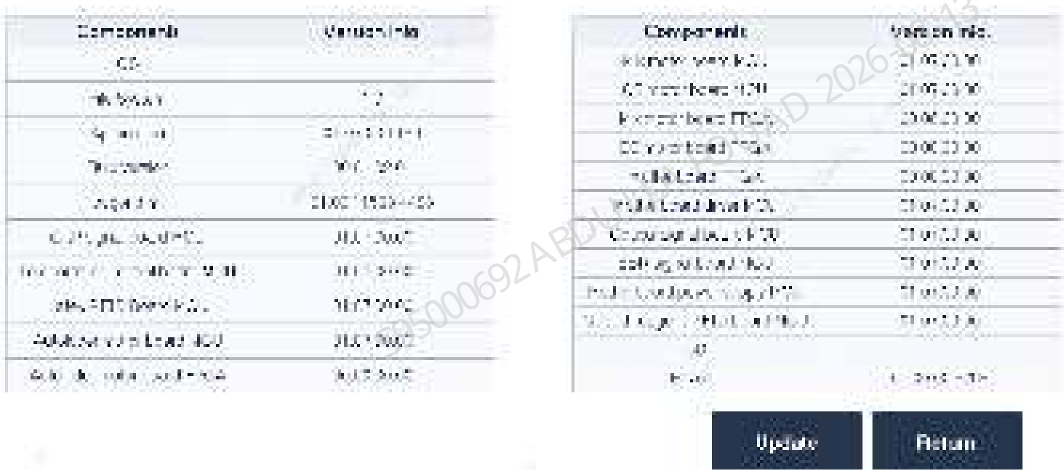
Version Info.

1. Click **Menu>>Status>>Version**.
2. Check analyzer version.

Figure3–24 Version Info.

3. Click *Details* to view:

Figure3–25 Detailed Version Information



4. Check in the "Version" screen whether it is the latest version. If not, upgrade it accordingly.

Confirmation of Temperature and Pressure Statuses

Confirm the temperature status

1. Click **Menu>>Status>>Temperature & Pressure**.

2. Check whether all temperature values of the analyzer are within specification. If not, click "Remove Error" to eliminate the error. If the flag is not cleared, follow the troubleshooting mechanism.

Confirmation of Pressure Status

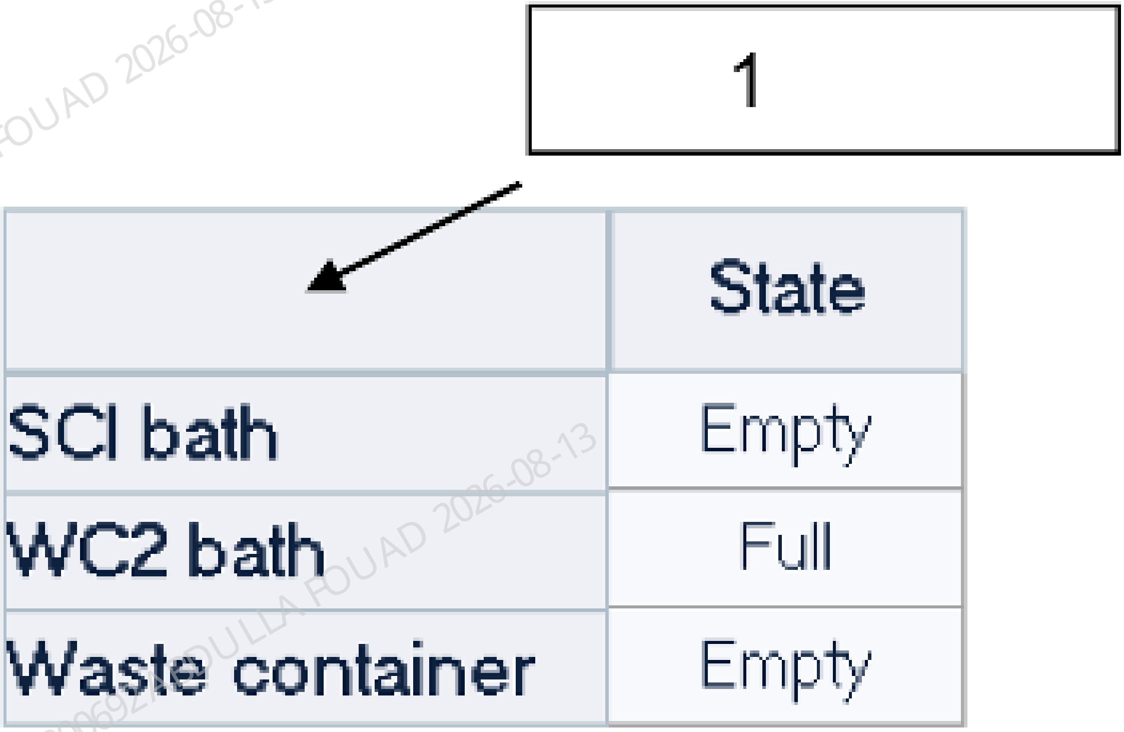
1. Click **Menu>>Status>>Temperature & Pressure**.
2. Check all pressure values of the analyzer, make sure there are no pressure related flags.
3. In case of red flag for pressure during installation, click Rise button to rise the pressure. If the flag remains, follow the troubleshooting mechanism.

Figure3–26 Temperature & Pressure

Confirmation of Other Statuses

1. Click **Menu>>Status>>Floater Status**, and make sure all statuses are consistent with the figure below.

Figure3–27 Floater Status



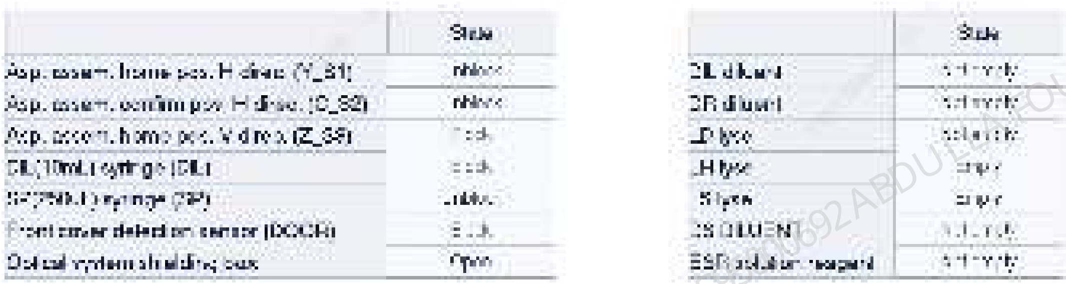
1 Make sure the statuses are consistent with those shown in the diagram

2. Click **Menu**>>**Status**>>**Sensor Status**>>**Fluorescent Reagent Sensor**. Make sure the current voltage of dye sensor is within specification and the statuses are all **Loaded**.

Figure3–28 Fluorescent Reagent Sensor Status

3. Click **Menu**>>**Status**>>**Sensor Status**>>**Motherboard Sensor**, and make sure all sensor statuses are consistent with the figure below.

Figure3–29 Motherboard Sensor Status



4. Click **Menu**>>**Status**>>**Voltage & Current**, and make sure all statuses are within specification.

Figure3–30 Voltage & Current Statuses

	State	Range
SPMT voltage	25.0	[22.00,33.00]
Optical black volt.	0.352V	[0.000,0.400]
HGB blank volt.	3.7	[3.20,4.80]
Laser drive current	12.81mA	[0.00,999.00]
Laser output power	52000	[0.00,99.00]
ESR background intensity	5.30	[6000,60000]
Motherboard Digital +D5V	10.56	[4.48,5.52]
Motherboard Analog +A12V	-9.30	[10.48,13.16]
Motherboard Analog -A12V	-9.30	[-17.10,-7.20]
Motherboard Analog +A5V	5.23	[4.58,5.62]
Motherboard Analog -A5V	-5.12	[-7.61,-2.39]
Motherboard main power(ACDC_24V)	OK	/
Motherboard power (P12V)	OK	/
Motherboard power (P24V)	OK	/

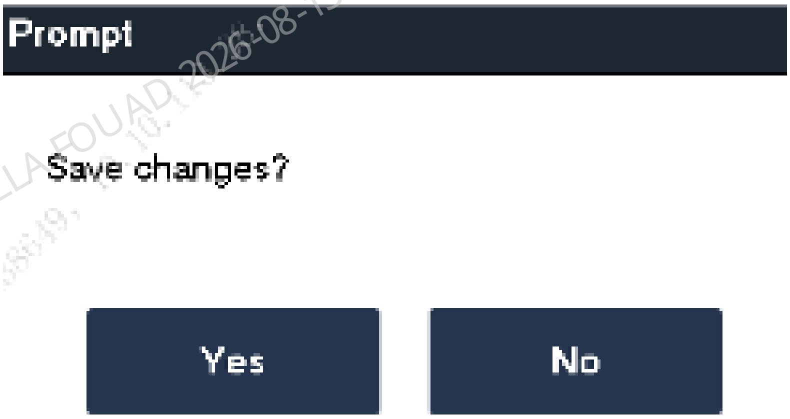
5. View the related analyzer statuses according to the path above. During installation, follow the troubleshooting mechanism to eliminate any flags relating to an abnormal status.

3.14 Setup

Save Changes

1. Follow the steps below to save the updated or changed instrument settings.
2. Click **Menu>>Setup**, and select the settings to be changed.
3. Make necessary changes on the related setup screen.
4. Click another button on the software screen.
5. A dialog box displays asking if you want to save the change.
6. Click **Yes**.

Figure3–31 Save the changes?



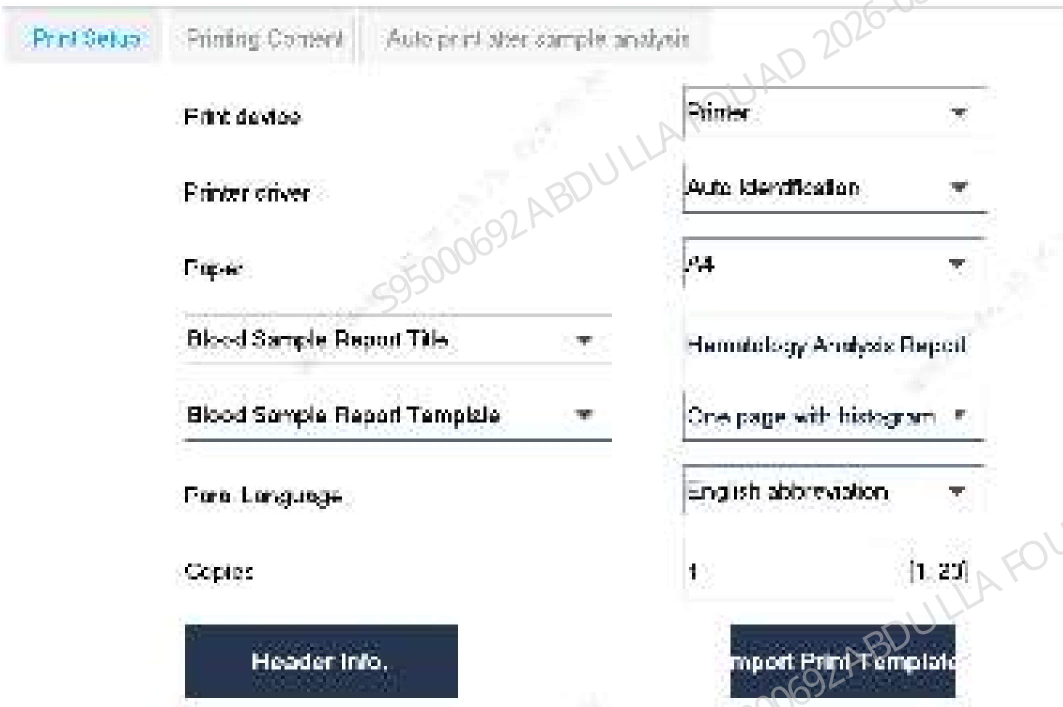
7. The new settings are saved.

System Setup

Print Setup

Print setup options (**Menu**>>**Setup**>>**System Setup**>>**Print Setup**)
Print settings are adjustable on the **Print Setup** screen.
Follow the steps below.
1. Click **Menu**>>**Setup**>>**System Setup**>>**Print Setup** to access the **Print Setup** screen.

Figure3–32 Print Setup



2. Adjust the print settings as required.

See below for setting descriptions:

Printer	Select the printer in the network from the pull-down list.	Administrator permission settings. For questions about printers and printer drive settings, contact the service personnel of the vendor.
Printer drive	Select the applicable printer drive from the pull-down list. • Automatic recognition • PCL6 • Raster print	
Paper type	Select the desired paper type from the pull-down list. Two paper types are supported: • A4 • A5	
Blood/body fluid report title	Select the blood/body fluid report title from the pull-down list. Enter the desired title in the edit box. The blood/body fluid report title entered will appear on the report printed.	
Blood/body fluid report template	Select the blood/body fluid report template from the pull-down list. • Select the desired template format from the corresponding pull-down list • Full page with graph • Full page without graph	If **Full Page with Graph** is selected, the printed results will include the parameter results and graphic results of the sample. If **Full Page without Graph** is selected, the printed results will include the parameter results of the sample only, with the graphic results not printed
Parameter language	Select the parameter language from the pull-down list: • Chinese • Chinese/English	If **Chinese** is selected, the parameters on the report printed will be in Chinese. If **Chinese/English** is selected, the parameters on the report printed will be in both Chinese and English.
Number of copies	Click **Print**, and enter the number of copies to be printed. The default quantity is 1; the set range is **1-20**.	

Print content	Check the corresponding options and select the report information to be printed: • Print the tags of edit results • Print the tags of high and low flags • Print the tags of suspect flags • Print the tags of modification • Print the tags of out-of-linearity-range flags • Print the flags • Print the reference range • Monochromatic printing • Print the RUO parameters	If **Monochromatic Printing** is selected, the report printed is in single color.
Automatic printing	Select this option to enable Automatic Printing: • Automatic printing after sample analysis • Automatic printing after review • Automatic printing after QC count	Select **Automatic Printing after Sample Analysis**, and the analyzer will automatically print the sample results after each analysis Select **Automatic Printing after Review**, and the analyzer will automatically print the sample results reviewed Select **Automatic Printing after QC Count**, and the analyzer will automatically print the QC count results after each QC count

Comm. Setup

Refer to the following for the LIS related introduction and connection of labXper system. Communication setup options (**Menu>>Setup>>System Setup>>Communication**)

NOTICE

The default IP address of analyzer is 10.0.0.12. The IP address of analyzer may be set by user as required.

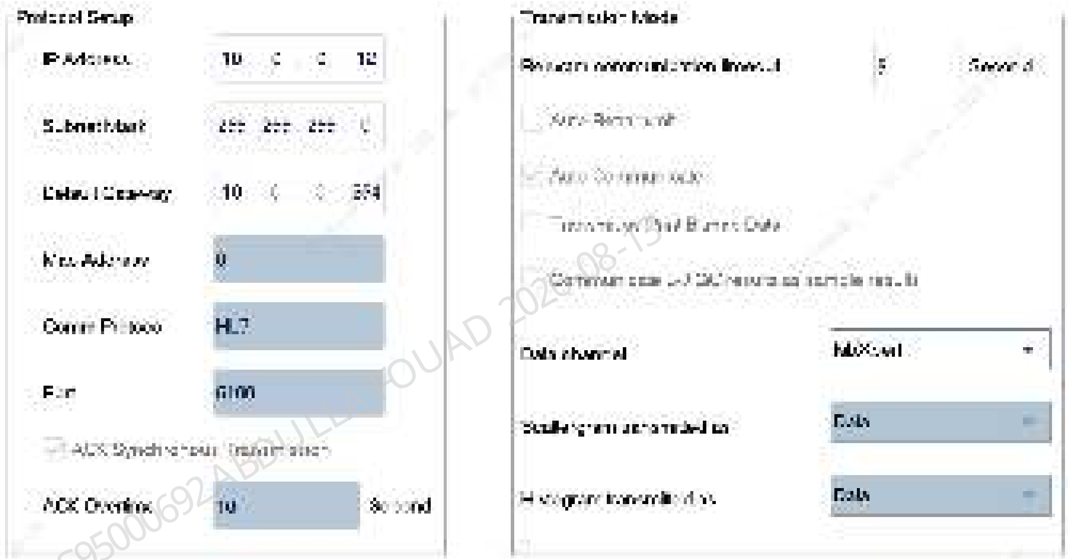
A user with the permissions of administrator level is allowed to set up the following communication settings:

- Protocol Setup
- Transmission Mode

Protocol Setup

1. Click **Menu**>>**Setup**>>**System Setup**>>**Communication** to access the **Communication** screen.
2. Set up the communication protocol according to the practical needs of the laboratory.

Figure3–33 Comm. Setup



See below for setting descriptions:

IP address	Enter the correct IP address	
Subnet mask	Enter the correct subnet mask	
Default gateway	Enter the correct gateway	
ACK synchronous transmission	Check **ACK Synchronous Transmission** to enable the ACK synchronous transmission.	After enabling **ACK ACK Synchronous Transmission**, enter the ACK overtime (10 seconds by default) in the **ACK Overtime** field.

Transmission Mode

1. Click **Menu**>>**Setup**>>**System Setup**>>**Communication** to access the **Communication** screen.
2. Set up the transmission mode according to the practical needs of the laboratory.

See below for setting descriptions:

Transmission Mode	Check to enable one or more transmission functions: • Auto retransmit • Auto communication • Transmission of bitmap data printed • Communication of L-J QC results in sample format **Re-exam Communication Overtime** shall be set if the analyzer is connected with Mindray's labXpert software.	Only when **ACK Synchronous Transmission** is enabled, the **Auto Retransmit** is available.
Data channel	Select from the pull-down list: • LIS • labXpert	When you are using Mindray's labXpert software, select **labXpert** as the data channel.
Scattergram transmitted as /Histogram transmitted as	Select from the pull-down list: • Not to be transmitted • Bitmap transmission • Data transmission	

Built-in Barcode Setup

(**Menu**>>**Setup**>>**System Setup**>>**Built-in Barcode**) Administrators may set up barcode system as needed.

The following barcode types are supported:

- CODE128
- CODE39
- ITF
- CODE93
- CODABAR
- UPC/EAN/JAN

1. Click **Menu>>Setup>>System Setup>>Built-in Barcode** to enter the interface of **Built-in Barcode Setup**.
2. Tap the desired code type to go to the corresponding setup screen.

3. Check the **Apply** box to apply the code type.
4. Select the number of digits used on site in the **Digits** area.
5. (Optional) If you are using CODE39, ITF, and CODABAR with check bit, check the **Check bit** box.

NOTICE

When you are using COD, ITF, and CODABAR with check bit, actual checked barcode digits length should be the length of barcode information plus 1 Check Bit If you are using 9 digits barcode, and check the Check bit box, you should select 10 for barcode length.

6. (Optional) Set other code types if needed.

NOTICE

For code types supporting check bit, use check bit in barcode labels if possible to reduce the rate of misreading. The code types and length limits set on the analyzer shall be those used in your laboratory. Do not select the code types that are not used, which may increase the rate of misreading. Barcodes longer than 20 digits will not be read correctly. Barcodes longer than 20 digits will not be read correctly.

Date/Time Setup

(**Menu**>>**Setup**>>**System Setup**>>**Date/Time Setup**)

Users may set up date and time on the **Date/Time** screen.

1. Click **Menu**>>**Setup**>>**Date/Time** to access the **Date/Time Setup** screen.

Figure3–34 Date/Time Setup

2. Set the date and time.
   See below for setting descriptions:

Date	Enter the current date	
Time	Enter the current time	24-hour system is adopted
Date Format	Select the desired date format from the pull-down list	

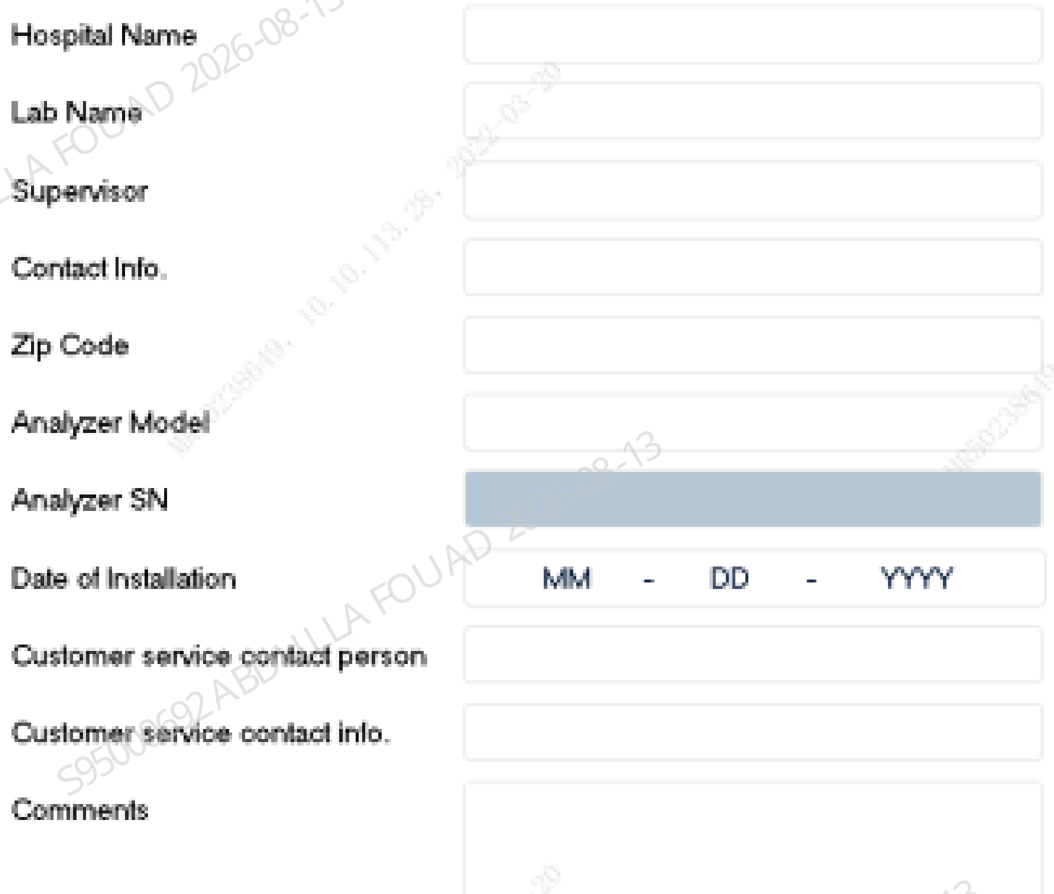
Lab Info. Setup

(**Menu**>>**Setup**>>**System Setup**>>**Lab Info. Setup**)

Users may enter the lab information on the **Lab Info. Setup** screen.

1. Click **Menu>>Setup>>Lab Info. Setup** to access the **Lab Info. Setup** screen.
2. Enter the desired lab information.

Figure3–35 Lab Info. Setup



Set Up User Account and Password

(**Menu**>>**Setup**>>**User Management**)

The "User Management" screen displays all the user accounts registered.

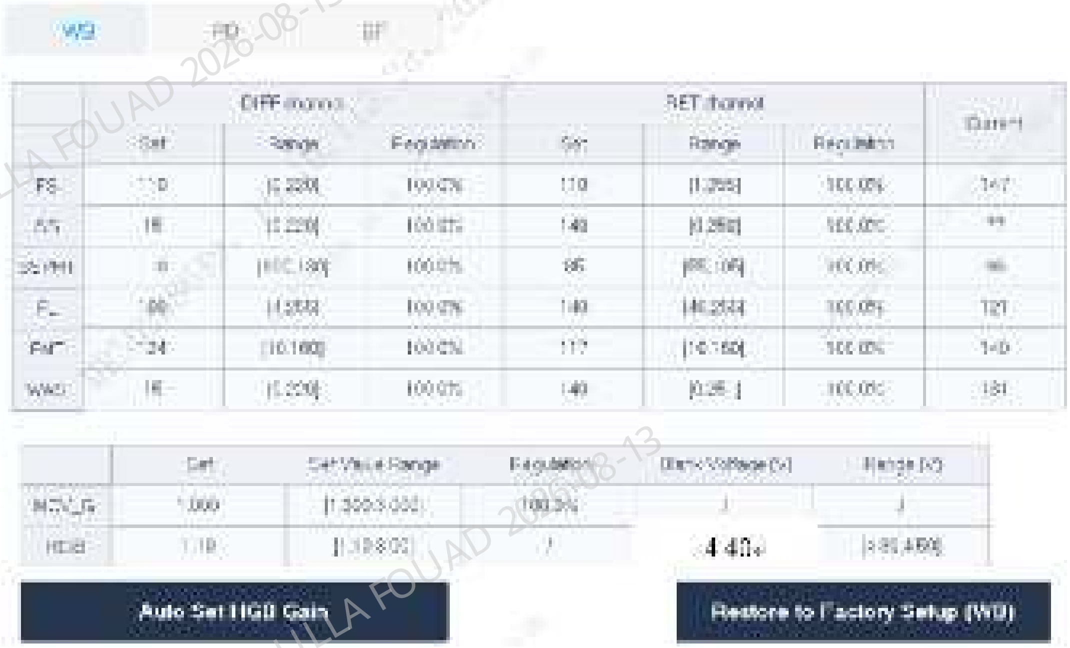
Figure3–36 User Management

Gain Setup

HGB Gain Setup

1. Click **Menu**>>**Setup**>>**Gain Setup**>>**WB** to access the **WB Gain Setup** screen.

Figure3–37 Gain Setup



2. Adjust the HGB default gain in the HGB **Default** text box, until the HGB background voltage is in the range of **4.30, 4.50**.

NOTICE

HGB background voltage will vary with the set HGB.

3.15 Mechanical Mechanism Confirmation and Adjustment

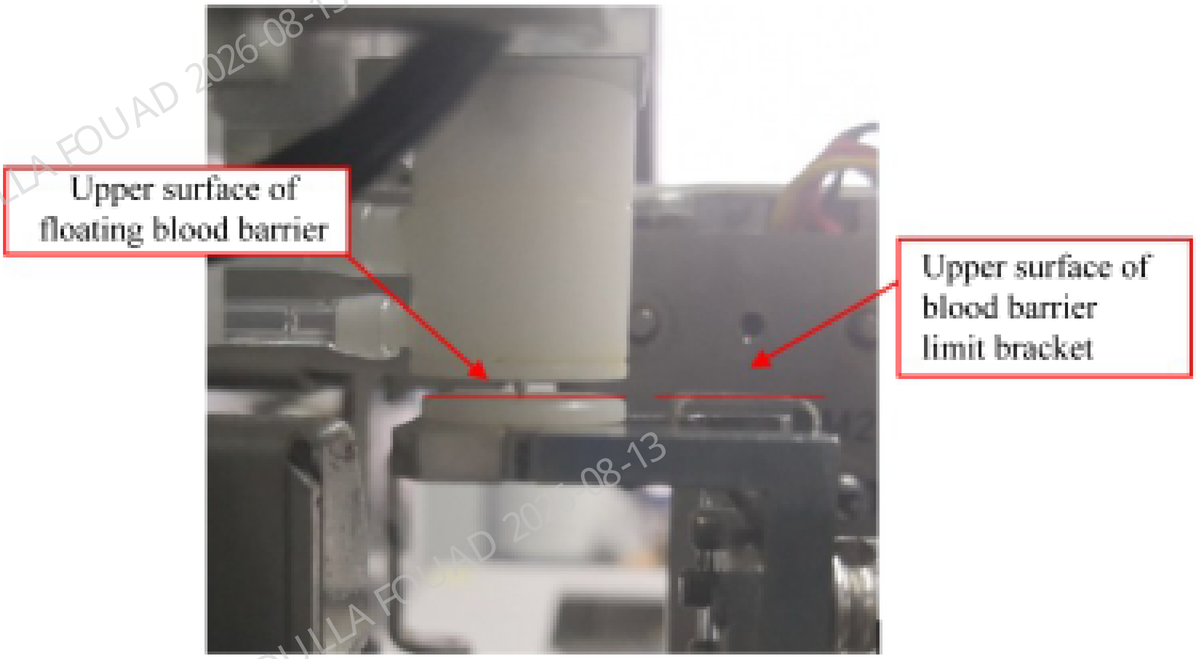
If the position of the sample probe is confirmed to be normal after startup, the content with reference to Blood Barrier Limit Bracket Verification and Confirming/Adjusting Autoloader Pos. can be skipped. If the mixing position is confirmed to be normal after startup, the content in Confirming/Adjusting the Position of Mixing Mechanism can be skipped.

Blood Barrier Limit Bracket Verification

Verify the relative height between the blood barrier limit bracket and the floating blood barrier according to the steps below:

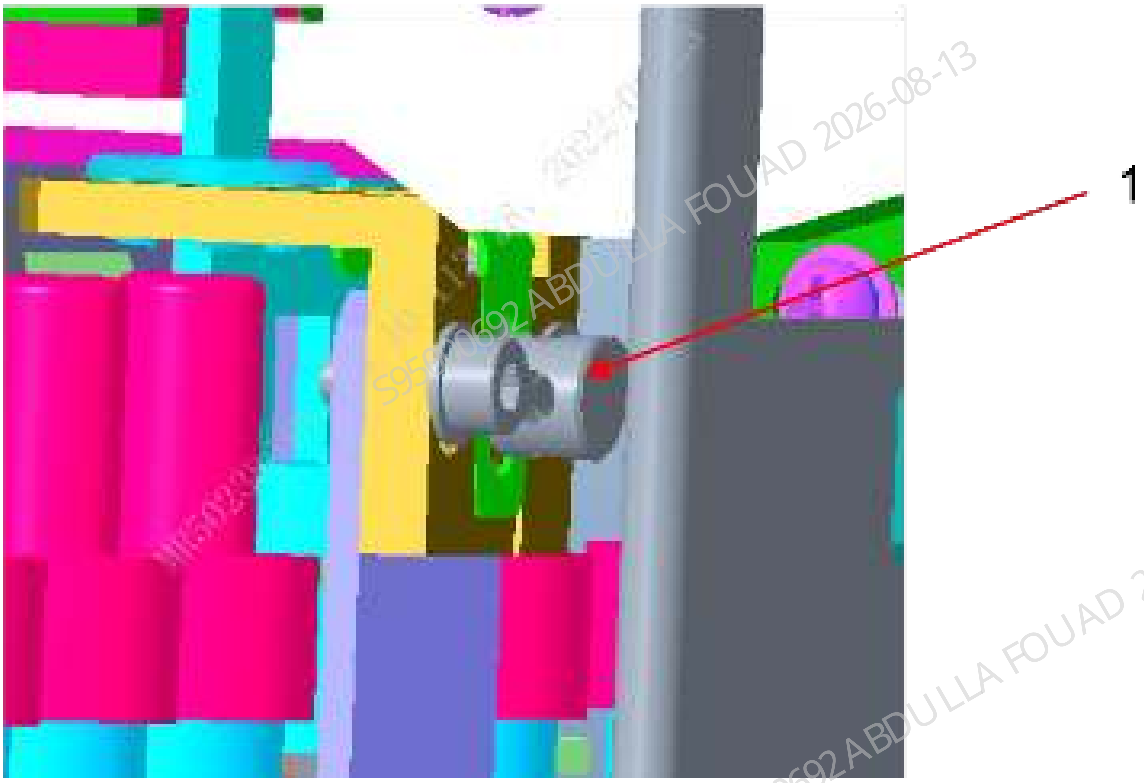
1. Lift the upper cover of the analyzer.
2. Confirm that the upper surface of the blood barrier limit bracket is not lower than the upper surface of the floating blood barrier.

Figure3–38 Blood Barrier Limit Bracket Position Verification



3. If the requirement of step 2 is not met, loosen the fixing screw, lift the blood barrier limit bracket up to the position that meets the requirement of step 2, and then fix it.

Figure3–39 Fixing Screw

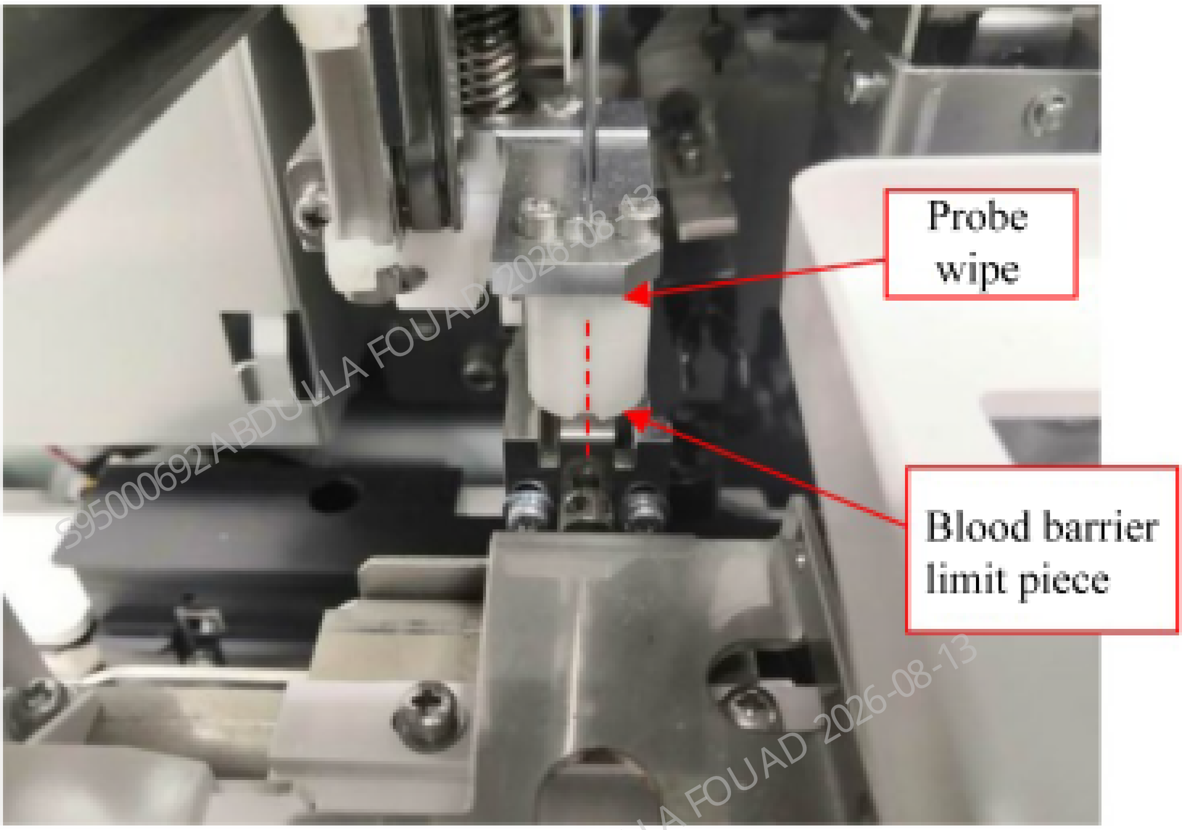


1	Fixing Screw

Confirming/Adjusting Autoloader Pos.

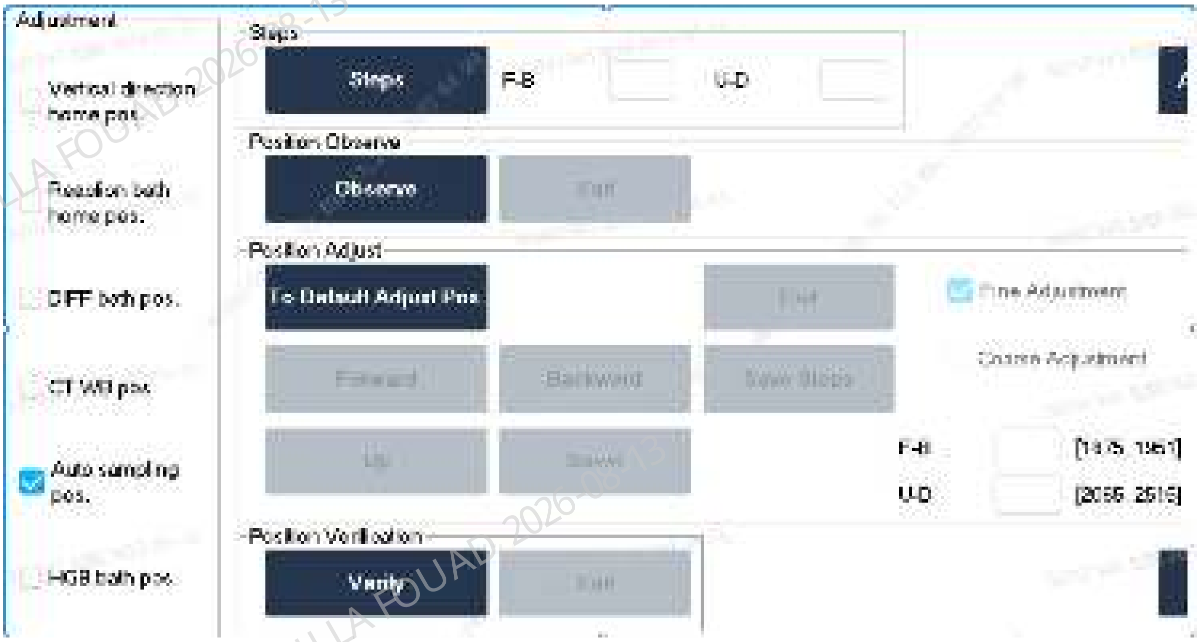
Confirm the relative position of the main unit and the autoloader according to the steps below:

1. Select **Menu>>Status>>Sensor Status** and enter the motor release torque status.
2. Pull the sampling assembly to the front position above the floating blood barrier.
3. Observe whether the center of the sampling assembly probe wipe groove aligns with the center of the floating blood barrier limit piece (≤ ±0.5 mm).



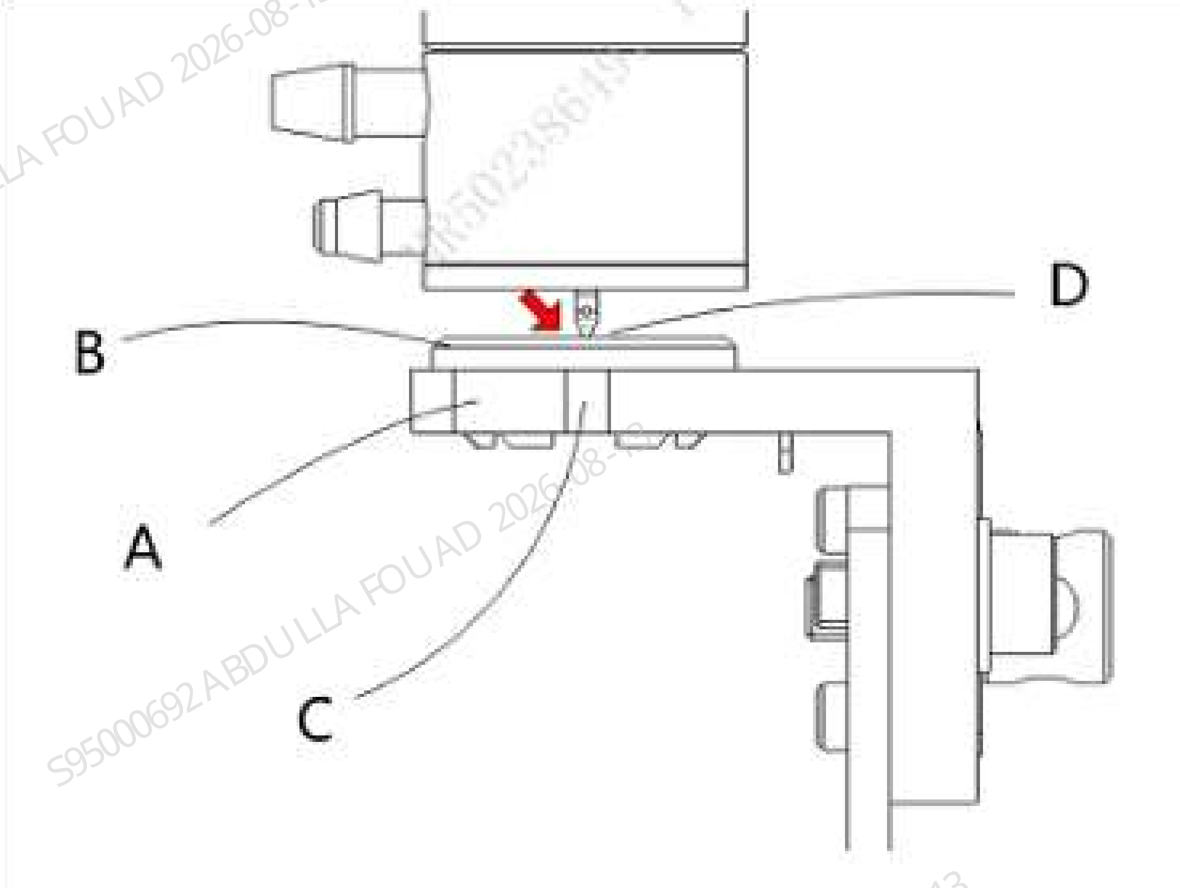
4. Select **Menu>>Service>>Commissioning & Self-Test>>Adj. Sample Probe Pos.>>Auto-sampling Position** to enter the interface for adjusting auto-sampling position.

Figure3–40 Interface for Adjusting Auto-Sampling Position



5. Click **Verify**, look vertically from a side of the instrument towards the side, and observe and confirm that the sample probe is exactly aligned with the notch of the retaining ring seat body, as shown in position C in the figure.

Figure3–41 Observing Sample Probe Position



A	Blood barrier bracket
B	Blood barrier
C	Gap on the blood barrier bracket
D	Sampling Probe

6. If the requirement of step 5 cannot be met, refer to the auto-sampling position commissioning requirements for the autoloader FRU.

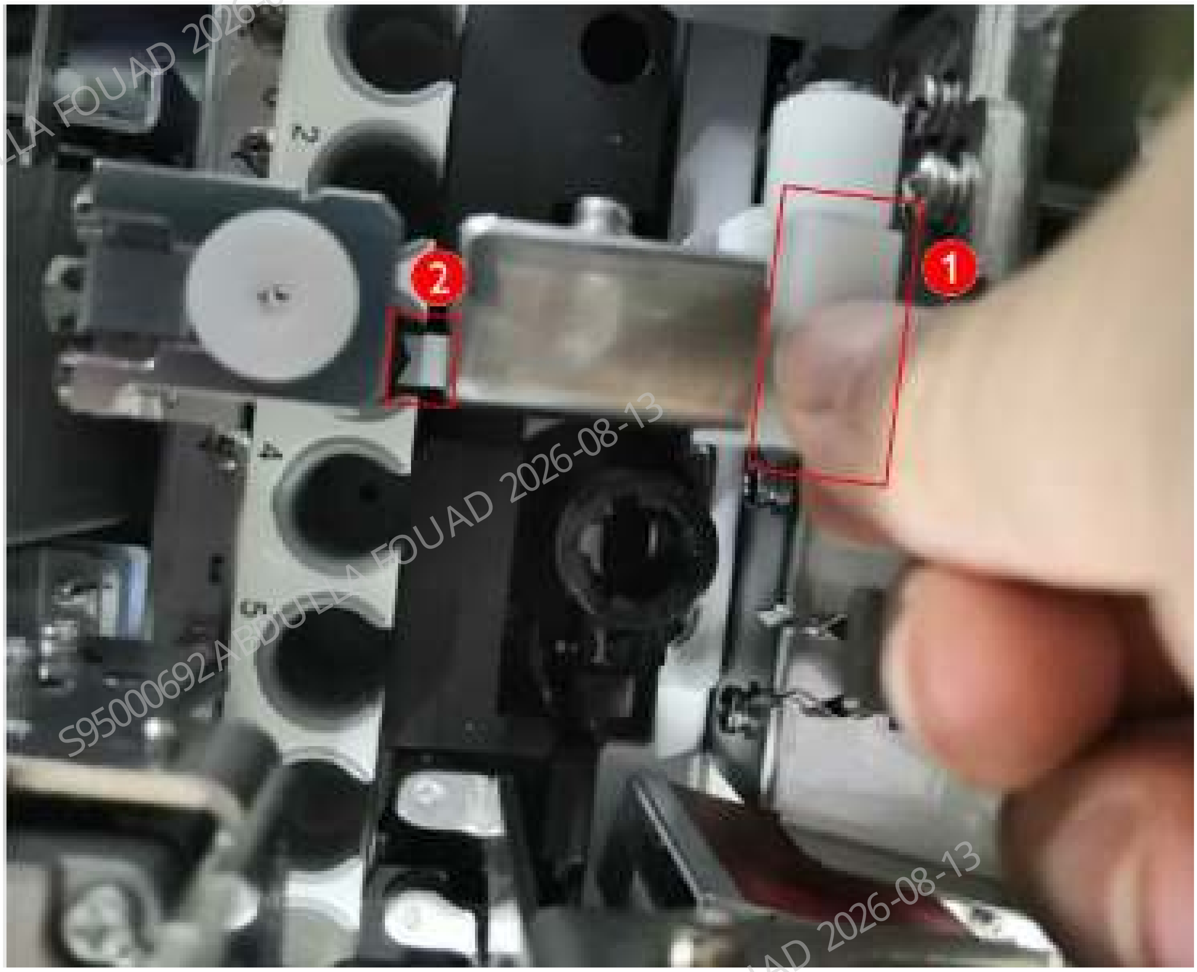
Confirming/Adjusting Position of Tube Hold Assembly

Confirm the position of the tube hold assembly for the analyzer according to the steps below:

1. Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Autoloading Assembly** to enter the autoloader self-test interface.
2. Click the **Assembly Initialization** button.
3. Put a row of test tube rack with an empty tube in tube position No.4 on the loading tray.
4. Click the **Feed** button 5 times (leave an interval after each clicking operation, and then click the button after the mechanism completes the action), and send the first position of the tube rack to the piercing position.

5. Press the roller fixing plate of the hold assembly by hand (as shown in Figure 1 below), and observe whether the finger pressure block of the hold assembly (as shown in Figure 2 below) is roughly aligned with the center of the opening in the front of the tube rack.

Figure3–42 Ideal Position of the Finger Pressure Block of Hold Assembly Relative to Tube Rack



6. If the left and right sides of the finger pressure block of hold assembly are close to the opening of the test tube (≤ 1 mm), the position of the hold assembly needs to be readjusted. About the method to adjust the position of the hold assembly, see the commissioning description in for the test tube hold assembly FRU.

Confirming/Adjusting the Rotary Compressing Assembly Position

After the hold assembly position is confirmed, do not take out the tube rack, but continue to confirm the rotary compressing assembly position according to the steps below:

1. Click the **Fix Tube** button.

Figure3–43 Autoloader Assembly Self-Test Screen

2. Observe whether the two pinch rollers are close to the test tube. If not, readjust the left and right positions of the mechanism (if they are not close to the test tube, a false alarm will be sent by the autoloader counter).

Figure3–44 Ideal Position of the Pinch Roller of Rotary Compressing Assembly Relative to Tube Rack

3. Click **Release Tube**.

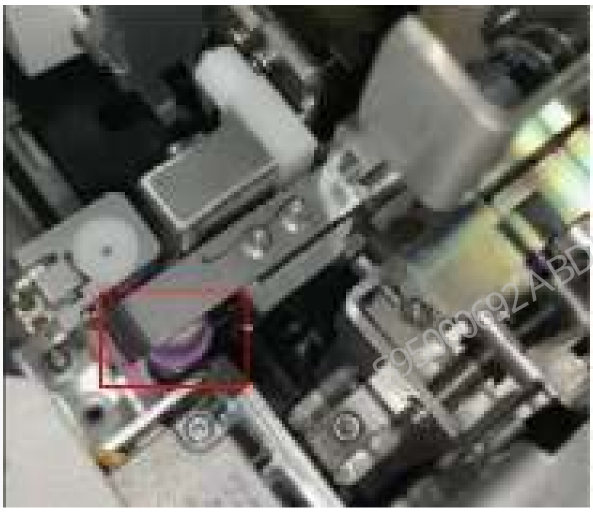
Confirming/Adjusting the Position of Mixing Mechanism

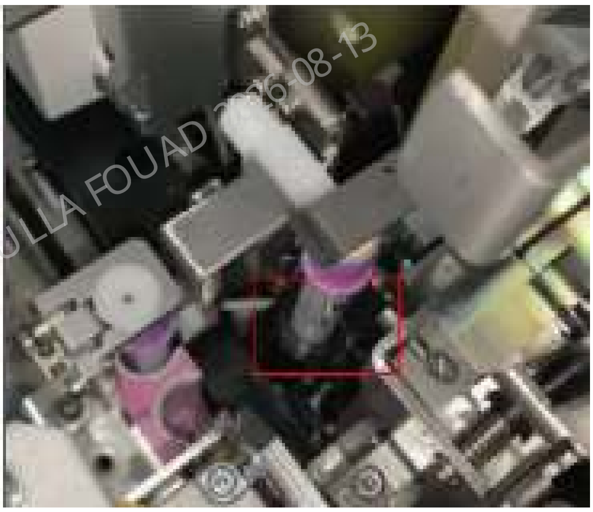
After the rotary compressing assembly position is confirmed, do not take out the tube rack, but continue to confirm the mixing mechanism position.

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.**>>**Mixing Gripper (Middle Position)** to enter the mixing gripper middle position adjustment interface.

Figure3–45 Path of Entering Autoload Position Adjustment

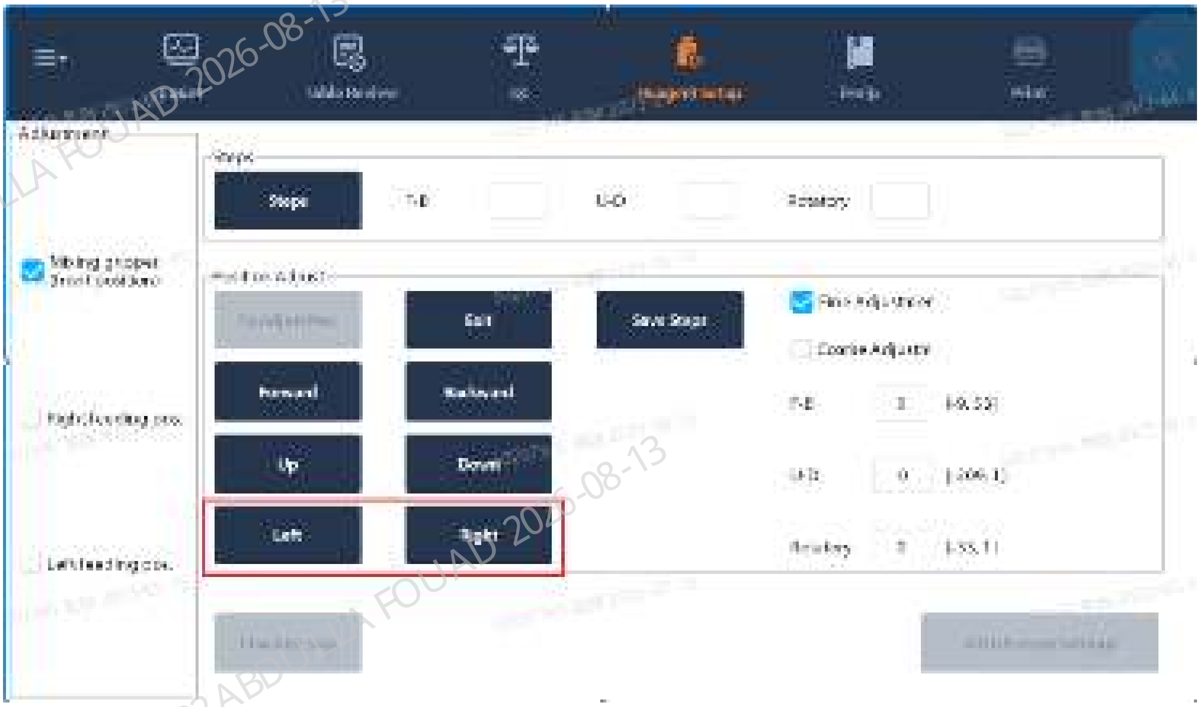
2. Click **Check by step**, click **Feed**, and then click **A&M**.
3. Observe whether there is no scratch between the test tube and the capillary blood mixing assembly (depending on the configuration) and the tube rack during the sampling and mixing action and during the process of placing the test tube into the capillary blood mixing assembly and the tube rack; if any scratch is found, make adjustment according to the subsequent steps.





4. If the scraping position is the left/right position of the tube rack or the capillary blood mixing assembly: Enter the **Mixing gripper (front position)** adjustment interface, click **To Adjust Pos.**, and then click **Left** and **Right** to make adjustment.

Figure3–46 Mixing Gripper (Front Position) Adjustment Interface



5. If the scraping position is the front/back position of the tube rack: Enter the **Mixing gripper (front position)** adjustment interface, click **To Adjust Pos.**, and then click **Front** and **Back** to make adjustment.

Figure3–47 Mixing Gripper (Front Position) Adjustment Interface

6. If the scraping position is the front/back position of the capillary blood mixing assembly: Enter the **Mixing Gripper (Middle Position)** adjustment interface, click **To Adjust Pos.**, and then click **Front** and **Back** to make adjustment (applicable when the capillary blood function is selected).

Confirming/Adjusting the Position of Capillary Blood Mixing Assembly (applicable when capillary blood mixing is configured)

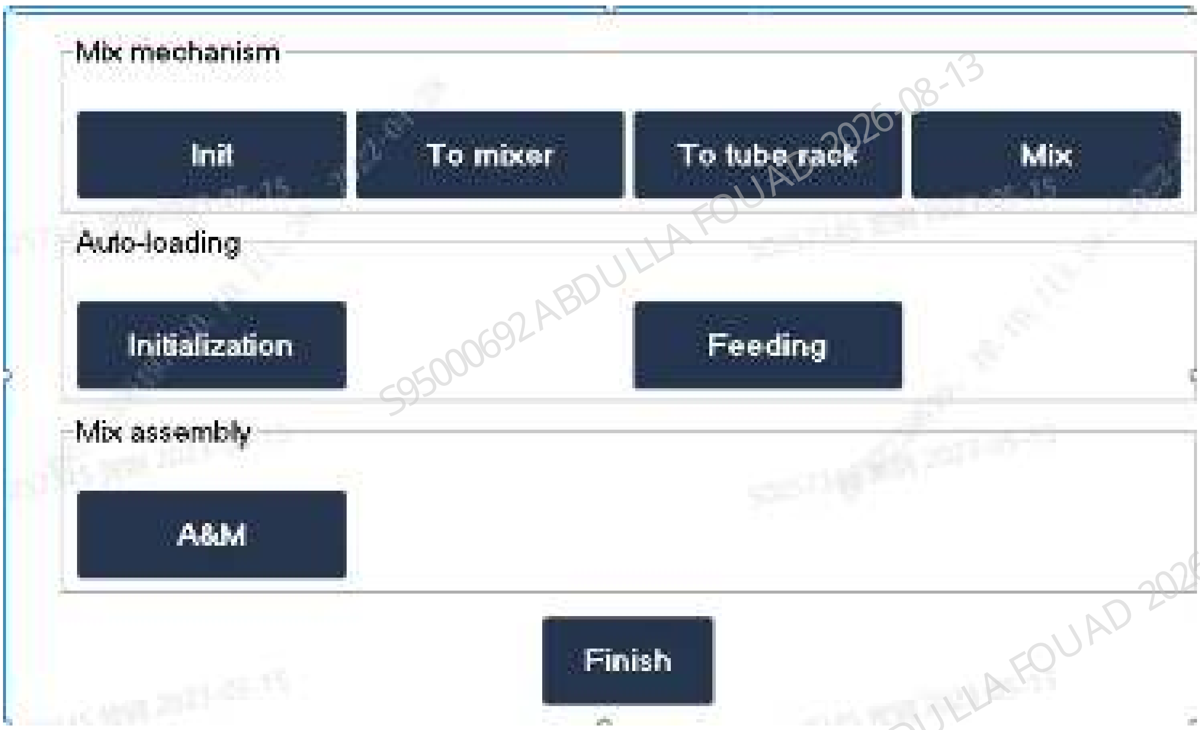
Perform the operation of this step for the analyzer equipped with capillary blood mixing.

1. Prepare a WB tube rack with WB tubes.
2. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.** to enter the mixing mechanism adjustment interface, and select **Mixing Gripper (Middle Position)**.

Figure3–48 Capillary Blood Position Adjustment Interface of Mixing Gripper

3. Place the tube rack in the test tube loading position, click **Check by step**, push the test tube rack to the feeding position, and then click **Assembly Initialization>>Feed** to feed the test tube on the test tube rack to the clamping position of the blending assembly.

Figure3–49 Check by Step Interface



4. Click **To Mixer** and **To tube rack**, check whether the gripper can clamp the test tube between the tube rack and the capillary blood mixer smoothly, and whether there is interference between the test tube and the capillary blood mixer.

1) If the clamping action is not smooth or the test tube interferes with the capillary blood mixer, confirm whether there is a problem with the left/right or front/back position of the capillary blood mixer.

2) If the left/right position of the capillary blood mixer is improper, loosen the fixing screw that fixes the capillary blood mixer and adjust the mixer left and right.

3) If there is a problem with the front/back position of the capillary blood mixer, exit the **Check by step** interface and adjust the position forward and backward on the commissioning interface.

Confirming Position of the Scanner

If the user does not need the sample bar code scanning function, skip this section.

1. Select **Menu**>>**Setup**>>**Auxiliary Setup**, and enter the sample information getting method setting interface. Confirm that the **Auto-Scan rack No.** option has been selected.

2. Use the code scanning software or machine bar code recognition function to identify the bar code type used by the client.
3. Select **Menu**>>**Setup**>>**System Setup**>>**Built-in Bar Code Setup** to enter the built-in bar code setup interface. Check the correct bar code type, length and check bit. Try to avoid checking the bar code type and length not used by the client.

4. After completing the above setup items, select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**, and select **Autoloading Assembly** to enter the autoloader self-test interface.
5. After clicking **Assembly Initialization**, put a test tube with bar code in the rotary scan position (the test tube is placed in the tube rack).
6. Click **Auto-Read Bar Code**, check whether the bar code content display box displays the bar code content and whether the bar code recognition rate display box displays 100%. If not, first confirm whether the red indicator of the scanner is turned on when you click the **Auto-Read Bar Code** button. If the indicator has been turned on but the bar code cannot be scanned, adjust the position according to the commissioning description for the bar code scanner FRU.

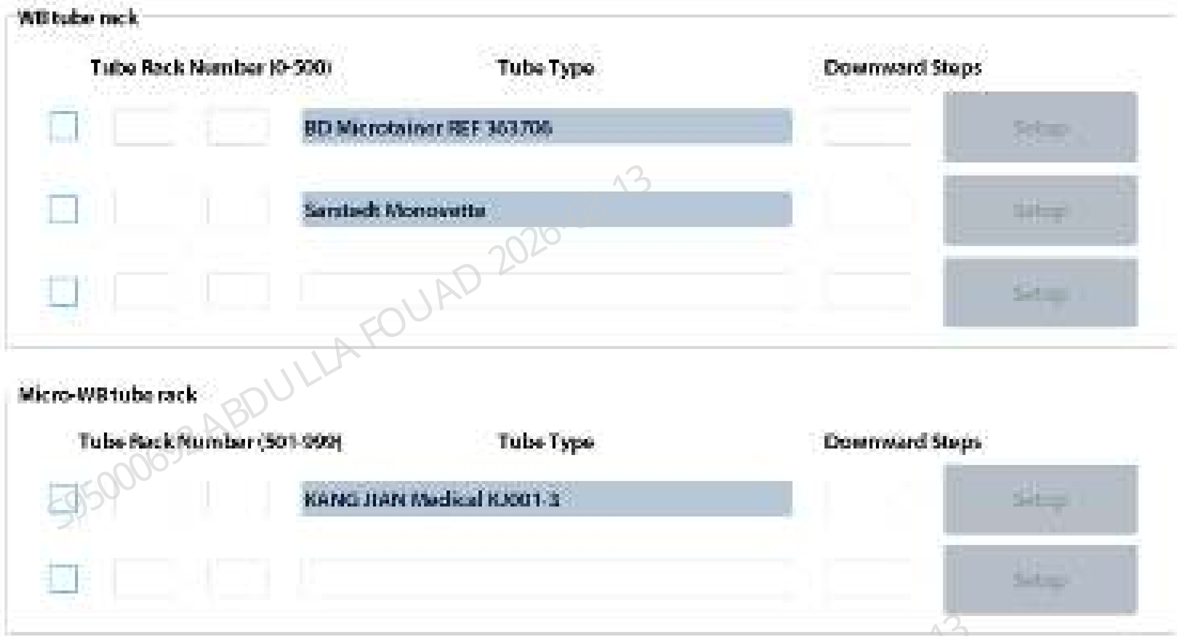
Figure3–50 Autoloader Assembly Self-Test Screen

Adjusting Downward Distance of the Special Micro WB Tube Rack (Optional)

The capillary blood mixer is optional. If it is equipped, the downward distance of the micro WB tube rack shall be adjusted as follows.

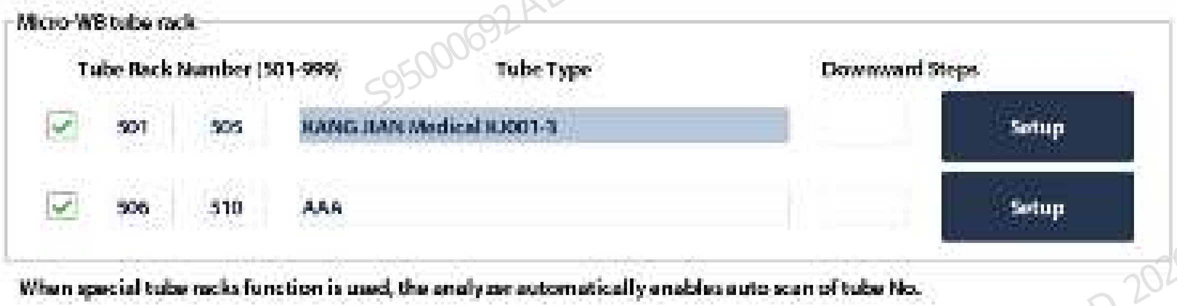
1. Click **Menu>>Setup>>System Setup>>Tube Rack Type Setup** to access the adjustment screen.

Figure3–51 Tube Rack Type Setup Screen



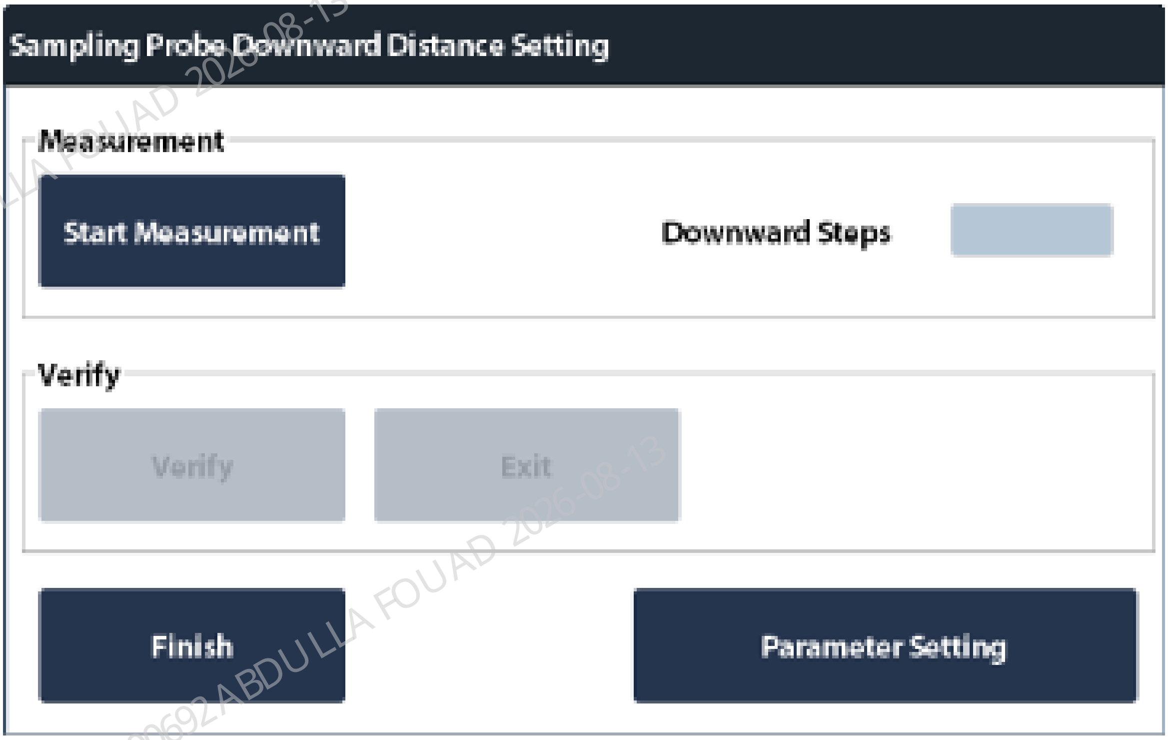
2. Select the first column of the micro WB tube rack. You can select the default tube type or customize a new type according to the specific type of the micro WB tube rack used by the customer. Fill in the appropriate range of tube rack numbers.

Figure3–52 Micro WB Tube Rack Setup Screen



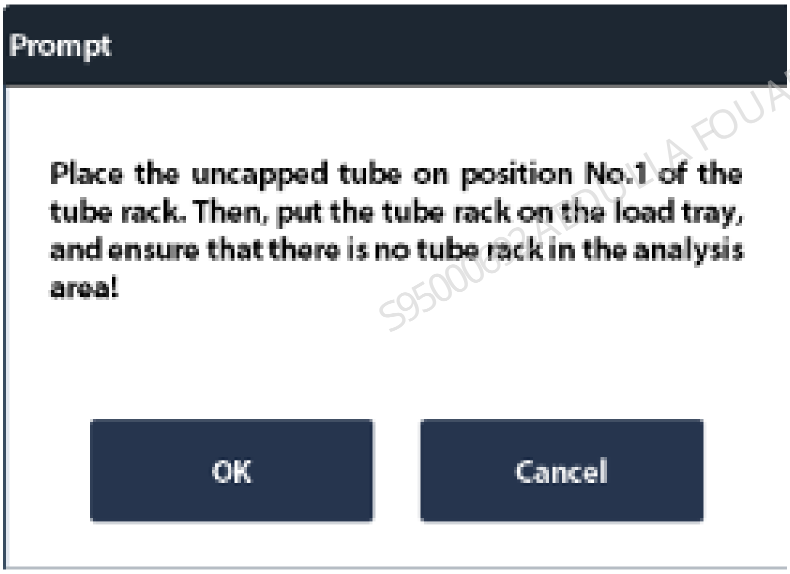
3. Click **Setup** to enter the downward distance setup screen.

Figure3–53 Tube Downward Distance Setup Screen



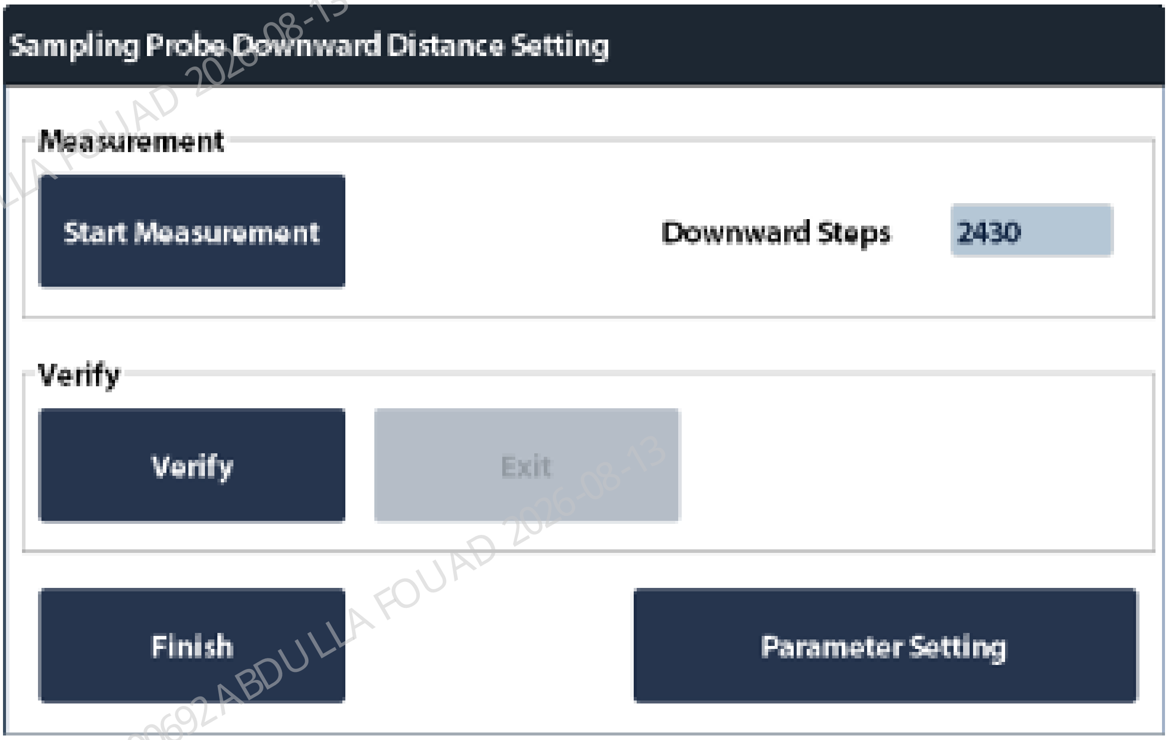
4. Click **Start**, and the software prompts to place a tube rack. Please place an uncapped tube in the tube rack position 1 and place it in the loading area. Then, click **OK** to start commissioning.

Figure3–54 Prompts



5. After the downward distance commissioning is done automatically, the new **Downward Steps** will be displayed. Click **Verify** to verify the current downward steps. If the downward distance is appropriate, click **Done** to save it.

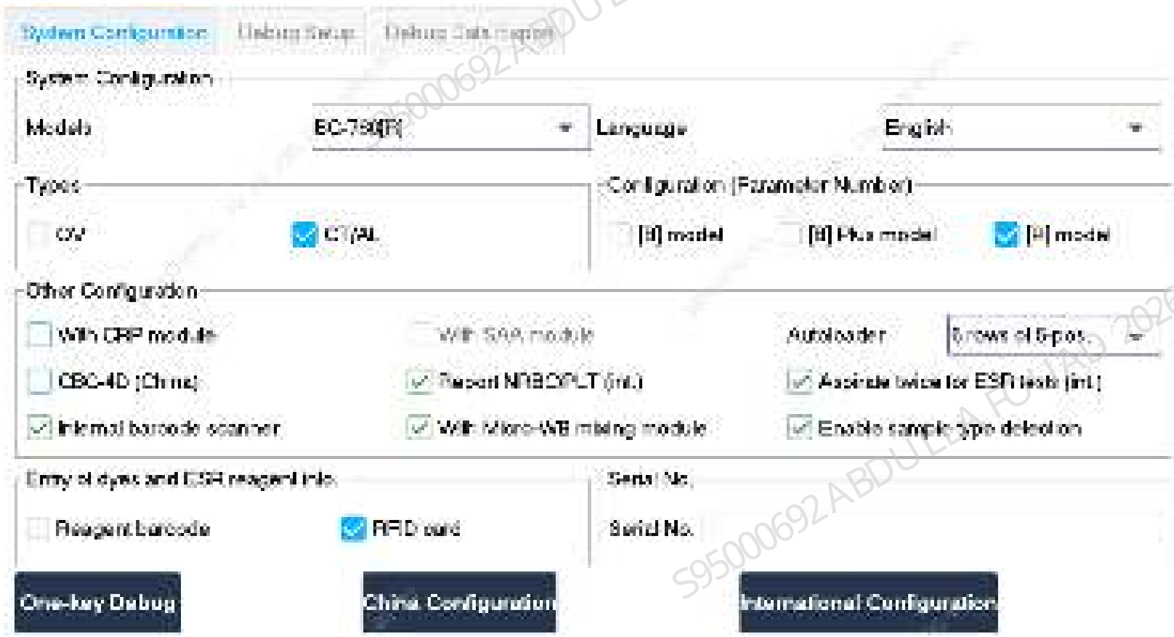
Figure3–55 New Downward Steps



Verifying the Sample Type Detection Sensor (only if the sensor is equipped)

The sample type detection sensor is optional. If it is equipped, confirm whether the detection is enabled: Select Menu **Service>Advanced Toolbox>System Configuration**, select **Enable Sample Type Detection**, and save the configuration. Then, verify the sample type detection sensor.

Figure3–56 Advanced Toolbox_System Configuration Screen



1. Fill a tube rack with micro WB tubes containing about 1 mL of water or DS diluent and venous blood tubes containing about 1 mL of water or DS diluent, and place the tube rack on the loader.
2. Tap **Menu** > **Service** > **Debug & Self−Test** > **Self−Test** > **Autoloading Assembly** to enter the autoloading assembly self-test interface.

Figure3–57 Autoloader Assembly Self-Test Screen

3. Tap **Initialization**. Then, tap **OK** in the popup dialog.
4. Tap **Sample Type Detection**, and the analyzer would detect the sample type automatically.
5. After the detection is completed, check whether the order of results on the interface is consistent with the order of tubes in the tube rack. The following are the meanings of symbols on the **Sample Type Detection** interface:

1) C: capillary blood, represents capillary blood sample, namely Micro-WB tube;

2) V: venous blood, represents venous blood sample, namely evacuated blood collection tube.

Figure3–58 Sample Type Detection Results Screen

6. If the order of results on the interface is consistent with the order of tubes in the tube rack, the verification is completed. If the results are inconsistent, refer to Sample Type Detection Sensor -> to adjust the sensor.

3.16 Performance Confirmation

Sample Mixture and Use

Controls and Calibrators for Blood Routine

1. Withdraw the controls and calibrators from the refrigerator, and warm them up at room temperature (15-30℃) for about 15min.
2. Rub the tubes back and forth vertically for 20 times with both palms, then rub them back and forth horizontally for 20 times with both palms.
3. Repeat Step 2 until no coagulum is attached on the tube wall with the tube upside down. Mix the solution well before use after a long-term storage.

Figure3–59 View of Well Mixed Solution

4. Turn it up and down for 8-10 times before testing.
5. Calibrators are for single use only. No reuse is permitted.
6. The minimum sample volume of control/calibrator shall be 1ml. It is forbidden to use any control or calibrator with a sample volume less than 1ml.
7. For each use, the storage period of a control at room temperature shall be less than 1h.

Blood Sample

- **Vacuum Blood Collection Tube**

1. With the midpoint of tube's central axis as the pivot point of rotation, hold both tube ends with both hands, slowly mix the solution upside down for 10-15 times (while rolling the test tube) at a rotation frequency of about 40 times/minute and a rotation angle of about ±20°.

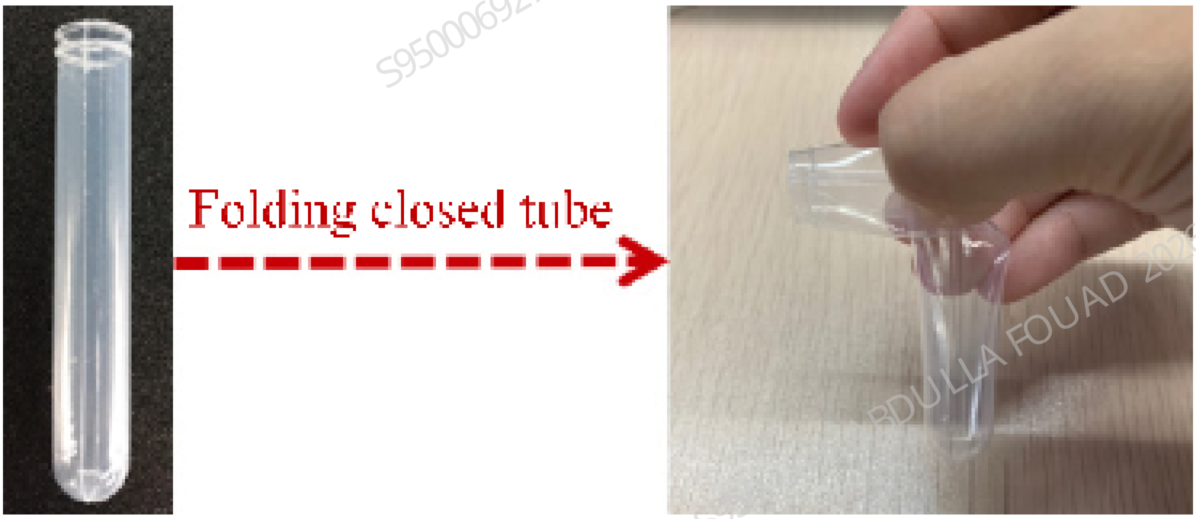
Figure3–60 View of Well Mixed Content in Vacuum Blood Collection Tube

2. After mixing for 10-15 times, observe whether the content in the tube is evenly suspended: slowly turn the tube upside down and observe whether there is any coagulum attached to the bottom or wall of the tube. If any, repeat the previous step.

- Flexible Tube with Round Bottom

1. Fold the top end of the tube, close the tube, and slowly turn it up and down for 10 times, with the folded part clamped during the turning process to keep the tube closed and avoid sample leakage. In case of any leakage, repeat this step with a new clean tube.

Figure3–61 Tube Folding



2. After turning the tube up and down for 10 times, observe whether the content in the tube is evenly suspended: slowly turn the tube upside down and observe whether there is any coagulum attached to the bottom or wall of the tube. If the cells are not completely suspended, continue to turn the tube up and down.

- Centrifugal Tube
  After loading the blood sample, cap the centrifugal tube and keep it upright, grasp the upper part of the tube with one hand, and turn the tube up and down for 20 times while flicking the lower part of the tube with the fingers of the other hand.

Figure3–62 EP Tube

Prediluted Sample

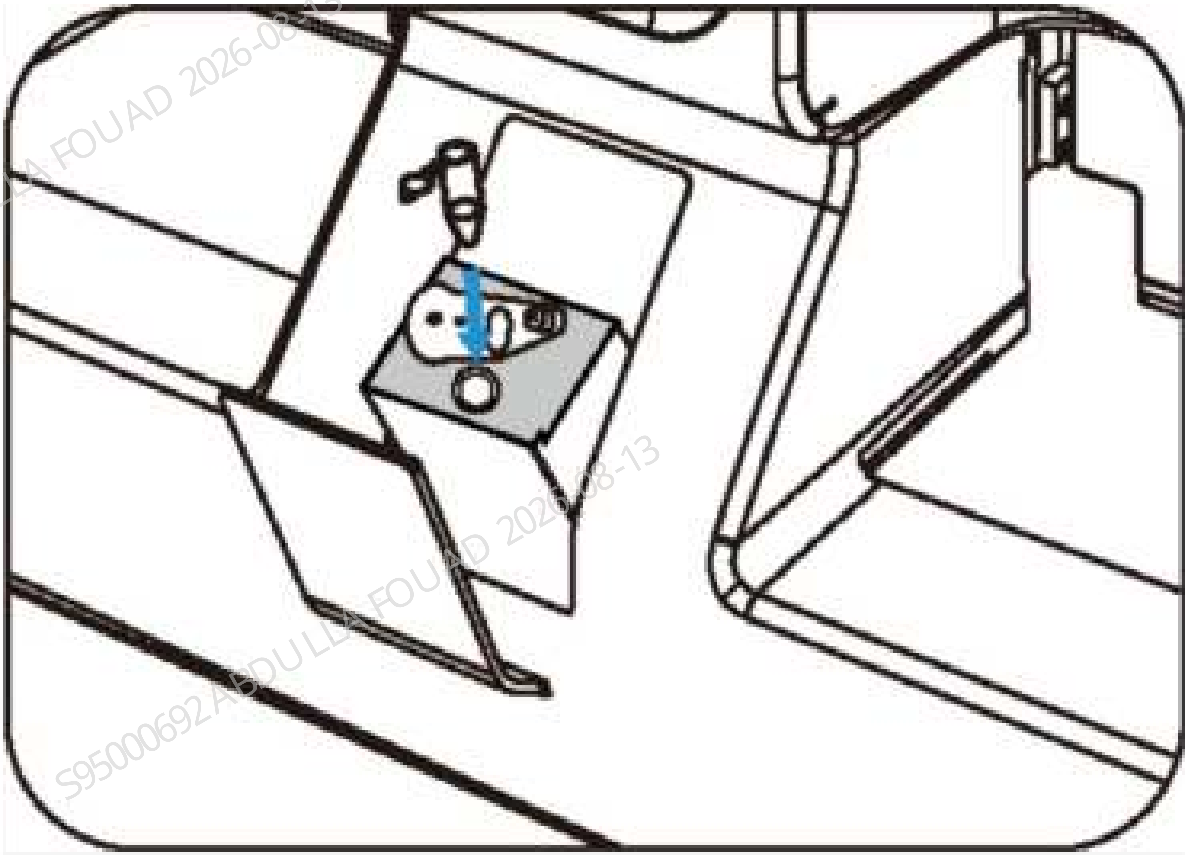
- **Sample Preparation**

Predilute the samples strictly as per a ratio of 20:100 (Whole Blood : Diluent) to guarantee the accuracy of analysis results. A ratio of 200μL : 1000μL is generally followed to prepare the samples for repeatability test (repeat 10 times), in order to guarantee that the sufficient samples are available. The analyzer dispenses 100 μL of diluent per time.

Steps (200μL : 1000μL)

1. Click **Mode** on the Sample Analysis screen.
2. Click the shortcut button **Diluent** to get ready for diluent dispensing.
3. Have the centrifugal tube in place, with the sample probe upright and in contact with the bottom of centrifugal tube.
4. Prepare the centrifugal tube.
5. Closed-sampling type analyzer: Take a clean centrifugal tube, and place it into the Mirco-WB tube position after uncapping.

Figure3–63 Diluent dispensing for closed-sampling type analyzer

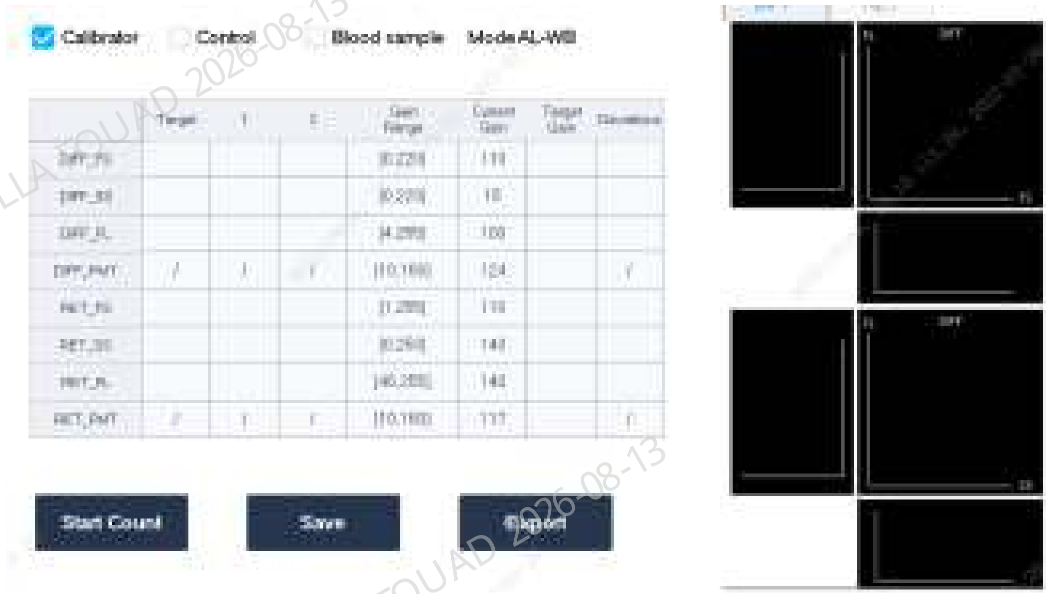


6. Press the Aspirate key or Count key to dispense the diluent. Click the key for 10 times until the dialog box of **Cancel (10)** appears. Then, click **Cancel (10)** to complete diluent dispensing.
7. Pipette 200 μL sample to the centrifugal tube loaded with diluent, and then cap it.
8. Well mix the sample by shaking according to the requirements of 11.16.1.2.

Optical gain verification and adjustment

1. Log in at the service access level.
2. Click **Menu>>Calibration>>Optical Gain Calibration** to access the Optical Gain Calibration screen.
3. Select **Calibrator** as the work sample.
4. Take one PLUS calibrator, and input the corresponding targets in the fields.

Figure3–64 Target Input



5. Take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader), put the PLUS calibrator at the No. 1 tube position, and perform a counting operation in this mode.
6. Check the data in the deviation column. Where, the DIFF deviations in all directions shall be ≤5%, and the RET deviations in all directions shall be ≤10%. No subsequent regulation is required if the deviations are within specifications. In case of any out-of-specification deviation, regulate the optical gain according to the subsequent steps.
7. Click **Save** after completing each counting, until all data in the deviation column is within specification, with the gain factors recorded in table.

Background verification

- **Introduction**

Test the background results of the analyzer, make sure the results meet the requirements of the analyzer specification.

NOTICE

Use diluent for background test.

The background requirements for the analyzers of BC-760[B], BC-760[R] and BC-780[R] series are as below:

Table 3–5 Background Specifications for Blood Sample Tests

Parameters	Background and Blank Counts	Model
WBC	$\leq 0.1 \times 10^9$ / L	All models
RBC	$\leq 0.02 \times 10^{12}$/ L	All models

Table 3–5 Background Specifications for Blood Sample Tests(continued)

Parameters	Background and Blank Counts	Model
HGB	≤ 1 g / L	All models
PLT	≤ 3 × 10^9 / L	All models

Steps

1. Log in at the service access level.
2. Click **Menu**>>**Performance**>>**Background** to access the Background screen.
3. Click **Mode**, and the mode selection interface appears as shown in the table below.

Analyzer model	Sample introduction mode	Material mode	Analysis mode
[R] model	AL-WB	Blood sample	CDR
[B] model	AL-WB	Blood sample	CD

4. Load a vacuum blood collection tube with about 1.5mL clean diluent.
5. Click **Menu**>>**Performance**>>**Background** to access the Background screen. Click **Mode** to select blood sample mode, and select **AL-WB**. Take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader), and place the vacuum tube with about 1.5mL clean diluent at No. 1 tube position. After selecting the corresponding analysis mode, click OK, and then click to start counting.
6. When the test finishes, the analyzer will evaluate the background results and report **Pass** or **Fail** at the bottom of the screen.
7. After completing the test, click **Export** to export the data into a USB drive.

Figure3–65 Background Screen

Carryover verification

Verification of Carryover in Blood Routine

- **Steps**

1. Log in at the service access level.
2. Prepare a few high-value fresh blood samples and a few low-value fresh blood samples (each sample shall have one or more conforming parameters, provided that, however, each parameter shall go through the carryover test to the satisfaction).

Table 3–6 Fresh Blood Samples

Parameters	Parameters of high-value samples	Parameters of low-value samples
WBC	>90.0×10^9/L	0.0-3.0×10^9/L
RBC	>6.20×10^{12}/L	0.0-1.5×10^{12}/L
HGB	>180g/L	0.0-50.0g/L
HCT	>54.0%	0.0-18.0%
PLT	>900×10^9/L	0.0-30×10^9/L

3. Mix the samples according to the requirements of Blood Sample. Click **Menu** > **Performance** > **Carryover** to access the Carryover screen.

Figure3–66 Carryover Screen

4. Click **Mode**, select **AL-WB**, **QC**, and select the following analysis mode and sample mode for testing.

Analyzer model	Sample introduction mode	Analysis mode
[R] model	AL-WB	CDR
[B] model	AL-WB	CD

5. Take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader). Put the high-value sample at the No. 1 tube position, and the low-value sample at No. 2 tube position, and then click **Start Counting** to complete the carryover test.
6. After testing, the system will automatically calculate the carryover of each parameter per the test results, and report the results of Pass or Fail.

Repeatability test

Accept to conduct the repeatability test with either normal-value control or fresh blood samples.

Repeatability Test with Control

- **Steps**

1. Have 1-2 tubes of normal-value control in place.
2. Click **Menu**>>**Performance**>>**Repeatability** to access the Repeatability screen.
3. Select **QC**, and test the blood routine normal-value QC under the analysis modes respectively selected for specific models.

Sample introduction mode		Analysis mode	Remarks
[R] model	AL-WB	CD (6D normal-value) / RET (RET normal-value)	Mandatory
	CT-PD	CD (6D normal-value) / RET (RET normal-value)	Optional according to client's requirements
[B] model	AL-WB	CD (6D normal-value)	Mandatory
	CT-PD	CD (6D normal-value)	Optional according to client's requirements

4. Repeatability test

1) Input the lot number, and take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader). Put the sample at the No. 1 tube position, and then click "Start Counting" to automatically complete the repeatability test.

Figure3–67 Repeatability Screen

2) Wait for the completion of test.

- CT-PD

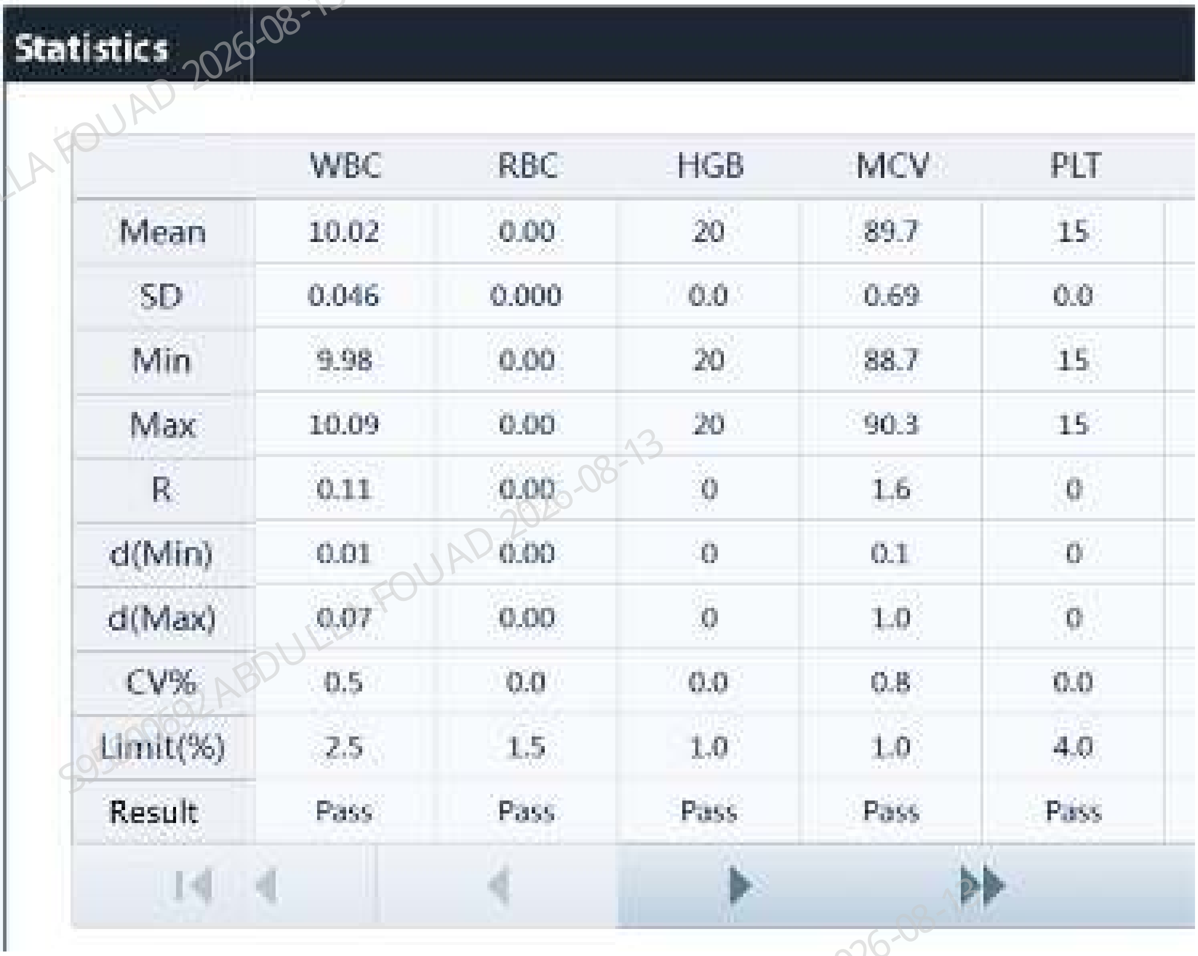
1) Prepare the prediluted samples according to the requirements of 11.15.1.3.

2) Select **CT-PD**, select **QC**, and input the QC lot numbers.

3) Run the counting operation for 10 consecutive times. Mix the samples according to the requirements of 11.15.1.2 before each counting.

5. Then select a sample, its reproducibility test result of each parameter (Pass/Fail) will be displayed on the screen, click the single or double arrow button to review the parameter results.
6. To view the repeatability result of each parameter, click **Statistics** to check the specific CV of repeatability.

Figure3–68 CV of Repeatability


Statistics

	WBC	RBC	HGB	MCV	PLT
Mean	10.02	0.00	20	89.7	15
SD	0.046	0.000	0.0	0.69	0.0
Min	9.98	0.00	20	88.7	15
Max	10.09	0.00	20	90.3	15
R	0.11	0.00	0	1.6	0
d(Min)	0.01	0.00	0	0.1	0
d(Max)	0.07	0.00	0	1.0	0
CV%	0.5	0.0	0.0	0.8	0.0
Limit(%)	2.5	1.5	1.0	1.0	4.0
Result	Pass	Pass	Pass	Pass	Pass

Blood Sample Repeatability Test

- **Steps**

1. Select 1-2 normal fresh blood sample. Only one sample shall be tested under one mode.

Table 3–7 Normal Fresh Blood Sample

Parameters	Parameter range	Parameters	Parameter range
WBC	4-12(10^9/L)	Mono%	0-10%
RBC	3.5-5.5(10^{12}/L)	Eos%	0-5%
HGB	100-160(g/L)	Baso%	0-2%
MCV	80-100(fL)	Ret%	0-2%
PLT	100-350(10^9/L)	Others	No WBC/RBC/PLT flag
Neut%	45%-75%		EDTA2K anticoagulant
Lymph%	15%-45%		Within 8h after being collected

2. Click **Menu>>Performance>>Repeatability** to access the Repeatability screen.
3. Select **Blood Sample**, and test the samples under the analysis modes respectively selected for specific models.

Sample introduction mode		Analysis mode	Remarks
[R] model	AL-WB	CDR	Mandatory
	CT-PD	CDR	Optional according to client's requirements
[B] model	AL-WB	CD	Mandatory
	CT-PD	CD	Optional according to client's requirements

4. Repeatability test

1) Take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader). Put the sample at the No. 1 tube position, and then click "Start Counting" to automatically complete the repeatability test.

Figure3–69 Repeatability Screen

2) Wait for the completion of test.

- CT-PD

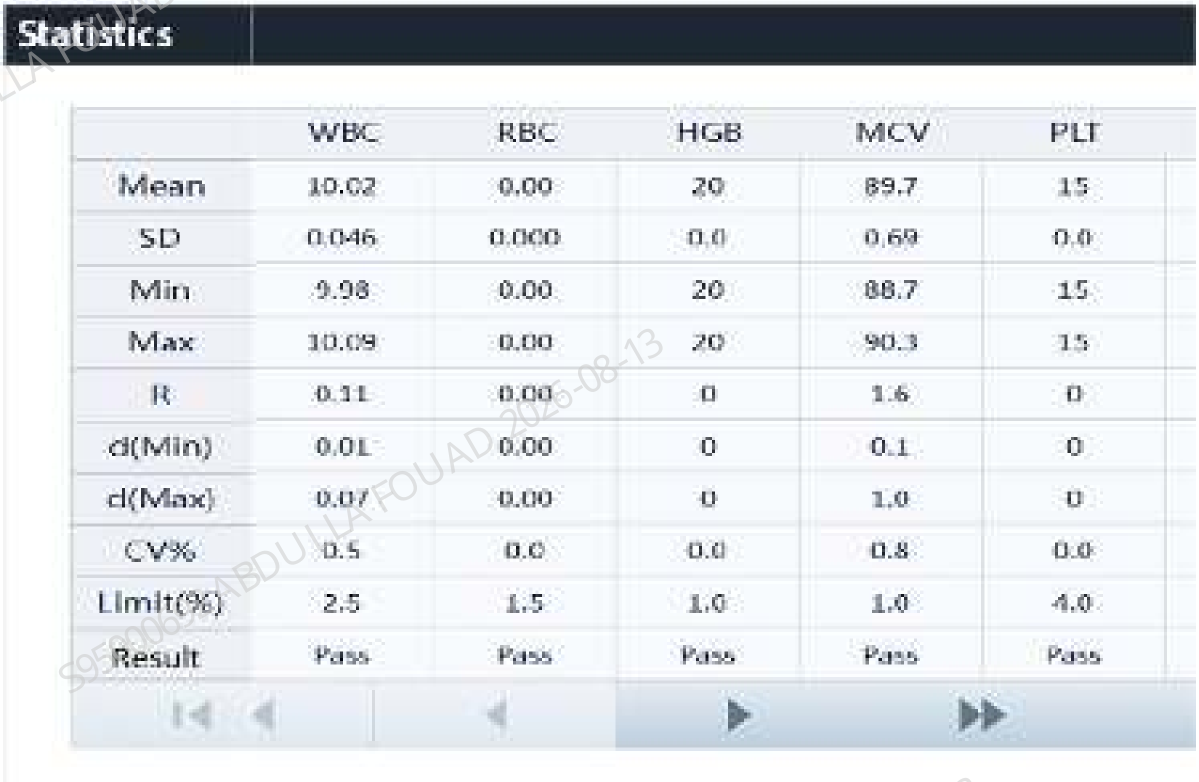
1) Prepare the prediluted samples according to the requirements of 11.15.1.3.

2) Select **CT-PD**, **Blood Sample**, and run the counting operation for 10 consecutive times. Mix the samples according to the requirements of 11.15.1.2 before each counting.

5. Then select a sample, its reproducibility test result of each parameter (Pass/Fail) will be displayed on the screen, click the single or double arrow button to review the parameter results.

6. To view the repeatability result of each parameter, click **Statistics** to check the specific CV of repeatability.

Figure3–70 CV of Repeatability


Statistics

	WBC	RBC	HGB	MCV	PLT
Mean	10.02	0.00	20	89.7	15
SD	0.046	0.000	0.0	0.69	0.0
Min	9.98	0.00	20	88.7	15
Max	10.09	0.00	20	90.3	15
R	0.11	0.00	0	1.6	0
d(Min)	0.01	0.00	0	0.1	0
d(Max)	0.07	0.00	0	1.0	0
CV%	0.5	0.0	0.0	0.8	0.0
Limit(%)	2.5	1.5	1.0	1.0	4.0
Result	Pass	Pass	Pass	Pass	Pass

ESR Repeatability Test

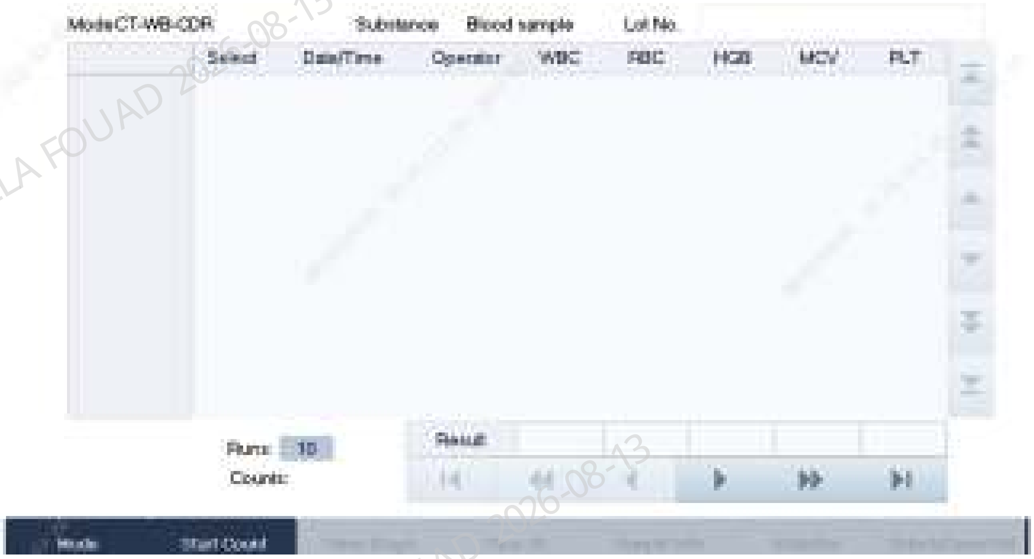
Requirements:

Test Item	Range	Acceptance Criteria
ESR	0~20mm/h	≤1.0(SD)

Steps:

1. Have 1 tube of normal-value control in place.
2. Click **Menu>>Performance>>Repeatability** to access the Repeatability screen. Input the QC lot number, and select **AL-WB**, **QC**, **ESR**

Figure3–71 Repeatability Screen



3. Input the lot number, and take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader). Put the sample at the No. 1 tube position, and then click "Start Counting" to automatically complete the repeatability test.
4. Wait for the test results.
5. After testing, select the corresponding samples, and the repeatability results (Pass or Fail) will appear on the software screen.

WB Calibration Qualification and Calibration

WB Calibration Qualification

- Steps

1. Select a tube of fresh WB calibrator, and mix it according to the requirements of 11.15.1.2.
2. On the Sample Analysis screen, click **Mode** to access the Mode Selection screen.
3. Select OV-WB, CD mode, and click **OK**.
4. Count the calibrators for 11 consecutive times on the Sample Analysis screen.
5. Click **Table Review**, select 10 group counting, click **CV**.
6. Compare the averages and targets of the tested parameters. The deviation of the parameters and targets in the table below shall be within specification.

Table 3–8 Deviation Range of WB Calibration Parameters and Targets

Parameters	Deviation range
WBC	±1.5%
RBC	±1.0%
HGB	±1.0%
HCT	±2.0%

Table 3–8 Deviation Range of WB Calibration Parameters and Targets(continued)

Parameters	Deviation range
MCV	±1.0%
PLT	±3.0%

7. In case of any out-of-specification deviation or as required by client, follow the requirements under WB Calibration.

WB Calibration

- **Steps**

1. Click **Menu>>Calibration>>Calibrator Calibration** to access the Calibrator Calibration screen.

Figure3–72 Calibrator Calibration Screen of [R] Model

Figure3–73 Calibrator Calibration Screen of [B] Model

2. Take a new PLUS calibrator, input its lot number, expiration date and target respectively in the corresponding fields.
3. Take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader). Put the calibrator at the No. 1 tube position, and then click "Start Counting" to automatically complete the calibration test.

NOTE

Select WB CDR screen for [R] model and WB CD screen for [B] model.

4. After calibration, repeat the steps under WB Calibration Qualification to confirm that the calibration qualification meets the requirements.

Background Verification for Body Fluid Samples (On-demand)

The background verification required by a client shall be performed according to the following steps.

This section provides instruction of background verification for body fluid samples. The background requirements for body fluid samples are as follows:

Table 3–9 Background Requirements for Body Fluid

Parameters	Background and Blank Counts
WBC-BF	≤ 0.001×10^9 / L
RBC-BF	≤ 0.003×10^{12} / L

Steps

1. Click **Menu**>>**Performance**>>**Background** to access the Background screen.

Figure3–74 Background Screen

2. Click **Mode**, select**CT-BF**mode, and click **OK**.

Figure3–75 Background Mode Selection Screen

3. Take a tube rack (the type of tube rack depends on the autoloader type, 10-position tube rack for 10X5 autoloader, and 5-position tube rack for 5X6 autoloader). Put the tube containing diluent at the No. 1 tube position, and then click "Start Counting" to automatically complete the test.
4. After testing, the analyzer automatically reports the background results. Make sure all the results are **Pass**.

Carryover Verification for Body Fluid Samples (On-demand)

The carryover verification required by a client shall be performed according to the following steps.

Requirements

The requirements for the body fluid samples used in the carryover tests, and the requirements for the carryover results are listed as below:

Table 3–10 Carryover Requirements for Body Fluid Samples

Sample type	**Parameters**	**Unit**	**High-value sample**	**Low-value sample**
Cerebrospinal fluid	WBC-BF RBC-BF	$\times10^9$/L	>0.200	< 0.01
		$\times10^{12}$/L	> 0.010	< 0.005
Hydrothorax, ascites, synovial fluid	WBC-BF RBC-BF	$\times10^9$/L	> 1.000	< 0.05
		$\times10^{12}$/L	> 0.100	< 0.005

Table 3–11 Carryover Specifications for Body Fluid Samples

Parameters	Carryover
WBC-BF	≤0.3% or ≤0.001×10^9/L
RBC-BF	≤0.3% or ≤0.003×10^{12}/L

NOTE

Prepare the high-value sample by mixing the control with diluent. Refer to the steps for details. Use diluent as low-value samples.

Steps

1. Click **Menu>>Performance>>Carryover** to access the Carryover screen.
2. Click **Mode** to select **CT-BF** mode.
3. Load an empty tube with 3ml diluent and 500ul normal-value control. Mix the sample well.
4. Run the OV counting for the mixed samples for 3 times to acquire the high-value test results. Manually mix the samples before each counting.
5. Run the OV counting for the diluent for 3 times to acquire the low-value test results.
6. After testing, the analyzer will automatically calculate the carryover which shall be within specification.
7. WBC-BF≤0.3% or 0.001×10^9/L, RBC-BF≤0.3% or 0.003×10^{12}/L.

Figure3–76 Carryover Test for Body Fluid Samples

Repeatability Verification for Body Fluid Samples (On-demand)

The repeatability verification required by a client shall be performed according to the following steps.

Requirements

The repeatability requirements for body fluid samples are listed in the following table.

Table 3–12 Repeatability Requirements for Body Fluid Samples

Parameters	Range	CV/range
WBC-BF	(0.015-0.100)×10^9/L	≤30%
RBC-BF	(0.003-0.050)×10^{12}/L	≤40% or ≤7000/μL (d)

Steps

1. Click **Menu**>>**Performance**>>**Repeatability** to access the Repeatability screen.
2. Click **Mode** to select **OV-BF** mode.
3. Load an empty tube with 4 mL diluent and 20 μL normal-value control. Mix the sample well.
4. Run the OV counting for the prepare sample for 10 times. Manually mix the samples before each counting.
5. After testing, locate WBC-BF and RBC-BF parameters, and make sure the results are **Pass**.

Figure3–77 Repeatability Test for Body Fluid Samples

3.17 Analyzer Information Export

Exporting Logs

Logs record all activities of the analyzer. It contributes significantly to searching for operation history and troubleshooting the analyzer.

Operators are allowed to export the logs of a specific period to the USB drive. Before exporting logs, make sure that the USB drive is under no security risks and is properly connected to the USB port on the side of the analyzer.

Follow below instructions:

1. Click **Menu>>Service>>Log** to access the "Log" screen.
2. Select **All Logs**, **General Logs**, **Parameter Modification**, **Error Information** or **Sequential Operation** as required.
3. Click **Export**, and the **Export Dialog Box** pops up.

Figure3–78 "Logs Export" Dialog Box

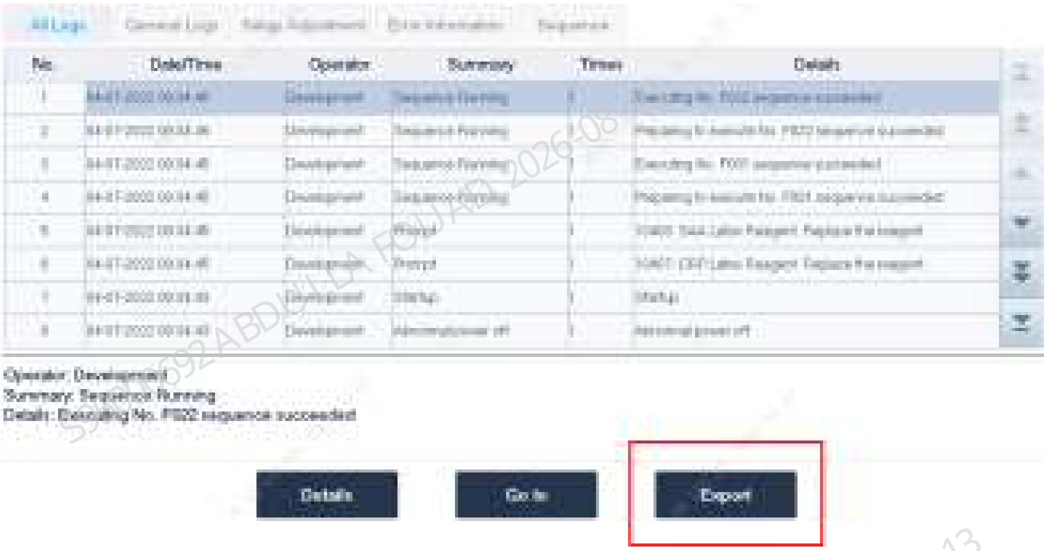
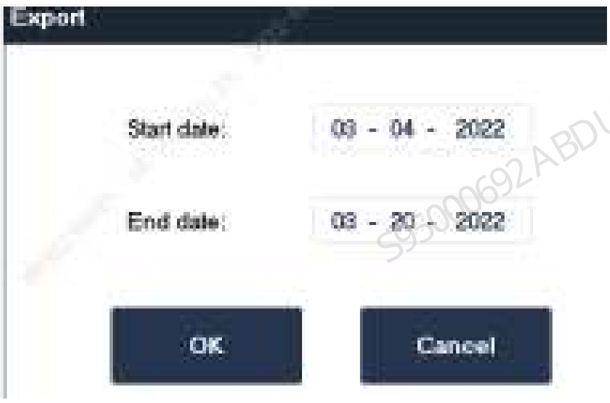


Figure3–79 "Export" Dialog Box



4. Enter the dates in the edit boxes of **Start Date** and **End Date** to specify the period for log export, then click **OK**.
5. After export, the screen displays the **Export succeeded**" dialog box.

Figure3–80 "Export succeeded" Dialog Box

Exporting Commissioning Data

The commissioning data records all about the analyzer and is significant for troubleshooting by maintenance personnel.

Operators are allowed to export the commissioning data of a specific period to the USB drive. Before exporting commissioning data, make sure that the USB drive is under no security risks and is properly connected to the USB port on the side of the analyzer.

Follow below instructions:

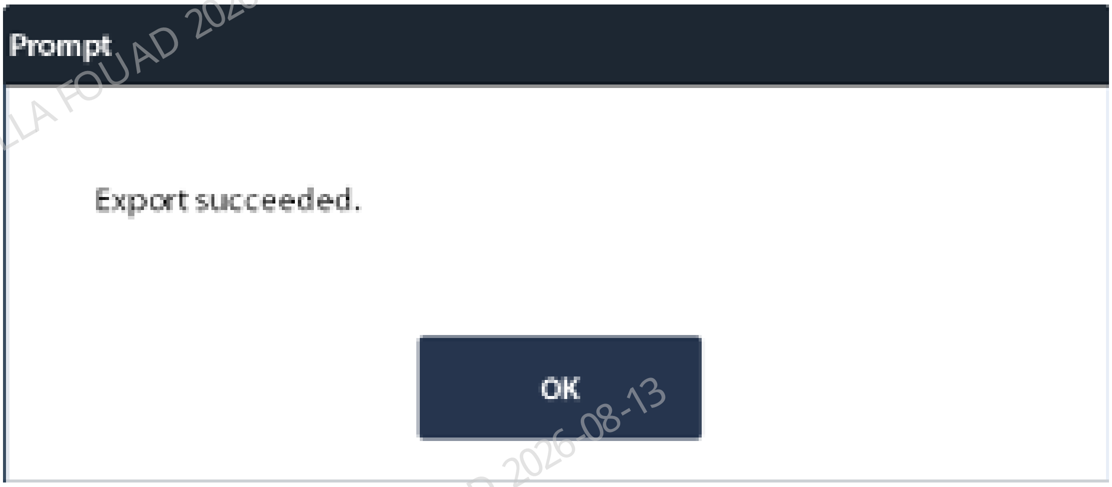
1. Click **Menu**>>**Service**>>**Advanced Toolbox**>>**commissioning Data Export** to access the "commissioning Data Export" screen.

Figure3–81 Commissioning Data Export Screen

2. Click to choose to export **Instrument information**, **SW commissioning data**, **inf file** or **Sample data** as required, where it is available to export **inf file** of a specific period.
3. Click **Export** and wait for the analyzer to export data.

4. After a successful data export, the dialog box of **Export succeeded** pops up. Click **OK**.

Figure3–82 "Export succeeded" Dialog Box



3.18 Installation Checklist

No. NO.	Items Inspection Item		Description Description	Reference Reference	Result Result	Re-marks
1	Operating environment	Ambient temperature	Within the normal operating temperature range	10℃□35℃	□Pass □Fail	
2		Humidity	Within the normal operating humidity range	30%□85%	□Pass □Fail	
3		Atmospheric pressure	Within the normal operating atmospheric pressure range	70KPa□ 106KPa	□Pass □Fail	
4		Electromagnetic interference	Away from electromagnetic interference source	/	□Pass □Fail	
5	Space Requirements		Space	Meet the space requirements	□Pass □Fail	

No. NO.	Items Inspection Item	Description Description	Reference Reference	Result Result	Re- marks
6	Heat dissipation	Space requirement for radiator fan	Distance to wall: ≥250mm	□Pass □Fail	
7	Power requirements of the main unit	Power supply voltage	100V-240V~ (allowance ±10%)	□Pass □Fail	
8		Zero-to-earth voltage	< 5v	□Pass □Fail	

No. NO.	Items Inspection Item	Description Description	Reference Reference	Result Result	Re- marks
1.	The package box and appearance of the main unit	Package and appearance	Package and appearance are in good conditions	□Pass □Fail	
2.	Package and appearance of the autoloader	Package and appearance	Package and appearance are in good conditions	□Pass □Fail	
3.	Packing list	Items in packing list	All items included	□Pass □Fail	
4.	Removal of fixing parts for analyzer transport	Sampling assembly and blending assembly	Cut the wire ties	□Pass □Fail	
5.		Fluorescent reagent container	Cut the wire ties	□Pass □Fail	
6.		Rotary Compressing Assembly	Cut the wire ties	□Pass □Fail	
7.		Sample Compartment	Cut the wire ties	□Pass □Fail	
8.		Peripheral Blood Blending Assembly	Cut the wire ties	□Pass □Fail	
9.	HGB blank voltage	HGB background voltage values	4.3-4.5 V	□Pass □Fail	

No. NO.	Items Inspection Item	Description Description	Reference Reference	Value Vallue	Result Result	Re- marks
1.	Calibration factors	WBC	75□125		□Pass □Fail	
		RBC	75□125		□Pass □Fail	
		HGB	75□125		□Pass □Fail	
		MCV	95□105		□Pass □Fail	
		PLT	75□125		□Pass □Fail	
		RBC-O Whether to run □Yes □No	75□125		□Pass □Fail	
		PLT-O Whether to run □Yes □No	75□125		□Pass □Fail	
		PLT-H Whether to run □Yes □No	75□125		□Pass □Fail	
2.	Background (OV-WB)	WBC	$\leq 0.1 \times 10^{9}$ / L		□Pass □Fail	
		RBC	$\leq 0.02\times 10^{12}$/ L		□Pass □Fail	
		HGB	≤ 1 g / L		□Pass □Fail	
		PLT	$\leq 3 \times 10^{9}$ / L		□Pass □Fail	
3.	Optical gain confirmation	DIFF-FS	≤5.0%		□Pass □Fail	
		DIFF-SS	≤5.0%		□Pass □Fail	
		DIFF-FL	≤5.0%		□Pass □Fail	
		RET-FS	≤10.0%		□Pass □Fail	

No. **NO.**	**Items** **Inspection Item**	**Description** **Description**	**Reference** **Reference**	**Value** **Vallue**	**Result** **Result**	**Re-** **marks**
		Whether to run □Yes □No				
		RET-SS Whether to run □Yes □No	≤10.0%		□Pass □Fail	
		RET-FL Whether to run □Yes □No	≤10.0%		□Pass □Fail	
4.	Carryover in blood routine	WBC	≤1.0□		□Pass □Fail	
		RBC	≤1.0□		□Pass □Fail	
		HGB	≤1.0□		□Pass □Fail	
		HCT	≤1.0□		□Pass □Fail	
		PLT	≤1.0□		□Pass □Fail	
5.	Repeatability (OV-WB)	WBC	≤2.5%		□Pass □Fail	
		RBC	≤1.5%		□Pass □Fail	
		HGB	≤1.0%		□Pass □Fail	
		MCV	≤1.0%		□Pass □Fail	
		HCT	≤1.5%		□Pass □Fail	
		PLT	≤4.0%		□Pass □Fail	
		ESR	≤1.0(SD)		□Pass □Fail	
		RET# Whether to run	≤15%		□Pass □Fail	

No. NO.	Items Inspection Item	Description Description	Reference Reference	Value Vallue	Result Result	Re- marks
		□Yes □No				
		RET% Whether to run □Yes □No	≤15%		□Pass □Fail	
6.	Repeatability (OV-PD) Whether to run □Yes □No	WBC	≤3.5%		□Pass □Fail	
		RBC	≤2.0%		□Pass □Fail	
		HGB	≤2.0%		□Pass □Fail	
		MCV	≤3.0%		□Pass □Fail	
		HCT	≤3.0%		□Pass □Fail	
		PLT	≤8.0%		□Pass □Fail	
		RET# Whether to run □Yes □No	≤30.0%		□Pass □Fail	
		RET% Whether to run □Yes □No	≤30.0%		□Pass □Fail	
7.	Background (OV-BF) Whether to run □Yes □No	WBC-BF	≤0.001× 10^9 / L		□Pass □Fail	
		RBC-BF	≤0.003× 10^{12}/ L		□Pass □Fail	
8.	Carryover (OV-BF) Whether to run □Yes □No	WBC-BF	≤0.3%		□Pass □Fail	
		RBC-BF	≤0.3%		□Pass □Fail	
9.	Repeatability (OV-BF) Whether to run □Yes □No	WBC-BF	≤30%		□Pass □Fail	
		RBC-BF	≤40% or ≤7000/μL		□Pass □Fail	

4 Software setup and update

4.1 Screen Introduction

The software screen is as follows:

Figure4–1 Software Screen

Menu  Click the **Menu** button  at the top left corner of the software screen to access the system menu.

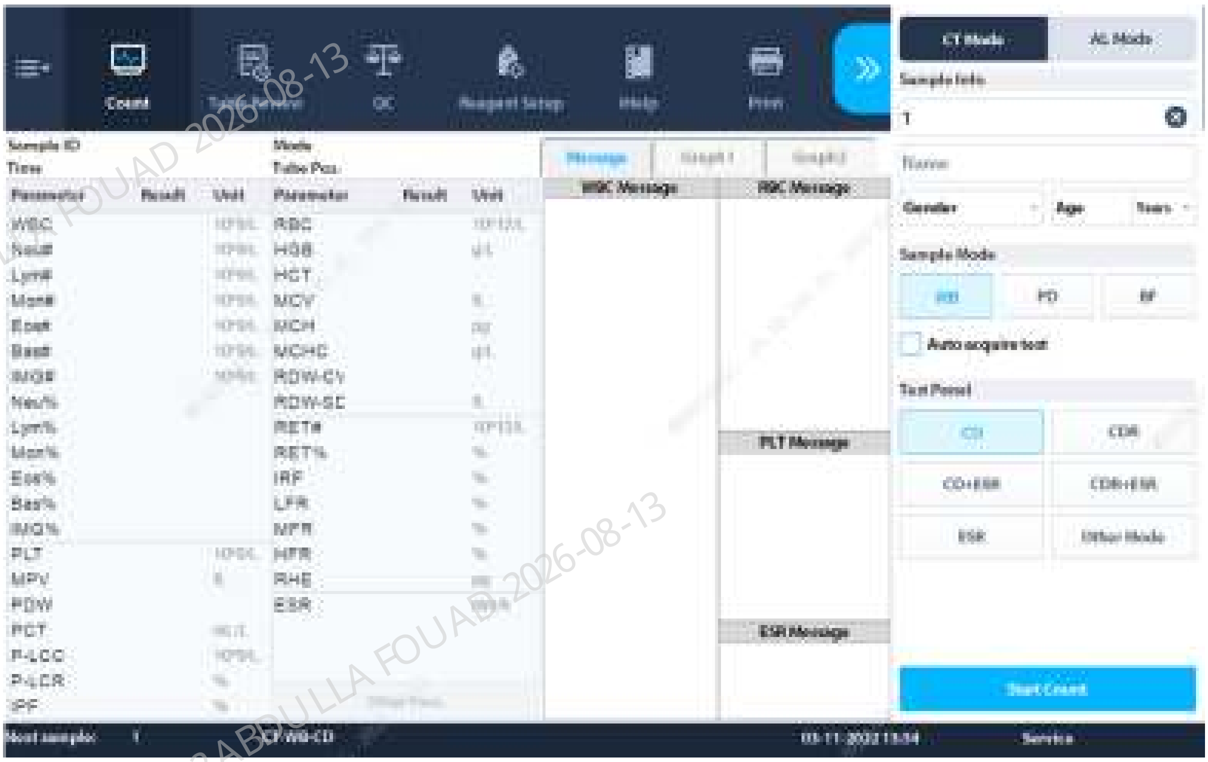
Area	Content
(1) Utility button area	Access to common functions of menu bar
(2) Operation area	Displays the content of the corresponding screen For example, on the **Sample Analysis** screen, the area displays the corresponding keys for sample analysis operations and sample analysis results.

Area	Content
(3) Auxiliary information area	Display of current screen related auxiliary information
(4) Sample information	For example, the number and mode of next sample are displayed on the **Sample Analysis** screen; the position and total number of current sample are displayed on the Result Review screen
(5) Other information	Display of system time and analyzer error information (if any)

Introduction to buttons in operation area:

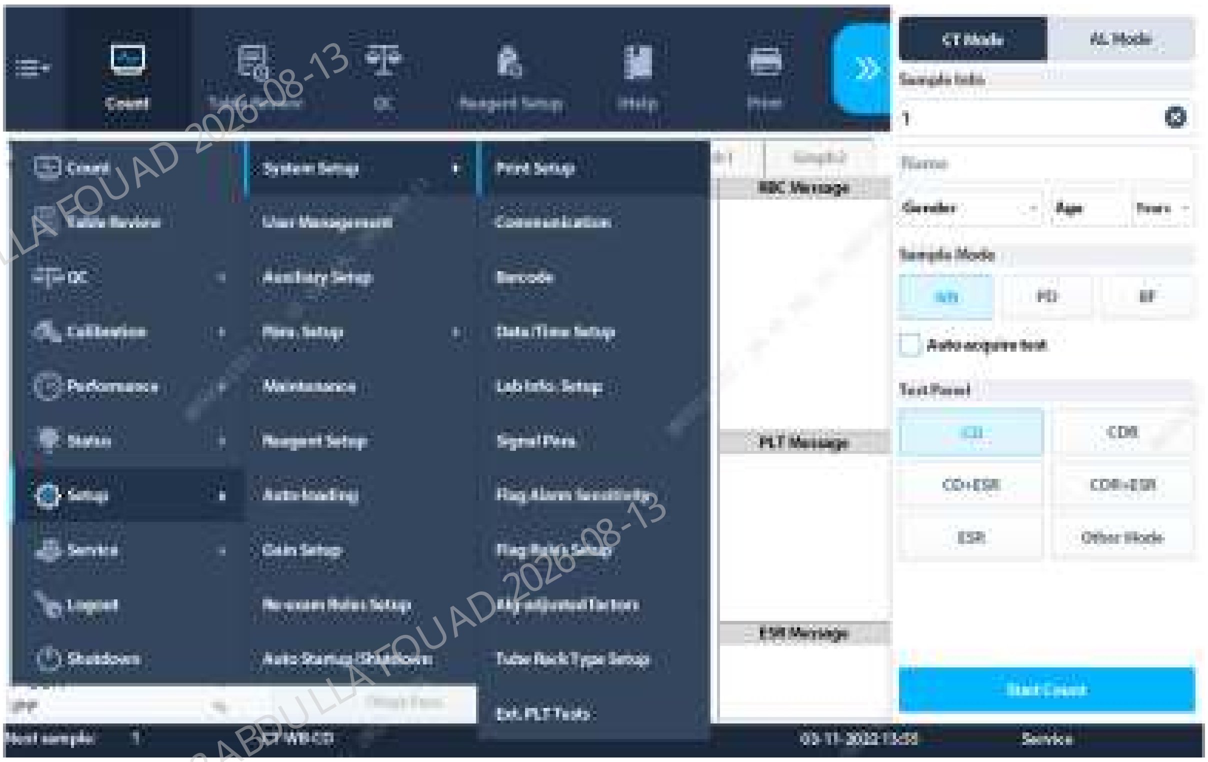
Name	Functions
Count	Set the sample related information (e.g. mode selection) and run the sample analysis procedure
Table Review	Browse, screen, delete and export the data after sample analysis
QC	Add, edit and delete QC files, and conduct corresponding QC test
Reagent Setup	Test the effectiveness and consumption of reagent management
Guidance Video	View the related videos in Guidance Video, deal with analyzer issues
Print	Print the data on sample analysis, QC and calibration screens, etc.
<<	Expand the suspension analysis window

Introduction to suspension analysis window:



Area	Functions
Sample information	Enter the information including sample number, patient name, gender and age
Sample mode	Select sample mode: WB, PD, BF
Auto acquisition mode	Inquire analysis mode settings of expert system/LIS side
Analysis mode area	Select analysis mode
Diluent dispensing	Add diluent

Introduction to buttons of special menu for service level



Main menu	Function Description	Submenu level 1	Function Description	Submenu level 2	Function Description
Calibration	Calibrate analyzer	ESR calibrator calibration	Calibrate ESR calibrator	N/A	N/A
Performance	Analyze performance	Background	Run analyzer performance - background analysis	N/A	N/A
		Repeatability	Run analyzer performance - repeatability analysis	N/A	N/A
		Carryover	Run analyzer performance - carryover analysis	N/A	N/A
		Aging	Run instrument performance - aging analysis	N/A	N/A

Main menu	Function Description	Submenu level 1	Function Description	Submenu level 2	Function Description
Status	Display analyzer status	Baseline status	Display analyzer's baseline status	N/A	N/A
Service	Provide service functions	Commissioning & Self-Test	Commissioning all components	Adj. Sample Probe Pos	Adjust sample probe position
				Heating Commissioning	Commissioning instrument heater
				Pressure Commissioning	Commissioning pressure assembly
				Sensor Commissioning	Commissioning sensor assembly
				Optical Commissioning	Commissioning optical assembly
		Advanced Toolbox	View analyzer's advanced toolkit	N/A	N/A

4.2 Login Password

- Service level user ID: **Service**
  Password: **Se s700** (Note there is a space between Se and s700).
- Administrator level user ID: **Admin**
  Password: **Admin**

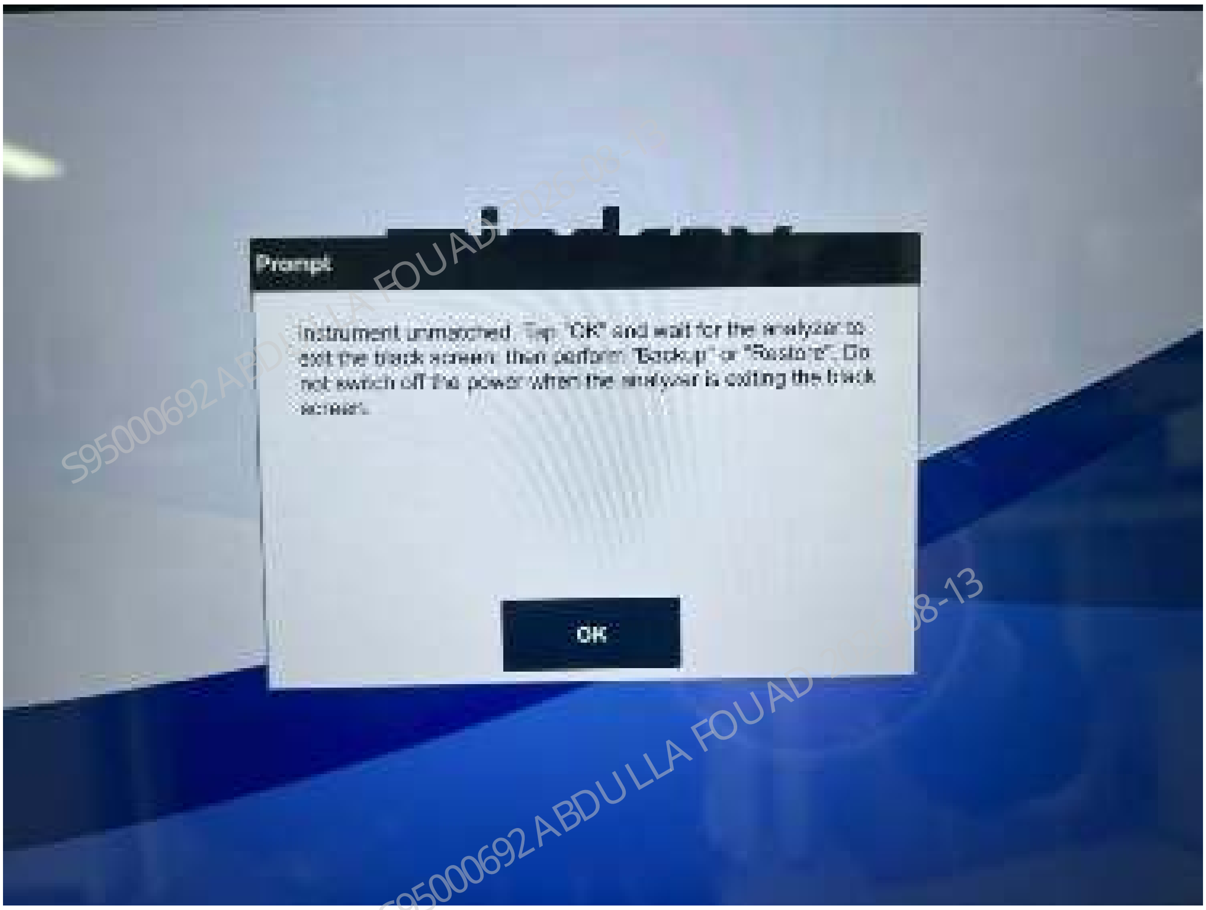
NOTICE

- **Password is case sensitive.**

4.3 System Self-Test

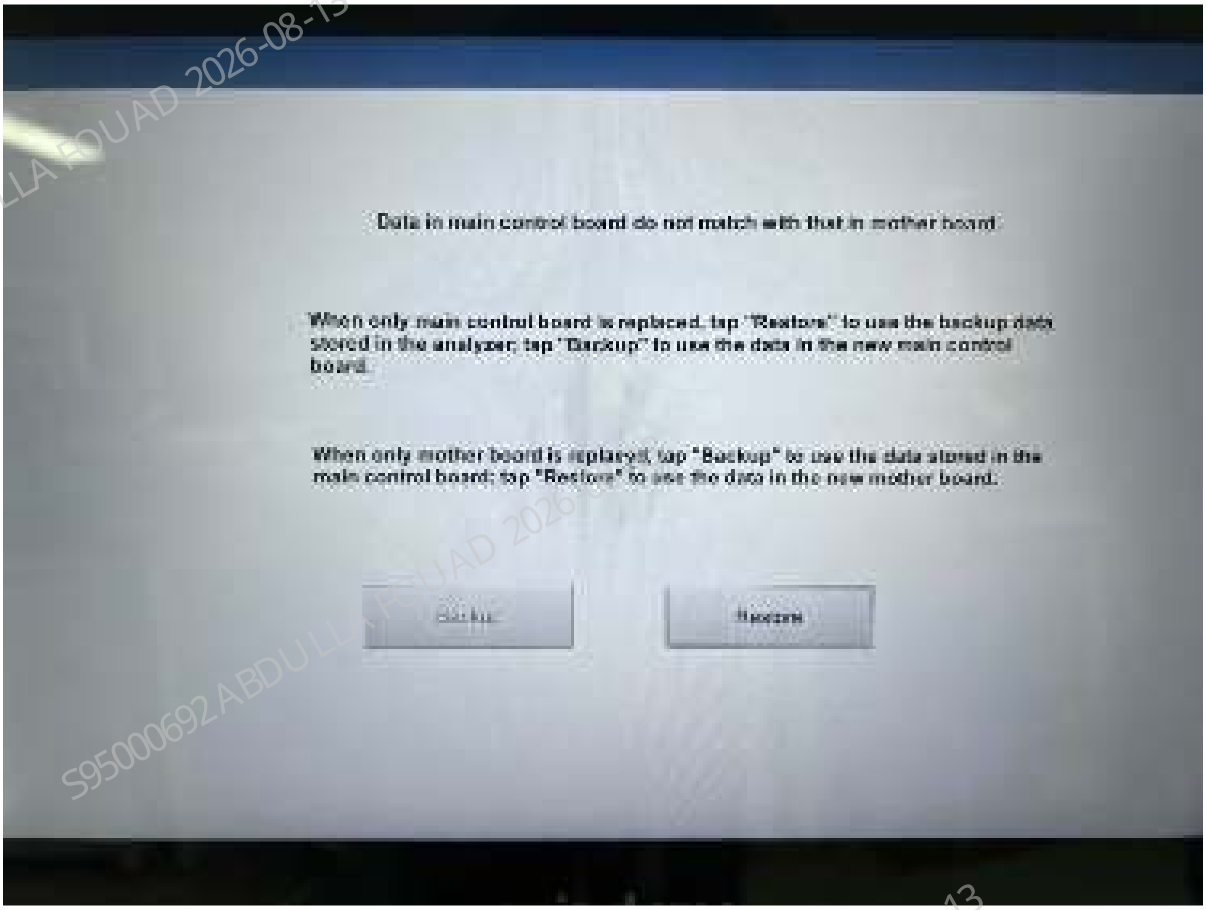
When a user logs in at service level, the analyzer will automatically test the matching rate of motherboard backup data and main control board data. If the motherboard or main control board has been replaced or if the configuration has been modified and any abnormal shutdown has been identified, a prompt will appear after login at service level to remind the user of data restore or backup.

Figure4–2 System Self-Test Prompt



Click **OK** to access the "Backup and Restore Data" screen. Perform data backup and restore according to the prompts.

Figure4–3 Backup and Restore Data Screen



If a new main control board is installed, select **Restore** to restore the key parameters to the main control board according to the follow-up instructions. If a new motherboard is installed, select **Backup** to back up the data to the new motherboard according to the follow-up instructions.

NOTICE

- **Before installing a new main control board or motherboard, it is required to run the startup/shutdown procedure for the software to automatically back up the data.**

4.4 Access Checklist

Table 4–1 Different Access for Each Function

No.	Level 1	Level 2	Level 3
1	Count	N/A	
2	Table Review	N/A	
3	QC	L-J QC	Setup

Table 4–1 Different Access for Each Function(continued)

No.	Level 1	Level 2	Level 3
			Count
		X-B QC	Setup
			Graph
4	Calibration	Manual Calibration	
		ESR Calibration (Administrator, Service)	
		ESR Calibrator Calibration (Administrator, Service)	
		Calibrator Calibration (Administrator, Service)	
		Fresh Blood Calibration (Administrator, Service)	
		Gain Calibration	CBC Gain Calibration (Service)
		Calibration History (Administrator, Service)	Optical Gain Calibration (Service)
5	Performance	Background (Service)	
		Repeatability (Service)	
		Carryover (Service)	
		Aging (Service)	
6	Status	Statistics	
		Temperature & Pressure (Administrator, Service)	
		Floater Status (Administrator, Service)	
		Sensor Status (Administrator, Service)	
		Voltage & Current (Administrator, Service)	
		Baseline Status (Service)	
		Version Info.	
7	Setup	System Setup	Print Setup
			Communication (Administrator, Service)
			Barcode (Administrator, Service)

Table 4–1 Different Access for Each Function(continued)

No.	Level 1	Level 2	Level 3
			Date/Time Setup
			Lab Info. Setup
			Signal Para. (Service)
			Flag Alarm Sensitivity (Administrator, Service)
			Flag Rules Setup (Administrator, Service)
			Alg-Adjusted Factors (Service)
			Tube Rack Type Setup
			Re-exam Setup (Administrator, Service)
		User Management	
		Auxiliary Setup	
		Para. Setup	Parameter Unit Setup
			Reference Range Setup
			Microscopic Para. Setup (Administrator, Service)
		Maintenance (Administrator, Service)	
		Reagent Setup	
		Auto-loading (Administrator, Service)	
		Gain Setup (Administrator, Service)	
		Re-exam Rules Setup (Administrator, Service)	
		Auto Startup/Shutdown (Administrator, Service)	
8	Service	Commissioning & Self-Test	Adj. Sample Probe Pos. (Service)
			Adjust Autoload Pos. (Service)
			Self-Test
			Heating Commissioning(Service)
			Pressure Commissioning(Service)
			Sensor Commissioning(Service)

Table 4–1 Different Access for Each Function(continued)

No.	Level 1	Level 2	Level 3
			Optical Commissioning (Service)
		Accurate Diagnosis (Administrator, Service)	
		Guidance Video	
		Maintenance	
		Screen Cal.	
		Logs	
		Advanced Toolbox (Service)	
9	Logout		
10	Shutdown		

Differences Between Various Access on the Same Function Screen

Calibration - Manual Calibration

- A user with the permissions of service and above level is allowed to manually modify the WB manufacturer calibration factor (MC1). The UC1 corresponding to the modified MC1 will be automatically updated to 100%.
- A user with the permissions of service and above level is allowed to manually modify the predilution transfer factor (TC3), WB transfer factor (TC16, TC18), and low-value WBC transfer factor (TC17).
- A user with the permissions of service and above level is allowed to click **Factory Reset** button to restore the factory calibration factor, user calibration factor and transfer factor.
- A user with the permissions of service and above level is allowed to export the manual calibration configuration data to the root directory of USB drive, with a file named algparaconfig.xml.
- A user with the permissions of service and above level is allowed to import the manual calibration configuration data (file name is algparaconfig.xml) from the root directory of USB drive to the instrument of the same model (it is not supported to import the data to the instrument of the different model or with the different configuration or software version).
- A user with the permissions of service and above level is allowed to click **Help** button to view the transfer relations among all factory calibration factors, user calibration factors and transfer factors.

Value	Code	Meaning	Calculation	Permission for Manual Setup	Permission for Auto Setup	Initial Value
Original value	N/A	Raw measurements without multiplying by any calibration or transfer factors	N/A	N/A	N/A	N/A
Manufacturer calibration factor	MC1	Calibration factor for seven parameters under WB mode	Manufacturer target/original value of WB basic mode (Basic/standard model of blood routine: CD, high-end model: CDR)	Service	Service	100%
Manufacturer transfer factor	TC3	Transfer factor for seven parameters of predilution and WB	Original value of basic mode/ original value of predilution CDR (Low-level model, basic mode CDR: CD)	Service	Service	100%
	TC16	Transfer factor for CDR (CD) of ESR measurement mode and basic mode	Original value of basic mode/ original value of ESR measurement (Low-level model, basic mode changed from CDR to CD)	Service	Service	100%
	TC17	Transfer factor for WB CDR (CD) modes of low-value WBC3x mode and basic mode	Original value of basic mode/ original value of low-value WBC3x mode	Service	Service	100%

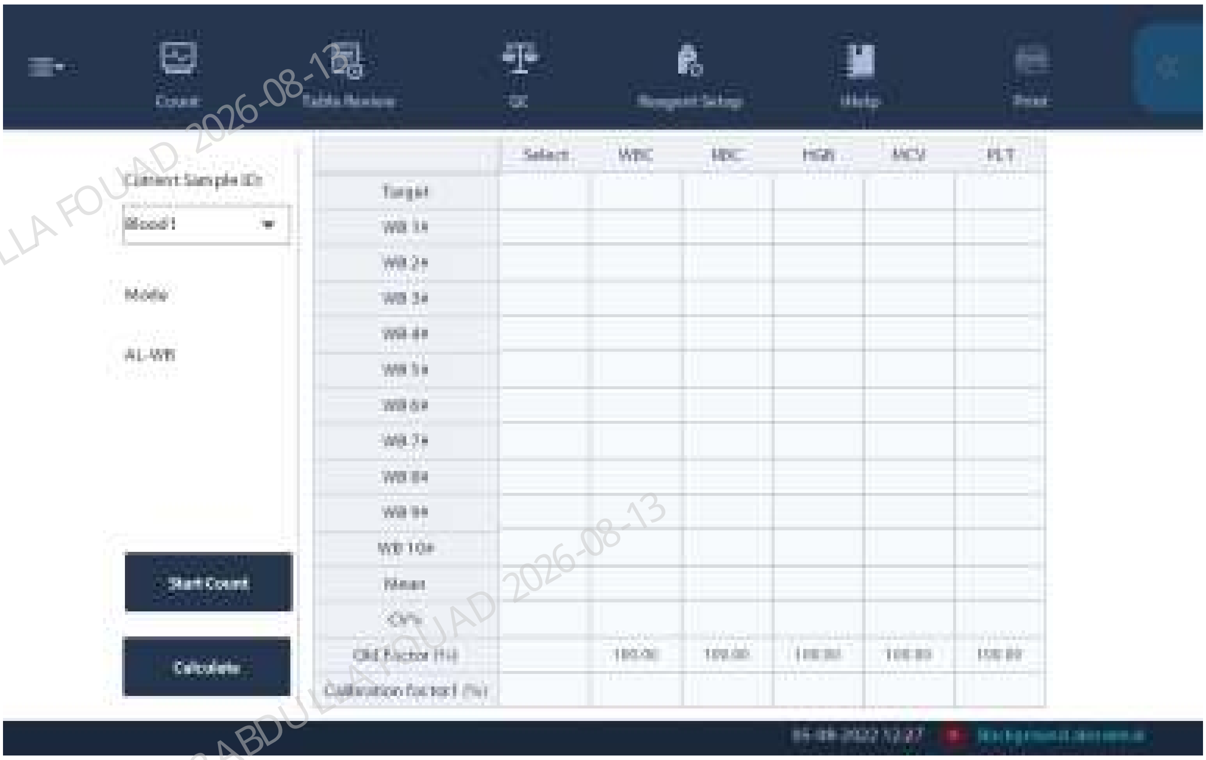
Value	Code	Meaning	Calculation	Per-mis-sion for Man-ual Setup	Per-mis-sion for Auto Setup	Initial Value
			(Low-level model, basic mode changed from CDR to CD)			
	TC18	Transfer factor for WB CDR (CD) modes of CBC/CD mode and basic mode	Original value of basic mode/ original value of CD mode (Low-level model, basic mode changed from CDR to CD) applicable to R model	Serv ice	Serv ice	100%
User calibra tion factor	UC1	Calibration factor for five parameters (generated by user) under WB mode	Results of user target/WB CDR after manufacturer calibration (Low-level model, mode changed from CDR to CD)	Admin istrator	Admin istrator	100%
	UC2	Calibration factor for five parameters (input by user) under predilution mode	Manual modification by user	Admin istrator	N/A	100%

The specific screen differences are as below: The first is for administrator level, and the following is for service level.

Calibration - Calibrator Calibration

- A user with the permissions of service and above level is allowed to access predilution and WB mode calibration.
- A user with the permissions of service and above level is allowed to delete items.
- A user with the permissions of service and above level is allowed to check the box of **Control (Others)**.

A user with the permissions of service and above level is allowed to view RBC-O, PLT-O, PLT-H and HCT parameters. The specific screen differences are as below: The first is for administrator level, and the second is for service level.



Status - Temperature & Pressure

A user with the permissions of service and above level is allowed to rise and release pressure. The specific screen differences are as below: The first is for administrator level, and the second is for service level.

Status - Floater Status

A user with the permissions of service and above level is allowed to run the floater self-test. The specific screen differences are as below: The first is for administrator level, and the second is for service level

Setup - System Setup - Comm. Setup.

- A user with the permissions of service and above level is allowed to change the data channel settings.

The specific screen differences are as below: The first is for administrator level, and the second is for service level.

Setup - Maintenance Setup

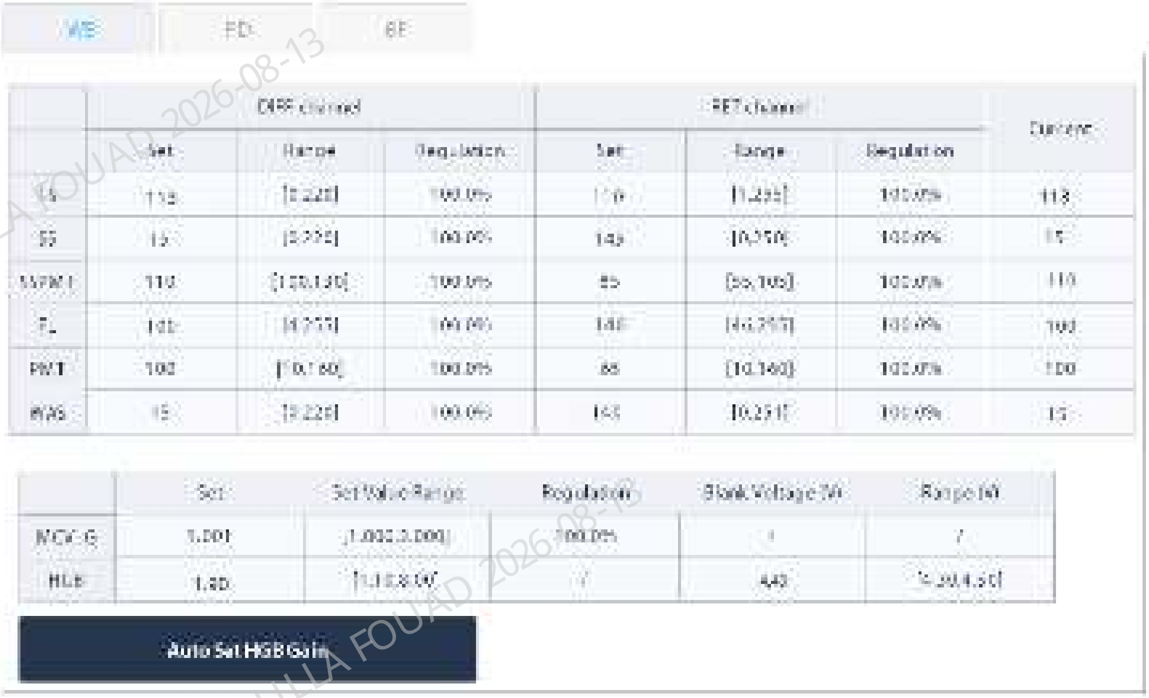
- A user with the permissions of service and above level is allowed to set the standby time, schedule the probe cleanser maintenance time, remind time, interval days of probe cleanser maintenance, and probe cleanser maintenance threshold.
- After modification, a prompt of saving changes appears when the user intends to quit the screen. Save the changes and reboot for the changes to take effect.

The specific screen differences are as below: The first is for administrator level, and the second is for service level.

Setup - Gains Setup

- A user with the permissions of service and above level is allowed to restore the factory gain settings in case of any gain setting error.
- The HGB gains may be modified by **Auto Setting of HGB Gains**.

The specific screen differences are as below: The first is for administrator level, and the second is for service level.



4.5 Setup Screen

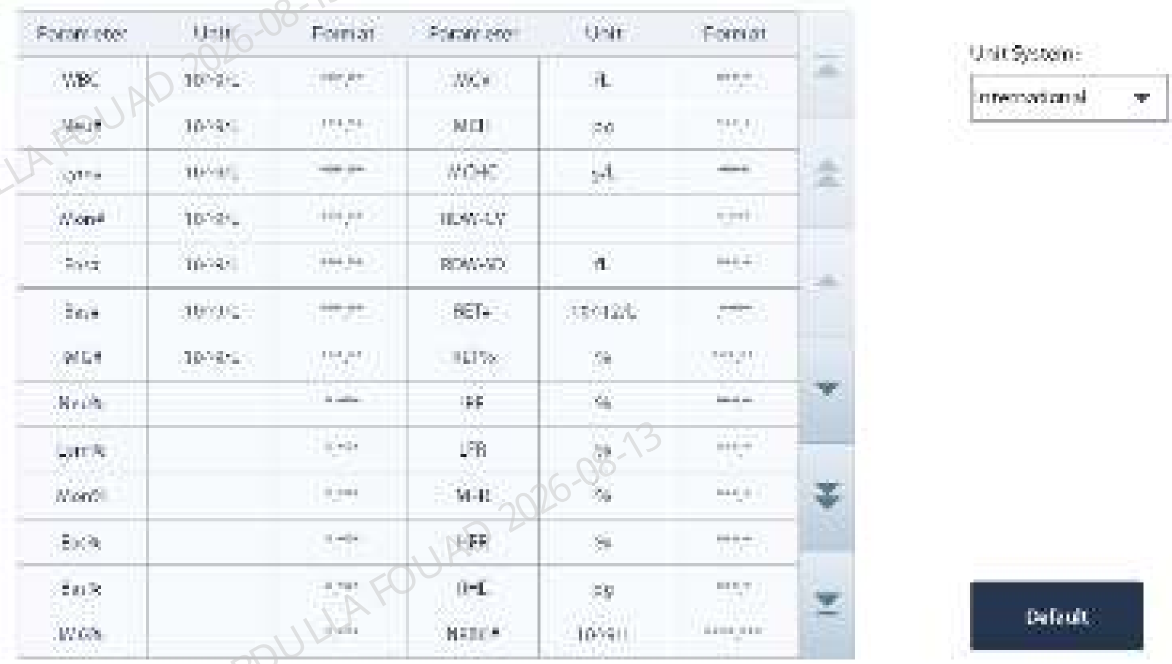
Para. Setup

Set Parameter Unit

- This screen is for setting the units of report parameters and RUO parameters.

1. Click **Menu**>>**Setup**>>**Para. Setup**>>**Parameter Unit Setup** to access the **Parameter Unit Setup** screen.

Figure4–4 Parameter Unit Setup Screen



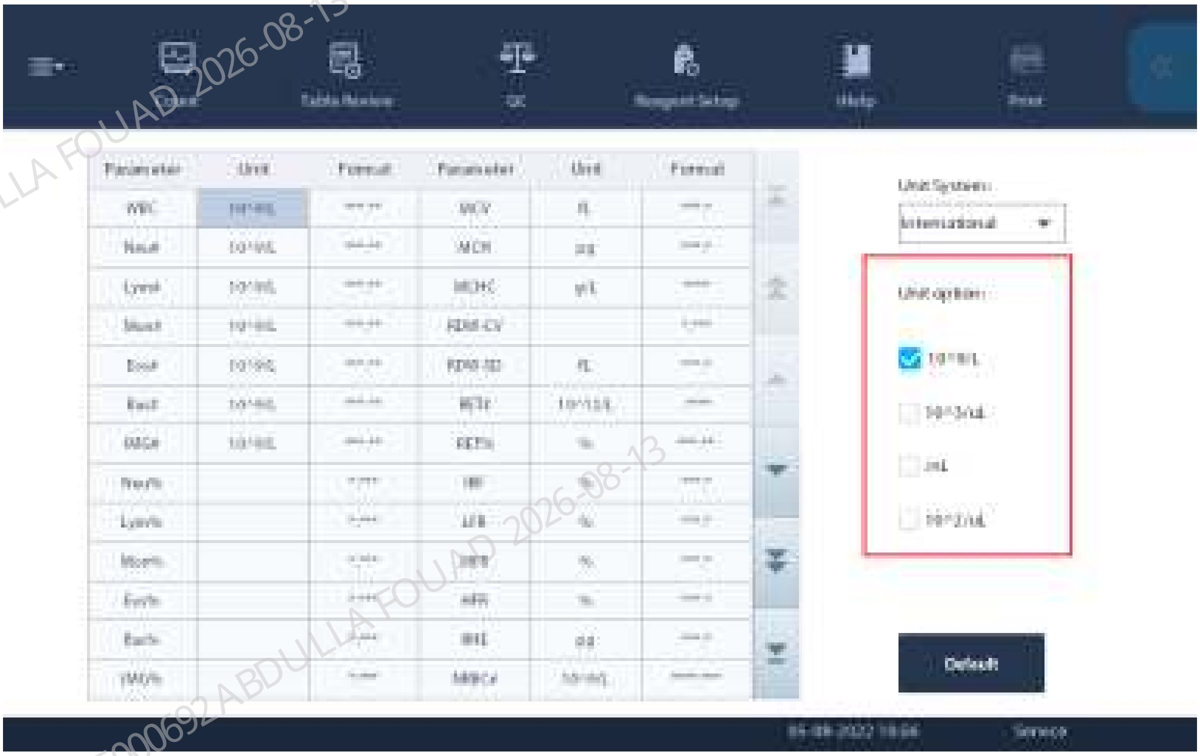
2. Select the desired unit system from the **Unit System** pull-down list.
3. Click the **Unit** column of the parameter of which the unit is to be modified.

Figure4–5 Select Unit Screen

Parameter	Unit	Format	Parameter	Unit	Format
WBC	10^9/L	***.**	MCV	fL	***.*
Neu#	10^9/L	***.**	MCH	pg	***.*
Lym#	10^9/L	***.**	MCHC	g/L	****
Mon#	10^9/L	***.**	RDW-CV		*.***
Eos#	10^9/L	***.**	RDW-SD	fL	***.*
Bas#	10^9/L	***.**	RET#	10^12/L	*.****
IMG#	10^9/L	***.**	RET%	%	***.**
Neu%		*.***	IRF	%	***.*
Lym%		*.***	LFR	%	***.*
Mon%		*.***	MFR	%	***.*
Eos%		*.***	HFR	%	***.*
Bas%		*.***	RHE	pg	***.*
IMG%		*.***	NRBC#	10^9/L	****.***

4. The available units for the parameter appear on the right side of the screen. Select the desired unit, and the content in the corresponding parameter unit column will update accordingly.

Figure4–6 Unit Selection Screen



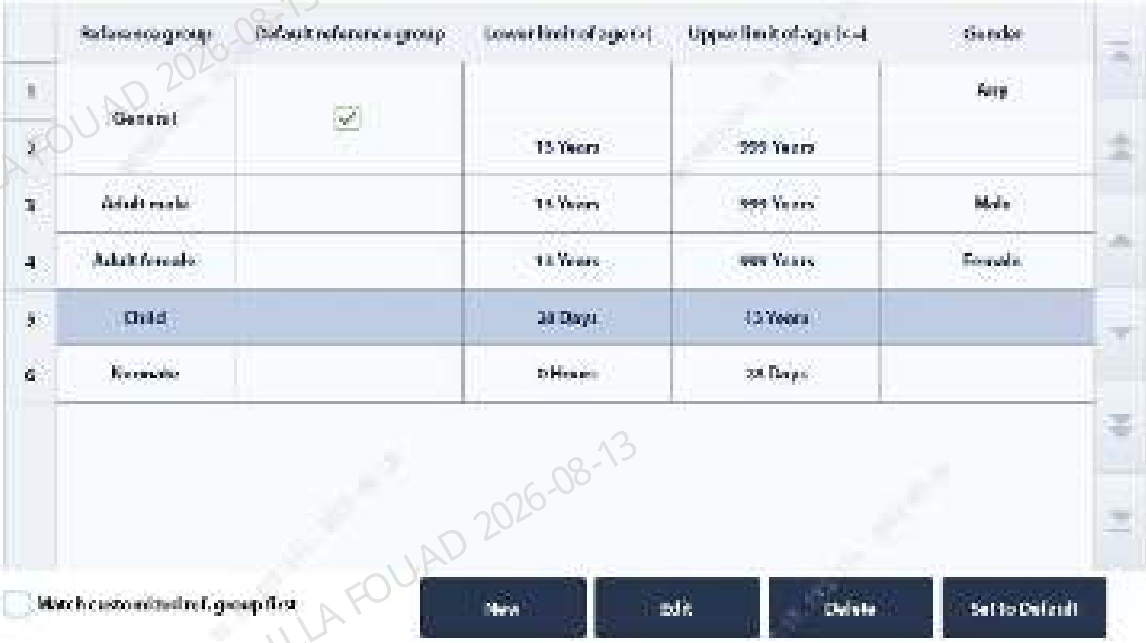
5. A prompt of saving changes appears when the user intends to quit the screen

Set Reference Range

There are five factory reference groups and 10 custom reference groups on the **Reference Range** screen. A user with the permissions of administrator level is allowed to select and set the reference range and reference group.

1. Click **Menu>>Setup>>Reference Range Setup** to access the **Reference Range Setup** screen.

Figure4–7 Reference Range Setup Screen



2. Add a new reference group, edit or delete reference groups, or set default groups as needed. Click the **Edit** button to set the upper and lower limits of parameters, as shown in the figure below:

Figure4–8 Upper/Lower Limit Setup Screen

Set Microscopic Parameters

The **Microscopic Parameter Setup** screen displays 22 microscopic parameters of blood cell analysis by default.

A user with the permissions of administrator level is allowed to add new parameters, edit or delete existing microscopic parameters on the "Microscopic Para. Setup" screen.

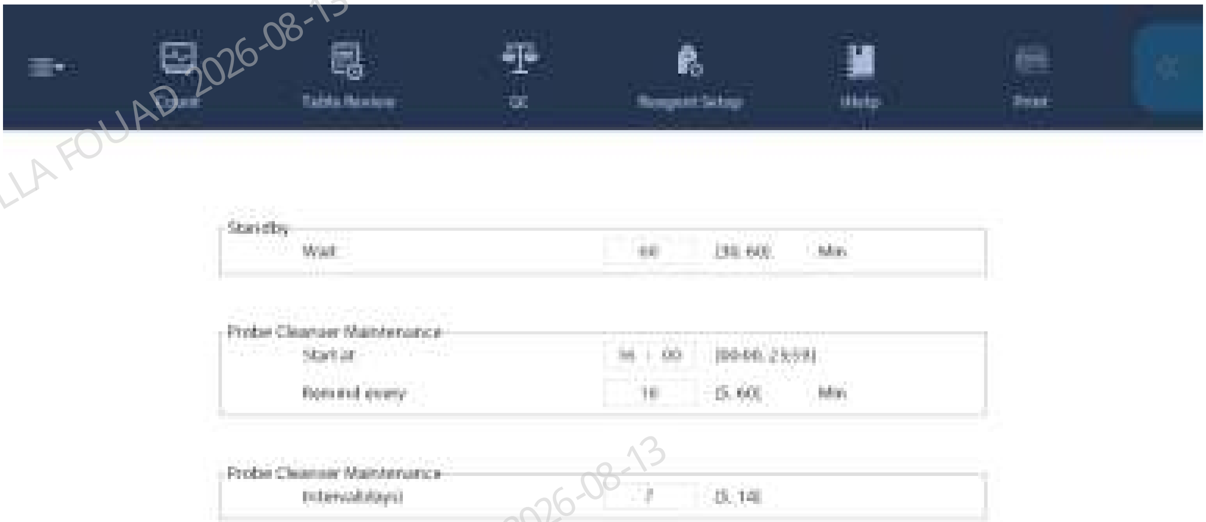
1. Click **Menu>>Setup>>Para. Setup>>Microscopic Para. Setup** to access the **Microscopic Para. Setup** screen.
2. Add new microscopic parameters, or edit or delete existing microscopic parameters as needed.

Figure4–9 Microscopic Para. Setup Screen

Maintenance

1. Click **Menu>>Setup>>Maintenance** to access the "Setup" screen, where it is allowed to.
2. Set the wait time before the analyzer entering standby.
3. Set the scheduled probe cleanser time.
4. Interval days (How long to call different probe fluid maintenance sequences after setup. For example, the setting in below graph requires an interval of 7 days, the more complicated Probe Cleanser procedure will be performed during analyzer shutdown or during auto maintenance)

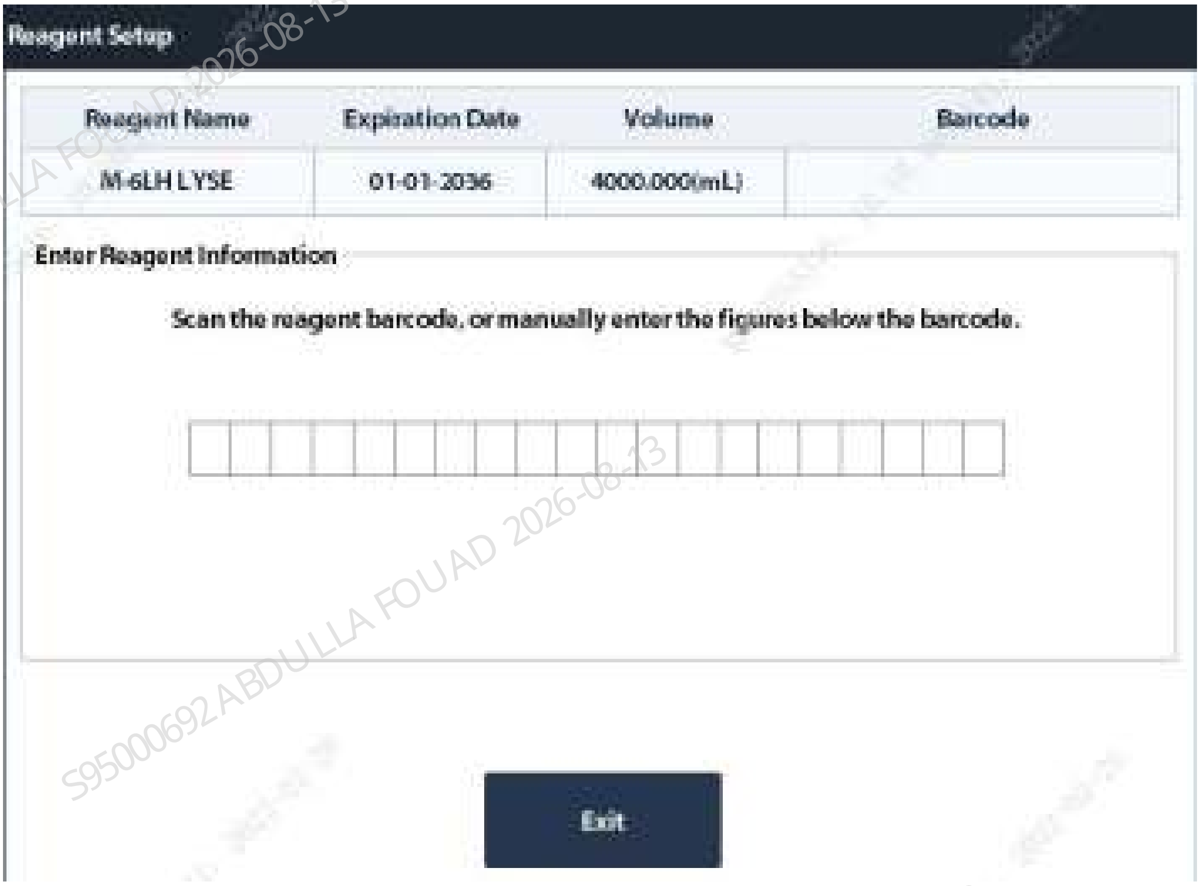
Figure4–10 Maintenance Setup Screen



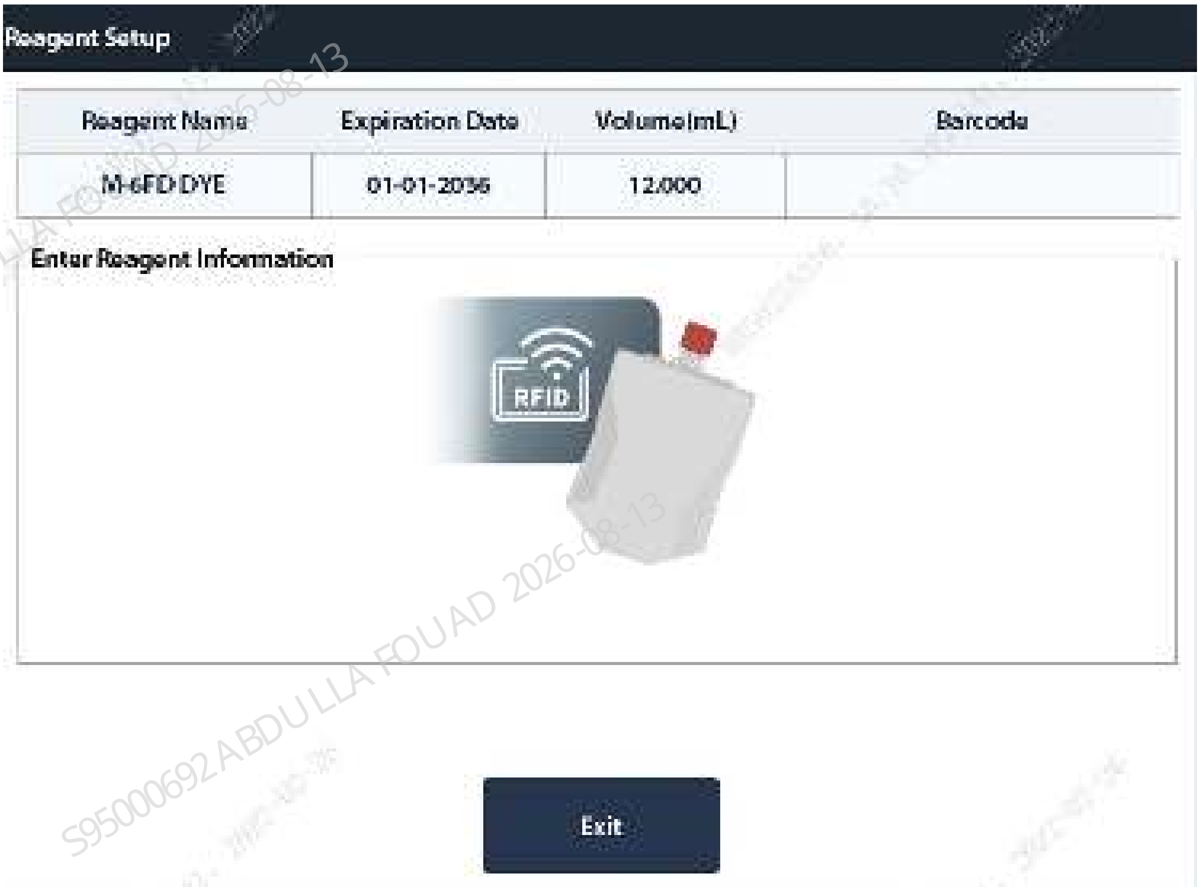
Reagent Setup

1. In case of no reagent related flags, check the reagent(s) to be replaced.
2. Scan or manually enter the barcode.
3. If the barcode is correct, the software will report the error indicating that the reagent is not replaced, and remind the user of reagent replacement.
4. After completing all reagent settings, click **Replace Now** button (**Automatic Replacement After Enabling Reagent Settings** disabled), or automatically replace reagent after counting down (**Automatic Replacement After Enabling Reagent Settings** enabled).

Figure4–11 Reagent Setup Screen



5. The dye and ESR cleanser replacement screen of the products for global market is as below.


Reagent Setup
Reagent Name
Expiration Date
Volume(mL)
Barcode
M-6FD DYE
01-01-2036
12.000
Enter Reagent Information
RFID
Exit


Reagent Setup
Reagent Name
Expiration Date
Volume(mL)
Barcode
ESR Solution Reagent
01-01-2036
999.989
Enter Reagent Information
RFID
ESR
Exit

Comm. Setup

Figure4–12 Comm. Setup Screen

- **Data channel**
  LIS or labXpert is optional.
  Please note that the QC files cannot be set on the instrument if labXpert is selected, in which case, only the QC files on labXpert can be used.
- **Protocol Setup**
  IP address: The IP for the analyzer, which is 10.0.0.12 by default.
  Subnet mask: Generally 255.255.255.0 for most scenarios of the analyzer.
  Default Gateway: Gateway IP address
  MAC address: The MAC address of the analyzer is fixed by manufacturer and cannot be changed.
- **Communication Protocol:**
  Click the pull down list to select a suitable communication protocol.
- **Synchronous Transmission:**
  Check the **ACK synchronous transmission** check box to enable the function.
  When the function is enabled, ACK overtime is **10** seconds by default.

NOTICE

- **The IP address for the analyzer is assigned statically. Ask your network administrator for the IP address to avoid IP conflict.**
- **If the communication involves more than one subnet, ask your network administrator for correct subnet mask and gateway.**

Special Information

Click **Table Review**, **Graph Review** and then **Special Information**, the window below will pop up:

Figure4–13 Special Information Screen

Table 4–2 Meanings of Special Information

HGB blank voltage	The voltage when diluent is added to the HGB reaction bath
HGB measurement voltage	The voltage when blood sample and lyses are added to the HGB reaction bath
DIFF channel particle number	Total particle count at the DIFF channel
RET channel particle number	Total particle count at the RET channel
RBC particle number	Total RBC particle count measured at the impedance channel

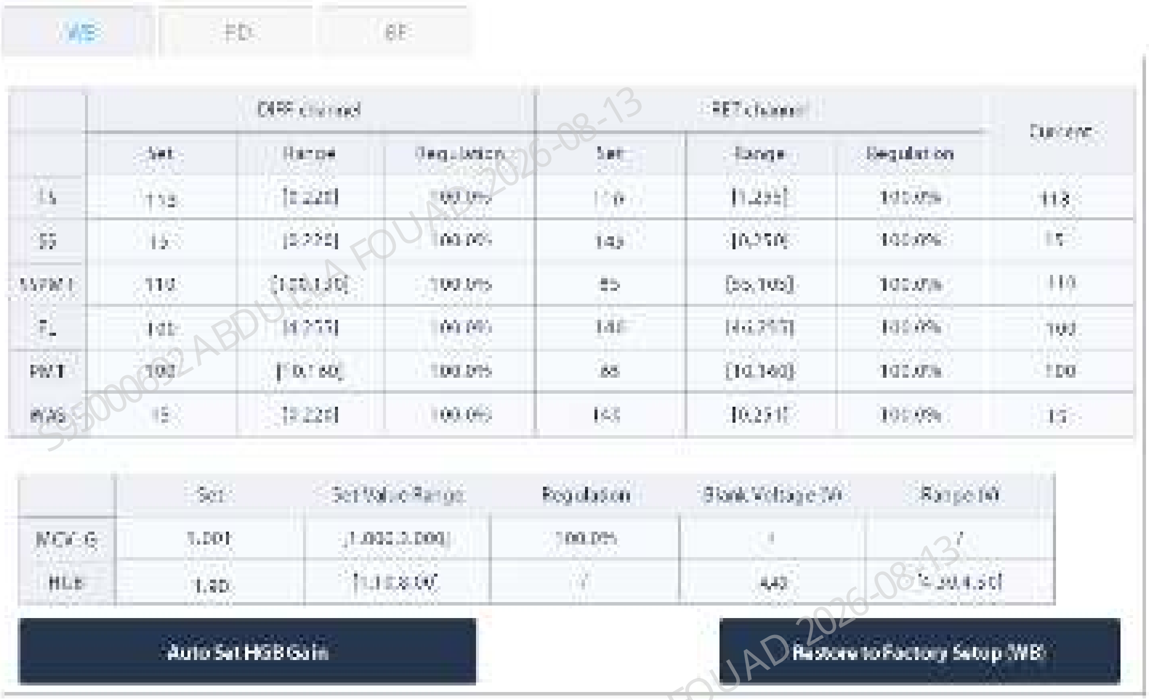
Table 4–2 Meanings of Special Information(continued)

PLT particle number	Total PLT particle count measured at the impedance channel
Ambient temperature	Ambient temperature

Gain Setup

Click **Menu>Setup>Gain Setup** to access the setup screen:

Figure4–14 Gain setup screen



FS, SS, FL, PMT, WAS and SSPMT gains corresponding to DIFF and RET can be set under WB, predilution and body fluid modes. MCV_G, HGB gain set under WB mode is applicable to all modes.

When the analyzer reports the abnormal HGB background voltage, which cannot be eliminated by pressing the reset/mute key, the user may adjust the background voltage through **Auto Setting of HGB Gains**.

NOTICE

- **Carefully set the gains for the effectiveness of test results will be affected.**

Commissioning Setup

Mechanical position adjustment includes sample probe position adjustment and autoload position adjustment.

Adj. Sample Probe Pos

There will be certain system deviation during the production and assembly of sampling mechanism. Sample probe position adjustment is used to eliminate such deviation to guarantee that sample probe can travel to the accurate position.

- Steps: Inquire the forward, backward, upward and downward steps of current sample probe.
- Position observation: Move the sample probe to the observation position by clicking the "Observation" button.
- Position adjustment: Click the "Help" button on the screen to learn how to adjust.

Adjust Autoload Pos.

There will be certain system deviation during the production and assembly of autoloading mechanism. Autoloading position adjustment is used to eliminate such deviation to guarantee that the loading mechanism can travel to the accurate position.

Built-in Barcode Setup

The following barcode types are supported:

CODE128

CODE39

ITF

CODE93

CODABAR

UPC/EAN/JAN

Before setting the built-in barcode system, please confirm: log in as the administrator account.

1. Click **Menu> Setup>System Setup> Built-in Barcode Setup** to access the **Built-in Barcode Setup** screen.
2. Tap the desired code type to go to the corresponding setup screen.

Figure4–15 Built-in Barcode Setup Screen

3. Check the **Apply** box to apply the code type.
4. Select the number of digits used on site in the **Digits** area.
5. (Optional) If you are using CODE39, ITF, and CODABAR with check bit, check the **Check bit** box.
6. (Optional) Set other code types if needed.

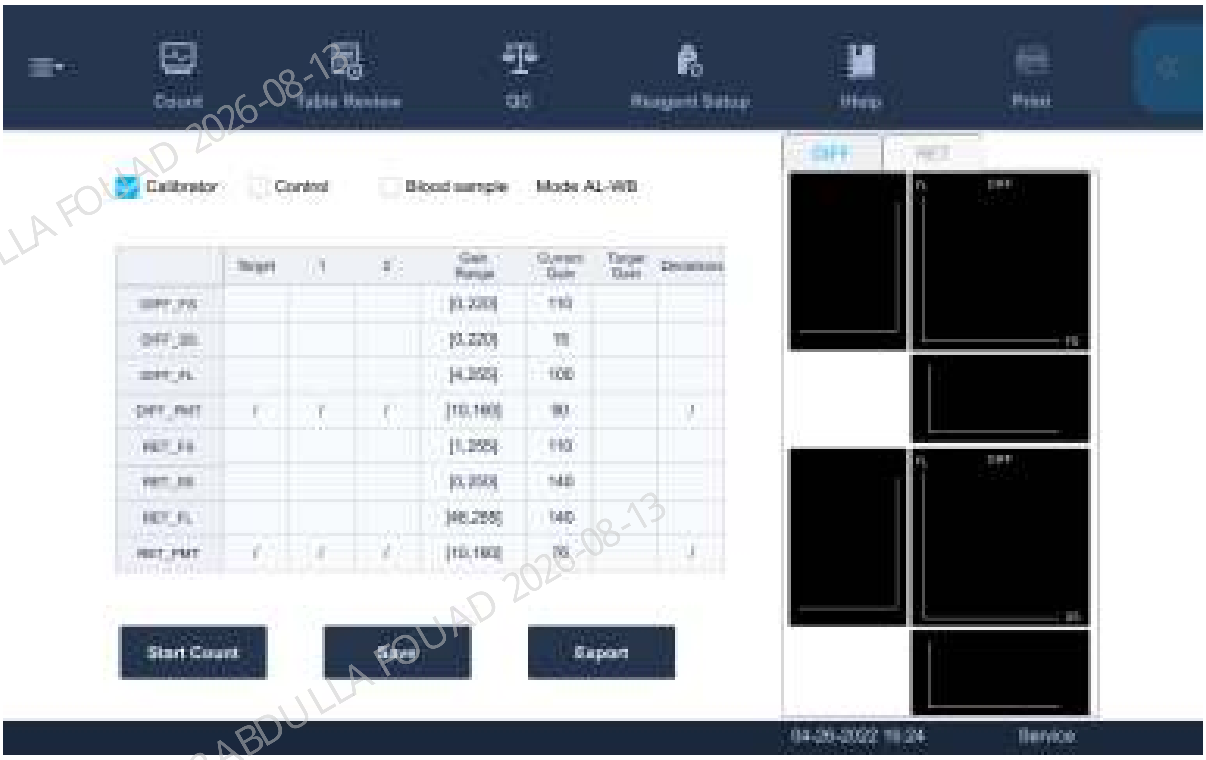
Advanced

Calibration - CBC Gain Calibration

There is deviation among the detection sensors of CBC channels. Service engineers shall eliminate the deviation thorough gain adjustment to achieve the satisfying consistency. The screen is shown below:

Calibration - Optical Gain Calibration

There is deviation among the detection sensors of optical channels. Service engineers shall eliminate the deviation thorough gain adjustment to achieve the satisfying consistency. Refer to the optical gain commissioning section for the methods, steps, acceptance criteria and precautions of gain commissioning. The screen is shown below:



4.6 Service

Maintenance

The analyzer supports various maintenance functions for easy maintenance to guarantee the accurate and smooth performance of the analyzer.

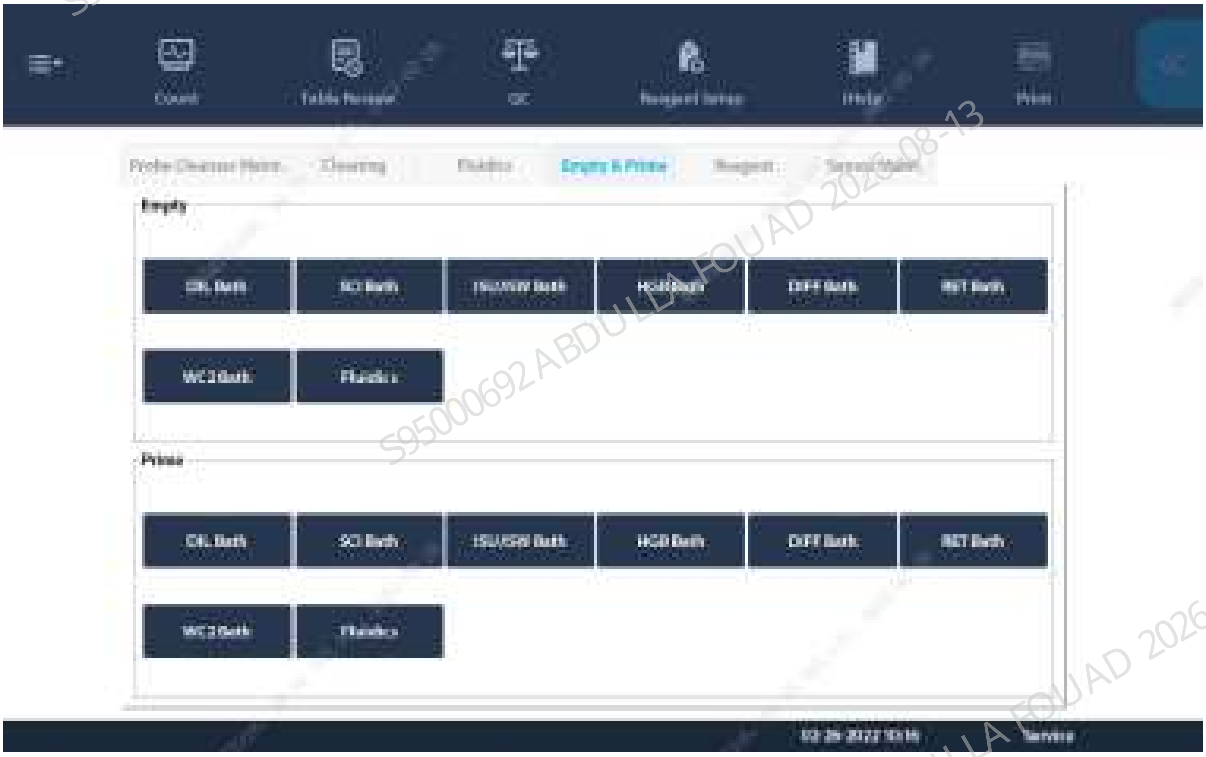
Access the **Service**>**>Maintenance** screen, and the available maintenance functions include:

- Probe cleanser maintenance

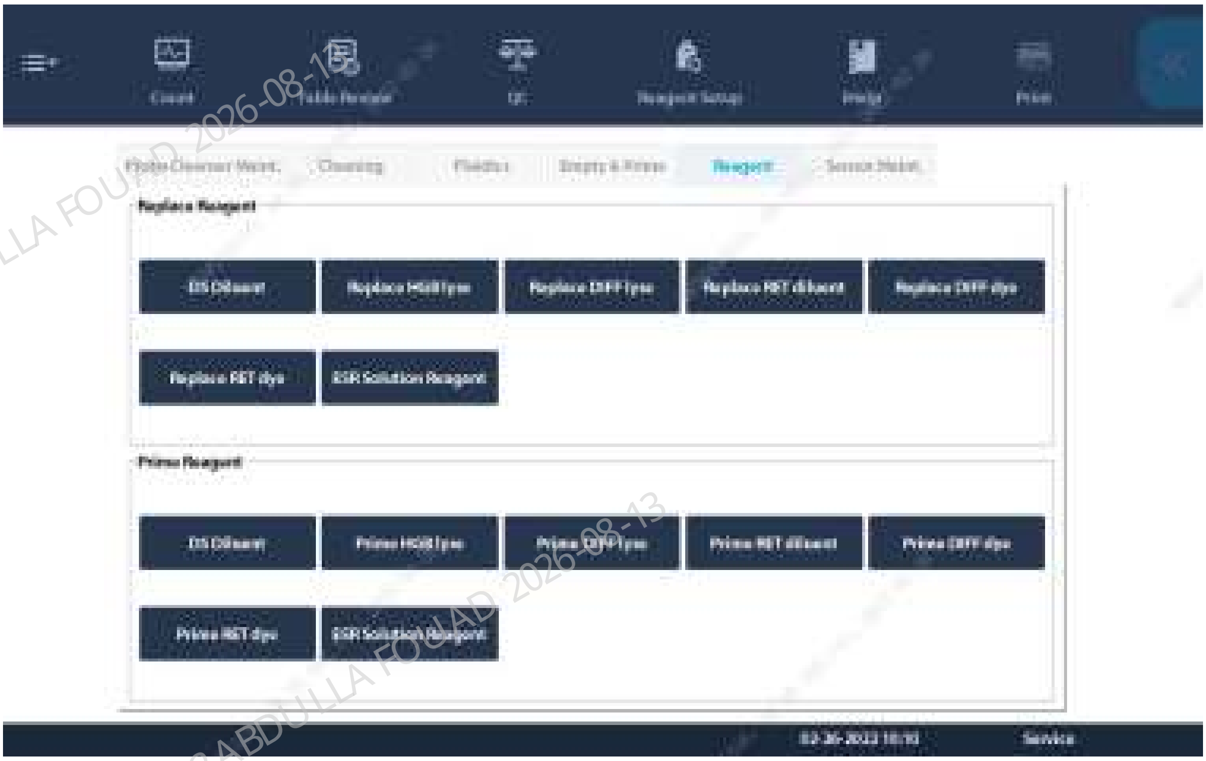
- Clean

- Fluidics

- Reagent draining and priming



- Reagent replacement and priming



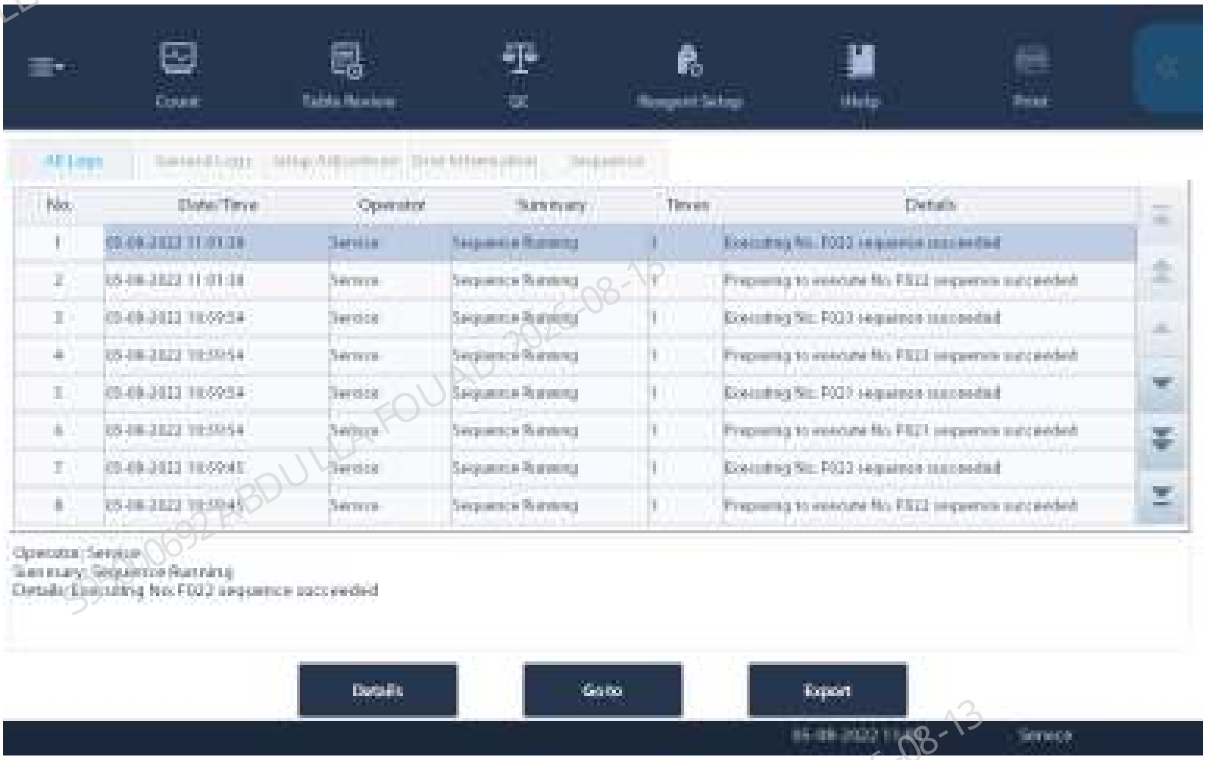
- Sensor maintenance

Logs

Logs record all activities of the analyzer. It contributes significantly to searching for operation history and troubleshooting the analyzer. The analyzer can store logs of the recent two years. If number of logs exceeds the upper limit, the latest log will overwrite the oldest one.

Click **Menu>>Service>>Log** to access the "Log" screen:

Figure4–16 Log Screen



The categories include All Logs, General Logs, Parameter Modification, Error Information, Sequential Operation.

Click **Go to** to review the logs at specified date range.

Click **Export** to export the logs in specified time range to the USB device.

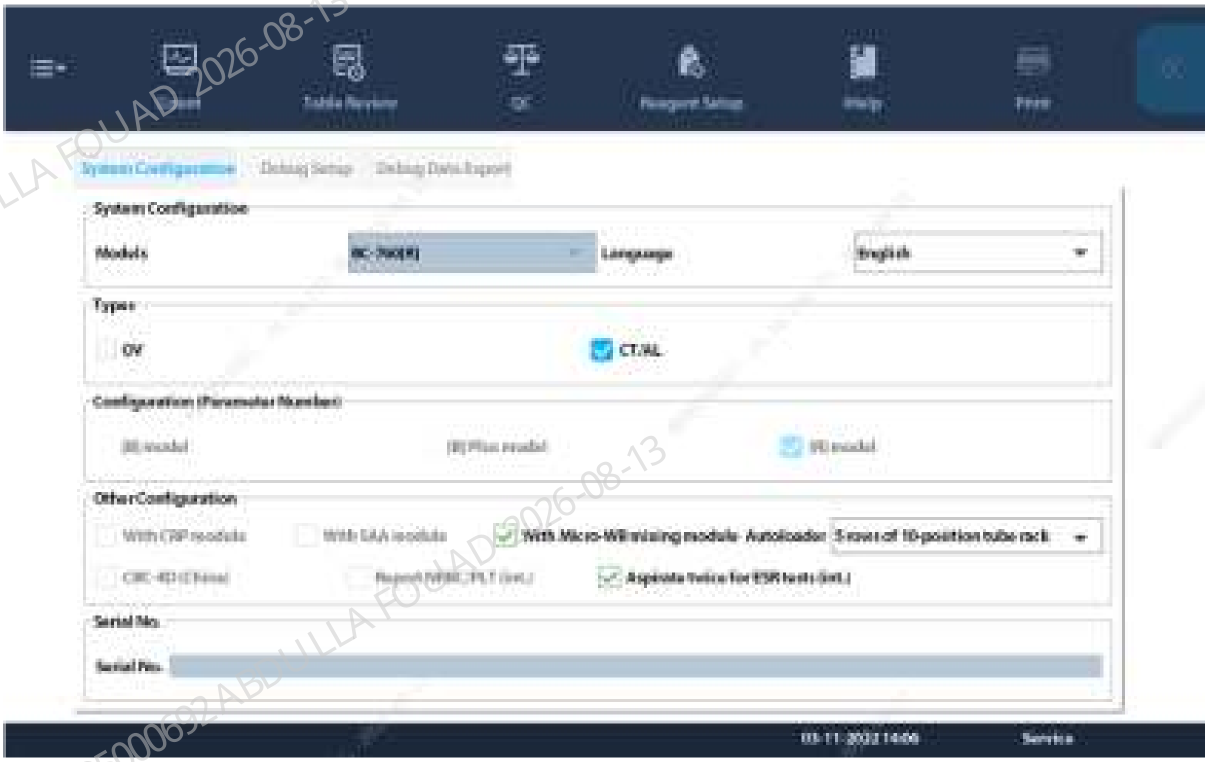
NOTICE

- **The USB should have been formatted to FAT32;**
- **Recommended USB models: Kingston 8/16 G, SanDisk 8/16 G, Maxell 4/8 G.**
- **Make sure there is enough free space on the USB flash drive. (USB 2.0 or USB 3.0).**

Advanced Toolbox

Click **Menu>>Service>>Advanced Toolbox** to access the "Advance Toolbox" screen:

Figure4–17 Advanced Toolbox Setup Screen



The following settings are supported:

- **System Configuration**

  The system configuration varies with the machine type.

- **Commissioning Setup**

  Data erase supported. The screen is shown below:

Exporting Commissioning Data

In consideration that the export function is available for each module, the one-click export function is equipped for easily exporting all instrument data desired. The screen is shown in the figure:

Exported item	Exported *content*
Instrument Info.	Instrument related information
Software commissioning information	Software related log
inf file	Sample inf file, waveform file
Sample data	Sample data table review

NOTE

In case of any error, the Instrument Information and Software Commissioning Information shall be exported for analyzing the root cause.

Commissioning & Self-Test

Adj. Sample Probe Pos

Adjustable items: vertical initial position, reaction bath initial position, DIFF bath position, closed whole blood position, automatic sampling position, specific protein bath position, and HGB bath position.

Adjust Autoload Pos.

Adjustable items: gripper (front), gripper (middle), right feeding position, left feeding position.

Self-Test

Self-test items: valve, fluidics commissioning, autoload assembly, gripper and other components. The screen is shown below:

Heating Commissioning

The curve data of five channels (i. e. preheating, DIFF, environment, ESR and instrument) is recorded, with data export supported.
The screen is shown below:

Pressure Commissioning

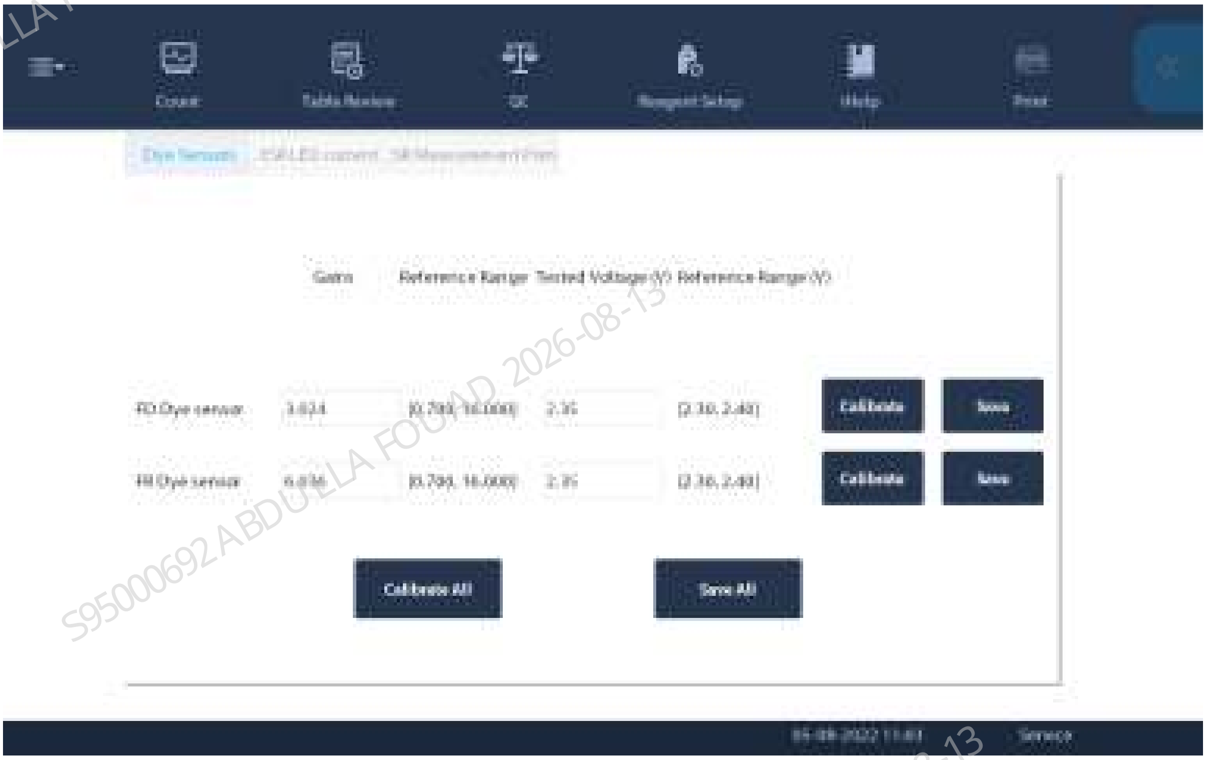
Pressure Commissioning supported: +50kPa, +40kPa and -40kPa, with pressure data export and erase supported.

The screen is shown below:

Sensor Commissioning

- FD dye sensor calibration, FR dye sensor calibration and calibration data save supported.
- Click *Calibration*. If the test voltage is within the reference range, click *Save*.

The screen is shown below:



Optical Commissioning

Click **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Optical Commissioning** to access the "Optical Commissioning" screen: optical assembly commissioning supported and graphic demonstration of measurement data supported.

The screen is shown below:

4.7 Indicators

The indicator on the front cover of the analyzer may be on in 3 colors and supports a blinking frequency of 2 seconds. The indicator color changes with the instrument status, as shown in the table below:

Table 4–3 Analyzer Status Indicator

Analyzer status	Indicators	Remarks
Ready	Normally on in green	Waiting for sequential operation
Running Status	Blinking in green	Performing sequential operation
Running with error	Blinking in red	Running with error
Error and not running	Normally on in red	Any error occurs and the analyzer is not running
No error, but fluidic actions are not allowed	Normally on in yellow	Startup initialization or standby, involving no fluidic actions
Enter/exit standby	Blinking in yellow	Enter/exit standby

4.8 Software Update

Preparation Before Upgrade

Copy the update.tar.gz file to the root directory of the formatted USB.

NOTICE

- **The USB should have been formatted to FAT32;**
- **Only the update.tar.gz file is in the root directory of the USB. You must not modify the file name. You must not unzip the file.**

Upgrade Procedure

1. Insert the USB to one of the USB ports on the analyzer to start the update.
2. **Version>>Details>>Upgrade**:

Figure4–18 Version Info.: Upgrade

Upgrade includes two steps.

- Step 1: Update the guide and operation system.
- Step 2: Upgrade the software module.

If the upgrade guide and the operating system are also needed to be upgraded, the system will prompt you to restart the analyzer between step 1 and step 2; if only the software module is to be upgraded, the upgrade will start from step 2 directly.

CAUTION

- **Do not unplug the USB or disconnect power during the update; otherwise the analyzer may not start.**

NOTICE

- **The update usually takes about 10 minutes but depends on the number of modules to be updated. Do not leave the analyzer as the process requires user operation.**

Upgrade Validation

Restart the device after the upgrade is completed. Click **Status>>Version Info**, and check whether the version number is consistent with the upgrade version number. If yes, the upgrade is successful.

Fixing Problems in Upgrade

If the update fails, try again.

5 Commissioning

5.1 Adjusting Mechanical Positions

Table 5–1 Adjustment item correspondence table

Assembly to be replaced/ disassembled	Positions to be adjusted/ checked	Index sections (in order of adjustment)
Sampling assembly	Sampling probe position	Refer to the sampling assembly **8.13.2** Disassembly and Assembly
Reaction bath assembly	Sampling probe position	Refer to the reaction bath assembly **8.36.2** Disassembly and Assembly
HGB assembly	Sampling probe position	Refer to the HGB bath assembly **8.41.2** Disassembly and Assembly
Rotary Compressing Assembly	Position of rotary compressing assembly	Refer to the rotary compressing assembly **8.25.2** Disassembly and Assembly
Leuze fixed barcode scanner CCD 700 scan/s	Scanner position	Refer to the scanner **8.26.2** Disassembly and Assembly
Tube Clamp	Tube clamp position	Refer to the tube clamp **8.24.2** Disassembly and Assembly
Blending Assembly	Positions of tube clamp and blending assembly	Refer to the blending assembly **8.11.2** Disassembly and Assembly
Peripheral Blood Blending Assembly	Position of peripheral blood blending assembly	Refer to the rotary compressing assembly **8.28.2** Disassembly and Assembly
Hold Assembly	Position of stabilizing assembly	Refer to the stabilizing assembly **8.23.2** Disassembly and Assembly
10X5 Closed-sampling Autoloader	10X5 Closed-sampling Autoloader Position	Refer to the rotary compressing assembly **8.35.2** Disassembly and Assembly
5X6 Closed-sampling Autoloader	5X6 Closed-sampling Autoloader Position	Refer to the rotary compressing assembly **8.28.2** Disassembly and Assembly

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
1	115-061264-00	0.5mL diaphragm pump (same direction as EPDM connector)	Confirm that the dosing pump can work normally after the replacement, that is, the reagent can be replaced normally		Refer to 0.5 mL dosing pump (same direction as EPDM connector) **8.52.3** Commissioning and Verification
2	115-011901-00	10 mL lead screw driving syringe assembly	Self-Test		Refer to 10 mL lead screw driving syringe assembly **8.48.3** Commissioning and Verification
3	0033-30-74615	Mindray 10 mL syringe	Self-Test		Refer to Mindray 10 mL syringe **8.49.3** Commissioning and Verification
4	115-005240-00	20 µL Dosing Pump Assembly	Confirm that the dosing pump can work normally		Refer to 20 µL dosing pump assembly **8.66.3** Commissioning and Verification
5	115-071527-00	250 µL syringe assembly (43F4J)	Self-Test		Refer to 250 µL syringe assembly (43F4J) **8.50.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
6	115-027479-00	Mindray 250 μL syringe (with connector)	Self-Test		Refer to Mindray 250 μL syringe (with connector) 8.51.3 Commissioning and Verification
7	051-003263-00	9202 Switch Key Board PCBA	Check that the key board can work normally after replacement		Refer to 9202 switch key board PCBA 8.17.3 Commissioning and Verification
8	115-071234-00	HGB bath assembly	1. Adjust the position of the sampling probe relative to the HGB reaction bath. 2. Adjust the HGB gain.		Refer to the HGB bath assembly 8.41.3 Commissioning and Verification
9	115-081655-00	KNF Liquid Pump, Dia phragm	Self-Test		Refer to KNF liquid pump, diaphragm pump, 12VDC 8.59.3 Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
		Pump, 12VDC			
10	115-061171-00	LVM fluid valve body	1. Perform a blank count to determine if there is a leak. 2. Use the calibrator/ quality control test to confirm that the DIFF/RET scatter-gram of the corre-sponding channel and its measured value are normal.		Refer to LVM fluid valve body
11	011-000021-00	PHOTO ELEC Optical Sensor IP55	1. Check whether the sensor is normal.		Refer to PHOTOELEC Optical Sensor IP55 940nm 15cm **8.19.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
		940nm 15cm	Perform the self-test.		
12	115-061156-00	SCI bath assembly	1. Prime the fluidics and check the corresponding floater status.		Refer to the SCI bath assembly **8.46.3** Commissioning and Verification
13	115-047895-00	WC2 cistern assembly	1. Prime and empty the WC2 cistern and check the corresponding floater status		Refer to WC2 cistern assembly **8.43.3** Commissioning and Verification
14	115-071221-00	Sampling manifold assembly	1. Perform a blank count to determine if there is a leak. 2. Use the calibrator/		Refer to sampling manifold assembly **8.53.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			quality control test to confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.		
15	115-081839-00	Sampling assembly	1. Adjust the relative relationship between the vertical initial position of the sampling probe, the reaction bath position,		Refer to the sampling assembly **8.13.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			and the HGB bath position. 2. Perform commissioning on the ESR background voltage and measurement point position, and perform ESR calibration.		
16	115-081656-00	Air pump 12VDC	1. Check the pressure status.		Refer to air pump 12VDC **8.58.3** Commissioning and Verification
17	011-000247-00	Optical switch COJ-MY03-04 (Customized)	1. Dye liquor sensor gain adjustment		Refer to optical switch COJ-MY03-04 (Customized) **8.65.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
18	115-081652-00	Reaction bath assembly	1. Adjust the position of the sampling probe relative to the position of the reaction bath. 2. Confirm the temperature status. 3. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the		Refer to the reaction bath assembly **8.36.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			corresponding channel and its measured value are normal.		
19	041-023727-00	Probe Wash Device 2	1. Adjust the vertical home position of sampling probe. 2. Clean the sampling probe.		Refer to probe wipe 2 (closed-sampling) **8.21.3** Commissioning and Verification
20	3003-20-34949	Isolation chamber	1. Check the pressure status. 2. Waste pump self-test.		Refer to the isolation chamber **8.47.3** Commissioning and Verification
21	011-000297-00	Photoelectric Position Sensor	1. Self-Test		See FRU-Photoelectric sensor (10 mL lead screw driving syringe assembly), FRU-Photoelectric sensor (250 μL syringe assembly) FRU-Photoelectric position sensor **8.20.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
22	115-081654-00	Optical system	1. Perform optical gain calibration. 2. Perform the calibrator calibration.		Refer to the optical system **8.42.3** Commissioning and Verification
23	115-011660-00	Filter assembly (injection molding)	1. Perform overall cleaning.		Refer to the filter assembly (injection molding) **8.62.3** Commissioning and Verification
24	115-078179-00	Back sheath isolation bath assembly (3120)	Prime and empty the ISU/ISW bath to confirm that it is as expected.		Refer to the back sheath isolation bath assembly (3120) **8.39.3** Commissioning and Verification
25	115-069029-00	Piercing Needle	1. Perform commissioning on the vertical home position of		Refer to the piercing needle **8.22.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			sampling probe. 2. Perform commissioning on the position of ESR measurement point. 3. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured		

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			value are normal.		
26	115-071230-00	Two-way LVM valve assembly (10R6 with valve board, 3120)	1. Self-Test 2. Overall cleaning		Refer to the two-way LVM valve assembly (10R6 with valve board, 3120) **8.57.3** Commissioning and Verification
27	115-061170-00	Two-position ways LVM valve assembly (with valve board and connec tor)	1. Self-Test 2. Overall cleaning		Refer to the two-way LVM valve assembly (with valve board and connector)
28	115-061170-00	Two-position ways NC LVM valve body	1. Check whether there is any leakage after the replacement is		Refer to the two-way NC LVM valve body **8.55.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			completed.		
29	115-033289-00	Two-position ways valve (PEIZH)	1. Self-Test 2. Overall cleaning		Refer to the two-way valve (PEIZH) **8.5.3** Commissioning and Verification
30	115-046257-00	Pressure-proof three-position ways valve	1. Self-Test 2. Overall cleaning		
31	115-071241-00	Air tank assembly	1. Pressure status detection		Refer to the air tank assembly **8.45.3** Commissioning and Verification
32	115-071243-00	Sheath fluid count bath assembly	1. Perform overall cleaning 2. Calibrate CBC gain		Refer to the sheath fluid count bath assembly **8.38.3** Commissioning and Verification
33	115-081653-00	Sheath fluid impe dance assembly	1. Perform overall cleaning 2. Calibrate CBC gain		Refer to the sheath fluid impedance assembly **8.37.3** Commissioning and Verification
34	115-010567-00	Sheath fluid filter assembly (3201)	1. Perform a back-ground test to		Refer to the sheath fluid filter assembly (3201) **8.60.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			check for leaks		
35	115-081657-00	Fully-fitted Screen Assembly	1. Screen commissioning 2. Indicator board commissioning		Refer to the fully-fitted screen assembly 8.18.3 Commissioning and Verification
36	115-061171-00	Three-position ways LVM valve assembly (with valve board and connector)	1. Self-Test 2. Overall cleaning		Refer to the three-way LVM valve assembly (with valve board and connector) 8.54.3 Commissioning and Verification
37	115-069699-00	Diluent heating assembly	1. Check that the diluent can fill the heating bath		Refer to the diluent heating assembly 8.44.3 Commissioning and Verification
38	115-072035-00	ESR Measurement Assembly	1. Perform commissioning on the	/	Refer to the ESR measurement assembly 8.12.3 Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
			ESR LED current. 2. Perform commissioning on the position of ESR measurement point.		
39	115-033322-00	Fluid pressure detection assembly (3109)	1. View th fluid pressure status		Refer to the fluid pressure detection assembly (3109) **8.68.3** Commissioning and Verification
40	115-082029-00	Trumpet Assembly	1. Self-Test		Refer to the trumpet assembly **8.16.3** Commissioning and Verification
41	115-069363-00	Hold Assembly	1. Position Adjustment		Refer to the stabilizing assembly **8.23.3** Commissioning and Verification
42	115-052164-00	Tube Clamp	1. Position Adjustment		Refer to the tube clamp **8.24.3** Commissioning and Verification
43	115-052166-00	Blending Assembly	1. Position Adjustment		Refer to the blending assembly **8.11.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
44	115-075024-00	Rotary Com pressing Assembly	1. Position Adjustment		Refer to the rotary compressing assembly **8.25.3** Commissioning and Verification
45	115-085180-00	10X5 Closed-sampling Autoloader	1. Position Adjustment		Refer to 10X5 closed-sampling autoloader **8.35.3** Commissioning and Verification
46	115-085181-00	5X6 Closed-sampling Autoloader	1. Position Adjustment		Refer to 5X6 closed-sampling autoloader **8.28.3** Commissioning and Verification
47	115-059469-00	Peripheral Blood Blending Assembly	1. Position Adjustment		Refer to peripheral blood blending assembly **8.28.3** Commissioning and Verification

No.	FRU Code	FRU Name	Commissioning item_1 after replacement	Commissioning item_2 after replacement	Index sections (in order of adjustment)
48	115-081840-00	Fluorescence Dye Pump Assembly (Without RET)	1. Confirmation of sensor voltage		Refer to dye pump assembly (without RET) **8.67.3** Commissioning and Verification
49	115-074579-00	Fluorescence Dye Pump Assembly (With RET)	1. Confirmation of sensor voltage		Refer to dye pump assembly (with RET) **8.64.3** Commissioning and Verification

5.2 Adjusting Sampling Probe Positions

Refer to the **8.13.3** Commissioning and Verification of sampling assembly

5.3 Adjusting Rotary Compressing Positions

Refer to the **8.25.3** Commissioning and Verification of rotary compressing assembly

5.4 Adjusting Blending Assembly Positions

See **8.11.3** Commissioning and Verification for the blending assembly.

5.5 10X5 Adjusting Closed-sampling Autoloader Positions

See **8.35.3** Commissioning and Verification for autoloader, closed-sampling model (10 x 5).

5.6 5X6 Adjusting Closed-sampling Autoloader Positions

See **8.28.3** Commissioning and Verification for autoloader, closed-sampling model (5 x 6).

5.7 Hold Assembly

See **8.23.3** Commissioning and Verification for the hold assembly.

5.8 Optical Gain Commissioning

Generally, no manual modification is required for PMT gain. If the target gain of FL channel is out of specification, however, it is required to manually modify the PMT gain and restart counting, as shown in Figure 18.

The optical gain commissioning steps are as below:

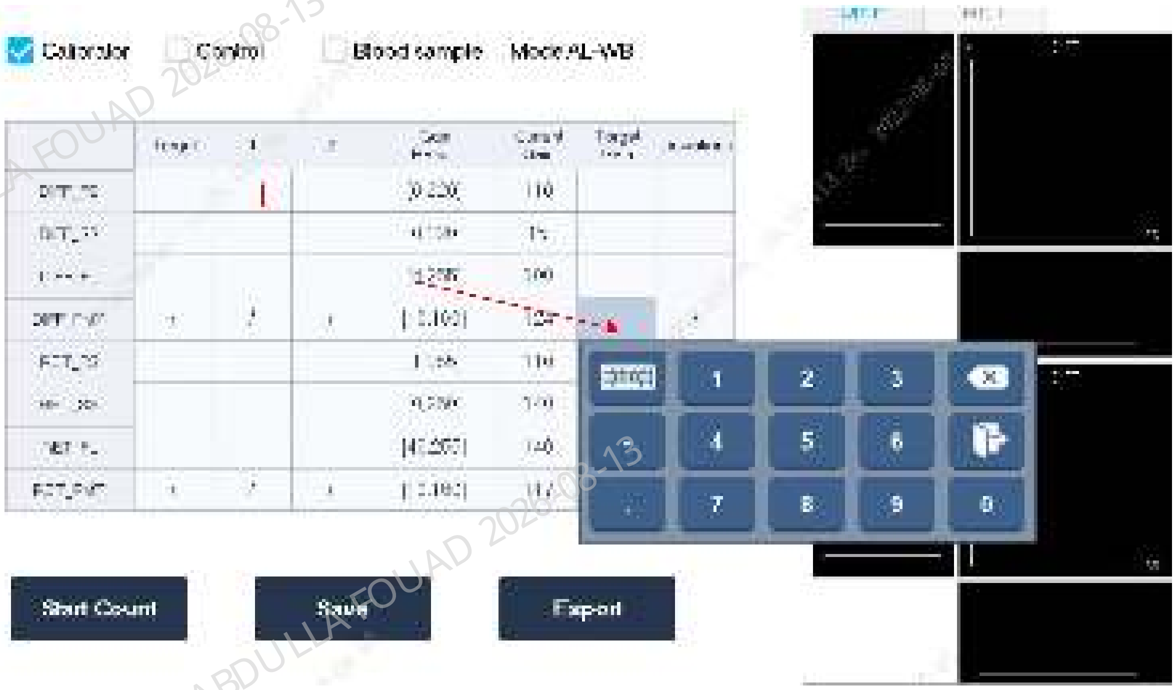
1. Click **Menu>>Calibration>>Optical Gain Calibration** to access the Optical Gain Calibration screen (as shown in the Figure below). Select **Calibrator** to enter the optical target of calibrator in the target column.

Figure5–1 Optical Gain Calibration screen

A Optical target of calibrator B Current gain deviation

2. Place a calibrator at the position 1# of the tube rack, and tap **Start Count**. After measurement, a prompt of saving current counts appears. In case of normal counting, click **Yes** and continue the following automatic counts.
3. After continuous measurement, if the deviation column satisfies ≤2.0%, click **Save** and **Export**.
4. Click **Menu>>Setup>>Gain Setup** to check whether the **Set Value** of optical gain has been updated to the one saved in Step 3.

Figure5–2 PMT Modification



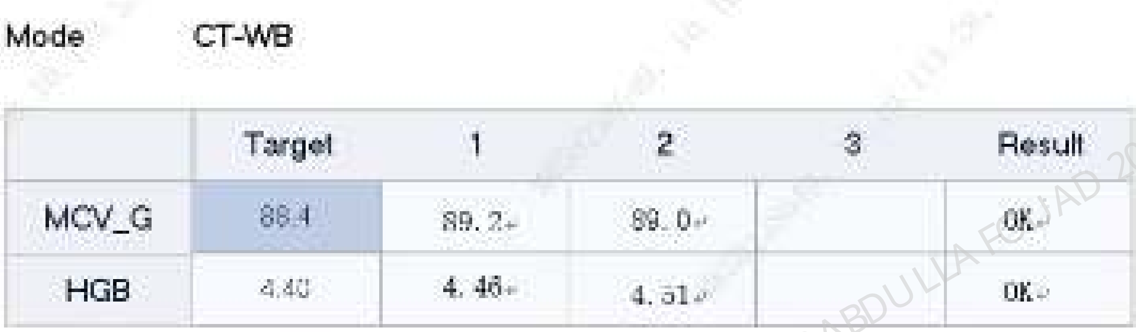
1 In the Target Gain column, the gain value of PMT channel can be changed manually. When the gain of FL channel exceeds the range, the PMT gain value can be adjusted manually (within the gain range).

5.9 Impedance Gain Commissioning

- Materials to be prepared
  One calibrator
- Commissioning procedures:

1. Tap the **Menu**>**>Calibrate**>**>CBC Gain Calibration** to enter the CBC Gain Calibration screen. Enter the MCV_G target value for the batch of calibrator used.

Figure5–3 CBC Gain Calibration screen


Mode CT-WB

	Target	1	2	3	Result
MCV_G	88.4	89.2	89.0		OK
HGB	4.40	4.46	4.51		OK

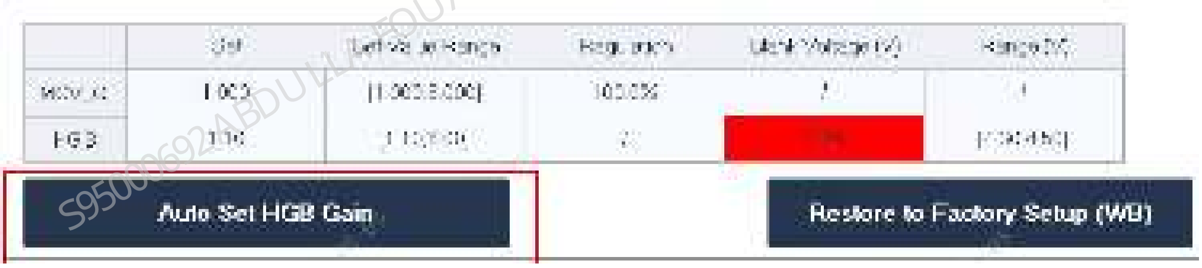
2. Perform three times of counting continuously until OK is displayed in the result columns of both MCV_G and HGB.
3. When exiting the CBC Gain screen, click **Yes** on the pop-up dialog box, save the gain and complete the commissioning process.

5.10 HGB Gain Commissioning

You can perform the HGB gain commissioning in two methods.

- Method 1
  Refer to the impedance gain commissioning above. If OK is displayed in the HGB column, the commissioning is complete.
- Method 2
  Click **Menu**>>**Setup**>>**Gain Setup**>>**WB** to enter the Gain Setup screen.

Figure5–4 Gain setup screen



In the HGB setting value column, click **Auto set HGB gain** until the value in the HGB background voltage column is within the range. When exiting the CBC Gain screen, click **Yes** on the pop-up dialog box, save the gain and complete the commissioning process.

5.11 Position Commissioning of Measurement Point

1. On the main unit software, click **Service**>>**Commissioning and Self-Test**>>**Sensor Commissioning**>>**Position of ESR Measurement Point**.

If the position of the measurement point is not within the range, click **Calibrate**, and the analyzer prompts that **the calibration is successful**.

6 Maintenance

6.1 Overview

6.1.1 Overview

The following maintenance content are the suggested maintenance items based on the product and for references. During onsite service, engineers can make adjustment based on the specific situation of the client.

6.2 Replace wearing parts

6.2.1 Tool List

No.	Tool Names	Quantity	Remarks
1	Cross-headed screwdriver	1	
2	Flat-headed screwdriver	1	
3	M2.5 Allen wrench	1	
4	Diagonal pliers	1	
5	Tweezers	1	

6.2.2 Wearing Parts Replacement List

Maintenance Object	Frequency	Maintenance Items	Applicable model	Remarks
Piercing needle	120,000 times	Replacement	BC-760[B] BC-760[R] BC-760[B]CS BC-760[R]CS	Execute the wearing parts replacement validation procedure [item 1] after replacement.
Filter (DIFF waste channel)	3 years/60000 times	Replacement	BC-760[B] BC-760[R] BC-760[B]CS BC-760[R]CS	\

Maintenance Object	Frequency	Maintenance Items	Applicable model	Remarks
Filter (HGB waste channel)	3 years/60000 times	Replacement	BC-760[B] BC-760[R] BC-760[B]CS BC-760[R]CS	\
Filter (RET waste channel)	3 years/60000 times	Replacement	BC-760[R] BC-760[R]CS	\
Filter (CRP waste channel)	3 years/60000 times	Replacement	BC-760[B]CS BC-760[R]CS	\
Filter (SAA waste channel)	3 years/60000 times	Replacement	BC-760[B]CS BC-760[R]CS	\
Probe wipe filter	3 years/60000 times	Replacement	BC-760[B] BC-760[R] BC-760[B]CS BC-760[R]CS	\

6.2.3 Validation After Replacement of Wearing Parts

Item 1	Maintenance	Procedure	Remarks
Replace the piercing probe	Verify the vertical home position of the sample probe in the reaction bath.	Choose **Service>Commissioning & Self-Test>Adj. Sample Probe Pos.**, adjust the vertical home position, HGB bath position, DIFF bath position, and reaction bath home position (RET position), validate the position of the piercing probe, and observe whether the sample probe in the vertical home position aligns with the swab at the bottom and the piercing probe is in the middle of the HGB bath, DIFF bath, and reaction bath home position (RET position). (For details, see Help.)	

6.3 Customer Maintenance Items

6.3.1 Customer Maintenance Items

Maintenance	Inspection/Maintenance Criteria	When
Pre-maintenance Preparations	Wearing: Wear gloves during the processing and wash hands with the cleanser or the disinfectant after the processing.	Before Maintenance
Probe cleanser maintenance	Execute probe cleanser maintenance for the system.	Alarm on abnormal measurement results
Sensor maintenance	Tap **Menu>Service>Maintenance>Sensor Maintenance** and execute **Unclog Aperture** and **Flow Cell Cleaning**.	Alarm on abnormal measurement results
Clean probe wipe	Clear the reagent crystals and stain on the sample probe wipe.	As needed

Maintenance	Inspection/Maintenance Criteria	When
Clean the autoloader.	Clean the autoloader desk panel and stain on the desk panel.	As needed
Clean the instrument cover.	Clear the stain on the analyzer cover.	As needed

6.3.2 Maintenance Report of Mindray Auto hematology analyzer

See maintenance report
on the next page.

Maintenance Report of Mindray Auto hematology analyzer

Customer Name		Device Model	
Analzyer SN		Maintenance Order No.	
Analyzer Status	□ Normal □ Faulty □ Stopped		
Maintenance Personnel		Maintenance Time	

No.	Maintenance	Inspection and Maintenance Criteria	Execution result
1	Probe cleanser maintenance	Execute probe cleanser maintenance for the system.	□ Done
2	Sensor maintenance (advanced toolbox)	Tap **Menu>Service>Maintenance>Sensor Maintenance** and execute **Unclog Aperture** and **Flow Cell Cleaning**.	□ Done
3	Clean probe wipe	Clear the reagent crystals and stain on the sample probe wipe.	□ Done
4	Clean the autoloader desk panel.	Clean the autoloader desk panel and sensor on the desk panel.	□ Done
5	Clean the cover.	Clear the stain on the analyzer cover.	□ Done
6	Piercing needle	120,000 times	□ Done
7	Filters	□ DIFF waste channel □ HGB waste channel □ Wipe filter □ RET waste channel □ CRP waste channel □ SAA waste channel	□ Done

Engineer description	Customer description

Signature	Signature
Time	Time

7 Update (Optional Modules)

7.1 Overview

The upgrade guide is a normative document used to guide engineers to upgrade the analyzer software functions and optional modules. The following content is prepared according to the upgrade project.

7.2 Sample Tube Type Detection Upgrade Scenario

7.2.1 Preparation Before Upgrade

Upgrade Time

The estimated upgrade time is 30 minutes

Upgrade Tool

Cross-headed screwdriver
Allen wrench
M1 slotted head screwdriver (one is configured for each sensor material package, so it does not need to be prepared)
Blood routine tube rack
Evacuated blood collection tube
Capillary blood tube

Upgrade Material Source

115-090340-00 sample type detection upgrade package

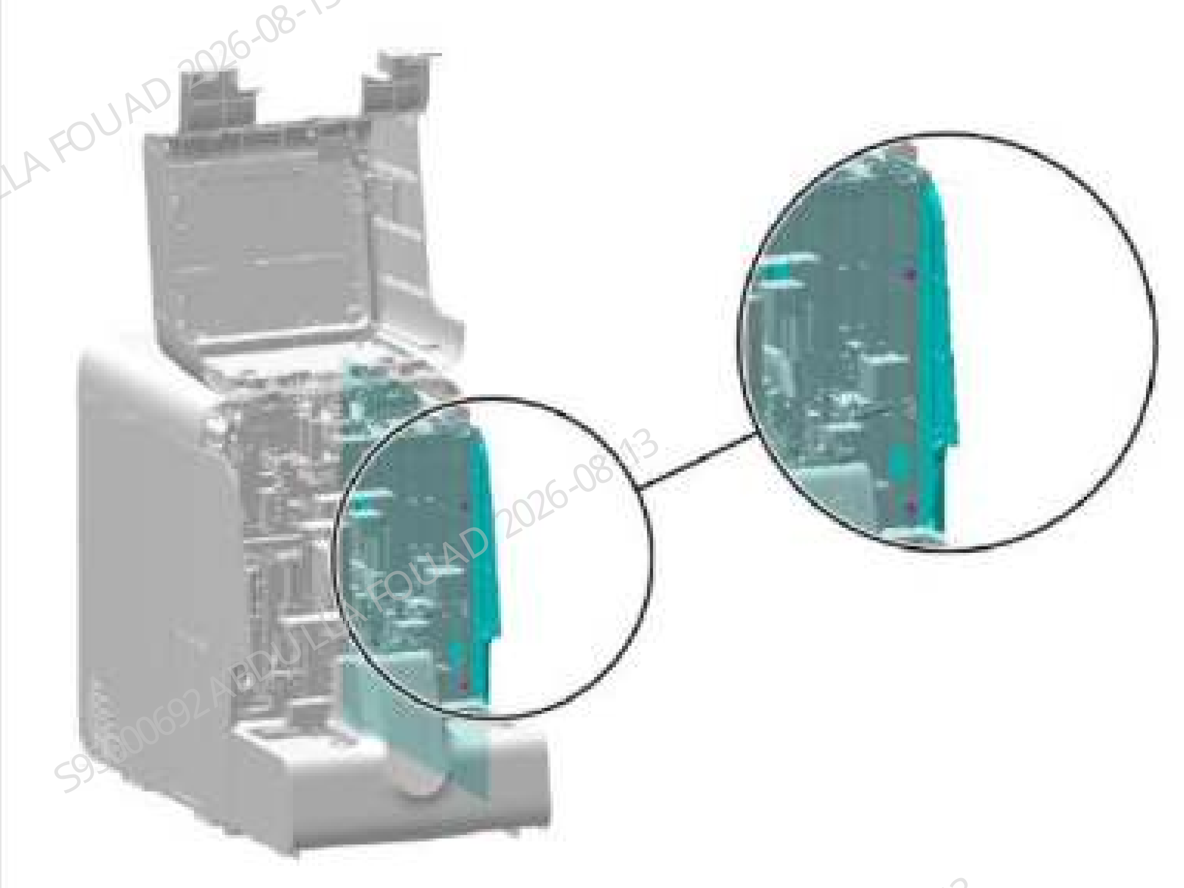
Upgrade Precautions

Before upgrade, turn off the analyzer

7.2.2 Procedure

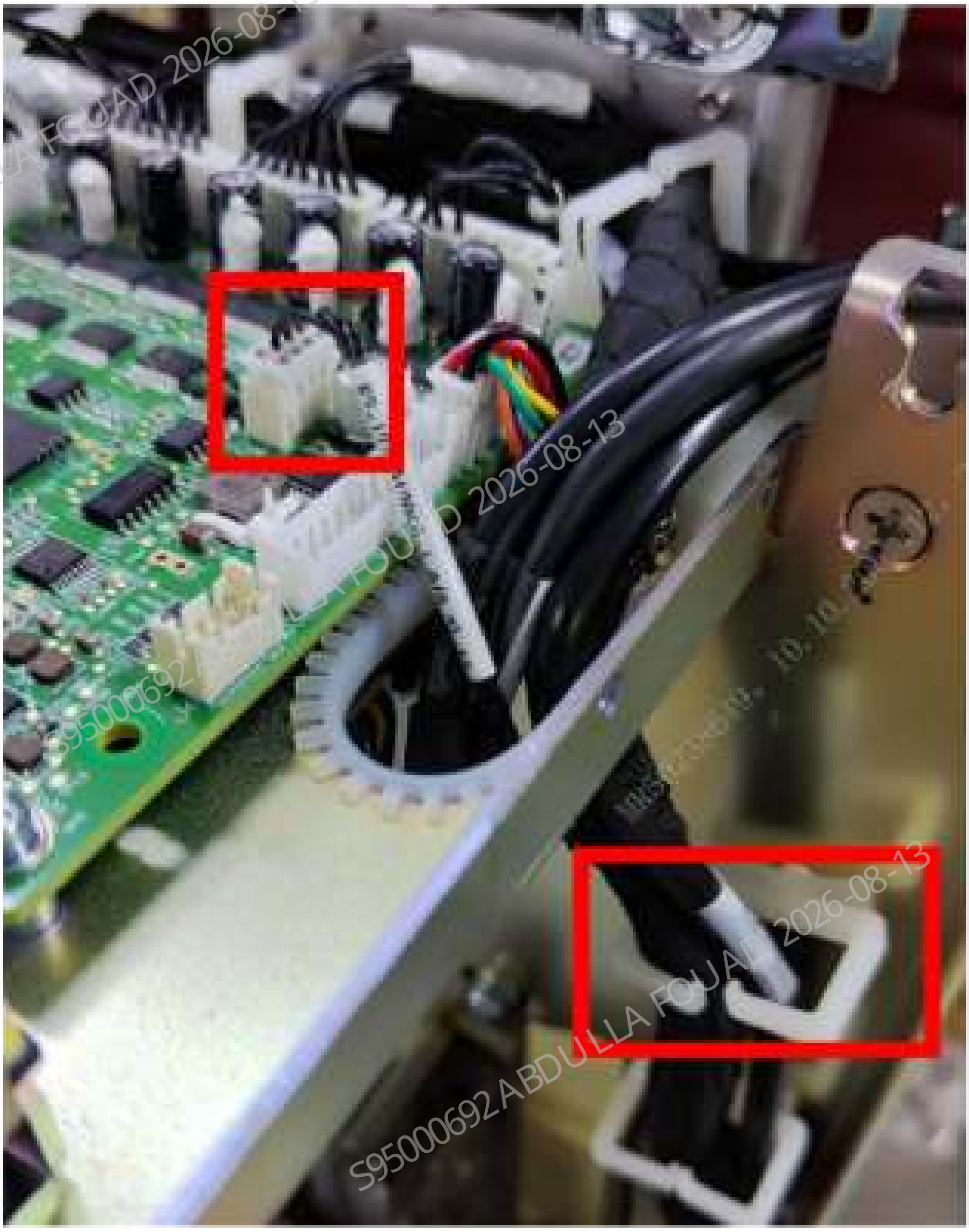
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure7–1 Diagram of opening the front cover



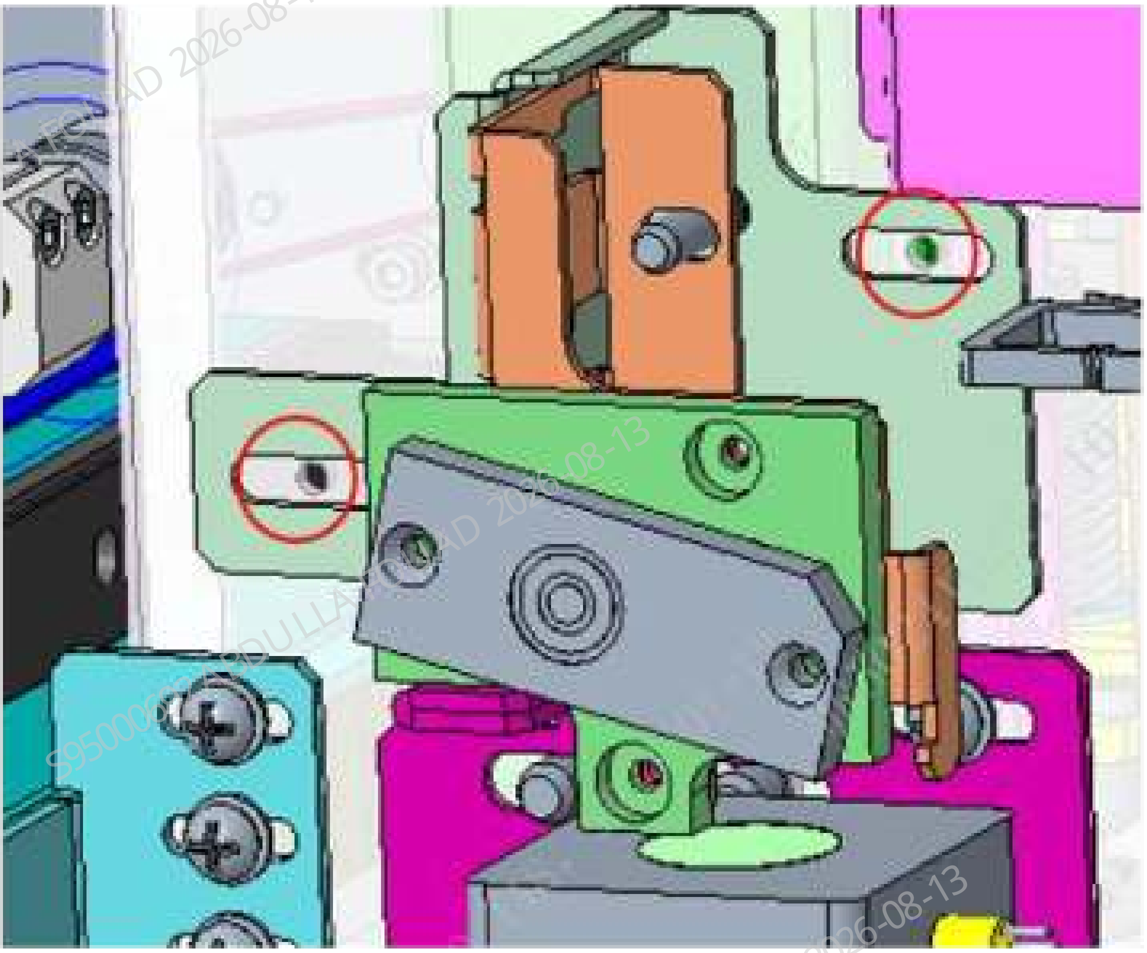
2. Lead the connecting wire at the board end and sample type identification sensor (009-012959-00) through the cabling hole on the top board, and then plug it into the J2 connector of the mixing motor board; lead the wire at the other end through the three wire harness retainers, as shown in the figure below.

Figure7–2 Wire connection diagram



3. Install the sample type detection assembly on the front plate of the main unit, locate it through the pin of the blending assembly, and then fix it using two screws (note that the two screws only need to be pre-tightened), as shown in the figure below.

Figure7–3 Assembly pre-lock diagram



4. Connect the terminal of the capacitive sensor to the connecting wire and the connector of the sample type identification sensor (009-012959-00), as shown in the following figure.

Figure7–4 Connector splicing diagram

1 Capacitive sensor wire plug

5. Complete the functional debugging and verification of the sample type detection assembly.
6. Restore the right cover and front cover.

7.2.3 Upgrade Validation

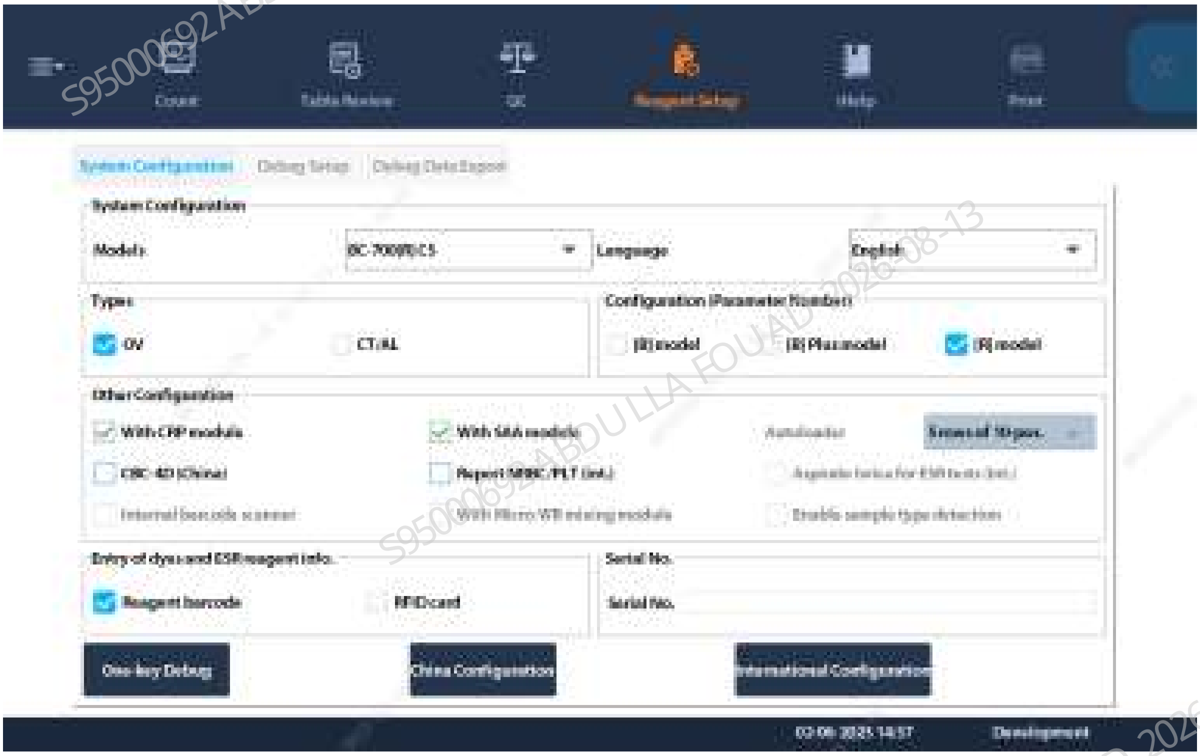
Version Confirmation after Upgrade

Connect the power supply of the analyzer and confirm whether the green light of the capacitive sensor is on (the green light indicates the connected status)

Post-upgrade Verification Procedure

1. Copy the supporting software and upgrade the analyzer.
2. After the upgrade is completed, select **Service>Advanced Toolbox** in the menu, and check **Enable sample type detection** in **Analyzer Configuration**, as shown in the following figure.

Figure7–5 Diagram of checking the sample type detection configuration option



3. Select **Service>Commissioning & Self-Test>Sensor Commissioning** in the menu, and select the **Sample Type Detection Sensor** interface, as shown in the following figure.

Figure7–6 Sample type detection interface

Figure7–7 Sample type detection interface

4. Select **Start**, and operate according to the following figure. The gripper will clamp the evacuated blood collection tube to the sample detection sensor.

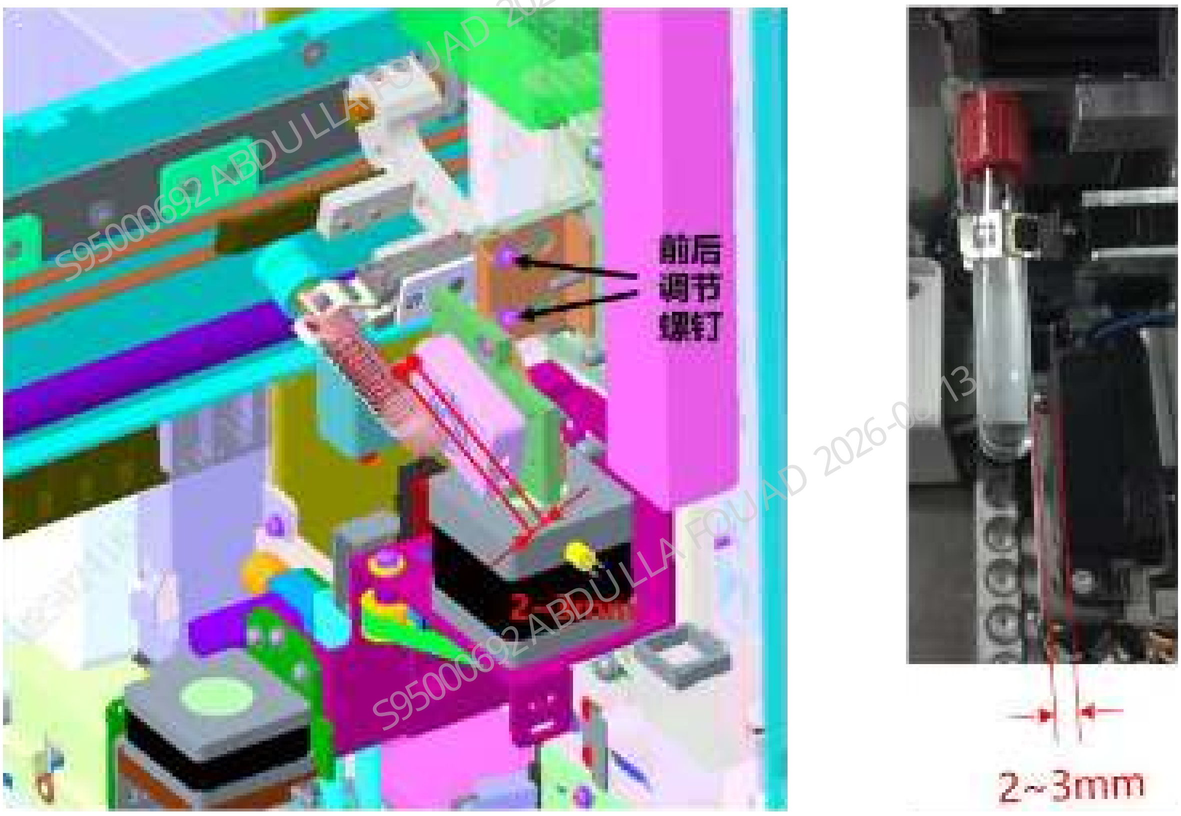
NOTICE

Here use the venous blood tube commonly used for the client

Figure7–8 Diagram of prompting to place the tube

5. As shown in the figure below, adjust the front-back adjustment screw fixing the capacitive sensor, and adjust the front-back position of the sensor to ensure that the gap between the test tube and the capacitive sensor is 2 to 3 mm.

Figure7–9 Front-back position commissioning diagram



6. As shown in the figure below, push the evacuated blood collection tube containing 0.5 ml of water or diluent up to the mixing gripper pressing plate (Note: do not swing the gripper during pushing; the green light indicates the connected status, and the orange light is an indicator), adjust the sample type detection assembly in the left and right directions, ensuring that the circular mark on the sensor is aligned with the bottom of the tube, and tighten the two fixing screws of the sample type detection assembly on the front plate of the main unit after adjustment.

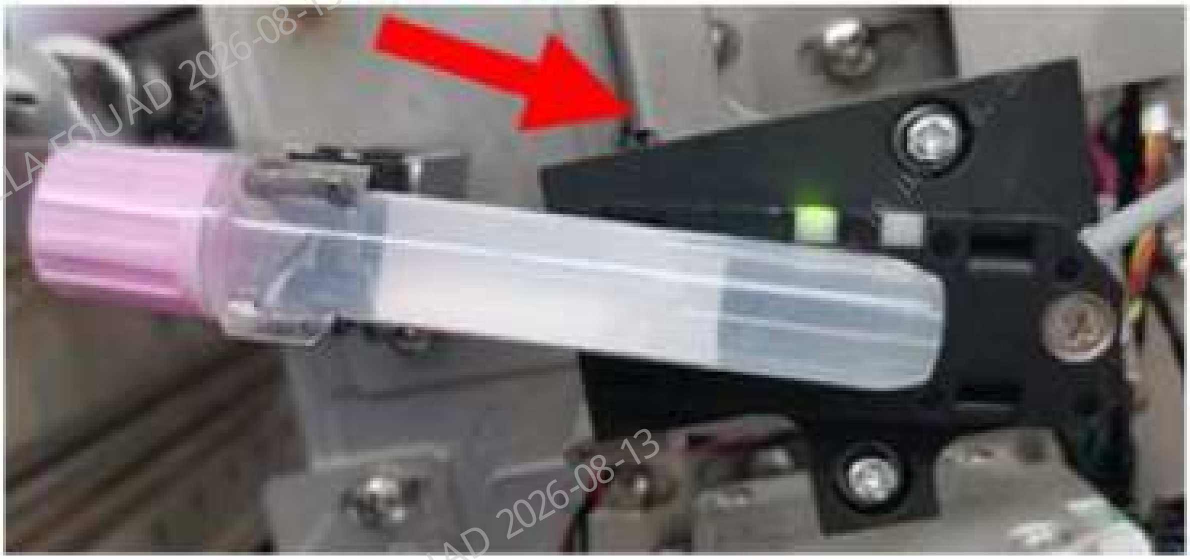
Figure7–10 Tube position commissioning diagram

7. As shown in the figure below, use the slotted head screwdriver provided with the sensor to adjust the sensor knob. If the orange indicator of the sensor is on in the initial state, rotate the screwdriver anticlockwise to turn the indicator off first. Then, adjust the knob clockwise until the orange indicator of the sensor is turned on, and then turn it by 90°.

Figure7–11 Sensitivity adjustment diagram of the venous blood tube

8. Replace the evacuated blood collection tube with an AL Micro-WB tube filled with 0.5 ml water or diluent. Then, push the AL Micro-WB tube downward to the limit, and observe whether the orange indicator is on.

Figure7–12 Capillary blood tube detection board



9. If the orange indicator is turned on after the last step, as shown in the figure below, use the slotted head screwdriver to adjust the sensor knob so that the orange indicator of the sensor goes out.

Figure7–13 Sensitivity adjustment of the capillary blood tube

10. Make sure that the orange indicator on the sensor meets the requirements in step 4 and step 5 when a vacuum blood collection tube is used, and meets the requirements in steps 6 and 7 when an AL Micro-WB tube is used.
11. Tap Exit. Pull out the autoloader after the instrument stops action.
12. Select to enter the **Service>Commissioning and Self-Test>Self-Test>Autoloading Assembly** interface, place the venous blood tube and the capillary blood tube in the tube rack, put the tube rack on the loading tray, perform **Sample Type Detection**, and confirm that the detection results meet the actual situation, as shown in the following figure.

NOTICE

V is the abbreviation of venous blood and indicates the venous blood sample, namely, the evacuated blood collection tube; C is the abbreviation of capillary blood, which indicates the capillary blood sample, namely, the Micro-WB tube; (Empty) - it means there is no tube at this position

Figure7–14 Sample Type Detection Result Display

Figure7–15 Sample Type Detection Result Display

Introduction to Exception Prompt during Upgrade

N/A

Exception Handling Measures during Upgrade

N/A

7.2.4 Overview

The upgrade guide is a normative document used to guide engineers to upgrade the analyzer software functions and optional modules. The following content is prepared according to the upgrade project.

8 FRU Replacement

8.1 Optical Sensor

8.1.1 General Information

Figure8–1

Table 8–1 Optical Sensor

FRU Code	FRU Name	Lower-Level FRU Code	Lower-level FRU Name
011-000022-00	Optical Sensor	XXX	XXX

Functions:

8.1.2 Disassembly and Assembly

N/A

8.1.3 Commissioning and Verification

N/A

8.2 ROUTER 4G USB

8.2.1 General Information

N/A

8.2.2 Disassembly and Assembly

N/A

8.2.3 Commissioning and Verification

N/A

8.3 BC-700 series annual maintenace kit

8.3.1 General Information

N/A

8.3.2 Disassembly and Assembly

N/A

8.3.3 Commissioning and Verification

N/A

8.4 3-way valve (PEIZH)

8.4.1 General Information

Figure8–2 Photo of 3-way Valve (PEIZH)

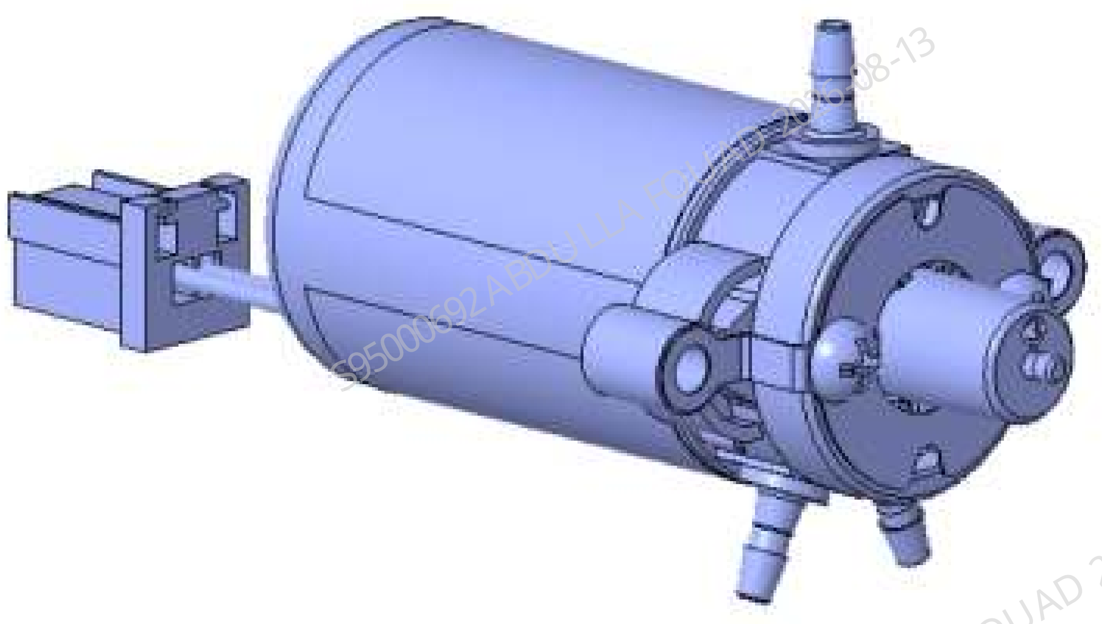


Table 8–2 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-033286-00	Three-position ways valve (PEIZH)	N/A	N/A

Function:
Turn on/off pneumatics and fluidics.

8.4.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

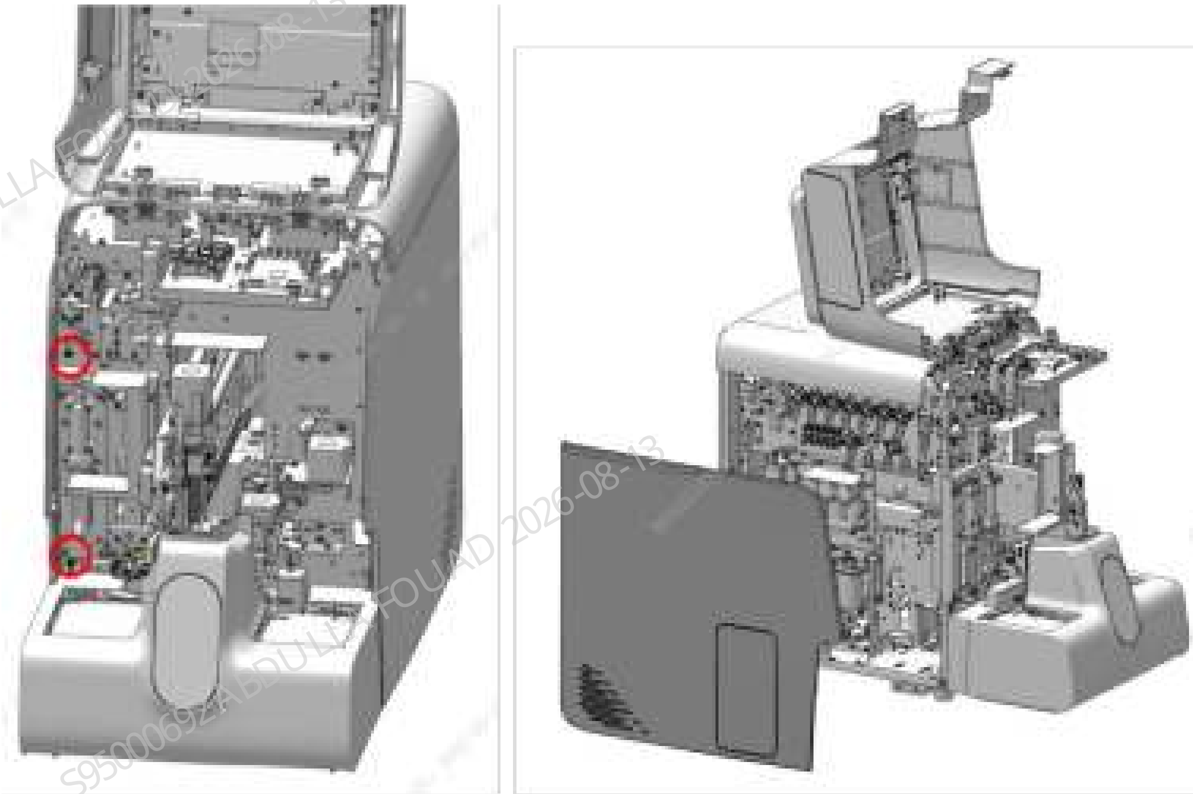
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Flip the front cover up and dock on the top cover, and remove the left cover, as shown in the figure below.

Figure8–3 Removing the covers



2. Find the location of the three-way valve (PEIZH), as shown in Figure A below, pull out the rubber tube on the three-way valve (PEIZH), loosen the two screws that fix the three-way valve (PEIZH), as shown in Figure B below, loosen the wiring terminals connected to the main unit, and remove the three-way valve (PEIZH).

Figure8–4 Diagram A for Location of Three-way Valve (PEIZH)

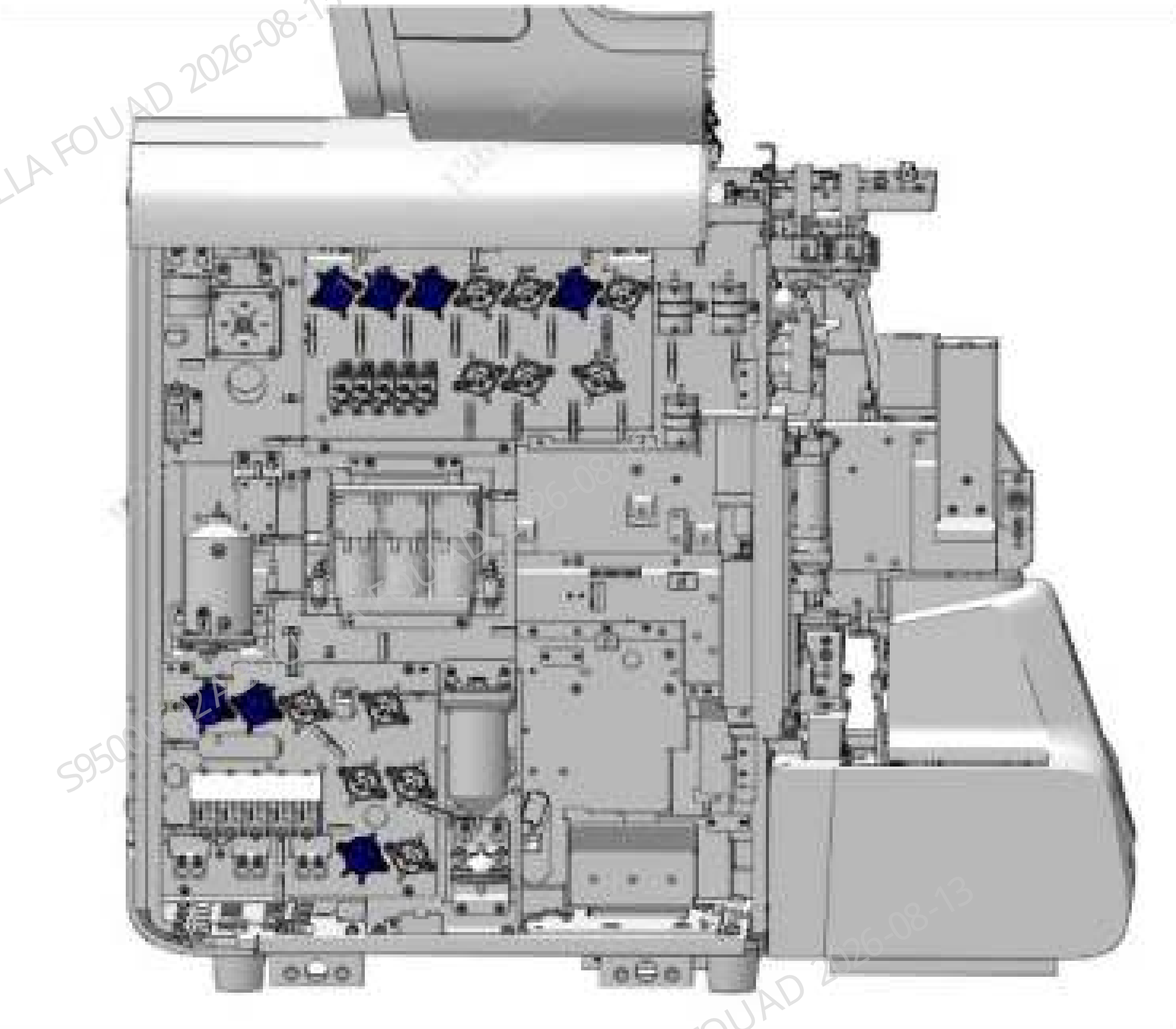
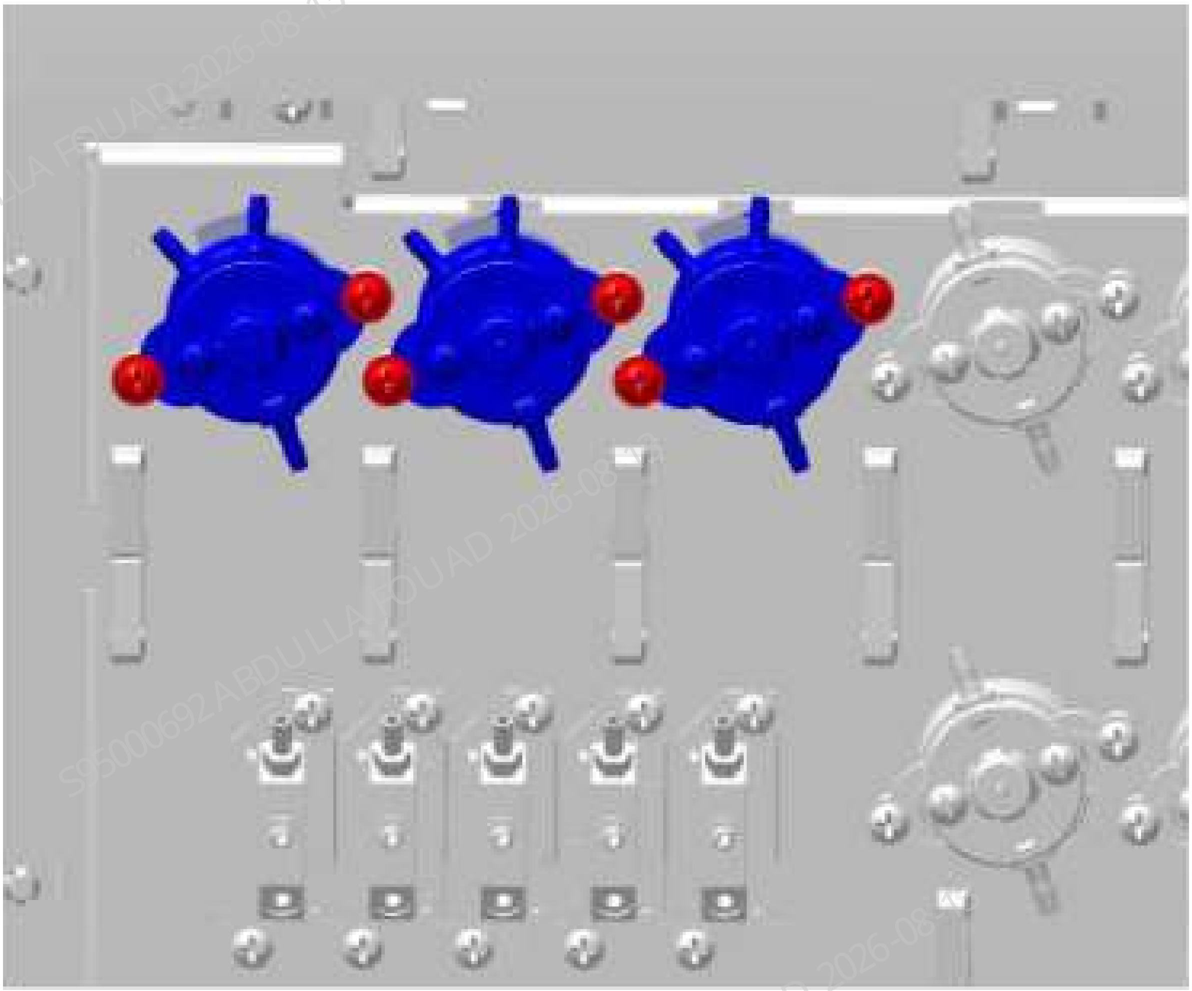


Figure8–5 Diagram B for Locations of Three-way Valve (PEIZH) Fixing Screws



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the three-way valve (PEIZH), and insert the terminal connected to the main unit.
2. Tighten the screws that fix the three-way valve (PEIZH).
3. Install the left cover back, and flip down the front cover to reset it.

8.4.3 Commissioning and Verification

Required tools

Cross-headed screwdriver

Procedures

1. Click**Service**>**>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. According to the replacement object, click *Self-Test* on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–6 Valve Self-Test screen

Verification:

1. Select **Service**>**>Maintenance>Cleaning**>**>Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance>>Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.5 2-way valve (PEIZH)

8.5.1 General Information

Figure8–7 Photo of two-way valve (PEIZH)

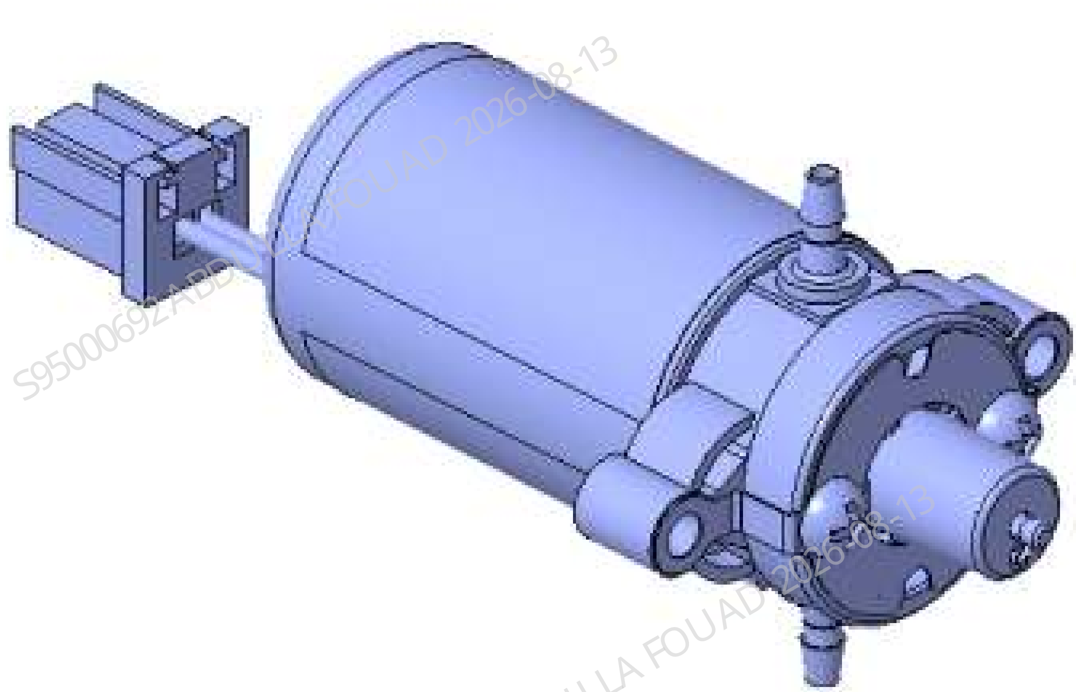


Table 8–3 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-033289-00	Two-position ways valve (PEIZH)	N/A	N/A

Function:
Turn on/off pneumatics and fluidics.

8.5.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the main unit, and unplug the power cord. Before replacing the valve assembly on the right hinge, first select **Status>Temperature and Pressure** to execute **Release Pressure**, and avoid spraying liquid when pulling out the tube.

Procedures

1. Flip the front cover up and lean it on the top cover, and remove the left cover, as shown in Figure A below. Remove the right cover, as shown in Figure B below. Find the position of two-way valve (PEIZH), as shown in Figure C below.

Figure8–8 Diagram A of Removing the Left Cover

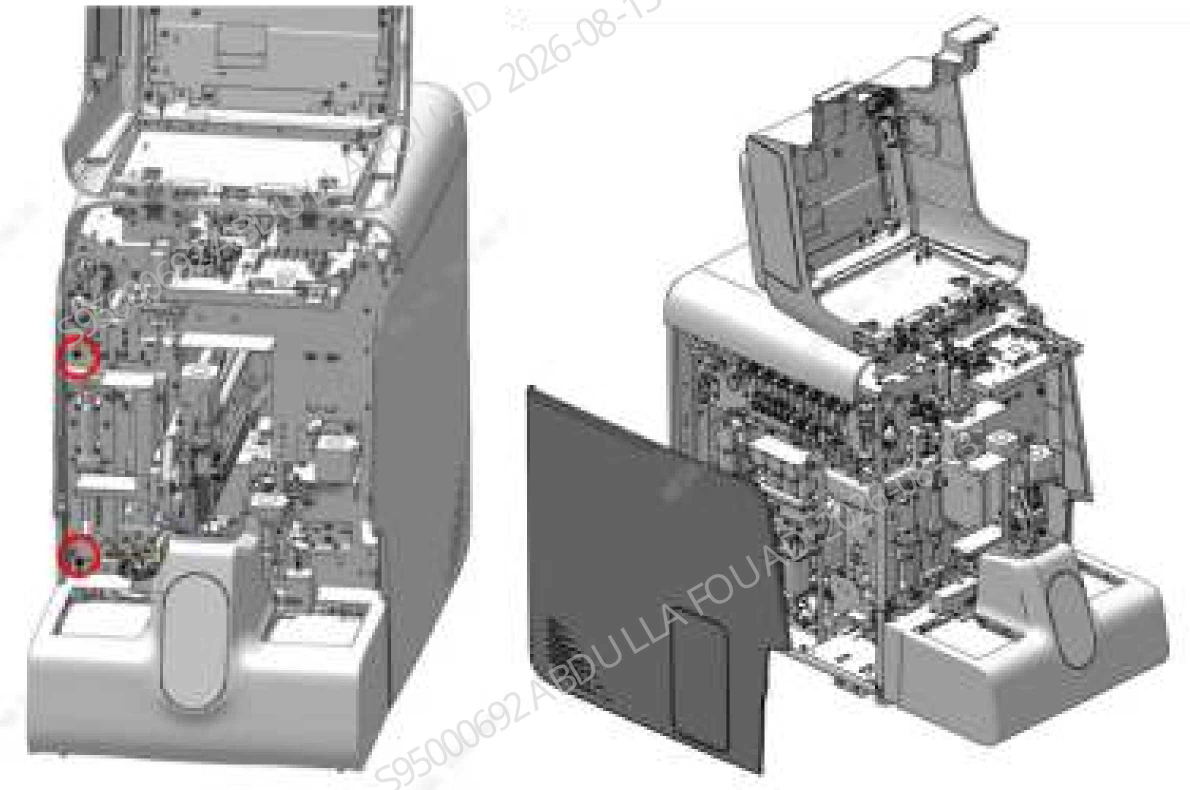


Figure8–9 Diagram B of Removing the Right Cover

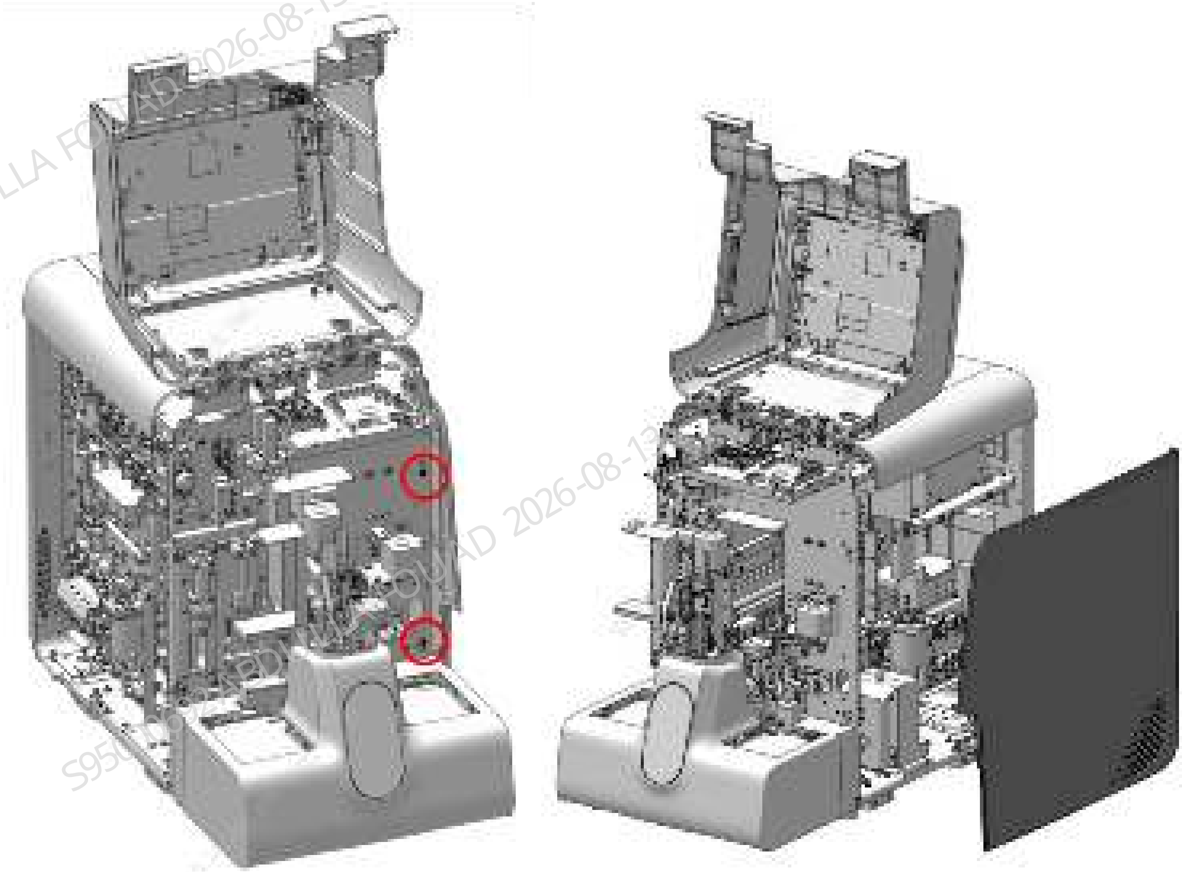
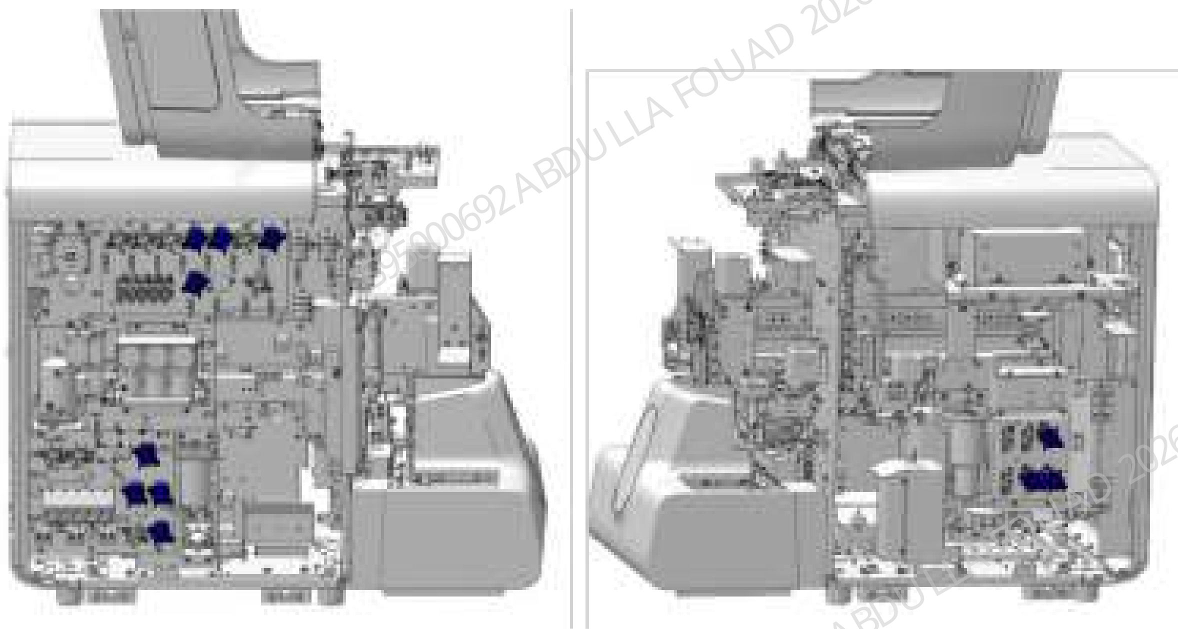
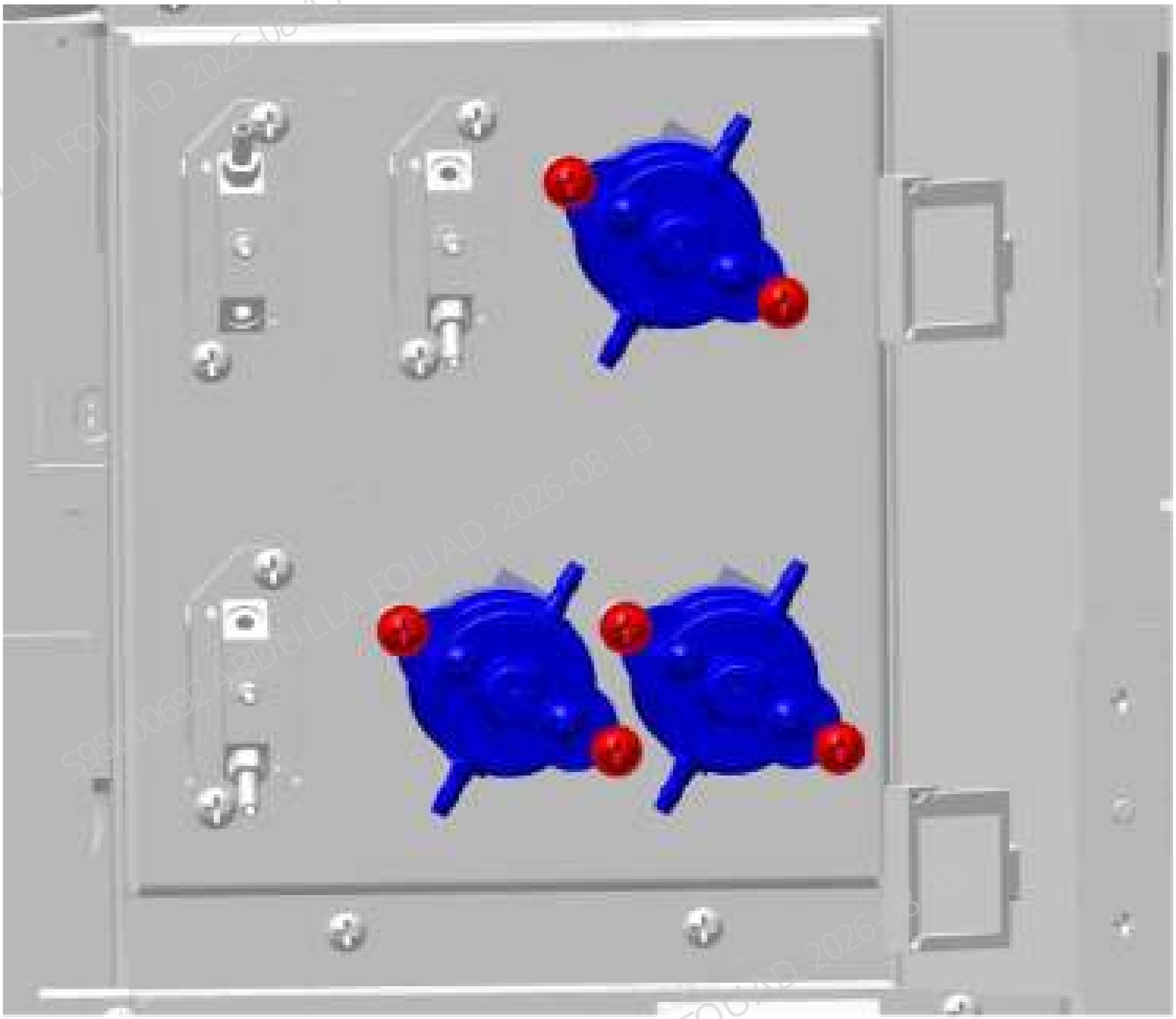


Figure8–10 Diagram C of Location of Two-way Valve (PEIZH)



2. Pull out the rubber tube on the two-way valve (PEIZH), loosen the two screws that fix the two-way valve (PEIZH), as shown below, loosen the wiring terminals connected to the main unit, and remove the two-way valve (PEIZH).

Figure8–11 Positions of set screws of two-way valve (PEIZH)



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the two-way valve (PEIZH), and insert the terminal connected to the main unit.
2. Tighten the screws that fix the two-way valve (PEIZH).
3. Install the left and right covers back, and flip down the front cover for resetting.

8.5.3 Commissioning and Verification

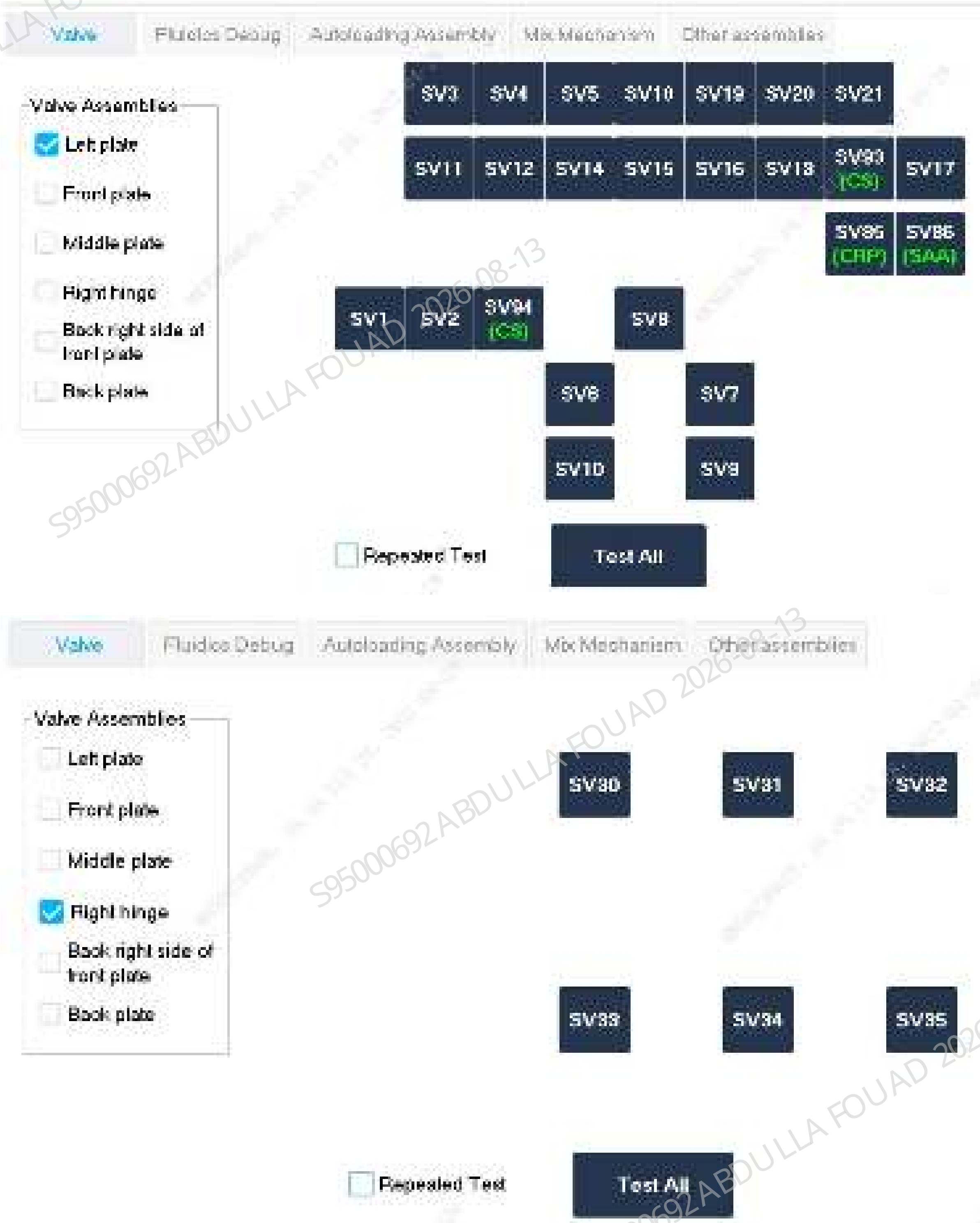
Required tools

Cross-headed screwdriver

Procedures

1. Click**Service>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. Click **SV08** Self-Test on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–12 Valve Self-Test screen



Verification:

1. Select **Service>Maintenance>Cleaning>Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance>Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.6 High pressure 2-way Valve(PEIZH)

8.6.1 General Information

Figure8–13 Photo of two-way valve (PEIZH)

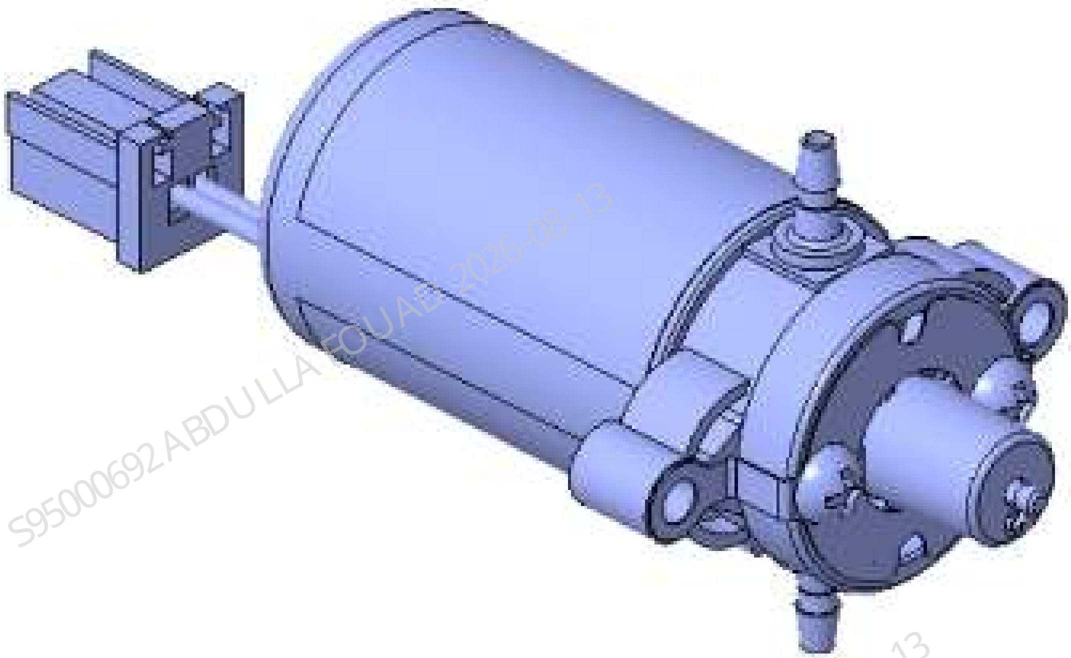


Table 8–4 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-033289-00	Two-position ways valve (PEIZH)	N/A	N/A

Function:

Turn on/off pneumatics and fluidics.

8.6.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the main unit, and unplug the power cord. Before replacing the valve assembly on the right hinge, first select **Status>Temperature and Pressure** to execute **Release Pressure**, and avoid spraying liquid when pulling out the tube.

Procedures

1. Flip the front cover up and lean it on the top cover, and remove the left cover, as shown in Figure A below. Remove the right cover, as shown in Figure B below. Find the position of two-way valve (PEIZH), as shown in Figure C below.

Figure8–14 Diagram A of Removing the Left Cover

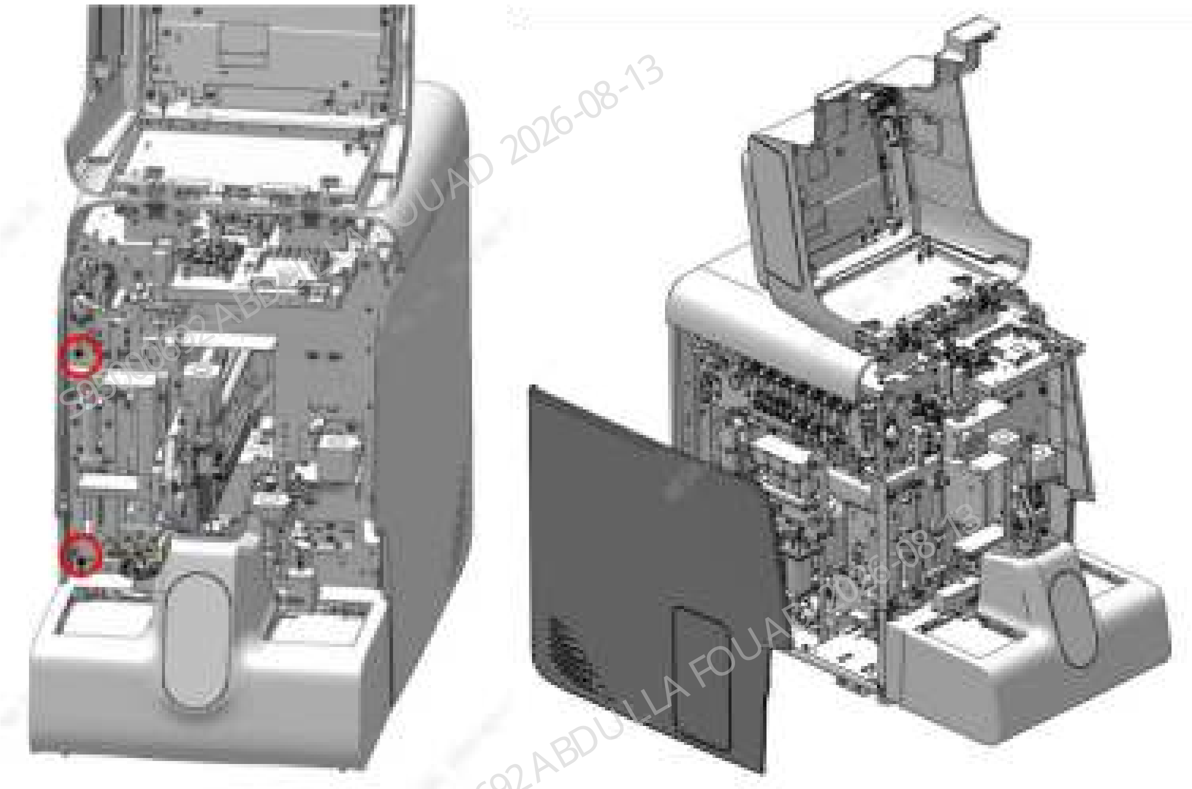


Figure8–15 Diagram B of Removing the Right Cover

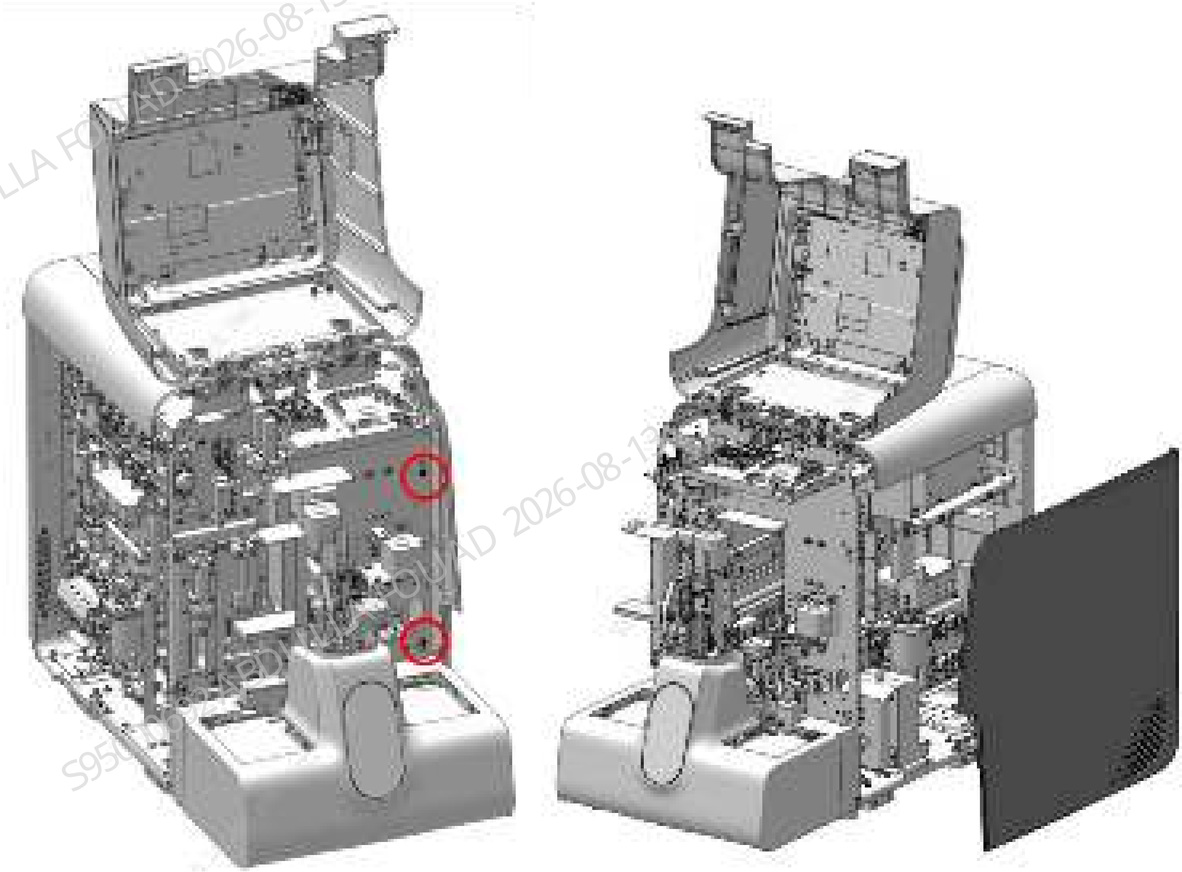
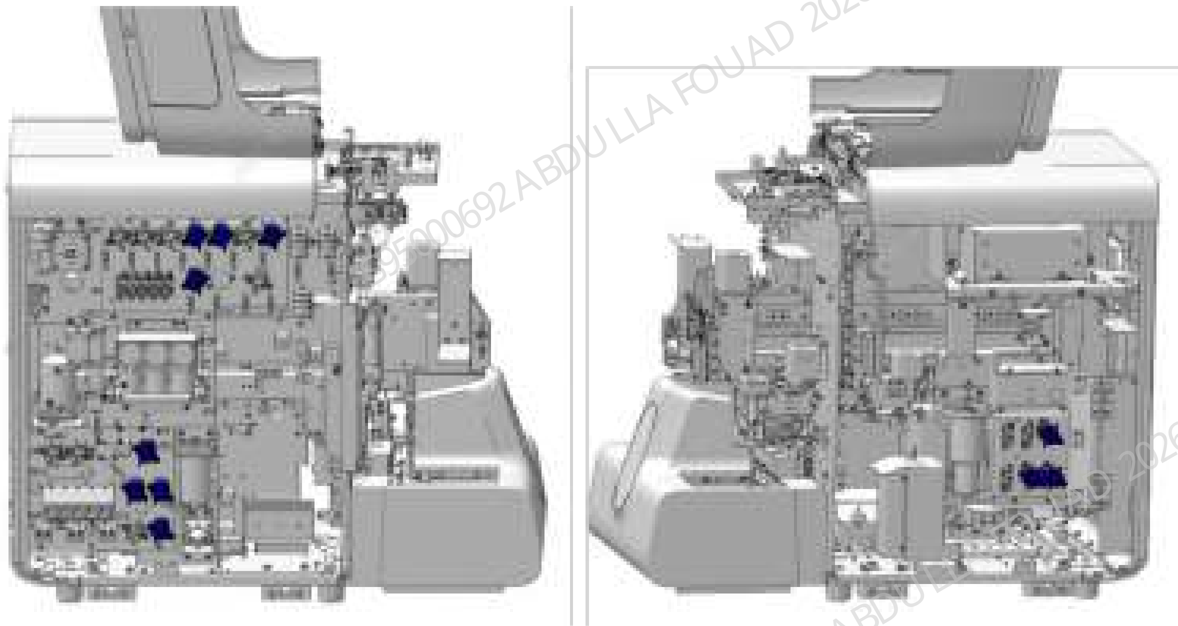
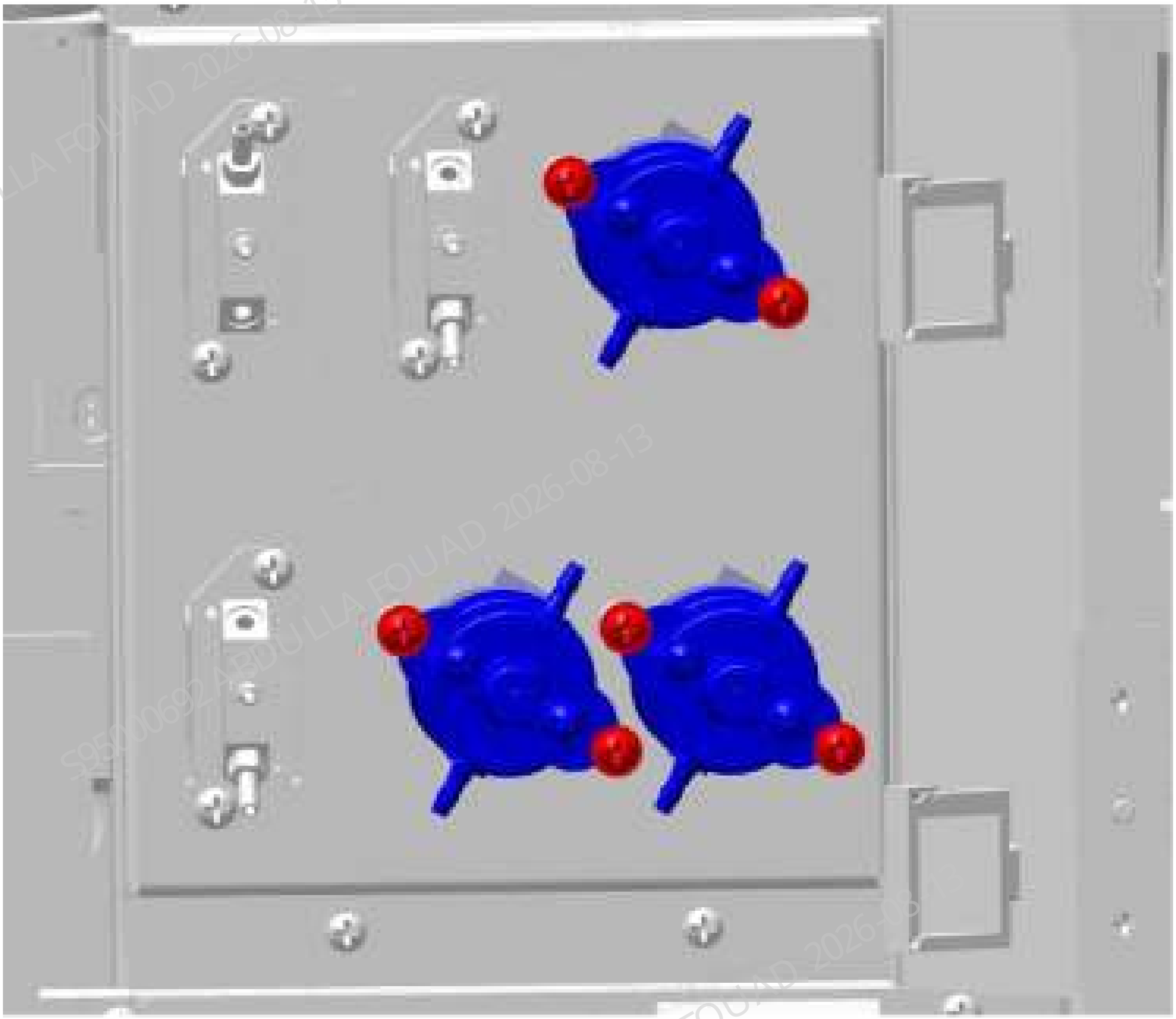


Figure8–16 Diagram C of Location of Two-way Valve (PEIZH)



2. Pull out the rubber tube on the two-way valve (PEIZH), loosen the two screws that fix the two-way valve (PEIZH), as shown below, loosen the wiring terminals connected to the main unit, and remove the two-way valve (PEIZH).

Figure8–17 Positions of set screws of two-way valve (PEIZH)



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the two-way valve (PEIZH), and insert the terminal connected to the main unit.
2. Tighten the screws that fix the two-way valve (PEIZH).
3. Install the left and right covers back, and flip down the front cover for resetting.

8.6.3 Commissioning and Verification

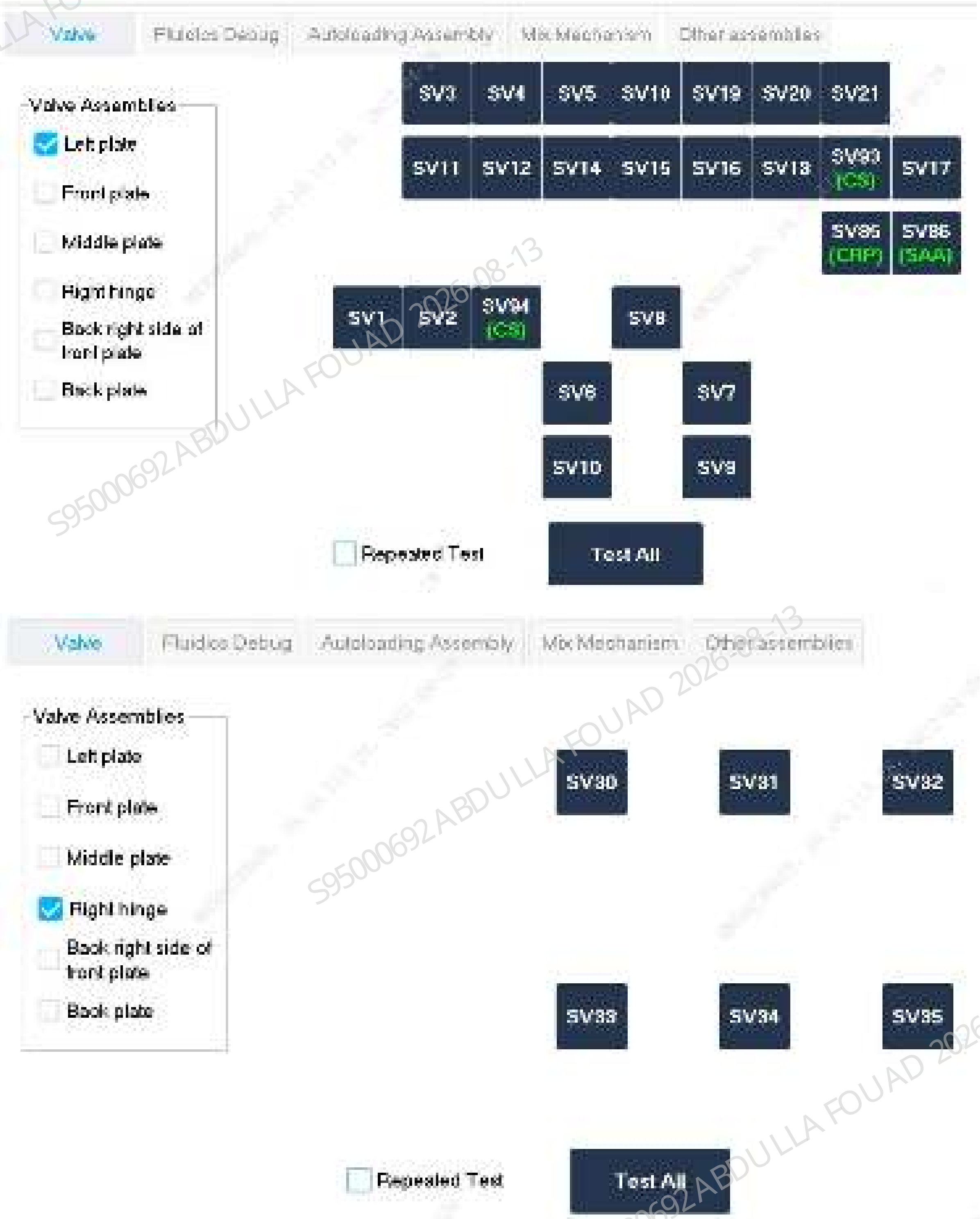
Required tools

Cross-headed screwdriver

Procedures

1. Click**Service>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. Click **SV08** Self-Test on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–18 Valve Self-Test screen



Verification:

1. Select **Service>Maintenance>Cleaning>Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance>Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.7 high-pressure 3 way value

8.7.1 General Information

Figure8–19 Photo of high-pressure 3 way value

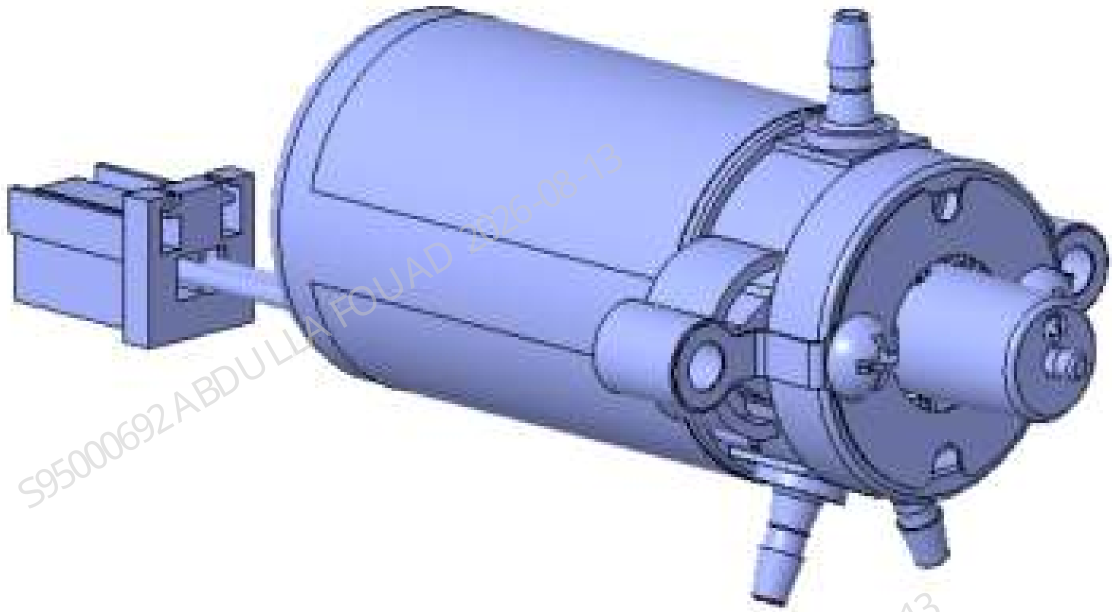


Table 8–5 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-046257-00	high-pressure 3 way value	N/A	N/A

Function:

Controlling connection/disconnection of pneumatics and fluidics, with the maximum withstand pressure of 250 Kpa.

8.7.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

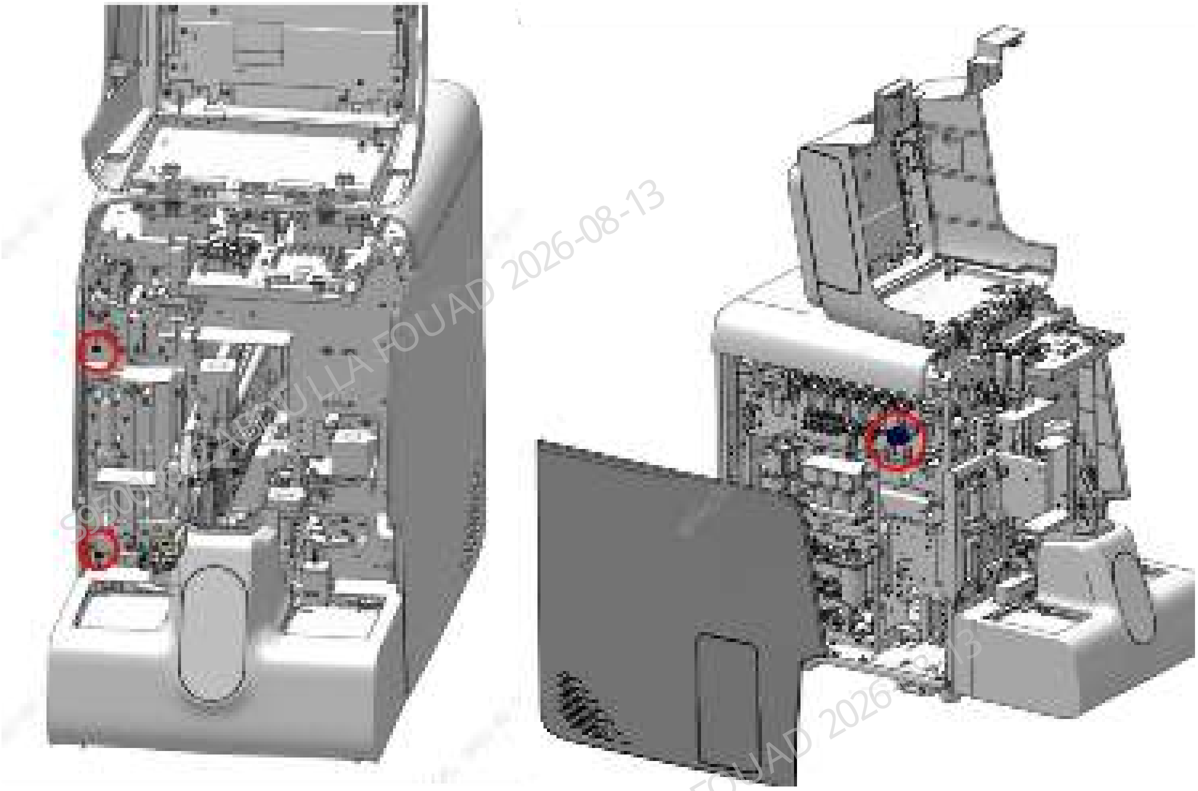
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

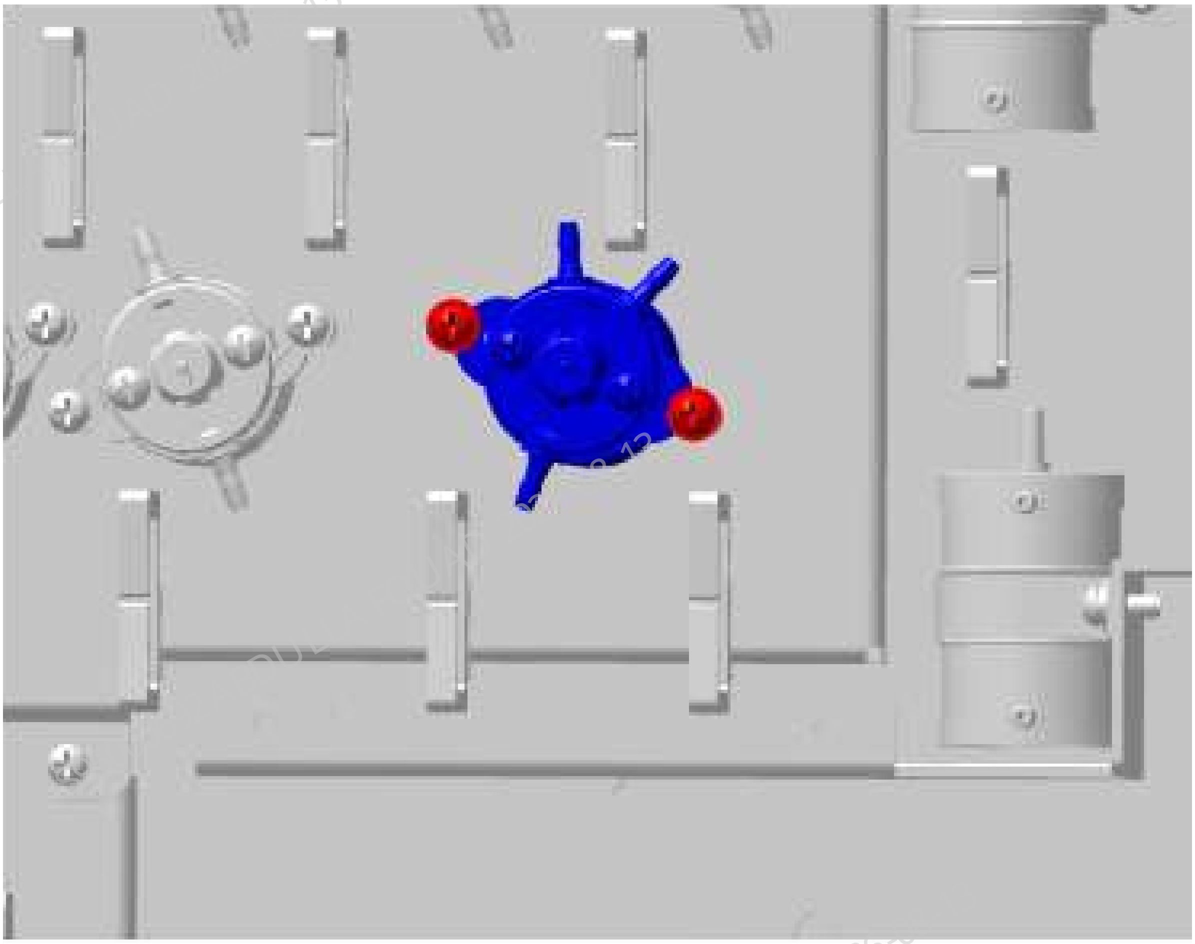
1. Flip the front cover up and dock on the top cover, and remove the left cover, as shown in the figure below.

Figure8–20 Removing the covers



2. Pull out the rubber tube on the high-pressure 3 way value, loosen the two screws that fix the high-pressure 3 way value, as shown blow, loosen the wiring terminals connected to the main unit, and remove the high-pressure 3 way value.

Figure8–21 Positions of the set screws of the high-pressure 3 way value



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the high-pressure 3 way value, and insert the terminal connected to the main unit.
2. Tighten the screws that fix the high-pressure 3 way value.
3. Install the left cover back, and flip down the front cover to reset it.

8.7.3 Commissioning and Verification

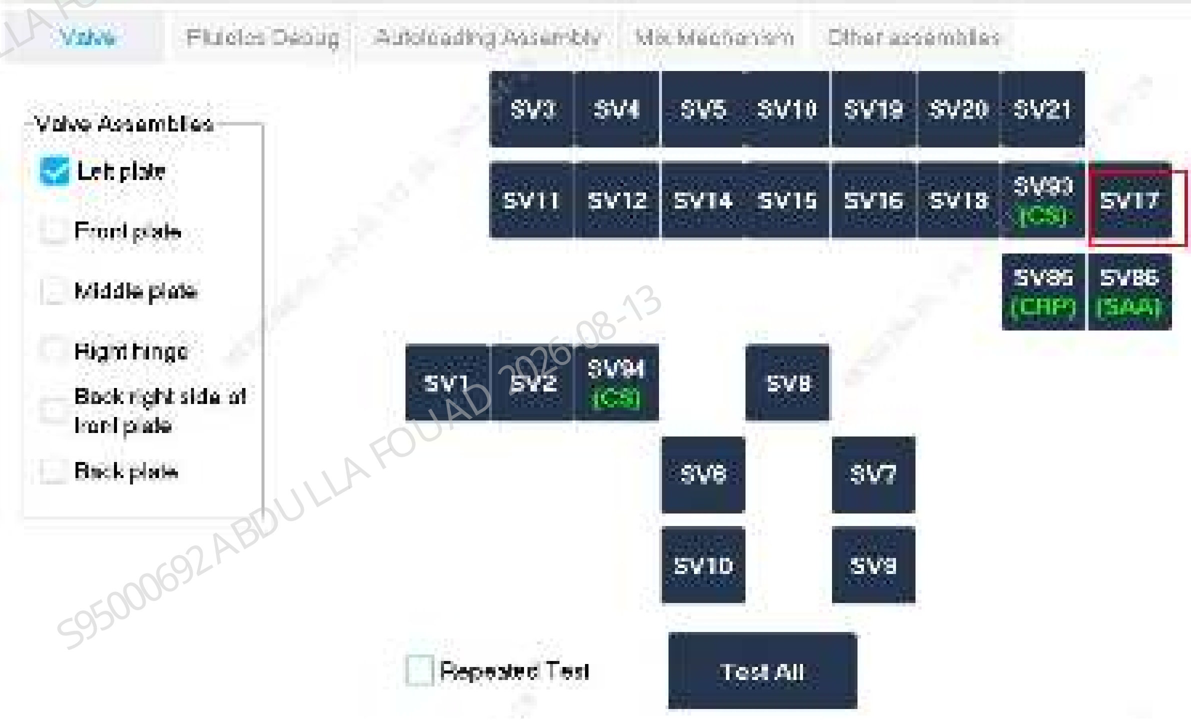
Required tools

Cross-headed screwdriver

Procedures

1. Click**Service>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. Click **SV08** Self-Test on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–22 Valve Self-Test screen



Verification:

1. Select **Service>Maintenance>Cleaning>Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance>Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.8 Picture of the PCBA (FRU), Small Fluorescent Main Control Bo

8.8.1 General Information

Figure8–23 Picture of the PCBA (FRU), Small Fluorescent Main Control Board

PCBA 3120 main control board

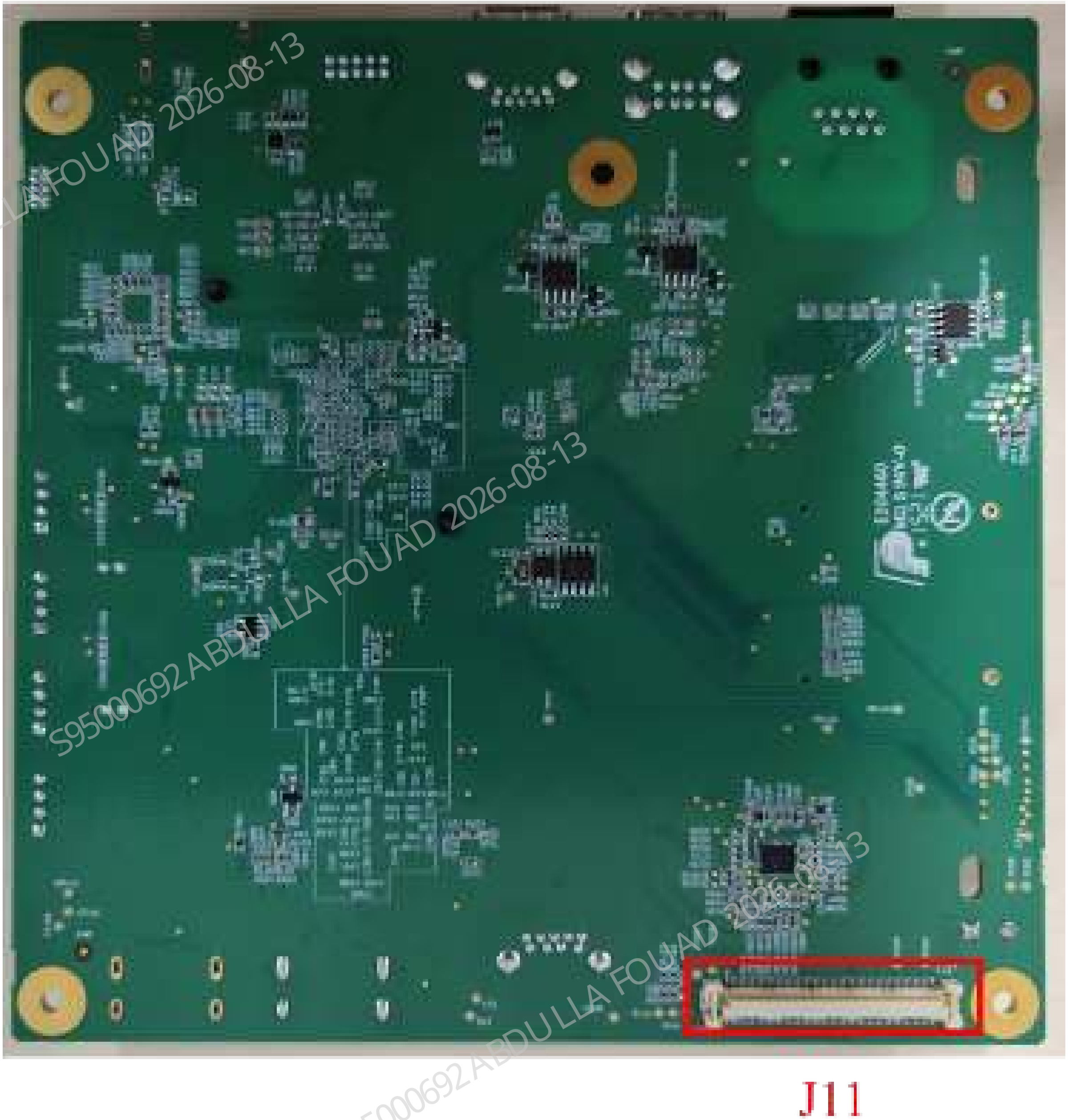
Table 8–6 List of Code, Name and Lower-Level FRU Code

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-094647-00	Picture of the PCBA (FRU), Small Fluorescent Main Control Board	N/A	N/A

Functions:

Providing various sockets, including the communication socket, display socket, control socket, display screen socket, touch screen socket, audio socket, USB 3.0/2.0 socket, motherboard communication socket, WiFi module socket, 4G module socket.

Overview of Connectors



Location of key sockets on the main control board

Socket	Functions	Pin No.	Pin Name	Description
J14	Touch screen	1,4,7,14,15,16	GND	Ground, 0V
		2	VDD_3V3	Power supply, 3.3V
		3	VDD_5V	Power supply, 5V
		5	I2C3_SCL	I2C communication
		6	I2C3_SDA	
		8	TS_RST#	Reset signal

Socket	Functions	Pin No.	Pin Name	Description
		9	TS_INT#	Touch screen interrupt signal
		10	NC	N/A
		12	RS232_TX3	Serial port communication
		13	RS232_RX3	
J4	Backlight	1,2,3,4	VPP_12V	Backlight power supply, 12V
		5,6,7,8,11,12	GND	Ground, 0V
		9	BLT_PWM	Backlight brightness control
		10	BLT_EN	Backlight enabling control
J1	USB3.0 connector			
J13	Two USB 2.0 connectors			
J12	Internal USB2.0 sockets (2)			
J27	Internal USB2.0 sockets, connect with the side or front USB2.0 socket of the analyzer			
J6	Network port			
J8	LVDS display signal output connector			
J20	4G module socket			
J10	12 V power supply connector			
J5	Audio connector			
J11	Mother board communication connector			

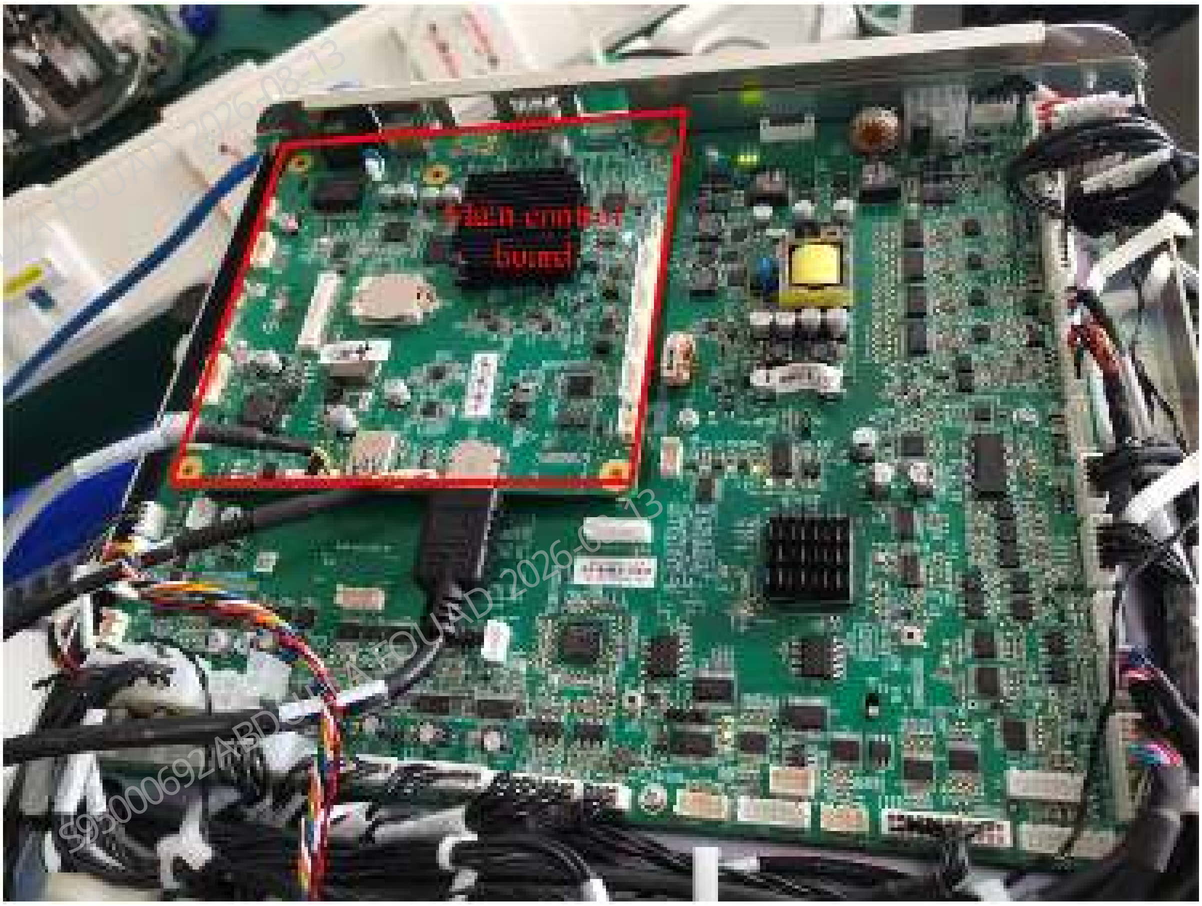
Indicators

Indicator Position Number	Meaning	Normal State
D9	MCU indicator	White, steady on
D27	Ethernet connection status indicator	Yellow, flashing

Test Points

Test Point Position Number	Meaning	Normal Range
TP35	D5V	4.983V - 5.240V
TP37	D3.3V	3.241V - 3.400V
TP64	D1.8V	1.746V - 1.854V
TP23, TP24	+12V	+(11.4V - 12.6V)
TP138, TP139	GND	0 – 0.1V

Board Position



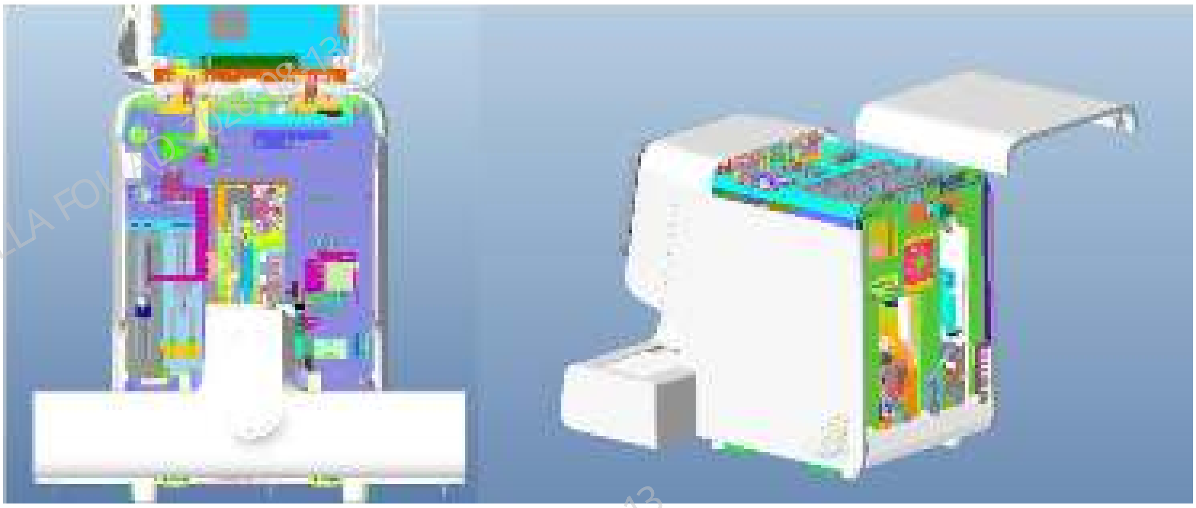
8.8.2 Disassembly and Assembly

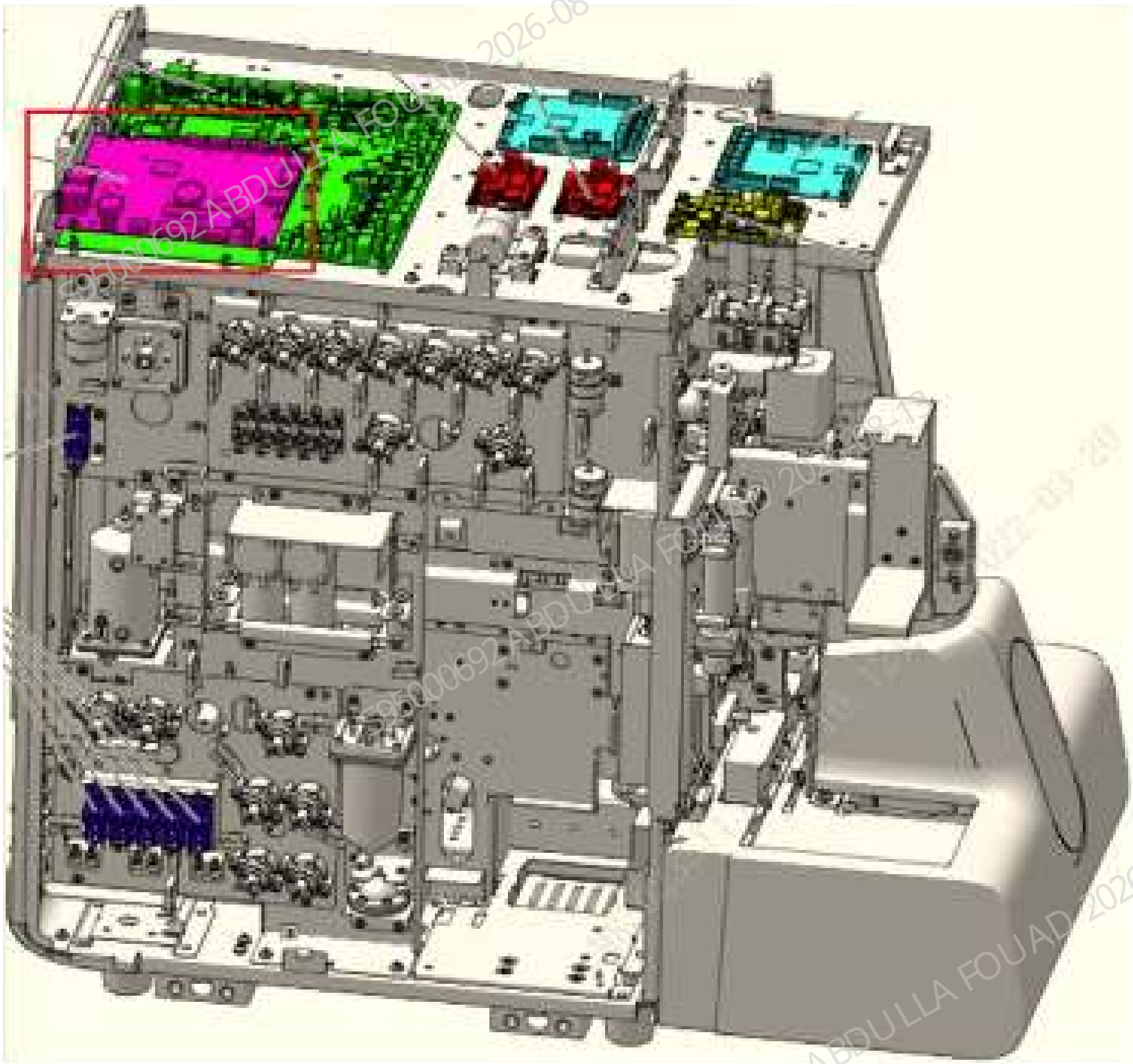
Required tools

Cross-headed screwdriver
Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Remove the top cover: Turn up the front cover, loosen the three fixing screws of the top cover, and remove the top cover.



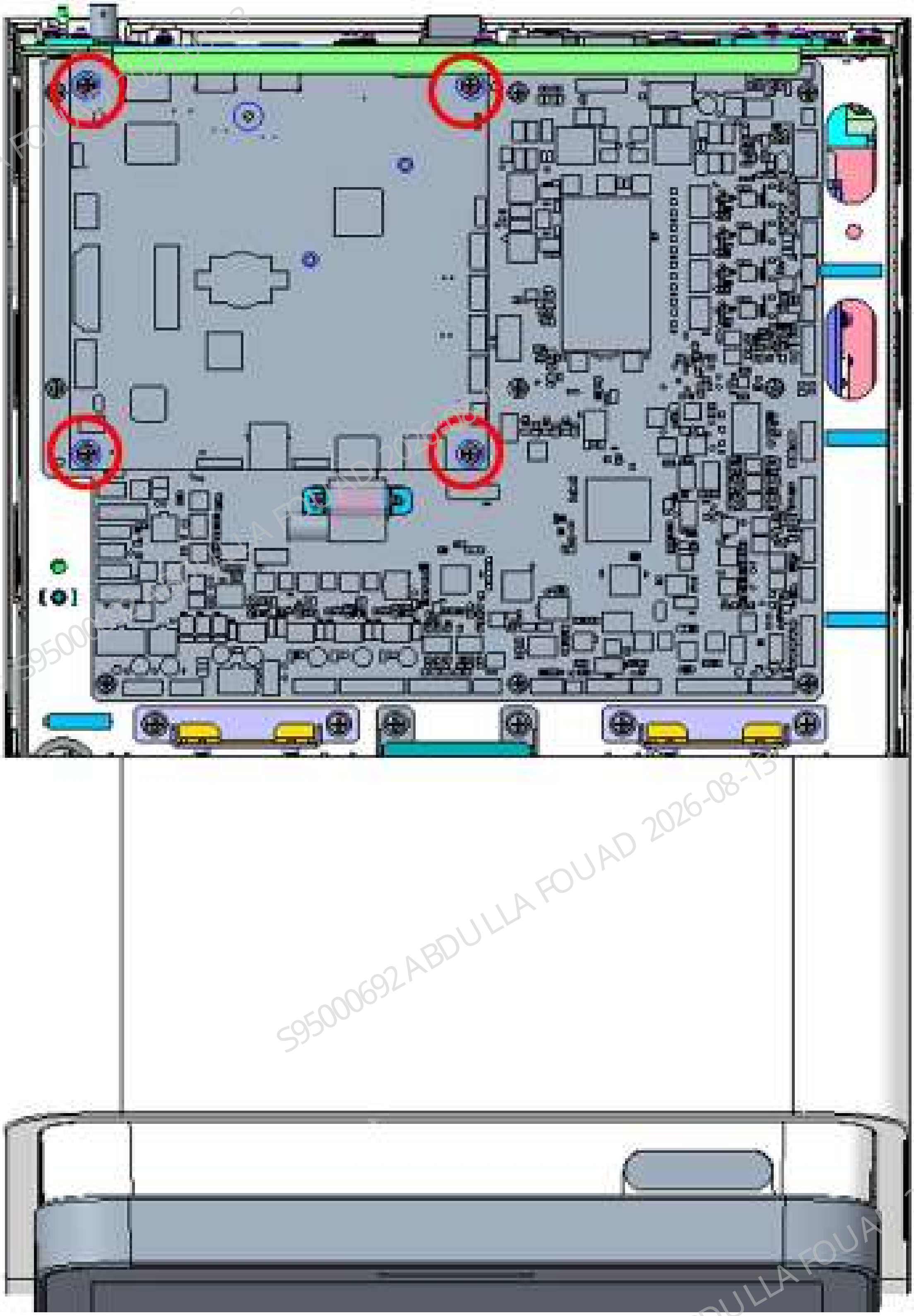


3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Disassembly procedure:

1. Remove the four screws on the rear cover of the analyzer, and take off the rear cover.
2. Remove the two screws behind the upper cover of the analyzer, and remove the upper cover.
3. Confirm that the main control board is identified, and pull out the wires from all the sockets in turn.
4. Remove all the four fixing screws in sequence.
5. At the fixing hole of MH1, hold the lower edge of the board with your fingers, and gently press toward the top of the analyzer to remove the main control board from the carrier board.



Assembling procedure:

1. Confirm that the analyzer is shut down.

2. Identify the correct orientation of the board and identify the connection position between the main control board and the carrier board.
3. Place the board to the installation position. Make sure that all the wire plugs are not pressed by the board.
4. Above the connector position of the main control board-carrier board, press down gently to connect the main control board to the carrier board through the connector.
5. Install all the four fixing screws.
6. Place the upper cover in the correct position and install the two screws of the upper cover.
7. Place the rear cover in the correct position and install the four screws of the rear cover.

8.8.3 Commissioning and Verification

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Commissioning procedures:

1. Replace the board, and power on the analyzer.
2. Observe the operation interface. If no upgrade prompt is displayed, ignore this step. If an upgrade prompt is displayed on the analyzer screen, insert the USB flash drive that contains the software upgrade package (the same version before replacement), and perform the upgrade operation. After the analyzer display prompts successful upgrade, shut down and restart the analyzer.
3. After restart, the analyzer should enter the startup initialization process. Log in with a service account. Follow system prompts to back up parameters (be sure to select **Resume**, and **Resume** means to transfer the data from the motherboard to the main control board. **Backup** is the opposite, and backup means to transfer the data from the main control board to the motherboard).
4. Select **Setup**>>**System Setup**>>**Date/Time Setup** to set the current system time and save the settings.

Verification Procedure:

1. Power on the analyzer, and check whether any alarm is reported during the startup.
2. Connect the analyzer to a PC using the network wire, and confirm that the PC can be connected to the analyzer.
3. Take a blank tube, select the **OV-WB-CD** mode, and after startup, confirm whether the test process alarms.

Subsequent Requirements

Table 8–7 Judging Whether the PCBA, Main Control Board, iMX8 Is Faulty

No.	Error phenomenon	Troubleshooting
1	The PC cannot communicate with the analyzer	1. In the case of ensuring that the PC network port works normally, check if the network wire connector gets loose. If yes, reconnect the wire and see whether the problem is solved; if not, go to step 2. 2. Open the upper cover of the analyzer and check the Ethernet connection status indicator D27. If D27 flashes, the network connection of the main control board is normal, and you need to consider the possible problems in other parts of the analyzer; if D27 does not flash, the main control board is faulty. Replacement is required.
2	The LCD screen becomes black.	1. Open the upper cover of the analyzer and check the status of the MCU indicator D9. If it is not always on and the power indicator of the carrier board is in normal state, the main control board is faulty and replace the main control board; if D9 is always on after the analyzer is powered on, go to step 2. 2. Use a multimeter to check the voltage of each pin of the J4 connector. If the pin voltage is abnormal, the main control board is faulty.

Table 8–7 Judging Whether the PCBA, Main Control Board, iMX8 Is Faulty(continued)

No.	Error phenomenon	Troubleshooting
		If the pin voltage is normal, go to step 3. 3. Replace the corresponding wire of the J4 connector. If the screen does not recover abnormally, the screen assembly is considered faulty and replace the screen assembly.

8.9 Picture of PCBA (FRU), Small Fluorescent Mother Board

8.9.1 General Information

Figure8–24 Picture of the Small Fluorescent Mother Board

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-094642-00	Picture of PCBA (FRU), Small Fluorescent Mother Board	N/A	N/A

Functions:

1. Completing the 24V DC power conversion of the purchased power supply, and providing power for the main control board and various parts.
2. Providing optical signal board communication socket, RFID board communication socket, indicator board control socket, valve drive board control socket, air pressure detection board socket, and liquid detection board socket.
3. Driving motors, valves, pumps, heating, fans etc., and detect the temperature, pressure, reagents, hydraulic pressure, sensor, floater, and micro-switches.
4. Driving the count bath, HGB sensor, optical system, ESR system, and CRP system and detect signals.
5. Providing analog/digital conversion and digital processing of optical signals, impedance signals, HGB signals, and pulse identification and post-amplification & conditioning of optical signals.

Connectors

Area	Position No.	Connector Function and Definition
D	J1	Chassis door open detection micro switch socket
	J2	Indicator board signal socket
	J4	Switch key board socket
	J5	ESR signal sockets

Area	Position No.	Connector Function and Definition
	J20	Valve driver board (F1) communication socket
	J21	WC2 waste pump, probe wipe waste pump, positive pressure pump signal socket
	J23	Valve driver board (F1) and power socket
	J22	Motor board A power supply and communication socket
	J24	Motor board B power and communication connector
	J26	Motor board C power supply and communication socket
	J25	Valve driver board (F2) and power socket
C	J28	Socket to heaters
	J29	Socket to 10 mL syringe motor
	J30	Socket to 250 μL syringe motor
	J31	Sockets to aspiration module motors in the horizontal and vertical directions
	J32	Socket to sensor signals
	J33	4 floater signal socket
	J34	Scanner signal socket
	J35	Signal sockets to RFID boards 1 and 2
	J48	Optical system power socket
	J53	Temperature control board communication signal socket
	J36	Socket to level signal processing board
	J37	Signal socket to liquid detection board
	J38	Optical system control signal socket

Area	Position No.	Connector Function and Definition
	J39	CRP board power supply and communication signal sockets
B	J40	Air-pressure detection board socket
	J41	Socket to dye detection sensor 1
	J42	Socket to dye detection sensor 2
	J43	Temperature detection socket
	J44	Socket to impedance detection module
	J45	Laser control socket
	J46	Detection socket to fluid pressure sensor
	J47	HGB sensor socket
	J49	Optical system signal socket
	J50	24V power socket to MP400 power module power
	J54	5V power socket to MP400 power module power
A	J51	Sockets to system fan and power fan
	J52	Temperature control version power socket
	J56	Power_mcu version programming socket

Pin Number	Connector Function and Definition	Pin Number	Connector Function and Definition
J50/1,2	+24V	J26/1,3	+24V
J50/3,4	GND	J26/2,4,5,6,9	GND
J54/1	+5V	J22/1,3	+24V
J54/2	GND	J22/2,4,5,6,9	GND
J1/1,5	+5V	J23/3,4	+12V
J1/4,8	GND	J23/1,2	GND
J28/4,5,6	+24V	J25/3,4	+12V

Pin Number	Connector Function and Definition	Pin Number	Connector Function and Definition
J21/6,7,8,9	+12V	J25/1,2	GND
J24/1,3	+24V	J20/6	+5V
J24/2,4,5,6,9	GND	J20/8	GND
J34/1,4	+5V	J2/1	+5V
J34/2,7,10	GND	J2/2,3,8	GND
J37/3,4,9,10,15,16,21	+5V	J18/6	+3.3V
J37/5,6,11,12,17,18,20,22~24	GND	J18/1,2,8	GND
J49/2,4,6,8,10,22~22	GND	J16/1	+3.3V
J49/11	+3.3V	J16/8	GND
J32/1,5,9,13,17	+5V	J17/9,10	+5V
J32/3,4,7,8,11,12,15,16,19,20	GND	J17/1,2	GND
J35/3,4	+5V	J36/5,7,8	GND
J35/1,2,13,14	GND	J36/1	+3.3V
J38/2	GND	J36/3	+5V
J39/6	+3.3V	J40/11	+3.3V
J39/5,7,8,13,14	GND	J40/13,15,16	GND

Communication signal type

Commissioning sockets

Position No.	Network Name	Meaning
J17	STM32_uart_debug	MCU serial port debugging
J16	STM32_JTAG	Downloading and upgrading the mcu_boot version
J18	FPGA_JTAG	Downloading and upgrading the logical version
J58	LPC_JTAG	Downloading and upgrading the power management MCU version

Communication sockets

Master equipment	Slave equipment	Communication sockets
Mother board	iMX8 main control board	1. Serial port, RS232, RS232 serial port for ISP upgrade and reuse 2. PCIe
Mother board	RFID Control Board	RS232 serial port
Mother board	Optical signal processing board	RS232 serial port
Mother board	Air pressure detection board	SPI socket, TTL level
Mother board	Valve drive board	SPI socket, TTL level
Mother board	ESR signal board	RS232 serial port
Mother board	Refrigeration temperature control board	RS232 serial port
Mother board	Motor board	RS232 serial port
Mother board	CRP signal processing board	RS232 serial port
Mother board	Indicator board	GPIO port

Indicator

Position No.	Network silkscreen	Meaning
D2	D5V	It is always on when power on, which means the 5V power supply of the motherboard is normal.
D30	P24V	It is always on when power on, which means the 24V power supply of the motherboard is normal.
D41	D12V	It is always on when power on, which means the 12V power supply of the motherboard is normal.
D71	VIN_24V	It is always on when power on, which means the 24V input power supply of the motherboard is normal.
D72	A5V_P	It is always on when power on, which means the 5V analog power supply of the motherboard is normal.

Position No.	Network silkscreen	Meaning
D73	A5V_N	It is always on when power on, which means the 5V analog power supply of the motherboard is normal.
D5,D6,D7	MCU_LED_1/2/3	The three indicators flash slowly (1 time/second), which means the MCU version is loaded properly.
D3,D4	LED_FPGA0/1	The two indicators flash slowly (LED0: 2–3 times/ second; LED1: 1 time/ second), which means the FPGA version is loaded OK.
D60	PWR_LED0	The green indicator flashes once every 1 second, which means the MCU power management is normal.
D61	PWR_LED1	Any power failure: Always on; when the fault is removed: Off.
D12	PWR_LED2	ACDC_BC (in-position 24V detection) Power failure: Always on; fault removed: Off.

Test Points

Test Point Position Number	Meaning	Test Point Position Number	Meaning
TP100/TP12/TP4	GND	TP14/TP8/TP97	GND
TP112~TP124	GND	TP105	P12V
TP31	D3V3	TP104	P24V
TP108	A12V_P	TP106	D12V
TP109	A12V_N	TP107	P5V
TP110	A5V_P	TP105	P12V
TP111	A5V_N	TP29	D5V
TP33	D1V2	TP32	D2V5

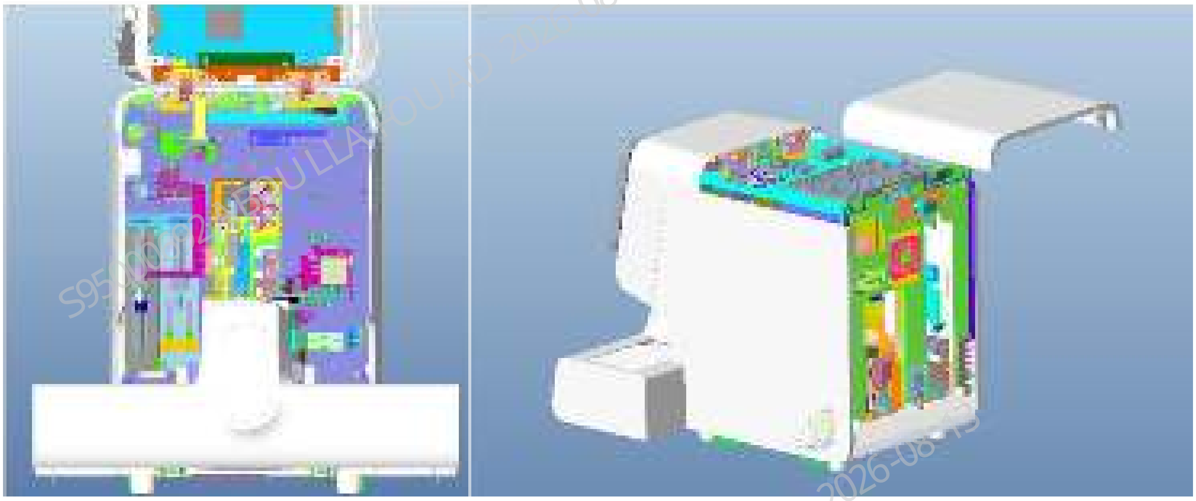
8.9.2 Disassembly and Assembly

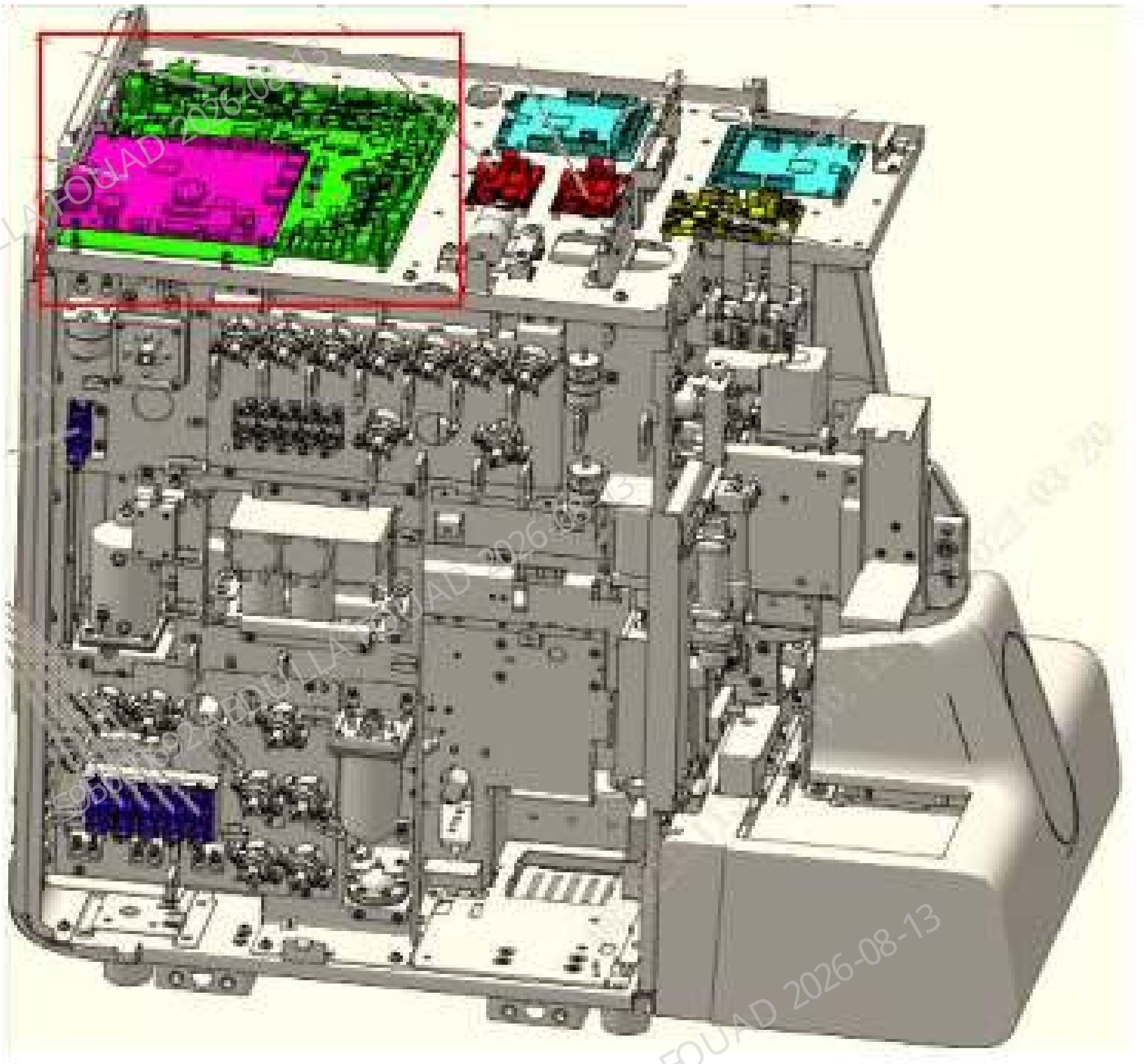
Required tools

Cross-headed screwdriver
Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Remove the top cover: Turn up the front cover, loosen the three fixing screws of the top cover, and remove the top cover.

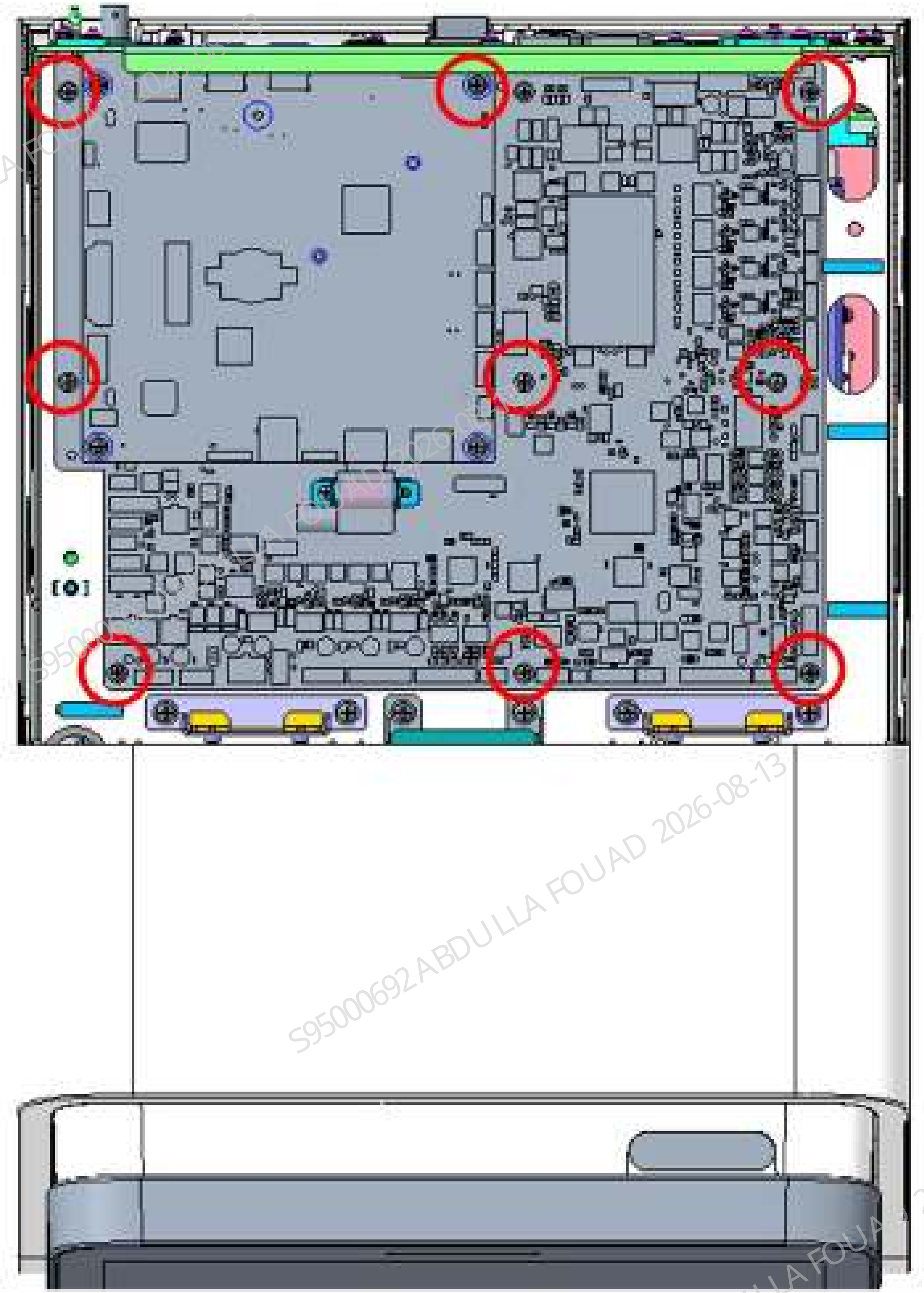




3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.
4. During disassembly and assembly, take out and place the board with caution to avoid any collision or scratch with the bolts.
5. When disassembling, first unplug the socket wire and then remove the set screws; when installing, install the set screws first, and then insert the wire.
6. Hold the wire end and unplug it out with even force to avoid pulling out the lead terminals of the wire connector. During assembly, make sure that the lead terminals of all wire connectors are not pulled out, and connectors are inserted correctly.
7. Make sure the wire silkscreens are on the right position on the PCB board.

Procedures

1. Confirm that the main control board (the board above the motherboard) is identified, and pull out all wires connected to the main control board; remove the set screws (four in total) of the main control board, and remove the main control board.
2. Confirm that the motherboard is identified, and pull out all wires; remove all the set screws (nine in total) of the motherboard in sequence, and take out the motherboard to complete the disassembly and assembly of the motherboard.



Subsequent Requirements

After disassembly and assembly, power on the analyzer for verification, and ensure that the disassembly process does not affect the analyzer.

8.9.3 Commissioning and Verification

Note

1. Before starting the analyzer, check whether the screws are tightened, and whether the wires of each connector interface on the side of the board are missing or loose.
2. Make sure that other useless tools (metals such as screws) will not be scattered on the board, which may otherwise cause a short circuit.

Required tools

Software update package

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

1. Start the analyzer normally
2. Log in the software at the service access level.
3. Calibrate the CBC gain. Select **Calibrate-Gain Calibration-CBC Gain Calibration**, enter the MCV_G target value of the batch of calibrators used, and **enter the MCV_G target value of the calibrator** as shown in the figure. Perform OV-WB counting continuously with the calibrators until OK is displayed in the **Results** column for both MCV and HGB, as shown in Figure **CBC Gain Calibration Results**. When exiting the CBC gain calibration screen, select **Yes** to save the gain in the pop-up prompt box.

Mode CT-WB

	Target	1	2	3	Result
MCV_G					
HGB	4.40				

Subsequent Requirements

1. Start the analyzer and check whether an alarm is generated during startup.
2. Take a tube with fresh blood, select the **OV-WB-CD** mode, and after startup, confirm whether the measurement process alarms.
3. Take the same tube with fresh blood on the reference machine to compare whether the measurement results are consistent with the measurement results of the reference machine.

8.10 Sample Type Detection Sensor

8.10.1 General Information

Figure8–25 Capacitive sensor

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
009-014500-00	Capacitive sensor wire	N/A	N/A

Functions:
Identify and distinguish venous blood tube and capillary blood tube.

8.10.2 Disassembly and Assembly

Note:

Before disassembly, power off and unplug the power supply.

Required tools:

Cross-headed screwdriver
Allen wrench

Required Tooling and Fixtures

/

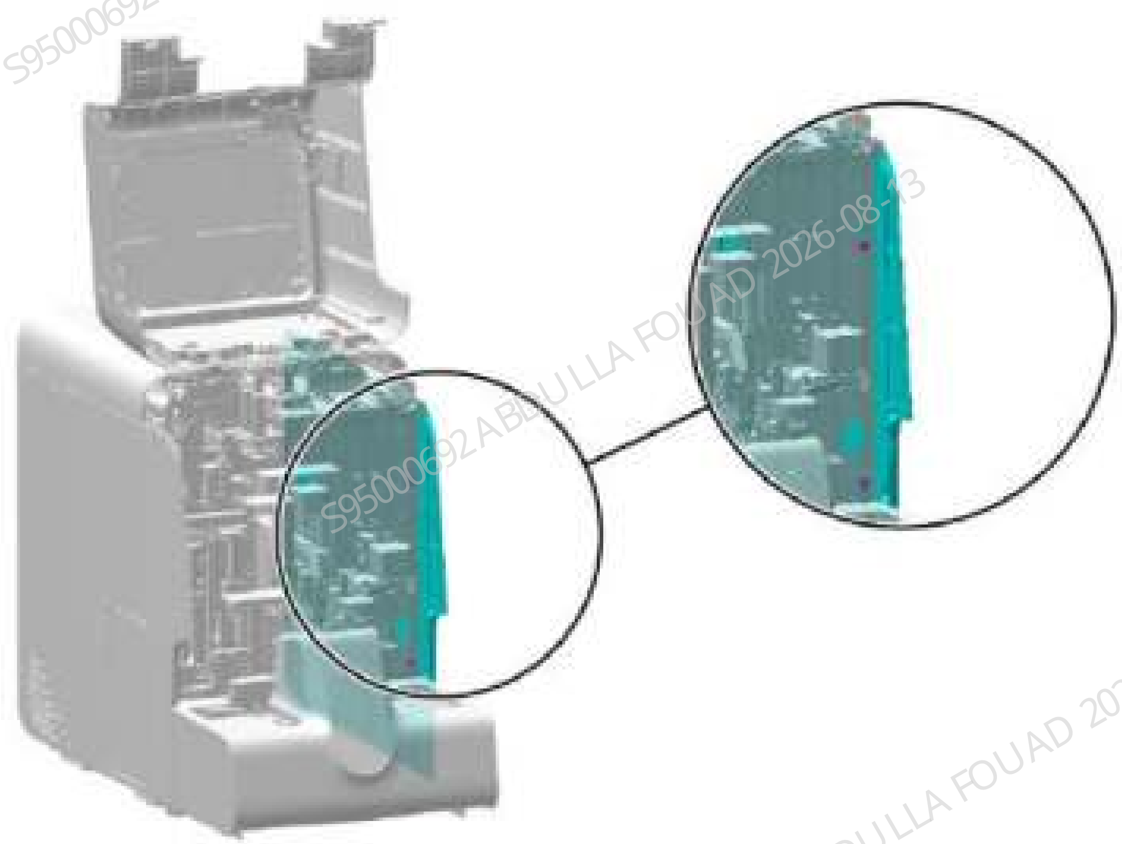
Pre-requirement

Power off the analyzer, and unplug the power cord.

Steps:

1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–26 Diagram of opening the front cover



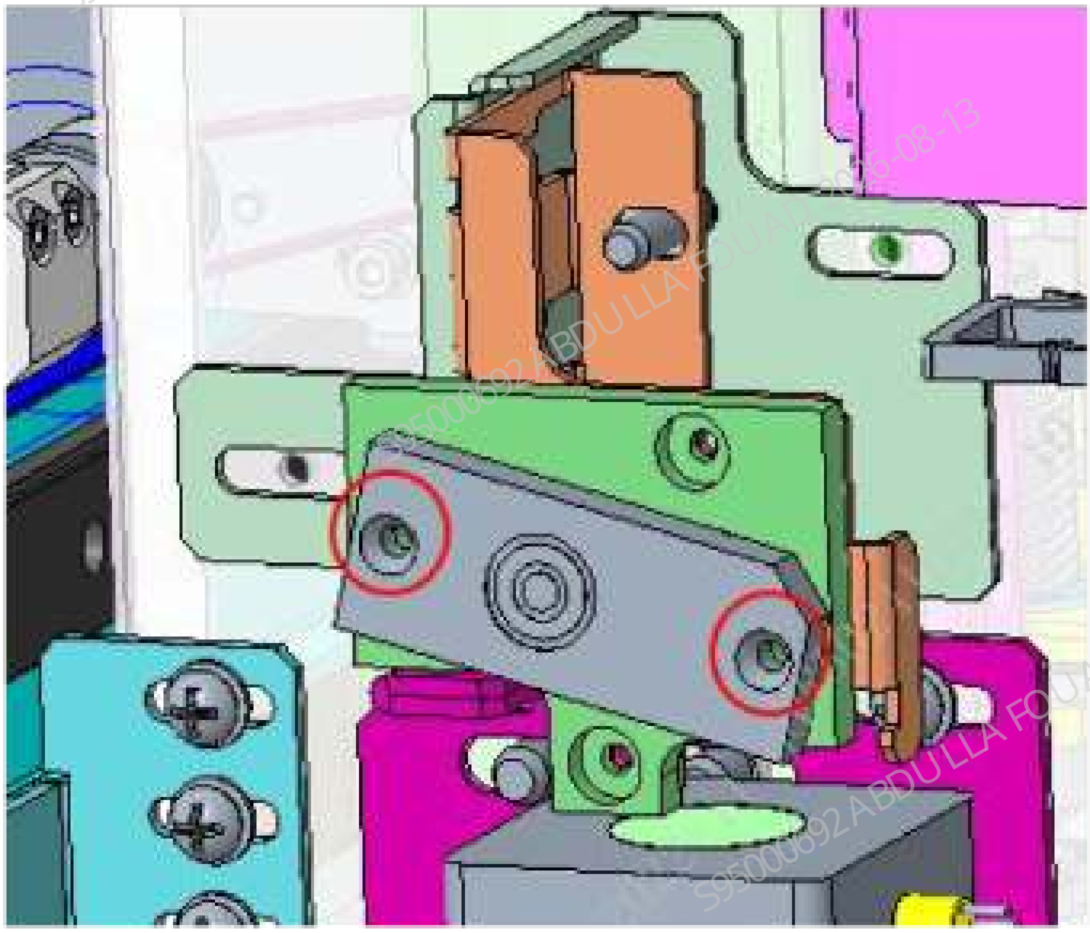
2. Unplug the wire from the capacitive sensor, as shown in the figure below.

Figure8–27 Position diagram of the capacitive sensor wire plug

1 Capacitive sensor wire plug

3. As shown in the figure below, remove the two screws fixing the capacitive sensor and remove the capacitive sensor from the fixing plate.

Figure8–28 Diagram of removing the sensor screw



Subsequent Requirements

Installation procedure:

1. Install the new capacitive sensor on the fixing plate using two screws, as shown in the figure below.

Figure8–29 Diagram of the sensor locking screw

1 Screw

2. Connect the wire terminal of the capacitive sensor. For the connection position, see **Figure8–27**.
3. Power on the analyzer and confirm that the new capacitive sensor is in good conditions (refer to the capacitive sensor commissioning and confirmation sections for specific steps).
4. Restore the right cover and front cover.

8.10.3 Commissioning and Verification

Precautions

N/A

Required tools:

M1 slotted head screwdriver (one is configured for each sensor material package, so it does not need to be prepared)
Cross-headed screwdriver
Blood routine tube rack
Evacuated blood collection tube
Capillary blood tube

Prerequisites

N/A

Steps

1. Select**Service>Commissioning & Self-Test>Sensor Commissioning** in the menu, and select the **Sample Type Detection Sensor** interface, as shown in the following figure.

Figure8–30 Sample type detection interface

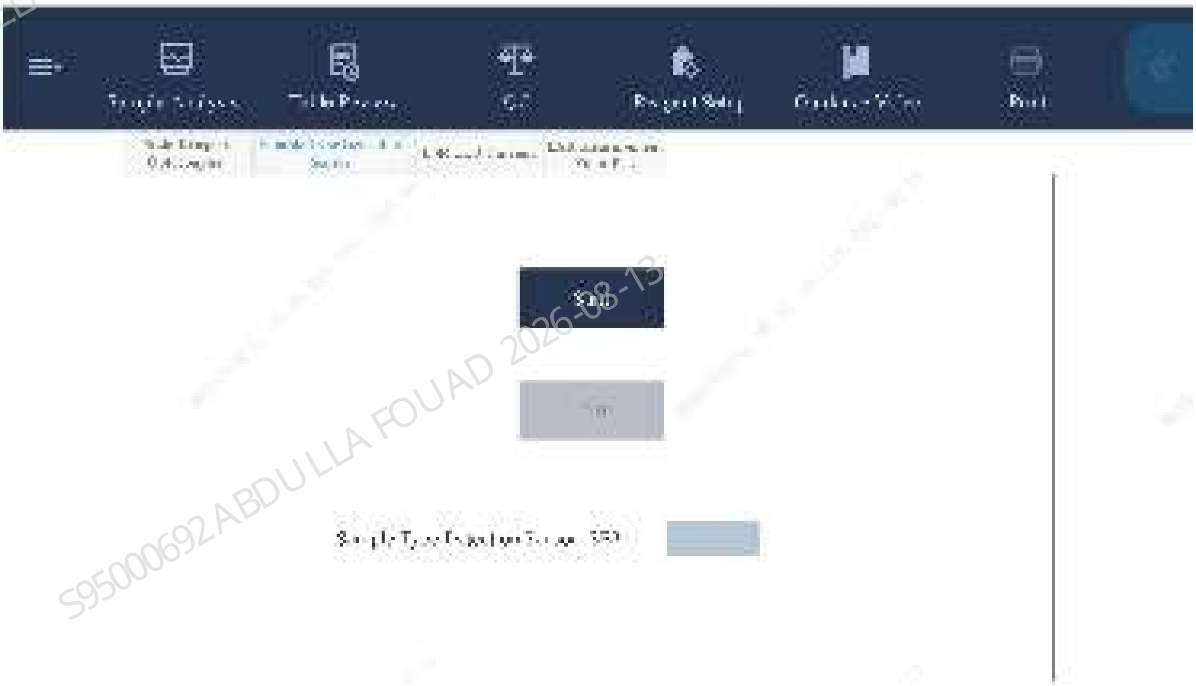


Figure8–31 Sample type detection interface

2. Select **Start**, and operate according to the prompt shown on the following figure. The gripper will clamp the evacuated blood collection tube to the sample detection sensor.

NOTICE

Here use the venous blood tube commonly used for the client

Figure8–32 Diagram of prompting to place the tube

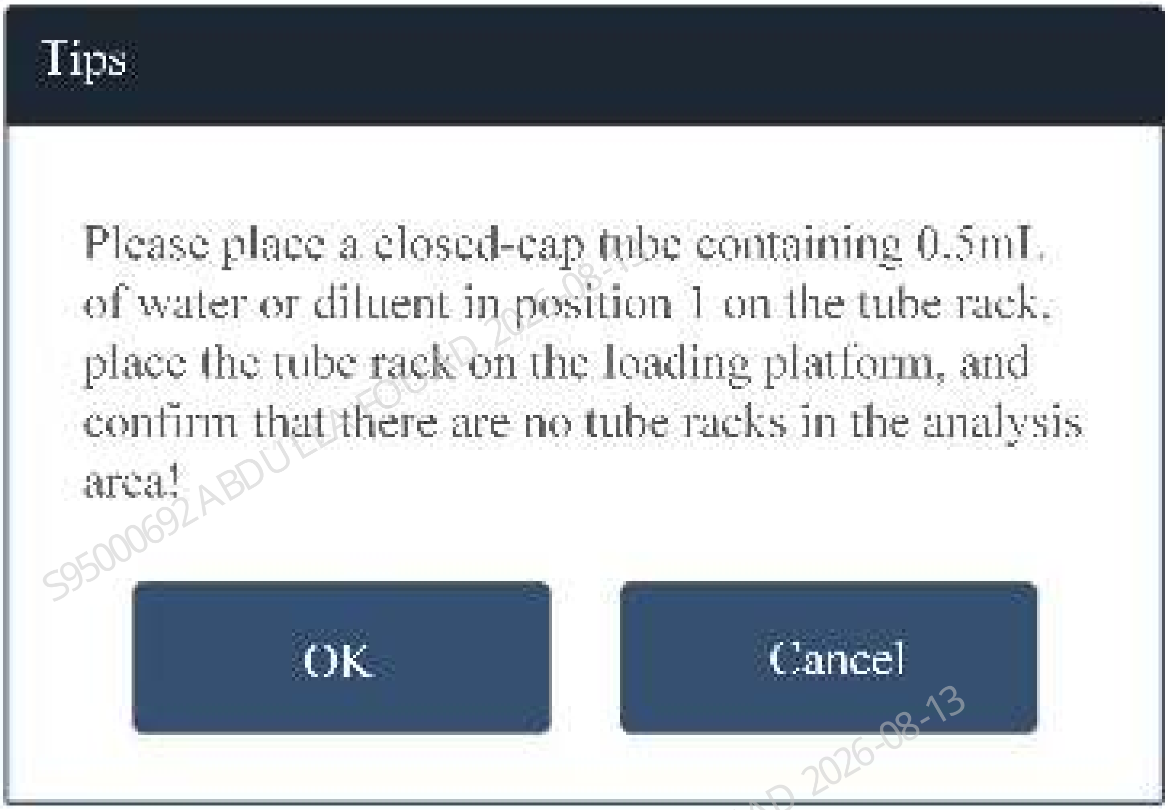
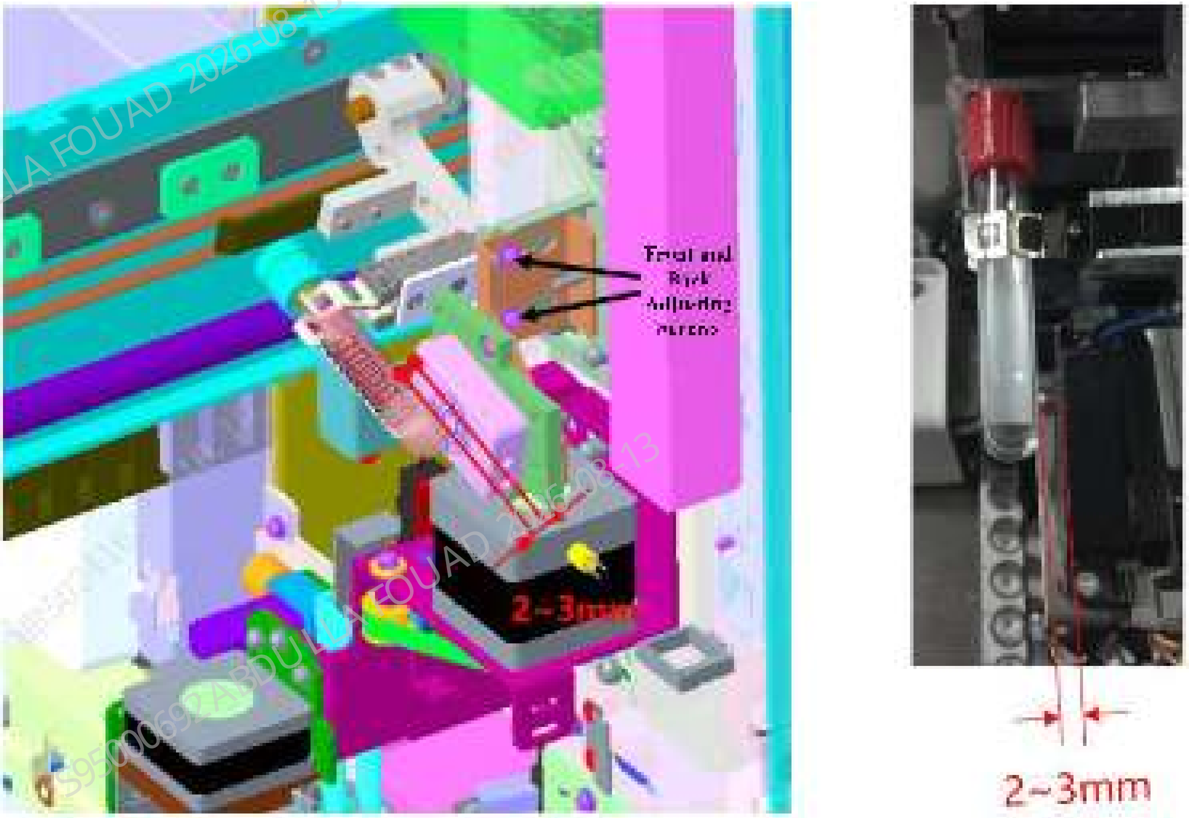


Figure8–33 Diagram of prompting to place the tube

3. As shown in the figure below, adjust the front-back adjustment screw fixing the capacitive sensor, and adjust the front-back position of the sensor to ensure that the gap between the test tube and the capacitive sensor is 2 to 3 mm.

Figure8–34 Front-back position commissioning diagram



4. As shown in the figure below, push the evacuated blood collection tube containing 0.5 ml of water or diluent up to the mixing gripper pressing plate.

NOTICE

Do not swing the gripper when pushing; the green light indicates the connected status, and the orange light is an indicator.

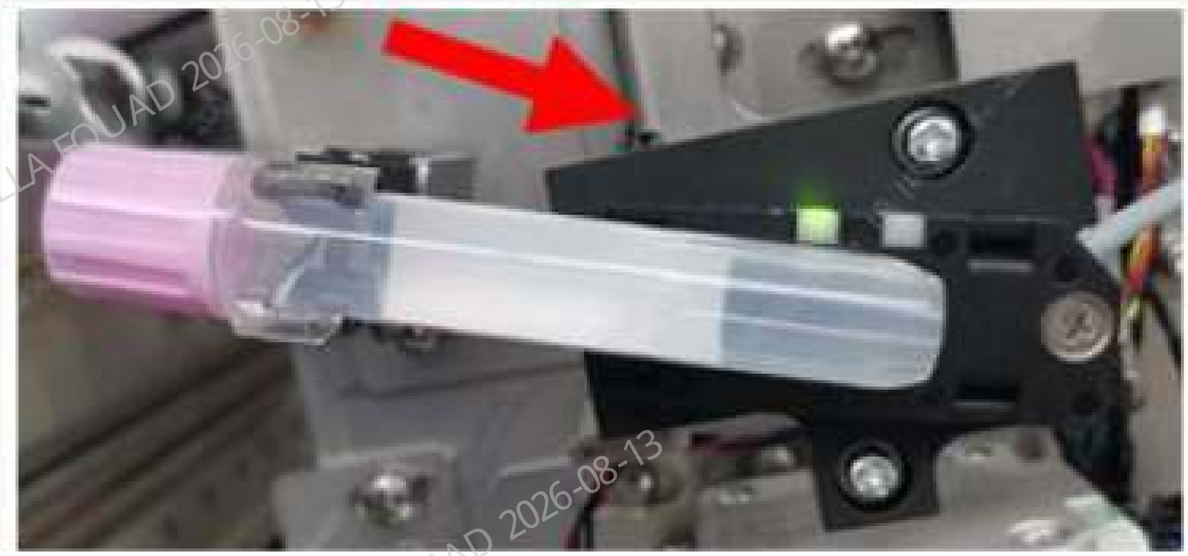
Figure8–35 Tube position commissioning diagram

5. As shown in the figure below, use the slotted head screwdriver provided with the sensor to adjust the sensor knob. If the orange indicator of the sensor is on in the initial state, rotate the screwdriver anticlockwise to turn the indicator off first. Then, adjust the knob clockwise until the orange indicator of the sensor is turned on, and then turn it by 90°.

Figure8–36 Sensitivity adjustment diagram of the venous blood tube

6. Replace the evacuated blood collection tube with an AL Micro-WB tube filled with 0.5 ml water or diluent. Then, push the AL Micro-WB tube downward to the limit, and observe whether the orange indicator is on.

Figure8–37 Capillary blood tube detection board



7. If the orange indicator is turned on after the last step, as shown in the figure below, use the slotted head screwdriver to adjust the sensor knob so that the orange indicator of the sensor goes out.

Figure8–38 Sensitivity adjustment of the capillary blood tube

8. Make sure that the orange indicator on the sensor meets the requirements in step 4 and step 5 when a vacuum blood collection tube is used, and meets the requirements in steps 6 and 7 when an AL Micro-WB tube is used.
9. Tap Exit. Pull out the autoloader after the instrument stops action.
10. Select to enter the **Service>Commissioning and Self-Test>Self-Test>Autoloading Assembly** interface, place the venous blood tube and the capillary blood tube in the tube rack, put the tube rack on the loading tray, perform **Sample Type Detection**, and confirm that the detection results meet the actual situation, as shown in the following figure.

NOTICE

V is the abbreviation of venous blood and indicates the venous blood sample, namely, the evacuated blood collection tube; C is the abbreviation of capillary blood, which indicates the capillary blood sample, namely, the Micro-WB tube; (Empty) - it means there is no tube at this position

Figure8–39 Sample Type Detection Result Display

Figure8–40 Sample Type Detection Result Display

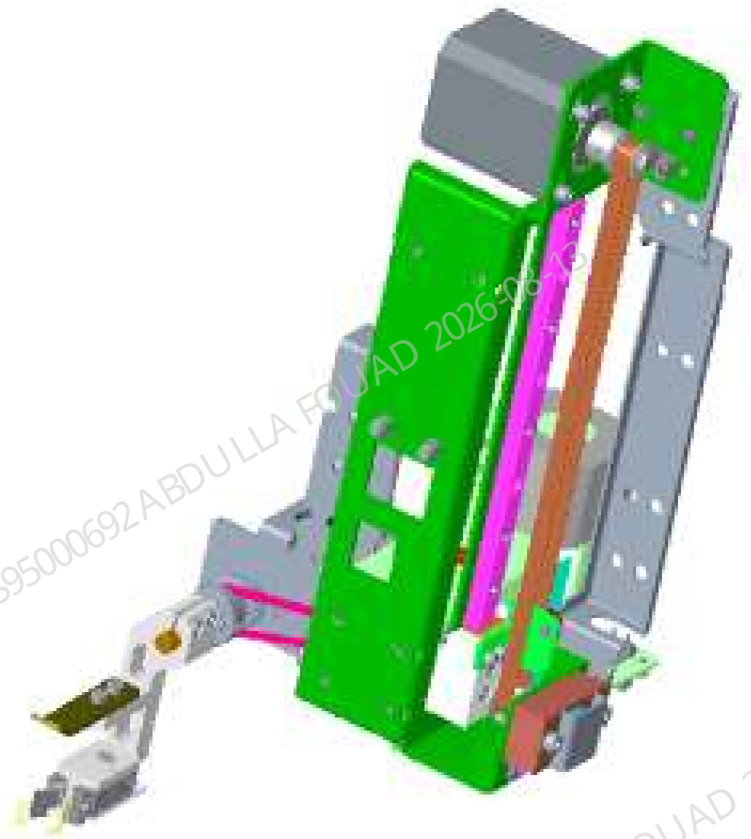
Subsequent Requirements

/

8.11 Sample mixing assembly

8.11.1 General Information

Figure8–41 Photo of Blending Assembly



FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-091723-00	Blending assembly (with the rotary scanning and blending motor photocoupler line)	115-052166-00	Photoelectric sensor wire terminal
		011-000021-00	PHOTOELEC Optical Sensor IP55 940nm 15cm
		011-000297-00	Photoelectric sensor wire terminal
		115-052164-00	Tube gripper
		024-000366-00	Motor Step 3.6V 1.8 Degree/Step Hold Torque 0.53Nm

Functions:

Mixing the blood sample to make the serum and blood cells mixed evenly.

8.11.2 Disassembly and Assembly

Note:

Before disassembly, power off and unplug the power supply.
When unscrewing the last set screw of the mixing assembly, protect the mixing assembly from falling by accident by holding it with your hand.

Required tools:

Tweezer x 1
Allen wrench x 1

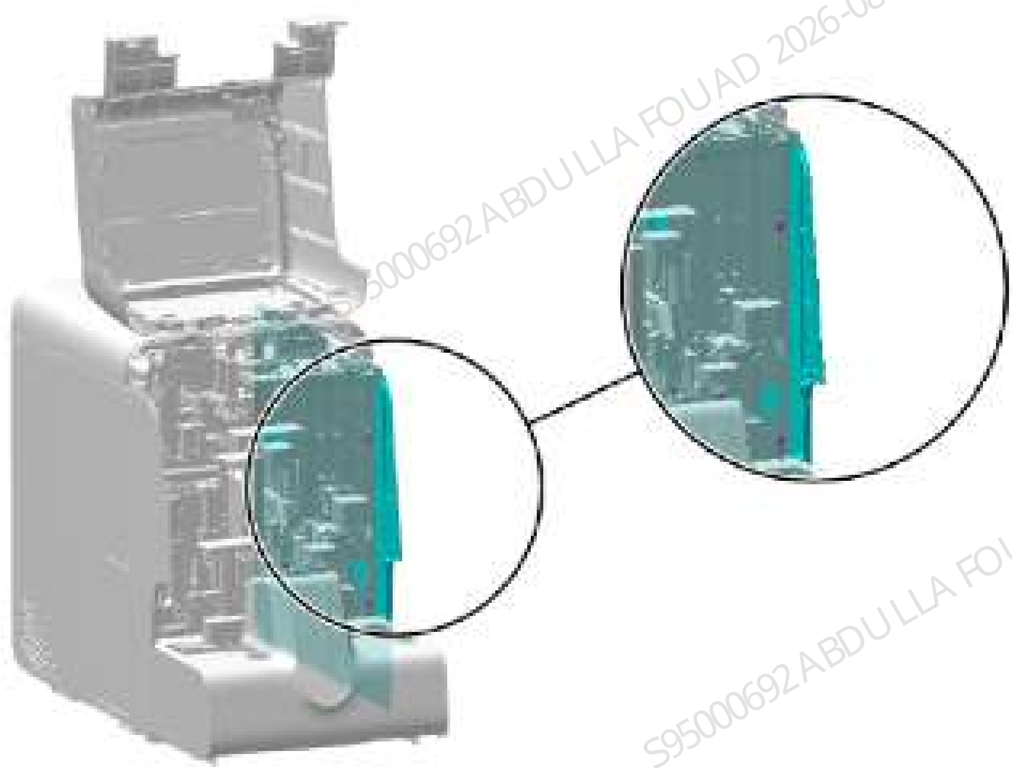
Pre-requirement

Power off the analyzer, and unplug the power cord.

Steps:

1. Flip the front cover up, and put it against the top cover. Unscrew the two set screws of the right cover, as shown in the figure below.

Figure8–42 Removing the right cover

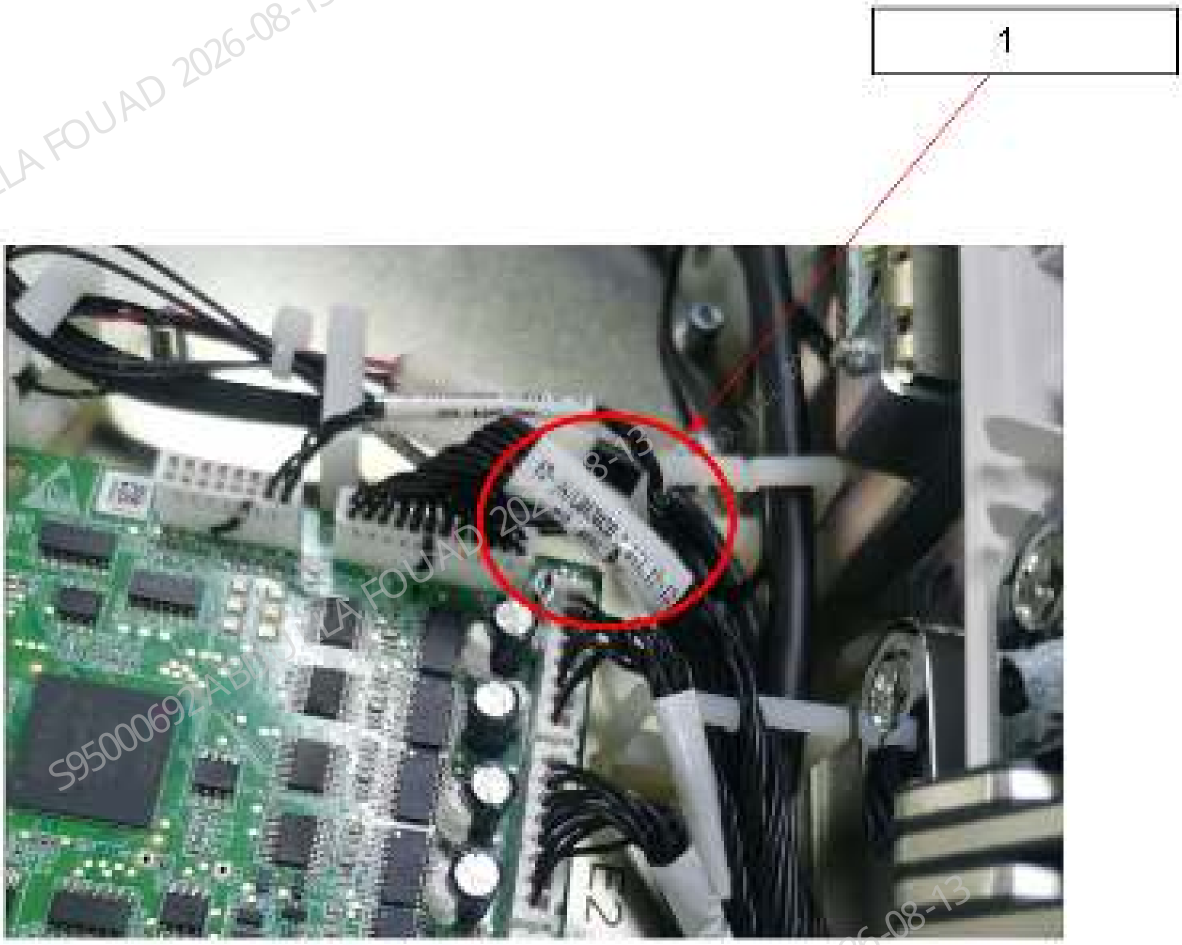


2. Unplug the wires from the mixing assembly, as shown in the figure below.

Figure8–43 Removing wires from mixing assembly

3. Check the No. of the wire connected at the board end. If the wire No. is C013020-00, remove this wire from the board end, as shown in the figure below. If the wire No. is C014879-00, the wire does not need to be removed.

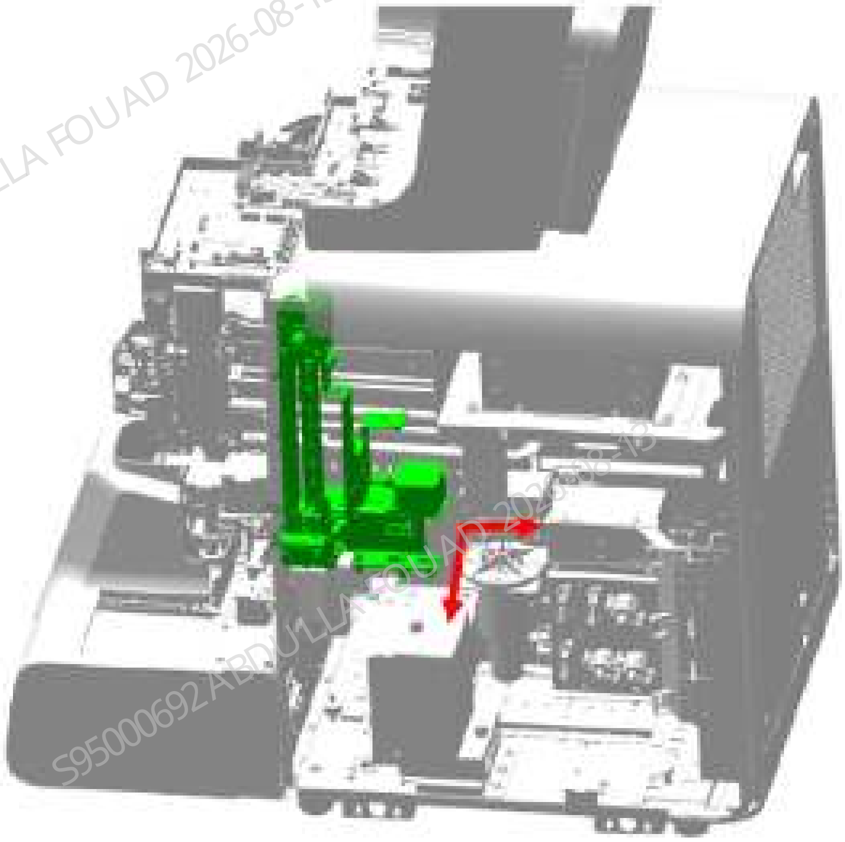
Figure8–44 Check the FRU code of the paired wire, AL board to CB motor, M end



1 Check the FRU code of wire

4. Unscrew the four set screws of the mixing assembly with a cross screwdriver.
5. Move the mix assembly backward in the direction of the arrow shown in the figure below, and remove the assembly from the right side of the analyzer.

Figure8–45 Schematic diagram 2 of removing the mixing assembly

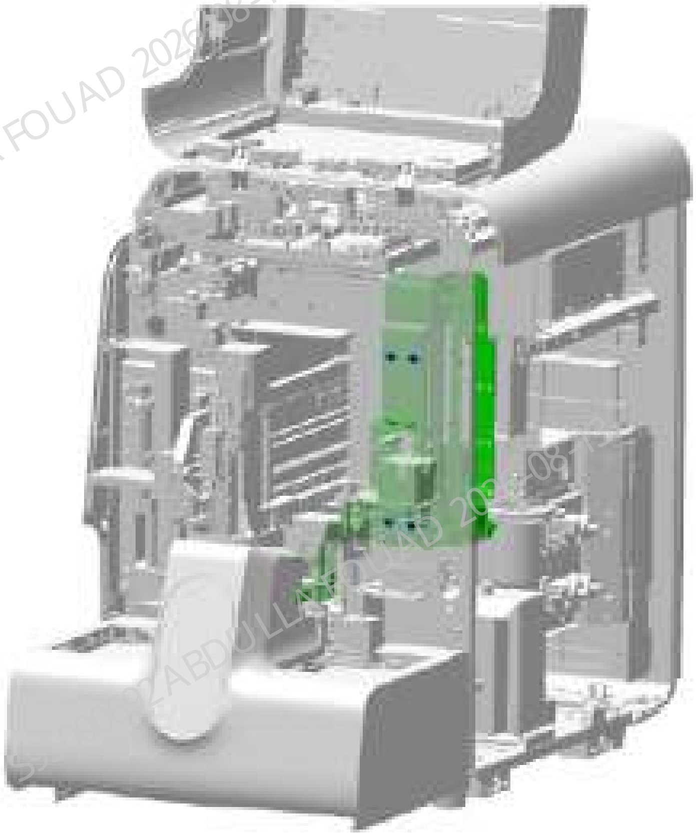


Subsequent Requirements

Installation procedure:

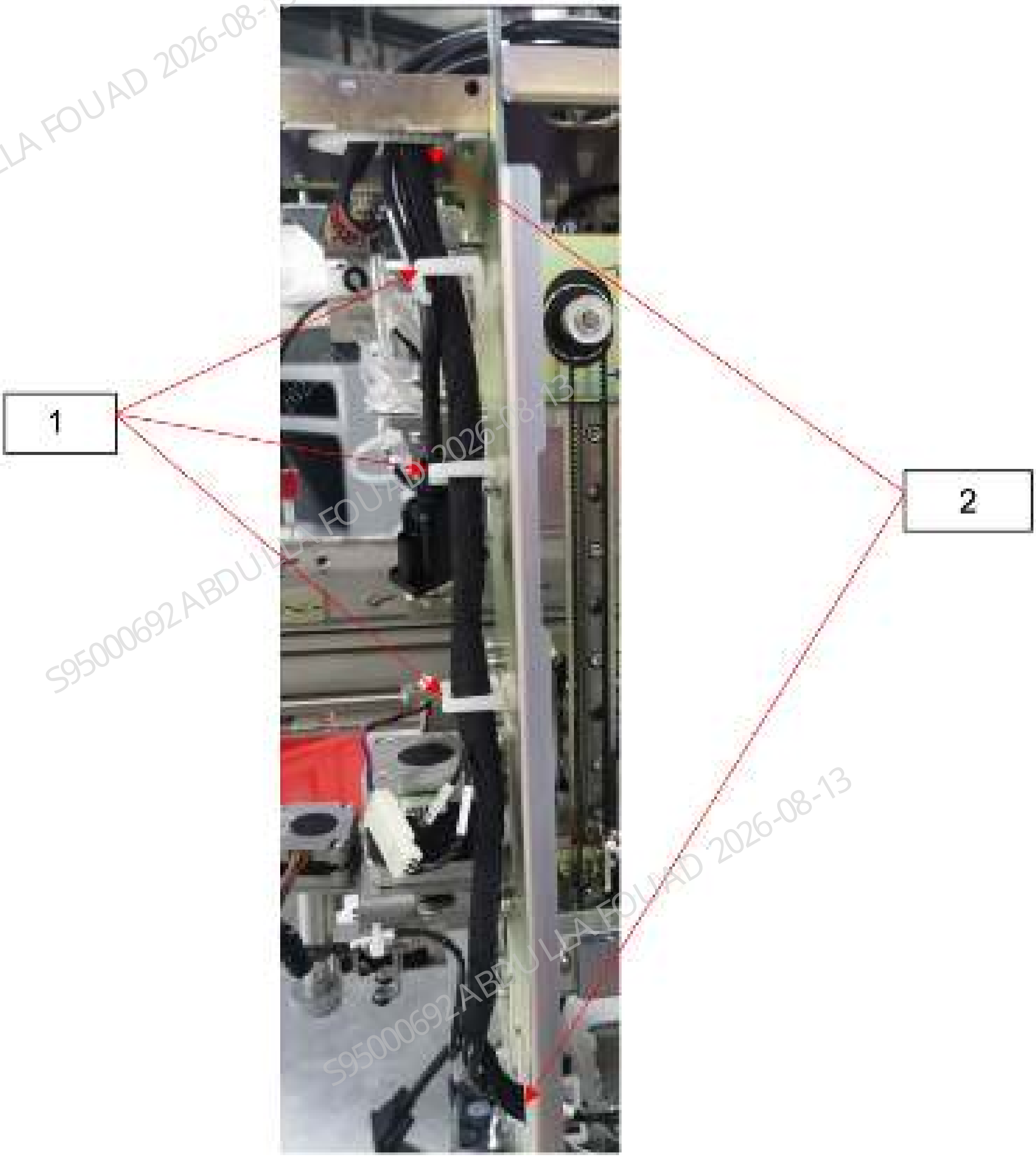
1. Pre-tighten the mixing assembly to the analyzer with four screws at the position shown in the figure below.

Figure8–46 Schematic diagram 1 of removing the mixing assembly



2. If the code of the paired wire, AL board to CB motor, M end, is C013020-00, you need to replace the wire with C014879-00 and arrange the wire correctly according to the original wiring method. As shown in the following figure.

Figure8–47 Cabling diagram



1 Wire harness retainer

2 Cabling hole

3. Splice the blending assembly wire with the board end wire.
4. Refer to the commissioning section for the commissioning method. After commissioning, tighten the four set screws of the mixing assembly.
5. Install the right cover to the analyzer.

8.11.3 Commissioning and Verification

Required tools:

One cross-headed screwdriver
Allen wrench x 1

Steps:

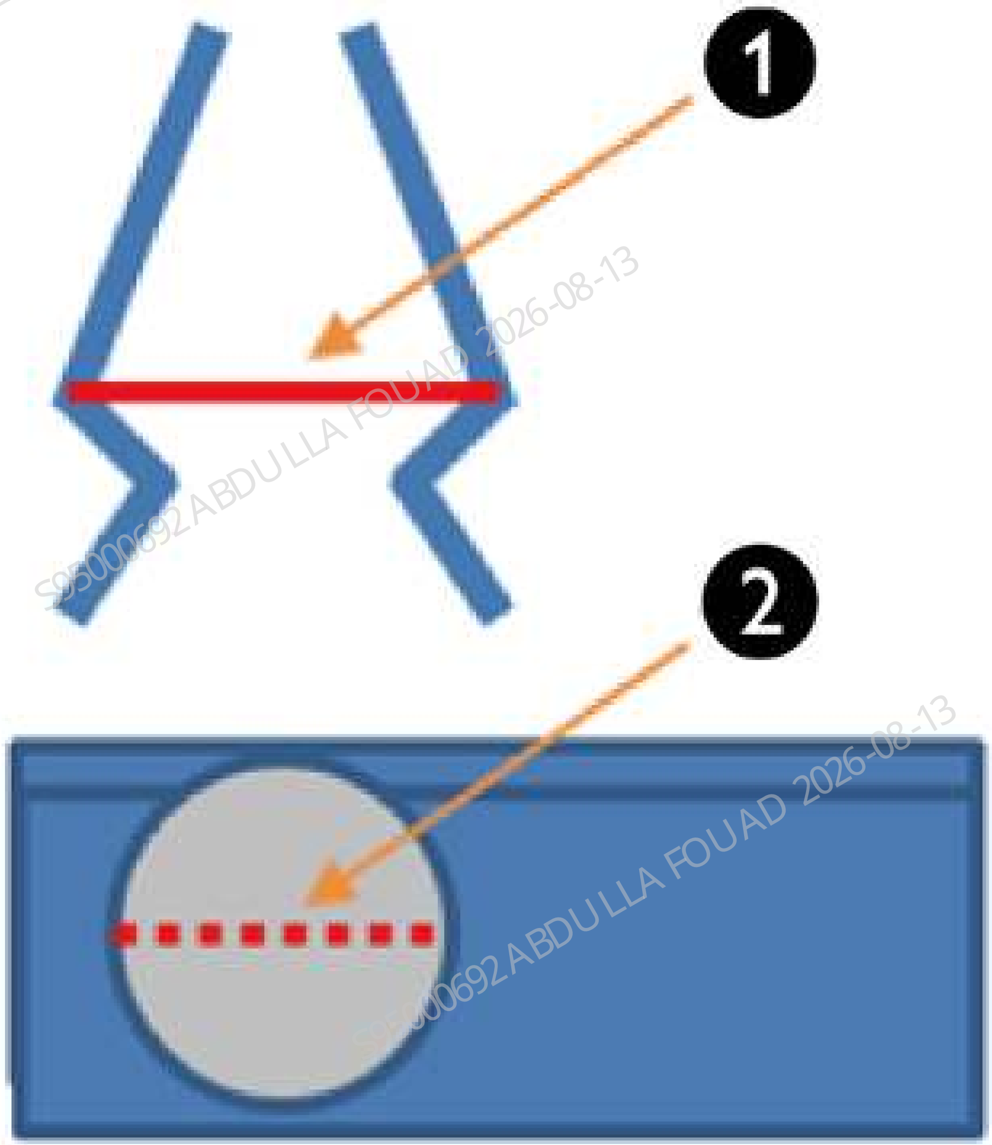
After removing and installing the blending assembly again, readjust the positions of the gripper relative to the front, back, left, right, and R directions of the tube rack and its height. Meanwhile, adjust the front and back positions of the gripper relative to the capillary blood mixing position and the gap between the gripper and the anti−collision block. Adjust the mixing gripper relative to the tube rack

- Adjust the mixing gripper relative to the tube rack
  Put an empty tube rack into the lateral feeding area, and make the first test tube position coincide with the rotary scanning position, select **Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.**>>**Mixing gripper (front position)**, click **Check by step**, click **Feeding**, feed test tube position No.1 of the tube rack to the mixing position, and click **Done**.

1. Click **To Adjust Pos.** to observe whether the gripper is vertical. If not, click **Left** and **Right** to make it parallel and complete the R-direction adjustment of the gripper relative to the tube rack.

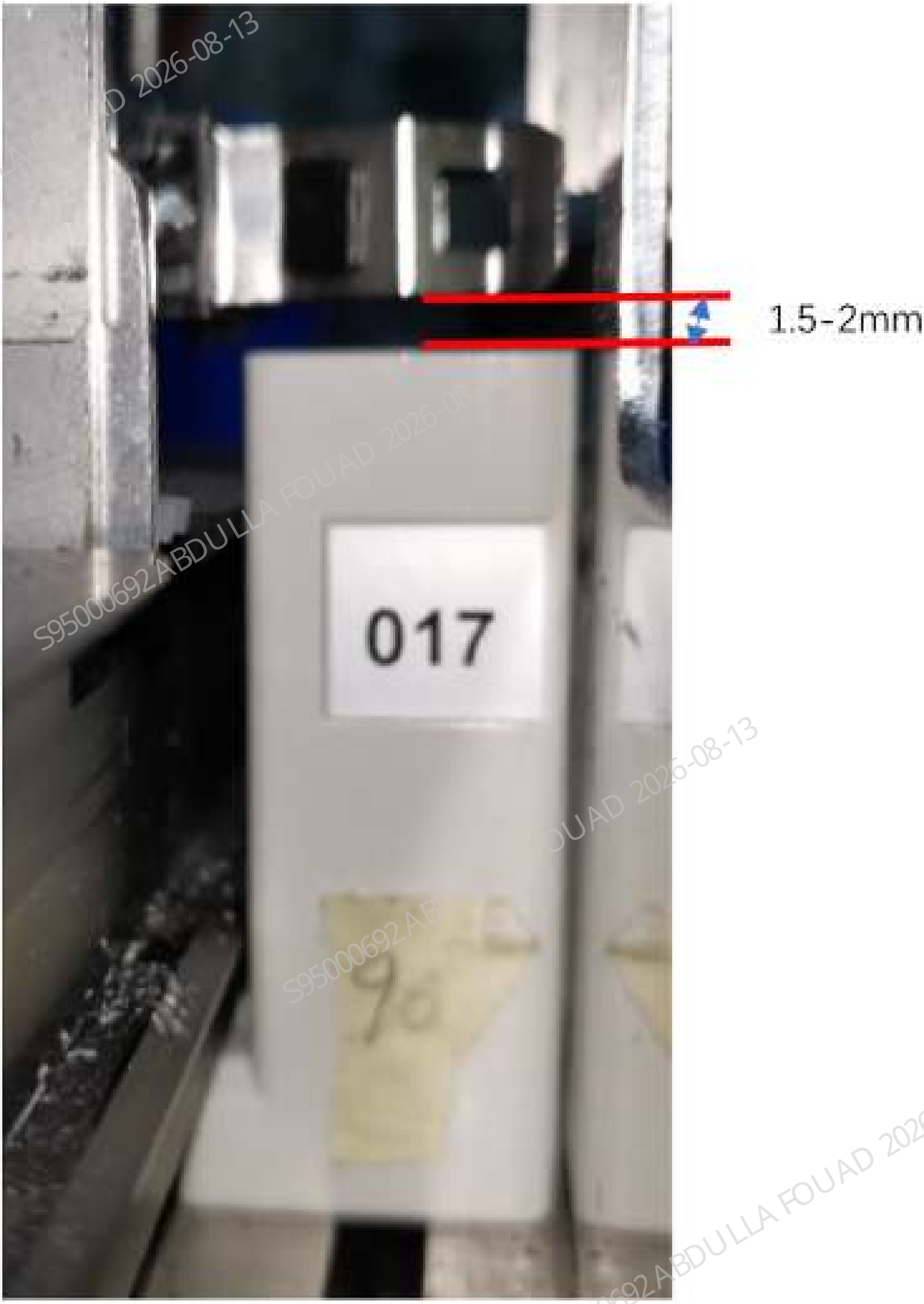
2. 1. After the adjustment in the R direction is completed, adjust the installation position of the blending assembly as a whole so that the clamping area of the gripper is located in the center of the test tube position in the tube rack (as shown in the above figure). After the gripper relative to the left and right positions of the tube rack is adjusted, lock the fixing screws of the blending assembly.
3. Observe whether the virtual connecting line in the clamping area of the two gripper pieces coincides with the hole diameter of the tube rack, as shown in the following figure (the red solid line should coincide with the red dotted line). If not, click **Front** and **Back** to make them coincide, and complete adjustment of the position of the mixing gripper relative to the front and

back of the tube rack. If the requirement is not met yet after the adjustment exceeds the limits of **Front** and **Back**, loosen the fixing screws of the mixing gripper and adjust the front and back positions of the gripper. During commissioning, the R direction position of the gripper may change. After repeatedly confirming that the R direction and the front and back positions meet the requirements, lock the fixing screws of the mixing gripper.



1	Virtual connecting wire of gripper	2	Diameter of the test tube hole in X direction

4. Looking at the feeding area from the unloading area, you can see the size of the lower edge of the gripper from the upper surface of the tube rack, click **Up** and **Down** to make the lower edge of the gripper 1.5 to 2 mm from the upper surface of the tube rack, as shown in the figure below, and complete the Z-direction adjustment of the front position of the mixing gripper.

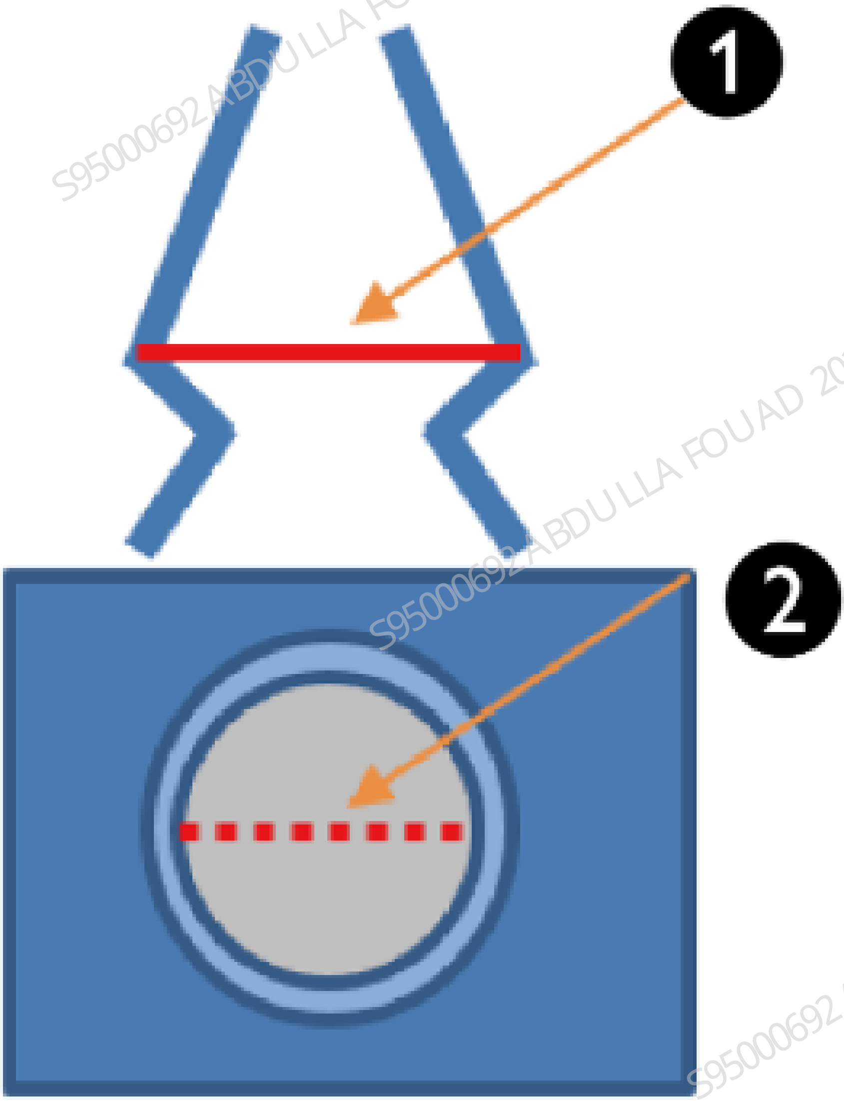


5. Select **Save the number of steps**>>**Exit** to complete the position adjustment of the gripper relative to the tube clamp.

6. Place the allowed test tube in the test tube position of the tube rack that is not fed into the rotary scanning area, click **Check by step** to feed the test tube to the mixing position, click **A&M**, and check whether the test tube interferes with the tube rack and other parts in the process of being clamped, mixed, and put back by the gripper. If yes, make adjustment again.

- Adjusting the mixing gripper at the position of capillary blood blending assembly (applicable when capillary blood mixing is configured)

1. Put an empty tube rack into the lateral feeding area, and make the first test tube position coincide with the rotary scanning position, select **Service>>Commissioning & Self-Test>>Adjust Autoload Pos.>>Mixing gripper (middle position)**, click **Check by step**, click **Feeding**, feed test tube position No.1 of the tube rack to the mixing position, and click **Done**.
2. Click **To Adjust Pos.**, observe whether the virtual connecting line of the two gripper pieces coincides with the X-direction diameter of the micro blood mixing chamber (the red solid line should coincide with the red dotted line), as shown below. If not, click **Front** and **Back** to make them coincide.

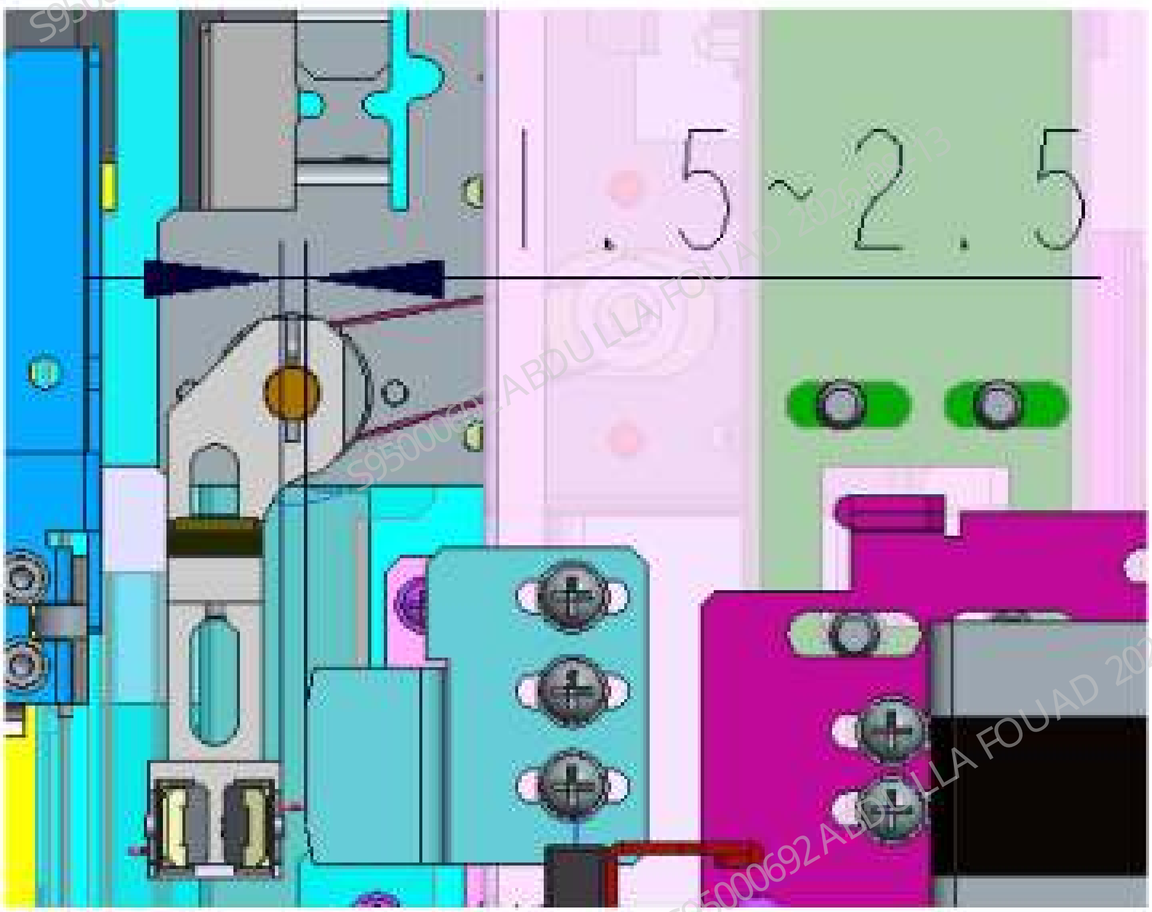


1	Virtual connecting wire of gripper	2	X-direction diameter of the micro blood mixing chamber

3. Select **Save the number of steps**>>**Exit** to complete adjustment of the mixing gripper at the position of capillary blood blending assembly.
4. Place the allowed micro tube in the test tube position of the tube rack that is not fed into the rotary scanning area, click **Check by step** to feed the test tube to the mixing position, click **A&M**, and observe whether the test tubes can be taken out of the tube rack and the compartment of capillary blood blending assembly position stably, and whether there is any interference in the process of putting the tubes into the rack. If it is unstable or there is interference, make adjustment again.

- Gap adjustment between the mixing gripper and anti-collision block

1. Select **Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.**>>**Mixing gripper (front position)**, click **Check by step**, and select **Initialize**>>**Gripper Forward**>>**To Center**. As shown in the following figure, adjust the gap between the mixing gripper and the anti-collision block by 1.5 to 2.5 mm (two slides can be clamped between the gripper and the anti-collision block for auxiliary adjustment).



2. After commissioning is completed, lock the anti-collision block screws, click **Initialize** to reset the gripper.

8.12 ESR horizontal assembly

8.12.1 General Information

Figure8–48 Photo of the ESR horizontal assembly

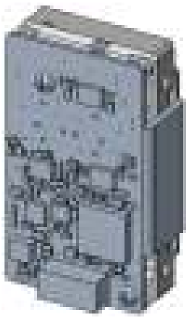


Table 8–8 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-090546-00	ESR horizontal assembly	N/A	N/A

Functions:
Measure the erythrocyte sedimentation rate of the whole blood sample in the sampling tube.

8.12.2 Disassembly and Assembly

Note:

N/A

Required tools:

One cross-headed screwdriver

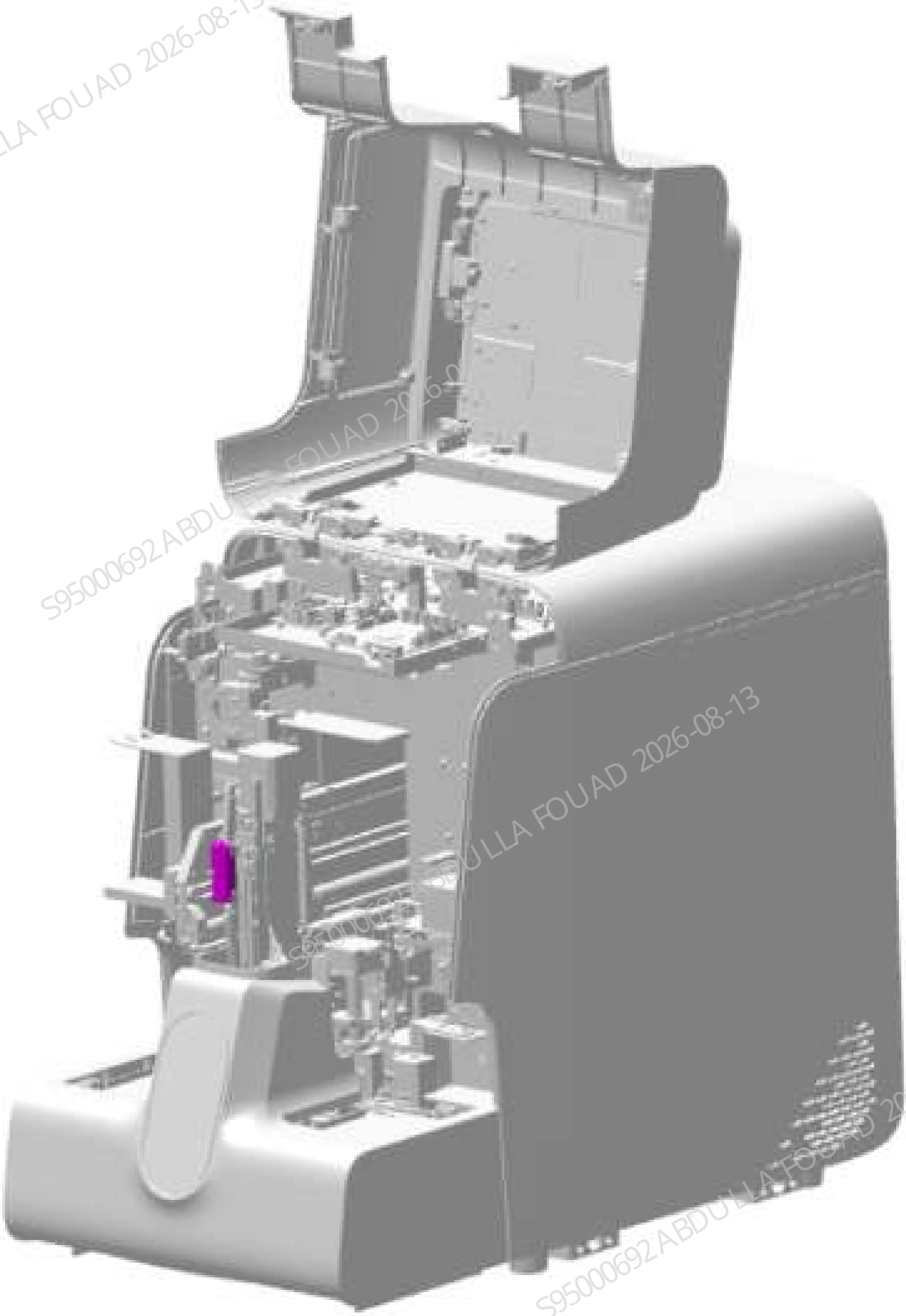
Pre-requirement

Power off the instrument and remove the power plug.

Steps:

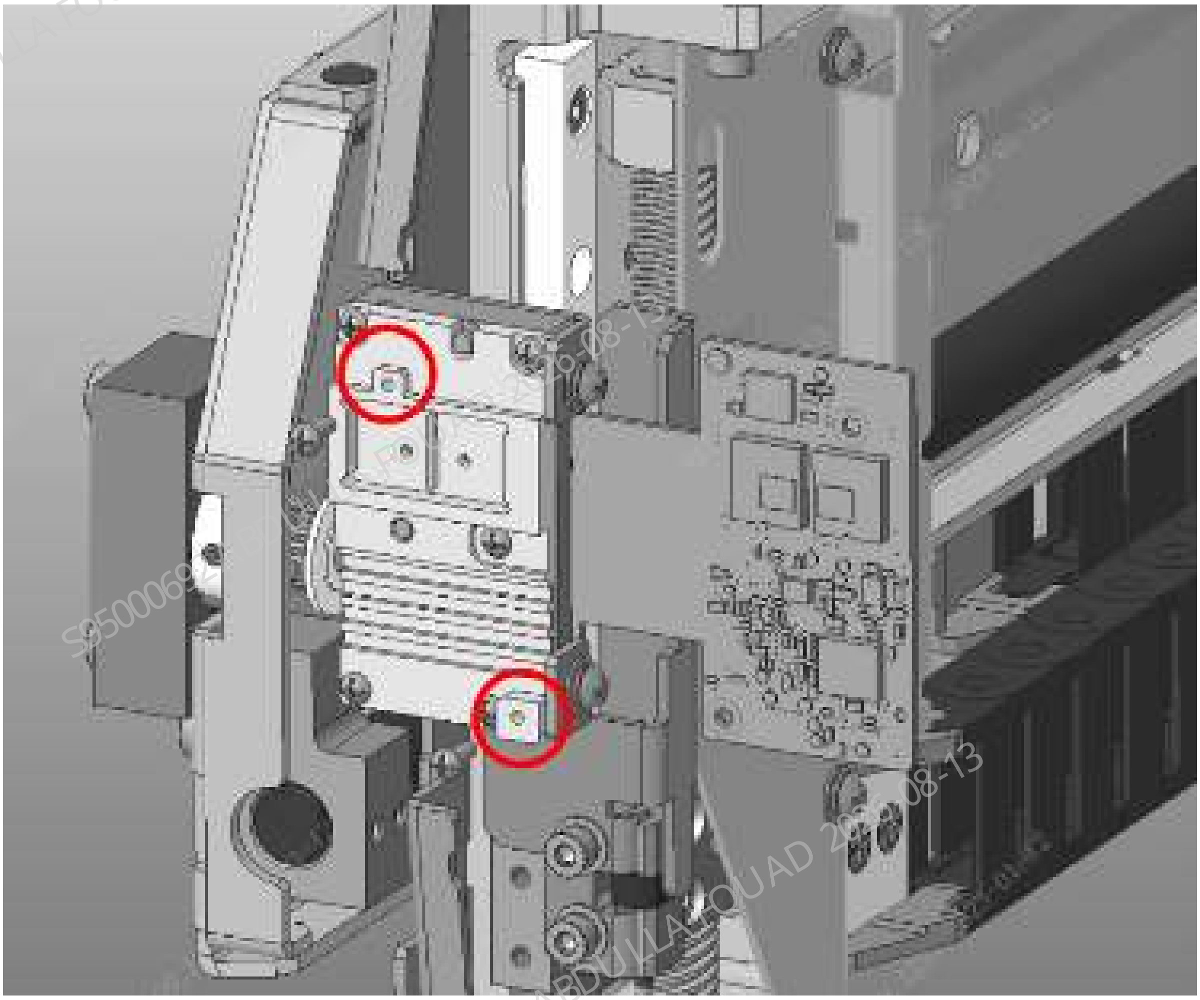
1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–49 Position of ESR measurement assembly



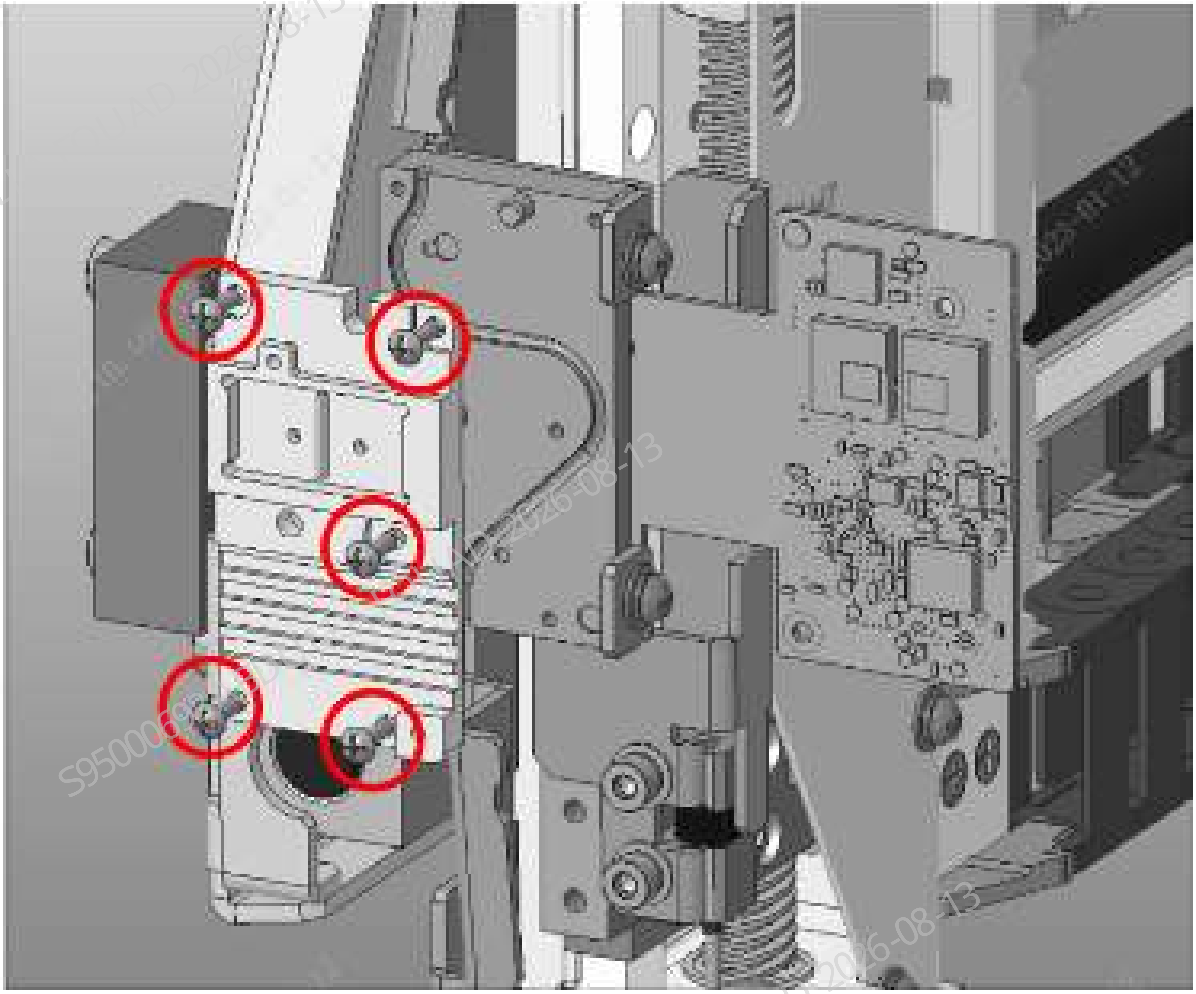
2. Pull out the wire plug on the ESR flexible board and loosen the set screws of the ESR flexible board. As shown in the figure below, unfold the flexible board to expose the ESR PD fixation block.

Figure8–50 Positions of set screws of ESR flexible board



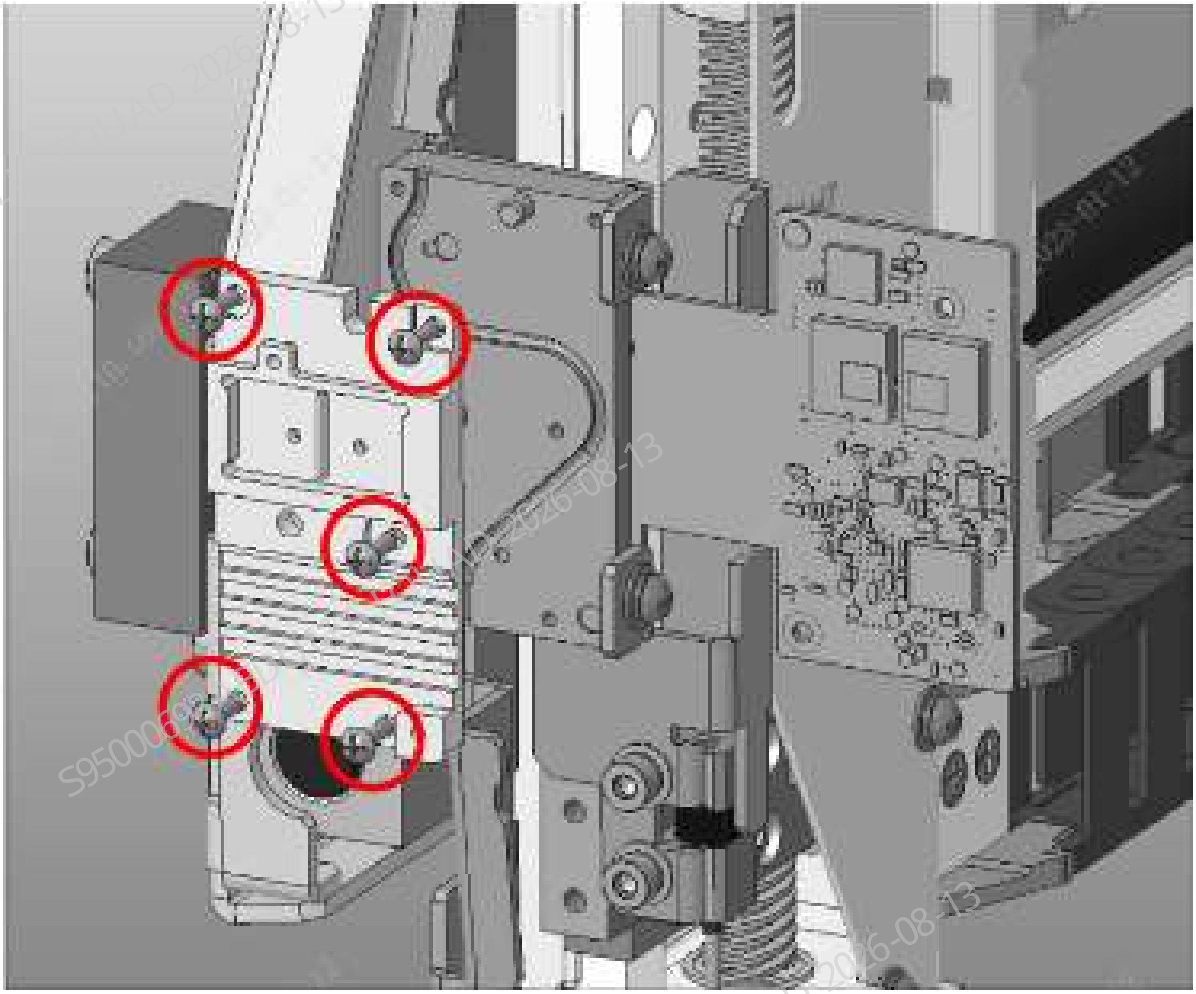
3. Loosen the set screws of the ESR PD fixation block, as shown in the figure below, and remove the PD fixation block.

Figure8–51 Positions of set screws of ESR PD fixation block



4. Take the sampling tube out of the ESR LED fixation block, loosen the screw of the ESR LED fixation block, and remove the rest of the ESR measurement assembly, as shown in Figure **Figure8–50**.

Figure8–52 Positions of set screws of ESR PD fixation block

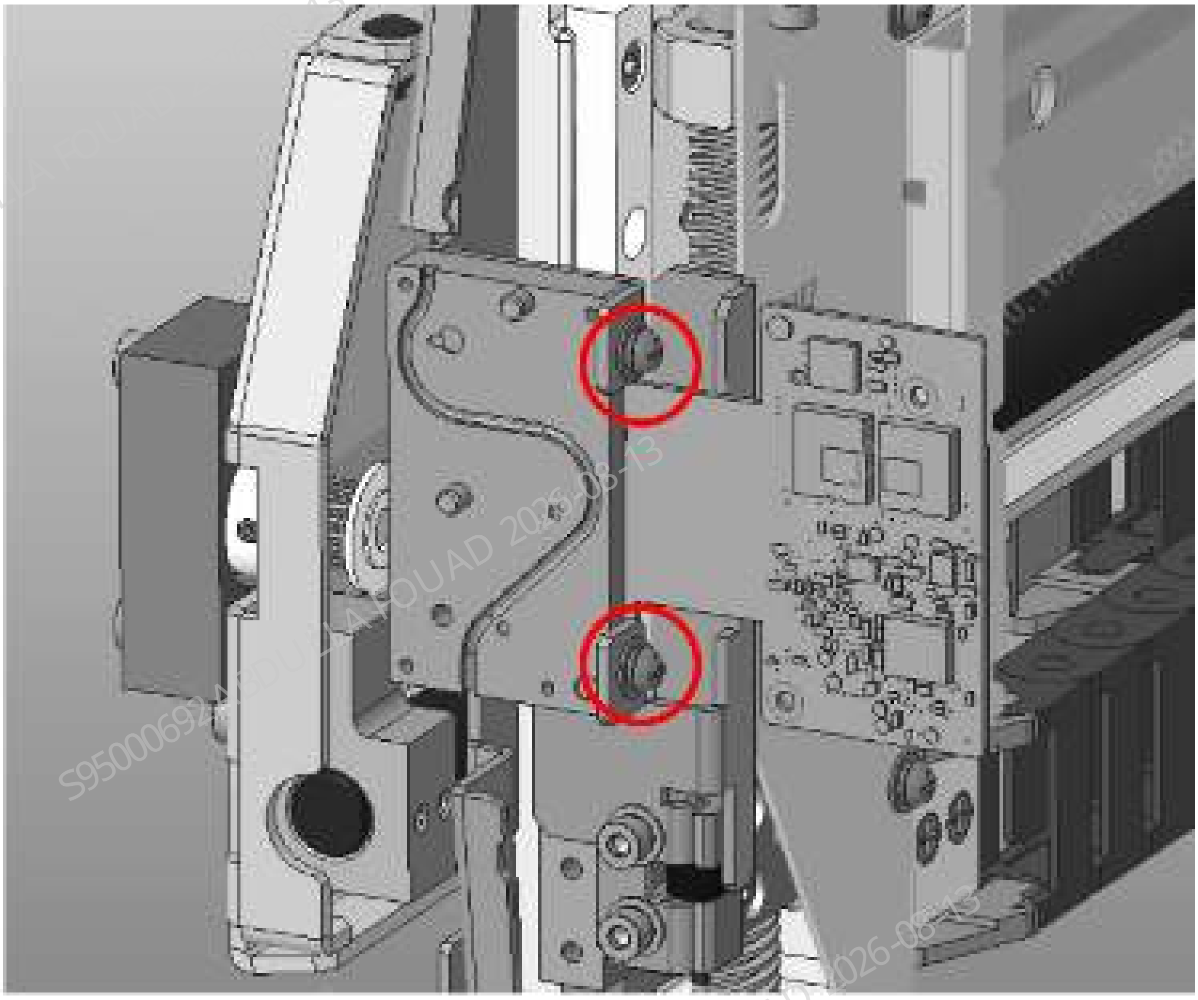


Subsequent Requirements

Installation procedure:

1. Fix the ESR LED fixation block and the flexible board to the bracket with two screws, as shown in the figure below.

Figure8–53 Positions of set screws of ESR LED fixation block



2. Install the sampling tube into the slot of the LED fixation block.
3. Install the ESR PD fixation block onto the LED fixation block and fix it with five screws.
4. Secure the flexible board to the PD fixation block using two screws, and connect the wires.
5. Flip down the front cover to reset it.

8.12.3 Commissioning and Verification

Note:

N/A

Required tools:

One cross-headed screwdriver

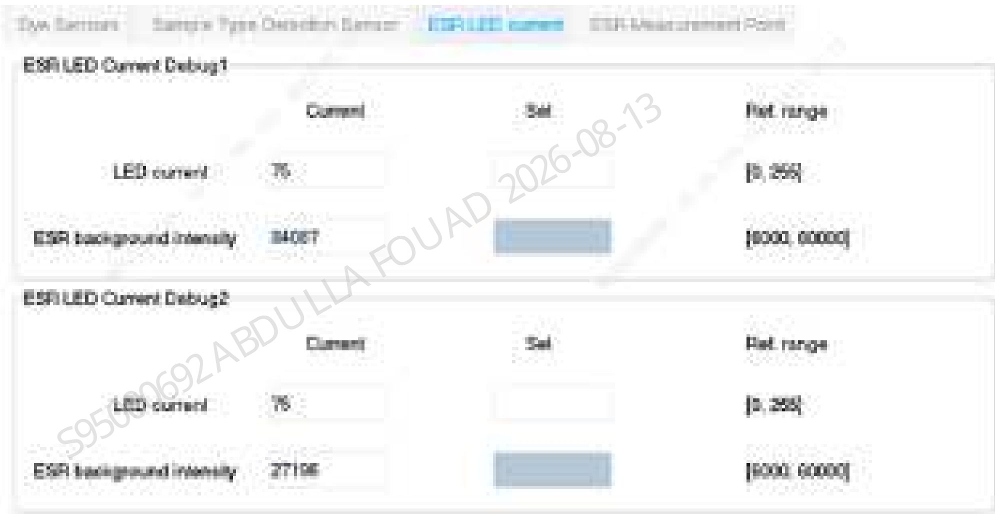
Pre-requirement

N/A

Steps:

1. After the assembly is completed, power on the analyzer, select **Status**>>**Voltage & Current**, and check whether the value corresponding to **ESR background intensity** is within the reference range. If not, select **Service**>>**Commissioning & Self-Test**>>**Sensor Commissioning**, select the **ESR LED Current** page, and enter the value in the set value column of corresponding **LED**, ensuring that the current value of ESR background intensity is within the reference range, as shown below.

Figure8–54 ESR Background Intensity Adjustment



2. Then select **Position of ESR Measurement Point**, and click the **Calibrate** button. The **Test Value** should be within the reference range. Then, click the **Save** button. As shown in the following figure.

Figure8–55 Position commissioning diagram of the measurement point

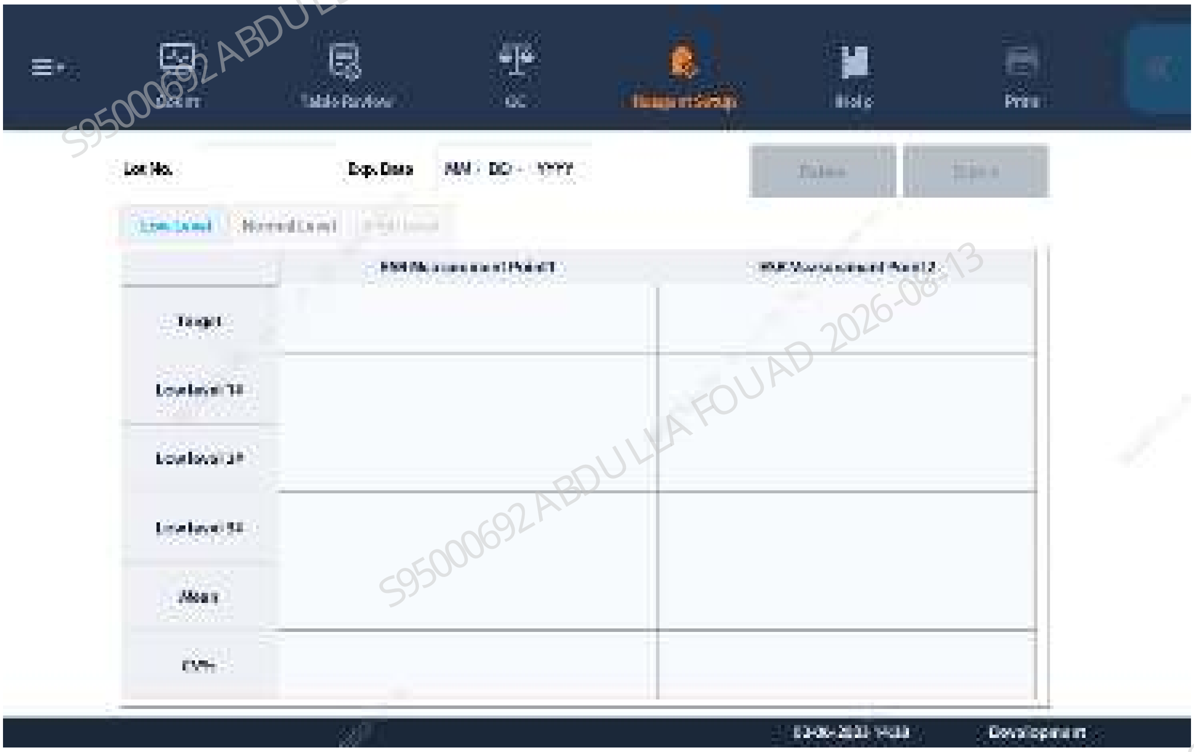
ESR calibration

1. Calibration Scenario
   During the analyzer production or after the ESR assembly is replaced due to a failure, the ESR assembly needs to be calibrated using a PLUS calibrator.
2. ESR parameter calibration process (double measuring points)

1) Prepare the unopened PLUS calibrator, query the target value according to the lot No., and mix it after warming.

2) Select **Calibrate>ESR Calibrator Calibration**, enter the lot No. and validity period of PLUS calibrator, tap the **Low Value** page, and fill the ESR low target value in the **Target** column (both measuring points need to be set with targets and the targets are consistent). Then use PLUS calibrator to perform counting three times. If the test result is normal, the software will prompt **The low value calibration succeeds**. If the test result is abnormal, the software will clear the data and repeat the counting 3 times until it succeeds. As shown in the following figure.

Figure8–56 ESR Calibration Target Value Input - Low Value



3) Tap the **Normal Value** page and fill the target value of ESR normal value in the **Target** column (both measuring points require entering the target and the targets are consistent). Then use PLUS calibrator to perform counting three times. If the test result is normal, the software will prompt **The normal value calibration succeeds**. If the test result is abnormal, the software will clear the data and repeat the counting 3 times until it succeeds. As shown in the following figure.

Figure8–57 ESR Calibration Target Value Input - Target Value

4) The **High Value** page can be tapped after the **low value** and **normal value** are calibrated successfully. Fill the target value of ESR high value in the **Target** column (both measuring points require a target and the targets are consistent). Then use the PLUS calibrator to perform counting three times. If the test result is normal, the software will prompt **The high value calibration succeeds**. If the test result is abnormal, the software will clear the data and repeat the counting 3 times until it succeeds. As shown in the following figure. Save calibration factors.

Figure8–58 ESR Calibration Target Value Input - High Value

5) Take a BC-6D normal value quality control, query the target value according to the lot No., and then make measurement in the CD+ESR counting mode to confirm that the ESR measurement result is within the allowable deviation range of the control target value.

Subsequent Requirements

/

8.13 Sampling assembly (CS AL)

8.13.1 General Information

Figure8–59 FRU photo of the 115-086250-00 aspiration module (CS AL)

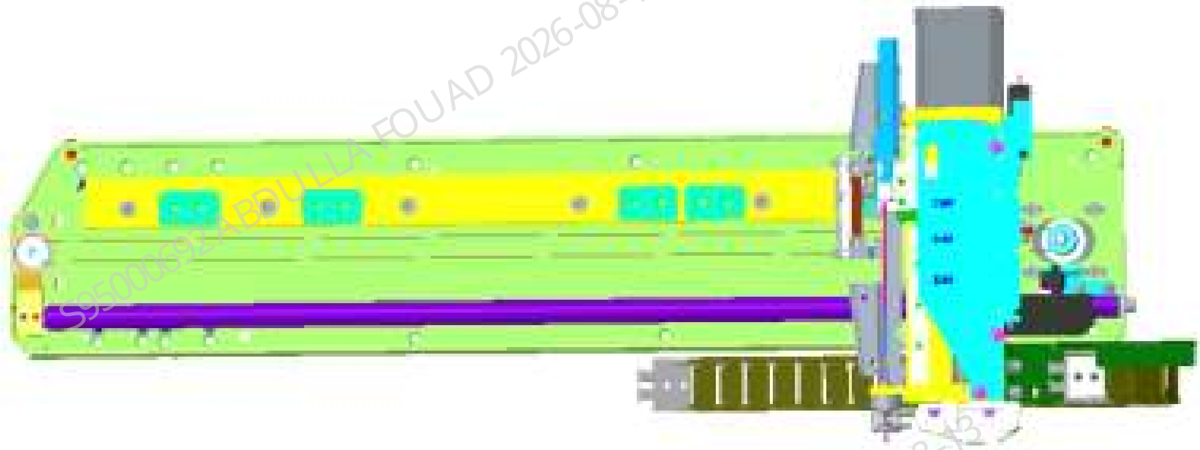


Table 8–9 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-086250-00	Aspiration Module (CS AL) FRU	115-090546-00	ESR horizontal assembly
		115-069029-00	Piercing needle
		041-023727-00	Probe wash device 2
		011-000021-00	PHOTOELEC Optical Sensor IP55 940nm 15cm
		011-000297-00	Photoelectric sensor wire terminal

Functions:

Aspirating the samples in the CT sample aspiration position and auto-sampling position, and dispensing the collected sample into each reaction bath.

8.13.2 Disassembly and Assembly

Precautions

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.
3. Be careful not to damage the connector when removing the tube from the syringe.

Required tools:

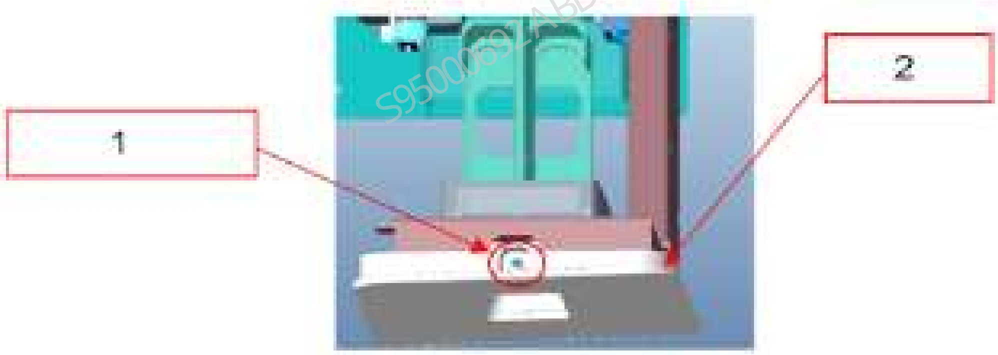
One cross-headed screwdriver
Allen wrench x 1
Cutting nipper x 1

Prerequisites

Before disassembly, lift the latex reagent probe and remove the latex reagent compartment door before shutting down the analyzer. There may be leakage when removing the sampling probe. Please lay a tissue under the probe.

Steps

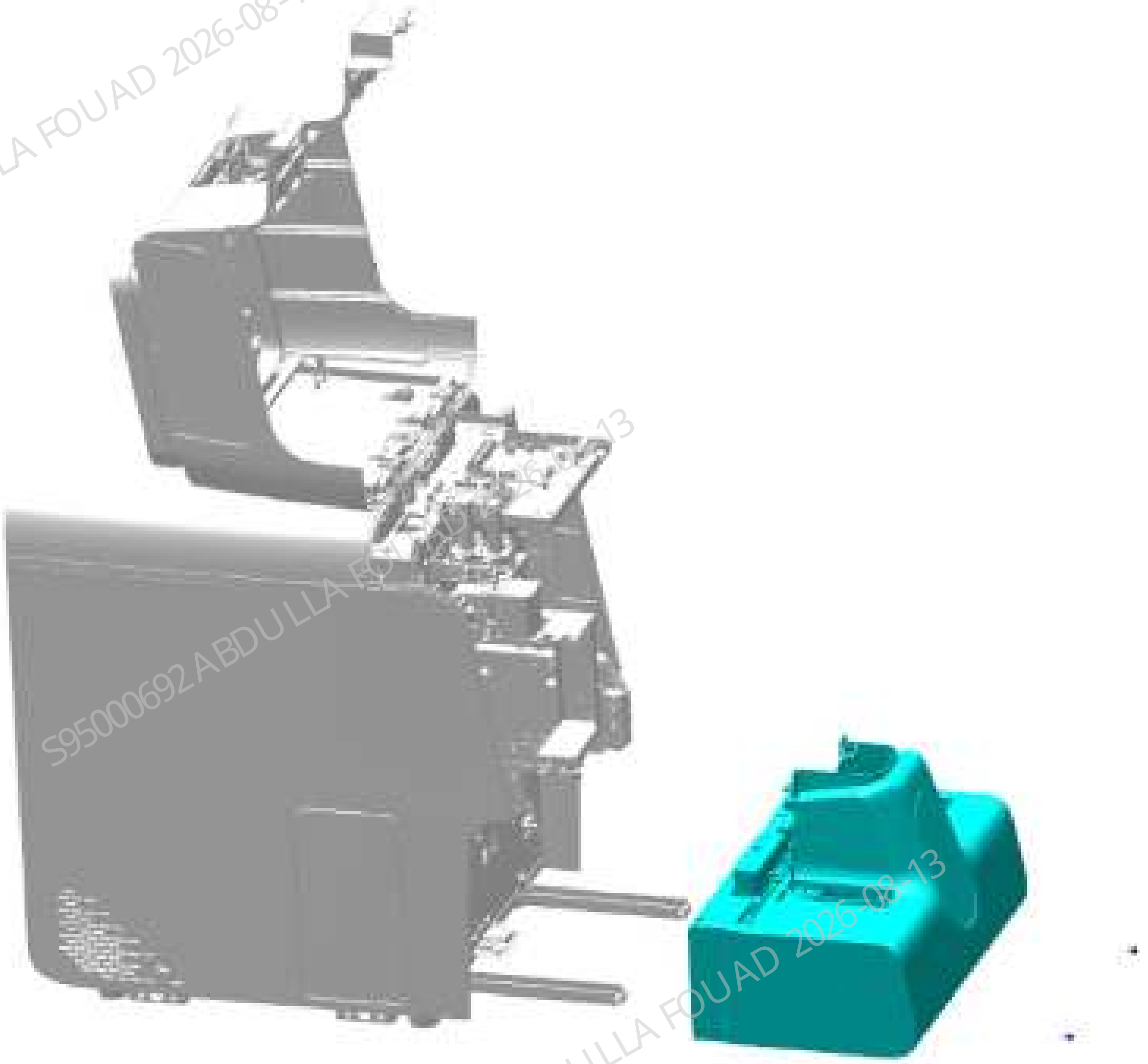
1. Select **Service>Maintenance>Reagent** on the software interface, tap **Replace CRP Latex**, pull out the latex reagent compartment door according to the prompt on the software interface, take out the latex, loosen the M3*8 cross-recessed pan head screw, and take down the latex reagent compartment door, as shown in the following figure. After the compartment door is removed, push the cold chamber door into the analyzer, and then perform the power-off operation.



1	Loosen the M3*8 screw
2	Latex reagent compartment door

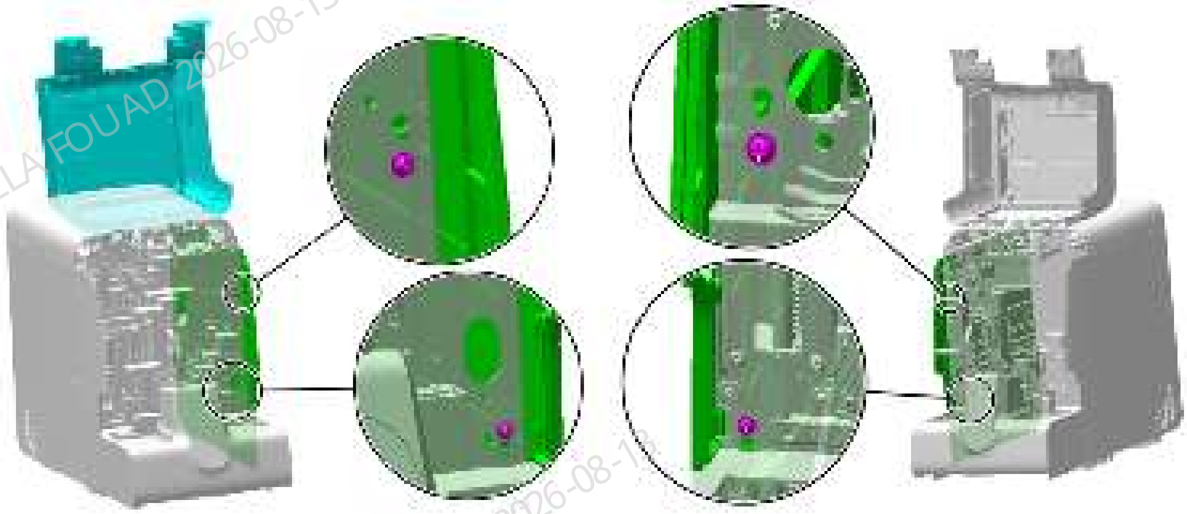
2. Pull the autoloader to an appropriate distance towards the operator, and disconnect the wires connecting the analyzer and the autoloader.
3. Pull the autoloader out of the autoloader crossbeam. As shown in the following figure.

Figure8–60 Diagram 2 of autoloader removing procedure



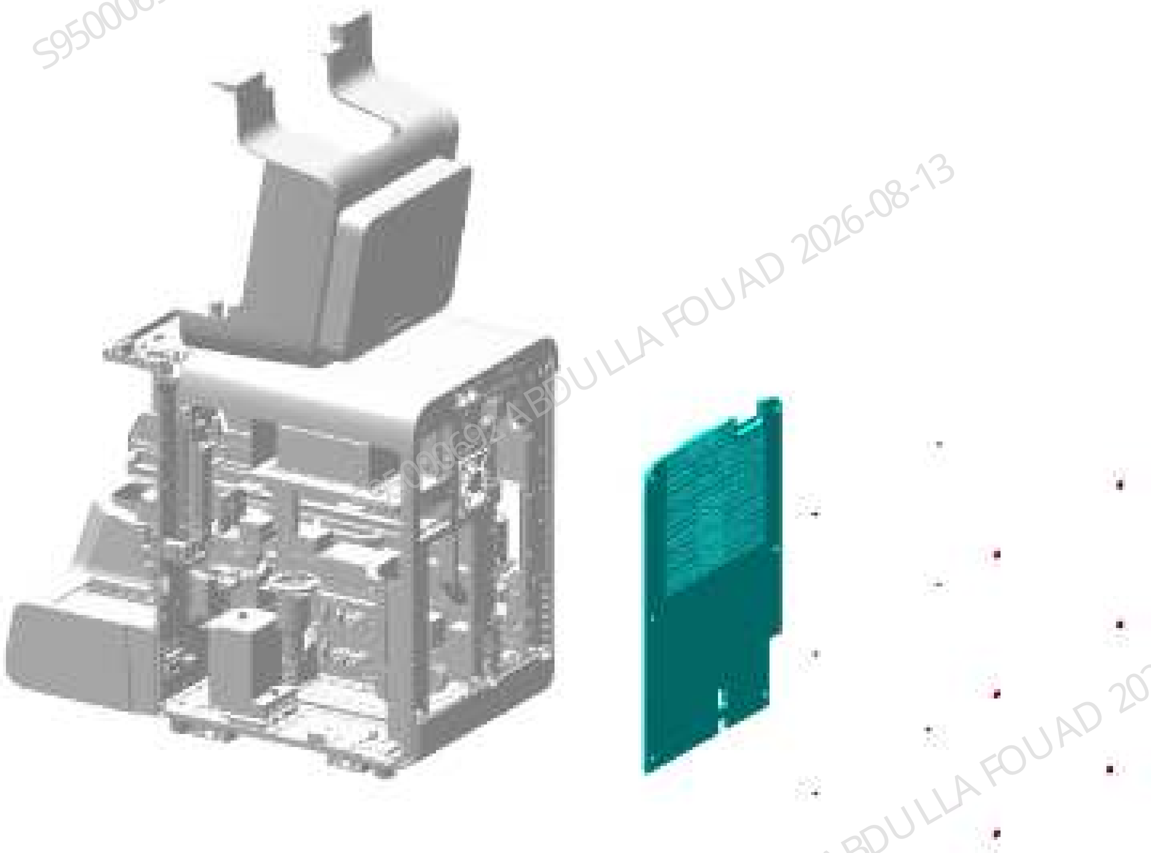
4. Loosen the two screws that fix the left and right covers, and remove the left and right covers, as shown in the figure below.

Figure8–61 Diagram of left and right cover removing procedure



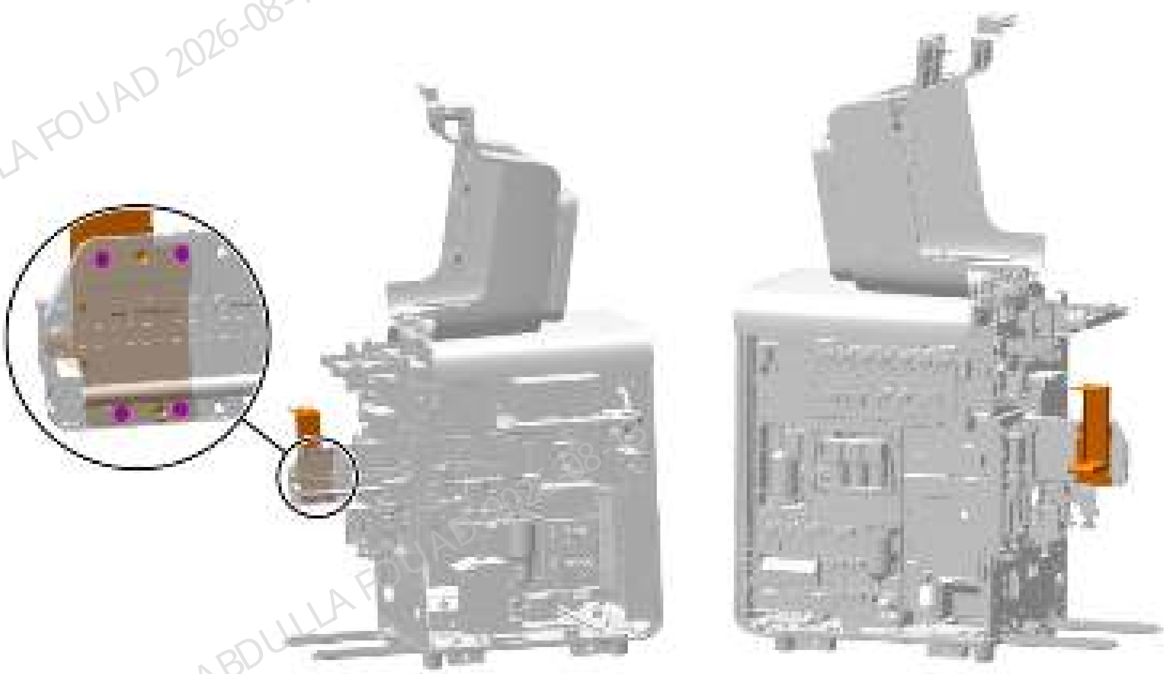
5. As shown in the figure below, unscrew the set screws of the rear panel, and remove the rear panel.

Figure8–62 Removing the rear cover



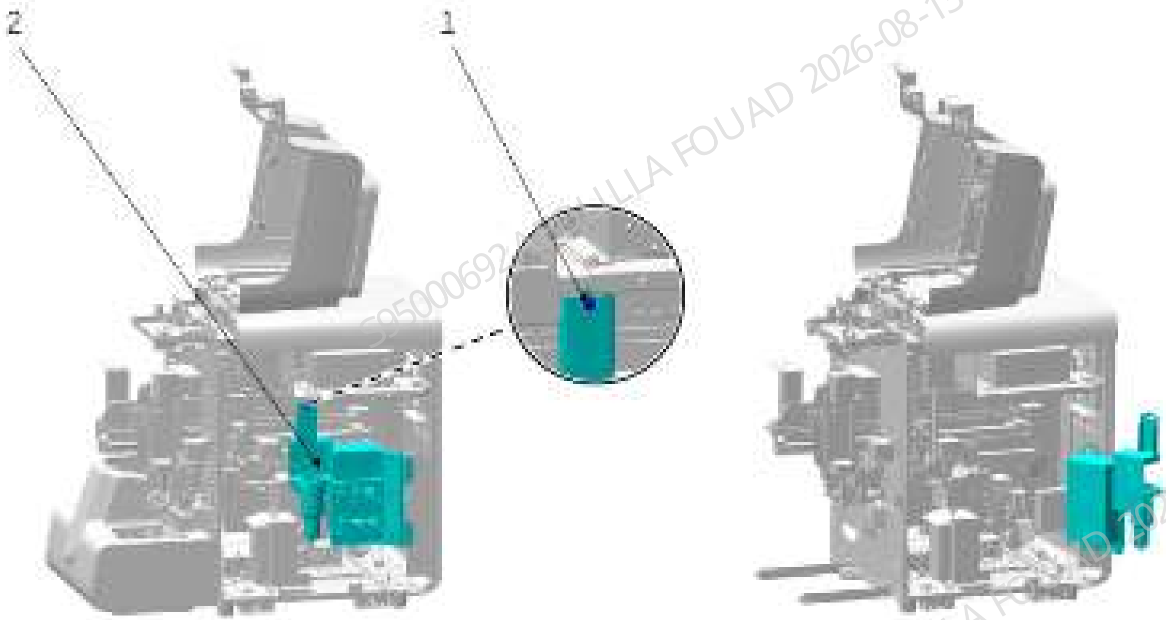
6. As shown in the figure below, unscrew the four countersunk screws securing the dye bag assembly, and remove the dye bag assembly.

Figure8–63 Diagram of dye bag assembly removing procedure



7. Unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–64 Opening the right panel assembly



8. Remove the snap connector on the sampling manifold at the upper left of the analyzer's front panel, as shown in the figure. As shown in the following figure.

Figure8–65 Removing snap connector from sampling manifold

9. As shown in the figure below, lift the Y-axis assembly of the mixing assembly to the upper limit position, and fix it well with ties.

Figure8–66 Diagram of the Y-axis assembly on the tying mixing assembly

10.Unplug the sampling Y-axis motor wire and the home position optocoupler cable from the back of the analyzer, as shown in the figure below.

Figure8–67 Diagram of disconnecting wires from the motor and sensor of sampling Y-axis assembly

11.As shown in the figure below, disconnect the wires from the sampling Z-axis assembly from the right side of the analyzer.

Figure8–68 Disconnecting wires from sampling Z-axis assembly

12.Remove the chain baffle, as shown in the figure below.

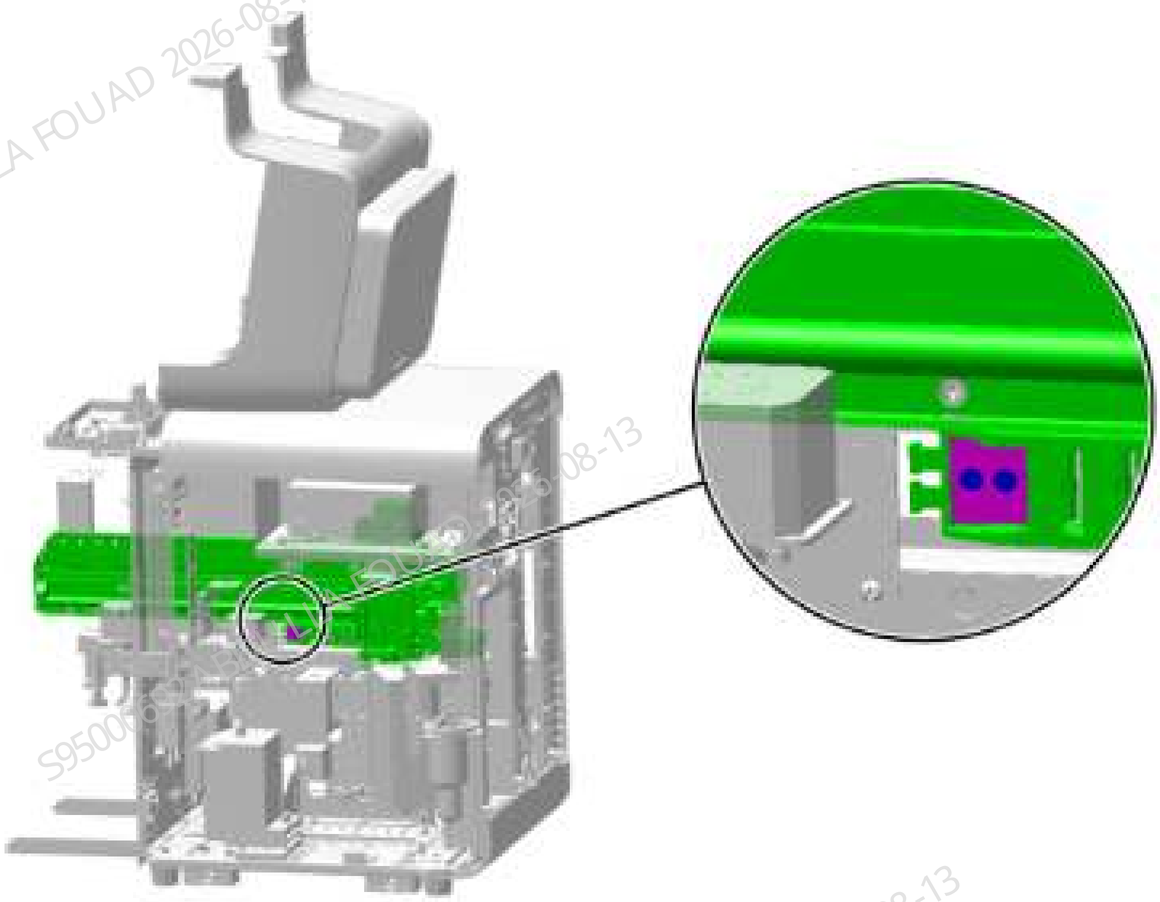
Figure8–69 Removing the chain baffle

13.Use tweezers to separate the two I-connectors fixed at the end of the tank chain of the aspiration module from the two rubber tubes at the analyzer, as shown in the figure below.

Figure8–70 Removing the sampling tube

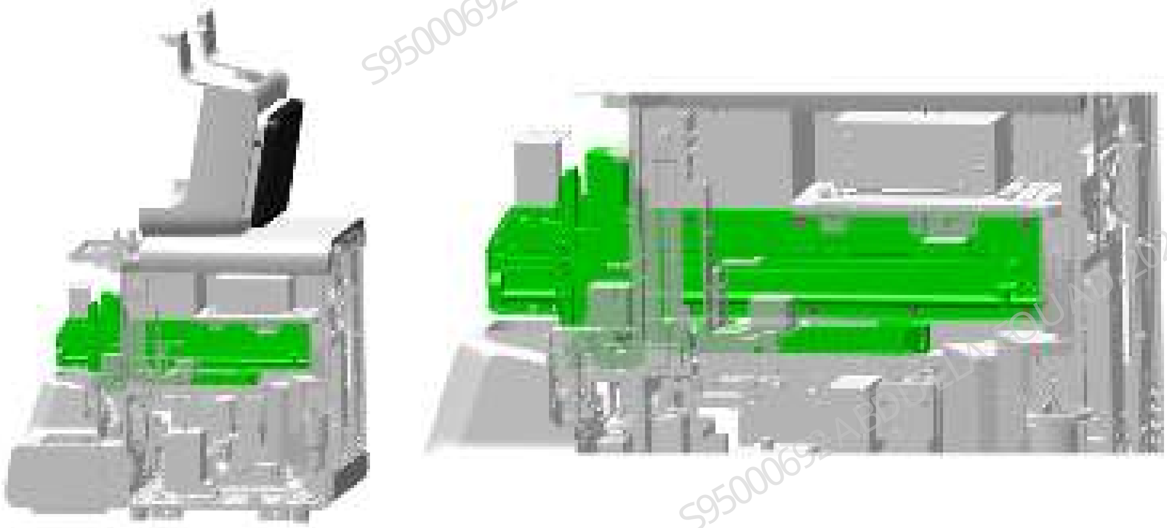
14.As shown in the figure below, remove the tank chain end at the middle plate.

Figure8–71 Removing the sampling chain end



15.Use a Phillips screwdriver to unscrew the screws securing the aspiration module, as shown in the figure below, and remove the aspiration module (hold the aspiration module with your hand when unscrewing the last screw to prevent it from falling suddenly).

Figure8–72 Removing the aspiration module



Subsequent Requirements

Installation procedure:

1. Fix the aspiration module on the analyzer with eight M4 countersunk head screws.
2. Connect the wires of the aspiration module to the corresponding terminals one by one.
3. Connect the two I-type connectors at the end of the tank chain of the aspiration module to the rubber tubes on the two analyzers that were disassembled (Note: The two tubes are of different tube types, and the tubes at both ends of the I-type connectors should be of the same tube type). Secure the snap connector of the sampling tube to the sampling manifold.
4. Connect the power supply, turn it on, and align the position of the sample probe.
5. Install the right bath subassembly using one M4 combination screw.
6. Install the left and right covers and the rear cover, and place the front shell assembly to complete the installation.

8.13.3 Commissioning and Verification

Required tools:

One cross-headed screwdriver

Pre-requirement

After completing the assembly, upgrade the analyzer software version to V01.12.00.1566 or above.

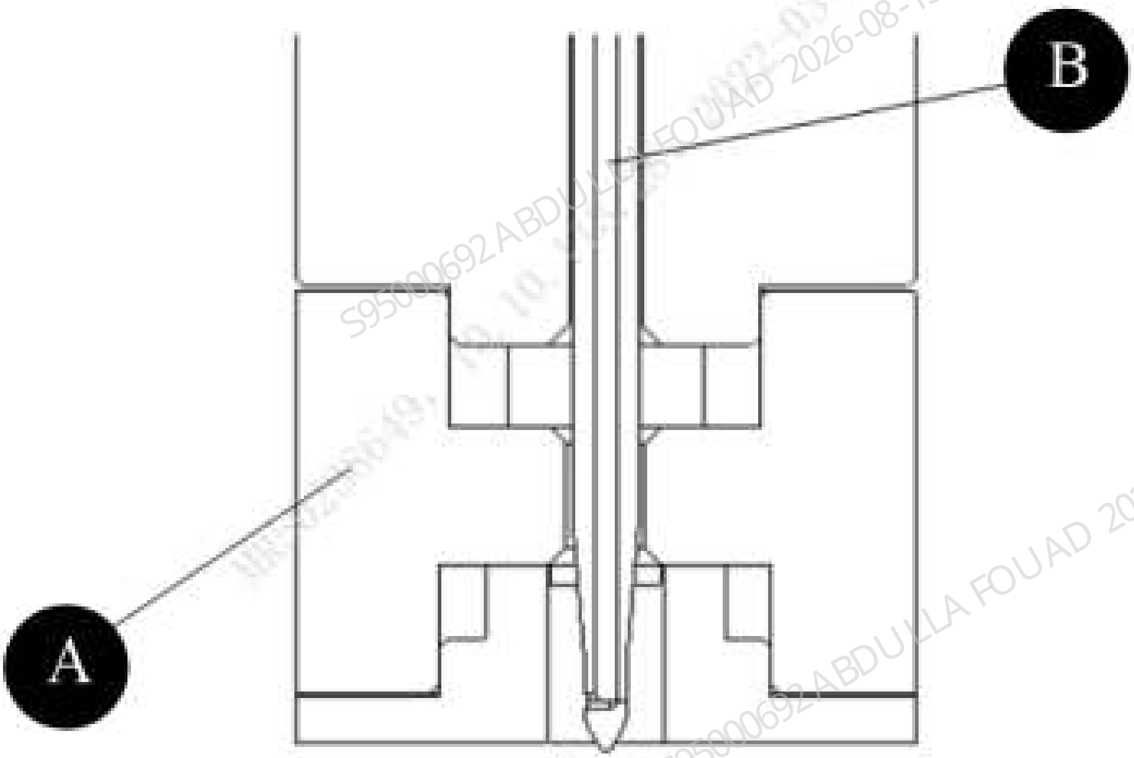
Steps:

1. Start the analyzer, and log in the software at the service access level.
2. Select **Service**>**>Commissioning & Diagnosis**>**>Adj. Sample Probe Pos.** in the menu bar, enter the screen shown in the figure below, and lift the front shell to ensure that the wipe can be seen.

Figure8–73 Adjusting interface of the vertical initial position of sampling probe

3. Select **Vertical initial position** and click **To Default Adj. Pos**, as shown in the following figure.

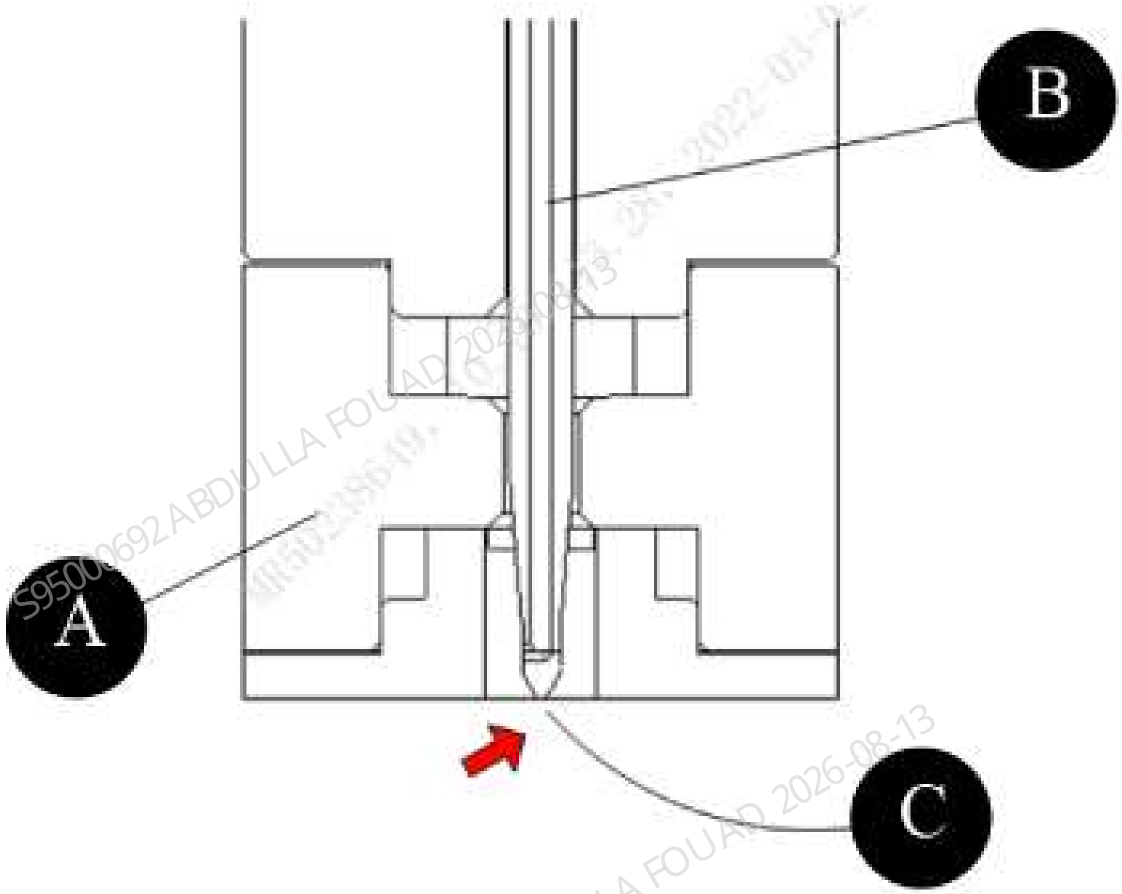
Figure8–74 Sampling Probe to Default Adjust Pos.



A	Probe wipe	B	Sampling Probe

4. Click the **Down** button to adjust the probe wipe and adjust the piercing probe to be flush with the bottom of the probe wipe, as shown in the figure below.

Figure8–75 Sampling probe tip Pos. (Adjustment target position/observation position)



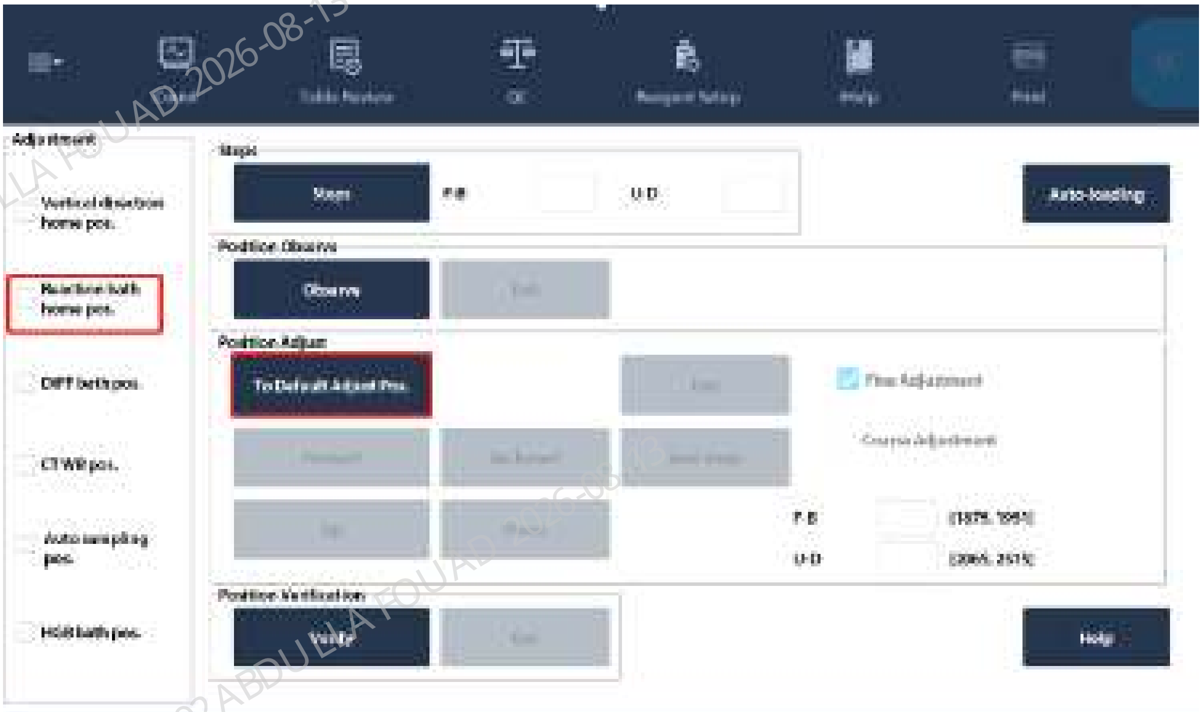
C	The probe tip is flush with the bottom of the probe wipe

After finishing, click **Save the number of steps**, and then click the **Observe** button to confirm that the adjustment position is correct (The probe tip is flush with the bottom of the probe wipe), and then click **Exit the observation**; if the position is not correct, repeat the above adjustment steps.

- **Reaction bath commissioning**

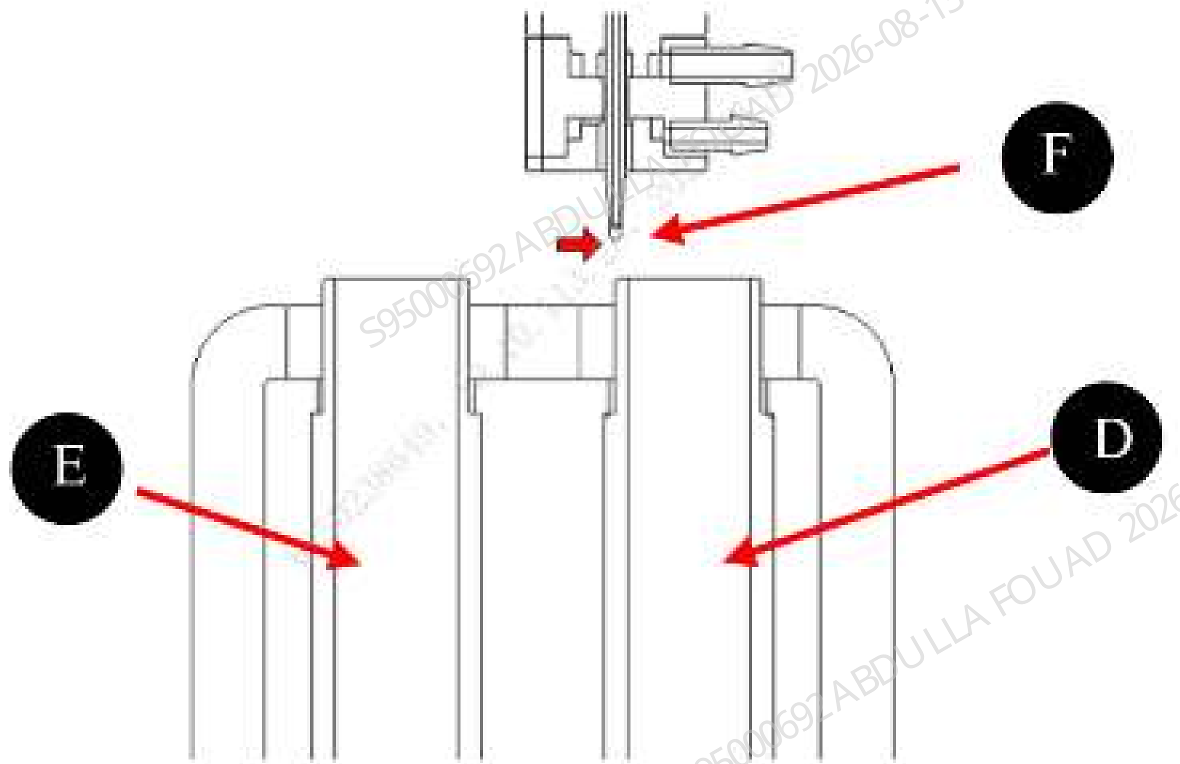
5. Remove the right door and at the same time remove the screws from the lugs of the right bath subassembly to open the right bath subassembly. Select **Service** >>**Commissioning & Diagnosis**>>**Adj. Sample Probe Pos.** in the menu bar, and select **Reaction bath initial position** to enter the following screen.

Figure8–76 Adjust the reaction bath initial position of sampling probe



6. Click **To Default Adj. Pos**, and move the piercing probe to the upper left part of the reaction bath 1 hole position, as shown in the following figure.

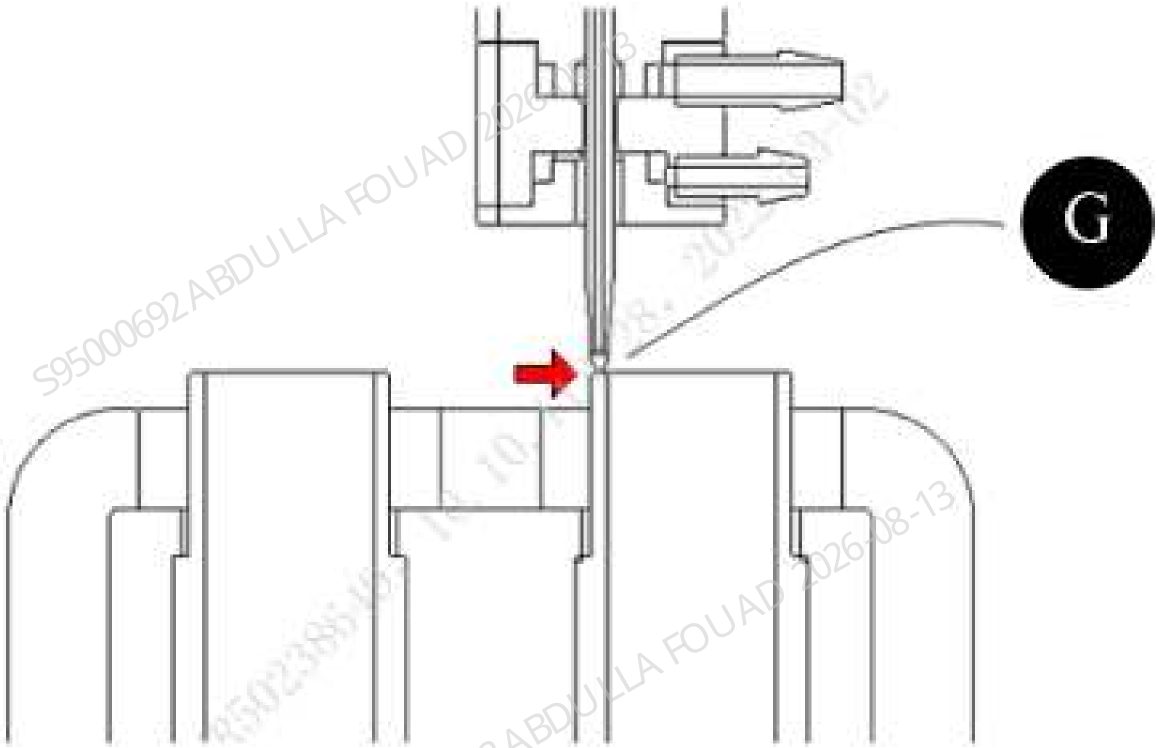
Figure8–77 Sampling probe To Default Adjust Pos. of reaction bath



D	Reaction bath 1	E	Reaction bath 2	F	The sampling probe is about 3mm above the reaction bath 1

7. Click the **front**, **back**, **up**, and **down** buttons to adjust the front and rear positions and vertical height of the piercing probe (the vertical height position can be confirmed with a piece of paper) and meet the requirements of the following figure. (The sampling probe in the horizontal direction is centered from the bath walls, and the probe tip in the vertical direction is just in contact with the upper end face of the bath wall), then adjust it in place.

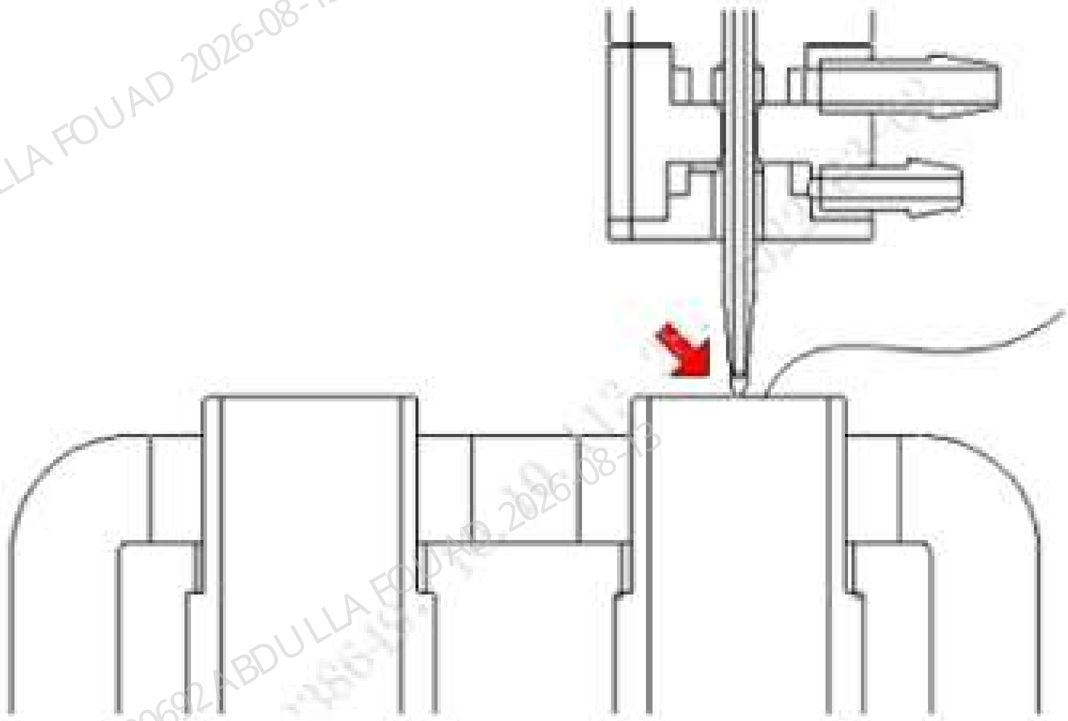
Figure8–78 Adjustment instructions of sampling probe to the reaction bath front and rear positions and vertical height



G	The sampling probe in the horizontal direction is centered from the bath walls, and the probe tip in the vertical direction is just in contact with the upper end face of the bath wall

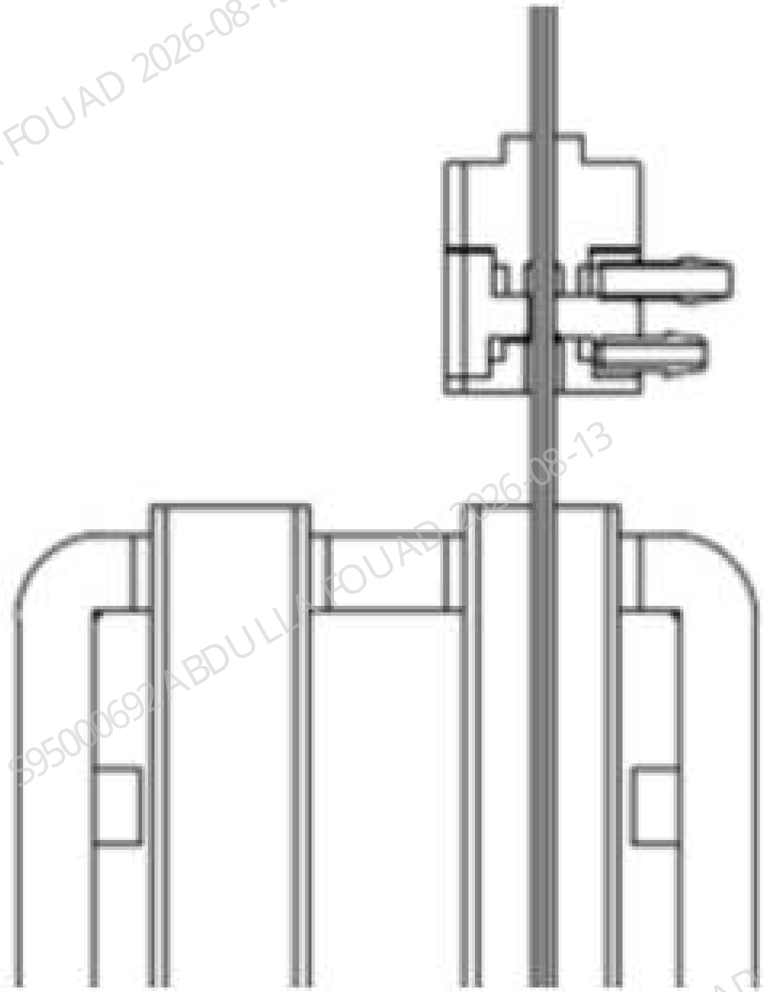
After the adjustment is completed, tap **Save the number of steps**, and then tap the **Observe** button, the sampling probe will move horizontally to the horizontal center position of reaction bath 1, and the probe tip will move vertically to the position of the top surface of the bath (if the front position is adjusted accurately). Visually confirm the adjustment results with the **Observe** button. If the sampling probe is aligned with the center of the bath in the horizontal direction, and the probe tip in the vertical direction is aligned with the top surface of the bath, then the position has been adjusted. As shown in the following figure.

Figure8–79 Observation position of the sampling probe relative to the reaction bath



Finally, click the **Check** button to confirm that the adjustment position is correct, and then click **Exit Check**. (Note: After the **Check** button is pressed, the piercing probe will reach the last position of the count, at this time the piercing probe does not interfere with the bath wall); as shown in the figure below. If the position is not correct, repeat the above adjustment steps.

Figure8–80 Check position of the sampling probe relative to the reaction bath



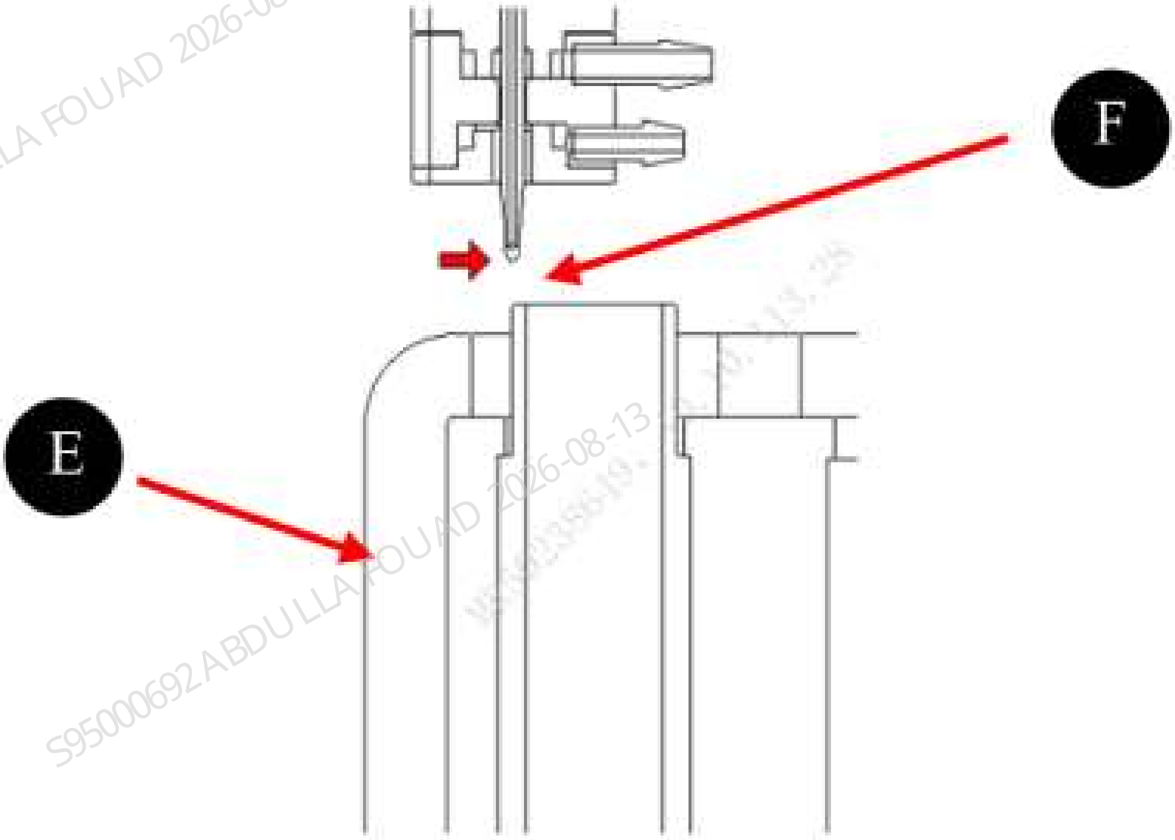
- **Adjusting the DIFF bath position**

  Select **DIFF bath initial position** to enter the following interface.

Figure8–81 Interface of Adjusting the DIFF Bath Initial Position of Sampling Probe

8. Click **To Default Adj. Pos**, and move the piercing probe to the upper left part of the reaction bath 2 hole position, as shown in the following figure.

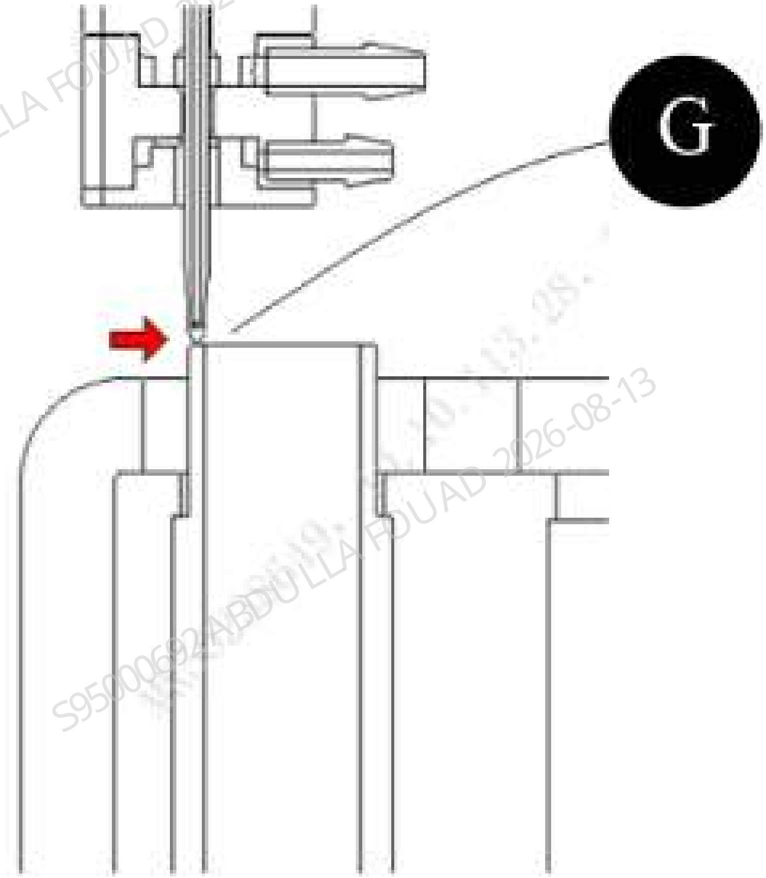
Figure8–82 Sampling probe To Default Adjust Pos. of reaction bath



E	Reaction bath 2	F	The sampling probe is about 3mm above the reaction bath 1

9. Click the **front**, **back**, **up**, and **down** buttons to adjust the front and rear positions and vertical height of the piercing probe (the vertical height position can be confirmed with a piece of paper) and meet the requirements of the following figure. (The sampling probe in the horizontal direction is centered from the bath walls, and the probe tip in the vertical direction is just in contact with the upper end face of the bath wall), then adjust it in place.

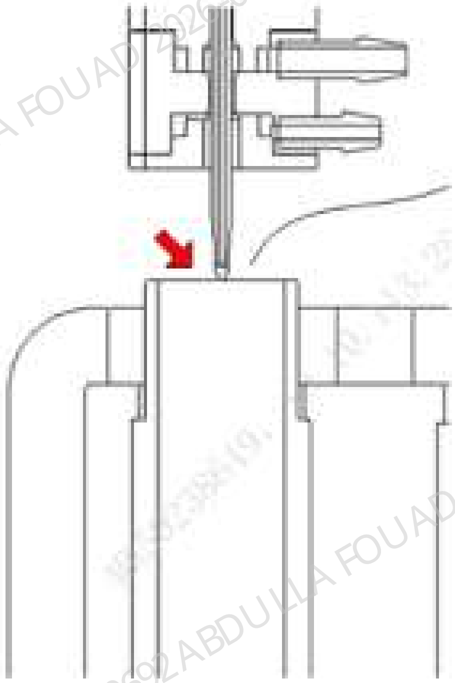
Figure8–83 Adjustment Instructions of Sampling Probe to the DIFF Bath Front and Rear Positions and Vertical Height



G	The sampling probe in the horizontal direction is centered from the bath walls, and the probe tip in the vertical direction is just in contact with the upper end face of the bath wall

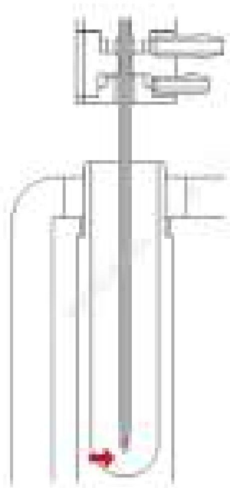
After the adjustment is completed, click **Save the number of steps**, and then click the **Observe** button, the sampling probe will move horizontally to the horizontal center position of reaction bath 2, and the probe tip will move vertically to the position of the top surface of the bath (if the front position is adjusted accurately). Visually confirm the adjustment results with the **Observe** button. If the sampling probe is aligned with the center of the bath in the horizontal direction, and the probe tip in the vertical direction is aligned with the top surface of the bath, then the position has been adjusted. As shown in the following figure.

Figure8–84 Observation Position of the Sampling Probe Relative to the DIFF Bath



Finally, click the **Check** button to confirm that the adjustment position is correct, and then click **Exit Check**. (Note: After the **Check** button is pressed, the piercing probe will reach the last position of the count, at this time the piercing probe does not interfere with the bath wall); as shown in the figure below. If the position is not correct, repeat the above adjustment steps.

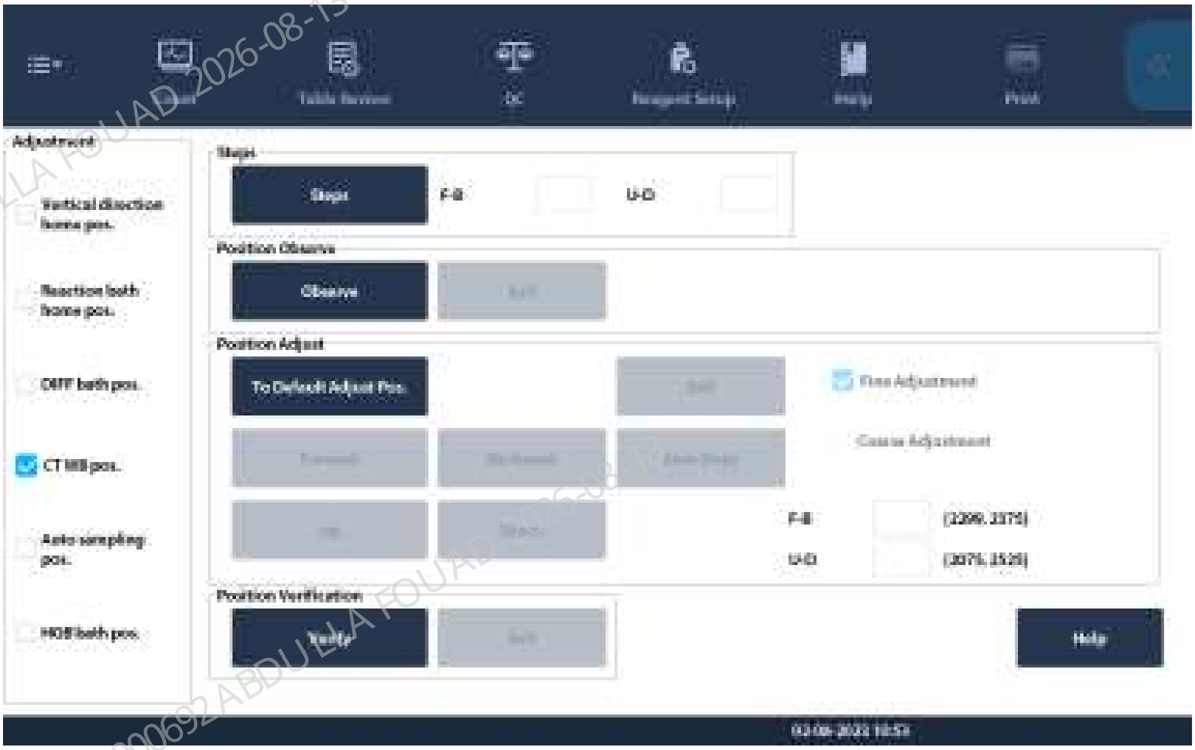
Figure8–85 Check position of the sampling probe relative to the HBG bath



- ## CT−WB Commissioning

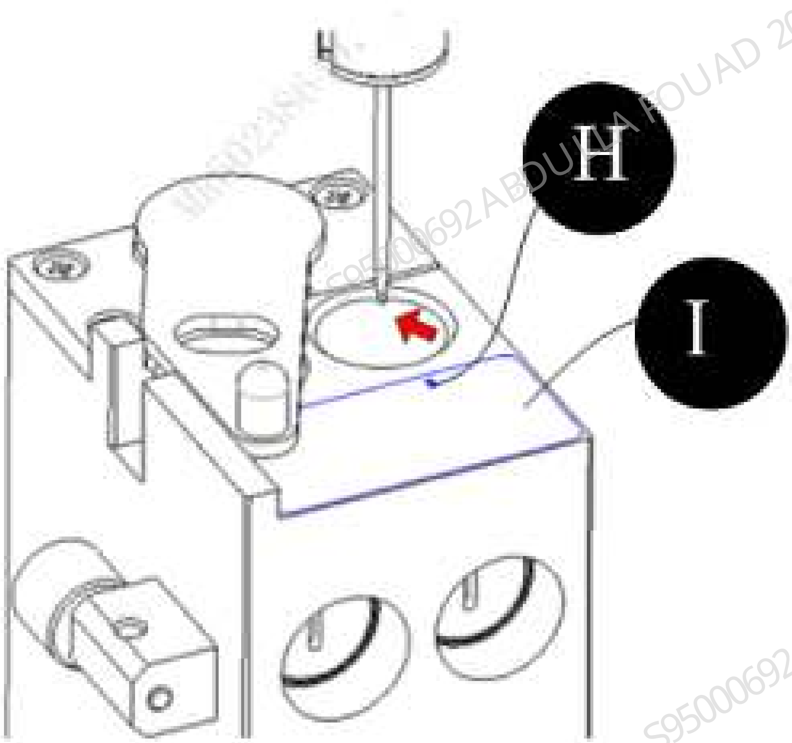
1) Tap **CT-WB Pos.** to enter the interface shown in Figure A, tap the **To Default Adj. Pos** button, tap **CT-WB** to enter the interface shown in the figure below, and tap the **To Default Adj. Pos** button.

Figure8–86 Adjusting Interface of the Initial Position of Sampling Probe CT-WB Pos.



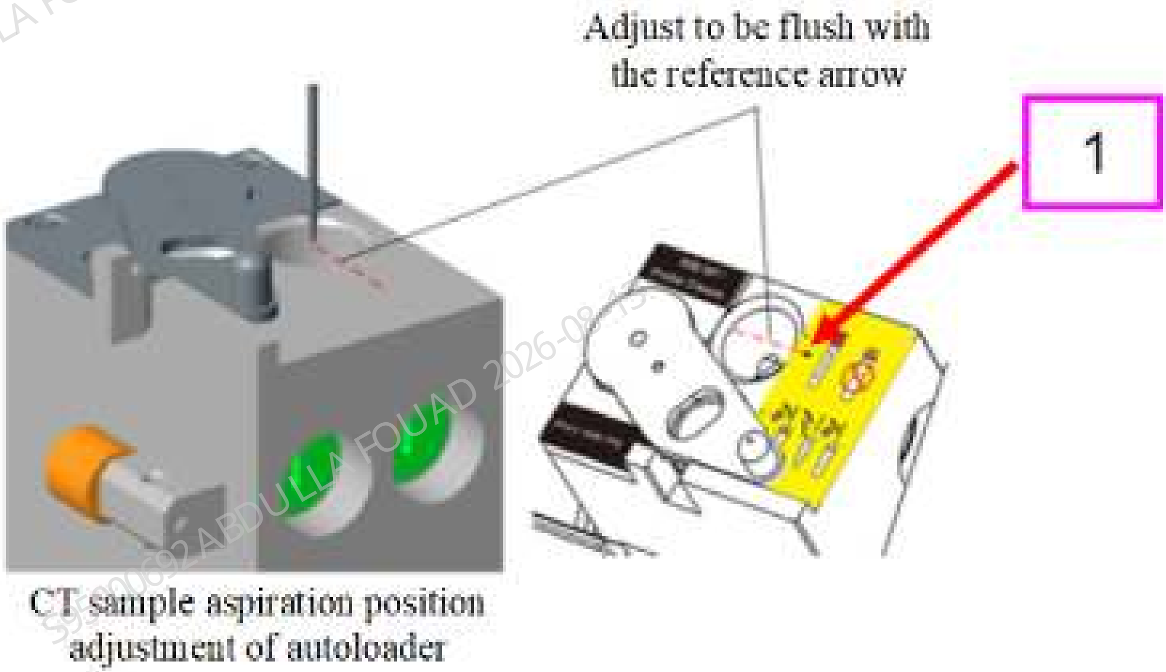
2) Tap **To Default Adj. Pos**, and move the piercing probe to the upper part of the CT aspiration position, as shown in the following figure.

Figure8–87 Default Adjust Pos. of Sampling Probe CT-WB Pos.



3) Adjust the **Front**, **Back**, **Up**, and **Down** buttons to make the position of the sampling probe relative to the CT-WB aspiration position meet the following requirement.

Figure8–88 Diagram of the the front-to-back position requirement of sampling probe acceptance criteria



1 Aligned with the label arrow

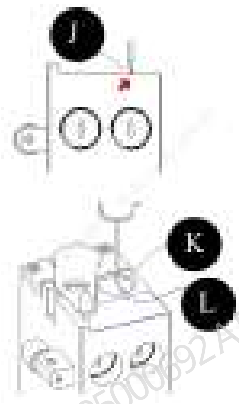
Figure8–89 Diagram of the Up-to-Down Position Requirement of Sampling Probe Acceptance Criteria

□ In the up-to-down direction, if the tip of the sampling probe is flush with the top surface, the sampling probe up-to-down position is good.

□ In the front-to-back direction, if the tip of the sampling probe is aligned with the arrow on the tube position label, the sampling probe front-to-back position is good.

4) After the adjustment is completed, save the adjustment steps, exit the adjustment interface, and then tap the observation button. The observation position needs to meet the following requirements.

Figure8–90 Diagram of the Observation Position



L	Arrow	J, K	The horizontal direction is flush with the arrow, and the probe tip in the vertical direction is flush with the top surface of the sample compartment

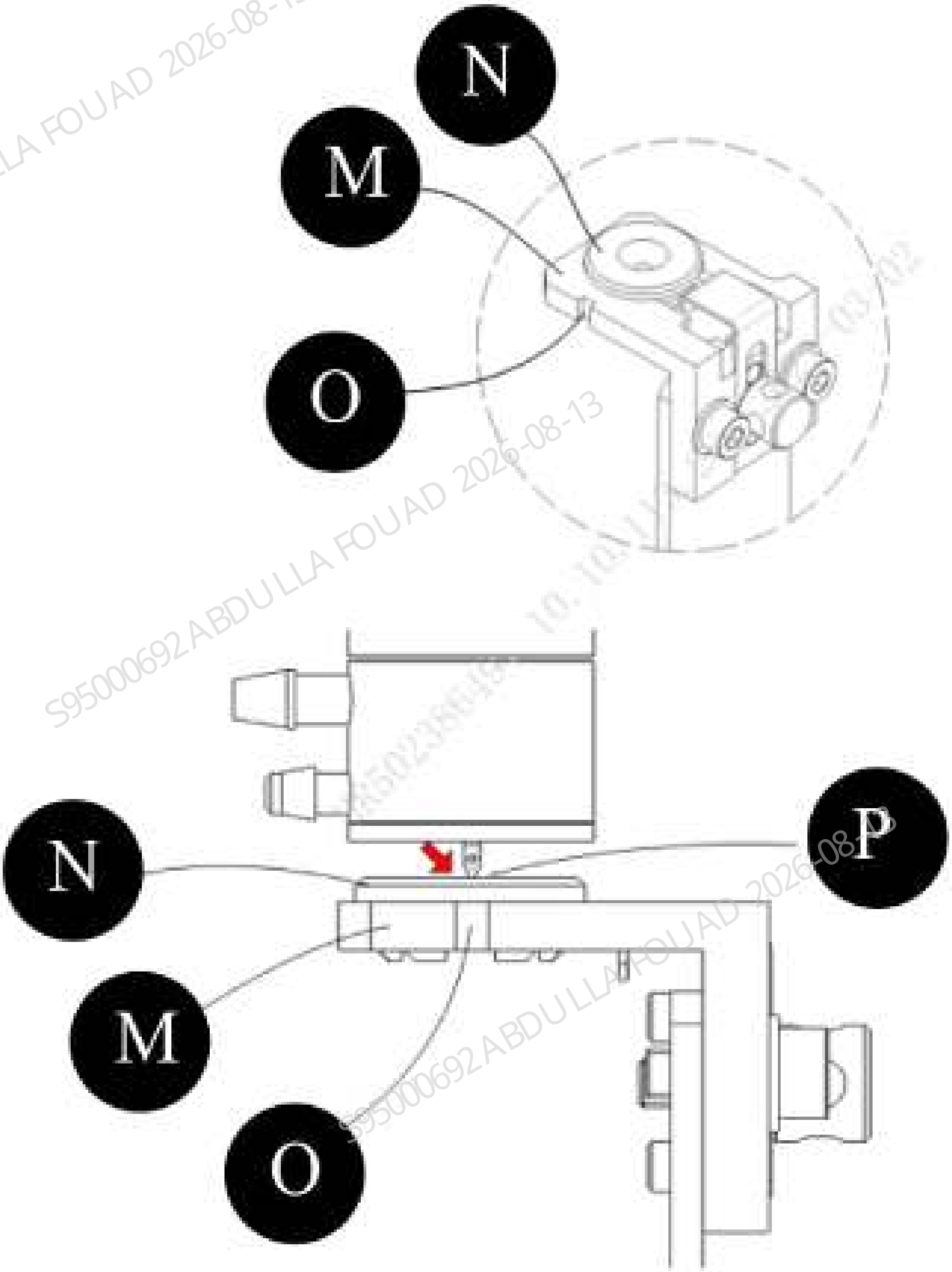
- **Auto−sampling Position Commissioning**

1) Select **Auto-Sampling Position**, as shown in the following figure.

Figure8–91 Interface of Adjusting AL Aspiration Position

2) Tap **To Default Adj. Pos**, and adjust the **Front**, **Back**, **Up**, and **Down** buttons to make the position of the sampling probe relative to the AL aspiration position meet the following requirement.

Figure8–92 Diagram of Adjustment Acceptance Criteria of AL asp. Pos.



M	Blood barrier bracket	N	Blood barrier	O	Gap on the blood barrier bracket	P	The sampling probe is aligned with the gap center in the horizontal direction, and flush with the top surface of the blood barrier in the vertical direction.

After the above requirement is met, tap **Save Steps** and then exit the adjustment interface.

3) Prepare one tube rack, place an empty test tube at position 1, and tap **Autoloader** on the interface. After the assembly is initialized, put the tube rack near the main unit end in the autoloader, tap **Feeding** to feed the test tube to the auto-sampling position, and tap the **Finish** button. Then tap **Check**. The sampling probe can penetrate into the test tube without passing through the blood barrier.

Figure8–93 Autoloader Button

Figure8–94 Tube Feeding Control Interface

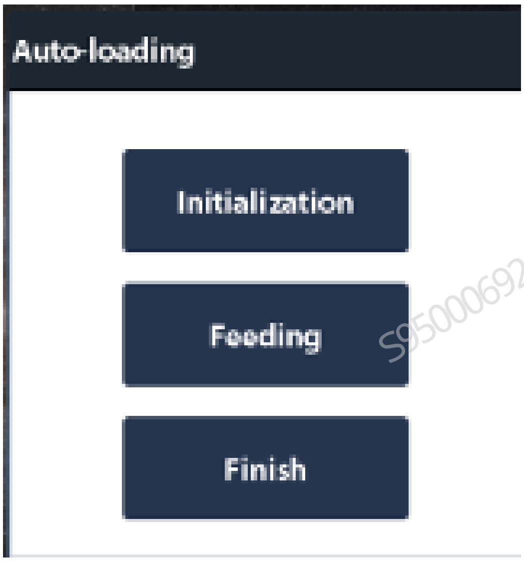
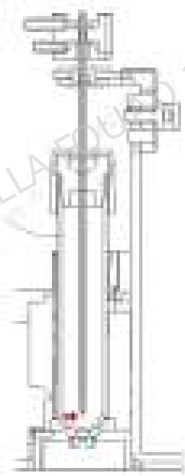


Figure8–95 Calibration position of the AL aspiration position



- ## Adjusting the HGB bath position

10.Remove the right door and at the same time remove the screws from the lugs of the right bath subassembly to open the right bath subassembly. Remove the HGB shielding cover, and select **Service** >>**Commissioning & Self-Test**>>**Adj. Sample Probe Pos.** on the menu bar, and select **HGB bath position** to enter the HGB bath adjustment screen, as shown in Figure **Figure8–96**. Tap **To Default Adjust Pos**, and move the piercing probe to the upper left part of the HGB bath hole position, as shown in **Figure8–97**.

Figure8–96 HGB bath adjustment screen

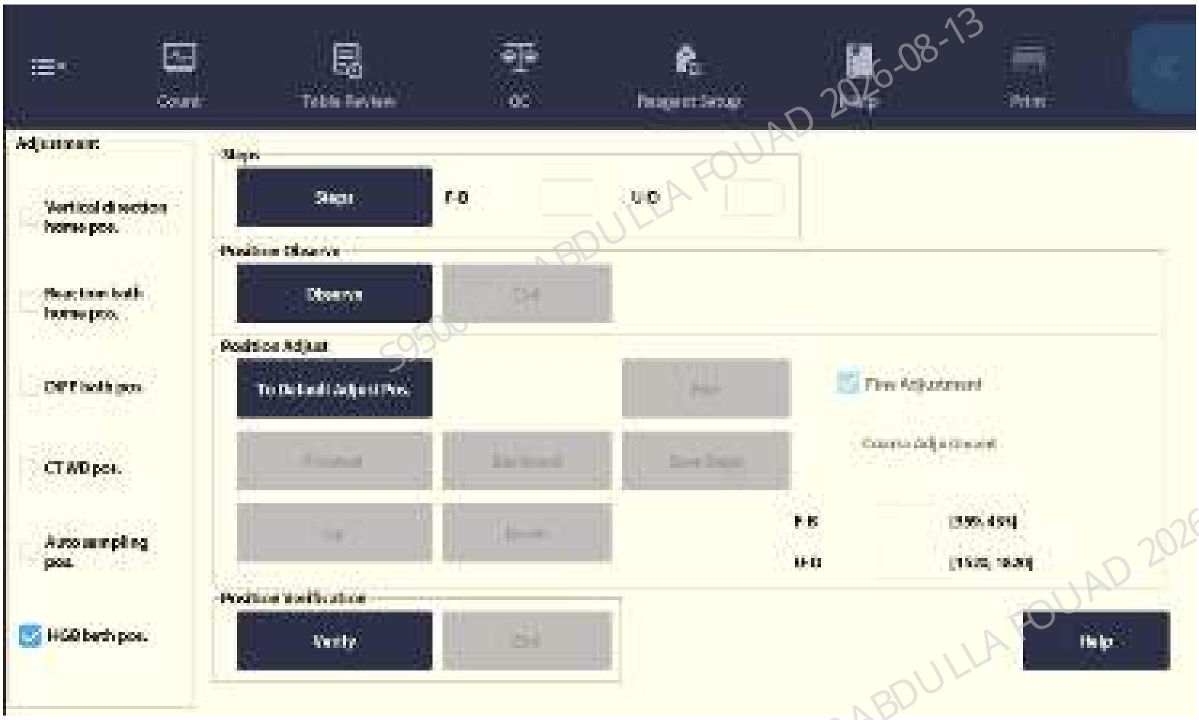
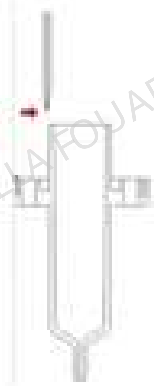
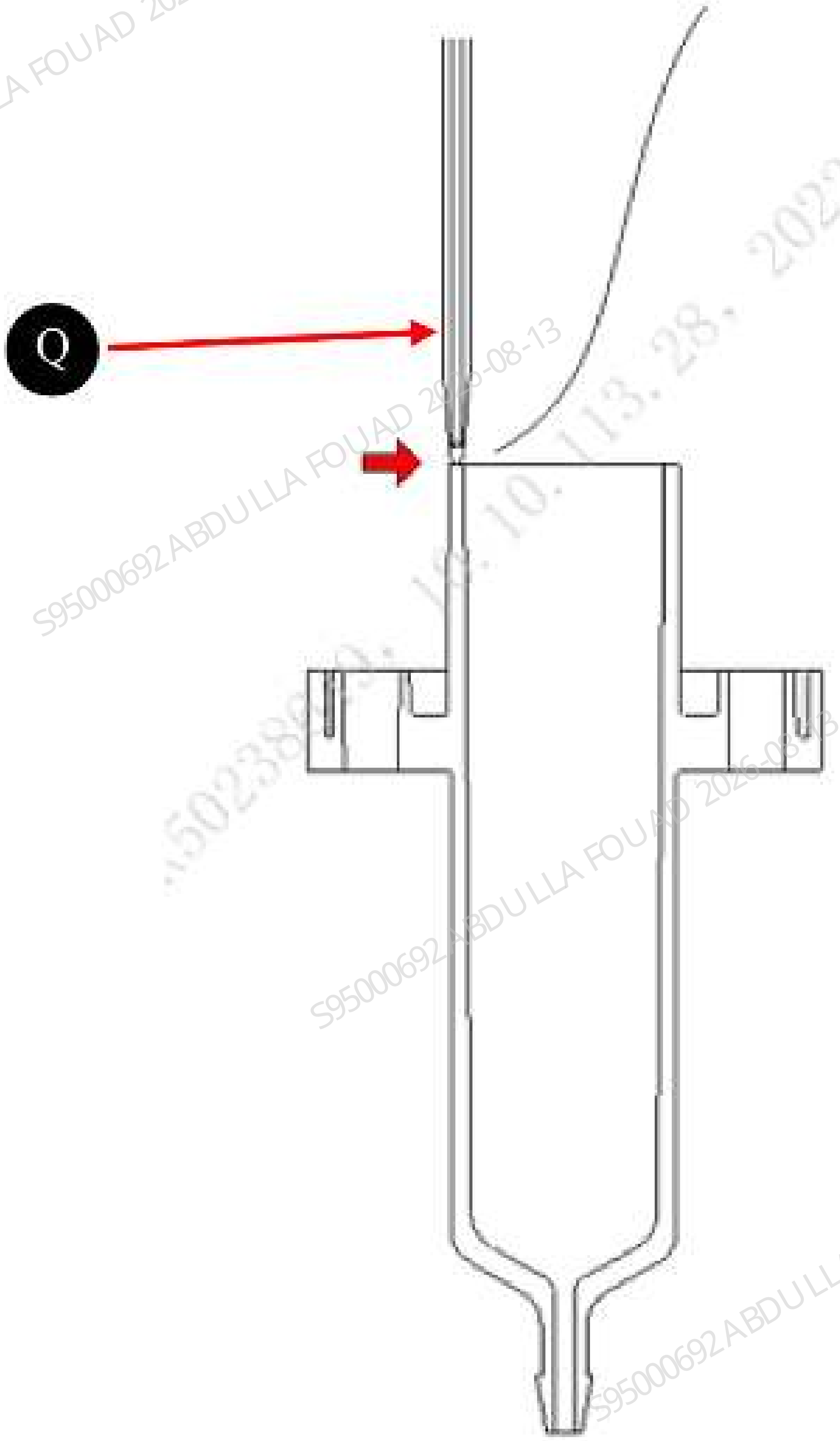


Figure8–97 Sampling probe To Default Adjust Pos. of HGB bath



11.Click the **front**, **back**, **up**, and **down** buttons to adjust the front and rear positions and vertical height of the piercing probe (the vertical height position can be confirmed with a piece of paper) and meet the requirements of the following figure (The sampling probe stops at the upper left part of the bath hole, the sampling probe is horizontally aligned in the center of the bath, and the probe tip is flush with the top surface of the bath wall), then the adjustment is in place.

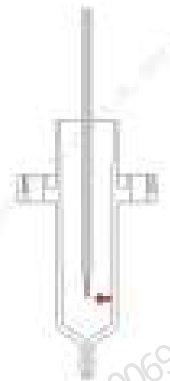
Figure8–98 Adjustment instructions of sampling probe to the HGB bath front and rear positions and vertical height



Q	The sampling probe stops at the upper left part of the bath hole, the sampling probe is horizontally aligned in the center of the bath, and the probe tip is flush with the top surface of the bath wall

12.After the adjustment is completed, click **Save Steps**, click the **Check** button to confirm that the adjustment position is correct, and then click **Exit Check**; install the shielding cover back (Note: After the **Check** button is clicked, the piercing probe cannot interfere with the bath wall); as shown in the figure below, if the position is not correct, repeat the above adjustment steps.

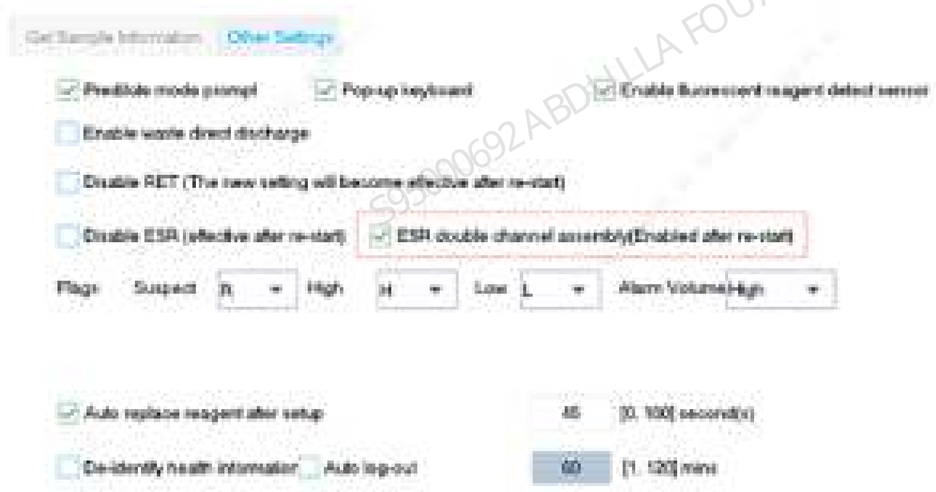
Figure8–99 Check position of the sampling probe relative to the HBG bath



- **ESR commissioning**

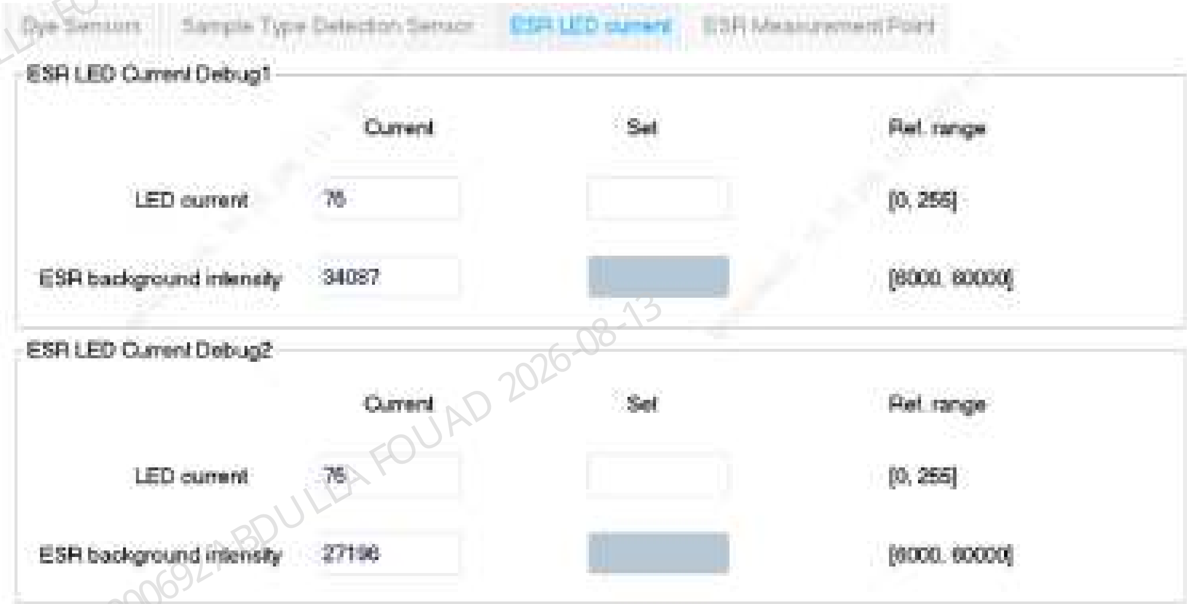
13.Select **Setup>Auxiliary Setup** on the menu bar of the on the menu bar, select **Other Settings**, and verify that the configuration item of activating **ESR 2-channel Assembly** has been checked.

Figure8–100 ESR 2-channel Configuration Setup



14.On the main unit software, tap **Service**>>**Commissioning and Self-Test>>Sensor Commissioning>>ESR LED Current**. Check whether the **ESR background intensity** at two positions in the figure below is within the reference range; if it exceeds the reference range, set the appropriate **LED current value** in the **set value** area of the corresponding **LED current** and save it. The ESR background intensity automatically changes accordingly and needs to be adjusted in the range of [50000, 60000]. The higher the LED current value, the higher the ESR background intensity

Figure8–101 Setting ESR Background Intensity



15.Select **Service>Commissioning and Self-Test>Position of ESR Measurement Point** in the main unit software, and tap the **Calibrate** button. The analyzer automatically calibrates the ESR measurement point position, and saves it after the screen displays a prompt.

Figure8–102 Diagram of adjusting the ESR measurement point position

- **ESR calibration**

1. **Calibration Scenario**

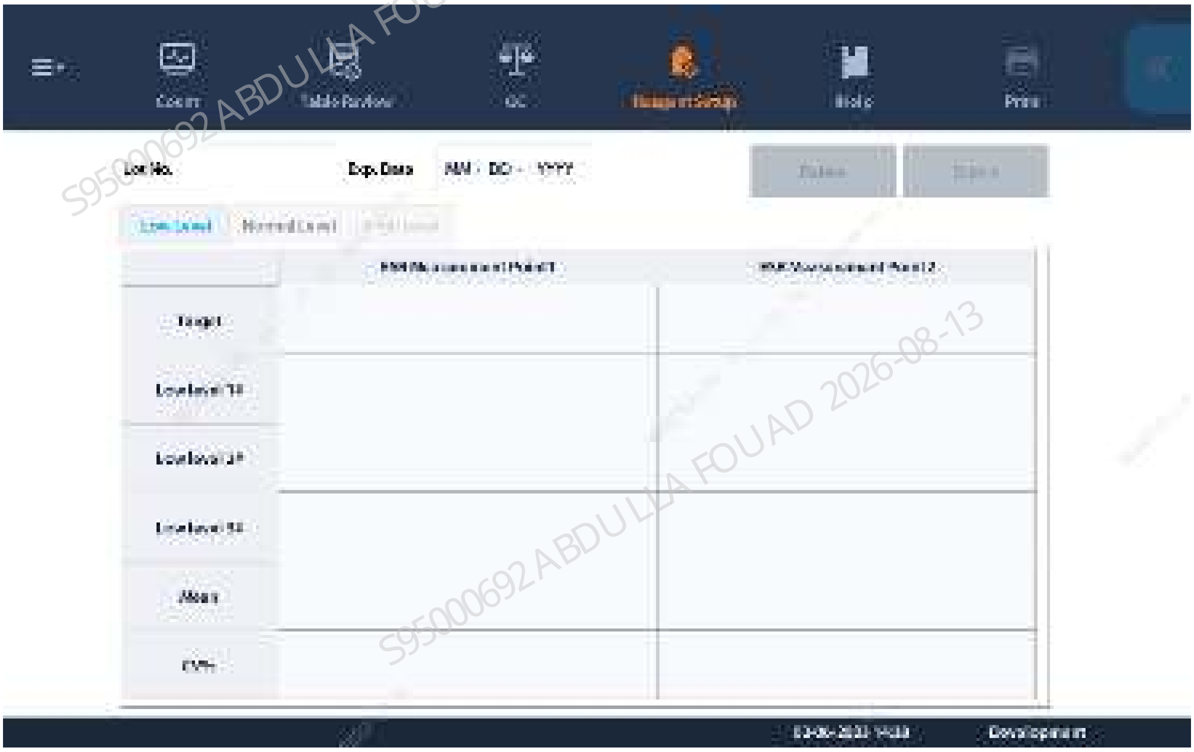
During the analyzer production or after the ESR assembly is replaced due to a failure, the ESR assembly needs to be calibrated using a PLUS calibrator.

2. ESR parameter calibration process (double measuring points)

1) Prepare the unopened PLUS calibrator, query the target value according to the lot No., and mix it after warming.

2) Select **Calibrate>ESR Calibrator Calibration**, enter the lot No. and validity period of PLUS calibrator, tap the **Low Value** page, and fill the ESR low target value in the **Target** column (both measuring points need to be set with targets and the targets are consistent). Then use PLUS calibrator to perform counting three times. If the test result is normal, the software will prompt **The low value calibration succeeds**. If the test result is abnormal, the software will clear the data and repeat the counting 3 times until it succeeds. As shown in the following figure.

Figure8–103 ESR Calibration Target Value Input - Low Value



3) Tap the **Normal Value** page and fill the target value of ESR normal value in the **Target** column (both measuring points require entering the target and the targets are consistent). Then use PLUS calibrator to perform counting three times. If the test result is normal, the software will prompt **The normal value calibration succeeds**. If the test result is abnormal, the software will clear the data and repeat the counting 3 times until it succeeds. As shown in the following figure.

Figure8–104 ESR Calibration Target Value Input - Target Value

4) The **High Value** page can be tapped after the **low value** and **normal value** are calibrated successfully. Fill the target value of ESR high value in the **Target** column (both measuring points require a target and the targets are consistent). Then use the PLUS calibrator to perform counting three times. If the test result is normal, the software will prompt **The high value calibration succeeds**. If the test result is abnormal, the software will clear the data and repeat the counting 3 times until it succeeds. As shown in the following figure. Save calibration factors.

Figure8–105 ESR Calibration Target Value Input - High Value

5) Take a BC-6D normal value quality control, query the target value according to the lot No., and then make measurement in the CD+ESR counting mode to confirm that the ESR measurement result is within the allowable deviation range of the control target value.

8.14 Reagent detection tube

8.14.1 General Information

N/A

8.14.2 Disassembly and Assembly

N/A

8.14.3 Commissioning and Verification

N/A

8.15 Paired wire, 5-position and 6-row closed-sampling autoloader

8.15.1 General Information

N/A

8.15.2 Disassembly and Assembly

N/A

8.15.3 Commissioning and Verification

N/A

8.16 Trumpet Assembly

8.16.1 General Information

Figure8–106 Photo of 115-082029-00 Trumpet Assembly

Figure8–107 Diagram of Location of Trumpet Assembly in the Main Unit

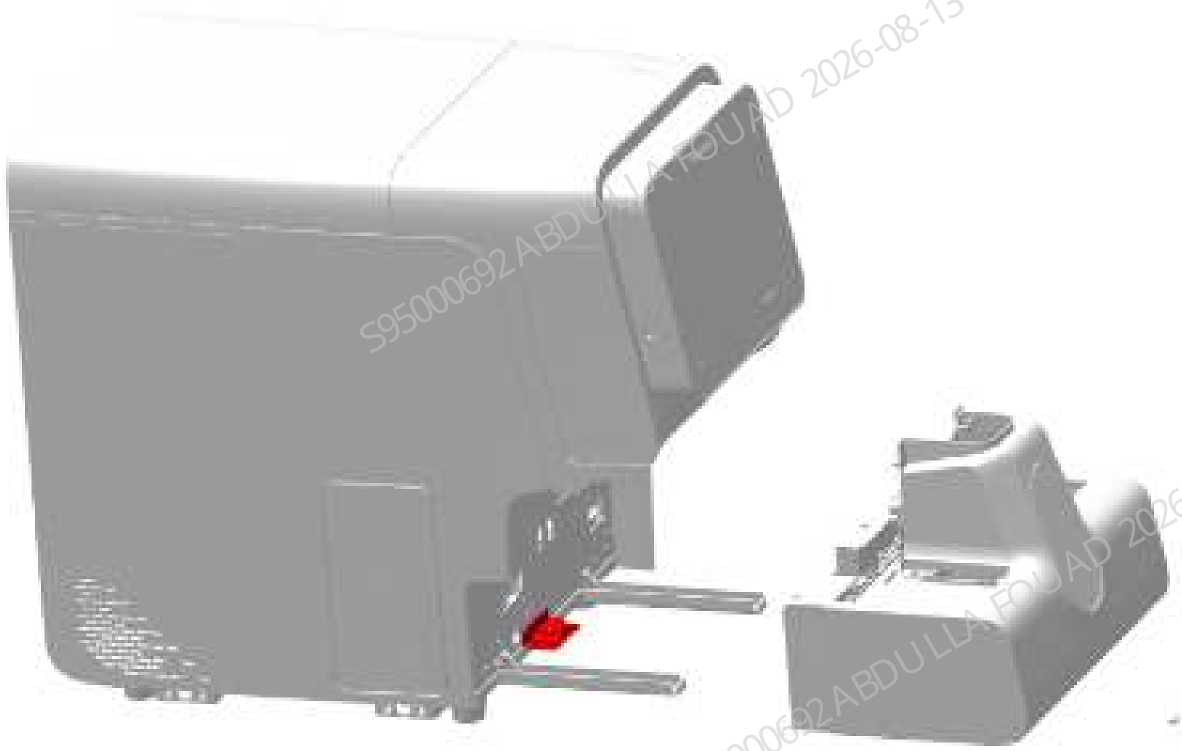


Table 8–10 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-082029-00	Trumpet Assembly	N/A	N/A

Function:
Providing alarm tones and beeps.

8.16.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. Remove the two M4x8 cross-recessed combination screws (No. 1) fixing the autoloader on the autoloader crossbeam.
2. Drag the autoloader (No. 2) to an appropriate distance in the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
3. Pull the autoloader completely out of the autoloader crossbeam (No. 3) along the direction of the arrow.
4. Use cross-headed screwdriver to remove the two screws (No. 4) fixing the trumpet assembly and pull out the wire of the trumpet assembly. Then, remove the trumpet assembly (No. 5).

Figure8–108 Diagram 1 for Maintenance Procedure of Trumpet Assembly

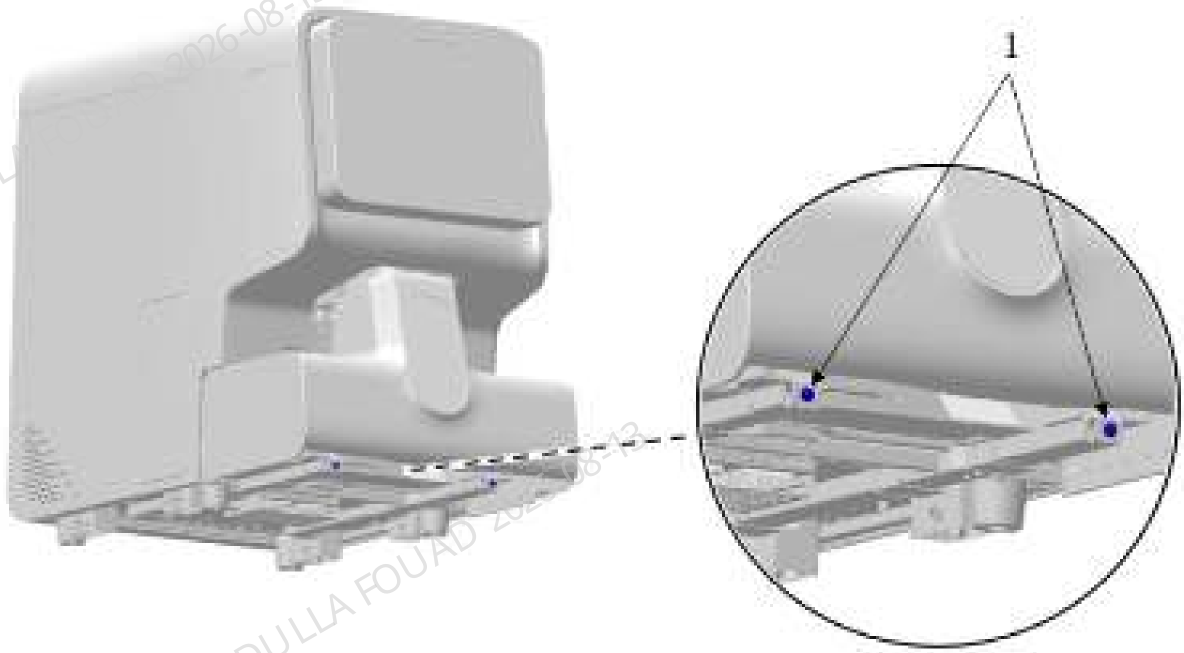


Figure8–109 Diagram 2 for Maintenance Procedure of Trumpet Assembly

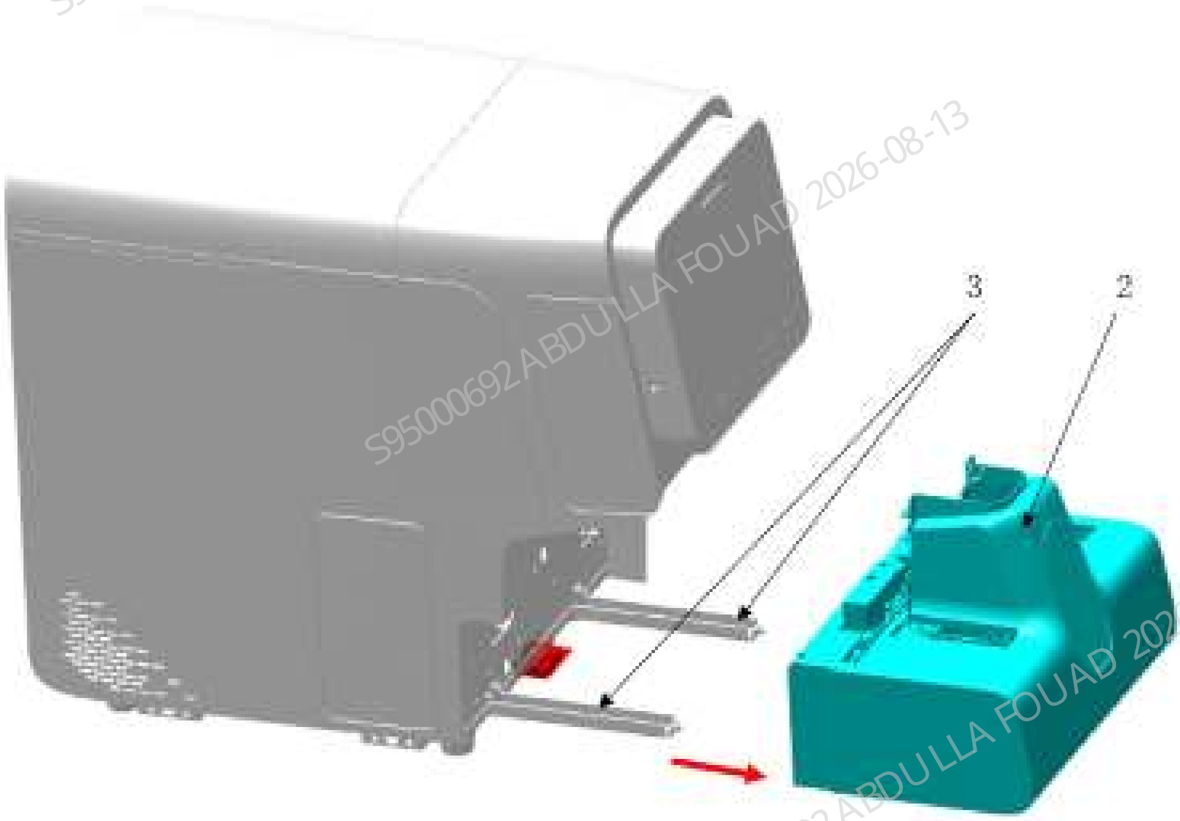
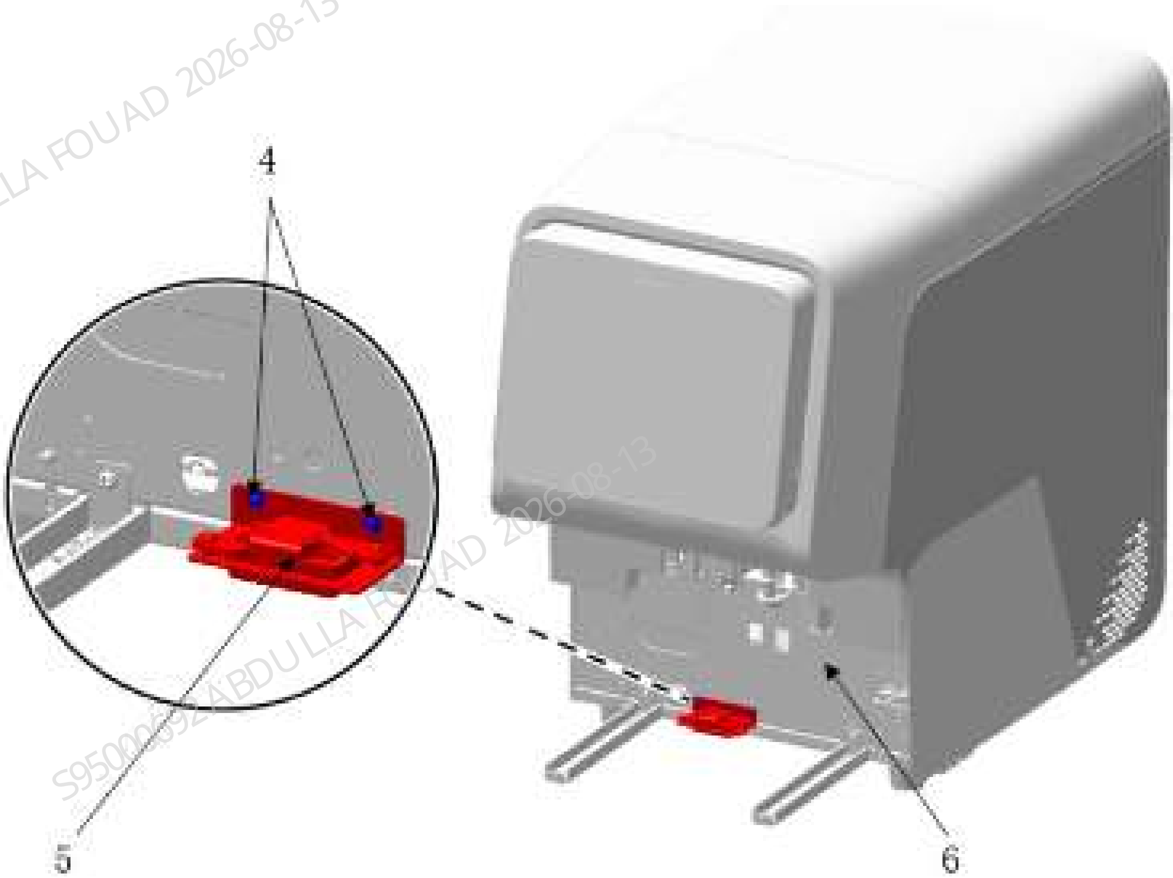


Figure8–110 Diagram 3 for Maintenance Procedure of Trumpet Assembly



Subsequent Requirements

Installation procedure:

1. Use two screws (No. 4) to lock the trumpet assembly (No. 5) on the front panel of the main unit (No. 6), and insert the trumpet assembly wire.
2. Connect the wire between the autoloader and the main unit, and then push the autoloader (No. 2) along the autoloader crossbeam (No. 3) to fit the front panel of the main unit (No. 6).
3. Use two screws (No. 1) to lock the autoloader (No. 2) on the autoloader crossbeam.

8.16.3 Commissioning and Verification

Required tools

Cross-headed screwdriver

Procedures

Commissioning is not required.

Verification:

1. Check whether the analyzer will send an alarm sound. If so, the replacement is normal. Click on the error area and click **Eliminate Error** on the pop-up error elimination screen to confirm that the error can be eliminated.

8.17 9202 Switch Key Board PCBA-T

8.17.1 General Information

Figure8–111 Photo of 9202 switch key board PCBA-T

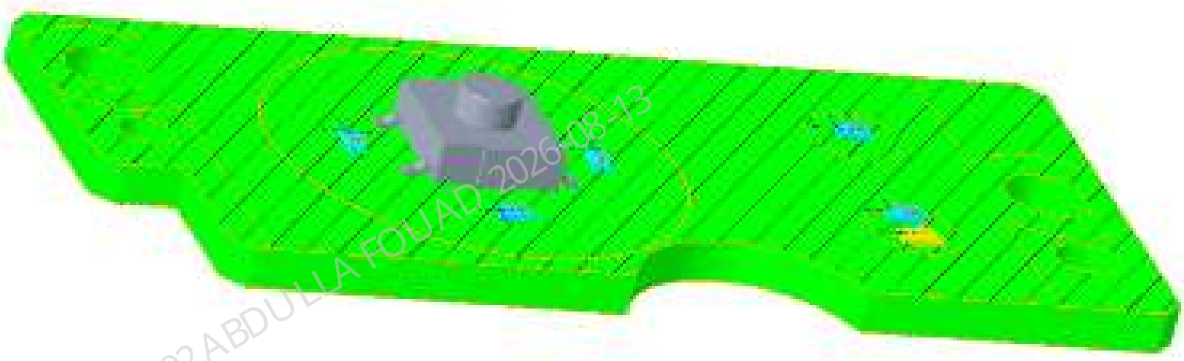


Figure8–112 Position of Switch Key Board PCBA-T on the Overall Unit

Table 8–11 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
051-004842-00	9202 Switch Key Board PCBA-T	N/A	N/A

Functions:

Providing a soft switch of the analyzer.

8.17.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. Open the front cover assembly (No. 1).
2. Use a cross-headed screwdriver to remove the two screws fixing the power switch board PCBA, and then pull out the wire.

Figure8–113 Figure 1 - Maintenance Procedure of the Switch Key Board PCBA

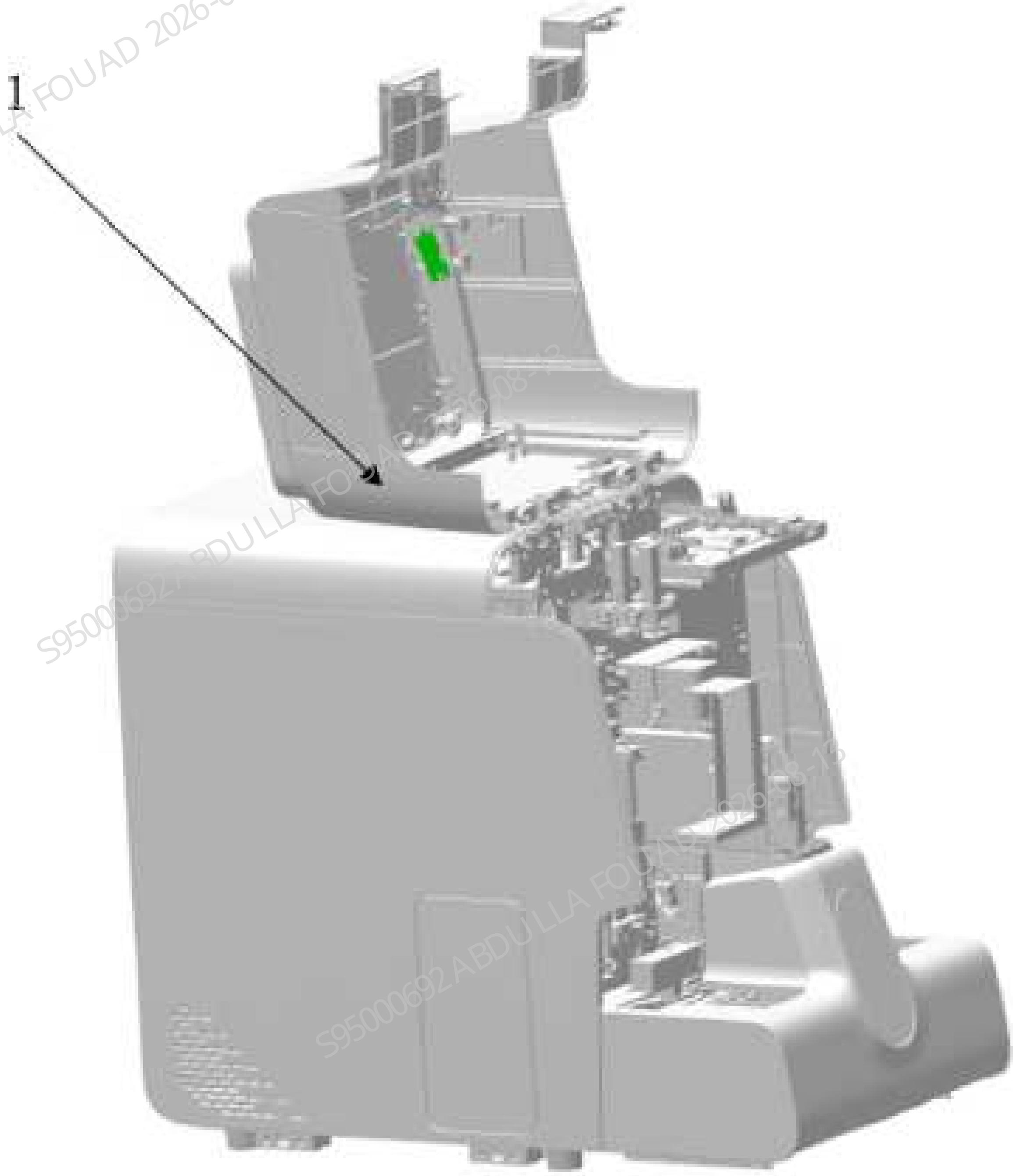
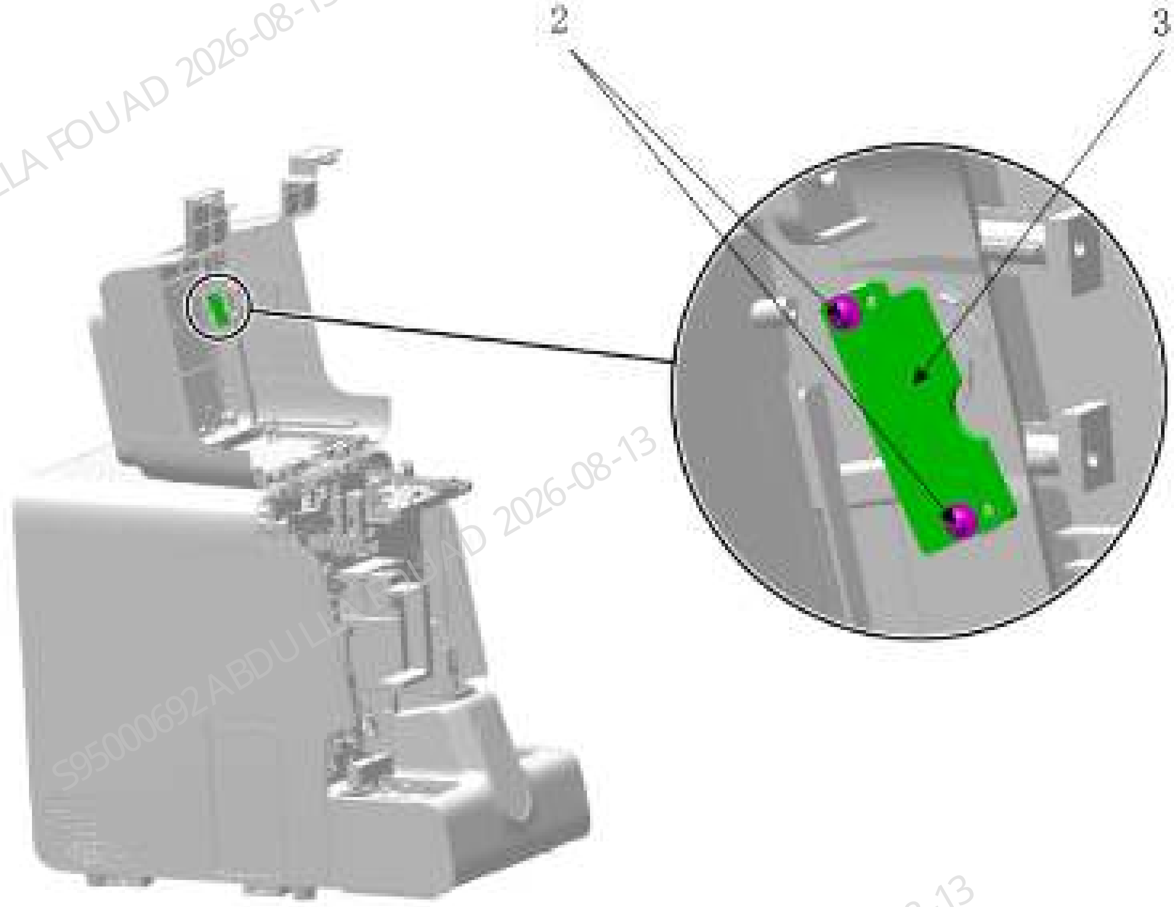


Figure8–114 Figure 2 - Maintenance Procedure of the Switch Key Board PCBA



Subsequent Requirements

Installation procedure:

1. Use two screws (No. 2) to lock the power switch board PCBA (No. 3) on the front cover assembly (No. 1), and insert the power switch board PCBA wire.
2. Close the front cover assembly (No. 1)

8.17.3 Commissioning and Verification

Required tools

Cross-headed screwdriver

Procedures

Commissioning is not required.

Verification:

1. Check whether the soft switch of the analyzer is normal. If it is normal, the replacement is normal. Click on the error area and click **Eliminate Error** on the pop-up error elimination screen to confirm that the error can be eliminated.

8.18 Fully-fitted Screen Assembly

8.18.1 General Information

Figure8–115 Photo of display assembly

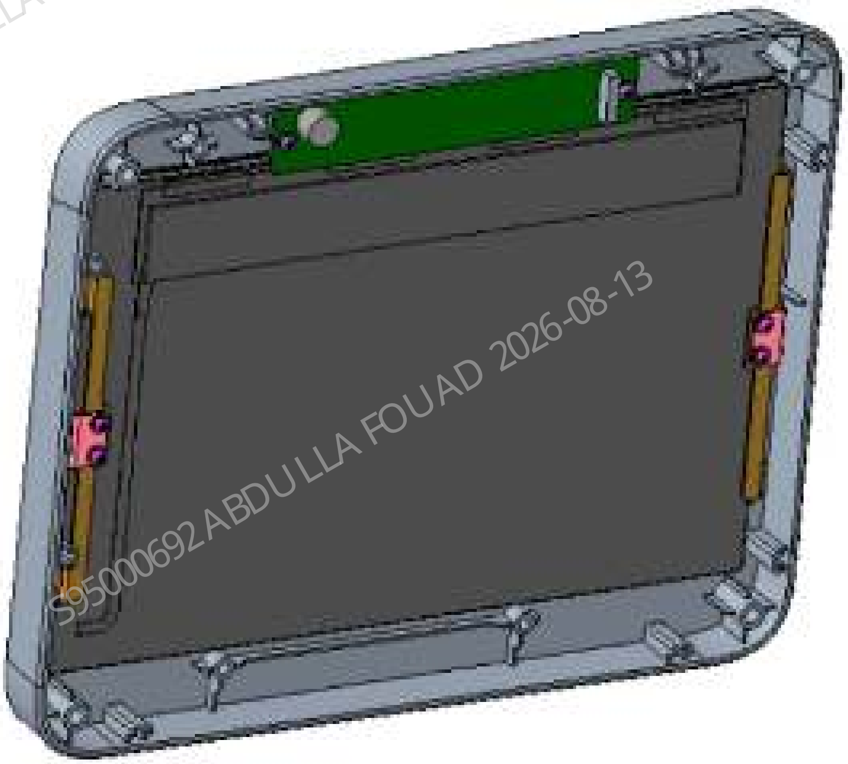


Table 8–12 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081657-00	Display assembly	N/A	N/A

Function:
Full screen for user interaction and its mounted cover bracket.

8.18.2 Disassembly and Assembly

Required fixtures

Cross-headed screwdriver
One pair of cutting nippers

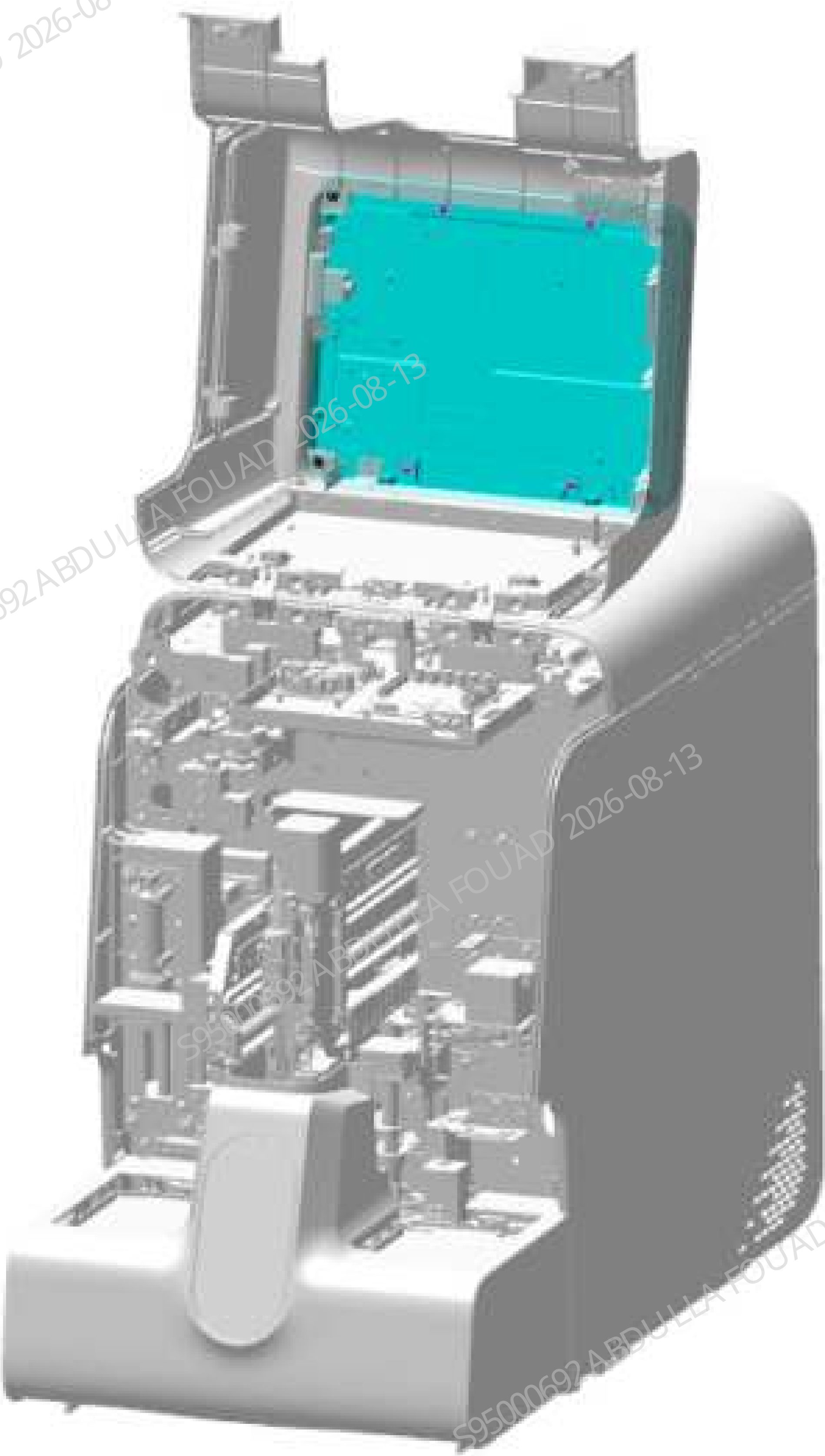
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

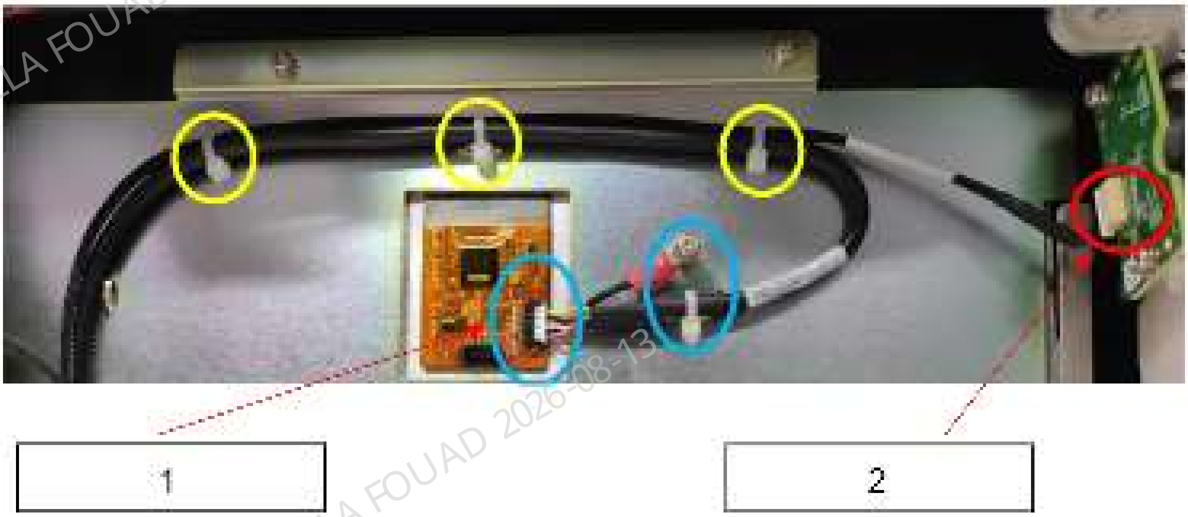
1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–116 Opening the front cover



2. Disconnect the touch screen cable and the switch key cable, and remove the cable tie at the same time.

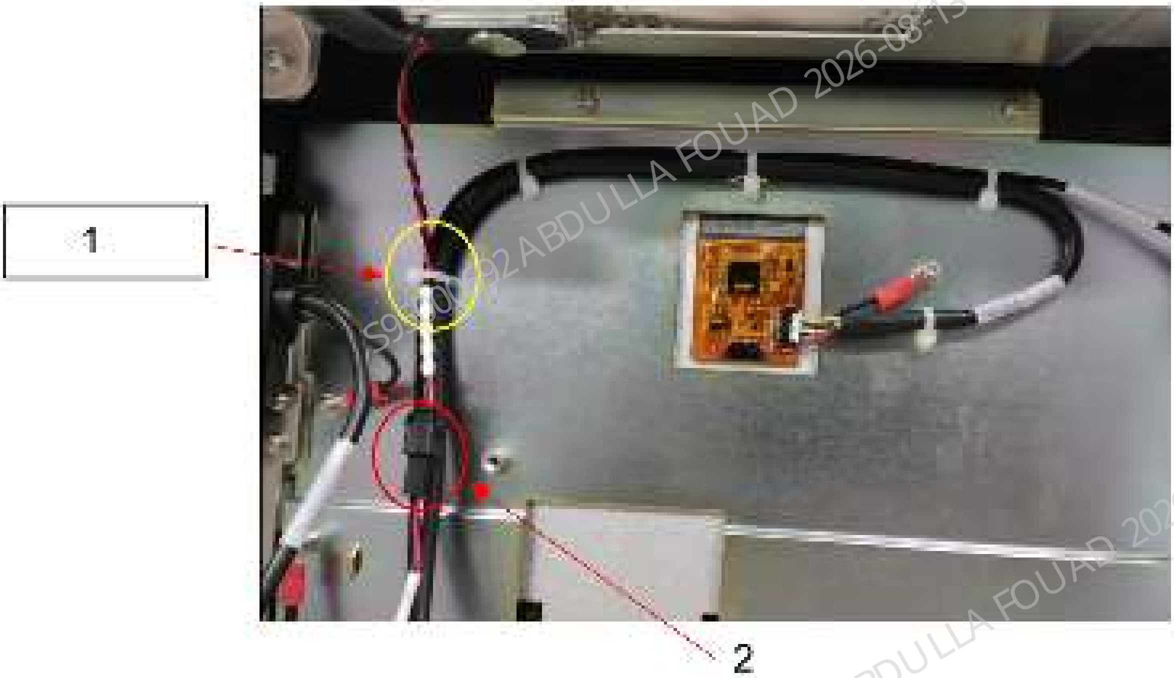
Figure8–117 Touch screen cable and switch key cable



1	Touch screen cable
2	Switch key cable

3. Remove the cable ties securing the speaker wire while disconnecting the connector.

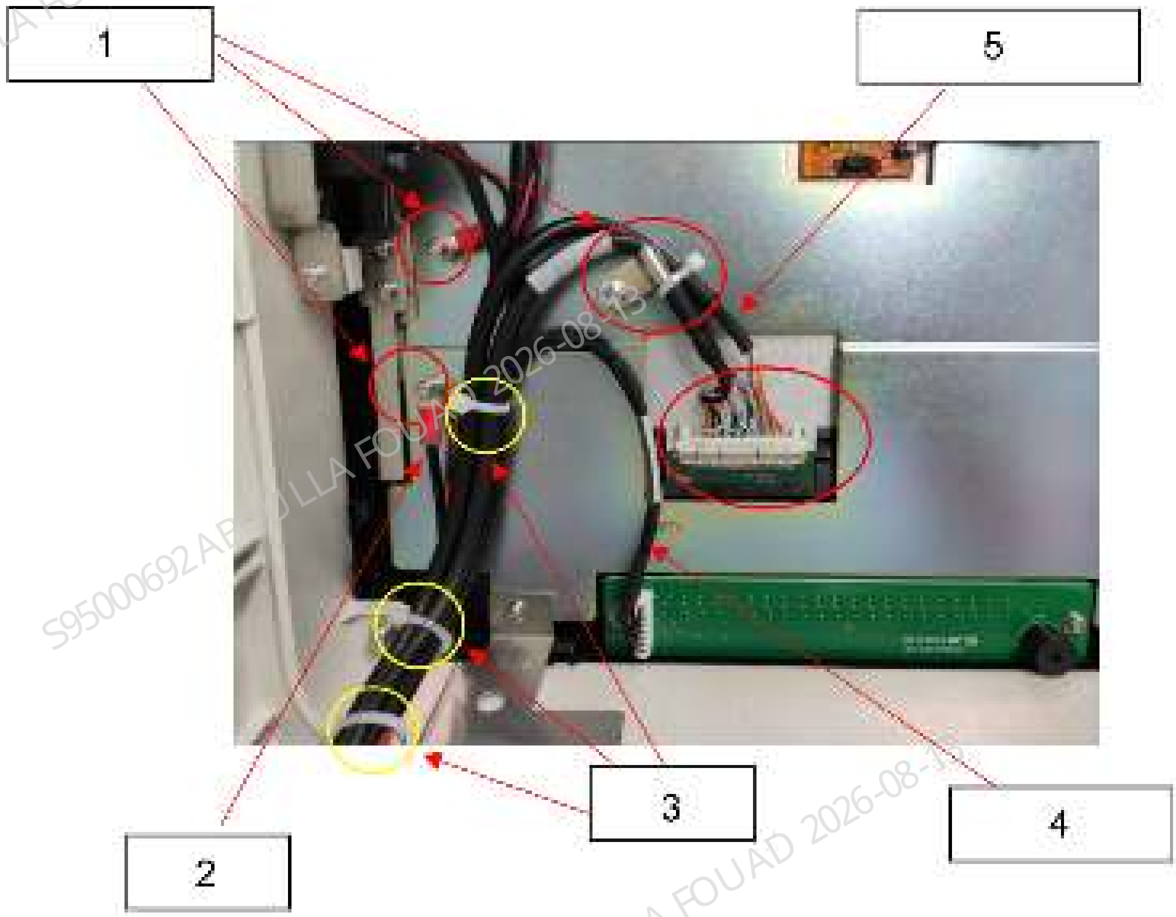
Figure8–118 Speaker wires



1	Wire tie
2	Speaker wires

4. Remove the three grounding screws, remove the three cable ties marked by the yellow circles, and unplug the display backlight cable and indicator board cable.

Figure8–119 Display backlight cable, indicator board cable, and grounding cable



1	Grounding screw
2	Grounding cable
3	Wire tie
4	Indicator board cable
5	Screen backlight cable

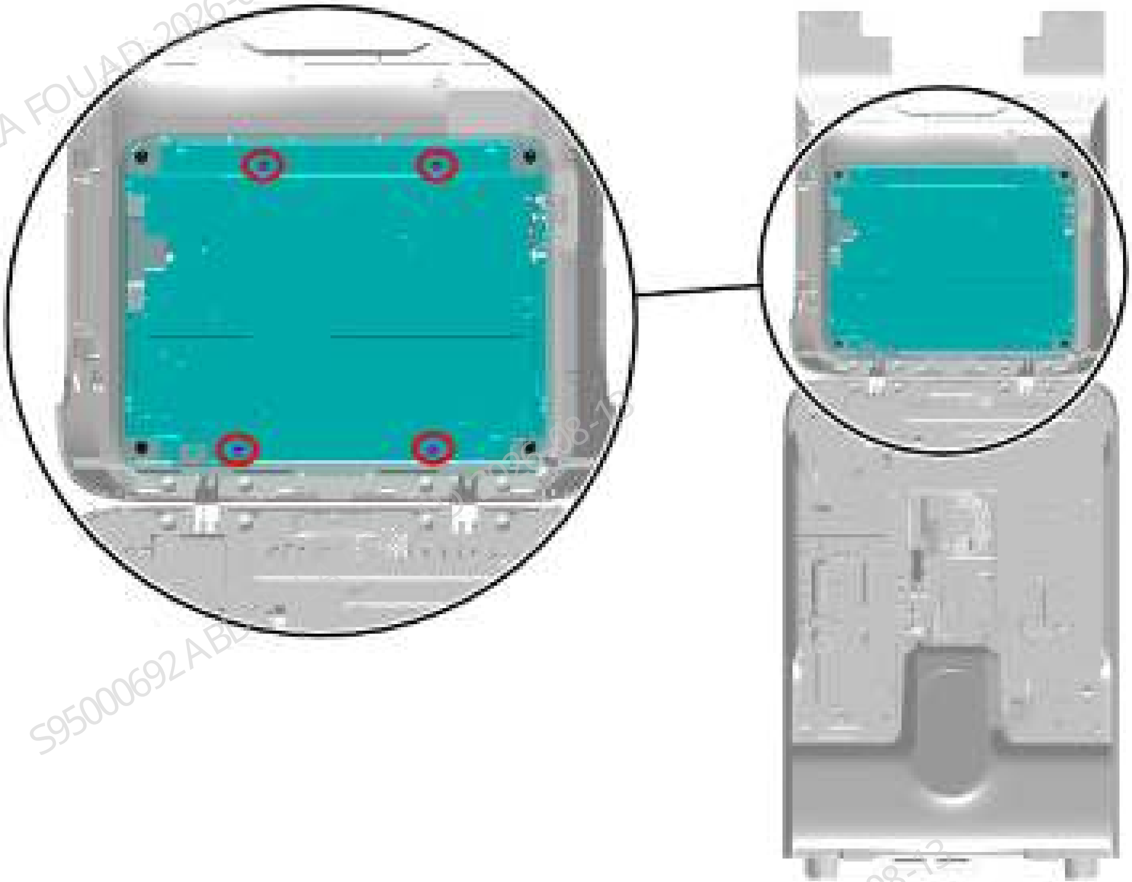
5. Remove the USB assembly.

Figure8–120 USB assembly

1	Remove the two cross small pan head screw assemblies M3X8 and take out the USB assembly

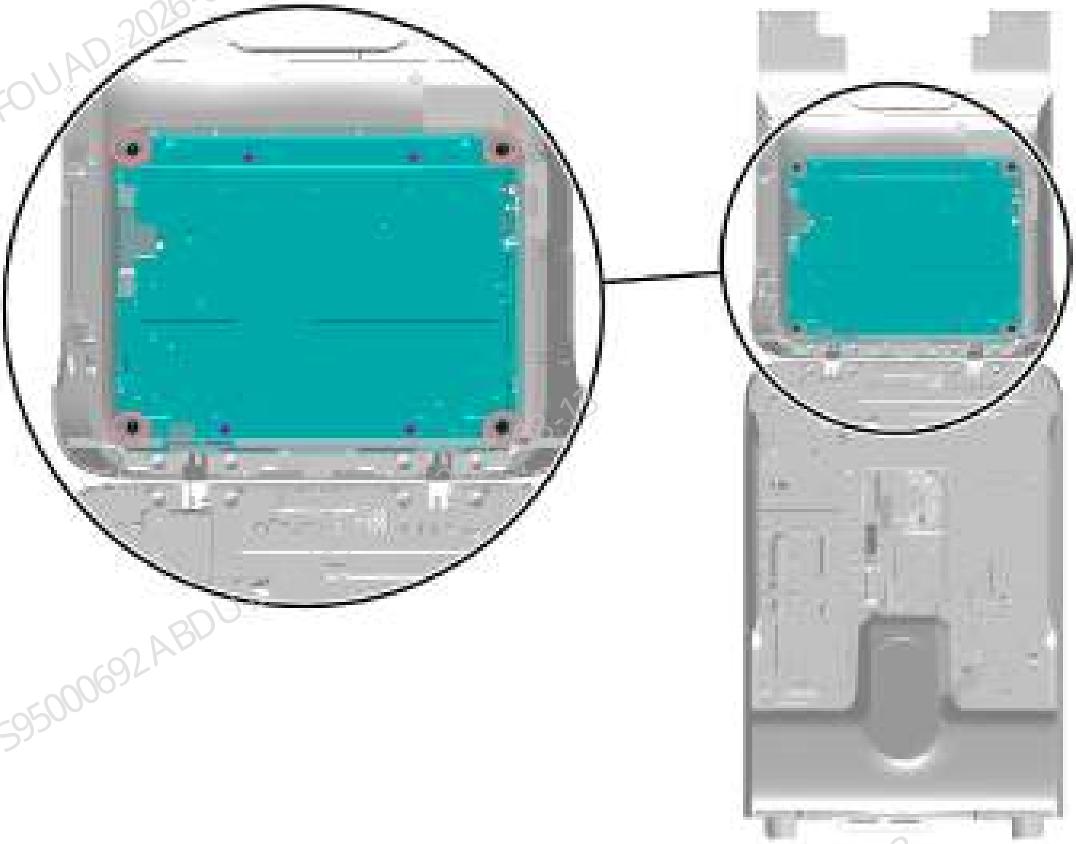
6. Loosen the four screws securing the screen cable cover, as shown in the figure below, and remove the cover from the fully-fitted screen assembly.

Figure8–121 Positions of the set screws of the screen cable cover



7. Loosen the screen cable plug, loosen the four screws securing the fully-fitted screen assembly, as shown in the figure below, and remove the fully-fitted screen assembly.

Figure8–122 Positions of the set screws of the fully-fitted screen assembly



Subsequent Requirements

Required tools:

Cross-headed screwdriver

One pair of cutting nippers

Several wire ties

Installation procedure:

1. Install the fully-fitted screen assembly (FRU) on the front cover and fix it using four screws.
2. Fasten the screen cable cover to the fully-fitted screen assembly (FRU) using four screws.
3. Connect the screen cable plug to the fully-fitted screen according to Figures **Figure8–117**, **Figure8–118**, **Figure8–119**, and **Figure8–120**, and fix it in the corresponding position using wire ties.

NOTICE

1 All wires here need to be tied on the left side of the sheet metal

4. Flip down the front cover to reset it.

8.18.3 Commissioning and Verification

Procedures

1. Connect the power supply, start the operation, and log in with the service access level.
2. Select **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Other parts**. Select **Front cover assembly**, click the **Display**button, (as shown in the figure below), observe whether the display screen shows white, black, red, green and blue stripes in sequence.
   Check whether the indicator board shows red, green, and yellow in sequence, and whether the buzzer sounds.

Figure8–123 Commissioning screen of the front cover assembly

3. Press and hold the switch button on the right for 10s and observe whether the analyzer screen is turned off.

8.19 PHOTOELEC Optical Sensor IP55 940nm 15cm

8.19.1 General Information

For the general information in the case of Y-direction initial position sensor of sampling, see Y-direction Initial Position Sensor of Sampling.

For the general information in the case of Z-direction initial position sampling sensor of the sampling assembly, see Z-direction Initial Position Sampling Sensor of the Sampling Assembly.

For the general information in the case of Y-direction initial position sensor of the sampling assembly, see Y-direction Initial Position Sensor of Sampling Assembly.

For the general information in the case of rotary compressing assembly sensor, see Rotary compressing assembly sensor.

Y-direction Initial Position Sensor of Sampling

Figure8–124 Photo of the Sensor

Figure8–125 Diagram of Position of the Y-direction Initial Position Sensor in the Sampling Assembly

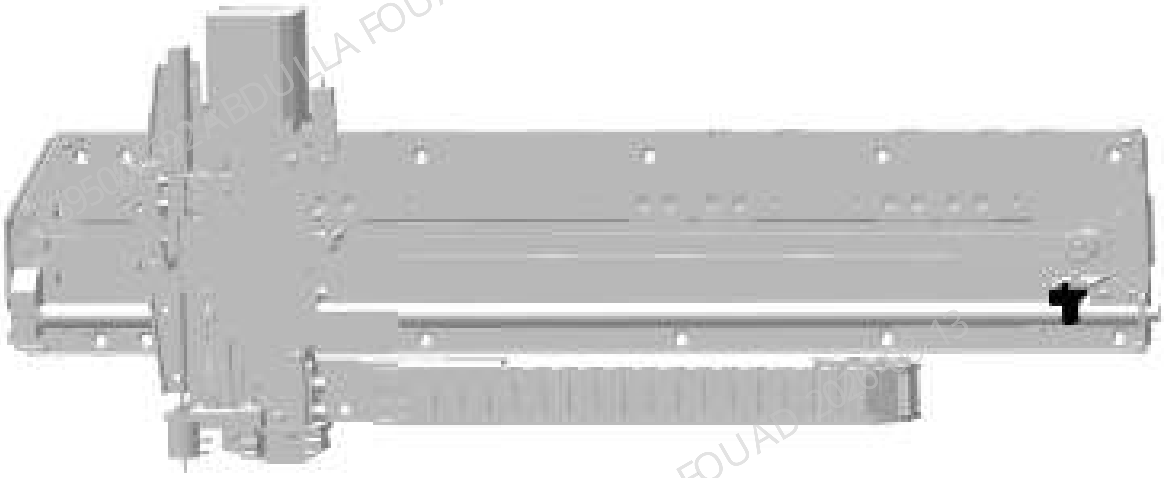


Table 8–13 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000021-00	Sensor	N/A	N/A

Function:

Used to detect the initial position in the Y direction of the sampling assembly.

Z-direction Initial Position Sampling Sensor of the Sampling Assembly

Figure8–126 Photo of the Sensor

Figure8–127 Diagram of Position of the Sampling Notch Detection Sensor in the Sampling Assembly

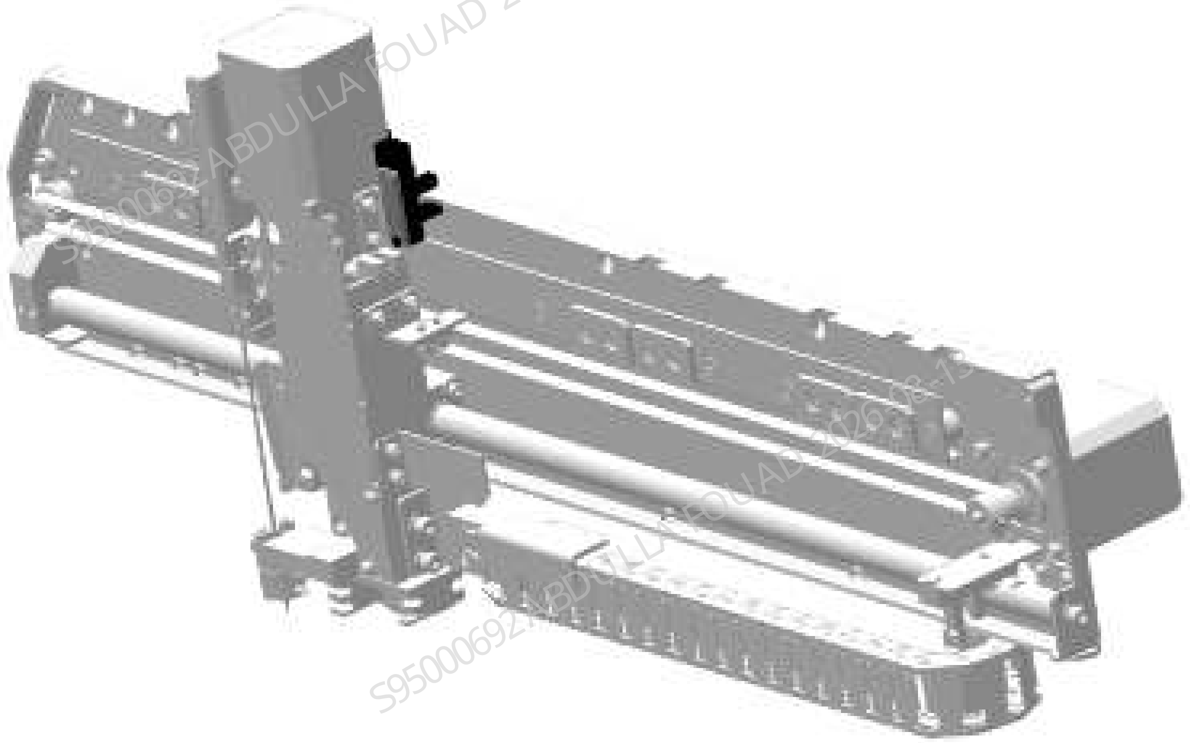


Table 8–14 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000021-00	PHOTOELEC Optical Sensor IP55 940nm 15cm	N/A	N/A

Function:
Used to detect the initial position in the Z direction of the sampling assembly.

Y-direction Initial Position Sensor of Sampling Assembly

Figure8–128 Photo of the Detection Sensor

Figure8–129 Photo of Position of the Detection Sensor in the Sampling Assembly

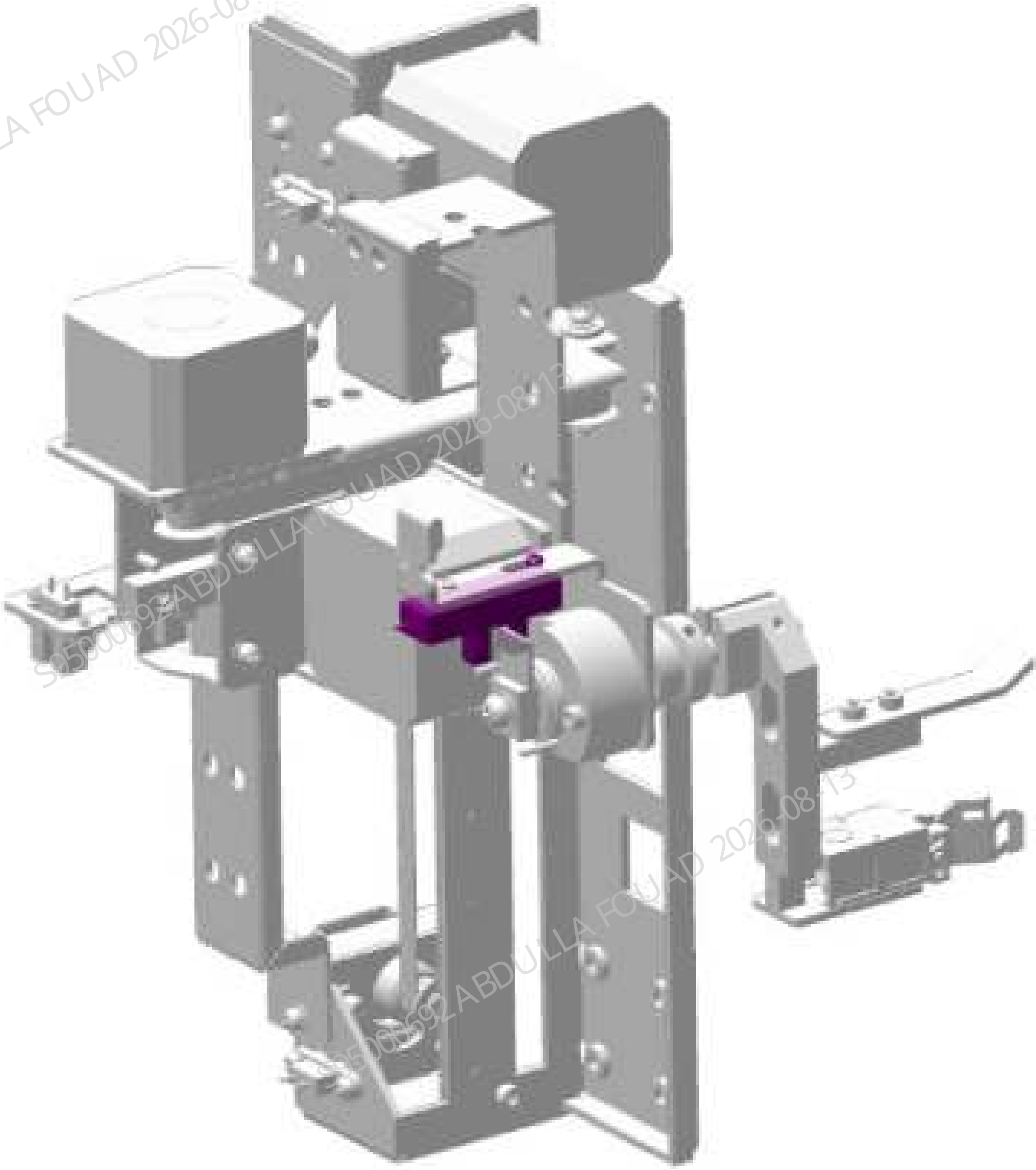


Table 8–15 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000021-00	Tube detecting sensors	N/A	N/A

Function:

Used for home position detection of the syringe.

Rotary compressing assembly sensor

Figure8–130 Photo of the Sensor

Figure8–131 Diagram of Position of the Sensor in the Rotary Compressing Assembly

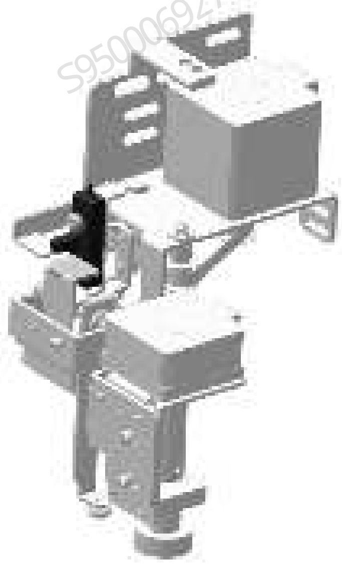


Table 8–16 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000021-00	PHOTOELEC Optical Sensor IP55 940nm 15cm	N/A	N/A

Function:
Used to detect the front-rear position of the two pinch rollers on the rotary scanning assembly.

8.19.2 Disassembly and Assembly

For the disassembly and assembly in the case of Y-direction initial position sensor of sampling, see Y-direction Initial Position Sensor of Sampling.

For the disassembly and assembly in the case of Z-direction initial position sampling sensor of the sampling assembly, see Z-direction Initial Position Sampling Sensor of the Sampling Assembly.

For the disassembly and assembly in the case of Y-direction initial position sensor of the sampling assembly, see Y-direction Initial Position Sensor of Sampling Assembly.

For the disassembly and assembly in the case of rotary compressing assembly sensor, see Rotary compressing assembly sensor.

Y-direction Initial Position Sensor of Sampling

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench

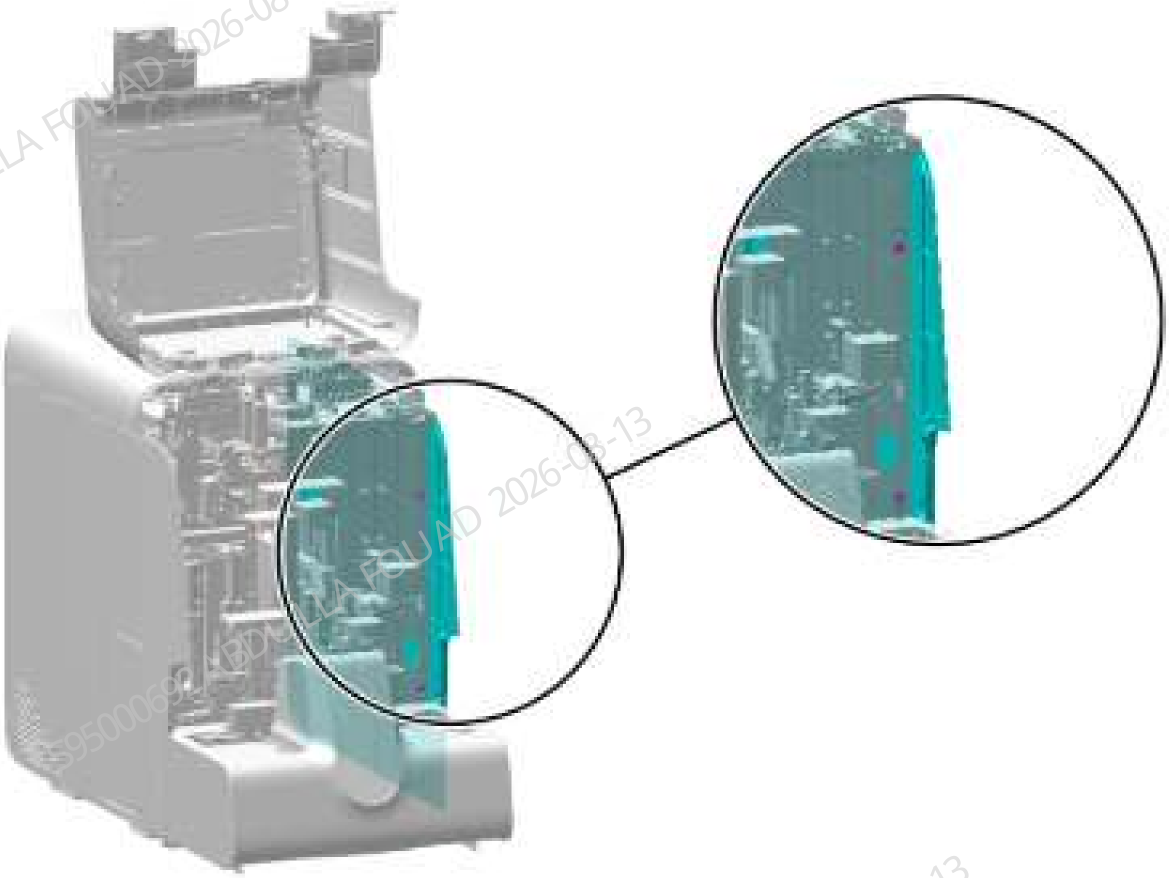
Pre-requirements

N/A

Procedures

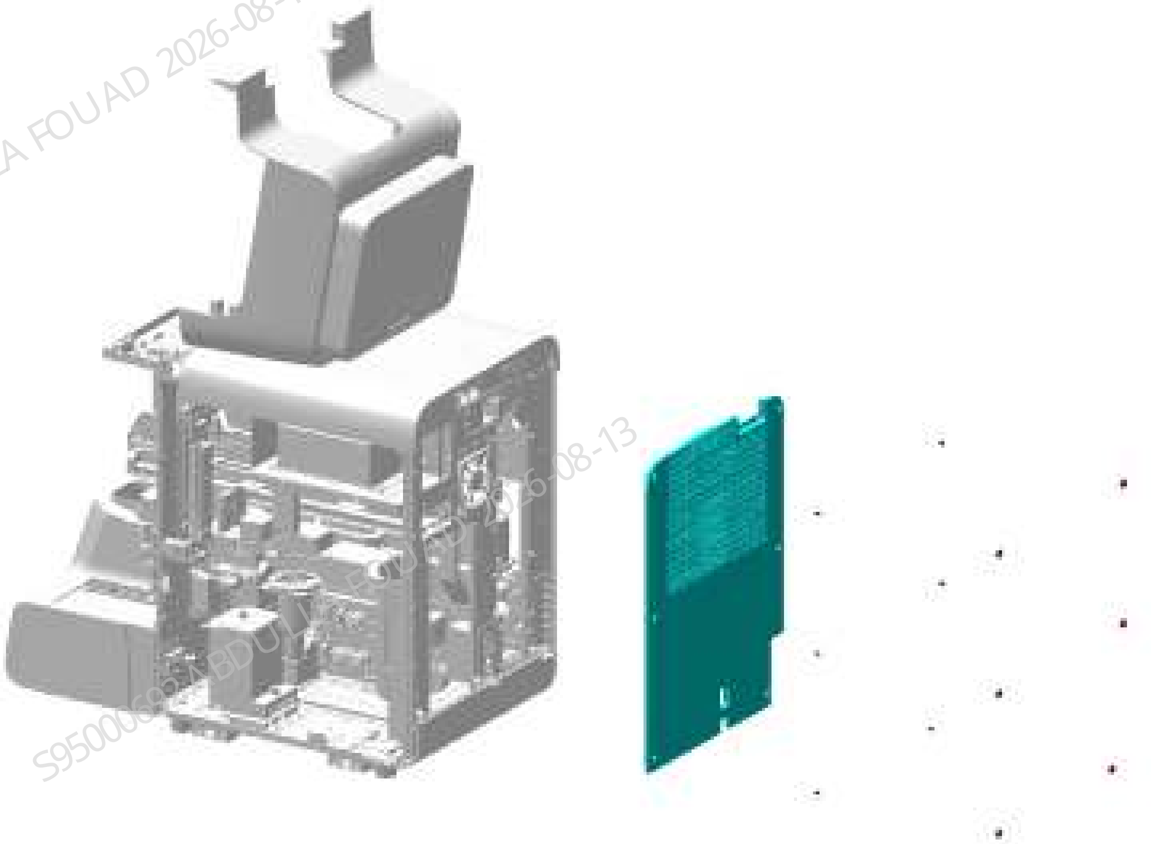
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–132 Removing the right cover



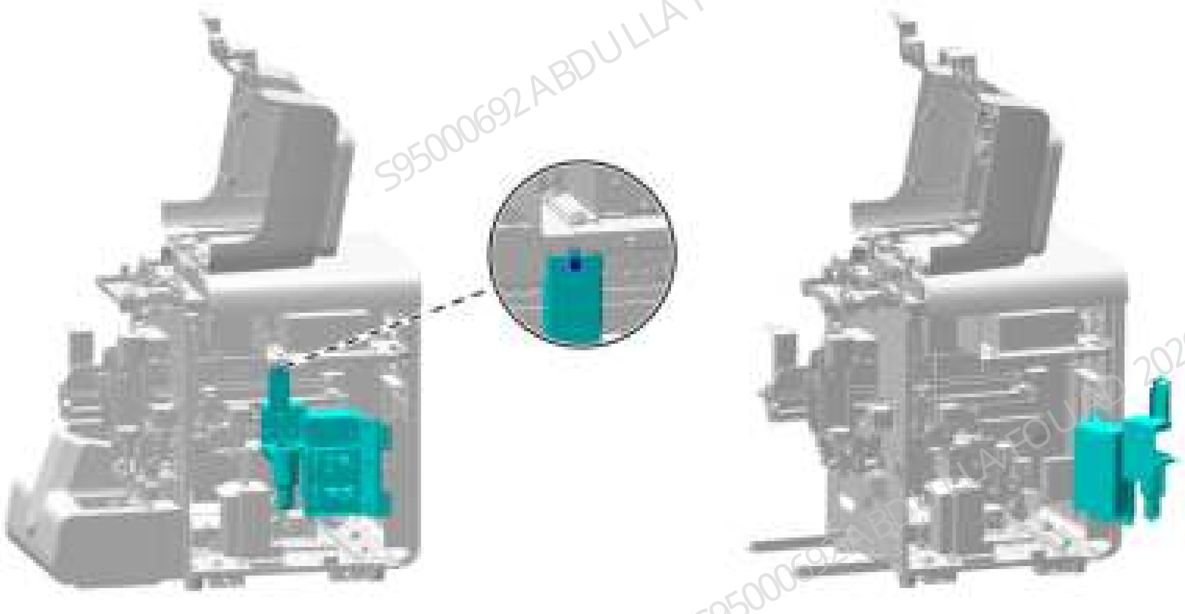
2. As shown in the figure below, remove the screws fixing the rear panel, and remove the rear panel.

Figure8–133 Removing procedure of the rear panel



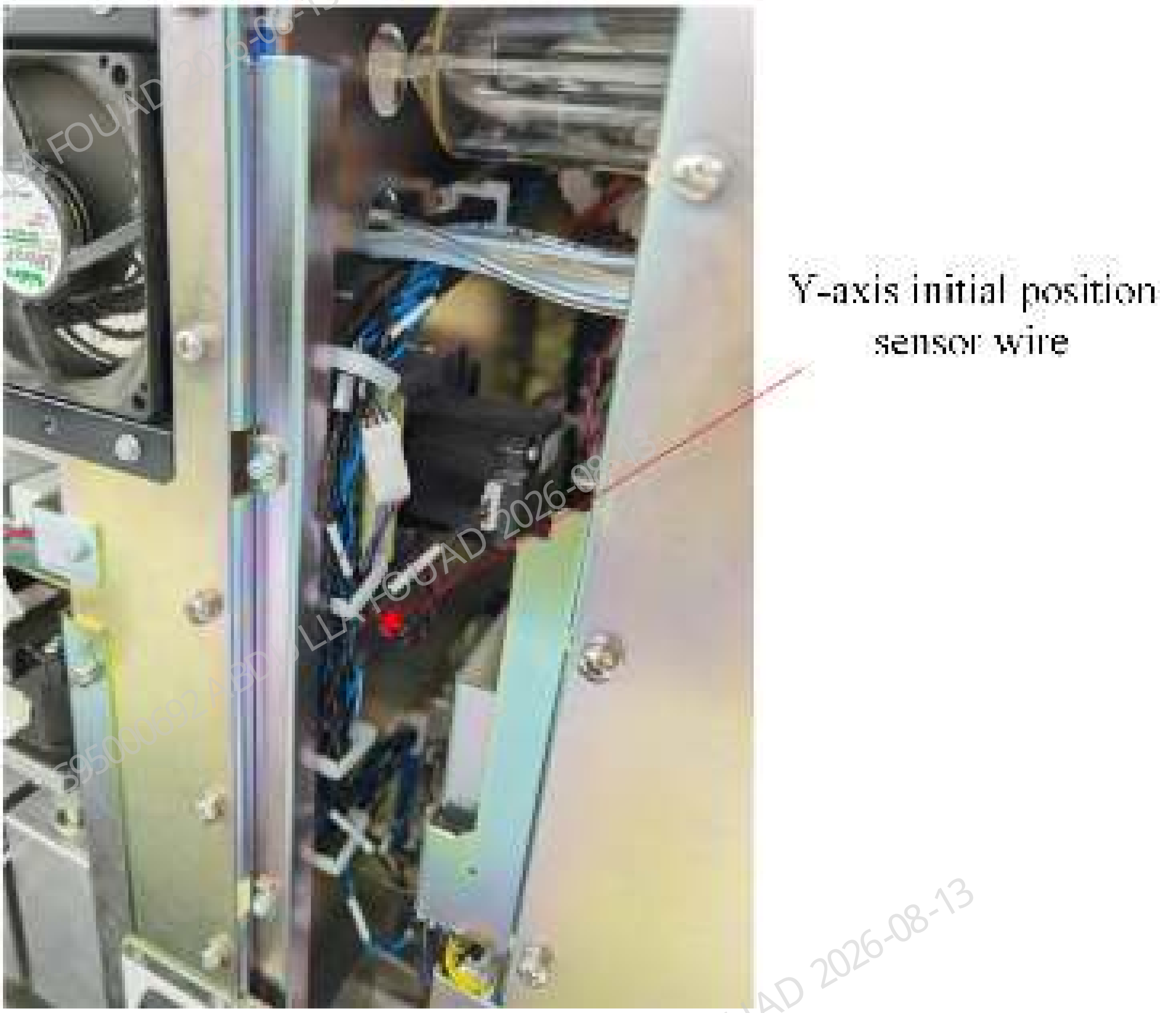
3. As shown below, unscrew the one screw that fixes the right bath subassembly; as shown in the figure, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–134 Opening the right cover assembly



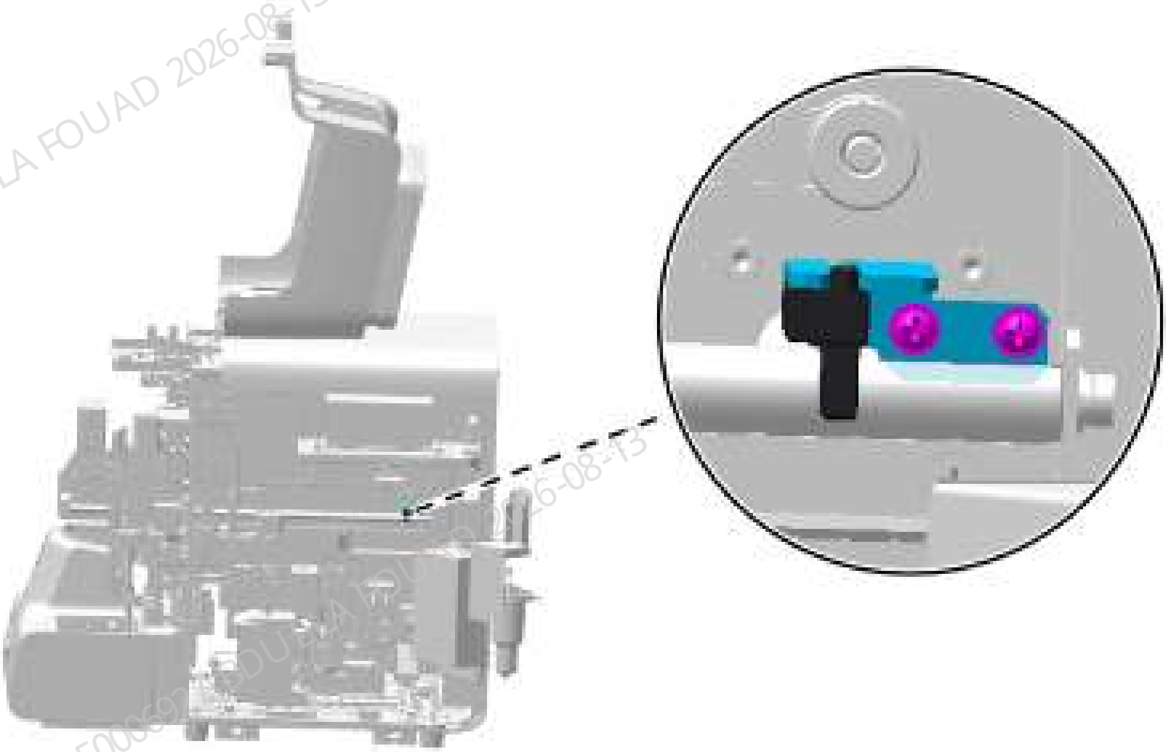
4. As shown in the following figure, remove the sensor wire.

Figure8–135 Removing the Y-direction initial position sensor wires of sampling



5. As shown in the figure below, remove the two screws fixing the Y-direction sensor fixing plate from the right side of the analyzer.

Figure8–136 Removing the small assembly of Y-direction initial position sensor



6. As shown below, use a screwdriver to remove the one screw fixing the sensor on the Y-direction sensor fixing plate, and take down the sensor.

Figure8–137 Removing the Y-direction initial position sensor

Subsequent Requirements

Installation procedure:

1. Install the sensor on the Y-direction sensor fixing plate at the position shown in the above figure.
2. Install the sensor fixing plate on the sampling assembly and insert the sensor wire.
3. Turn the right side panel to its original position and fix it using one screw.
4. Restore the right, rear, and front covers.

Z-direction Initial Position Sampling Sensor of the Sampling Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One Allen wrench

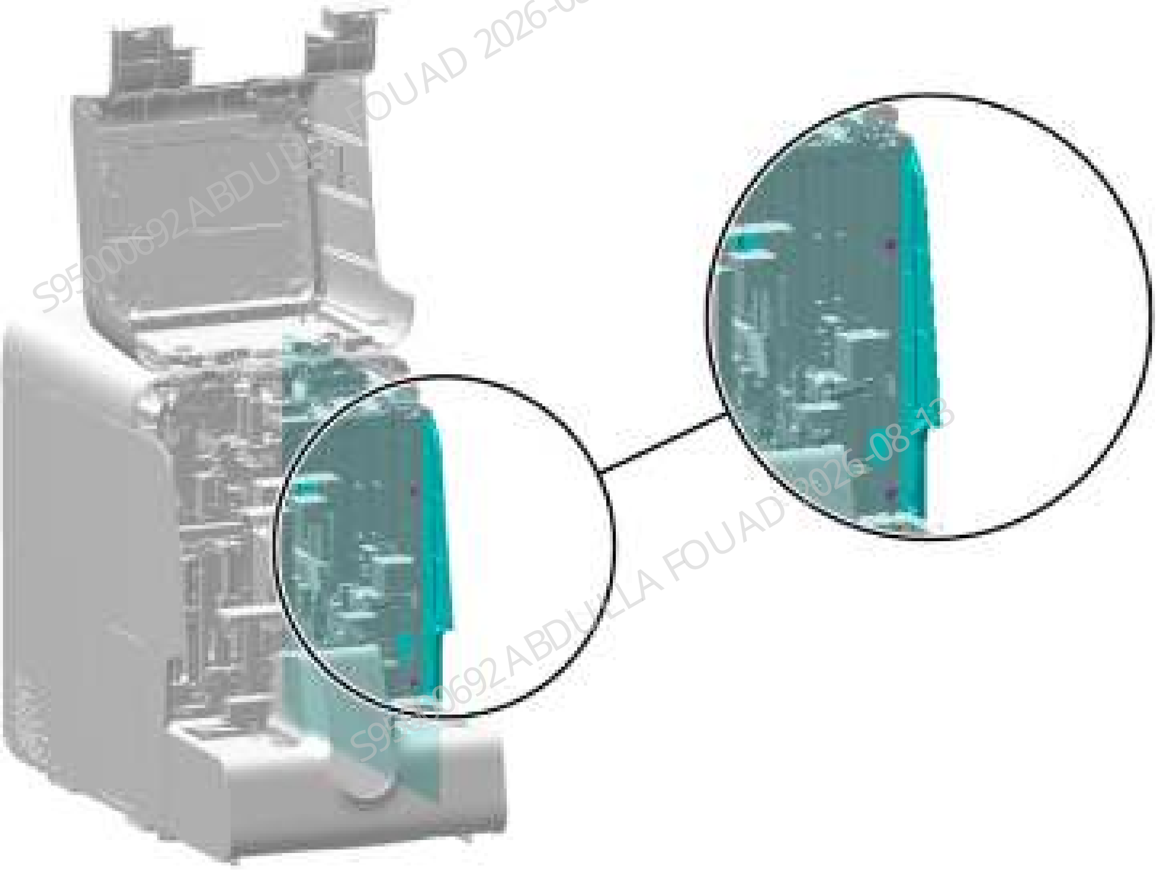
Pre-requirements

N/A

Procedures

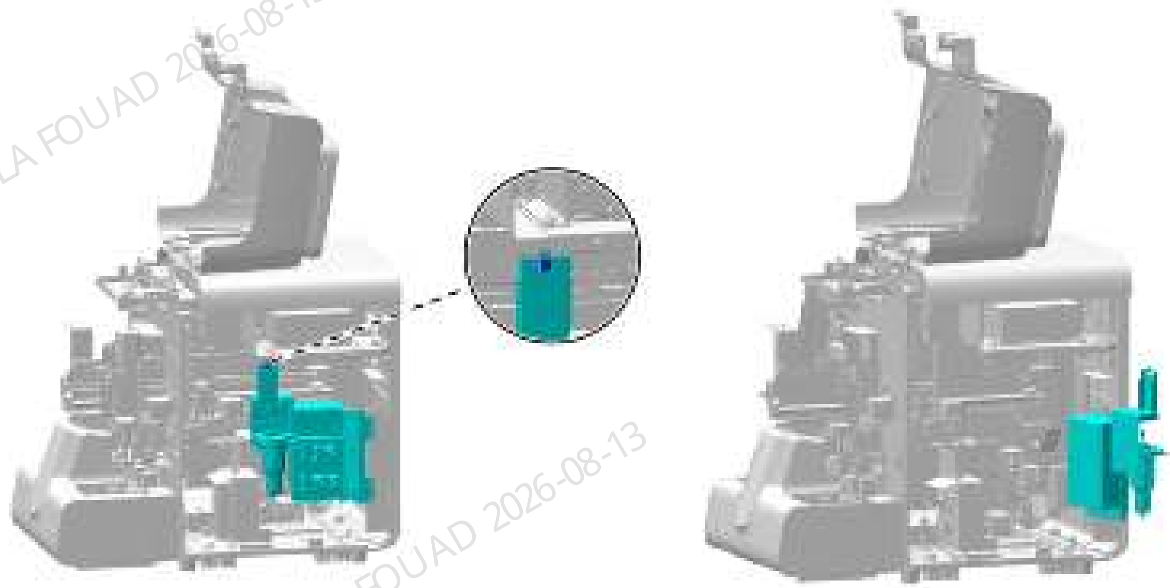
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–138 Removing the right cover



2. As shown below, unscrew the one screw that fixes the right bath subassembly; as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–139 Opening the right cover assembly

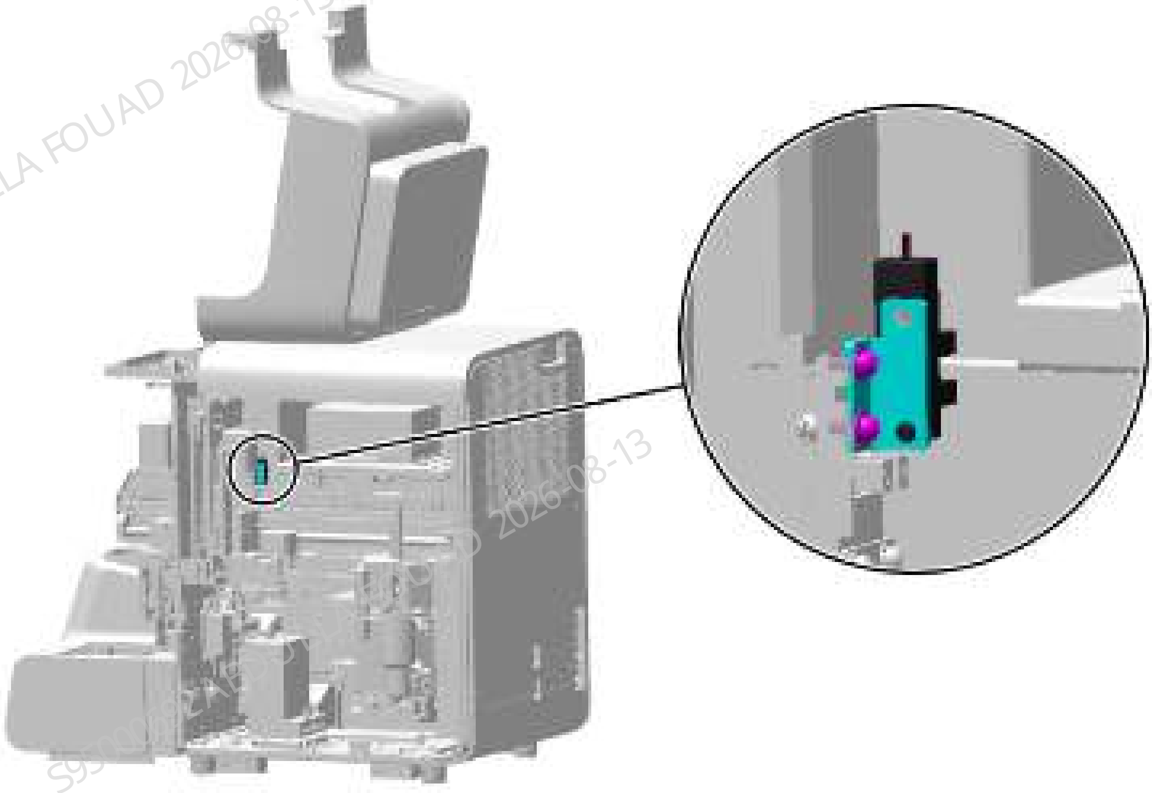


3. As shown in the following figure, remove the sensor wire.

Figure8–140 Removing the sampling notch detecting sensor wire

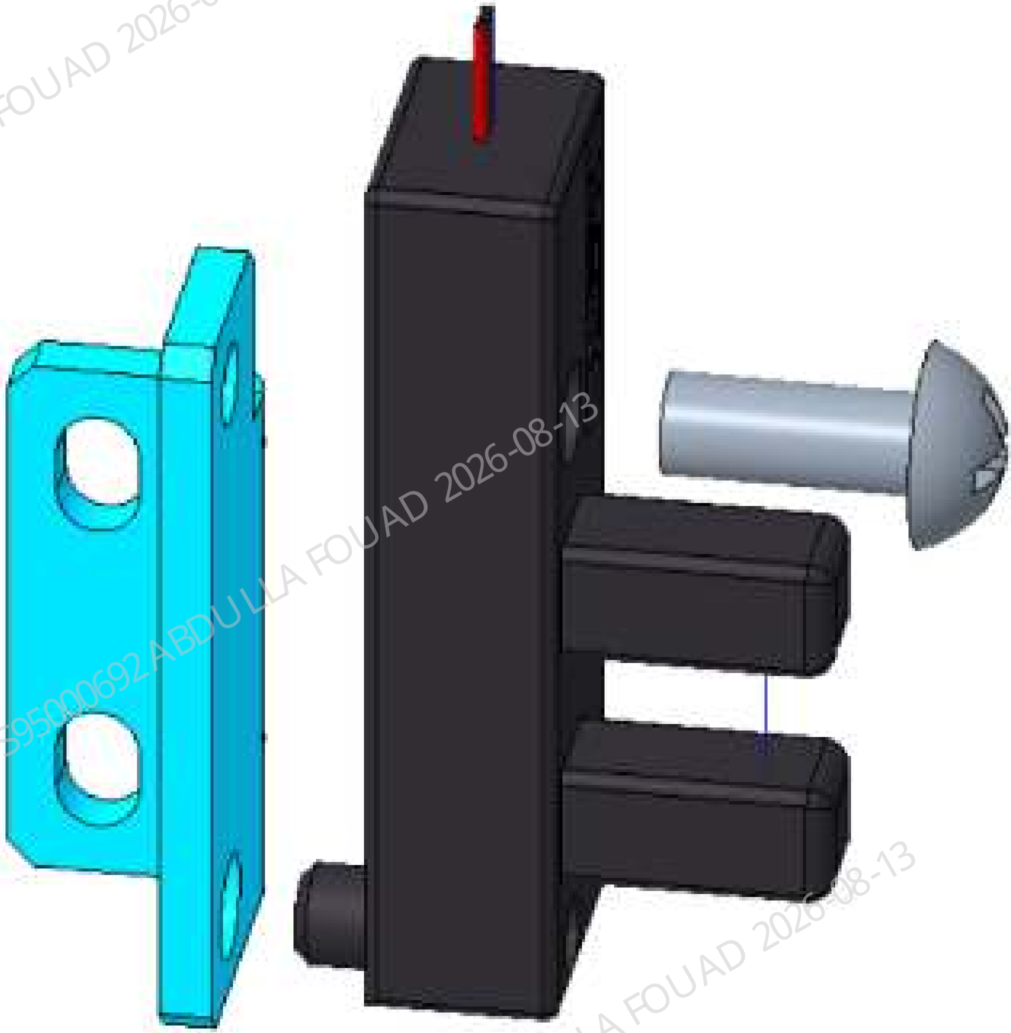
4. As shown in the figure below, remove the two screws fixing the sampling notch detecting sensor fixing plate from the right side of the analyzer.

Figure8–141 Removing the sensor fixing plate



5. As show below, loosen the set screws of the sensor and remove the sensor.

Figure8–142 Removing the detecting Sensor



Subsequent Requirements

Installation procedure:

1. Mount the new sensor to the sensor bracket.
2. Connect the sensor to the main unit wire, fix the sensor bracket on the sampling Z-direction assembly using two screws, and insert the sensor wire.
3. Turn the right side panel to its original position and fix it using one screw.
4. Restore the right, rear, and front covers.

Y-direction Initial Position Sensor of Sampling Assembly

Note

N/A

Required tools

Cross-headed screwdriver

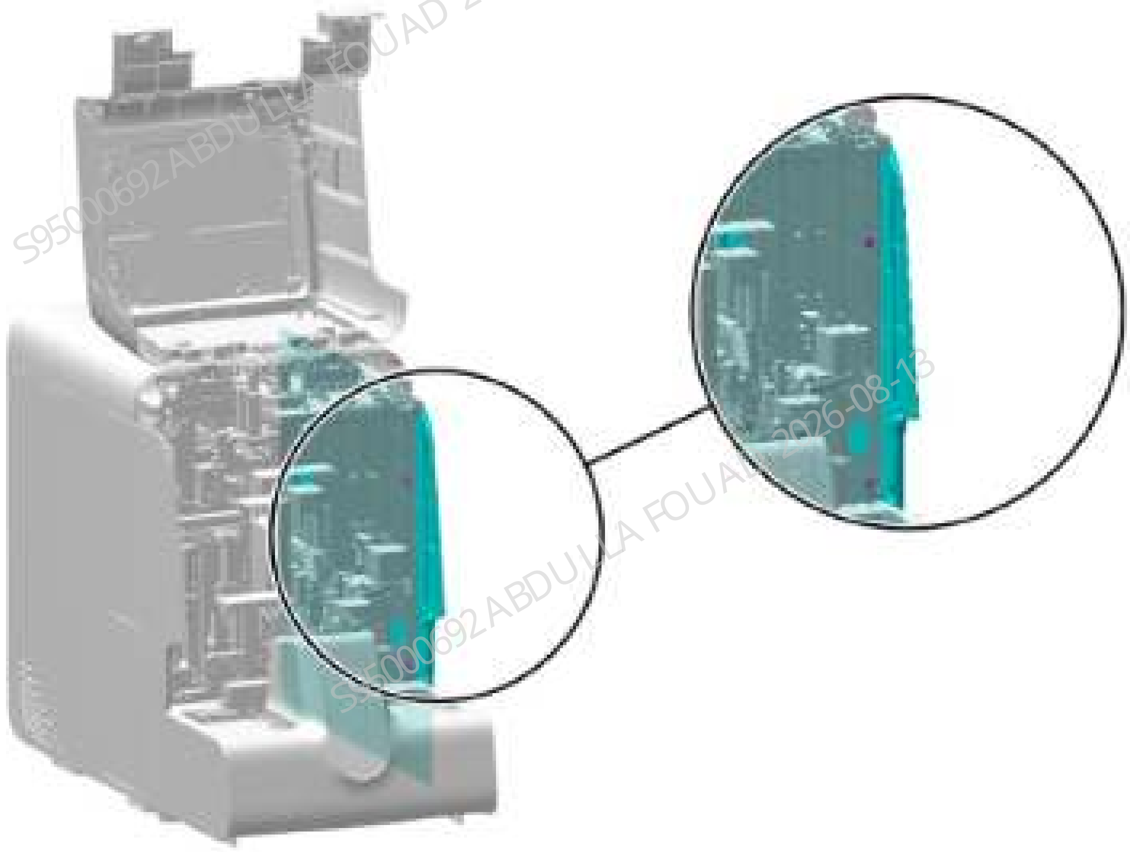
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

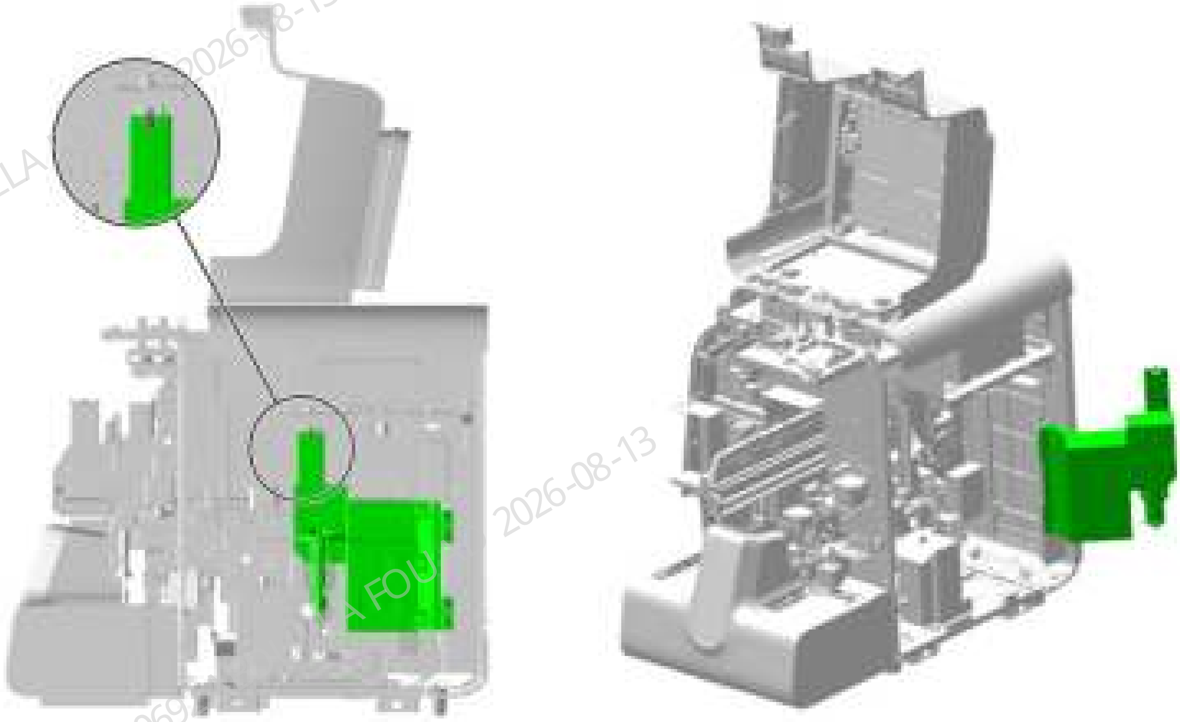
1. Open the front cover and lean it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–143 Removing the right cover



2. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right cover assembly upward, disengage it, and turn it outwards.

Figure8–144 Opening the right cover assembly

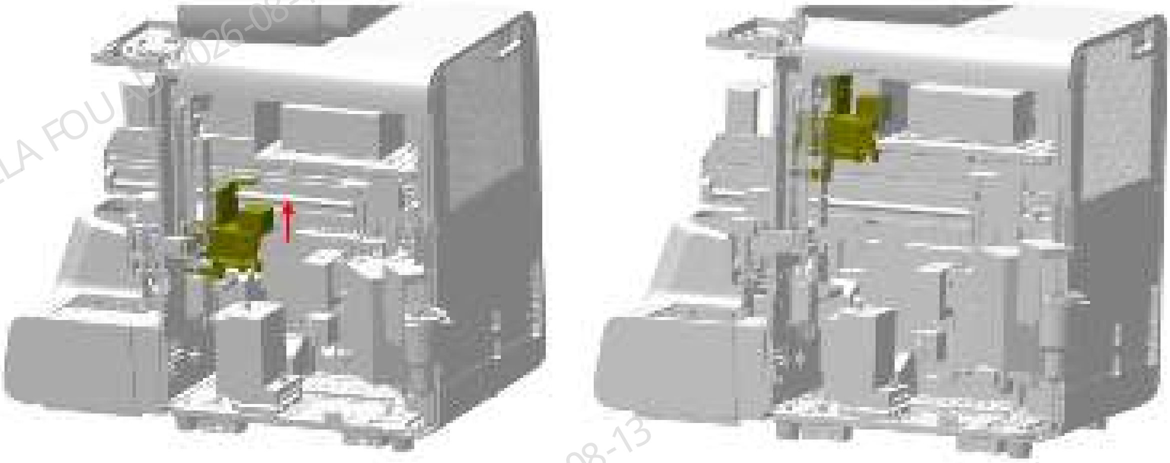


3. As shown in the following figure, remove the detecting sensor wire.

Figure8–145 Removing the detecting sensor wire

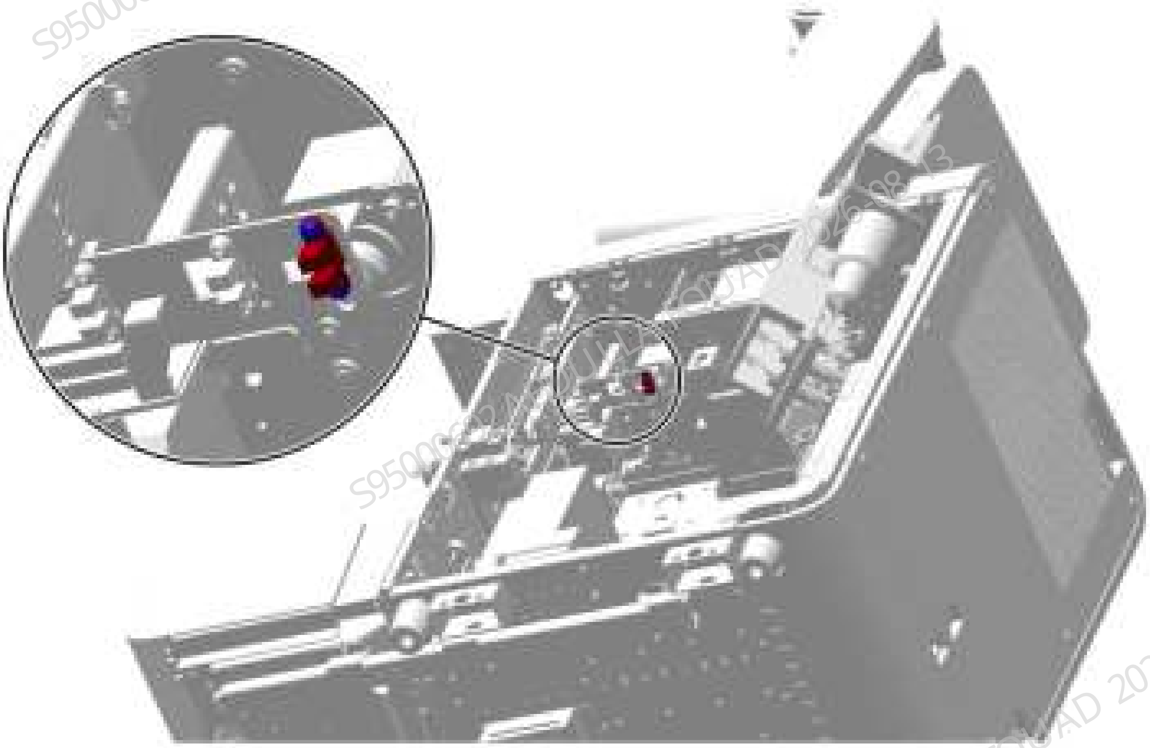
4. As shown in the following figure (left), pull the mix Y-axis assembly up in the direction of the arrow to the upper limit position as shown in the following figure (right), and then bind it using wire tape.

Figure8–146 Binding after lifting the mix Y-direction Assembly



5. As shown in Figure **Figure8–145**, remove the detecting sensor wire.
6. As show below, remove the two screws fixing the detecting sensor and remove the detecting sensor.

Figure8–147 Positions of the set screws of the detecting sensor



Subsequent Requirements

Installation procedure:

1. Use two screws to fix the detecting sensor to the mix assembly and insert the wire.

2. Cut off the wire tie binding the Y-direction assembly, and slowly drop the Y-direction assembly to the lowest position.
3. Install back the right cover, and put down the front cover for resetting.

Rotary compressing assembly sensor

Note

Before disassembly, power off and unplug the power supply.

Required tools

One Allen wrench
One cross-headed screwdriver with short handle

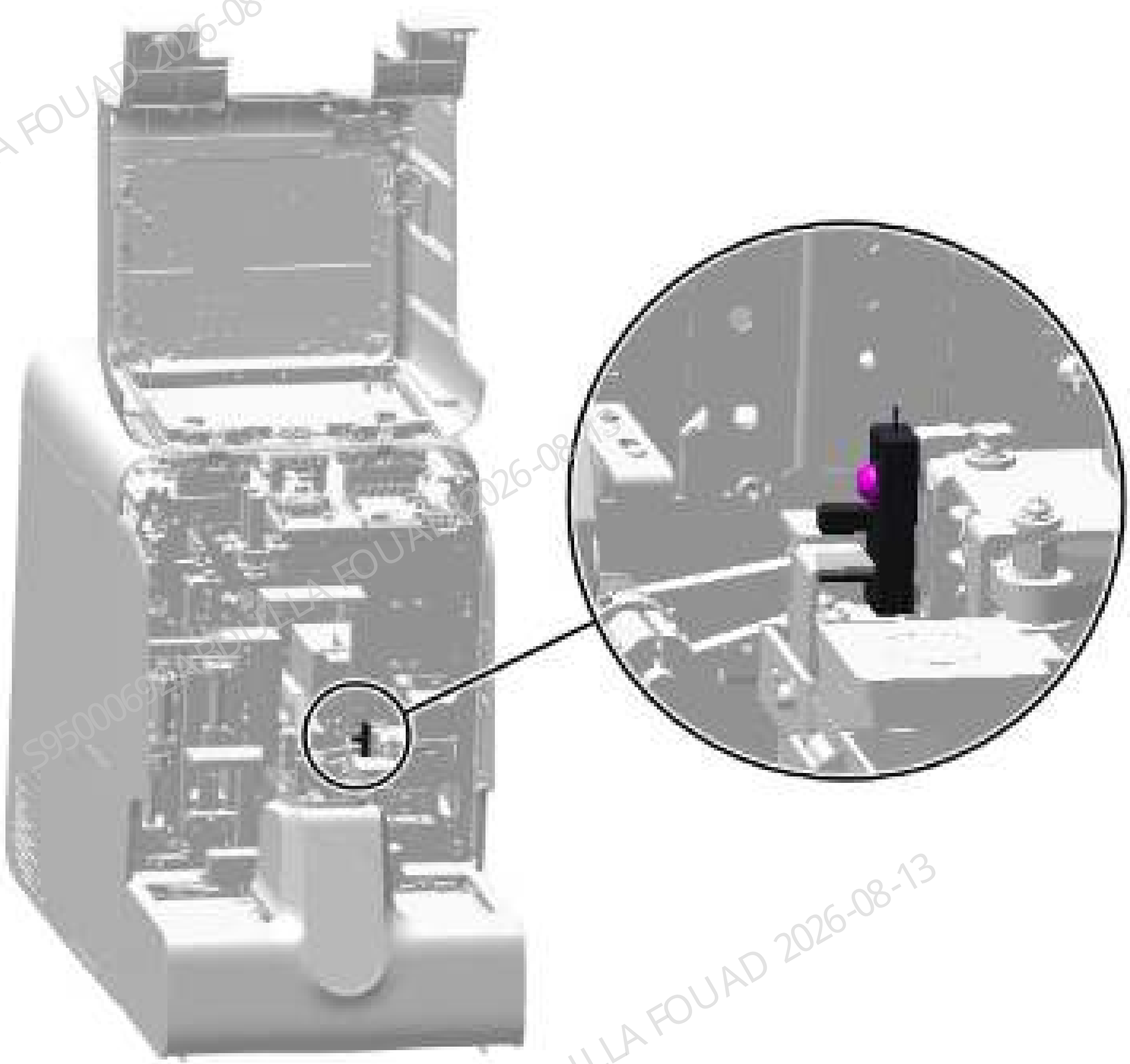
Pre-requirements

N/A

Procedures

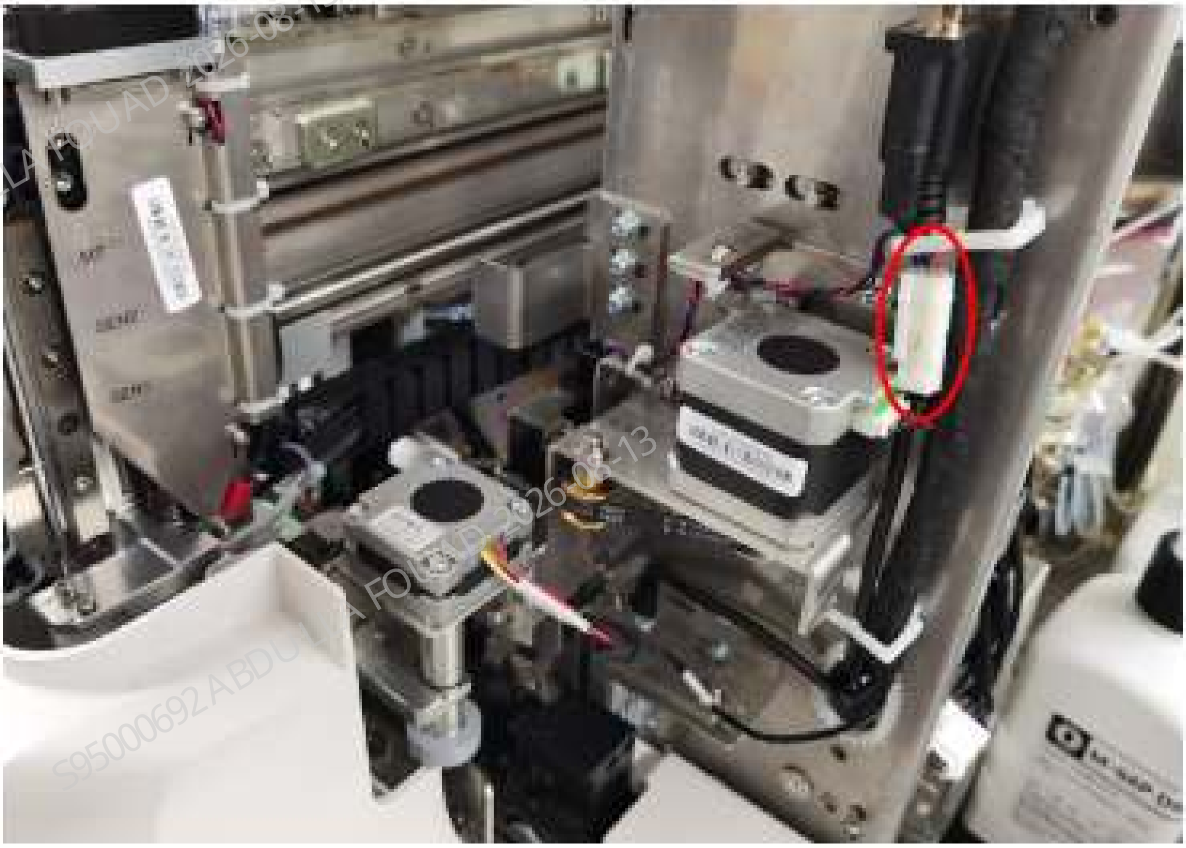
1. Flip the front cover up and dock on the top cover, as shown in the following figure.
2. As shown below, use a cross-headed screwdriver with short handle to loosen the screw that fixes the sensor on the rotary compressing assembly, and remove the rotary compressing assembly.

Figure8–148 Removing the sensor



3. As shown in the following figure, unplug the sensor wire to remove the sensor.

Figure8–149 Removing the sensor



Subsequent Requirements

Installation procedure:

1. Lock the sensor on the rotary scanning assembly.
2. Insert the wire plug.
3. Put down the front cover.

8.19.3 Commissioning and Verification

For the commissioning and verification in the case of Y-direction initial position sensor of sampling, see Y-direction Initial Position Sensor of Sampling.

For the commissioning and verification in the case of Z-direction initial position sampling sensor of the sampling assembly, see Z-direction Initial Position Sampling Sensor of the Sampling Assembly.

For the commissioning and verification in the case of Y-direction initial position sensor of the sampling assembly, see Y-direction Initial Position Sensor of Sampling Assembly.

For the commissioning and verification in the case of rotary compressing assembly sensor, see Rotary compressing assembly sensor.

Y-direction Initial Position Sensor of Sampling

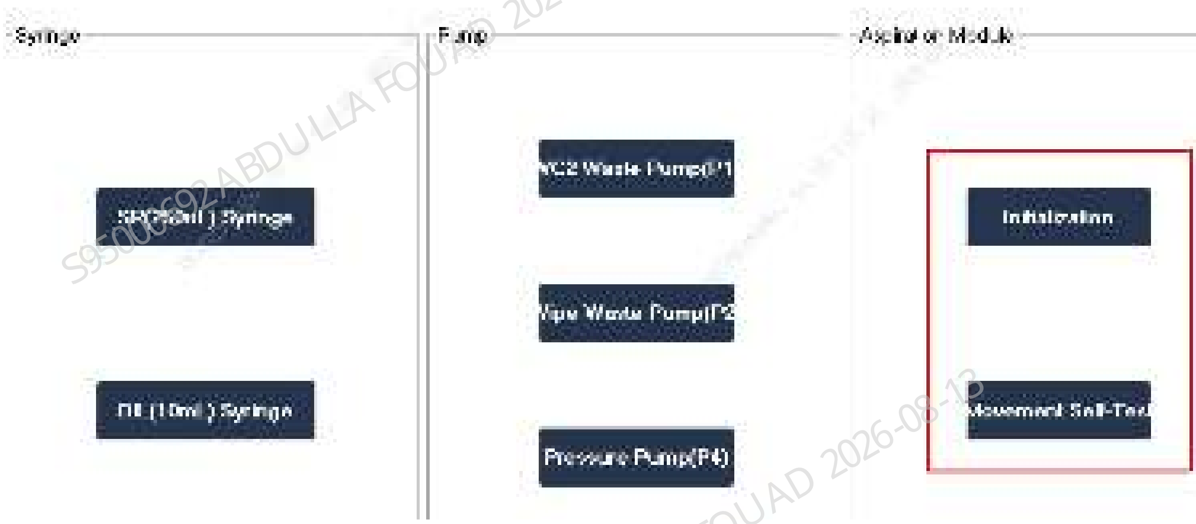
Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning is not required.
Verification:

1. Click **Service**>>**Commissioning & Self-Test**>>**Fluidics Commissioning**, and click **Assembly Initialization**. After the self-test succeeds, click *OK*. Then click **Assembly Movement Self-Test** to see if the self-test is successful. If the self-test is successful, the sensor has been replaced successfully. Otherwise, it is unsuccessful, and you need to check the tube connection.



Z-direction Initial Position Sampling Sensor of the Sampling Assembly

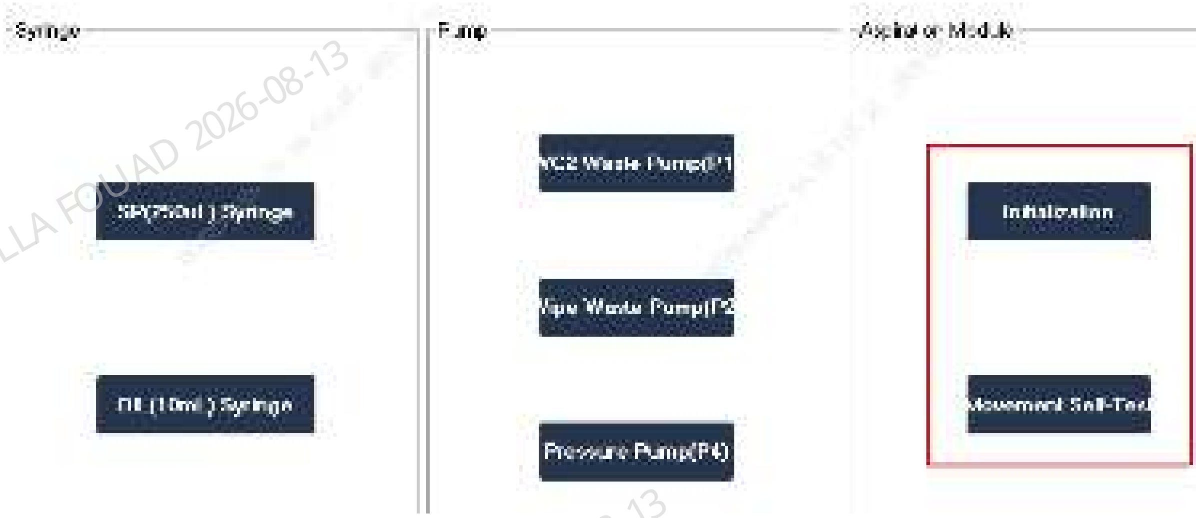
Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning is not required.
Verification:

1. Click **Service**>>**Commissioning & Self-Test**>>**Fluidics Commissioning**, and click **Assembly Initialization**. After the self-test succeeds, click *OK*. Then click **Assembly Movement Self-Test** to see if the self-test is successful. If the self-test is successful, the sensor has been replaced successfully. Otherwise, it is unsuccessful, and you need to check the tube connection.



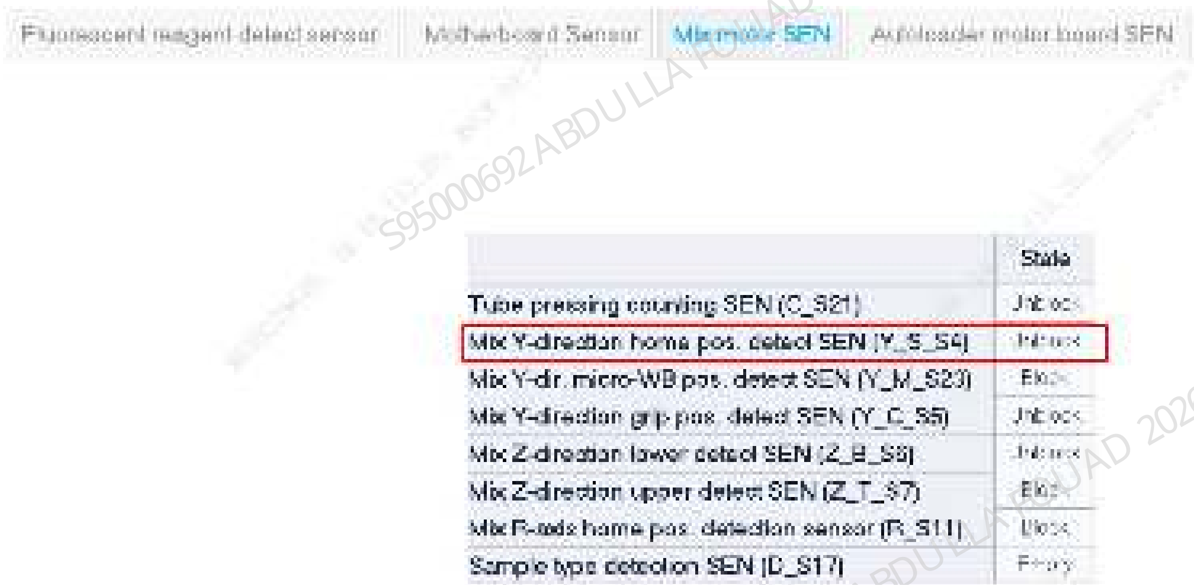
Y-direction Initial Position Sensor of Sampling Assembly

Required tools

N/A

Procedures

1. After logging in at the service access level, click **Status>>Sensor Status>>Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful Otherwise, check whether the wire is properly connected.



Rotary compressing assembly sensor

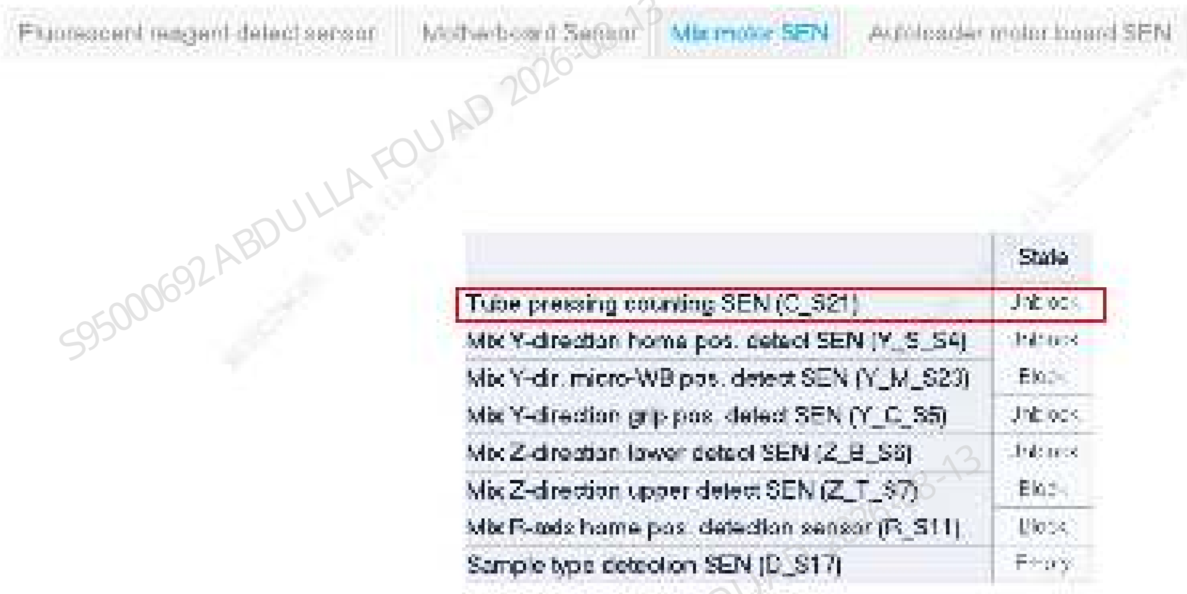
Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning not required
Verification:

1. After logging in at the service access level, click **Status>>Sensor Status>>Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful



8.20 Photoelectric sensor wire terminal

8.20.1 General Information

For the general information in the case of photoelectric position sensor, see Photoelectric Position Sensor.

For the general information in the case of photoelectric position sensor in Z direction, see Sampling Z-direction Photoelectric Position Sensor.

For the general information in the case of 250 μL syringe assembly-photoelectric position sensor, see 250ul syringe assembly-photoelectric position sensor.

For the general information in the case of mix assembly SEN4-photoelectric position sensor, see Mix Assembly SEN4-Photoelectric Position Sensor.

For the general information in the case of mix assembly SEN5-photoelectric position sensor, see Mix Assembly SEN5-Photoelectric Position Sensor.

For the general information in the case of mix assembly SEN6-photoelectric position sensor, see Mix Assembly SEN6-Photoelectric Position Sensor.

For the general information in the case of mix assembly SEN7-photoelectric position sensor, see Mix assembly SEN7-photoelectric position sensor.

For the general information in the case of mix assembly SEN23-photoelectric position sensor, see Mix assembly SEN23-photoelectric position sensor.

Photoelectric Position Sensor

Figure8–150 Photo of photoelectric position sensor

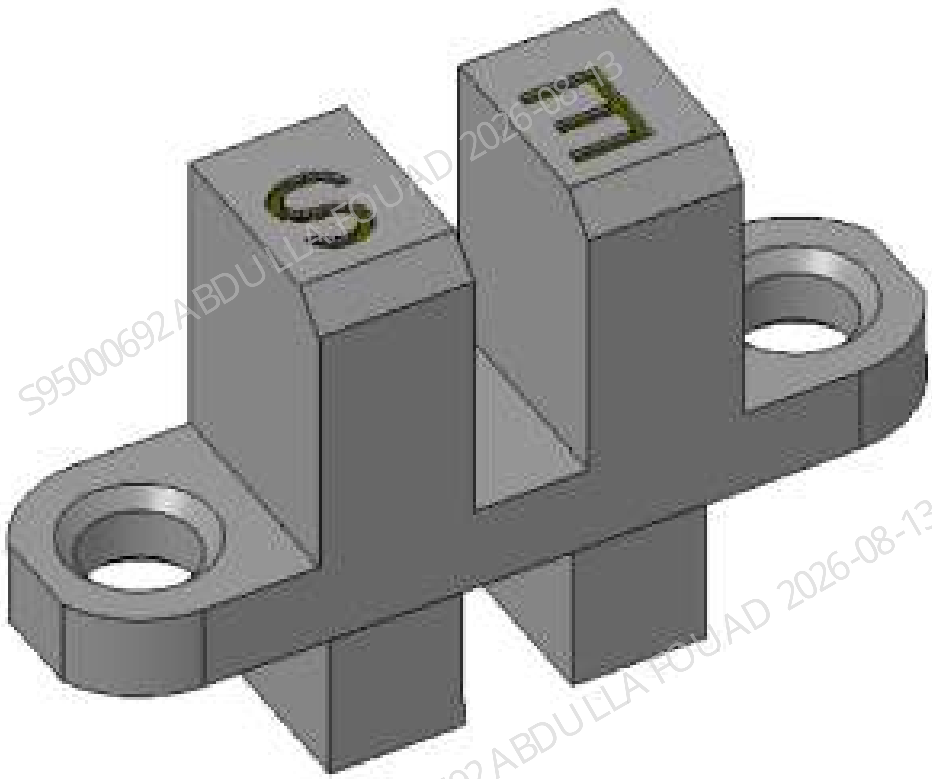


Table 8–17 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric Position Sensor	N/A	N/A

Function:

Used for home position detection of the syringe.

Sampling Z-direction Photoelectric Position Sensor

Figure8–151 Photo of photoelectric position sensor

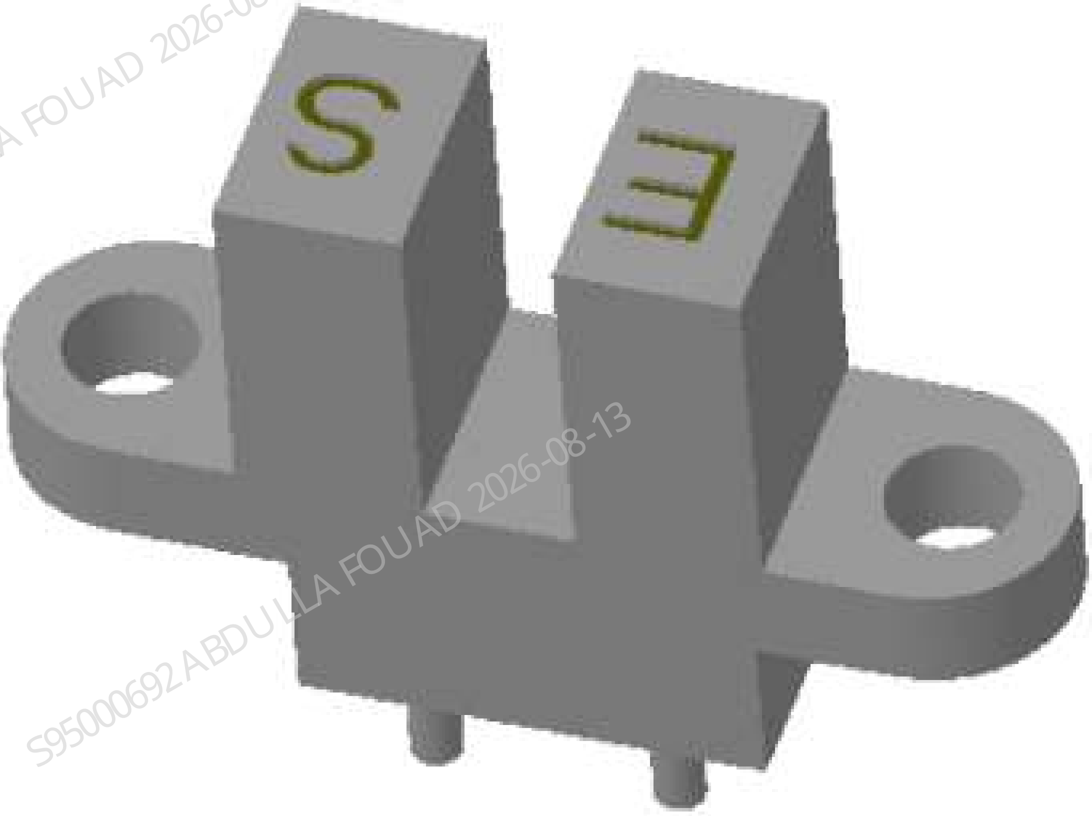


Figure8–152 Diagram of Position of the Sampling Notch Detection Sensor in the Sampling Assembly

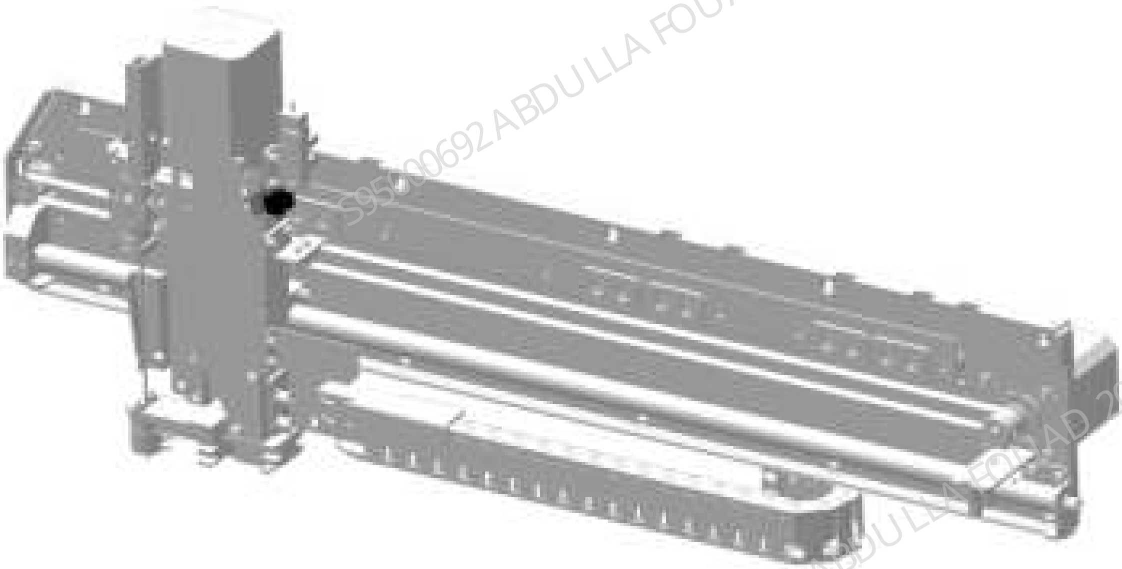


Table 8–18 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric Position Sensor	N/A	N/A

Function:
Used to detect the pierce height of the sample probe.

250ul syringe assembly-photoelectric position sensor

Figure8–153 Photo of photoelectric position sensor

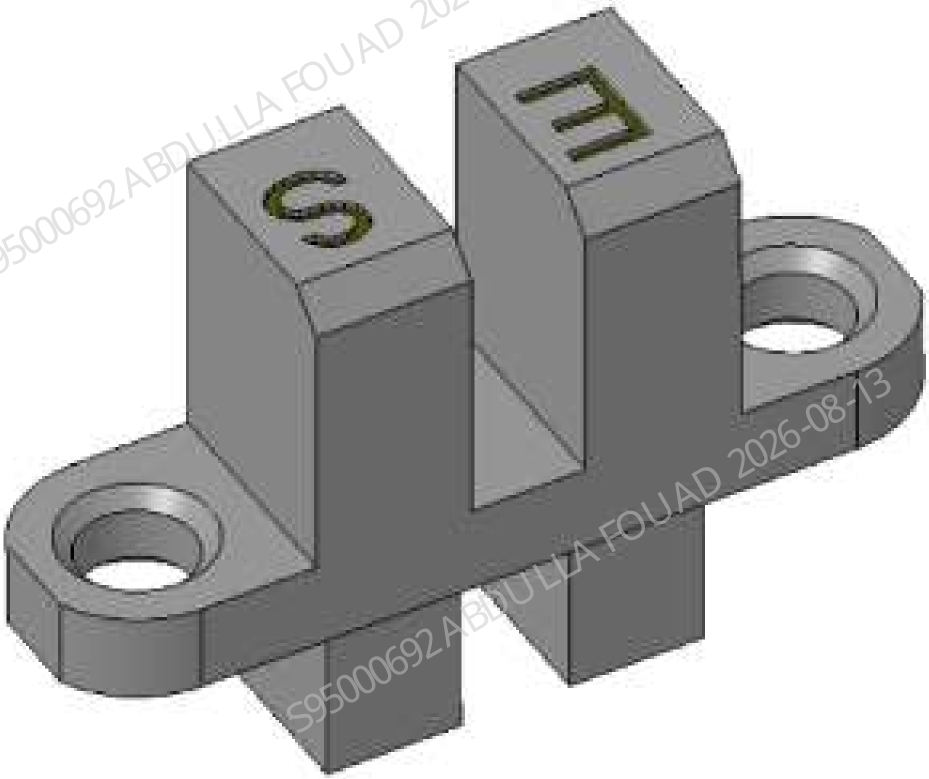


Table 8–19 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric Position Sensor	N/A	N/A

Function:
Used for home position detection of the syringe.

Mix Assembly SEN4-Photoelectric Position Sensor

Figure8–154 Photo of photoelectric position sensor

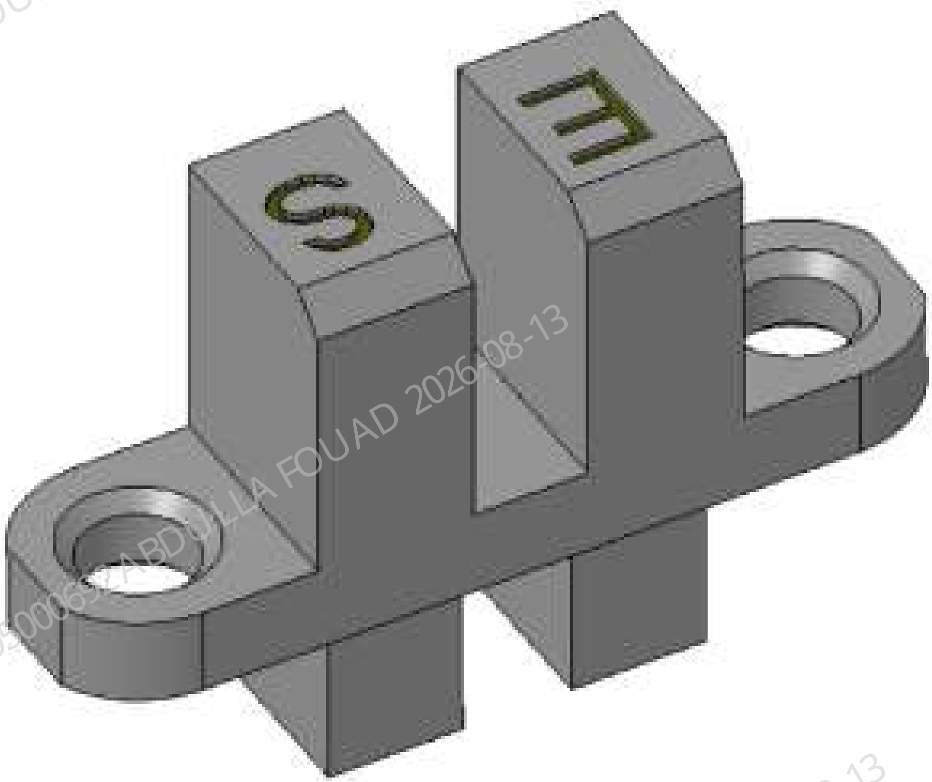


Figure8–155 Photo of Location of Photoelectric Position Sensor (SEN4) in Mix Assembly

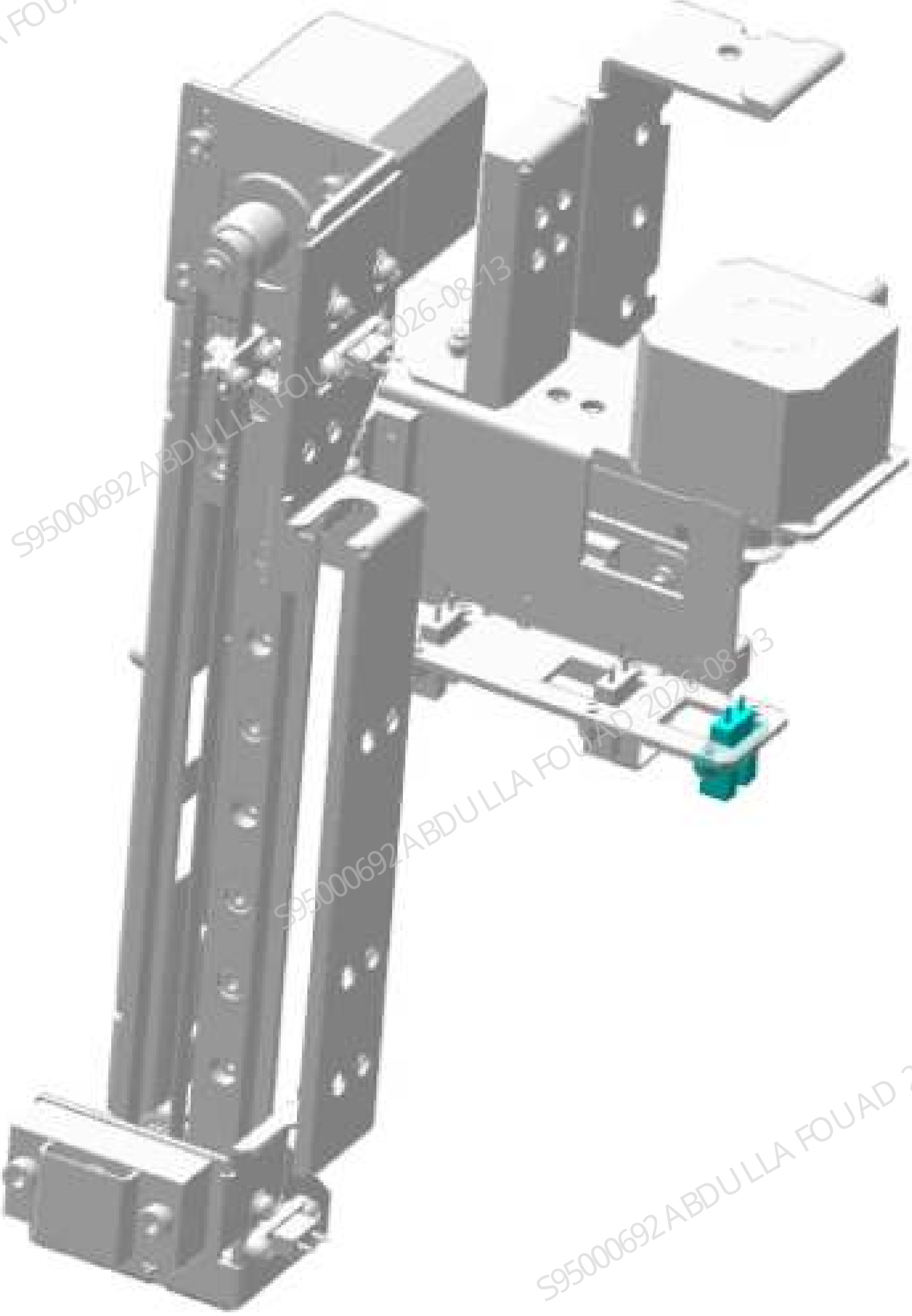


Table 8–20 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric position sensor (SEN4)	N/A	N/A

Function:
Used for home position detection of the syringe.

Mix Assembly SEN5-Photoelectric Position Sensor

Figure8–156 Photo of photoelectric position sensor

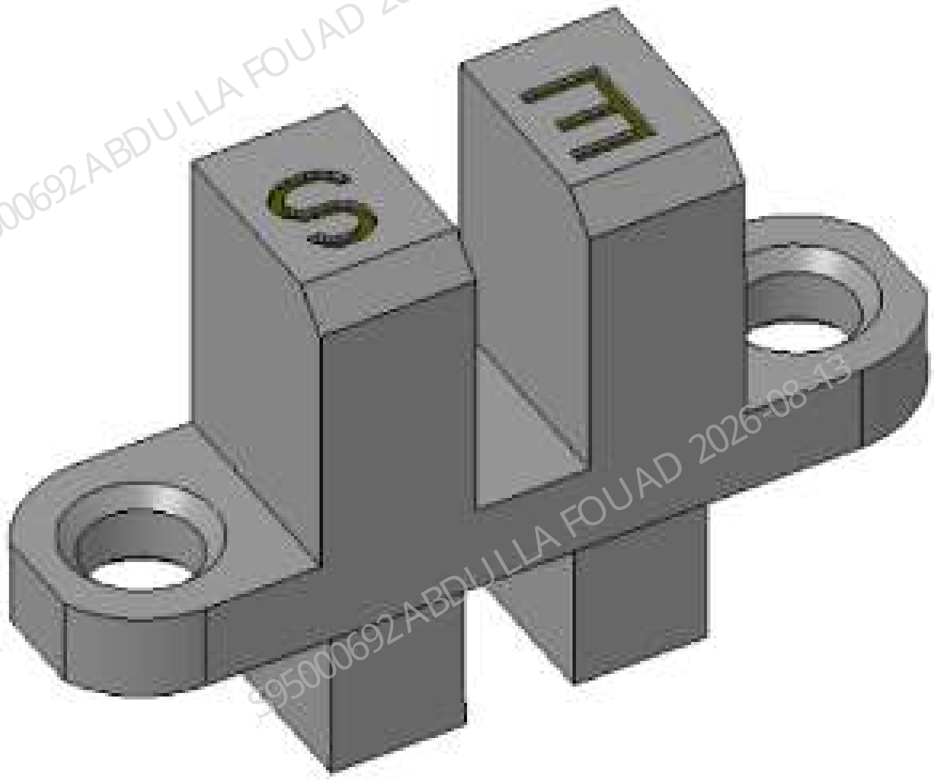


Figure8–157 Photo of Location of Photoelectric Position Sensor (SEN5) in Mix Assembly

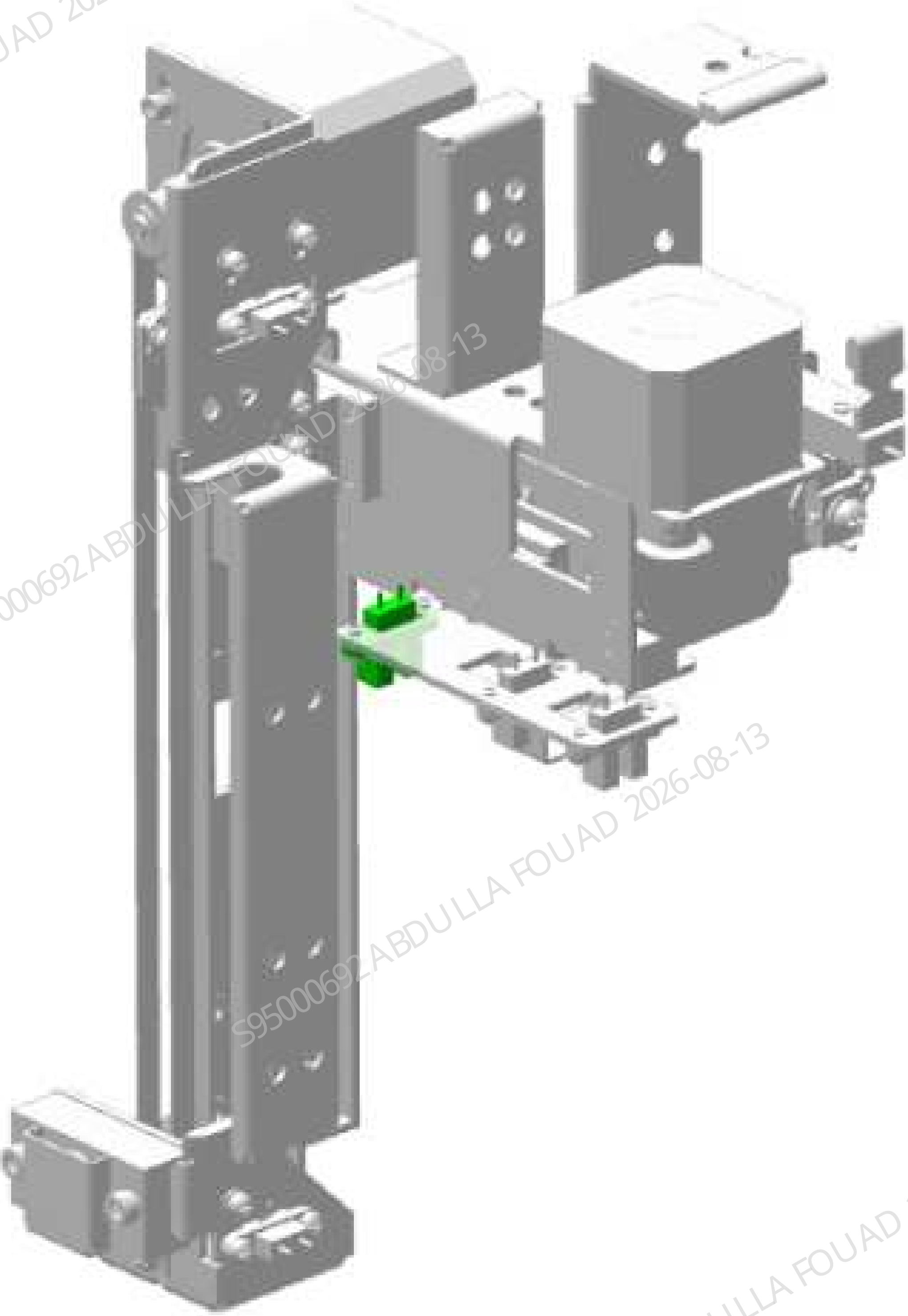


Table 8–21 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric position sensor (SEN5)	N/A	N/A

Function:

Used for home position detection of the syringe.

Mix Assembly SEN6-Photoelectric Position Sensor

Figure8–158 Photo of photoelectric position sensor

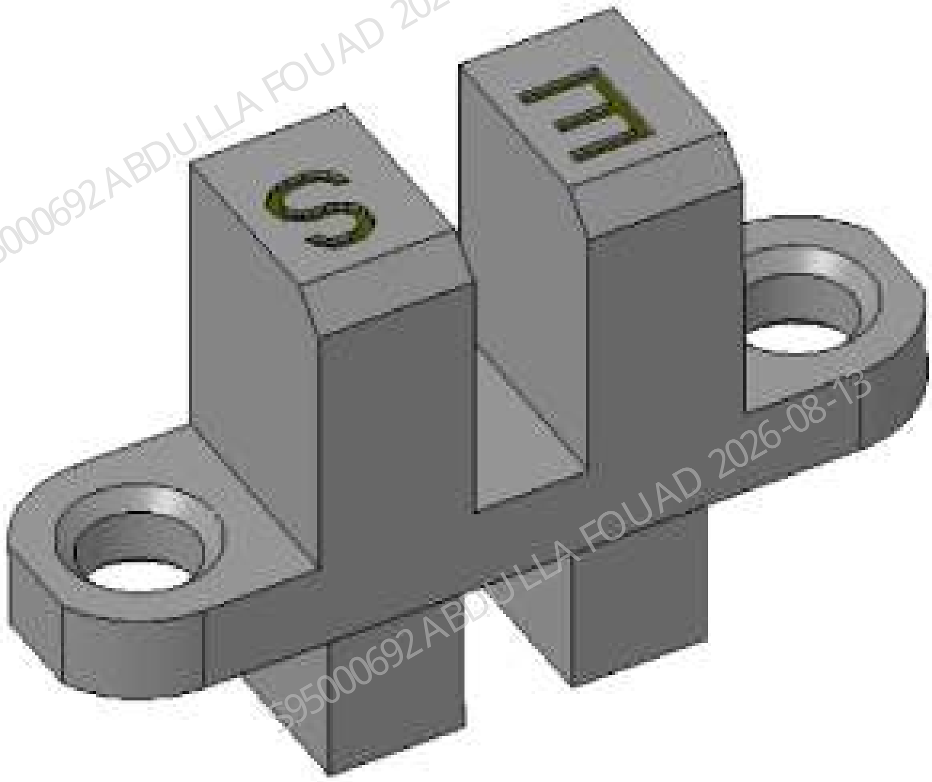


Figure8–159 Photo of Location of Photoelectric Position Sensor (SEN6) in Mix Assembly

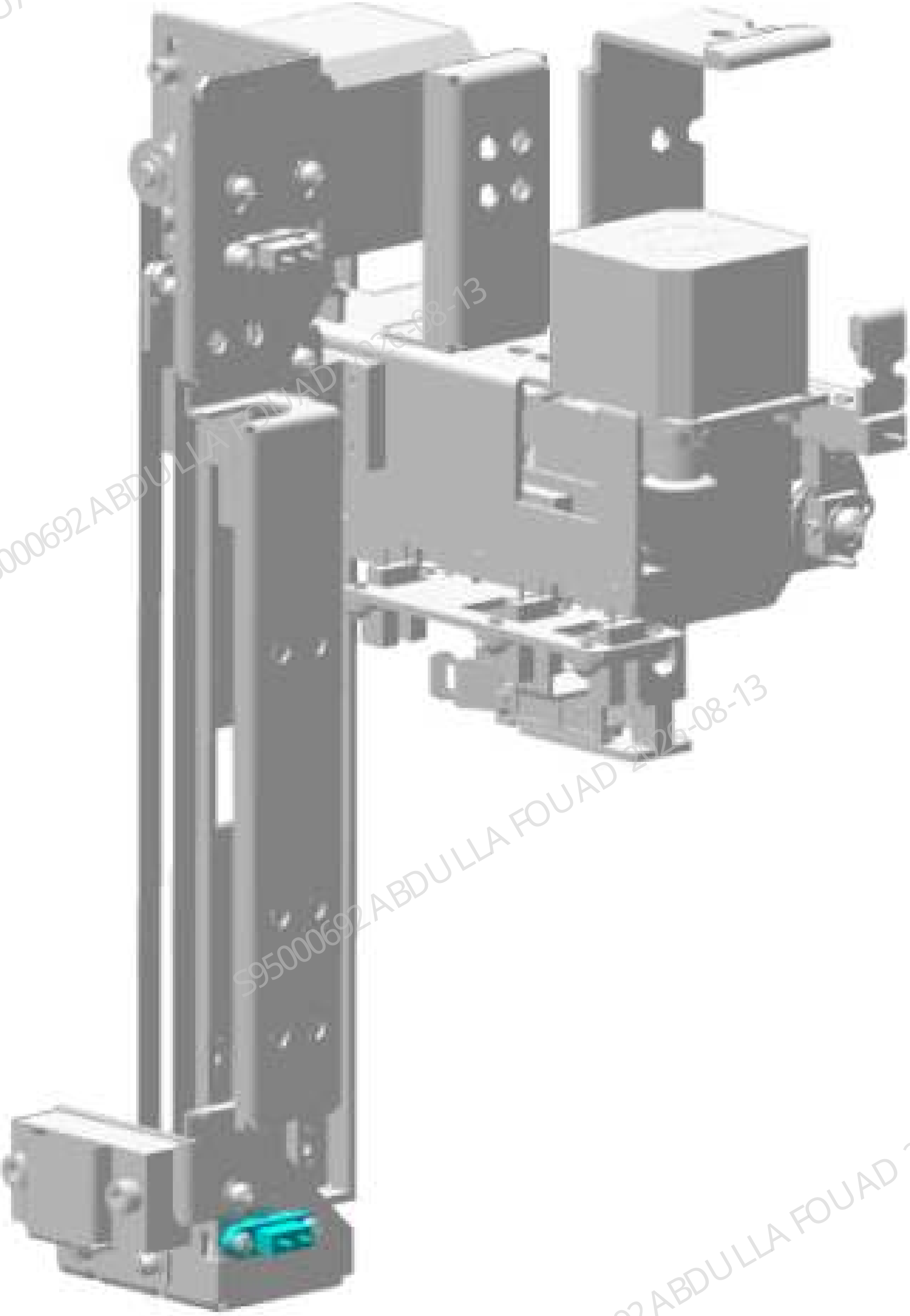


Table 8–22 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric position sensor (SEN6)	N/A	N/A

Function:
Used for home position detection of the syringe.

Mix assembly SEN7-photoelectric position sensor

Figure8–160 Photo of photoelectric position sensor

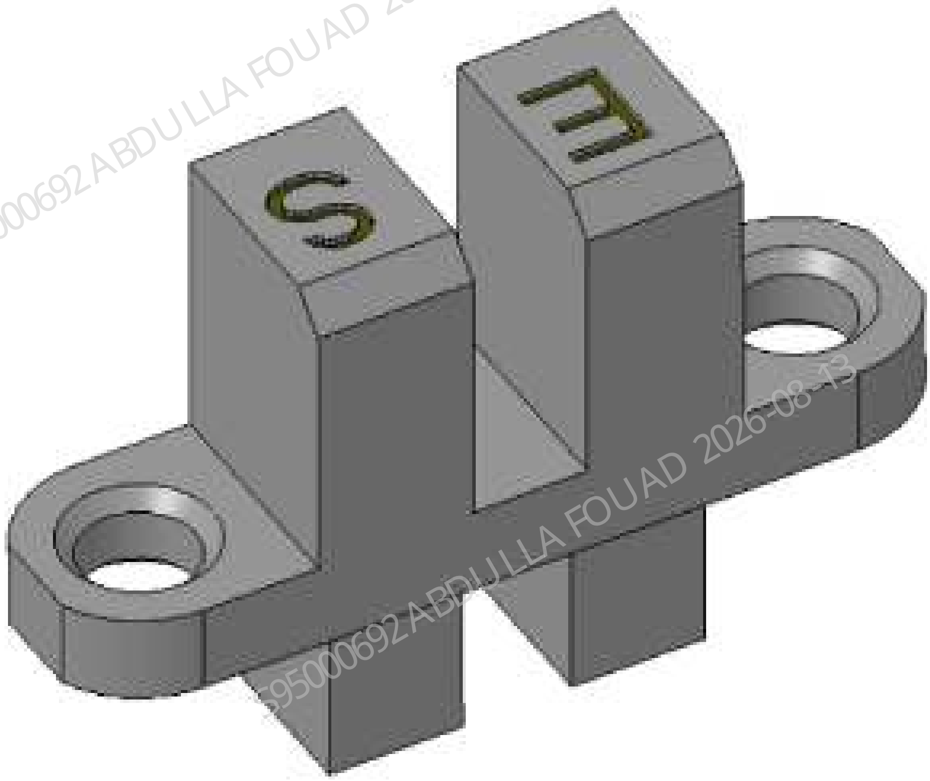


Figure8–161 Photo of Location of Photoelectric Position Sensor (SEN7) in Mix Assembly

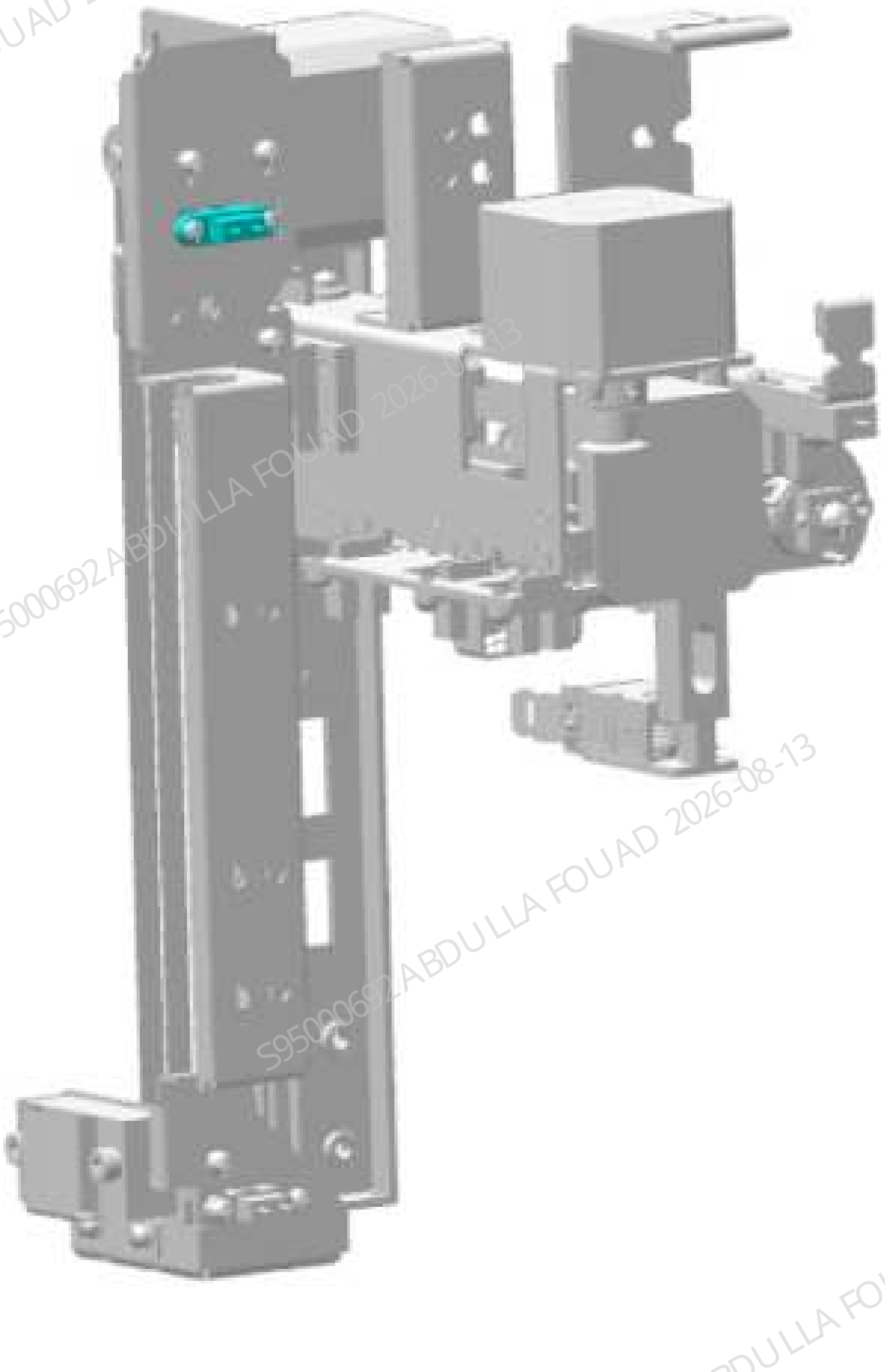


Table 8–23 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric position sensor (SEN7)	N/A	N/A

Function:

Used for home position detection of the syringe.

Mix assembly SEN23-photoelectric position sensor

Figure8–162 Photo of photoelectric position sensor

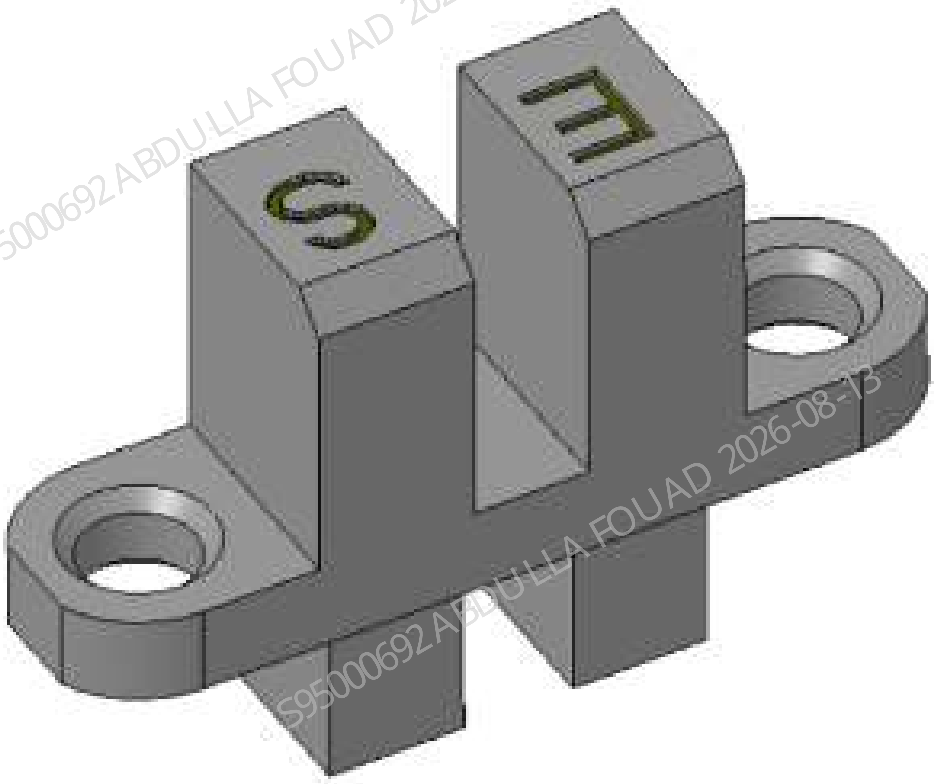


Figure8–163 Photo of Location of Photoelectric Position Sensor (SEN23) in Mix Assembly

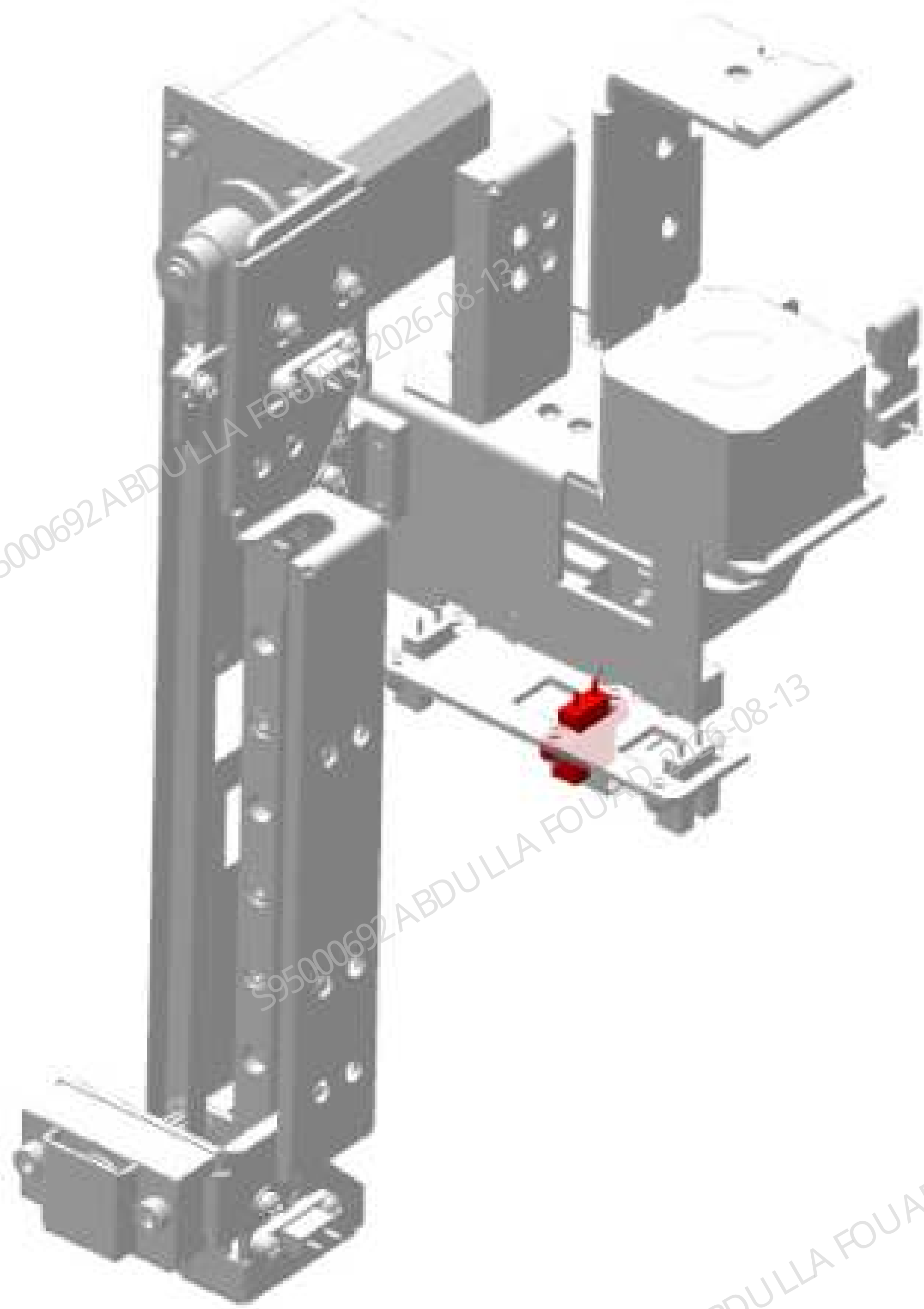


Table 8–24 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000297-00	Photoelectric position sensor (SEN23)	N/A	N/A

Function:
Used for home position detection of the syringe.

8.20.2 Disassembly and Assembly

For the disassembly and assembly in the case of photoelectric position sensor, see Photoelectric Position Sensor.
For the disassembly and assembly in the case of photoelectric position sensor in Z-direction, see Sampling Z-direction Photoelectric Position Sensor.
For the disassembly and assembly in the case of 250ul syringe assembly-photoelectric position sensor, see 250ul syringe assembly-photoelectric position sensor.
For the disassembly and assembly in the case of mix assembly SEN4-photoelectric position sensor, see Mix Assembly SEN4-Photoelectric Position Sensor.
For the disassembly and assembly in the case of mix assembly SEN5-photoelectric position sensor, see Mix Assembly SEN5-Photoelectric Position Sensor.
For the disassembly and assembly in the case of mix assembly SEN6-photoelectric position sensor, see Mix Assembly SEN6-Photoelectric Position Sensor.
For the disassembly and assembly in the case of mix assembly SEN7-photoelectric position sensor, see Mix Assembly SEN7-Photoelectric Position Sensor.
For the disassembly and assembly in the case of mix assembly SEN23-photoelectric position sensor, see Mix Assembly SEN23-Photoelectric Position Sensor.

Photoelectric Position Sensor

Note

N/A

Required tools

Cross-headed screwdriver

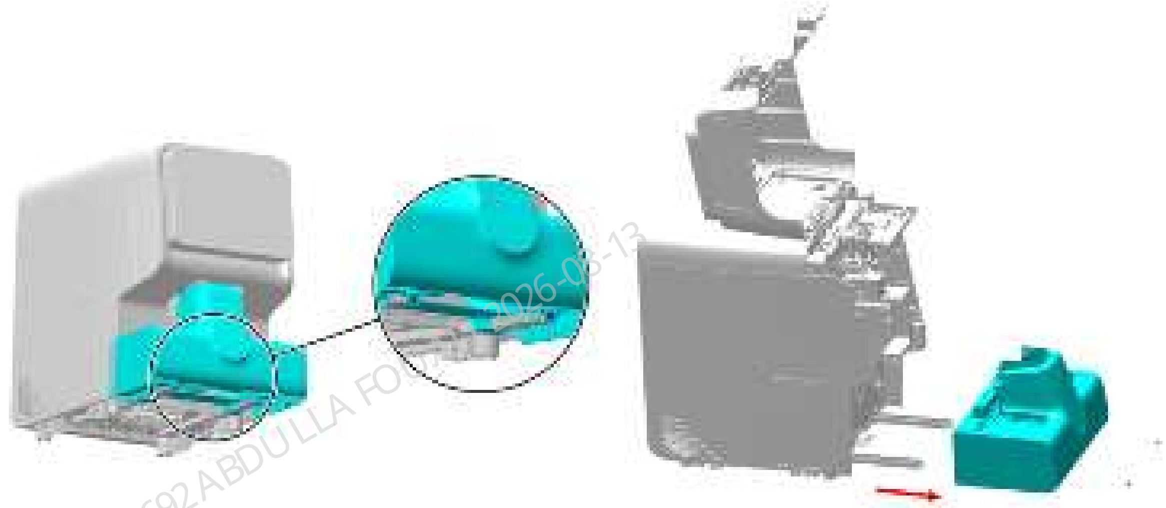
Pre-requirements

Dispense the liquid, power off the main unit, and unplug the power cord.

Procedures

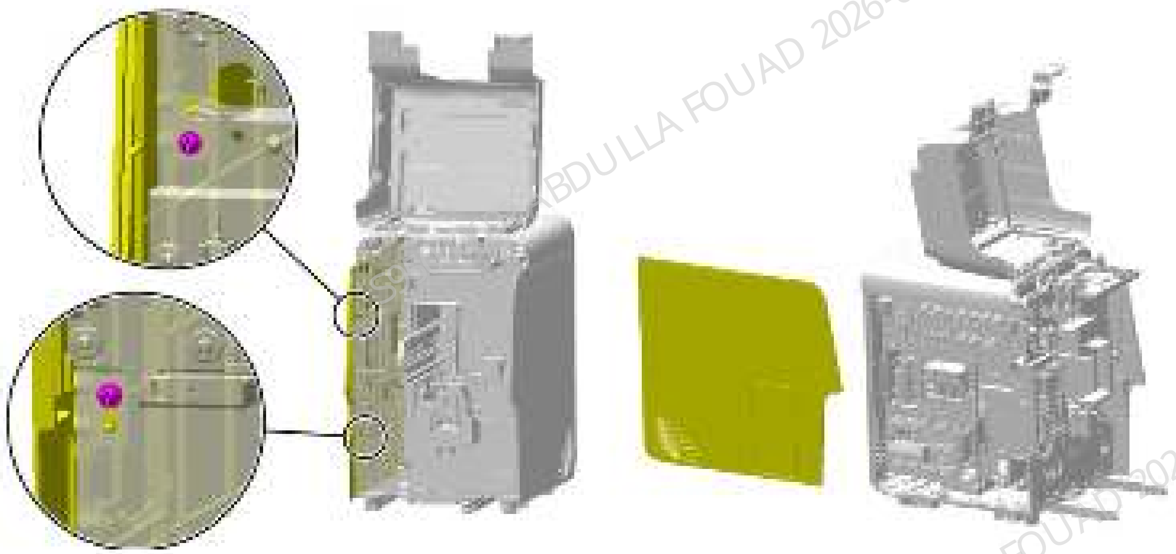
1. Loosen the two screws fixing the autoloader, open the front cover, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in the figure below.

Figure8–164 Removing the autoloader



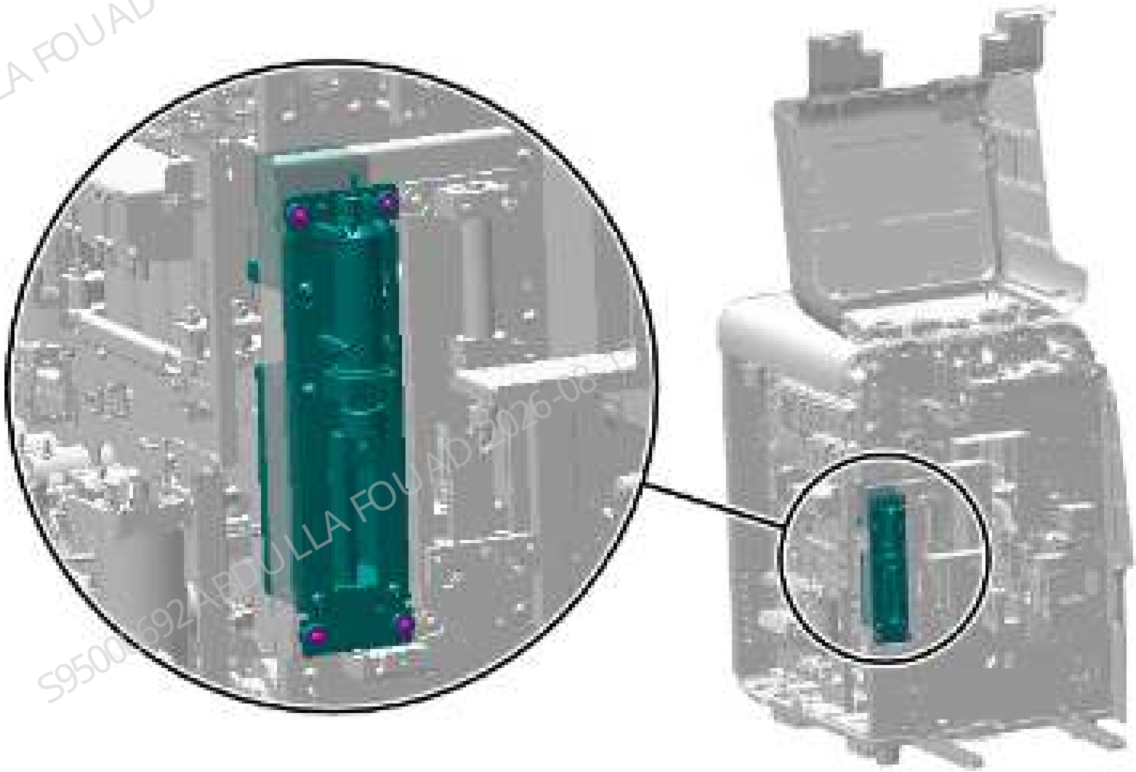
2. Loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–165 Removing the left cover



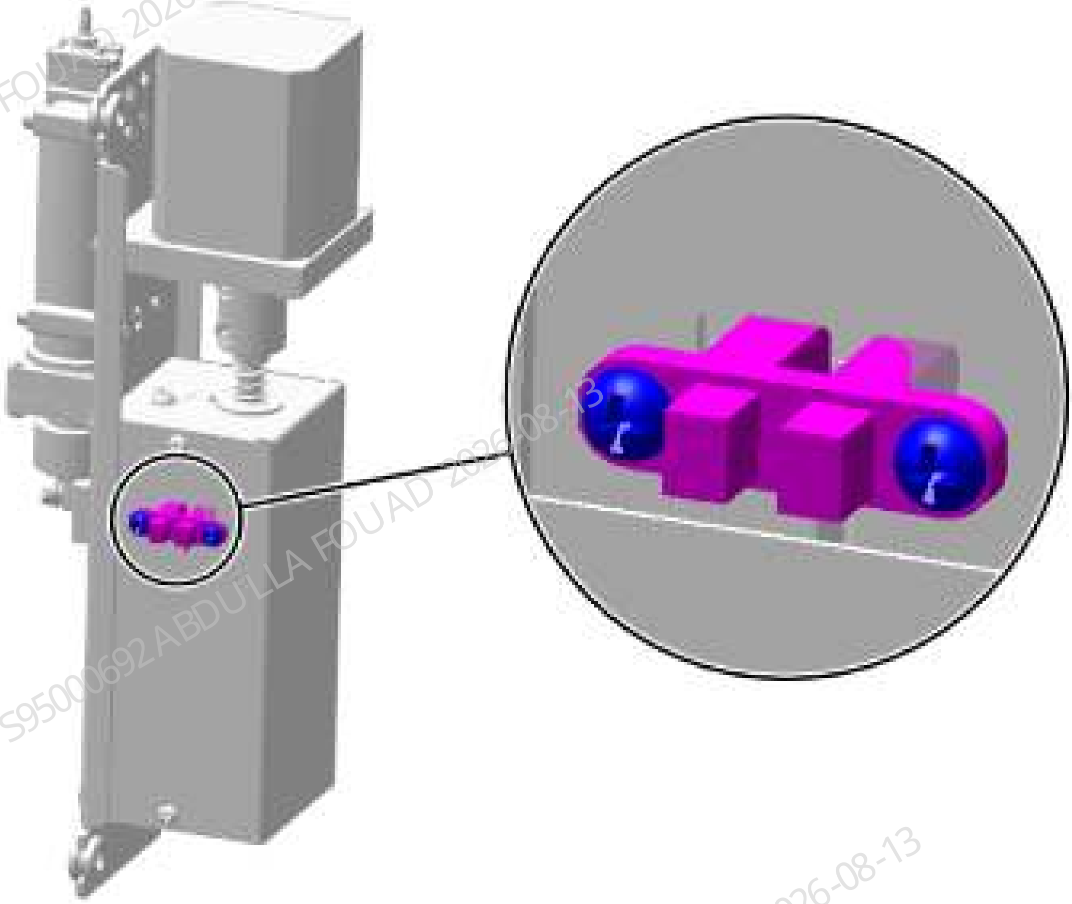
3. Remove the tube and ground wire from the 10ml lead screw driving syringe assembly.
4. Remove the rubber tube and ground wire from the 10ml lead screw driving syringe assembly, as shown in the following figure, pull out the assembly terminal, and finally remove the entire assembly.

Figure8–166 Positions of the set screws of the 100ul lead screw driving syringe assembly



5. As shown in the figure below, unplug the photoelectric position sensor wire, remove the set screws of the photoelectric position sensor, and remove the photoelectric position sensor.

Figure8–167 Positions of the set screws of the Photoelectric Position Sensor



Subsequent Requirements

Installation procedure:

1. Attach the photoelectric position sensor to the syringe assembly and secure it with two screws.
2. Insert the terminal block of the syringe assembly, put it back in the original position of the analyzer, and fix it with four screws.
3. Connect the tube and ground wire of the syringe assembly.
4. Install the left cover, and flip down the front cover to reset it.
5. Restore the autoloader.

Sampling Z-direction Photoelectric Position Sensor

Note

Before disassembly, power off and unplug the power supply.

Required tools

One Allen wrench

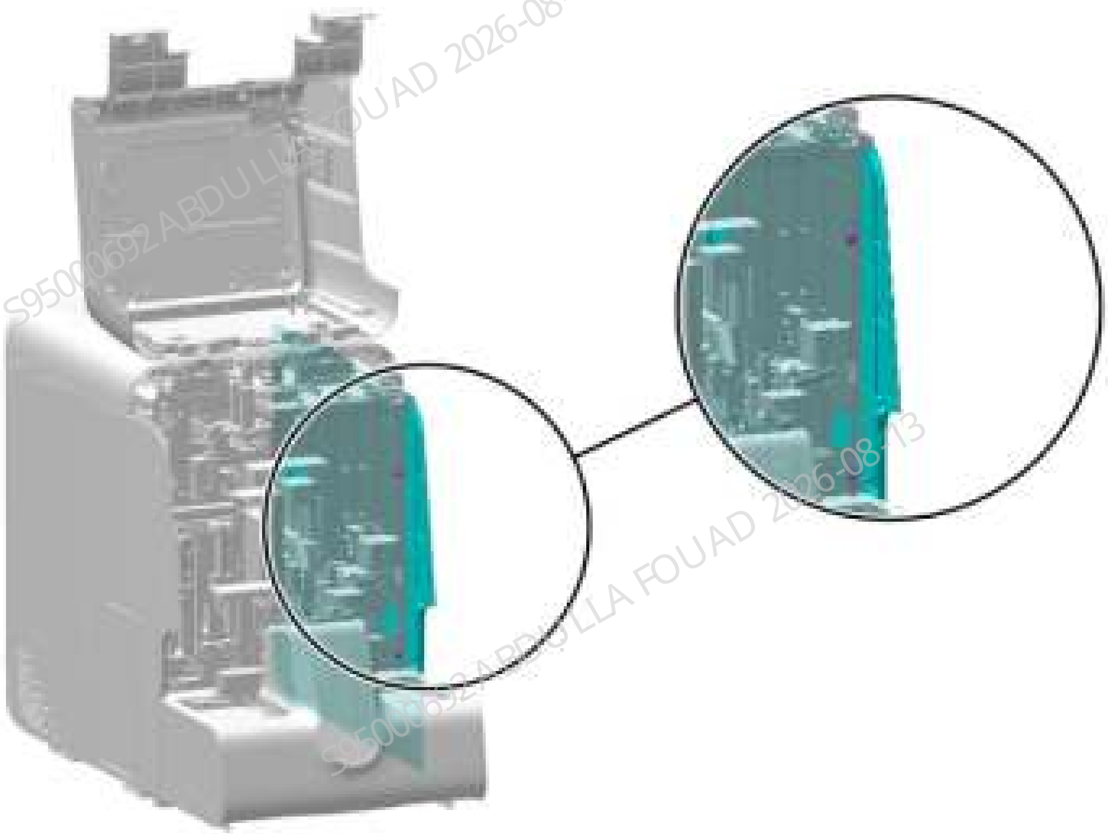
Pre-requirements

N/A

Procedures

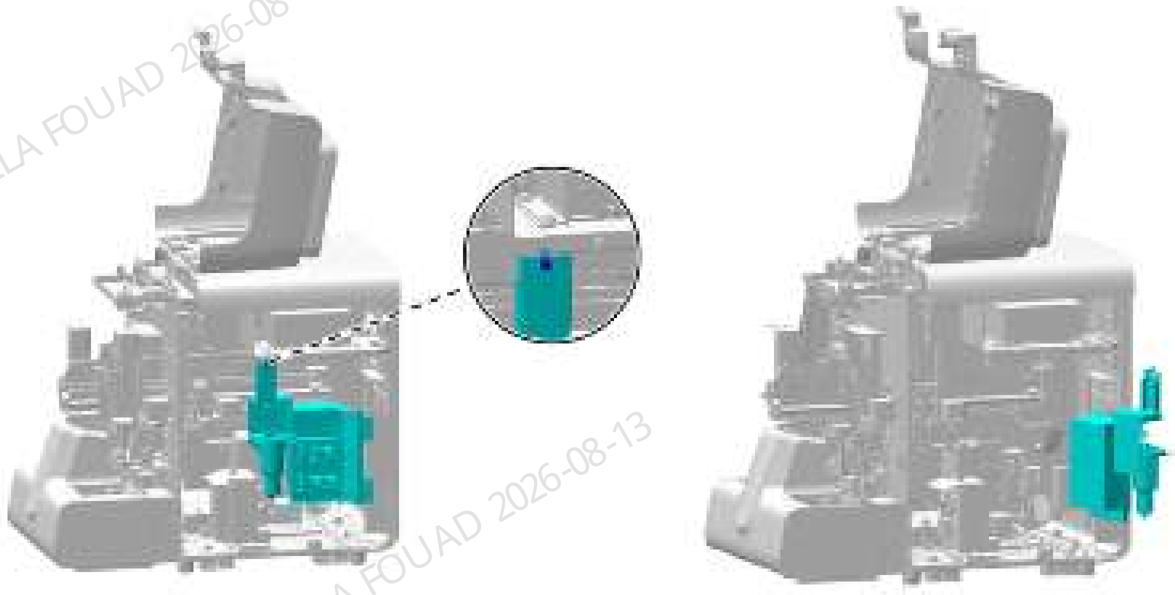
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–168 Removing the right cover



2. As shown below, unscrew the one screw that fixes the right bath subassembly; as shown in the figure, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–169 Opening the right cover assembly

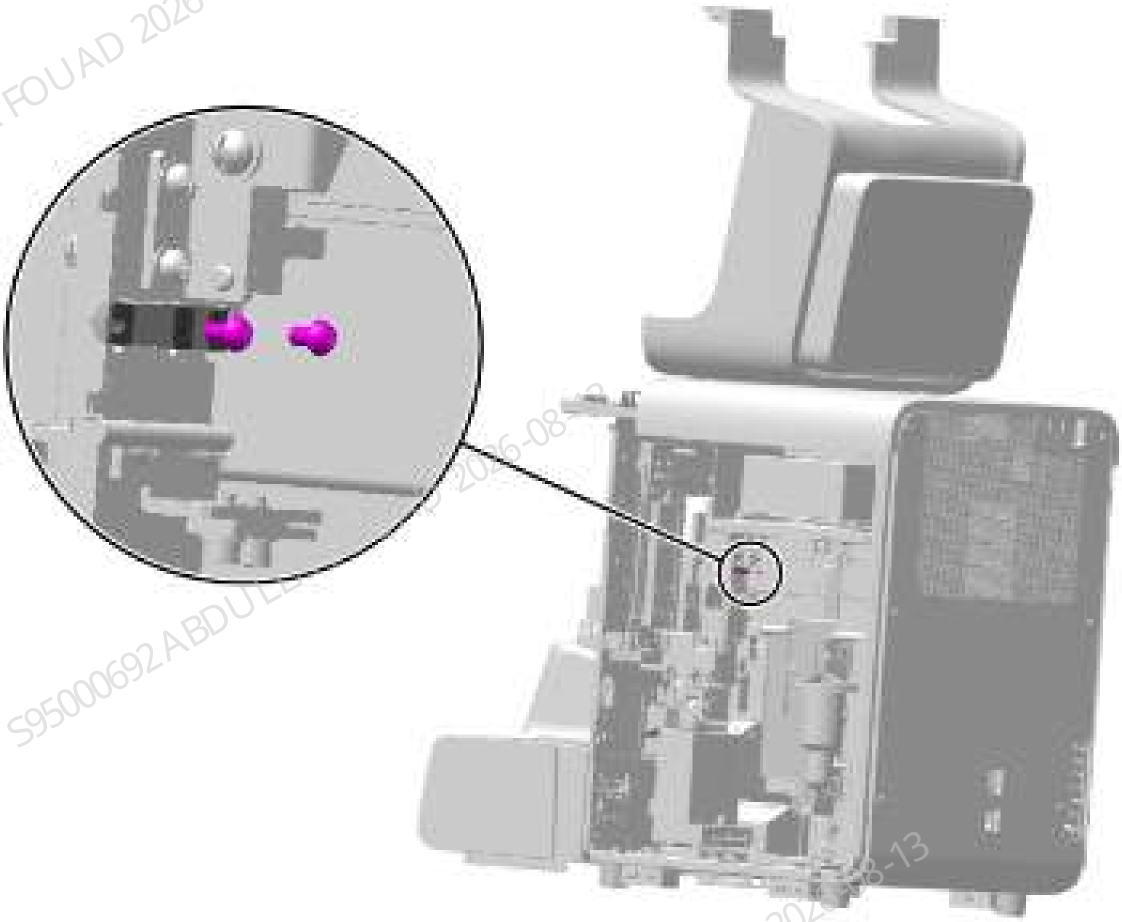


3. As shown in the following figure, remove the sensor wire.

Figure8–170 Removing the photoelectric position sensor wire

4. As shown in the figure below, remove the two screws fixing the photoelectric position sensor from the sampling Z-axis assembly on the right side of the analyzer, and remove the photoelectric position sensor.

Figure8–171 Removing the sensor fixing plate



Subsequent Requirements

Installation procedure:

1. Fix the new photoelectric position sensor on the sampling Z-axis assembly.
2. Insert the photoelectric position sensor wire.
3. Turn the right side panel to its original position and fix it using one screw.
4. Restore the right, rear, and front covers.

250ul syringe assembly-photoelectric position sensor

Note

N/A

Required tools

Cross-headed screwdriver

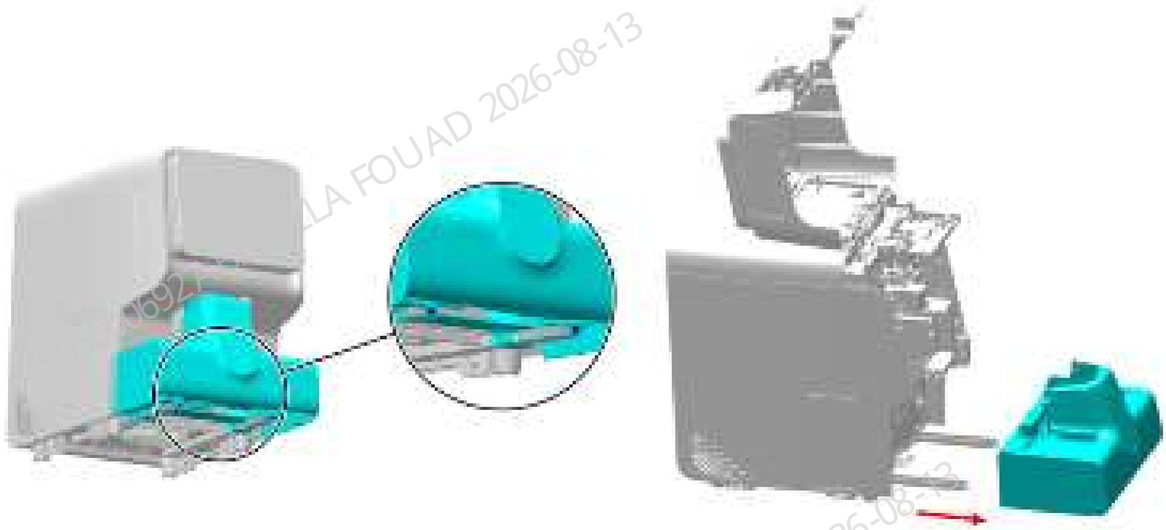
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

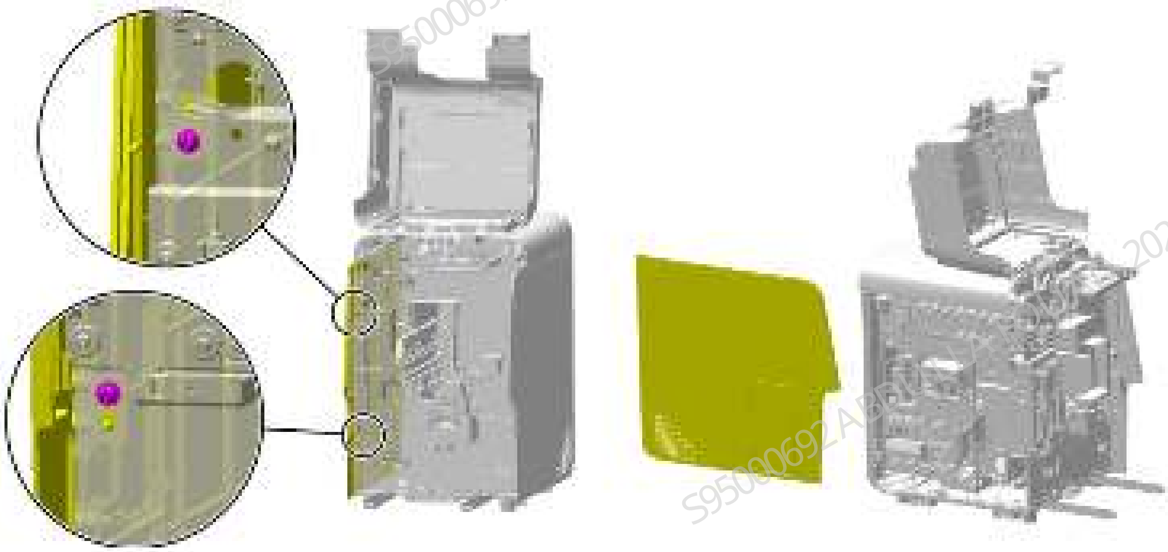
1. Loosen the two screws fixing the autoloader, open the front cover, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in the figure below.

Figure8–172 Removing the autoloader



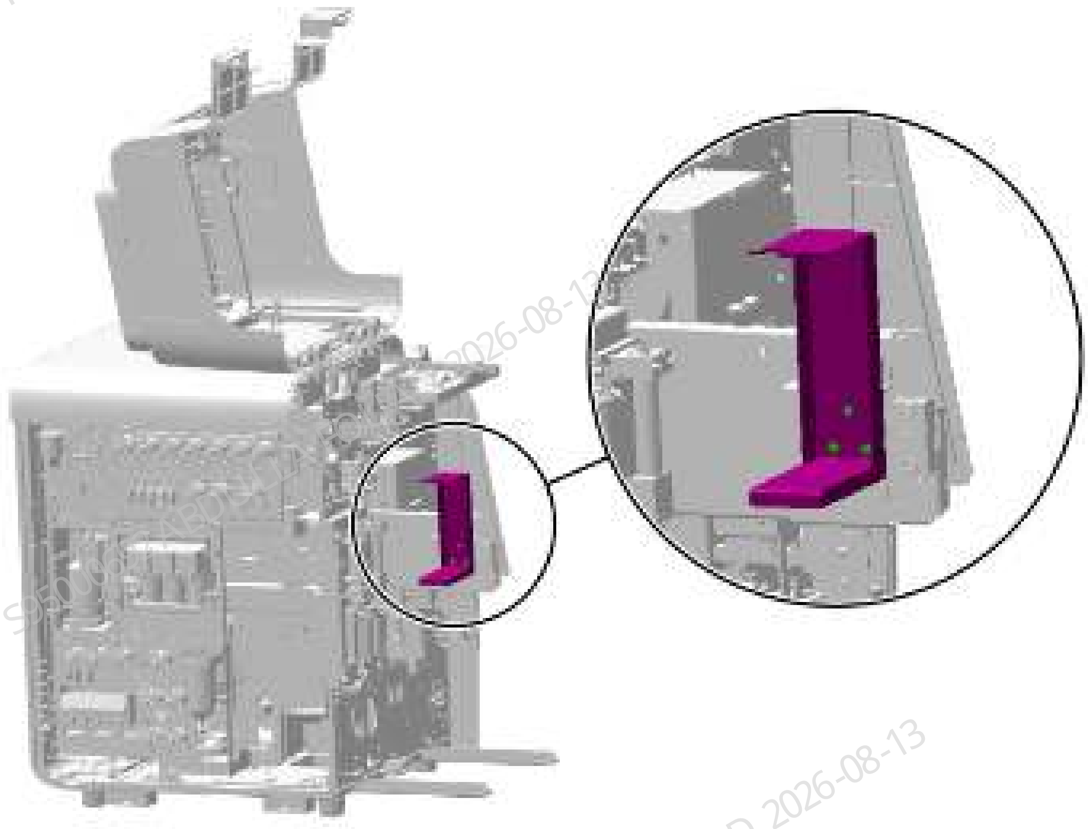
2. Loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–173 Removing the left cover



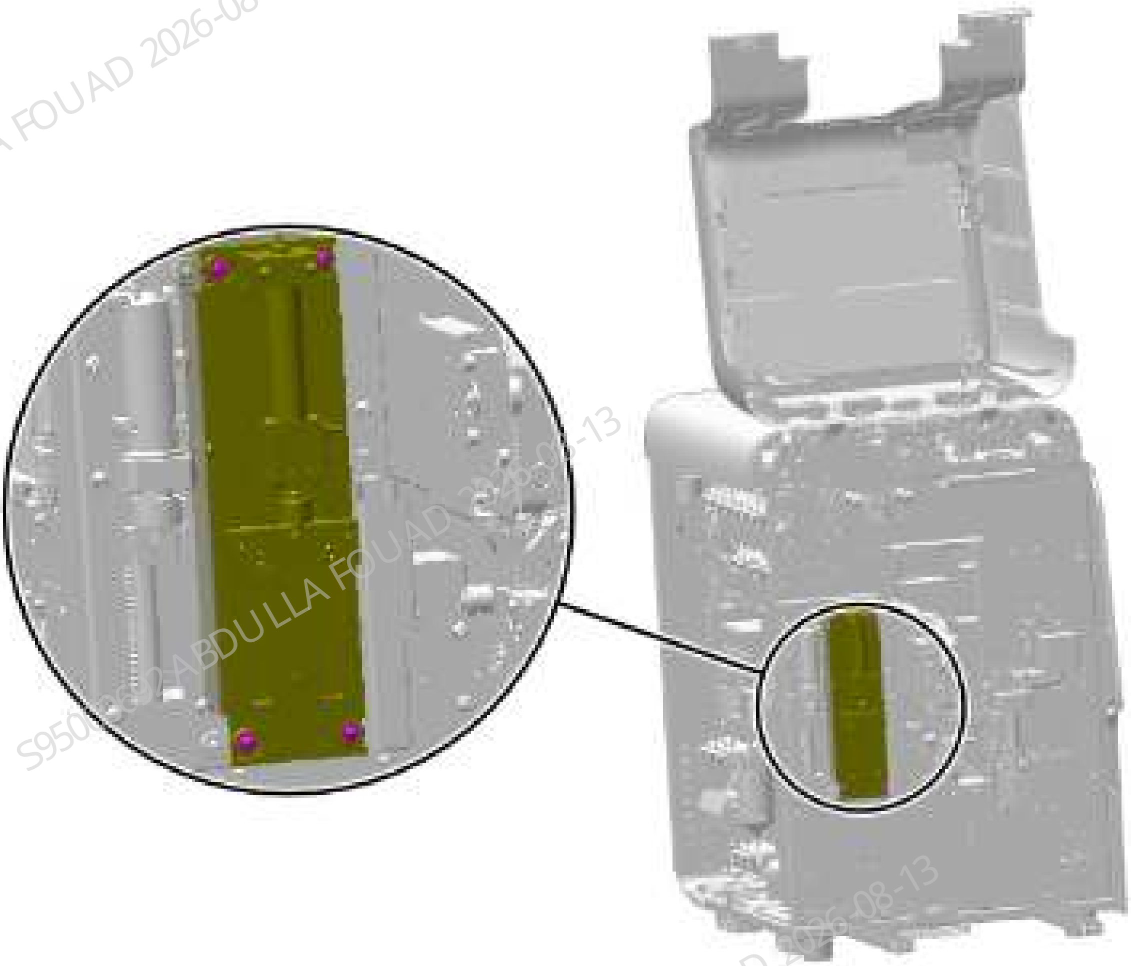
3. Remove the rubber tube and ground wire from the 250ul syringe assembly.
4. As shown in the figure below, remove the three countersunk screws fixing the dye bag support and remove the dye bag support.

Figure8–174 Removing the dye bag support



5. Remove the four set screws of the 250ul syringe assembly (43F4), as shown in the figure below, pull out the assembly terminal, and remove the entire assembly.

Figure8–175 Positions of the set screws of the 250ul syringe assembly



6. Removing the set screws of the photoelectric position sensor bracket.

Figure8–176 Removing the set screws of photoelectric position sensor bracket

7. Remove the set screws of the photoelectric position sensor and remove the sensor.

Figure8–177 Removing the set screws of photoelectric position sensor

Subsequent Requirements

Installation procedure:

1. Fix the photoelectric position sensor on the sensor bracket.
2. Fix the sensor bracket on the syringe assembly.
3. Insert the terminal block of the syringe assembly, put it back in the original position of the analyzer, and fix it with four screws.
4. Connect the tube and ground wire of the syringe assembly.
5. Install the dye bag support back to the original position.
6. Install the left cover, and flip down the front cover to reset it.
7. Restore the autoloader.

Mix Assembly SEN4-Photoelectric Position Sensor

Note

N/A

Required tools

Cross-headed screwdriver

Pre-requirements

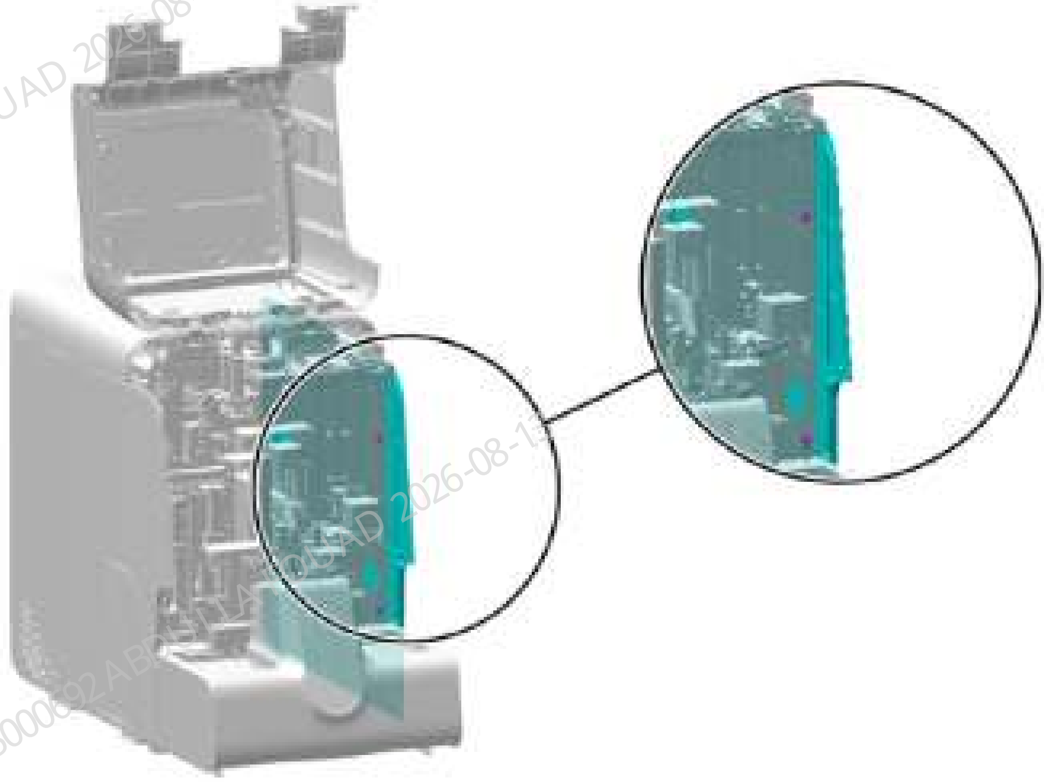
Power off the main unit, and unplug the power cord.

Procedures

Procedures

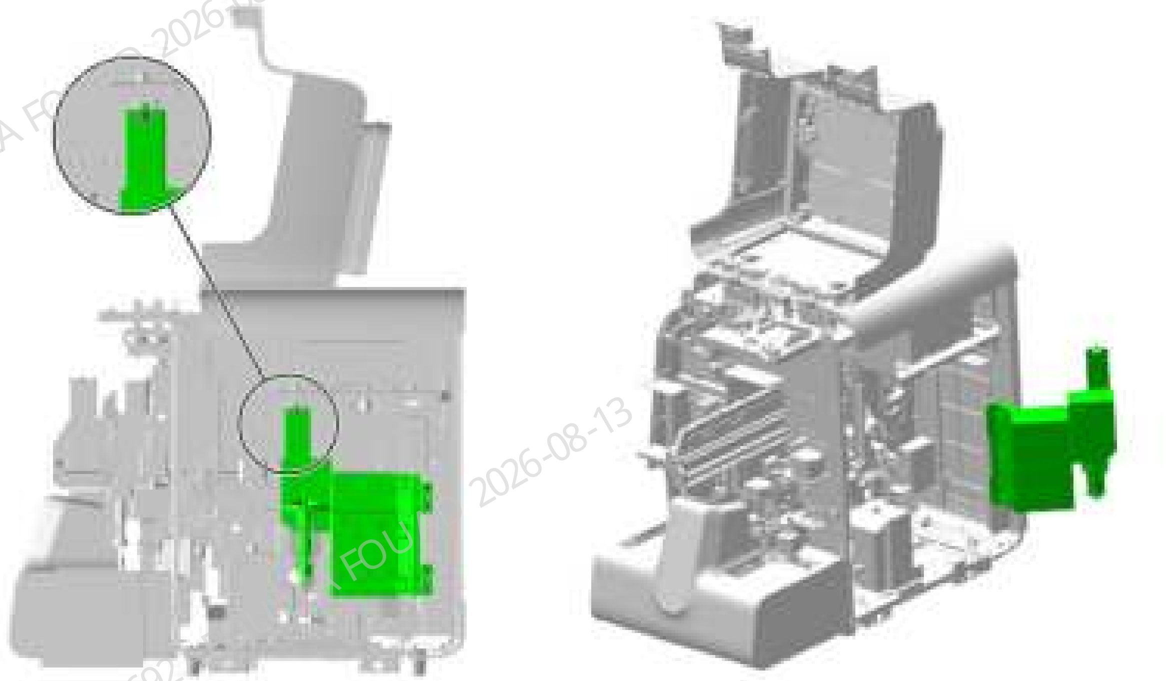
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–178 Removing the right cover



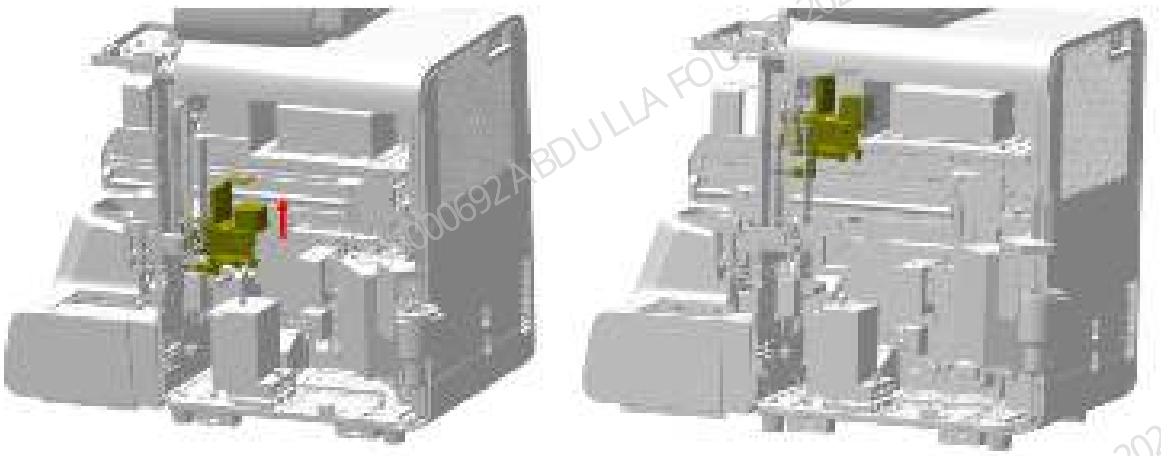
2. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right cover assembly upward, disengage it, and turn it outwards.

Figure8–179 Opening the right cover assembly



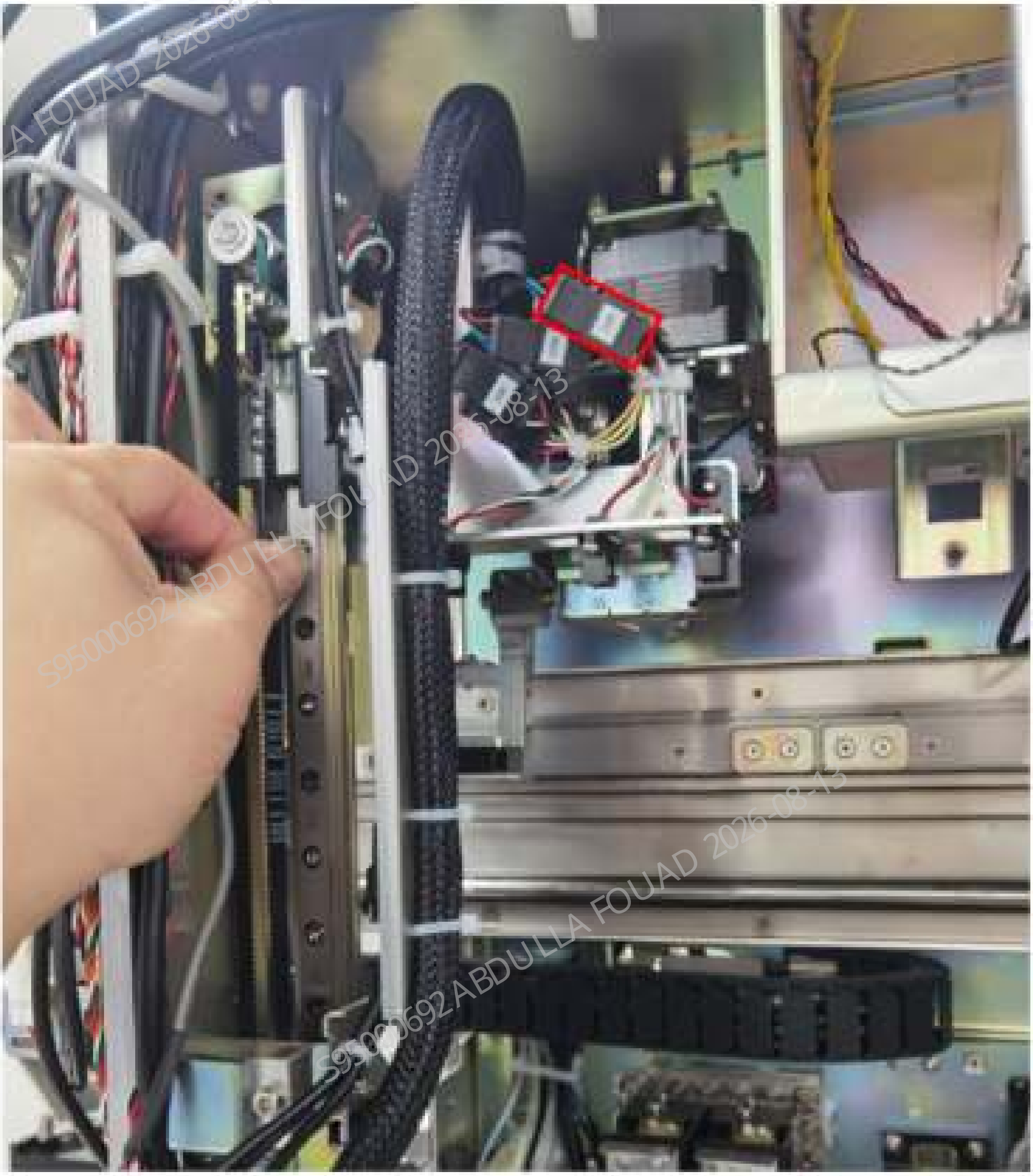
3. As shown in the following figure (left), pull the mix Y-axis assembly up in the direction of the arrow to the upper limit position as shown in Figure 5 (right), and then bind it using wire tape.

Figure8–180 Binding after lifting the mix Y-direction Assembly



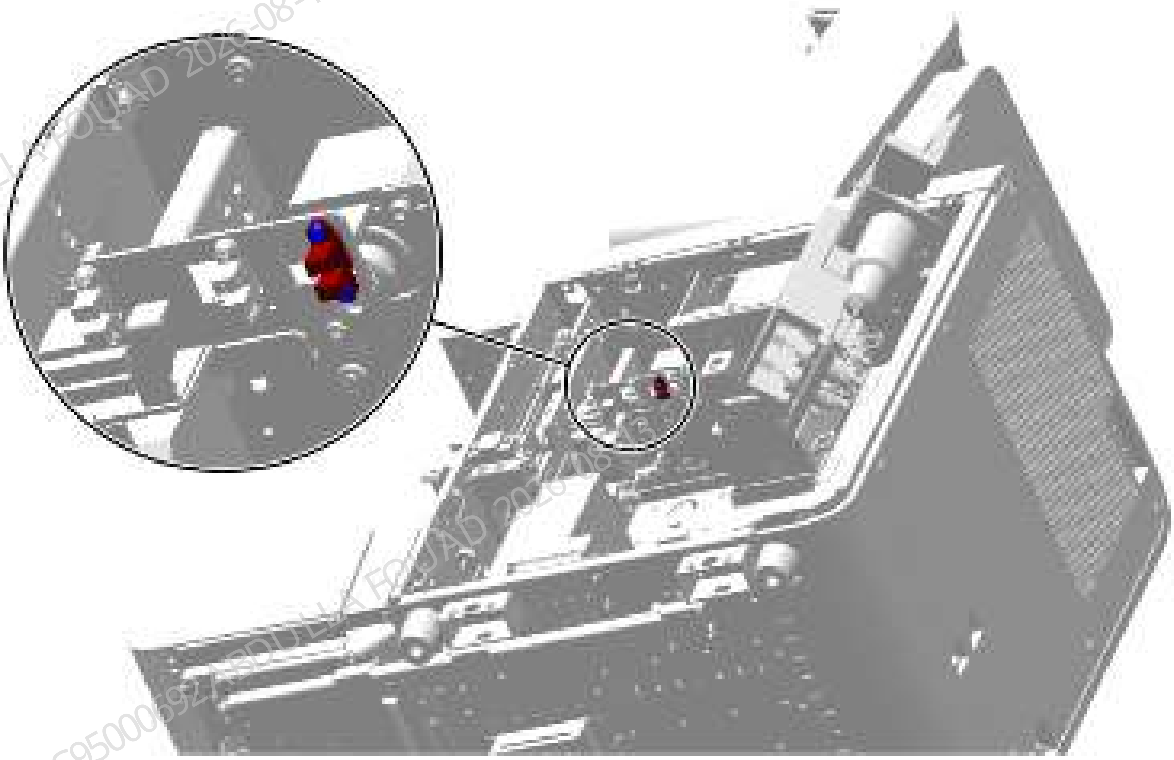
4. As shown in the following figure, unplug the wire of the photoelectric position sensor (SEN4).

Figure8–181 Removing the photoelectric position sensor (SEN4) wire



5. As shown below, remove the two screws fixing the photoelectric position sensor (SEN4) and remove the sensor.

Figure8–182 Positions of the set screws of the photoelectric position sensor (SEN4)



Subsequent Requirements

Installation procedure:

1. Use two screws to fix the photoelectric position sensor to the mix assembly and insert the wire.
2. Cut off the wire tie binding the Y-direction assembly, and slowly drop the Y-direction assembly to the lowest position.
3. Install back the right cover, and put down the front cover for resetting.

Mix Assembly SEN5-Photoelectric Position Sensor

Note

N/A

Required tools

Cross-headed screwdriver

Pre-requirements

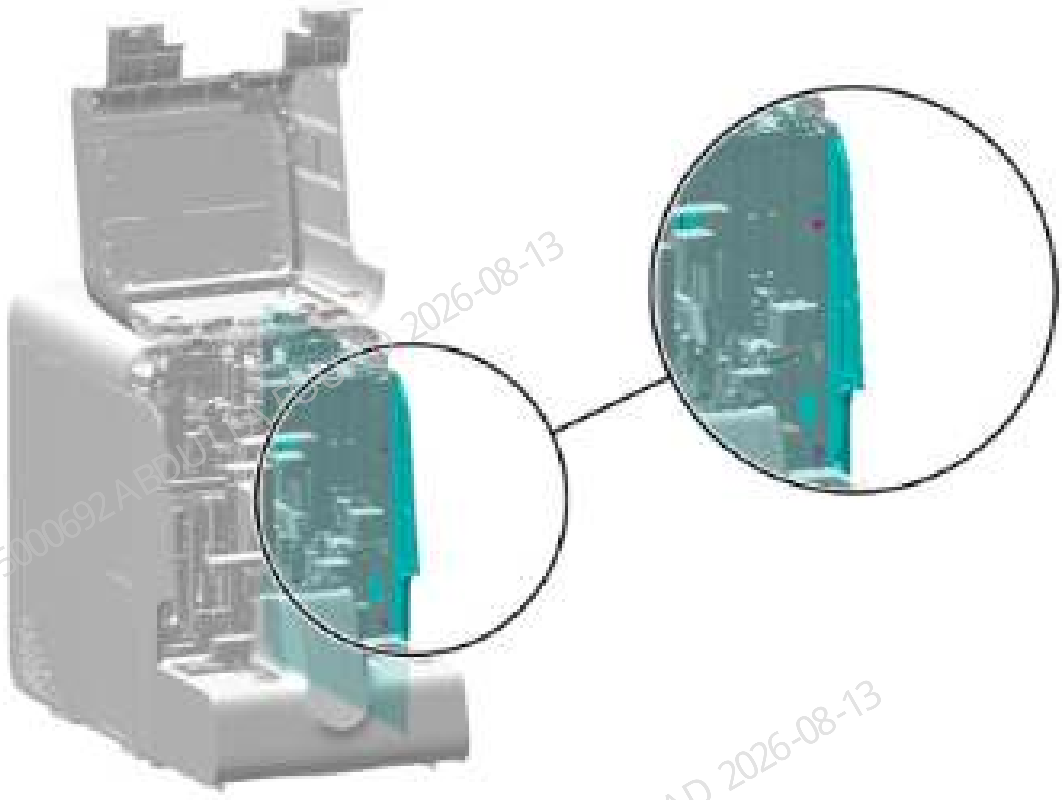
Power off the main unit, and unplug the power cord.

Procedures

Procedures

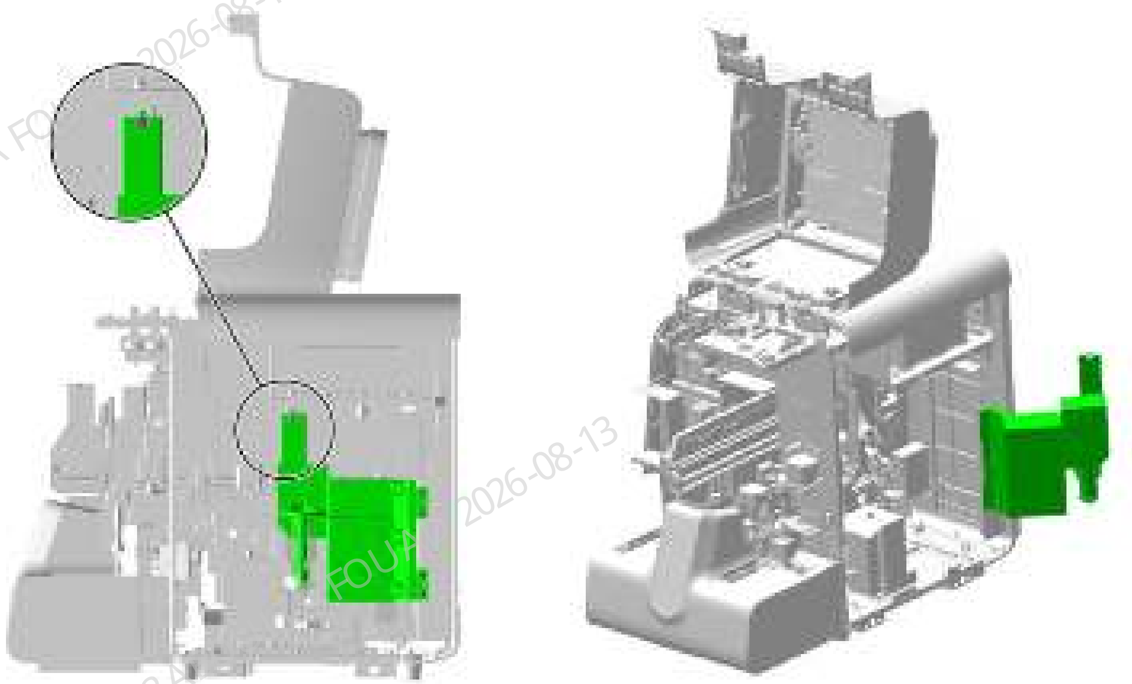
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–183 Removing the right cover



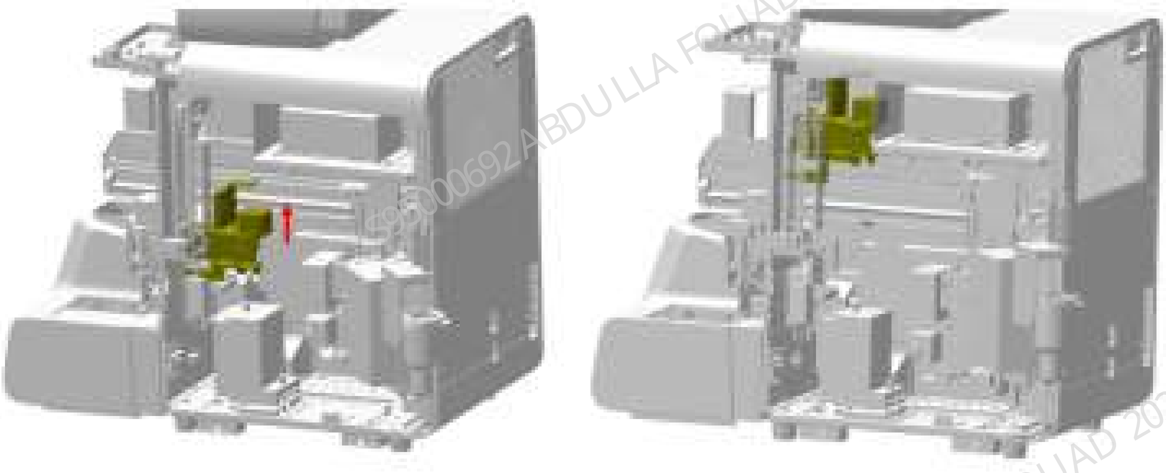
2. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right cover assembly upward, disengage it, and turn it outwards.

Figure8–184 Opening the right cover assembly



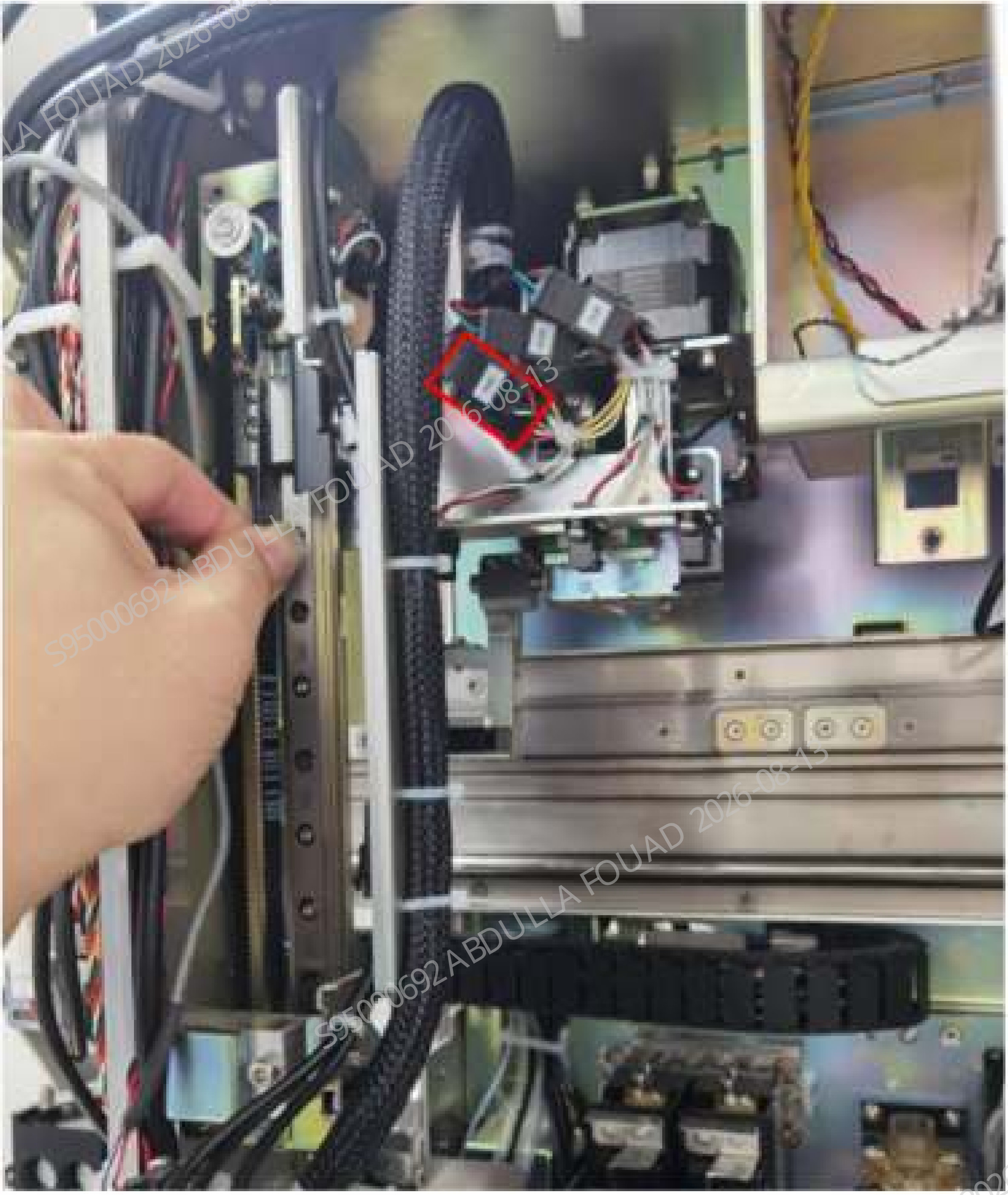
3. As shown in the following figure (left), pull the mix Y-axis assembly up in the direction of the arrow to the upper limit position as shown in the following figure (right), and then bind it using wire tape.

Figure8–185 Binding after lifting the mix Y-direction Assembly



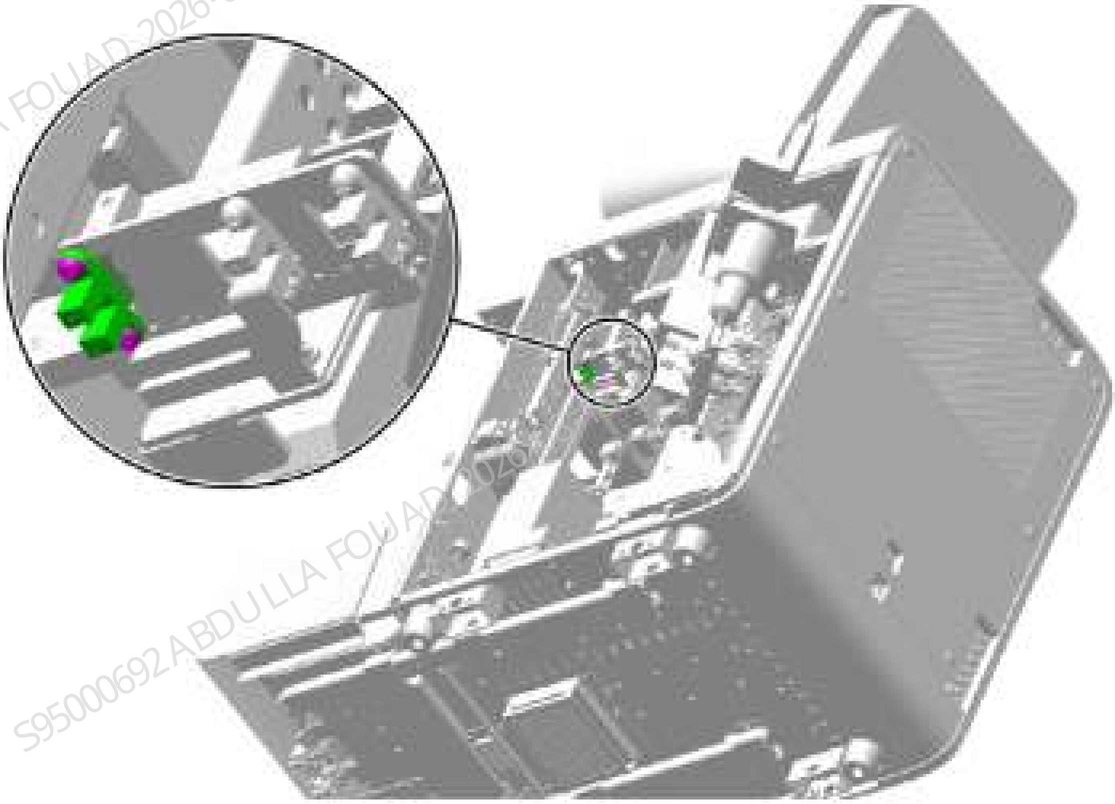
4. As shown in the following figure, unplug the wire of the photoelectric position sensor (SEN5).

Figure8–186 Removing the photoelectric position sensor (SEN5) wire



5. As shown below, remove the two screws fixing the photoelectric position sensor (SEN5) and remove the sensor.

Figure8–187 Positions of the set screws of the photoelectric position sensor (SEN5)



Subsequent Requirements

Installation procedure:

1. Use two screws to fix the photoelectric position sensor to the mix assembly and insert the wire.
2. Cut off the wire tie binding the Y-direction assembly, and slowly drop the Y-direction assembly to the lowest position.
3. Install back the right cover, and put down the front cover for resetting.

Mix Assembly SEN6-Photoelectric Position Sensor

Note

N/A

Required tools

Cross-headed screwdriver

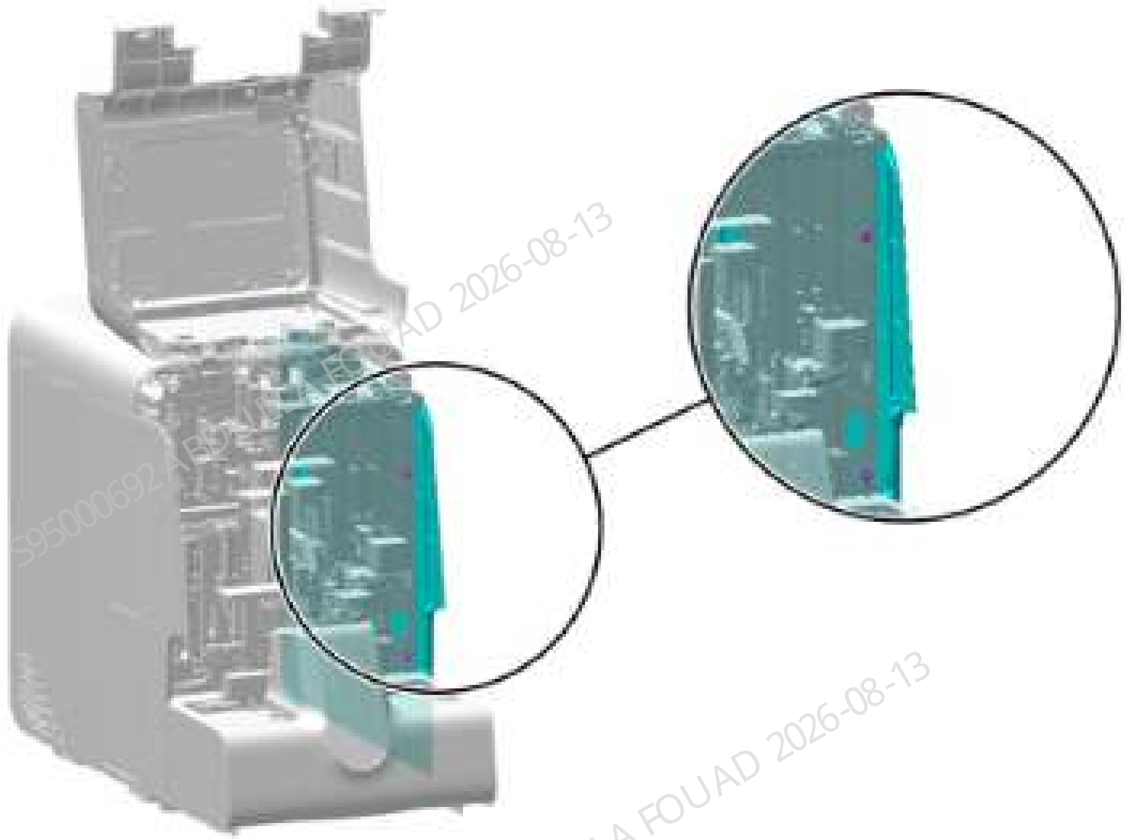
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

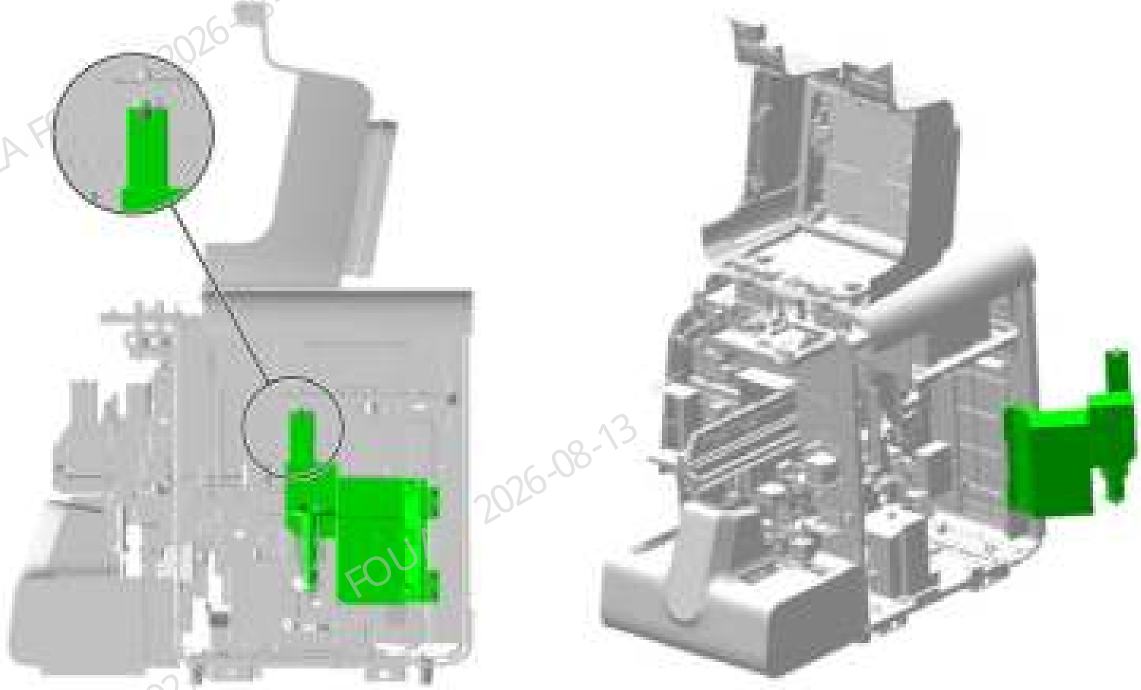
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–188 Removing the right cover



2. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right cover assembly upward, disengage it, and turn it outwards.

Figure8–189 Opening the right cover assembly

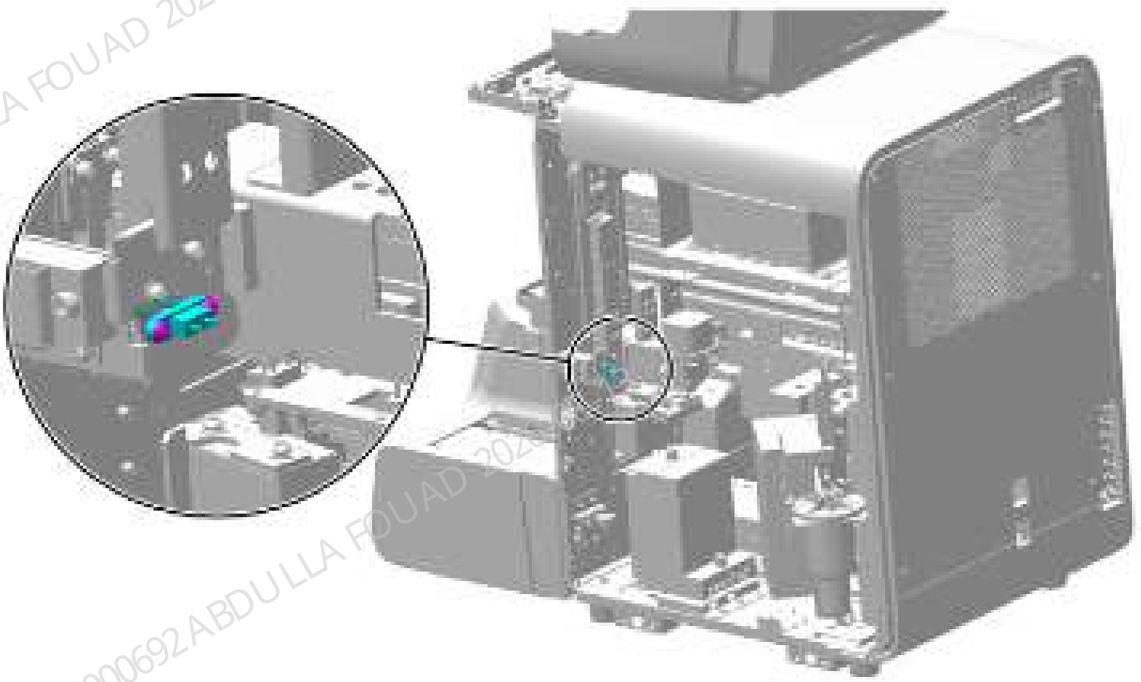


3. As shown in the following figure, unplug the wire of the photoelectric position sensor (SEN6).

Figure8–190 Removing the photoelectric position sensor (SEN6) wire

4. As shown below, remove the two screws fixing the photoelectric position sensor (SEN6) and remove the sensor.

Figure8–191 Positions of the set screws of the photoelectric Pposition sensor (SEN6)



Subsequent Requirements

Installation procedure:

1. Use two screws to fix the photoelectric position sensor to the mix assembly and insert the wire.
2. Install back the right cover, and put down the front cover for resetting.

Mix Assembly SEN7-Photoelectric Position Sensor

Note

N/A

Required tools

Cross-headed screwdriver

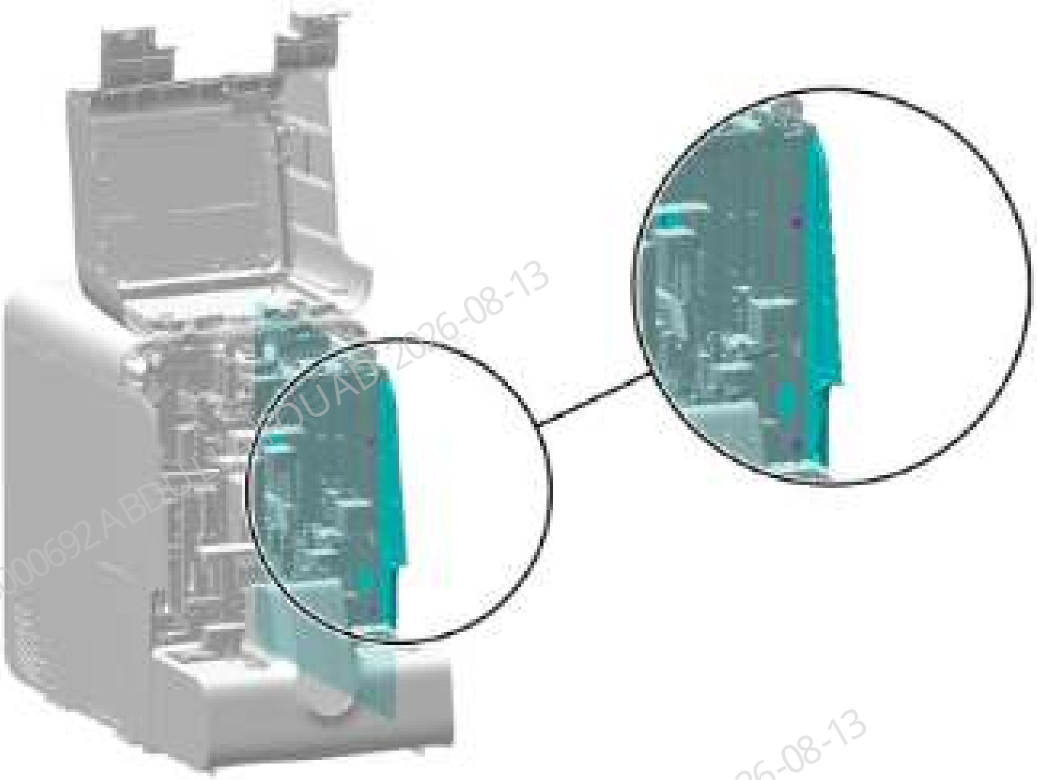
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

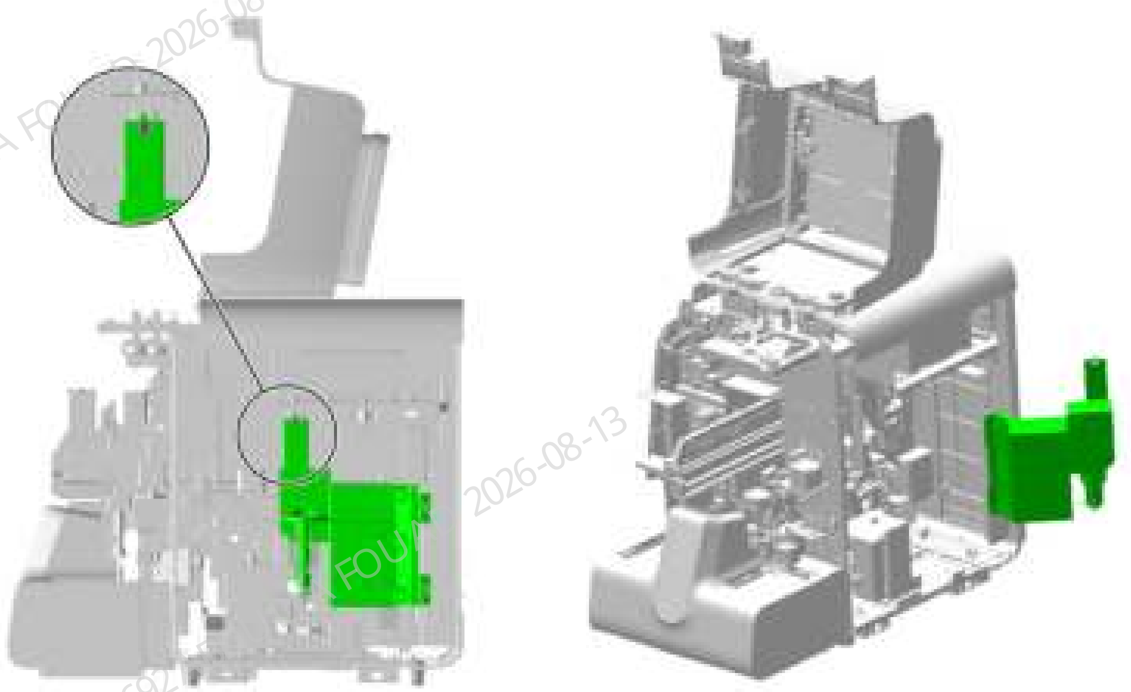
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–192 Removing the right cover



2. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right cover assembly upward, disengage it, and turn it outwards.

Figure8–193 Opening the right cover assembly

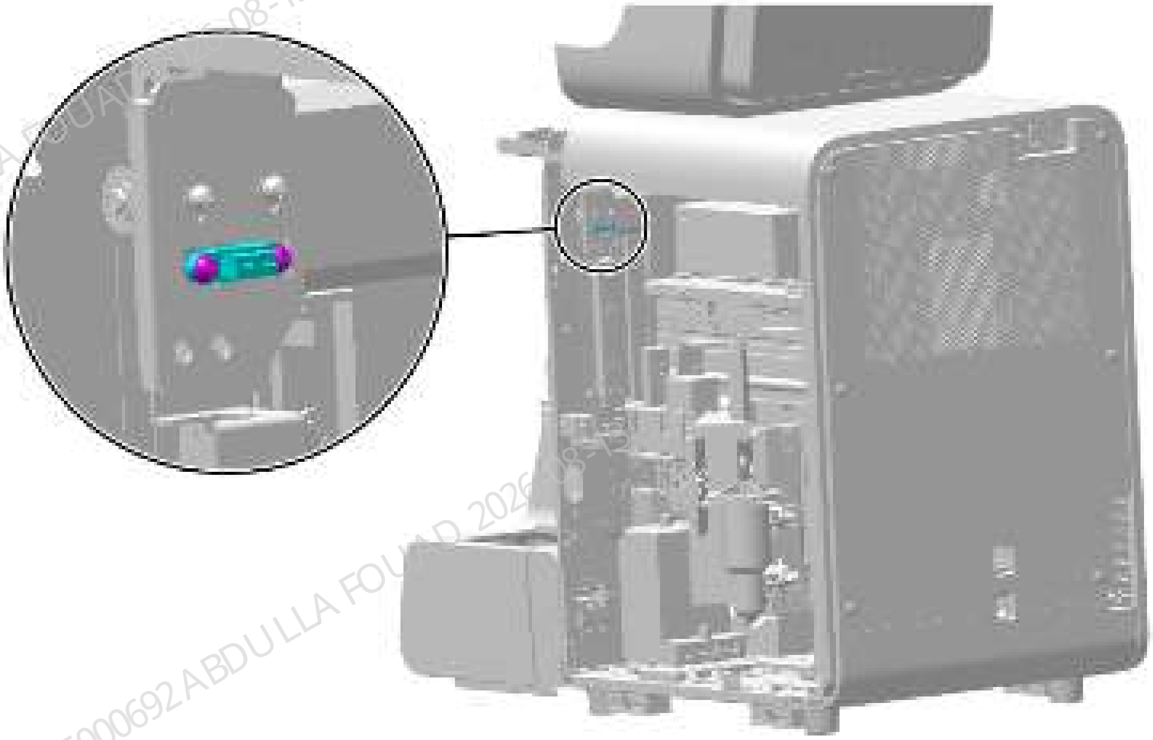


3. As shown in the following figure, unplug the wire of the photoelectric position sensor (SEN7).

Figure8–194 Removing the photoelectric position sensor (SEN7) wire

4. As shown below, remove the two screws fixing the photoelectric position sensor (SEN7) and remove the sensor.

Figure8–195 Positions of the set screws of the photoelectric position sensor (SEN7)



Subsequent Requirements

Installation procedure:

1. Use two screws to fix the photoelectric position sensor to the mix assembly and insert the wire.
2. Install back the right cover, and put down the front cover for resetting.

Mix Assembly SEN23-Photoelectric Position Sensor

Note

N/A

Required tools

Cross-headed screwdriver

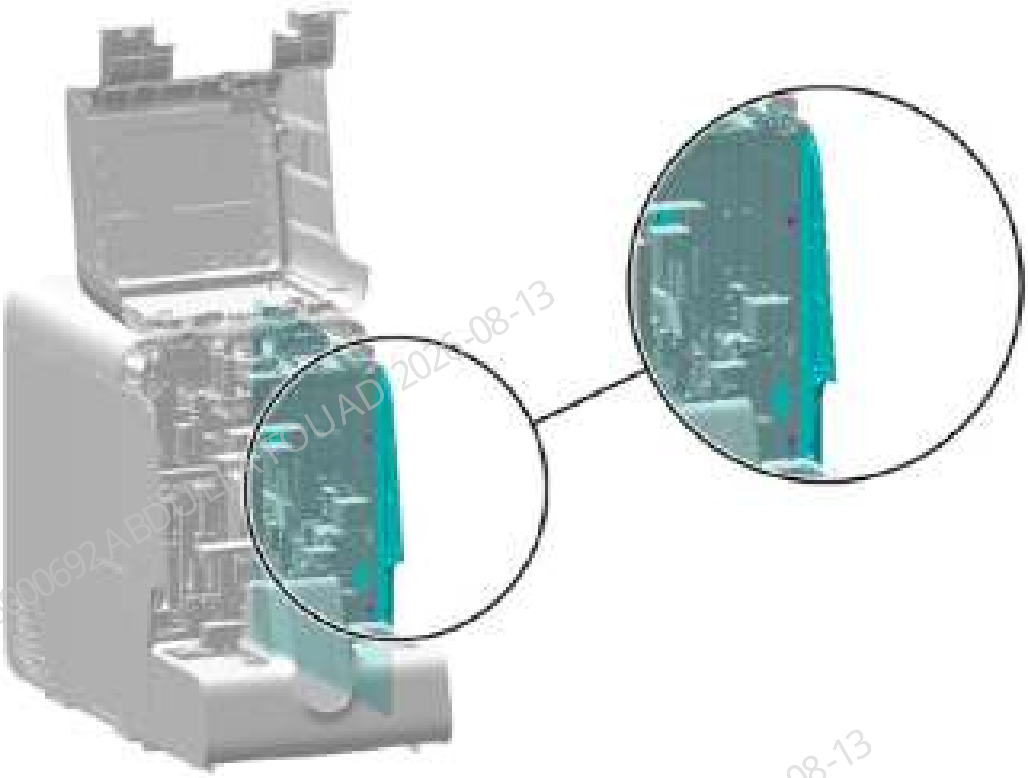
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

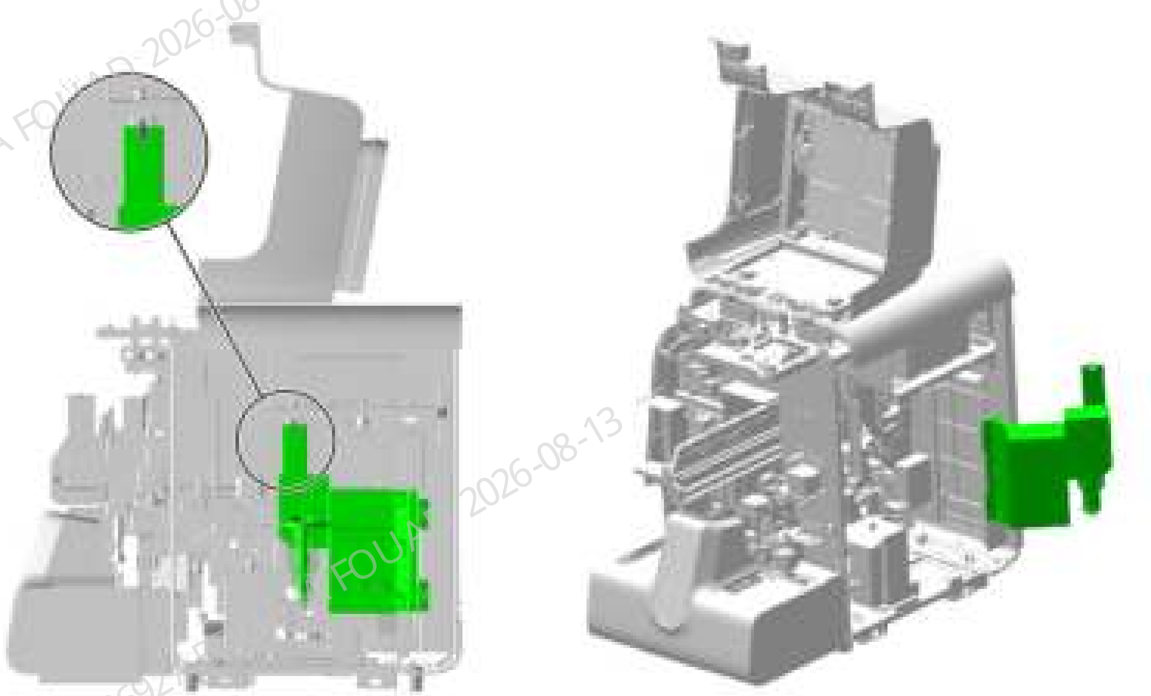
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–196 Removing the right cover



2. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right cover assembly upward, disengage it, and turn it outwards.

Figure8–197 Opening the right cover assembly



3. As shown in the following figure (left), pull the mix Y-axis assembly up in the direction of the arrow to the upper limit position as shown in the following figure (right), and then bind it using wire tape.

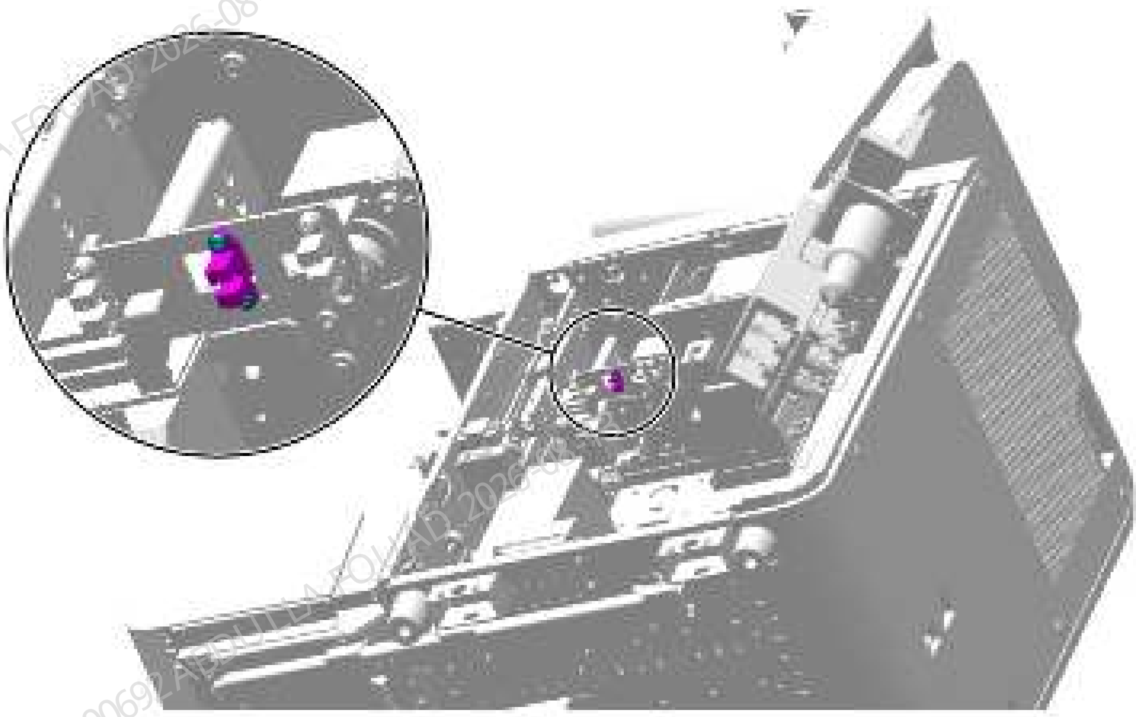
Figure8–198 Binding after lifting the mix Y-direction Assembly

4. As shown in the following figure, unplug the wire of the photoelectric position sensor (SEN23).

Figure8–199 Removing the photoelectric position sensor (SEN23) wire

5. As shown below, remove the two screws fixing the photoelectric position sensor (SEN23) and remove the sensor.

Figure8–200 Positions of the set screws of the photoelectric position sensor (SEN23)



Subsequent Requirements

Installation procedure:

1. Use two screws to fix the photoelectric position sensor to the mix assembly and insert the wire.
2. Install back the right cover, and put down the front cover for resetting.

8.20.3 Commissioning and Verification

For the commissioning and verification in the case of photoelectric position sensor, see Photoelectric Position Sensor.

For the commissioning and verification in the case of photoelectric position sensor in Z direction, see Sampling Z-direction Photoelectric Position Sensor.

For the commissioning and verification in the case of 250 μL syringe assembly-photoelectric position sensor, see 250ul syringe assembly-photoelectric position sensor.

For the commissioning and verification in the case of mix assembly SEN4-photoelectric position sensor, see Mix Assembly SEN4-Photoelectric Position Sensor.

For the commissioning and verification in the case of mix assembly SEN5-photoelectric position sensor, see Mix Assembly SEN5-Photoelectric Position Sensor.

For the commissioning and verification in the case of mix assembly SEN6-photoelectric position sensor, see Mix Assembly SEN6-Photoelectric Position Sensor.

For the commissioning and verification in the case of mix assembly SEN7-photoelectric position sensor, see Mix Assembly SEN7-Photoelectric Position Sensor.

For the commissioning and verification in the case of mix assembly SEN23-photoelectric position sensor, see Mix Assembly SEN23-Photoelectric Position Sensor.

Photoelectric Position Sensor

Pre-requirements

Install the 10 mL lead screw driving syringe assembly.

Procedures

1. Log in the software at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**DIL (10 mL) Syringe** in sequence.

Figure8–201 Self-test screen entrance

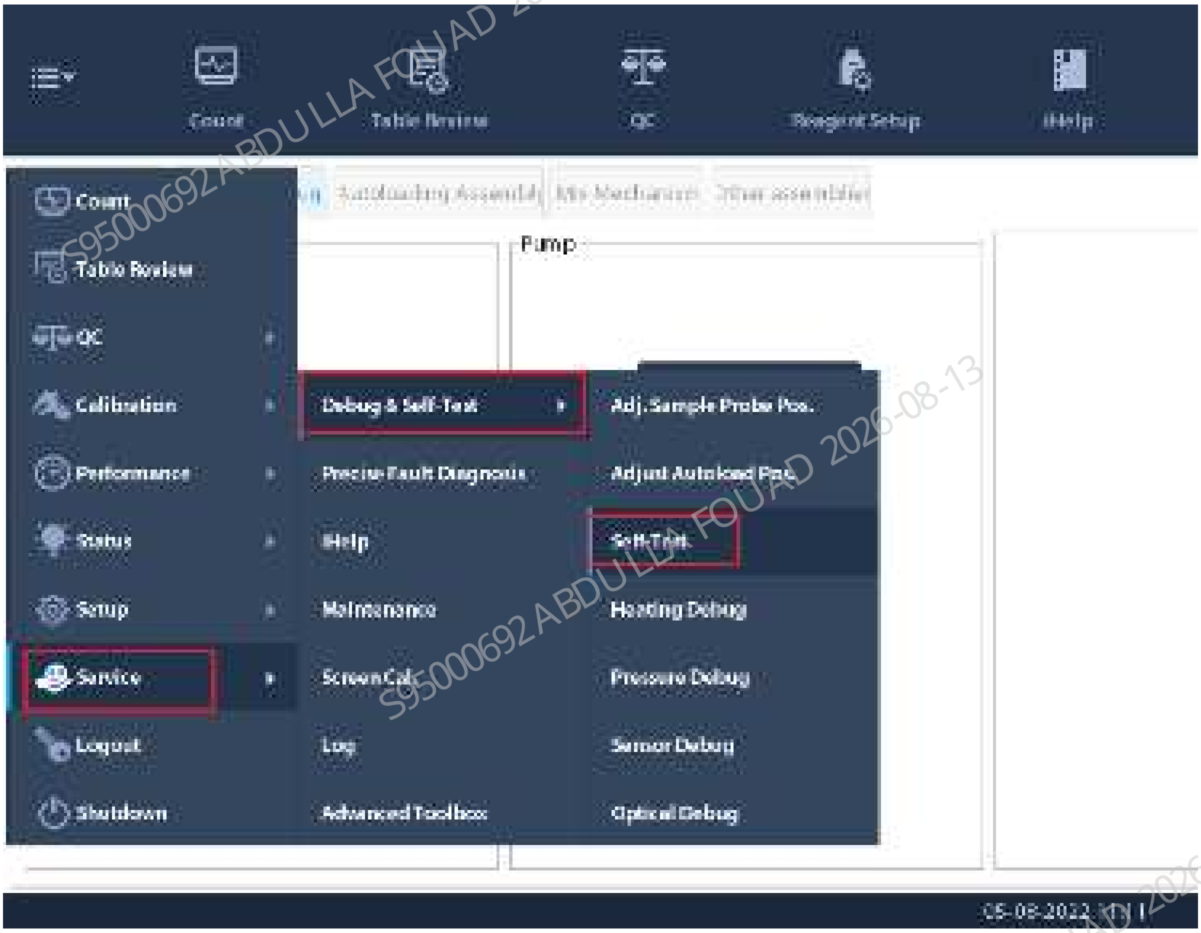


Figure8–202 Self-test screen of the 10 mL syringe

2. Confirm whether the 10 mL lead screw driving syringe assembly is initialized normally.
3. Check whether the syringe is leaking or splashing.
4. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

Sampling Z-direction Photoelectric Position Sensor

Required tools

Cross-headed screwdriver
One Allen wrench

Pre-requirements

N/A

Procedures

Commissioning not required

Verification:

1. After logging in at the service access level, click **Status**>>**Sensor Status**>>**Motherboard Sensor**, and check whether the status of the vertical initial position sensor Se3 is consistent with the actual situation.

	State
Asp. assem. home pos. H direc.(SE1)	Block
Asp. assem. confirm pos. H direc.(SE2)	Block
Asp. assem. home pos. V direc. (SE3)	Block
DIL(10mL) syringe (DIL)	Unblock
SP(250uL) syringe (SP)	Block
Front cover detection sensor (DOOR)	Block
Optical system shielding box	Open

2. In the sample analysis screen, perform three background counts. If there is no alarm related to the sampling probe assembly in the software screen, the replacement is successful.

250ul syringe assembly-photoelectric position sensor

Required tools

N/A

Pre-requirements

N/A

Procedures

1. Log in the software at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**SP(250 μL) Syringe** in sequence

Figure8–203 Self-test screen entrance

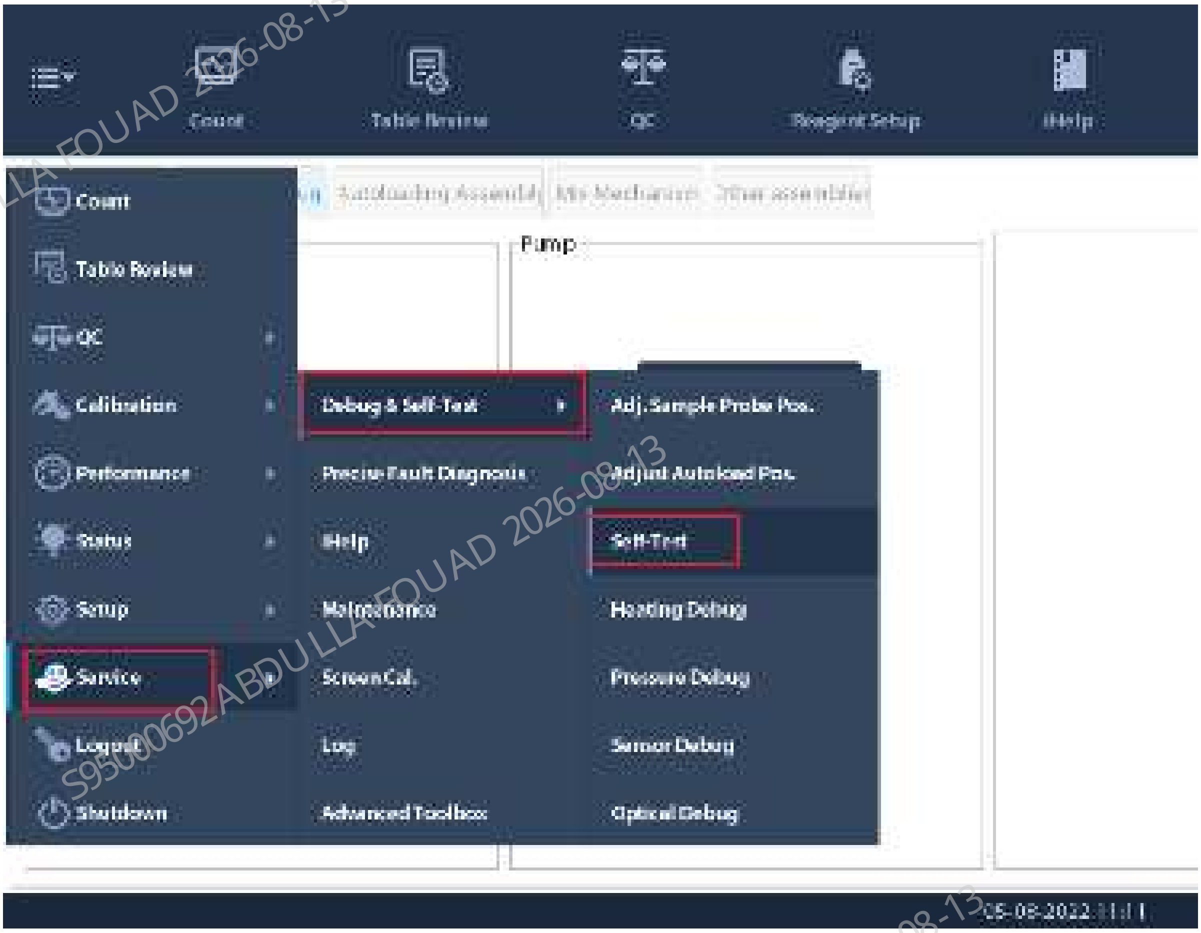
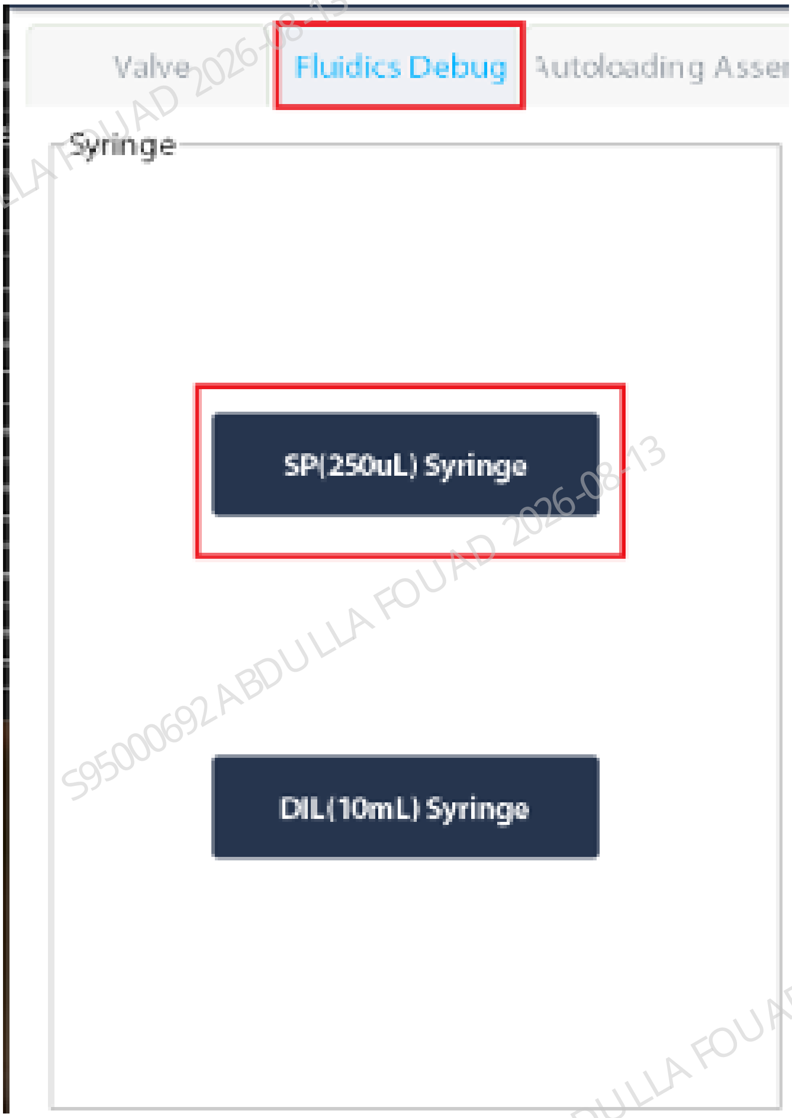


Figure8–204 Self-test Interface of the 250 μL Syringe



2. Confirm whether the 250 μL lead screw driving syringe assembly is initialized normally.
3. Check whether the syringe is leaking or splashing.

Verification:

Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

Mix Assembly SEN4-Photoelectric Position Sensor

Required tools

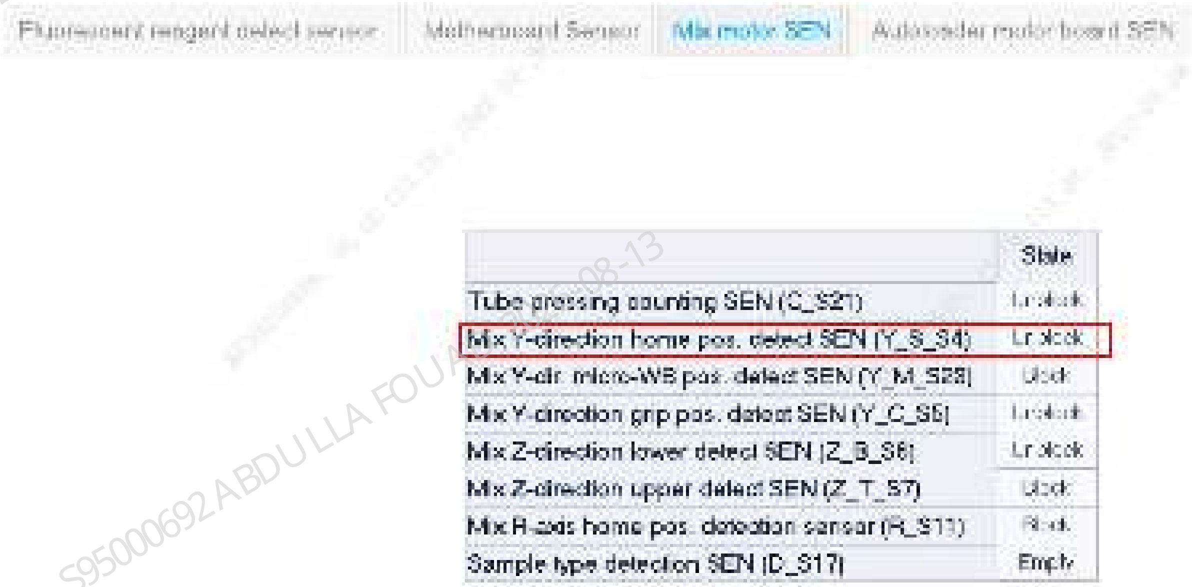
N/A

Pre-requirements

N/A

Procedures

After logging in at the service access level, click **Status**>>**Sensor Status**>>**Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful Otherwise, check whether the wire is properly connected.



Mix Assembly SEN5-Photoelectric Position Sensor

Required tools

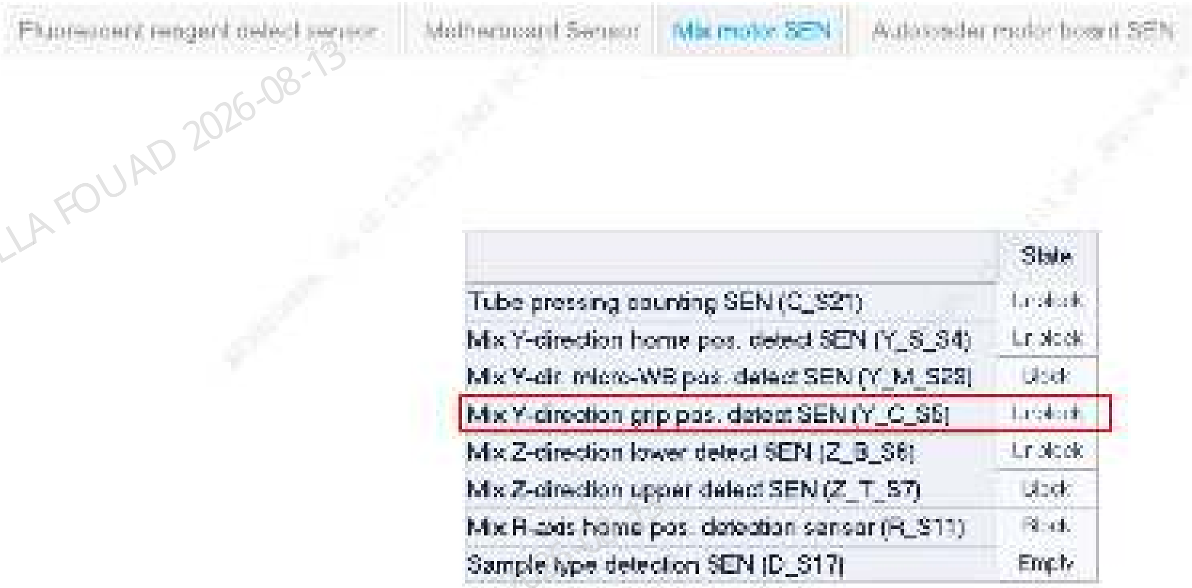
N/A

Pre-requirements

N/A

Procedures

After logging in at the service access level, click **Status**>>**Sensor Status**>>**Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful Otherwise, check whether the wire is properly connected.



	State
Tube pressing counting SEN (C_S21)	Unlock
Mix Y-direction home pos. detect SEN (Y_S_S4)	Unlock
Mix Y-dir. micro-WB pos. detect SEN (Y_M_S28)	Block
Mix Y-direction grip pos. detect SEN (Y_C_S5)	Unlock
Mix Z-direction lower detect SEN (Z_B_S6)	Unlock
Mix Z-direction upper detect SEN (Z_T_S7)	Block
Mix R-axis home pos. detection sensor (R_S11)	Block
Sample type detection SEN (D_S17)	Empty

Mix Assembly SEN6-Photoelectric Position Sensor

Required tools

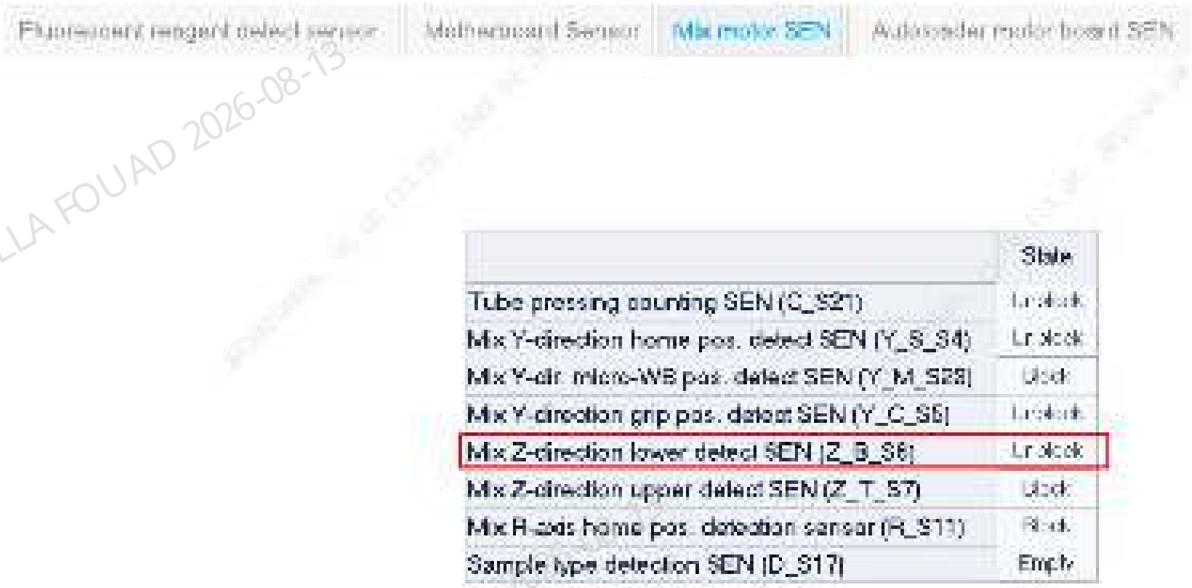
N/A

Pre-requirements

N/A

Procedures

After logging in at the service access level, click **Status>>Sensor Status>>Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful Otherwise, check whether the wire is properly connected.



Mix Assembly SEN7-Photoelectric Position Sensor

Required tools

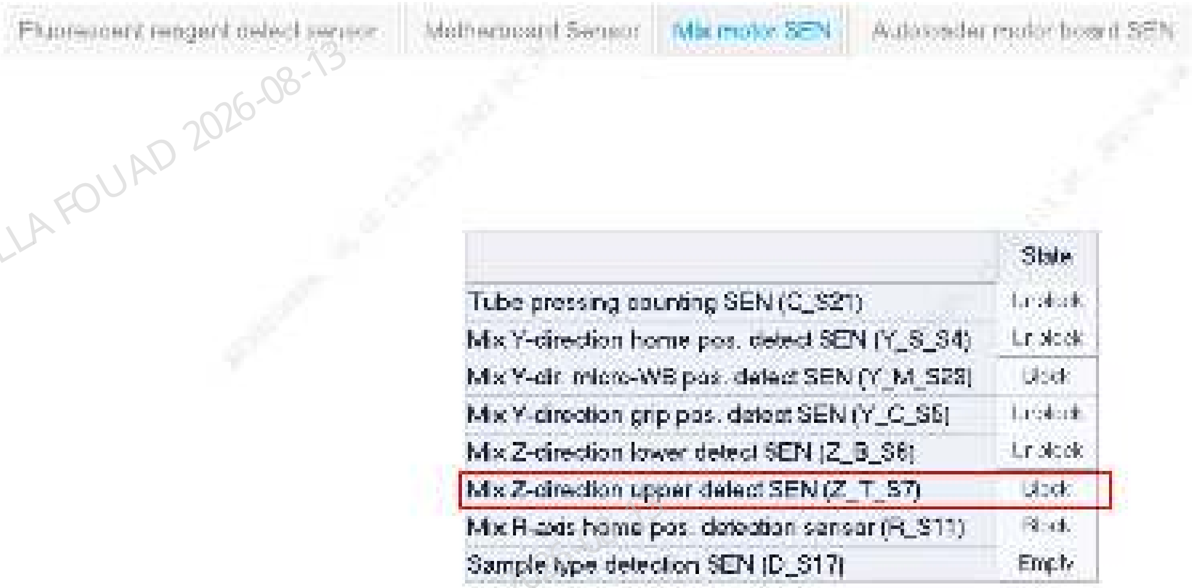
N/A

Pre-requirements

N/A

Procedures

After logging in at the service access level, click **Status>>Sensor Status>>Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful Otherwise, check whether the wire is properly connected.



Mix Assembly SEN23-Photoelectric Position Sensor

Required tools

N/A

Pre-requirements

N/A

Procedures

After logging in at the service access level, click **Status>>Sensor Status>>Mix Motor Board Sensor**, and check whether the status of the compressing counting sensor is consistent with the actual situation. If they are consistent, the replacement is successful Otherwise, check whether the wire is properly connected.

Fluorescent reagent detect sensor | Motherboard Sensor | Mix motor SEN | Autoloader motor board SEN

	State
Tube pressing counting SEN (C_S21)	Unblock
Mix Y-direction home pos. detect SEN (Y_S_S4)	Unblock
Mix Y-dir. micro-WB pos. detect SEN (Y_M_S28)	Block
Mix Y-direction grip pos. detect SEN (Y_C_S5)	Unblock
Mix Z-direction lower detect SEN (Z_B_S6)	Unblock
Mix Z-direction upper detect SEN (Z_T_S7)	Block
Mix R-axis home pos. detection sensor (R_S11)	Block
Sample type detection SEN (D_S17)	Empty

8.21 Probe wash device 2

8.21.1 General Information

Figure8–205 Photo of probe wipe 2 (closed-sampling)

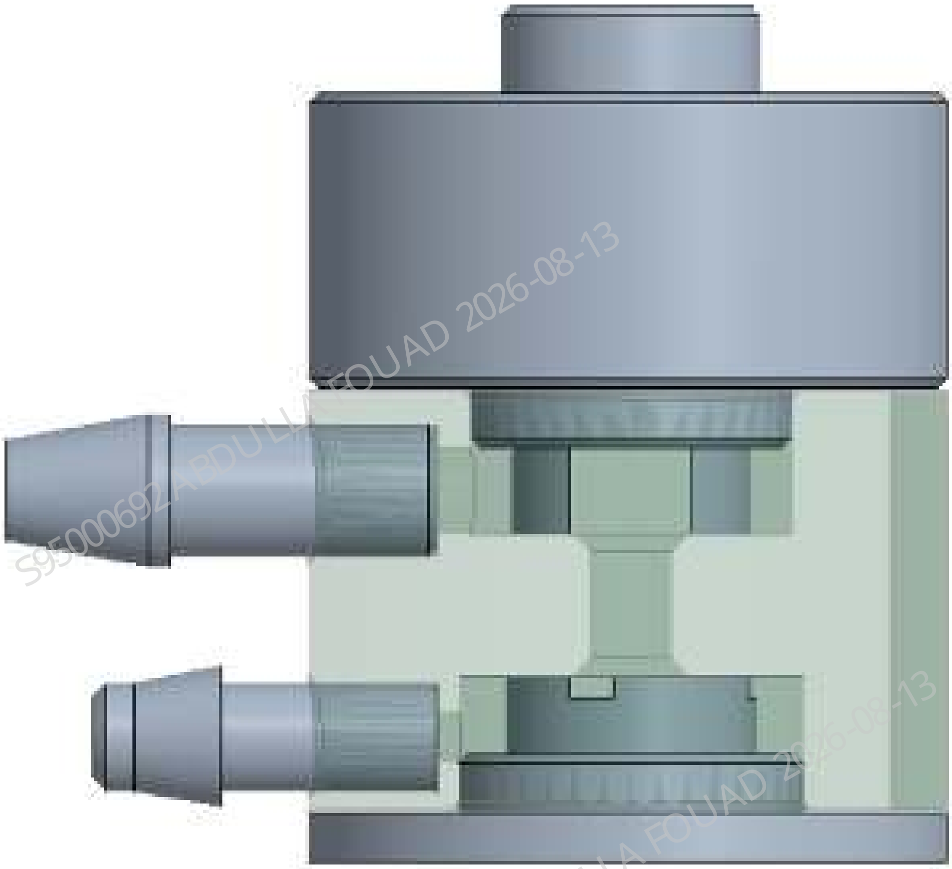


Figure8–206 Location of Probe Wipe 2 on the Sampling Assembly

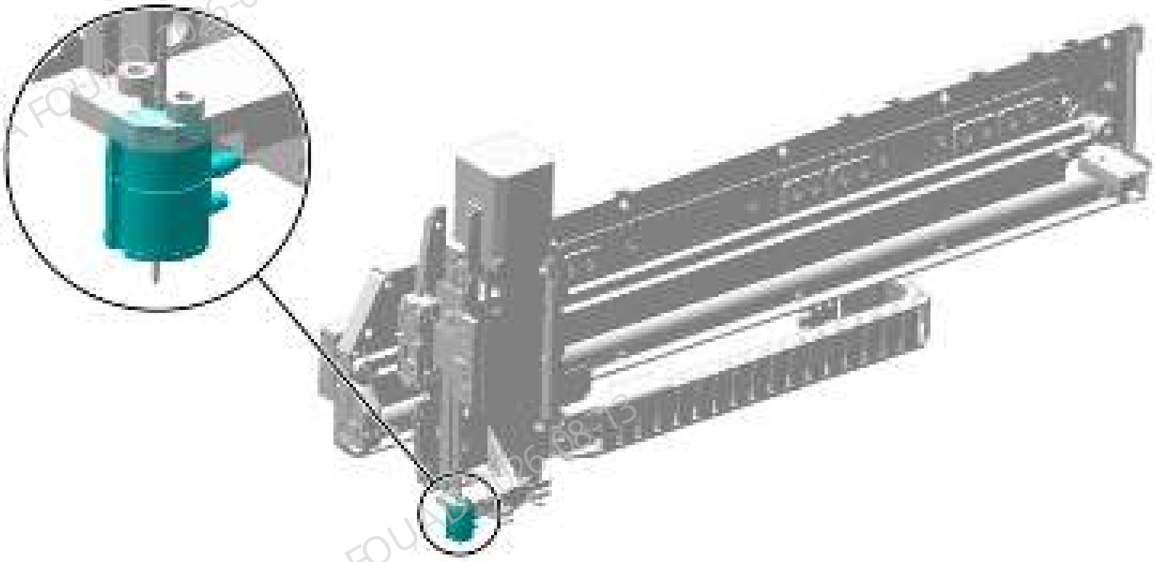


Table 8–25 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
041-023727-00	Probe Wash Device 2	N/A	N/A

Function:
Provide a cavity space to clean the exterior of the sample probe and collect the waste generated by the cleaning of the interior.

8.21.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench

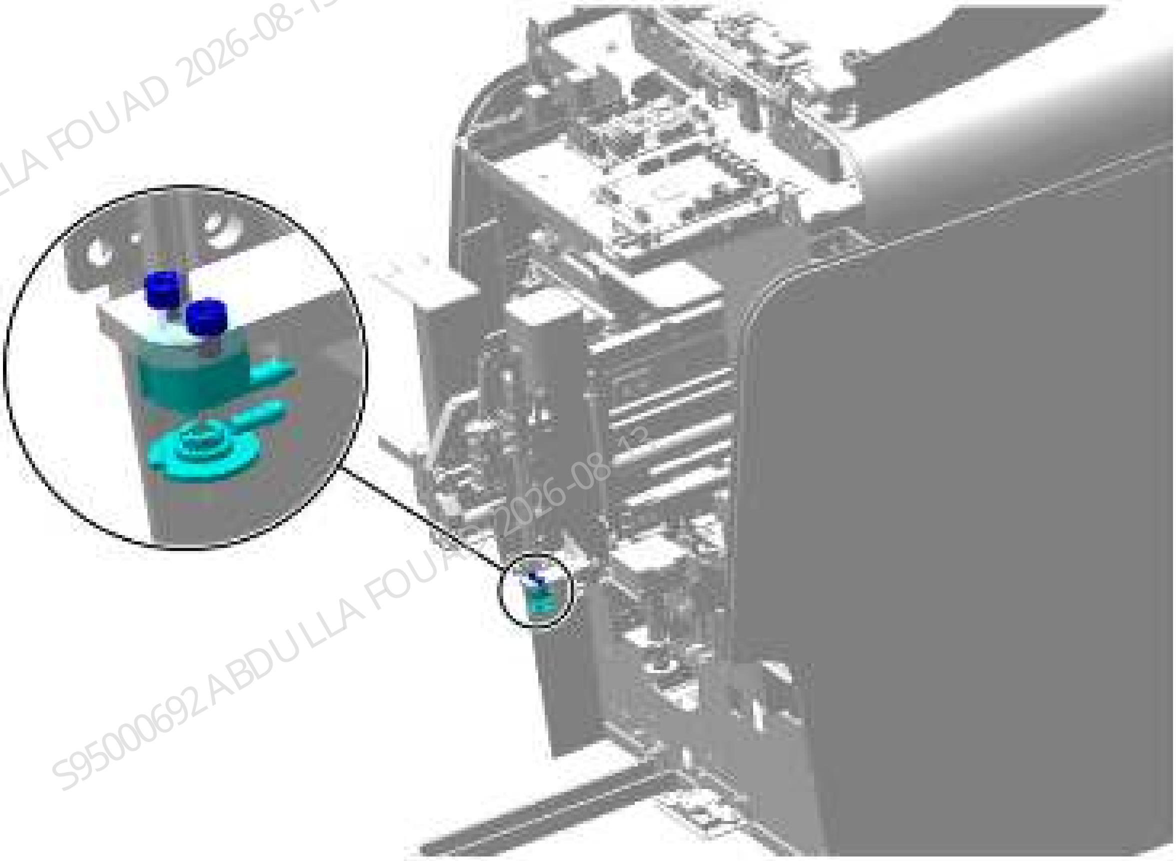
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

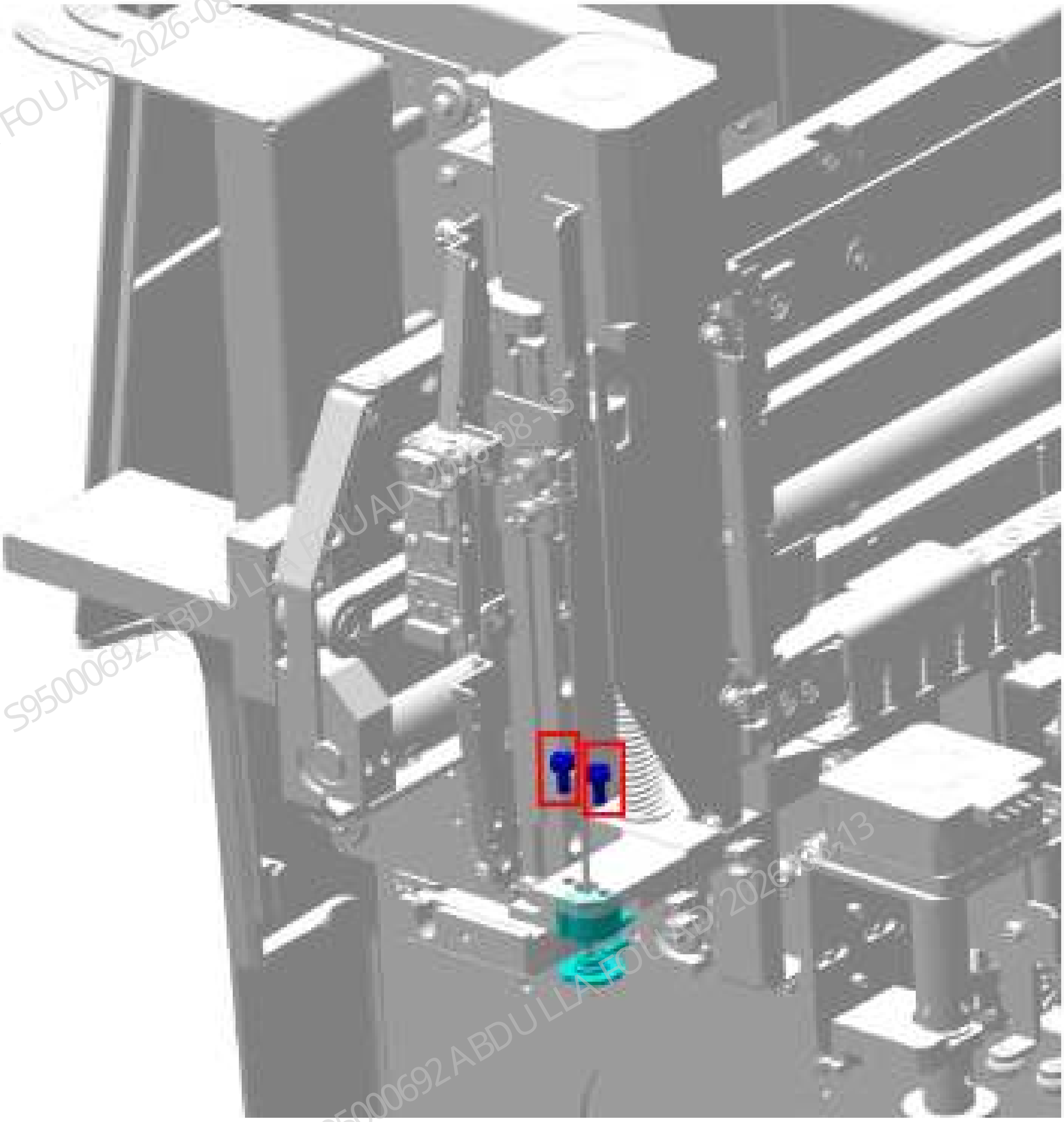
1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–207 Opening the front cover



2. Use an Allen key to loosen the two screws that secure the probe wipe (closed-sampling), and remove the probe wipe (closed-sampling), as shown in the figure below.

Figure8–208 Removing the probe wipe (closed-sampling)



3. Use tweezers to pull the tube from the probe wipe.

Subsequent Requirements

Installation procedure:

1. Insert the tube back into the new probe wipe 2 (closed-sampling), and connect the upper tube and lower tube of the probe wipe respectively according to the tube labels.
2. Install the probe wipe 2 (closed-sampling) back in the original position of the aspiration module, and tighten the screws to fix it.

8.21.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

1. After logging in at service access level, select **Service>>Commissioning & Self-Test>>Adj. Sample Probe Pos.**, and complete adjustment of the vertical initial position according to the adjustment requirements for the vertical initial position in the description of sampling assembly FRU.
2. Select **Service>>Maintenance>>Cleaning**, select **Sampling Probe Cleaning**, and observe whether there is any spillage or leakage of the probe wipe during the cleaning process.

Verification:

Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.22 Piercing needle

8.22.1 General Information

Figure8–209 Photo of Piercing Needle

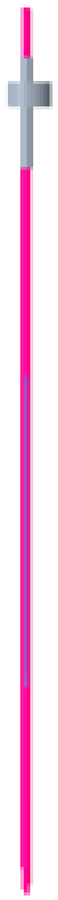


Table 8–26 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-069029-00	Piercing Needle	N/A	N/A

Function:
Aspirating sample after test tube piercing.

8.22.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench

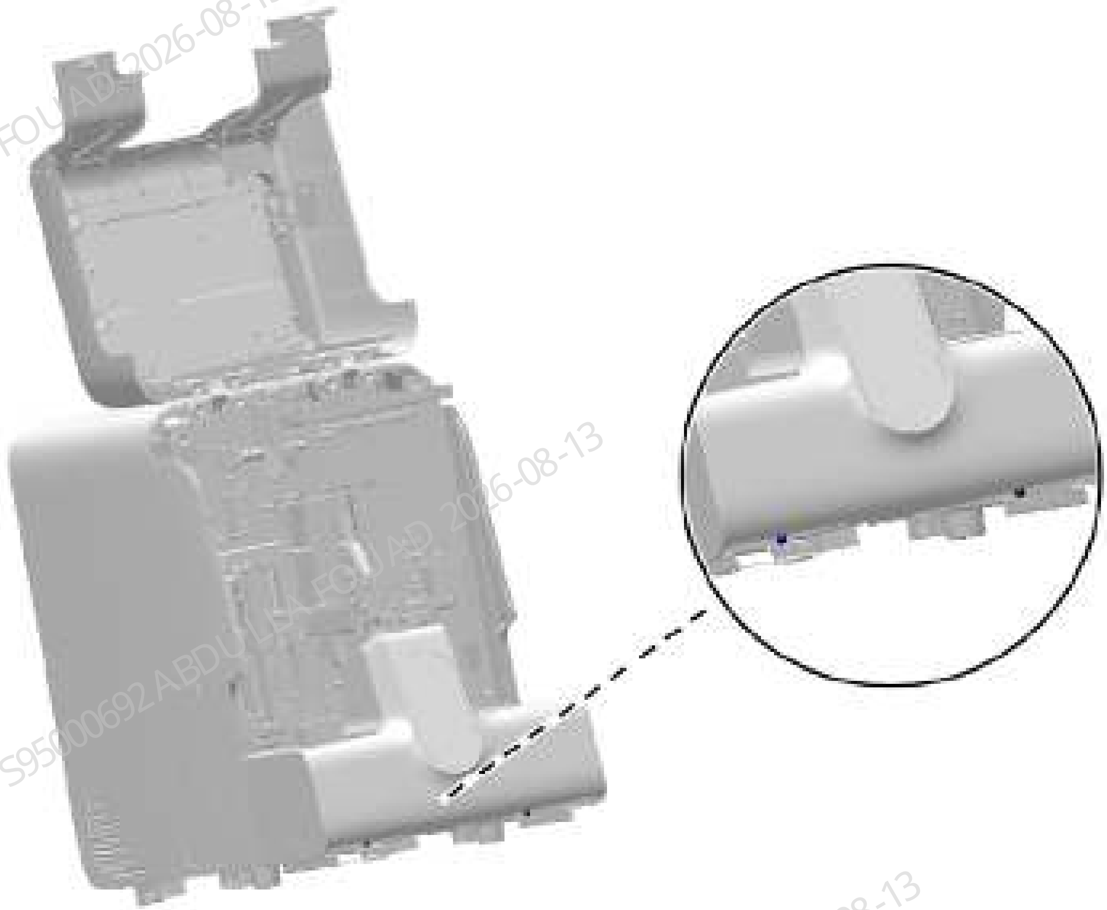
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

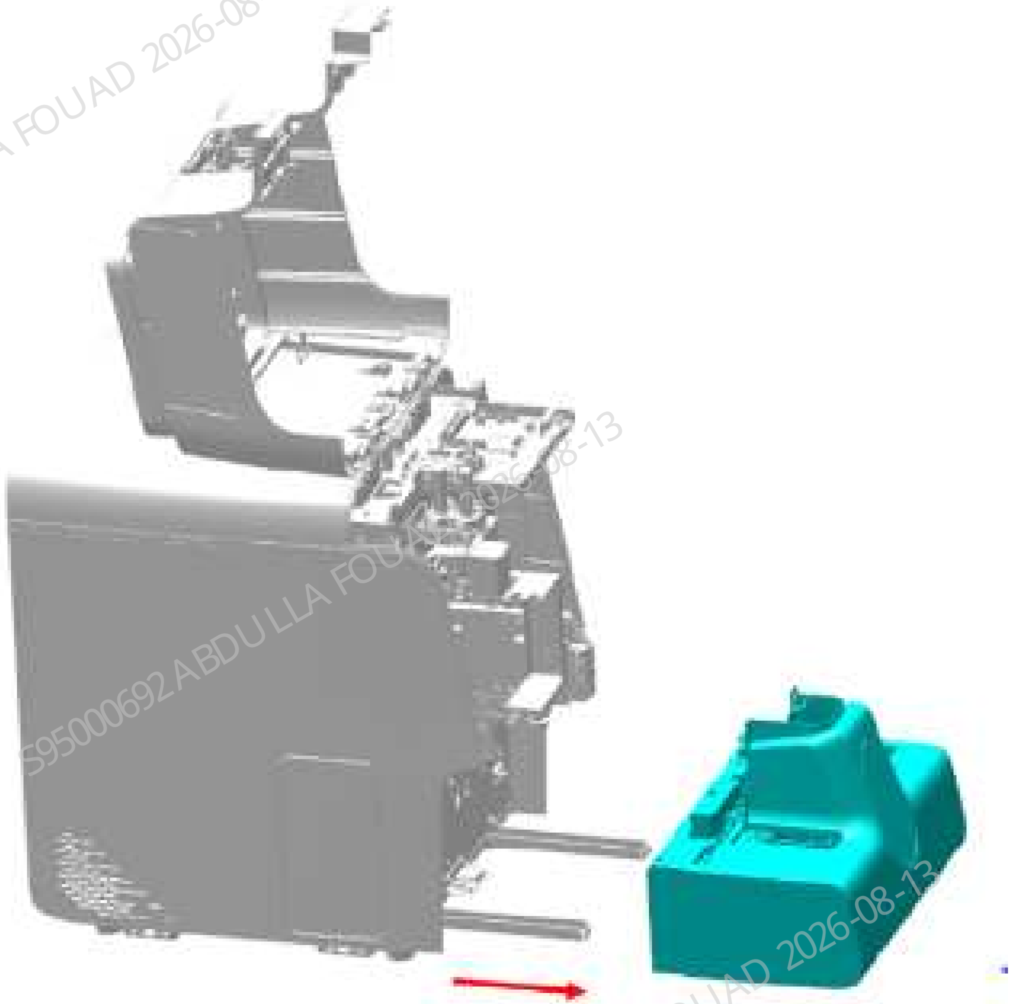
1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–210 Figure 1 - Removing the autoloader



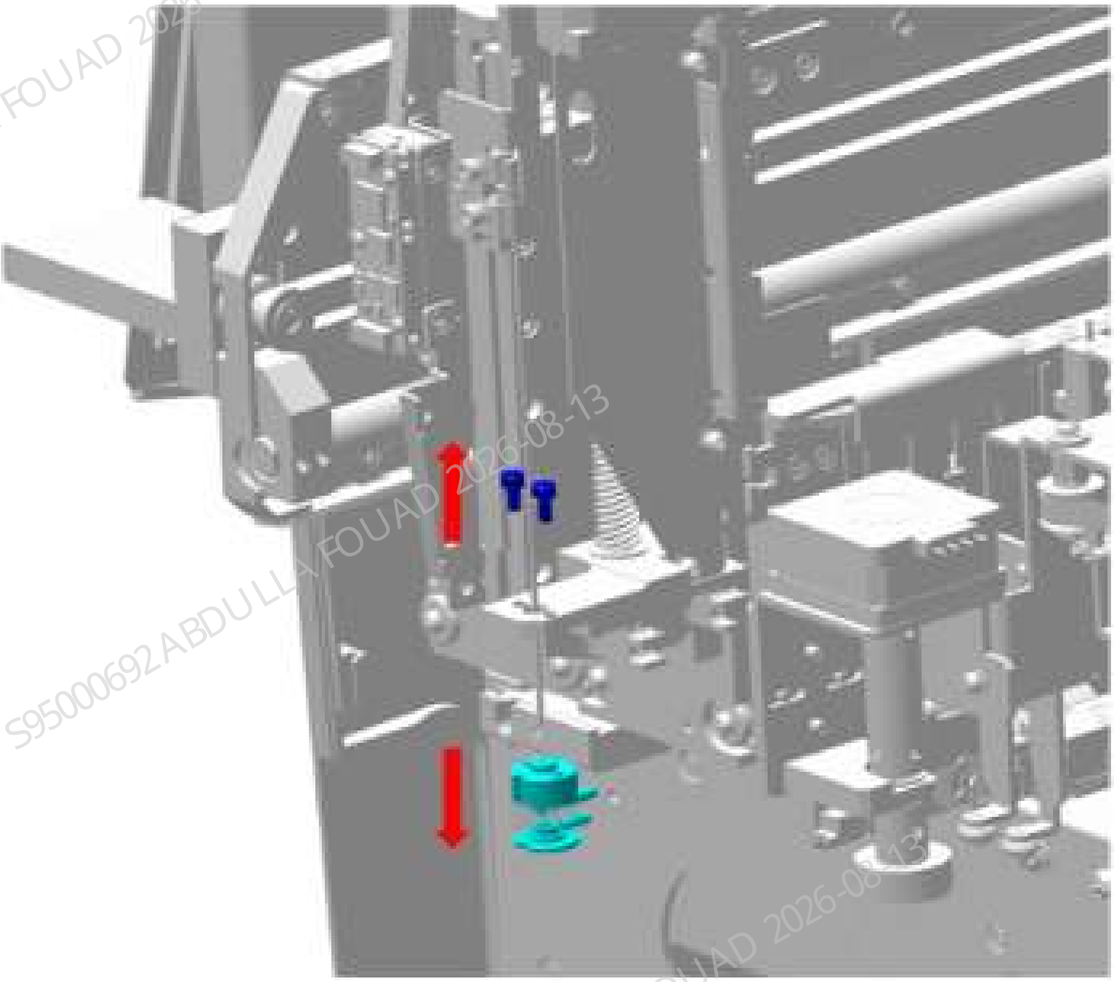
2. Loosen the two screws fixing the autoloader, and drag the autoloader to the appropriate position along the direction of the arrow, as shown in the figure below.

Figure8–211 Figure 2 - Removing the autoloader



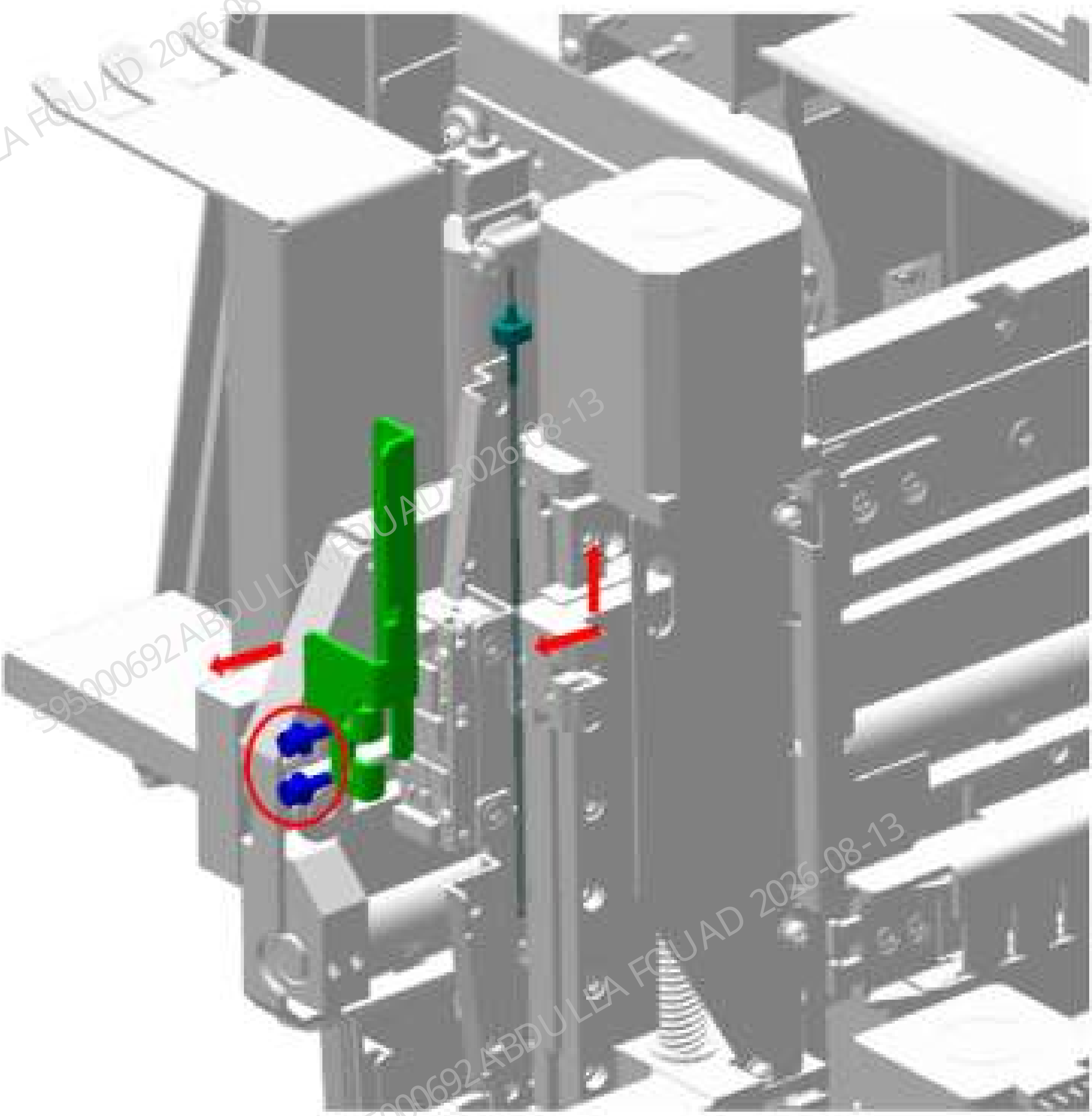
3. Use an Allen key to loosen the two screws that secure the probe wipe (closed-sampling), and remove the probe wipe (closed-sampling), as shown in the figure below.

Figure8–212 Removing the probe wipe (closed-sampling)



4. Use an Allen wrench to loosen the screw fixing the piercing needle, as shown in the figure below, use tweezers to remove the sampling tube, and remove the piercing needle.

Figure8–213 Removing the piercing probe



Subsequent Requirements

Installation procedure:

1. Install the piercing probe, connect the sampling tube, and righten the screw.
2. Install the probe wipe 2 (closed-sampling) back in the original position of the aspiration module, and tighten the screws to fix it.
3. Restore the autoloader, and turn over the front cover for resetting.

8.22.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

1. Adjust the vertical initial position of sampling probe. For details, see the adjustment requirements for the vertical initial position in the section about sampling assembly FRU.
2. Observe whether the sampling tube is smooth when the sample probe moves up and down, and whether the tubes are folded in the vertical position (confirm whether the running of the sampling tube is affected after the length of the sampling tube is changed).
3. Use the calibrator for calibration confirmation. If the calibration confirms the deviation from the target value, a recalibration is required.

Verification:

- Verify that the positions of the ESR measurement points are within range

1. On the main unit software, click **Service**>>**Commissioning and Self-Test**>>**Sensor Commissioning**>>**Position of ESR Measurement Point**.

2. If the position of the measurement point is not within the range, click **Calibrate**.

- Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.23 Holding assembly

8.23.1 General Information

Figure8–214 Photo of Hold Assembly

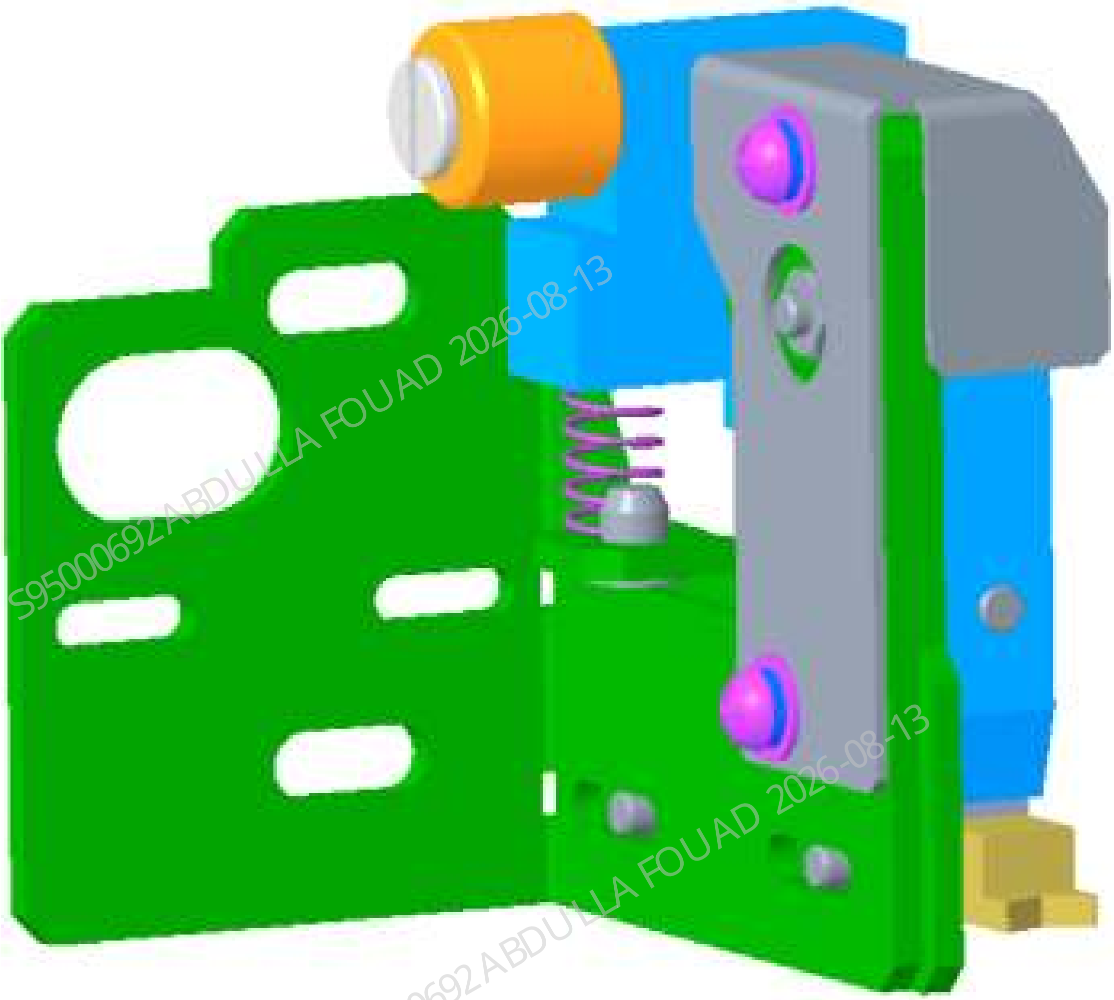


Table 8–27 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-069363-00	Hold Assembly	N/A	N/A

Function:
The hold assembly is used to eliminate the random oscillation of the tube in the tube rack during piercing and make the tube oscillate in a fixed direction, which can prevent the sample probe from sticking on the tube cap, then the probability of bending the probe can be reduced.

8.23.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench

Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Flip the front cover up and dock on the top cover, as shown in the following figure.
2. Loosen the two screws fixing the autoloader, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in Figures 1 and 2 below.

Figure8–215 Figure 1 - Removing the autoloader

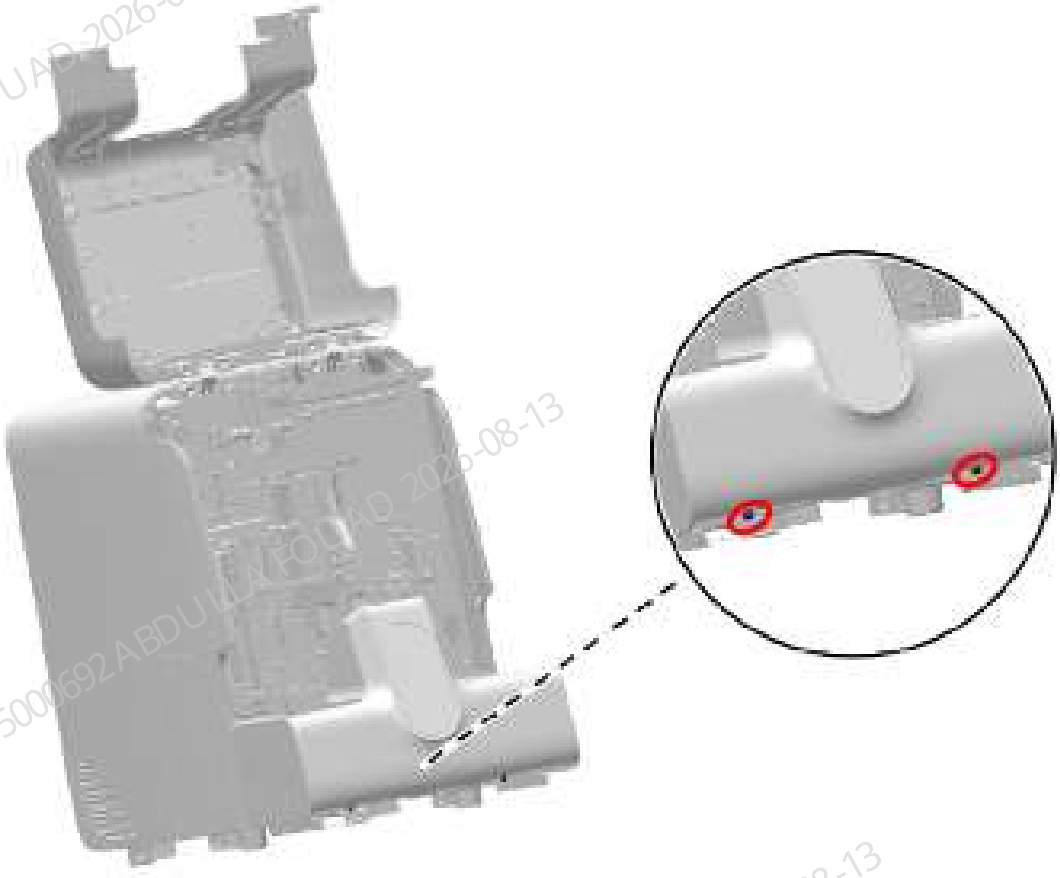
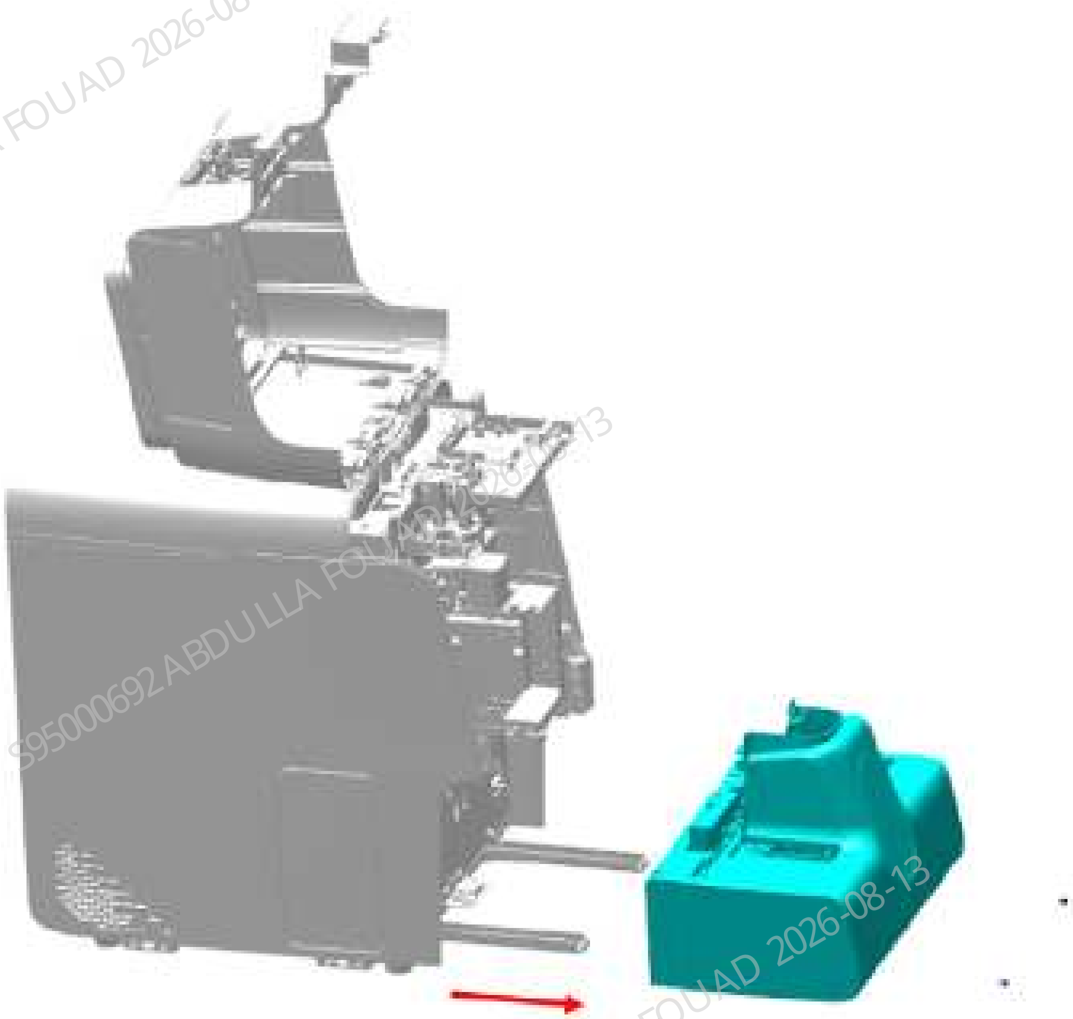
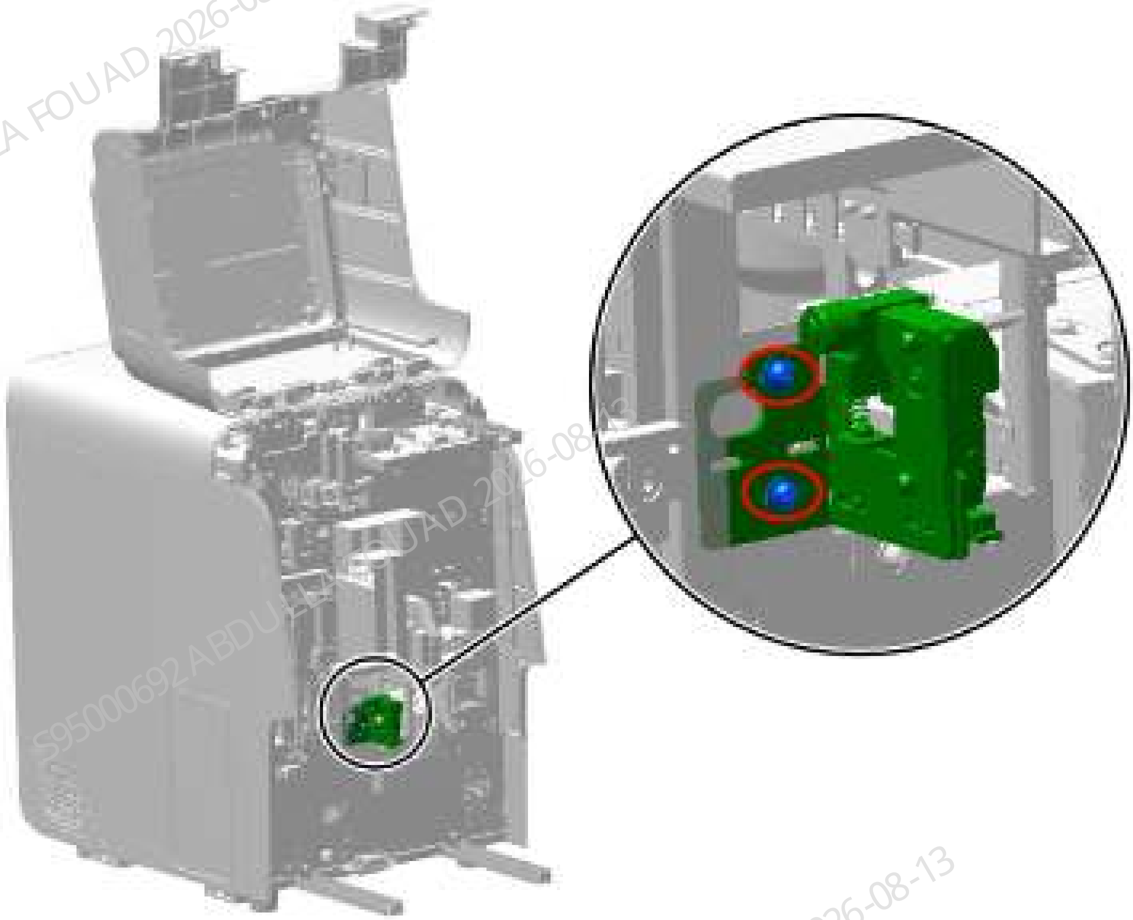


Figure8–216 Figure 2 - Removing the autoloader



3. Use a cross-head screwdriver to loosen the two screws fixing the hold assembly, and take down the stabilizing assembly, as shown below.

Figure8–217 Removing the stabilizing assembly



Subsequent Requirements

Installation procedure:

1. Use two screws to pre-tighten the hold assembly on the position shown in Figure **Figure8–217**.
2. Adjust it to the appropriate position. For the adjustment method, see Section 3.
3. After adjustment is completed, tighten and fix the two screws, and reset and fix the cover and autoloader.

8.23.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

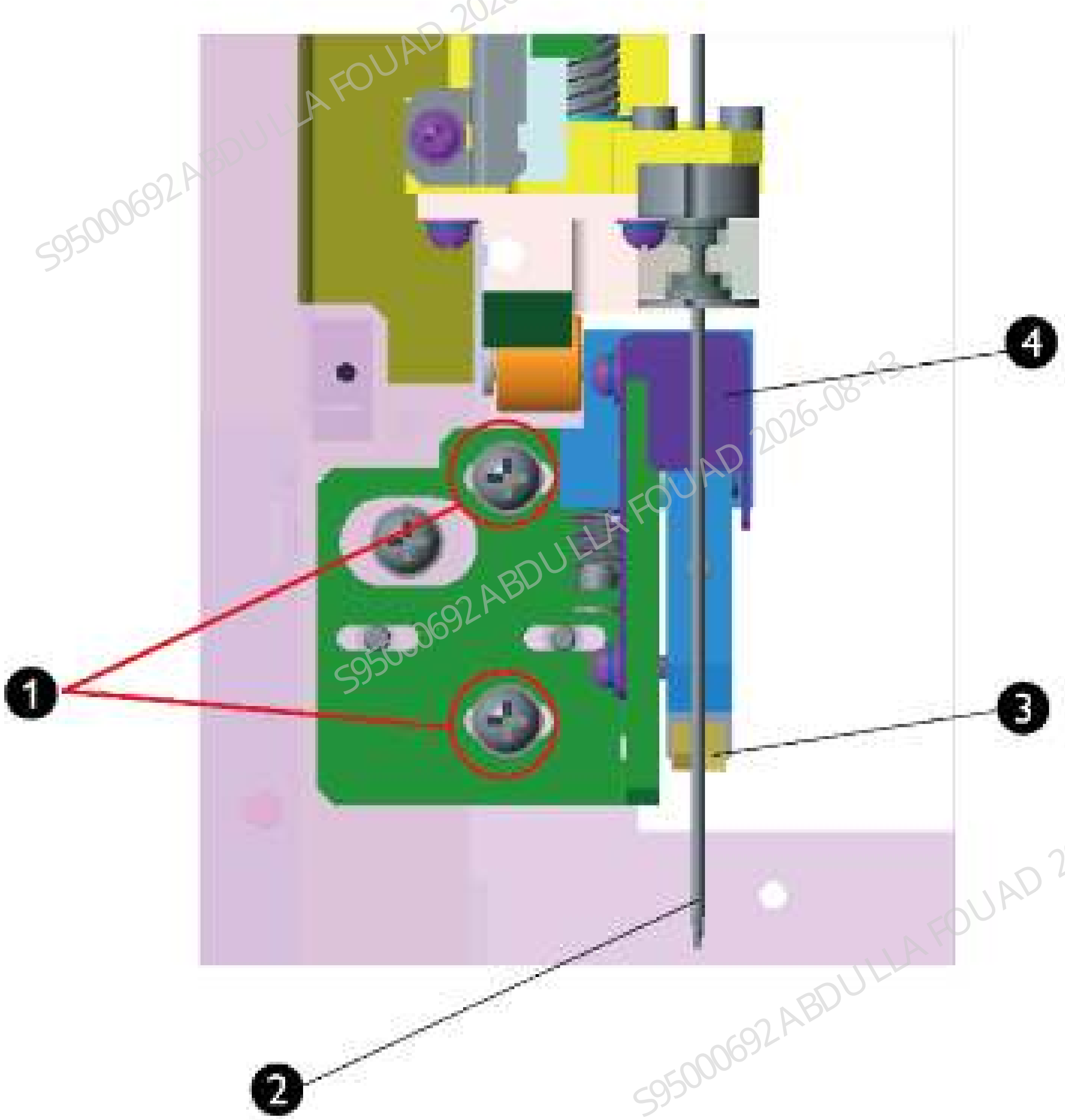
Procedures

1. First adjust the left and right positions of the hold assembly clamping block:

1) Before splicing the autoloader to the main unit, in case of a power failure, drag the sample probe horizontally near the autoloader position, and press the sample probe vertically downward to make the sample probe tip lower than the clamping block of the hold assembly.

2) Observe whether the sample probe is aligned with the center of the V-shaped clamping block of the hold assembly. If not, loosen the two screws of screw group 1 in the figure below, move horizontally the hold assembly left and right to align the center of the V-shaped clamping block with the sample probe, and then lock screw group 1.
After adjustment, lift the sample probe vertically up to the top and push the sample probe horizontally away from the autoloader position to avoid bending the sample probe in subsequent adjustment.

Figure8–218 Left/Right Position Adjustment of the Hold Assembly

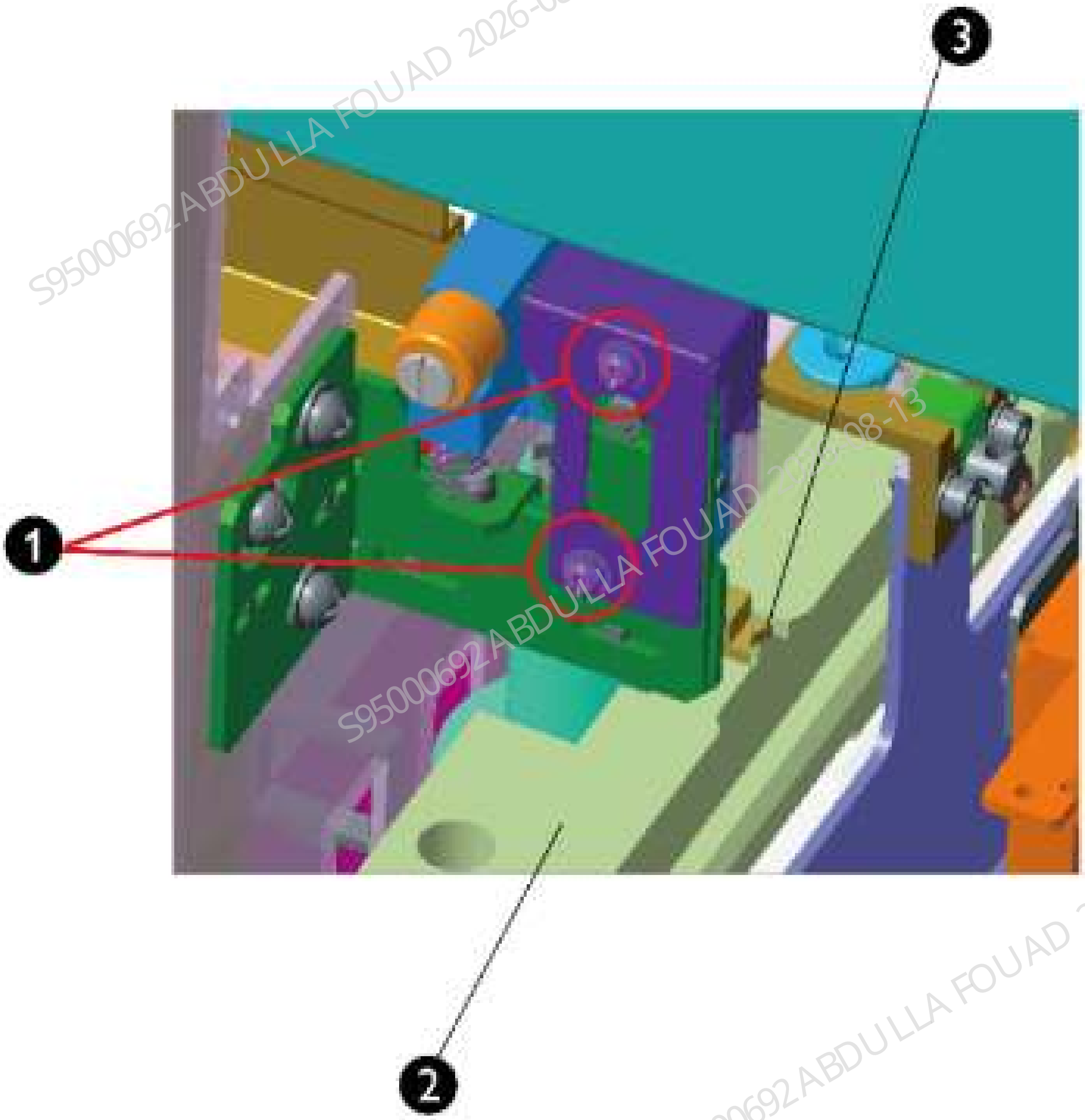


1	Screw group 1	2	Sampling probe
3	V-shaped clamping block	4	Hold Assembly

2. Then adjust the front and rear positions of the hold assembly clamping block.

1) Splice the autoloader to the main unit, and note to make the autoloader close to the front panel of the main unit. Observe whether the end of the V-shaped clamping block of the hold assembly is about 1 mm behind the end face of the back supporting board of the autoloader. If not, loosen screw group 2 in the figure below, move horizontally the hold assembly back and forth until the above requirement is met, and then lock screw group 2.

Figure8–219 Front/Rear Position Adjustment of the Hold Assembly



1	Screw group 2	2	Autoloader back supporting board
3	V-shaped clamping block		

NOTICE

The end of the V−shaped clamping block must not protrude from the end face of the autoloader back supporting board, otherwise the autoloading process easily causes a counter alarm of the autoloader.

Verification:

Procedures

1. On the **Service**>>**Commissioning & Diagnosis**>>**Self-test** interface, select autoloader assembly, and click **Assembly Initialization**.
2. Put one row of empty test tube rack on the loading tray, click "Load" on the interface, and then click "Feed" 5 times to send position 1 of the test tube rack to the automatic sample aspiration position.
3. Manually press the roller of the hold assembly, and observe whether the V-shaped clamping block of the hold assembly can be pressed into the groove at position 1 of the test tube rack, and there is a gap between it and both sides of the groove.
4. Loosen the roller of the hold assembly and observe whether the V-shaped clamping block of the hold assembly can retract smoothly.

8.24 Tube gripper

8.24.1 General Information

Figure8–220 Photo of Tube Gripper

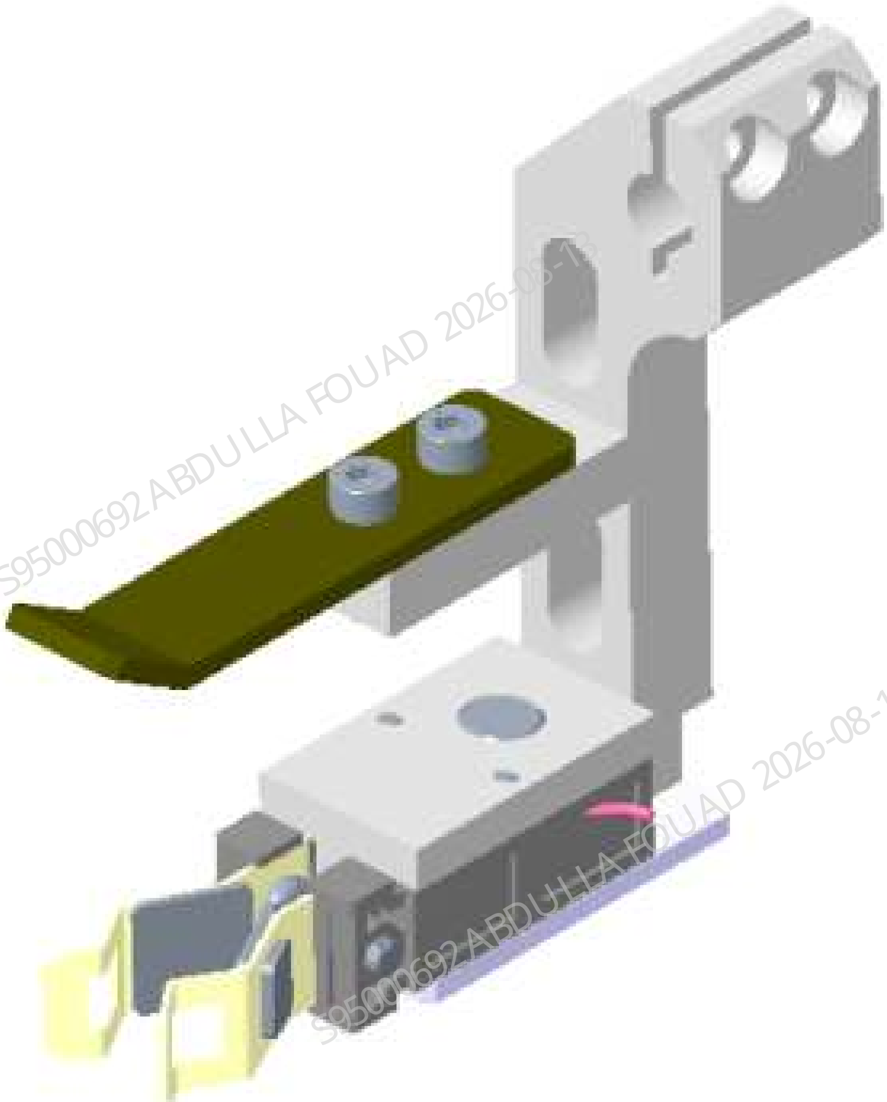


Figure8–221 Diagram of Location of Tube Gripper in the Main Unit

Figure8–222 Diagram of Location of Tube Gripper in the Main Unit

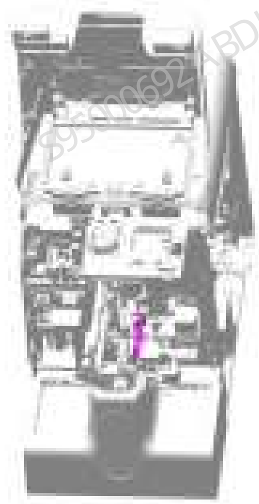


Table 8–28 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-052164-00	Tube Clamp	N/A	N/A

Function:

Used to grab the test tube when the blood sample in the test tube is mixed.

8.24.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench
One marking pen

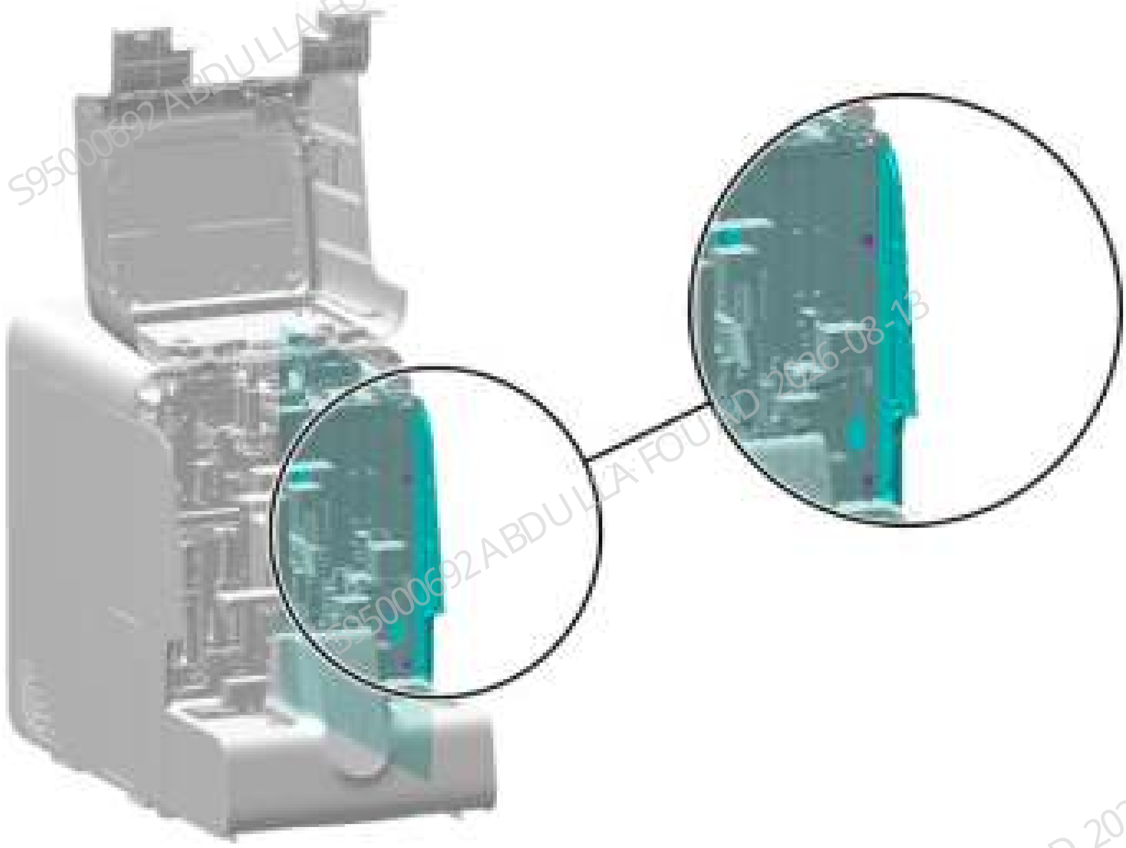
Pre-requirements

N/A

Procedures

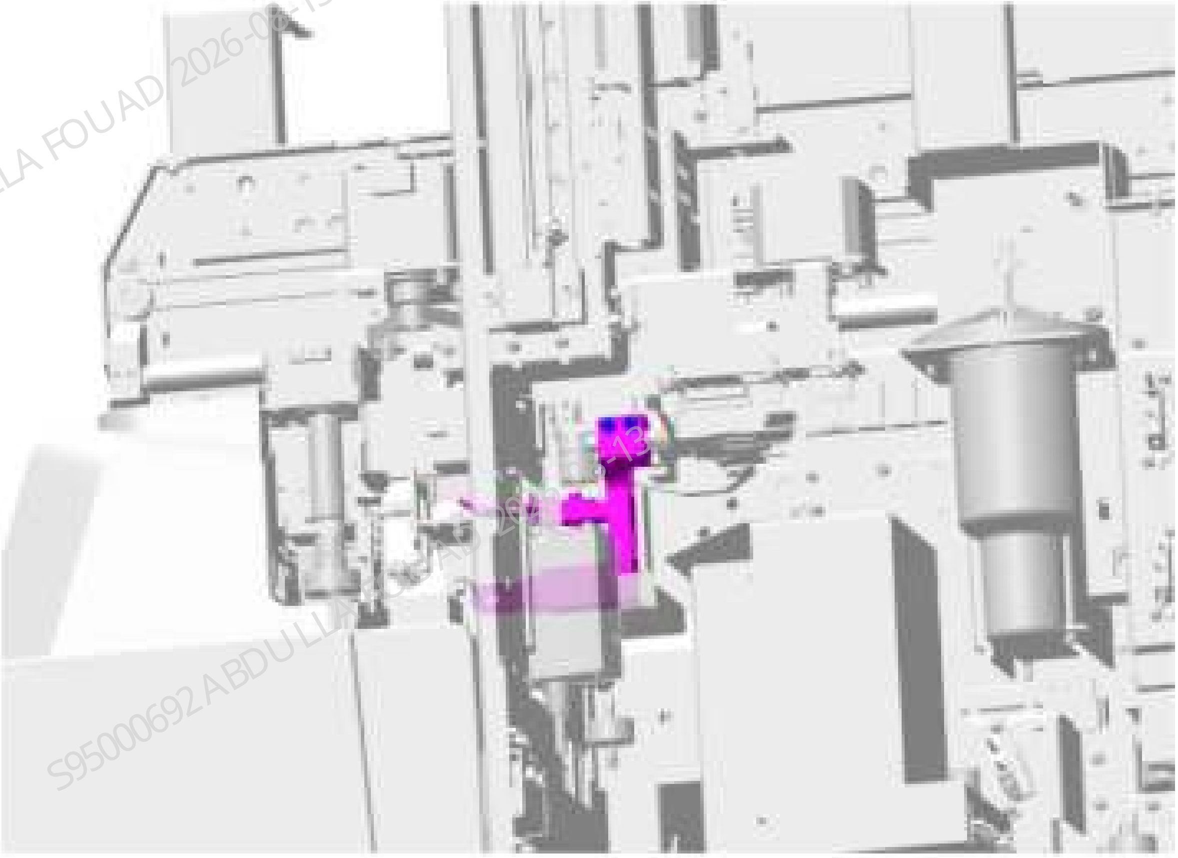
1. Turn up the front cover, lean it on the top cover, remove the two screws that fix the right cover, and remove the right cover. As shown in the following figure.

Figure8–223 Removing the right cover



2. **First use a marking pen to mark the position of the mixing gripper on the R−axis**, and then use an Allen wrench to loosen the two screws locking the tube gripper (do not remove the screws completely), as shown in the figure below.

Figure8–224 Figure 1 - Removing the tube gripper



3. As shown in Figure 2 below, push the Y-direction assembly (the part in the red box) on the mix assembly to the appropriate position in the direction of the arrow, and finally clamp and remove the test tube in the direction of the arrow as shown in Figure 3 below.

Figure8–225 Figure 2 - Removing the tube gripper

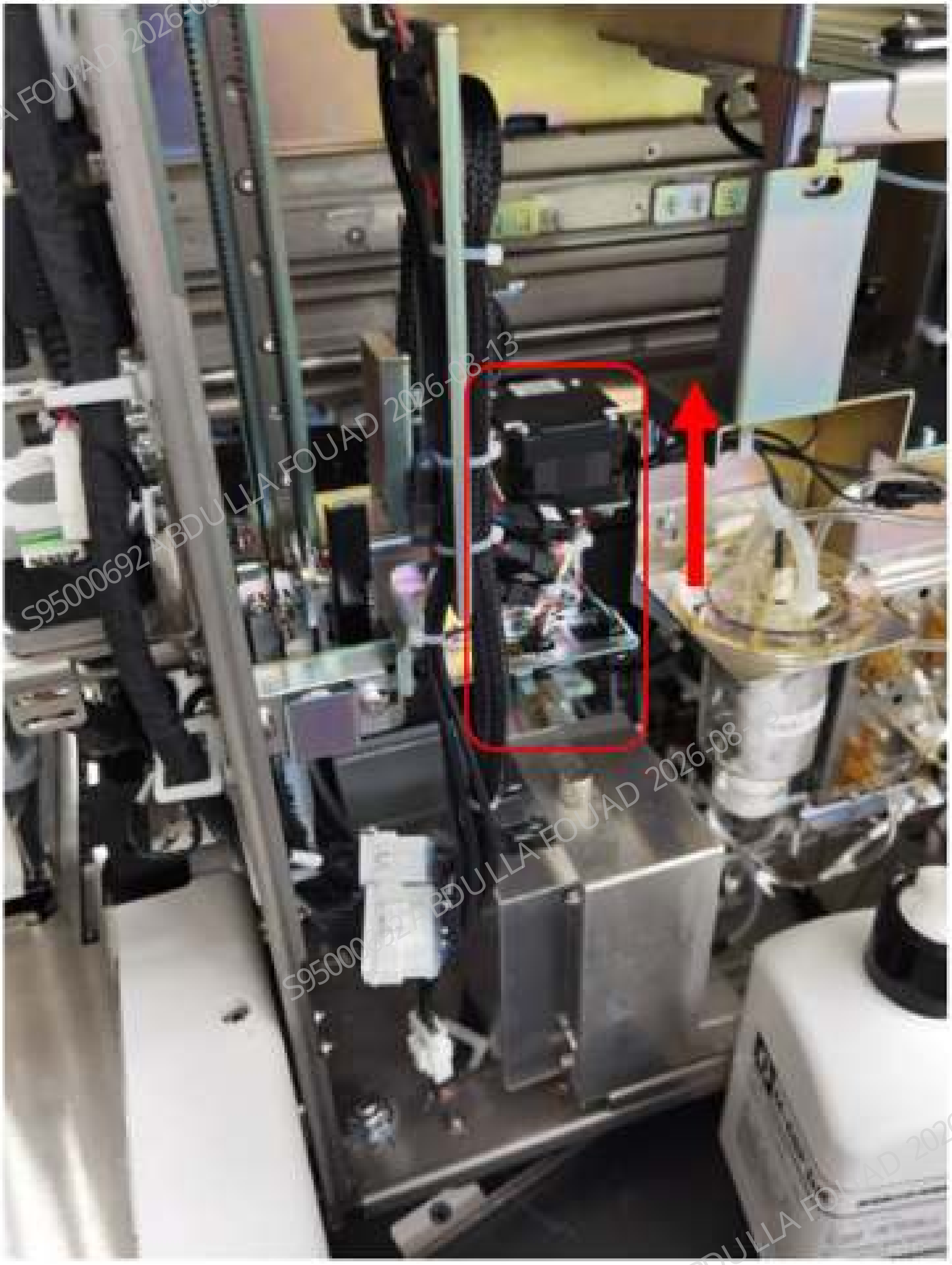
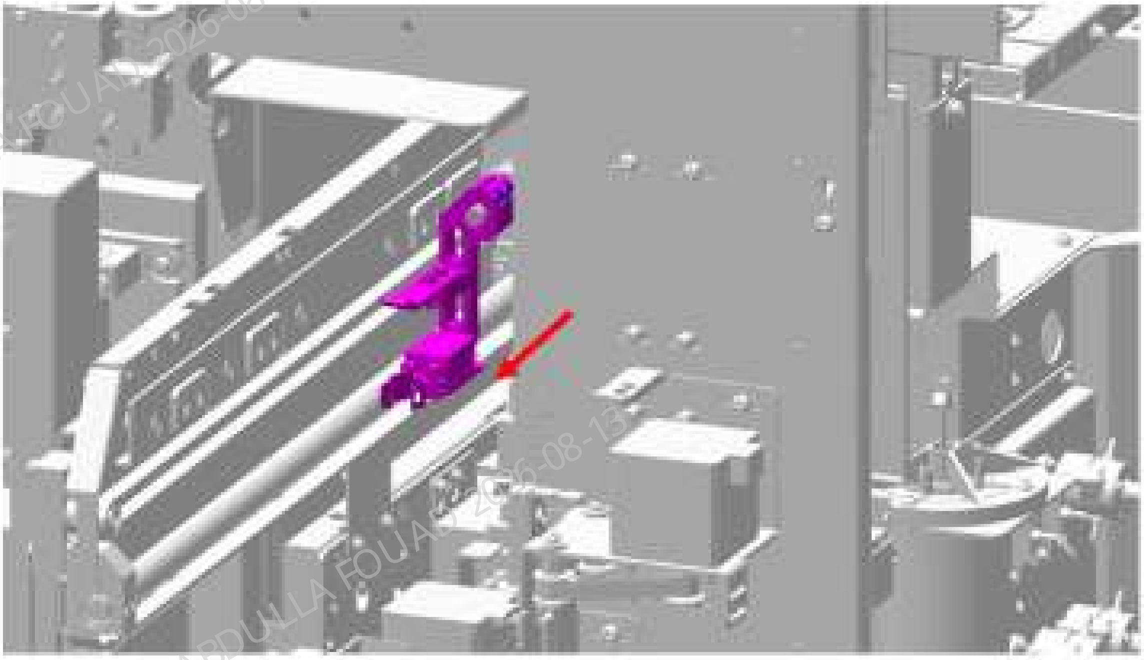


Figure8–226 Figure 3 - Removing the tube gripper



Subsequent Requirements

Installation procedure:

1. At the position shown in the figure above, install the new test tube gripper back on the R-axis and fix the two hexagon socket screws of the test tube gripper (**when installing the gripper, try to keep it vertical and coincide with the original position mark so as to reduce the adjustment amount of R direction and front−rear position**).
2. Restore the right cover and complete the adjustment and installation of the test tube gripper.

8.24.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning and verification:

1. Connect the power supply, log in at service access level, and first confirm the position of the gripper (the position and height of the gripper relative to the front and rear, left and right, and R direction of the test tube rack, the front/rear position of the gripper relative to the capillary blood mixing position, and the gap between the gripper and the anti-collision block). For the acceptance standard, see the mixing gripper adjustment description in the commissioning

section of the mix assembly FRU. If the requirements are not met, also see the mixing gripper adjustment description in the commissioning section of the mix assembly FRU.

8.25 Rotated Wheel Fixed Asm

8.25.1 General Information

Figure8–227 Photo of Rotary Compressing Assembly

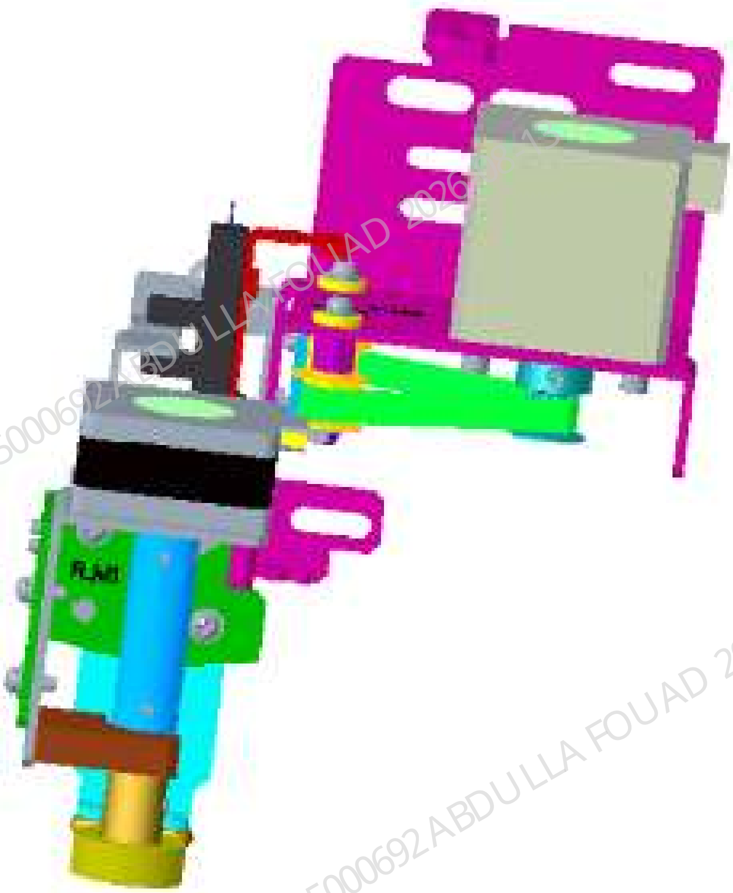


Table 8–29 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-075024-00	Rotary Compressing Assembly	011-000021-00	PHOTOELEC Optical Sensor IP55 940nm 15cm

Function:

When the analyzer automatically loads samples, it is used to compress and rotate the test tube so that the scanner can scan the barcode information on the test tube.

8.25.2 Disassembly and Assembly

Note

N/A

Required tools

Cross-headed screwdriver

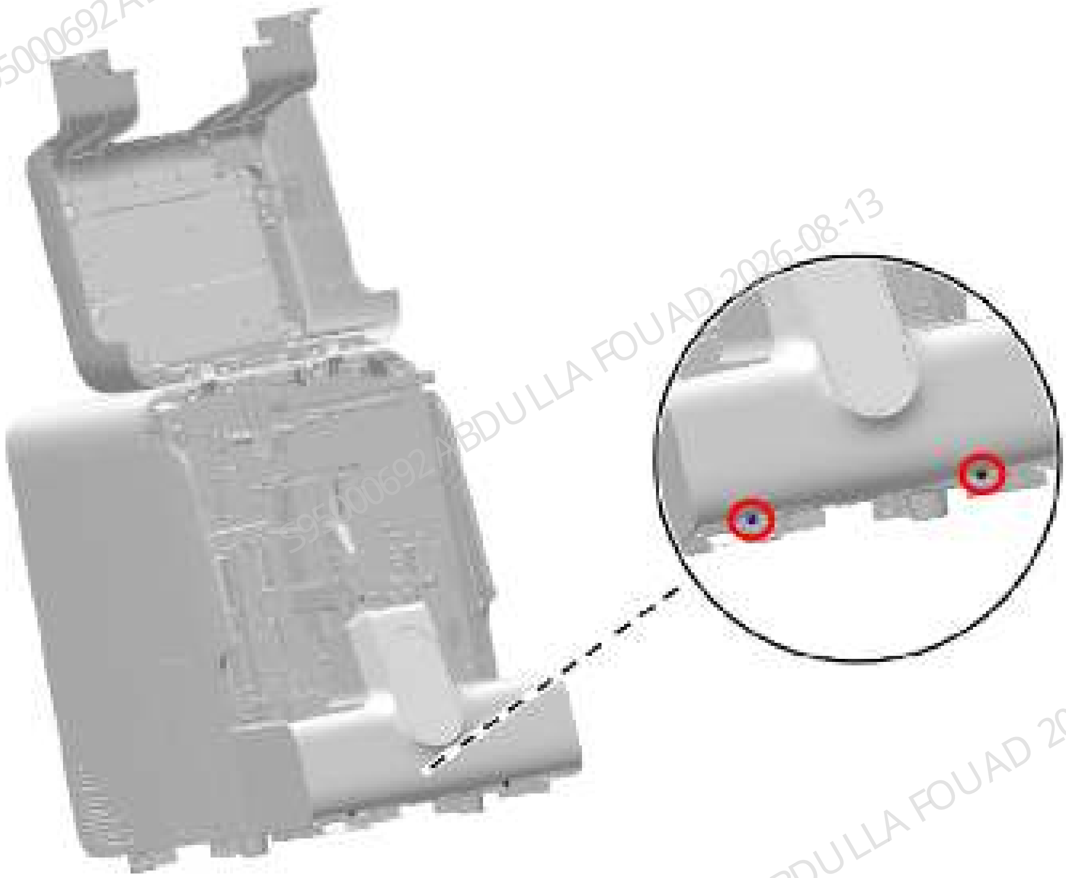
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–228 Figure 1 - Removing the autoloader



2. As shown in the figure above, loosen the two screws (in the red circle) fixing the autoloader, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire

between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in Figure **Figure8–229** and Figure **Figure8–230**.

Figure8–229 Figure 2 - Removing the autoloader

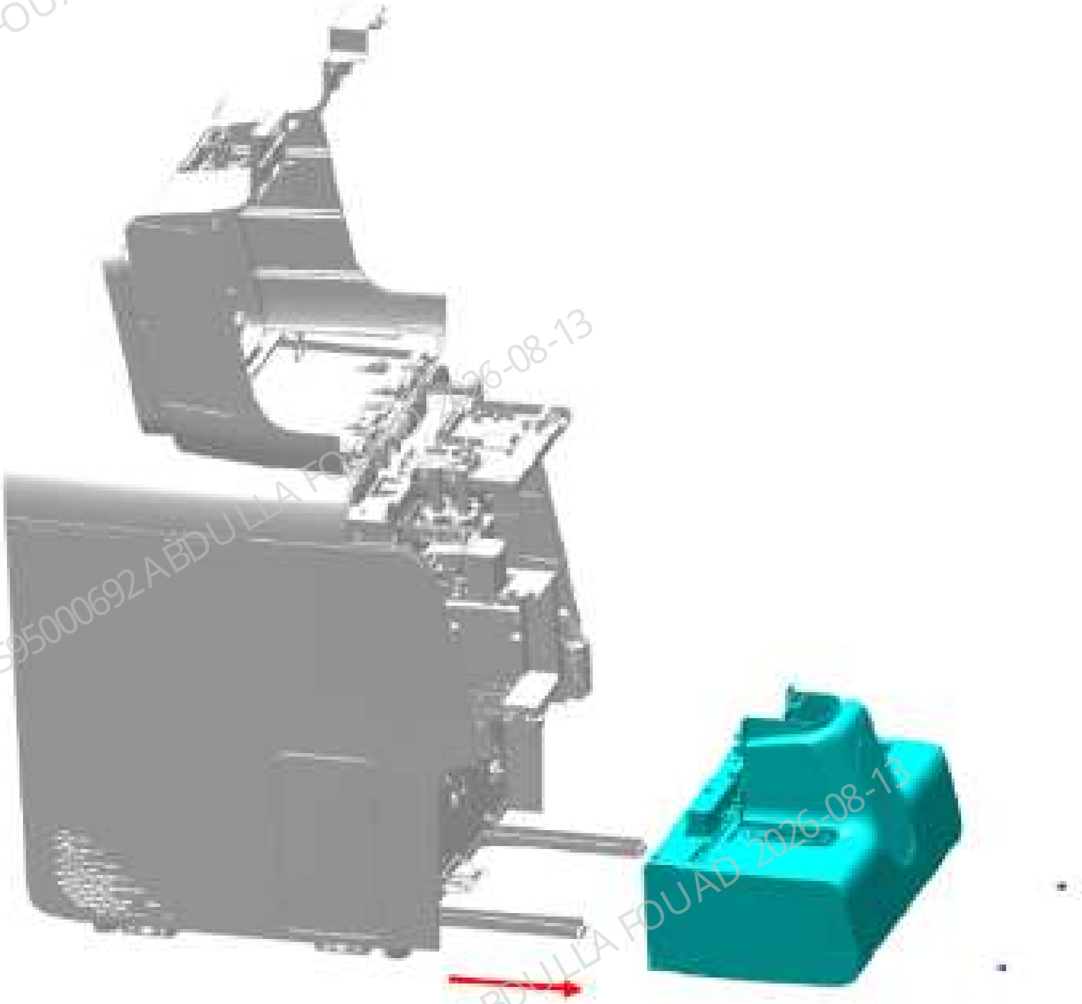
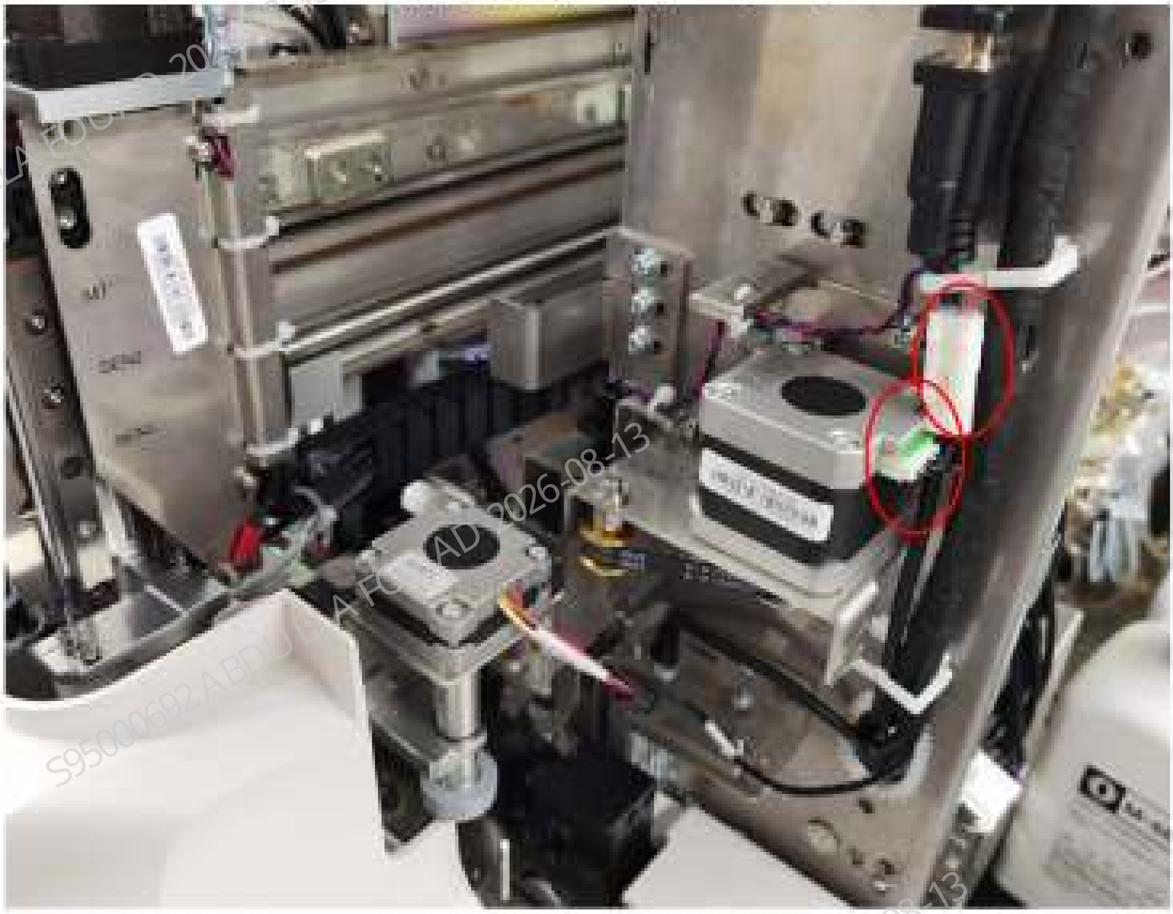
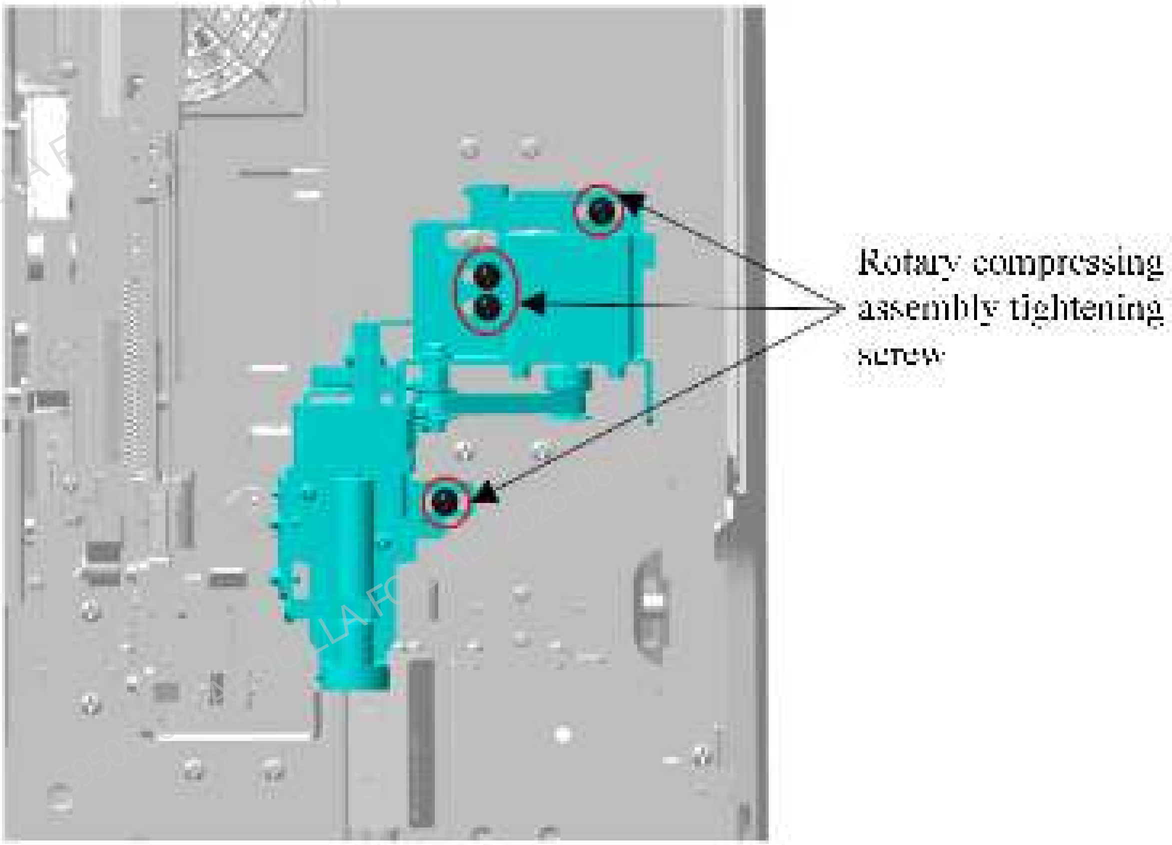


Figure8–230 Removing the wire for the rotary fixing assembly



3. As shown below, use a cross-head screwdriver to loosen the four screws (in the red circle) that fix the rotary fixing assembly, and remove the rotary fixing assembly.

Figure8–231 Removing the rotary fixing assembly



Subsequent Requirements

Installation procedure:

1. Use four screws to pre-tighten the rotary fixing assembly on the main unit according to the position shown in the figure above.
2. Insert the wire for the rotary fixing assembly.
3. Install back the autoloader and restore the right cover of the main unit.

8.25.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Fixtures Required

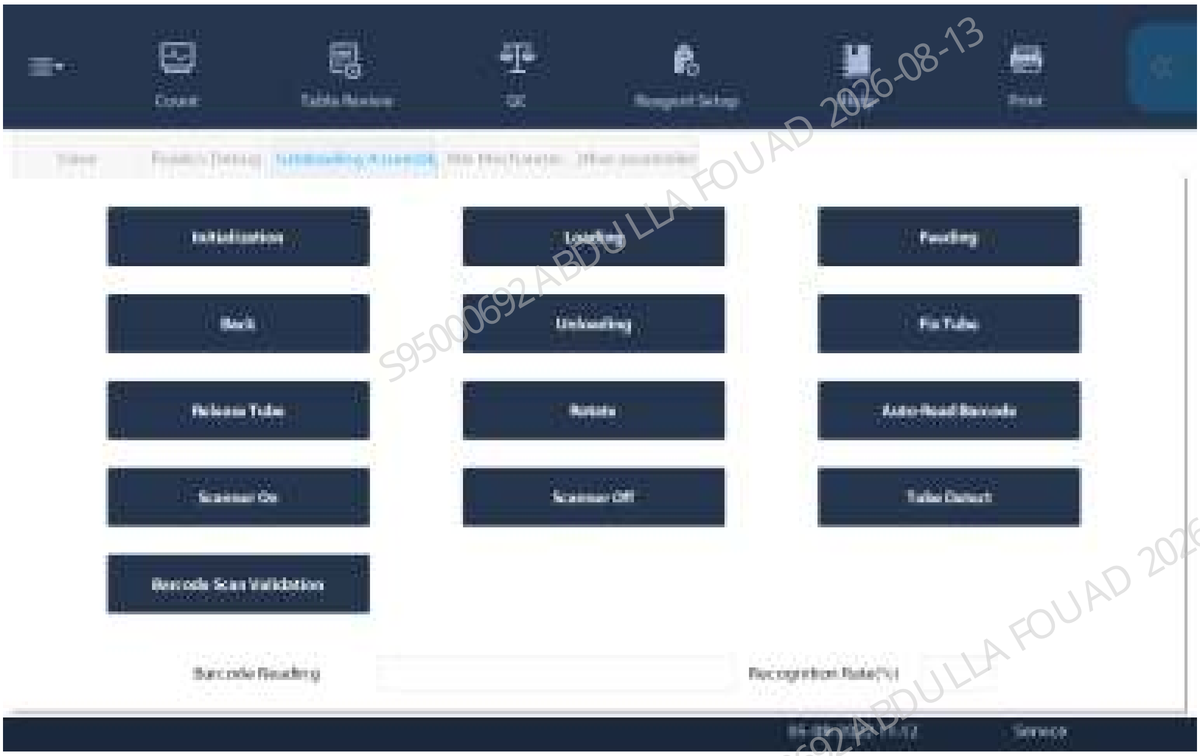
Venous blood test tube
White paper

Procedures

1. Cut a small piece of paper and wrap it around the outer circumference of the test tube, as shown in the figure below, so that the test tube is centered and fixed at a hole position in the test tube rack.

2. In the menu, click **Service**>>**Commissioning & Self-Test**>>**Self-test** in turn, enter the **Autoloader Assembly** interface, click **Assembly Initialization**>>**Load**>>**Feed** in turn, and feed the test tube under the rotary compressing assembly.

Figure8–232 Rotary Compressing Assembly Commissioning Interface



3. Click the **Fix Tube** and **Release Tube** buttons repeatedly and confirm as follows:

4. Check whether the lower end of the fixed shaft of pinch roller interferes with the upper surface of the test tube rack (if so, first consider whether the rotary compressing assembly is installed obliquely, loosen the screws, and adjust the whole assembly to keep the drive plate vertical).
5. Check whether there is left-right displacement of the test tube rack when checking the fixed tube (if yes, adjust the left-right position of the rotary compressing assembly, and re-feed the test tube rack from the initial position for confirmation).
6. When checking the fixed test tube or loosening the test tube, confirm whether the routing of the sensor is appropriate and whether the lead wire is pulled.
7. Put several empty test tubes commonly used for the client with barcodes in the test tube rack at random (there must be spaces in the test tube rack), select **Tube Detect**, and confirm that both the position with a test tube and the empty test tube position can be detected normally.
8. Click **Barcode Scan Verification**. All the barcodes can be scanned and regognized normally.

8.26 Leuze Fixed Barcode Scanner CCD 700scan/s

8.26.1 General Information

Figure8–233 Photo of Barcode Scanner

Figure8–234 Diagram of Location of the Barcode Scanner on the Scanner Assembly

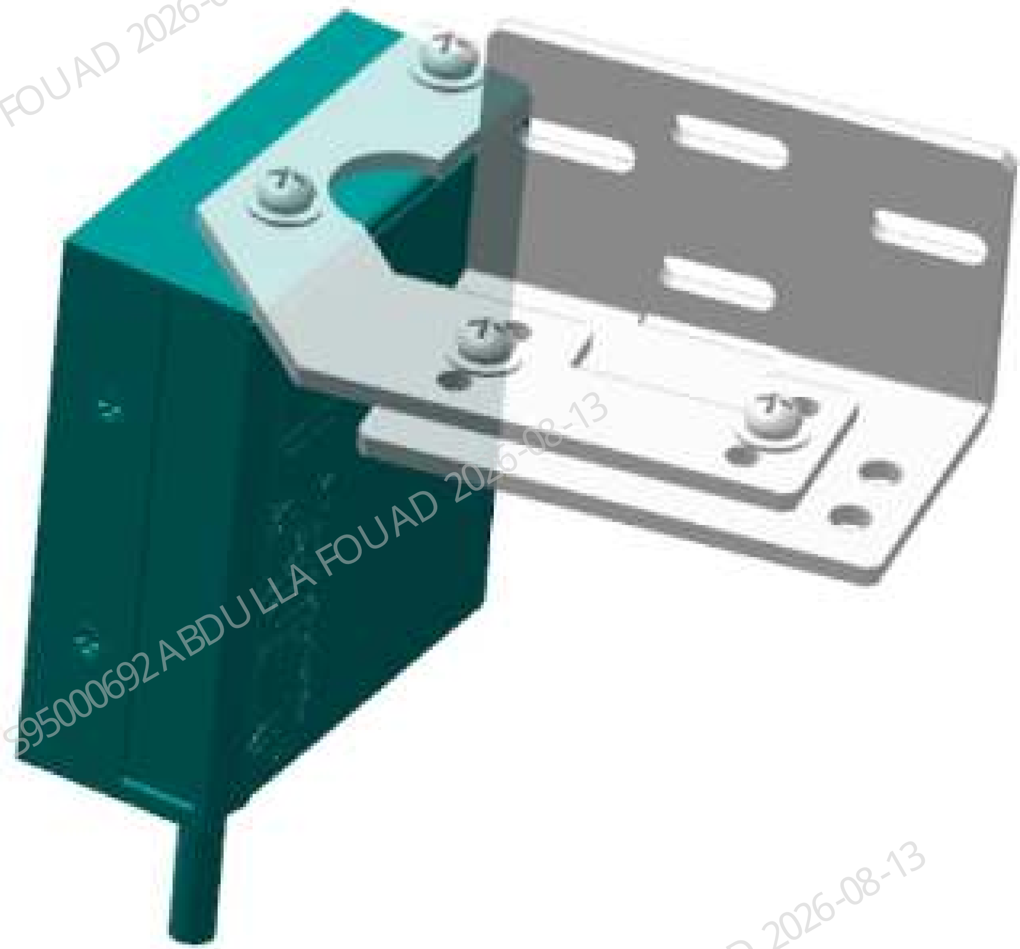


Table 8–30 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
023-001302-00	Barcode scanner	N/A	N/A

Function:

Used to scan the barcode information of the test tube.

8.26.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench

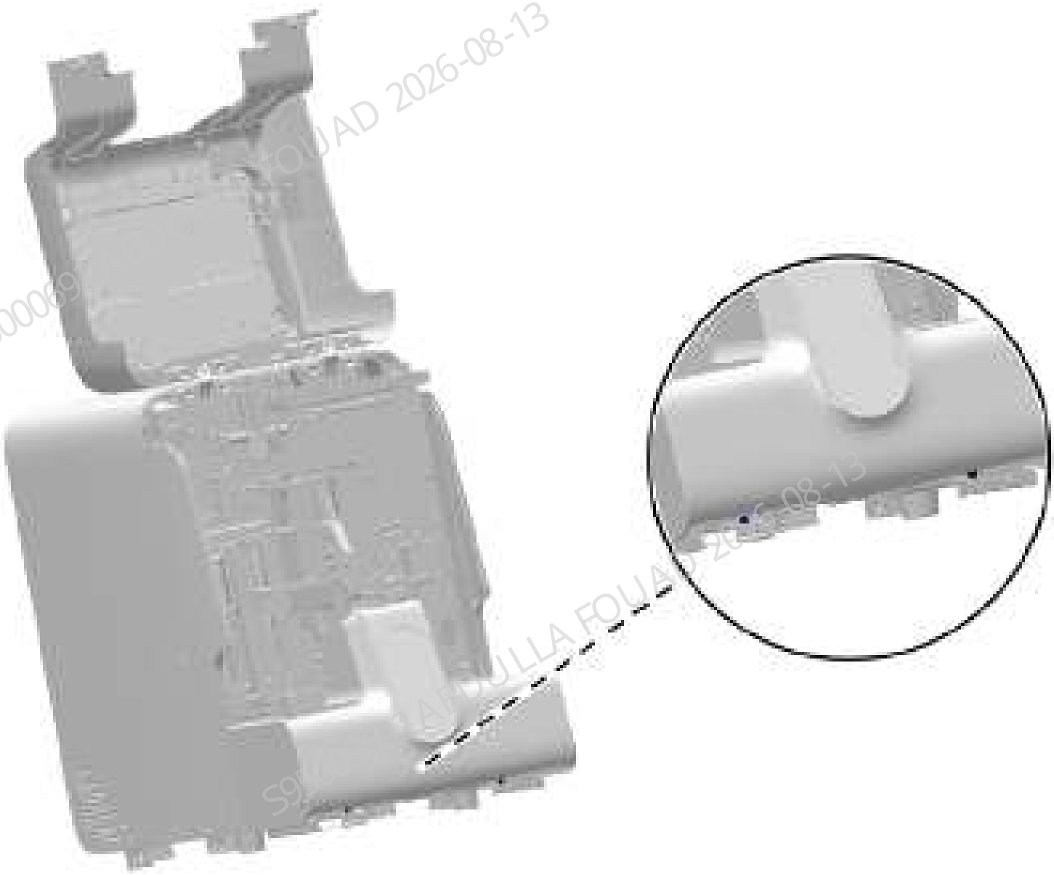
Pre-requirements

N/A

Procedures

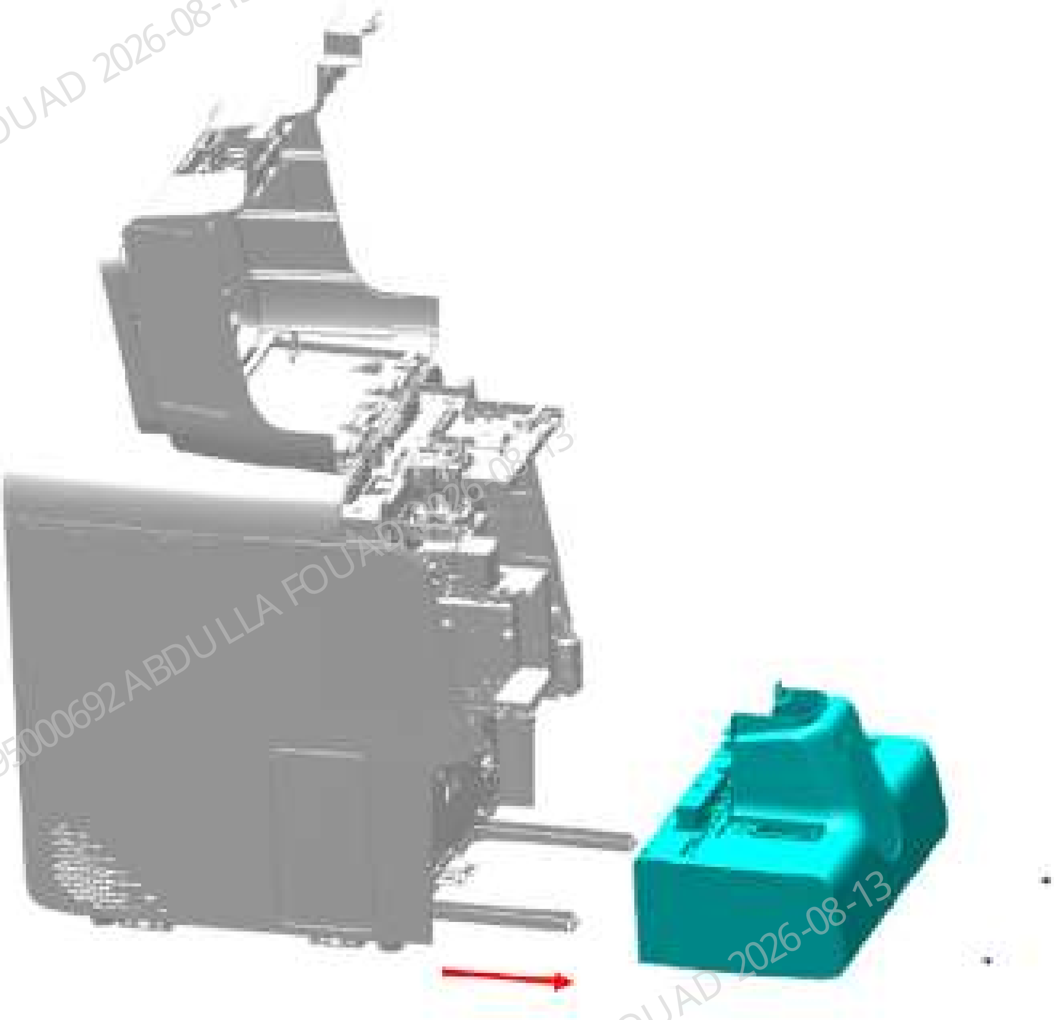
1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–235 Removing the right cover



2. Loosen the two screws fixing the autoloader, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in the figure below.

Figure8–236 Figure 1 - Removing the tube gripper



3. As shown in Figure **Figure8–237**, use a cross-head screwdriver to remove the two screws fixed on the scanner assembly, as shown in Figure **Figure8–238**, loosen the serial port cable connector in the red box, and remove the scanner assembly.

Figure8–237 Figure 2 - Removing the tube gripper

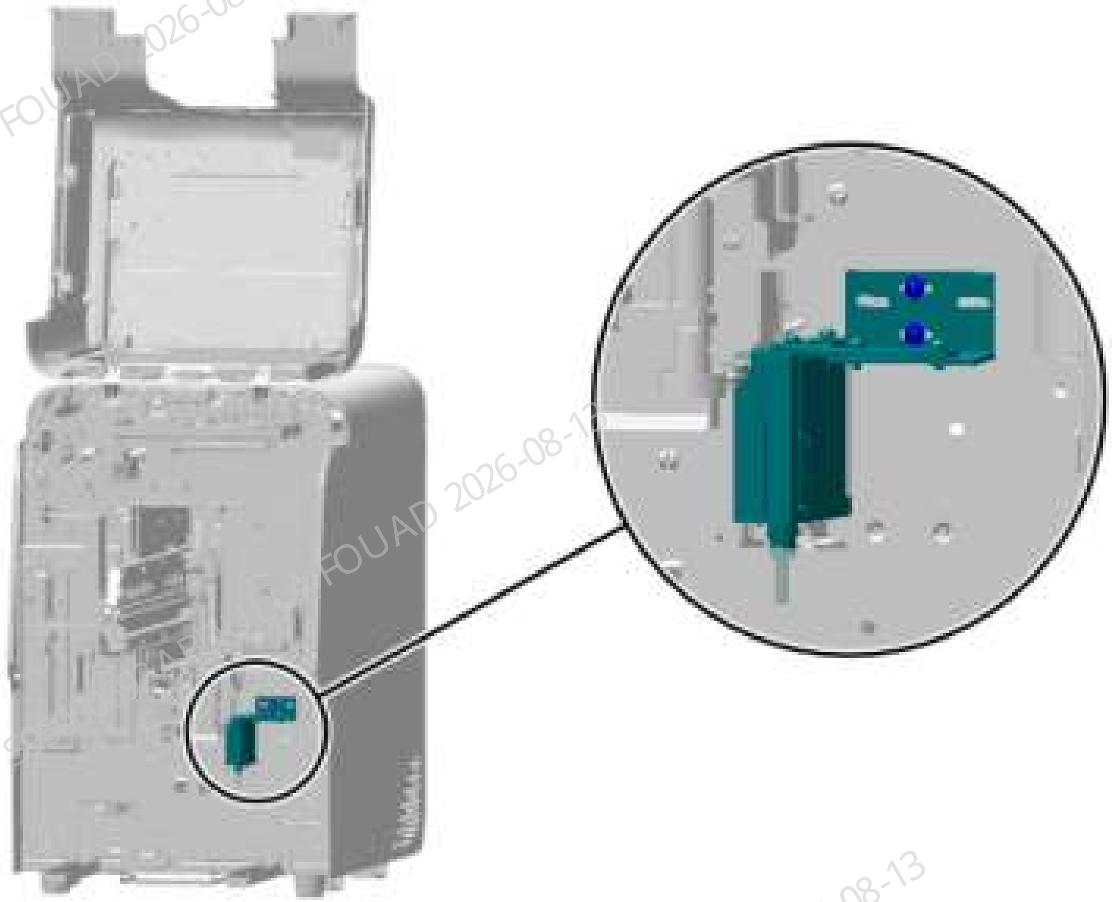
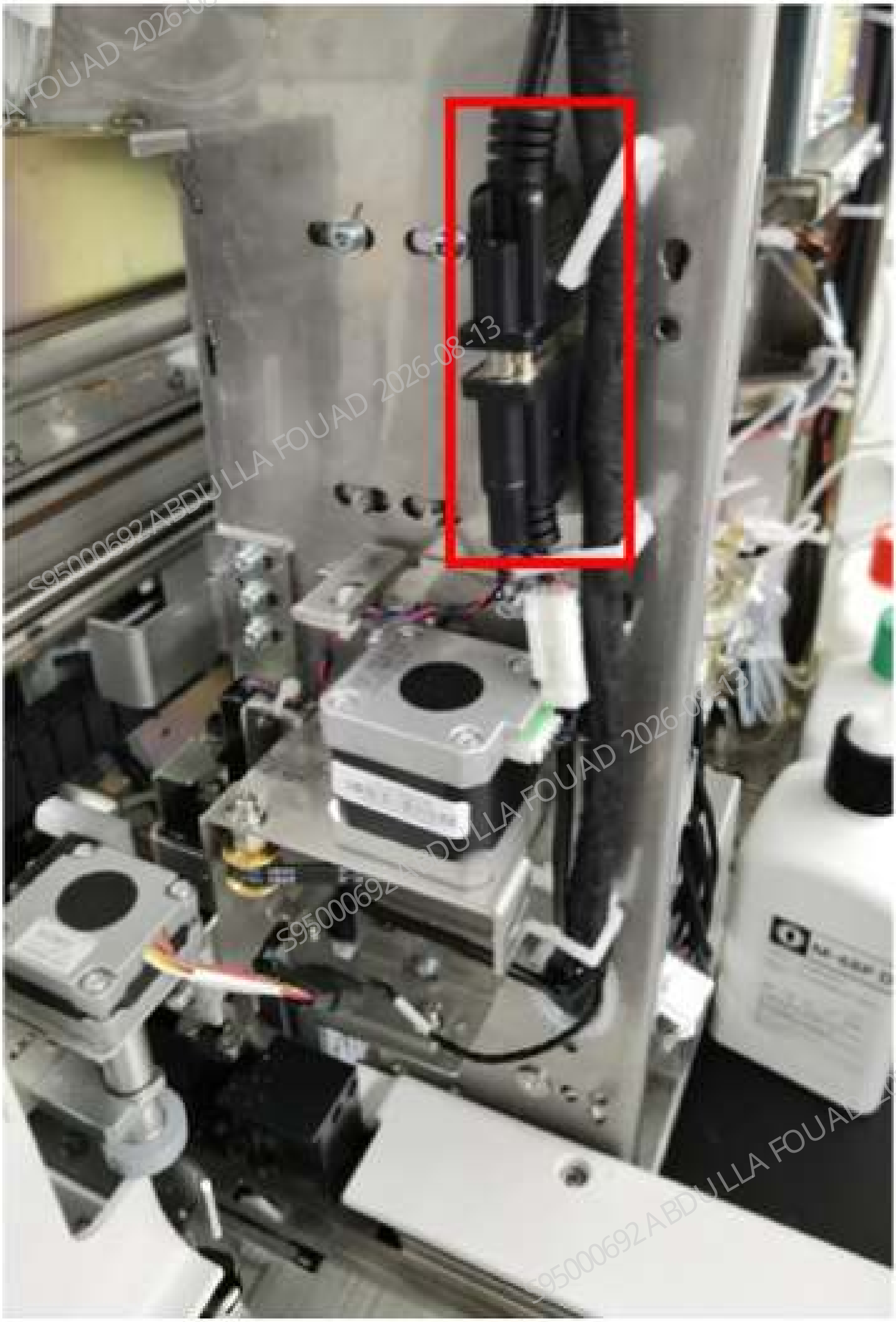
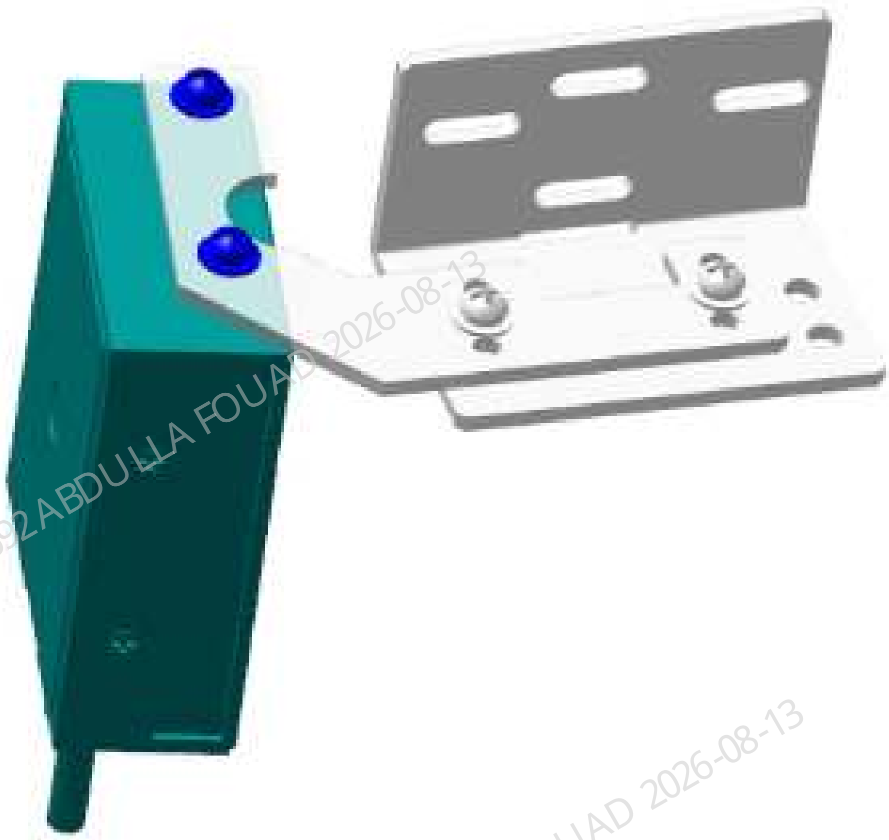


Figure8–238 Removing the scanner wire



4. Use a cross-head screwdriver to remove the two screws fixing the scanner, and take down the scanner, as shown below.

Figure8–239 Removing the scanner from the scanner assembly



Subsequent Requirements

Installation procedure:

1. As shown in Figure **Figure8–239**, use two screws to install the scanner on the scanner assembly.
2. As shown in Figure **Figure8–237** and Figure **Figure8–238**, use two screws to pre-tighten the scanner assembly on the analyzer and insert the scanner wire.
3. Adjust the scanner position (for the adjustment method, see Section 3 of this document).
4. After the adjustment is completed, lock the two screws fixing the scanner assembly.
5. Restore the autoloader and place the front cover.

8.26.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One blood routine test tube rack
Several vacuum blood collection tubes (test tubes)

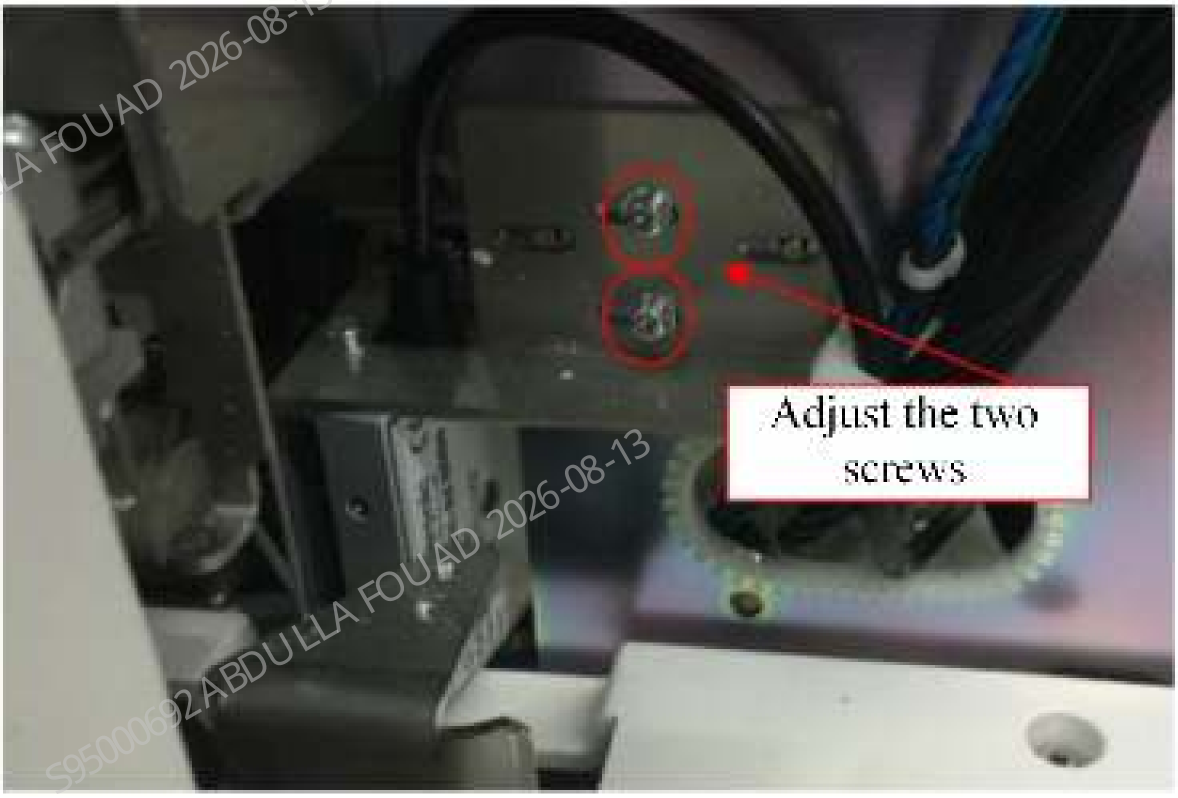
Procedures

1. Select **Menu>>Service>>Commissioning & Self-Test>>Self-test>>Autoloader Assembly** to enter the autoloader adjustment interface.

Figure8–240 Autoloader Adjustment Interface

2. Click **Assembly Initialization**.
3. After initialization ends, click **Scanner On**.
4. Check whether the scanner is turned on normally, and click **Scanner Off**. If the operation is normal, go to the next step. If the operation is abnormal, check the wire connection or replace the scanner.
5. Place the blood routine test tube rack with test tubes on the autoloader.
6. Click **Auto-Read Barcode**, and check whether the code is recognized correctly. If yes, the adjustment is completed; otherwise go to the next step.
7. Use a cross-head screwdriver to loosen the two screws fixing the scanner, adjust the scanner position, and then tighten the screws.

Figure8–241 Adjusting the Scanner Position

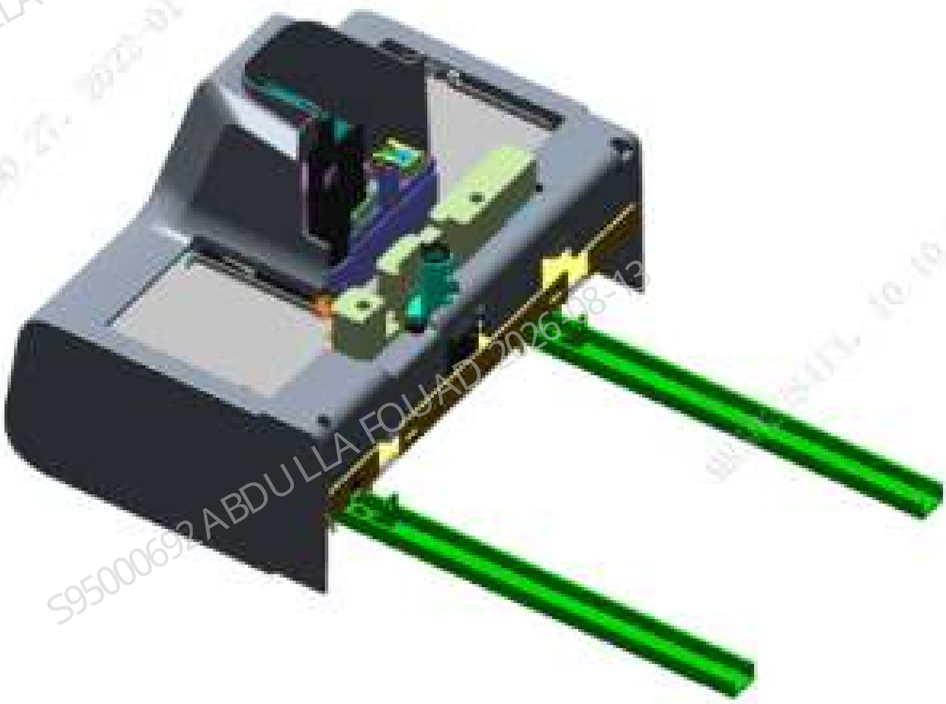


8. Repeat step 6.

8.27 Autoloader, closed-sampling model(5x6)

8.27.1 General Information

Figure8–242 115-084951-03 Autoloader, Closed-sampling Model (5x6) FRU - Photo



FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-084951-03	Autoloader, closed-sampling model(5x6)	115-059469-00	Capillary blood sample mixing assembly
		115-085181-00	5x6 Closed Autoloader (Excluding Capillary Blood Mixing) FRU

Function:

Batch process samples placed on a test rack. It facilitates automatic transport and distribution of samples, coordinates with functional components of the main unit for auto barcode recognition, sample auto-mixing, sample puncture, and sample suction.

8.27.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

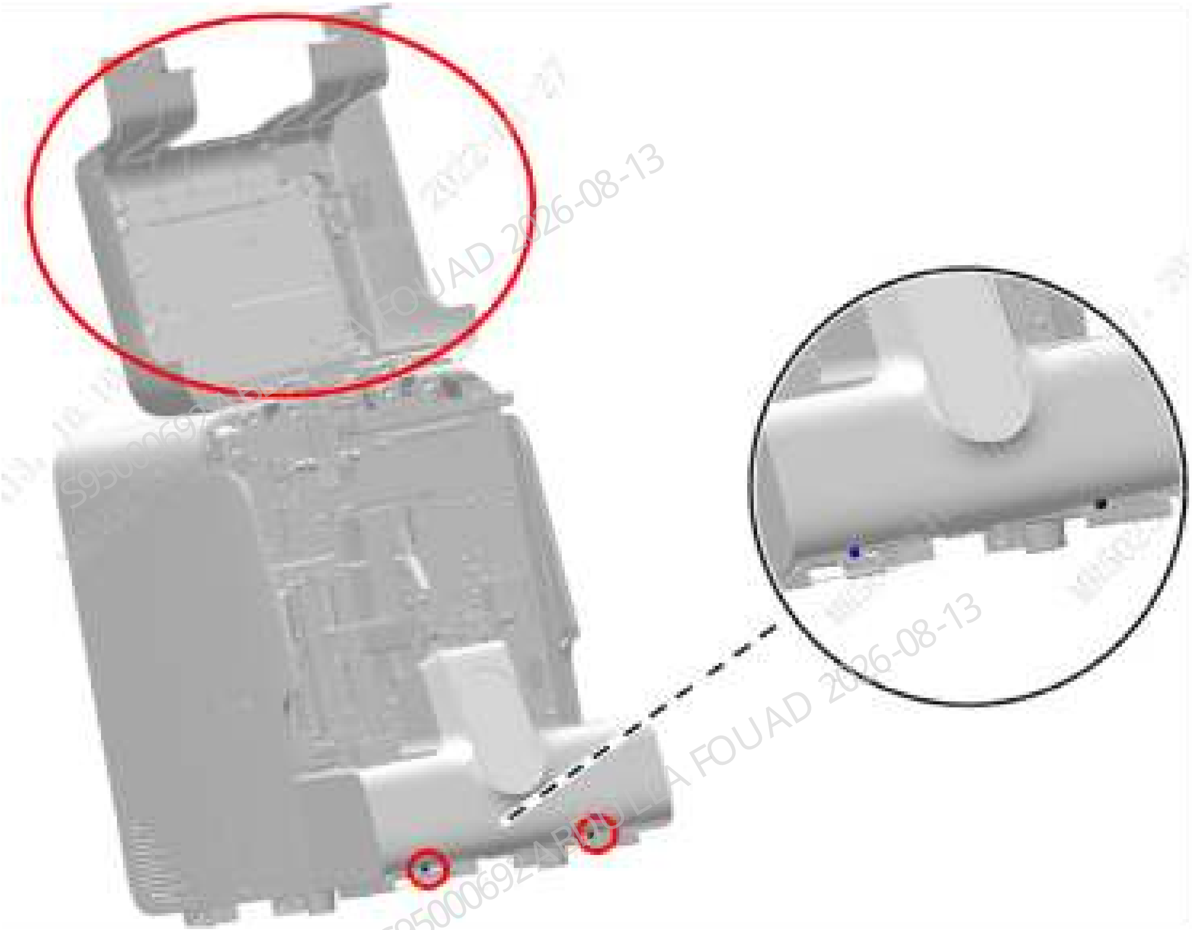
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

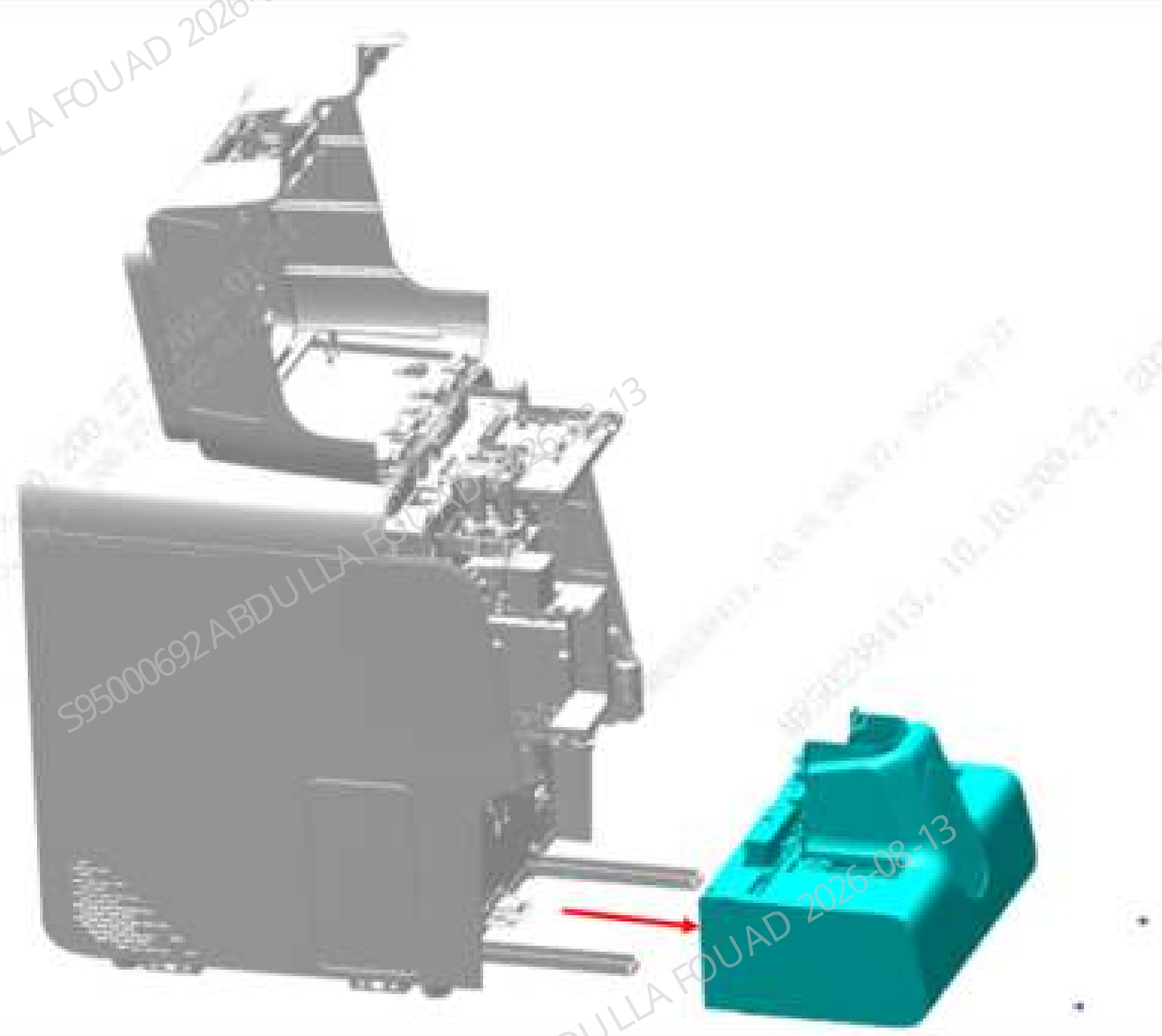
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–243 Opening the front cover



2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam, that is, remove the 5x6 closed autoloader (including capillary blood mixing) FRU, as shown in the figure below.

Figure8–244 Removing the 5x6 closed autoloader (including capillary blood mixing) FRU)



Subsequent Requirements

Installation procedure:

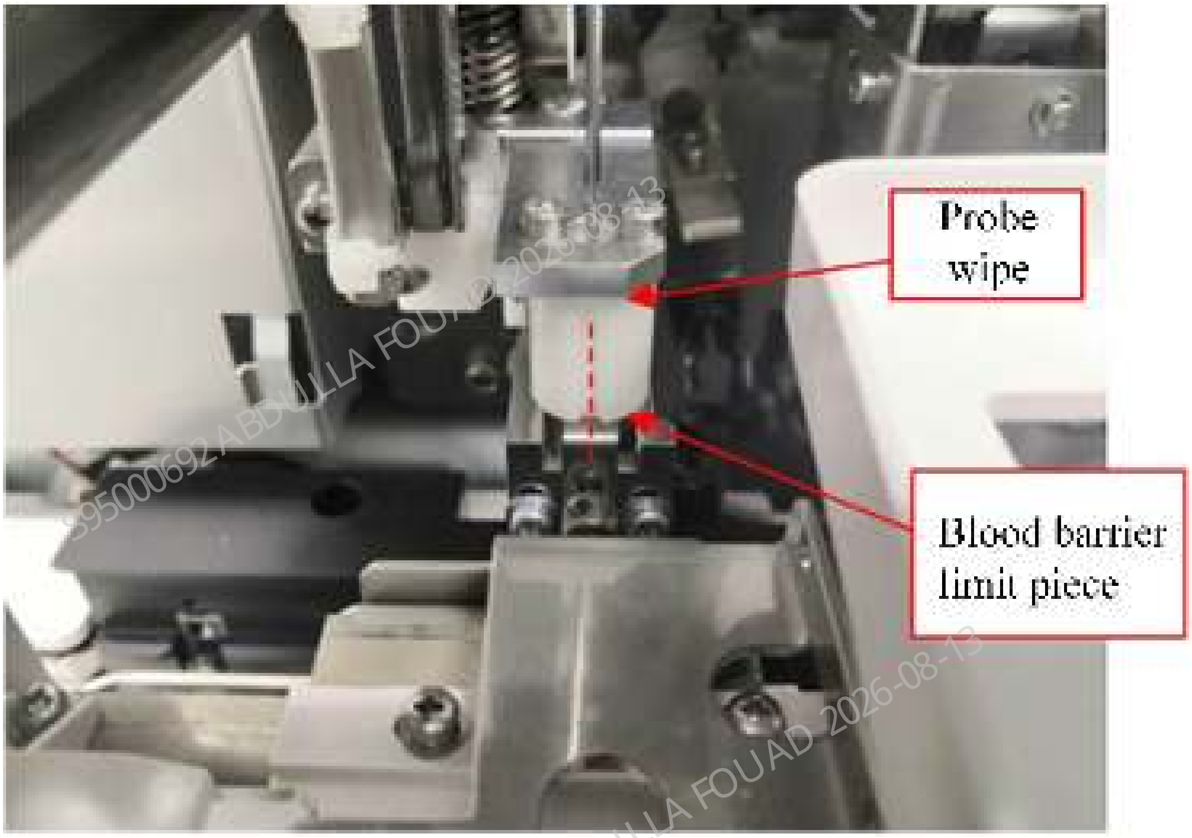
1. Place the autoloader on the autoloader crossbeam, and push it to an appropriate distance towards the front panel of the main unit.
2. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
3. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
4. Flip down the front cover.

8.27.3 Commissioning and Verification

Procedures

Commissioning and verification:

1. The left and right positions of the autoloader relative to the main unit have been adjusted in the assembly, so the FRU assembly does not require commissioning in the left and right positions.
2. Select **Menu**>**>Status**>**>Sensor Status** to enter the motor release torque status. Pull the sampling assembly to the front position a little ahead of the floating blood barrier. Observe whether the center of the sampling assembly probe wipe groove aligns with the center of the floating blood barrier limit piece (≤ ±0.5mm).

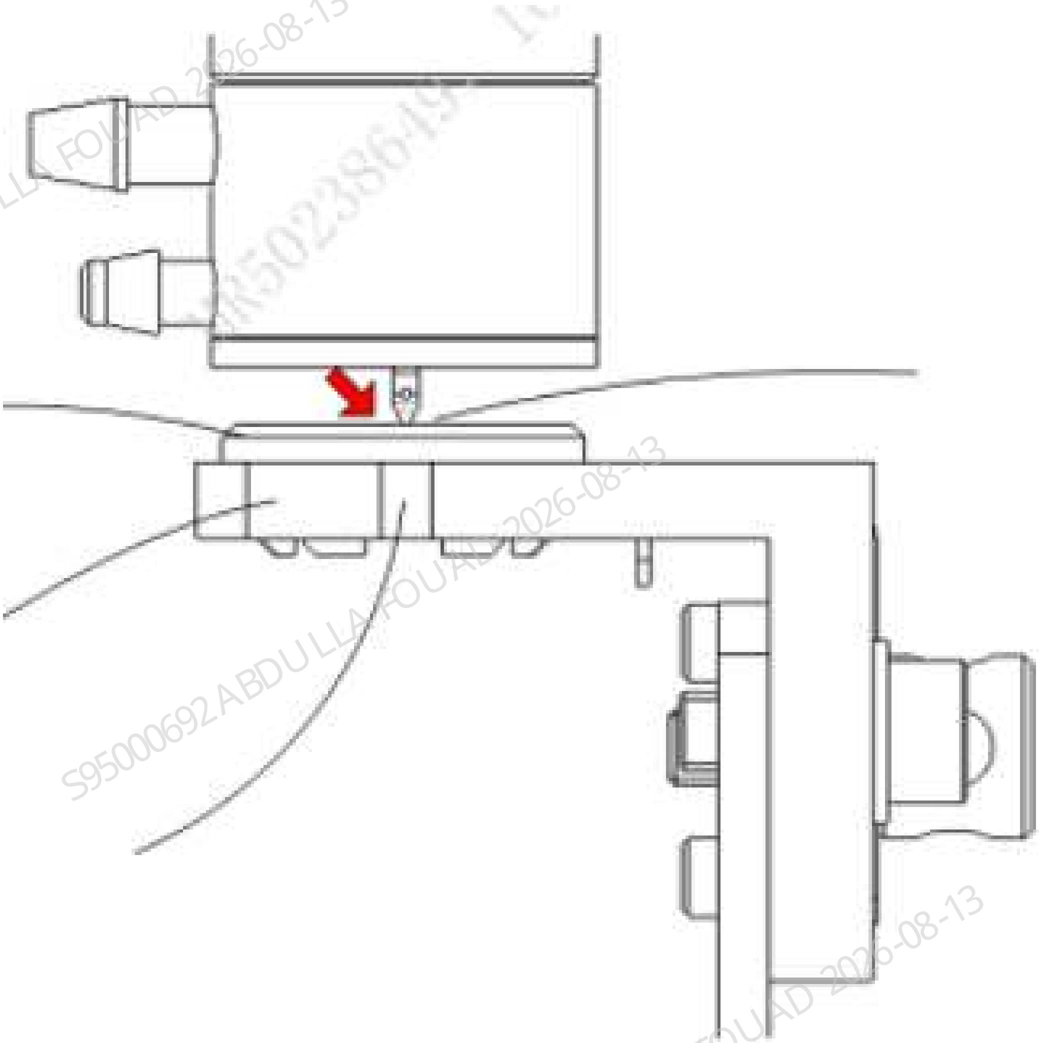


3. Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Adj. Sample Probe Pos.**>**>AL asp. Pos.** to enter the AL asp. Pos. adjustment interface.

Figure8–245 AL Aspiration Position Adjustment Interface

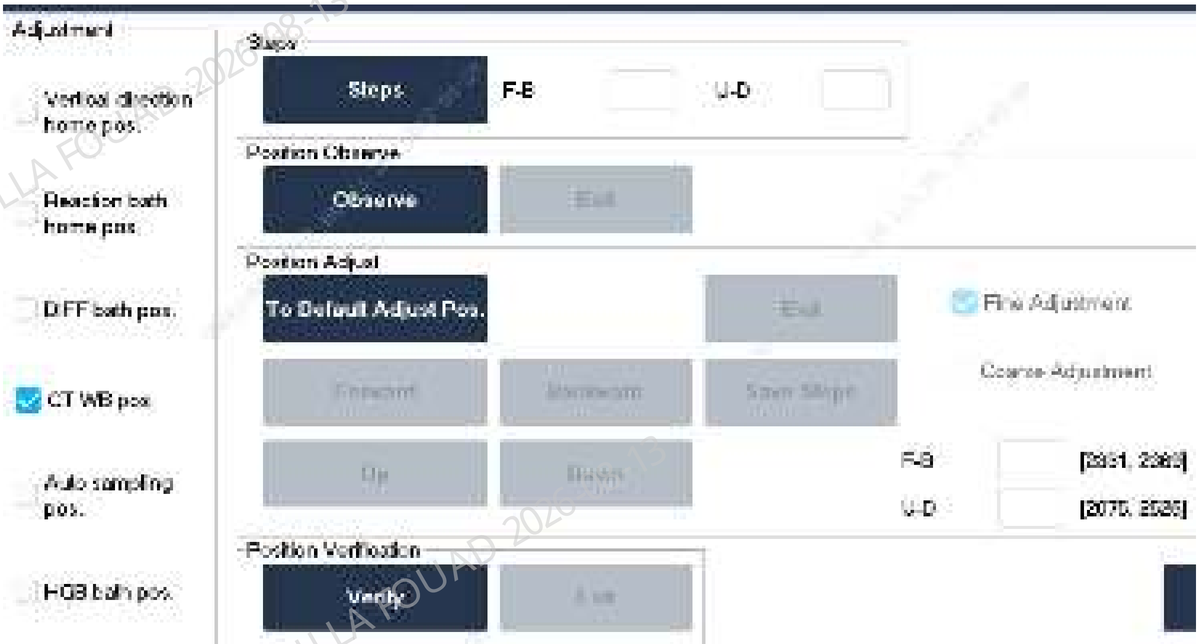
4. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sample probe is aligned with the groove of the blocking ring seat body.

Figure8–246 Observing Sampling Probe Position

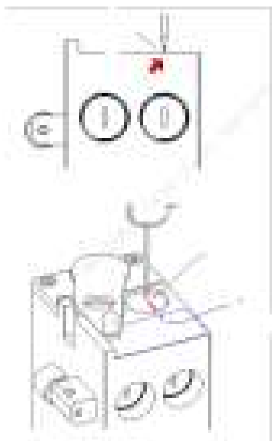


5. If the requirement of item 4 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the auto-sampling position.
6. Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Adj. Sample Probe Pos.**>**>CT-WB Position** to enter the CT-WB position adjustment interface.

Figure8–247 CT-WB Position Adjustment Interface



7. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sampling probe is aligned with the arrow on the cover opening label in the horizontal direction, and flush with the top surface of the sample compartment in the vertical direction.



8. If the requirement of item 7 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the CT-WB position.

Verification:

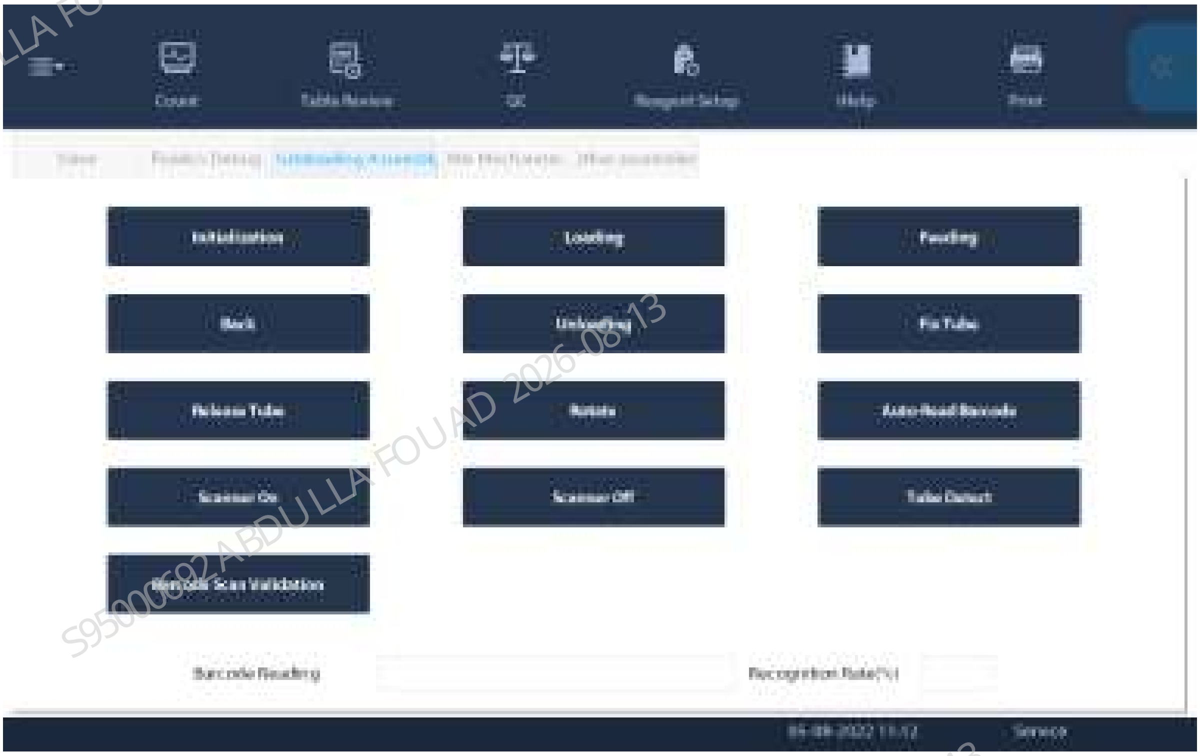
Confirming/Adjusting the Position of the Tube Hold Assembly

Confirm the position of the analyzer tube hold assembly. The steps are as follows:

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Autoloader Assembly** to enter the autoloader self-test interface.
2. Click the **Assembly Initialization** button.
3. Put one row of test tube rack with an empty test tube in No. 4 test tube position on the loading tray.

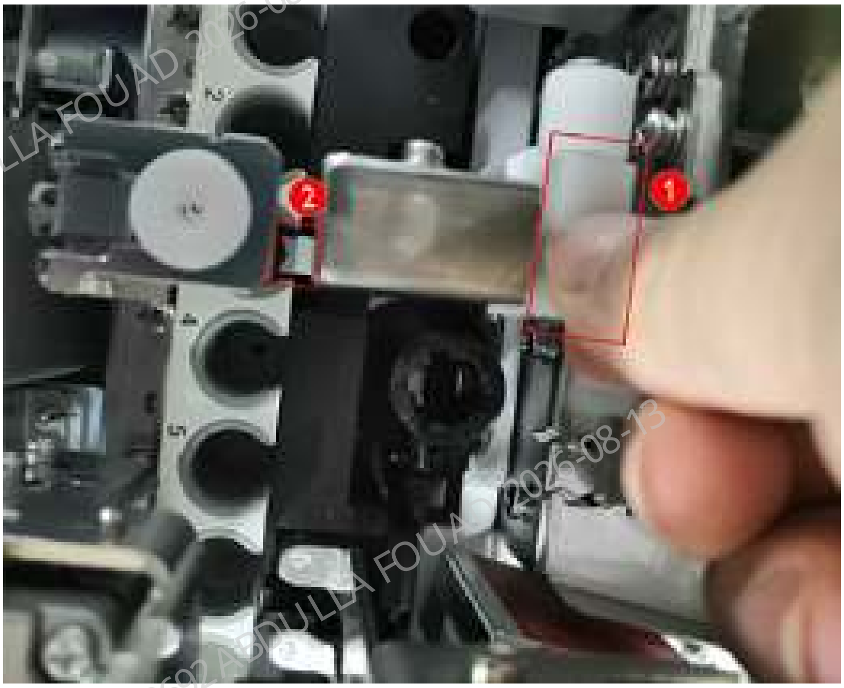
4. Click the **Feed** button (leave a time interval after each click, and then click the button next time after the mechanism action is completed), and send the first position of the test tube rack to the piercing position).

Figure8–248 Autoloader Self-test Interface



5. Press the roller fixing plate of the hold assembly by hand, and observe whether the finger pressure block of the hold assembly is approximate aligned with the center of the front opening of the test tube rack.

Figure8–249 Ideal Position of the Pincher Roller of the Hold Assembly Relative to the Tube Rack



6. If the left and right sides of the finger pressure block of the hold assembly are close to the opening of the test tube (≤ 1 mm), the left and right positions of the hold assembly need to be readjusted. For how to adjust the position of the hold assembly, see the section for adjusting the position of hold assembly FRU: **8.23.3** Commissioning and Verification.

Confirming/Adjusting the Position of the Rotatory Compressing Assembly

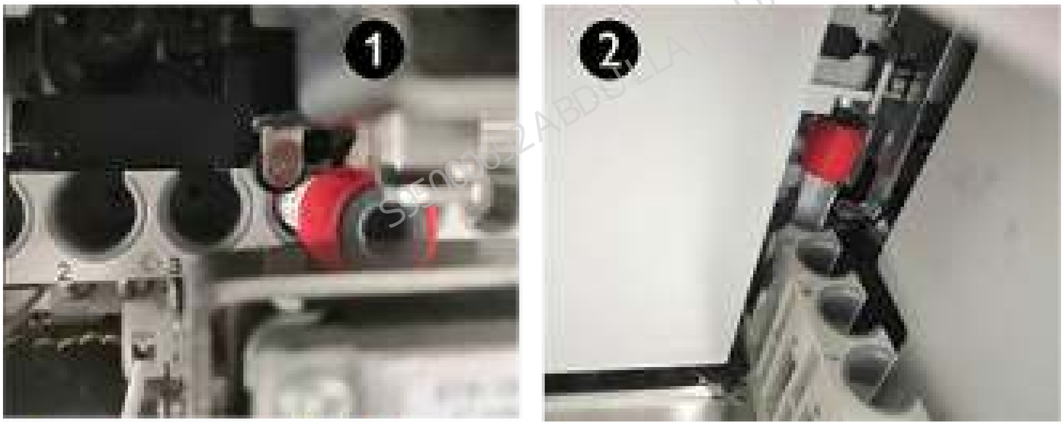
After the hold assembly position is confirmed, do not take out the tube rack, but continue to confirm the position of the rotatory compressing assembly. The steps are as follows:

1. Click the **Fix Tube** button.

Figure8–250 Autoloader Assembly Self-test Interface

2. Observe whether the two pinch rollers are close to the test tube. If they are not, readjust the left and right positions of the mechanism (failure to keeping them close will cause a false alarm of the autoloader counter).

Figure8–251 Ideal Position of the Pincher Roller of the Rotatory Compressing Assembly Relative to the Tube Rack



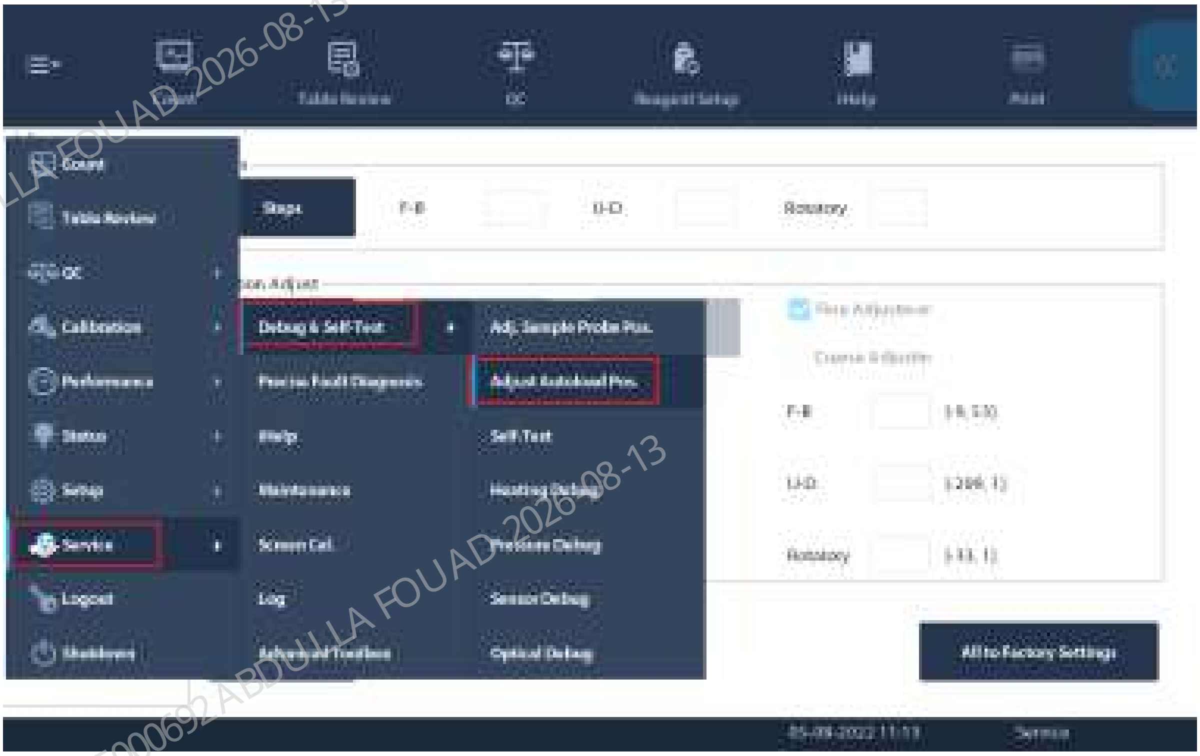
1	Left pinch roller	2	Right pinch roller

Confirming/Adjusting the Mix Mechanism Position

After the rotatory compressing assembly position is confirmed, do not take out the tube rack, but continue to confirm the mix mechanism position.

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.**, and select *Autoloader Assembly* to enter the mix mechanism self-test interface.

Figure8–252 Path of Entering Adjust Autoload Pos.



2. Select **Mixing gripper (front position)**, and click **Check by Step**.

Figure8–253 Mixing Gripper (Front Position) Interface

3. On the displayed box, click **Init.**, and then **Gripper Forward**. Observe whether the gripper is centered relative to the left and right positions of the test tube rack hole position. If it is centered, click **Gripper Backward**, place an empty tube into the mixing position, click the **A&M** button, and observe whether the tube squeezes the tube rack when the gripper is placing the tube.

Figure8–254 "Check by Step" Interface of Mixing Gripper (Front Position)

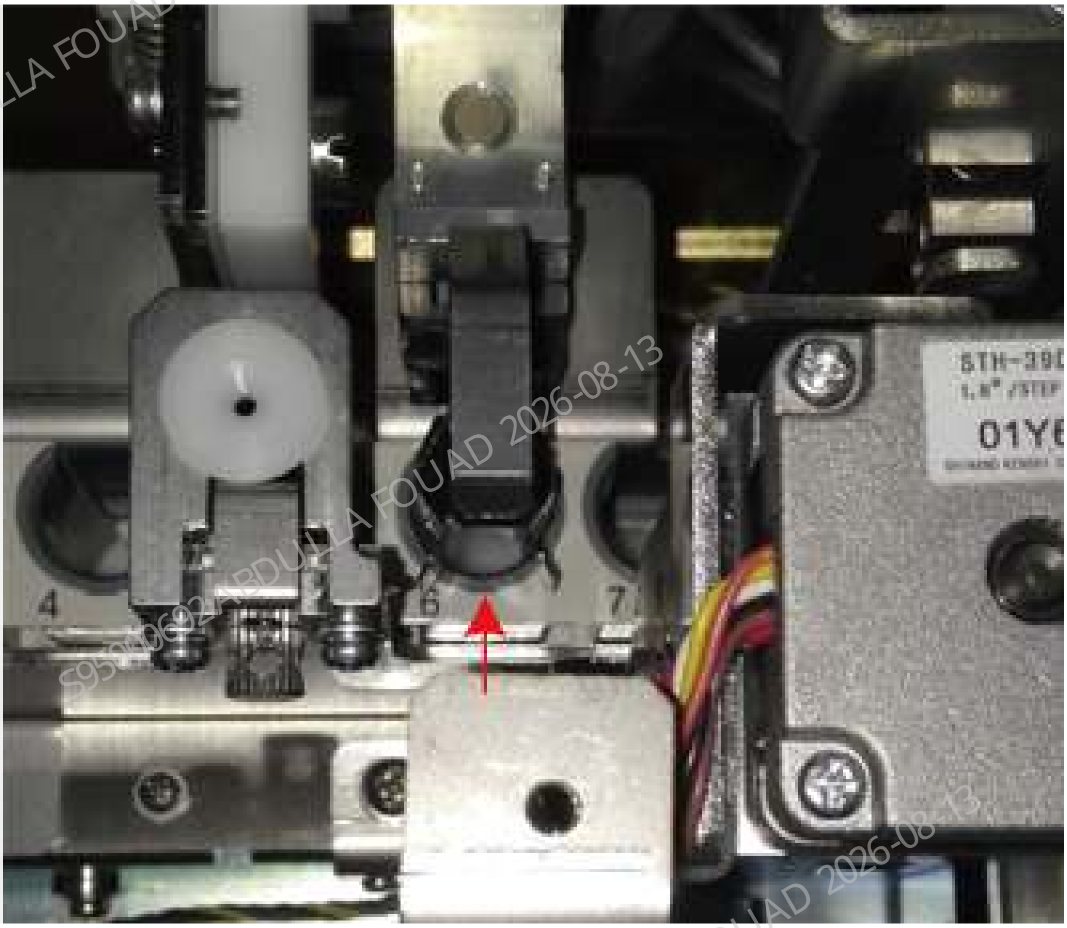
Figure8–255 Observing the Mixing Gripper Position

4. If the test tube squeezes the test tube rack, exit **Check by Step**, click **To Adjust Pos.**, and then press the **Forward**, **Backward**, **Left**, and **Right** buttons to fine adjust the gripper position.

Figure8–256 Observing the Mixing Gripper Position

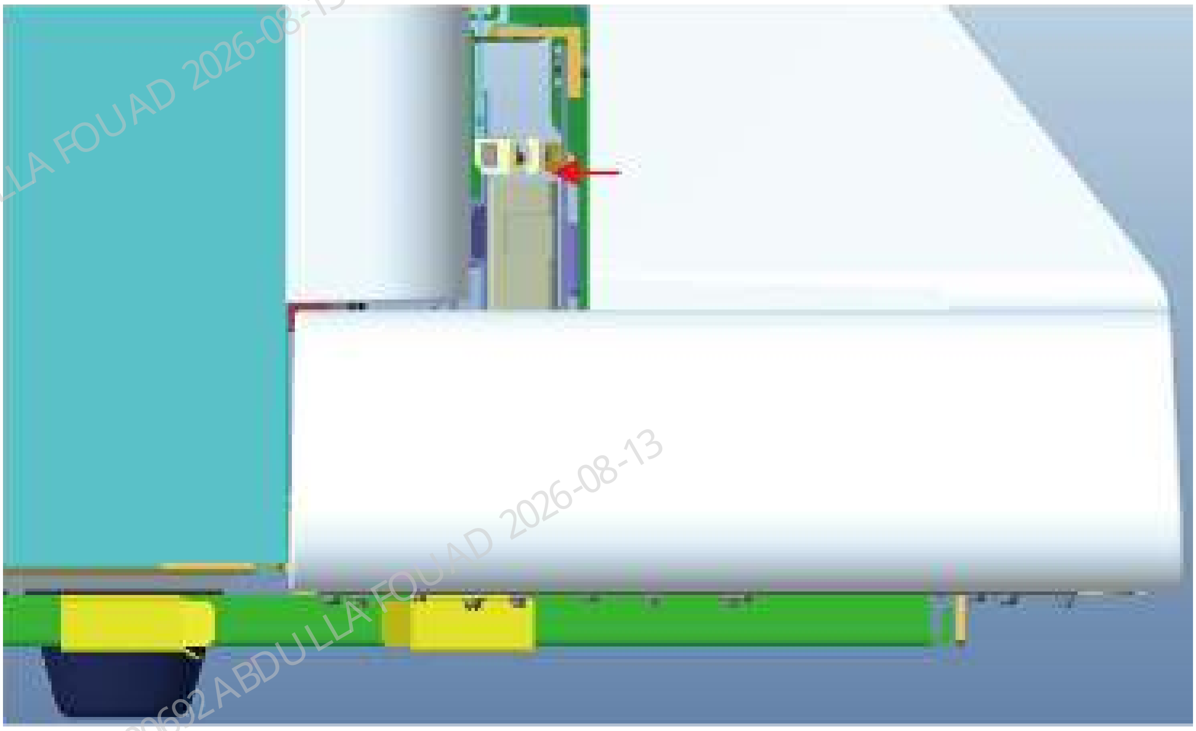
5. Finally enter **Check by Step** again, and observe again whether the tube squeezes the tube rack hole position when the gripper is placing the tube.

Figure8–257 Confirming the Mixing Mechanism Position in the Left-to-right Direction



6. After confirming the left, right, front, and back positions of the gripper relative to the test tube rack, confirm the upper and lower positions of the gripper. The distance between the bottom surface of the gripper and the top surface of the test tube rack needs to be controlled within the range of (2.0 ~ 2.5) mm. If slides are available, use two standard slices (thickness 1.2 mm) or one thick slice (nonstandard) as an auxiliary tool for test. If there are no slides, perform visual inspection. If the requirements are not met, make adjustment through **Up** and **Down** buttons of the gripper.

Figure8–258 Confirming the Mixing Gripper Height Position



7. Confirm the clearance between the gripper relative to the anti-collision bock. Click **Mixing gripper (front position)**, select **Check by Step**, click **Initialize**>**>To Center**, and observe whether the clearance between the gripper and the anti-collision bock is in the range of 1.5~2.5 mm. Use two slices as a tool for confirmation.

Figure8–259 Distance Requirements between the Mixing Gripper and Anti-Collision Bock

Confirming/Adjusting the Capillary Blood Mixing Position

Perform this operation for the analyzer configured with capillary blood mixing.

1. Prepare a WB test tube rack with WB test tubes.
2. Click **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.** to enter the mixing mechanism adjustment interface, and select **Mixing gripper (middle position)**.

Figure8–260 Capillary Blood Position Adjustment Interface of Tube Gripper

3. Place the test tube rack at the test tube loading position, click **Check by Step**, push the test tube rack to the feeding position, and then click **Assembly Initialization**>>**Feed**to feed the test tube on the test tube rack to the mix assembly gripping position.

Figure8–261 "Check by Step" Interface

4. Click **To Mixer** and **To tube rack**, check whether the gripping action of the gripper for gripping the test tube between the test tube rack and the capillary blood mixer is smooth, and whether there is interference between the test tube and the capillary blood mixer.

1) If the gripping action is not smooth or the test tube interferes with the capillary blood mixer, confirm whether there is a problem with the left/right or front/rear position of the capillary blood mixer.

2) If there is a problem with the left/right position of the capillary blood mixer, loosen the fixing screw fixing the capillary blood mixer and adjust the capillary blood mixer left and right.

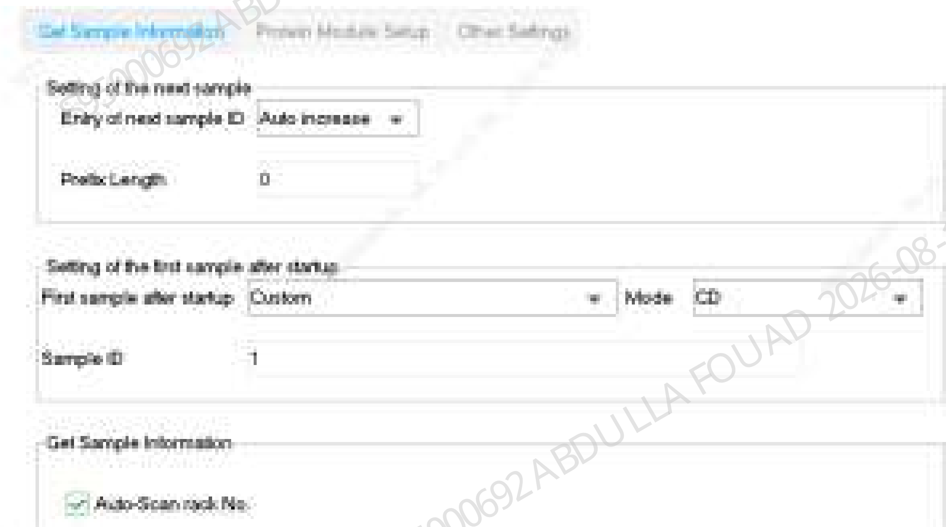
3) If there is a problem with the front/back position of the capillary blood mixer, exit the **Check by Step** interface and adjust the position forward and backward on the commissioning interface.

Confirming the Scanner Position

If the user does not need the sample barcode scanning function, this section can be skipped.

1. Click **Menu>>Setup>>Auxiliary Setup** to enter the interface for setting the sample information getting method. Confirm that the **Auto-Scan rack No.** option has been checked.

Figure8–262 Auto-Scan Setup Interface

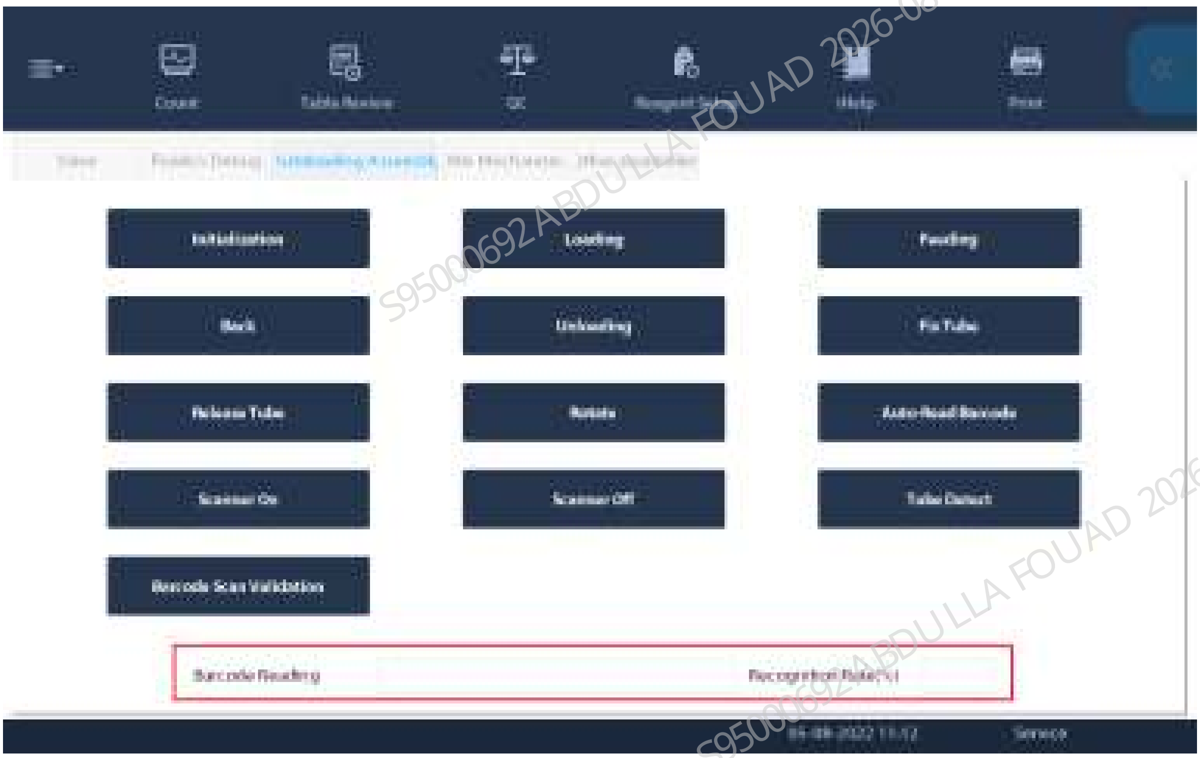


2. Use the barcode scanning software or machine barcode recognition function to identify the barcode system used by the client.
3. Click **Menu>>Setup>>System Setup>>Barcode** to enter the barcode setup interface. Check the corresponding code system, length, and check bit. Do not check the code system and length not used for the client.

Figure8–263 Built-in Barcode Setup Screen

4. After completing the above setup items, click **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**, and select **Autoloader Assembly** to enter the autoloader self-test interface.
5. After clicking **Assembly Initialization**, put a test tube with barcode in the rotating scanning position (the test tube is placed in the test tube rack).
6. Click **Auto-Read Barcode** to check whether the barcode content display box displays the barcode content and whether the barcode recognition rate display box shows 100%. If not, first confirm whether the red light of the scanner is turned on when you click the Auto-Read Barcode button. If the light has been turned on but the barcode cannot be scanned, adjust the position according to the position confirmation description in the chapter about scanner FRU.

Figure8–264 Autoloader Assembly Self-test Interface



8.28 Autoloader, closed-sampling model(5x6)

8.28.1 General Information

Figure8–265 Photo of Autoloader, closed-sampling model(5x6)

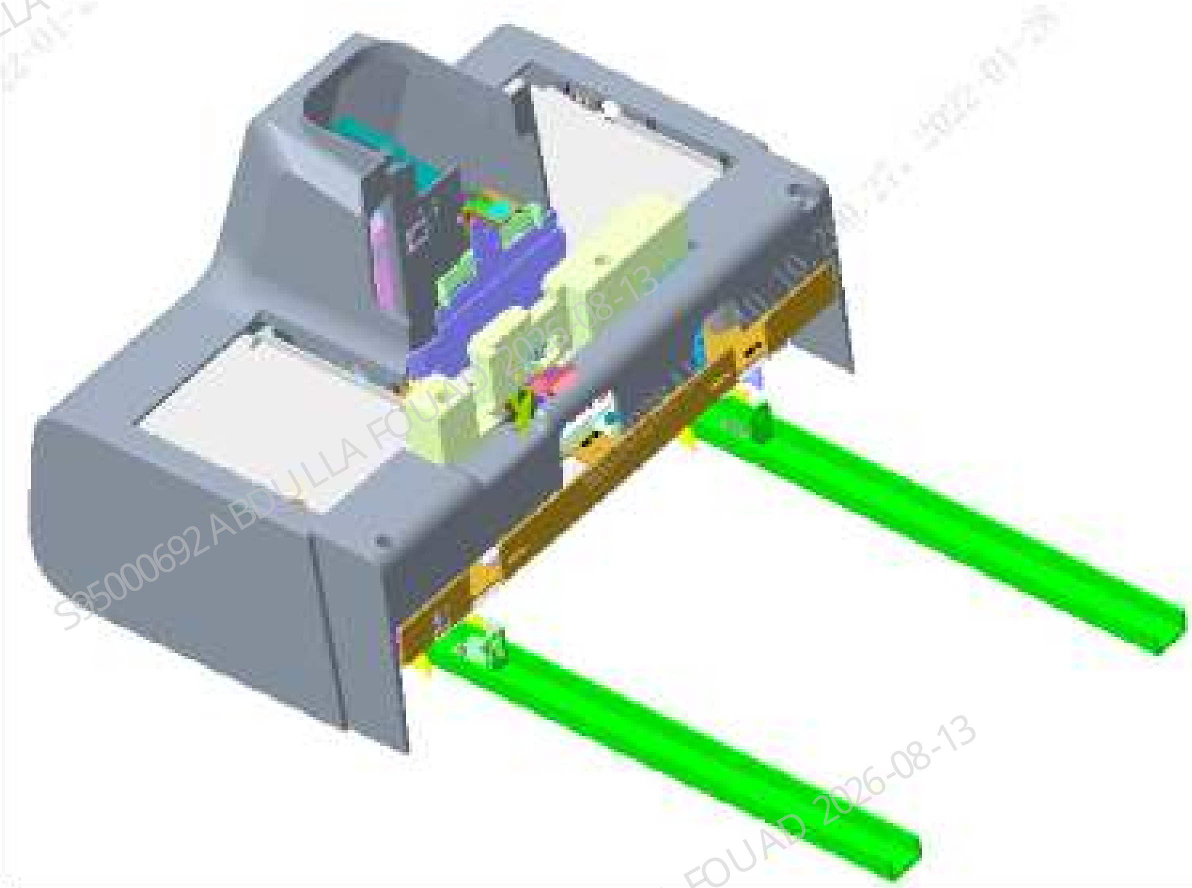


Figure8–266 Diagram of Location of Autoloader, closed-sampling model(5x6) in the Main Unit

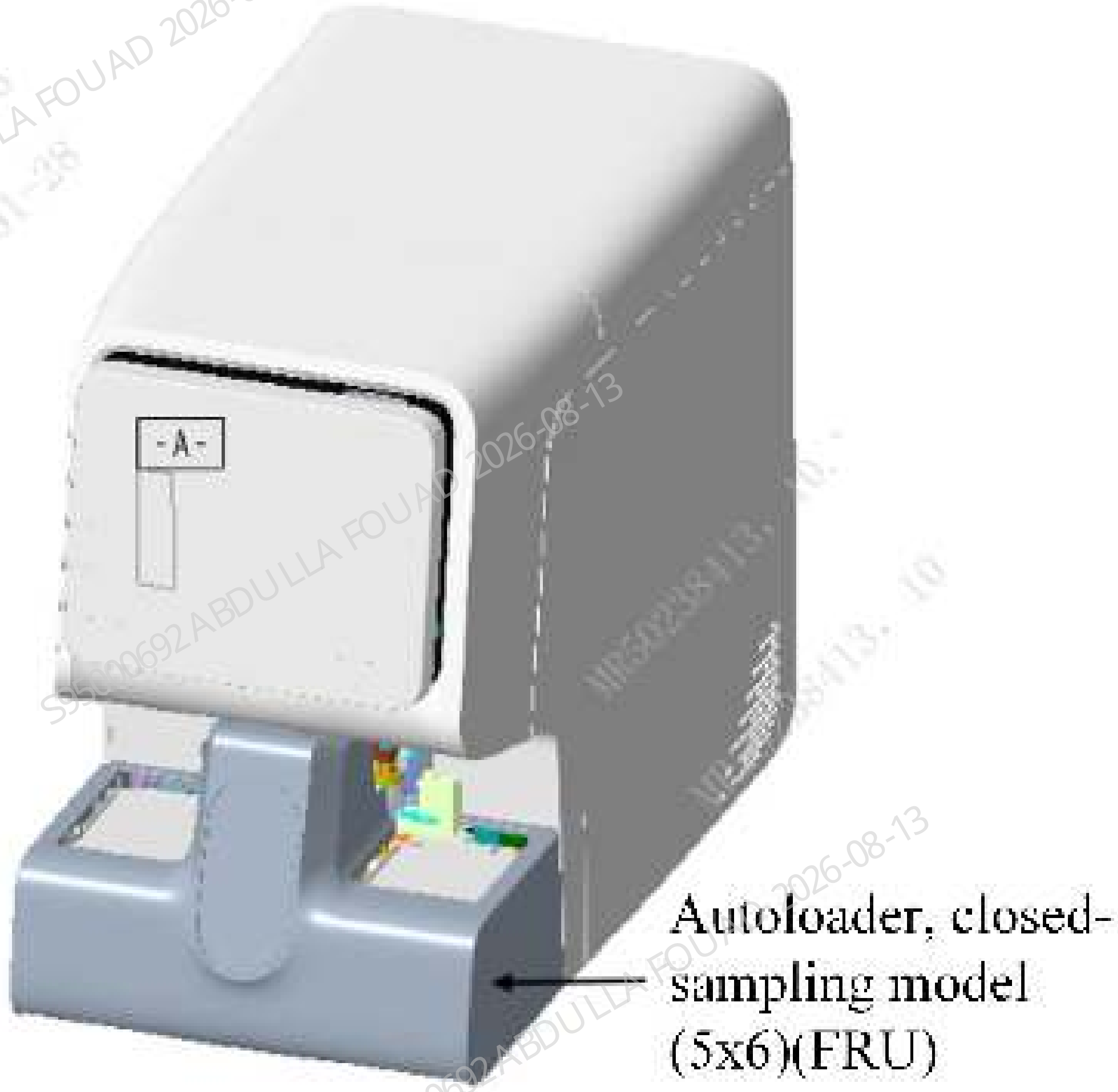


Table 8–31 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-085181-03	Autoloader, closed-sampling model(5x6)	N/A	N/A

Function:

Processing the samples placed on the test tube rack in batches, able to automatically transport and allocate the samples, and cooperate with the functional assembly of the main unit to automatically identify the sample barcode, automatically mix the samples, and pierce and aspirate samples.

8.28.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

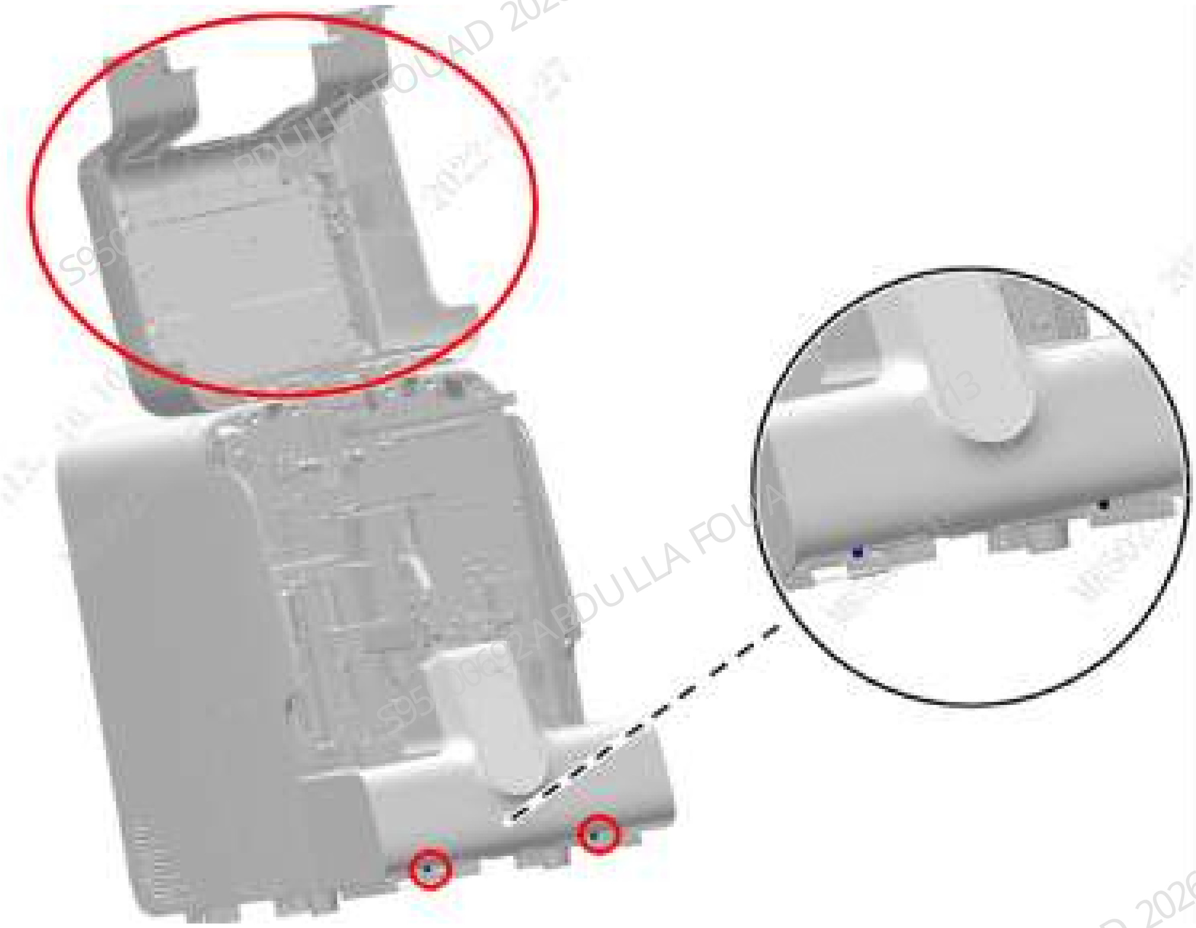
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

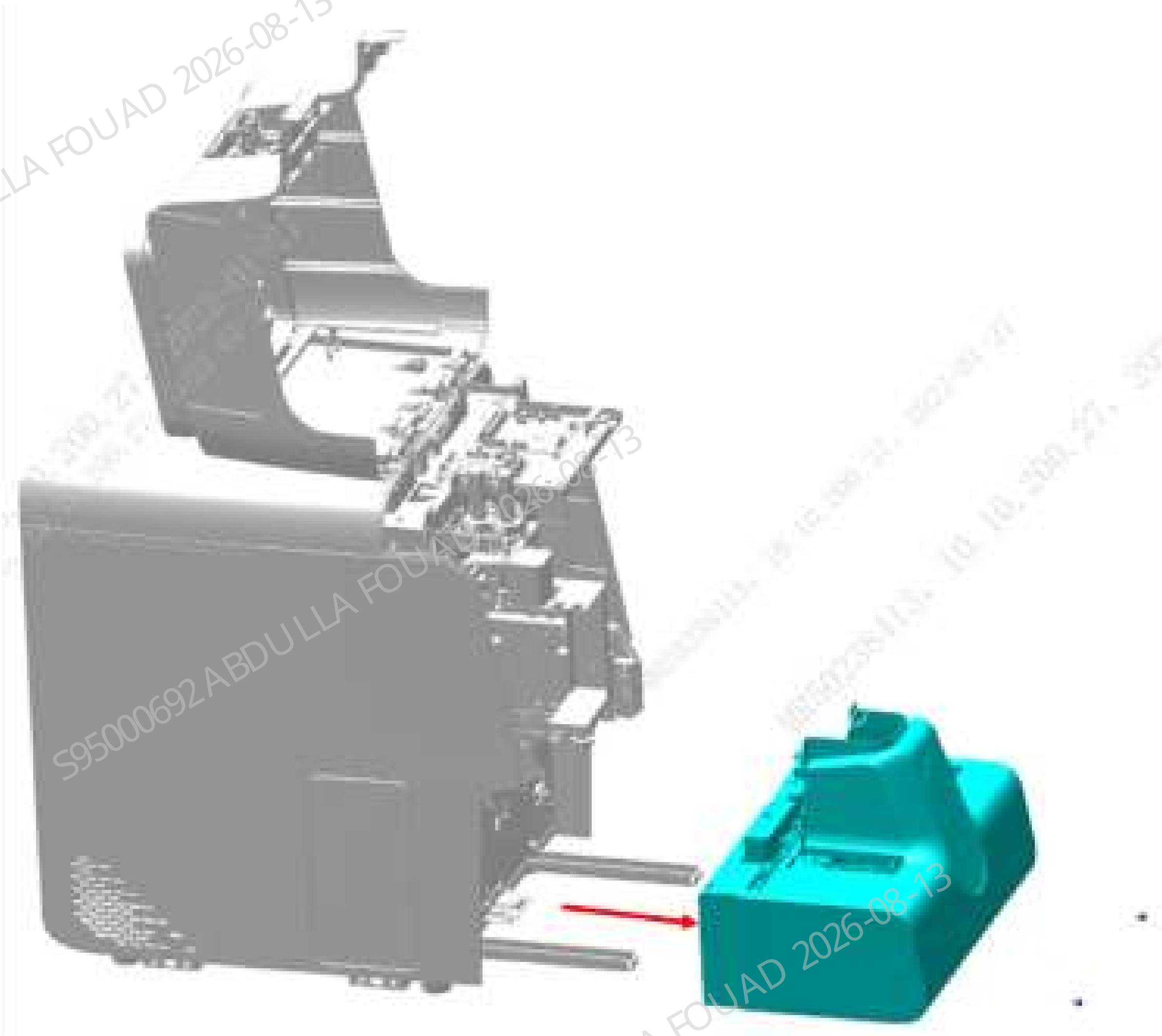
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–267 Opening the front cover



2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam, that is, remove the 5x6 closed autoloader (excluding capillary blood mixing) FRU , as shown in the figure below.

Figure8–268 Removing the autoloader assembly



Subsequent Requirements

Installation procedure:

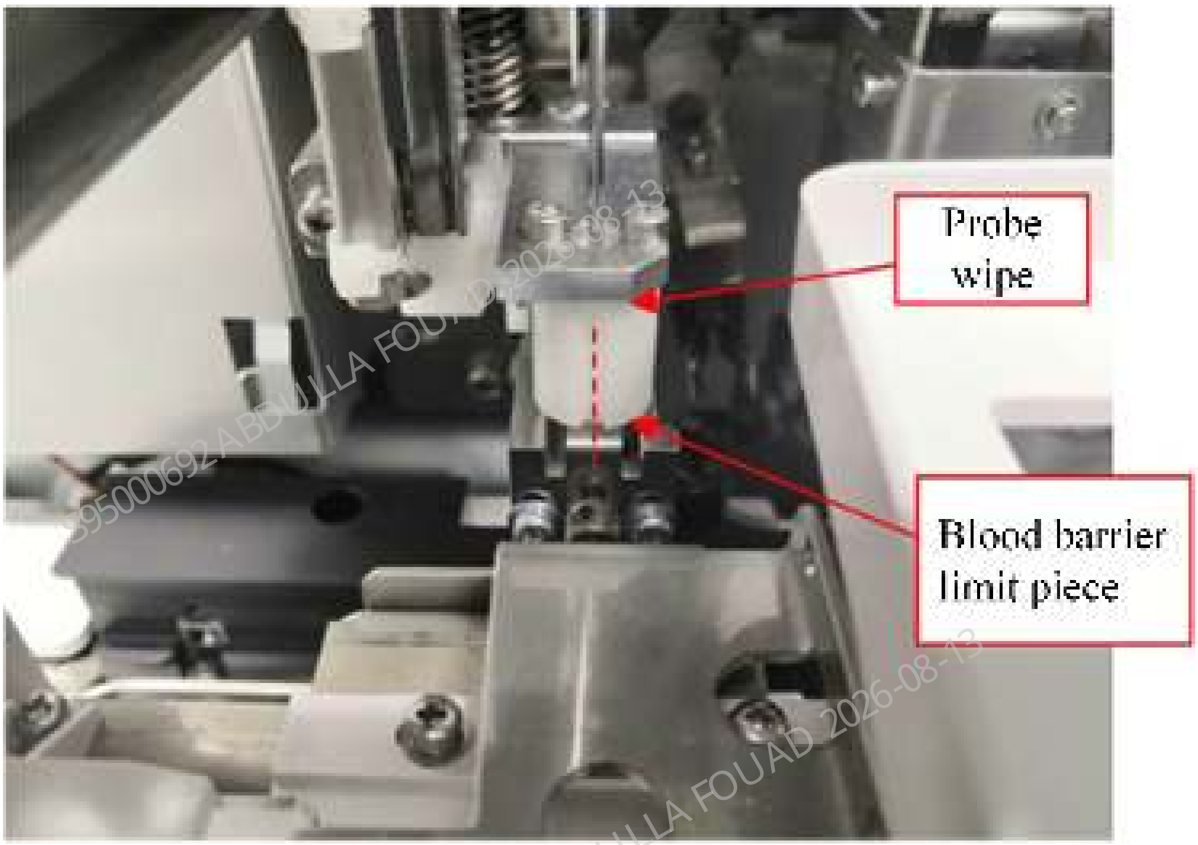
1. Place the 5x6 closed autoloader (excluding capillary blood mixing) FRU on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
2. Then push the 5x6 closed autoloader (excluding capillary blood mixing) FRU along the crossbeam direction of the autoloader to fit the front panel of the main unit.
3. Use two M4x8 cross-recessed combination screws to lock the 5x6 closed autoloader (excluding capillary blood mixing) FRU on the autoloader crossbeam.
4. Flip down the front cover.

8.28.3 Commissioning and Verification

Procedures

Commissioning and verification:

1. The left and right positions of the autoloader relative to the main unit have been adjusted in the assembly, so the FRU assembly does not require commissioning in the left and right positions.
2. Select **Menu**>>**Status**>>**Sensor Status** to enter the motor release torque status. Pull the sampling assembly to the front position a little ahead of the floating blood barrier. Observe whether the center of the sampling assembly probe wipe groove aligns with the center of the floating blood barrier limit piece (≤ ±0.5mm).

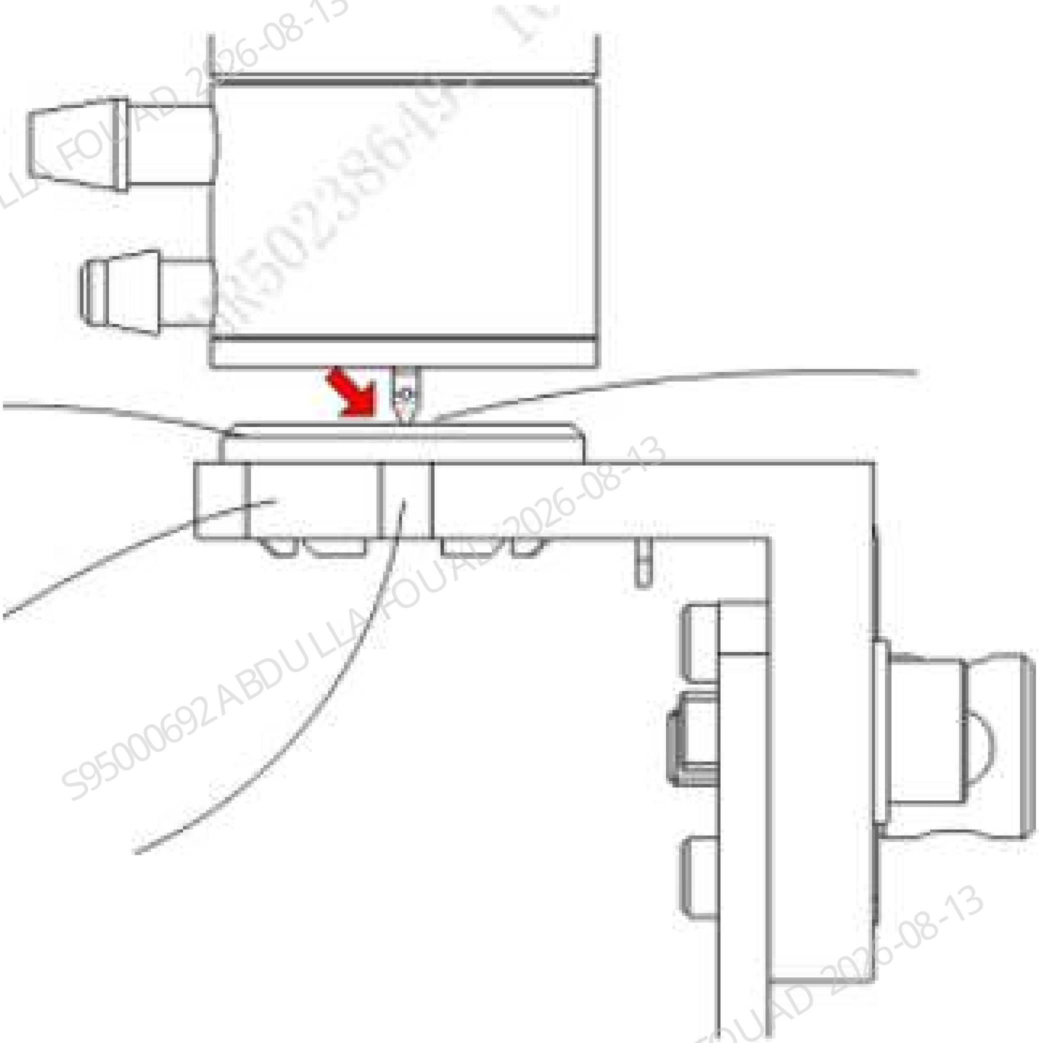


3. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adj. Sample Probe Pos.**>>**AL asp. Pos.** to enter the AL asp. Pos. adjustment interface.

Figure8–269 AL Aspiration Position Adjustment Interface

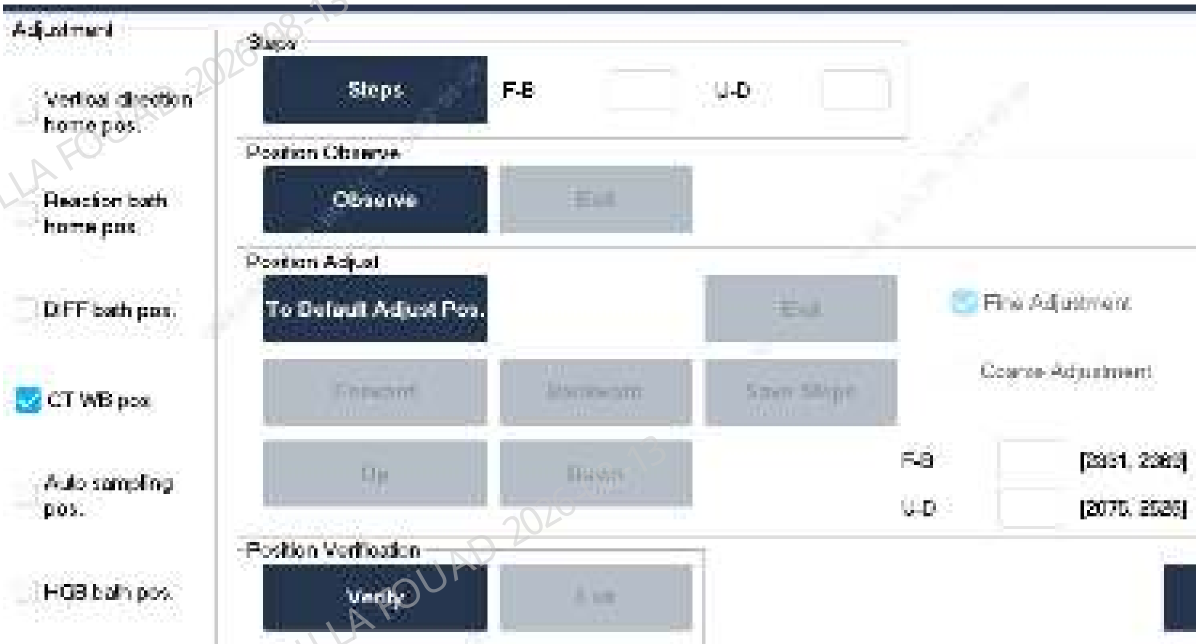
4. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sample probe is aligned with the groove of the blocking ring seat body.

Figure8–270 Observing Sampling Probe Position

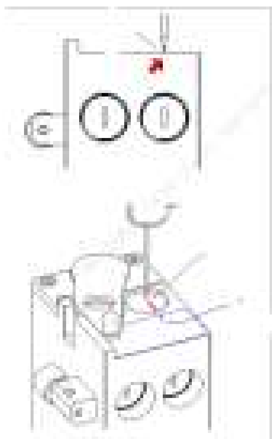


5. If the requirement of item 4 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the auto-sampling position.
6. Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Adj. Sample Probe Pos.**>**>CT-WB Position** to enter the CT-WB position adjustment interface.

Figure8–271 CT-WB Position Adjustment Interface



7. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sampling probe is aligned with the arrow on the cover opening label in the horizontal direction, and flush with the top surface of the sample compartment in the vertical direction.



8. If the requirement of item 7 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the CT-WB position.

Verification:

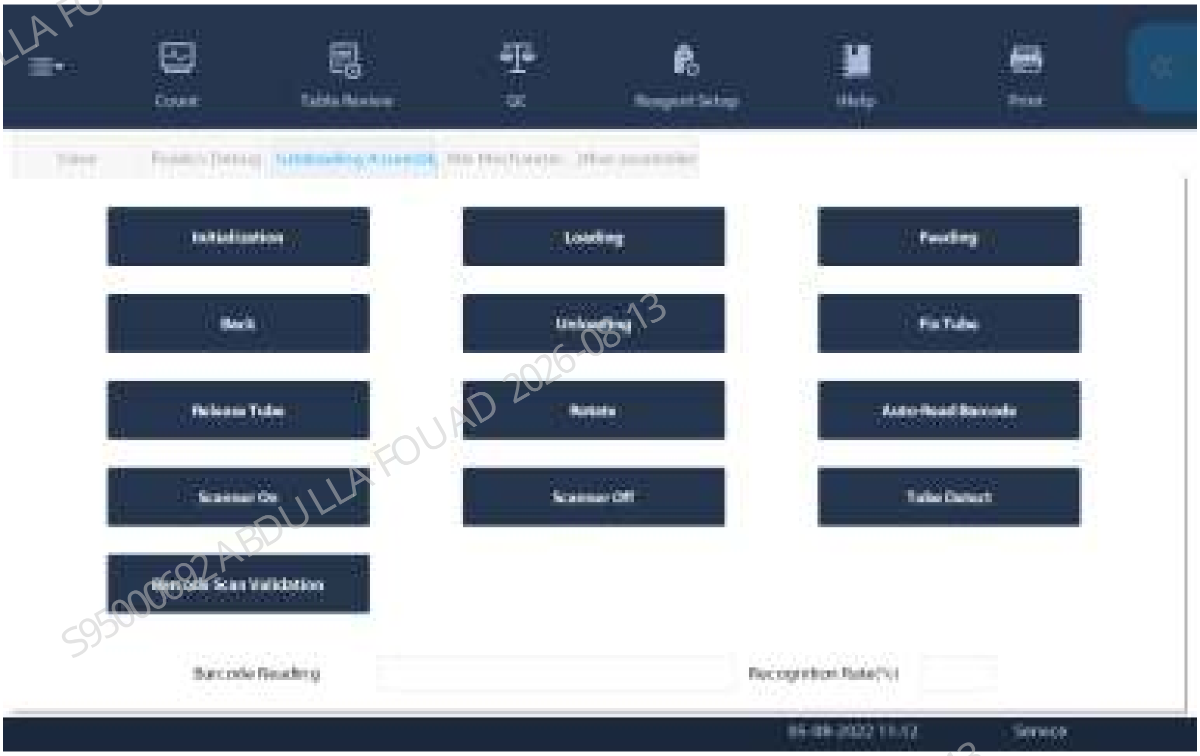
Confirming/Adjusting the Position of the Tube Hold Assembly

Confirm the position of the analyzer tube hold assembly. The steps are as follows:

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Autoloader Assembly** to enter the autoloader self-test interface.
2. Click the **Assembly Initialization** button.
3. Put one row of test tube rack with an empty test tube in No. 4 test tube position on the loading tray.

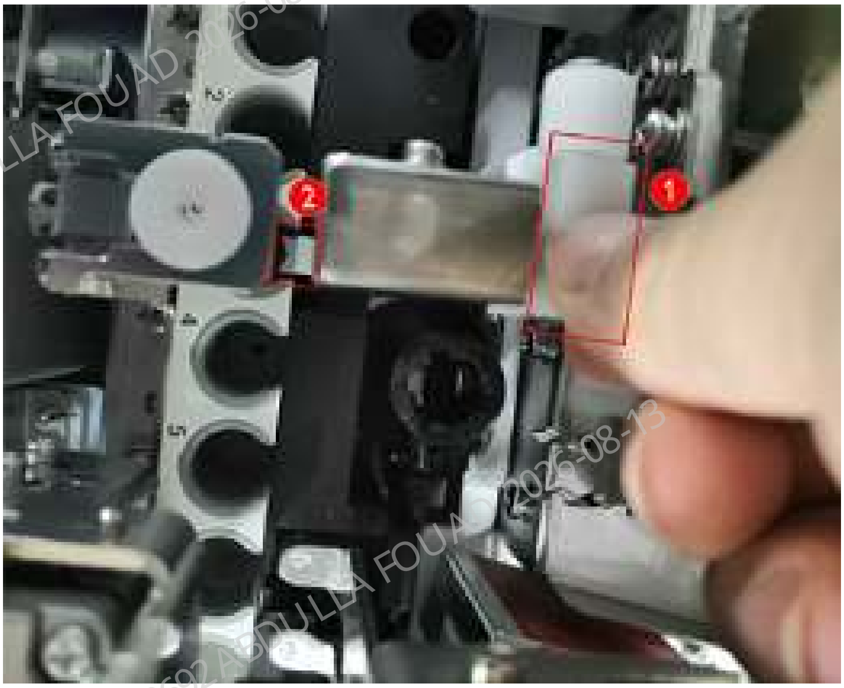
4. Click the **Feed** button (leave a time interval after each click, and then click the button next time after the mechanism action is completed), and send the first position of the test tube rack to the piercing position).

Figure8–272 Autoloader Self-test Interface



5. Press the roller fixing plate of the hold assembly by hand, and observe whether the finger pressure block of the hold assembly is approximate aligned with the center of the front opening of the test tube rack.

Figure8–273 Ideal Position of the Pincher Roller of the Hold Assembly Relative to the Tube Rack



6. If the left and right sides of the finger pressure block of the hold assembly are close to the opening of the test tube (≤ 1 mm), the left and right positions of the hold assembly need to be readjusted. For how to adjust the position of the hold assembly, see the section for adjusting the position of hold assembly FRU: **8.23.3** Commissioning and Verification.

Confirming/Adjusting the Position of the Rotatory Compressing Assembly

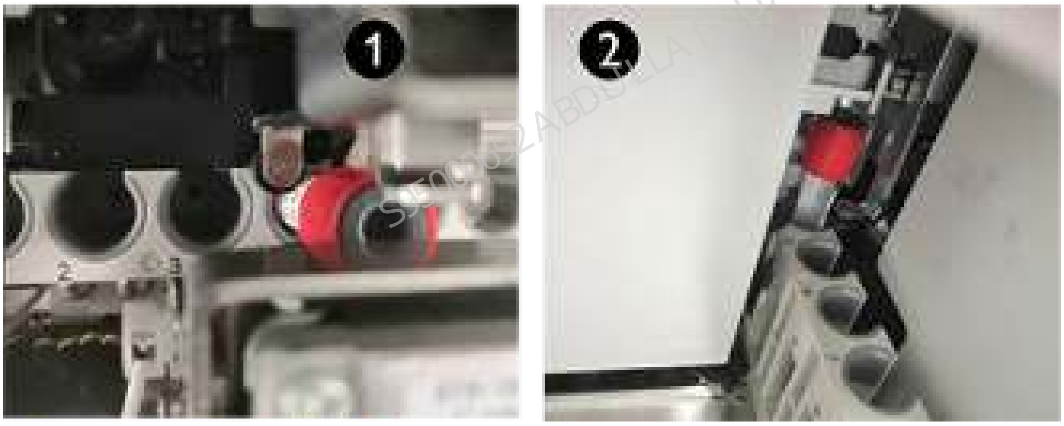
After the hold assembly position is confirmed, do not take out the tube rack, but continue to confirm the position of the rotatory compressing assembly. The steps are as follows:

1. Click the **Fix Tube** button.

Figure8–274 Autoloader Assembly Self-test Interface

2. Observe whether the two pinch rollers are close to the test tube. If they are not, readjust the left and right positions of the mechanism (failure to keeping them close will cause a false alarm of the autoloader counter).

Figure8–275 Ideal Position of the Pincher Roller of the Rotatory Compressing Assembly Relative to the Tube Rack



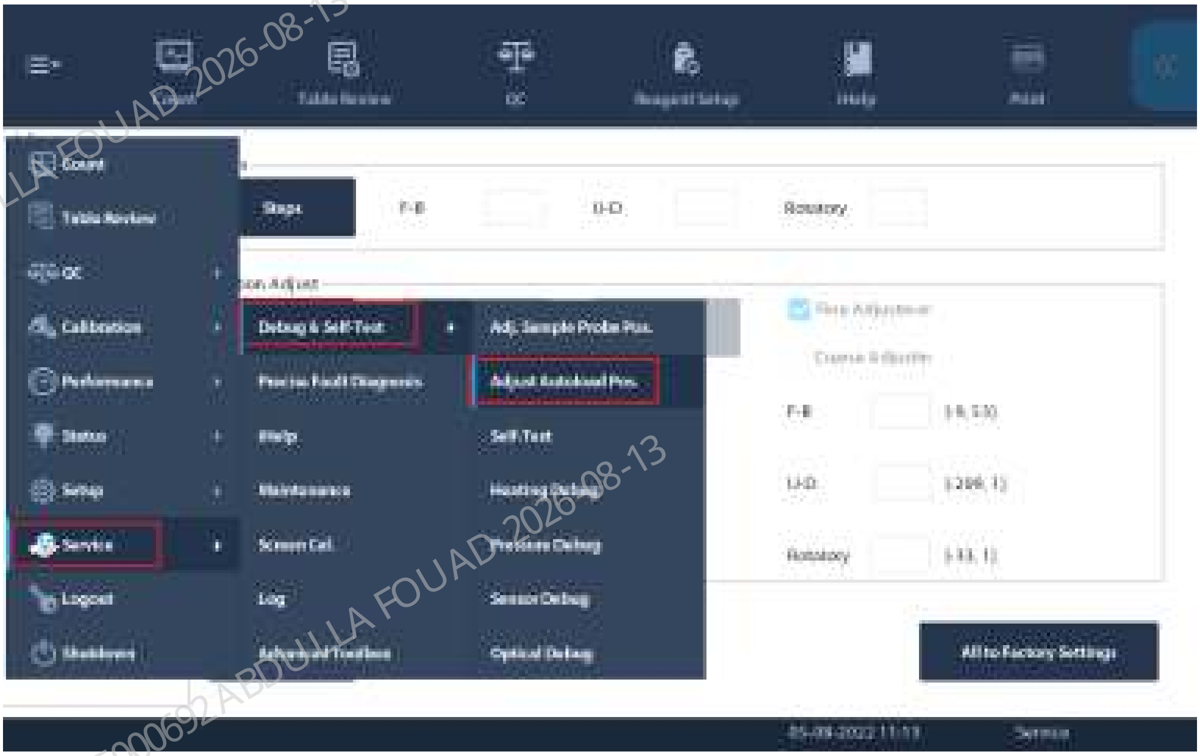
1	Left pinch roller	2	Right pinch roller

Confirming/Adjusting the Mix Mechanism Position

After the rotatory compressing assembly position is confirmed, do not take out the tube rack, but continue to confirm the mix mechanism position.

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.**, and select *Autoloader Assembly* to enter the mix mechanism self-test interface.

Figure8–276 Path of Entering Adjust Autoload Pos.



2. Select **Mixing gripper (front position)**, and click **Check by Step**.

Figure8–277 Mixing Gripper (Front Position) Interface

3. On the displayed box, click **Init.**, and then **Gripper Forward**. Observe whether the gripper is centered relative to the left and right positions of the test tube rack hole position. If it is centered, click **Gripper Backward**, place an empty tube into the mixing position, click the **A&M** button, and observe whether the tube squeezes the tube rack when the gripper is placing the tube.

Figure8–278 "Check by Step" Interface of Mixing Gripper (Front Position)

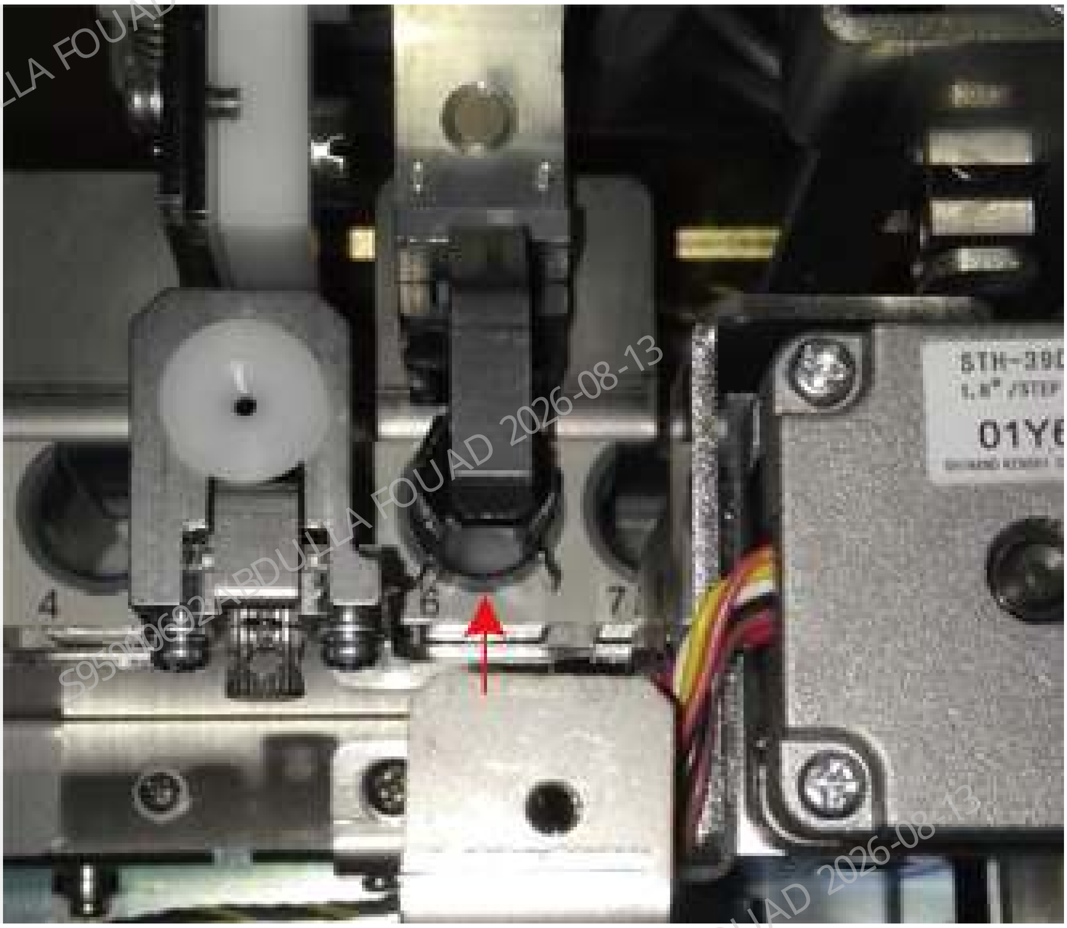
Figure8–279 Observing the Mixing Gripper Position

4. If the test tube squeezes the test tube rack, exit **Check by Step**, click **To Adjust Pos.**, and then press the **Forward**, **Backward**, **Left**, and **Right** buttons to fine adjust the gripper position.

Figure8–280 Observing the Mixing Gripper Position

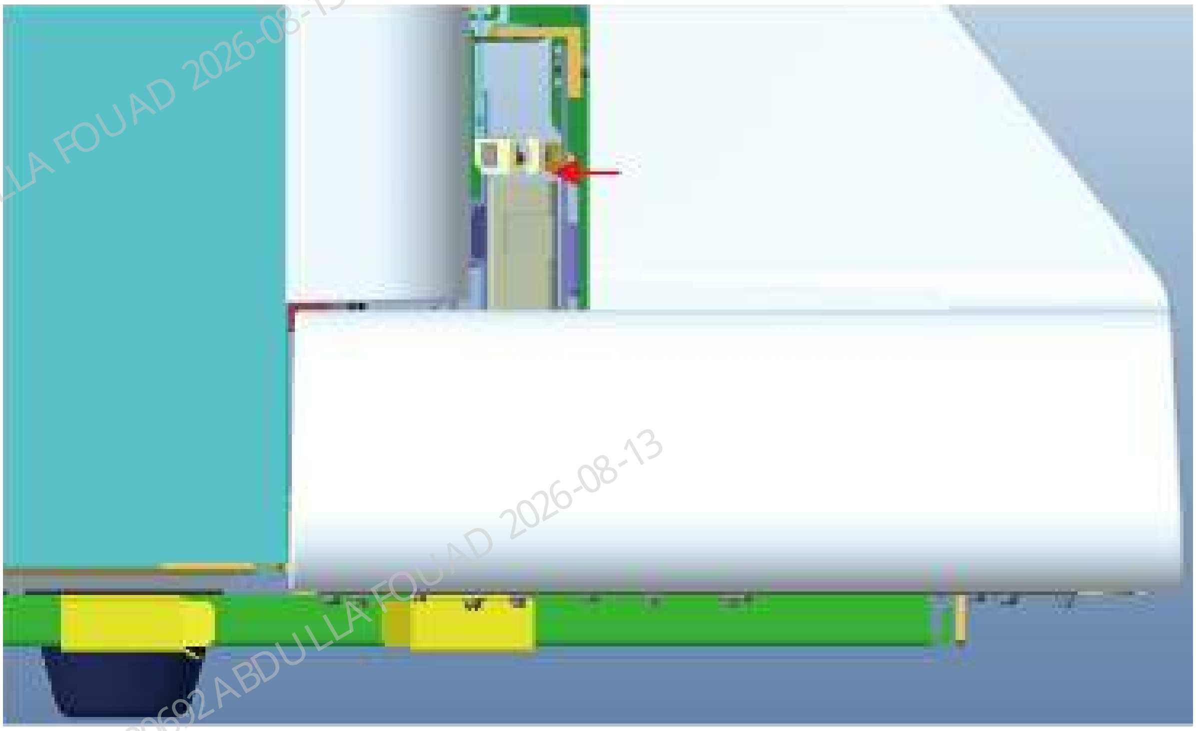
5. Finally enter **Check by Step** again, and observe again whether the tube squeezes the tube rack hole position when the gripper is placing the tube.

Figure8–281 Confirming the Mixing Mechanism Position in the Left-to-right Direction



6. After confirming the left, right, front, and back positions of the gripper relative to the test tube rack, confirm the upper and lower positions of the gripper. The distance between the bottom surface of the gripper and the top surface of the test tube rack needs to be controlled within the range of (2.0 ~ 2.5) mm. If slides are available, use two standard slices (thickness 1.2 mm) or one thick slice (nonstandard) as an auxiliary tool for test. If there are no slides, perform visual inspection. If the requirements are not met, make adjustment through **Up** and **Down** buttons of the gripper.

Figure8–282 Confirming the Mixing Gripper Height Position



7. Confirm the clearance between the gripper relative to the anti-collision bock. Click **Mixing gripper (front position)**, select **Check by Step**, click **Initialize**>**>To Center**, and observe whether the clearance between the gripper and the anti-collision bock is in the range of 1.5~2.5 mm. Use two slices as a tool for confirmation.

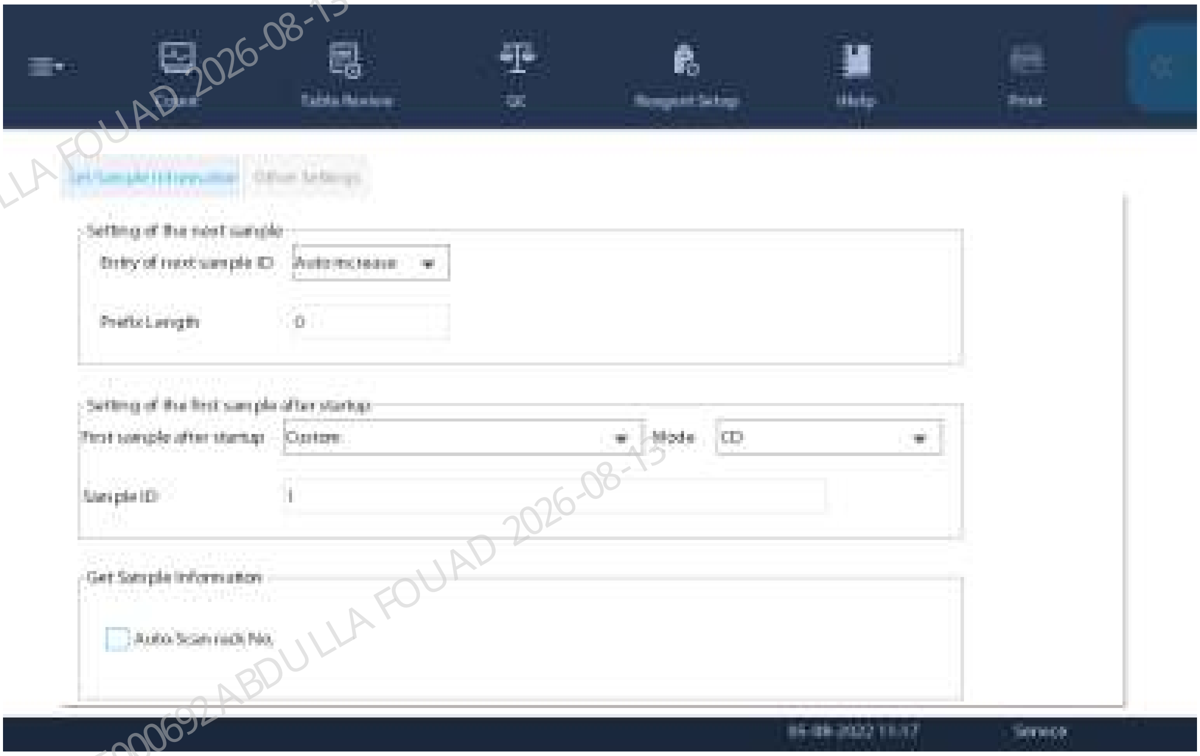
Figure8–283 Distance Requirements between the Mixing Gripper and Anti-Collision Bock

Confirming the Scanner Position

If the user does not need the sample barcode scanning function, this section can be skipped.

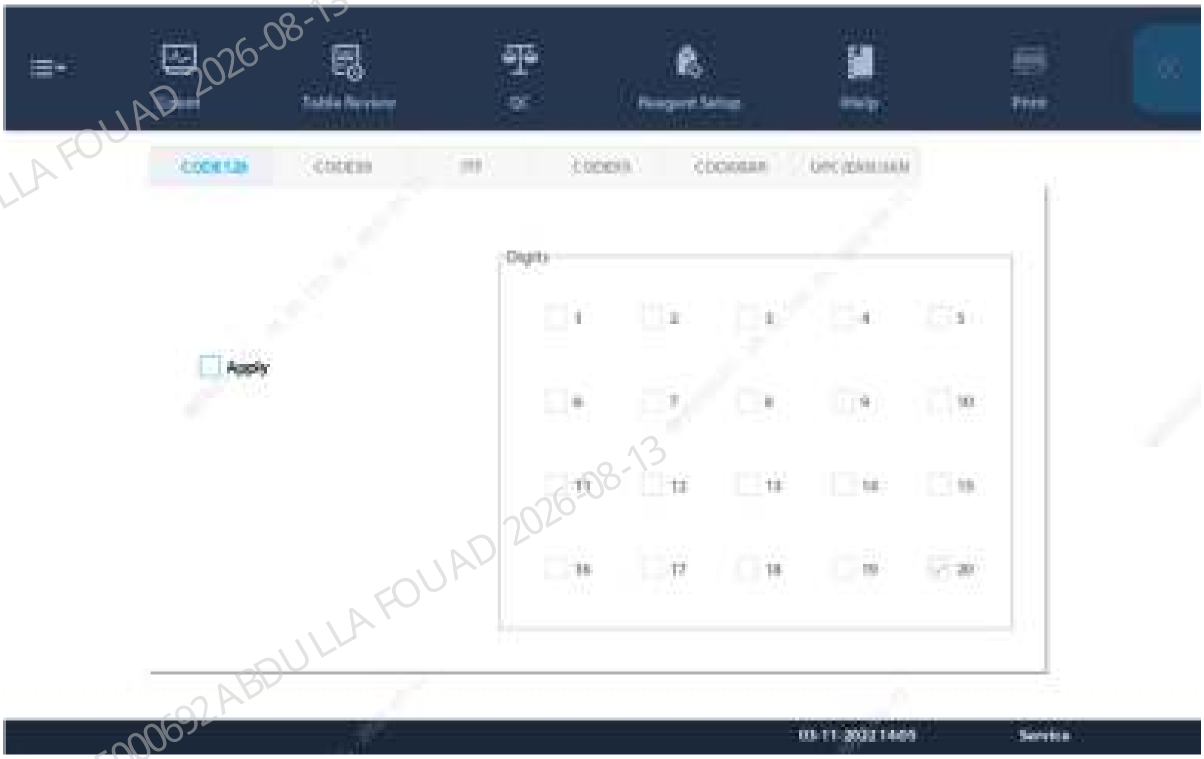
1. Click **Menu>>Setup>>Auxiliary Setup** to enter the interface for setting the sample information getting method. Confirm that the **Auto-Scan rack No.** option has been checked.

Figure8–284 Auto-Scan Setup Interface



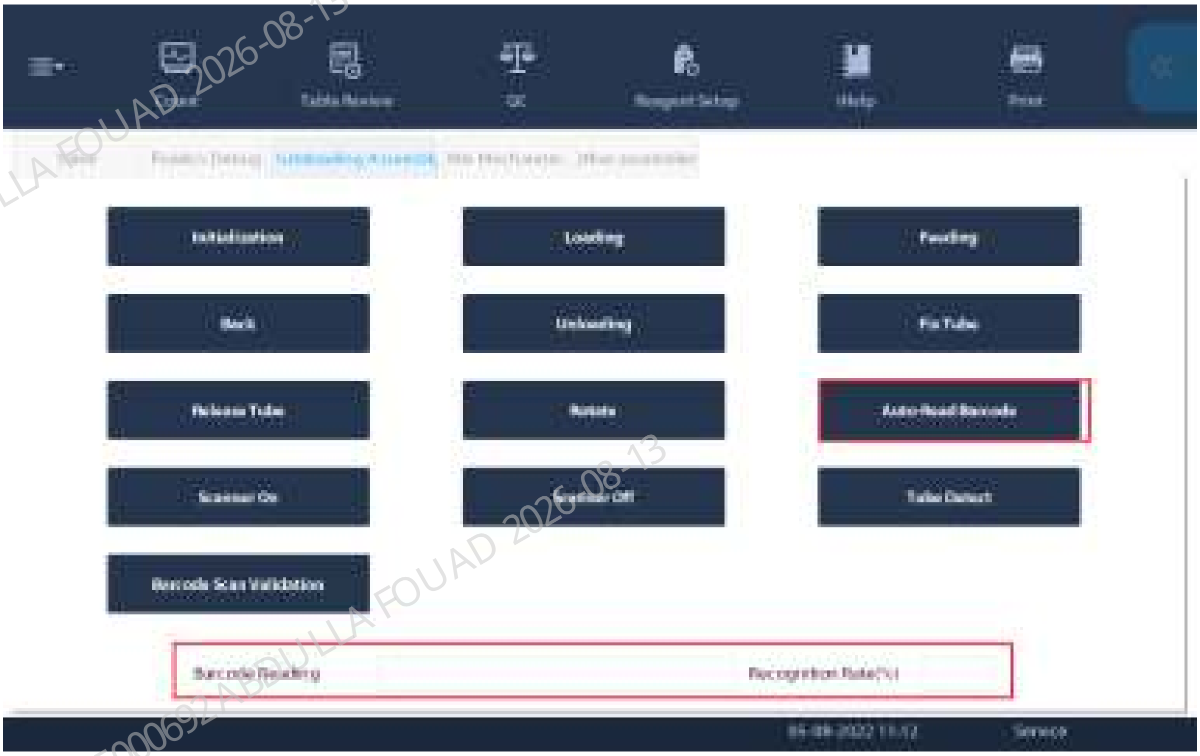
2. Use the barcode scanning software or machine barcode recognition function to identify the barcode system used by the client.
3. Click **Menu**>>**Setup**>>**System Setup**>>**Barcode** to enter the barcode setup interface. Check the corresponding code system, length, and check bit. Do not check the code system and length not used for the client.

Figure8–285 Built-in Barcode Setup Screen



4. After completing the above setup items, click **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-test**, and select **Autoloader Assembly** to enter the autoloader self-test interface.
5. After clicking **Assembly Initialization**, put a test tube with barcode in the rotating scanning position (the test tube is placed in the test tube rack).
6. Click **Auto-Read Barcode** to check whether the barcode content display box displays the barcode content and whether the barcode recognition rate display box shows 100%. If not, first confirm whether the red light of the scanner is turned on when you click the Auto-Read Barcode button. If the light has been turned on but the barcode cannot be scanned, adjust the position according to the position confirmation description in the chapter about scanner FRU.

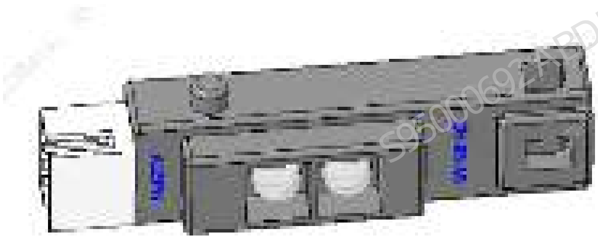
Figure8–286 Autoloader Assembly Self-test Interface



8.29 PHOTOELEC photosensor reflective 940nm

8.29.1 General Information

Figure8–287 PHOTOELEC photosensor reflective 940nm



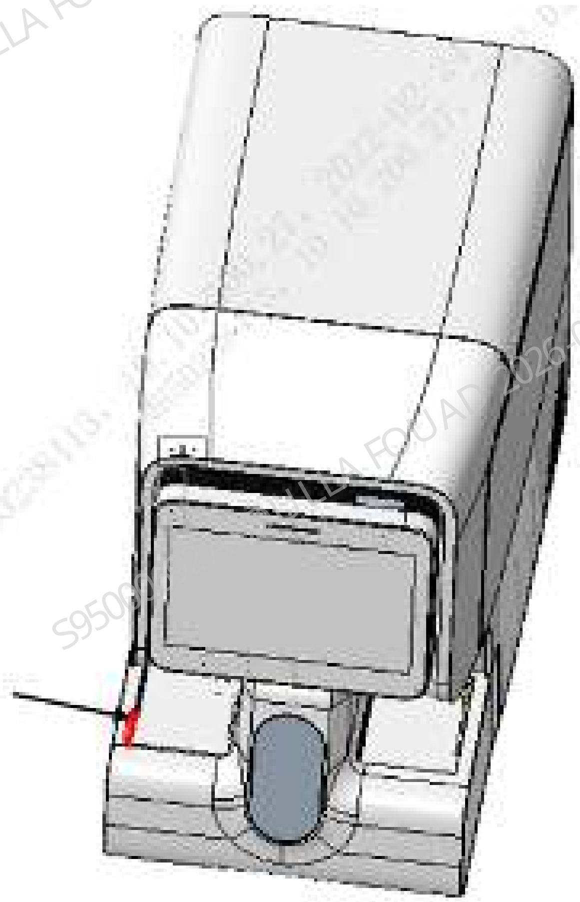
FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000049-00	PHOTOELEC photosensor reflective 940nm	N/A	N/A

Function: Detecting the position information of the test tube rack in the unloading area of the autoloader.

8.29.2 Disassembly and Assembly

Note

Figure8–288



Required tools

Cross-headed screwdriver

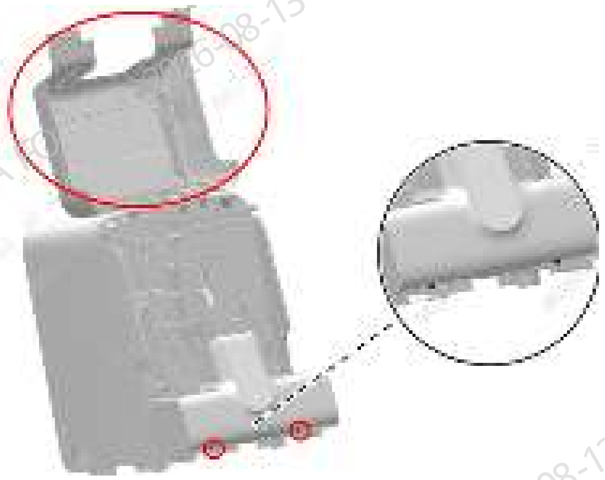
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. Open the front cover of the main unit and dock it on the top cover, as shown in **Figure8–289** below.

Figure8–289 Opening the front cover

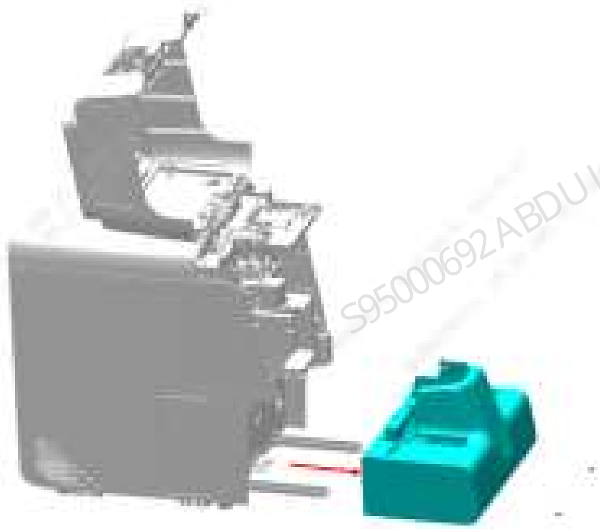


2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in **Figure8–289**.

3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.

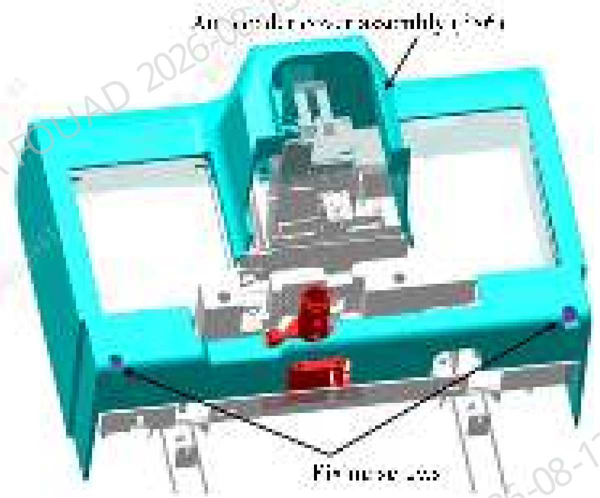
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader, closed-sampling model (5 x 6), as shown in **Figure8–292**.

Figure8–290 Removing the autoloader assembly



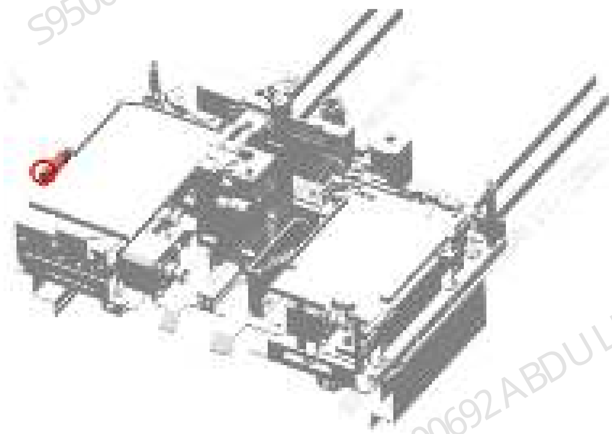
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (5 X 6), and then remove the autoloader front cover assembly (5 X 6), as shown in **Figure8–292**.

Figure8–291 Removing autoloader front cover assembly (5x6)



6. Remove the M3 x 14 screw fixing the PHOTOELEC photosensor reflective 940nm, pull out the wire on the PHOTOELEC photosensor reflective 940nm, and then remove the PHOTOELEC photosensor reflective 940nm, as shown in **Figure8–292**.

Figure8–292 Diagram of Removing PHOTOELEC Photosensor Reflective 940nm



Subsequent Requirements

1. Use one M3 x 14 screw to fix the PHOTOELEC photosensor reflective 940nm on theautoloader body, and insert the wire on the PHOTOELEC photosensor reflective 940nm.
2. Install the autoloader front cover assembly (5x6) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

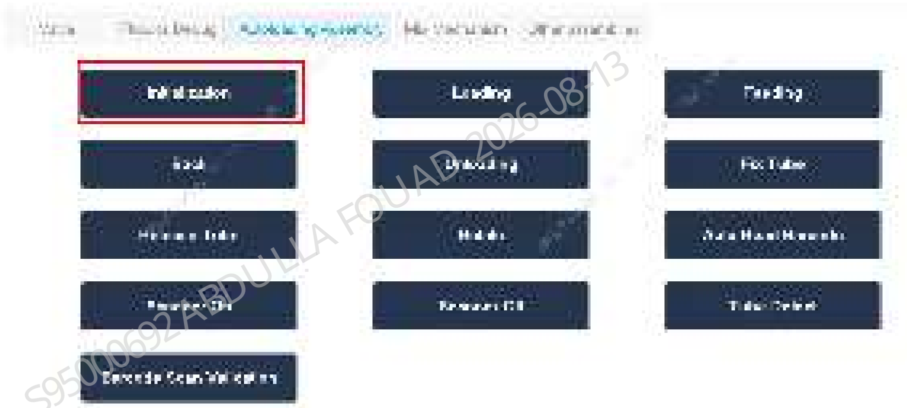
8.29.3 Commissioning and Verification

Procedures

Verification:

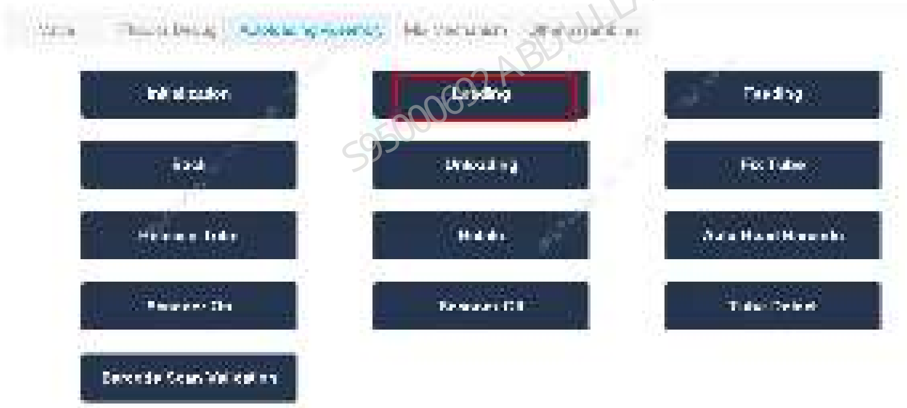
1. After the autoloader is connected to the main unit, power on the analyzer, log in at the service access level, select **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Autoloading Assembly**, execute assembly initialization, and check whether the self-test succeeds.

Figure8–293 Autoloader Self-test Interface



2. Place one test tube rack in the loading area, and execute the loading command. If the loading, feeding, and unloading operation succeeds and there are no relevant errors, the sensor is replaced successfully; otherwise it fails. Check whether the wire is connected correctly and in good contact.

Figure8–294 Autoloader Self-test Interface



8.30 PHOTOELEC emitter 940nm 400mm conn

8.30.1 General Information

Figure8–295 Photo of PHOTOELEC Emitter 940nm 400mm conn

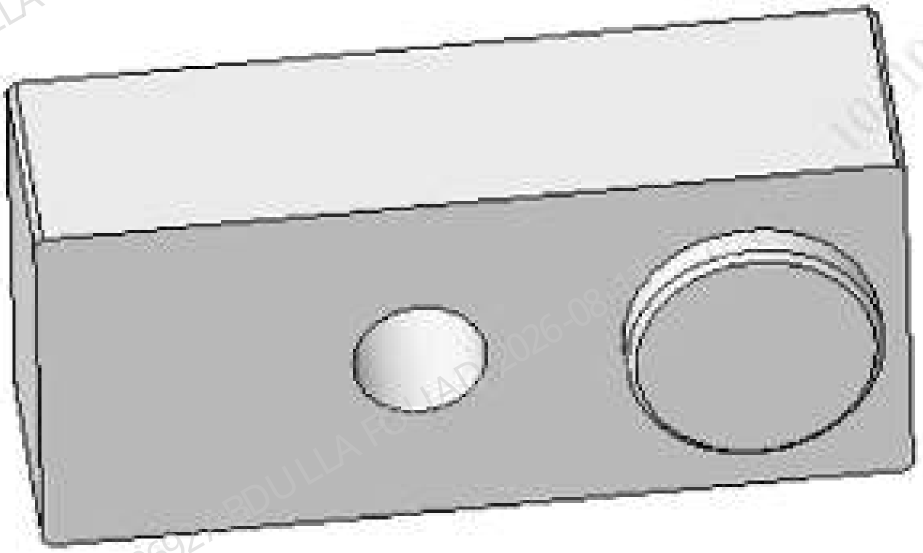


Figure8–296 Diagram of Location of PHOTOELEC Emitter 940nm 400mm conn in the Main Unit

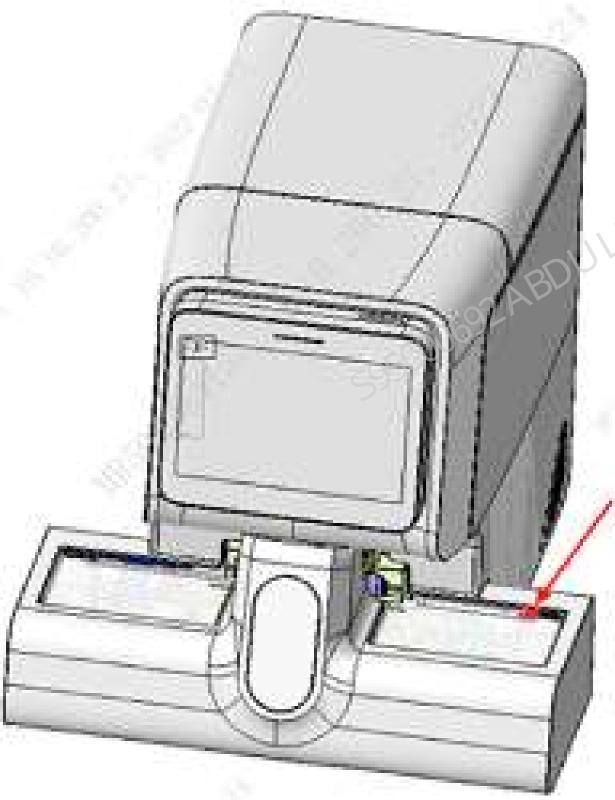


Figure8–297 Diagram of Location of PHOTOELEC Detector 900nm 400mm conn in the Main Unit

Table 8–32 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000334-00	PHOTOELEC emitter 940nm 400mm conn	N/A	N/A

Function:
Forming an transmissive sensor with PHOTOELEC detector 900nm 400mm conn in the loading area totransmit the test tube rack detection information signal on the loading tray.

8.30.2 Disassembly and Assembly

For the disassembly and assembly in the case of FRU-PHOTOELEC emitter 940nm 400mm conn (10x5 autoloader), see FRU-PHOTOELEC emitter 940nm 400mm conn (10x5 autoloader).
For the disassembly and assembly in the case of FRU-PHOTOELEC emitter 940nm 400mm conn (5x6 autoloader), see FRU-PHOTOELEC emitter 940nm 400mm conn (5X6 autoloader).

FRU-PHOTOELEC emitter 940nm 400mm conn (10x5 autoloader)

Required tools

Cross-headed screwdriver

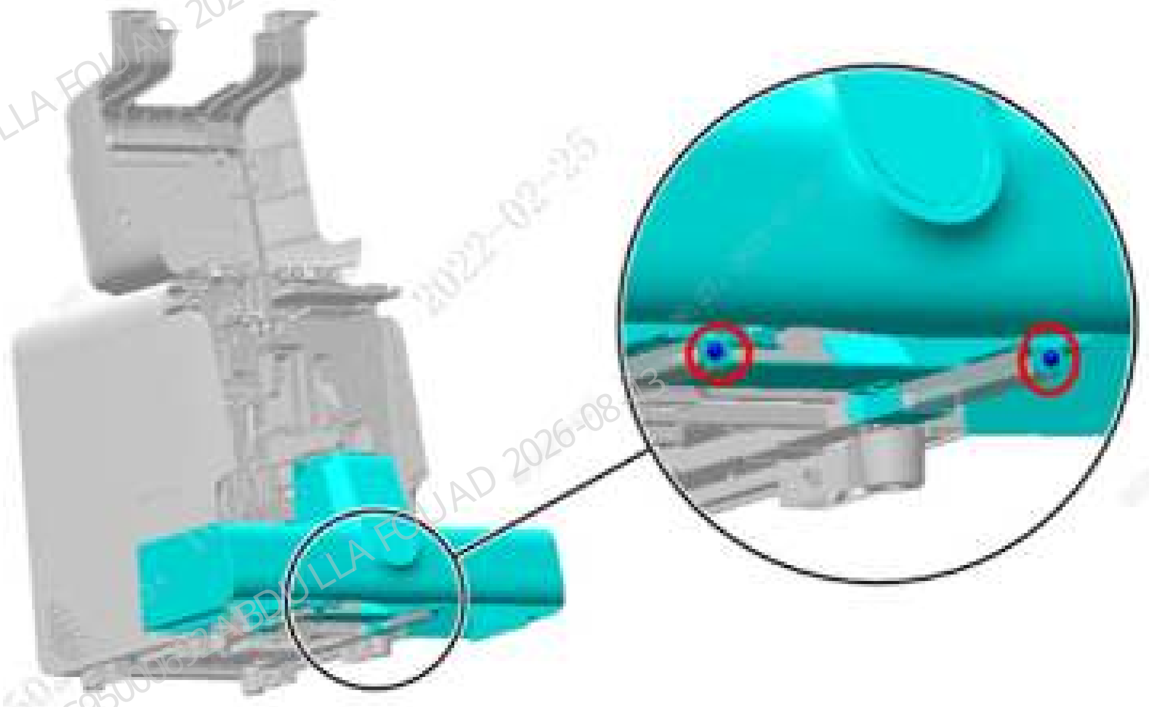
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

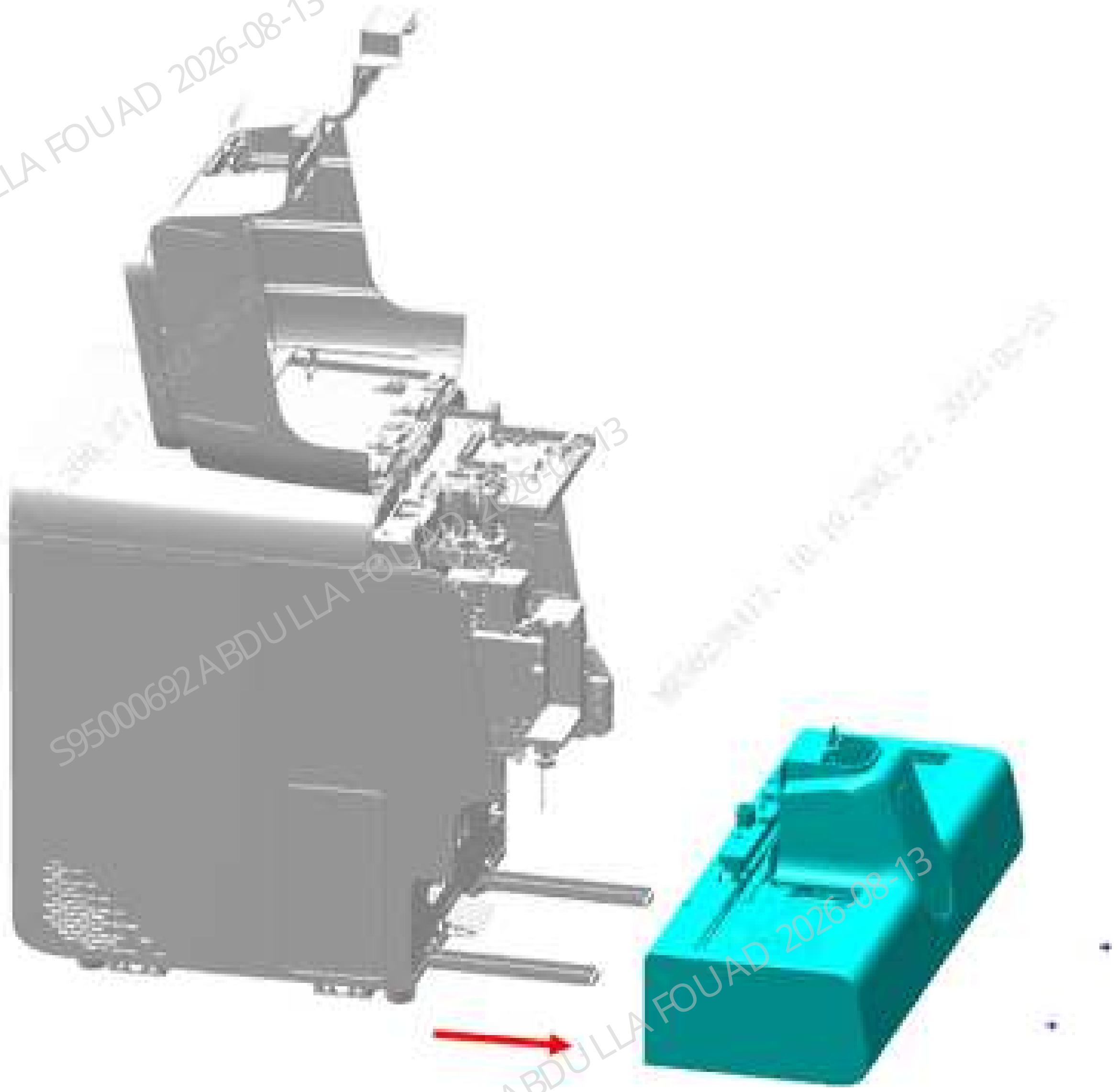
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–298 Opening the front cover



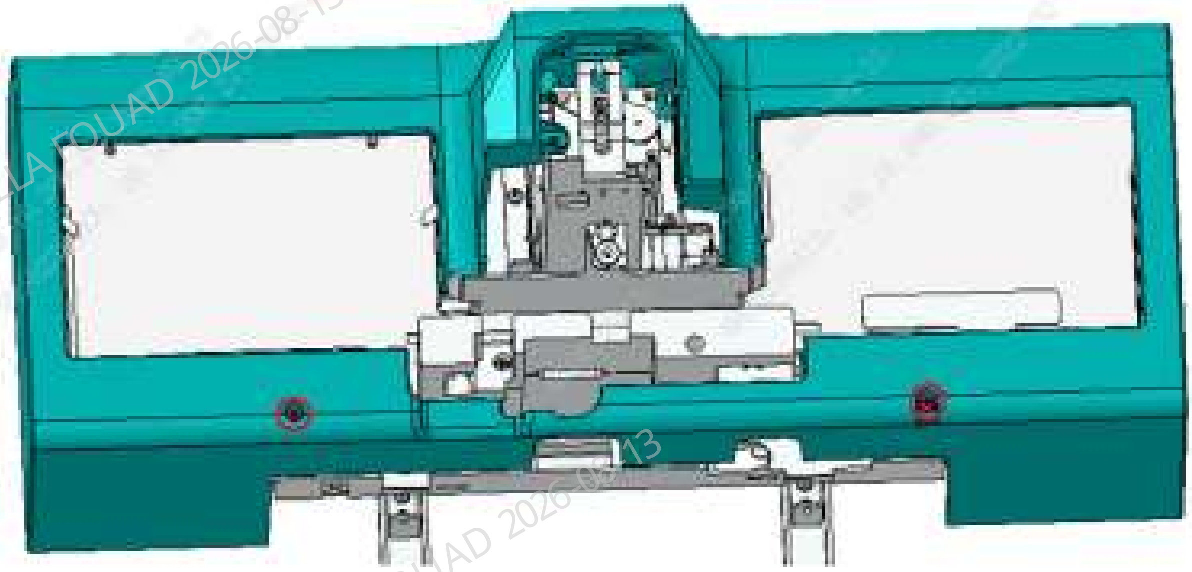
2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader assembly, as shown below.

Figure8–299 Removing the autoloader assembly



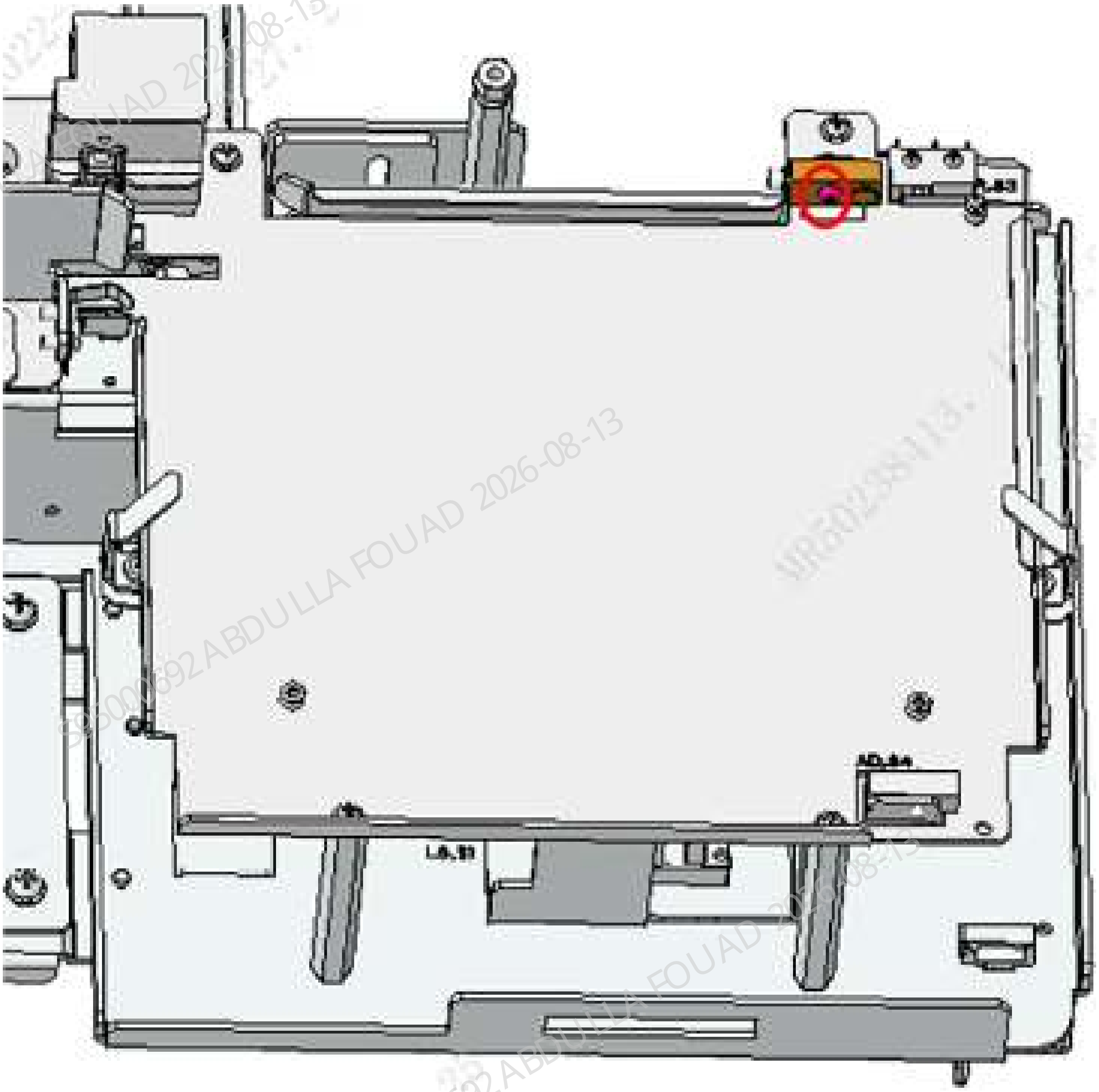
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (10x5), and then remove the autoloader front cover assembly (10x5), as shown the figure below.

Figure8–300 Removing autoloader front cover assembly (10x5)



6. Remove the M3x10 screw fixing PHOTOELEC emitter 940nm 400mm conn, remove the wire from the PHOTOELEC emitter 940nm 400mm conn, and then remove the PHOTOELEC emitter 940nm 400mm conn, as shown below.

Figure8–301 Removing the PHOTOELEC emitter 940nm 400mm conn



Subsequent Requirements

Installation procedure:

1. Use one M3x10 screw to fix the PHOTOELEC emitter 940nm 400mm conn on the autoloader body, and insert the wire on the PHOTOELEC emitter 940nm 400mm conn.
2. Install the autoloader front cover assembly (10x5) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.

5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

FRU-PHOTOELEC emitter 940nm 400mm conn (5X6 autoloader)

Required tools

Cross-headed screwdriver

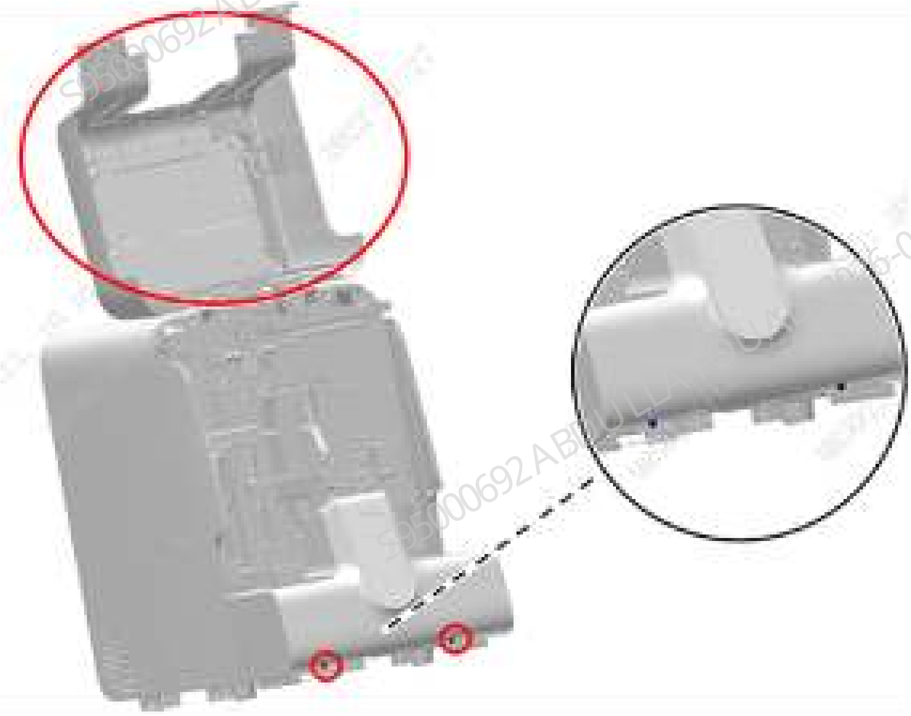
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

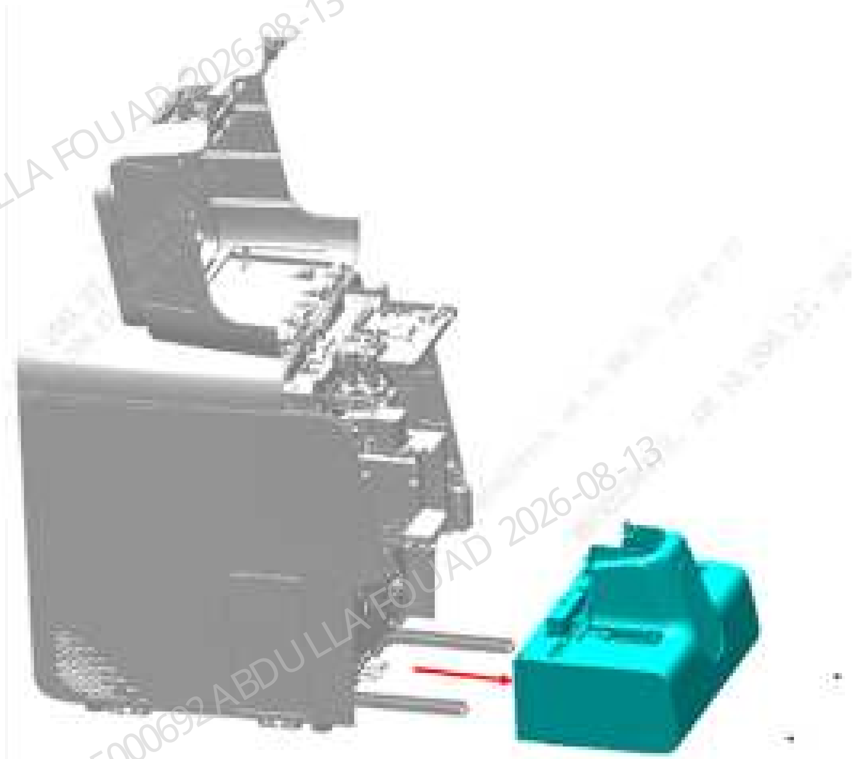
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–302 Opening the front cover



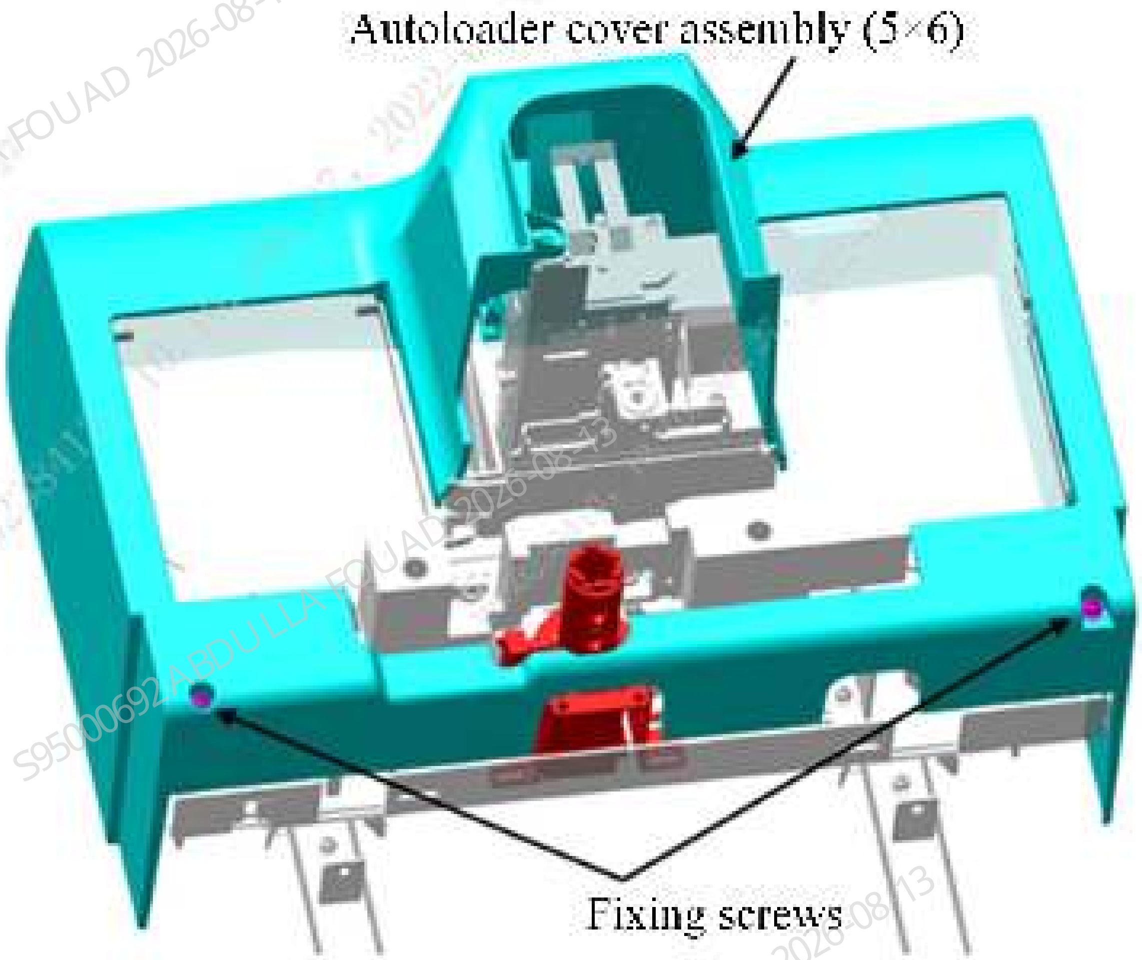
2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader, closed-sampling model (5 x 6), as shown below.

Figure8–303 Removing the autoloader assembly



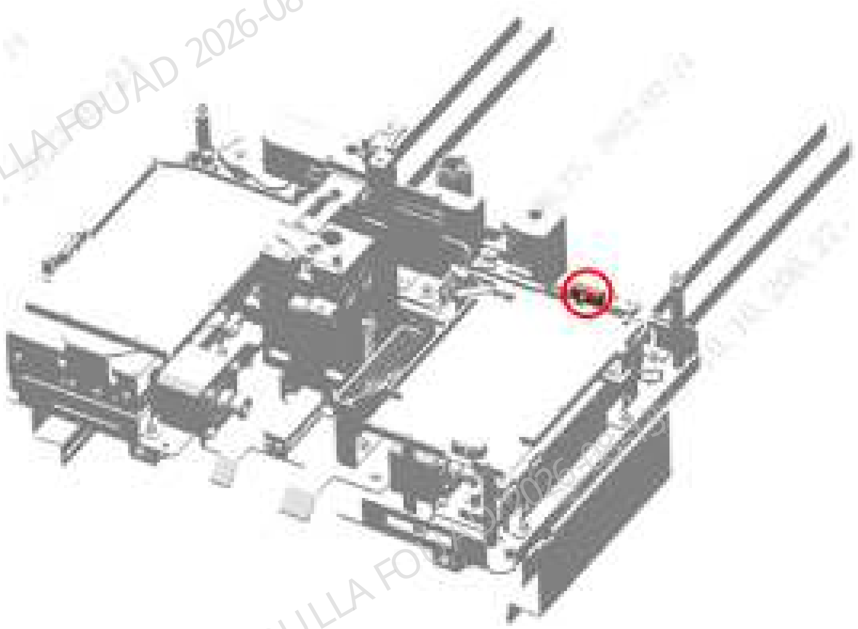
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (5x6), and then remove the autoloader front cover assembly (5x6), as shown in the figure below.

Figure8–304 Removing autoloader front cover assembly (10x5)



6. Remove the M3x10 screw fixing PHOTOELEC emitter 940nm 400mm conn, remove the wire from the PHOTOELEC emitter 940nm 400mm conn, and then remove the PHOTOELEC emitter 940nm 400mm conn, as shown below.

Figure8–305 Removing the PHOTOELEC emitter 940nm 400mm conn



Subsequent Requirements

Installation procedure:

1. Use one M3x10 screw to fix the PHOTOELEC emitter 940nm 400mm conn on the autoloader body, and insert the wire on the PHOTOELEC emitter 940nm 400mm conn.
2. Install the autoloader front cover assembly (5x6) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

8.30.3 Commissioning and Verification

For the commissioning and verification in the case of FRU-PHOTOELEC emitter 940nm 400mm conn (10X5 autoloader), see FRU-PHOTOELEC emitter 940nm 400mm conn (10x5 autoloader).
For the commissioning and verification in the case of FRU-PHOTOELEC emitter 940nm 400mm conn (5X6 autoloader), see FRU-PHOTOELEC emitter 940nm 400mm conn (5X6 autoloader).

FRU-PHOTOELEC emitter 940nm 400mm conn (10x5 autoloader)

Fixtures Required

1 tube rack

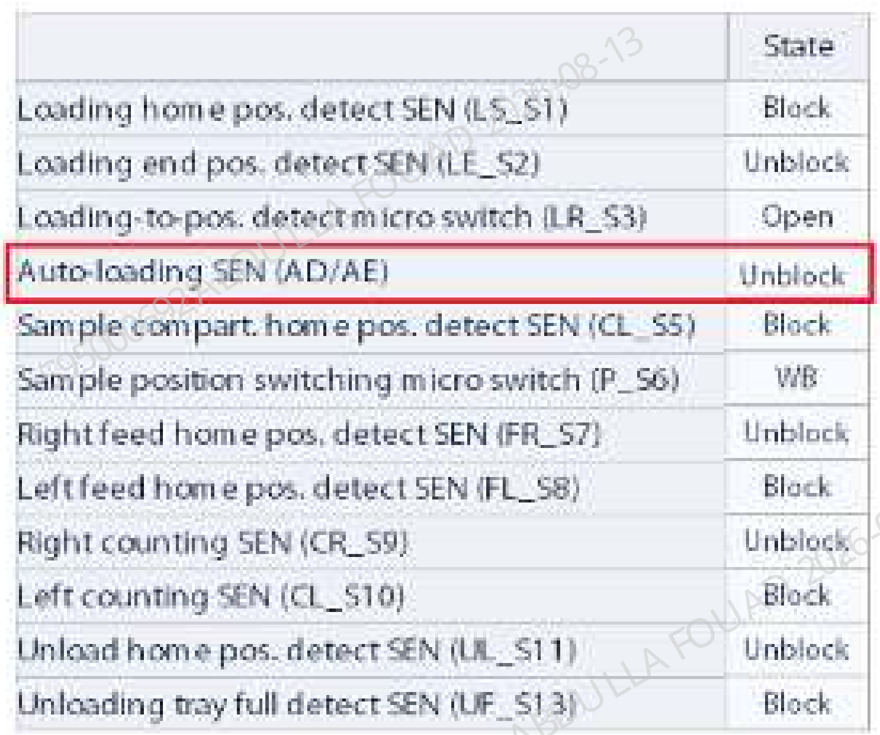
Procedures

Commissioning not required

Verification:

1. After connecting the autoloader to the main unit, power on the analyzer, log in at the service access level, select **Status>>Sensor Status>>Autoloader Motor Board Sensor**, and check whether the status of **Autoloader** is **Unshielded**.

Figure8–306 Sensor State Interface



	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Unblock
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

2. Place the test tube rack on the loading area and check whether the sensor is in the **Shielded** state now.

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Block
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

3. If the states of 1 and 2 are not satisfied, recheck the connection of sensor wire.

FRU-PHOTOELEC emitter 940nm 400mm conn (5X6 autoloader)

Fixtures Required

1 tube rack

Procedures

Commissioning not required

Verification:

1. After connecting the autoloader to the main unit, power on the analyzer, log in at the service access level, select **Status>>Sensor Status>>Autoloader Motor Board Sensor**, and check whether the status of **Autoloader** is **Unshielded**.

Figure8–307 Sensor State Interface

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Unblock
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

2. Place the test tube rack on the loading area and check whether the sensor is in the **Shielded** state now.

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Block
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

3. If the states of 1 and 2 are not satisfied, recheck the connection of sensor wire.

8.31 PHOTOELEC detector 900nm 400mm conn

8.31.1 General Information

Figure8–308 Photo of PHOTOELEC Detector 900nm 400mm conn

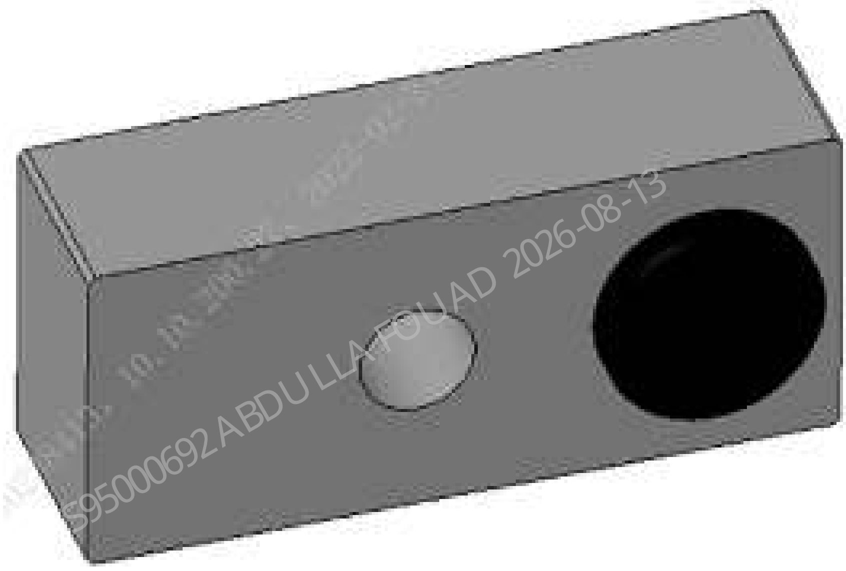


Figure8–309 Diagram of Location of PHOTOELEC Detector 900nm 400mm conn in the Main Unit (5X6 Autoloader)

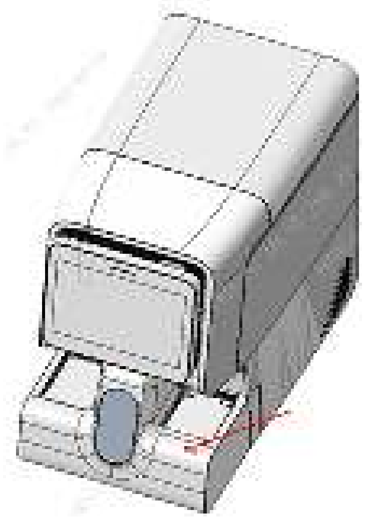


Figure8–310 Diagram of Location of PHOTOELEC Detector 900nm 400mm conn in the Main Unit (10X5 Autoloader)

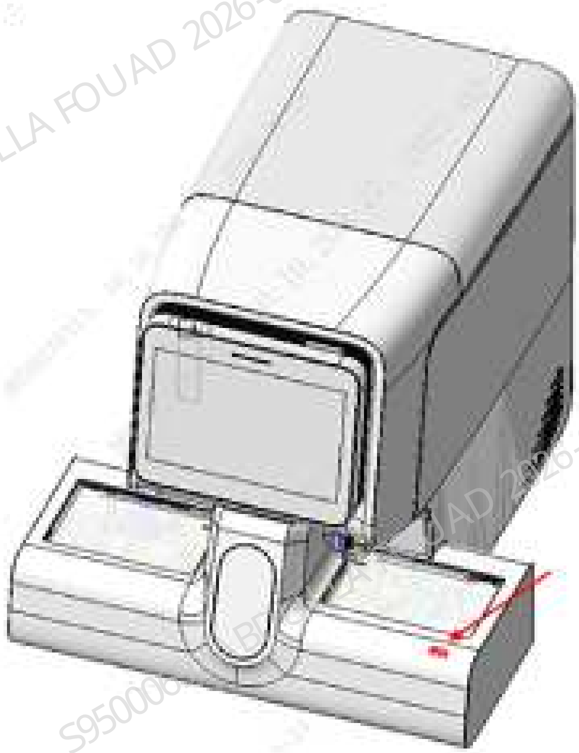


Table 8–33 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000335-00	PHOTOELEC detector 900nm 400mm conn	N/A	N/A

Function:
Forming an transmissive sensor with PHOTOELEC emitter 940nm 400mm conn in the loading area to receive the test tube rack detection information on the loading tray.

8.31.2 Disassembly and Assembly

For the disassembly and assembly in the case of FRU-PHOTOELEC detector 900nm 400mm conn (5x6 autoloader), see FRU-PHOTOELEC detector 900nm 400mm conn (5x6 autoloader).
For the disassembly and assembly in the case of FRU-PHOTOELEC detector 900nm 400mm conn (10x5 autoloader), see FRU-PHOTOELEC detector 900nm 400mm conn (10X5 autoloader).

FRU-PHOTOELEC detector 900nm 400mm conn (5x6 autoloader)

Required tools

Cross-headed screwdriver

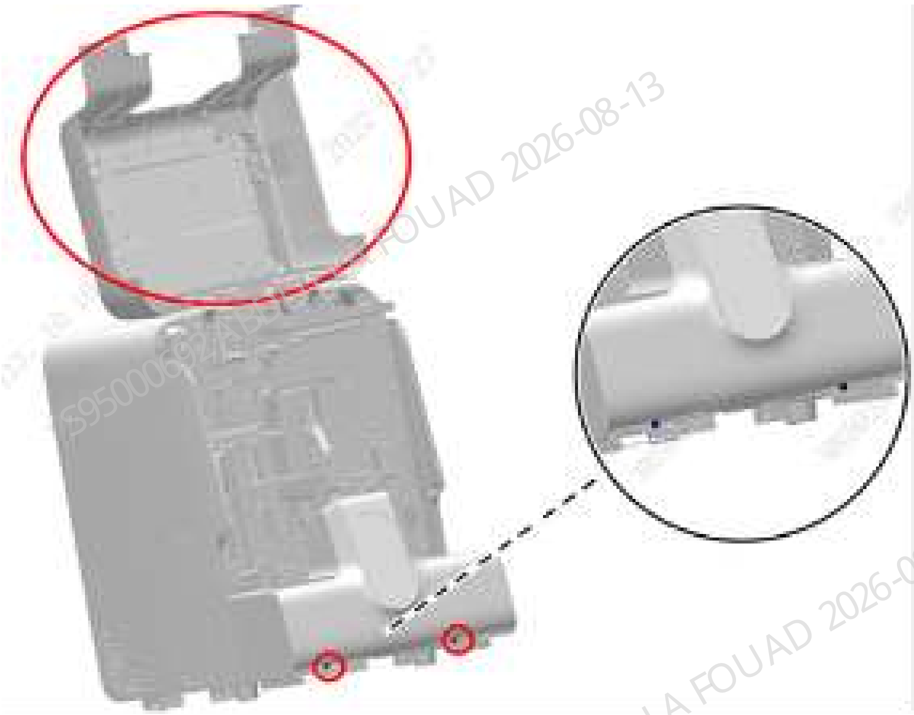
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

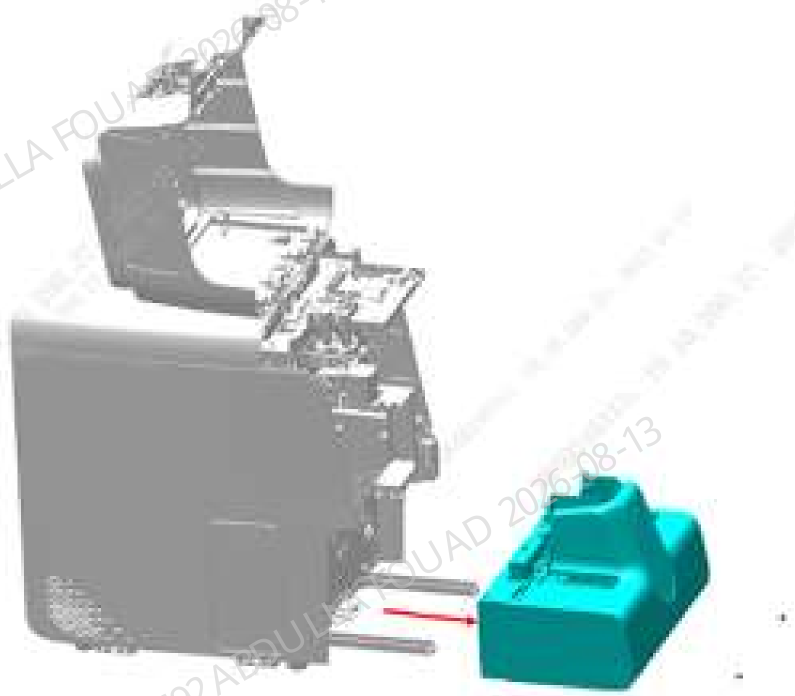
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–311 Opening the front cover



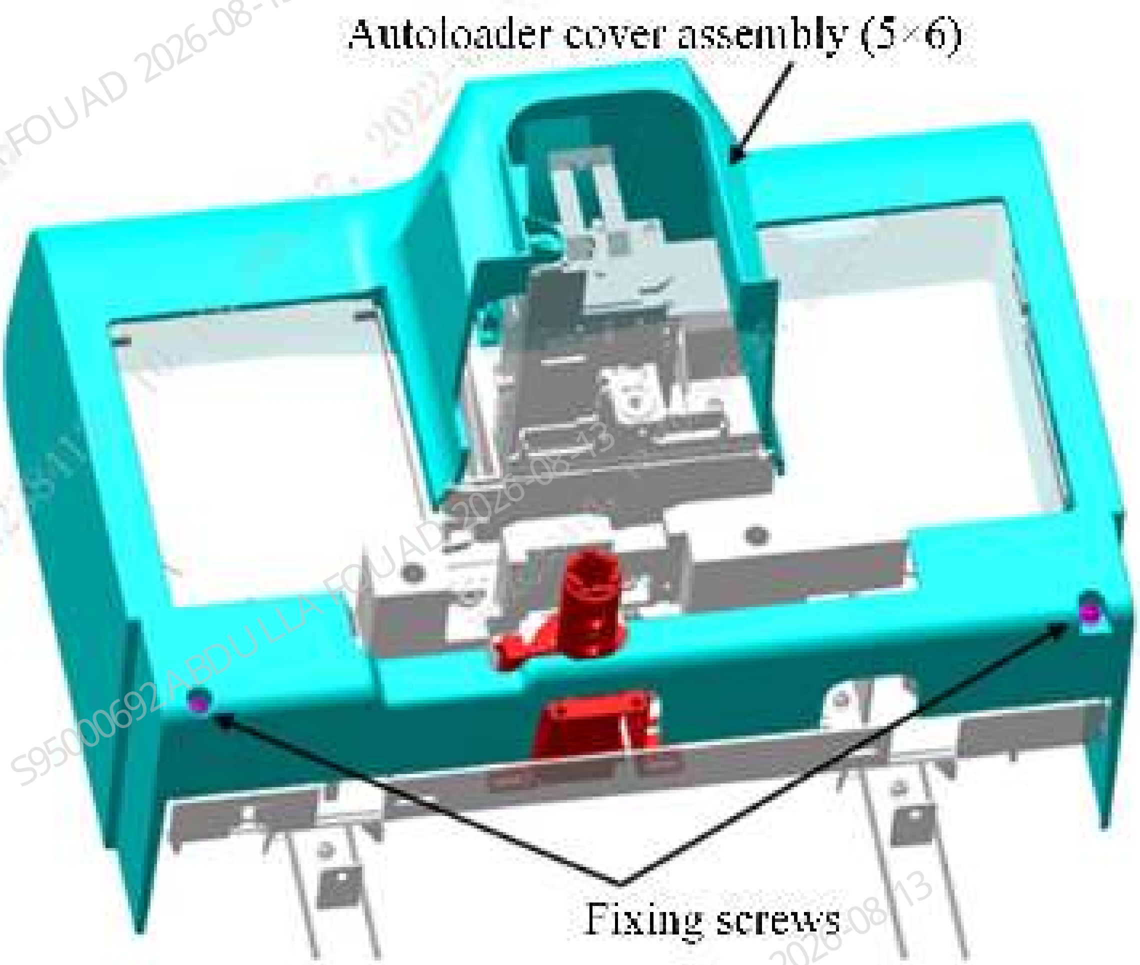
2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader, closed-sampling model (5 x 6), as shown below.

Figure8–312 Removing the autoloader assembly



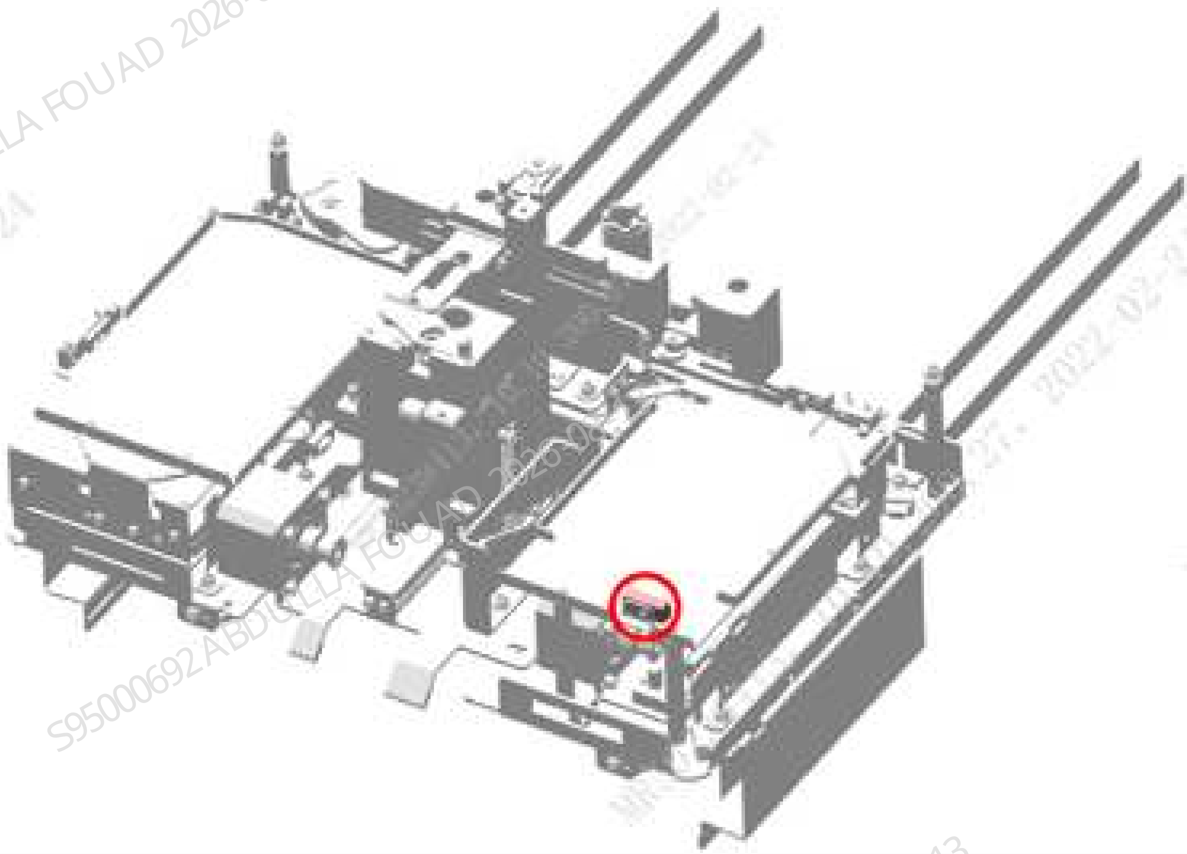
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (5x6), and then remove the autoloader front cover assembly (5x6), as shown in the figure below.

Figure8–313 Removing autoloader front cover assembly (5x6)



6. Remove the M3x10 screw fixing PHOTOELEC detector 900nm 400mm conn, remove the wire from the PHOTOELEC detector 900nm 400mm conn, and then remove the PHOTOELEC detector 900nm 400mm conn, as shown below.

Figure8–314 Removing the PHOTOELEC detector 900nm 400mm conn



Subsequent Requirements

Installation procedure:

1. Use one M3x10 screw to fix the PHOTOELEC detector 900nm 400mm conn on the autoloader body, and insert the wire on the PHOTOELEC detector 900nm 400mm conn.
2. Install the autoloader front cover assembly (5x6) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

FRU-PHOTOELEC detector 900nm 400mm conn (10X5 autoloader)

Required tools

Cross-headed screwdriver

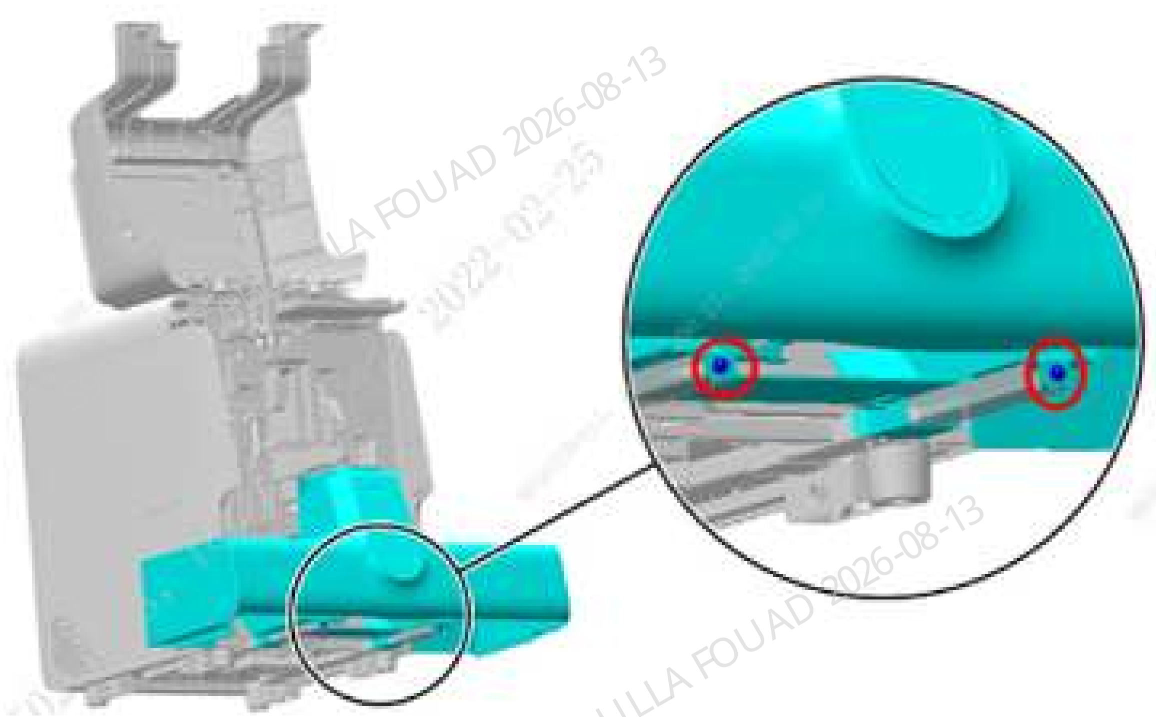
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

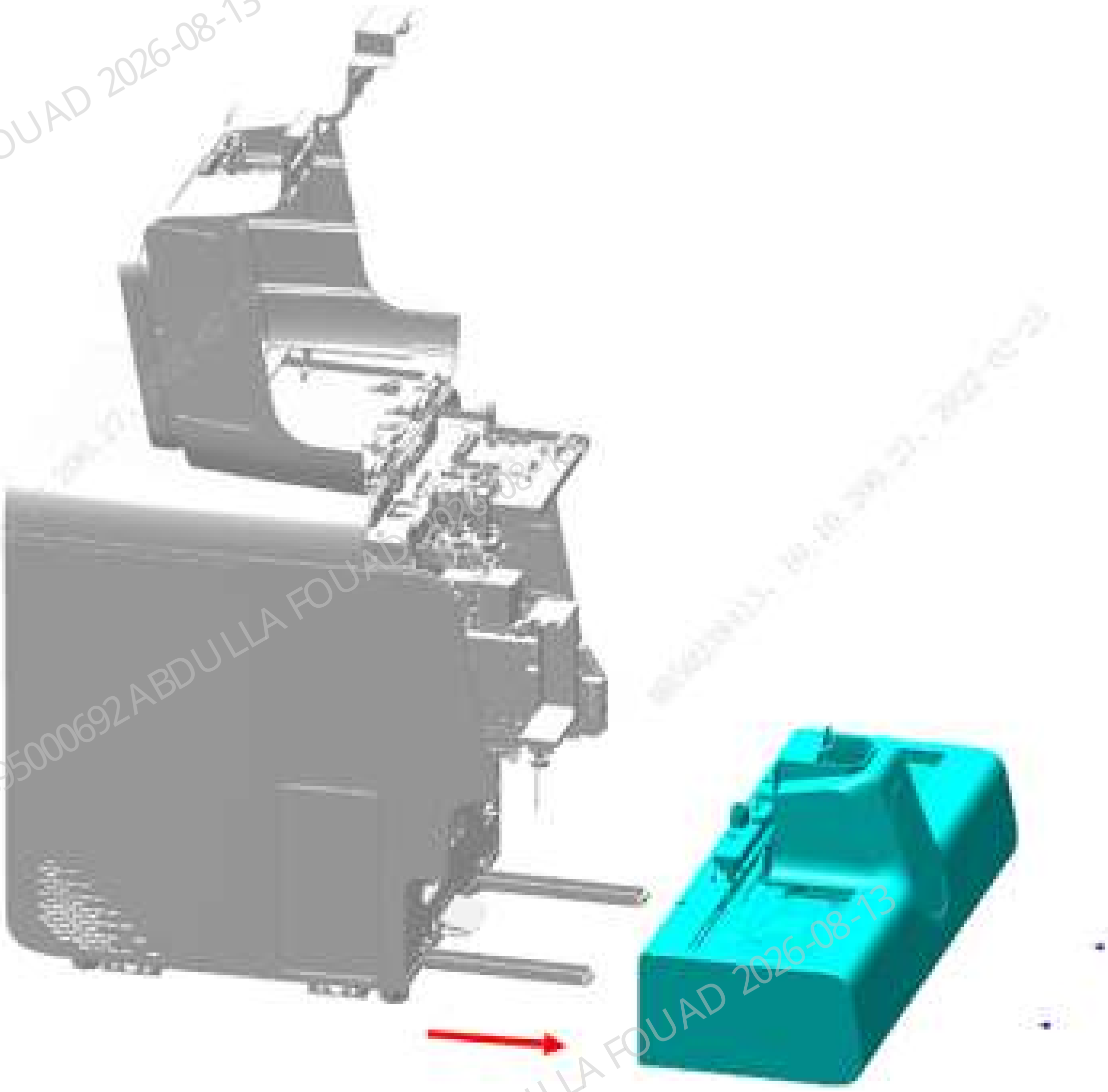
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–315 Opening the front cover



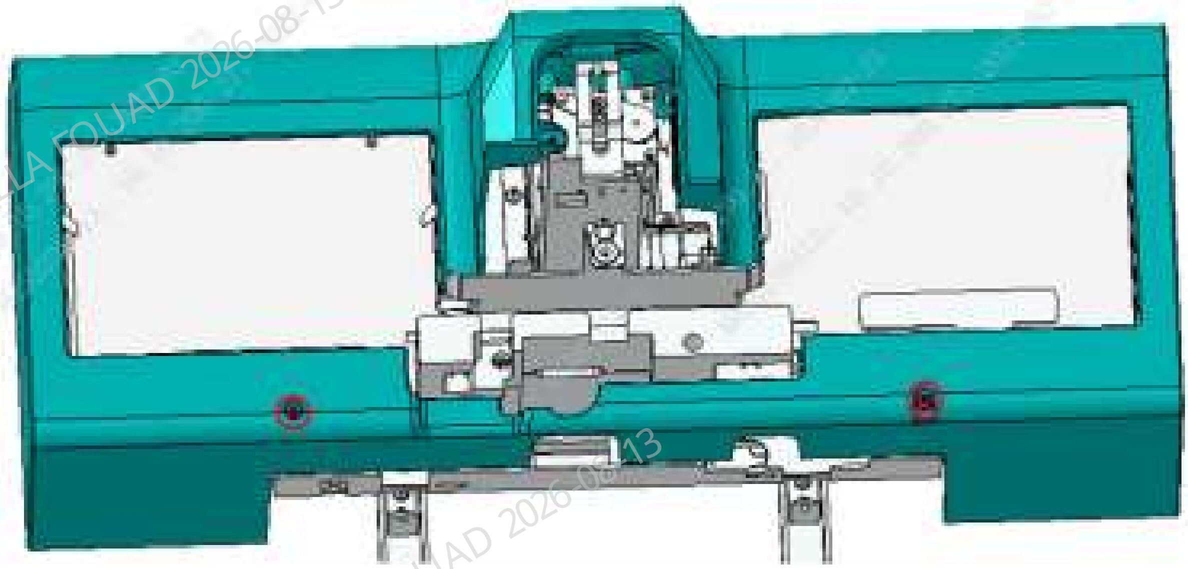
2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader assembly, as shown below.

Figure8–316 Removing the autoloader assembly



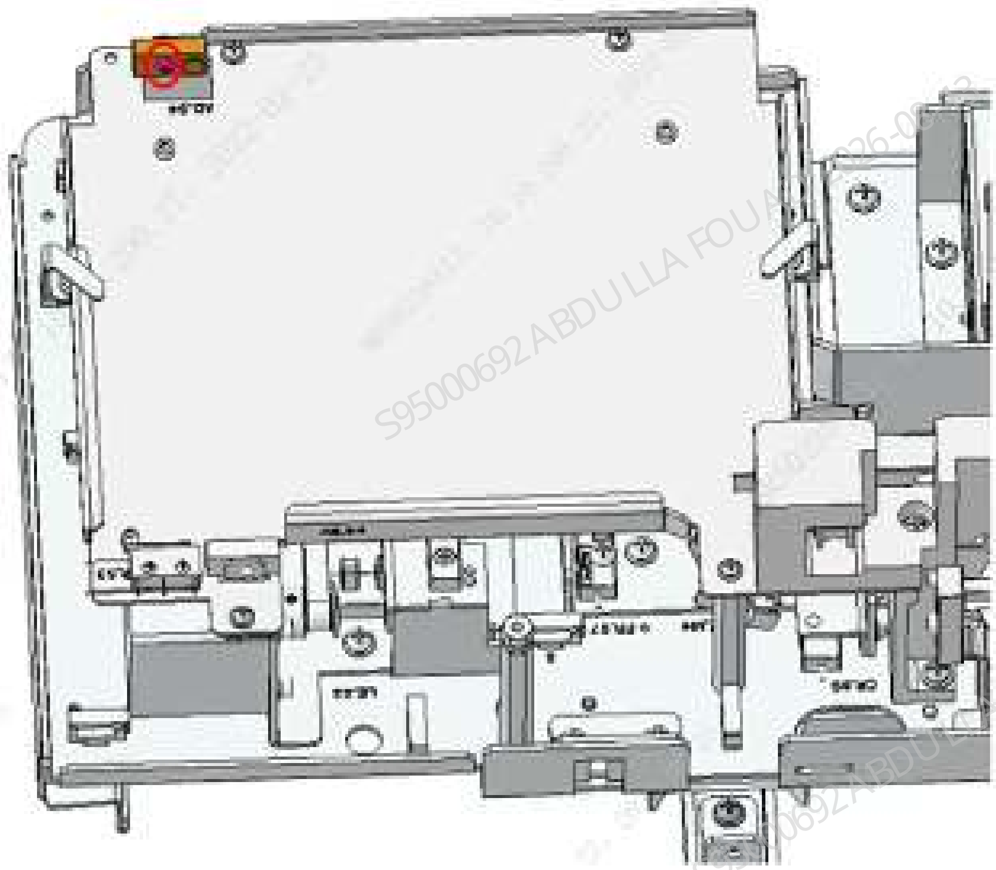
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (10x5), and then remove the autoloader front cover assembly (10x5), as shown the figure below.

Figure8–317 Removing autoloader front cover assembly (10x5)



6. Remove the M3x10 screw fixing PHOTOELEC detector 900nm 400mm conn, remove the wire from the PHOTOELEC detector 900nm 400mm conn, and then remove the PHOTOELEC detector 900nm 400mm conn, as shown below.

Figure8–318 Removing the PHOTOELEC detector 900nm 400mm conn



Subsequent Requirements

Installation procedure:

1. Use one M3x10 screw to fix the PHOTOELEC detector 900nm 400mm conn on the autoloader body, and insert the wire on the PHOTOELEC detector 900nm 400mm conn.
2. Install the autoloader front cover assembly (10x5) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

8.31.3 Commissioning and Verification

For the commissioning and verification in the case of FRU-PHOTOELEC detector 900nm 400mm conn (5X6 autoloader), see FRU-PHOTOELEC detector 900nm 400mm conn (5x6 autoloader).
For the commissioning and verification in the case of FRU-PHOTOELEC detector 900nm 400mm conn (10X5 autoloader), see FRU-PHOTOELEC detector 900nm 400mm conn (10X5 autoloader).

FRU-PHOTOELEC detector 900nm 400mm conn (5x6 autoloader)

Fixtures Required

1 tube rack

Procedures

Commissioning not required
Verification:

1. After connecting the autoloader to the main unit, power on the analyzer, log in at the service access level, select **Status**>>**Sensor Status**>>**Autoloader Motor Board Sensor**, and check whether the status of **Autoloader** is **Unshielded**.

Figure8–319 Sensor State Interface

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Unblock
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

2. Place the test tube rack on the loading area and check whether the sensor is in the **Shielded** state now.

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Block
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

3. If the states of 1 and 2 are not satisfied, recheck the connection of sensor wire.

FRU-PHOTOELEC detector 900nm 400mm conn (10X5 autoloader)

Fixtures Required

1 tube rack

Procedures

Commissioning not required

Verification:

1. After connecting the autoloader to the main unit, power on the analyzer, log in at the service access level, select **Status>>Sensor Status>>Autoloader Motor Board Sensor**, and check whether the status of **Autoloader** is **Unshielded**.

Figure8–320 Sensor State Interface

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Unblock
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

2. Place the test tube rack on the loading area and check whether the sensor is in the **Shielded** state now.

	State
Loading home pos. detect SEN (LS_S1)	Block
Loading end pos. detect SEN (LE_S2)	Unblock
Loading-to-pos. detect micro switch (LR_S3)	Open
Auto-loading SEN (AD/AE)	Block
Sample compart. home pos. detect SEN (CL_S5)	Block
Sample position switching micro switch (P_S6)	WB
Right feed home pos. detect SEN (FR_S7)	Unblock
Left feed home pos. detect SEN (FL_S8)	Block
Right counting SEN (CR_S9)	Unblock
Left counting SEN (CL_S10)	Block
Unload home pos. detect SEN (UL_S11)	Unblock
Unloading tray full detect SEN (UF_S13)	Block

3. If the states of 1 and 2 are not satisfied, recheck the connection of sensor wire.

8.32 SWITCH micro-switch minimum DIP

8.32.1 General Information

Figure8–321 Photo of Micro Limit Switch

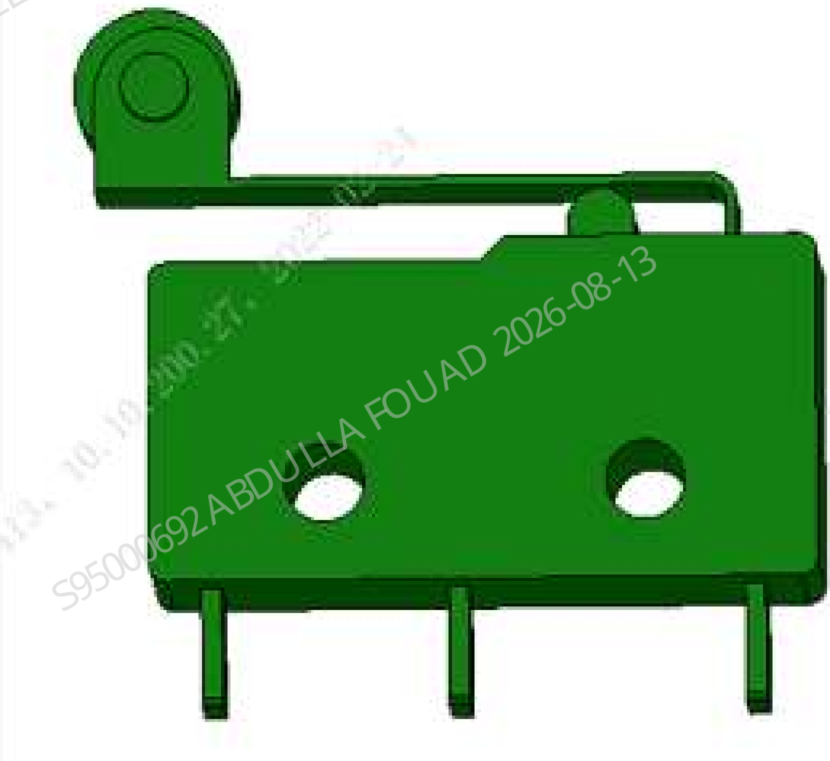


Figure8–322 Diagram of Location of Micro Limit Switch in the Main Unit (10X5 Autoloader)

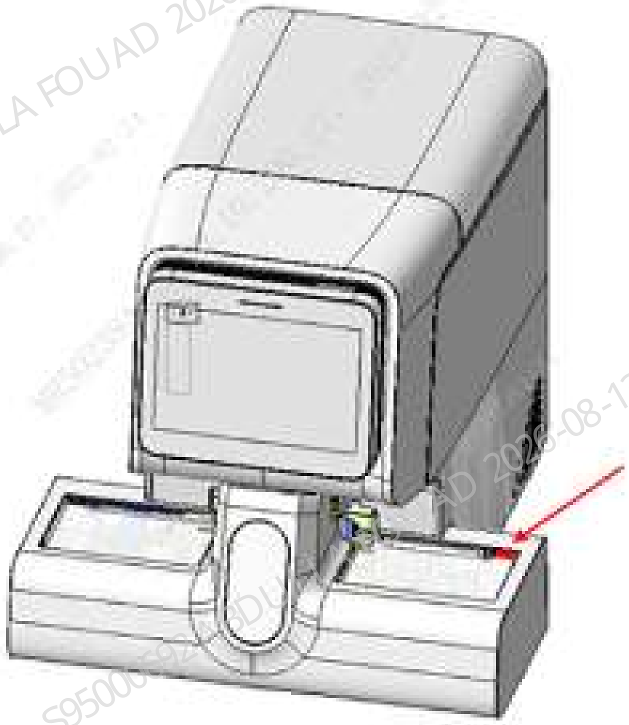


Figure8–323 Diagram of Location of Micro Limit Switch in the Main Unit (5X6 Autoloader)

Table 8–34 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
M07-00143S—	SWITCH Micro-switch Minimum DIP	N/A	N/A

Function:

Checking whether tube racks are loaded in place in the loading area.

8.32.2 Disassembly and Assembly

For the disassembly and assembly in the case of FRU-SWITCH micro switch minimum DIP (10x5 autoloader), see FRU-SWITCH micro switch minimum DIP (10x5 autoloader).
For the disassembly and assembly in the case of FRU-SWITCH micro switch minimum DIP (5x6 autoloader), see SWITCH Micro Switch minimum DIP (5x6 Autoloader).

FRU-SWITCH micro switch minimum DIP (10x5 autoloader)

Required tools

Cross-headed screwdriver

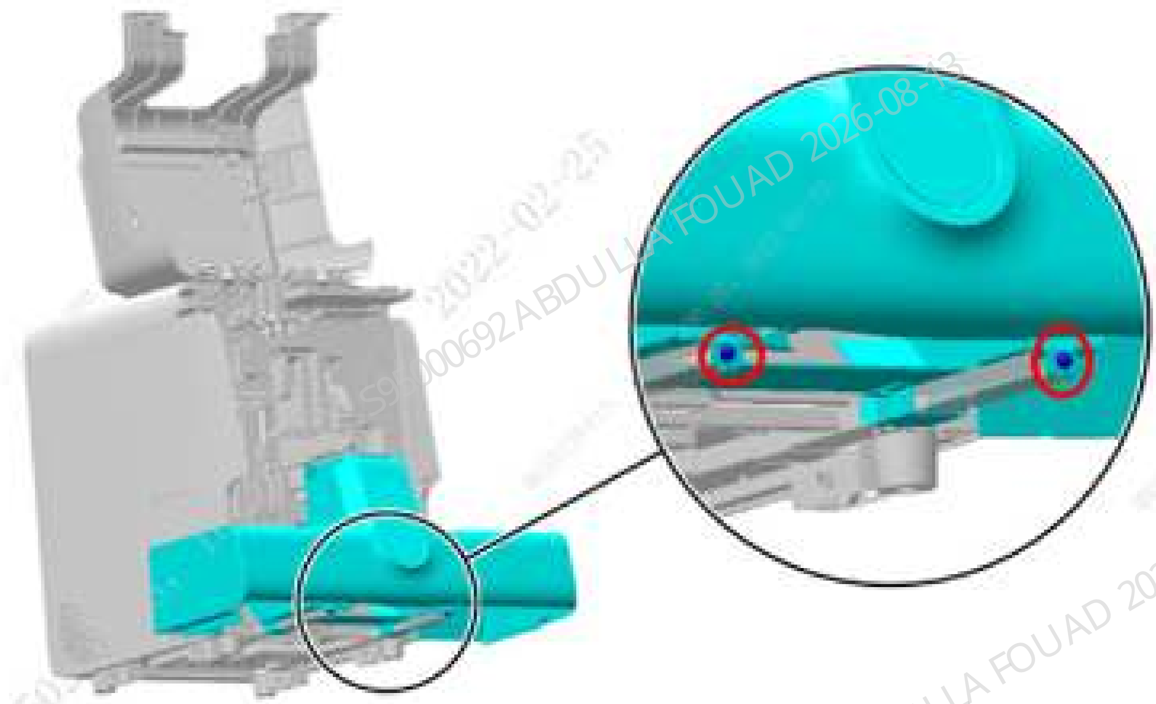
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

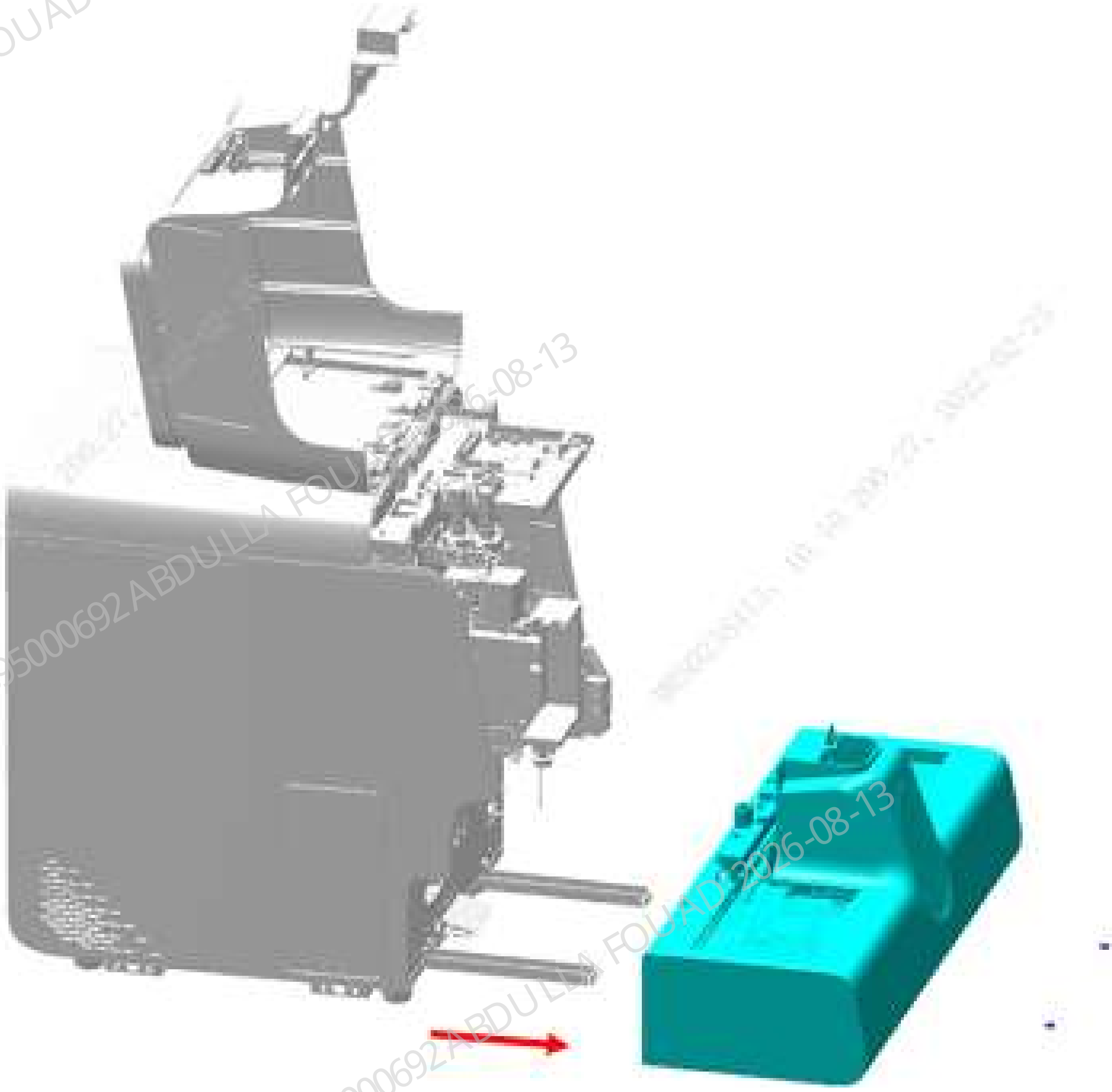
Figure8--324 Opening the front cover



2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.

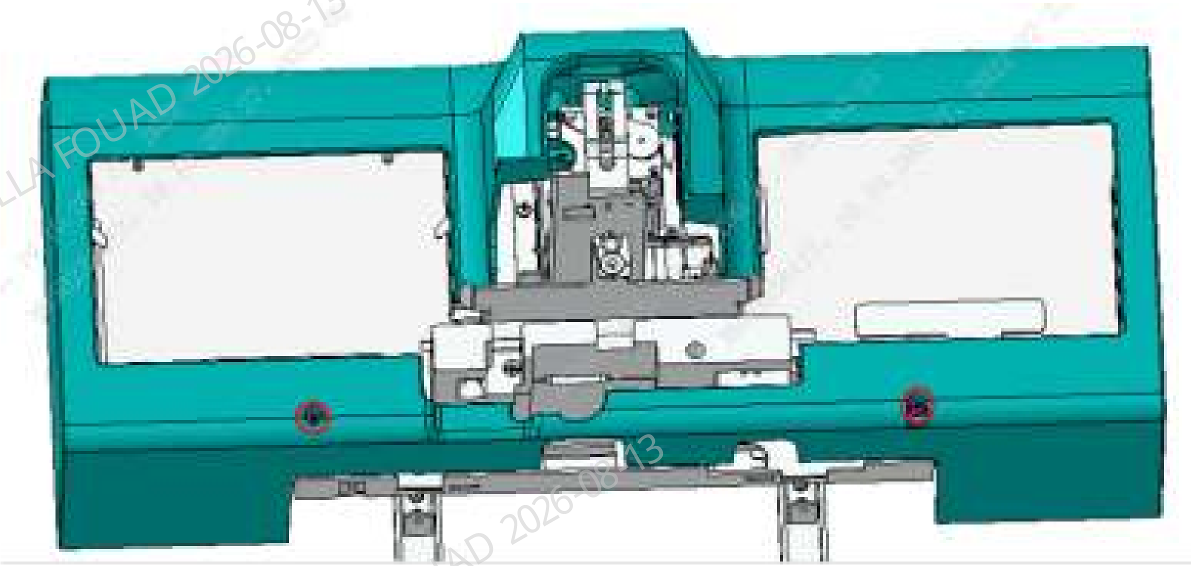
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader assembly, as shown below.

Figure8–325 Removing the autoloader assembly



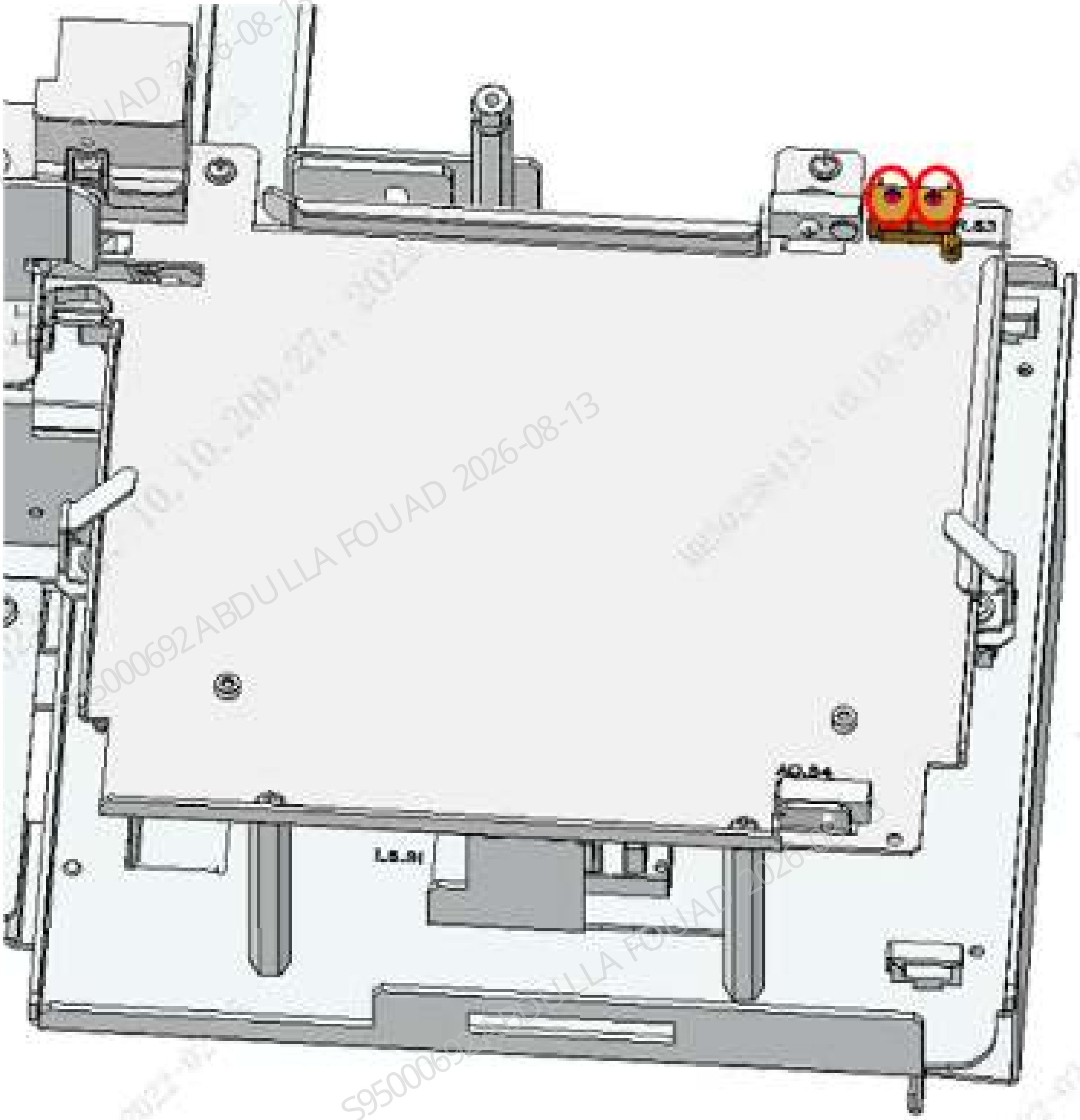
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (10x5), and then remove the autoloader front cover assembly (10x5), as shown the figure below.

Figure8–326 Removing autoloader front cover assembly (10x5)



6. Remove the two M2x10 screws fixing the SWITCH micro switch minimum DIP, pull out the wire of the SWITCH micro switch minimum DIP, and then remove the SWITCH micro switch minimum DIP, as shown in the figure below.

Figure8–327 Removing the SWITCH micro switch minimum DIP



Subsequent Requirements

Installation procedure:

1. Use two M2x10 screw to fix the SWITCH micro switch minimum DIP on the autoloader body, and insert the wire on the SWITCH micro switch minimum DIP.
2. Install the autoloader front cover assembly (10x5) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.

5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

SWITCH Micro Switch minimum DIP (5x6 Autoloader)

Required tools

Cross-headed screwdriver

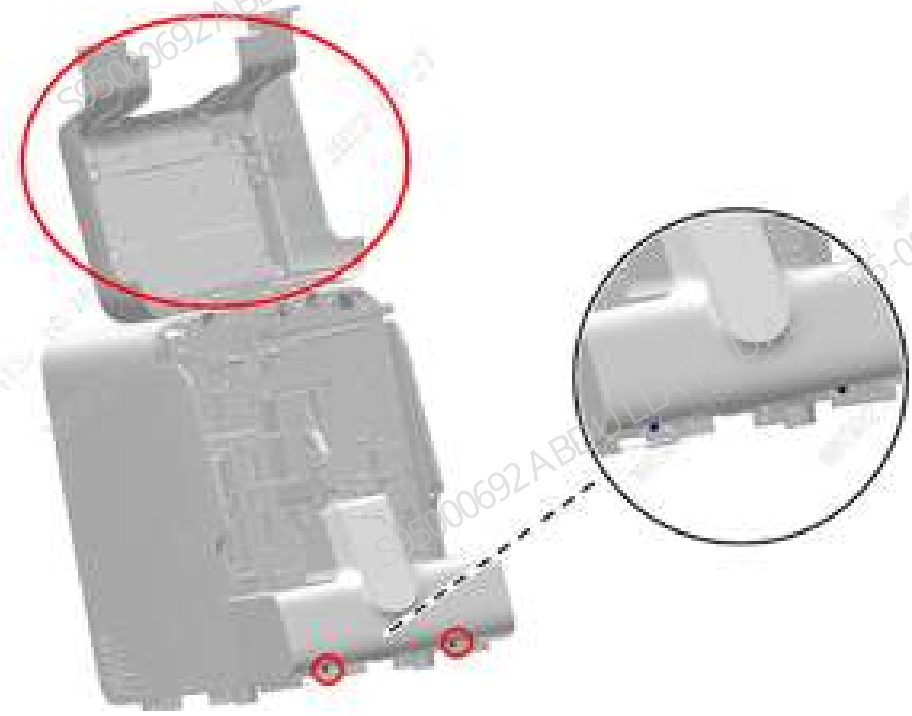
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

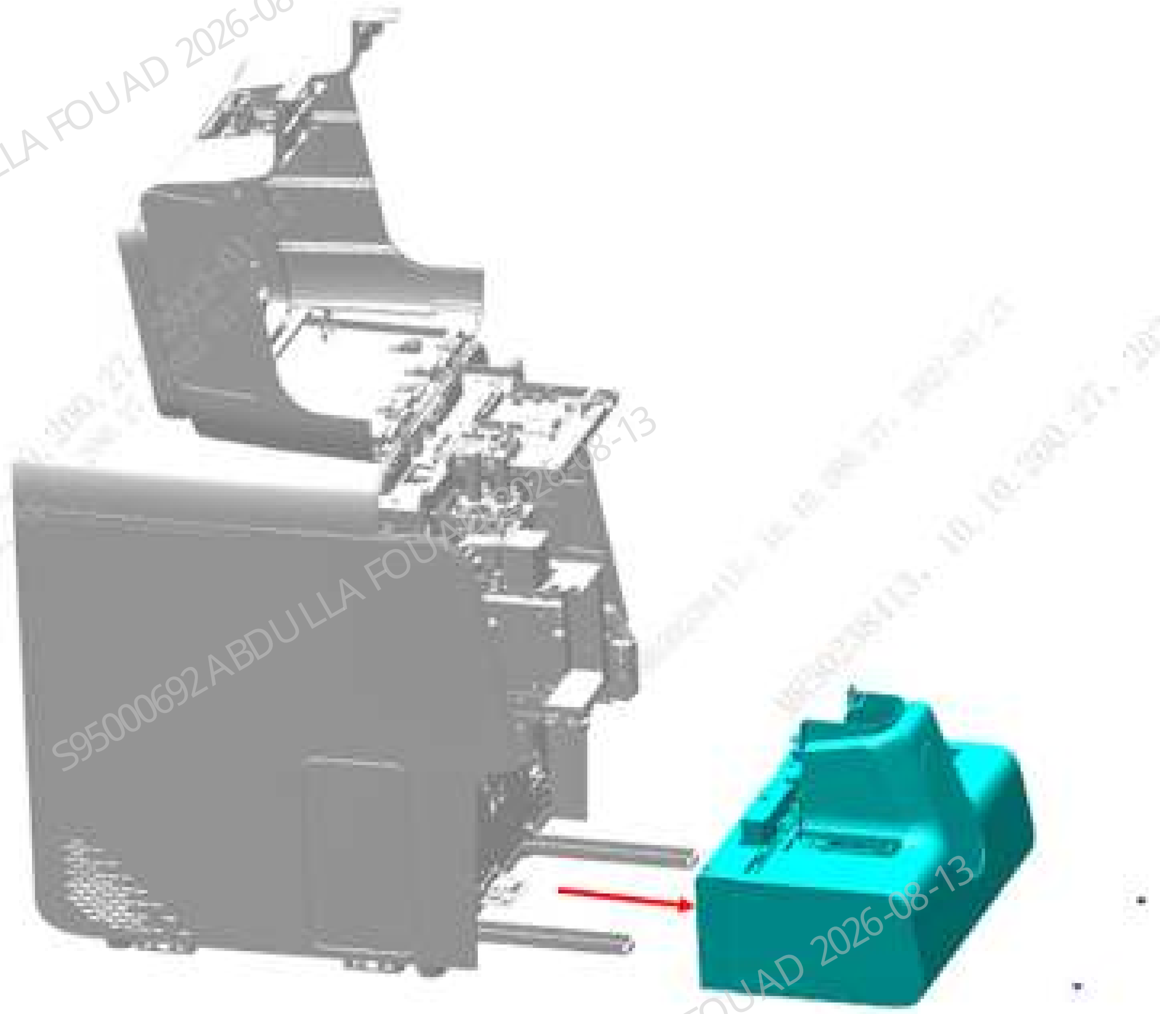
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–328 Opening the front cover



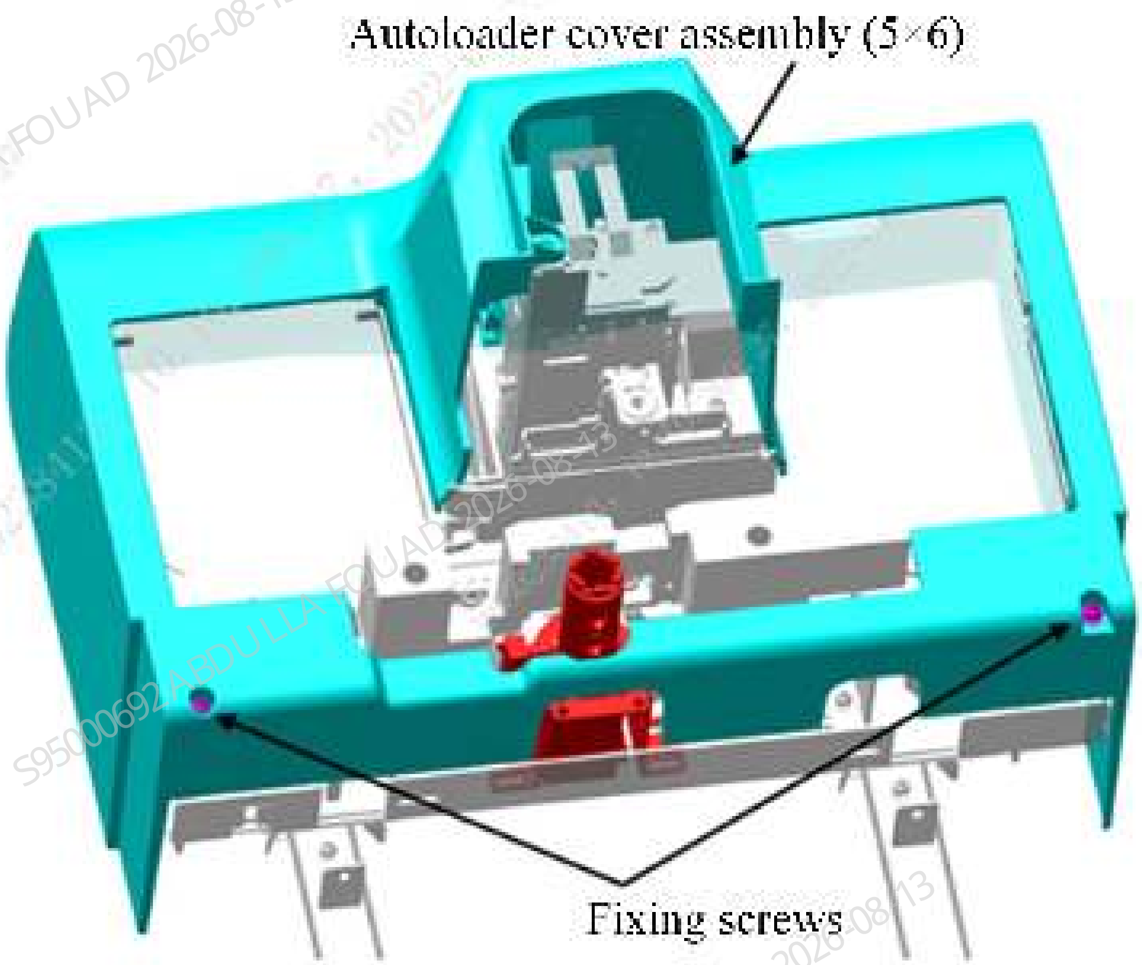
2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the autoloader, closed-sampling model (5 x 6), as shown below.

Figure8–329 Removing the autoloader assembly



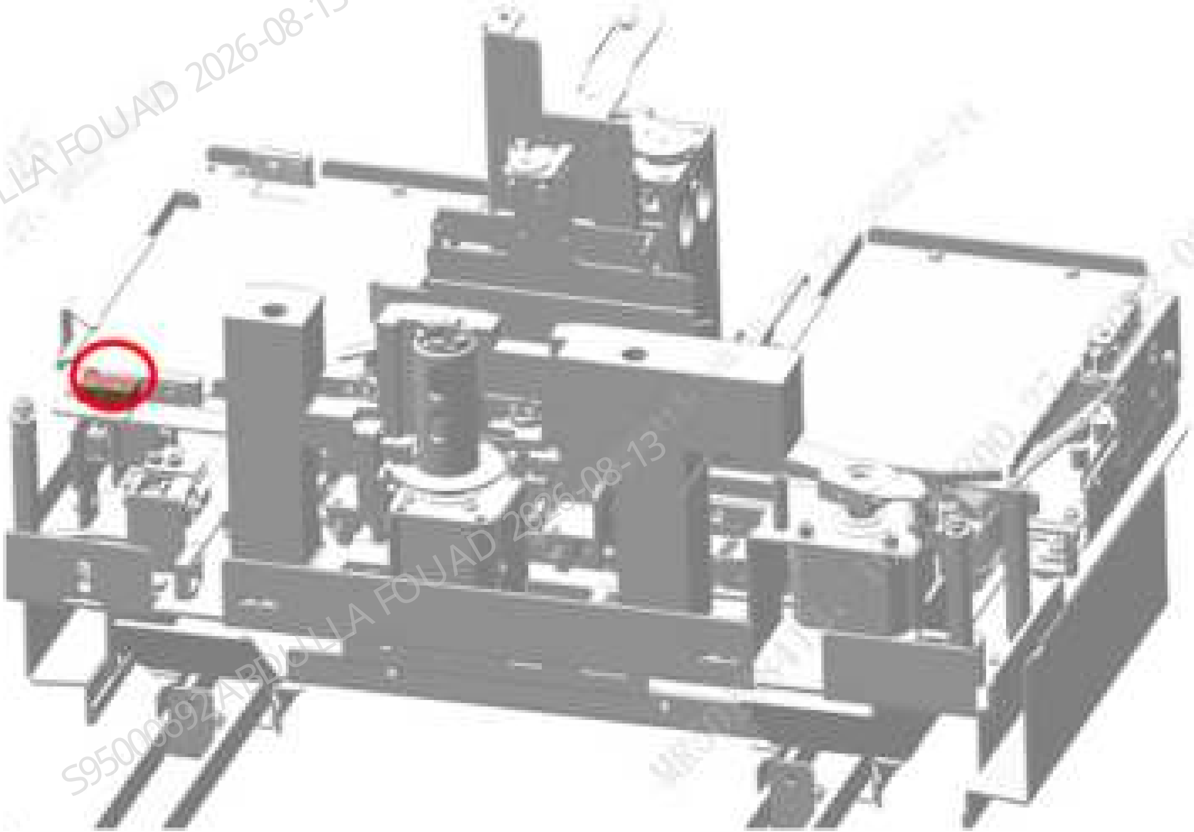
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (5x6), and then remove the autoloader front cover assembly (5x6), as shown in the figure below.

Figure8–330 Removing autoloader front cover assembly (5x6)



6. Remove the two M2x10 screws fixing the SWITCH micro switch minimum DIP, pull out the wire of the SWITCH micro switch minimum DIP, and then remove the SWITCH micro switch minimum DIP, as shown in the figure below.

Figure8–331 Removing the SWITCH micro switch minimum DIP



Subsequent Requirements

Installation procedure:

1. Use two M2x10 screw to fix the SWITCH micro switch minimum DIP on the autoloader body, and insert the wire on the SWITCH micro switch minimum DIP.
2. Install the autoloader front cover assembly (5x6) on the autoloader body.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

8.32.3 Commissioning and Verification

For the commissioning and verification in the case of FRU-SWITCH micro limit switch DIP (10X5 autoloader), see FRU-SWITCH micro switch minimum DIP (10x5 autoloader).

For the commissioning and verification in the case of FRU-SWITCH micro limit switch DIP (5X6 autoloader), see SWITCH Micro Switch minimum DIP (5x6 Autoloader).

FRU-SWITCH micro switch minimum DIP (10x5 autoloader)

Procedures

Commissioning not required

Verification:

1. After the autoloader is connected to the main unit, power on the analyzer, log in at the service access level, select **Service>>Commissioning & Self-Test>>Self-Test>>Autoloading Assembly**, execute the assembly initialization, and check whether the self-test succeeds.

Figure8–332 Autoloader Self-test Interface

2. Place one test tube rack in the loading area, and execute the loading command. If the loading and feeding operations succeed and there are no relevant errors, the sensor is replaced successfully; otherwise it fails. Check whether the wire is connected correctly and in good contact.

SWITCH Micro Switch minimum DIP (5x6 Autoloader)

Procedures

Commissioning not required

Verification:

1. After the autoloader is connected to the main unit, power on the analyzer, log in at the service access level, select **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Autoloading Assembly**, execute the assembly initialization, and check whether the self-test succeeds.

Figure8–333 Autoloader Self-test Interface

2. Place one test tube rack in the loading area, and execute the loading command. If the loading and feeding operations succeed and there are no relevant errors, the sensor is replaced successfully; otherwise it fails. Check whether the wire is connected correctly and in good contact.

8.33 Autoloader, closed-sampling model(10x5)

8.33.1 General Information

Figure8–334 Photo of 115-084952-04 Autoloader, closed-sampling model(10x5)

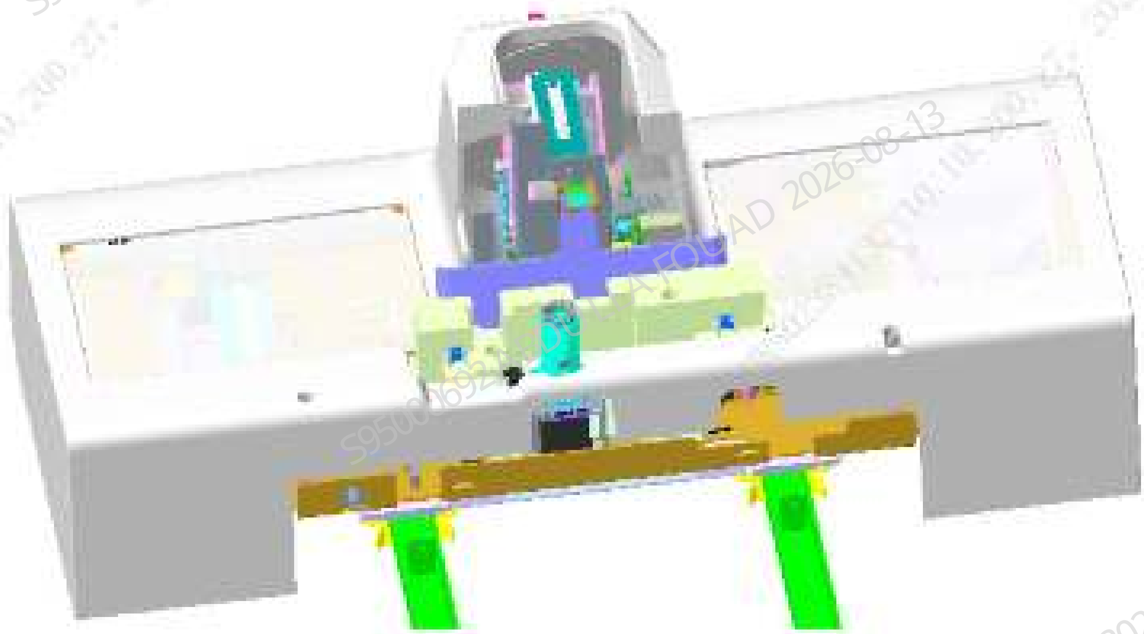


Figure8–335 Diagram of Location of Autoloader, closed-sampling model(10x5) in the Main Unit

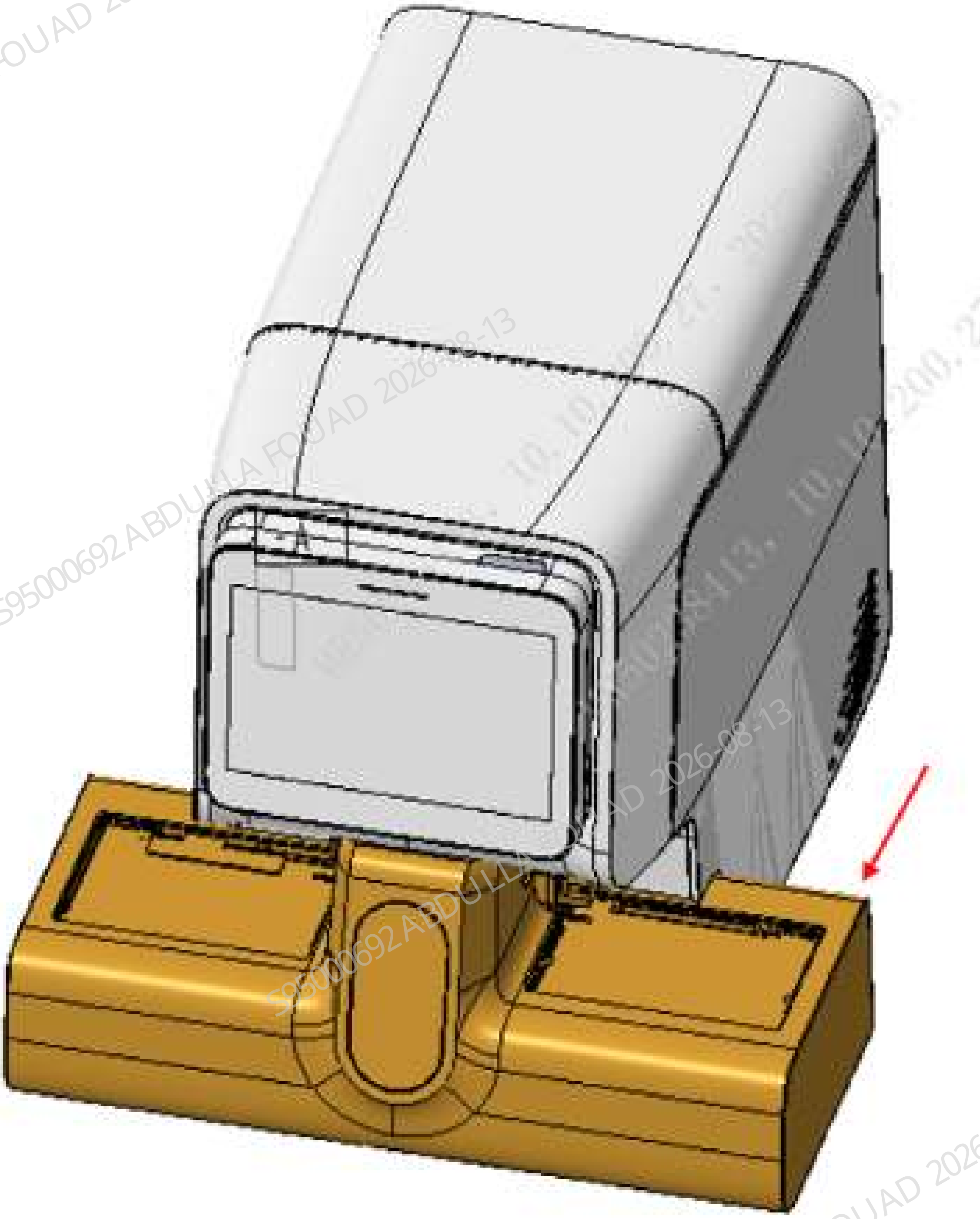


Table 8–35 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-084952-04	Autoloader, closed-sampling model(10x5)	115-059469-00	Peripheral Blood Blending Assembly
		115-085180-00	Autoloader, closed-sampling model (10x5) (excluding capillary blood mixing)

Function:

Providing a closed sample compartment, allowing placing and taking a single sample, able to automatically send the sample for sample collection, and support ordinary WB test tubes and micro blood test tubes. Providing the autoloading function at the same time, able to store and transport sample racks, transport the sample rack to be tested from the loading tray to the sample analysis area, and send the tested sample rack to the unloading tray. The loading tray is used to store the sample rack to be tested, and the unloading tray is used to store the tested sample rack.

8.33.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

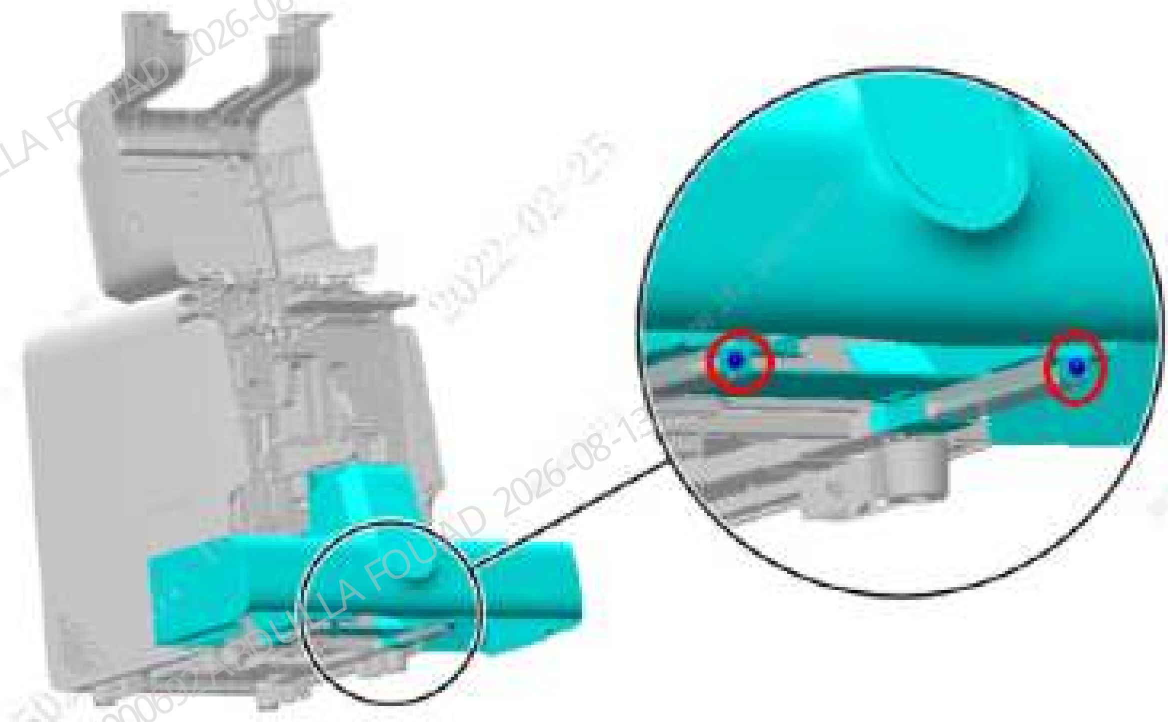
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

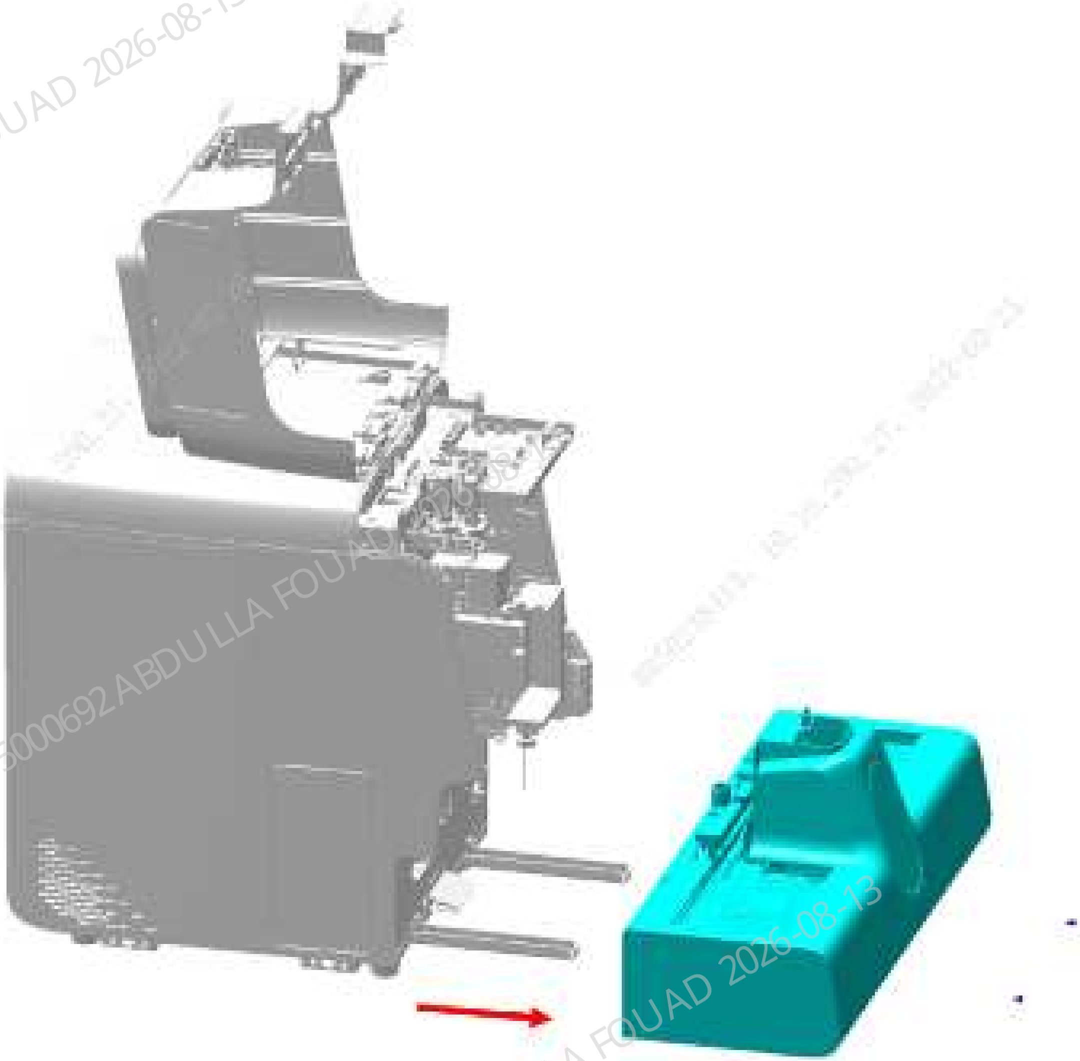
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–336 Opening the front cover



2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam, that is, remove the autoloader, closed-sampling model (10x5) FRU (including capillary blood mixing), as shown in the figure below.

Figure8–337 Removing the autoloader assembly



Subsequent Requirements

Installation procedure:

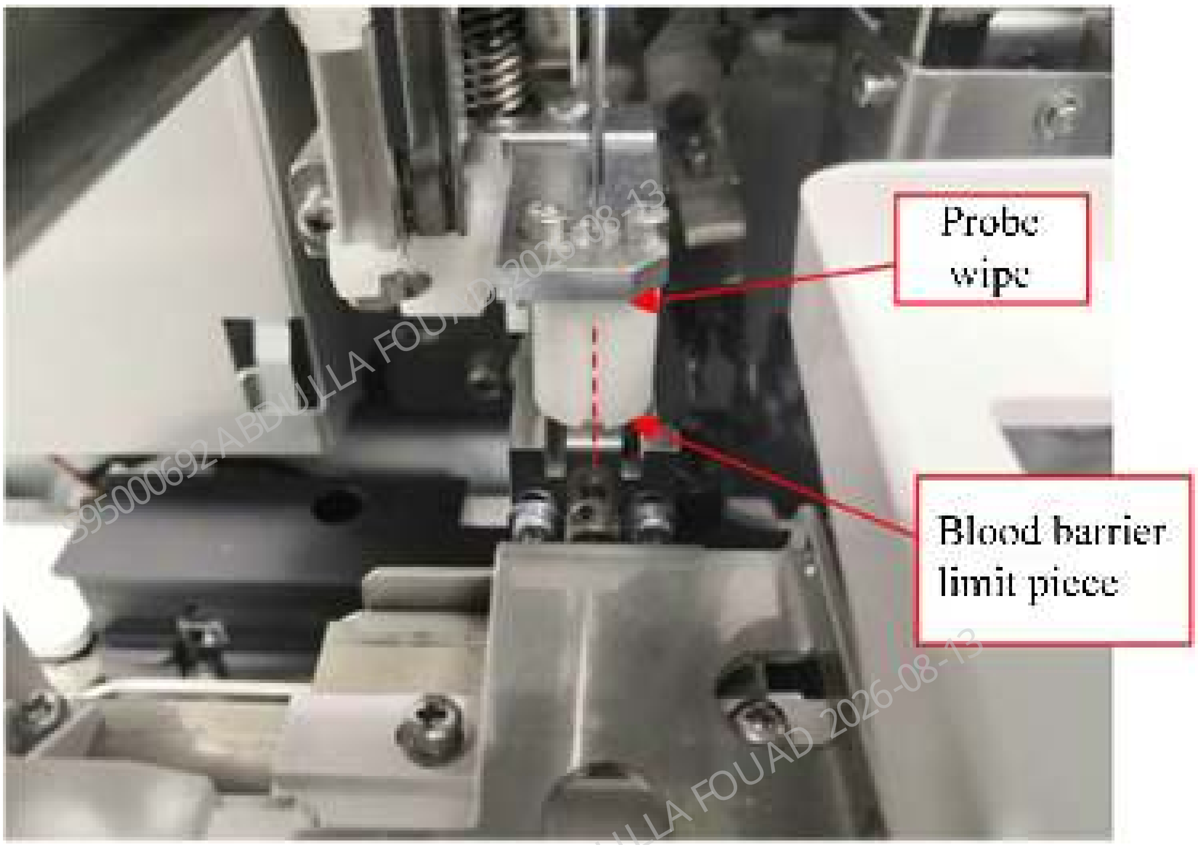
1. Place the autoloader, closed-sampling model (10x5) FRU (including capillary blood mixing) on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
2. Then push the autoloader, closed-sampling model (10x5) FRU (including capillary blood mixing) along the crossbeam direction of the autoloader to fit the front panel of the main unit.
3. Use two M4x8 cross-recessed combination screws to lock the autoloader, closed-sampling model (10x5) FRU (including capillary blood mixing) on the autoloader crossbeam.
4. Flip down the front cover.

8.33.3 Commissioning and Verification

Procedures

Commissioning and verification:

1. The left and right positions of the autoloader relative to the main unit have been adjusted in the assembly, so the FRU assembly does not require commissioning in the left and right positions.
2. Select **Menu**>>**Status**>>**Sensor Status** to enter the motor release torque status. Pull the sampling assembly to the front position a little ahead of the floating blood barrier. Observe whether the center of the sampling assembly probe wipe groove aligns with the center of the floating blood barrier limit piece (≤ ±0.5mm).

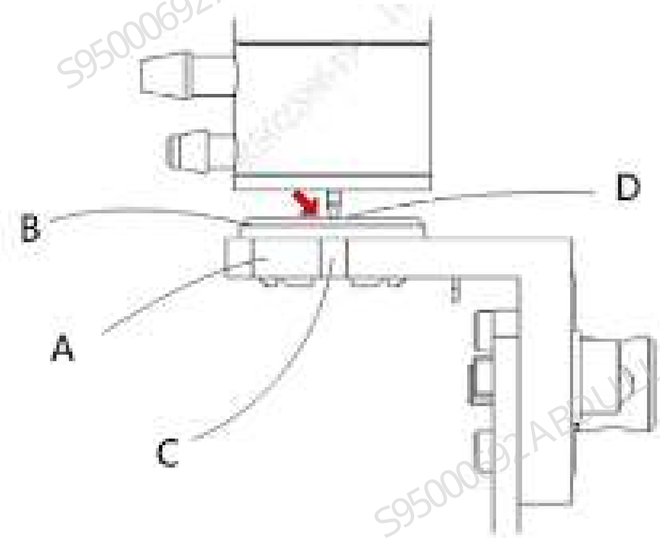


3. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adj. Sample Probe Pos.**>>**AL asp. Pos.** to enter the AL asp. Pos. adjustment interface.

Figure8–338 AL Aspiration Position Adjustment Interface

4. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sample probe is aligned with the groove of the blocking ring seat body.

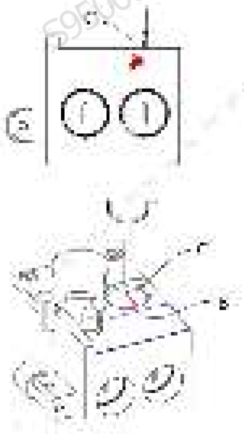
Figure8–339 Observing Sampling Probe Position



5. If the requirement of item 4 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the auto-sampling position.
6. Select **Menu>>Service>>Commissioning & Self-Test>>Adj. Sample Probe Pos.>>CT-WB Position** to enter the CT-WB position adjustment interface.

Figure8–340 CT-WB Position Adjustment Interface

7. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sampling probe is aligned with the arrow on the cover opening label in the horizontal direction, and flush with the top surface of the sample compartment in the vertical direction.



8. If the requirement of item 7 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the CT-WB position.

Verification:

Confirming/Adjusting the Position of the Tube Hold Assembly

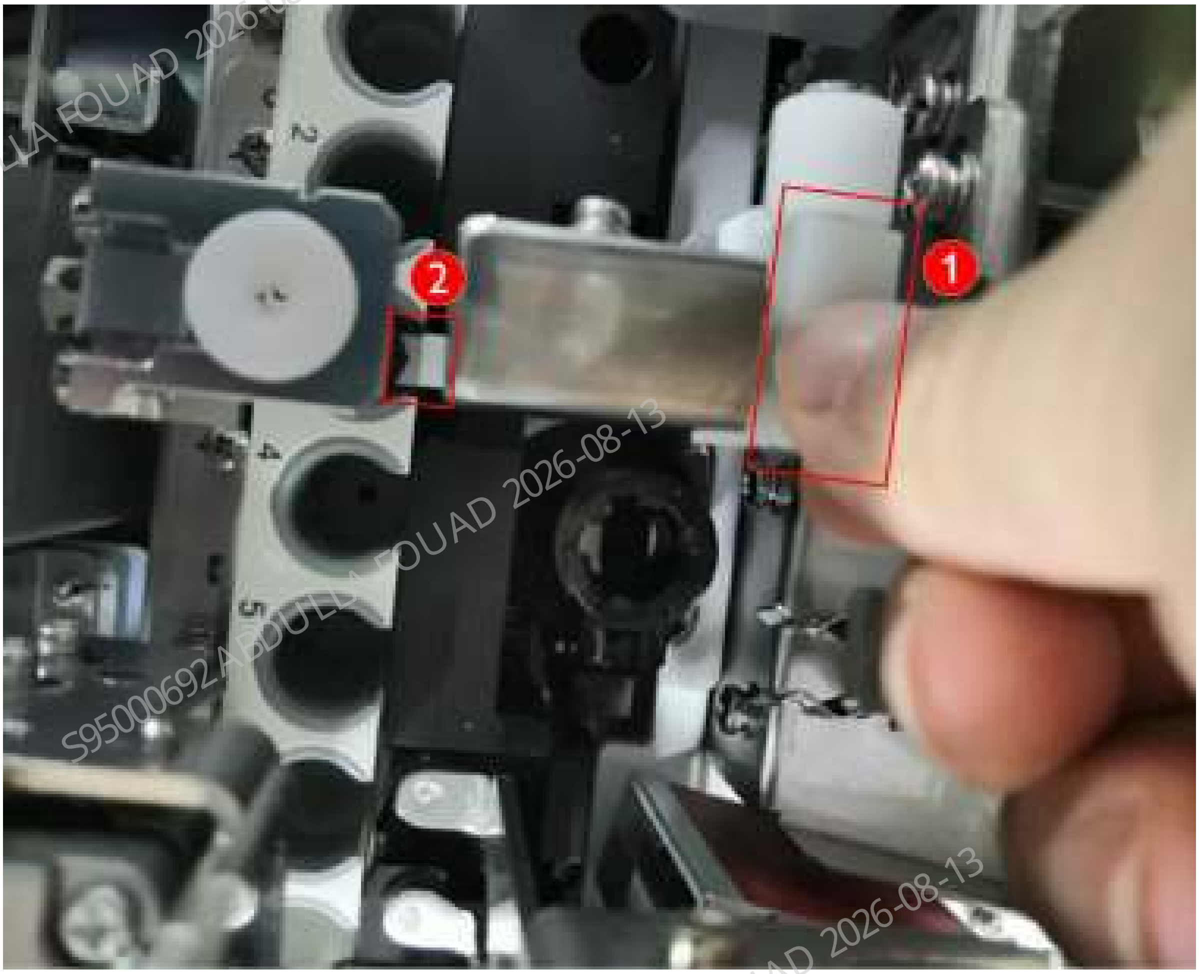
Confirm the position of the analyzer tube hold assembly. The steps are as follows:

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Autoloader Assembly** to enter the autoloader self-test interface.

Figure8–341 Autoloader Self-test Interface

2. Click the **Assembly Initialization** button.
3. Put one row of test tube rack with an empty test tube in No. 4 test tube position on the loading tray.
4. Click the **Feed** button (leave a time interval after each click, and then click the button next time after the mechanism action is completed), and send the first position of the test tube rack to the piercing position).
5. Press the roller fixing plate of the hold assembly by hand, and observe whether the finger pressure block of the hold assembly is approximate aligned with the center of the front opening of the test tube rack.

Figure8–342 Ideal Position of the Pincher Roller of the Hold Assembly Relative to the Tube Rack



6. If the left and right sides of the finger pressure block of the hold assembly are close to the opening of the test tube (≤ 1 mm), the left and right positions of the hold assembly need to be readjusted. For how to adjust the position of the hold assembly, see the section for adjusting the position of hold assembly FRU: **8.23.3** Commissioning and Verification.

Confirming/Adjusting the Position of the Rotatory Compressing Assembly

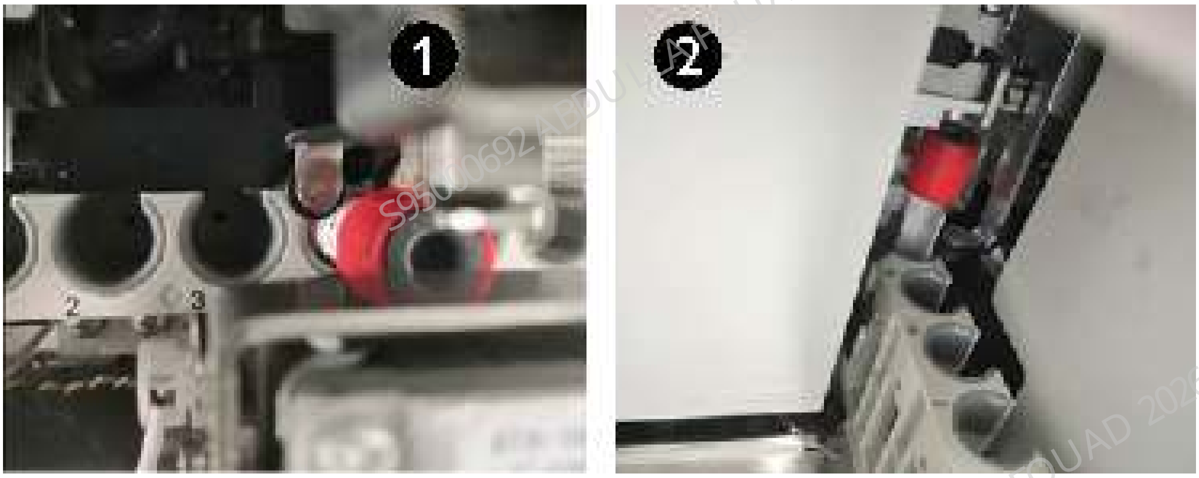
After the hold assembly position is confirmed, do not take out the tube rack, but continue to confirm the position of the rotatory compressing assembly. The steps are as follows:

1. Click the **Fix Tube** button.

Figure8–343 Autoloader Assembly Self-test Interface

2. Observe whether the two pinch rollers are close to the test tube. If they are not, readjust the left and right positions of the mechanism (failure to keeping them close will cause a false alarm of the autoloader counter).

Figure8–344 Ideal Position of the Pincher Roller of the Rotatory Compressing Assembly Relative to the Tube Rack



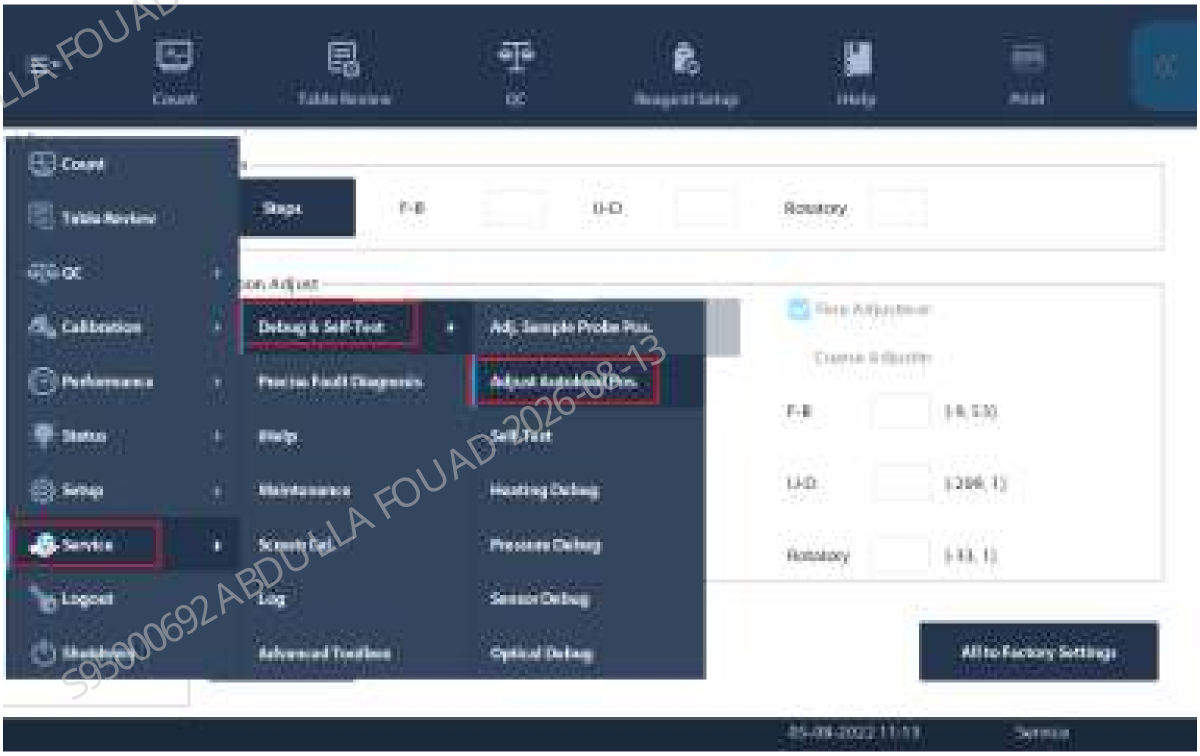
1	Left pinch roller	2	Right pinch roller

Confirming/Adjusting the Mix Mechanism Position

After the rotatory compressing assembly position is confirmed, do not take out the tube rack, but continue to confirm the mix mechanism position.

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.**, and select *Autoloader Assembly* to enter the mix mechanism self-test interface.

Figure8–345 Path of Entering Adjust Autoload Pos.



2. Select **Mixing gripper (front position)**, and click **Check by Step**.

Figure8–346 Mixing Gripper (Front Position) Interface

3. On the displayed box, click **Init.**, and then **Gripper Forward**. Observe whether the gripper is centered relative to the left and right positions of the test tube rack hole position. If it is centered, click **Gripper Backward**, place an empty tube into the mixing position, click the **A&M** button, and observe whether the tube squeezes the tube rack when the gripper is placing the tube.

Figure8–347 "Check by Step" Interface of Mixing Gripper (Front Position)

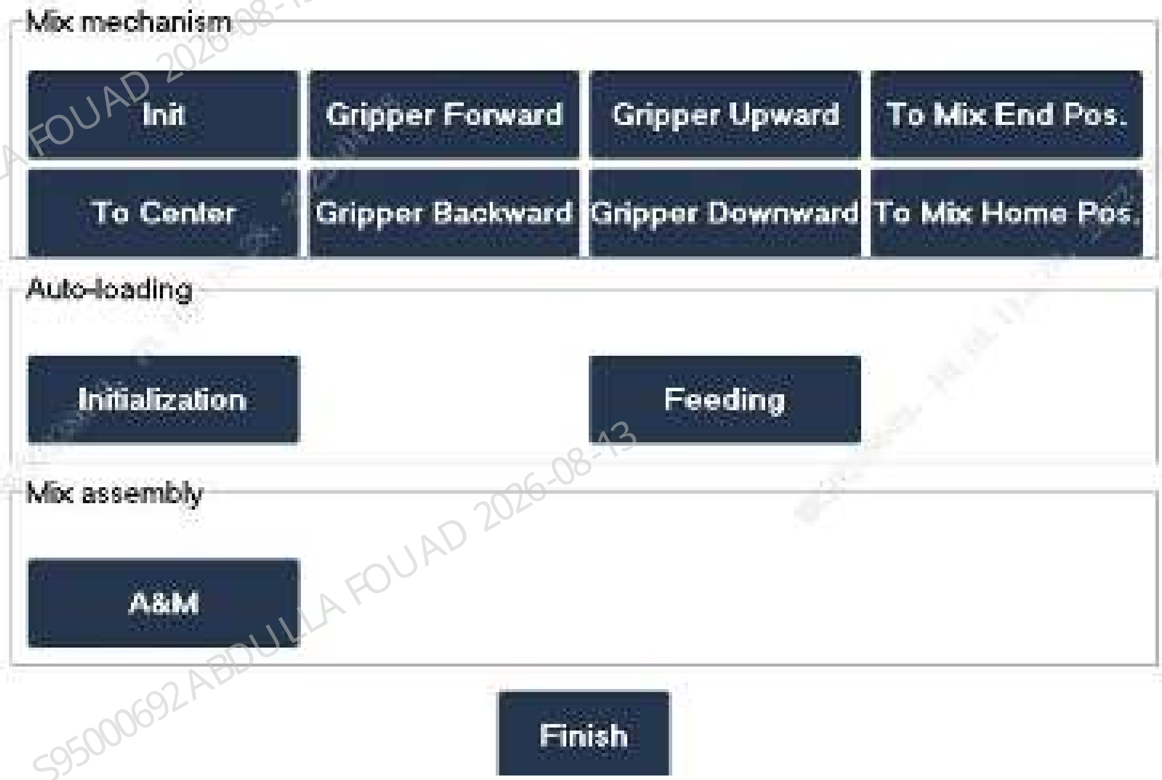


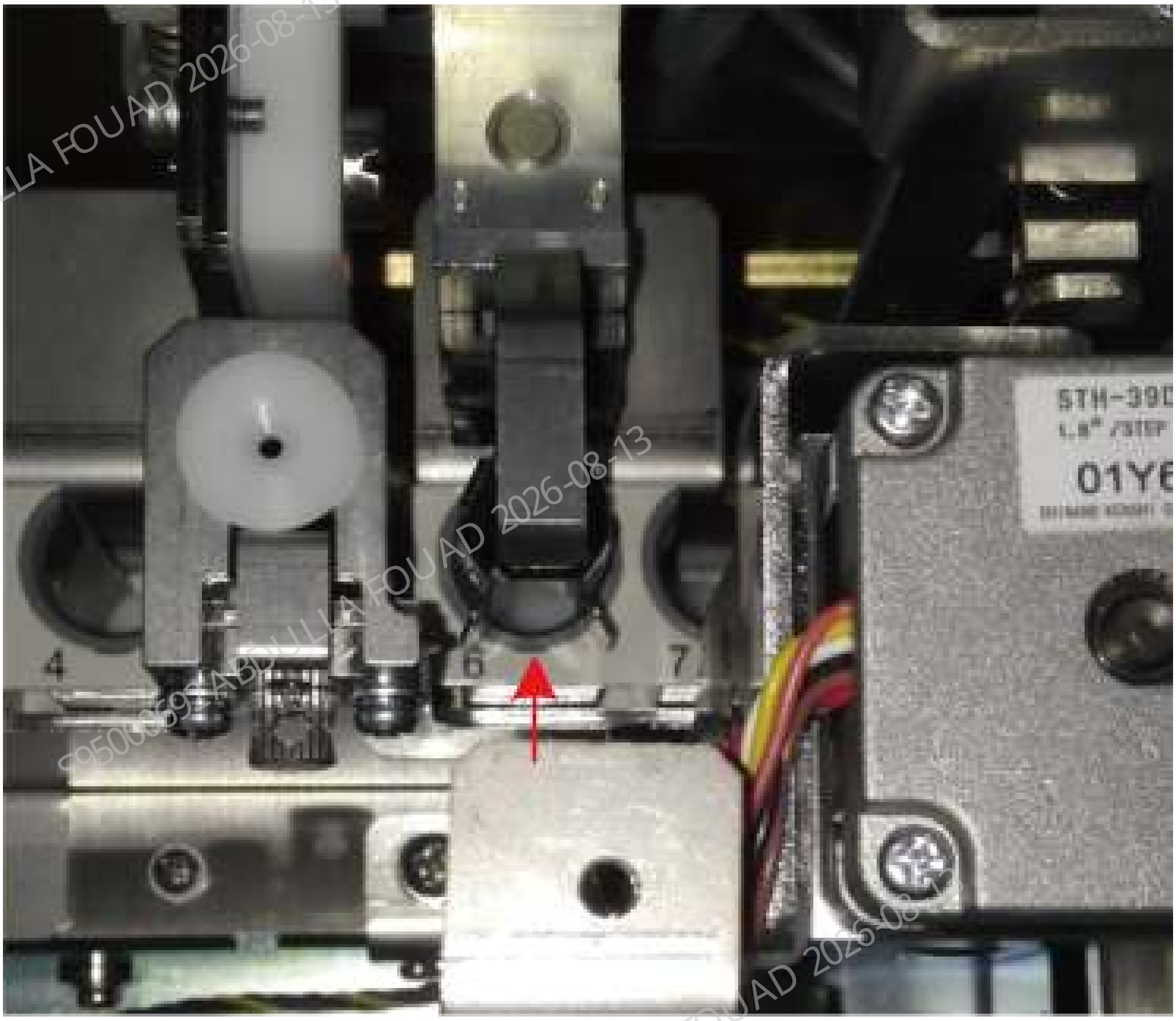
Figure8–348 Observing the Mixing Gripper Position

4. If the test tube squeezes the test tube rack, exit **Check by Step**, click **To Adjust Pos.**, and then press the **Forward**, **Backward**, **Left**, and **Right** buttons to fine adjust the gripper position.

Figure8–349 Observing the Mixing Gripper Position

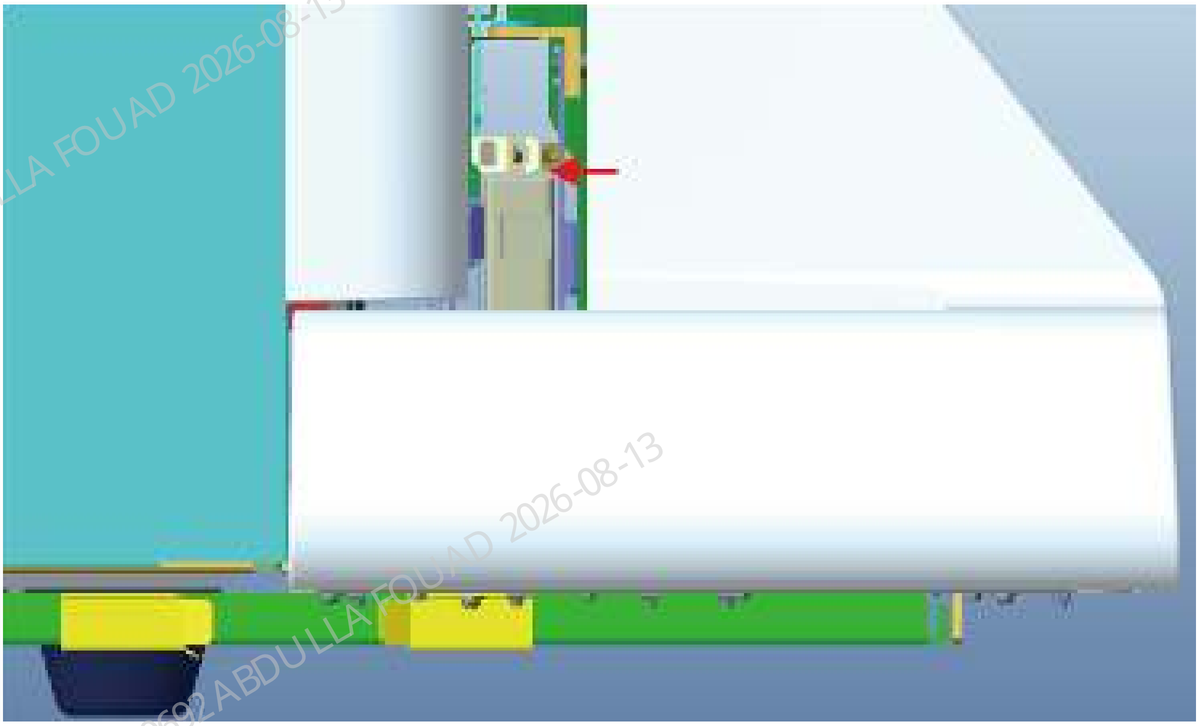
5. Finally enter **Check by Step** again, and observe again whether the tube squeezes the tube rack hole position when the gripper is placing the tube.

Figure8–350 Observing the Mixing Gripper Position



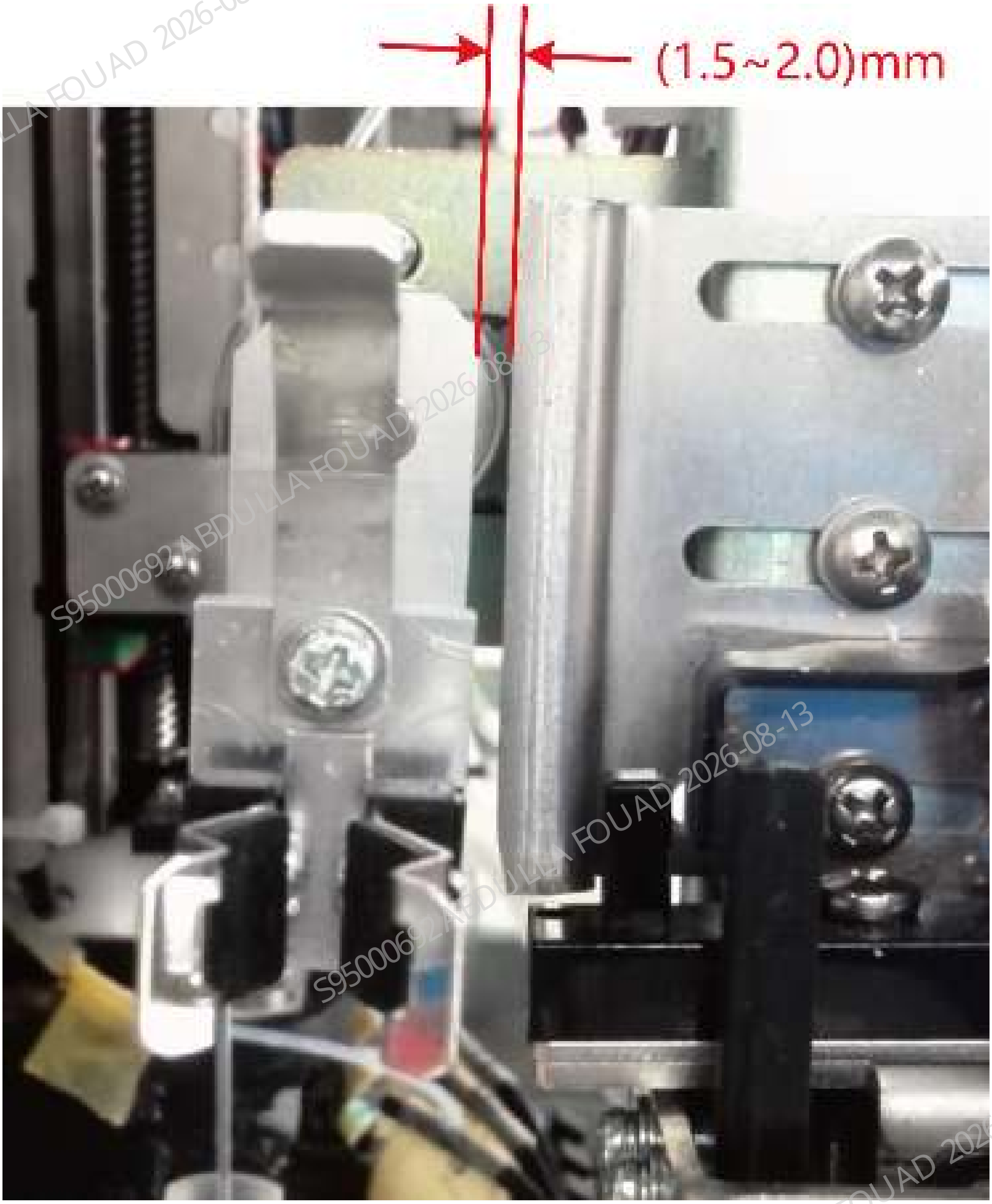
6. After confirming the left, right, front, and back positions of the gripper relative to the test tube rack, confirm the upper and lower positions of the gripper. The distance between the bottom surface of the gripper and the top surface of the test tube rack needs to be controlled within the range of (2.0 ~ 2.5) mm. If slides are available, use two standard slices (thickness 1.1~1.2 mm) or one thick slice (nonstandard) as an auxiliary tool for test. If there are no slides, perform visual inspection. If the requirements are not met, make adjustment through **Up** and **Down** buttons of the gripper.

Figure8–351 Confirming the Mixing Gripper Height Position



7. Confirm the clearance between the gripper relative to the anti-collision bock. Click **Mixing gripper (front position)**, select **Check by Step**, click **Initialize>To Center**, and observe whether the clearance between the gripper and the anti-collision bock is in the range of 1.5~2.5 mm. Use two slices as a tool for confirmation.

Figure8–352 Distance Requirements between the Mixing Gripper and Anti-Collision Bock



Confirming/Adjusting the Capillary Blood Mixing Position

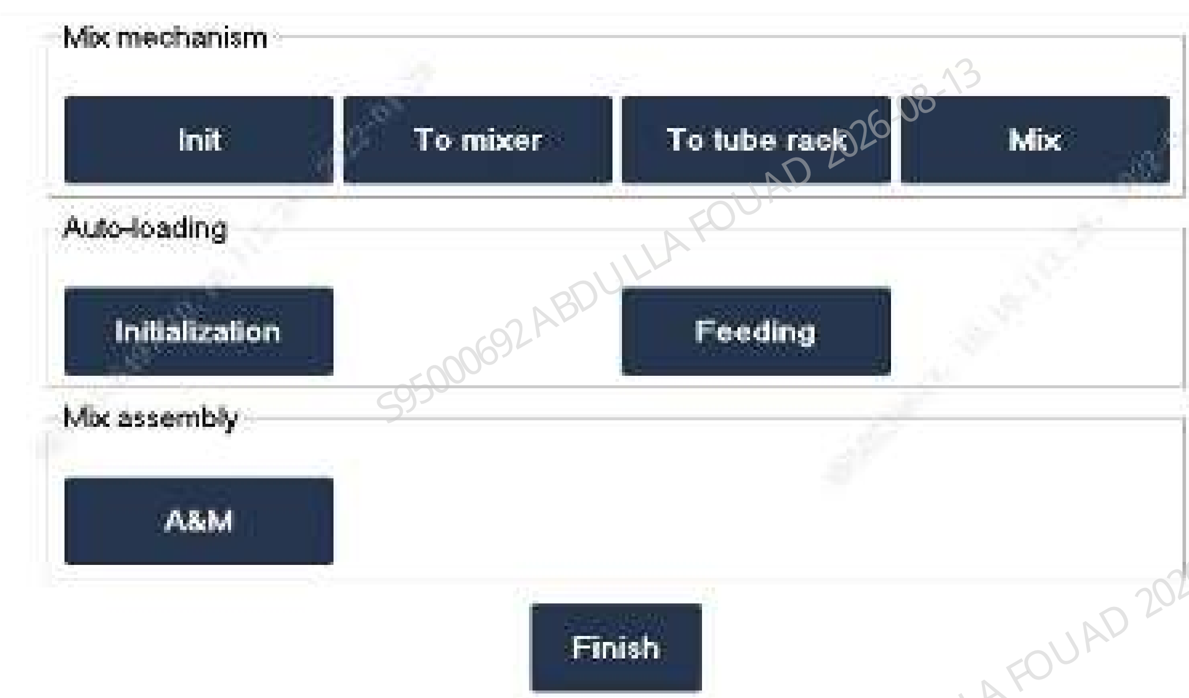
Perform this operation for the analyzer configured with capillary blood mixing.

1. Prepare a WB test tube rack with WB test tubes.
2. Click **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adjust Autoload Pos.** to enter the mixing mechanism adjustment interface, and select **Mixing gripper (middle position)**.

Figure8–353 Capillary Blood Position Adjustment Interface of Tube Gripper

3. Place the test tube rack at the test tube loading position, click **Check by Step**, push the test tube rack to the feeding position, and then click **Assembly Initialization>>Feed**to feed the test tube on the test tube rack to the mix assembly gripping position.

Figure8–354 "Check by Step" Interface



4. Click **To Mixer** and **To tube rack**, check whether the gripping action of the gripper for gripping the test tube between the test tube rack and the capillary blood mixer is smooth, and whether there is interference between the test tube and the capillary blood mixer.

1) If the gripping action is not smooth or the test tube interferes with the capillary blood mixer, confirm whether there is a problem with the left/right or front/rear position of the capillary blood mixer.

2) If there is a problem with the left/right position of the capillary blood mixer, loosen the fixing screw fixing the capillary blood mixer and adjust the capillary blood mixer left and right.

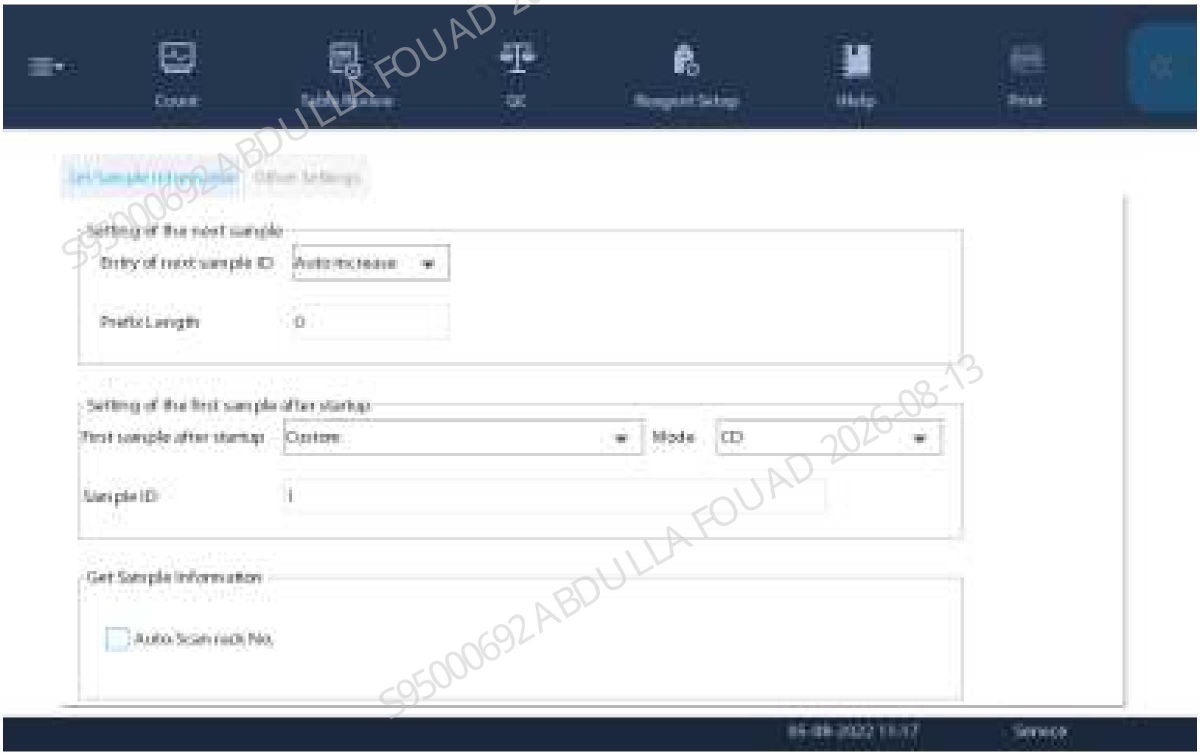
3) If there is a problem with the front/back position of the capillary blood mixer, exit the "Check by Step" interface and adjust the position forward and backward on the commissioning interface.

Confirming the Scanner Position

If the user does not need the sample barcode scanning function, this section can be skipped.

1. Click **Menu**>>**Setup**>>**Auxiliary Setup** to enter the interface for setting the sample information getting method. Confirm that the **Auto-Scan rack No.** option has been checked.

Figure8–355 Auto-Scan Setup Interface

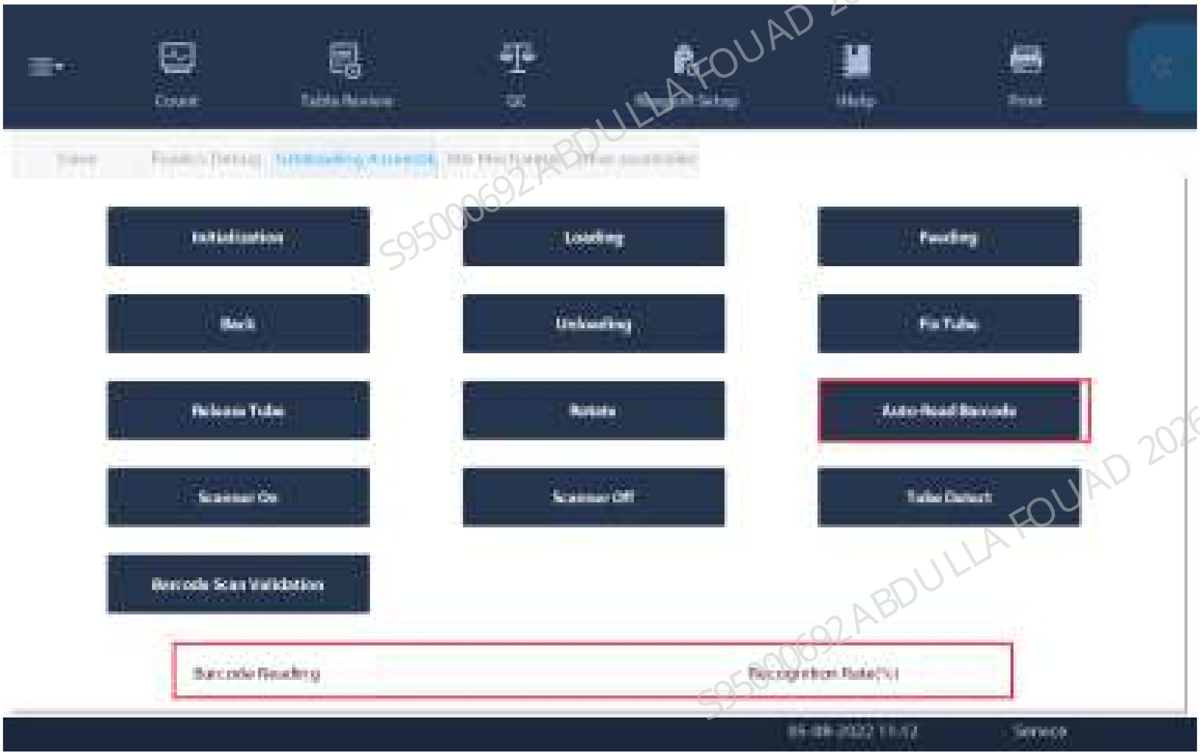


2. Use the barcode scanning software or machine barcode recognition function to identify the barcode system used by the client.
3. Click **Menu**>>**Setup**>>**System Setup**>>**Barcode** to enter the barcode setup interface. Check the corresponding code system, length, and check bit. Do not check the code system and length not used for the client.

Figure8–356 Built-in Barcode Setup Screen

4. After completing the above setup items, click **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**, and select **Autoloader Assembly** to enter the autoloader self-test interface.
5. After clicking **Assembly Initialization**, put a test tube with barcode in the rotating scanning position (the test tube is placed in the test tube rack).
6. Click **Auto-Read Barcode** to check whether the barcode content display box displays the barcode content and whether the barcode recognition rate display box shows 100%. If not, first confirm whether the red light of the scanner is turned on when you click the **Auto-Read Barcode** button. If the light has been turned on but the barcode cannot be scanned, adjust the position according to the position confirmation description in the chapter about scanner FRU.

Figure8–357 Autoloader Assembly Self-test Interface



8.34 Capillary blood sample mixing assembly

8.34.1 General Information

Figure8–358 Photo of Capillary Blood Mix Assembly

Figure8–359 Diagram of Location of Capillary Blood Mix Assembly in the Main Unit

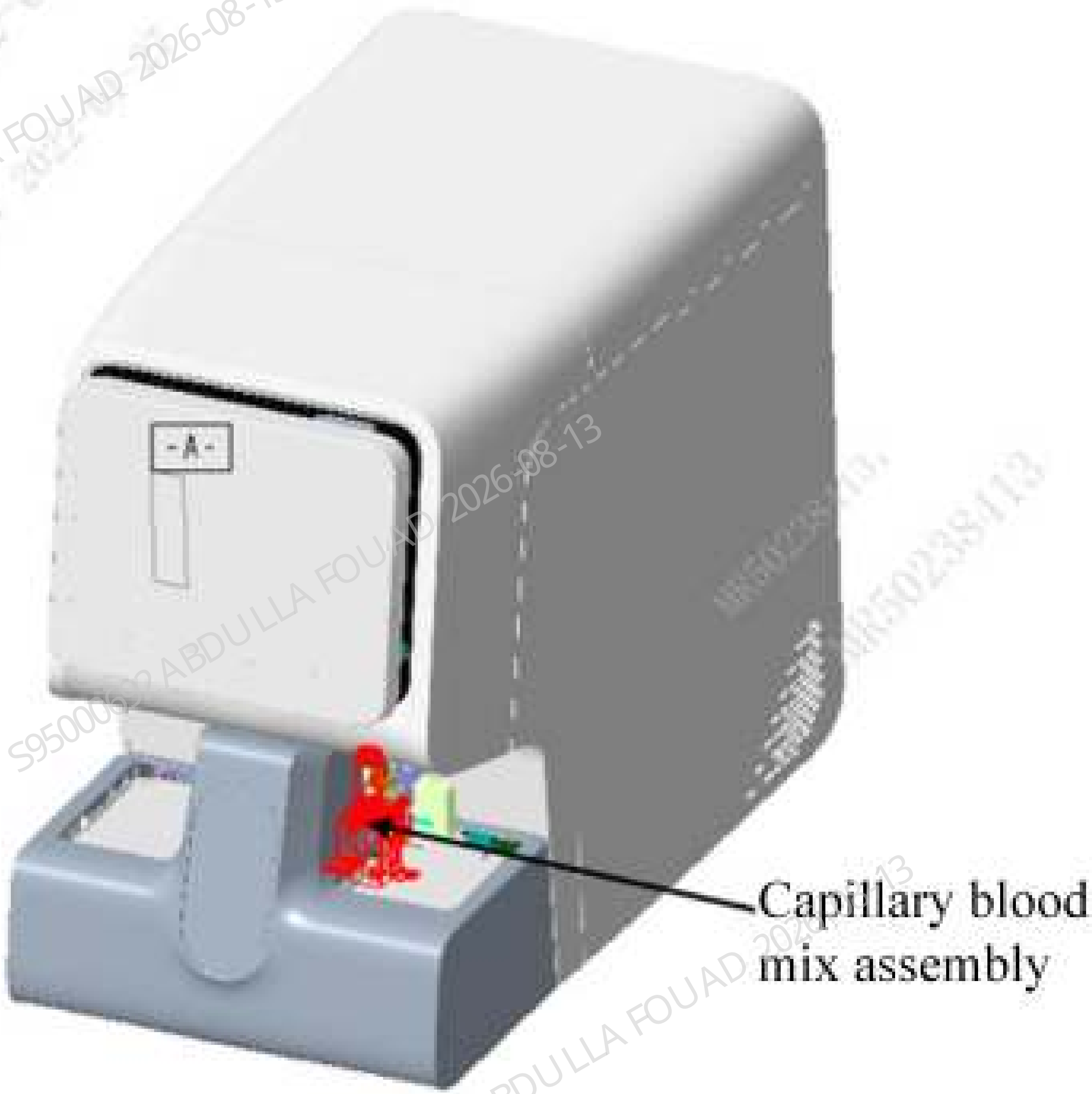


Table 8–36 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-059469-00	Peripheral Blood Blending Assembly	N/A	N/A

Function:
Used for capillary blood sample mixing.

8.34.2 Disassembly and Assembly

For the disassembly and assembly in the case of FRU-capillary blood mix assembly (10x5 autoloader), see FRU-capillary blood mix assembly (10x5 autoloader).

For the disassembly and assembly in the case of FRU-capillary blood mix assembly (5x6 autoloader), see FRU-capillary blood mix assembly (5x6 autoloader).

FRU-capillary blood mix assembly (10x5 autoloader)

Required tools

Cross-headed screwdriver

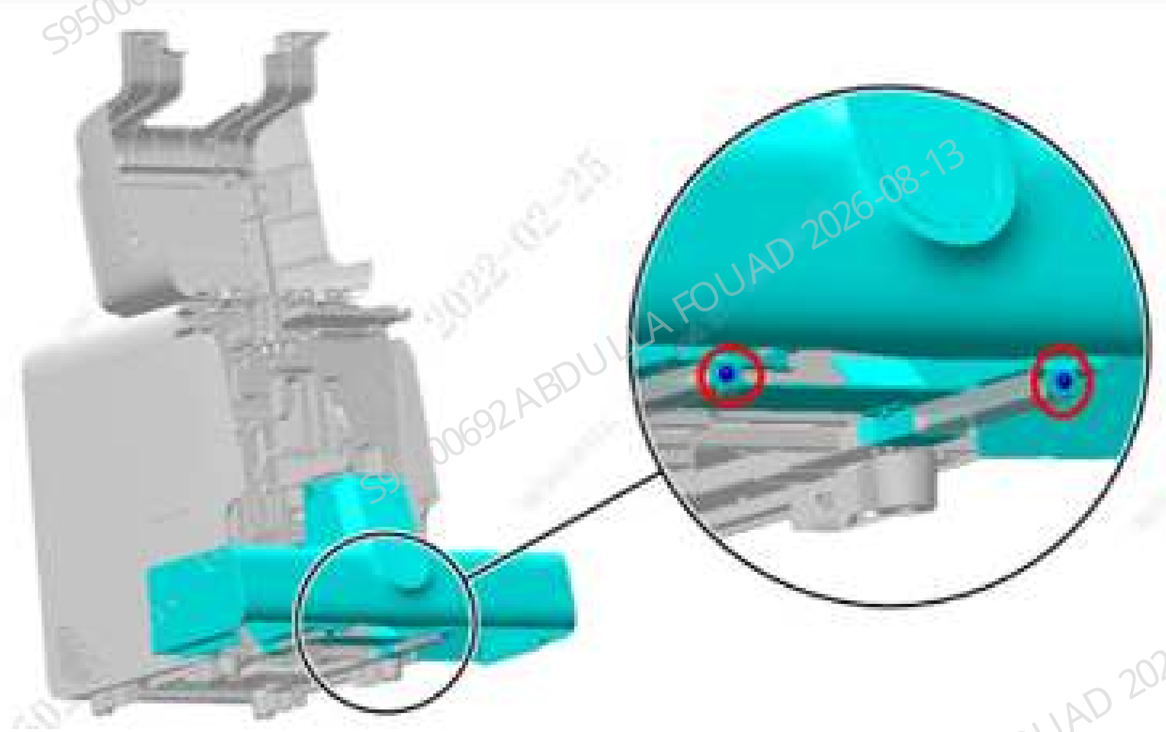
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

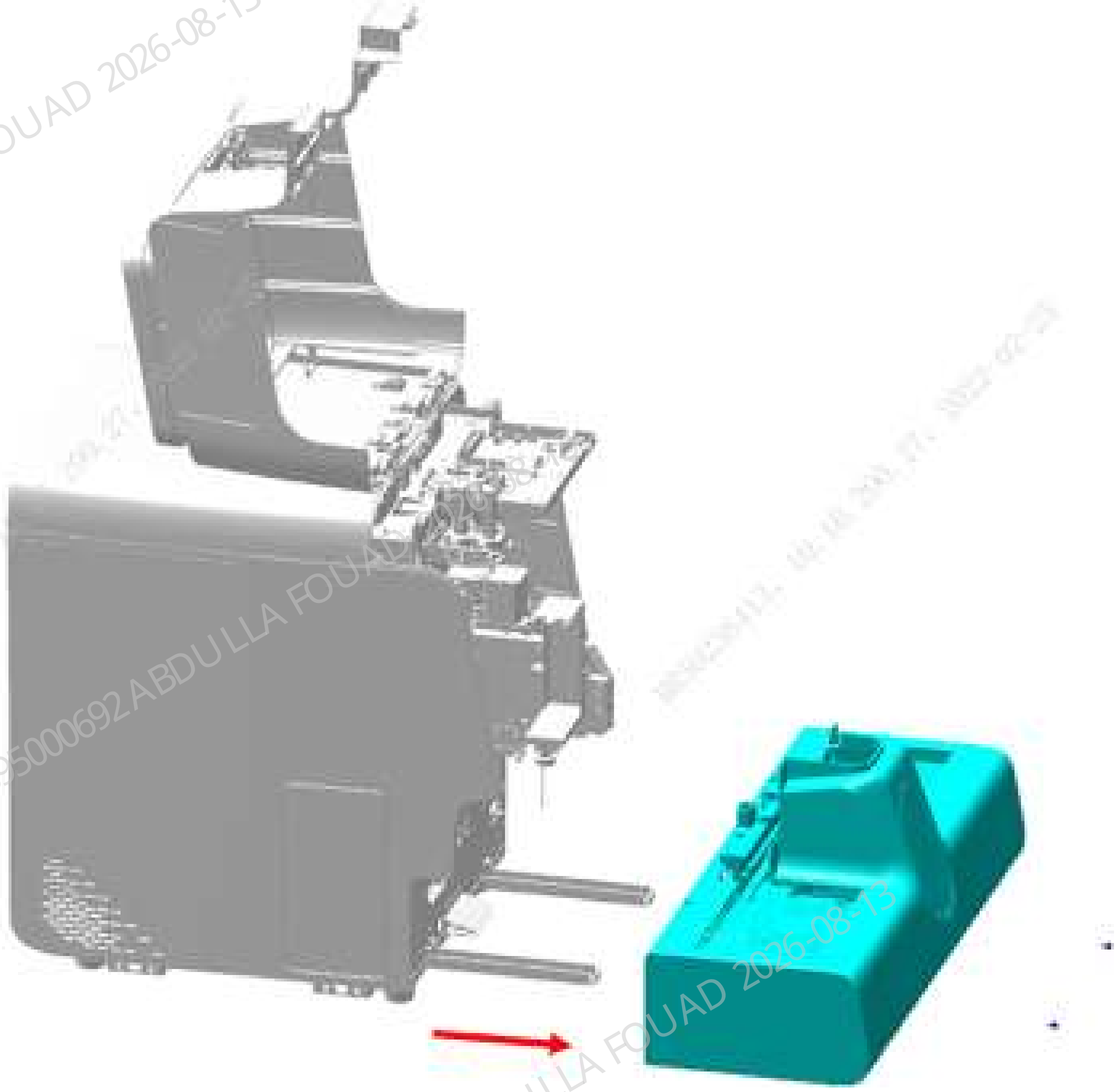
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.
2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown below.

Figure8–360 Opening the front cover



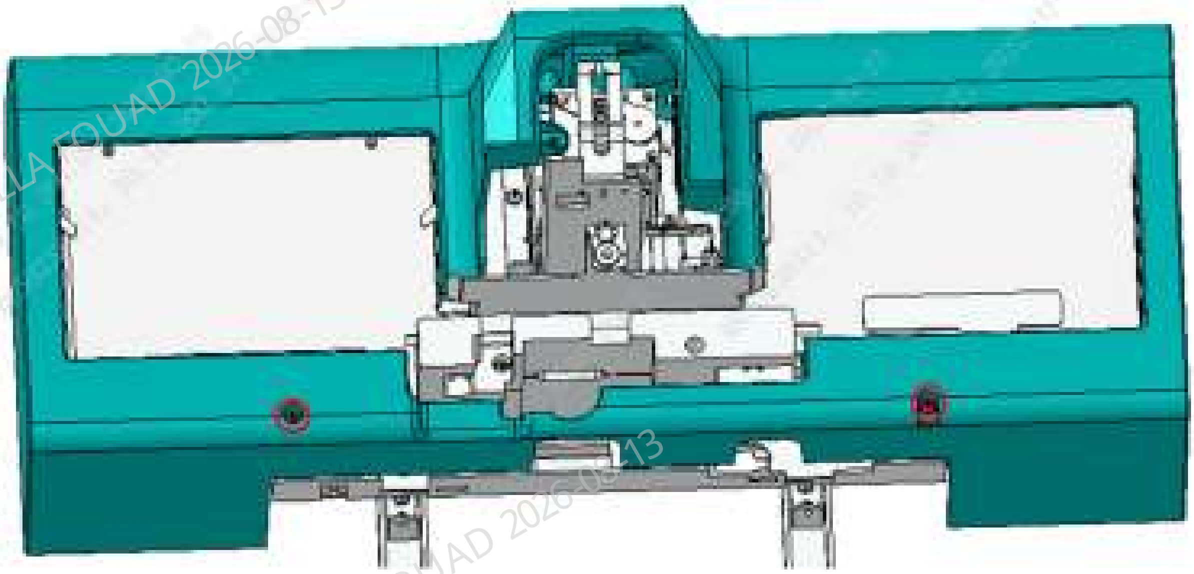
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the 10x5 closed autoloader (FRU) assembly, as shown below.

Figure8–361 Removing the 10x5 closed autoloader (FRU)



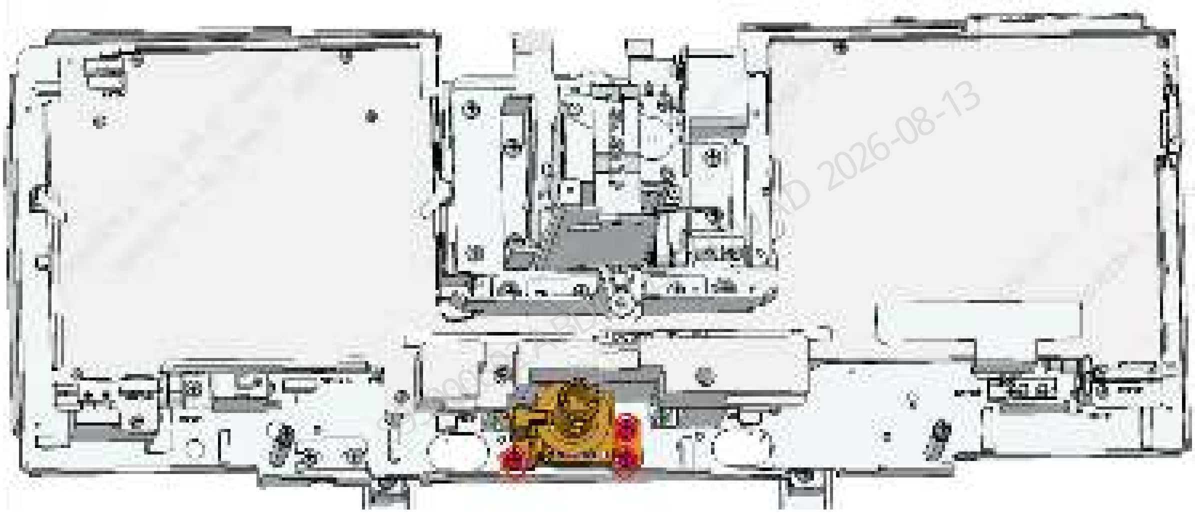
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (10x5), and then remove the autoloader front cover assembly (10x5), as shown the figure below.

Figure8–362 Removing autoloader front cover assembly (10x5)



6. Remove the three M4x8 cross-recessed combination screws fixing the capillary blood mix assembly, pull out the wire on the capillary blood mix assembly, and then remove the capillary blood mix assembly, as shown in the figure below.

Figure8–363 Removing the capillary blood mix assembly



Subsequent Requirements

Installation procedure:

1. Fix the capillary blood mix assembly to the autoloader using three M4x8 cross-recessed combination screws, and insert the wire on the capillary blood mix assembly (set the step for going to adjustment of the supplementary fixture).
2. Use two M4x8 cross-recessed combination screws to fix the autoloader front cover assembly (10x5) to the autoloader.

3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

FRU-capillary blood mix assembly (5x6 autoloader)

Required tools

Cross-headed screwdriver

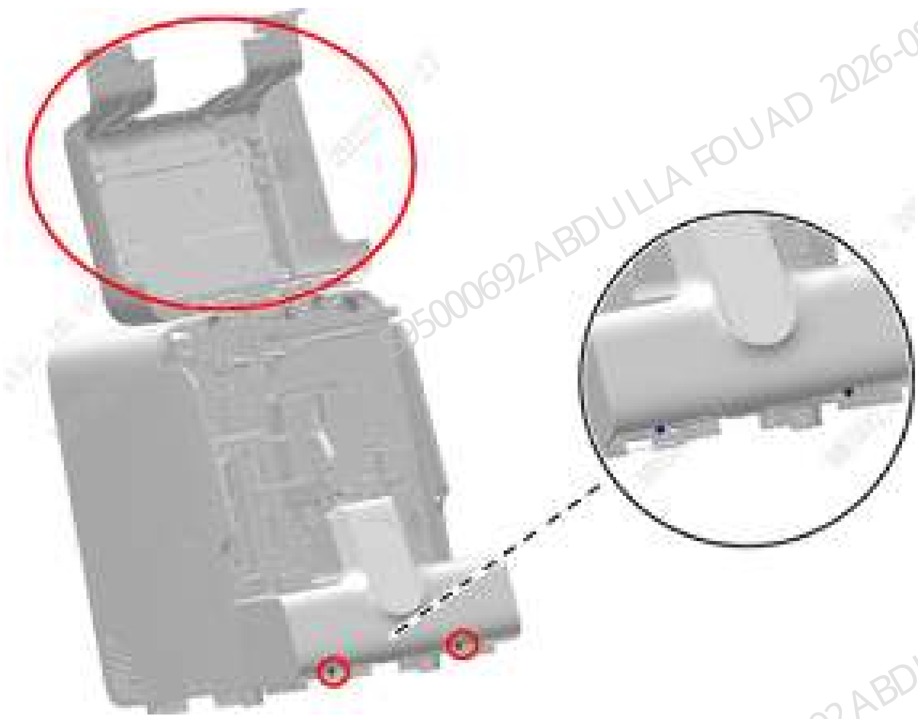
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

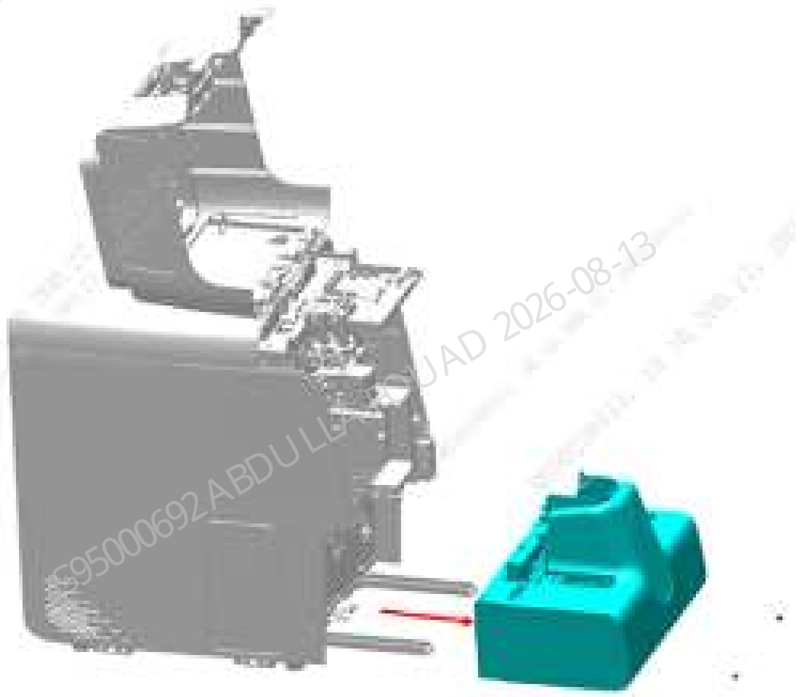
Figure8–364 Opening the front cover



2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.

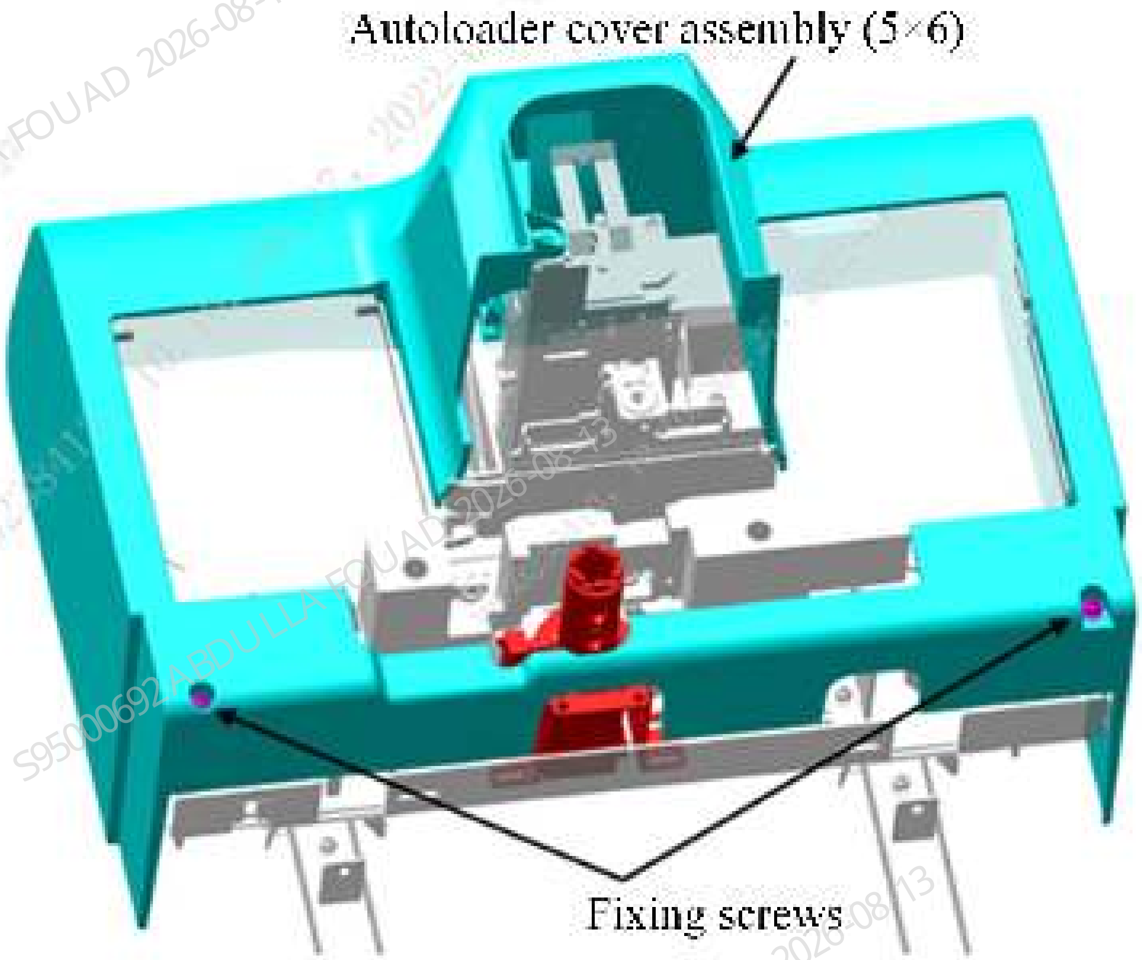
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam to remove the 5x6 closed autoloader (FRU) assembly, as shown below.

Figure8–365 Removing the 5x6 closed autoloader (FRU)



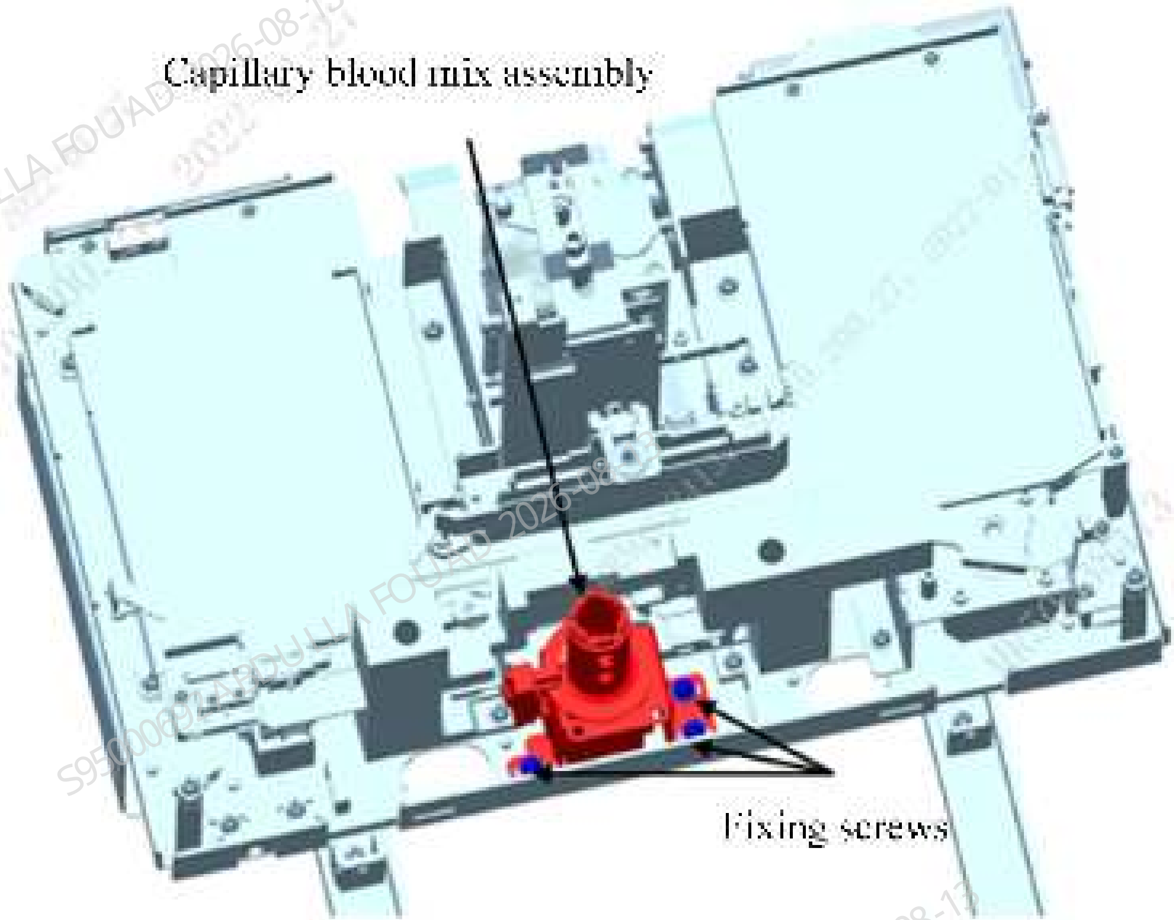
5. Remove the two M4x8 cross-recessed combination screws fixing the autoloader front cover assembly (5x6), and then remove the autoloader front cover assembly (5x6), as shown in the figure below.

Figure8–366 Removing autoloader front cover assembly (5x6)



6. Remove the three M4x8 cross-recessed combination screws fixing the capillary blood mix assembly, pull out the wire on the capillary blood mix assembly, and then remove the capillary blood mix assembly, as shown in the figure below.

Figure8–367 Removing the capillary blood mix assembly



Subsequent Requirements

Installation procedure:

1. Fix the capillary blood mix assembly to the autoloader using three M4x8 cross-recessed combination screws, and insert the wire on the capillary blood mix assembly (set the step for going to adjustment of the supplementary fixture).
2. Use two M4x8 cross-recessed combination screws to fix the autoloader front cover assembly (5x6) to the autoloader.
3. Place the autoloader on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
4. Then push the autoloader along the crossbeam direction of the autoloader to fit the front panel of the main unit.
5. Use two M4x8 cross-recessed combination screws to lock the autoloader on the autoloader crossbeam.
6. Flip down the front cover.

8.34.3 Commissioning and Verification

For the commissioning and verification in the case of FRU-capillary blood mix assembly (10X5 autoloader), see FRU-capillary blood mix assembly (10x5 autoloader).
For the commissioning and verification in the case of FRU-capillary blood mix assembly (5X6 autoloader), see FRU-capillary blood mix assembly (5x6 autoloader).

FRU-capillary blood mix assembly (10x5 autoloader)

Procedures

Commissioning:

1. After replacing the capillary blood mix assembly, do not install the autoloader front cover.
2. Splice the autoloader without front cover installed with the main unit, and push the autoloader to fit the front panel of the main unit (note that there is no gap between the autoloader and the main unit, and do not pinch the wire).
3. After startup, select **Service>>Commissioning & Self-Test>>Adjust Autoload Pos.** to enter the related interface:

Figure8–368 Adjust Autoload Pos. Interface

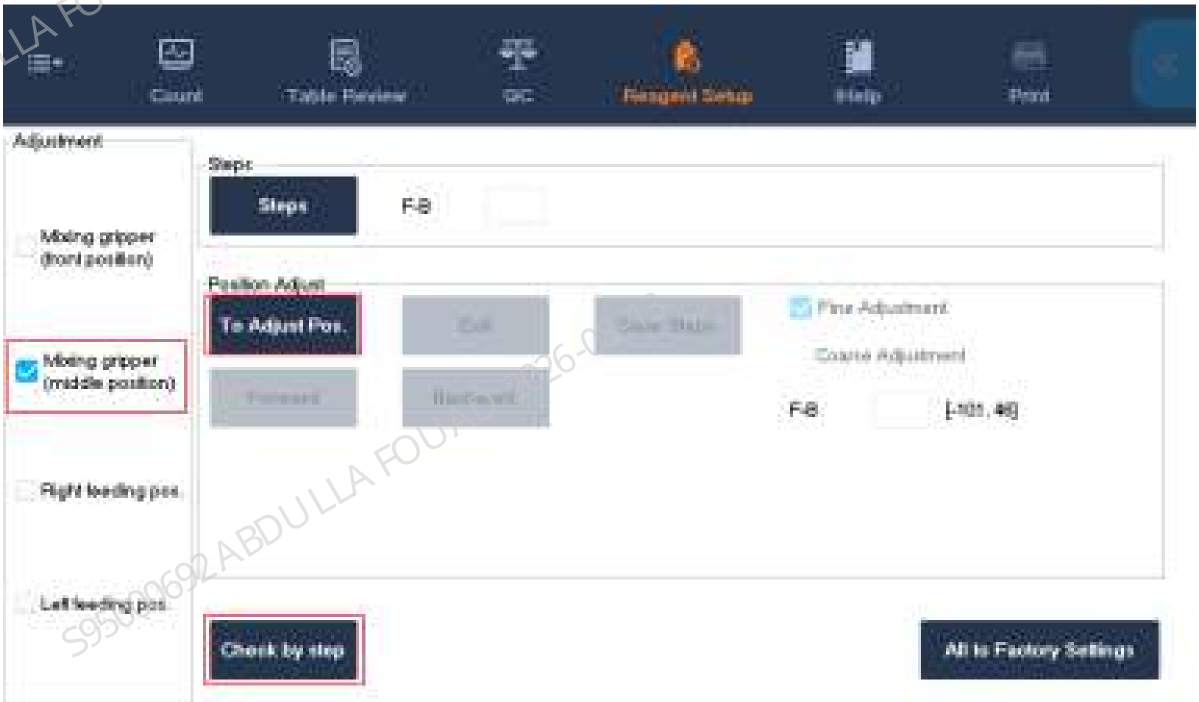
4. Select the mixing gripper (front position) and click **Check by Step**.

Figure8–369 "Check by Step" Interface of Mixing Gripper (Front Position)

5. Click **Assembly Initialization** of autoloader, place one row of test tube rack with one venous blood collection tube in the loading tray, click **Feed** several times to feed and send the venous blood collection tube to the mixing position. Click **A&M**, and check whether the bottom of the test tube can be just inserted into the fixing hole of the test tube rack without scratching the fixing hole when the venous blood collection tube is mixed and lowered. If not, first adjust the mechanical position or operating steps of the mixing gripper to make the gripper position meet the above requirement. If the requirement is met, take out the test tube rack and exit the "Check by Step" interface.
6. Select the mixing gripper (middle position), click **To Adjust Pos.**, and first check whether the left and right grippers are aligned with the fixing hole of the capillary blood mix assembly. If so, directly go to step (7). If not, drag the autoloader away from the front panel of the main unit (the connecting wire of the autoloader remains connected with the main unit wire), loosen the three fixing screws of the assembly,and then push the autoloader to fit the front panel of the main unit (note that there is no gap between the autoloader and the main unit, and do not pinch the wire); use a long screwdriver to gently push the capillary blood mix assembly from the side to align the gripper with the fixing hole of the capillary blood mix assembly in the left and right directions, and drag the autoloader away from the front panel of the main unit again, lock the three fixing screws of the capillary blood mix assembly, and then push the autoloader to the front panel of the main unit.

7. Press the **Forward** and **Backward** buttons to adjust and roughly align the gripper center with the front and rear directions of the capillary blood mix assembly fixing hole, and then click **Save Steps** and **Exit** in turn.

Figure8–370 Adjust Autoload Pos. Interface



8. Click **Check by Step**, click **Assembly Initialization** of autoloader, place one row of test tube rack with one capillary blood collection tube in the loading tray, click **Feed** several times to feed and send the capillary blood collection tube to the mixing position. Click **A&M**, observe whether the tube bottom of the test tube can be just inserted into the fixing hole of the capillary blood mix assembly without scratching the fixing hole when the capillary blood collection tube is transported and released. If the hole is scratched, return to step (6) for readjustment. If the hole is not scratched, take out the test tube rack and exit the "Check by Step" interface.

Figure8–371 "Check by Step" Interface of Mixing Gripper (Middle Position)

9. Drag the autoloader away from the front panel of the main unit again, install the front cover of the autoloader, and splice the autoloader back to the main unit (note that there is no gap between the autoloader and the main unit, and do not pinch the wire).

Verification content is included in commissioning, and no further verification is required.

FRU-capillary blood mix assembly (5x6 autoloader)

Procedures

Commissioning:

1. After replacing the capillary blood mix assembly, do not install the autoloader front cover.
2. Splice the autoloader without front cover installed with the main unit, and push the autoloader to fit the front panel of the main unit (note that there is no gap between the autoloader and the main unit, and do not pinch the wire).
3. After startup, select **Service>>Commissioning & Self-Test>>Adjust Autoload Pos.** to enter the related interface:

Figure8–372 Adjust Autoload Pos. Interface

4. Select the mixing gripper (front position) and click **Check by Step**.

Figure8–373 "Check by Step" Interface of Mixing Gripper (Front Position)

5. Click **Assembly Initialization** of autoloader, place one row of test tube rack with one venous blood collection tube in the loading tray, click **Feed** several times to feed and send the venous blood collection tube to the mixing position. Click **A&M**, and check whether the bottom of the test tube can be just inserted into the fixing hole of the test tube rack without scratching the

fixing hole when the venous blood collection tube is mixed and lowered. If not, first adjust the mechanical position or operating steps of the mixing gripper to make the gripper position meet the above requirement. If the requirement is met, take out the test tube rack and exit the "Check by Step" interface.

6. Select the mixing gripper (middle position), click **To Adjust Pos.**, and first check whether the left and right grippers are aligned with the fixing hole of the capillary blood mix assembly. If so, directly go to step (7). If not, drag the autoloader away from the front panel of the main unit (the connecting wire of the autoloader remains connected with the main unit wire), loosen the three fixing screws of the assembly,and then push the autoloader to fit the front panel of the main unit (note that there is no gap between the autoloader and the main unit, and do not pinch the wire); use a long screwdriver to gently push the capillary blood mix assembly from the side to align the gripper with the fixing hole of the capillary blood mix assembly in the left and right directions, and drag the autoloader away from the front panel of the main unit again, lock the three fixing screws of the capillary blood mix assembly, and then push the autoloader to the front panel of the main unit.
7. Press the **Forward** and **Backward** buttons to adjust and roughly align the gripper center with the front and rear directions of the capillary blood mix assembly fixing hole, and then click **Save Steps** and **Exit** in turn.

Figure8–374 Adjust Autoload Pos. Interface

8. Click *Check by Step*, click **Assembly Initialization** of autoloader, place one row of test tube rack with one capillary blood collection tube in the loading tray, click **Feed** several times to feed and send the capillary blood collection tube to the mixing position. Click **A&M**, observe whether the tube bottom of the test tube can be just inserted into the fixing hole of the capillary blood mix assembly without scratching the fixing hole when the capillary blood collection tube is

transported and released. If the hole is scratched, return to step (6) for readjustment. If the hole is not scratched, take out the test tube rack and exit the "Check by Step" interface.

Figure8–375 "Check by Step" Interface of Mixing Gripper (Middle Position)

9. Drag the autoloader away from the front panel of the main unit again, install the front cover of the autoloader, and splice the autoloader back to the main unit (note that there is no gap between the autoloader and the main unit, and do not pinch the wire).

Verification content is included in commissioning, and no further verification is required.

8.35 Autoloader, closed-sampling model(10x5)

8.35.1 General Information

Figure8–376 Photo of 115-085180-02 Autoloader, closed-sampling model(10x5)

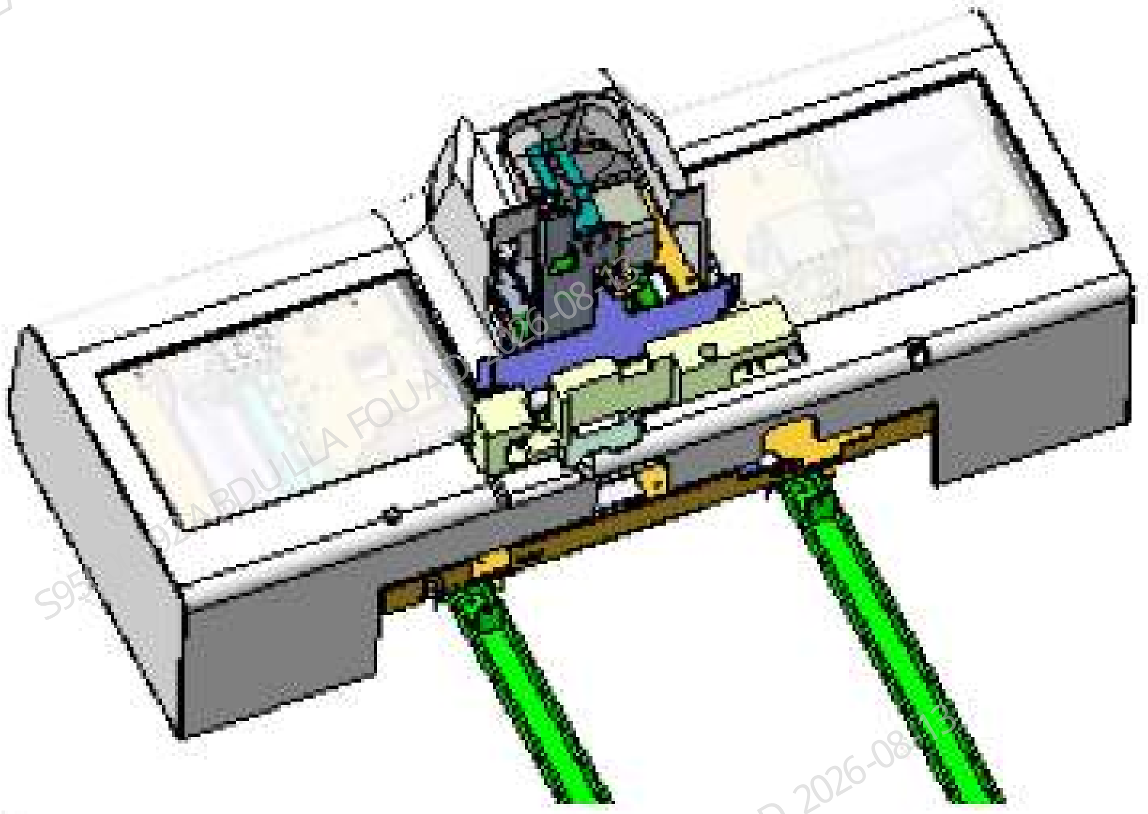


Figure8–377 Diagram of Location of Autoloader, closed-sampling model(10x5) in the Main Unit

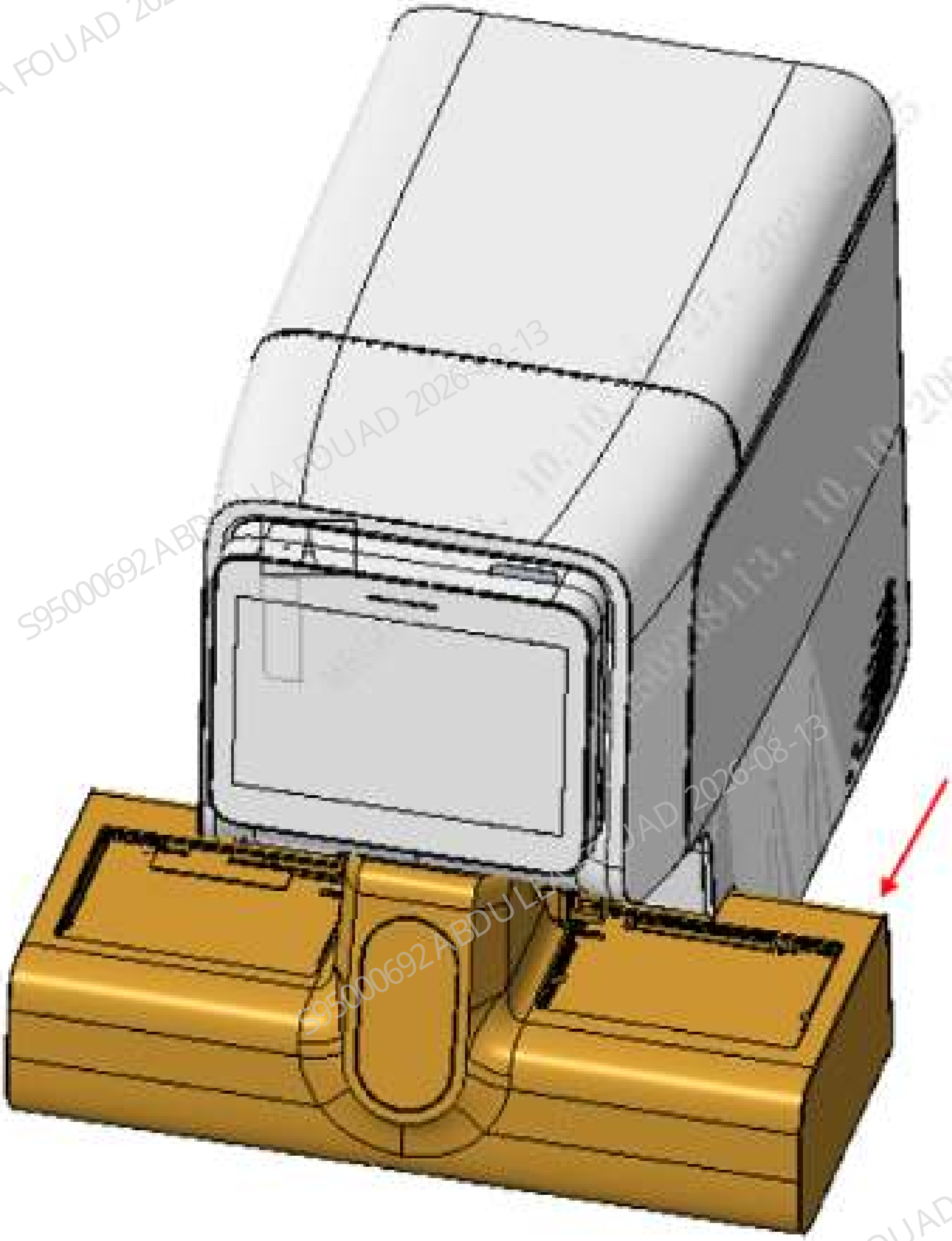


Table 8–37 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-085180-02	Autoloader, closed-sampling model(10x5)	N/A	N/A

Function:

Providing a closed sample compartment, allowing placing and taking a single sample, able to automatically send the sample for sample collection, and support ordinary WB test tubes and micro blood test tubes. Providing the autoloading function at the same time, able to store and transport sample racks, transport the sample rack to be tested from the loading tray to the sample analysis area, and send the tested sample rack to the unloading tray. The loading tray is used to store the sample rack to be tested, and the unloading tray is used to store the tested sample rack.

8.35.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

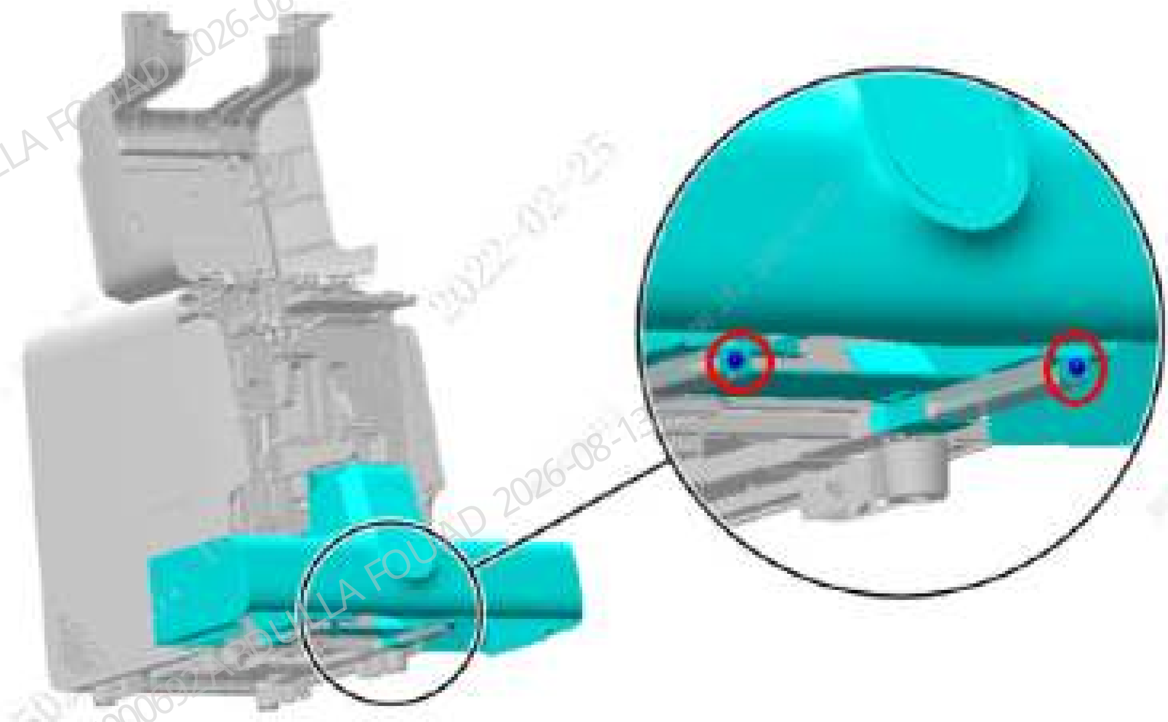
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

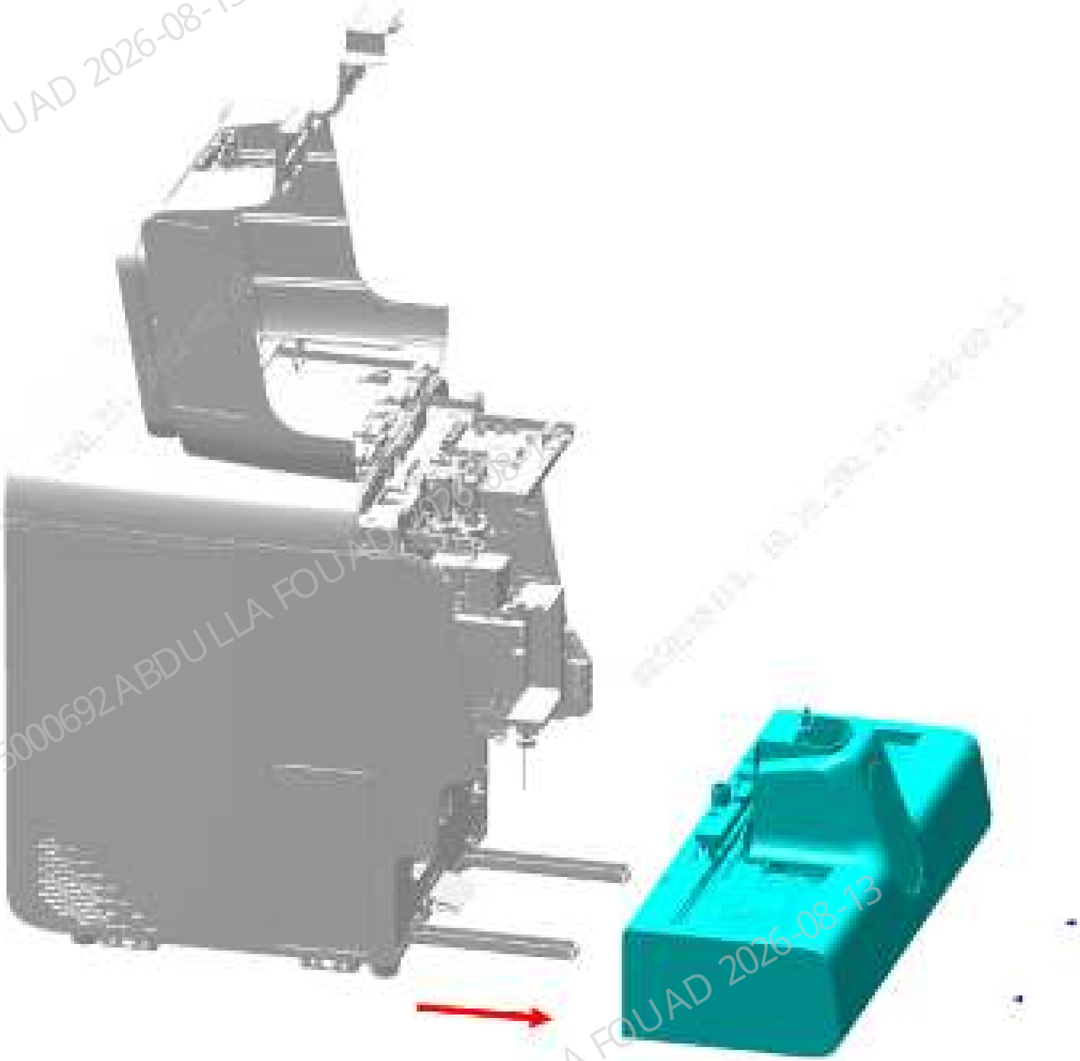
1. Open the front cover of the main unit and dock it on the top cover, as shown in the figure below.

Figure8–378 Opening the front cover



2. Remove the two M4x8 cross-recessed combination screws fixing the autoloader on the autoloader crossbeam, as shown in the above figure.
3. Drag the autoloader to an appropriate distance in the direction of the arrow and the direction of the operator, and loosen the connecting wire between the main unit and the autoloader.
4. Then pull the autoloader completely out of the autoloader crossbeam, that is, remove the autoloader, closed-sampling model (10x5) FRU, as shown in the figure below.

Figure8–379 Removing the autoloader assembly



Subsequent Requirements

Installation procedure:

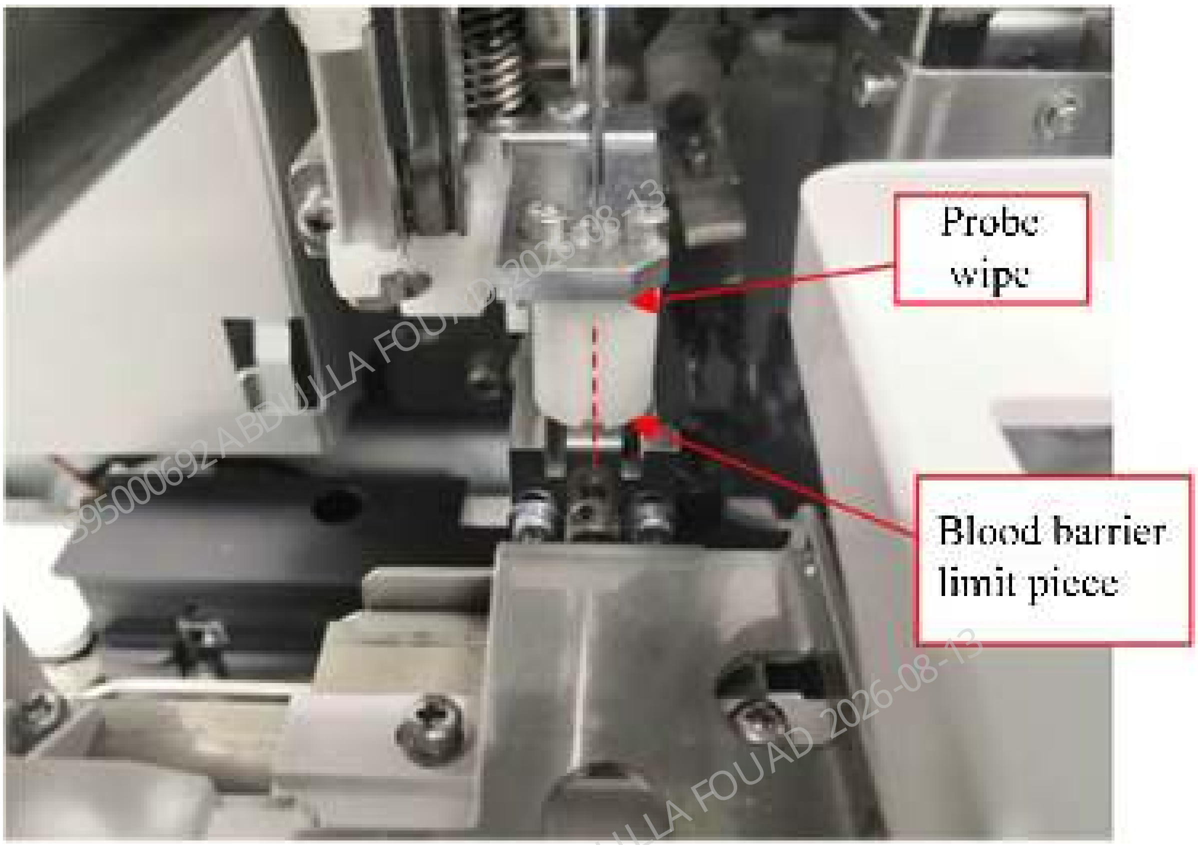
1. Place the autoloader, closed-sampling model (10x5) FRU on the autoloader crossbeam, push it to an appropriate distance towards the front panel of the main unit, and connect the connecting wire between the main unit and the autoloader.
2. Then push the autoloader, closed-sampling model (10x5) FRU along the crossbeam direction of the autoloader to fit the front panel of the main unit.
3. Use two M4x8 cross-recessed combination screws to lock the autoloader, closed-sampling model (10x5) FRU on the autoloader crossbeam.
4. Flip down the front cover.

8.35.3 Commissioning and Verification

Procedures

Commissioning and verification:

1. The left and right positions of the autoloader relative to the main unit have been adjusted in the assembly, so the FRU assembly does not require commissioning in the left and right positions.
2. Select **Menu>>Status>>Sensor Status** to enter the motor release torque status. Pull the sampling assembly to the front position a little ahead of the floating blood barrier. Observe whether the center of the sampling assembly probe wipe groove aligns with the center of the floating blood barrier limit piece (≤ ±0.5mm).

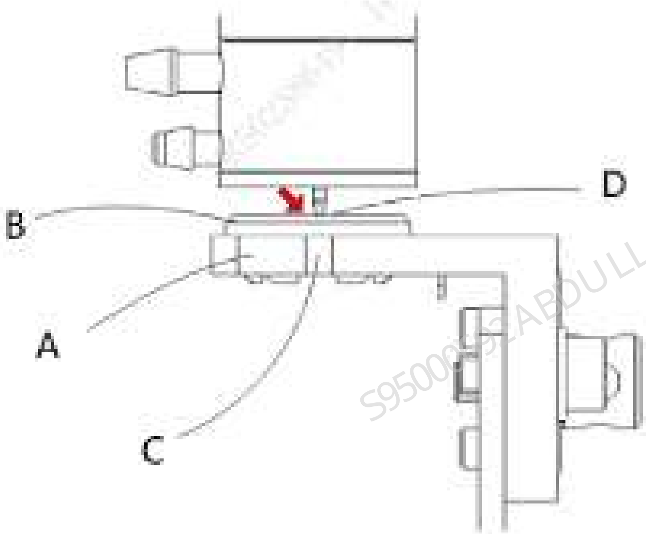


3. Select **Menu>>Service>>Commissioning & Self-Test>>Adj. Sample Probe Pos.>>AL asp. Pos.** to enter the AL asp. Pos. adjustment interface.

Figure8–380 AL Aspiration Position Adjustment Interface

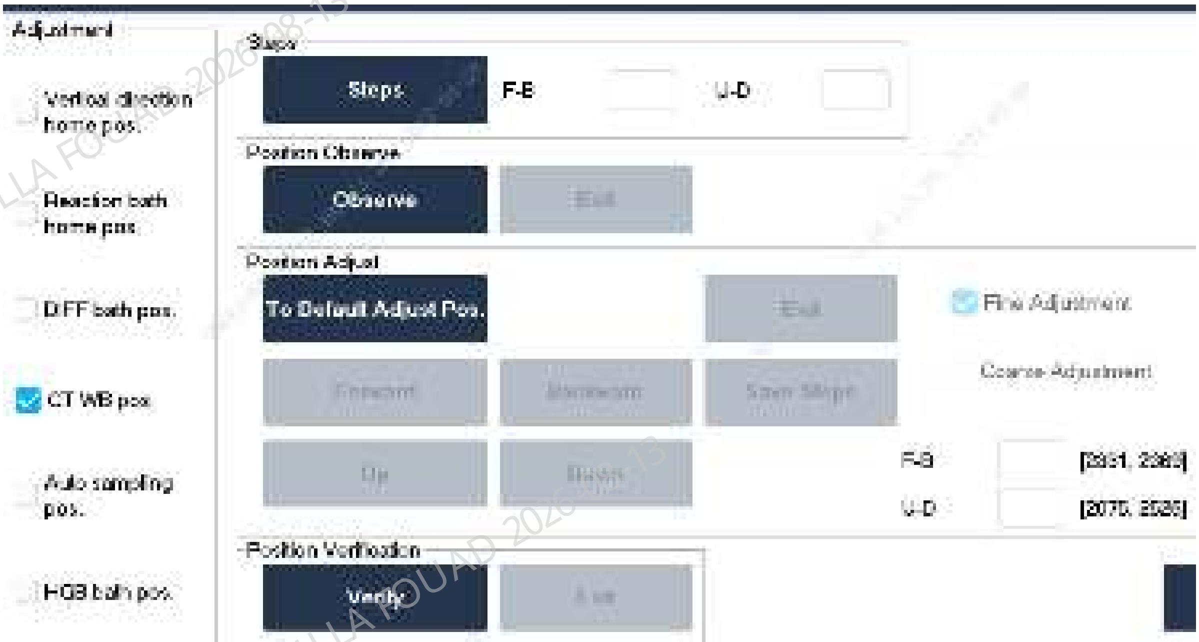
4. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sample probe is aligned with the groove of the blocking ring seat body.

Figure8–381 Observing Sampling Probe Position

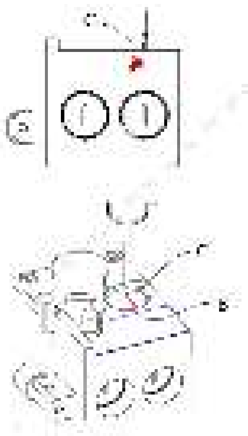


5. If the requirement of item 4 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the auto-sampling position.
6. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Adj. Sample Probe Pos.**>>**CT-WB Position** to enter the CT-WB position adjustment interface.

Figure8–382 CT-WB Position Adjustment Interface



7. Click **Observe**, look vertically from the side of the analyzer to the side of the analyzer, observe and confirm that the sampling probe is aligned with the arrow on the cover opening label in the horizontal direction, and flush with the top surface of the sample compartment in the vertical direction.



8. If the requirement of item 7 cannot be met, refer to the commissioning chapter for the sampling probe in the sampling assembly FRU relative to the CT-WB position.

Verification:

Confirming/Adjusting the Position of the Tube Hold Assembly

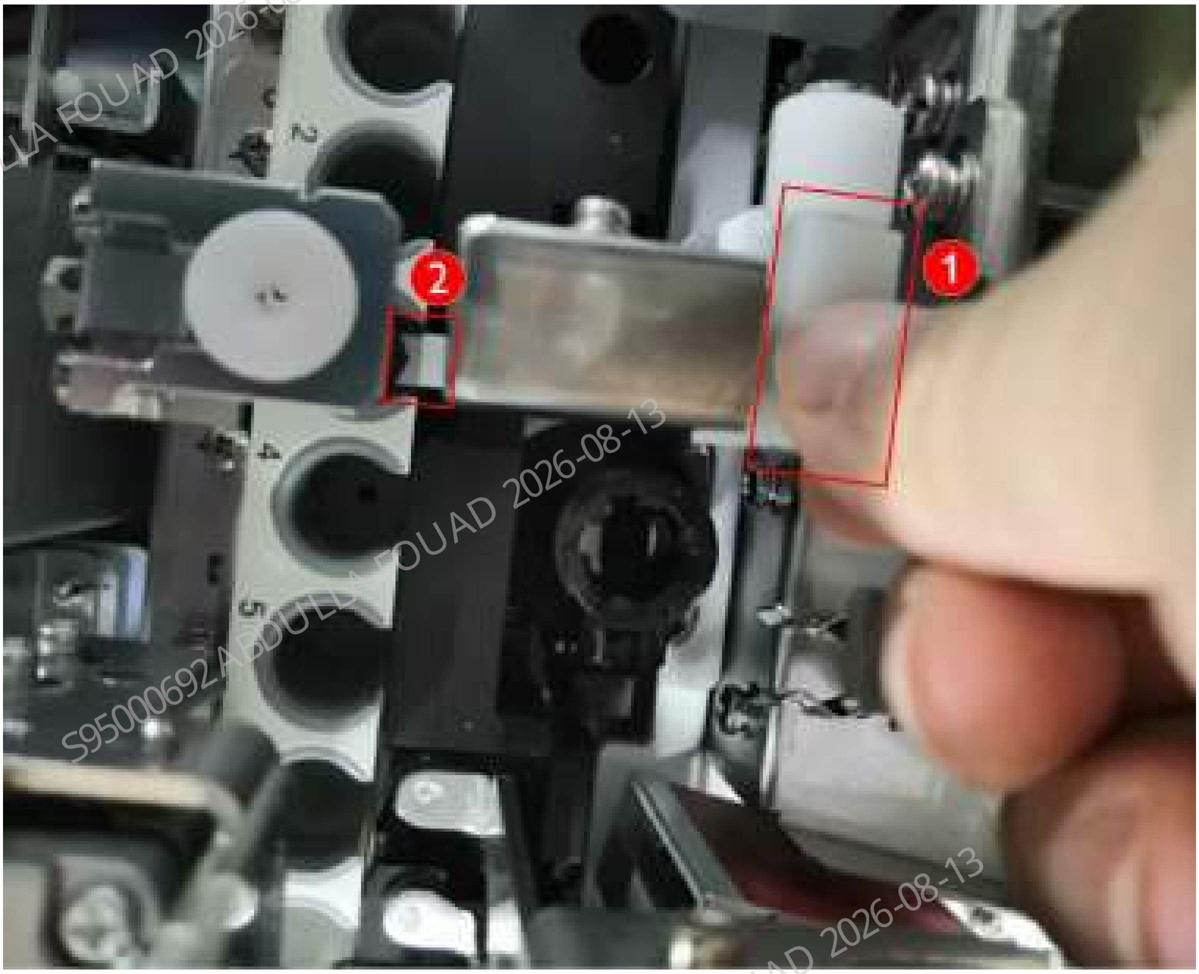
Confirm the position of the analyzer tube hold assembly. The steps are as follows:

1. Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Autoloader Assembly** to enter the autoloader self-test interface.

Figure8–383 Autoloader Self-test Interface

2. Click the **Assembly Initialization** button.
3. Put one row of test tube rack with an empty test tube in No. 4 test tube position on the loading tray.
4. Click the **Feed** button (leave a time interval after each click, and then click the button next time after the mechanism action is completed), and send the first position of the test tube rack to the piercing position).
5. Press the roller fixing plate of the hold assembly by hand, and observe whether the finger pressure block of the hold assembly is approximate aligned with the center of the front opening of the test tube rack.

Figure8–384 Ideal Position of the Pincher Roller of the Hold Assembly Relative to the Tube Rack



6. If the left and right sides of the finger pressure block of the hold assembly are close to the opening of the test tube (≤ 1 mm), the left and right positions of the hold assembly need to be readjusted. For how to adjust the position of the hold assembly, see the section for adjusting the position of hold assembly FRU: 8.23.3 Commissioning and Verification.

Confirming/Adjusting the Position of the Rotatory Compressing Assembly

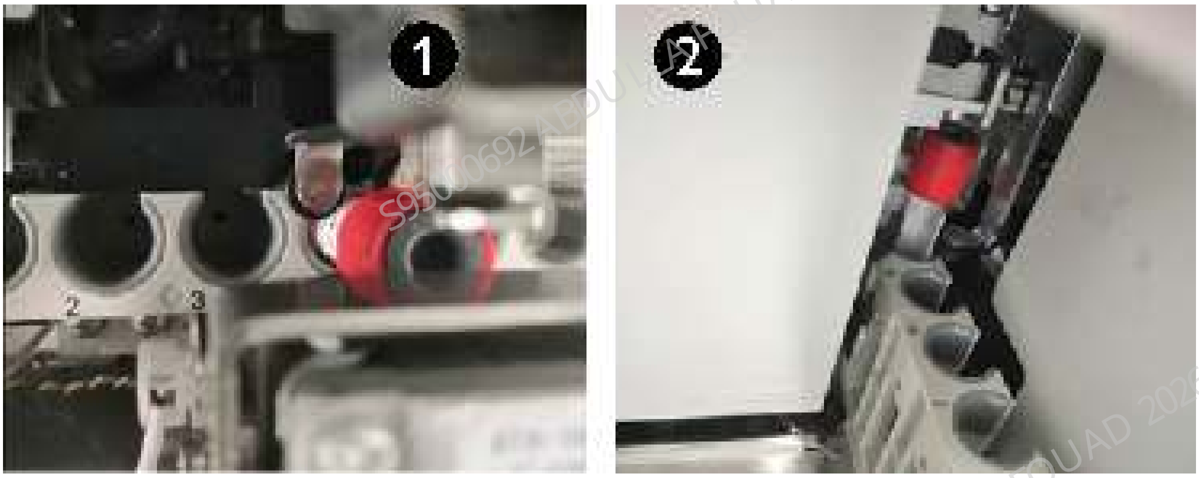
After the hold assembly position is confirmed, do not take out the tube rack, but continue to confirm the position of the rotatory compressing assembly. The steps are as follows:

1. Click the **Fix Tube** button.

Figure8–385 Autoloader Assembly Self-test Interface

2. Observe whether the two pinch rollers are close to the test tube. If they are not, readjust the left and right positions of the mechanism (failure to keeping them close will cause a false alarm of the autoloader counter).

Figure8–386 Ideal Position of the Pincher Roller of the Rotatory Compressing Assembly Relative to the Tube Rack



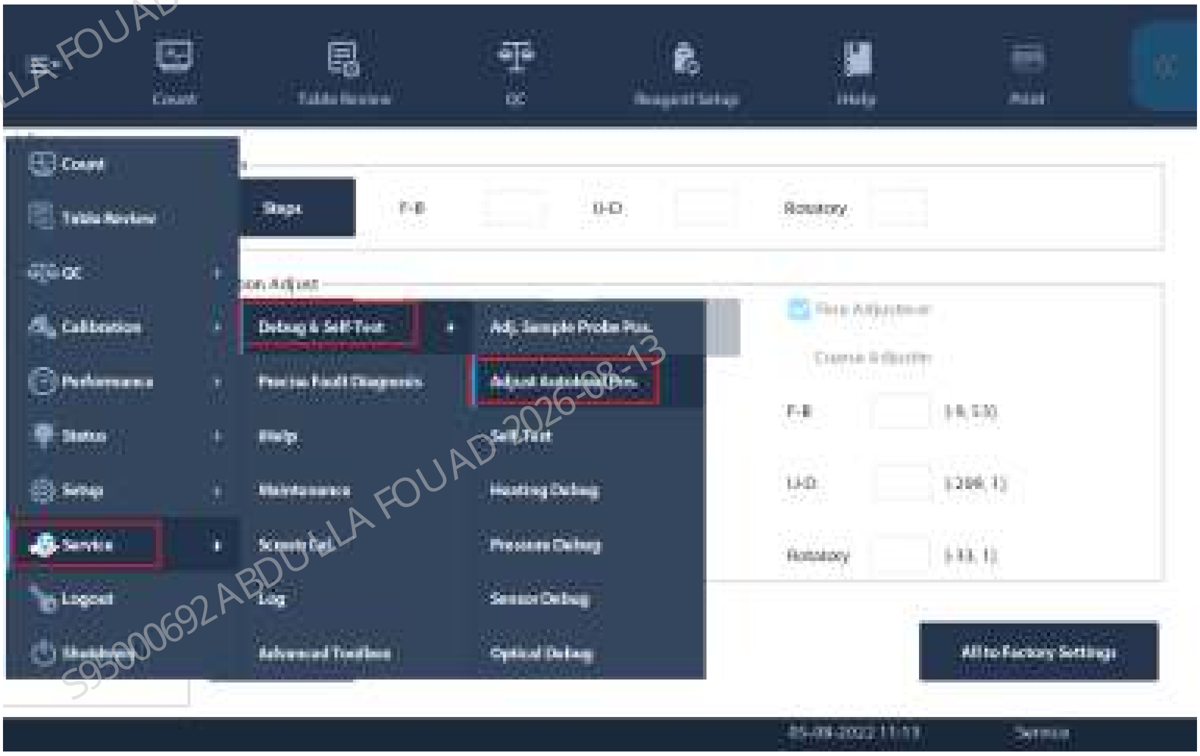
1	Left pinch roller	2	Right pinch roller

Confirming/Adjusting the Mix Mechanism Position

After the rotatory compressing assembly position is confirmed, do not take out the tube rack, but continue to confirm the mix mechanism position.

1. Select **Menu>>Service>>Commissioning & Self-Test>>Adjust Autoload Pos.**, and select *Autoloader Assembly* to enter the mix mechanism self-test interface.

Figure8–387 Path of Entering Adjust Autoload Pos.



2. Select **Mixing gripper (front position)**, and click **Check by Step**.

Figure8–388 Mixing Gripper (Front Position) Interface

3. On the displayed box, click **Init.**, and then **Gripper Forward**. Observe whether the gripper is centered relative to the left and right positions of the test tube rack hole position. If it is centered, click **Gripper Backward**, place an empty tube into the mixing position, click the **A&M** button, and observe whether the tube squeezes the tube rack when the gripper is placing the tube.

Figure8–389 "Check by Step" Interface of Mixing Gripper (Front Position)

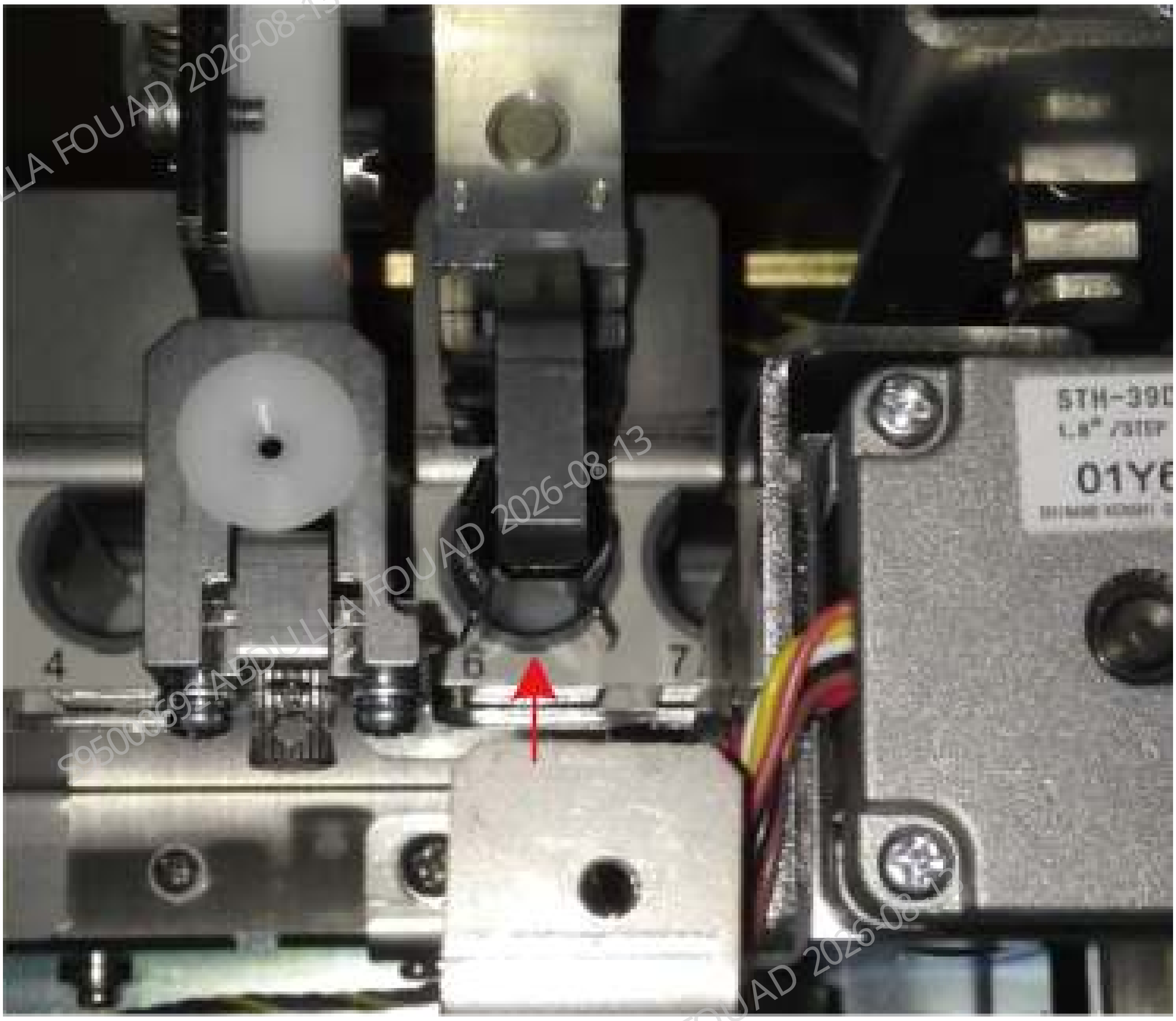
Figure8–390 Observing the Mixing Gripper Position

4. If the test tube squeezes the test tube rack, exit **Check by Step**, click **To Adjust Pos.**, and then press the **Forward**, **Backward**, **Left**, and **Right** buttons to fine adjust the gripper position.

Figure8–391 Observing the Mixing Gripper Position

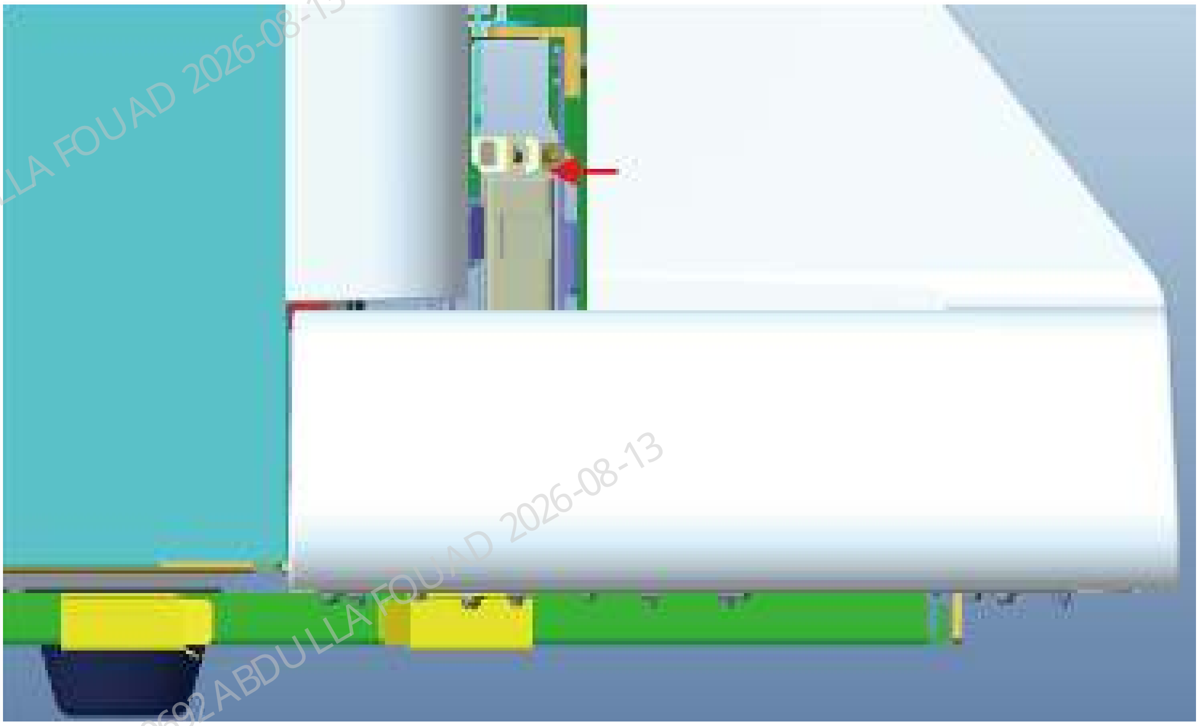
5. Finally enter **Check by Step** again, and observe again whether the tube squeezes the tube rack hole position when the gripper is placing the tube.

Figure8–392 Observing the Mixing Gripper Position



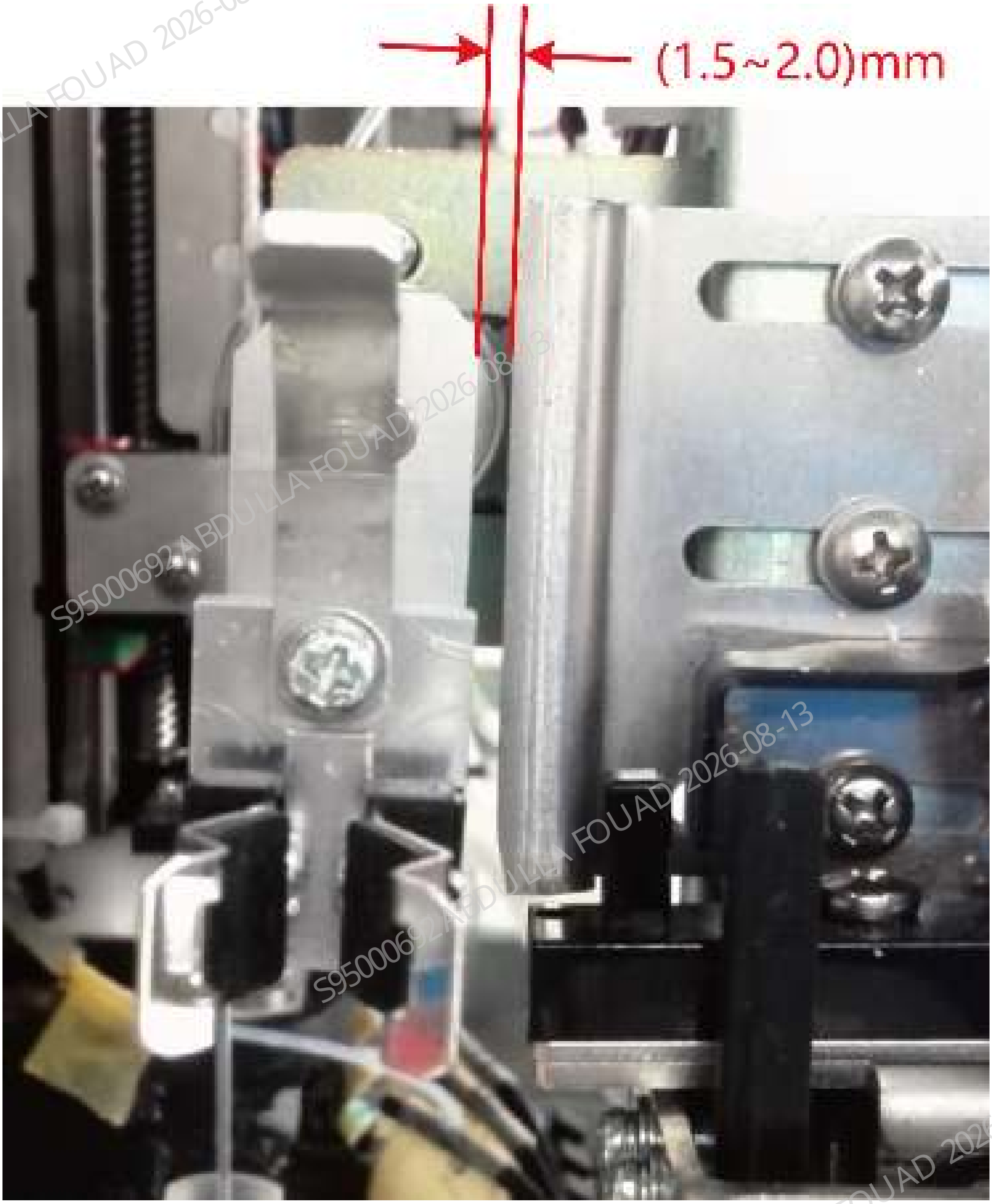
6. After confirming the left, right, front, and back positions of the gripper relative to the test tube rack, confirm the upper and lower positions of the gripper. The distance between the bottom surface of the gripper and the top surface of the test tube rack needs to be controlled within the range of (2.0 ~ 2.5) mm. If slides are available, use two standard slices (thickness 1.1~1.2 mm) or one thick slice (nonstandard) as an auxiliary tool for test. If there are no slides, perform visual inspection. If the requirements are not met, make adjustment through **Up** and **Down** buttons of the gripper.

Figure8–393 Confirming the Mixing Gripper Height Position



7. Confirm the clearance between the gripper relative to the anti-collision bock. Click **Mixing gripper (front position)**, select **Check by Step**, click **Initialize>To Center**, and observe whether the clearance between the gripper and the anti-collision bock is in the range of 1.5~2.5 mm. Use two slices as a tool for confirmation.

Figure8–394 Distance Requirements between the Mixing Gripper and Anti-Collision Bock

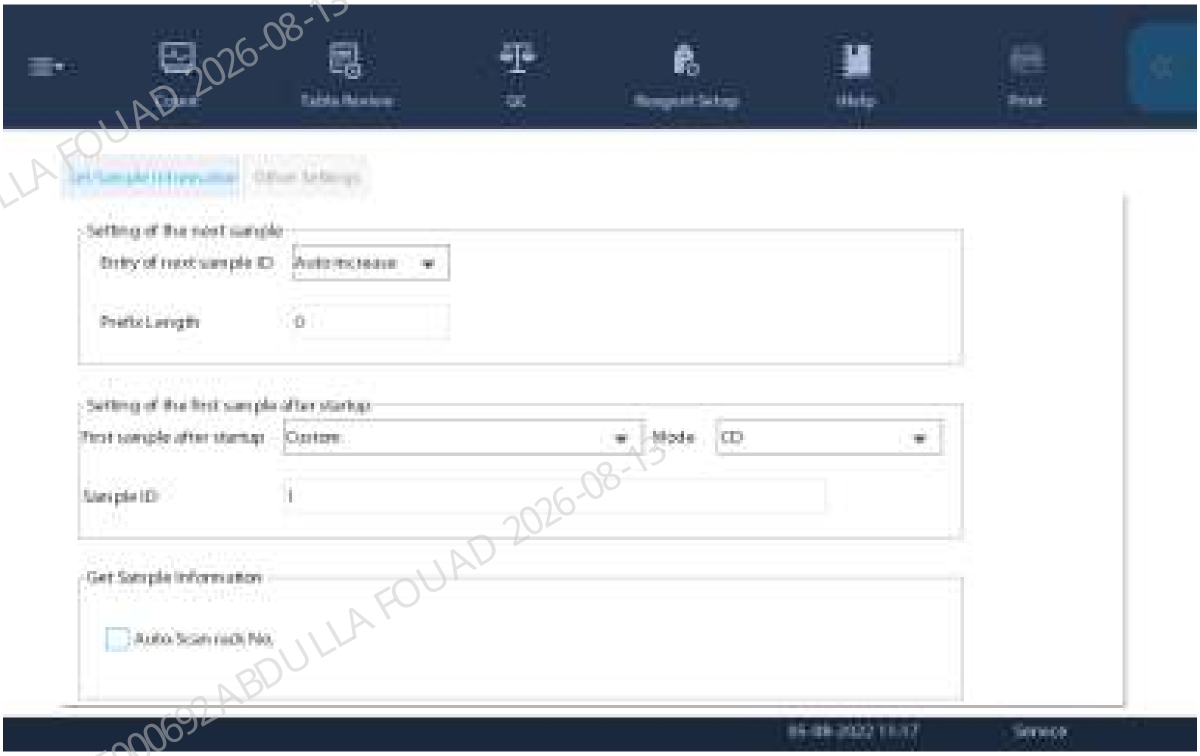


Confirming the Scanner Position

If the user does not need the sample barcode scanning function, this section can be skipped.

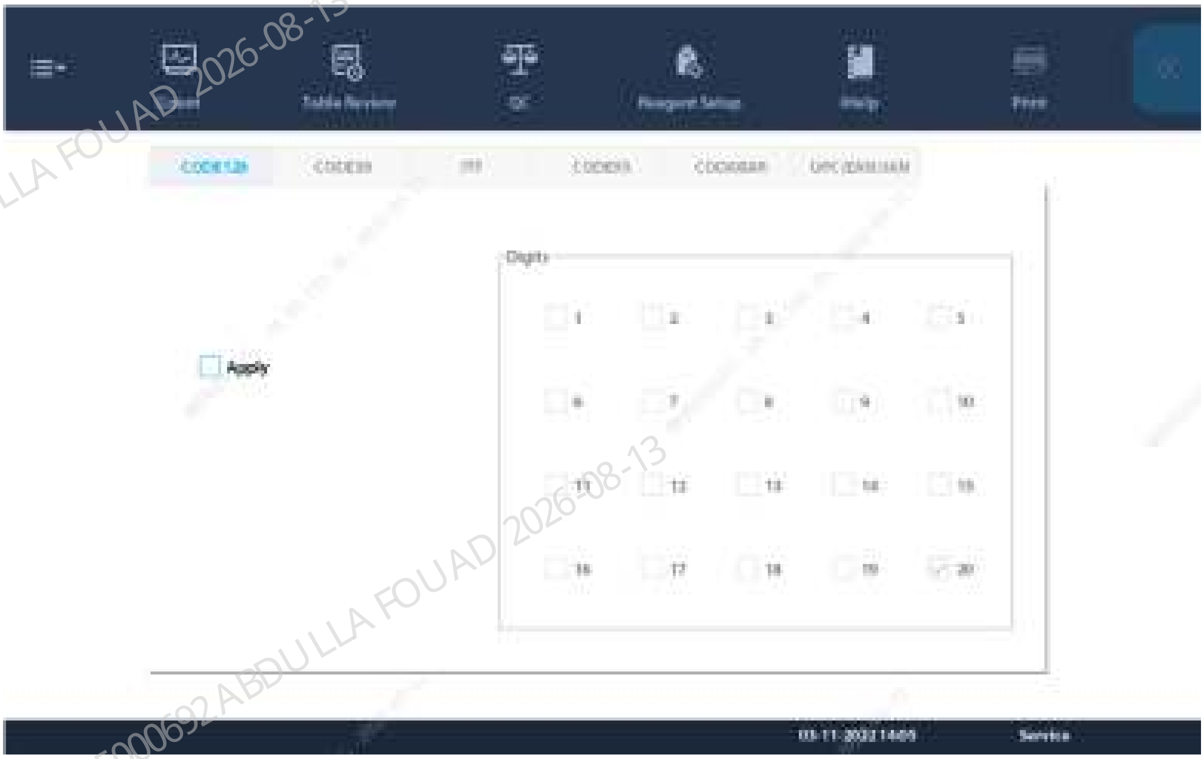
1. Click **Menu**>>**Setup**>>**Auxiliary Setup** to enter the interface for setting the sample information getting method. Confirm that the **Auto-Scan rack No.** option has been checked.

Figure8–395 Auto-Scan Setup Interface



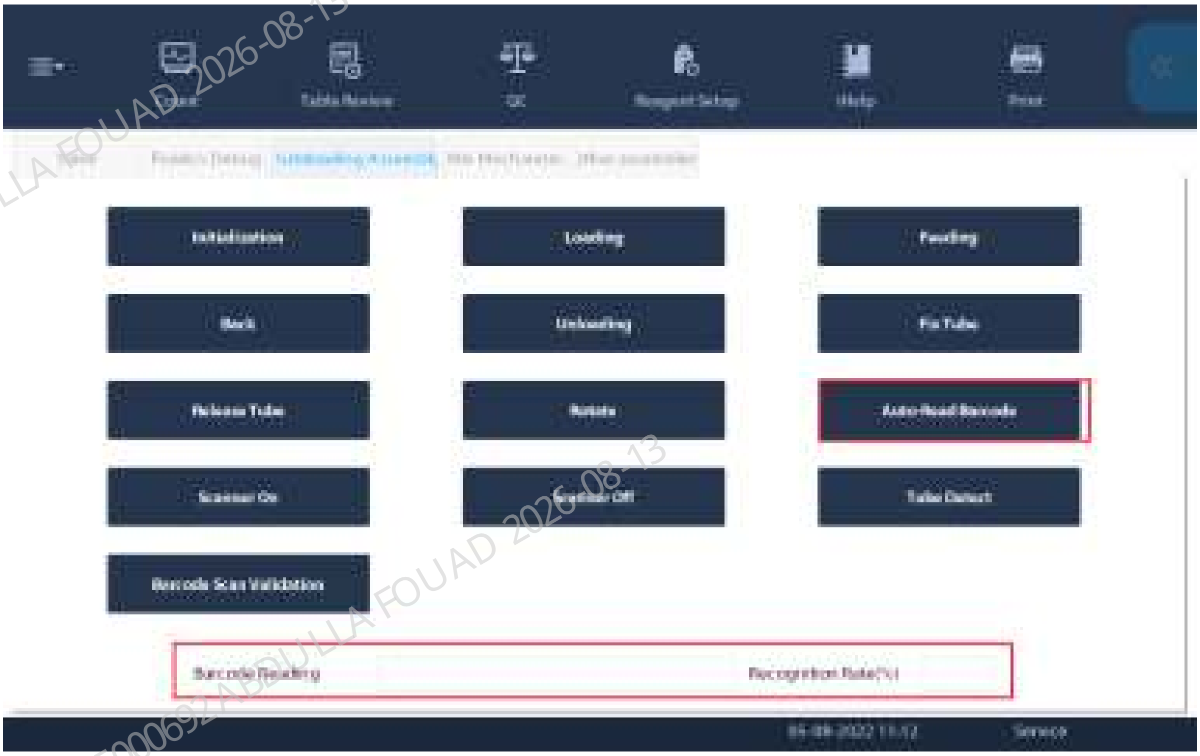
2. Use the barcode scanning software or machine barcode recognition function to identify the barcode system used by the client.
3. Click **Menu>>Setup>>System Setup>>Barcode** to enter the barcode setup interface. Check the corresponding code system, length, and check bit. Do not check the code system and length not used for the client.

Figure8–396 Built-in Barcode Setup Screen



4. After completing the above setup items, click **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**, and select **Autoloader Assembly** to enter the autoloader self-test interface.
5. After clicking **Assembly Initialization**, put a test tube with barcode in the rotating scanning position (the test tube is placed in the test tube rack).
6. Click **Auto-Read Barcode** to check whether the barcode content display box displays the barcode content and whether the barcode recognition rate display box shows 100%. If not, first confirm whether the red light of the scanner is turned on when you click the **Auto-Read Barcode** button. If the light has been turned on but the barcode cannot be scanned, adjust the position according to the position confirmation description in the chapter about scanner FRU.

Figure8–397 Autoloader Assembly Self-test Interface



8.36 Reaction bath assembly

8.36.1 General Information

Figure8–398 Photo of Bath Assembly (FRU)

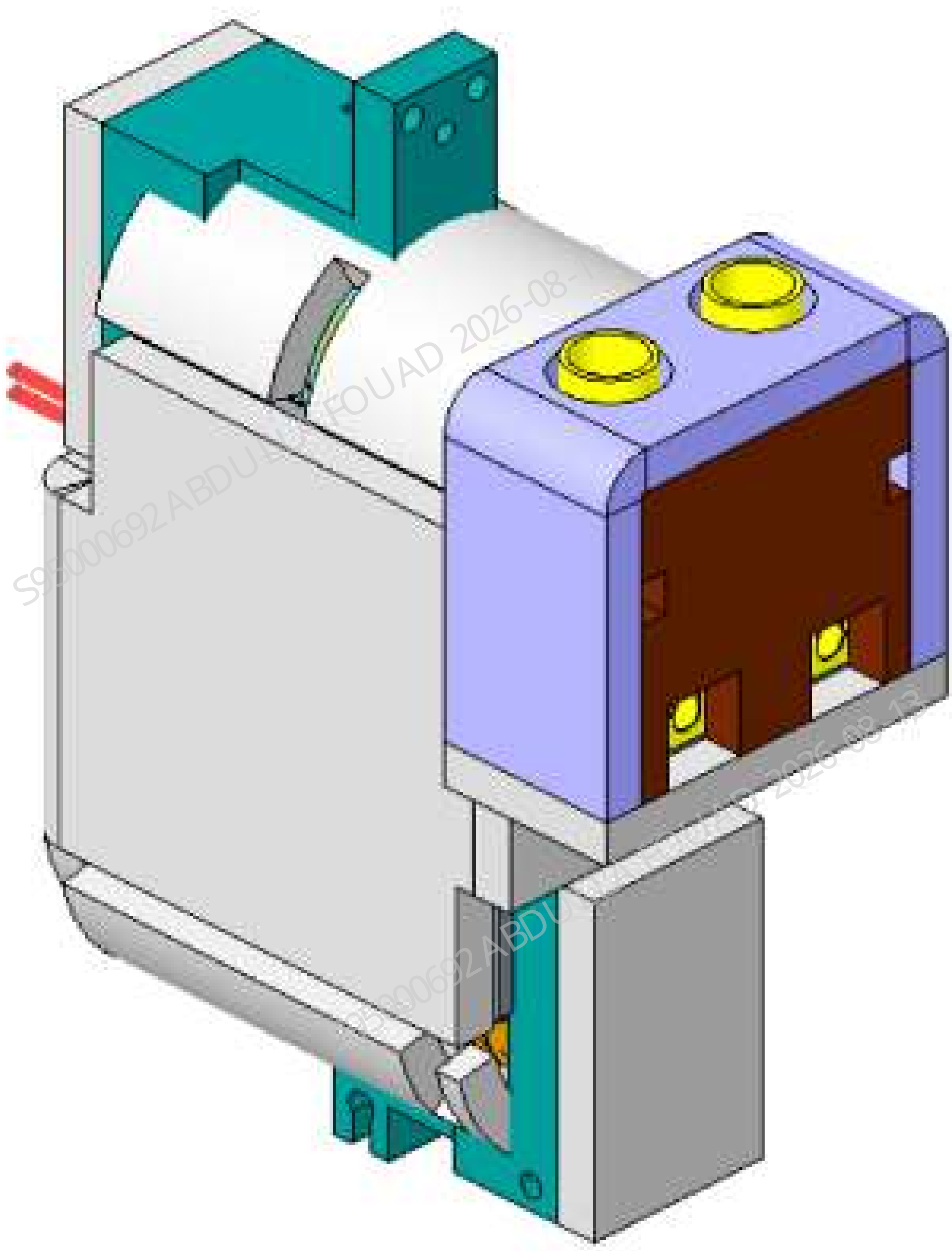


Table 8–38 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081652-00	Bath assembly (FRU)	N/A	N/A

Function:

Provide places for reaction and mixing of samples in the DIFF and RET channels, which can realize constant temperature incubation and provide the function of preheating LD and DR reagents.

8.36.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

One pair of tweezers
One Allen wrench

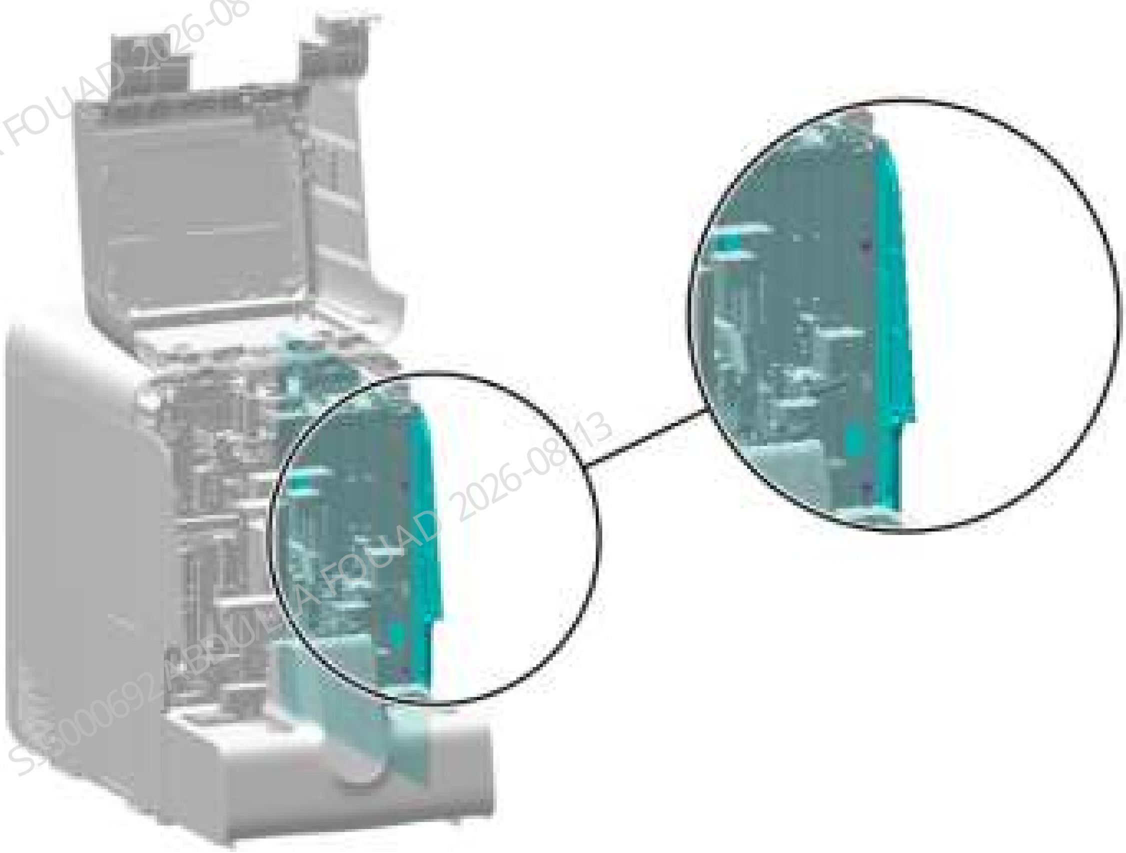
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

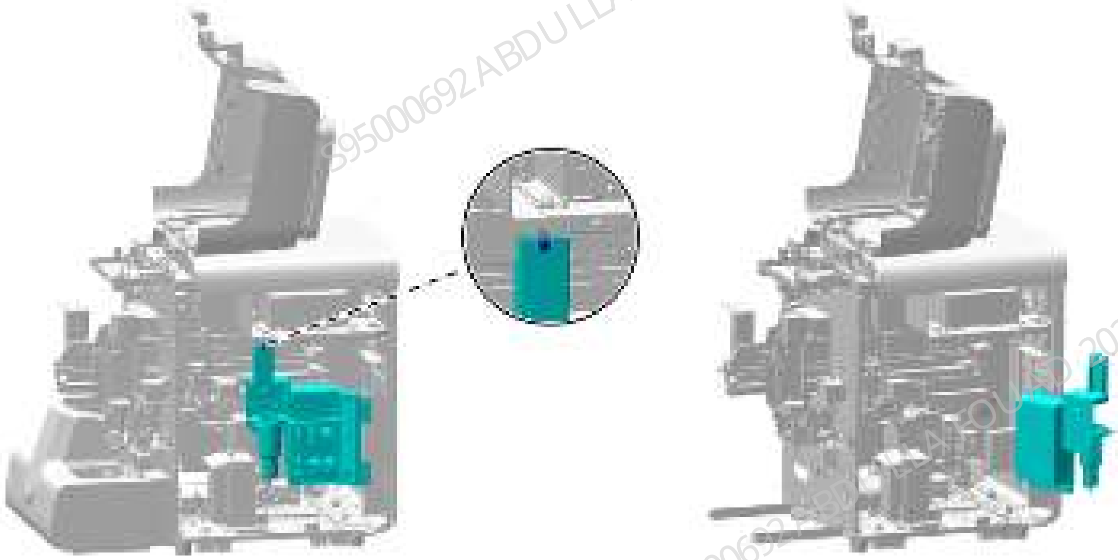
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–399 Removing the right cover



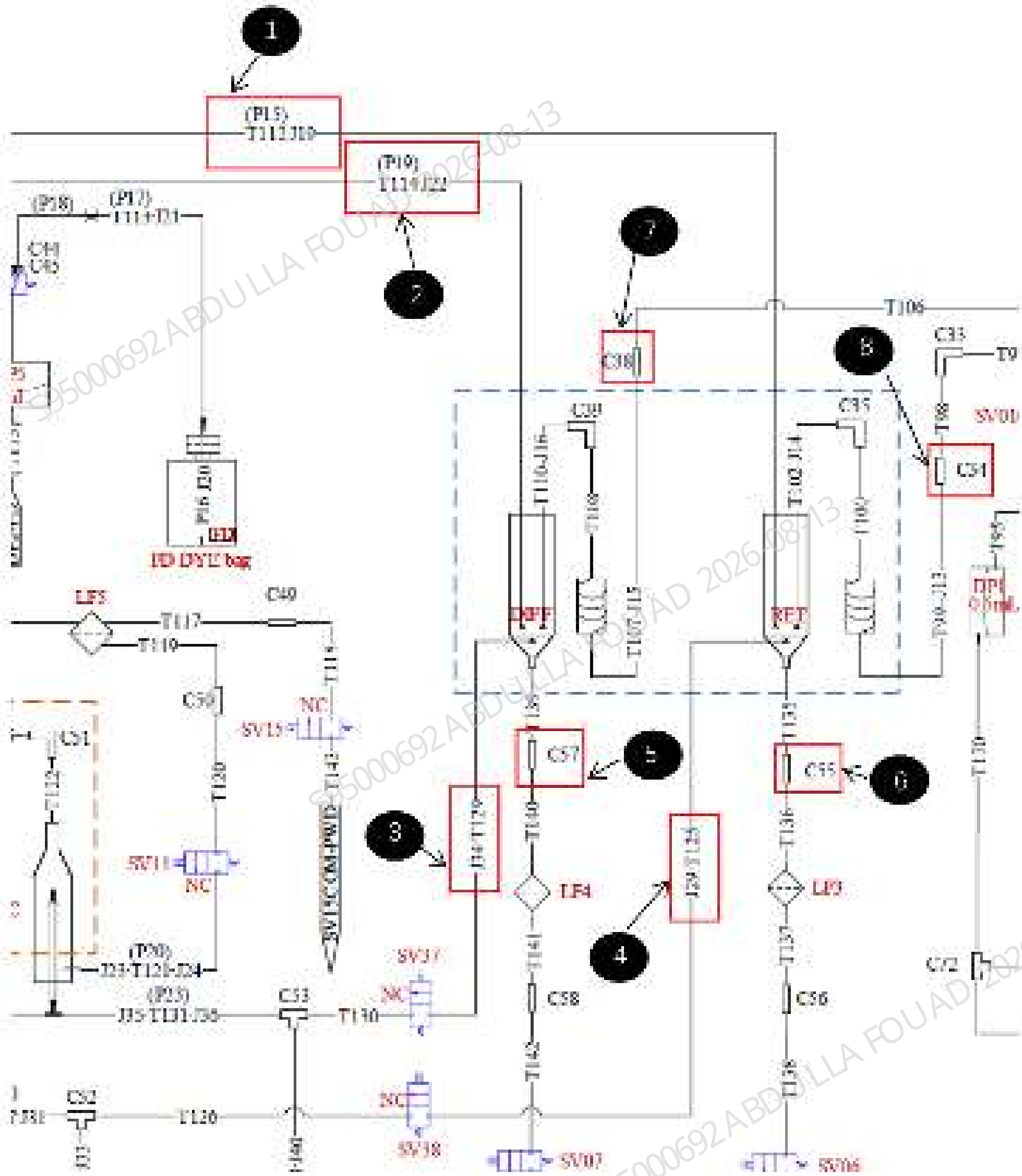
2. As shown below, unscrew the one screw that fixes the right bath subassembly; as shown in the figure, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–400 Opening the right cover assembly



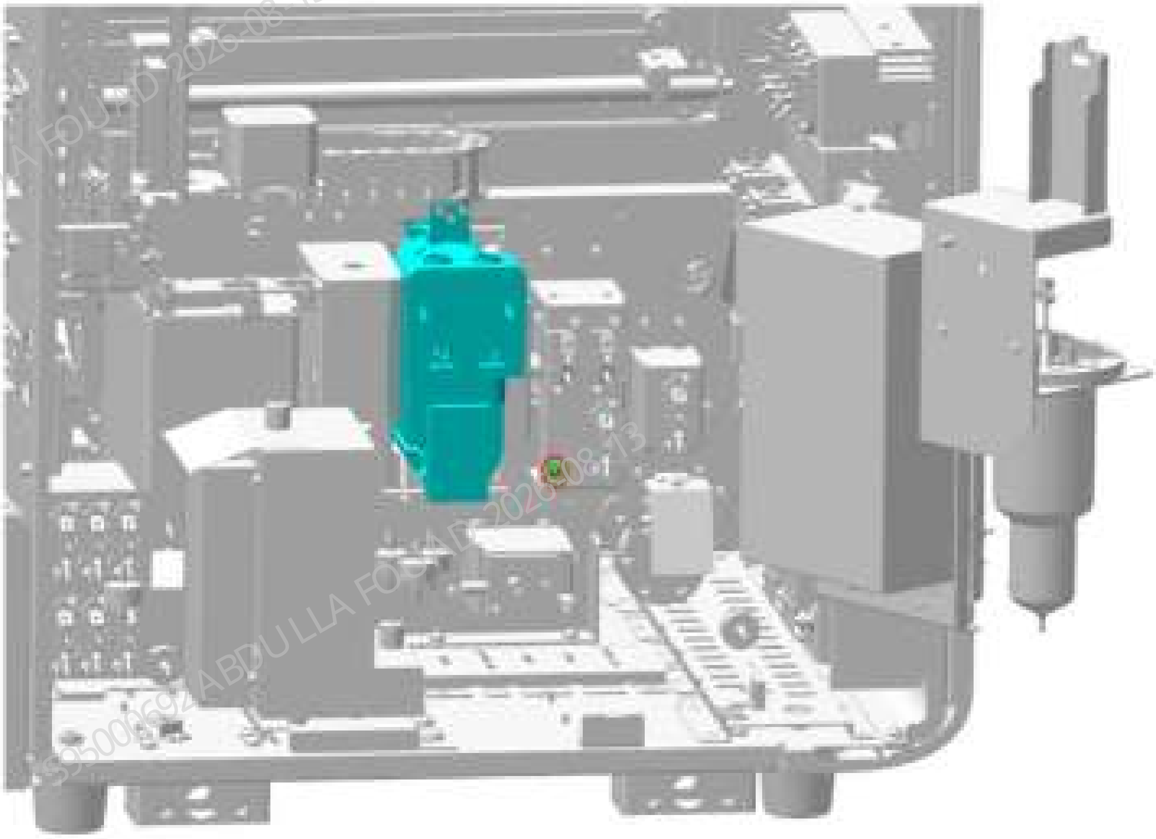
3. As shown below, unplug the eight tubes connected to the reaction bath. Wherein, the tube mark of T112-J9 at (1) is FR upper right. The tube mark of T114-J22 at (2) is FD upper left. Pull out of J34-T129 at (3) from the SV37, and the tub is marked as SV37-3. Pull out of J29-T125 at (4) from the SV37, and the tub is marked as SV38-3. At (5), first cut off the cable tie between C57 and T140, and then pull out C57. At (6), first cut off the cable tie between C55 and T136, and then pull out C55. Pull out C38 at (7), and the tube is marked as LD. Pull out C34 at (8), and the tube is marked as DR.

Figure8–401 Diagram of Reaction Bath Assembly Pipeline



4. Loosen the screw securing the ground wire as shown below.

Figure8–402 Diagram of Removing the Grounding Screws of Reaction Bath Assembly



5. As shown in Figure **Figure8–403**, loosen the three screws that fix the reaction bath assembly; as shown in Figure **Figure8–404**, pull the assembly upward so that the groove on the assembly is separated from the reaction bath fixing block, pull it outward, and disconnect the wires of heating rod, temperature sensor, and temperature switch on the assembly with the analyzer.

Figure8–403 Diagram of Removing the Fixing Screws of the Reaction Bath Assembly

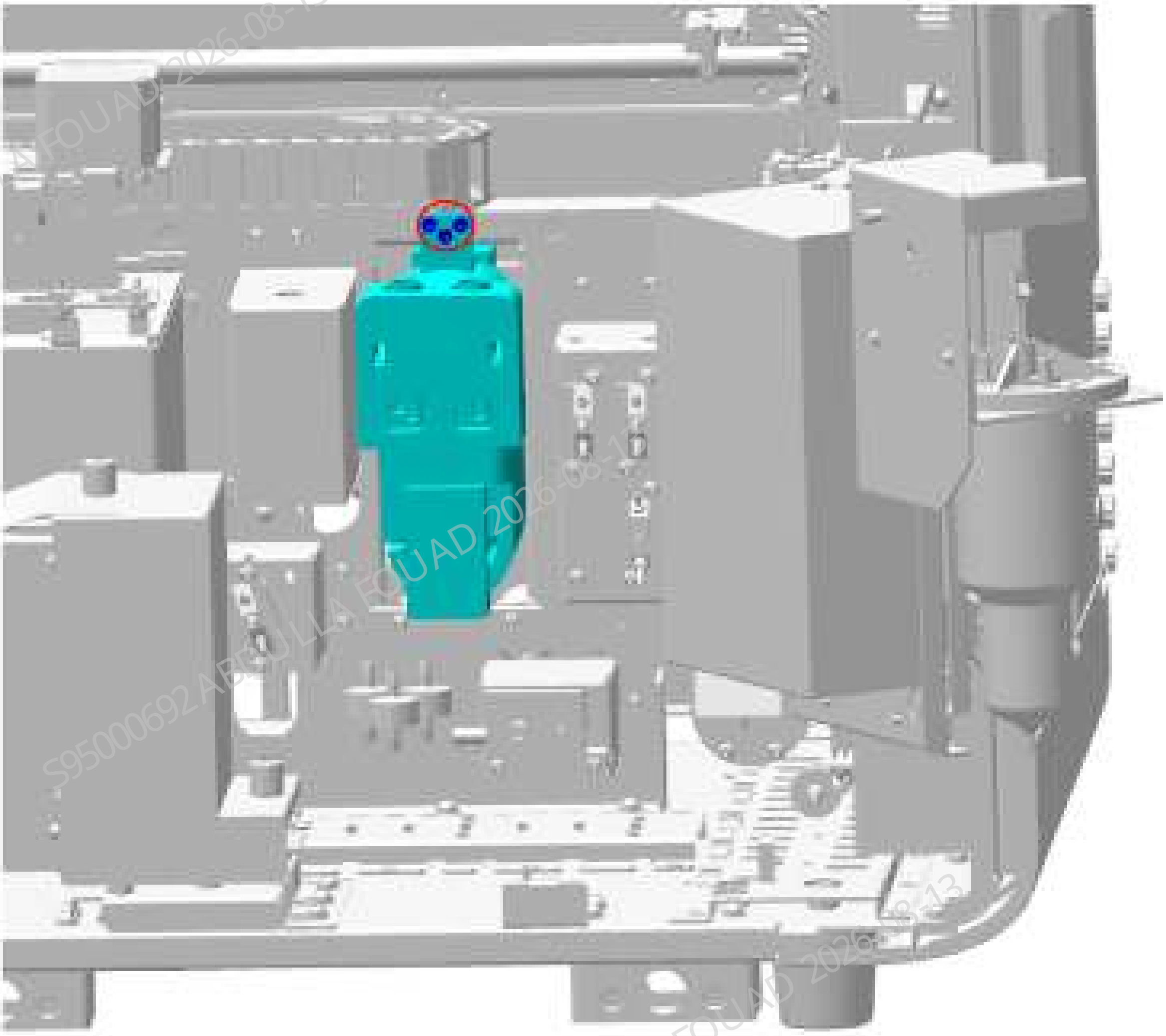
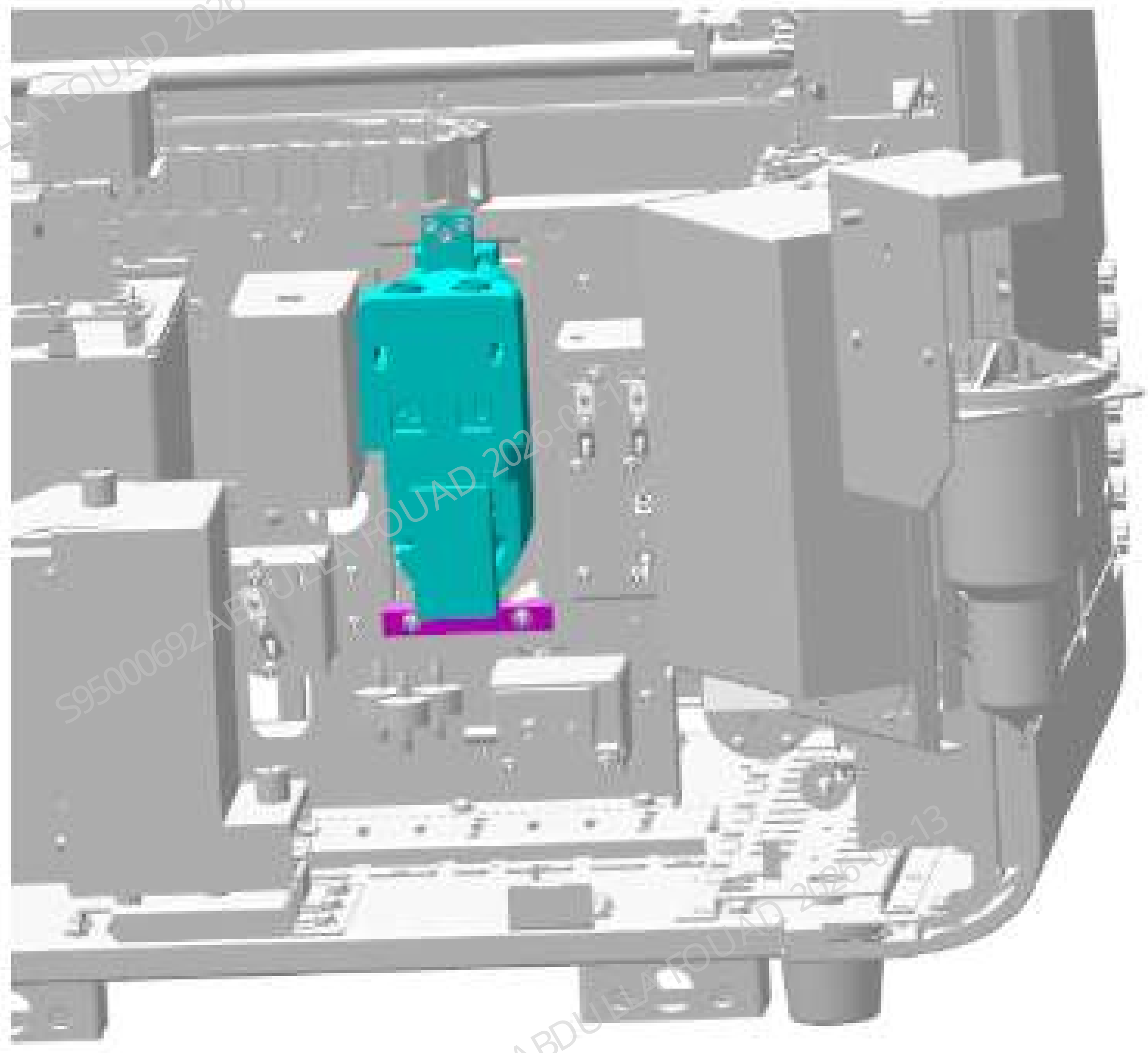


Figure8–404 Diagram of Separating the Reaction Bath Assembly from the Reaction Bath Fixing Block



Subsequent Requirements

Installation procedure:

1. Connect the pipeline on the reaction bath assembly according to Figure **Figure8–401**.
2. Plug in the reaction bath wire.
3. As shown in Figure **Figure8–404**, clamp the reaction bath assembly on the reaction bath fixing block.
4. As shown in Figure **Figure8–403**, use three screws to fix the reaction bath assembly on the main unit.
5. As shown in Figure **Figure8–402**, use screws to fasten the ground wire of the reaction bath assembly.
6. Turn the right side panel to its original position and fix it using one screw.
7. Restore the right, rear, and front covers.

8.36.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning:

1. Adjust the position of the sampling probe relative to the position of the reaction bath.
   For details on commissioning steps and effects, please refer to the content of reaction bath commissioning of aspiration module.
2. Confirm the temperature status
   Select **Status**>>**Temperature & Pressure** to see if the temperature of the **DIFF/RET reaction bath** is within the range.

Verification:

1. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.37 Sheath fluid impedance assembly

8.37.1 General Information

Figure8–405 Photo of Sheath Fluid Impedance Assembly (FRU)

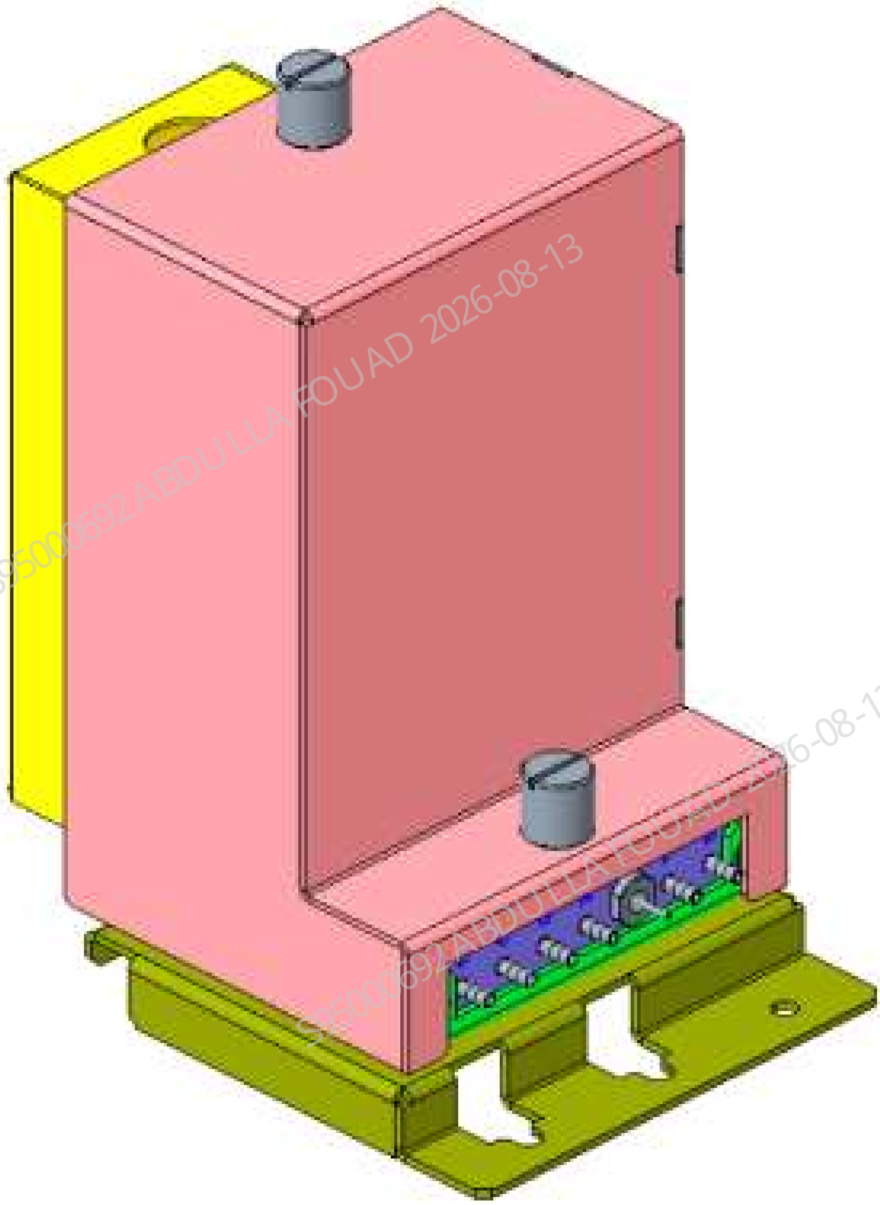


Table 8–39 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081653-00	Sheath fluid impedance assembly (FRU)	115-071243-00	Sheath fluid count bath assembly

Table 8–39 Information About the Assembly(continued)

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
		115-078179-00	Back sheath isolation bath assembly (3120)

Function:
Provide a test platform for the detection of RBC/PLT and its related parameters.

8.37.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.
3. Be careful not to damage the connector when removing the tube.

Required tools

Cross-headed screwdriver
One pair of tweezers

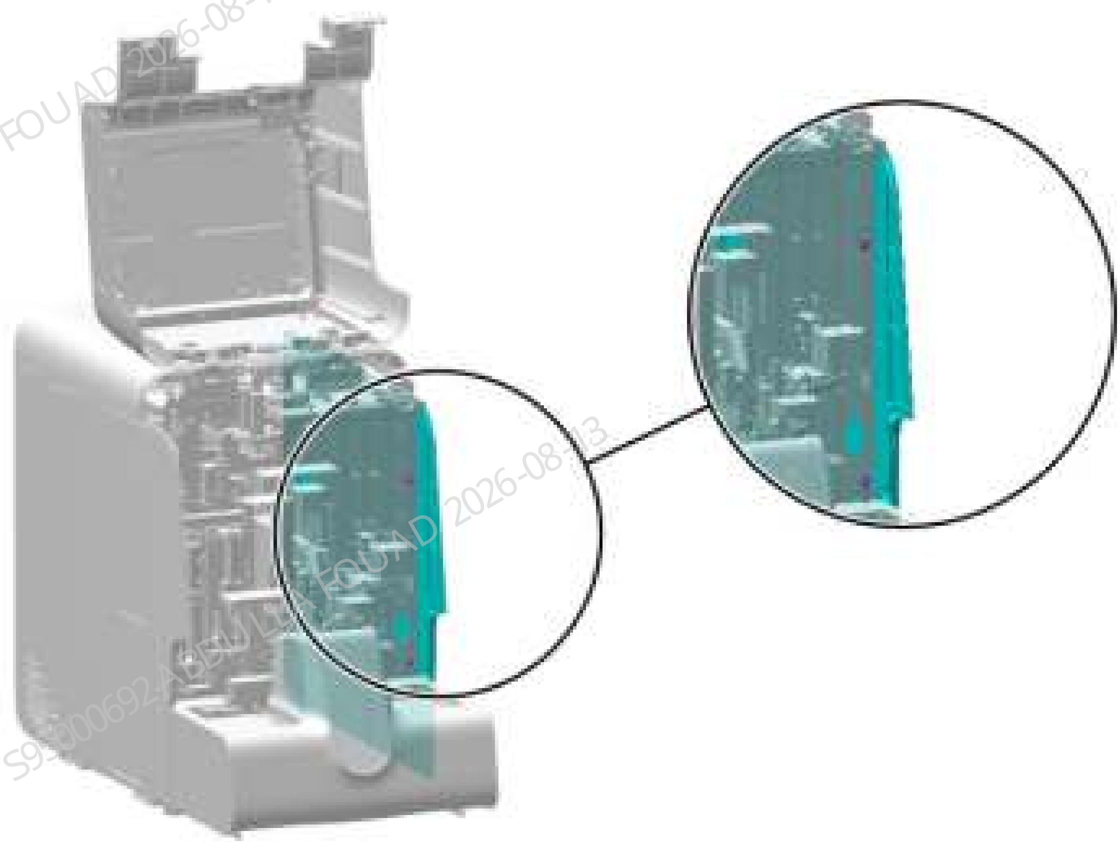
Pre-requirements

Select **Menu>>Service>>Maintenance>>Empty and Prime>>ISU/ISW Bath Emptying**, click **Menu>>Status>>Temperature and Pressure>>Release Pressure**, turn off the power, and unplug the power cord.

Procedures

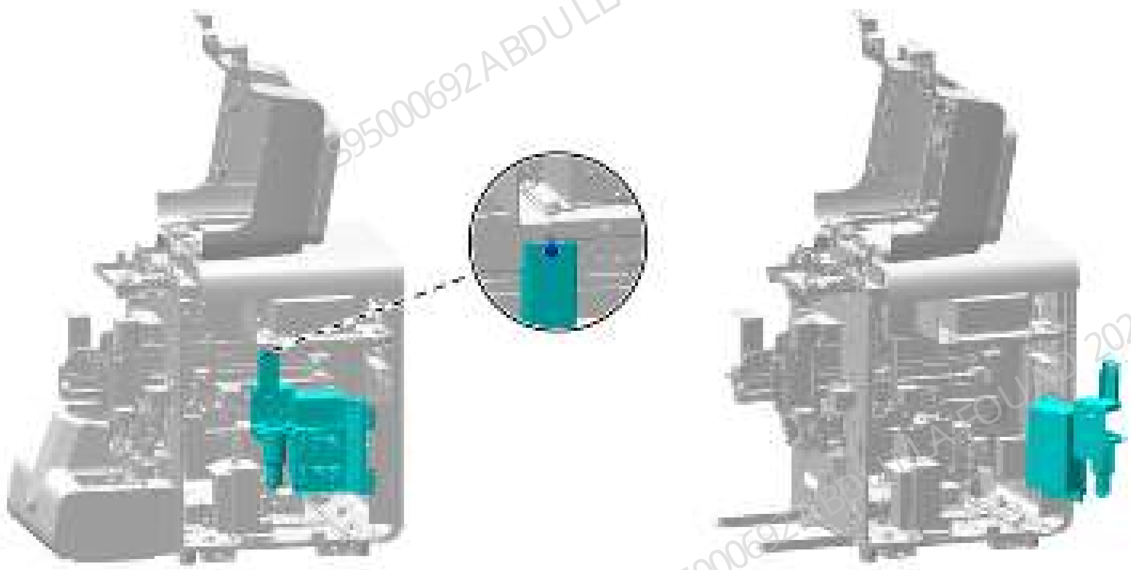
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–406 Removing the right cover



2. Unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

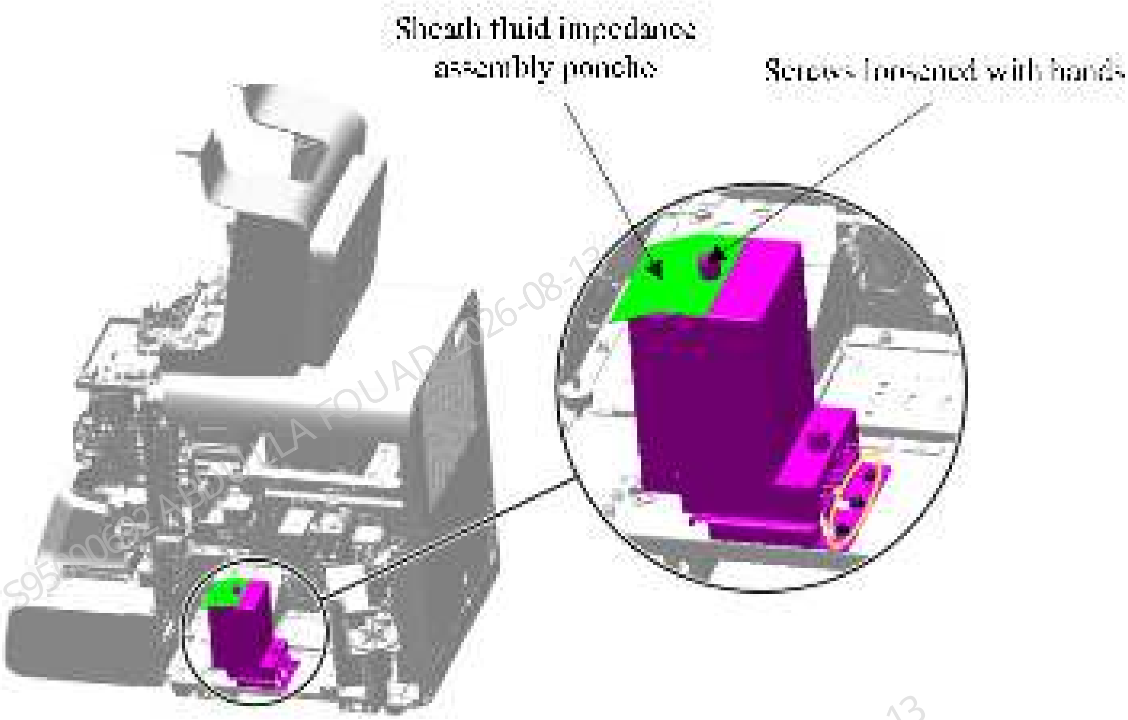
Figure8–407 Diagram of Opening the Right Cover



3. Disconnect the RBC signal wire from the impedance board.

4. Use tweezers to pull out the rubber tube on the sheath fluid impedance assembly (FRU) according to the tube marks "D-T77, C-82, G-T80, A-170, H-T71, F-T72, and B-T83".
5. As shown below, use a cross-head screwdriver to unscrew the three screws (in the yellow circle below) that fix the sheath fluid impedance assembly (FRU), and remove the assembly.

Figure8–408 Diagram of Removing the Sheath Fluid Impedance Assembly (FRU)



6. Unscrew the hand screw above the sheath fluid impedance assembly (FRU) and tear off the poncho.

Subsequent Requirements

Installation procedure:

1. Stick the torn poncho to the new sheath fluid impedance assembly (FRU).
2. Connect the RBC signal wire with the impedance board.
3. Use three screws to fix the sheath fluid impedance assembly (FRU) onto the base plate of the main unit.
4. Insert each tube back into the sheath fluid impedance connector.
5. Close the right cover assembly and re-secure it to the main unit.
6. Install the right cover to the main unit, and flip down the front cover.

8.37.3 Commissioning and Verification

Procedures

1. Tap the **Menu>>Calibrate>>CBC Gain Calibration**. Enter the MCV and HGB gain calibration screens, and enter the MCV_G target value of the batch of calibrators used; as shown in the figure below.

Figure8–409 MCV-G target value of calibrator

Mode CT-WB

	Target	1	2	3	Result
MCV_G	88.4				
HGB	4.40				

2. Perform OV-WB counting continuously with the calibrators until **OK** is displayed in the **Results** column for both MCV and HGB, as shown in the following figure.

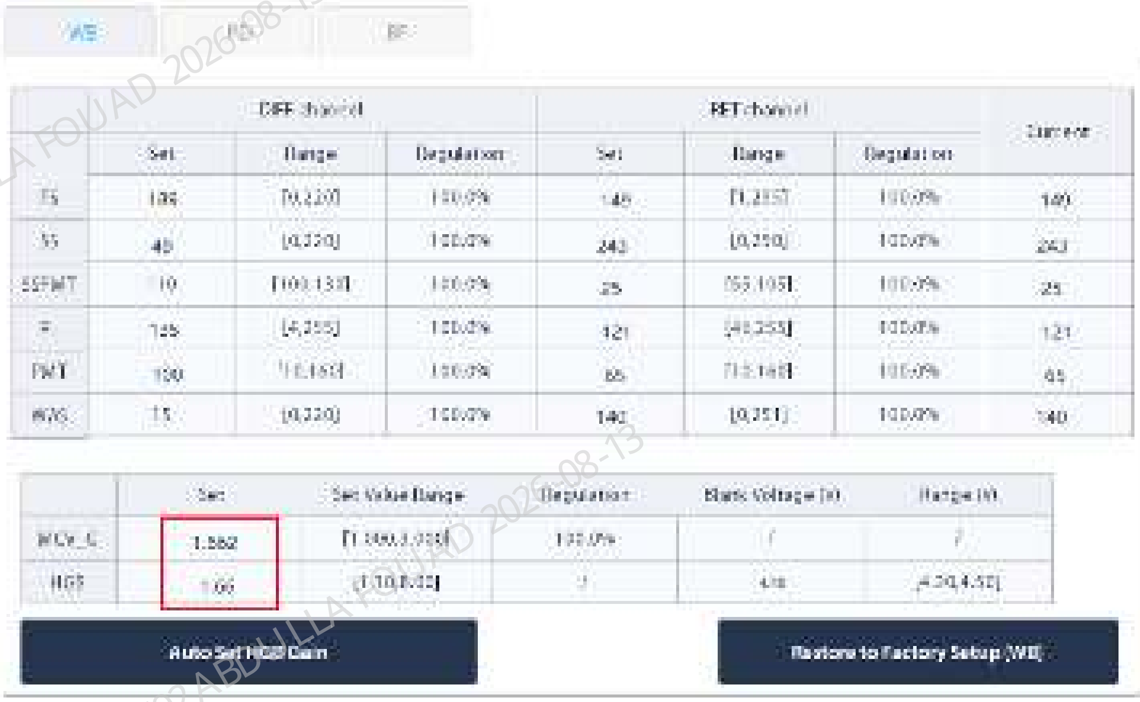
Figure8–410 MCV Results

Mode CT-WB

	Target	1	2	3	Result
MCV_G	88.4	89.2	89.0		OK
HGB	4.40	4.46	4.51		OK

3. When exiting the CBC gain calibration screen, select **Yes** to save the gain in the pop-up prompt box.
4. Click**Menu>>Setup>>Gain Setup** in turn to check whether the **setting values** of MCV and HGB gains have been updated (if not, the default gain of the system is adopted), as shown in the figure below.

Figure8–411 Gain setting value



8.38 Sheath fluid count bath assembly

8.38.1 General Information

Figure8–412 Photo of sheath fluid count bath assembly

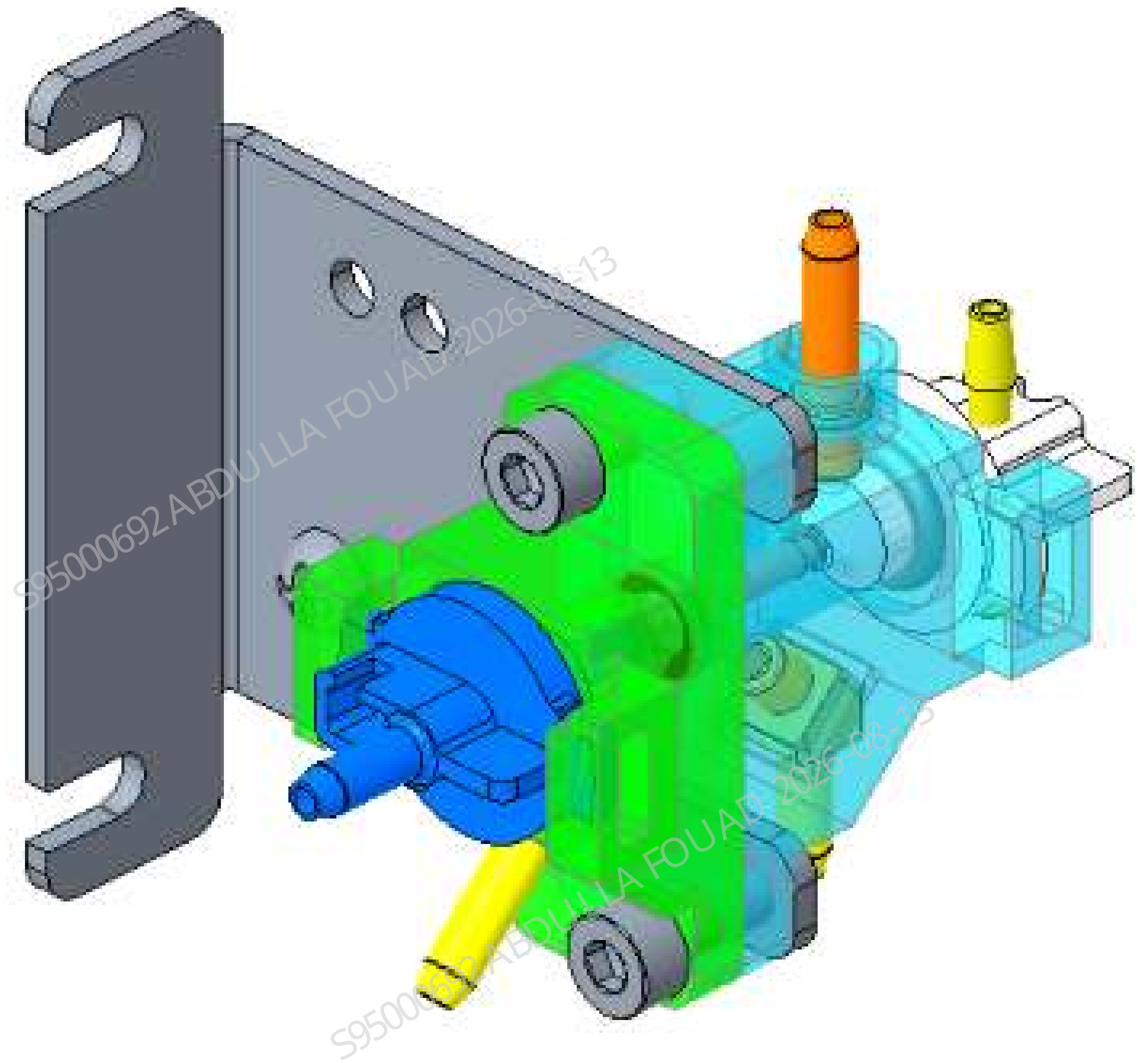


Table 8–40 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-071243-00	Sheath fluid count bath assembly	N/A	N/A

Function:

Used for measurement of RBC, PLT and other parameters.

8.38.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.
3. Be careful not to damage the connector when removing the tube.

Required tools

Cross-headed screwdriver
One pair of tweezers

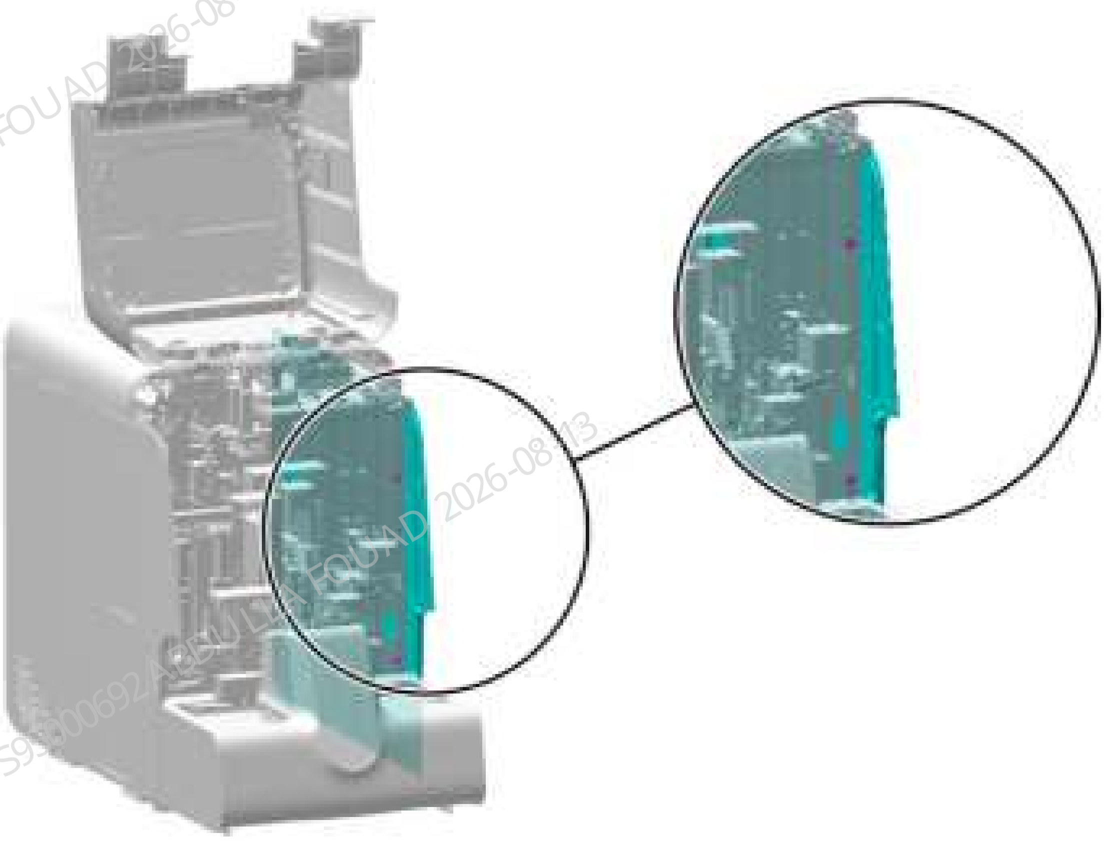
Pre-requirements

Click **Menu**>**>Service**>**>Maintenance**>**>Empty & Prime**>**>RET Bath Emptying > ISU/ISW Bath Emptying**, turn off the power, and unplug the power cord.

Procedures

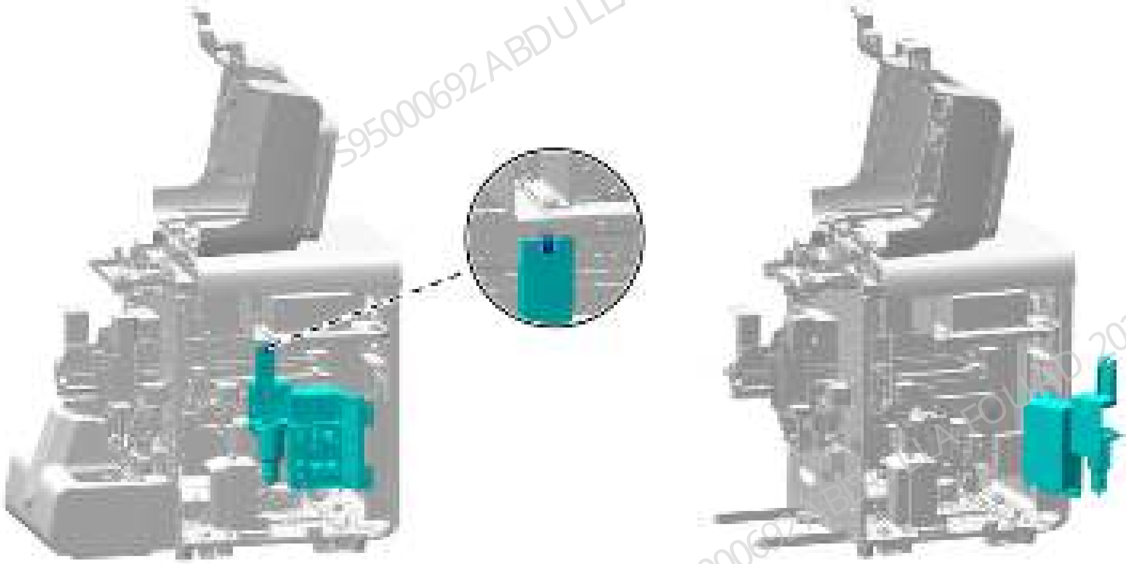
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–413 Removing the right cover



2. Unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

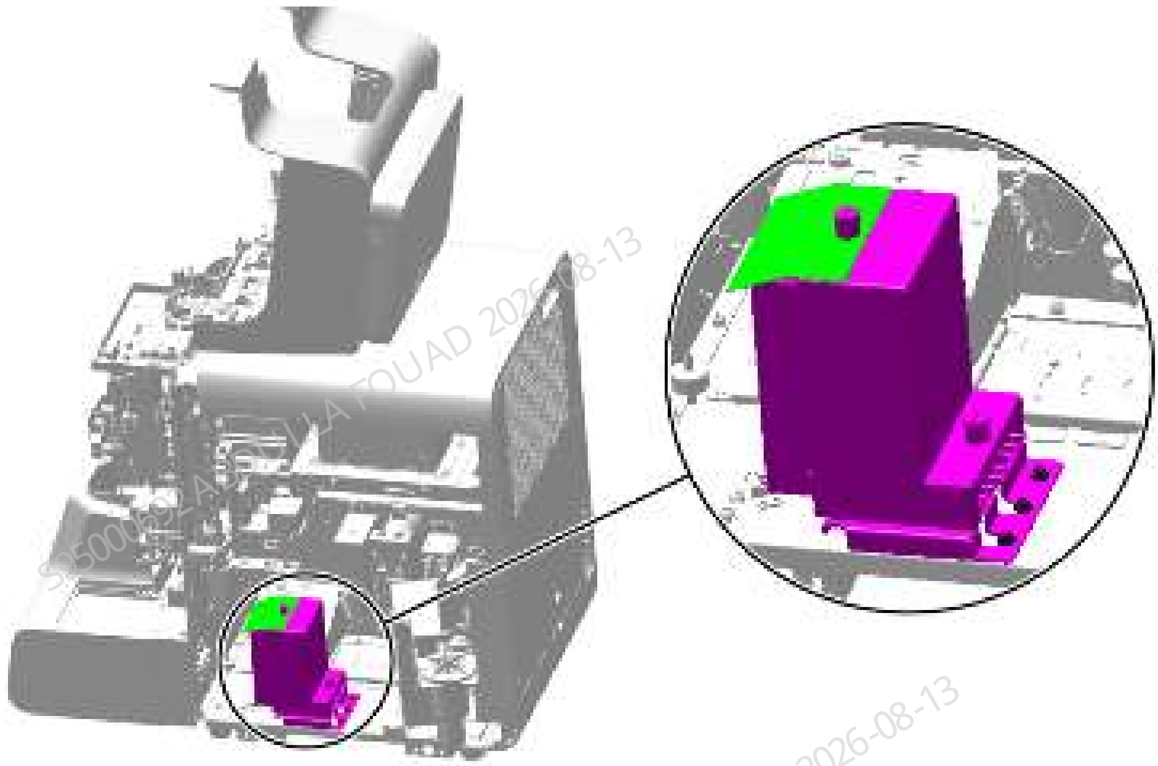
Figure8–414 Opening the right cover assembly



3. Disconnect the RBC signal wire from the impedance board.

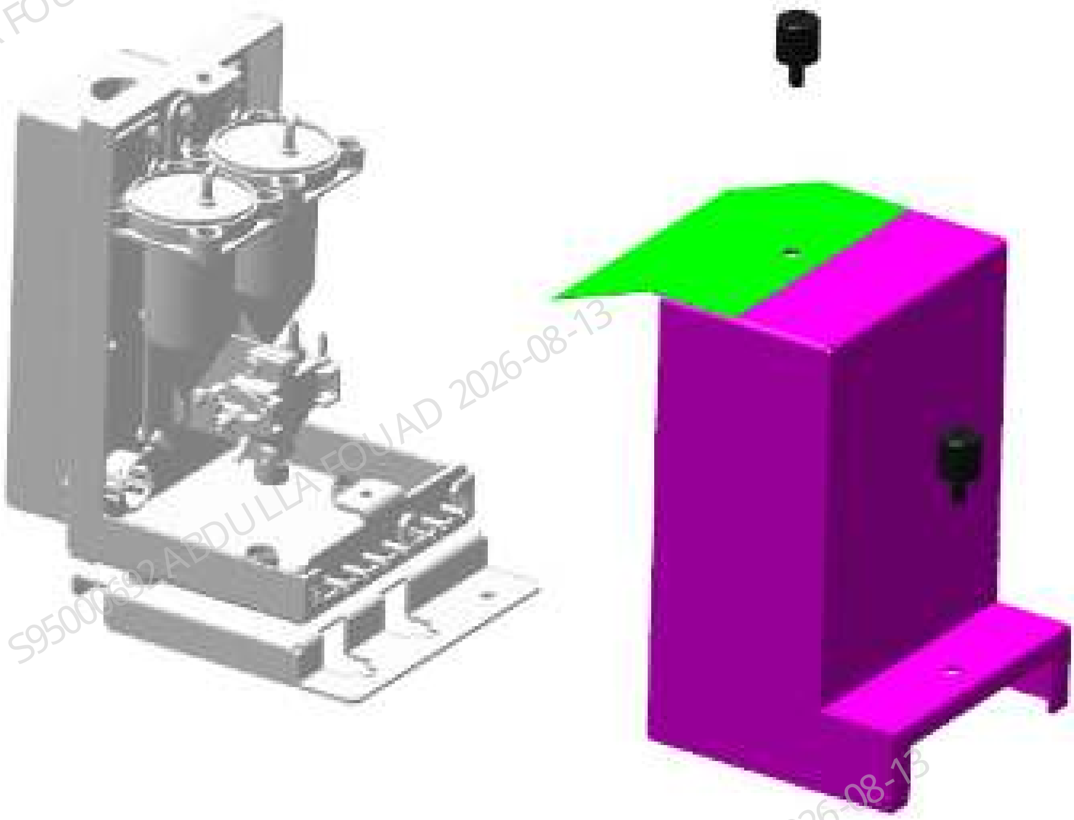
4. Use tweezers to pull out the rubber tube on the sheath fluid impedance assembly according to the tube marks "D-T77, C-82, G-T80, A-170, H-T71, F-T72, and B-T83".
5. Use a cross screwdriver to unscrew the three screws that fix the sheath fluid impedance assembly, as shown in the figure below, and remove the assembly.

Figure8–415 Removing the shielding cover of the sheath fluid impedance assembly



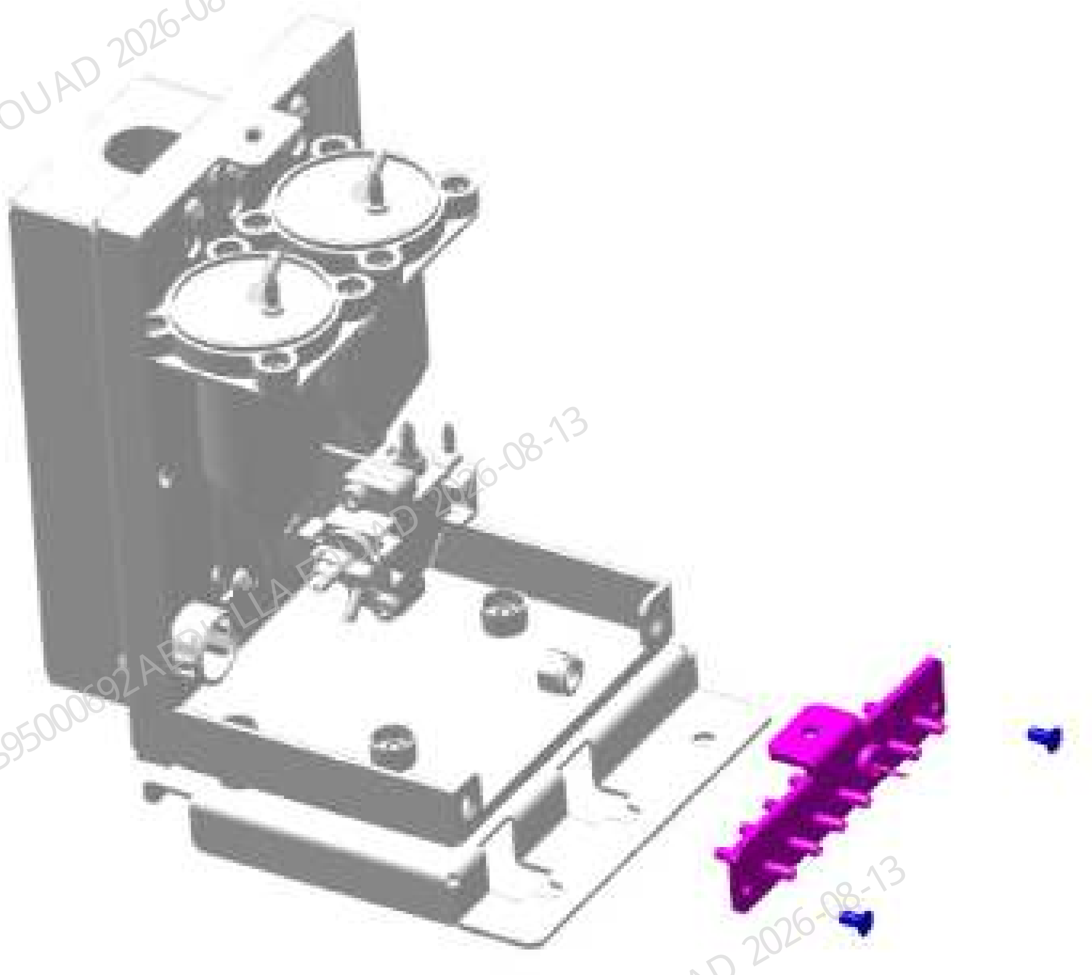
6. As shown below, loosen the two screws securing the shielding cover of the sheath fluid impedance assembly and remove the shielding cover.

Figure8–416 Removing the shielding cover of the sheath fluid impedance assembly



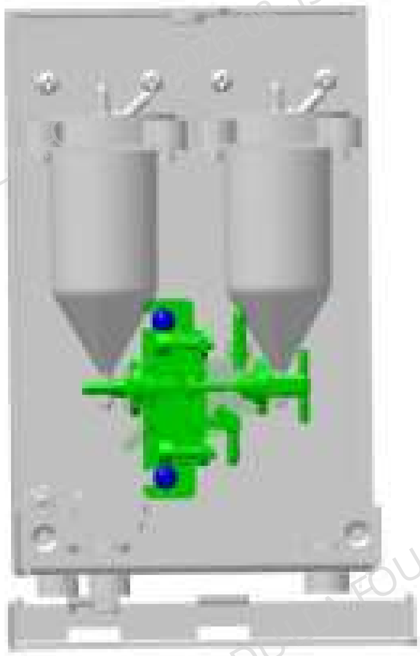
7. As shown below, loosen the two screws securing the connector mounting plate and remove the connector mounting plate.

Figure8–417 Removing the connector mounting plate



8. As shown below, remove the tube on the sheath fluid count bath assembly, loosen the two screws securing the sheath fluid count bath assembly, and remove the sheath fluid count bath assembly.

Figure8–418 Positions of the set screws of the sheath fluid count bath assembly



Subsequent Requirements

Installation procedure:

1. Fix the sheath fluid count bath assembly on the sheath fluid impedance assembly, connect the tube, and insert it into the corresponding connector.
2. Secure the connector mounting plate to the sheath fluid impedance assembly.
3. Install the sheath fluid impedance shielding cover back on the sheath fluid impedance assembly, and tighten the screws to fix it.
4. Connect the RBC signal wire with the impedance board.
5. Fix the sheath fluid impedance assembly onto the base plate of the main unit using three screws.
6. Insert each tube back into the sheath fluid impedance connector.
7. Close the right bath subassembly and re-secure it to the main unit.
8. Install the right cover to the main unit, and flip down the front cover.

8.38.3 Commissioning and Verification

Procedures

1. Tap the **Menu**>>**Calibrate**>>**CBC Gain Calibration**. Enter the MCV and HGB gain calibration screens, and enter the MCV_G target value of the batch of calibrators used; as shown in the figure below.

Figure8–419 MCV-G target value of calibrator

Mode CT-WB

	Target	1	2	3	Result
MCV_G	88.4				
HGB	4.40				

2. Perform OV-WB counting continuously with the calibrators until **OK** is displayed in the **Results** column for both MCV and HGB, as shown in the following figure.

Figure8–420 MCV Results

Mode CT-WB

	Target	1	2	3	Result
MCV_G	88.4	89.2	89.0		OK
HGB	4.40	4.46	4.51		OK

3. When exiting the CBC gain calibration screen, select **Yes** to save the gain in the pop-up prompt box.
4. Click**Menu**>**>Setup**>**>Gain Setup** in turn to check whether the **setting values** of MCV and HGB gains have been updated (if not, the default gain of the system is adopted), as shown in the figure below.

Figure8–421 Gain setting value

Verification:

1. Test controls or calibrators or random samples to confirm the accuracy of RBC and PLT.
   If there is any deviation, follow the steps in the Commissioning section to calibrate the MCV-G again.

8.39 Back sheath isolation bath assembly (3120)

8.39.1 General Information

Figure8–422 Photo of back sheath isolation bath assembly (3120)

Table 8–41 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-078179-00	Back sheath isolation bath assembly (3120)	N/A	N/A

Function:
ISU provides sheath fluid for the sheath fluid count bath, and ISW buffers the waste of the sheath fluid count bath.

8.39.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.
3. Be careful not to damage the connector when removing the tube.

Required tools

Cross-headed screwdriver
One pair of tweezers

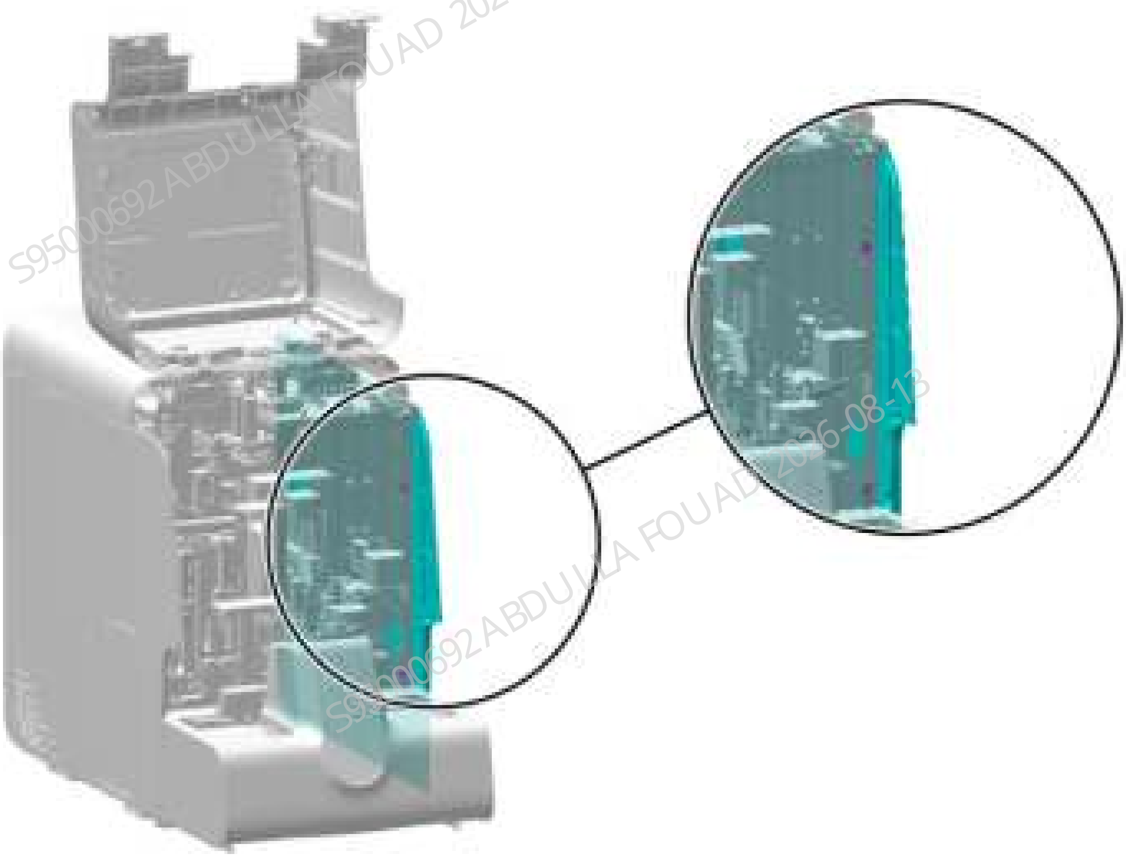
Pre-requirements

I. Select **Menu**>>**Service**>>**Maintenance**>>**Empty and Prime**>>**ISU/ISW Bath Emptying**, turn off the power, and unplug the power cord.

Procedures

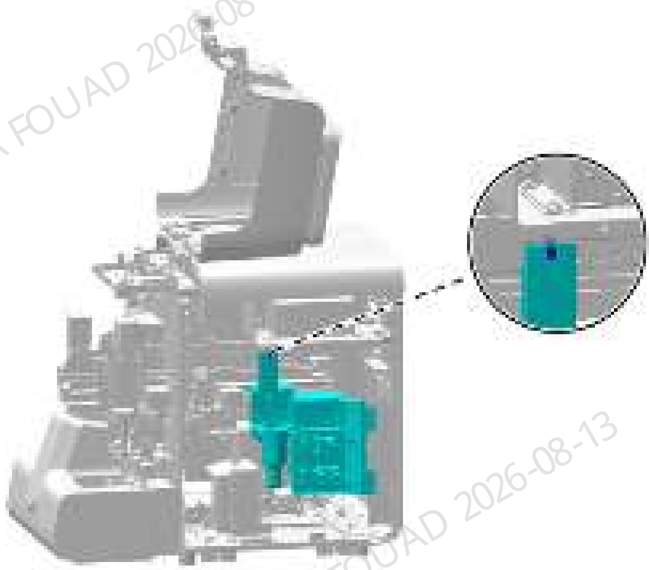
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–423 Removing the right cover



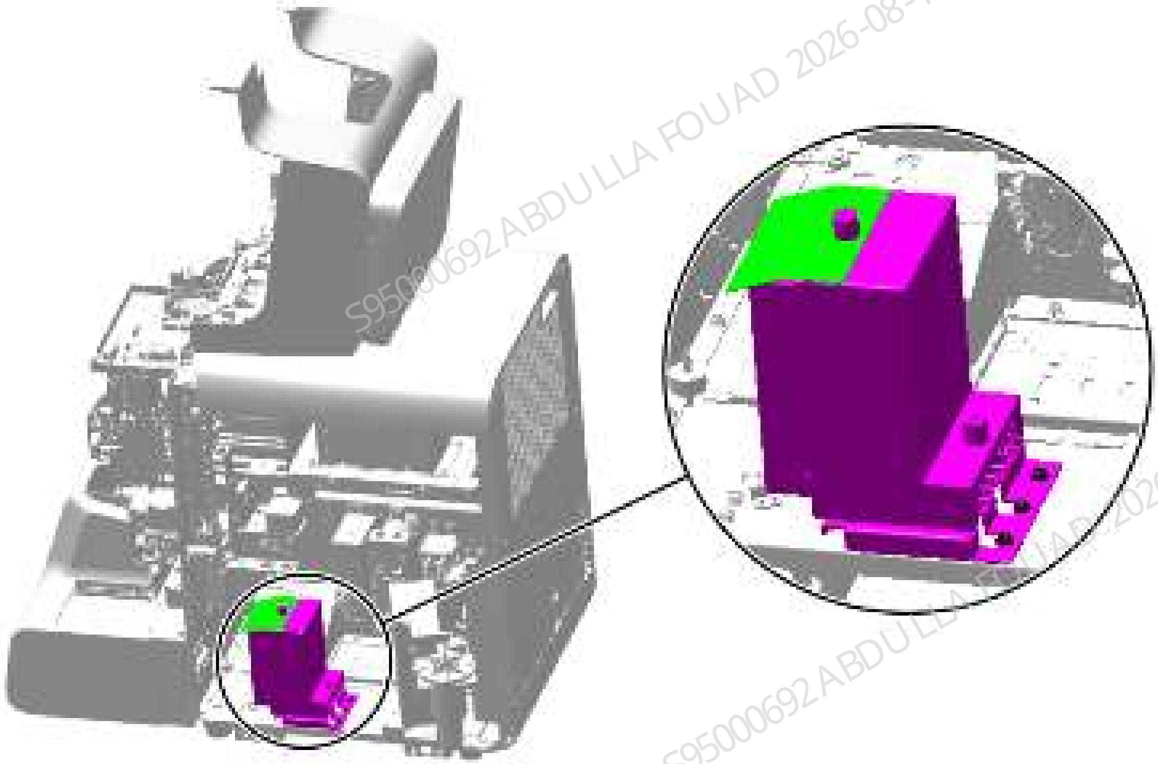
2. Unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–424 Opening the right cover assembly



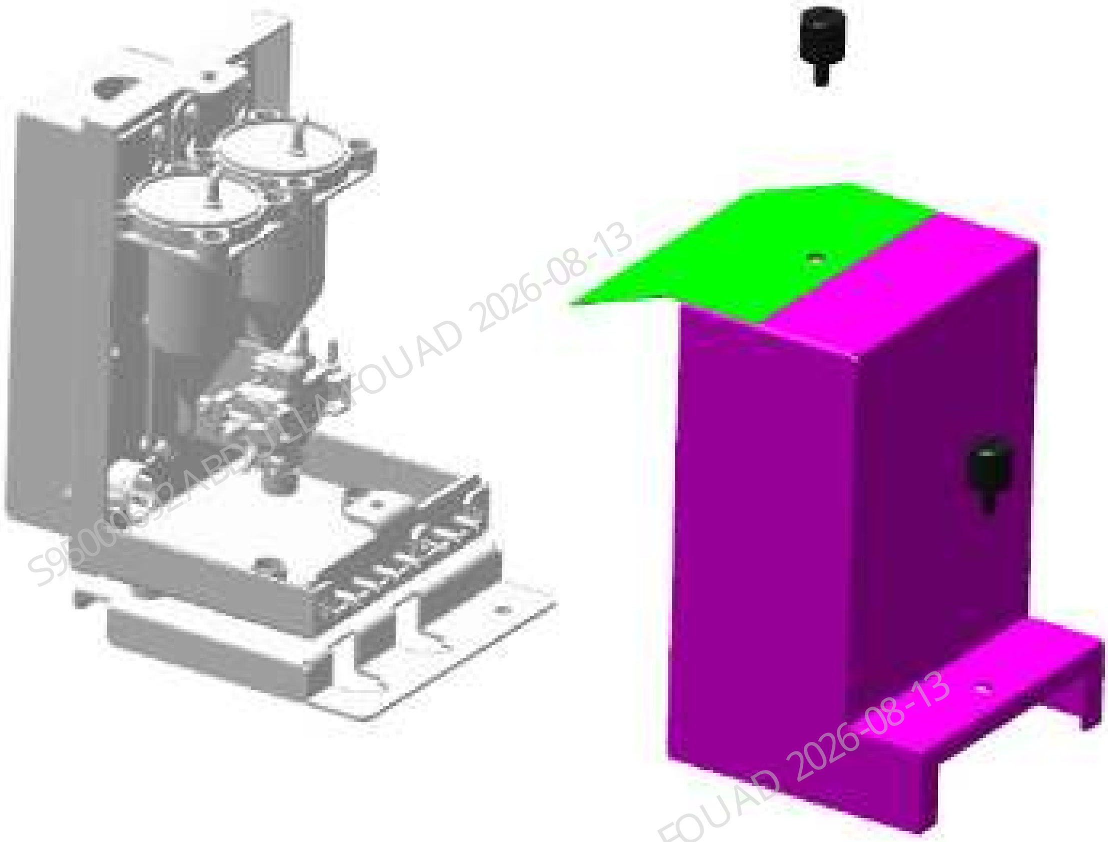
3. Disconnect the RBC signal wire from the impedance board.
4. Use tweezers to pull out the rubber tube on the sheath fluid impedance assembly according to the tube marks "D-T77, C-82, G-T80, A-170, H-T71, F-T72, and B-T83".
5. Use a cross screwdriver to unscrew the three screws that fix the sheath fluid impedance assembly, as shown in the figure below, and remove the assembly.

Figure8–425 Removing the shielding cover of the sheath fluid impedance assembly



6. As shown below, loosen the two screws securing the shielding cover of the sheath fluid impedance assembly and remove the shielding cover.

Figure8–426 Removing the shielding cover of the sheath fluid impedance assembly



7. Unplug the tube on the back sheath isolation bath assembly (3120), loosen the two screws securing the back sheath isolation bath assembly (3120), and remove the back sheath isolation bath assembly (3120), as shown in the figure below.

Figure8–427 Diagram for the Screw Positions of Back Sheath Isolation Bath Assembly (3120)

Subsequent Requirements

Installation procedure:

1. Fix the sheath fluid count bath assembly on the back sheath isolation bath assembly (3120), connect the tube, and insert it into the corresponding connector.
2. Fix the sheath fluid impedance assembly onto the base plate of the main unit using three screws.
3. Connect the lead wire on the sheath fluid impedance assembly with the lead wire on the board.
4. Insert each tube back into the sheath fluid impedance connector.
5. Install the sheath fluid impedance shielding cover back on the sheath fluid impedance assembly, and tighten the screws to fix it.
6. Close the right cover assembly and re-secure it to the main unit.
7. Install the right cover to the main unit, and flip down the front cover.

8.39.3 Commissioning and Verification

Pre-requirements

Perform the installation steps to step 4, do not install the sheath fluid shielding cover first, because the emptying and priming statuses of the ISU/ISW bath needs to be observed during the commissioning process.

Procedures

1. After powering on, select**Service**>**>Maintenance**>**>Emptying and Priming** in sequence on the screen, click **ISU/ISW bath priming** on the **Prime** screen, and observe whether the ISU/ ISW bath priming is normal. Check whether the software screen prompts that **the ISU/ISW bath priming is complete**
2. Then switch to the **Empty** screen, click **ISU/ISW bath emptying**, observe whether the ISU/ ISW bath emptying is normal, and whether the software screen prompts that **the ISU/ISW bath priming is complete**.

Verification:

Test controls or calibrators or random samples to confirm the accuracy of RBC and PLT values. If there is a deviation, recalibration is required.

8.40 PCBA, impedance signal board, analog

8.40.1 General Information

Figure8–428 Photo of back sheath isolation bath assembly (3120)

Table 8–42 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
051-004012-00	PCBA, impedance signal board, analog	N/A	N/A

Function:
Constant current source driving, signal processing, and signal detection circuit for sheath fluid impedance measurement.

8.40.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.

Required tools

Cross-headed screwdriver
One Allen wrench

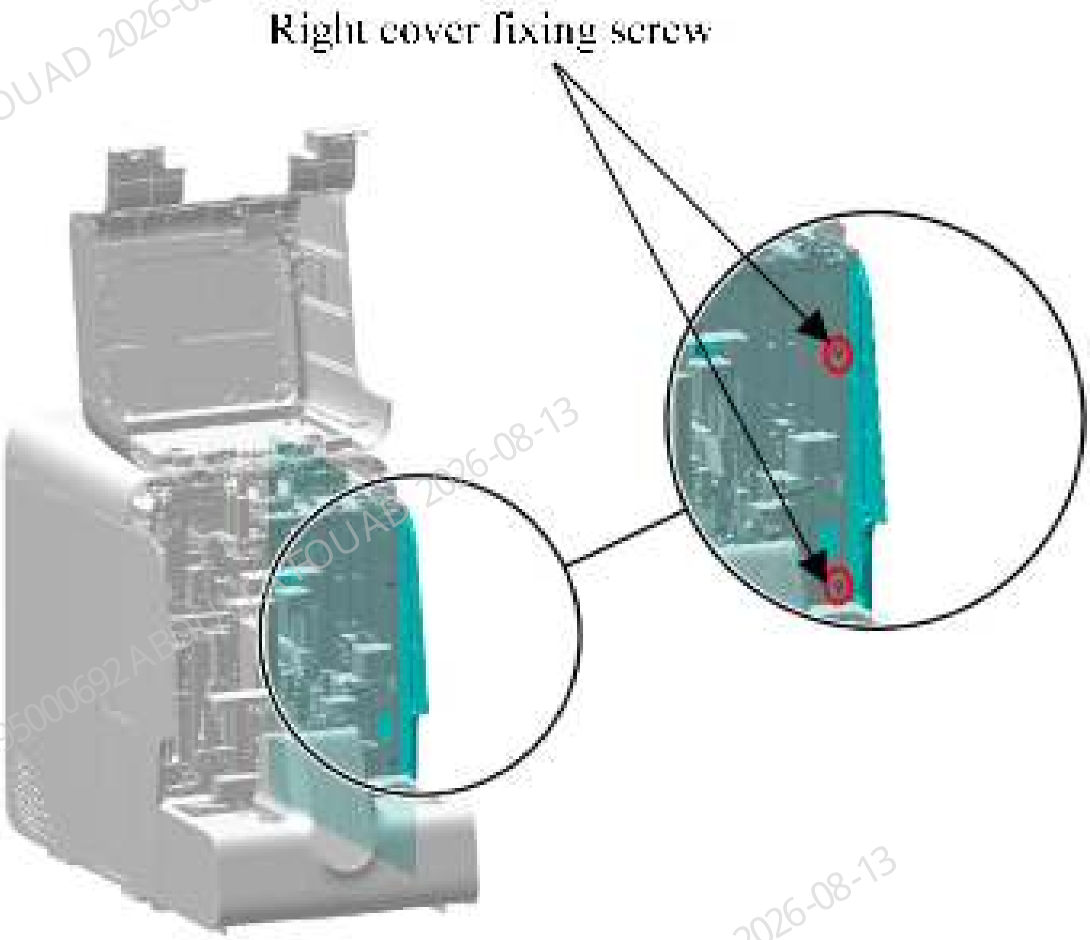
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

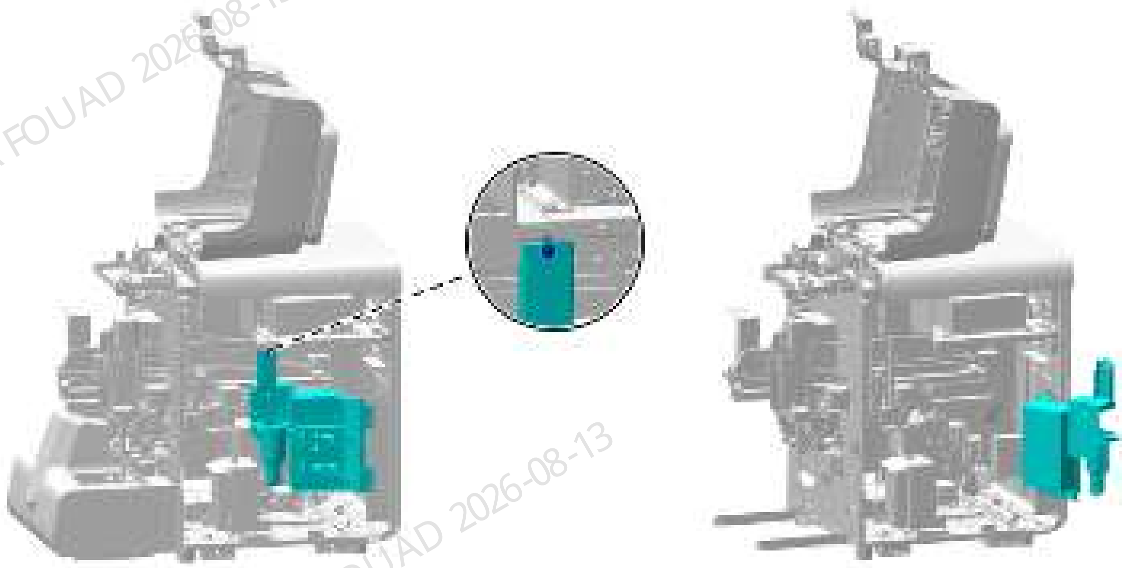
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–429 Removing the right cover



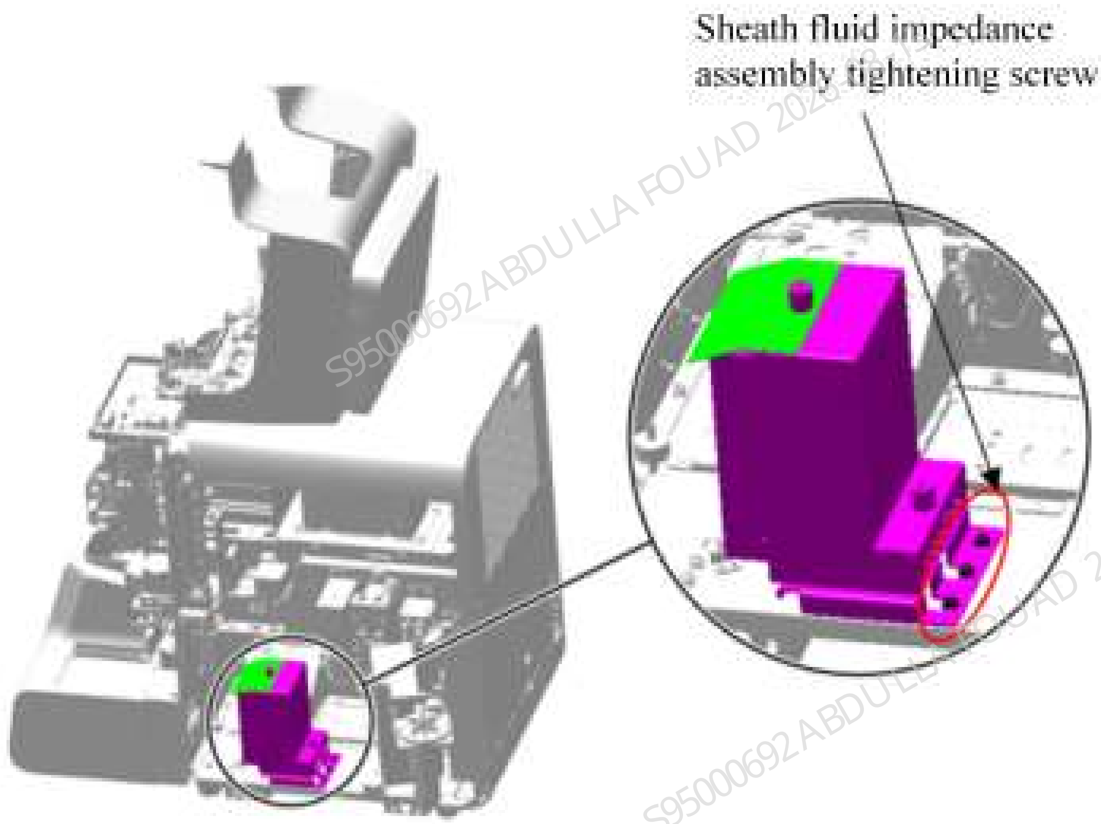
2. Unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–430 Opening the right cover assembly



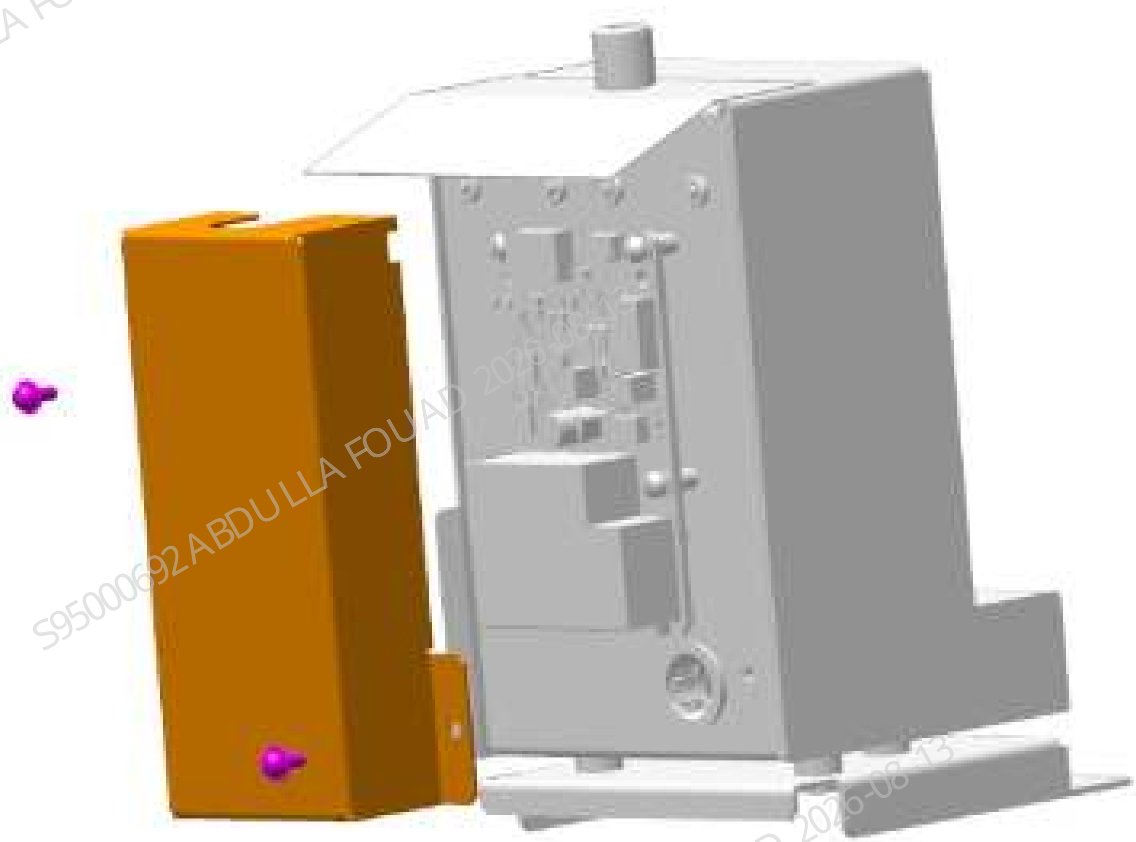
3. Disconnect the RBC signal wire from the impedance board.
4. Use a cross screwdriver to unscrew the three screws that fix the sheath fluid impedance assembly, as shown in the figure below, and remove the assembly.

Figure8–431 Removing the shielding cover of the sheath fluid impedance assembly



5. As shown below, loosen the two screws securing the shielding cover of the impedance signal board, and remove the shielding cover.

Figure8–432 Diagram of Removing the Shielding Cover of Impedance Signal Board



6. Unplug the wire of the impedance signal board, a cross screwdriver to remove the three screws fixing the impedance signal board, and remove the impedance signal board, as shown in the figure below.

Figure8–433 Diagram of Removing the Impedance Signal Board

Subsequent Requirements

Installation procedure:

1. Fix the impedance signal board on the sheath fluid impedance assembly, connect the internal wire of the sheath fluid impedance assembly, and plug it into the corresponding connector.

2. Fix the shielding cover of impedance signal board on the sheath fluid impedance assembly.
3. Fix the sheath fluid impedance assembly onto the base plate of the main unit using three screws.
4. Connect the lead wire on the sheath fluid impedance assembly with the lead wire on the board.
5. Close the right cover assembly and re-secure it to the main unit.
6. Install the right cover to the main unit, and flip down the front cover.

8.40.3 Commissioning and Verification

Pre-requirements

Perform the installation steps to step 4, do not install the sheath fluid shielding cover first, because the emptying and priming statuses of the ISU/ISW bath needs to be observed during the commissioning process.

Procedures

1. After powering on, select**Service**>**>Maintenance**>**>Emptying and Priming** in sequence on the screen, click **ISU/ISW bath priming** on the **Prime** screen, and observe whether the ISU/ ISW bath priming is normal. Check whether the software screen prompts that **the ISU/ISW bath priming is complete**.
2. Then switch to the **Empty** screen, click **ISU/ISW bath emptying**, observe whether the ISU/ ISW bath emptying is normal, and whether the software screen prompts that **the ISU/ISW bath priming is complete**.

Verification:

Test controls or calibrators or random samples to confirm the accuracy of RBC and PLT values. If there is a deviation, recalibration is required.

8.41 HGB assembly

8.41.1 General Information

Figure8–434 Photo of HGB bath assembly

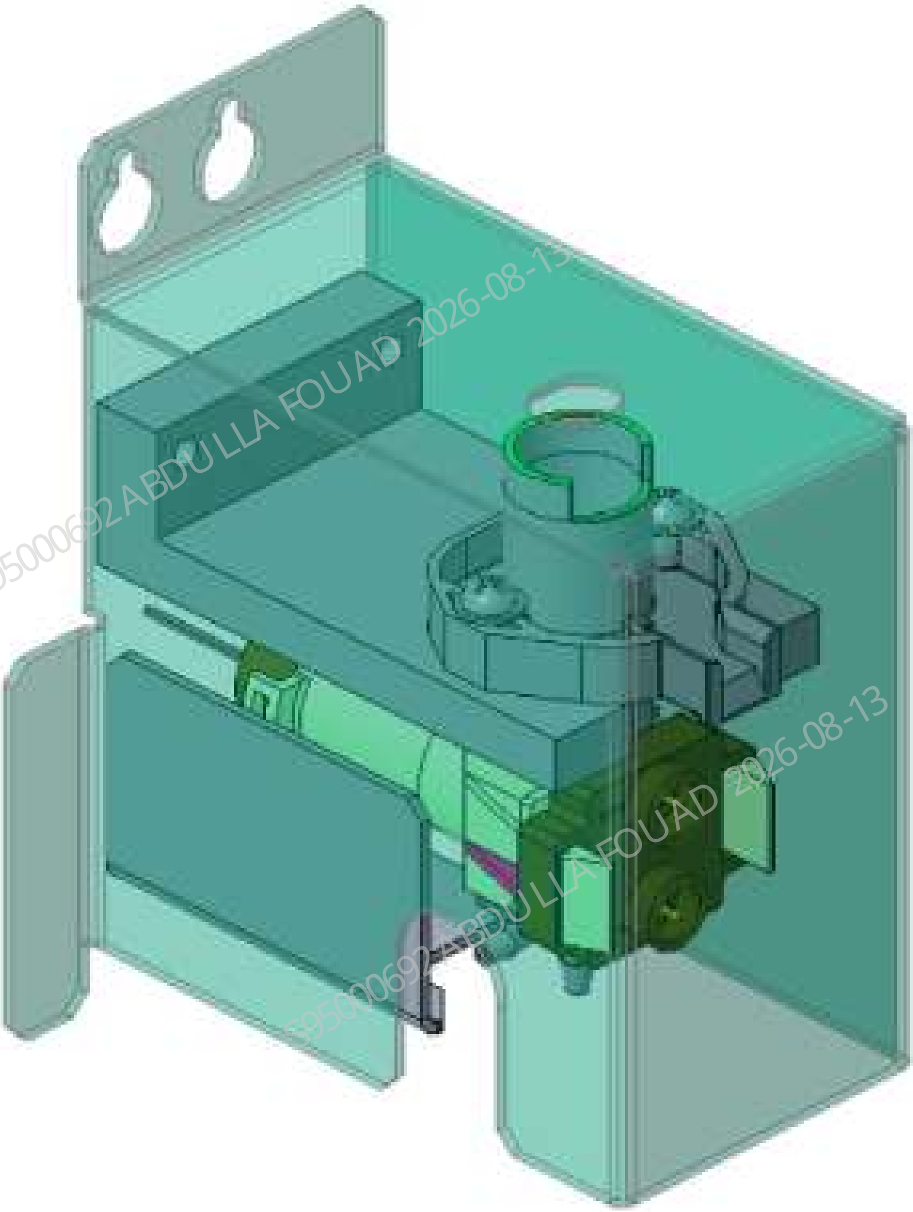


Table 8–43 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-071234-00	HGB bath assembly	N/A	N/A

Function:

Providing a place for HGB hemolysis and mixing, and measuring the HGB concentration of the sample by colorimetry.

8.41.2 Disassembly and Assembly

Note

1. Before disassembly, power off and unplug the power supply.
2. Prevent liquid from entering the terminals during disassembly.
3. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.

Required tools

Cross-headed screwdriver
One Allen wrench

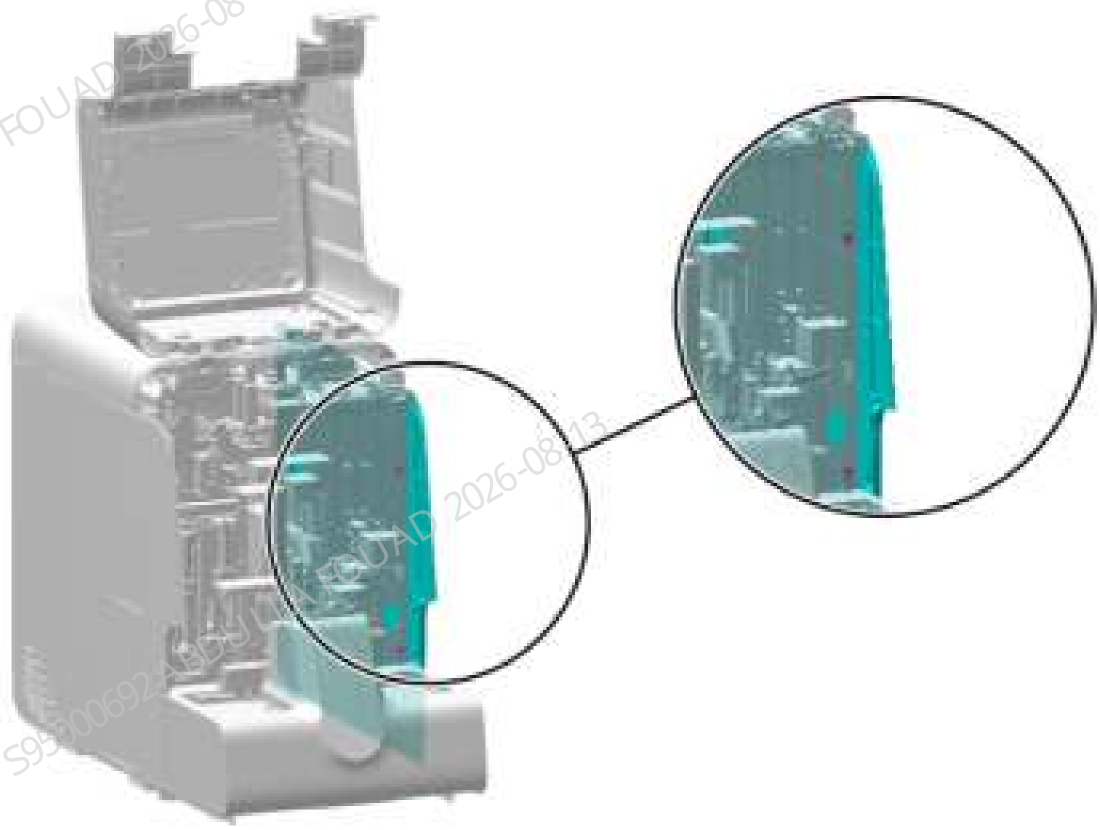
Pre-requirements

Enter the maintenance screen, perform HGB bath emptying, turn off the power, and unplug the power plug.

Procedures

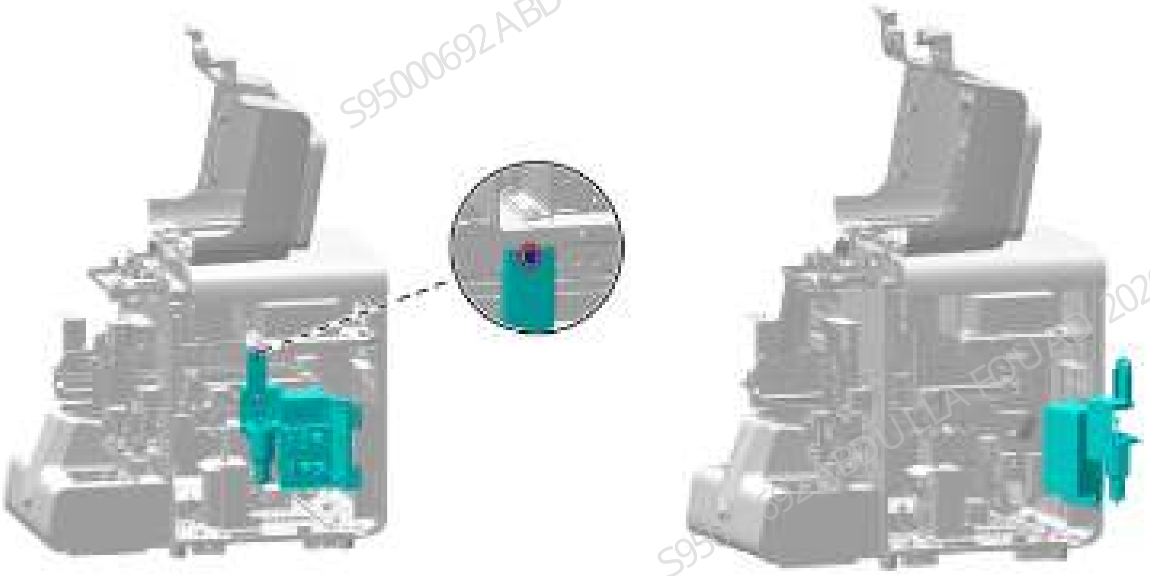
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the autoloader and the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–435 Removing the right cover



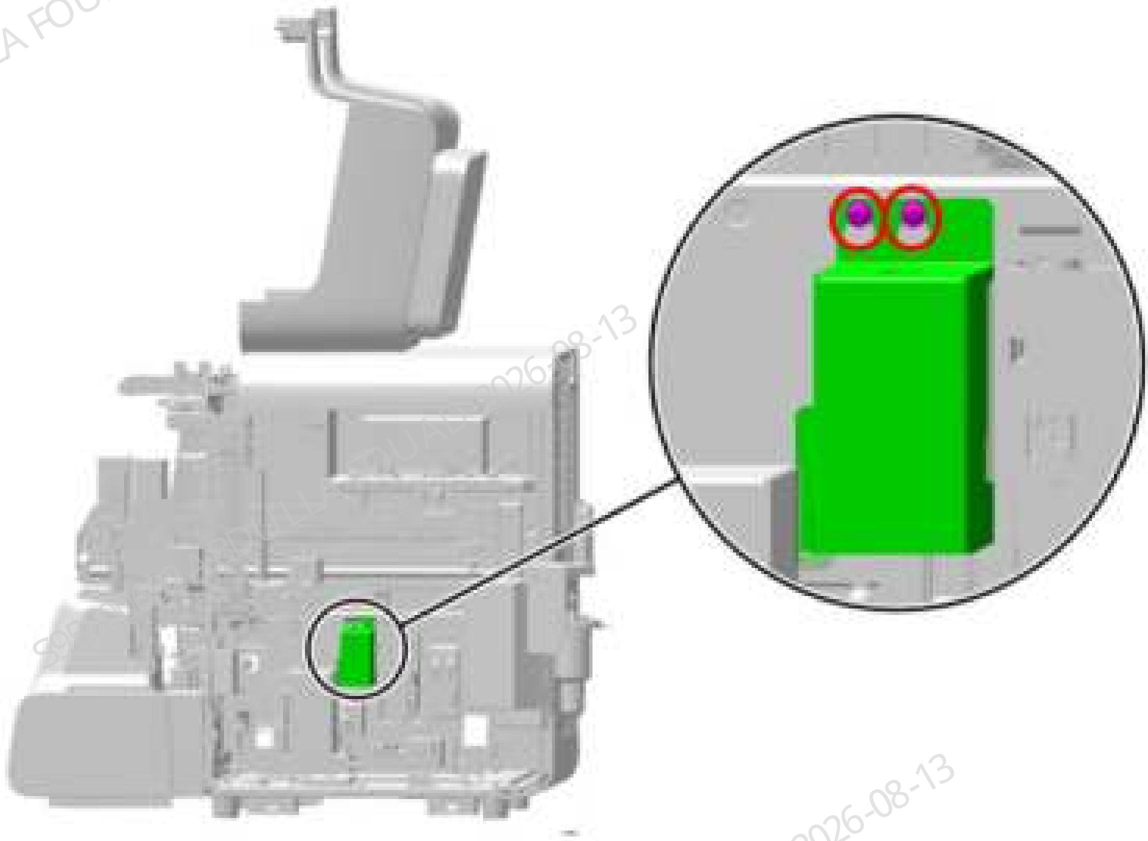
2. As shown below, unscrew the one screw (in the red circle) that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–436 Opening the right cover assembly



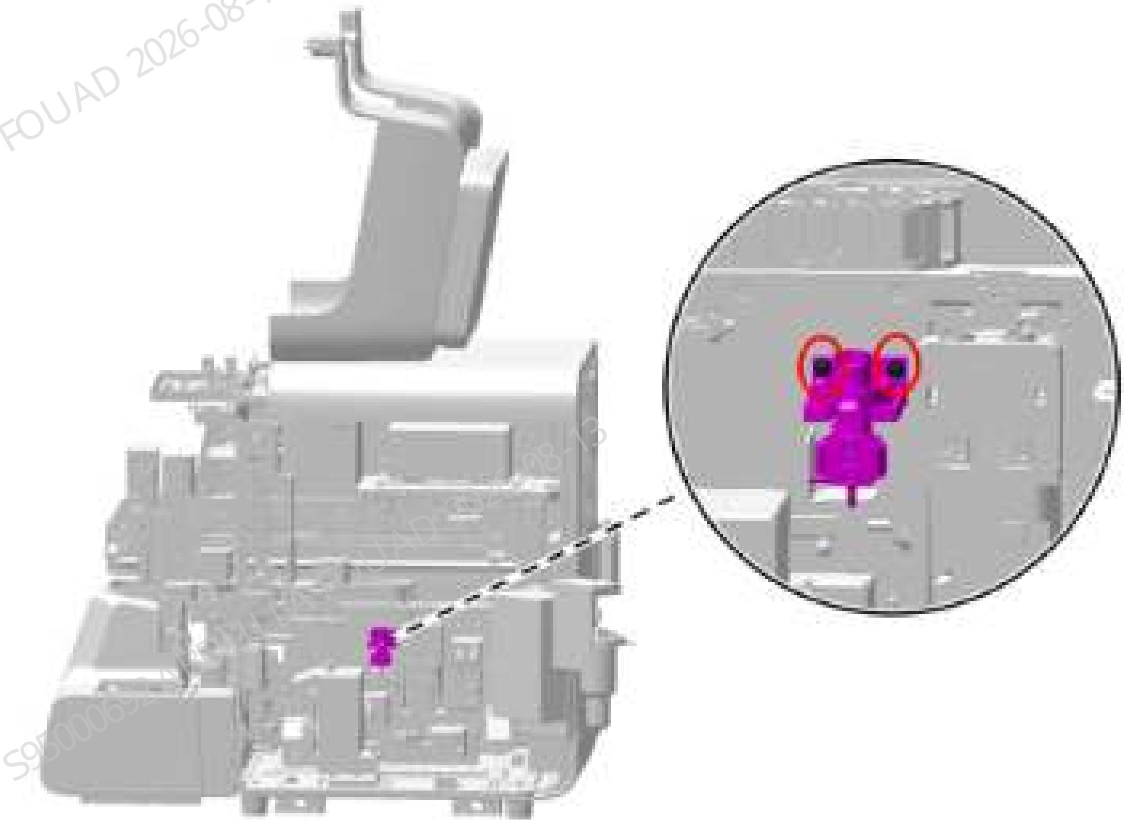
3. As show below, loosen the two screws (in the red circle) securing the shielding cover of the HGB bath assembly, and remove the shielding cover.

Figure8–437 Positions of the set screws of the HGB bath assembly shielding cover



4. As show below, loosen the two screws (in the red circle) securing the HGB bath assembly, and pull out the HGB bath assembly slightly.

Figure8–438 Positions of the set screws of the HGB cistern assembly



5. Use tweezers to pull out the two tubes connecting the HGB bath assembly, disconnect the HGB bath assembly from the signal cable connected to the board, and take out the HGB bath assembly.

Subsequent Requirements

Installation procedure:

1. Insert the tube into the corresponding plug of the HGB bath assembly, and connect the HGB signal cable to the lead wire of the board.
2. Secure the HGB bath assembly to the main unit with two screws.
3. Install the shielding cover of the HGB bath assembly back to its original position, and tighten the two set screws.
4. Install the right cover back, and flip down the front cover to reset it.

8.41.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Pre-requirements

Prepare a calibrator and leave it stand for 15 minutes.

Procedures

- **Adjusting the position of HGB bath**

— For details of the position adjustment steps and effects, please refer to the adjustment content of the sampling probe assembly relative to the HGB bath.

— Performance commissioning

1. Tap the **Menu**>>**Calibrate**>>**CBC Gain Calibration**. Enter the MCV and HGB gain calibration screens, and enter the MCV_G target value of the batch of calibrators used; as shown in the figure below.

Figure8–439 MCV-G target value of calibrator

Mode CT-WB

	Target	1	2	3	Result
MCV_G	[illegible]				
HGB	4.40				

2. Perform OV-WB counting continuously with the calibrators until **OK** is displayed in the **Results** column for both MCV and HGB, as shown in the following figure.

Figure8–440 MCV Results

Mode CT-WB

	Target	1	2	3	Result
MCV_G	88.4	89.2	89.0		OK
HGB	4.40	4.46	4.51		OK

3. When exiting the CBC gain calibration screen, select **Yes** to save the gain in the pop-up prompt box.

4. Click**Menu>>Setup>>Gain Setup** in turn to check whether the **setting values** of MCV and HGB gains have been updated (if not, the default gain of the system is adopted), as shown in the figure below.
   Perform OV-WB counting continuously with the calibrators until **OK** is displayed in the **Results** column for both MCV and HGB, as shown in the following figure.

Figure8–441 Gain setting value

Verification:

Test the control or calibrator or random sample to confirm the accuracy of HGB, if there is a deviation, calibrate it.

8.42 Optical system

8.42.1 General Information

Figure8–442 Photo of optical system

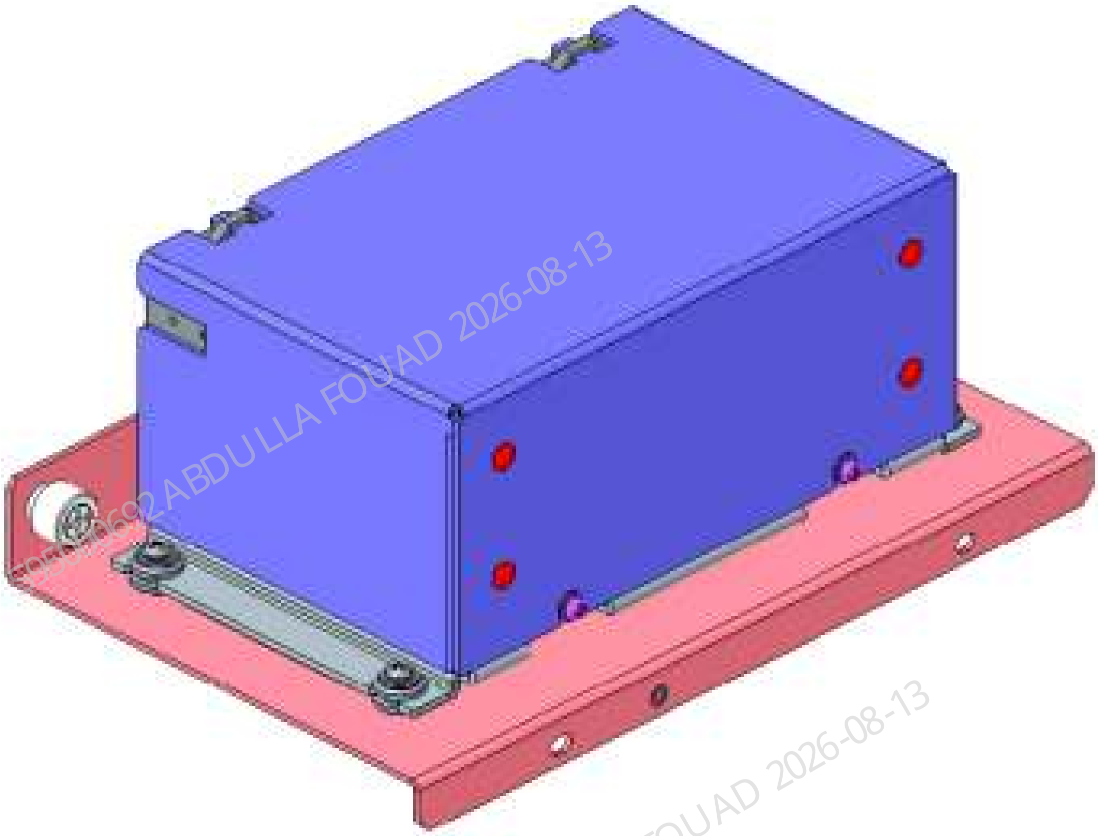


Table 8–44 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081654-00	Optical system	N/A	N/A

Function:
Analyzing and studying the blood cells in the tested blood sample through the light signal analysis processing.

8.42.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

Cross-headed screwdriver

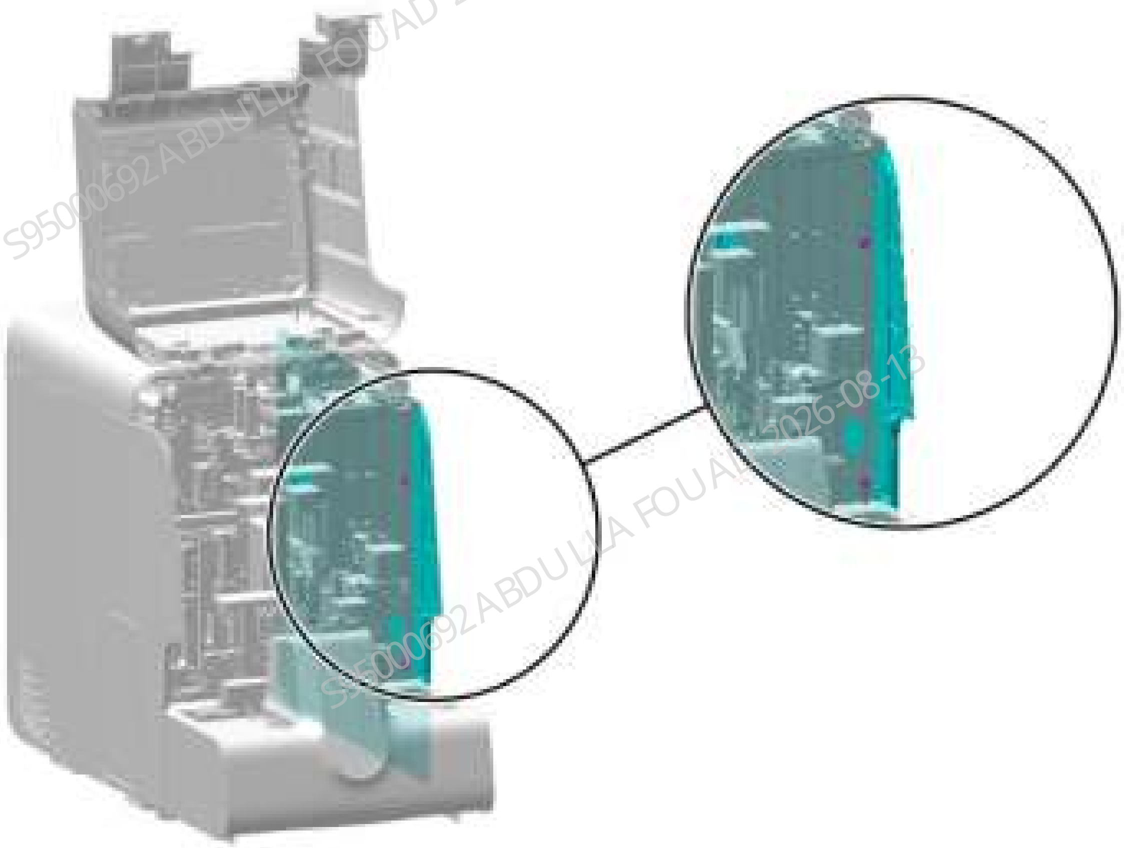
Pre-requirements

Power off the analyzer, and unplug the power cord. Liquid will leak when the tube is removed, so toilet paper should be laid below in advance.

Procedures

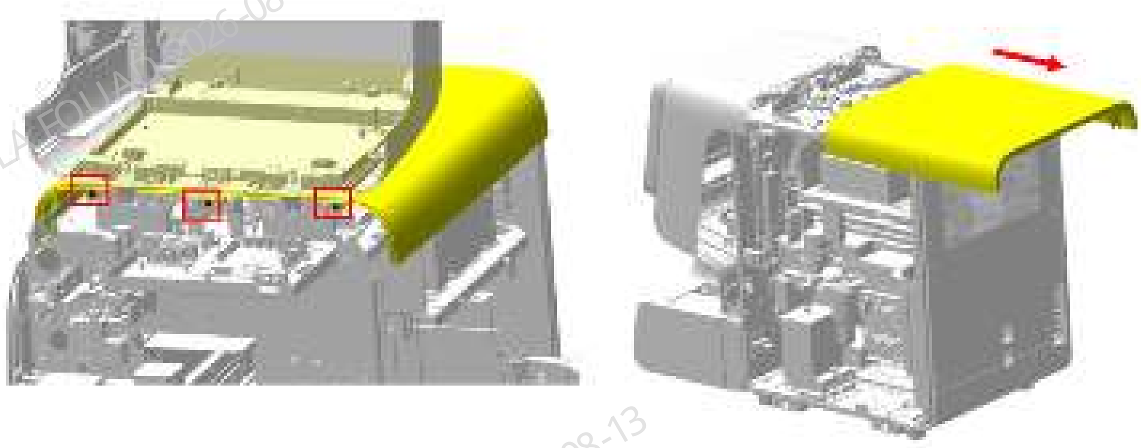
1. Loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–443 Removing the right cover



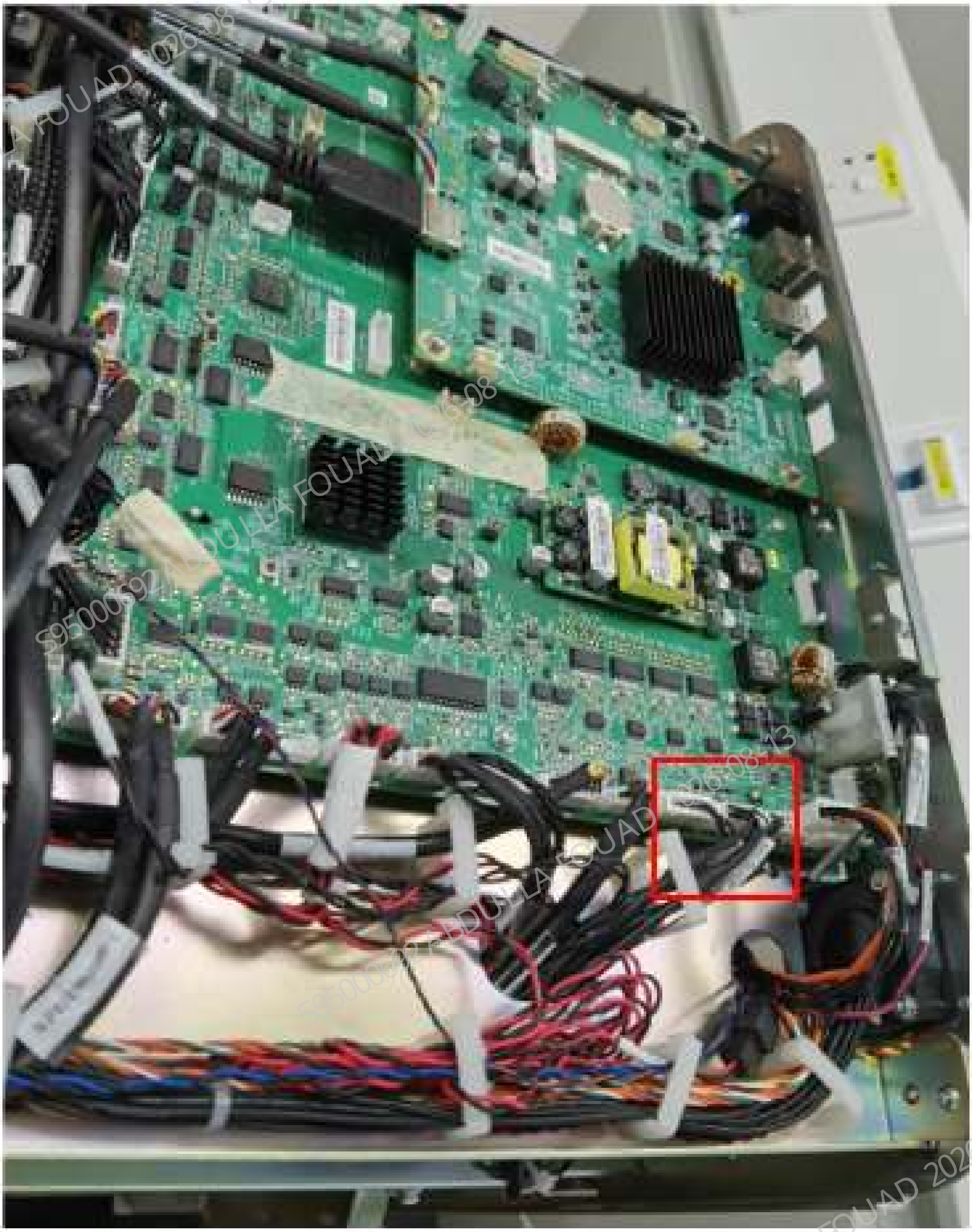
2. As shown in the figure below, unscrew the three screws fixing the top cover, reset the front cover for resetting, and then remove the top cover along the direction of the arrow.

Figure8–444 Diagram of Removing the Top Cover



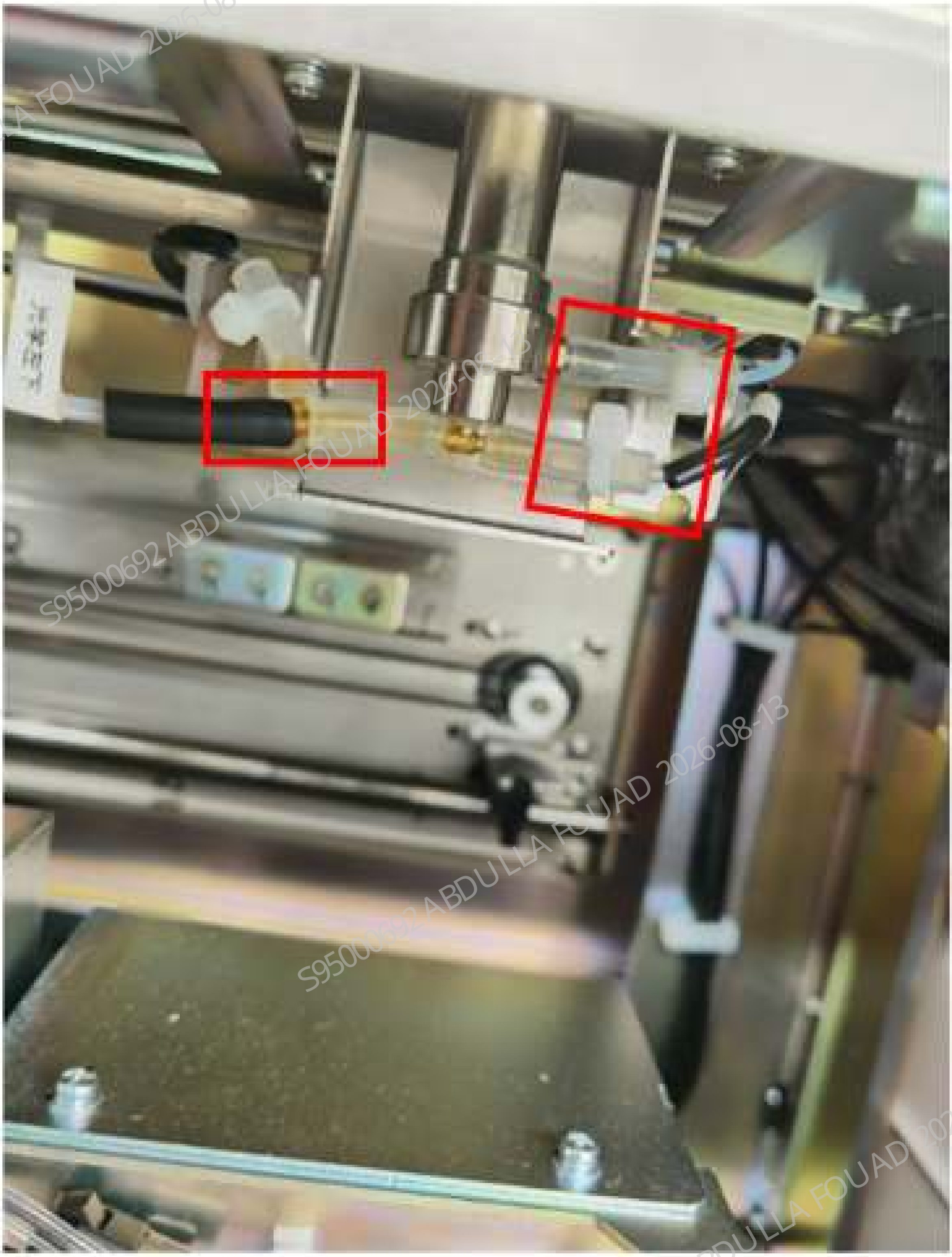
3. Pull the wire of the optical system from the top of the analyzer.

Figure8–445 Diagram of Removing the Optical System Wire



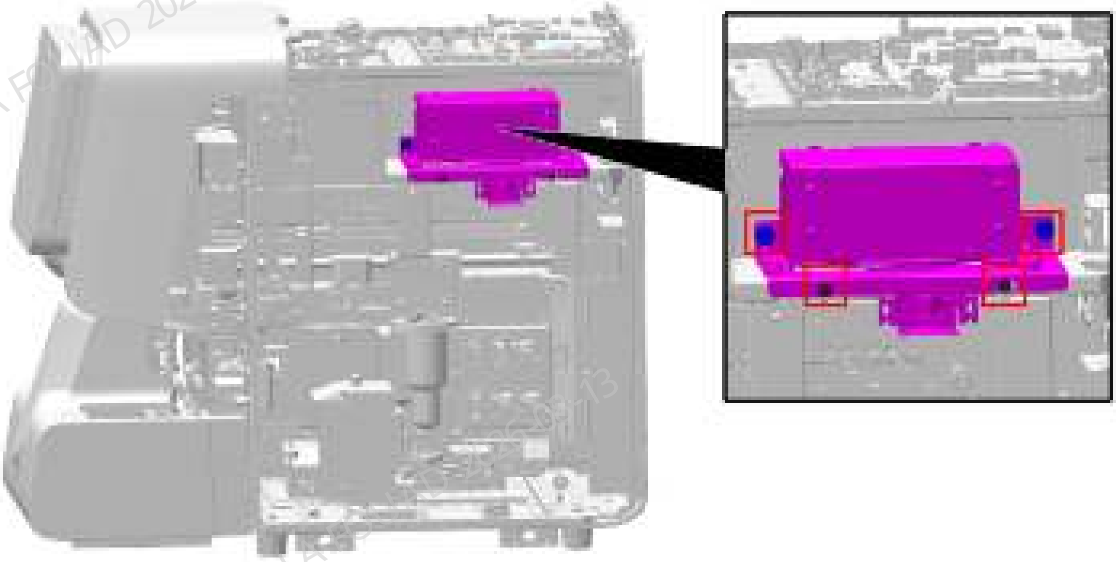
4. Remove the tube connecting the optical system flow cell.

Figure8–446 Diagram of Removing the Tube of the Optical System Flow Cell



5. Loosen the four screws that fix the optical system, as shown in the figure below, and remove the optical system.

Figure8–447 Diagram of Removing the Optical System



Subsequent Requirements

Installation procedure:

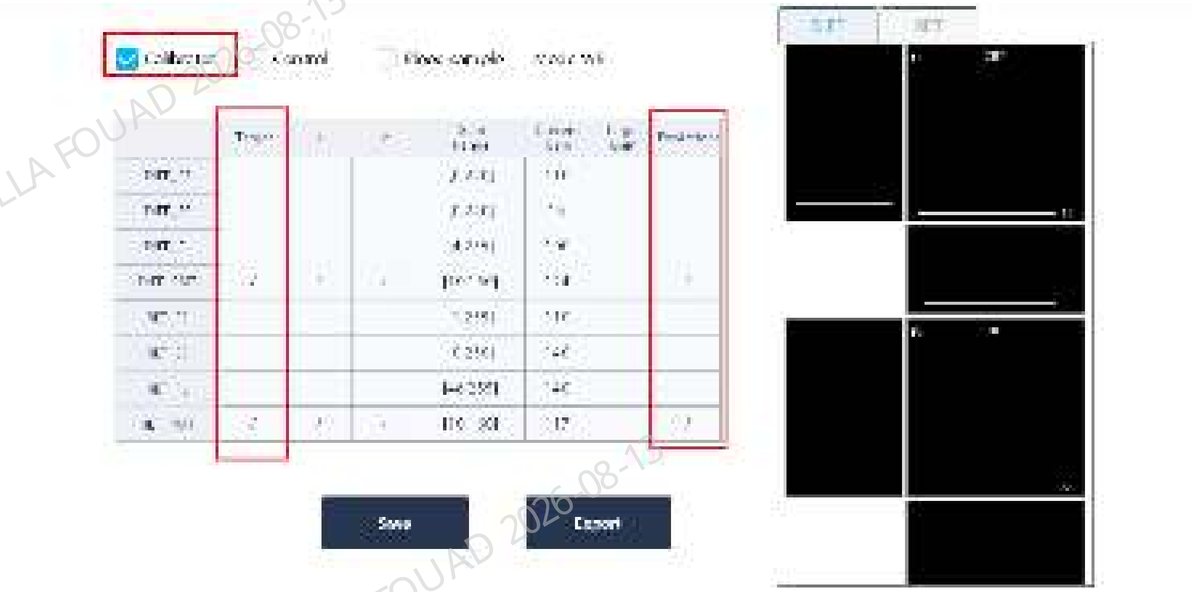
1. Fix a new optical system to the main unit.
2. Connect the optical system wires to the top board and connect the tubing back to the connector of optical system flow cell.
3. Install the right cover and top cover back on the main unit in sequence.

8.42.3 Commissioning and Verification

Procedures

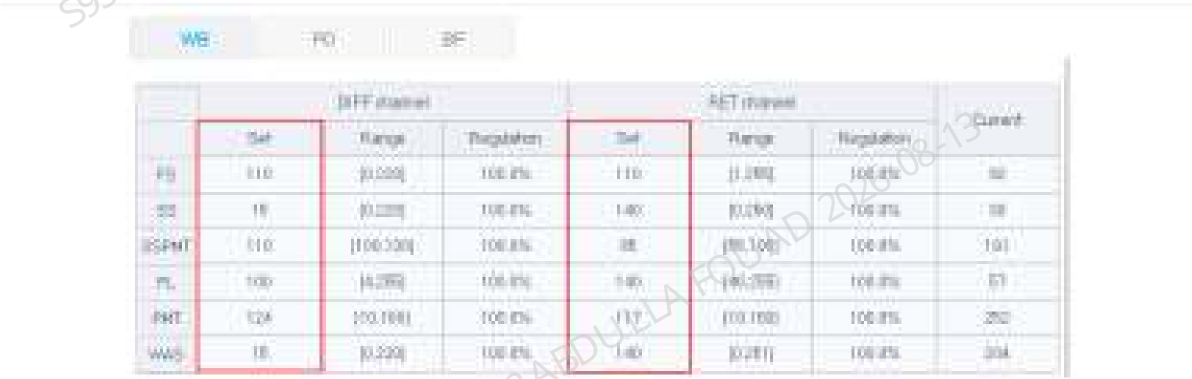
1. After powering on, log in at the service access level, select **Calibrate**>**>Gain Calibration**>**>Optical Gain Calibration**, select the **calibrator**, prepare the PLUS calibrator, and enter the optical target value of the calibrator used in the target value column. Perform one OV counting. After calibration, the analyzer software asks if you want to save the results. If the results are in expected ranges, tap **Yes** to start next auto counting. After continuous measurement, if the deviation column satisfies ≤2.0%, click **Save**, as shown in the figure below.

Figure8–448 Optical Gain Calibration screen



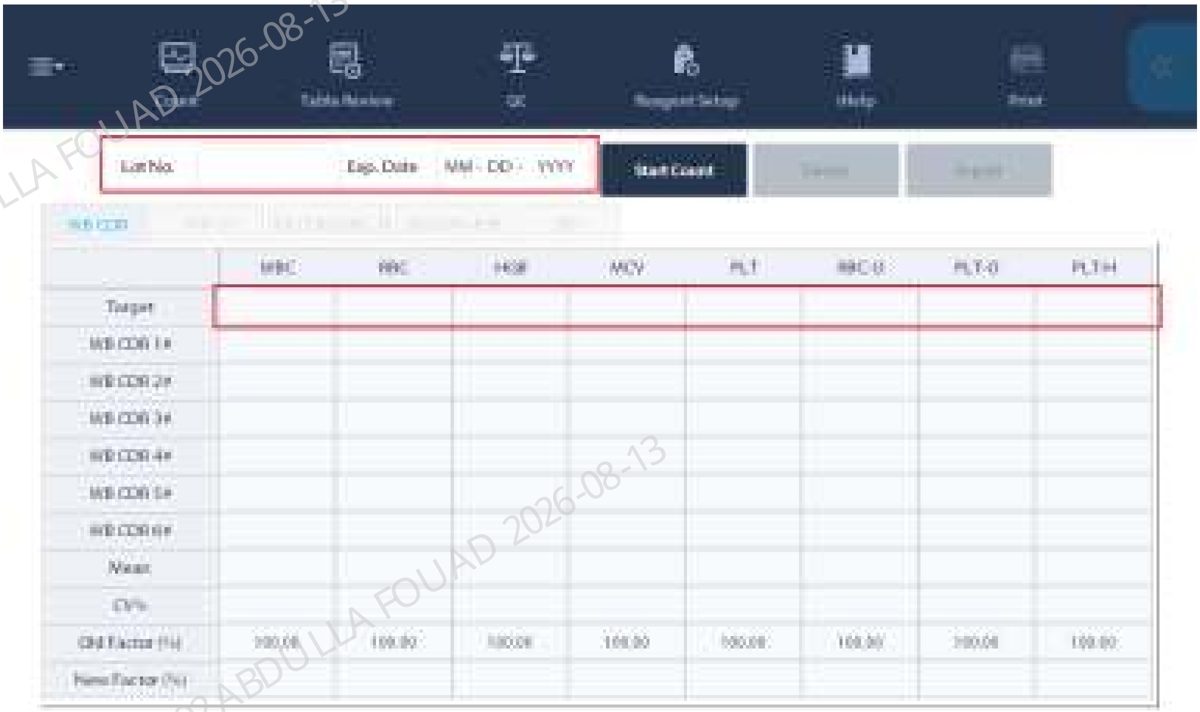
2. Click**Menu>>Setup>>Gain Setup** in turn to check whether the **setting value** of the optical gain has been updated, as shown in the figure below.

Figure8–449 Optical Gain Setup screen



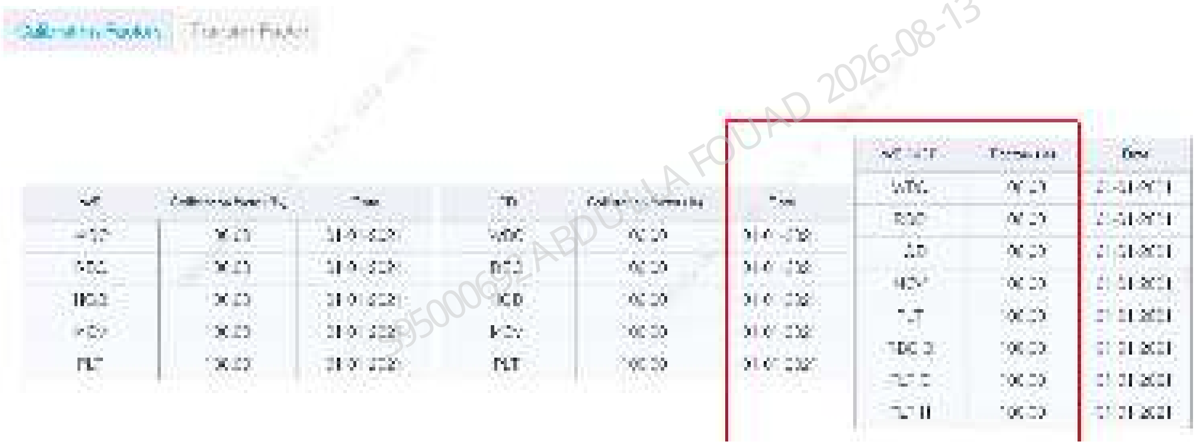
3. Select **Calibrate>>Calibrator Calibration**, enter the batch number of the PLUS calibrator in the **batch number** column, and enter the expiration date, and then enter the target values of WBC, RBC, HGB, MCV, PLT, RBC-O (depending on configuration), and PLT-O (depending on configuration). Perform six OV counts. After the calibration is completed, exit the calibration screen, and save the parameters, as shown in the figure below.

Figure8–450 Calibration screen



4. Select **Calibrate**>>**Manual Calibration**>>**Calibration Factor** to check whether the calibration factor is saved, as shown in the figure below.

Figure8–451 Viewing the calibration factor



8.43 WC2 Cistern

8.43.1 General Information

Figure8–452 Photo of WC2 waste cistern assembly

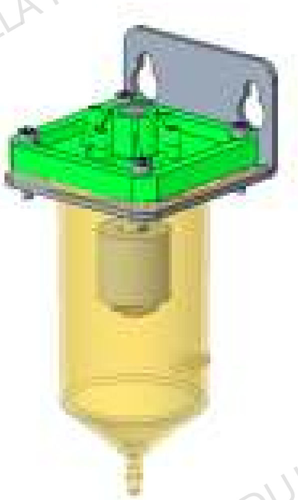


Table 8–45 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-047895-00	WC2 cistern assembly	N/A	N/A

Function:
Collecting waste and providing a negative pressure for the metering pump.

8.43.2 Disassembly and Assembly

Note

Before disassembly, power off and unplug the power supply.

Required tools

Cross-headed screwdriver
One pair of cutting nippers

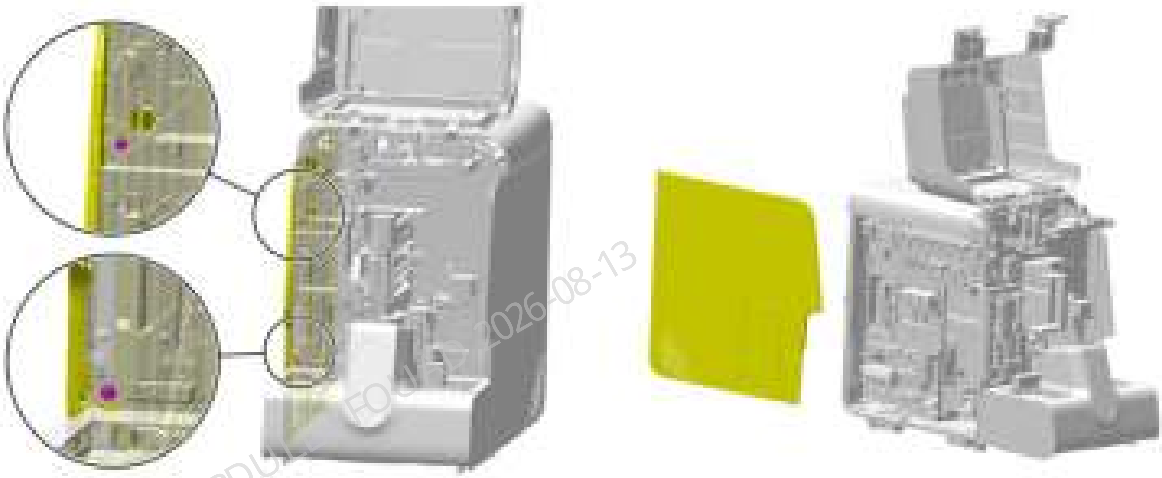
Pre-requirements

1. In the service access level, select **Service**>>**Maintenance**>>**Emptying and Priming**>>**WC2 Cistern Emptying** in sequence on the screen to empty the waste cistern.
2. Power off the main unit, and unplug the power cord.

Procedures

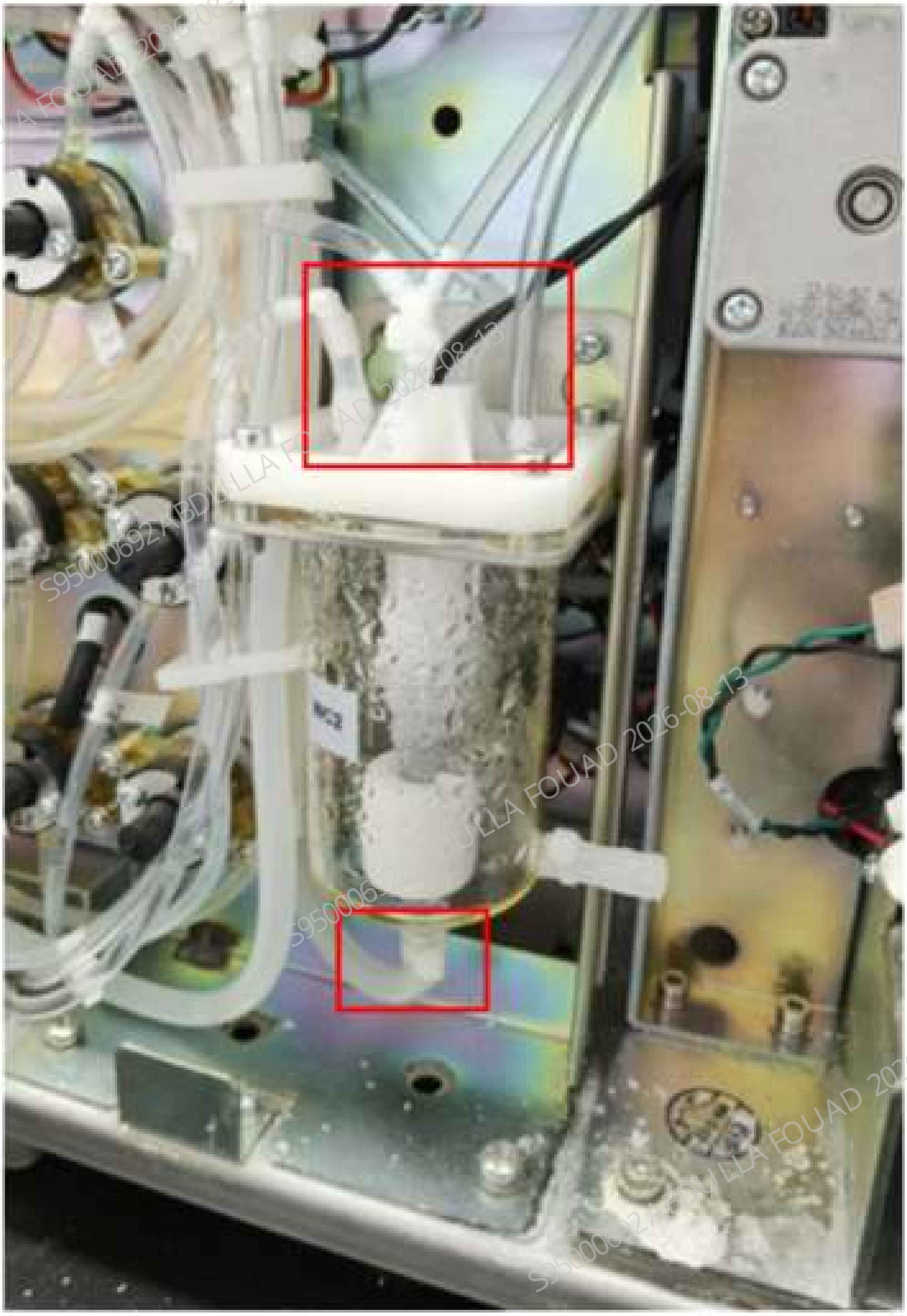
1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–453 Diagram of Removing the Left Cover



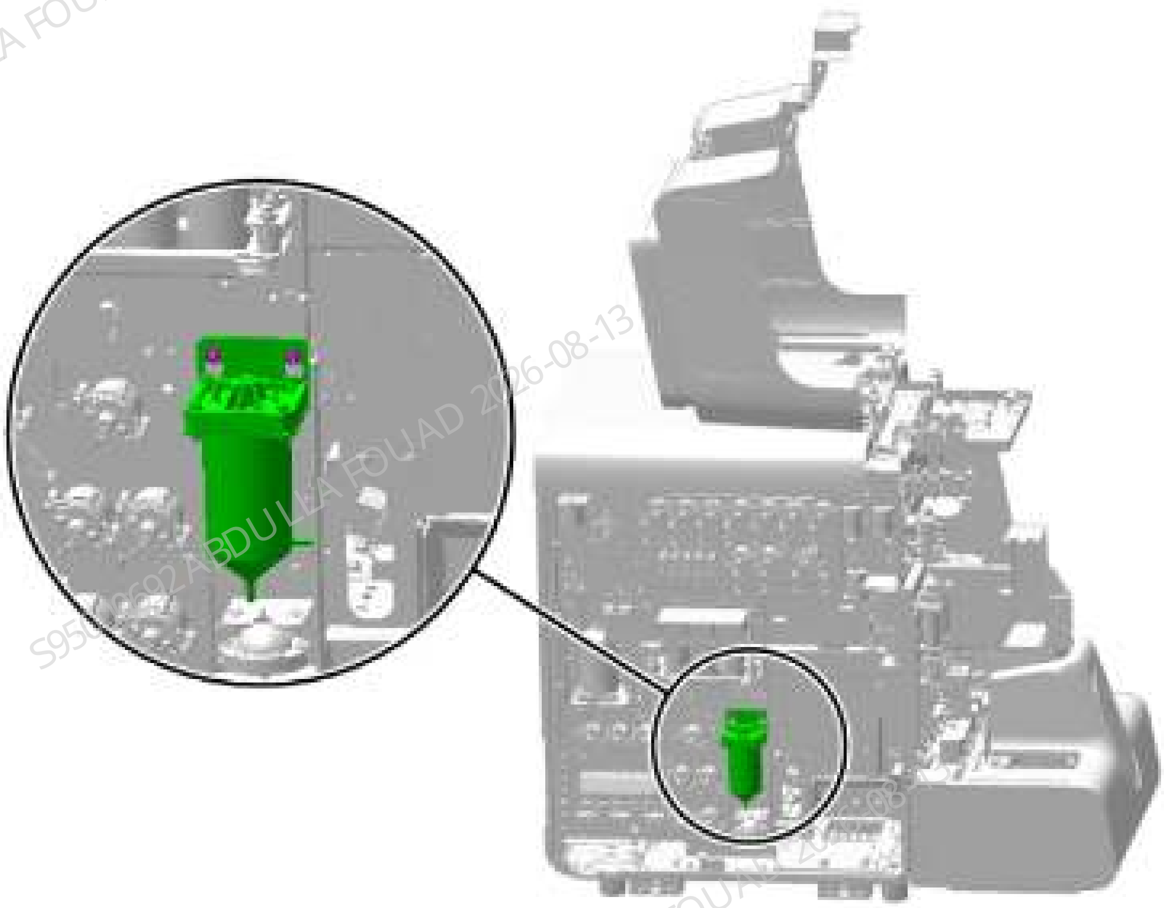
2. As shown in the figure below, unplug the pipeline of the waste cistern assembly.

Figure8–454 Diagram of the Pipeline of Waste Cistern Assembly



3. As shown in the figure below, loosen (do not remove them) the two screws fixing the waste cistern assembly and remove the waste cistern assembly.

Figure8–455 Diagram of Removing the Waste Cistern Assembly



4. Remove the wire plug of the waste cistern assembly.

Subsequent Requirements

Installation procedure:

1. Connect the tubes and terminals of the WC2 cistern assembly.
2. Install the WC2 cistern assembly back into the analyzer and tighten two screws to fix it.
3. Install the left cover back, and flip down the front cover to reset it.

8.43.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning:

1. After startup, log in at the service access level, select **Service>>Maintenance>>Empty and Prime>>WC2 Cistern Emptying** in sequence on the screen. As shown in the following figure.

Figure8–456 Enter the Maintenance screen

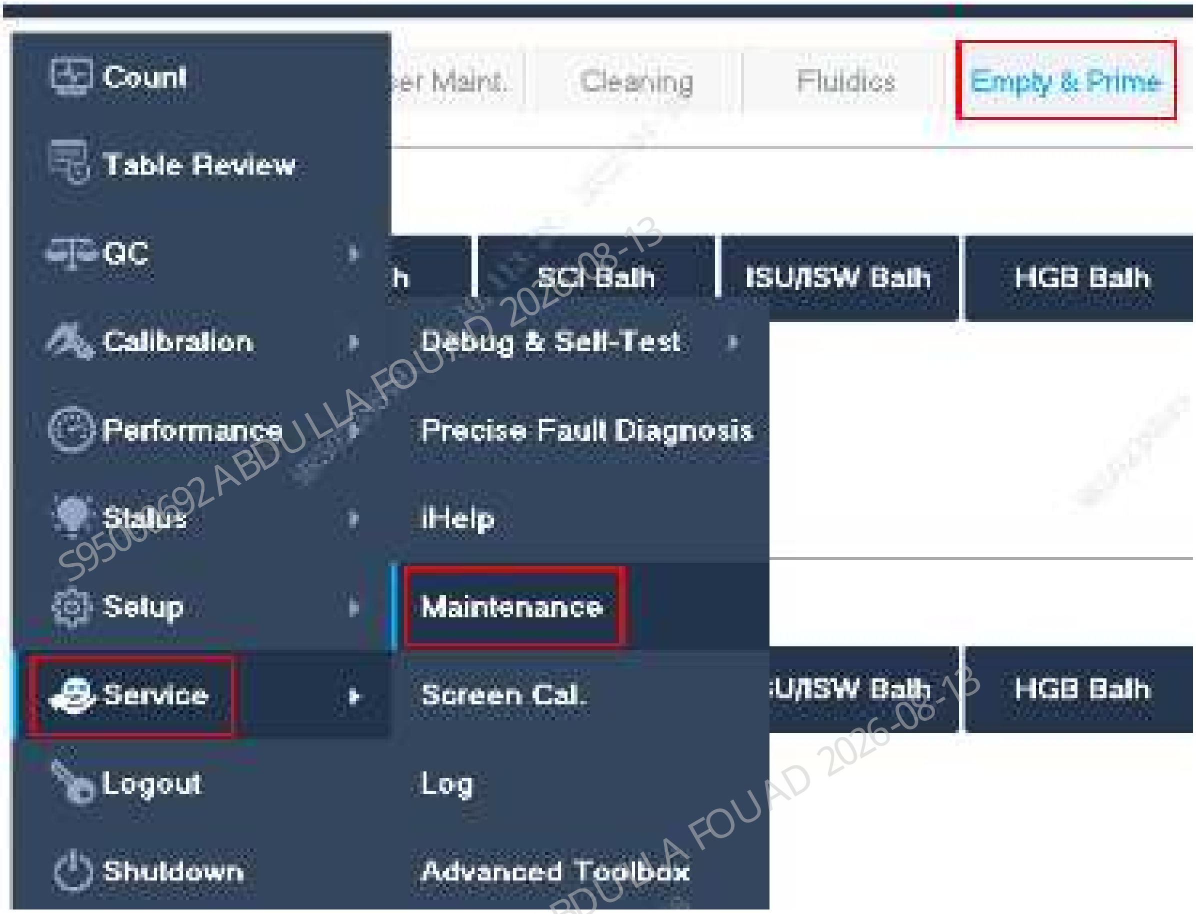
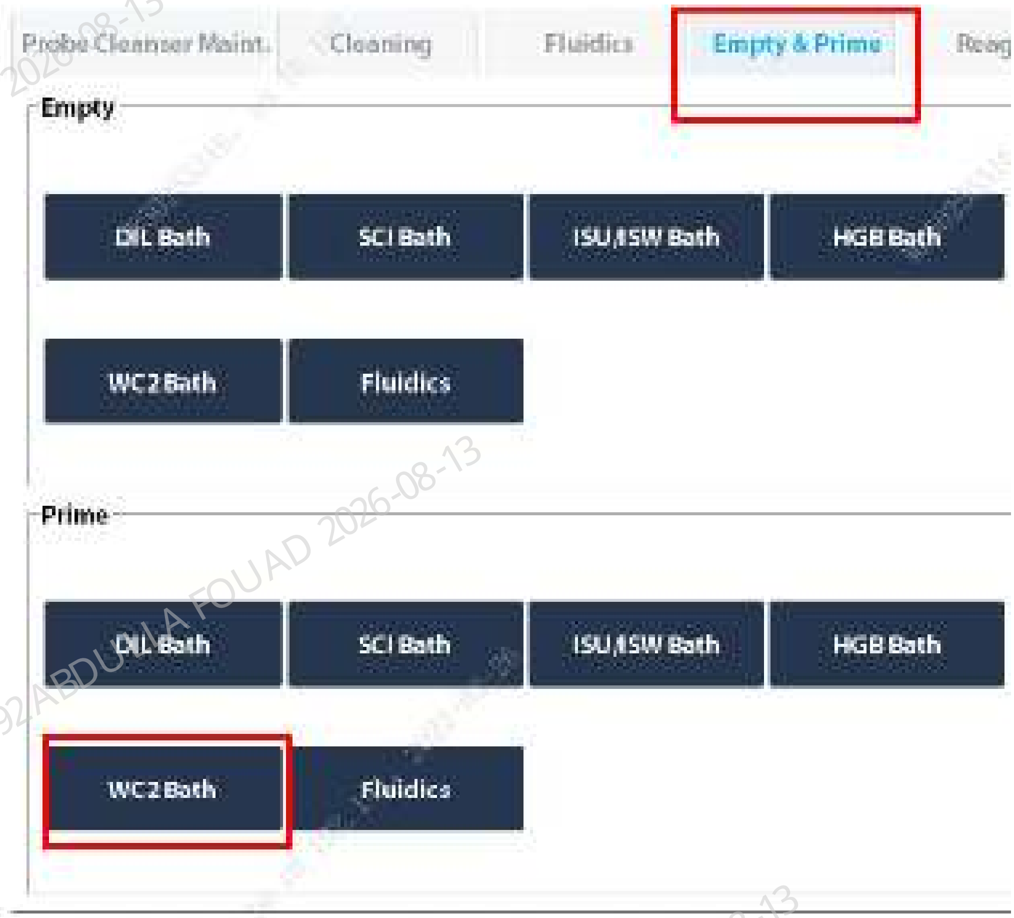


Figure8–457 Empty and Prime screen



2. After the analyzer prompts that the priming is complete, click **OK**.

Figure8–458 Priming Successful confirmation screen

3. Then click **Status**>>**Floater Status** in sequence on the software screen to check whether the status of the WC2 cistern is **full**.

Figure8–459 Querying the WC2 status

4. Then select **Service**>>**Maintenance**>>**Empty and Prime**>>**WC2 Cistern Emptying** in sequence on the screen. as shown in the following figure.

Figure8–460 Enter the Maintenance screen

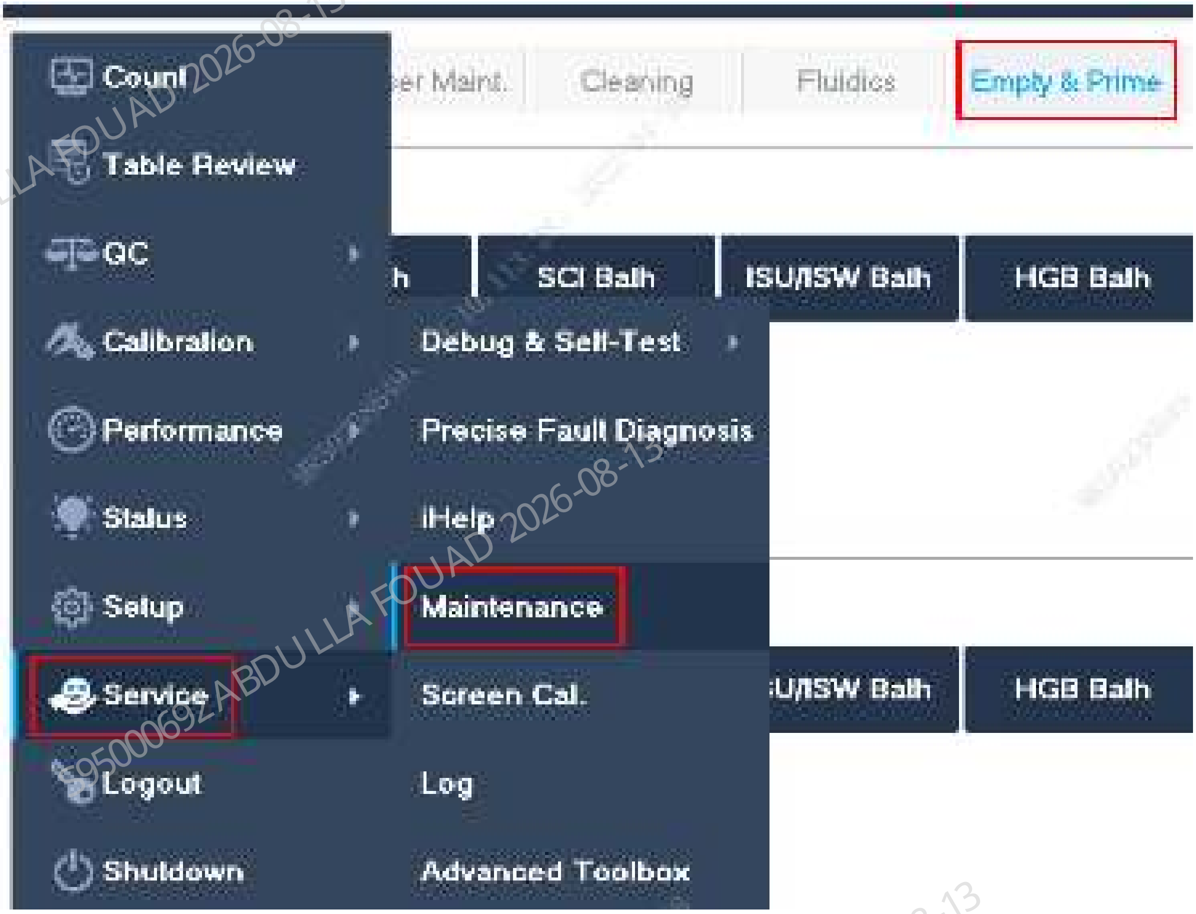
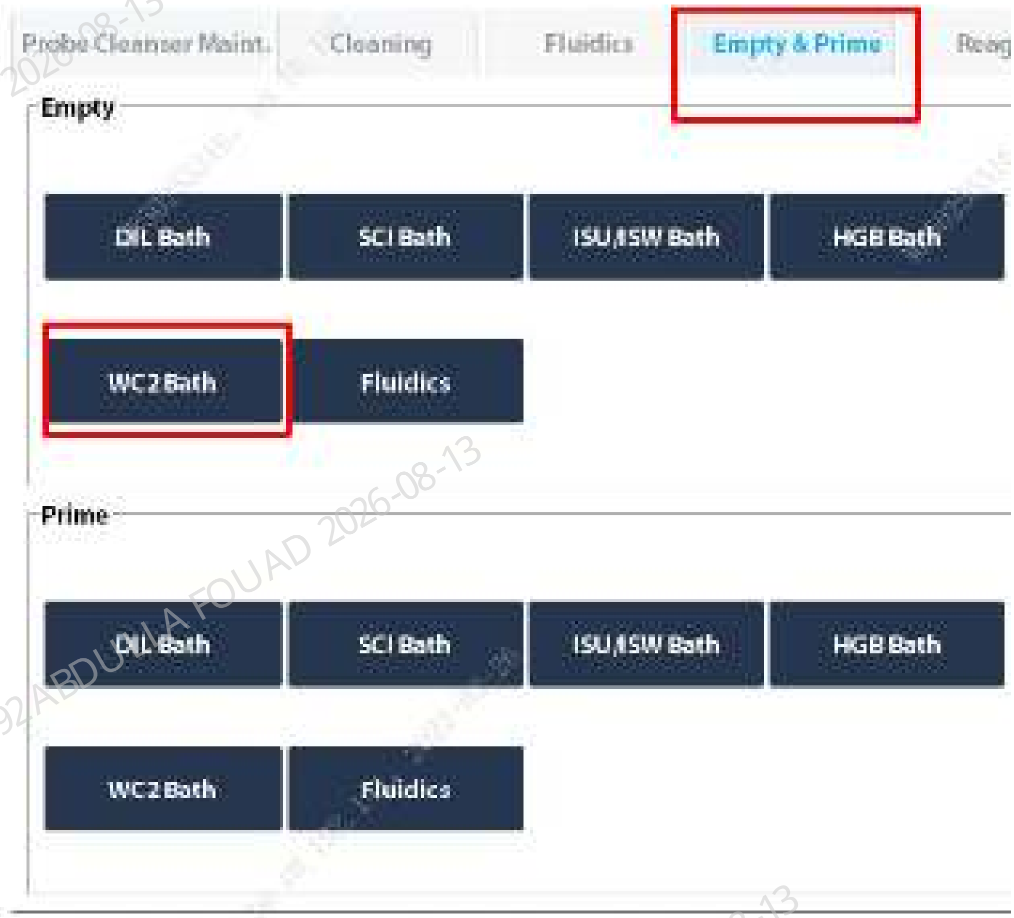
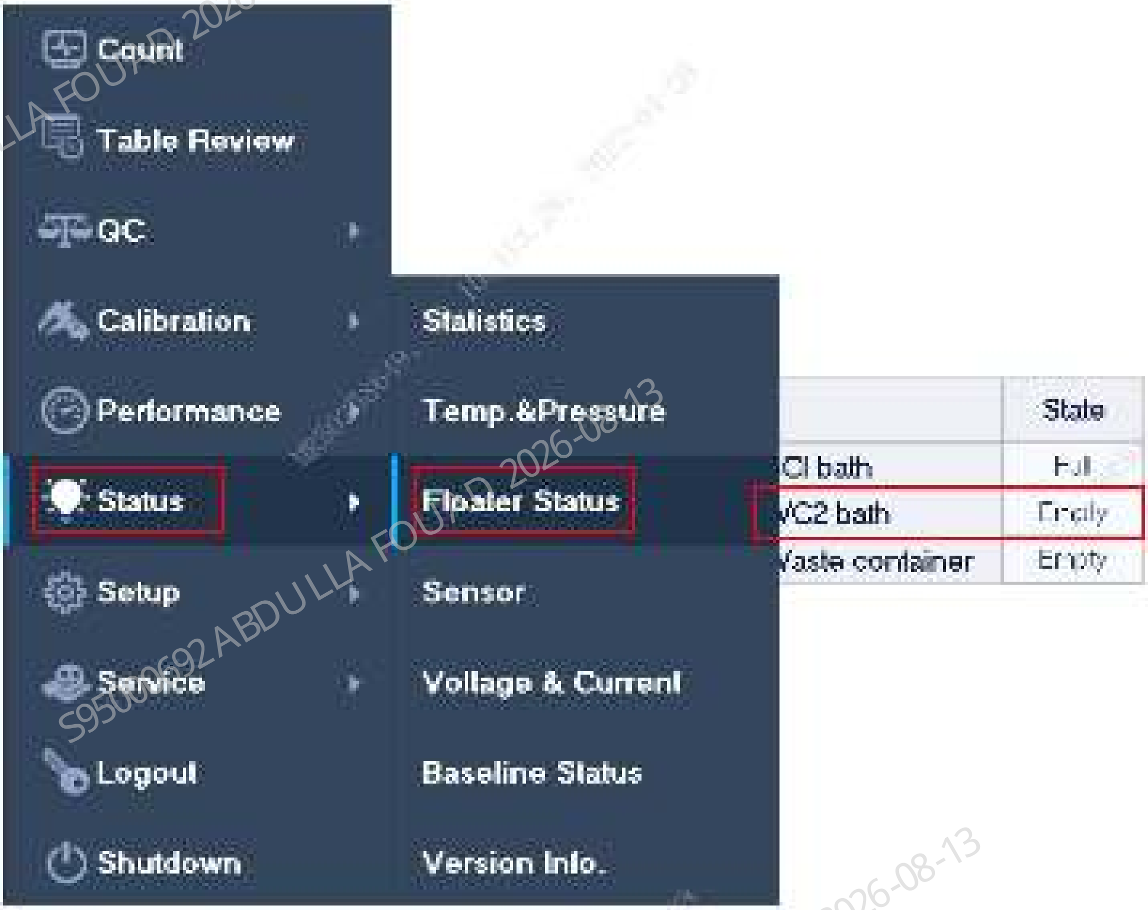


Figure8–461 Empty the WC2 cistern



5. After the analyzer prompts that the emptying is complete, click **OK**.

6. Then click **Status>>Floater Status** in sequence on the software screen to check whether the status of the WC2 cistern is **empty**.



Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.44 Diluent heating assembly (FRU)

8.44.1 General Information

Figure8–462 Photo of diluent heating assembly

Table 8–46 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-091025-00	Diluent heating assembly (FRU)	/	/

Functions:
Preheating the DS diluent.

8.44.2 Disassembly and Assembly

Note:

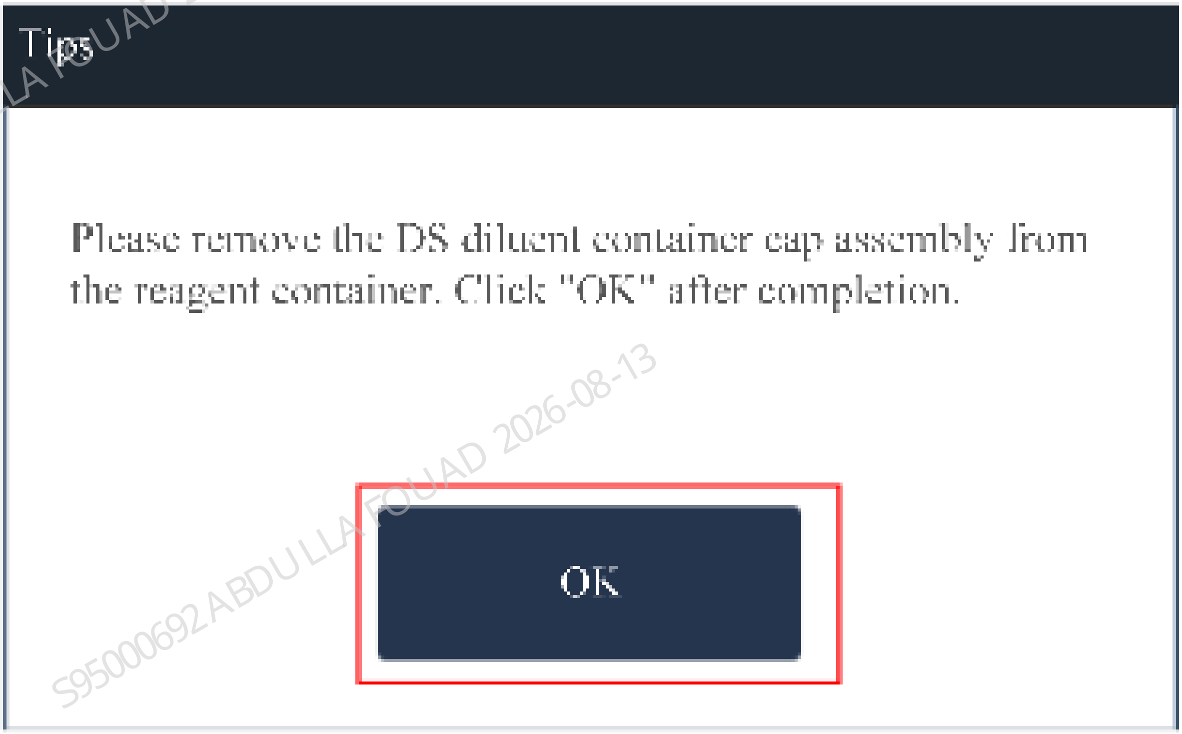
Prevent liquid from entering the terminals during disassembly.

Required tools:

One cross-headed screwdriver
Cutting nipper x 1

Pre-requirement

1. Select **Service>>Maintenance>>Emptying and Priming>>DIL Cistern Emptying** in sequence on the software screen, take out the DS container cap assembly from the reagent bottle according to the prompt, and click **OK** when finished.



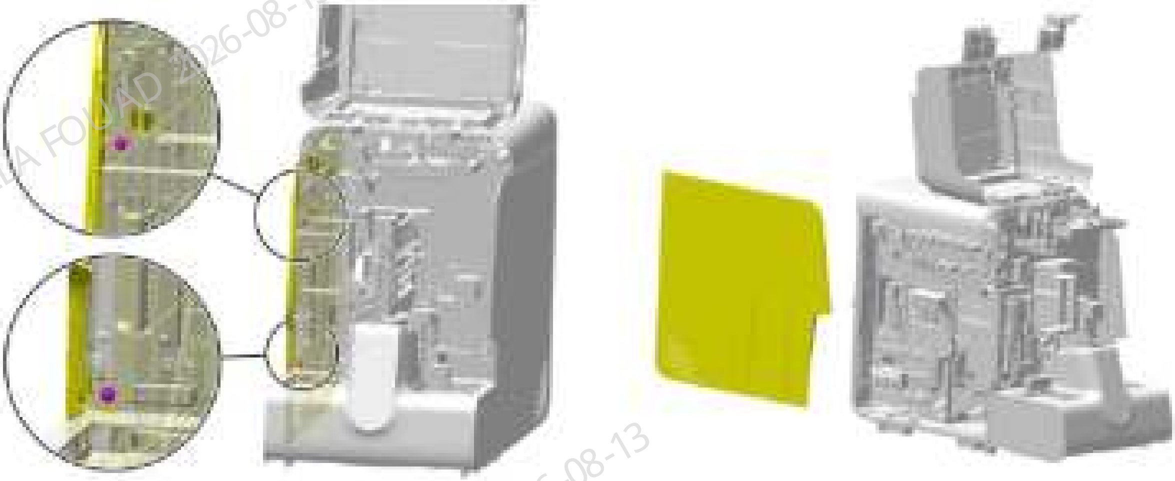
When the system prompts that DIL cistern has been emptied, click **OK**.

2. Power off the instrument and remove the power plug.

Steps:

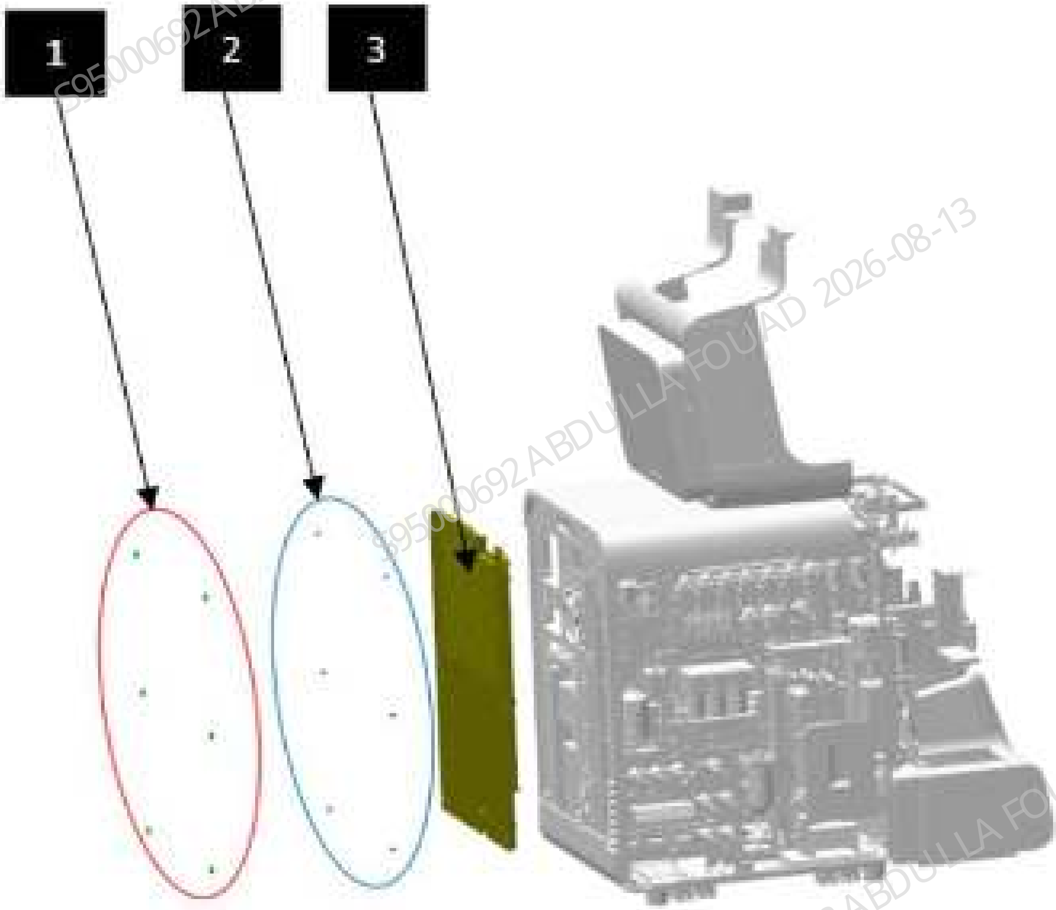
1. Flip the front cover up, and put it against the top cover. Unscrew the two set screws of the left cover, and remove the left cover, as shown in the figure below.

Figure8–463 Removing the left cover



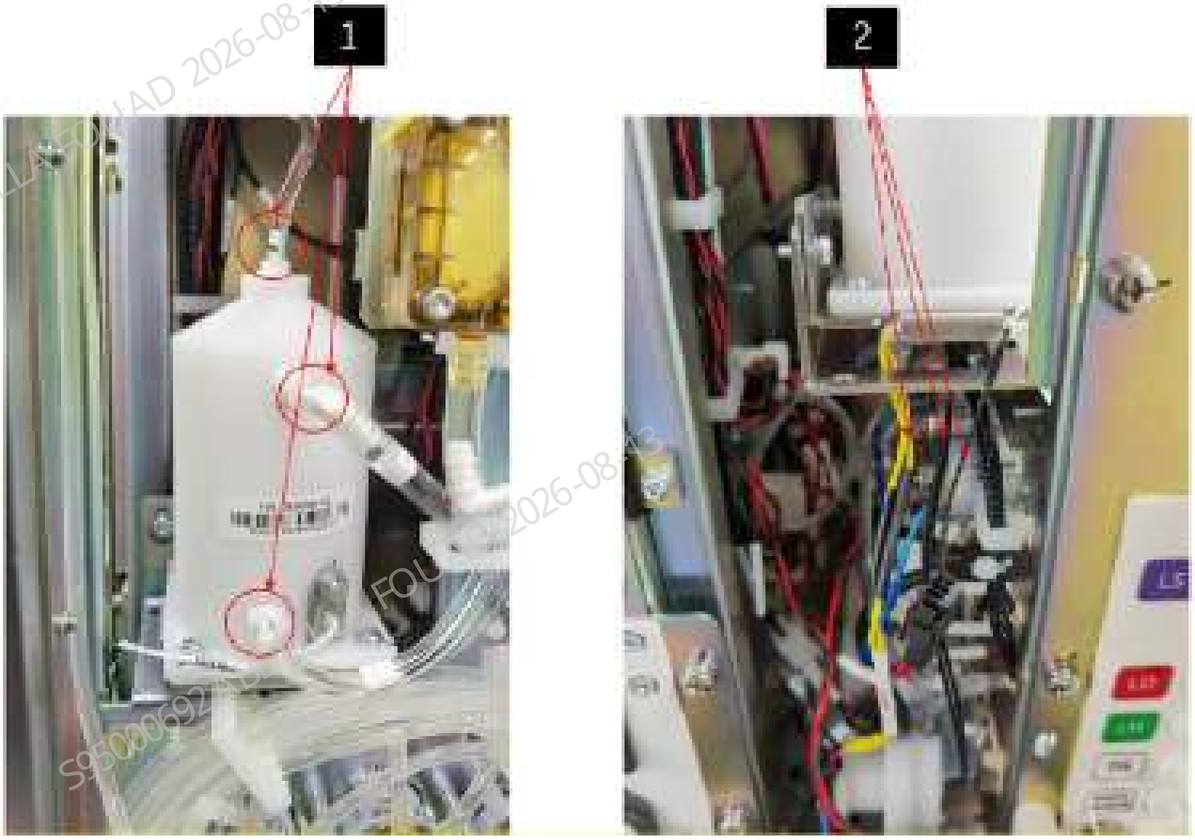
2. As shown in the figure below, manually remove the six screw plugs (#1), unscrew the six set screws (#2) of the rear cover with a cross screwdriver, and remove the rear cover (#3).

Figure8–464 Removing the rear cover



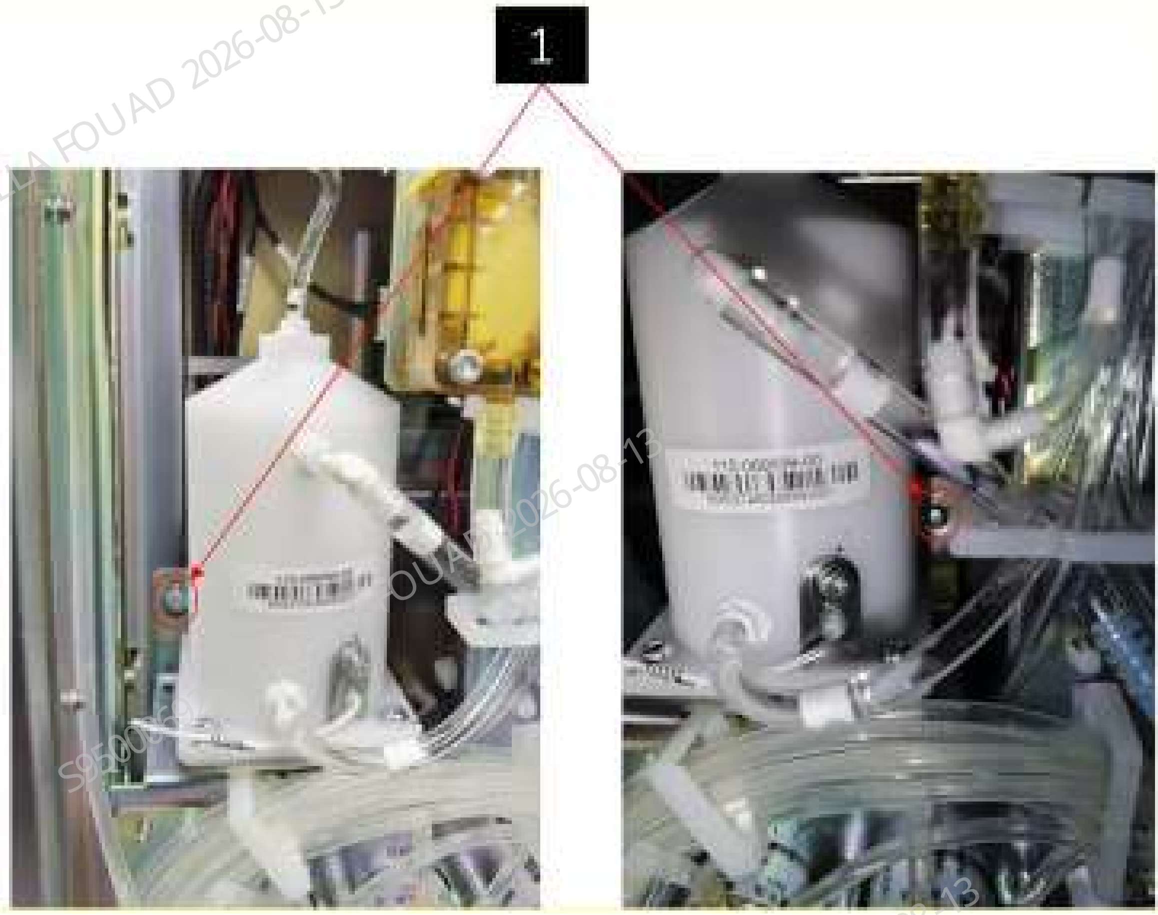
3. As shown in the figure below, disconnect the silicone tube (#1) from the assembly connector, and disconnect the terminal (#2) from the analyzer.

Figure8–465 Removing tubes and wires of diluent preheating bath assembly



4. As shown in the figure below, loosen the two set screws (#1) of the diluent preheating bath with a cross screwdriver, lift the diluent preheating bath assembly to disengage the hardy holes on the diluent preheating bath assembly from the set screws, and then remove the diluent preheating bath assembly.

Figure8–466 Removing set screws from diluent preheating bath assembly



5. Unplug the terminal connected to the main unit, unplug the tube on the assembly connector, and finally remove the diluent heating assembly.

Subsequent Requirements

NOTICE

The sold products may vary in the diluent heating assembly (New or Old). Please install them with attention to the difference in tubing locations.

Old version	New version
115-069699-00 Diluent Heating Assembly	115-091025-00 Diluent Heating Assembly (FRU)
	

Scenario 1: Take the scenario of "replacing the old version with the new version" as an example.
Installation procedure:

1. Connect the wiring terminals in the diluent heating assembly to the corresponding wiring terminals of the analyzer according to the wire markings.
2. Connect the silicone tube back to the diluent heating assembly. As shown in the figure below, connect the tubes according to the circled positions. Be careful not to make a mistake.

Figure8–467 Diagram of Tubing Positions of New FRU

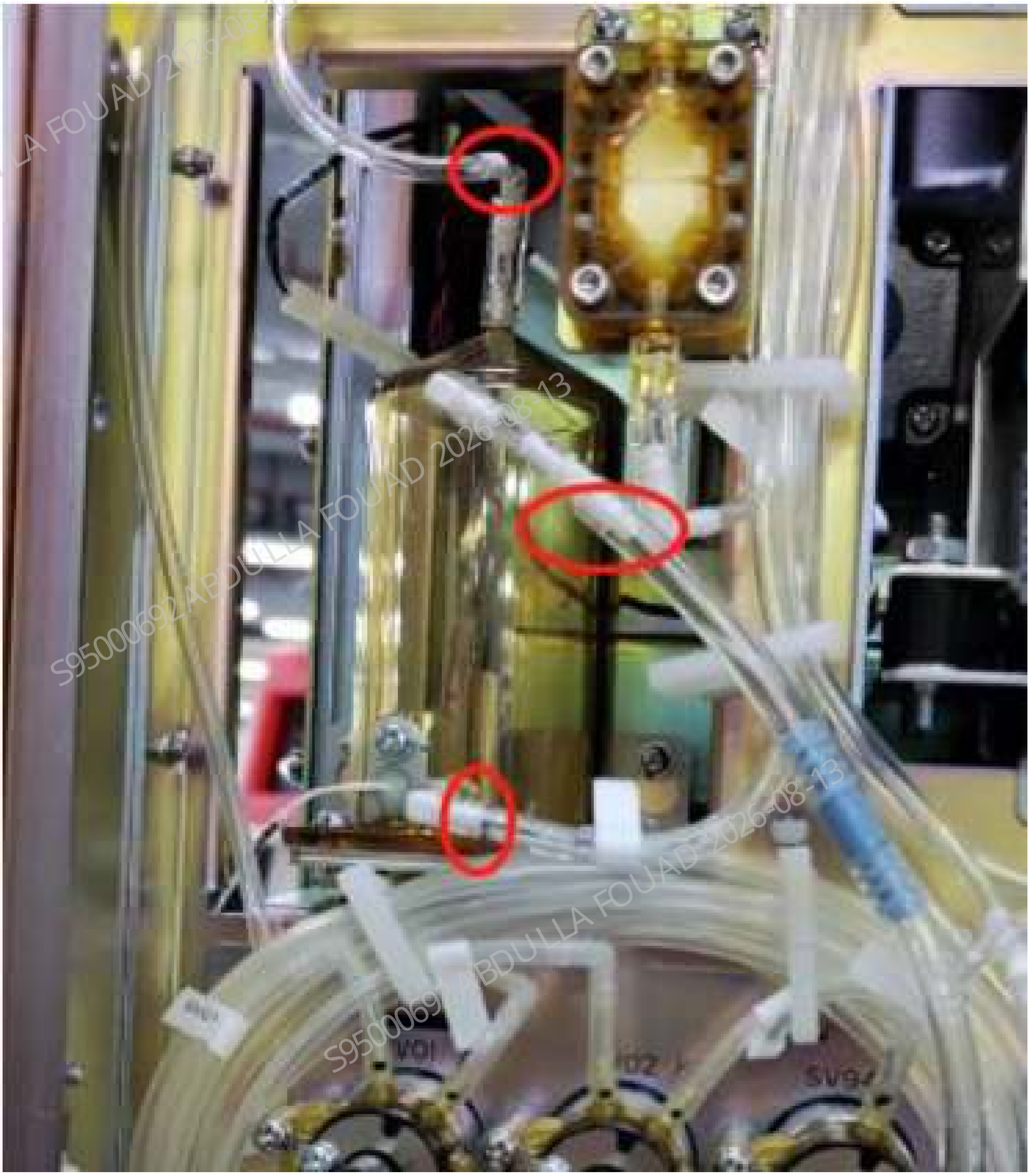
3. Install the diluent assembly back on the main unit, and tighten the two screws that fix the diluent assembly.
4. Install the left cover back, and flip down the front cover to reset it.

Scenario 2: Take the scenario of "replacing the new version with the new version" as an example.

Installation procedure:

1. Connect the wiring terminals in the diluent heating assembly to the corresponding wiring terminals of the analyzer according to the wire markings.
2. Connect the silicone tube back to the diluent heating assembly. As shown in the figure below, connect the tubes according to the circled positions. Be careful not to make a mistake.

Figure8–468 Diagram of Tubing Positions of New FRU



3. Install the diluent assembly back on the main unit, and tighten the two screws that fix the diluent assembly.
4. Install the left cover back, and flip down the front cover to reset it.

8.44.3 Commissioning and Verification

Pre-requirement

Place the DS container cap assembly into the reagent container.

Steps:

Commissioning:

1. Start the analyzer, and log in at the service access level. Click **Reagent Setup**, click **Setup**, scan the DS Diluent bar code into the analyzer, and click OK. Check whether **no diluent function** is reported in the software screen, if so, click **Eliminate Error**.

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.45 Air tank assembly

8.45.1 General Information

Figure8–469 Photo of air tank assembly

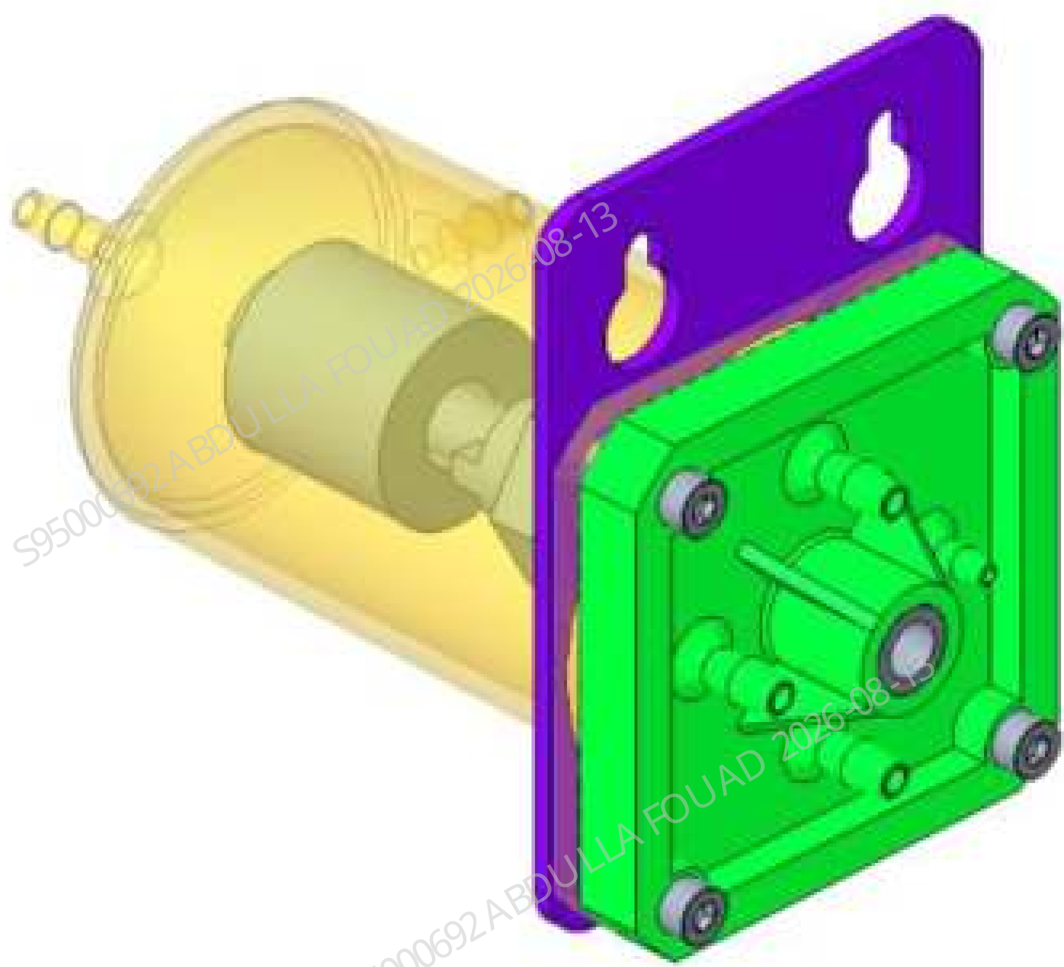


Table 8–47 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-071241-00	Air tank assembly	N/A	N/A

Function:
Providing a stable positive pressure for the metering pump.

8.45.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver
One pair of cutting nippers

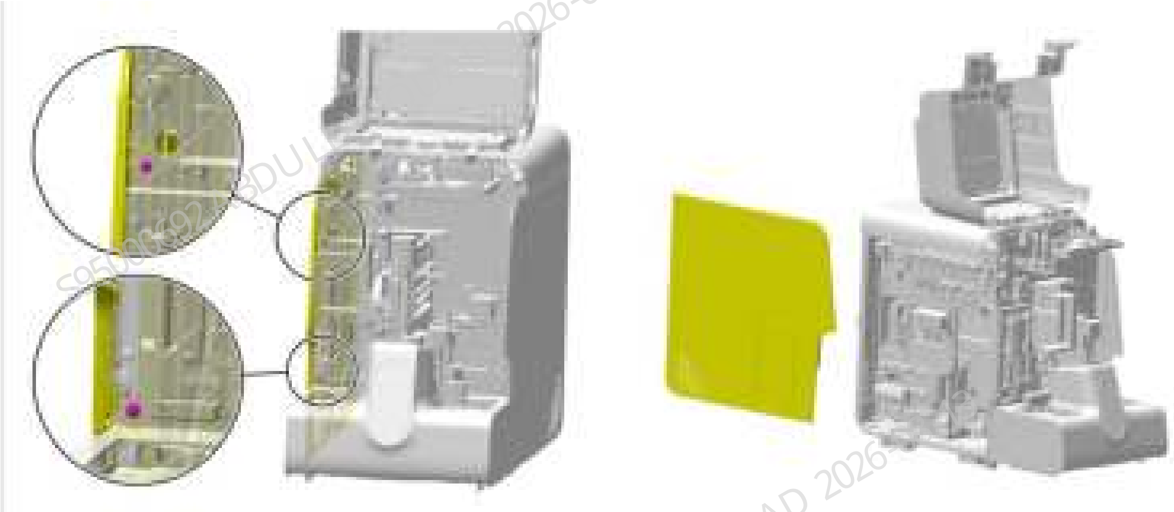
Pre-requirements

After performing the decompression operation, power off the main unit, and unplug the power cord.

Procedures

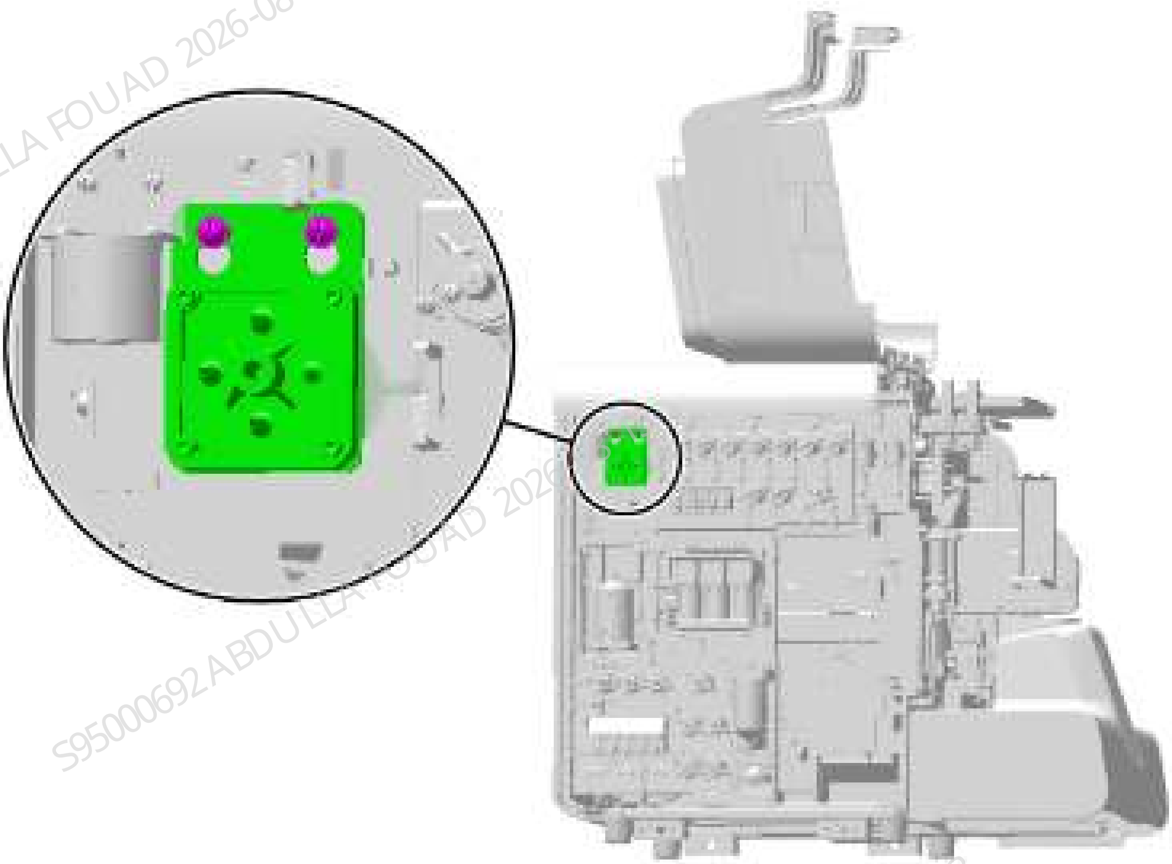
1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–470 Diagram of Removing the Left Cover



2. Unscrew the two set screws of the air tank assembly and remove the air tank assembly, as shown in the figure below.

Figure8–471 Positions of the set screws of the air tank assembly



3. Loosen the terminals of the pressure sensor on the air tank assembly and pull out the rubber tube.

Subsequent Requirements

Installation procedure:

1. Reconnect the rubber tubes and terminals of the air tank assembly.
2. Install the air tank assembly back into the analyzer and tighten the screws to fix it.
3. Power on the analyzer and confirm that the air tank replaced is in good condition.
4. Install the left cover back, and flip down the front cover to reset it.

8.45.3 Commissioning and Verification

Required tools

Cross-headed screwdriver

One Allen wrench

Procedures

1. Select **Status>>Temperature and Pressure**, and click the **Release Pressure** button to release the analyzer pressure.
2. Click the **Build Pressure** button to observe whether the air tank interface and its tube connections replaced during the pressure building process are leaking.
3. After the pressure is built completely, observe whether the channel pressure value of the air tank can remain stable after the replacement.

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.46 SCI bath assembly

8.46.1 General Information

Figure8–472 Photo of SCI bath assembly

Table 8–48 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-061156-00	SCI bath assembly	N/A	N/A

Function:
Provide sheath fluid to the sheath fluid impedance assembly through pressure transformation.

8.46.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver
One pair of tweezers

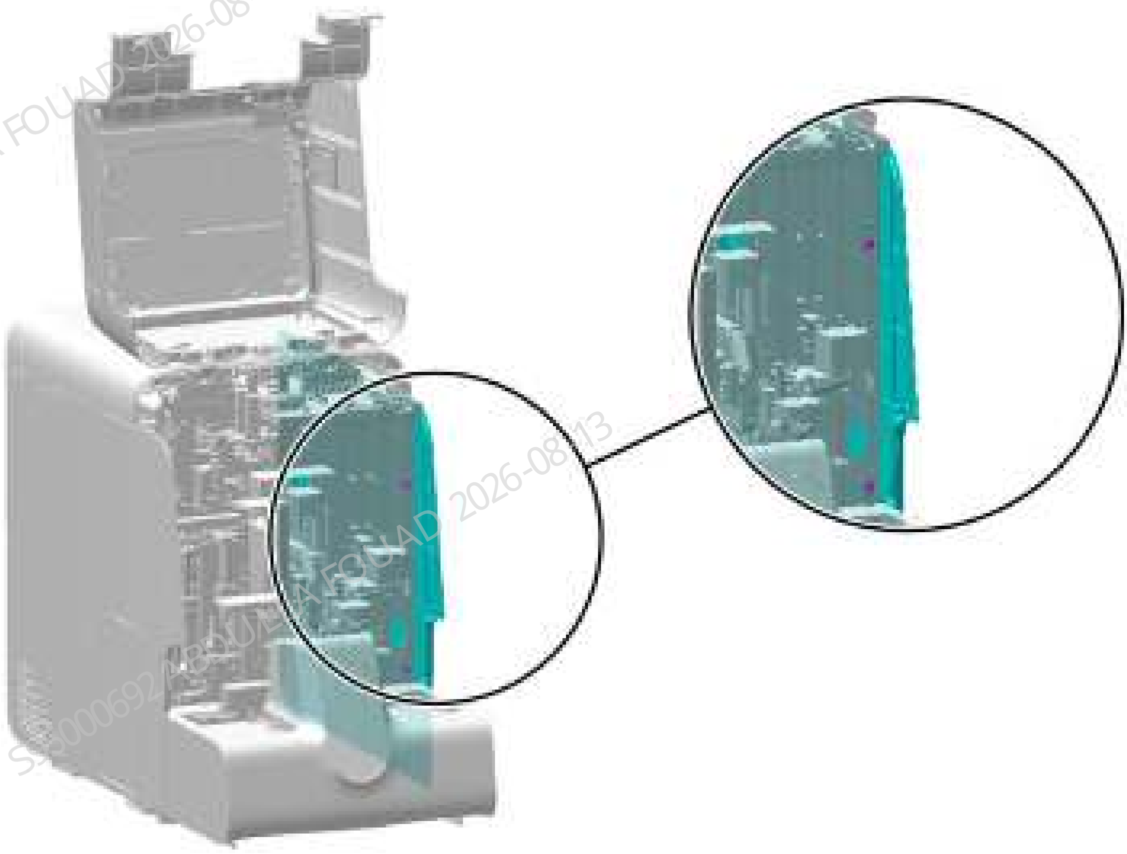
Pre-requirements

Empty the SCI cistern, select **Status**>>**Temperature and Pressure**, and click **Release Pressure** to release the pressure of the air tank assembly and SCI cistern. Turn off the power and unplug it.

Procedures

1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–473 Removing the right cover

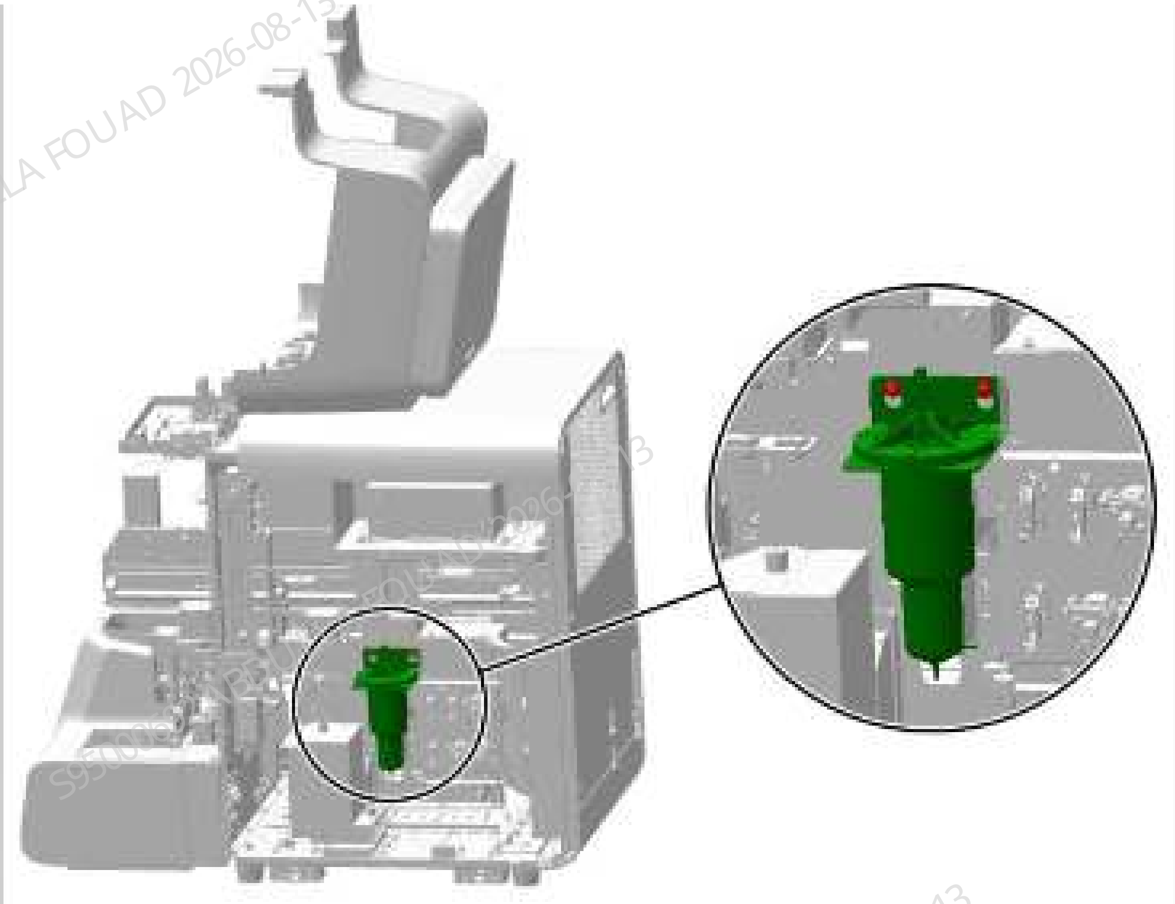


2. As shown below, loosen the terminals on the SCI cistern assembly and pull out the rubber tube.

Figure8–474 Diagram of Removing the Pipeline Wire of SCI Cistern Assembly

3. As shown below, loosen the two fixing screws of the SCI cistern assembly and remove the assembly.

Figure8–475 Positions of the set screws of the SCI cistern assembly



Subsequent Requirements

Installation procedure:

1. Connect the tubes and terminals of the SCI cistern assembly.
2. Install the SCI cistern assembly back into the analyzer and tighten two screws to fix it.
3. Install the right cover back, and flip down the front cover to reset it.

8.46.3 Commissioning and Verification

Procedures

Commissioning:

1. Power on normally, log in at service access level, and select **Service>>Maintenance>>Empty and Prime**, click **SCI Cistern Priming**, and wait for the priming to complete.

2. Then click **Service>> Status>> Floater Status** and check whether the status of the SCI cistern is **full**.

Verification:

Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.47 Isolation Chamber

8.47.1 General Information

Figure8–476 Photo of isolation chamber

Table 8–49 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-107763-00	Isolation chamber	N/A	N/A

Function:
Prevent liquid backflow from damaging the air pressure detection board.

8.47.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

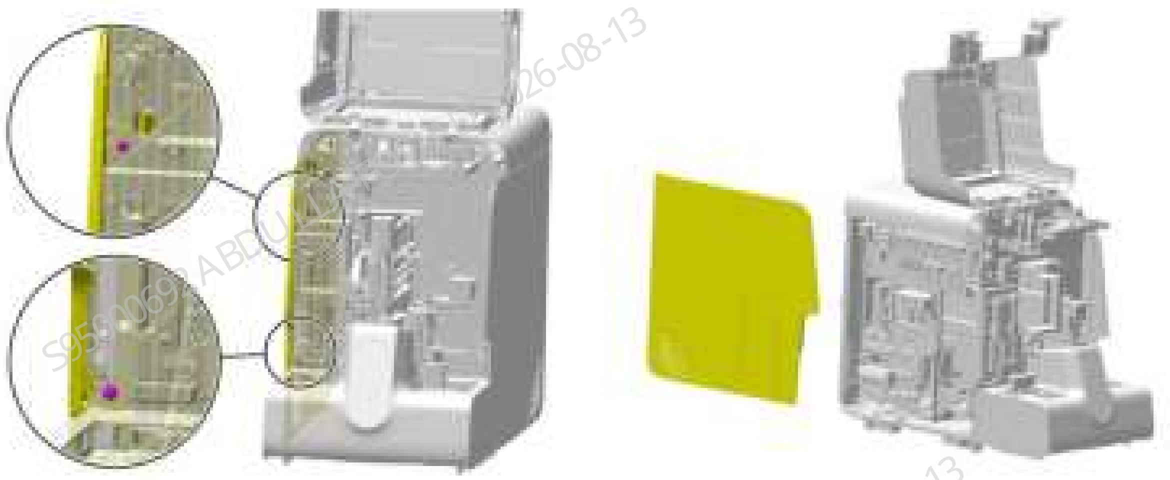
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

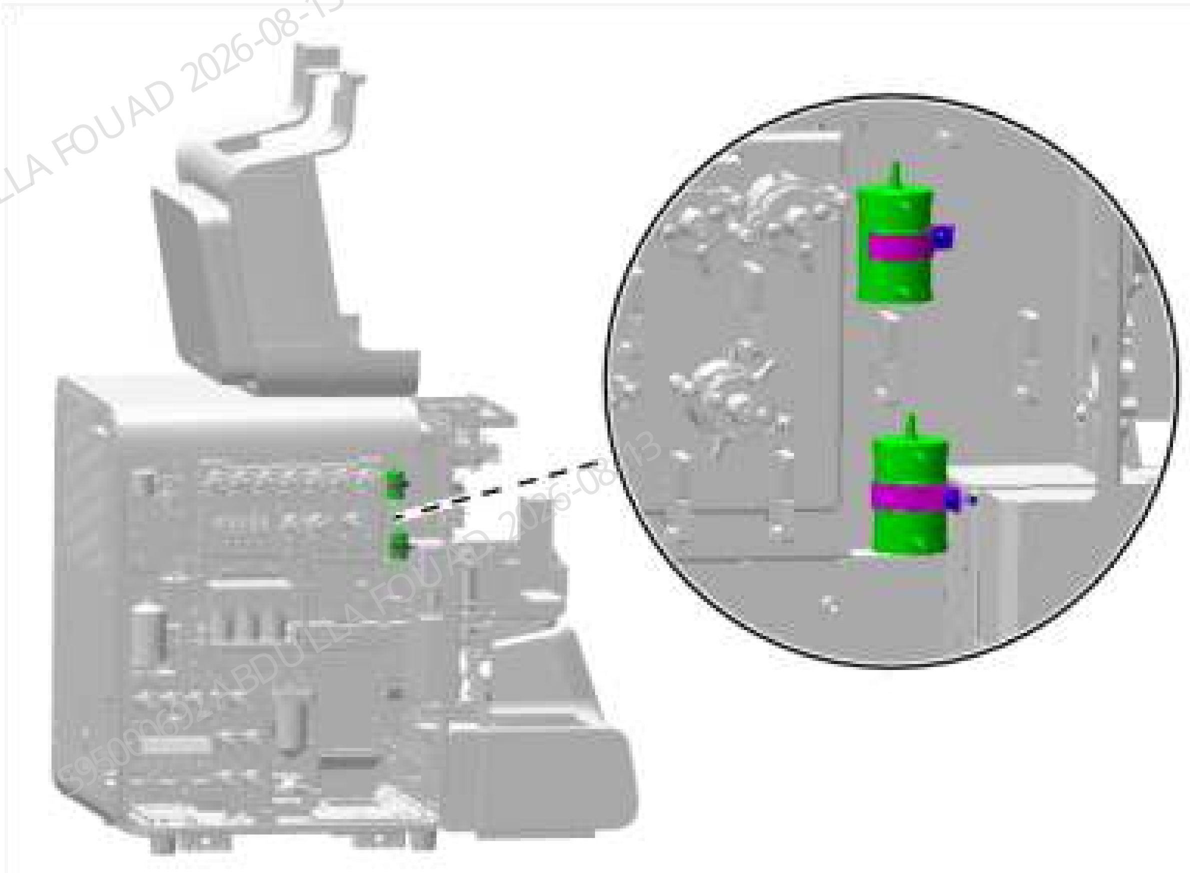
1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–477 Diagram of Removing the Left Cover



2. Pull out the rubber tube on the isolation chamber, (Note: the tube connecting the connector above the isolation chamber is marked TC1/2-up, and the tube connecting the connector below the isolation chamber is marked TC1/2-down), loosen the screws of the isolation chamber clamp as shown in the figure below, and remove the isolation chamber.

Figure8–478 Positions of the set screws of the isolation chamber



Subsequent Requirements

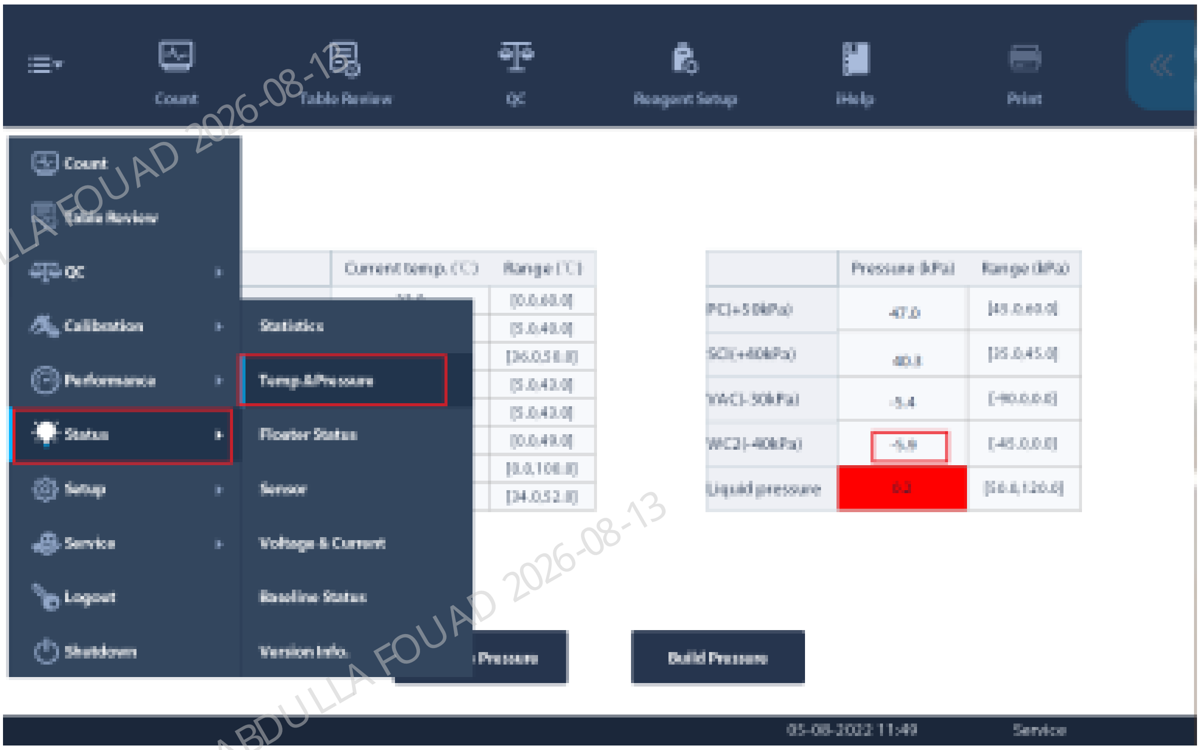
Installation procedure:

1. Insert the rubber tube to the corresponding connector of the isolation chamber according to the tube mark (refer to the instructions of step 2 in the disassembly procedure).
2. Tighten the screws that fix the isolation chamber clamp, and fix the isolation chamber through the isolation chamber clamp.
3. Install the left cover back, and flip down the front cover to reset it.

8.47.3 Commissioning and Verification

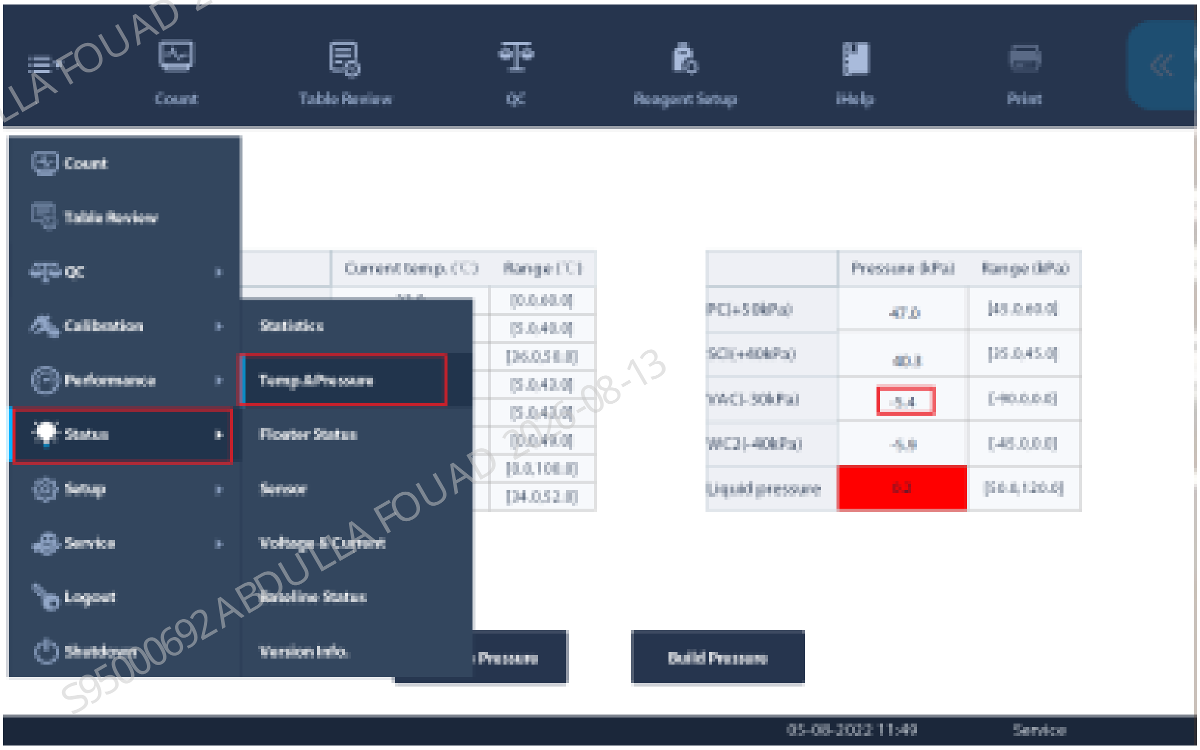
Procedures

1. Commissioning for TC1 replacement: After powering on, log in at the service access level, select **Status**>>**Temperature and Pressure** in turn, click **Build Pressure**, and check whether the pressure of WC2 (-40kPa) is within the range. If it is within the range, the replacement is successful.



2. Commissioning for TC2 replacement: After powering on, log in at the service access level, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and click **Probe Wipe Waste Pump (P2)**. Confirm if self-test succeeds.

3. Then select **Status>>Temperature and Pressure** in turn, click **Build Pressure**, and check whether the pressure of CBC_W (-30kPa) is within the range. If it is within the range, the replacement is successful.



Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.48 10ml syringe module

8.48.1 General Information

Figure8–479 Photo of 10 mL lead screw driving syringe assembly

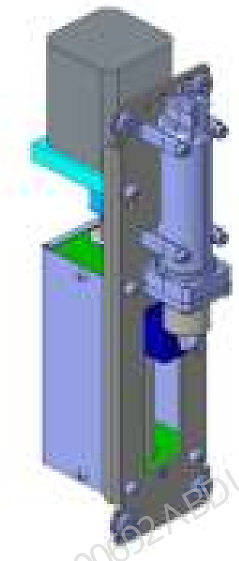


Table 8–50 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-011901-00	10 mL lead screw driving syringe assembly	0033-30-74615	Mindray 10 mL syringe
		011-000297-00	Photoelectric Position Sensor

Function:

Providing diluent for sample dilution of HGB bath, pipeline cleaning, SCI cistern priming, and sheath fluid supply to the flow cell; functioning as the power source for optical sample transfer, RBC sample transfer, and quantitative aspirating and dispensing of probe cleanser.

8.48.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

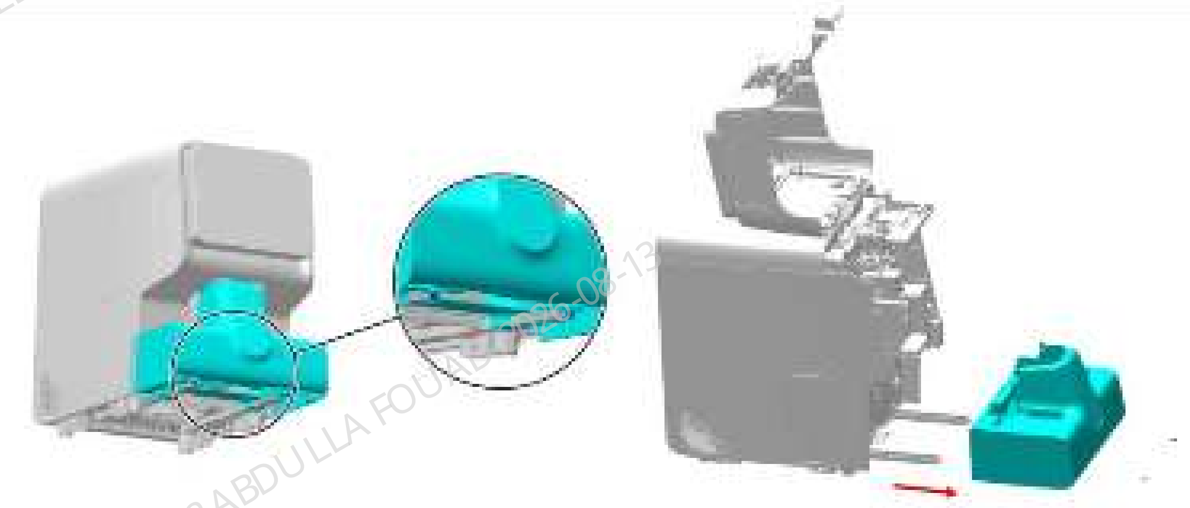
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

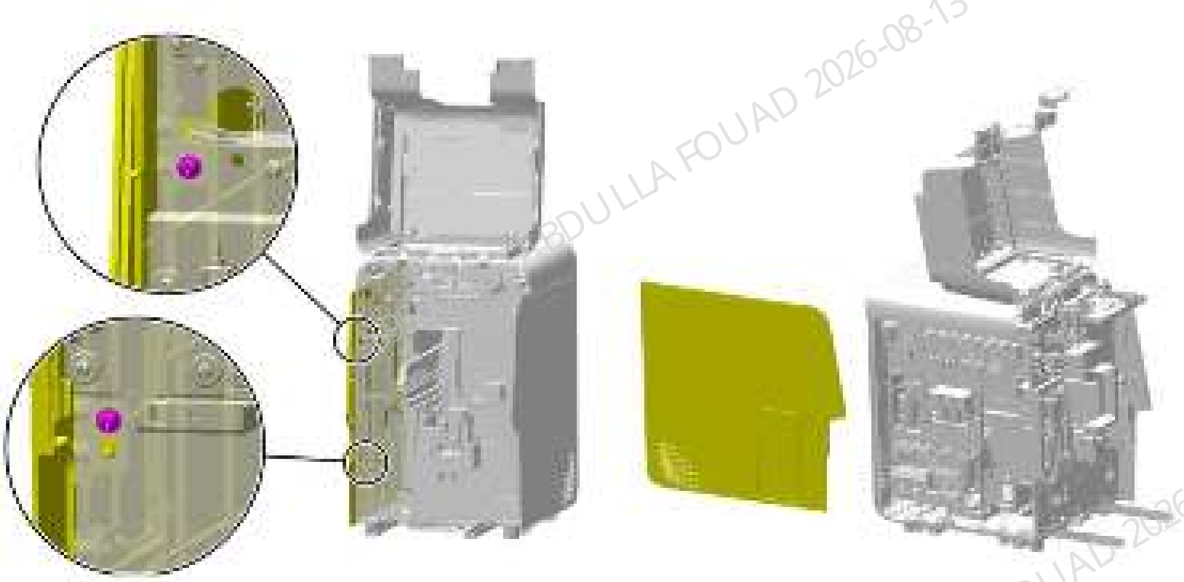
1. Loosen the two screws fixing the autoloader, open the front cover, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in the figure below.

Figure8–480 Removing the autoloader



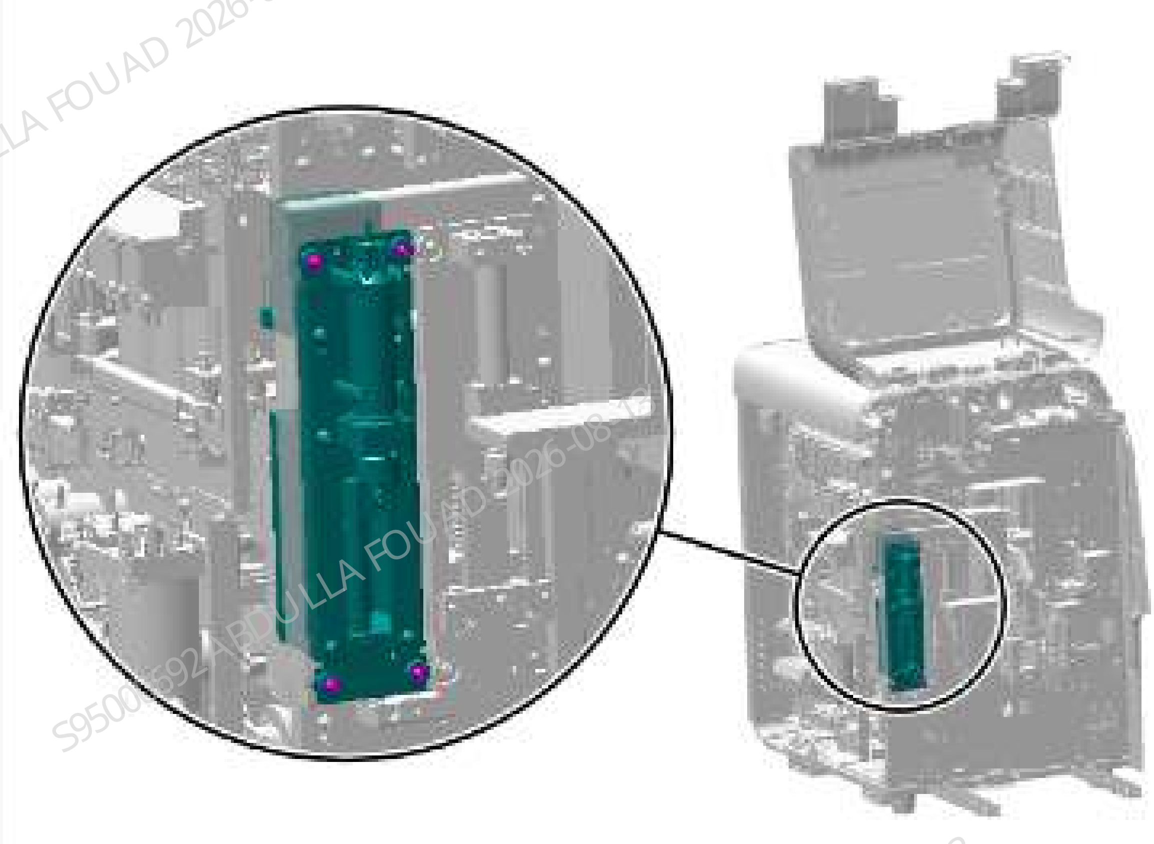
2. Loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–481 Removing the left cover



3. Remove the tube and ground wire from the 10ml lead screw driving syringe assembly.
4. Remove the four set screws of the 10 mL lead screw driving syringe assembly, as shown in the following figure, pull out the assembly terminal, and finally remove the entire assembly.

Figure8–482 Positions of the set screws of the 100ul lead screw driving syringe assembly



Subsequent Requirements

Installation procedure:

1. Insert the terminal block of the new assembly, put it back in the original position of the analyzer, and fix it with four screws.
2. Connect the tube and ground wire of the new assembly.
3. Power on the analyzer and confirm that the air tank replaced is in good condition.
4. Install the left cover, and flip down the front cover to reset it.
5. Restore the autoloader.

8.48.3 Commissioning and Verification

Pre-requirements

Install the 10 mL lead screw driving syringe assembly.

Procedures

1. Log in the software at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**DIL (10 mL) Syringe** in sequence.

Figure8–483 Self-test screen entrance

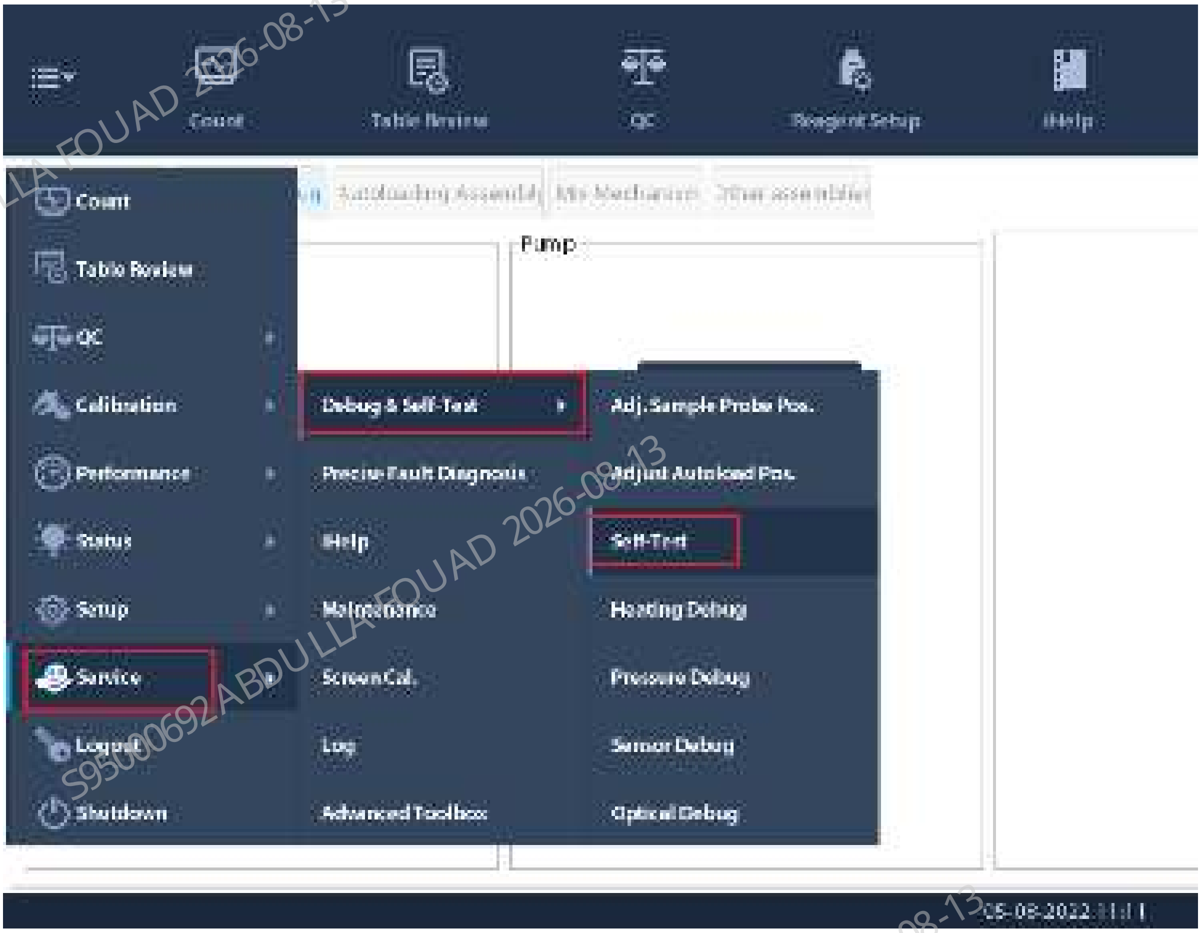


Figure8–484 Self-test screen of the 10 mL syringe

2. Confirm whether the 10 mL lead screw driving syringe assembly is initialized normally.
3. Check whether the syringe is leaking or splashing.

8.49 10ml self-made injector

8.49.1 General Information

Figure8–485 Photo of 10 mL lead screw driving syringe assembly

Table 8–51 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
0033-30-74615	Mindray 10 mL syringe	N/A	N/A

Function:

Providing diluent for sample dilution of HGB bath, pipeline cleaning, SCI cistern priming, and sheath fluid supply to the flow cell; functioning as the power source for optical sample transfer, RBC sample transfer, and quantitative aspirating and dispensing of probe cleanser.

8.49.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

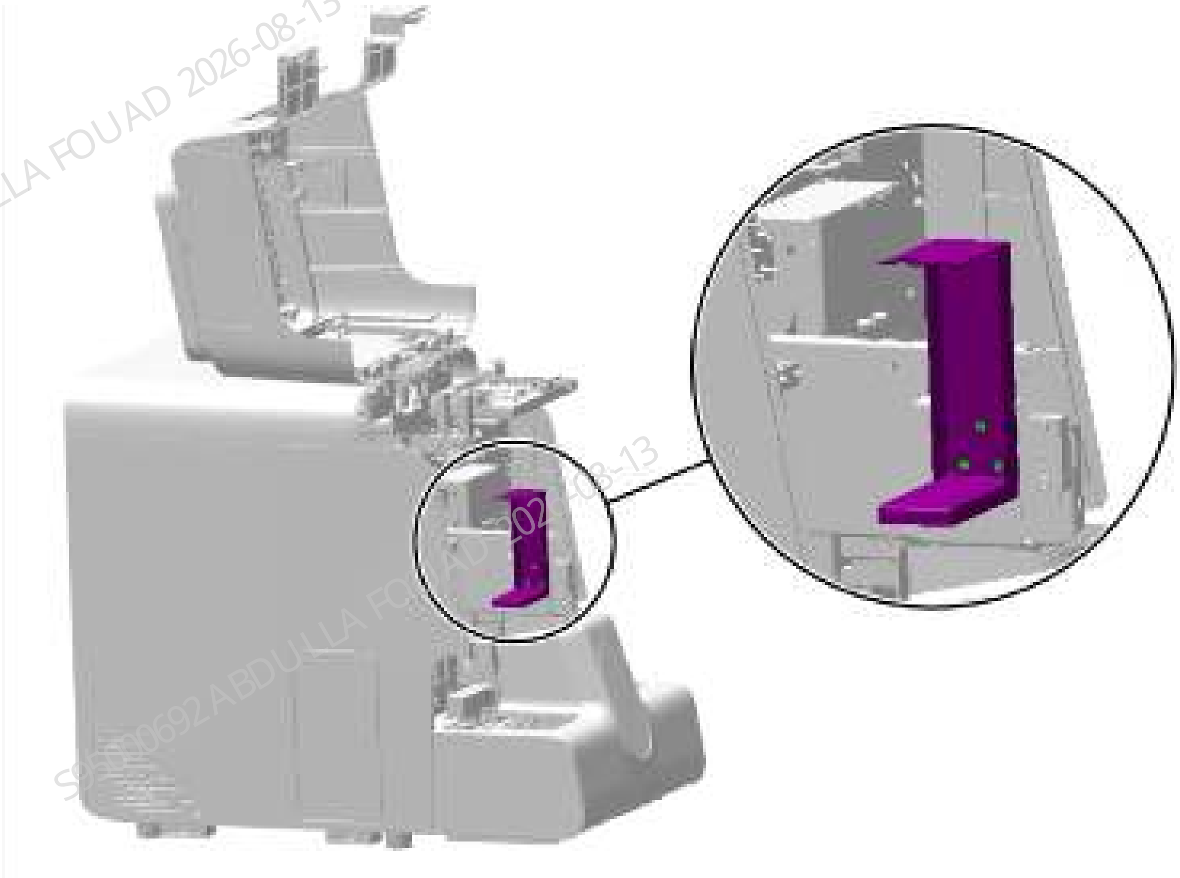
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

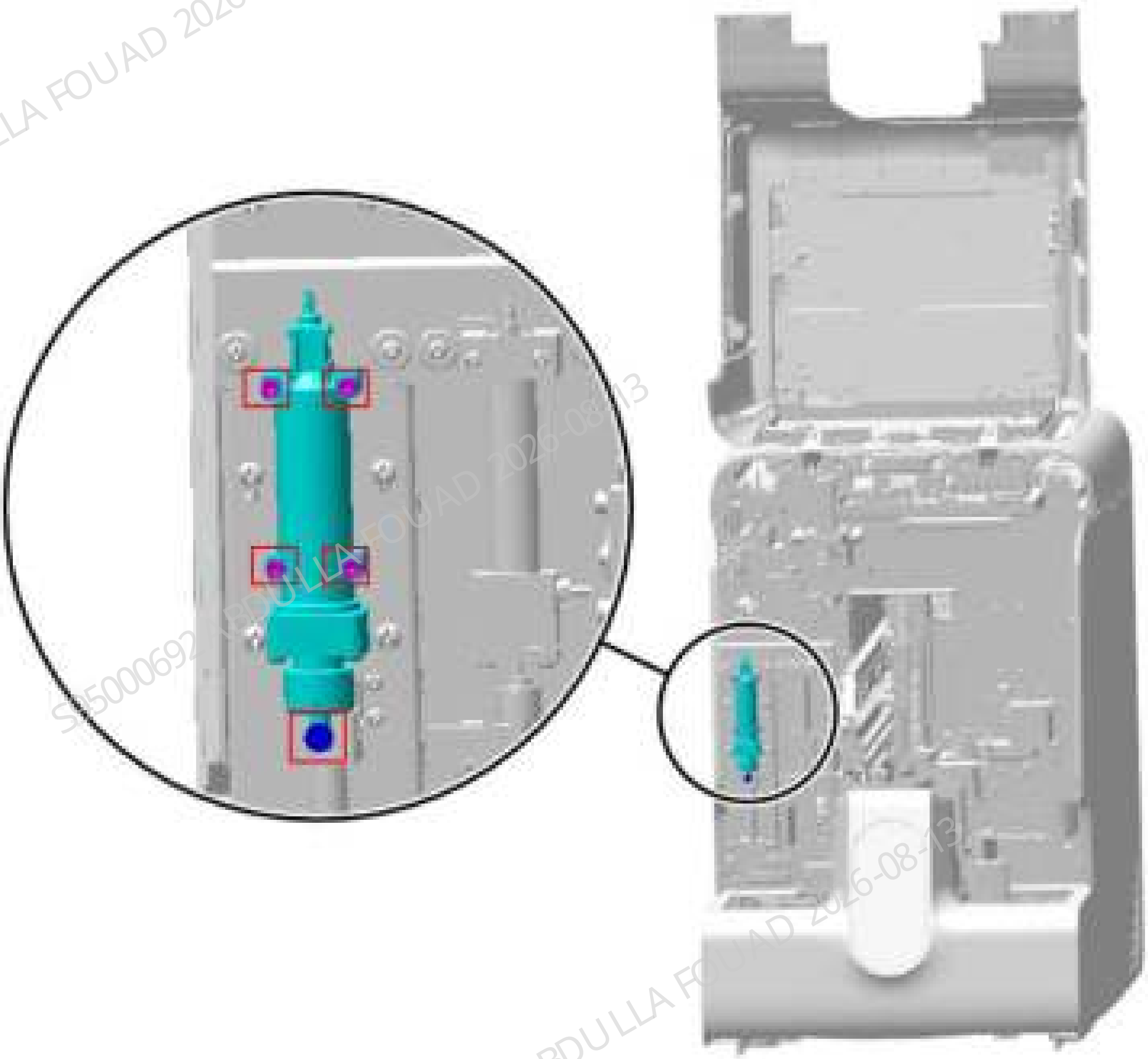
1. Turn up the front cover, lean it on the top cover, loosen the three countersunk screws fixing the dye bag support, and remove the dye bag support.

Figure8–486 Diagram of Removing the Dye Bag Support



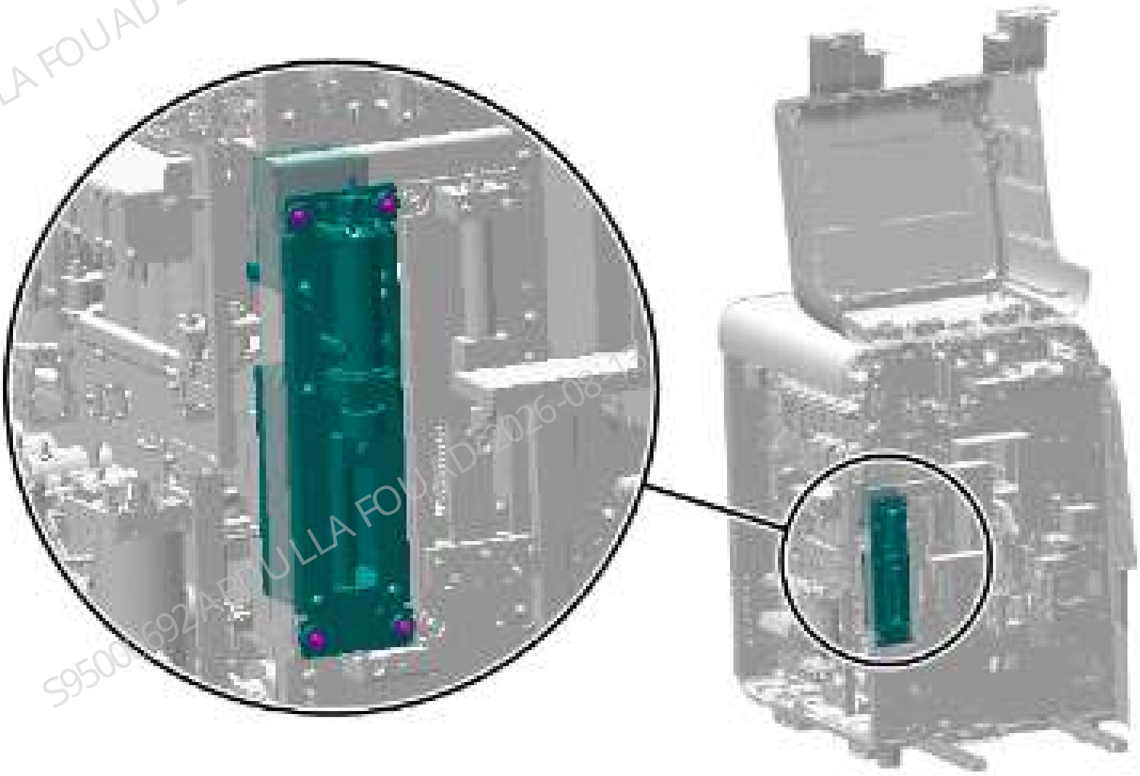
2. Remove the rubber tube on Mindray 10 mL syringe.

Figure8–487 Removing the left cover



3. Remove the set screws of Mindray 10 mL syringe, and remove the syringe.

Figure8–488 Positions of the set screws of the 100ul lead screw driving syringe assembly



Subsequent Requirements

Installation procedure:

1. Install Mindray 10 mL syringe back to the initial position of the analyzer and fix it with screws.
2. Connect the rubber tube of the new syringe.
3. Turn over the front cover to reset it.

8.49.3 Commissioning and Verification

Pre-requirements

Install Mindray 10 mL syringe.

Procedures

1. Log in the software at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**DIL (10 mL) Syringe** in sequence.

Figure8–489 Self-test screen entrance

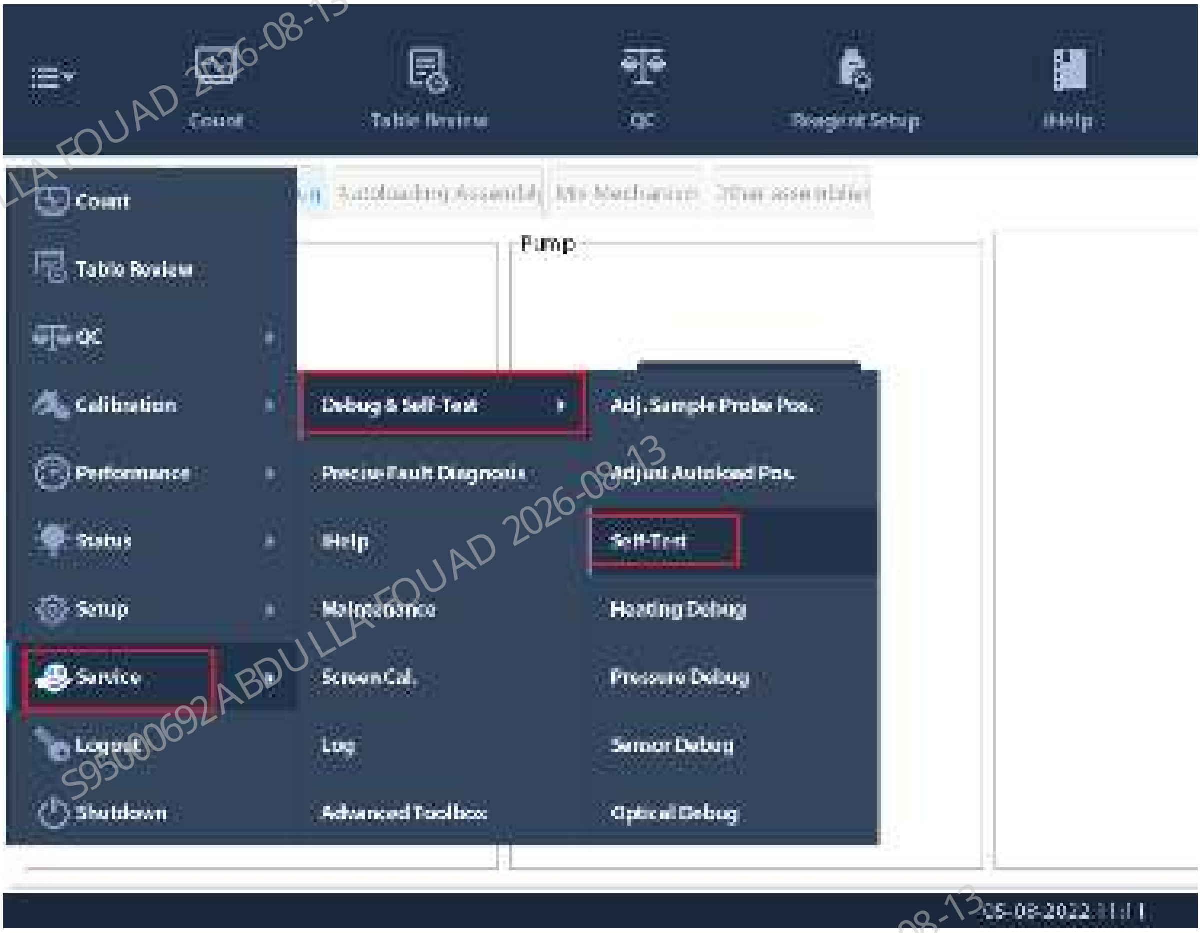


Figure8–490 Self-test screen of the 10 mL syringe

2. Confirm whether the 10 mL syringe assembly is initialized normally.
3. Check whether the syringe is leaking or splashing.

8.50 250 µL syringe assembly (43F4J)

8.50.1 General Information

Figure8–491 Photo of 250 µL Syringe Assembly

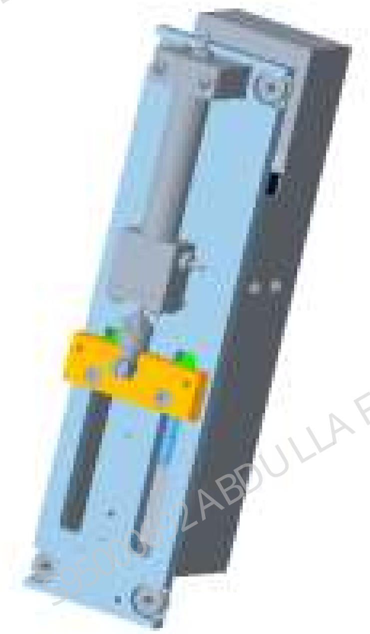


Table 8–52 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-071527-00	250 µL syringe assembly (43F4J)	115-027479-00	Mindray 250 µL syringe (with connector)
		011-000297-00	Photoelectric Position Sensor

Function:

Quantitatively absorb and distribute the samples; serve as the power source for optical sample flow formation, RBC sample flow formation, ESR measurement and diluent function.

8.50.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver
1 roll of toilet paper

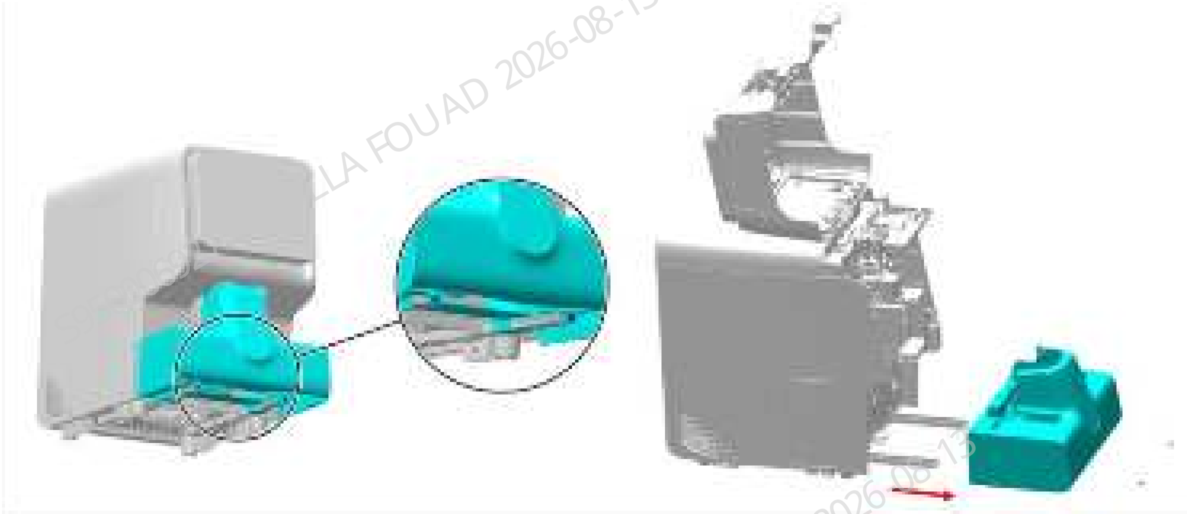
Pre-requirements

Power off the analyzer, and pull out the power cord. The sample probe will drop liquid when the tube is pulled out. You can pave toilet paper under the sample probe.

Procedures

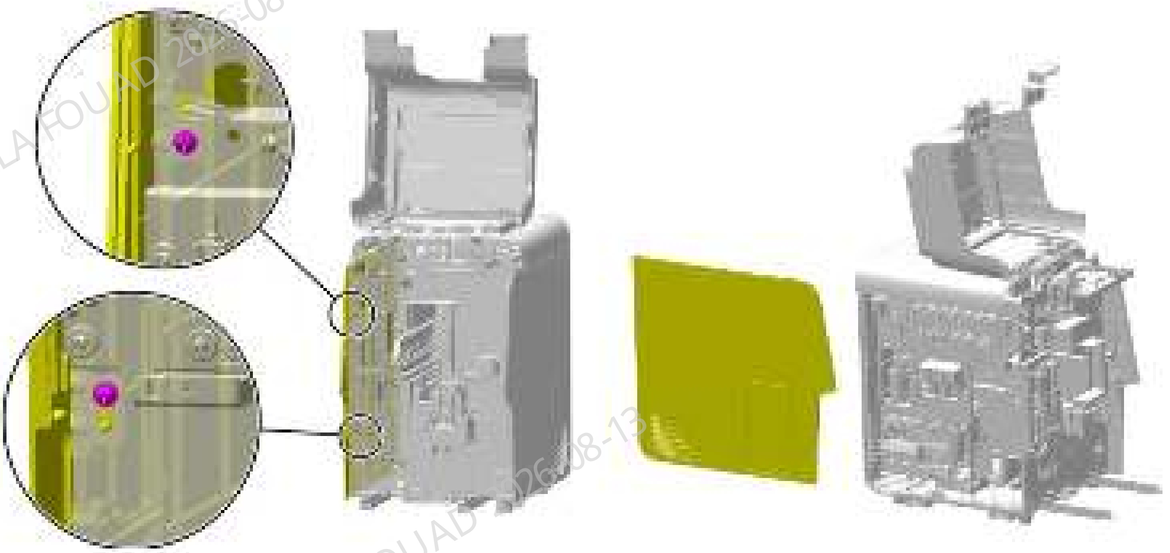
1. Loosen the two screws fixing the autoloader, open the front cover, drag the autoloader to the appropriate position along the direction of the arrow, pull out the wire between the autoloader and the main unit, and finally completely separate the autoloader from the main unit, as shown in the figure below.

Figure8–492 Removing the autoloader



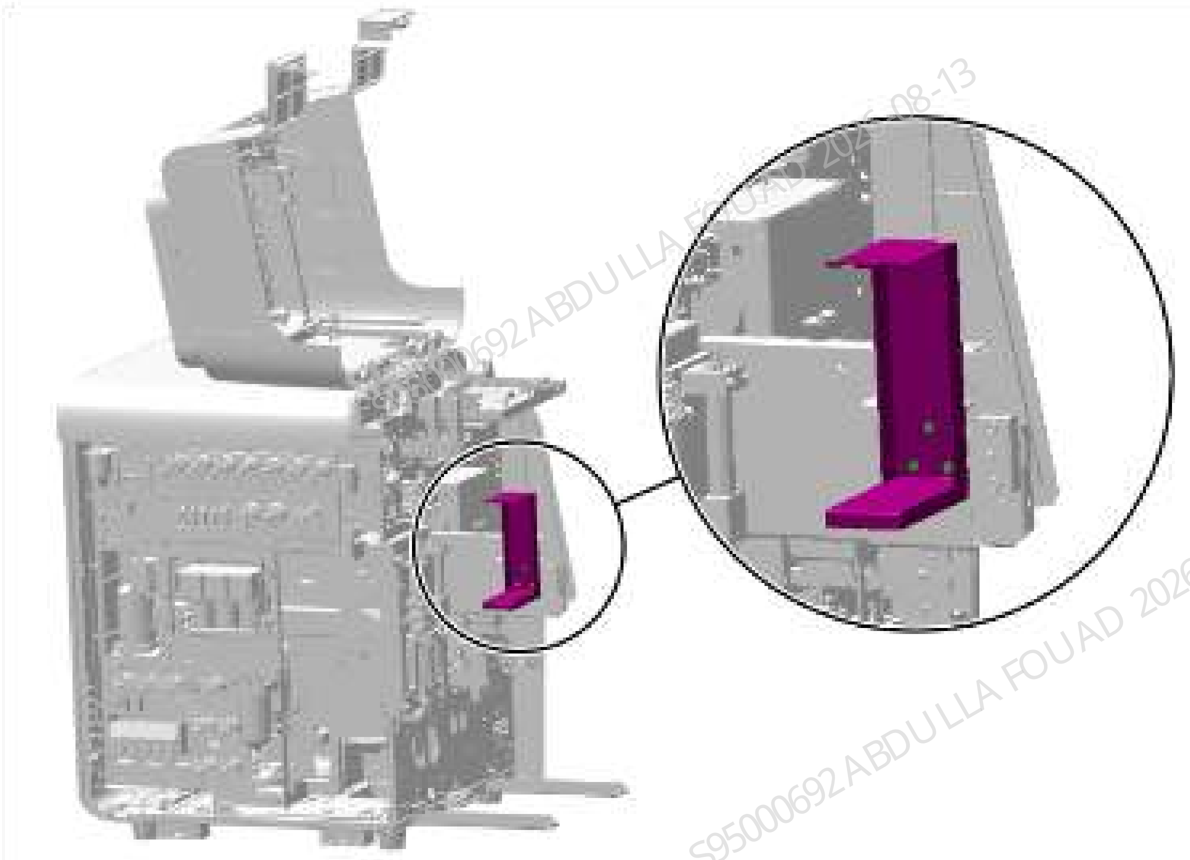
2. Loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–493 Removing the left cover



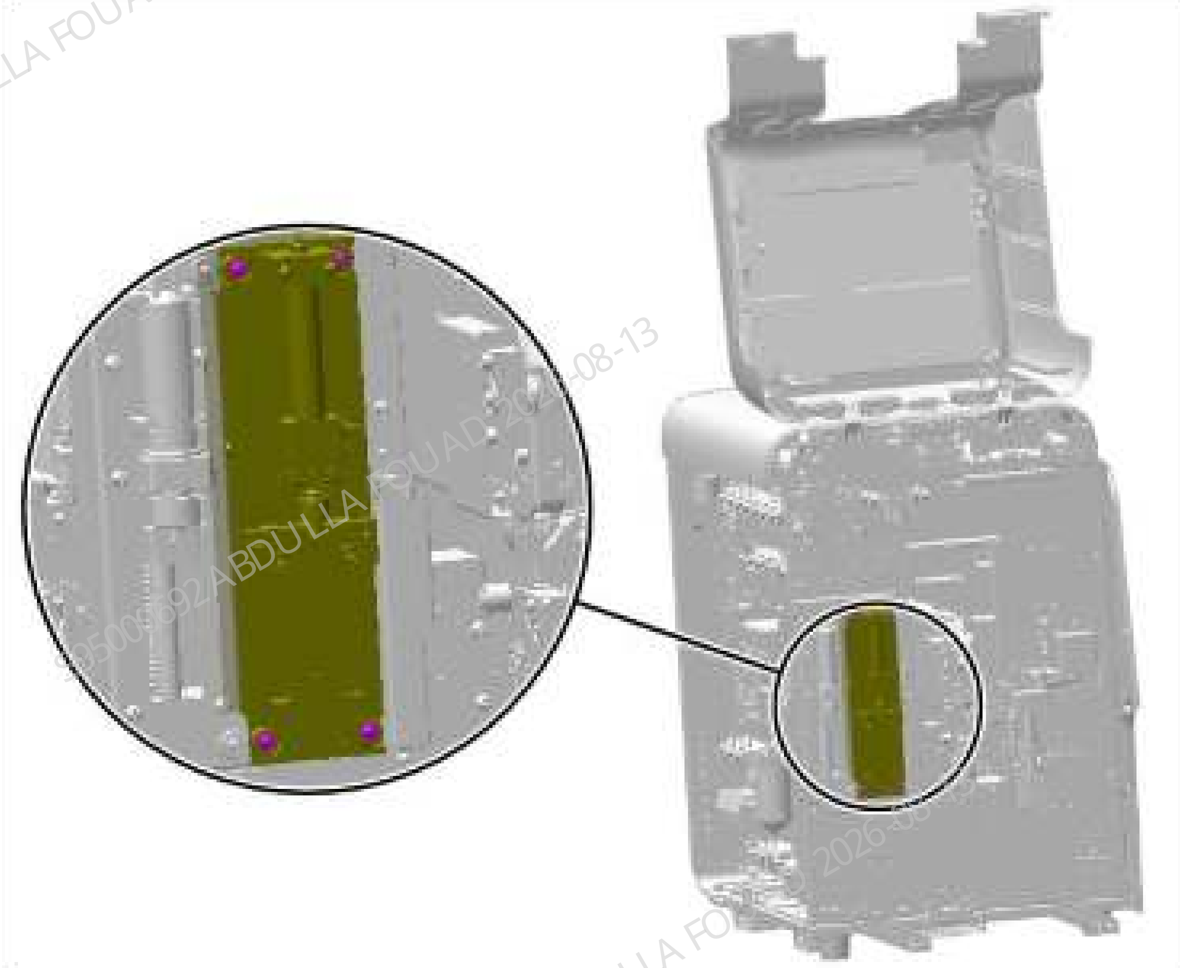
3. Remove the tube and ground wire from the 10ml lead screw driving syringe assembly.
4. As shown in the figure below, remove the three countersunk screws fixing the dye bag support and remove the dye bag support.

Figure8–494 Removing the dye bag support



5. Remove the four set screws of the 250ul syringe assembly (43F4), as shown in the figure below, pull out the assembly terminal, and remove the entire assembly.

Figure8–495 Positions of the set screws of the 250ul syringe assembly



Subsequent Requirements

Installation procedure:

1. Insert the terminal block of the new assembly, put it back in the original position of the analyzer, and fix it with four screws.
2. Connect the tube and ground wire of the new assembly.
3. Install the dye bag support back to the original position.
4. Install the left cover, and flip down the front cover to reset it.
5. Restore the autoloader.

8.50.3 Commissioning and Verification

Procedures

1. Log in the software at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**DIL (10 mL) Syringe** in sequence.

Figure8–496 Self-test screen entrance

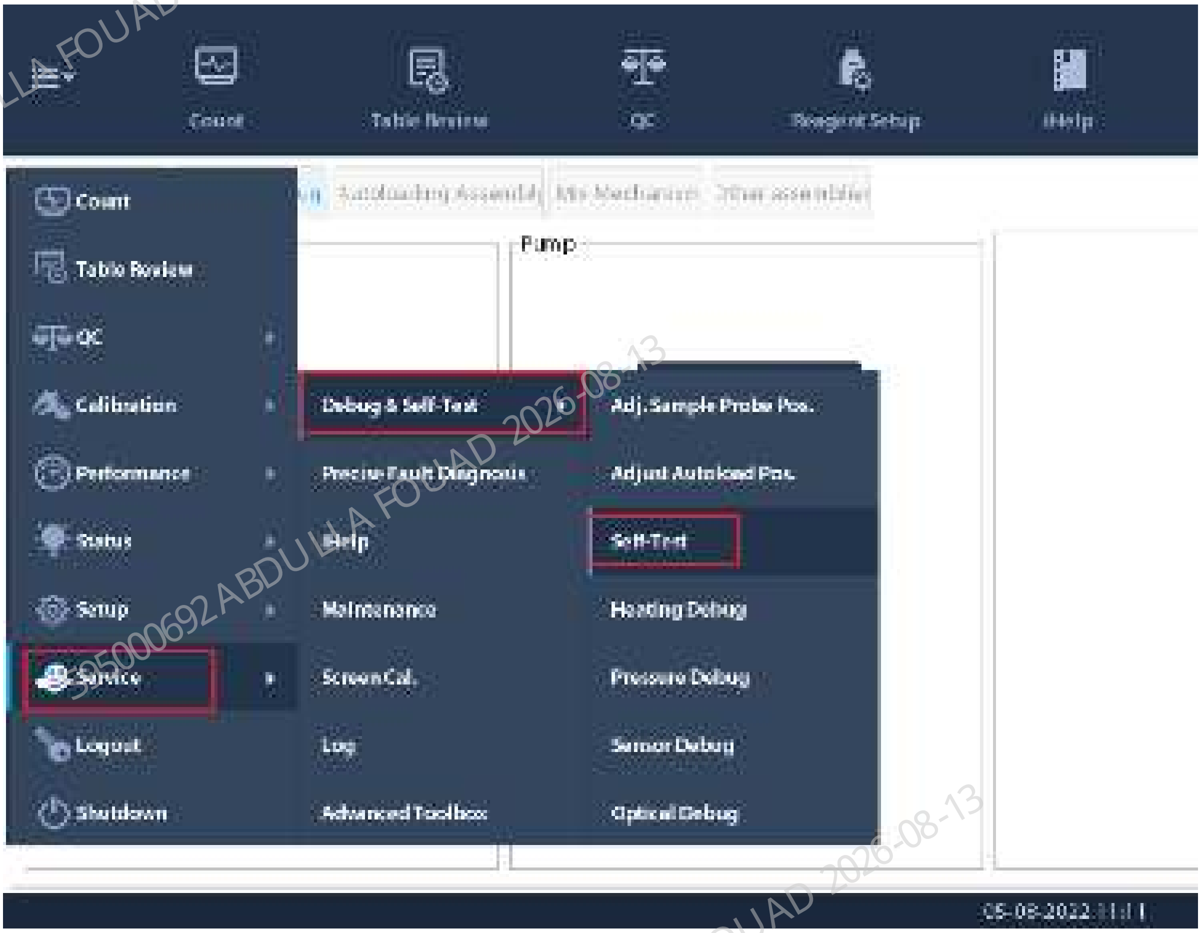


Figure8–497 Self-test Interface of the 250 μL Syringe

2. Confirm whether the 250 μL lead screw driving syringe assembly is initialized normally.
3. Check whether the syringe is leaking or splashing.

8.51 250ul Syringe (with Nozzle)

8.51.1 General Information

Figure8–498 Photo of 250 µL Syringe(with Nozzle)

Table 8–53 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-027479-00	Mindray 250 µL syringe (with connector)	N/A	N/A

Function:

Quantitatively absorb and distribute the samples; serve as the power source for optical sample flow formation, RBC sample flow formation, ESR measurement and diluent function.

8.51.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

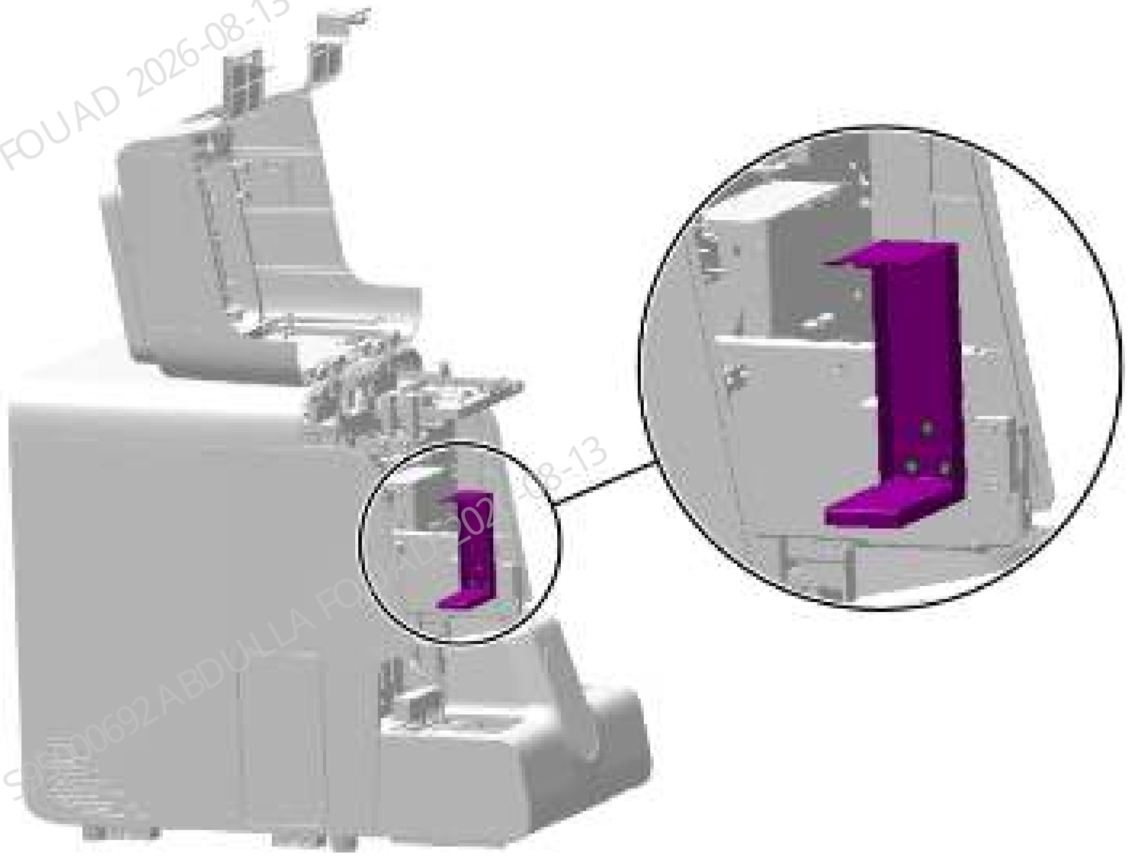
Pre-requirements

Power off the main unit, and unplug the power cord. The sample probe will drop liquid when the tube is pulled out. You can pave toilet paper under the sample probe.

Procedures

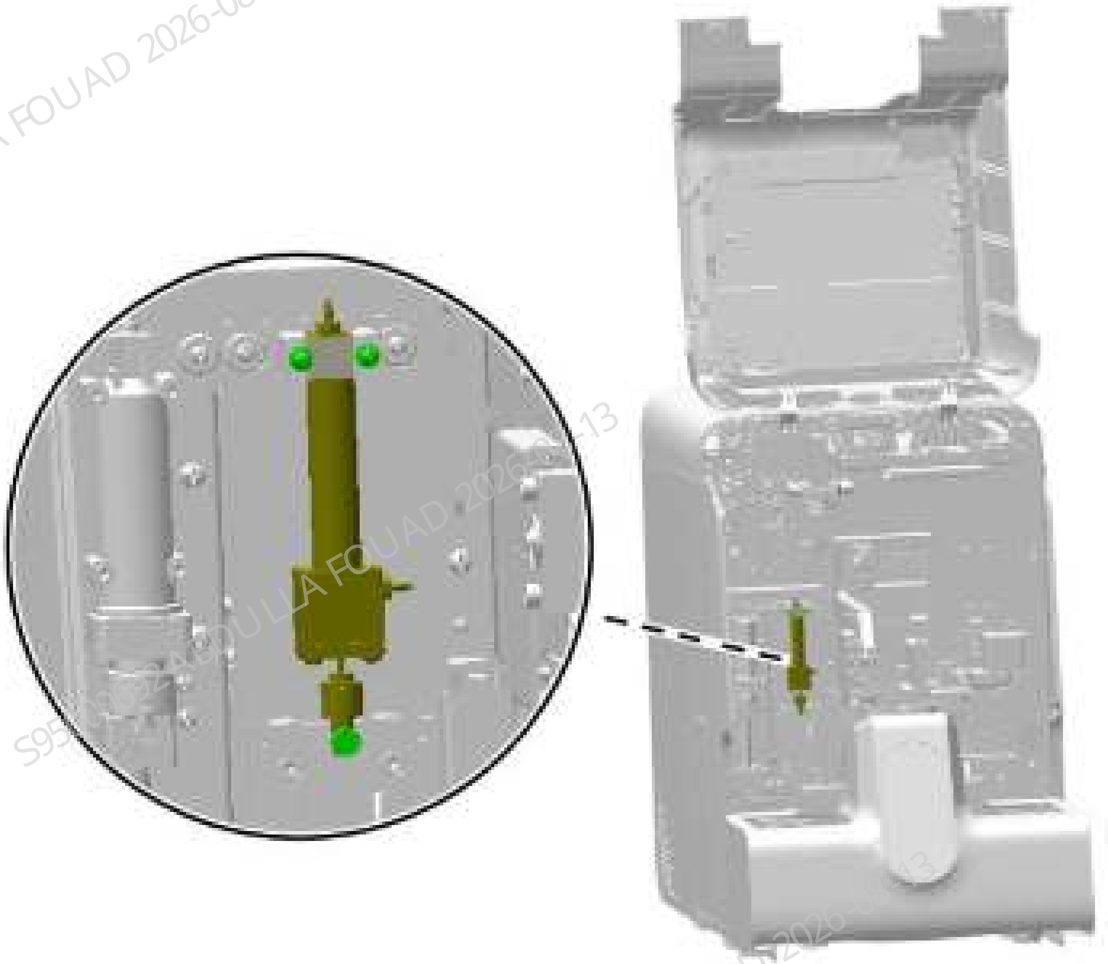
1. Turn up the front cover, lean it on the top cover, loosen the three countersunk screws fixing the dye bag support, and remove the dye bag support.

Figure8–499 Diagram of Removing the Dye Bag Support



2. Remove the rubber tube on Mindray 250 μL syringe (with connector).
3. Remove the set screws of Mindray 250 μL syringe (with connector), as shown in the figure below, and remove the syringe.

Figure8–500 Positions of set screws of Mindray 250 µL syringe (with connector)



Subsequent Requirements

Installation procedure:

1. Install Mindray 250 µL syringe (with connector) back to the initial position of the analyzer and fix it with screws.
2. Connect the rubber tube of Mindray 250 µL syringe (with connector).
3. Turn over the front cover to reset it.

8.51.3 Commissioning and Verification

Procedures

1. Log in the software at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**SP(250 µL) Syringe** in sequence.

Figure8–501 Self-test screen entrance

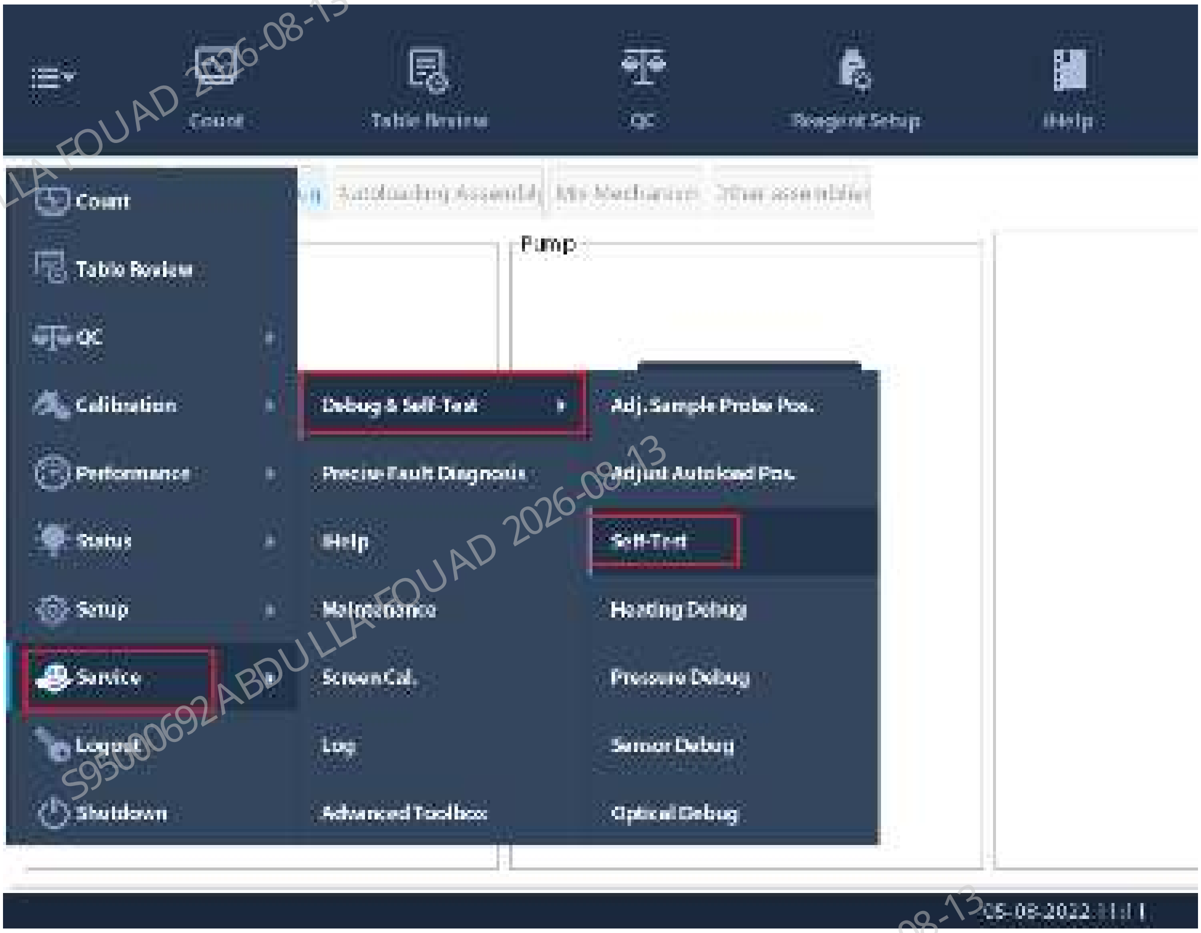


Figure8–502 Self-test Interface of the 250 μL Syringe

2. Confirm whether the 250 μL lead screw driving syringe assembly is initialized normally.
3. Check whether the syringe is leaking or splashing.

8.52 0.5mL diaphragm pump (EPDM conne.,1 direction)

8.52.1 General Information

Figure8–503 Photo of 0.5 mL Metering Pump with EPDM

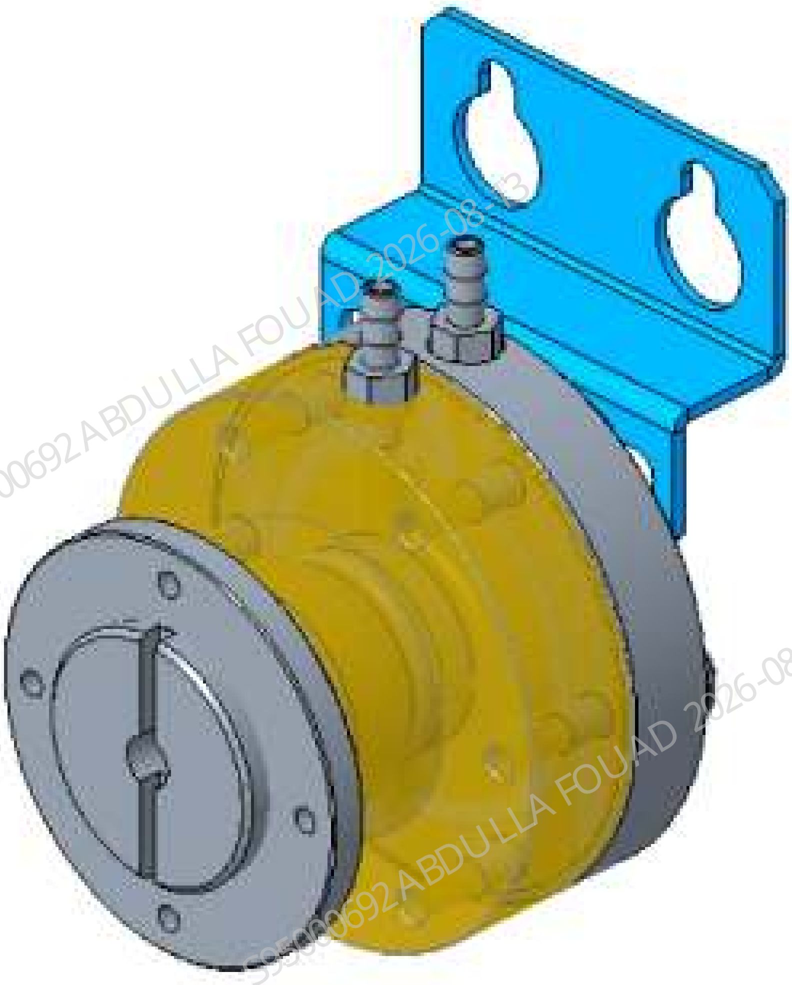


Figure8–504 Position of the 0.5 mL Metering Pump in the Analyzer

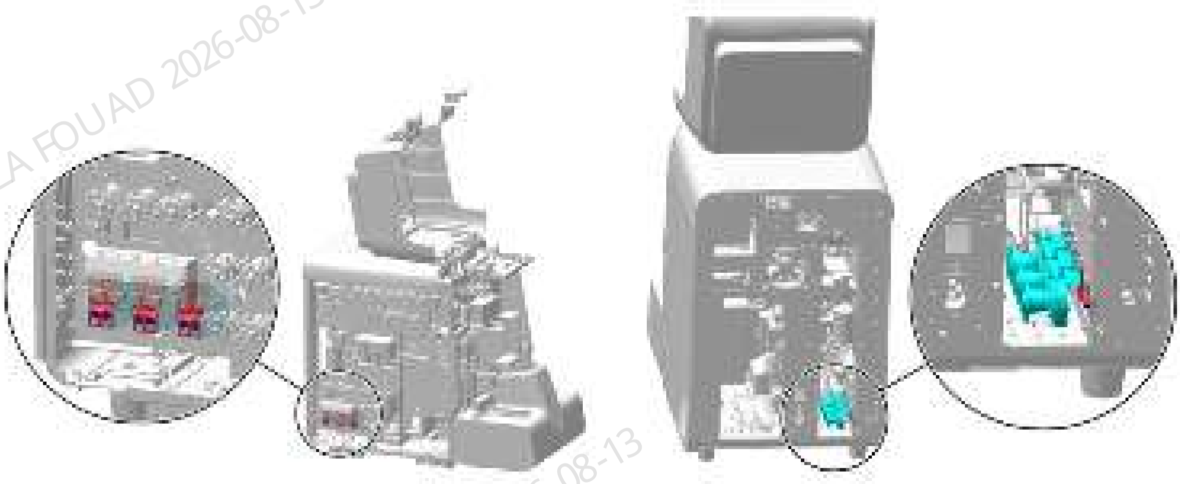


Table 8–54 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-061264-00	0.5mL diaphragm pump (same direction as EPDM connector)	N/A	N/A

Function:
Quantitatively aspirating the liquid such as LD lyse, LH lyse, and DR diluent.

8.52.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

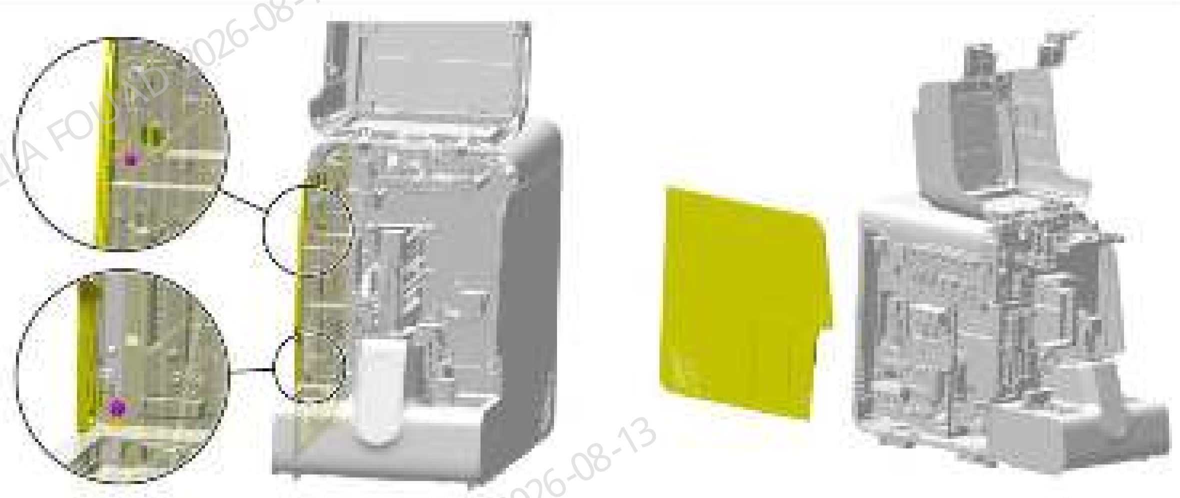
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

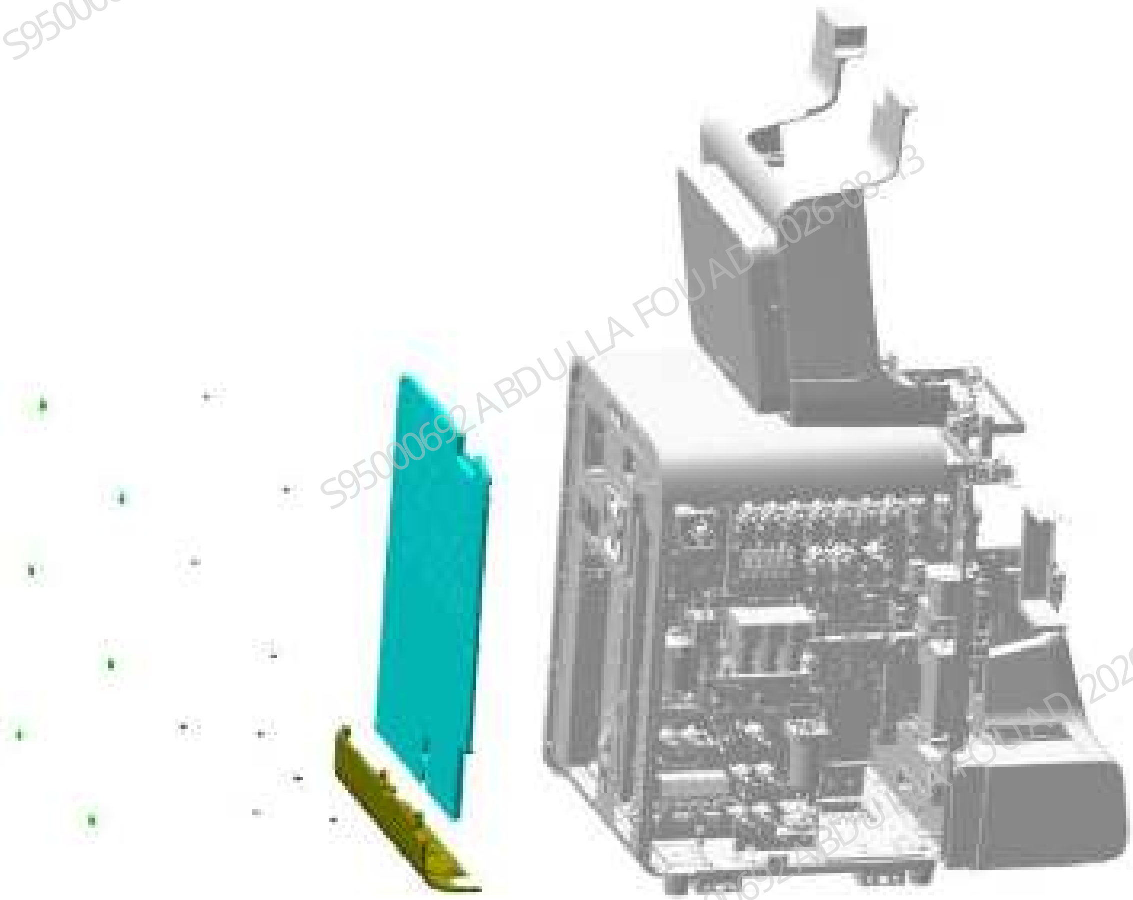
1. Open the front cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–505 Diagram of Removing the Left Cover



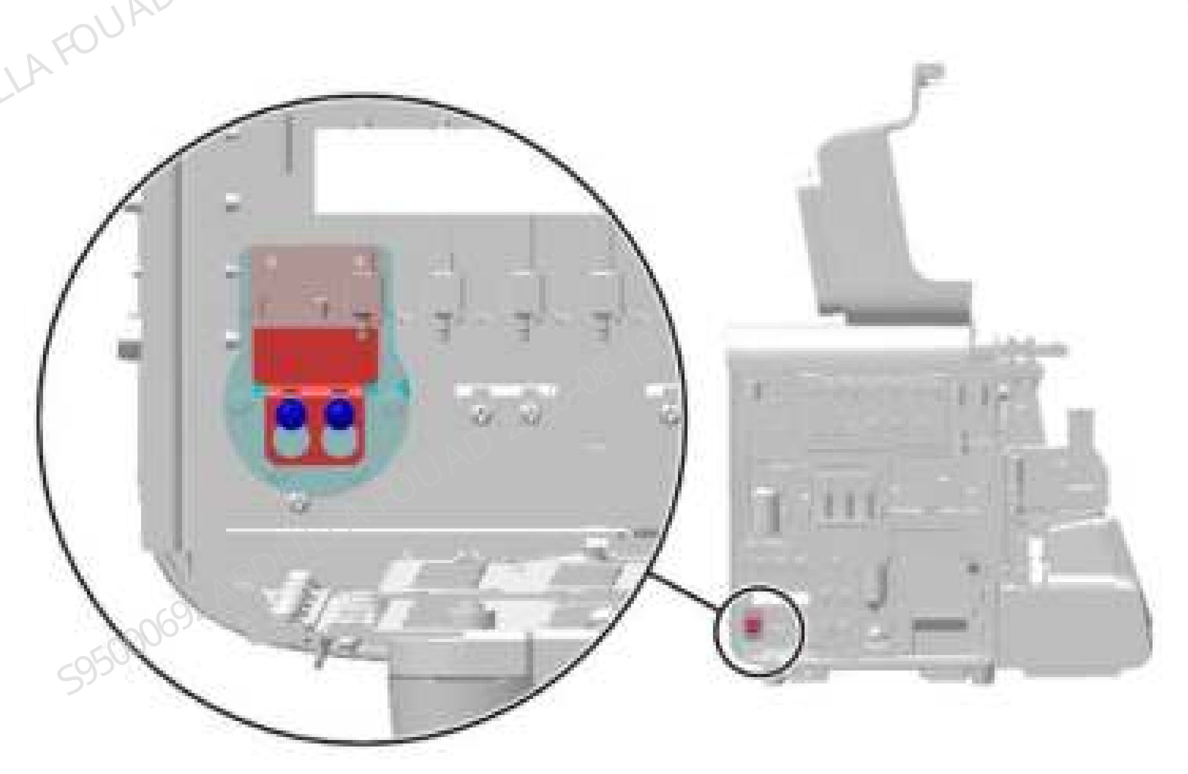
2. Pull out the rubber plug of the rear panel, remove the screws fixing the rear cover and bottom cover, and remove the rear cover and bottom cover, as shown in the figure below.

Figure8–506 Diagram of Removing the Rear Cover and Bottom Cover



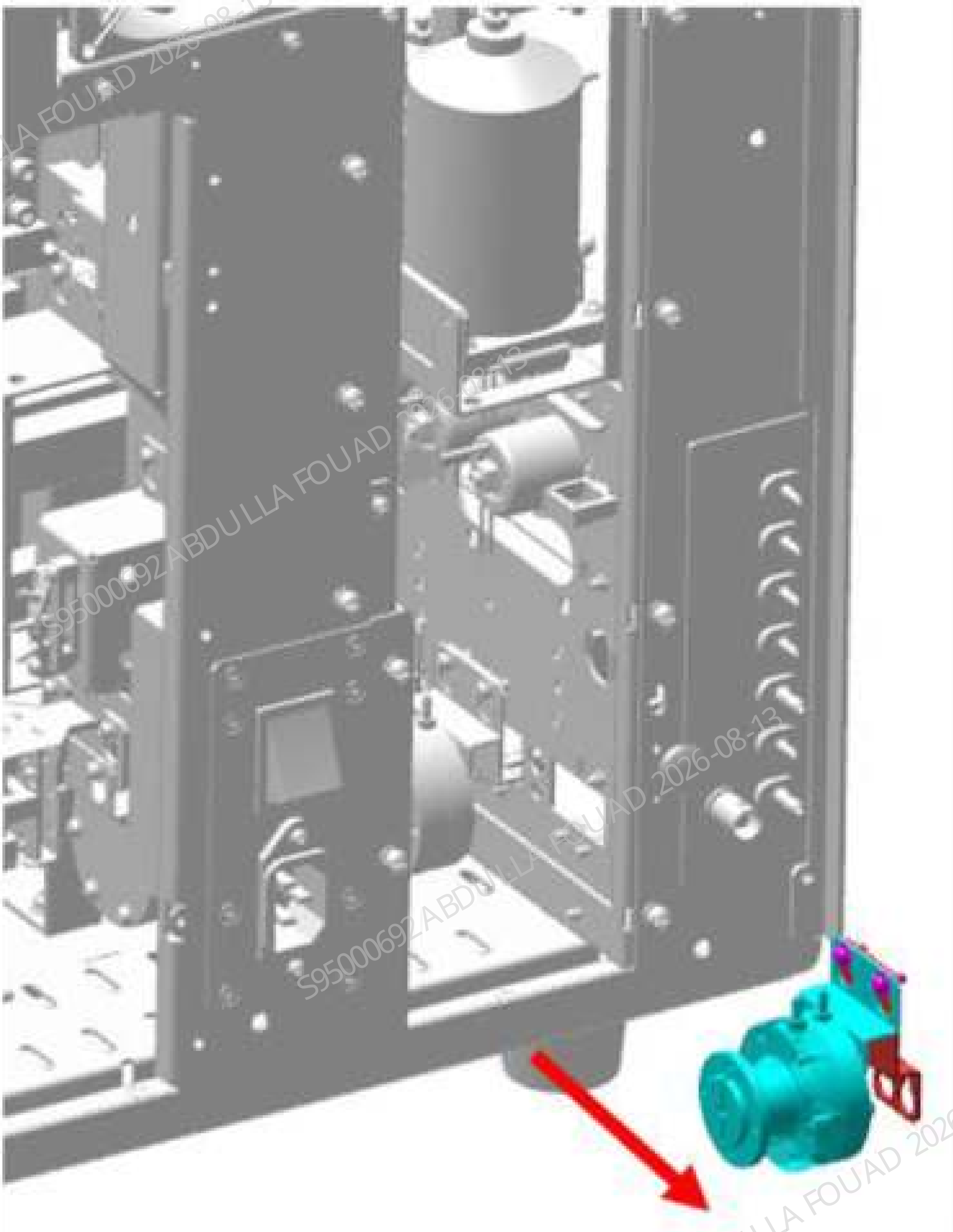
3. Loosen the two fixing screws for the metering pump adapter bracket, as shown in the figure below.

Figure8–507 Positions of set screws of the diaphragm pump adapter bracket



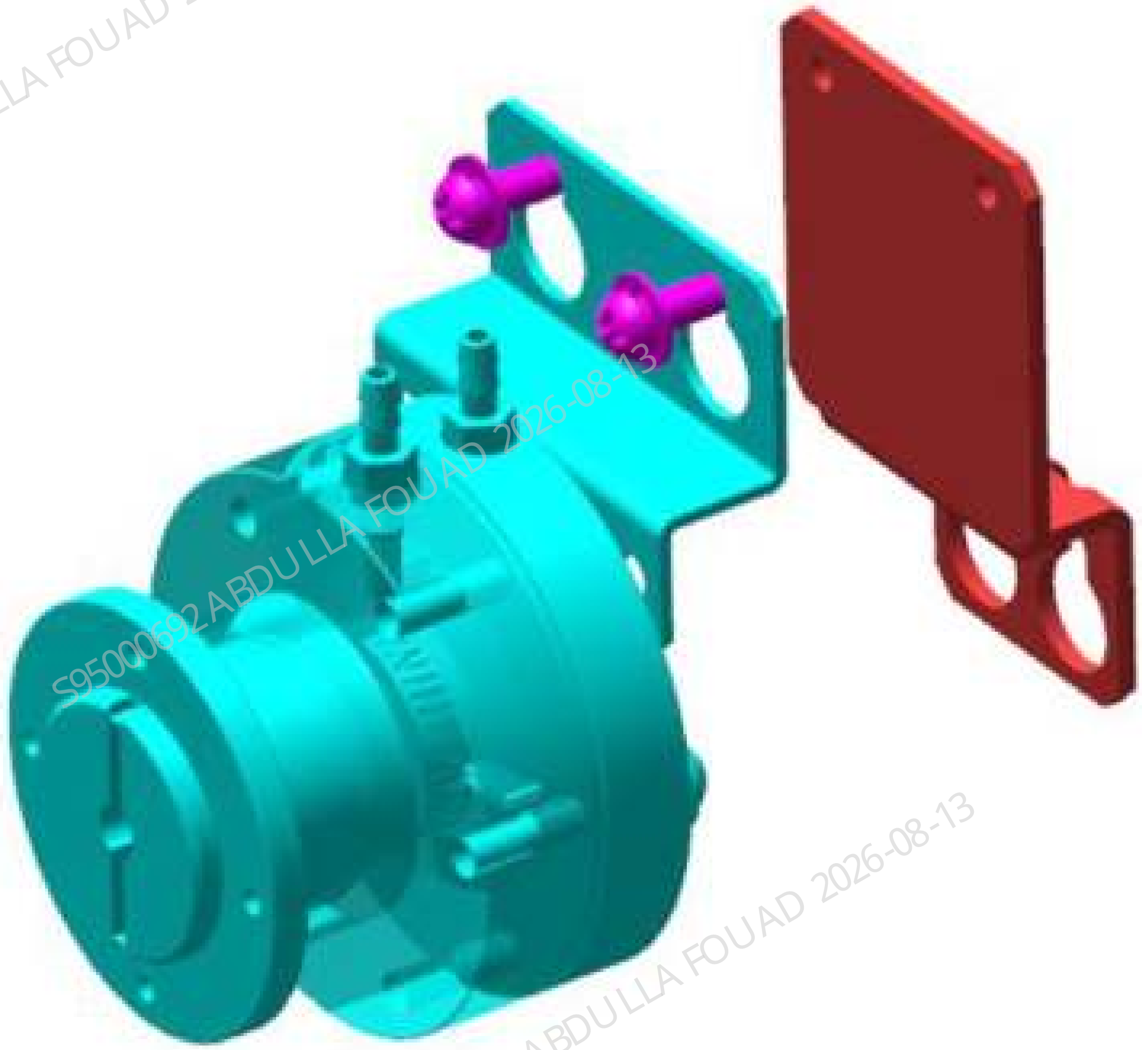
4. From the back of the analyzer, unplug the rubber tube on the connector of the metering pump, and take out the metering pump assembly, as shown in the figure below.

Figure8–508 Removing the metering pump assembly from the back



5. Loosen the screws between the metering pump and the adapter bracket, and remove the metering pump, as shown in the figure below.

Figure8–509 Removing the diaphragm pump



Subsequent Requirements

Installation procedure:

1. Secure the diaphragm pump with the adapter bracket using two screws.
2. Place the diaphragm pump assembly into the main unit from the back of the analyzer, match the hanging holes of the diaphragm pump adapter bracket with the screws and fix them, and insert the rubber tube into the connector of the diaphragm pump.
3. Install the left cover, bottom cover, and rear cover back on the main unit in sequence, and reset the front cover.

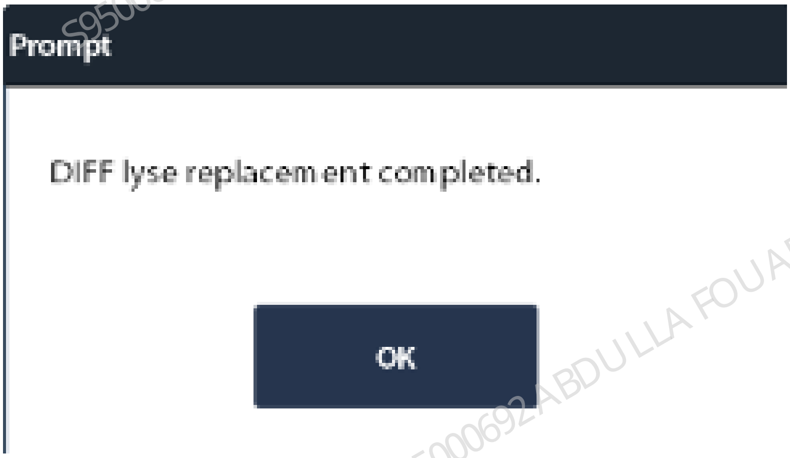
8.52.3 Commissioning and Verification

Procedures

1. Click **Service**>>**Maintenance**>>**Reagent**, and perform the corresponding reagent replacement operation according to the replaced diaphragm pump. For example, if the diaphragm pump of the LD lyse is replaced, click **Replace LD lyse**, and the lyse replacement complete is prompted.

Figure8–510 Reagent Replacement screen

Figure8–511 Reagent Replacement Complete screen



Verification method:

Select **Performance**>>**Background** on the software screen, select the **OV WB-Blood Sample-CD (CDR)** mode, and perform three background tests. The background measurement has no alarm and meets the index requirements, indicating that the replacement succeeds.

8.53 Sampling manifold board assembly

8.53.1 General Information

Figure8–512 Photo of sampling manifold assembly

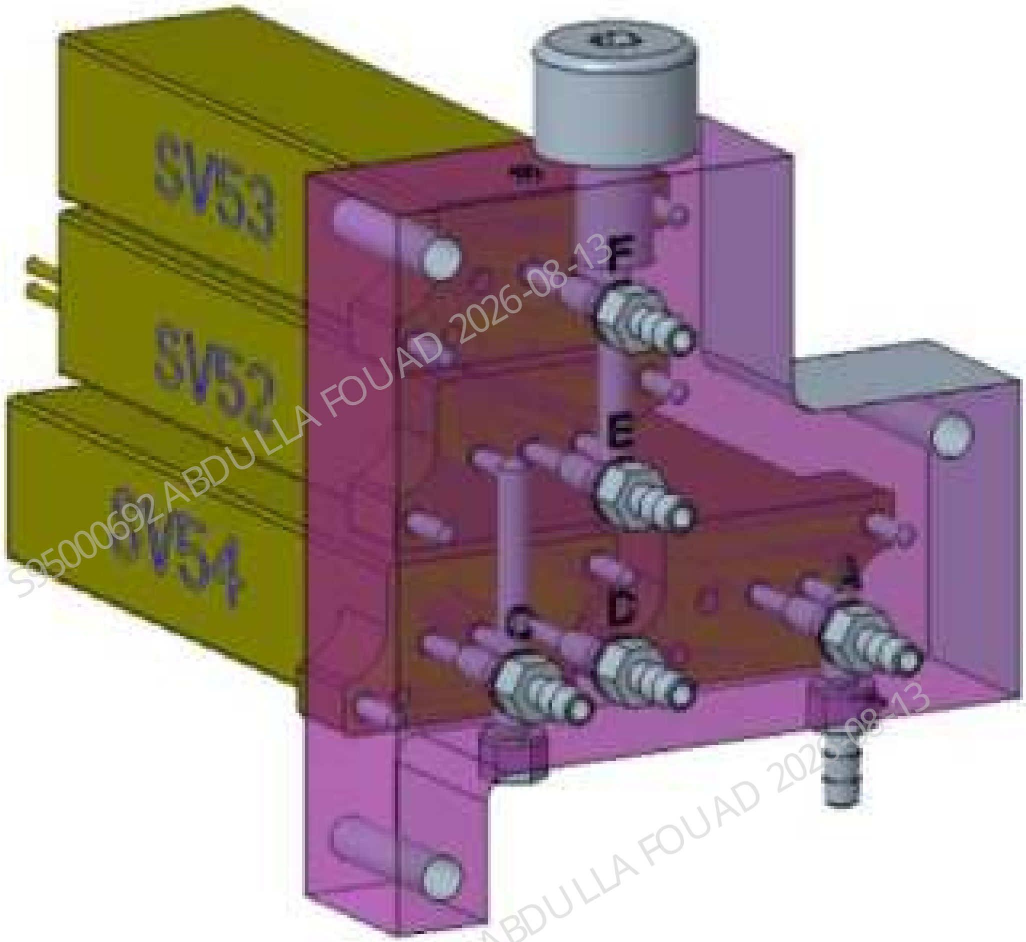


Table 8–55 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-071221-00	Sampling manifold assembly	N/A	N/A

Function:

Switch between sampling probe and 10 mL syringe assembly/250 μL syringe, and switch between 250 μL syringe and sheath fluid sample booster/optical sample booster.

8.53.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

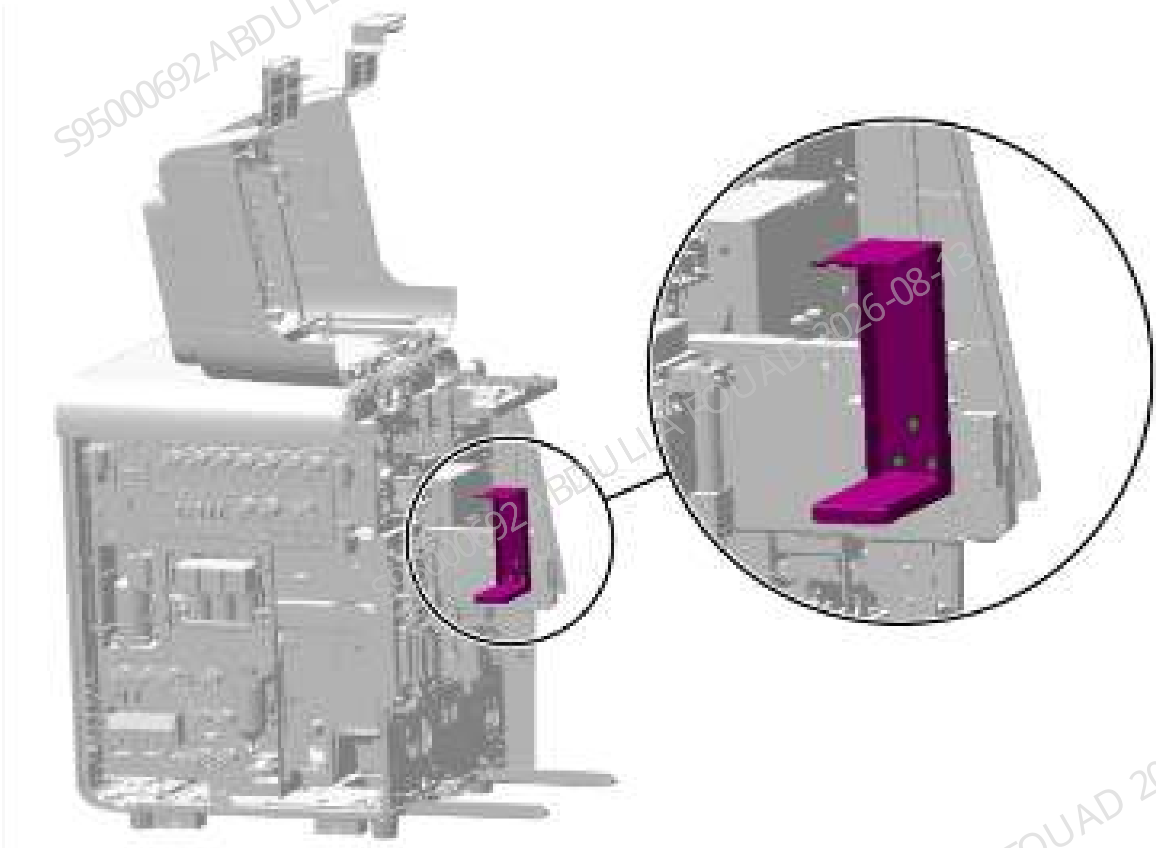
Pre-requirements

Power off the main unit, and unplug the power cord. A small amount of liquid will leak when the tube is pulled out. Wipe it clean using toilet paper. The sample probe will also leak liquid, so toilet paper should be laid under the sample probe in advance.

Procedures

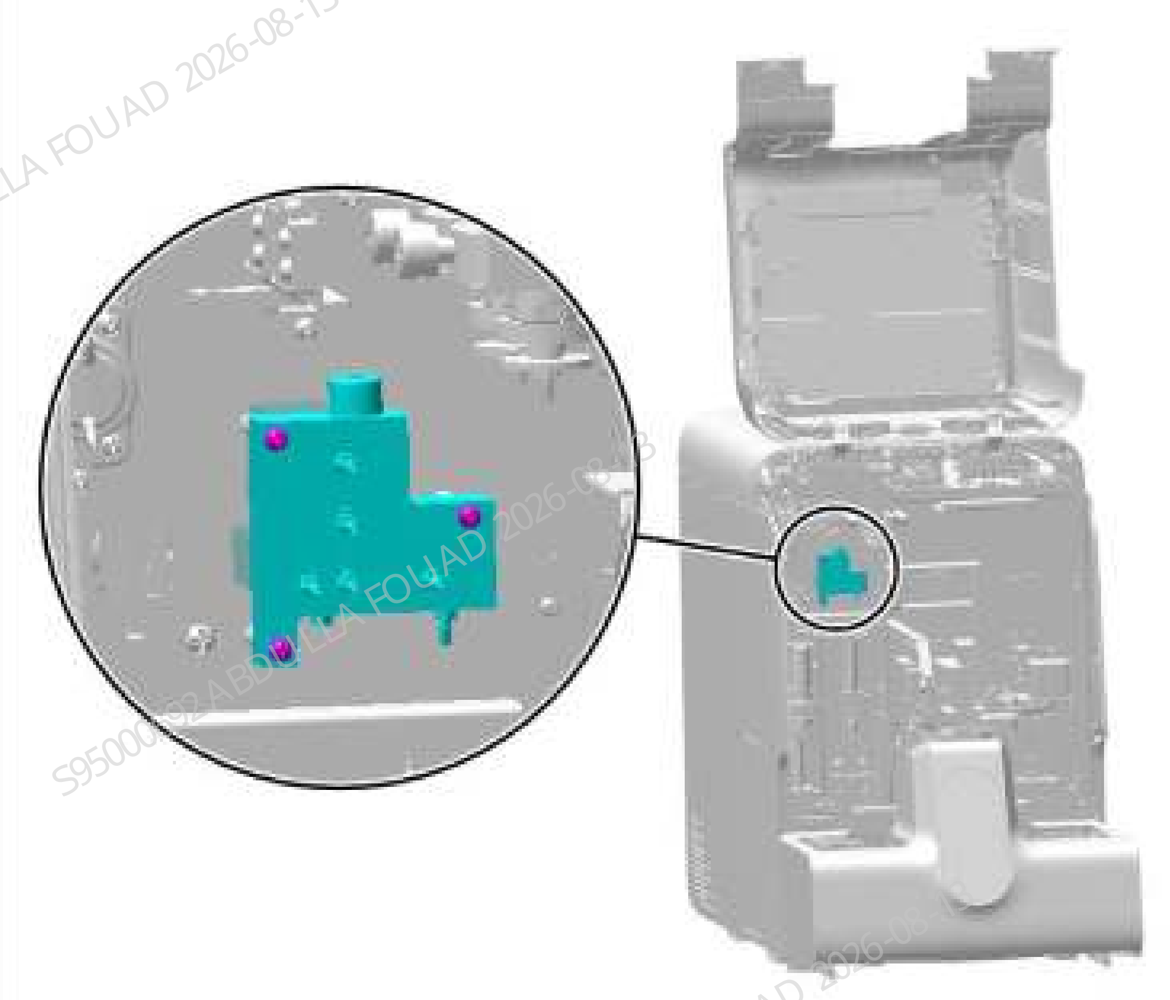
1. Turn up the front cover, lean it on the top cover, remove the three countersunk screws fixing the dye bag support, and remove the dye bag support, as shown in the figure below.

Figure8–513 Diagram of Removing the Left Cover



2. Unplug all rubber tubes attached to the connectors of the fluidic junction board.
3. Loosen the fixing screws of the fluidic junction board, as shown in the figure below, loosen the terminals connected to the main unit, and remove the assembly.

Figure8–514 Positions of the set screws of the air tank assembly



Subsequent Requirements

Installation procedure:

1. Reconnect the terminals of the sampling manifold assembly.
2. Install the sampling manifold assembly back into the analyzer, tighten three screws to fix it, and insert the corresponding tube into the tube connector of the assembly.
3. Flip down the front cover to reset it.

8.53.3 Commissioning and Verification

Procedures

1. Perform 10 counts in the WB CD mode.
2. Observe whether there is leakage or overflow of the sampling manifold assembly, and whether the analyzer reports an error.

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.54 Three-way LVM valve assembly (with valve board and connector

8.54.1 General Information

For the general information in the case of LVM fluid valve body, see LVM fluid valve body.
For the general information in the case of LVM valve assembly of three-way, see LVM Valve Assembly of three-way.

LVM fluid valve body

Figure8–515 Photo of LVM Fluid Valve Body

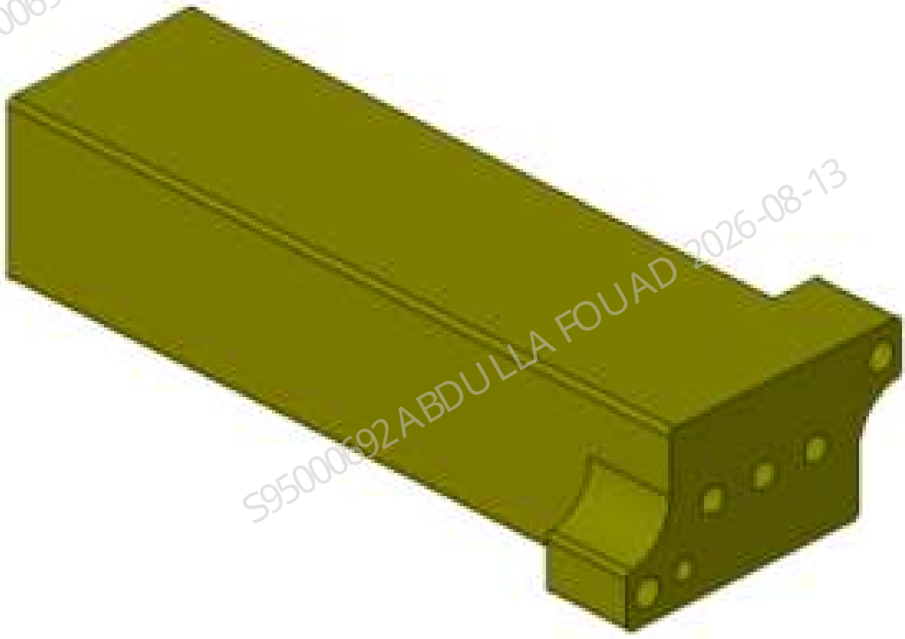


Table 8–56 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-061171-00	LVM fluid valve body	N/A	N/A

Function:
Turn on/off and switch the sampling tubes.

LVM Valve Assembly of three-way

Figure8–516 Photo of LVM valve assembly of three-way

Table 8–57 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-061171-00	Three-position ways LVM valve assembly (with valve board and connector)	N/A	N/A

Function:
Turn on/off pneumatics and fluidics.

8.54.2 Disassembly and Assembly

For the disassembly and assembly in the case of LVM fluid valve body, see LVM fluid valve body.
For the disassembly and assembly in the case of LVM valve assembly of three-way, see LVM Valve Assembly of three-way.

LVM fluid valve body

Required tools

Cross-headed screwdriver

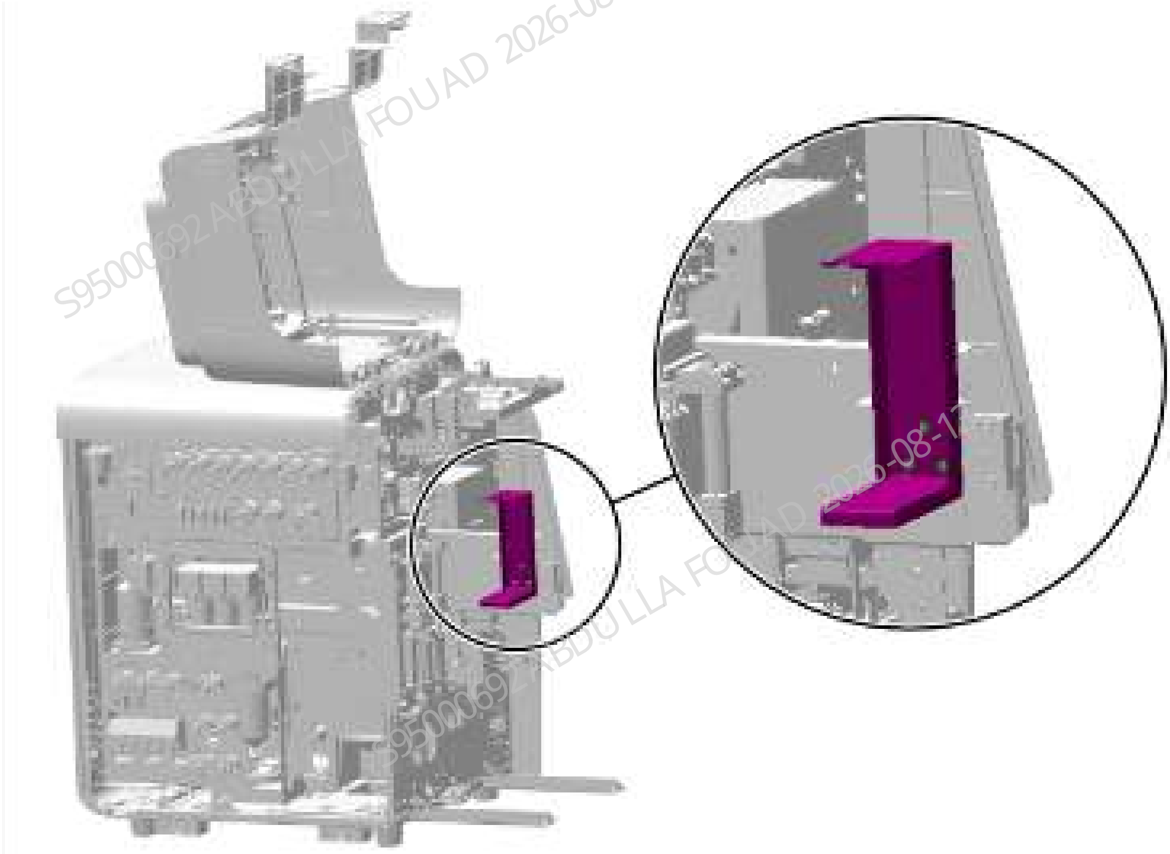
Pre-requirements

Empty the DIL and SCI cisterns and perform a depressurization operation.

Procedures

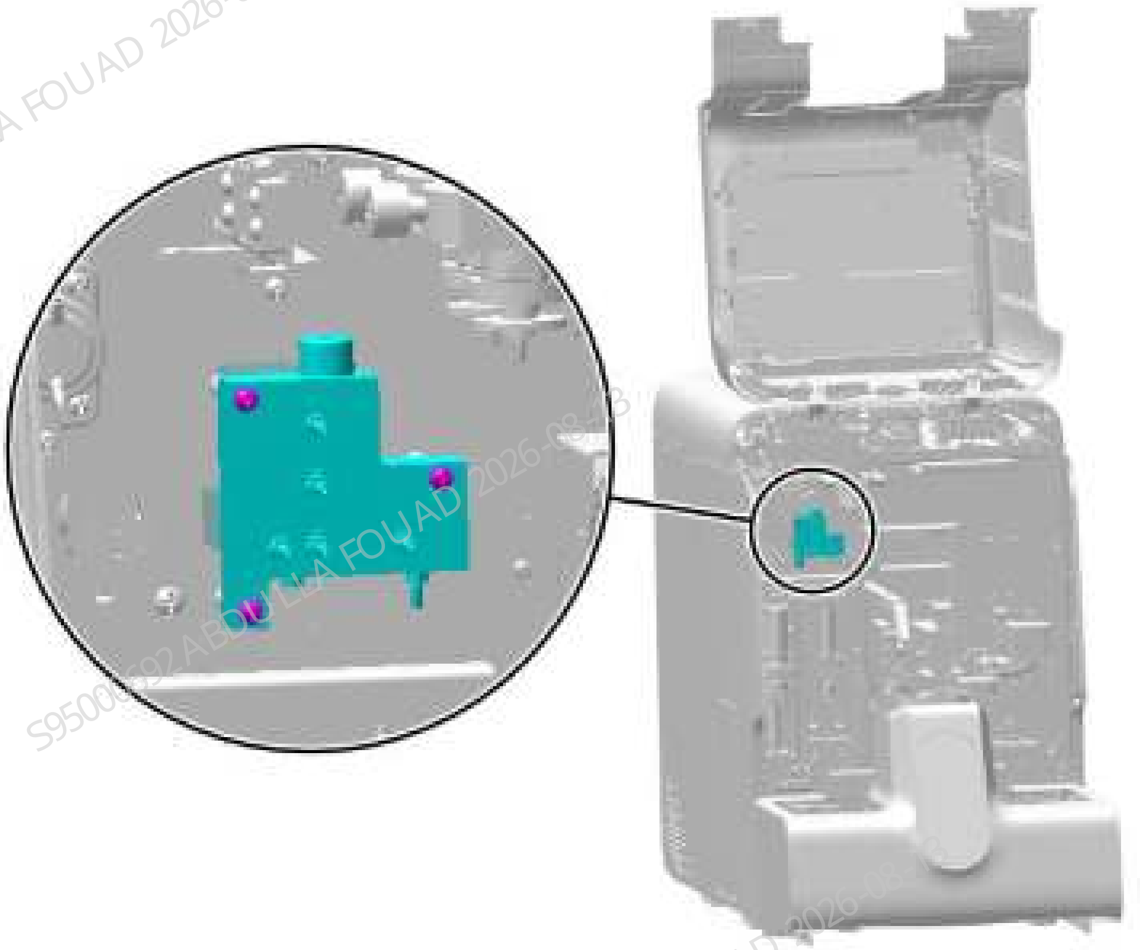
1. Turn up the front cover, lean it on the top cover, remove the three countersunk screws fixing the dye bag support, and remove the dye bag support, as shown in the figure below.

Figure8–517 Diagram of Removing the Left Cover



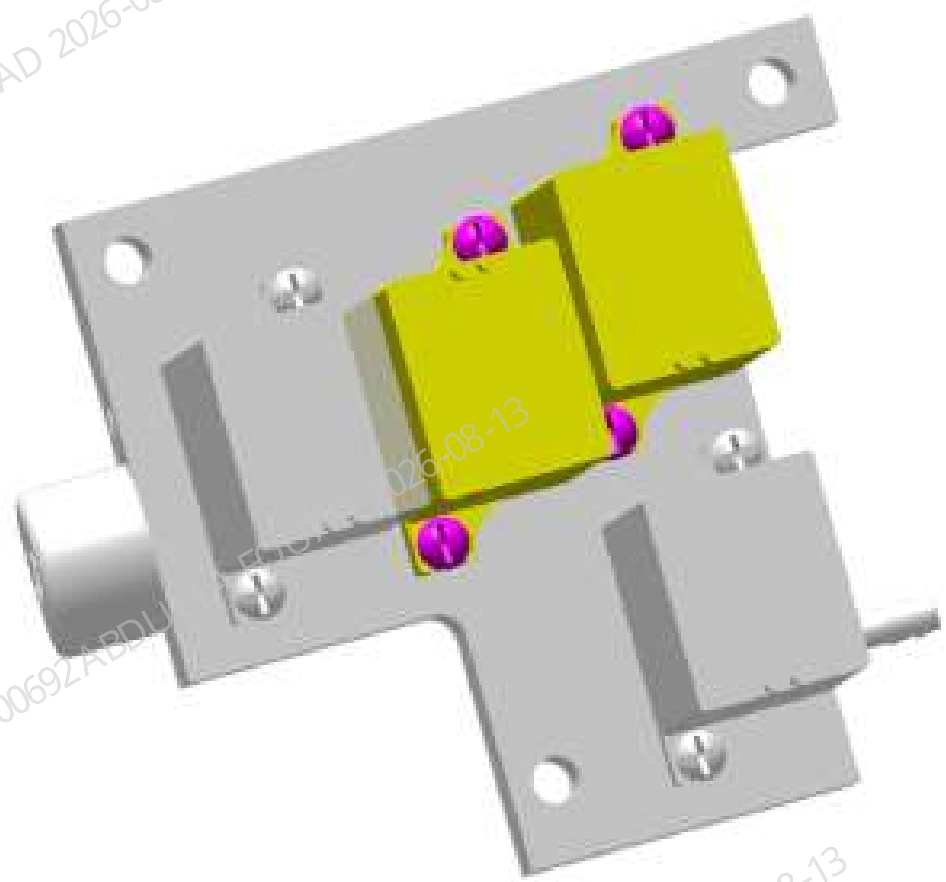
2. Unplug all rubber tubes attached to the connectors of the fluidic junction board.
3. Loosen the fixing screws of the fluidic junction board, as shown below, loosen the terminals connected to the main unit, and remove the fluidic junction board.

Figure8–518 Positions of the set screws of the air tank assembly



4. As shown in the figure below, two LVM fluid valve bodies are installed on each fluidic Junction board, and each LVM fluid valve body is fixed using two screws. To remove one LVM fluid valve body separately, loosen its fixing screws.

Figure8–519 Diagram of Removing the LVM Fluid Valve Body



Subsequent Requirements

Installation procedure:

1. Remove the valve board of the FRU (115-061171-00), and pay attention that the sealing ring cannot fall off during the process of removing the valve board.
2. Use two screws to fix the LVM fluid valve body to the fluidic junction board.
3. Reconnect the terminals of the sampling manifold assembly.
4. Install the sampling manifold assembly back into the analyzer, tighten the screws to fix it, and insert the corresponding tube into the tube connector of the assembly.
5. Use three countersunk screws to install the dye support back to the original position.
6. Turn over the front cover to reset it.

LVM Valve Assembly of three-way

Required tools

Cross-headed screwdriver

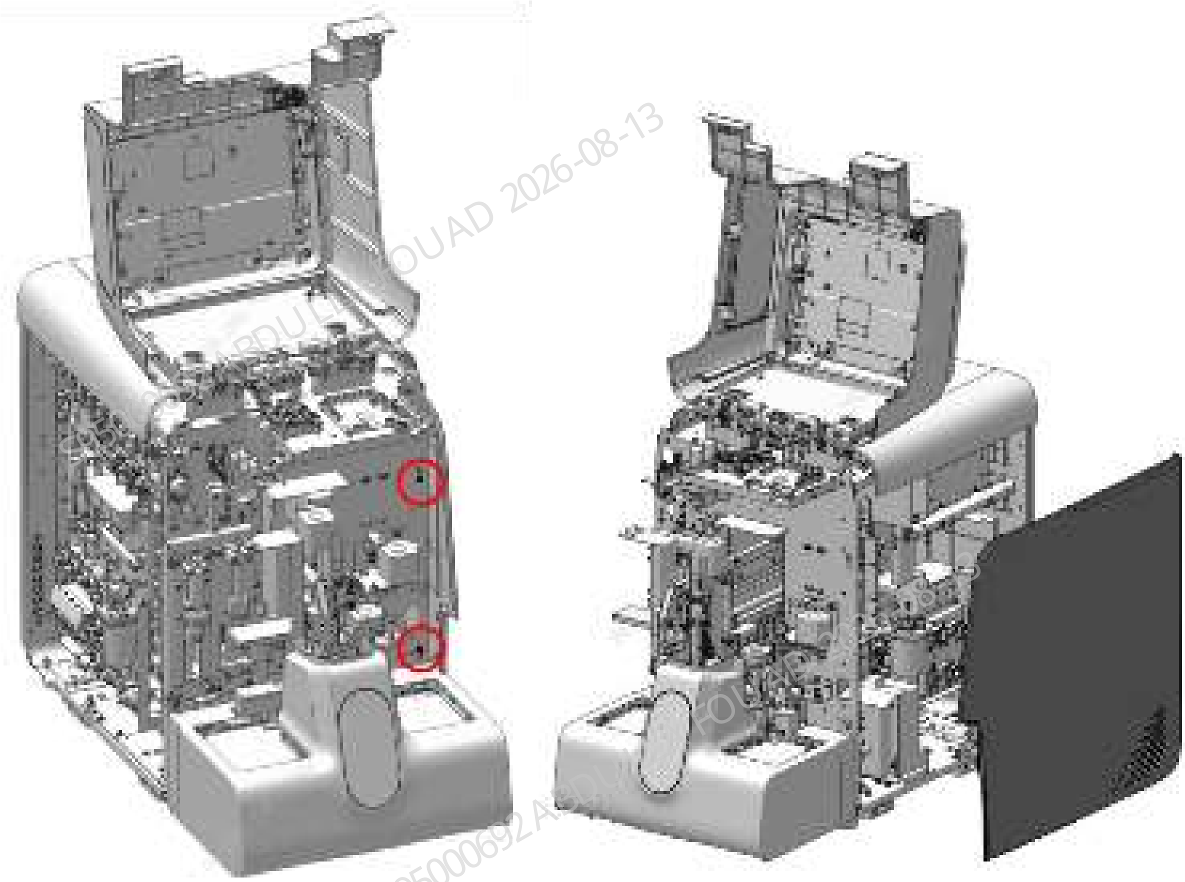
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Flip the front cover up and lean it on the top cover, and remove the right cover, as shown in the figure below.

Figure8–520 Diagram for Location of Removing the Covers



2. Loosen the fixing screws of the right hinge, as shown in Figure A below. Open the hinge and find the LVM valve assembly of three-way, as shown in Figure B below.

Figure8–521 Diagram A of Location of the Right Hinge Fixing Screw

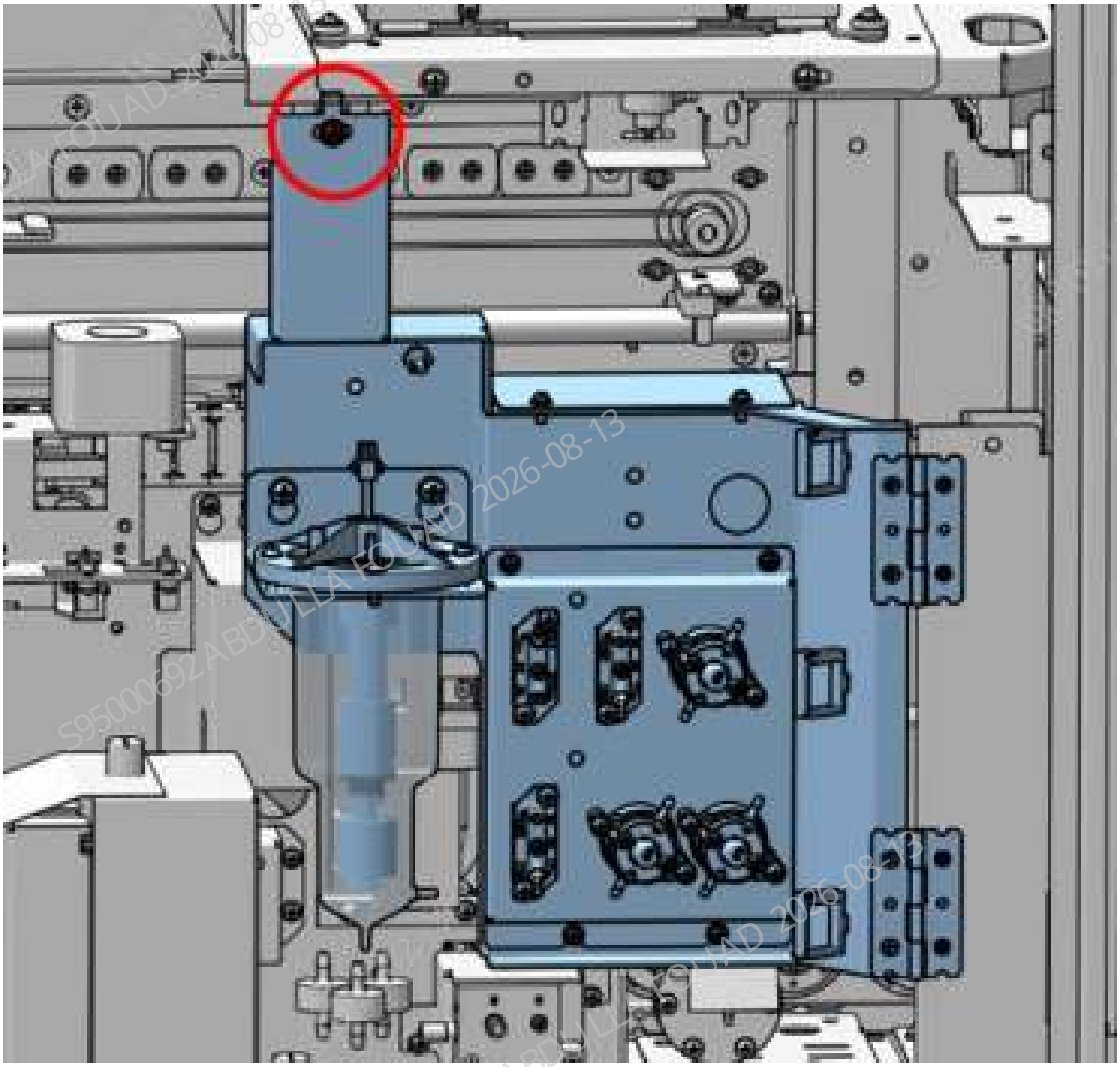
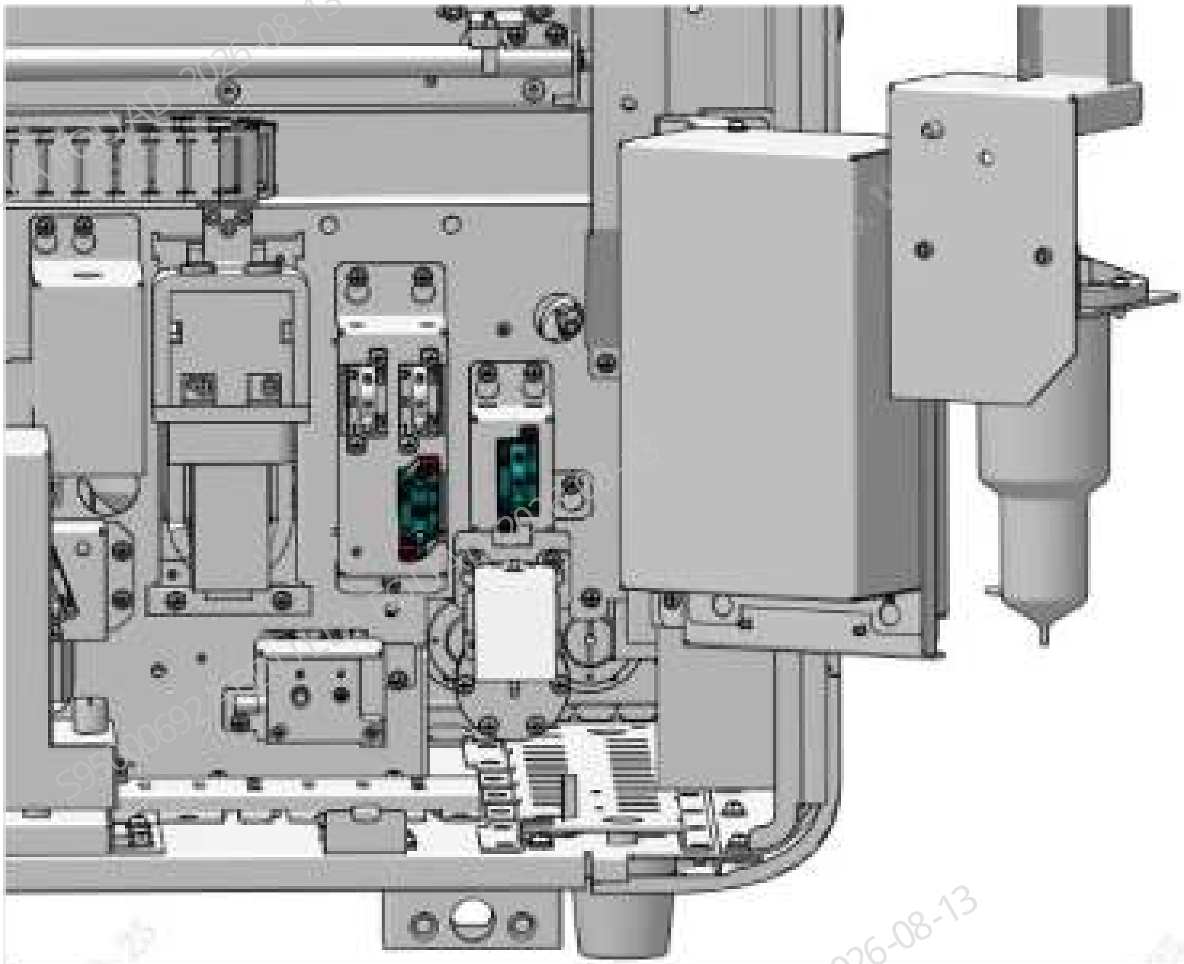
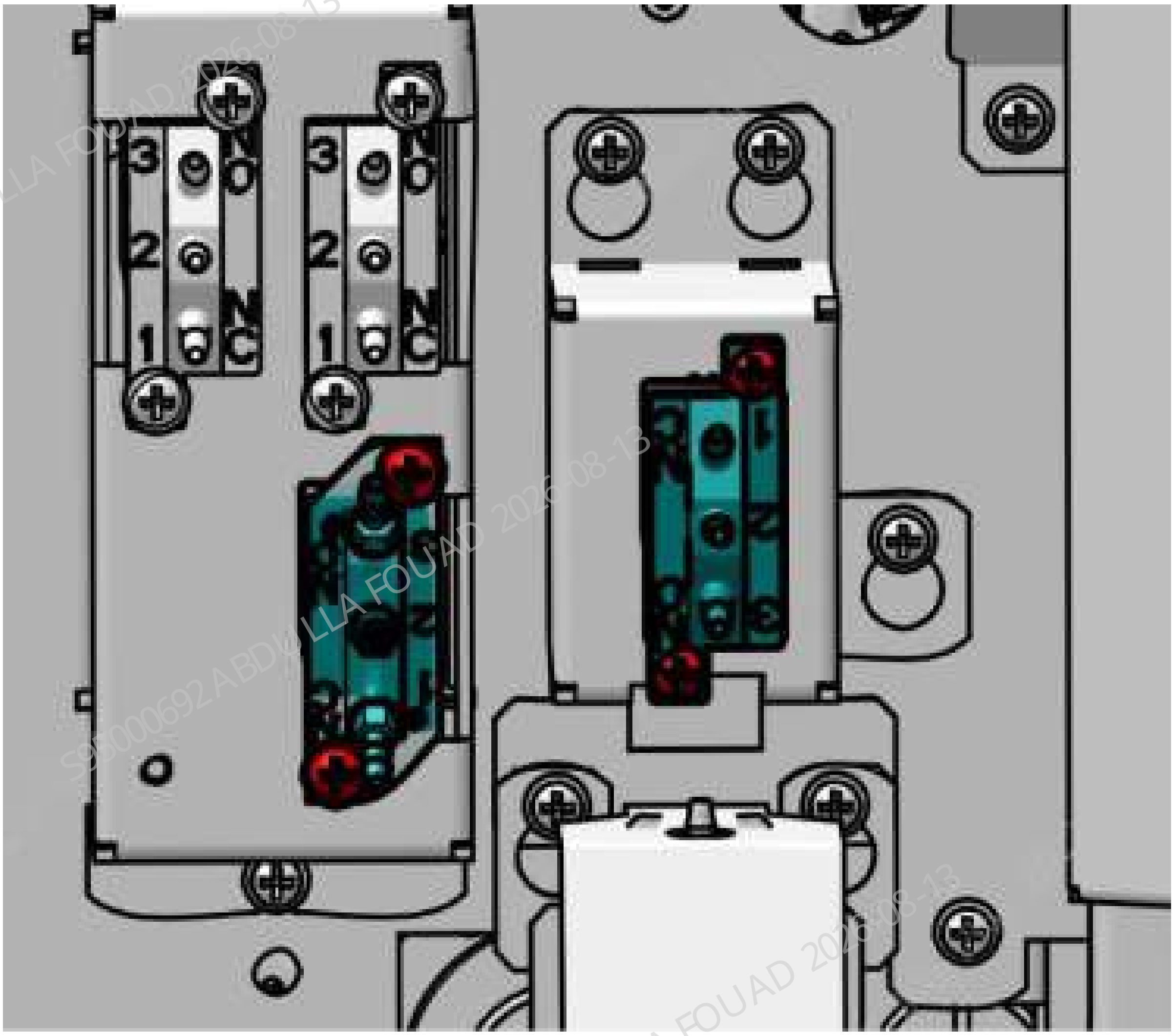


Figure8–522 Diagram B of Location of LVM Valve Assembly of three-way



3. Pull out the rubber tube on the LVM valve assembly of three-way, loosen the two screws that fix the LVM valve assembly of three-way, as shown below, loosen the wiring terminal connected to the main unit, and remove the LVM valve assembly of three-way.

Figure8–523 Positions of the set screws of the LVM valve assembly of three-way



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the LVM valve assembly of three-way, and insert the terminal connected to the main unit.
2. Tighten the screws that fix the LVM valve assembly of three-way.
3. Reset the hinge and use the locking screws to fix it. Install the right cover back, and flip down the front cover for resetting.

8.54.3 Commissioning and Verification

For the commissioning and verification in the case of LVM fluid valve body, see LVM fluid valve body.

For the commissioning and verification in the case of three-way LVM valve assembly (with valve board and connector), see LVM Valve Assembly of three-way.

LVM fluid valve body

Required tools

Cross-headed screwdriver

Procedures

1. Perform 10 counts in the WB CD mode.
2. Observe whether there is leakage or overflow of the sampling manifold assembly, and whether the analyzer reports an error.

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

LVM Valve Assembly of three-way

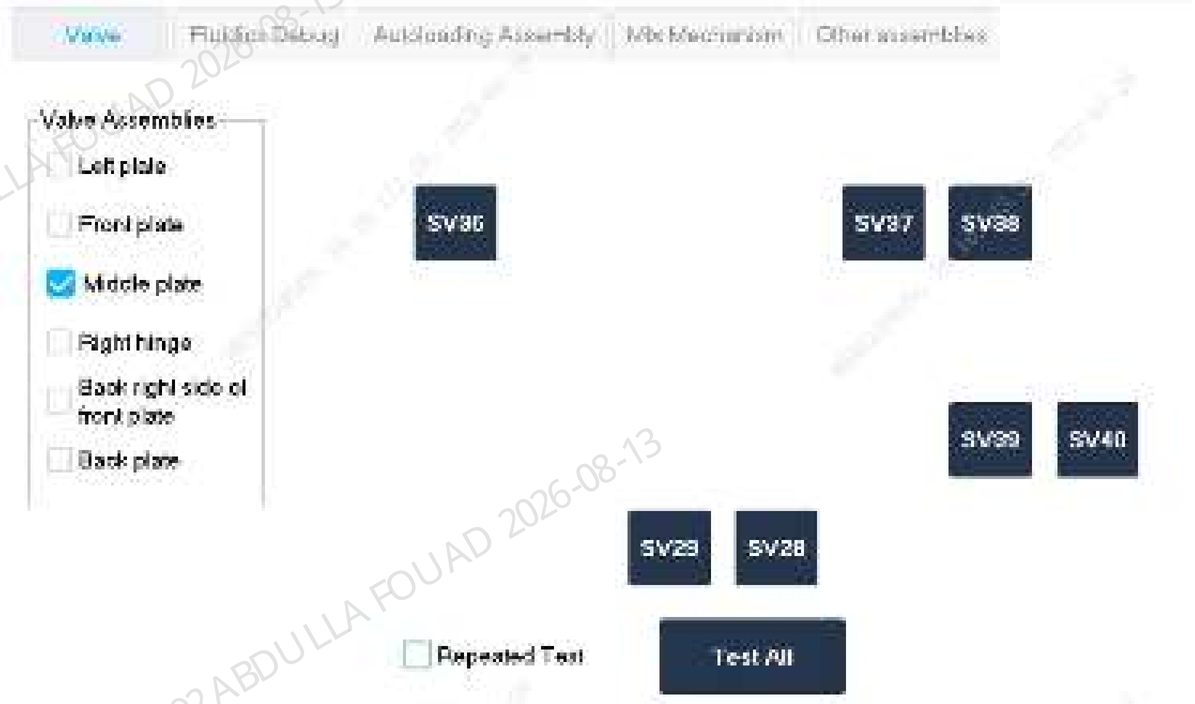
Required tools

Cross-headed screwdriver

Procedures

1. Click**Service**>**>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. According to the replacement object, click *Self-Test* on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–524 Valve Self-Test screen



Verification:

1. Select **Service**>>**Maintenance>Cleaning**>>**Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance**>>**Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.55 2-way LVM valve assembly (with valve board and connector)

8.55.1 General Information

For the general information in the case of two-way NC LVM valve body, see Two-position ways NC LVM valve body.

For the general information in the case of LVM valve assembly of two-way, see LVM Valve Assembly of two-way.

Two-position ways NC LVM valve body

Figure8–525 Photo of LVM valve assembly of two-way

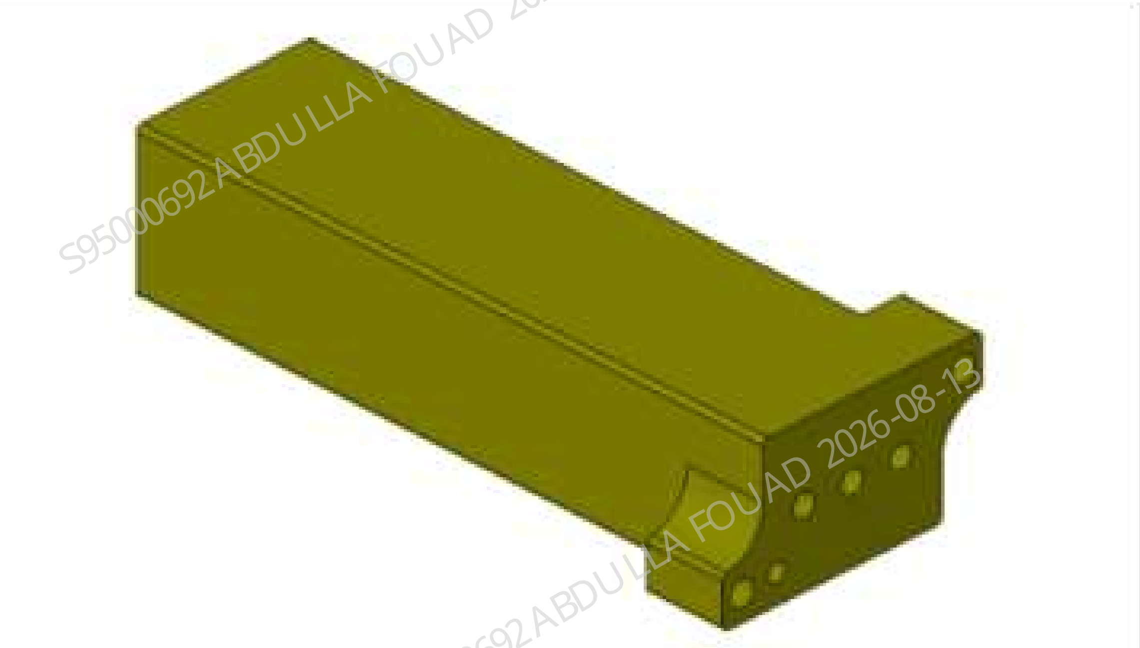


Table 8–58 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-061170-00	Two-position ways LVM valve assembly (with valve board and connector)	N/A	N/A

Function:

Turn on/off and switch the sampling tubes.

LVM Valve Assembly of two-way

Figure8–526 Photo of LVM valve assembly of two-way

Table 8–59 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-061170-00	Two-position ways LVM valve assembly (with valve board and connector)	N/A	N/A

Function:
Turn on/off pneumatics and fluidics.

8.55.2 Disassembly and Assembly

For the disassembly and assembly in the case of two-way NC LVM valve body, see Two-position ways NC LVM valve body.
For the disassembly and assembly in the case of LVM valve assembly of two-way, see LVM Valve Assembly of two-way.

Two-position ways NC LVM valve body

Required tools

Cross-headed screwdriver

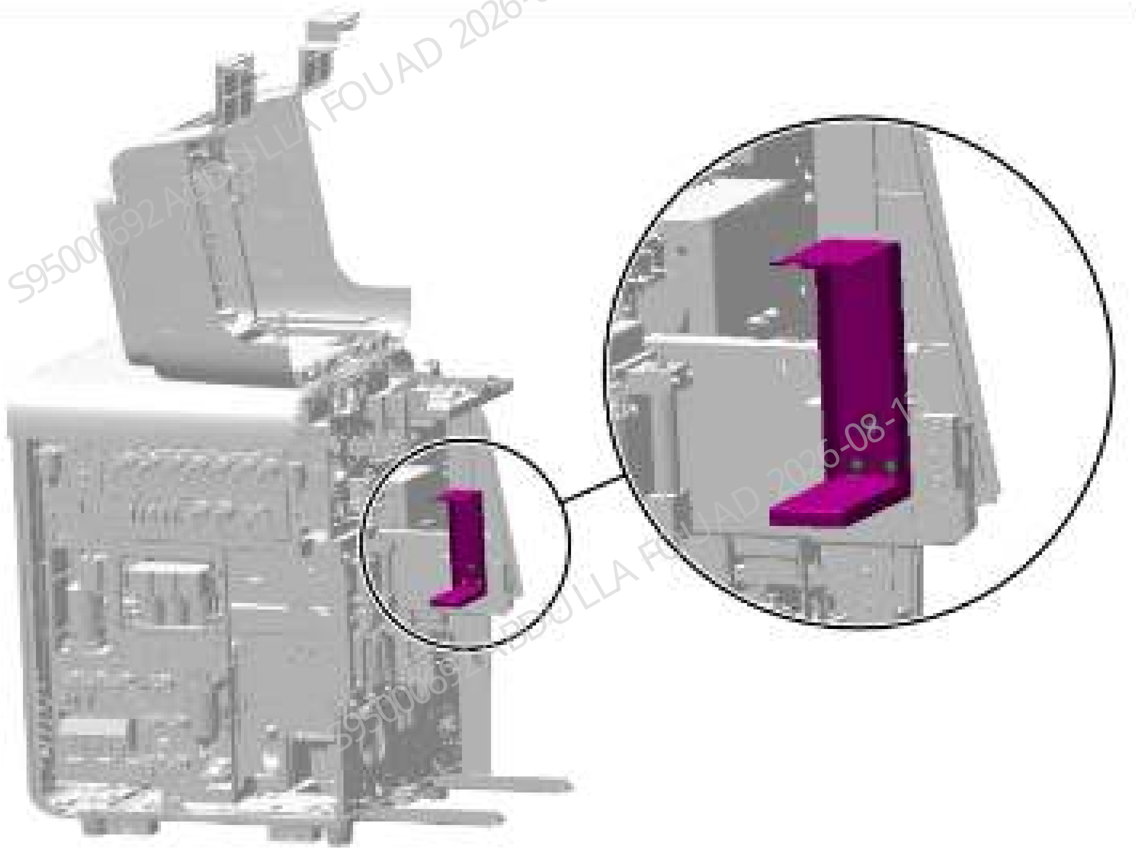
Pre-requirements

Power off the main unit, and unplug the power cord. A small amount of liquid will leak when the tube is pulled out. Wipe it clean using toilet paper. The sample probe will also leak liquid, so toilet paper should be laid under the sample probe in advance.

Procedures

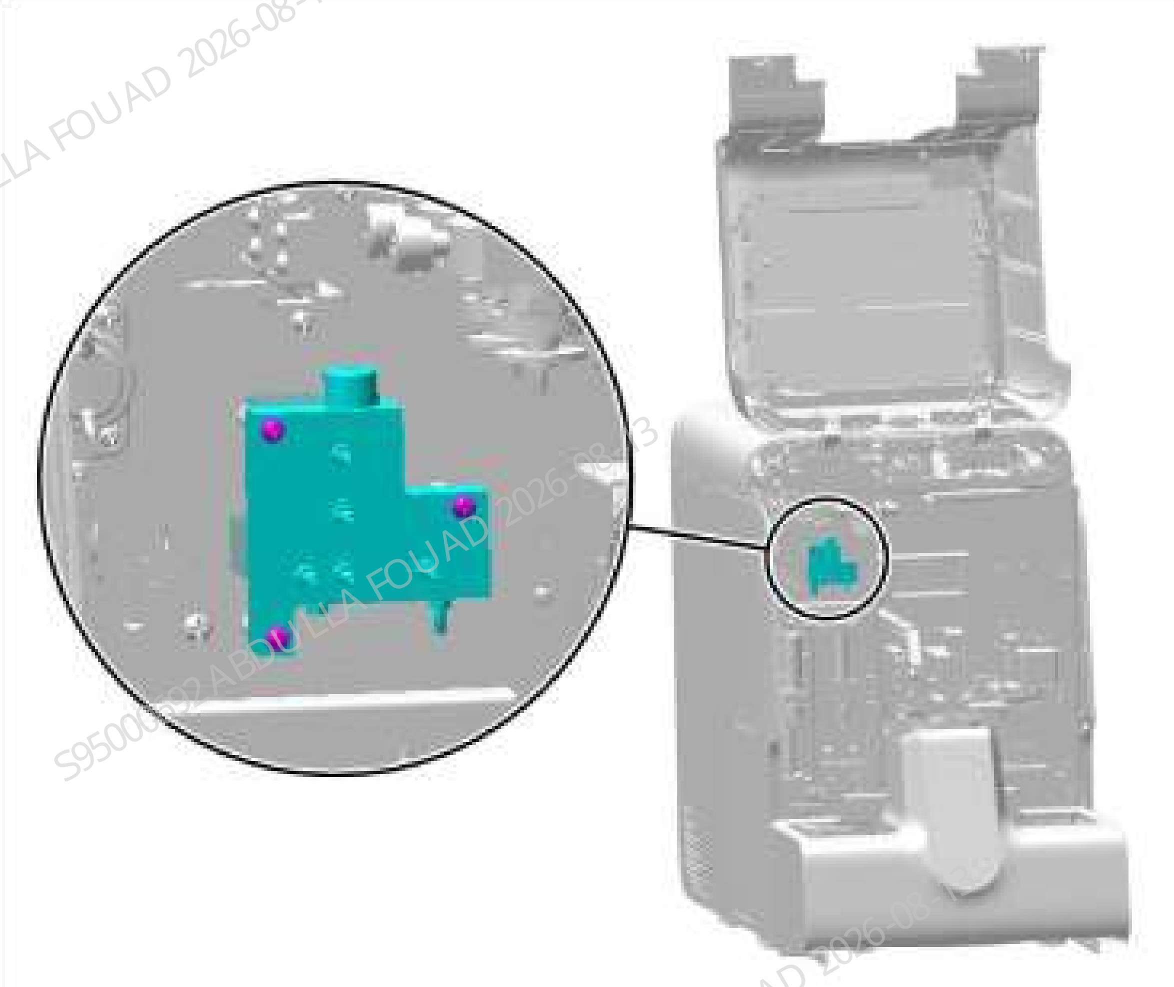
1. Turn up the front cover, lean it on the top cover, remove the three countersunk screws fixing the dye bag support, and remove the dye bag support, as shown in the figure below.

Figure8–527 Diagram of Removing the Left Cover



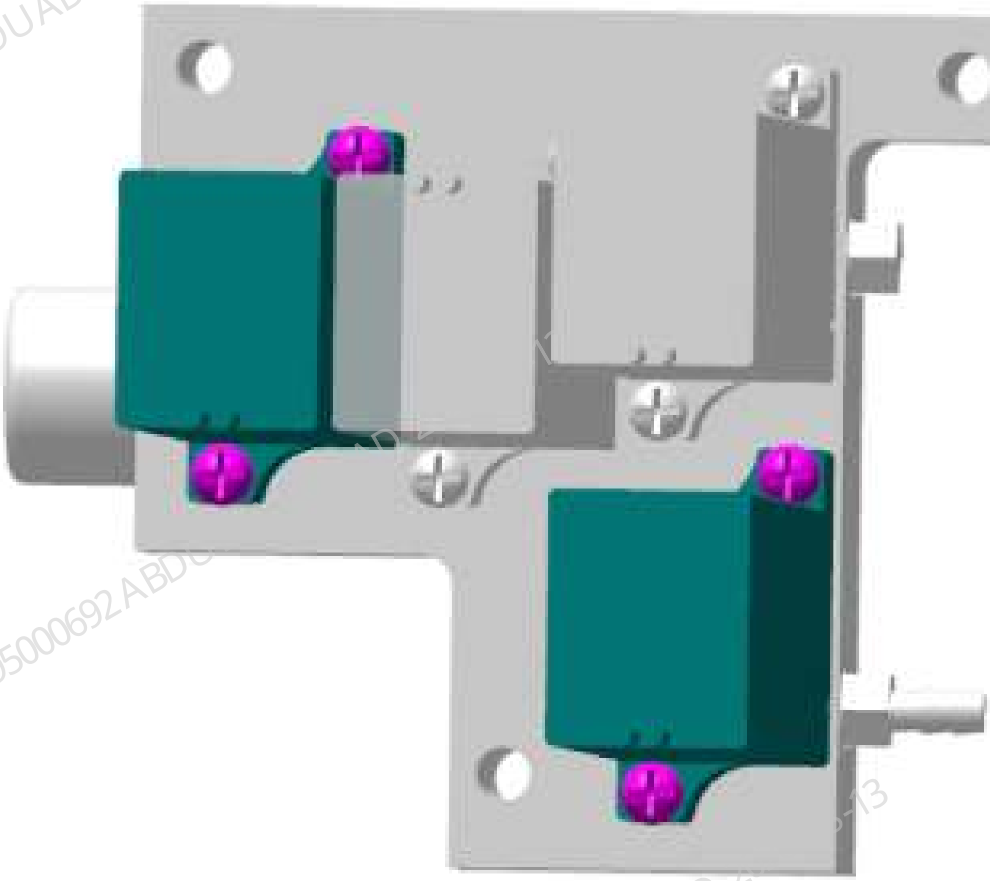
2. Unplug all rubber tubes attached to the connectors of the fluidic junction board.
3. Loosen the fixing screws of the fluidic junction board, as shown below, loosen the terminals connected to the main unit, and remove the fluidic junction board.

Figure8–528 Positions of the set screws of the air tank assembly



4. As shown in Figure 4, two LVM valve assemblies of two-way are installed on each fluidic junction board, and each LVM valve assembly of two-way is fixed using two screws. To remove one LVM valve assembly of two-way assembly (with valve board and connector) separately, loosen its fixing screws.

Figure8–529 Diagram of Removing the Two LVM Valve Assembly of two-way



Subsequent Requirements

Installation procedure:

1. Remove the valve board of the FRU (115-061171-00), and pay attention that the sealing ring cannot fall off during the process of removing the valve board.
2. Use two screws to fix the LVM valve assembly of two-way on the fluidic junction board.
3. Reconnect the terminals of the sampling manifold assembly.
4. Install the sampling manifold assembly back into the analyzer, tighten the screws to fix it, and insert the corresponding tube into the tube connector of the assembly.
5. Use three countersunk screws to install the dye support back to the original position.
6. Turn over the front cover to reset it.

LVM Valve Assembly of two-way

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the main unit, and unplug the power cord. Before replacing the valve assembly on the right hinge, first select **Status**>>**Temperature and Pressure** to execute **Release Pressure**, and avoid spraying liquid when pulling out the tube.

Procedures

1. Flip the front cover up and lean it on the top cover, and remove the left cover, as shown in Figure A below. Remove the right cover, as shown in Figure B below. Find the location of the LVM valve assembly of two-way, as shown in Figure C.

Figure8–530 Diagram A of Removing the Left Cover

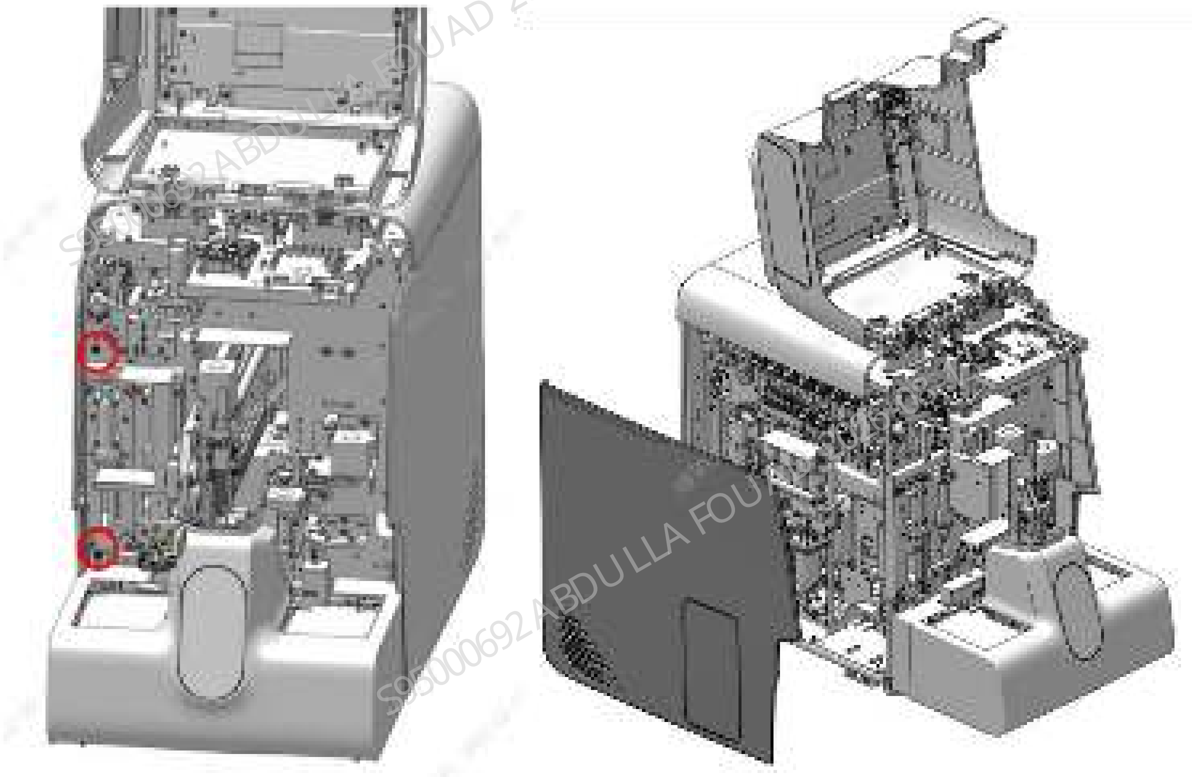


Figure8–531 Diagram B of Removing the Right Cover

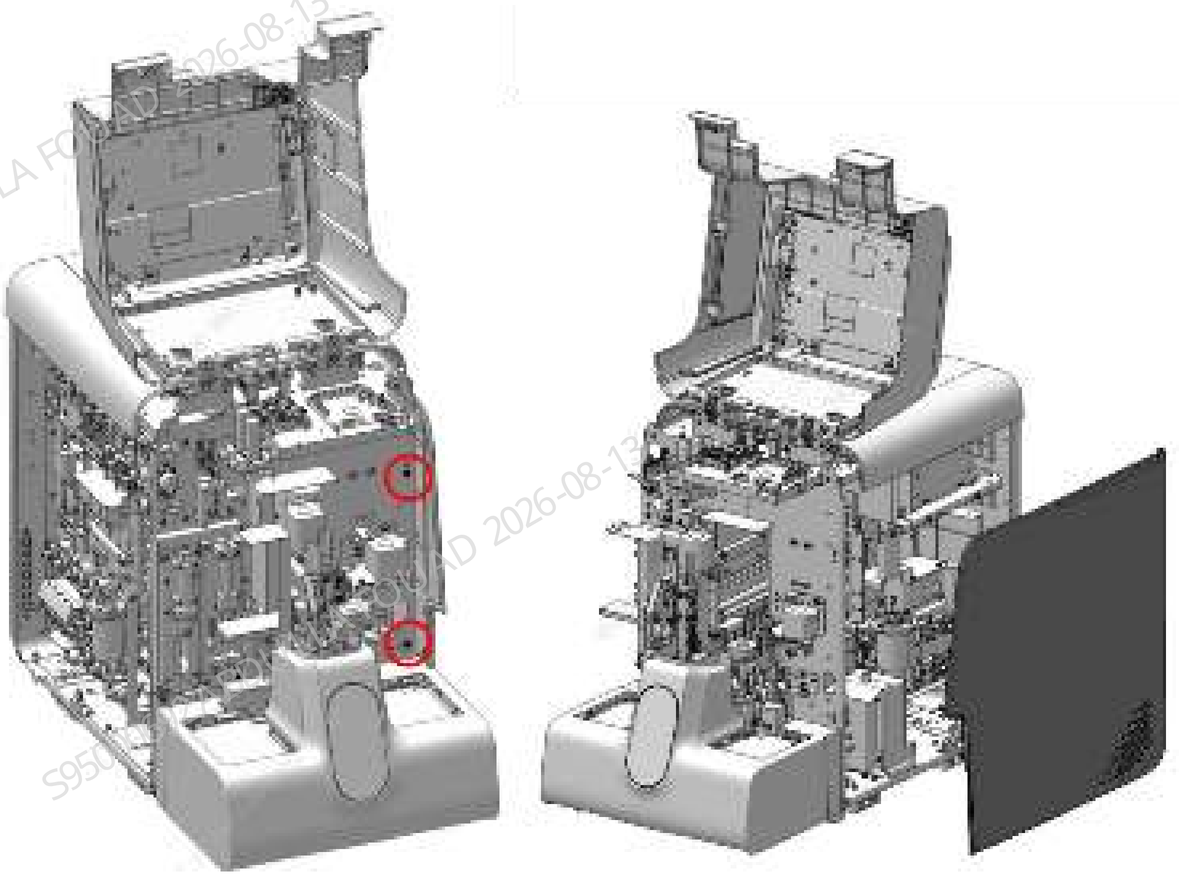
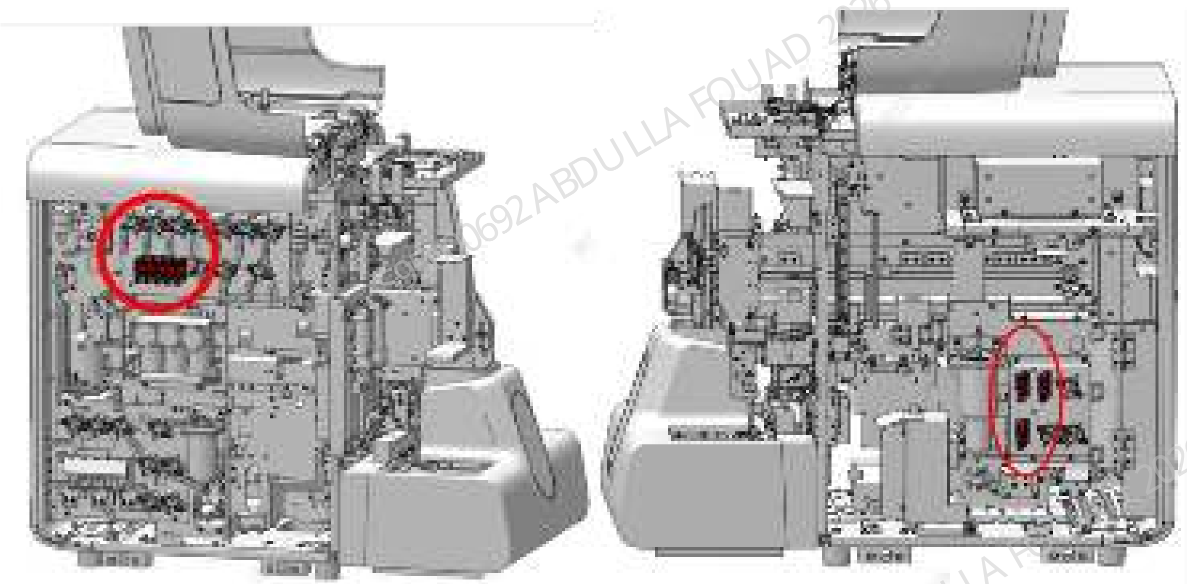
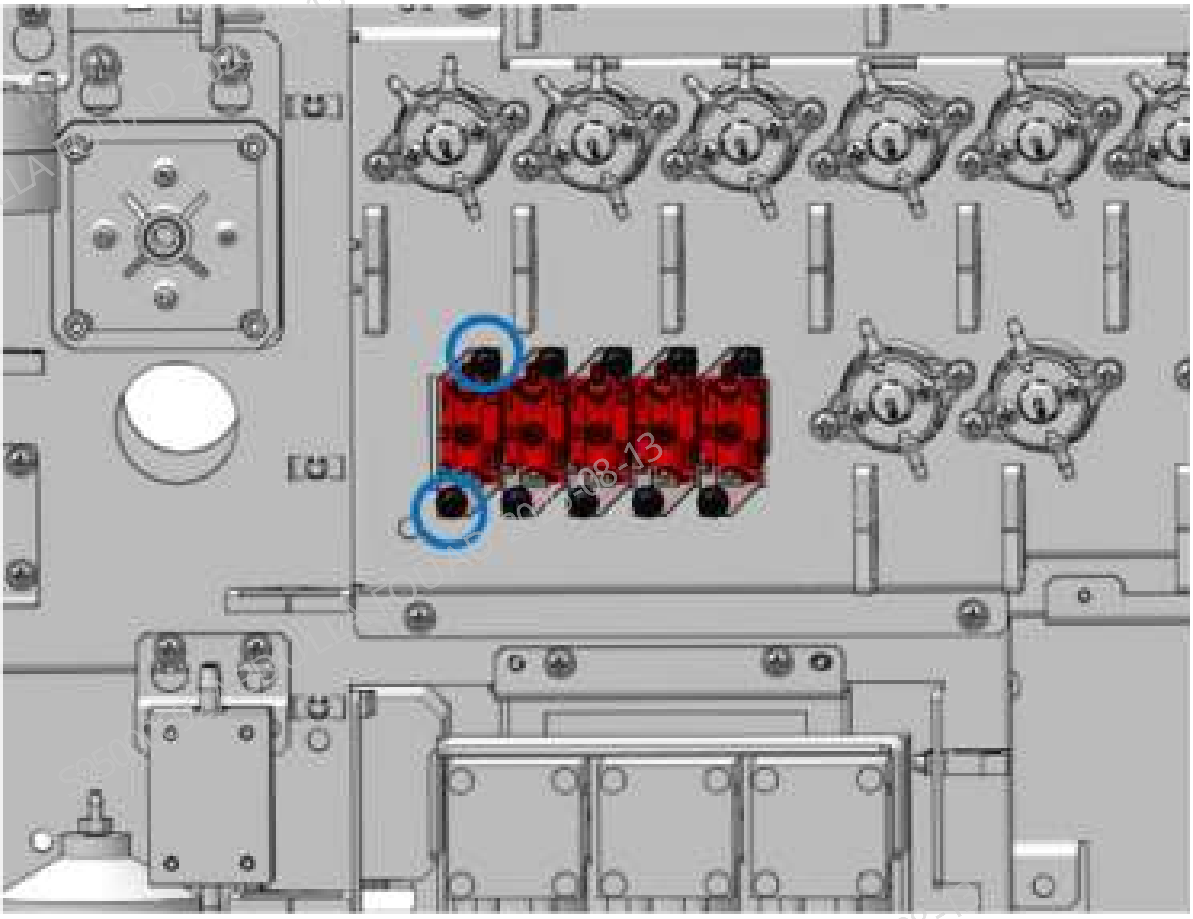


Figure8–532 Diagram C for Location of the LVM Valve Assembly of two-way



2. Pull out the rubber tube on the LVM valve assembly of two-way, loosen the two screws that fix the LVM valve assembly of two-way, as shown below, loosen the wiring terminal connected to the main unit, and remove the LVM valve assembly of two-way.

Figure8–533 Positions of the set screws of the LVM valve assembly of two-way



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the LVM valve assembly of two-way, and insert the terminal connected to the main unit.
2. Tighten the screws that fix the LVM valve assembly of two-way.
3. Install the left and right covers back, and flip down the front cover for resetting.

8.55.3 Commissioning and Verification

For the commissioning and verification in the case of two-way NC LVM valve body, see Two-position ways NC LVM valve body.

For the commissioning and verification in the case of two-way LVM valve assembly (with valve board and connector), see LVM Valve Assembly of two-way.

Two-position ways NC LVM valve body

Required tools

Cross-headed screwdriver

Procedures

1. Perform 10 counts in the WB CD mode.
2. Observe whether there is leakage or overflow of the sampling manifold assembly, and whether the analyzer reports an error.

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

LVM Valve Assembly of two-way

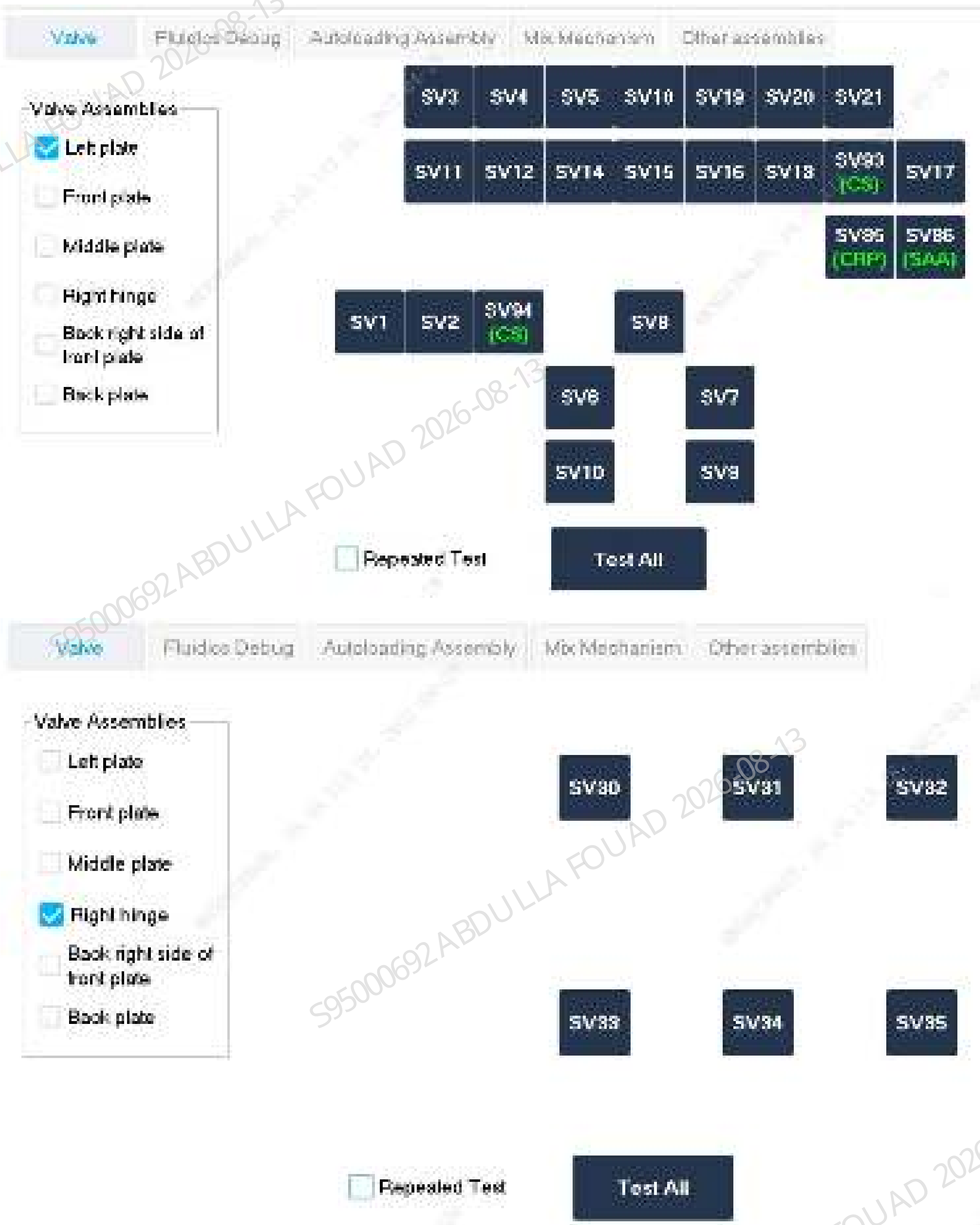
Required tools

Cross-headed screwdriver

Procedures

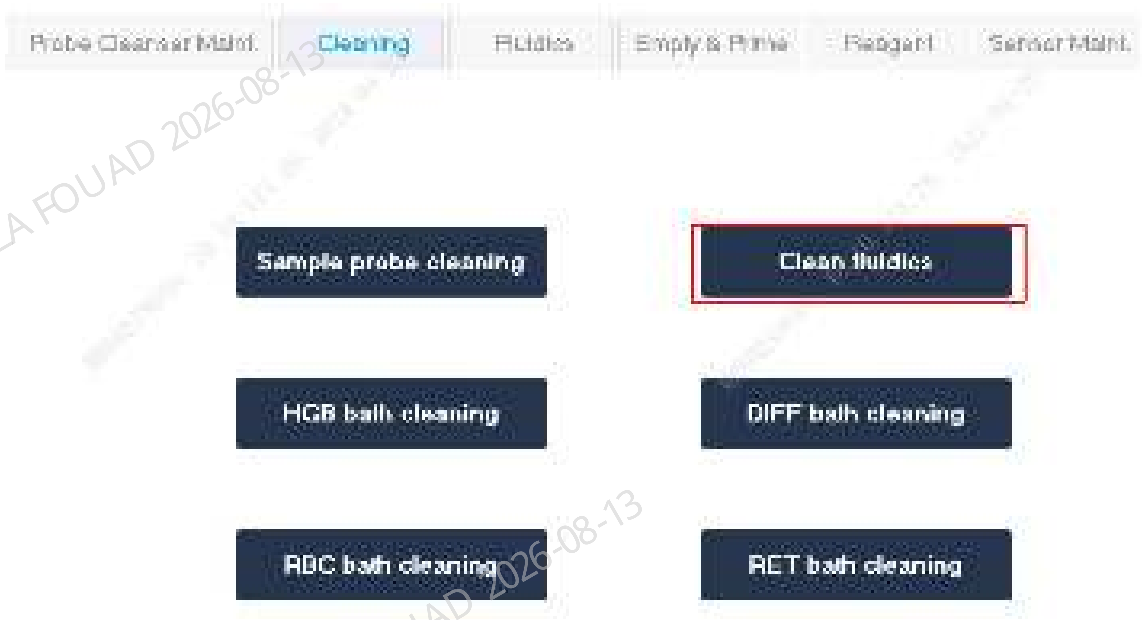
1. Click**Service**>**>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. According to the replacement object, click *Self-Test* on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–534 Valve Self-Test screen



Verification:

1. Select **Service>>Maintenance>Cleaning>>Overall cleaning**, and perform one overall cleaning action.



2. Select **Performance**>**>Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.56 Sheath Fluid Impedance Assembly

8.56.1 General Information

Figure8–535 Photo of Sheath Impedance Confluence Board Assem

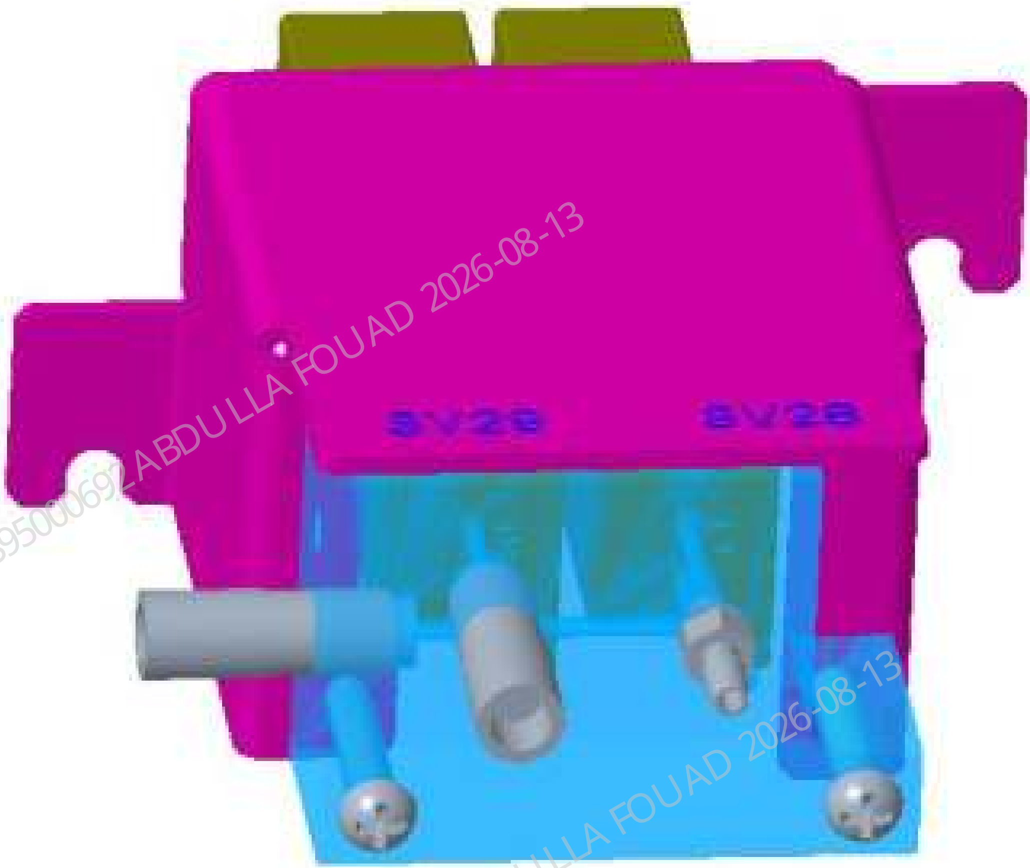


Table 8–60 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-085485-00	Sheath impedance confluence board assem (FRU)	115-061170-00	Two-position ways LVM valve assembly (with valve board and connector)

Function:
Controlling connection/disconnection of the RBC sample preparation tube.

8.56.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

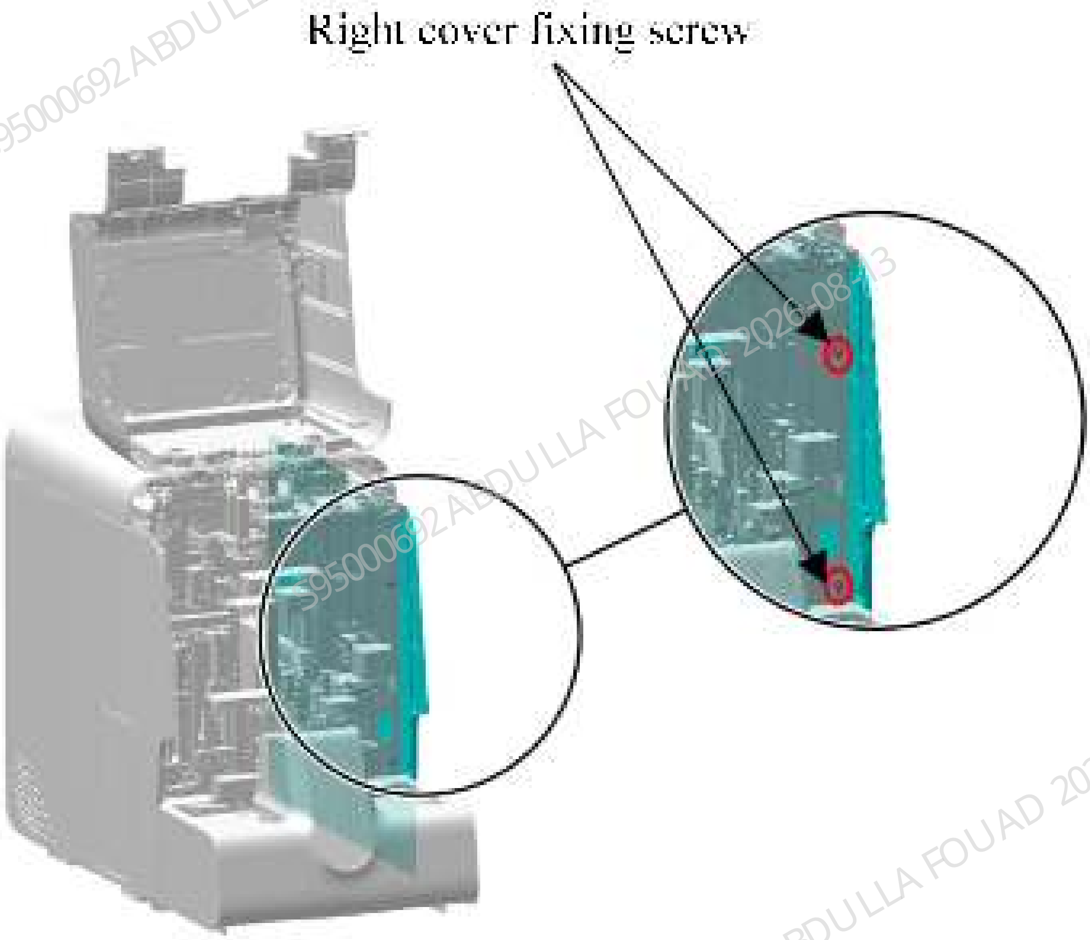
Pre-requirements

1. Power off the analyzer, and unplug the power cord.
2. Move the sample probe position to the INIT (RET bath) position.

Procedures

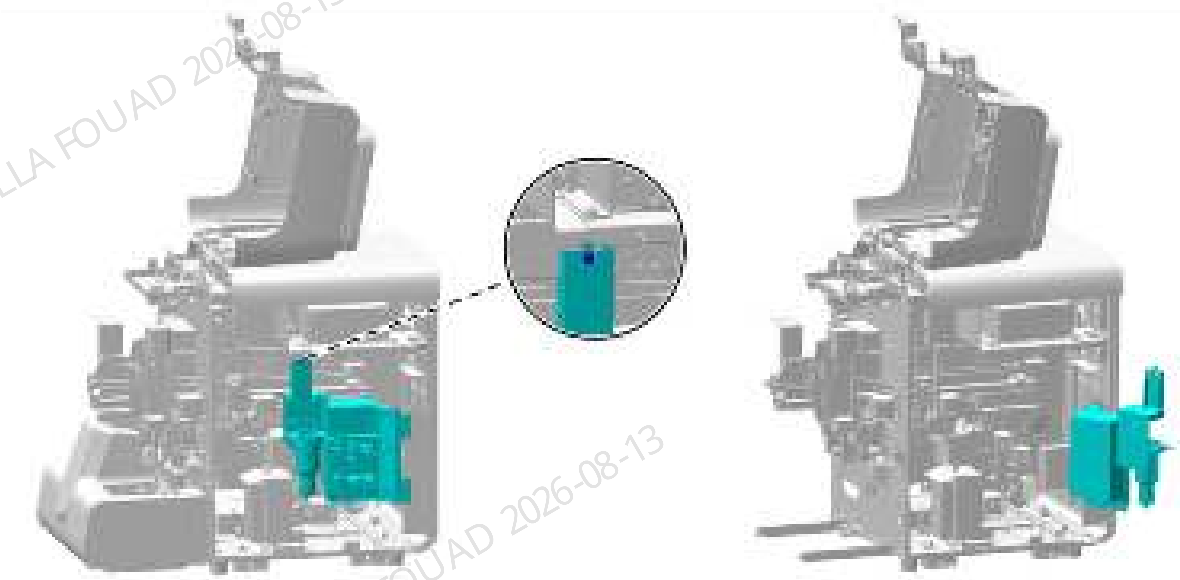
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–536 Diagram of Removing the Right Cover



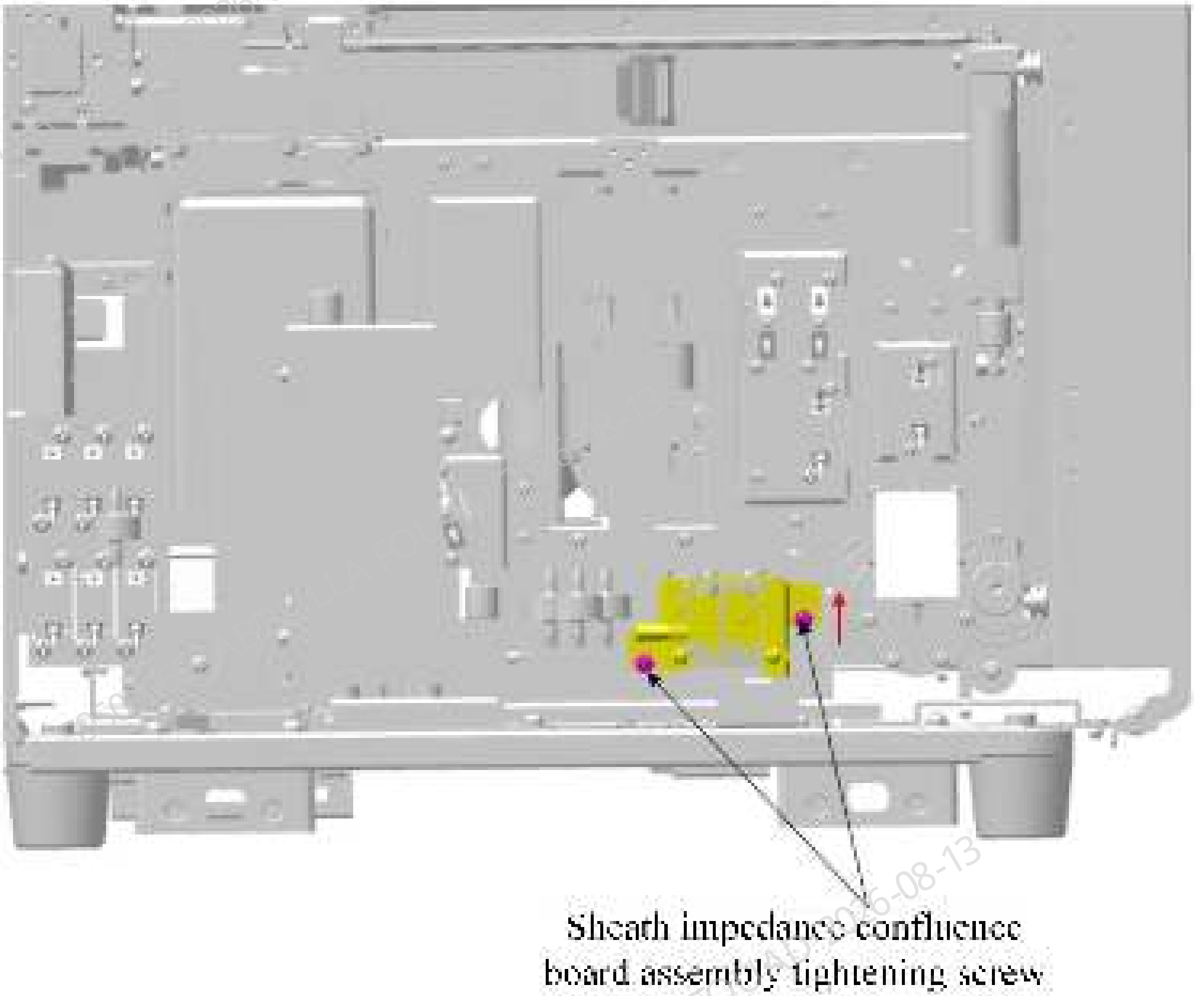
2. Unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–537 Opening the right cover assembly



3. Loosen the fastening screws of the sheath impedance confluence board assembly (FRU) (do not remove them completely). Lift the assembly up (in the direction of the red arrow) to take it out.

Figure8–538 Diagram of Removing the Sheath Impedance Confluence Board Assem (FRU)



Subsequent Requirements

Installation procedure:

1. Insert the pipeline connected to the sheath impedance confluence board assem (FRU).
2. Reset the sheath impedance confluence board assem (FRU) and lock the fastening screws.
3. Install the right cover back in the main unit and lock it.
4. Flip down the front cover to reset it.

8.56.3 Commissioning and Verification

Procedures

Commissioning:

1. After logging in at the service access level, click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve**, and select middle baffle, as shown in the figure below.

Figure8–539 Sheath Impedance Confluence Board Assembly (FRU) Commissioning Interface

2. Click "SV28" and touch the SV28 valve of the machine at the same time to confirm whether it operates. If it operates well, the status is normal. Then click "SV29" and touch the SV29 valve of the machine at the same time to confirm whether it operates. If it operates well, the status is normal.
3. Enter the sample analysis interface, take an empty test tube rack to start counting, and observe whether the parameter results comply with the background range. Count three times in sequence. The results of all the three times comply with the background range. (For the background range, refer to the background requirements of the installation guide.)
4. If the above two points are met, the commissioning and verification of sheath impedance confluence board assembly (FRU) is completed.
5. If the commissioning is abnormal, check whether the installation is correct. If it is correct, perform commissioning again.
6. If commissioning is still abnormal, replace the assembly and repeat the disassembly/assembly and commissioning methods.

8.57 Two-way LVM valve assembly (10R6 with valve board, 3120)

8.57.1 General Information

Figure8–540 Photo of LVM Valve Assembly of Two-way

Table 8–61 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-071230-00	Two-position ways LVM valve assembly (10R6 with valve board, 3120)	N/A	N/A

Function:
Turn on/off pneumatics and fluidics.

8.57.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

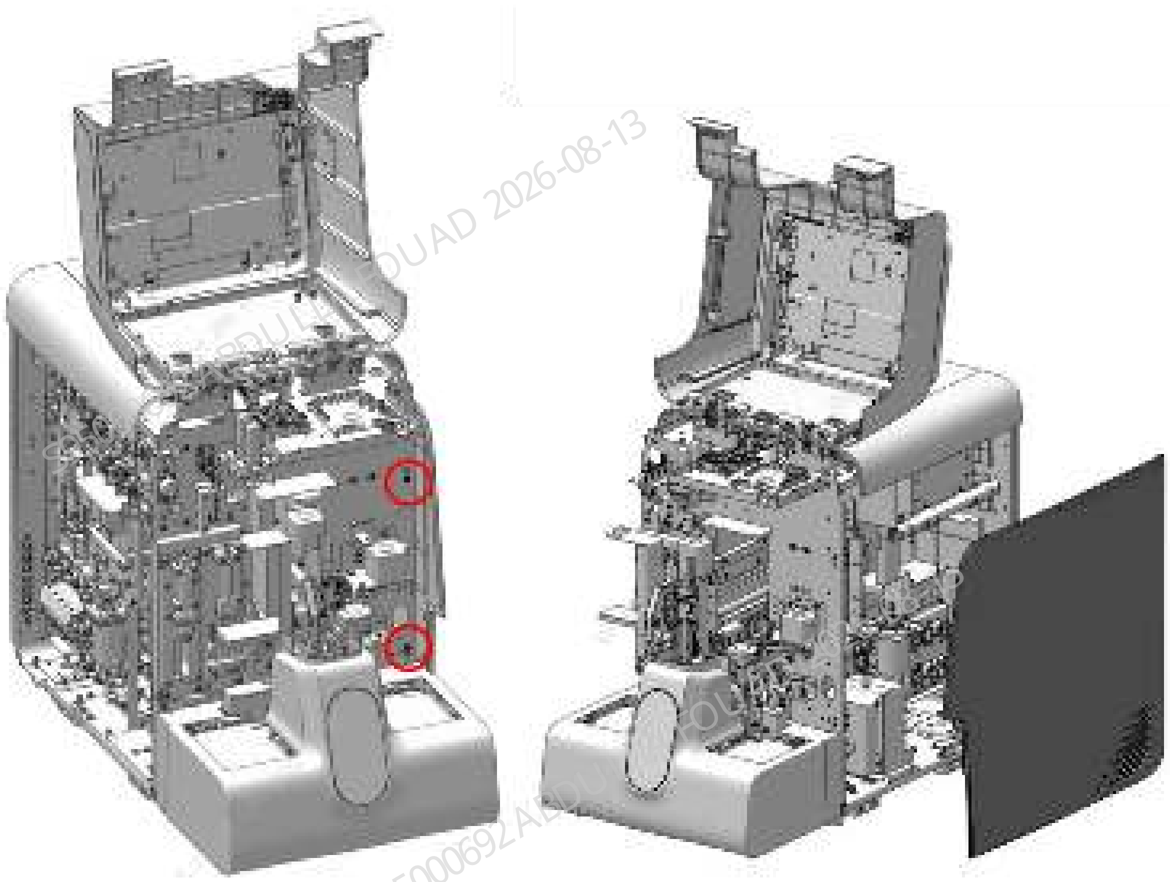
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Flip the front cover up and lean it on the top cover, and remove the right cover, as shown in the figure below.

Figure8–541 Removing the covers



2. Loosen the fixing screws of the right hinge, as shown in Figure A below. Open the hinge and find the LVM valve assembly of three-way, as shown in Figure B below.

Figure8–542 Diagram A for Locations of Right Bath Subassembly Fixing Screws

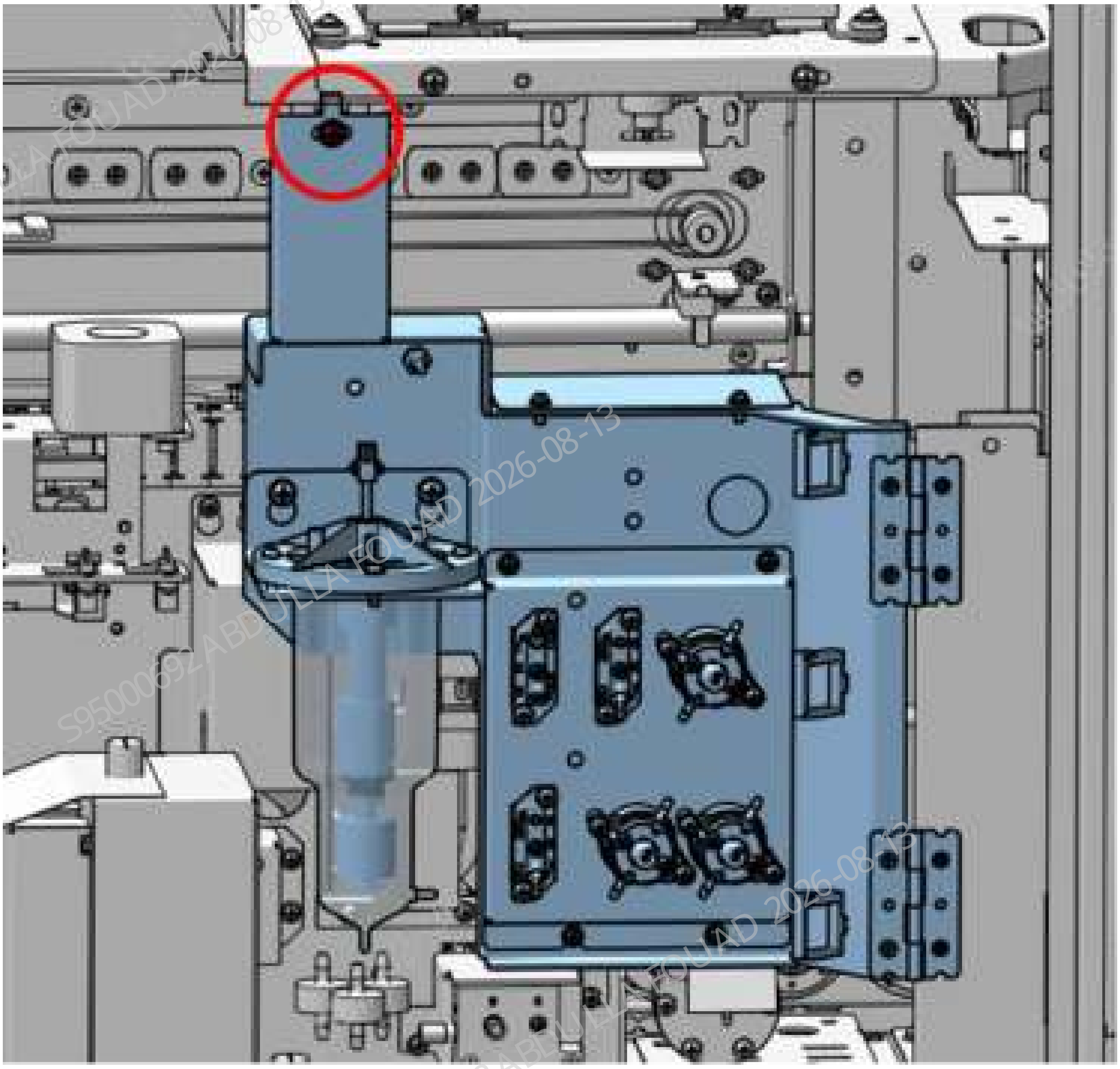
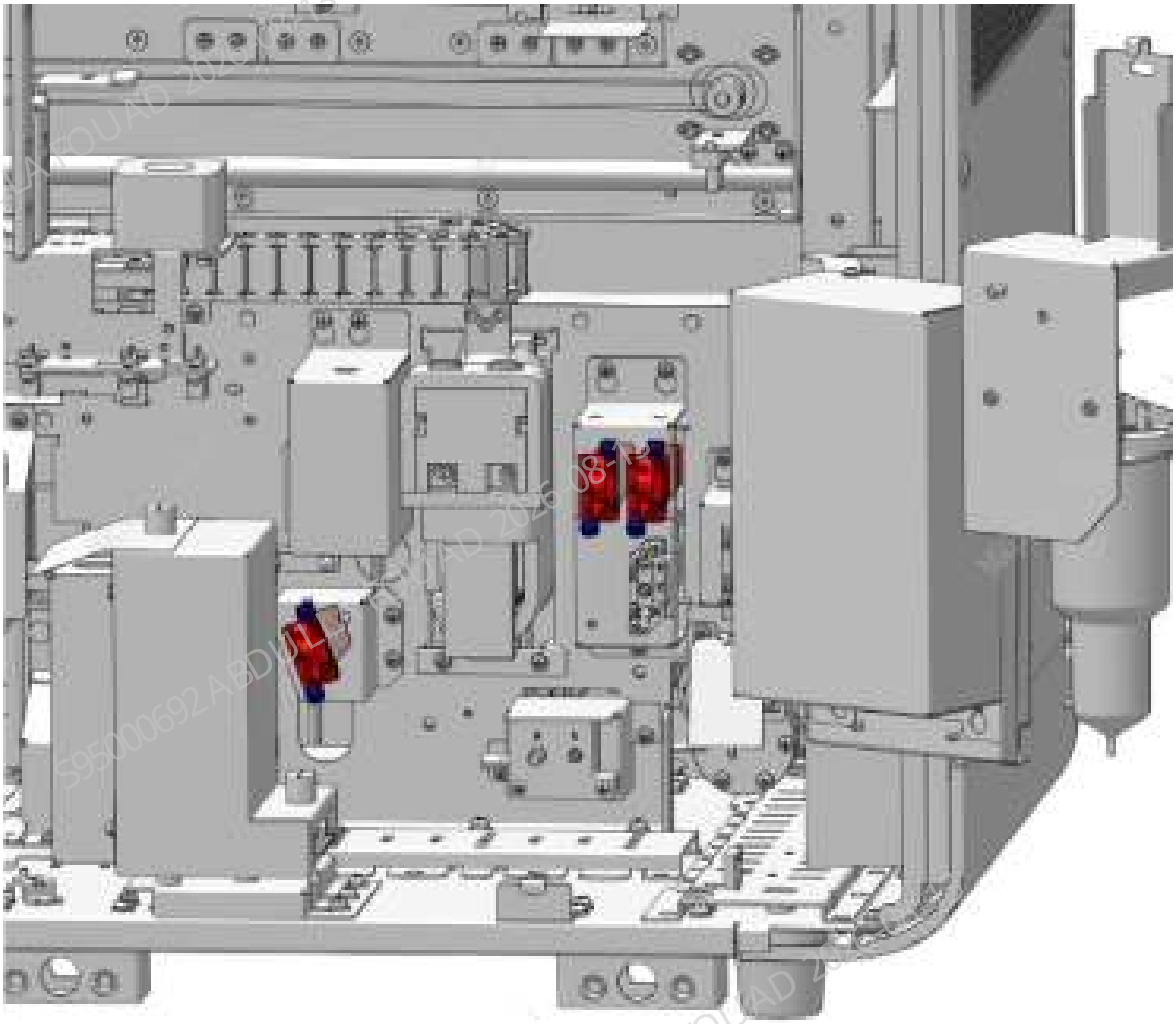
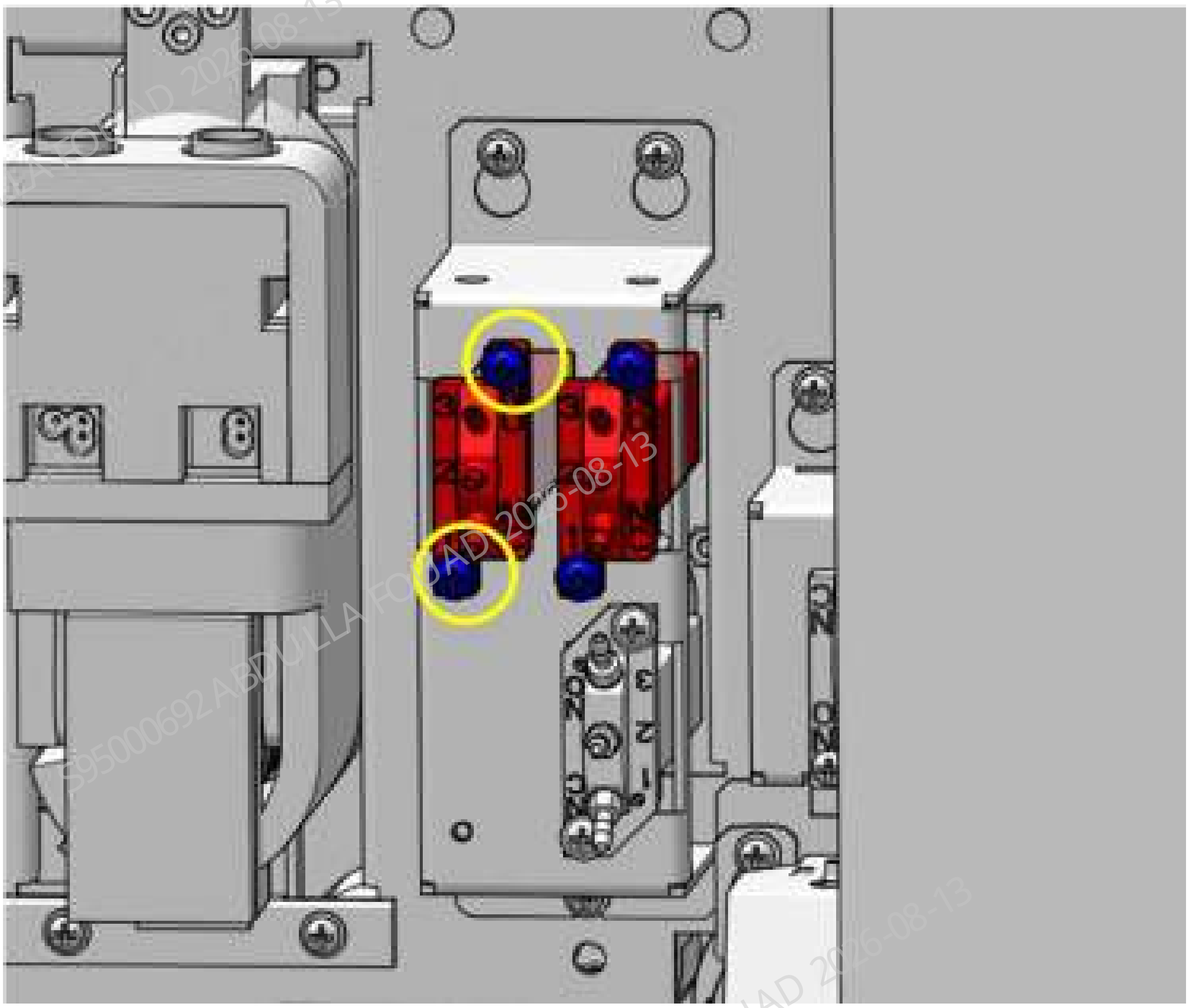


Figure8–543 Diagram B for Location of LVM Valve Assembly of two-way



3. Pull out the rubber tube on the LVM valve assembly of two-way, loosen the two screws that fix the two-way valve, as shown below, loosen the wiring terminals connected to the main unit, and remove the two-way valve.

Figure8–544 Diagram for Fixing Screws of LVM Valve Assembly of two-way



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube to the corresponding connector of the LVM valve assembly of two-way, and insert the terminal connected to the main unit.
2. Tighten the screws that fix the LVM valve assembly of two-way.
3. Reset the hinge and use the locking screws to fix it. Install the right cover back, and flip down the front cover for resetting.

8.57.3 Commissioning and Verification

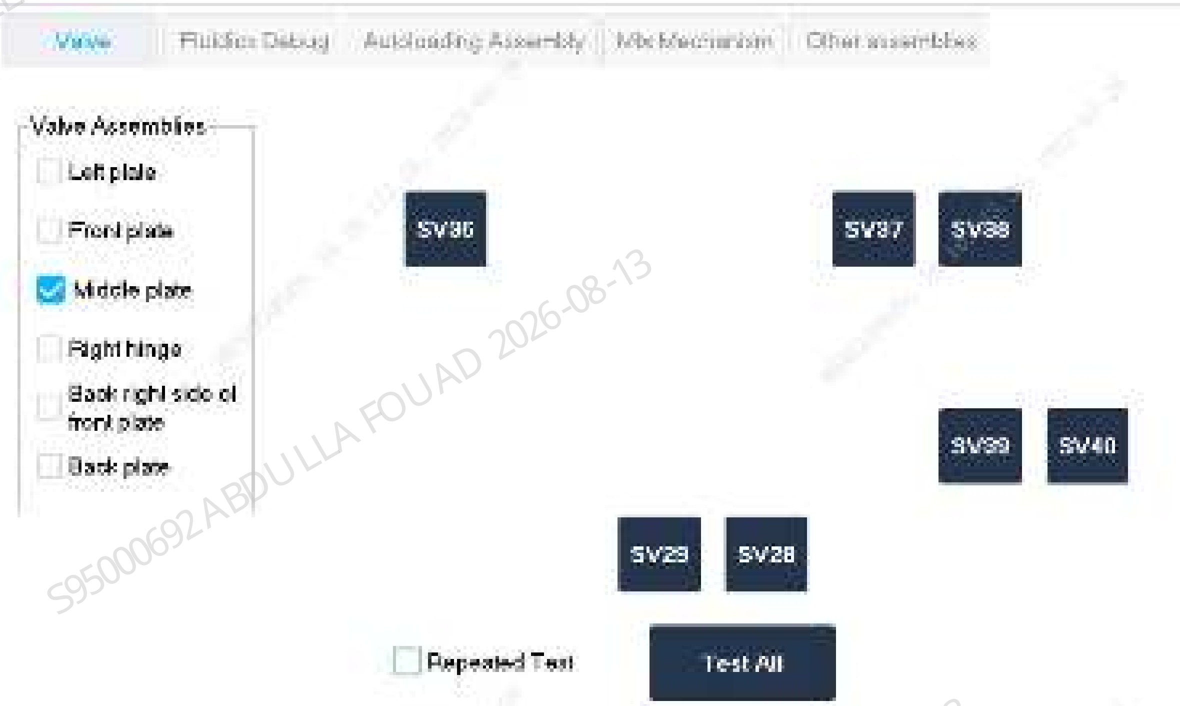
Required tools

Cross-headed screwdriver

Procedures

1. Click**Service>>Commissioning & Self-Test>Self-Test** to emter the **valve** self-test screen. According to the replacement object, click *Self-Test* on the software screen to check whether the valve action is normal, and at the same time see if there is a sound, if yes, the replacement is successful.

Figure8–545 Valve Self-Test screen



Verification:

1. Select **Service>>Maintenance>Cleaning>>Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance**>**>Background** on the software screen, select the **OV WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.58 Air Pump 12VDC

8.58.1 General Information

Figure8–546 Photo of Pump 12VDC (FRU)

Table 8–62 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081656-00	Pump 12VDC (FRU)	N/A	N/A

Function:
Building a positive pressure for the air tank assembly.

8.58.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

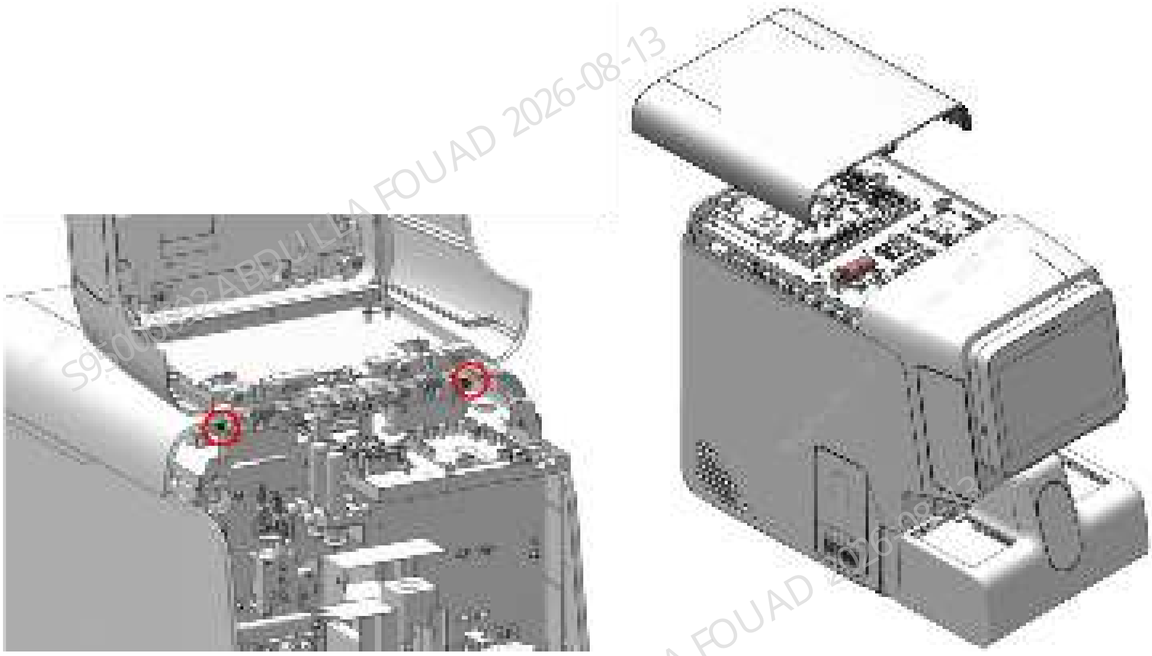
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

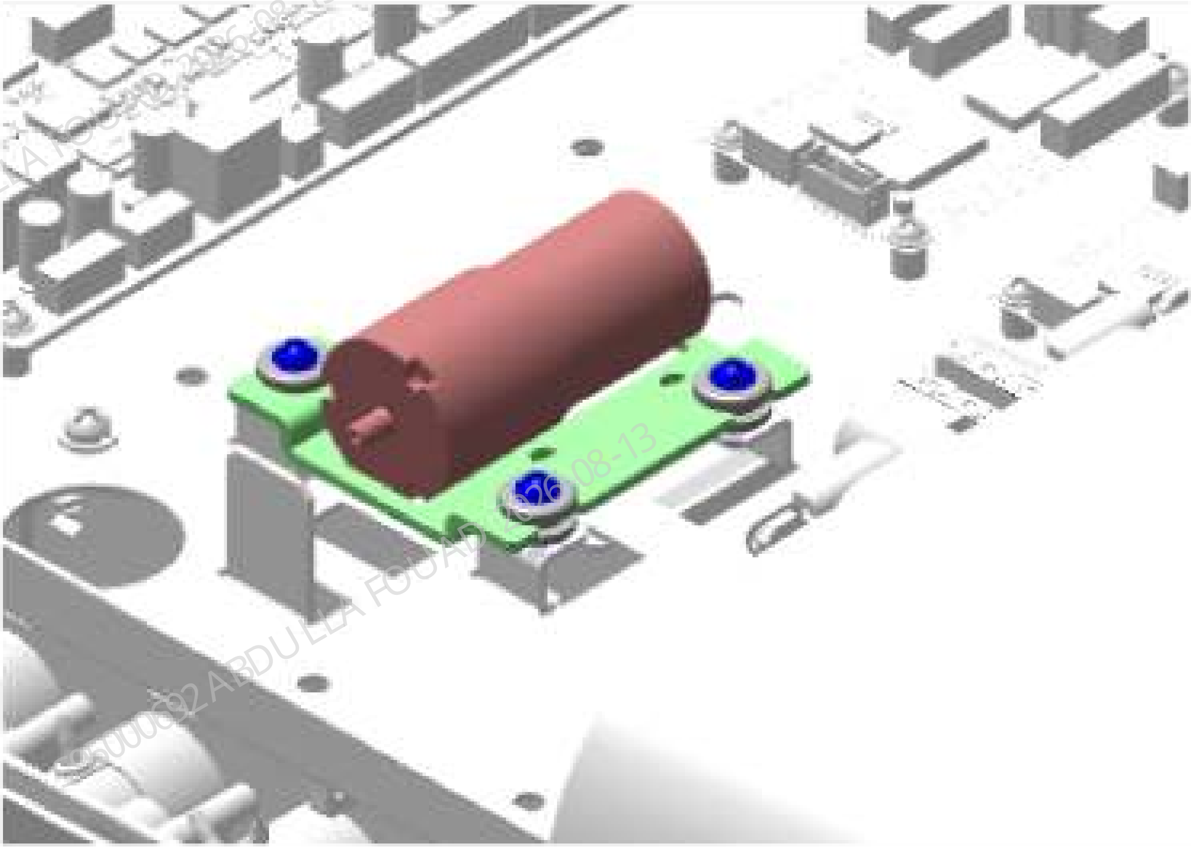
1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the top cover, and remove the top cover, as shown in the figure below.

Figure8–547 Removing the covers



2. Unplug the rubber tube connected to the connector of the air pump 12VDC, and disconnect the wire terminal connected to the main unit.
3. Loosen the three screws securing the air pump bracket, as shown in the figure below, and remove the air pump assembly.

Figure8–548 Positions of the set screws of the air pump



4. Cut off the cable ties that fix the air pump with cutting nippers, and remove the air pump 12VDC.

Subsequent Requirements

Installation procedure:
1. Secure the air pump 12VDC to the air pump bracket with cable ties.
2. Connect the air tube to the connector of the air pump 12VDC, and connect the terminals of the air pump to the main unit.
3. Use three screws to fix the air pump assembly on the top board of the main unit.
4. Fix the top cover, and turn over the front cover for resetting.

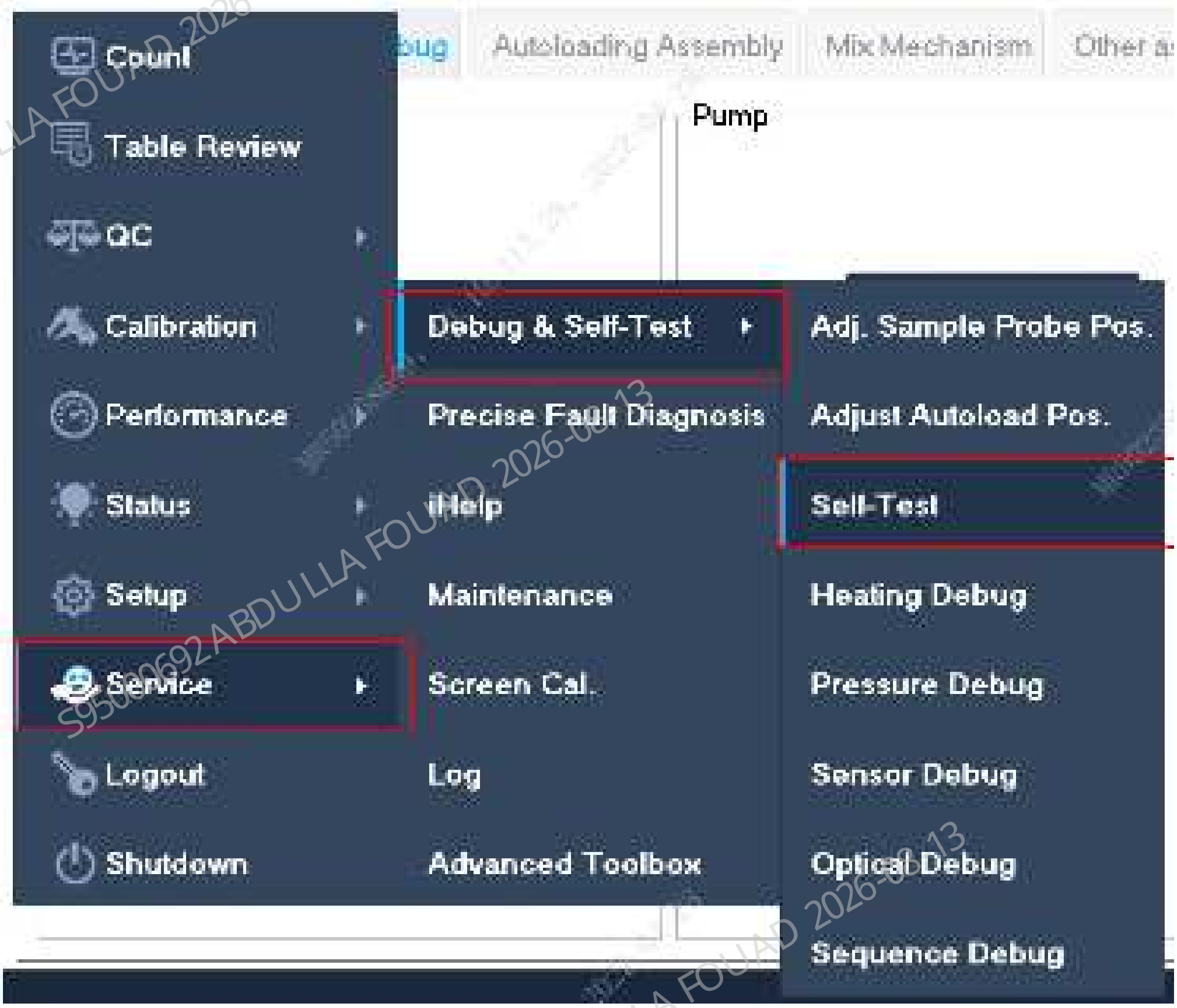
8.58.3 Commissioning and Verification

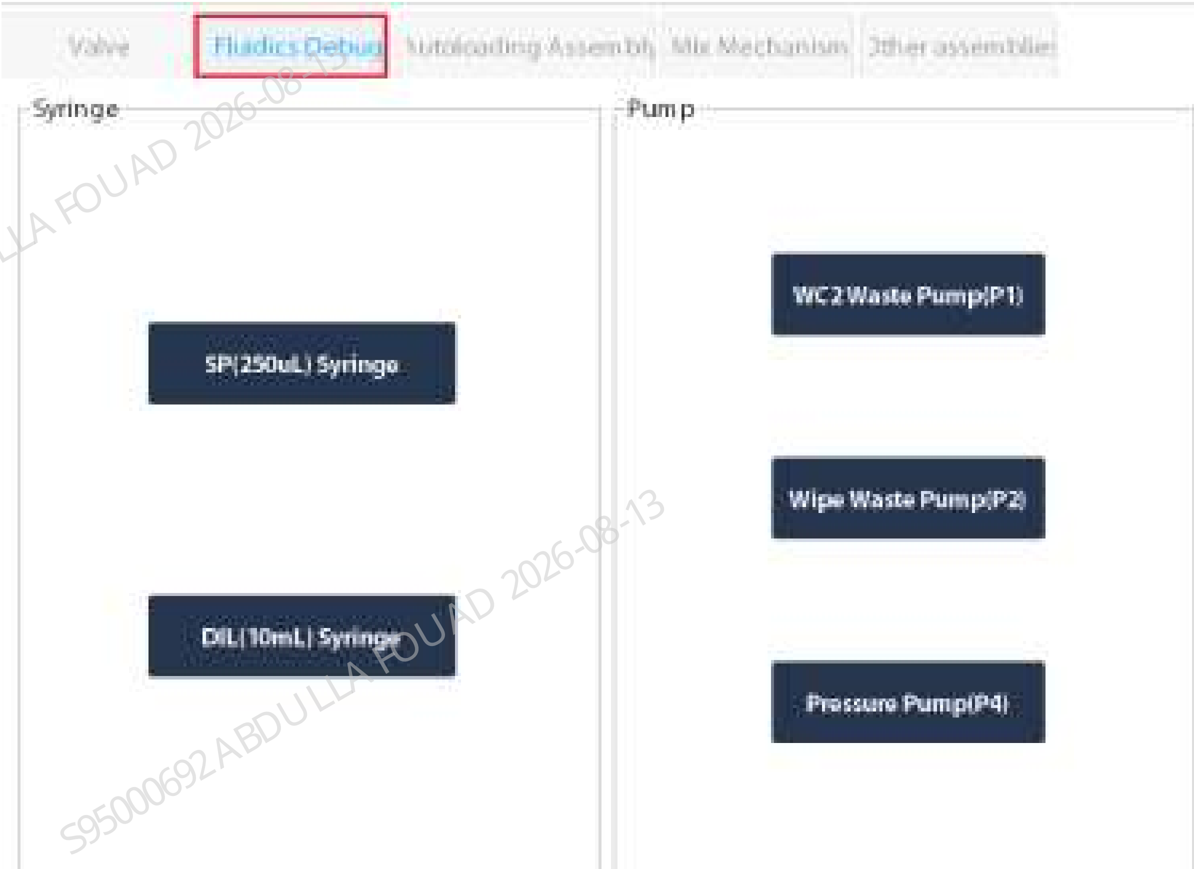
Required tools

Cross-headed screwdriver

Procedures

1. After startup, log in at the service access level. Click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Fluidics Commissioning**>>**Positive Pressure Pump (P4)** in sequence.





2. After confirming that the self-test is successful, click **OK**.

3. Then select **Status>>Temperature and Pressure**, and see whether the pressures of **PC (+50kPa)** and **SCI(+40kPa)** are within the range.

	Pressure (kPa)	Range (kPa)
PC(+50kPa)	24.8	[45.0,60.0]
SCI(+40kPa)	39.6	[35.0,45.0]
VAC(-30kPa)	-25.4	[-90.0,0.0]
WC2(-40kPa)	-19.1	[-45.0,0.0]
Liquid pressure	1.8	[50.0,120.0]

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.59 KNF Liquid Pump, Diaphragm Pump, 12VDC

8.59.1 General Information

Figure8–549 Photo of Liquid Diaphragm Pump of KNF 12VDC

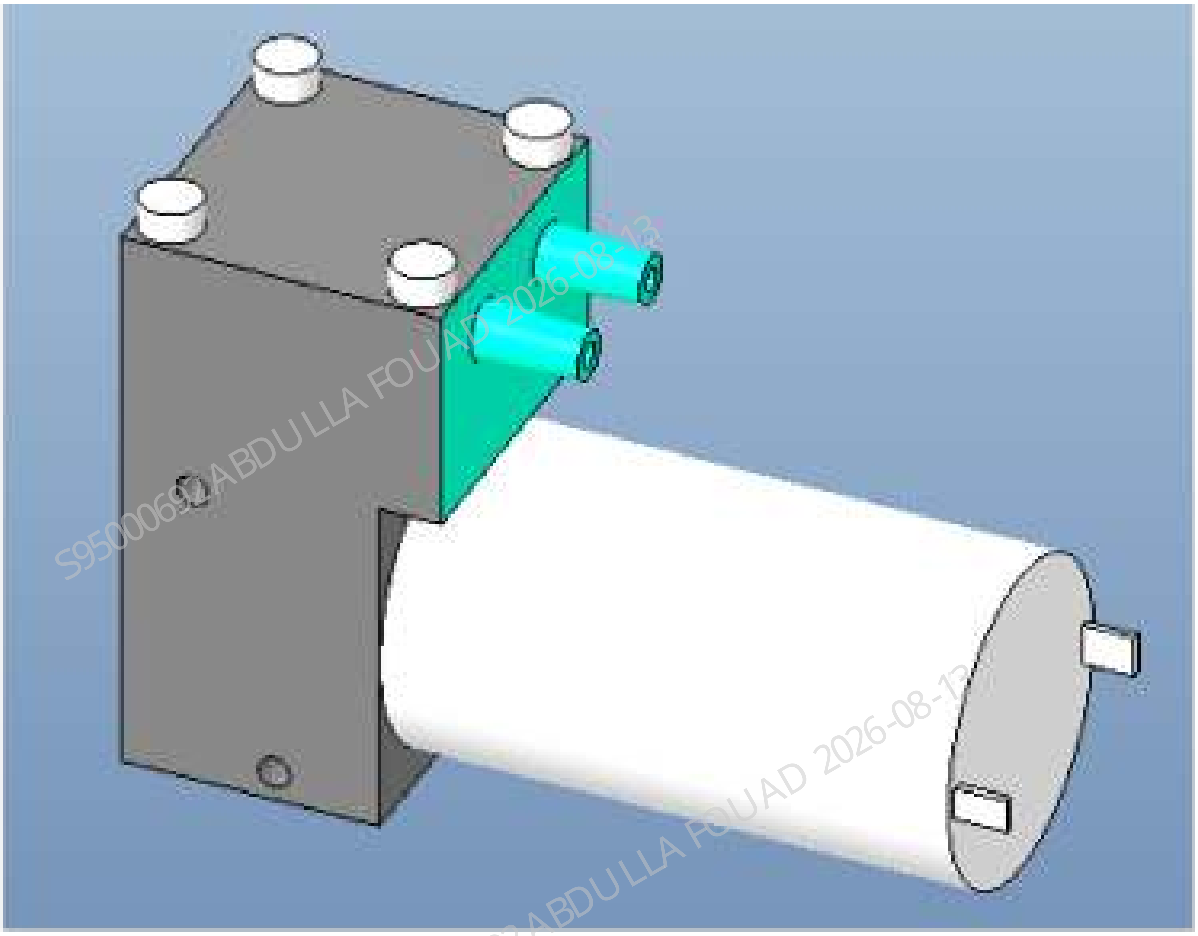


Table 8–63 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081655-00	Liquid Diaphragm Pump of KNF 12VDC (FRU)	N/A	N/A

Function:

Providing a negative pressure power source for the WC2 waste cistern and emptying the waste in the WC2 waste cistern. Recovering the cleaning waste of the probe wipe and the waste in the reaction bath.

8.59.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.

Required tools

Cross-headed screwdriver
One pair of tweezers

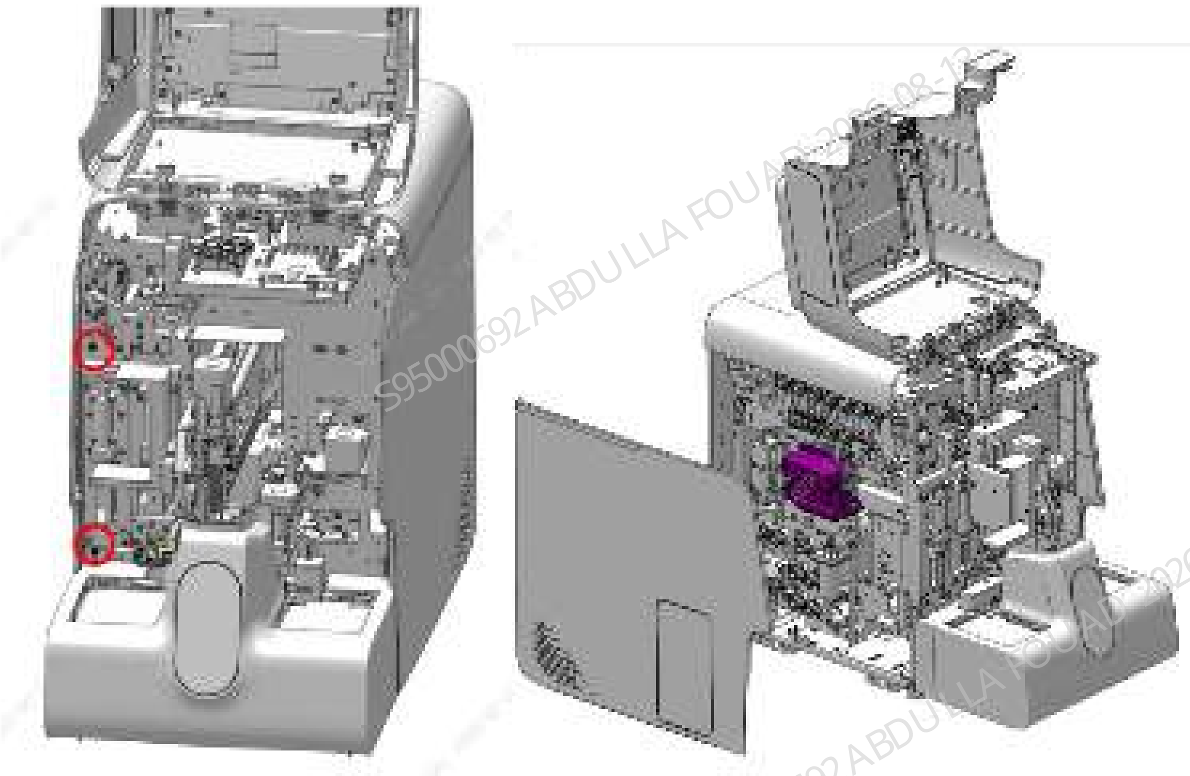
Pre-requirements

Empty the waste from the cistern: Select **Menu**>>**Service**>>**Maintenance**>>**Emptying & Priming**>>**WC2 Cistern Emptying**, power off the analyzer, and unplug the power cord.

Procedures

1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the left cover, remove the left cover, and find the waste pump assembly (without CRP), as shown in the figure below.

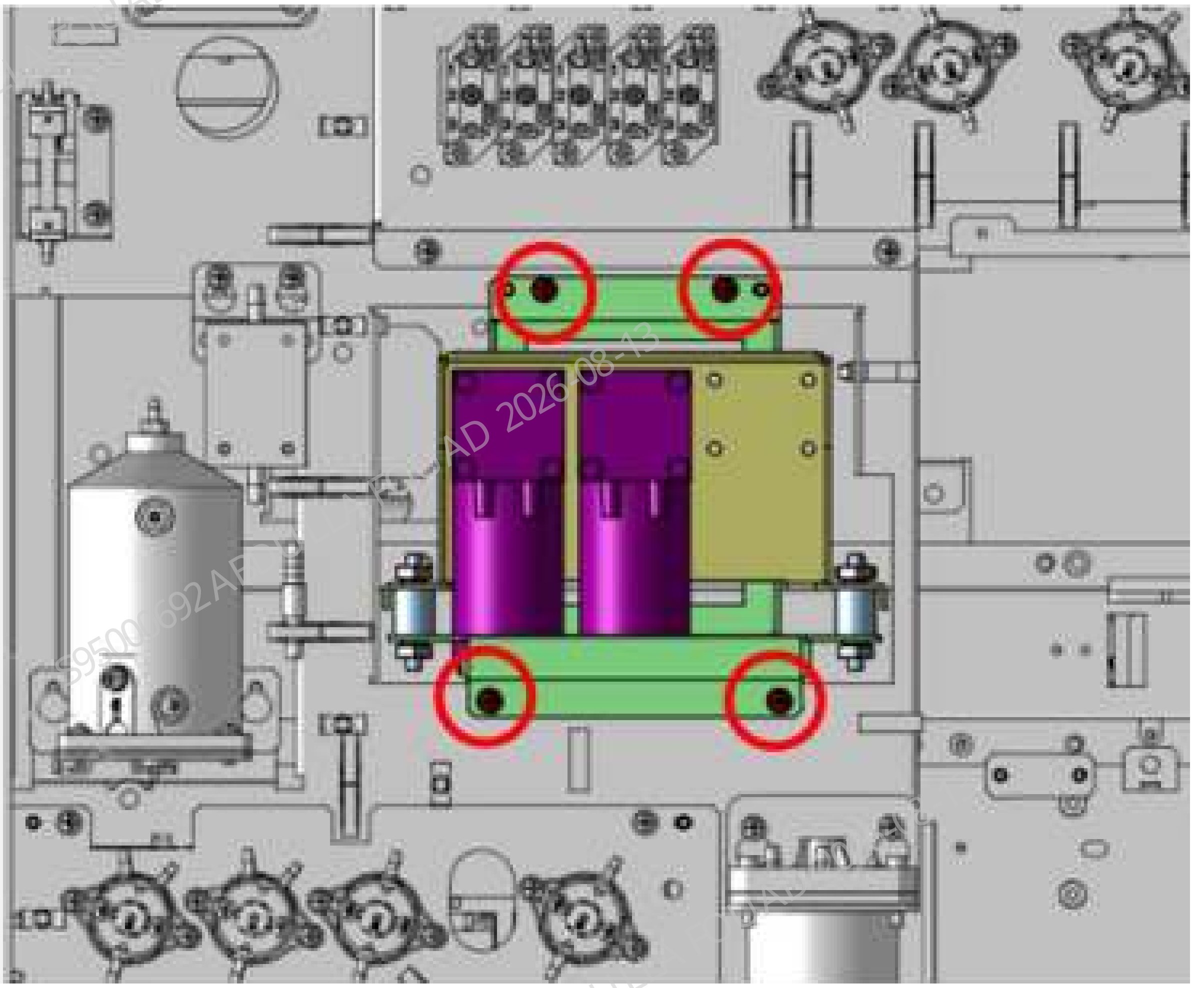
Figure8–550 Removing the covers



2. Unplug the four silicone tubes connected to the waste pump assembly, and mark them when they are unplugged to prevent incorrect tube installation.

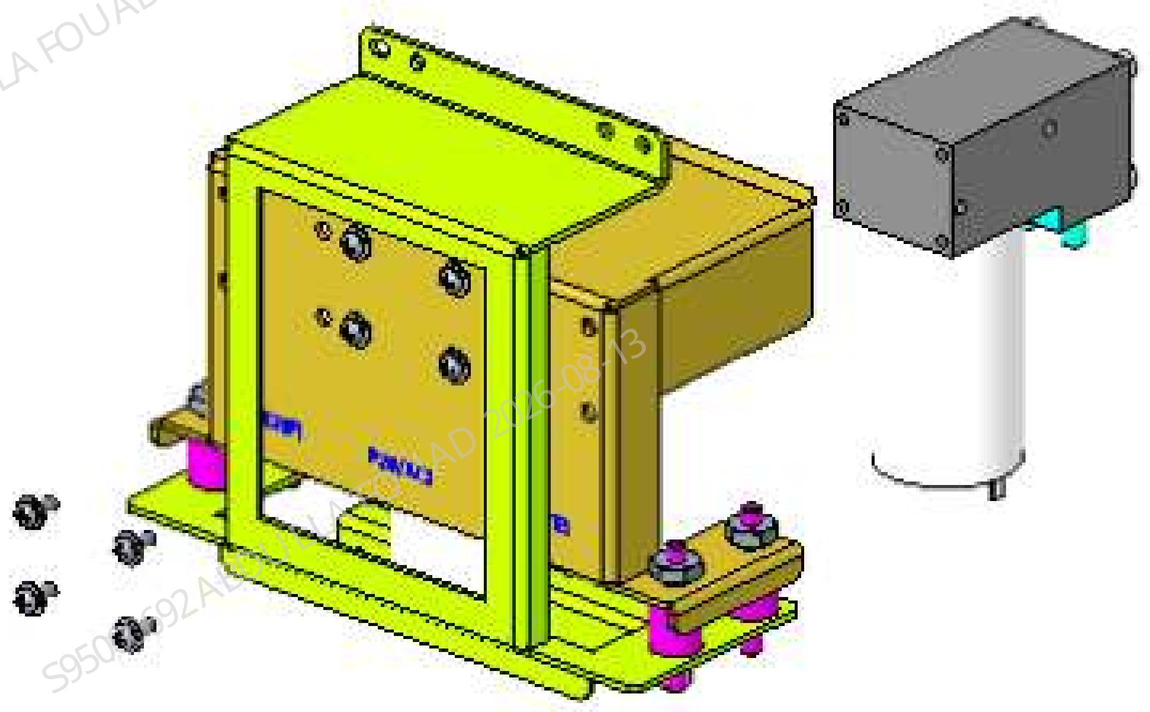
3. Unscrew the four screws that fix the waste pump assembly, as shown in the figure below, and pull the assembly out to a proper position for subsequent operations.

Figure8–551 Positions of the Fixing Screws of the Waste Pump Assembly



4. Unplug the terminal connected to the main unit, and then remove the waste pump assembly from the main unit.
5. Loosen the four screws securing the waste pump, as shown below, and remove the waste pump.

Figure8–552 Removing the waste pump



Subsequent Requirements

Installation procedure:

1. Secure the waste pump to the sheet metal bracket with four screws.
2. Connect the wiring terminals in the waste pump assembly to the corresponding wiring terminals of the main unit according to the wire markings.
3. Secure the waste pump assembly to the main unit with four combination screws.
4. Reconnect the four silicone tubes to the waste pump assembly.
5. Use four wire ties to fix the silicone tubes on the waste pump assembly, and use diagonal pliers to cut off the excess wire ties.
6. Install the left cover, flip down the front cover to reset it, and the installation is complete.

8.59.3 Commissioning and Verification

Procedures

Commissioning:

1. Select the CD mode in the sample analysis screen, and perform 20 background tests in a row.

2. During the test process and after the test is completed, if the **WC2 floater abnormal status** is not reported, the replacement and commissioning are successful.

Verification:

Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.60 Sheath filter assembly (3201)

8.60.1 General Information

Figure8–553 Photo of sheath fluid filter assembly (3201)

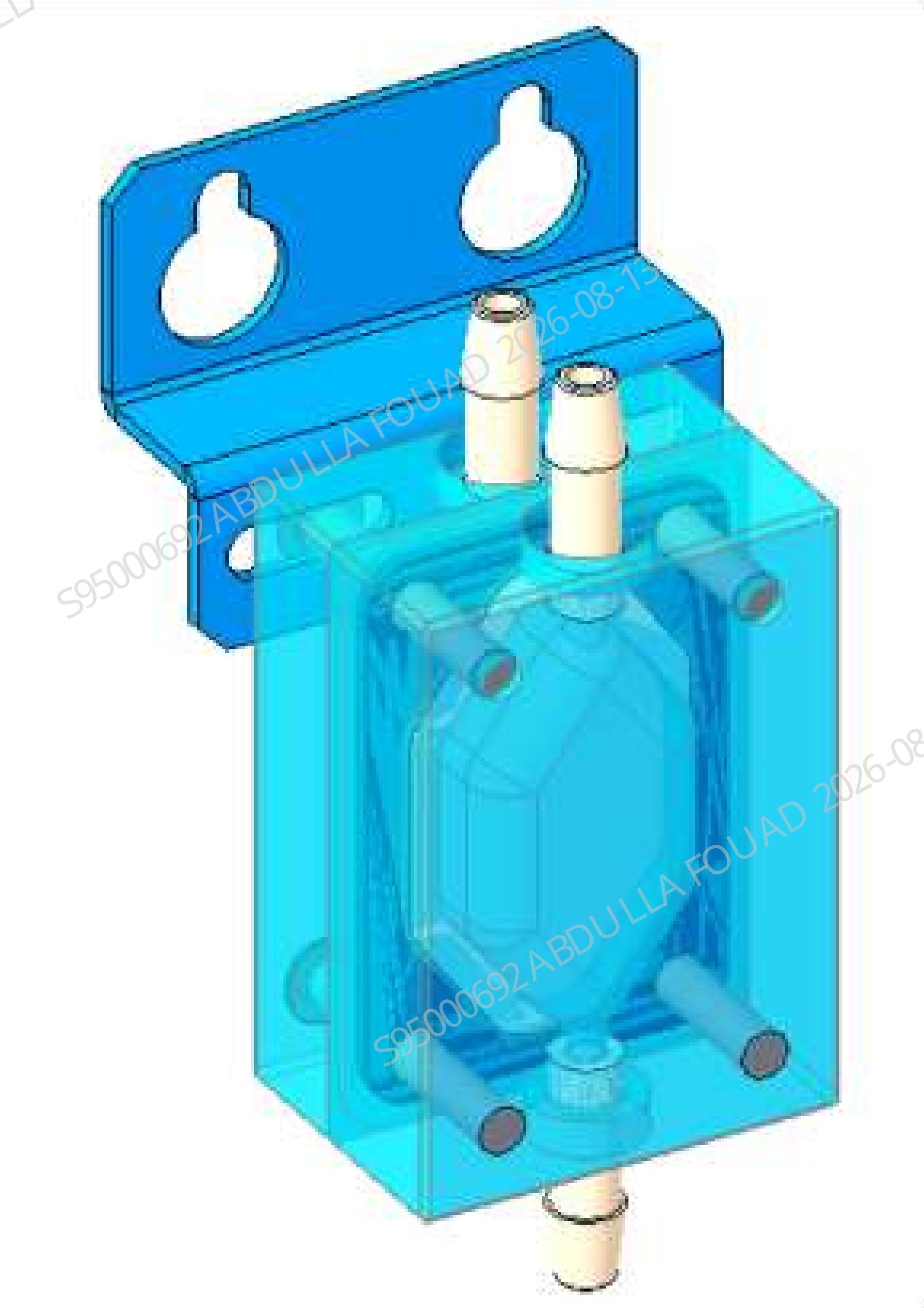


Table 8–64 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-010567-00	Sheath fluid filter assembly (3201)	N/A	N/A

Function:
Intercepting tiny bubbles in the diluent in the optical sheath fluid channel.

8.60.2 Disassembly and Assembly

Note

1. Be careful not to bend the tube.
2. Confirm that the tube is not bent
3. Confirm that the connecting tube is not leaking after replacement

Required tools

Cross-headed screwdriver
One pair of tweezers

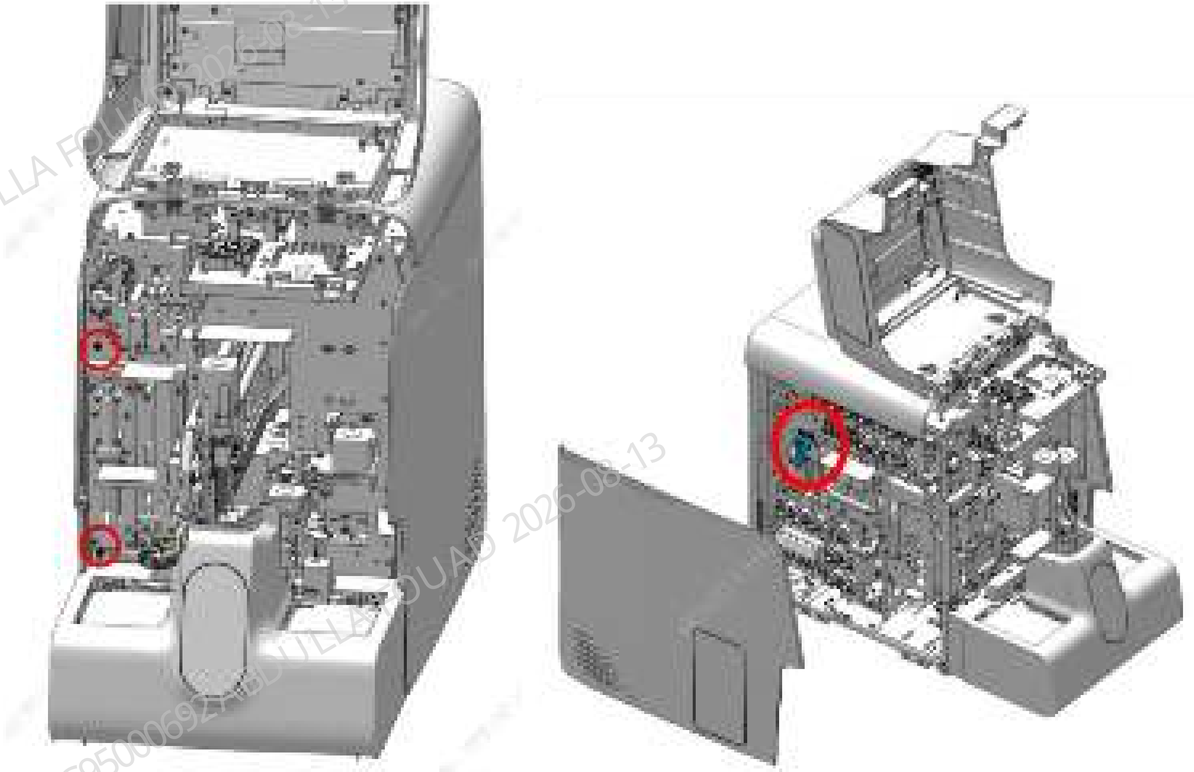
Pre-requirements

Enter the maintenance screen and empty the DIL cistern. Power off the main unit, and unplug the power cord.

Procedures

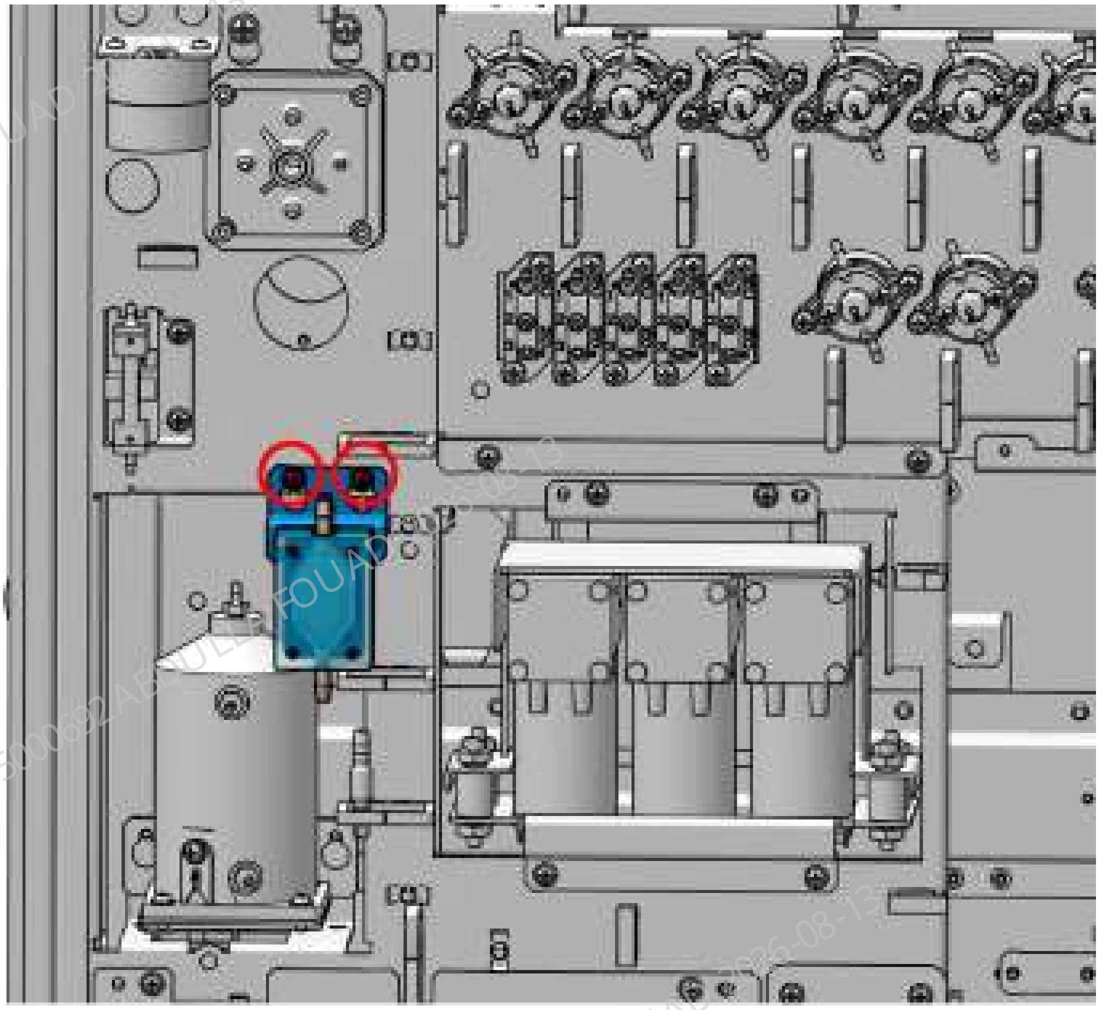
1. Turn up the front cover, dock it on the top cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–554 Removing the covers



2. Pull out the rubber tube on the sheath fluid filter. Take care not to bend the tube.
3. Use a screwdriver to loosen the screws securing the filter, as shown below, and remove the filter.

Figure8–555 Positions of the set screws of the sheath fluid filter assembly



Subsequent Requirements

Installation procedure:

1. Secure the new sheath fluid filter assembly to the main unit.
2. Replace the tube connected to the filter and insert it into the filter connector.
3. Install the left cover, flip down the front cover to reset it, and the installation is complete.

8.60.3 Commissioning and Verification

Procedures

1. Select the OV CD mode in the **Count** screen, test the background continuously for 10 times, check that the filter is not leaking, and the counting result is normal.

8.61 Filter.Inline Filter43um 1/8"I.D.Tubing

8.61.1 General Information

Figure8–556 Photo of Filter.Inline Filter43um 1/8"I.D.Tubing

Table 8–65 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
0010-10-12408	Filter.Inline Filter43um 1/ 8"I.D.Tubing	N/A	N/A

Function:
Responsible for filtering DS diluent flowing into the reagent pre-heater assembly.

8.61.2 Disassembly and Assembly

Note

Pay attention to the orientation of the filter when installing it. The filter provides English letter **FILTER**, and the filter flow direction is **F** to **R**.

Required tools

Cross-headed screwdriver
One pair of tweezers

Pre-requirements

Empty the DIL bath, power off the main unit, and unplug the power cord.

Procedures

1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–557 Diagram of Removing the Left Cover

2. Carefully remove the rubber tube from the filter, as shown in the figure below.

Figure8–558 Diagram of the Filter Location

Subsequent Requirements

Installation procedure:

1. Connect the new filter to the rubber tube.
2. Install the left cover, and flip down the front cover to reset it.

8.61.3 Commissioning and Verification

Pre-requirements

Install the Filter.Inline Filter43um 1/8"I.D.Tubing.

Procedures

1. Log in the software at the service access level. Click **Service**>>**Maintenance**>>**Reagent**>>**DS Diluent Replacement**. Confirm whether the filter leaks liquid and whether the DIL cistern can be filled.

Figure8–559 DS Diluent Replacement Interface Entrance

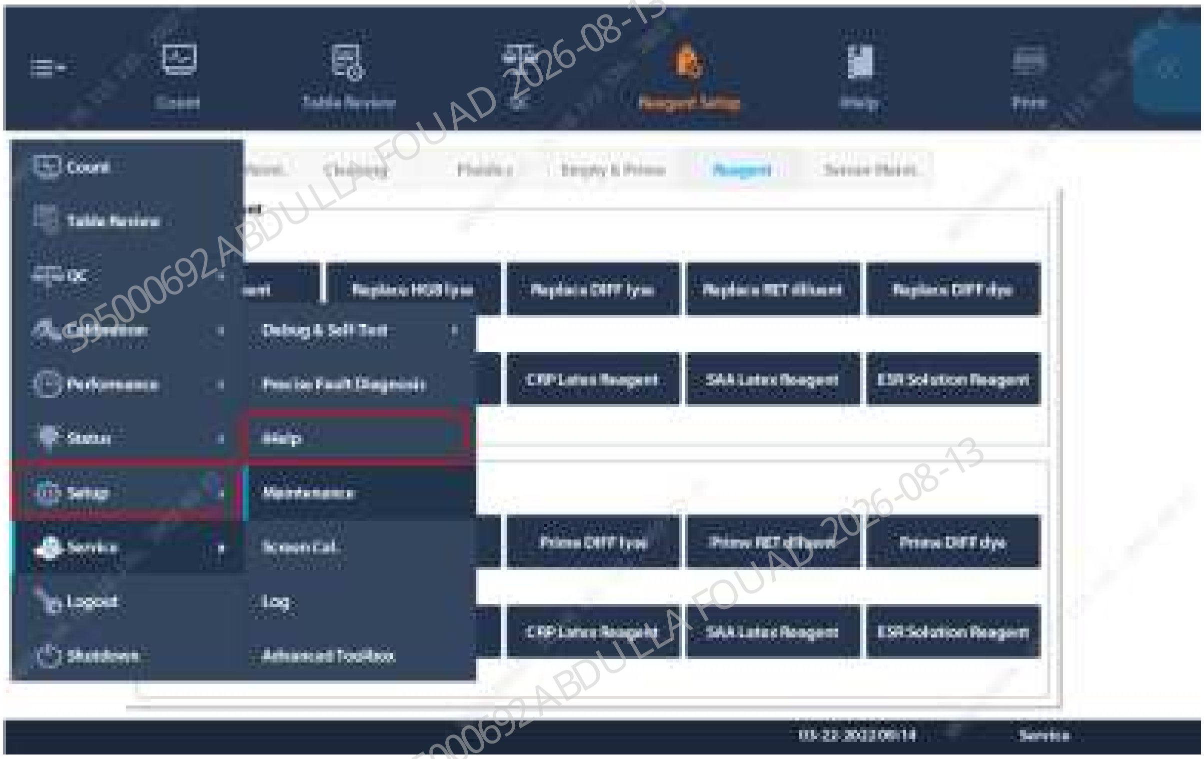
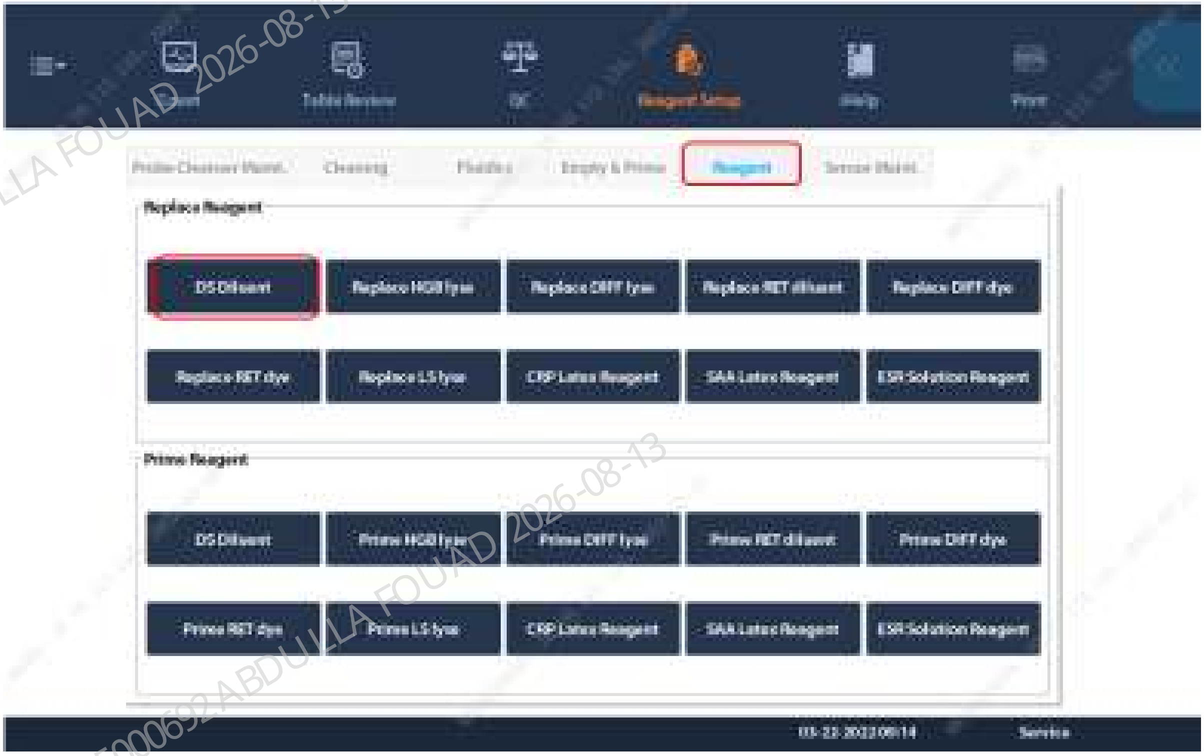


Figure8–560 DS Diluent Replacement Interface



2. Select the OV CD mode in the **Count** screen, test the background continuously for 10 times, check that the filter is not leaking, and the counting result is normal.

8.62 Filter

8.62.1 General Information

Figure8–561 Photo of filter assembly (injection molding)

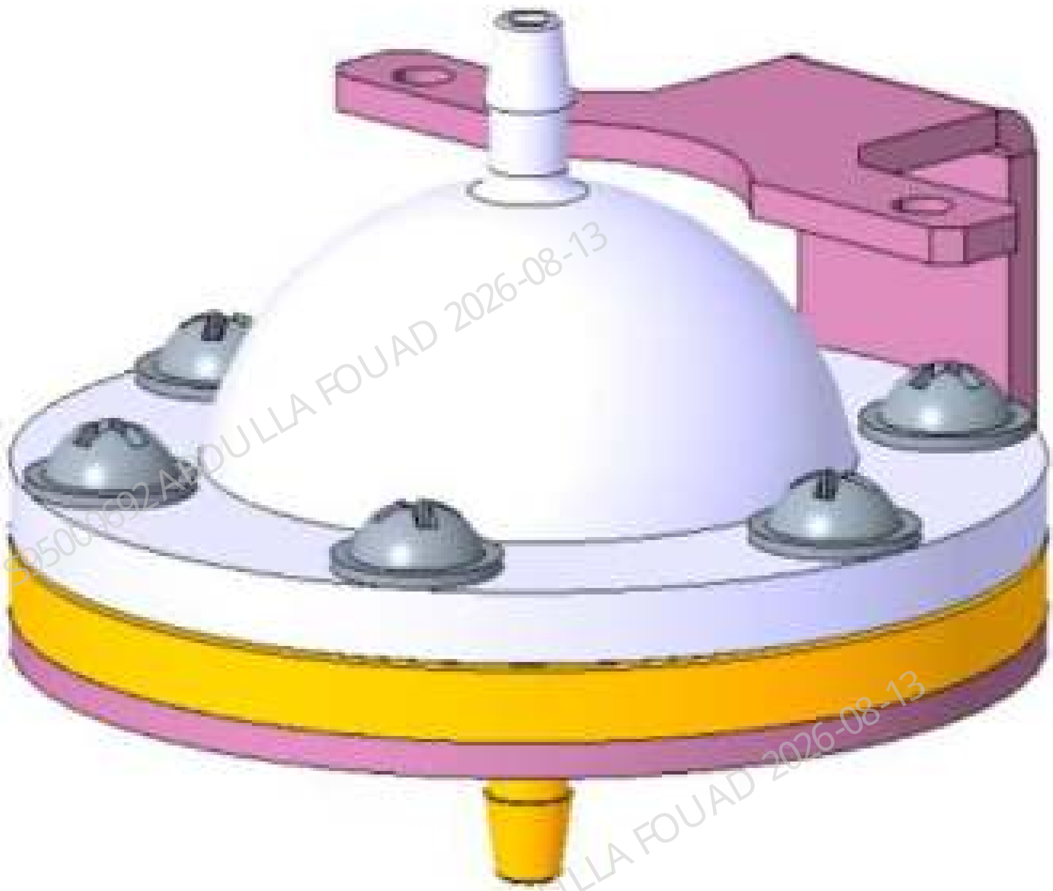


Table 8–66 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-011660-00	Filter assembly (injection molding)	N/A	N/A

Function:

Filtering the probe wipe cleaning waste.

8.62.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Turn up the front cover, lean it the top cover, and remove the left cover. The fixing screws of the left cover are shown in Figure A, and the location of the filter assembly (injection molding) is shown in Figure B.

Figure8–562 Diagram A of Location of the Left Cover Fixing Screw

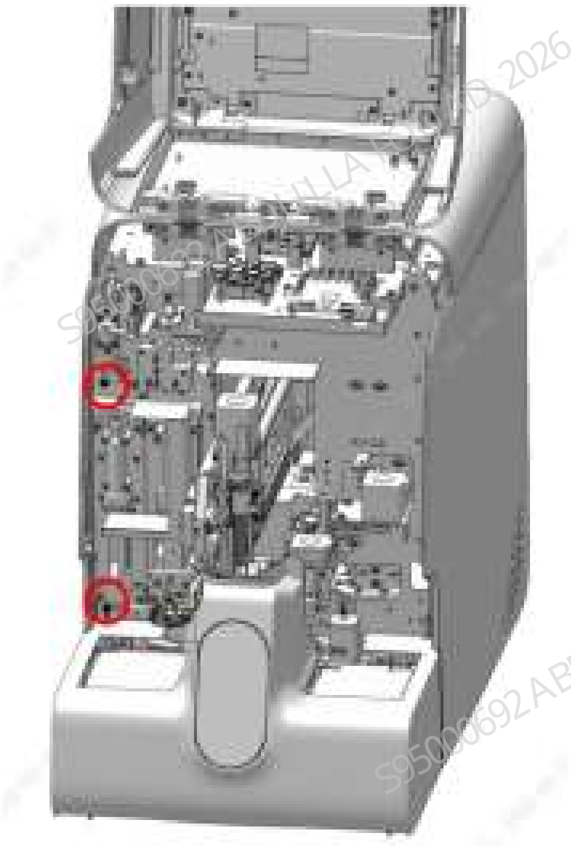
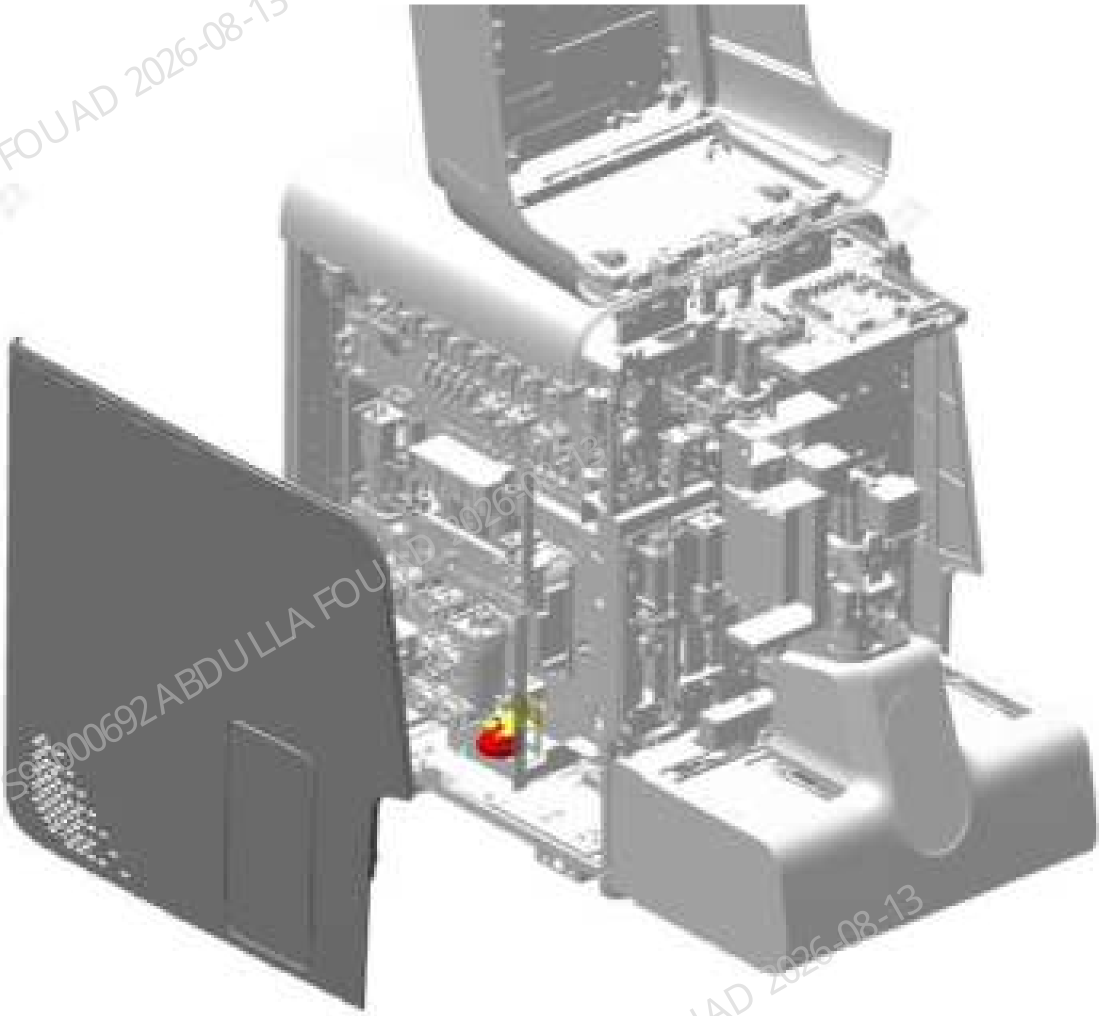
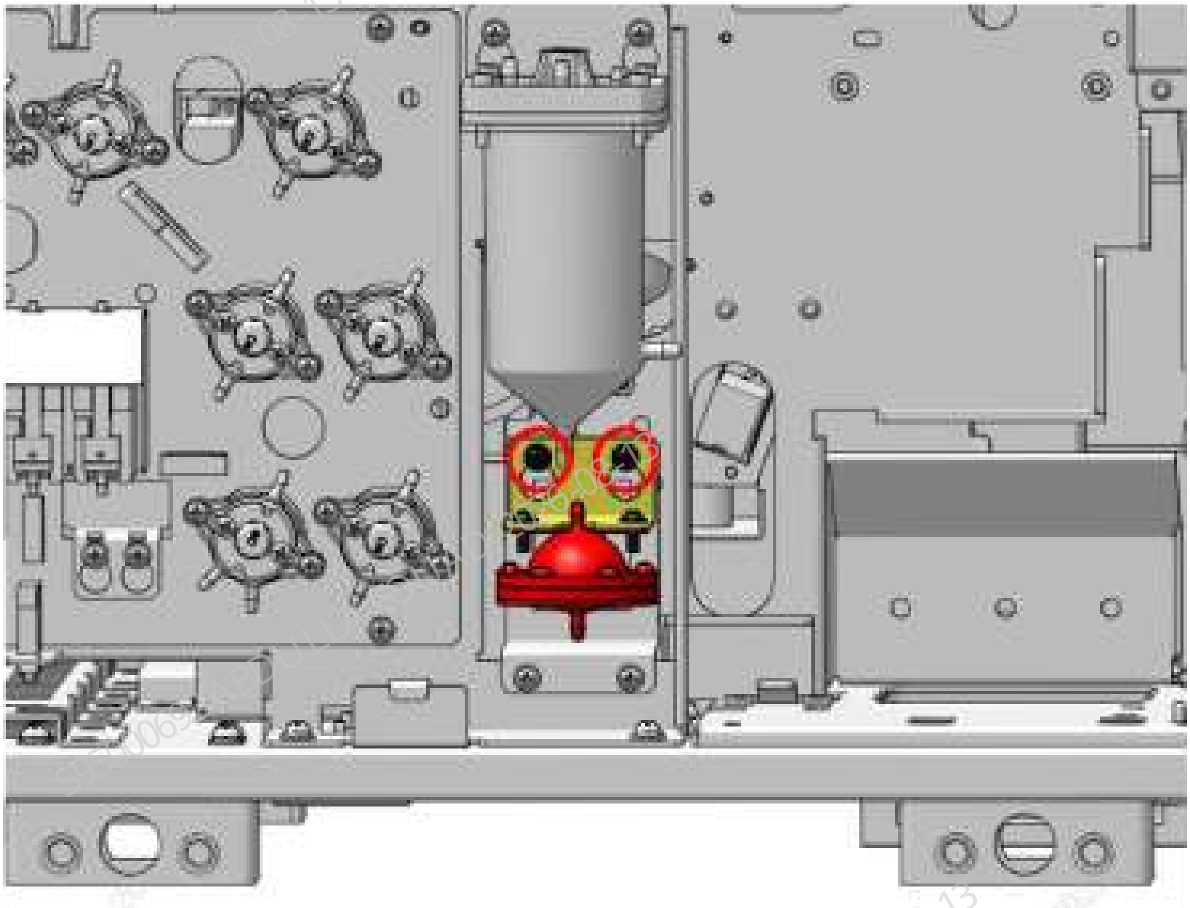


Figure8–563 Diagram B of Location of the Filter Assembly



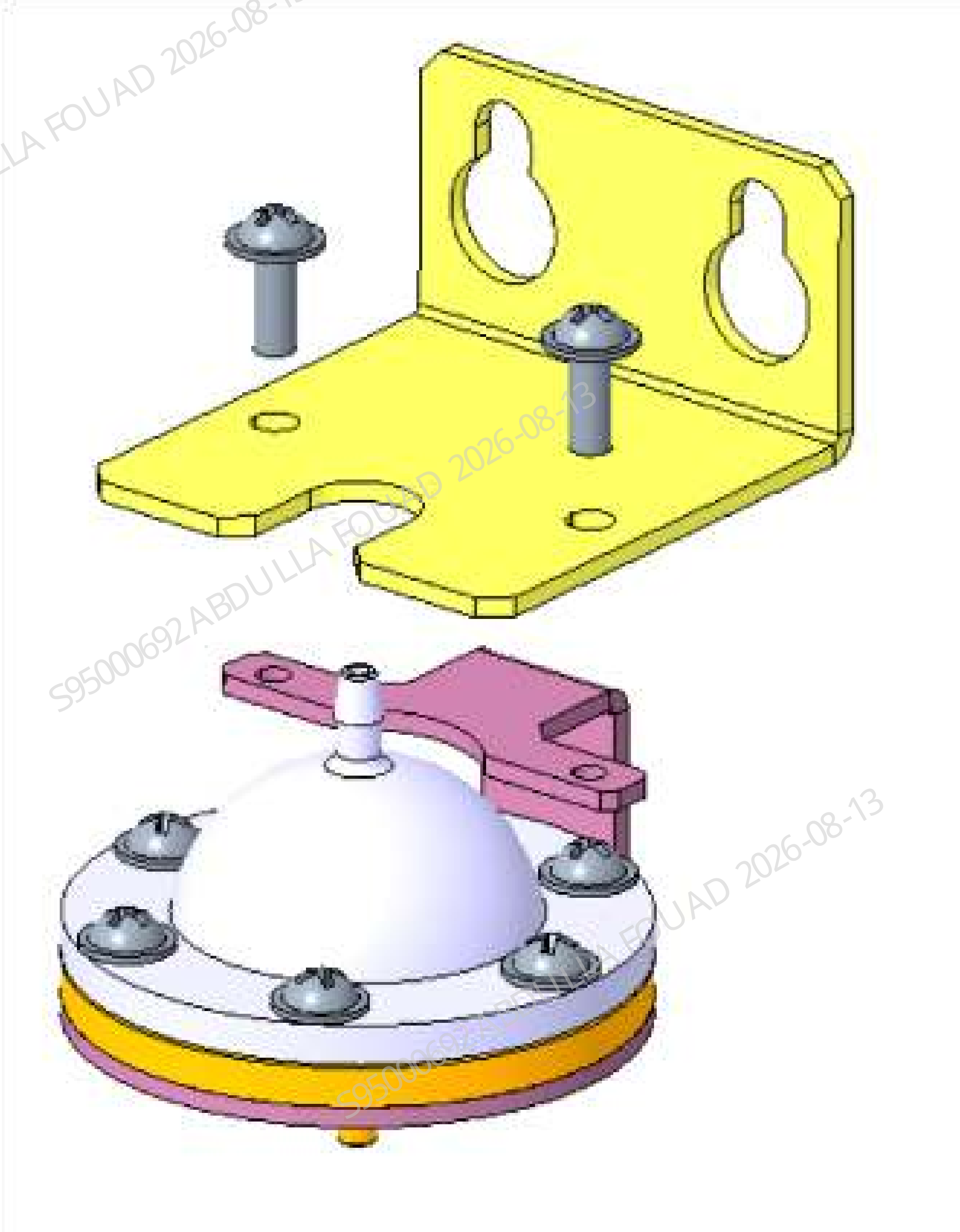
2. Loosen the two fixing screws of the filter assembly bracket, as shown below, and remove the filter assembly (including the bracket).

Figure8–564 Set screws of the filter assembly bracket



3. Loosen the two fixing screws of the filter assembly (injection), as shown below, unplug the rubber tube on the filter connector, and remove the filter assembly (injection).

Figure8–565 Removing the filter assembly (injection)



Subsequent Requirements

Installation procedure:

1. Insert the rubber tube into the connector of the filter and secure the filter to the bracket with two screws.
2. Secure the filter assembly (including bracket) to the main unit.
3. Install the left cover back, and flip down the front cover to reset it.

8.62.3 Commissioning and Verification

Required tools

Cross-headed screwdriver

Procedures

Commissioning is not required.

Verification:

1. Select **Service>>Maintenance>Cleaning>>Overall cleaning**, and perform one overall cleaning action.

2. Select **Performance>>Background** on the software screen, select the **AL WB-Blood Sample-CD** mode, and perform three background counts. The background measurement has no alarm and meets the index requirements.

8.63 Filter, 100um

8.63.1 General Information

Figure8–566 Photo of filter, 100 μm

Table 8–67 Information About the Assembly

FRU Code	**FRU Name**	**Lower-Level FRU Code**	**Lower-Level FRU Name**
082-003402-00	Filter, 100 μm	N/A	N/A

Function:
Responsible for filtering waste.

8.63.2 Disassembly and Assembly

Note

Pay attention to the orientation of the filter. There is an arrow on the filter that indicates the flow direction.

Required tools

Cross-headed screwdriver
One pair of tweezers
1 set of diagonal pliers
2 wire ties

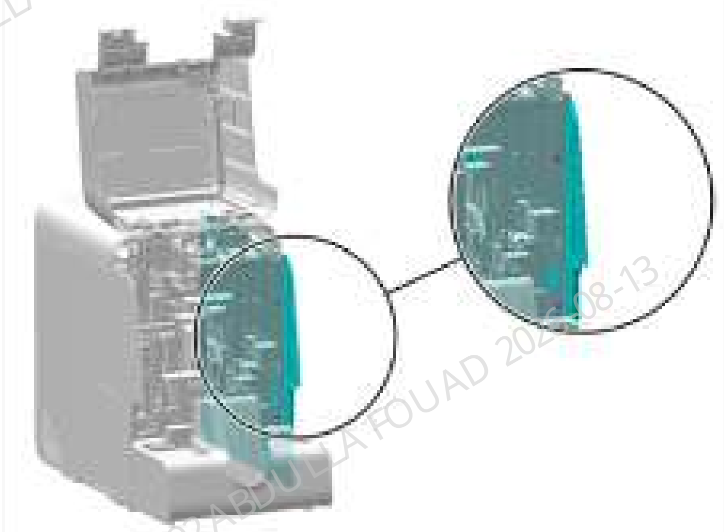
Pre-requirements

Drain the DIFF bath/RET bath/HGB bath according to the replacement needs, cut off the power supply of the analyzer, and unplug the power cord.

Procedures

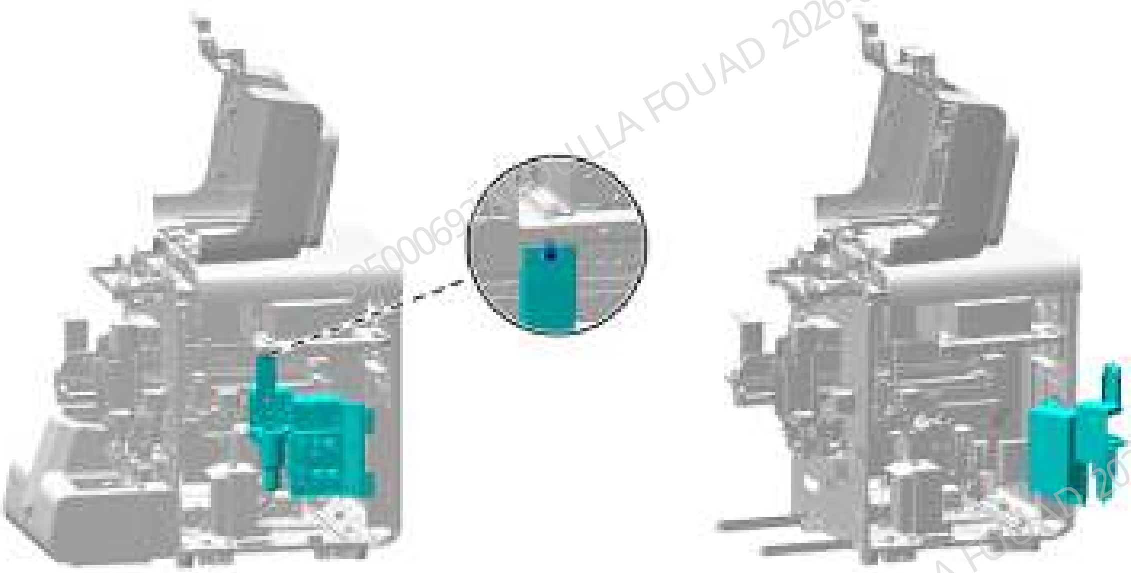
1. Turn up the front cover, lean it on the top cover, loosen the two screws that fix the left cover, and remove the left cover, as shown in the figure below.

Figure8–567 Removing the right cover



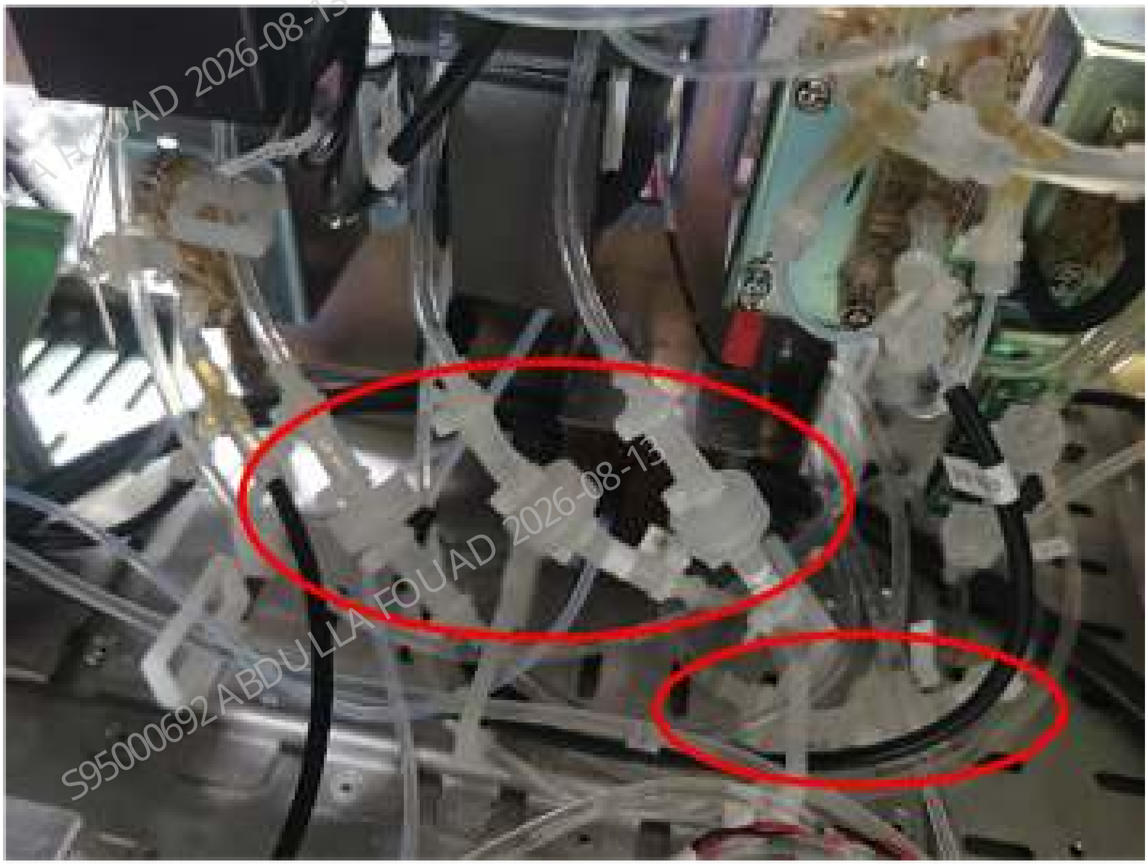
2. As shown below, unscrew the one screw that fixes the right bath subassembly; as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–568 Opening the right cover assembly



3. As shown in the figure below, remove the corresponding filter as required. From left to right there are the HGB bath waste filter, DIFF bath waste filter, RET bath waste filter, and RBC sample preparation waste filter.

Figure8–569 Opening the right cover assembly



Subsequent Requirements

Installation procedure:

1. Connect the rubber tube of the new filter, and bind the wire ties again.
2. Lift up and fasten the hook of the right bath subassembly, and tighten the screws.
3. Install the right cover, and flip down the front cover to reset it.

8.63.3 Commissioning and Verification

Pre-requirements

Install the filter, 100 um.

Procedures

Select the CD +ESR mode on the count interface, test the background continuously 10 times, check that the filter and reaction bath are not leaking, and the counting result is normal.

8.64 Dye Pump Assembly (Include RET)

8.64.1 General Information

Figure8–570 Photo of Flyorescence Dye Pump Asm (Include RET)

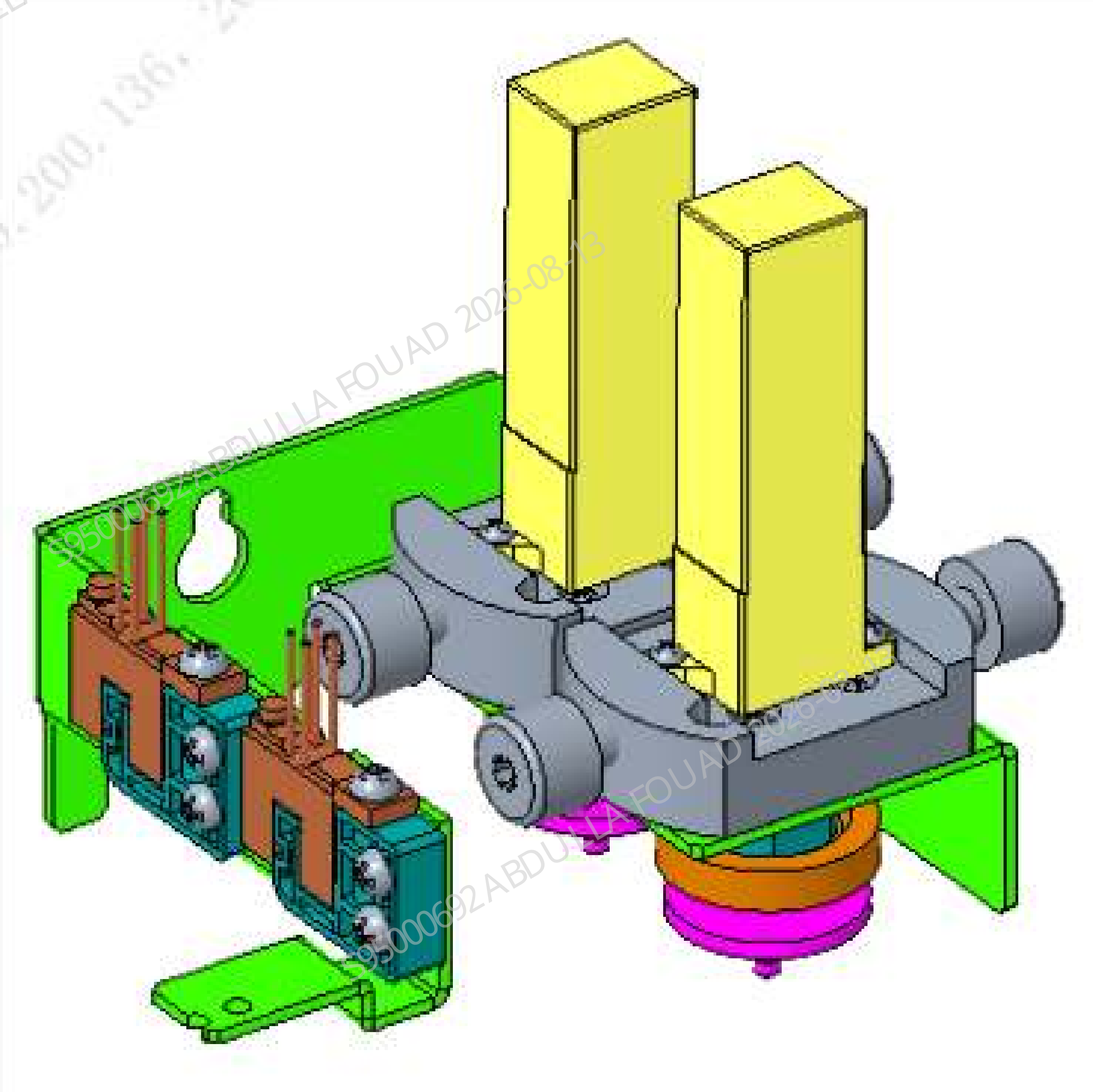


Table 8–68 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-074579-00	Fluorescence Dye Pump Assembly (With RET)	N/A	N/A

Function:
Absorbing dye reagent and detecting reagent.

8.64.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.
3. Be careful not to damage the connector when removing the tube.

Required tools

Cross-headed screwdriver
One pair of cutting nippers

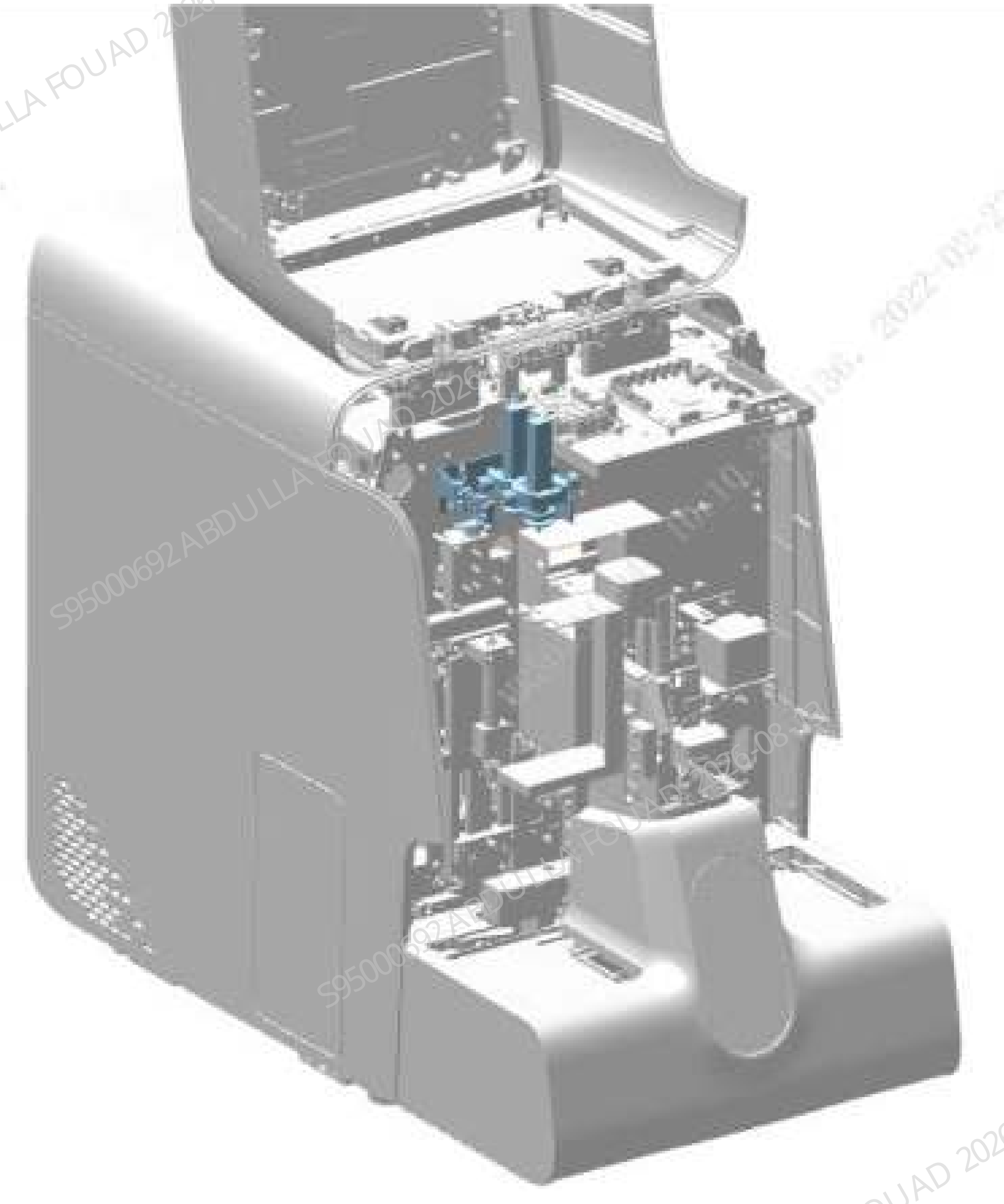
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

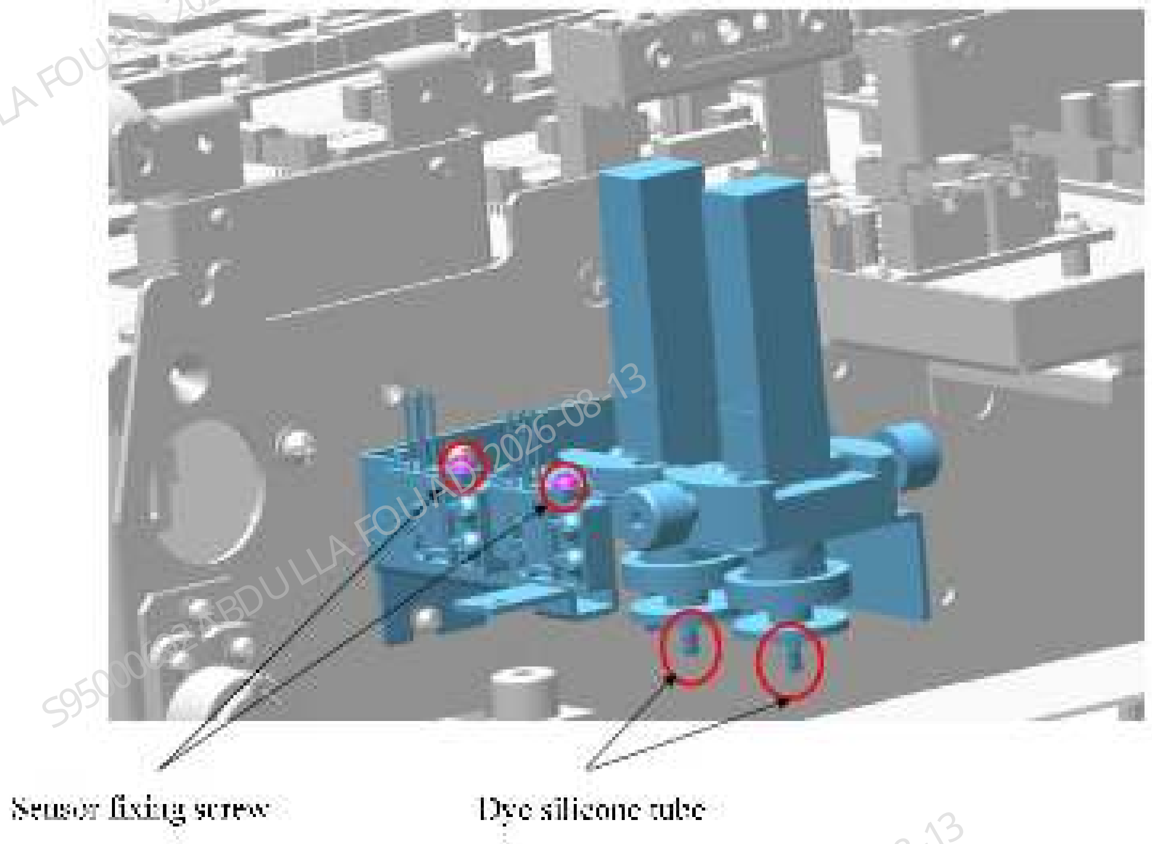
1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–571 Diagram of Removing the Flyorescence Dye Pump Assembly (Include RET)



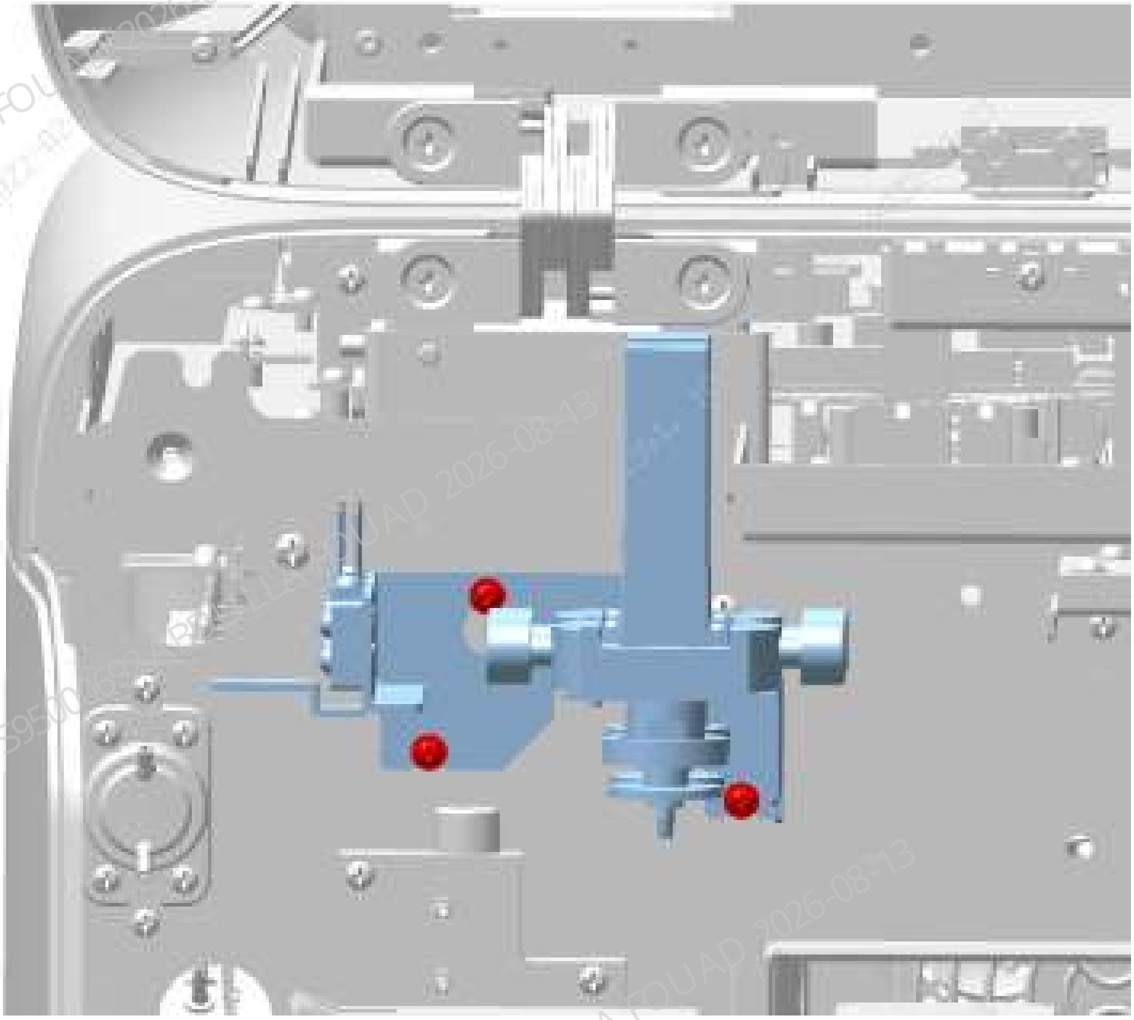
2. Loosen the wiring terminal of the sensor on the assembly, pull out the dye rubber tube connected to the reaction bath at the lower end of the metering pump, remove the M3X8 cross recessed panhead screws on the two dye liquor sensors respectively, and then take out the rubber tube connected to the dye reagent probe from the sensor, as shown in the figure below.

Figure8–572 Diagram of Removing the Rubber Tube of Flyorescence Dye Pump Asm (Include ERT)



3. Loosen the three screws fixing the assembly, as shown in the figure below, and remove the assembly outward. Note not to fold the tube.

Figure8–573 Diagram of Screw Fixing Location of the Flyorescence Dye Pump Asm (Include RET)



4. Remove the snap connector on the dye metering pump respectively.

Subsequent Requirements

Installation procedure:

1. Use a snap connector to connect the rubber tube connected to the dye reagent probe to the metering pump (Note: Connect them correctly according to the pipe label), clip the rubber hose into the groove of the dye sensor, and then tighten the M3x8 cross recessed panhead screws locking the dye sensor.
2. Install the assembly back into the analyzer and tighten the screws to fix it.
3. Connect the dye pipeline connected to the reaction bath to the metering pump.

NOTICE

Connect them correctly according to the pipe label.

4. Reset the front cover.

8.64.3 Commissioning and Verification

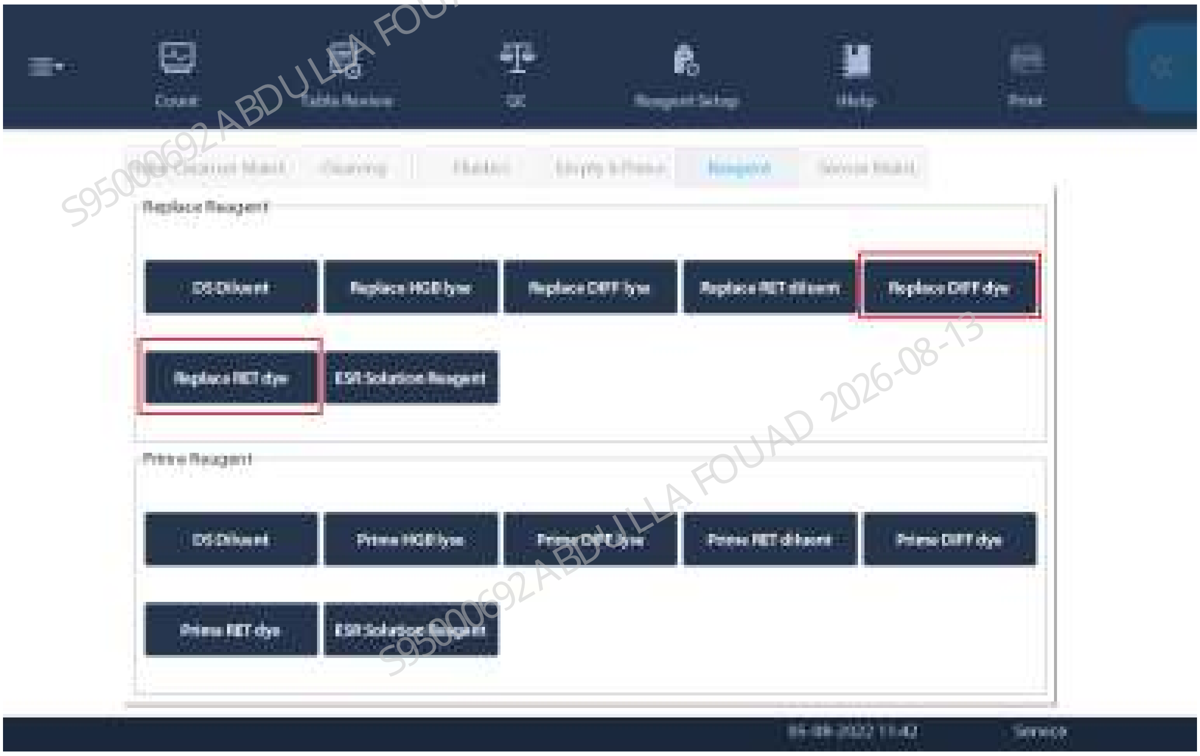
Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

1. Select **Service**>>**Maintenance**>>**Reagent**>>**Reagent Replacement**, click **FD Dye** and **FR Dye** (depending on the configuration) respectively, and complete the reagent replacement operation (as shown in the figure below).

Figure8–574 Reagent Replacement screen



2. Select **Service**>>**Commissioning & Self-Test**>>**Sensor Commissioning**>>**Dye Sensors**, click **Calibrate All**, check whether the test voltage is within the reference range. If yes, click **Save All**. (As shown in the following figure.)

Figure8–575 Dye Liquor Sensor Gain Calibration Interface

8.65 PHOTOELEC Optical Sensor 940nm

8.65.1 General Information

Figure8–576 Photo of PHOTOELEC Optical Sensor 940nm

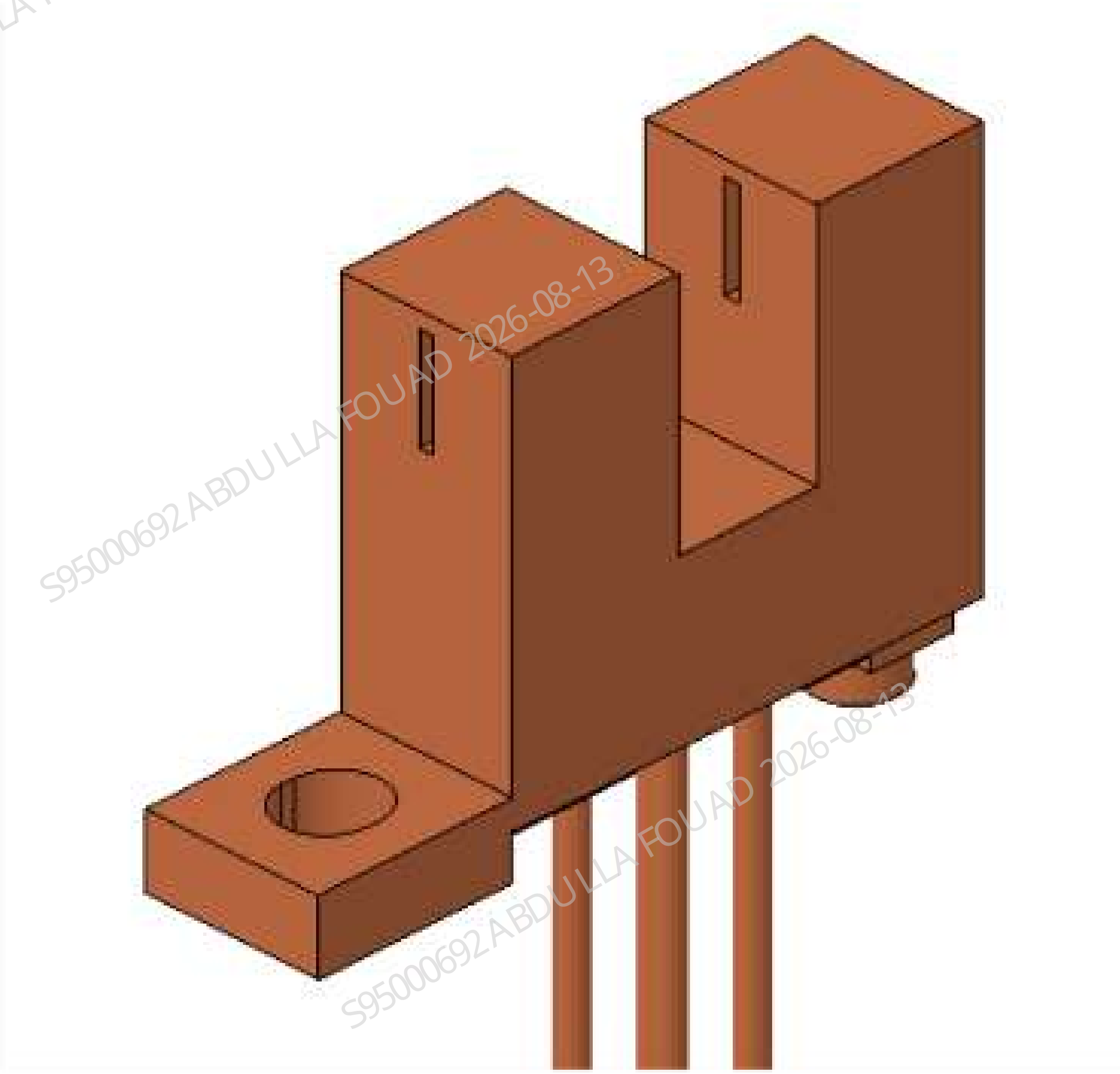


Table 8–69 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
011-000247-00	Optical switch COJ-MY03-04 (Customized)	N/A	N/A

Function:

Detecting the dye.

8.65.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.

Required tools

Cross-headed screwdriver
One pair of cutting nippers

Pre-requirements

After performing the decompression operation, power off the main unit, and unplug the power cord.

Procedures

1. Turn up the front cover, lean it on the top cover, and find the flyorescence dye pump assembly (for the one with RET, see Figure A below; for the one without RET, see Figure B below).

Figure8–577 Diagram A for Location of Flyorescence Dye Pump Assembly (Include RET)

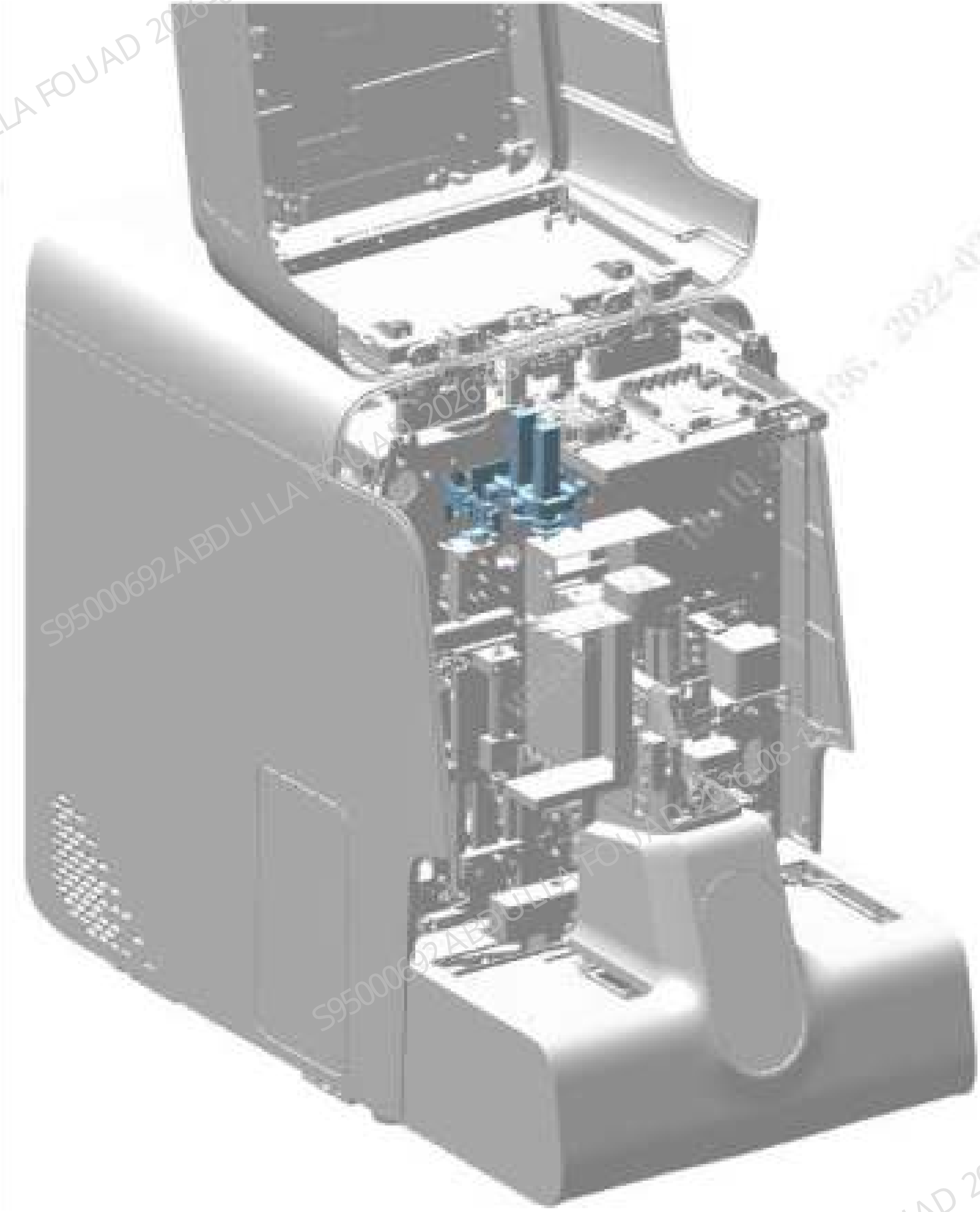
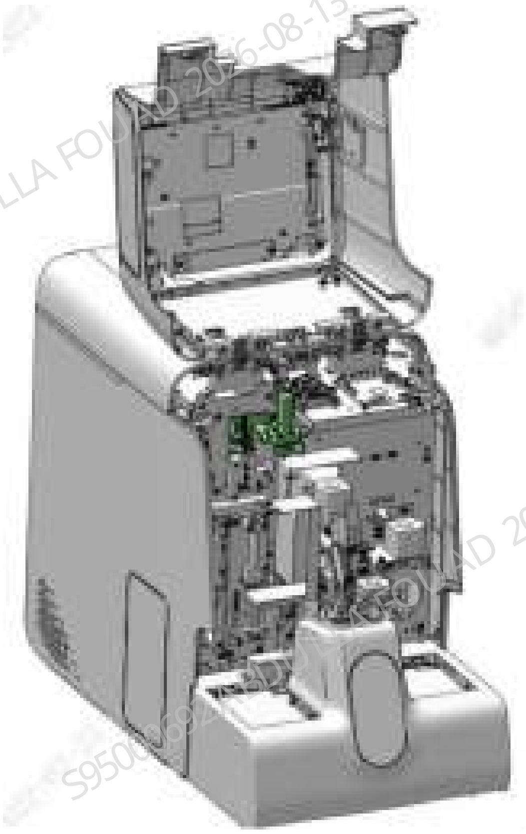


Figure8–578 Diagram of Location of the Flyorescence Dye Pump Asm (Exclude RET)



2. Loosen the three screws fixing the flyorescence dye pump assembly (for the one with RET, see Figure A below; for the one without RET, see Figure B below), and pull out the assembly slightly.

Figure8–579 Diagram of Screw Fixing Location of the Flyorescence Dye Pump Asm (Include RET)

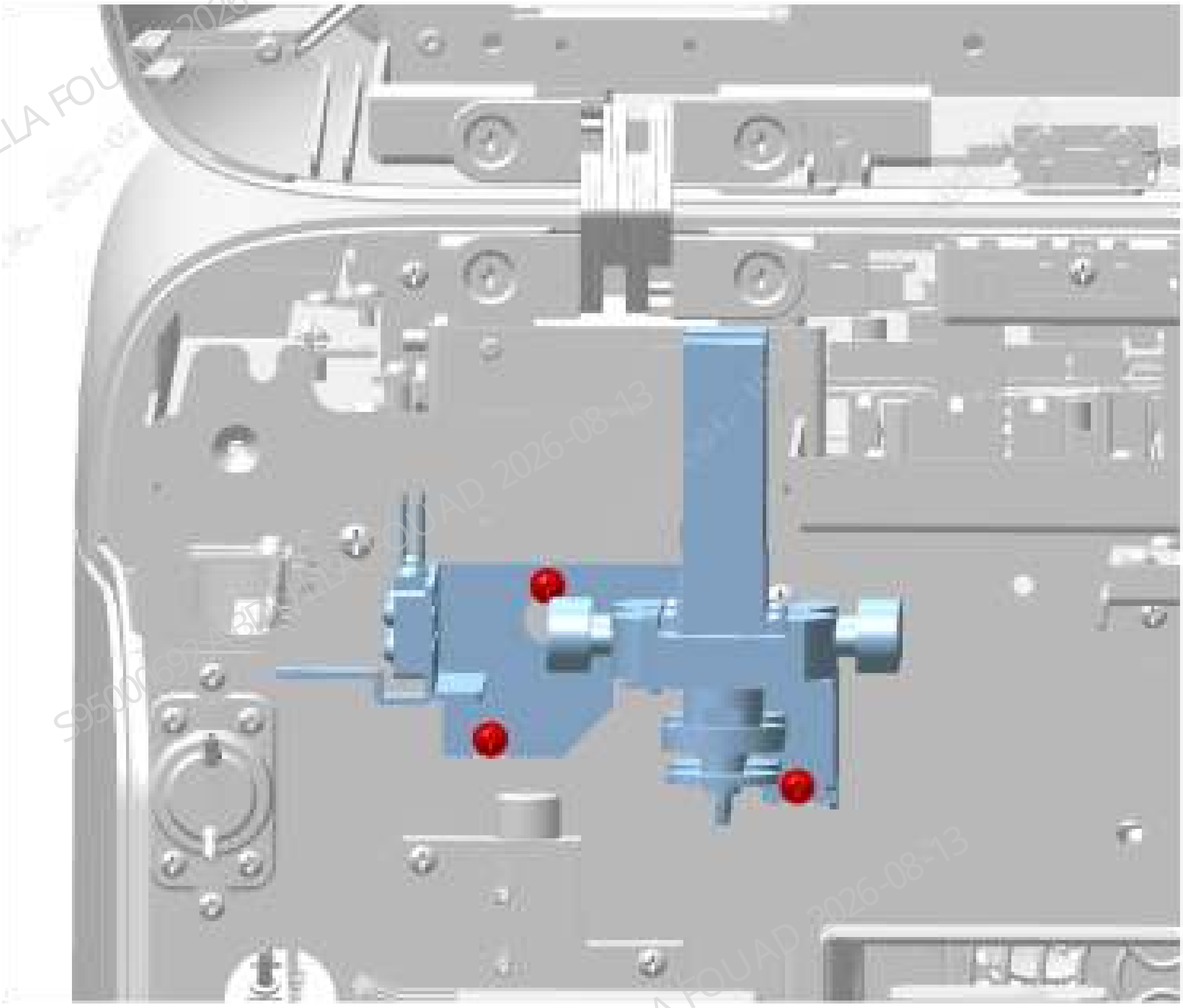
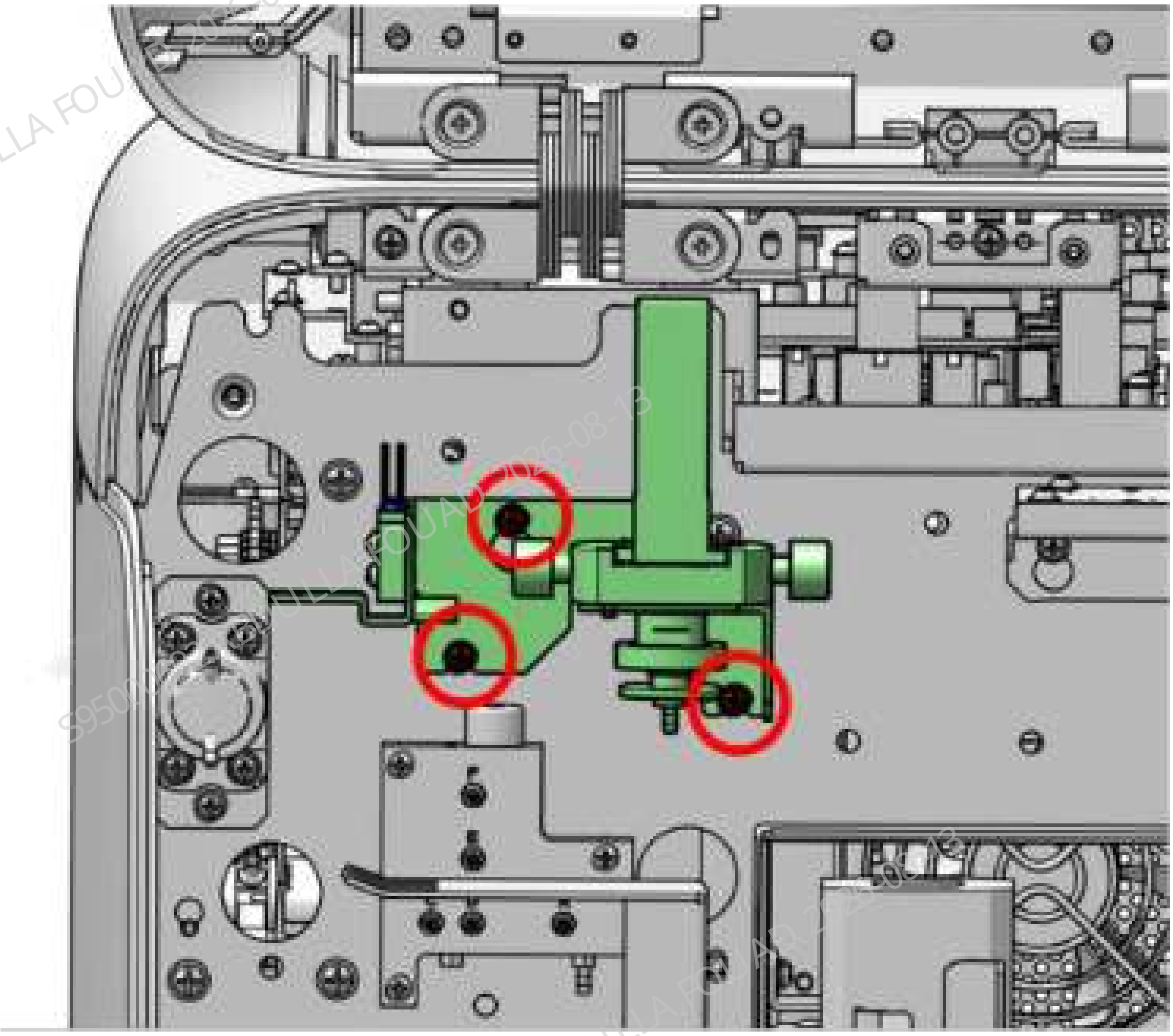


Figure8–580 Diagram of Screw Fixing Location of the Flyorescence Dye Pump Asm (Exclude RET)



3. Loosen the wiring terminals of the sensor on the assembly, loosen the screws that fix the optical switch, and remove the optical switch, as shown in the figure below.

Figure8–581 Diagram of Removing the Transmissive Sensor (Include RET)

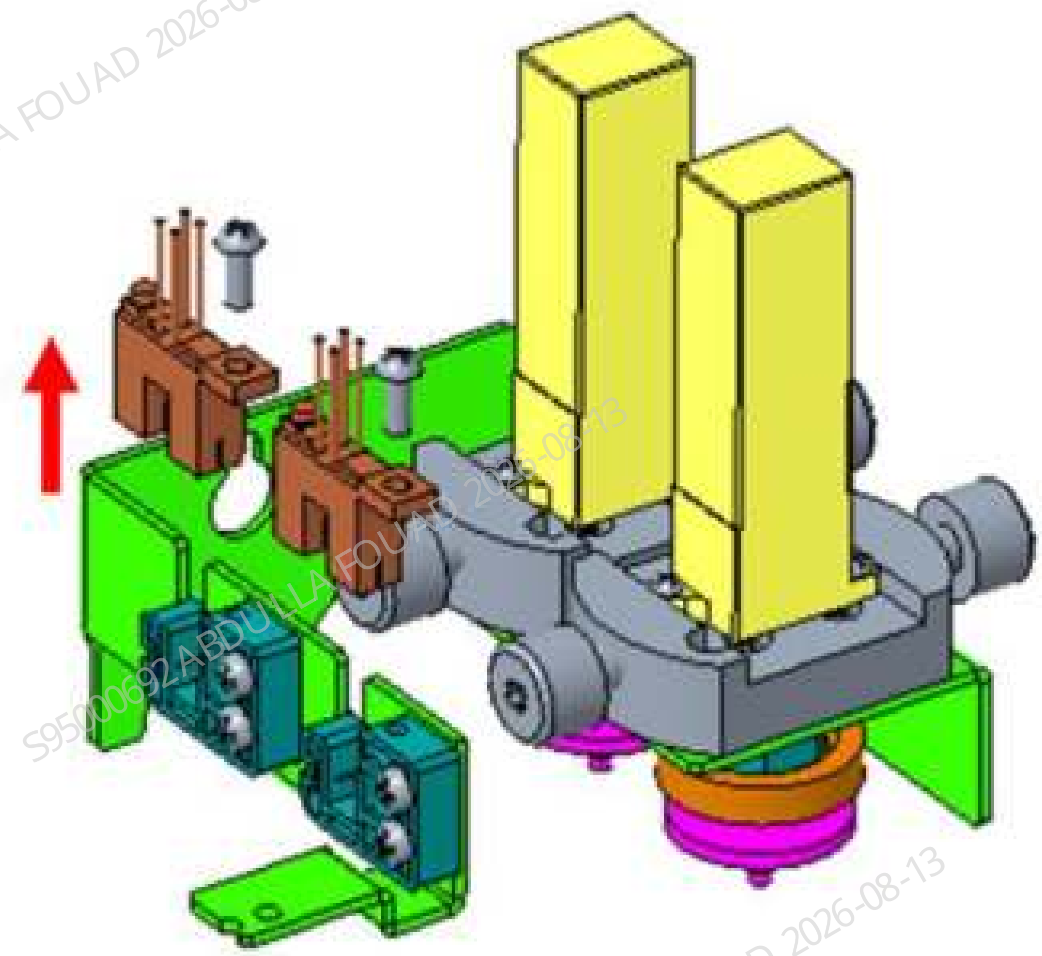
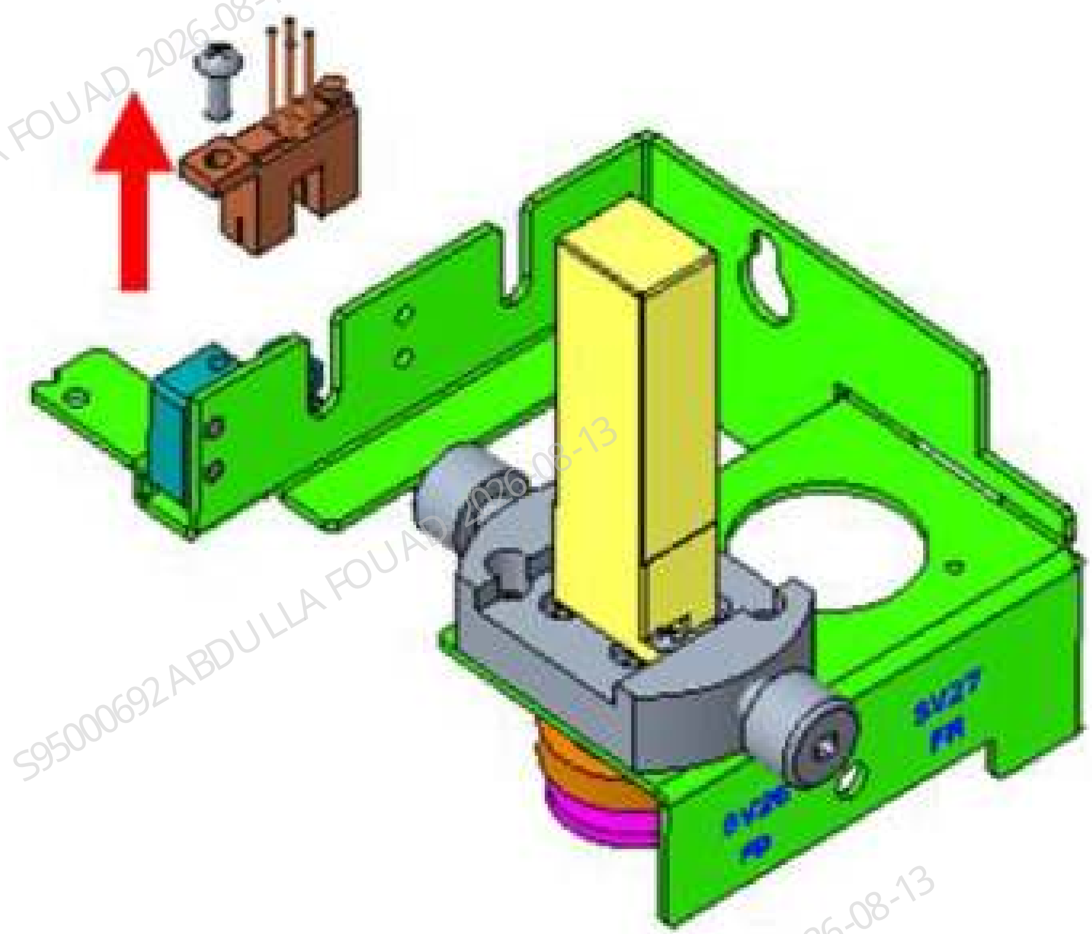


Figure8–582 Diagram of Removing the Transmissive Sensor (Exclude RET)



Subsequent Requirements

Installation procedure:

1. Reconnect the terminals of the optical switch, and tighten the screws to fix the optical switch.
2. Install the flyorescence dye pump assembly (include or exclude RET) back into the analyzer and tighten the three screws.
3. Turn over the front cover to reset it.

8.65.3 Commissioning and Verification

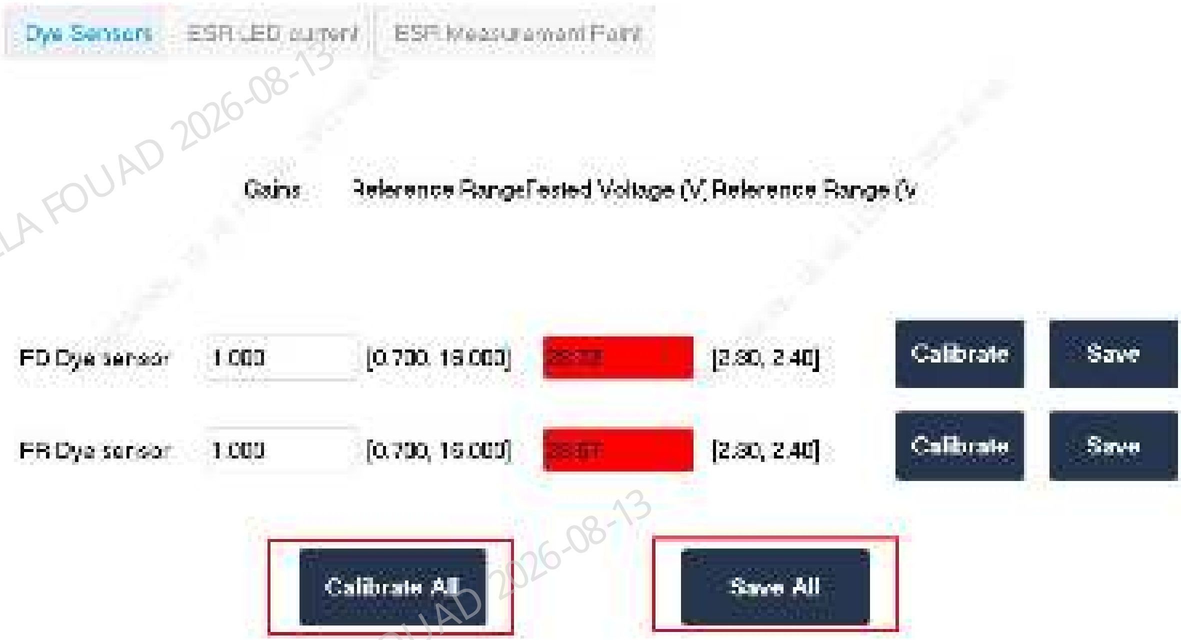
Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

1. After startup, log in at the service access level. Click **Service>>Commissioning & Self-Test>>Sensor Commissioning** on the software screen in sequence.

2. Select **Dye Sensors**. If the gain value and test voltage are displayed in red on the accessed interface, as shown in the figure below. Click **Calibrate All**, and follow the prompts to confirm that the **gain value** of the dye sensor and the **detection voltage** are within the range.



Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.66 20ul diaphragm pump assembly

8.66.1 General Information

Figure8–583 Photo of 20 μL dosing pump assembly

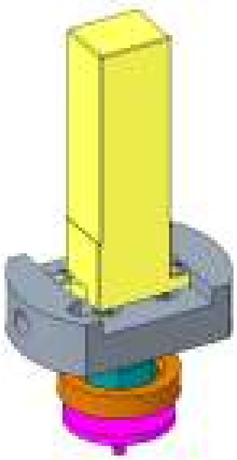


Table 8–70 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-005240-00	20 μL Dosing Pump Assembly	115-061171-00	Three-position ways LVM valve assembly (with valve board and connector)

Function:
Used to quantitatively inject FD dye or FR dye into the reaction bath.

8.66.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.

Required tools

Cross-headed screwdriver
One pair of cutting nippers

Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Turn up the front cover, lean it on the top cover, and find the flyorescence dye pump assembly (for the one with RET, see Figure A below; for the one without RET, see Figure B below).

Figure8–584 Diagram A for Location of Flyorescence Dye Pump Assembly (Include RET)

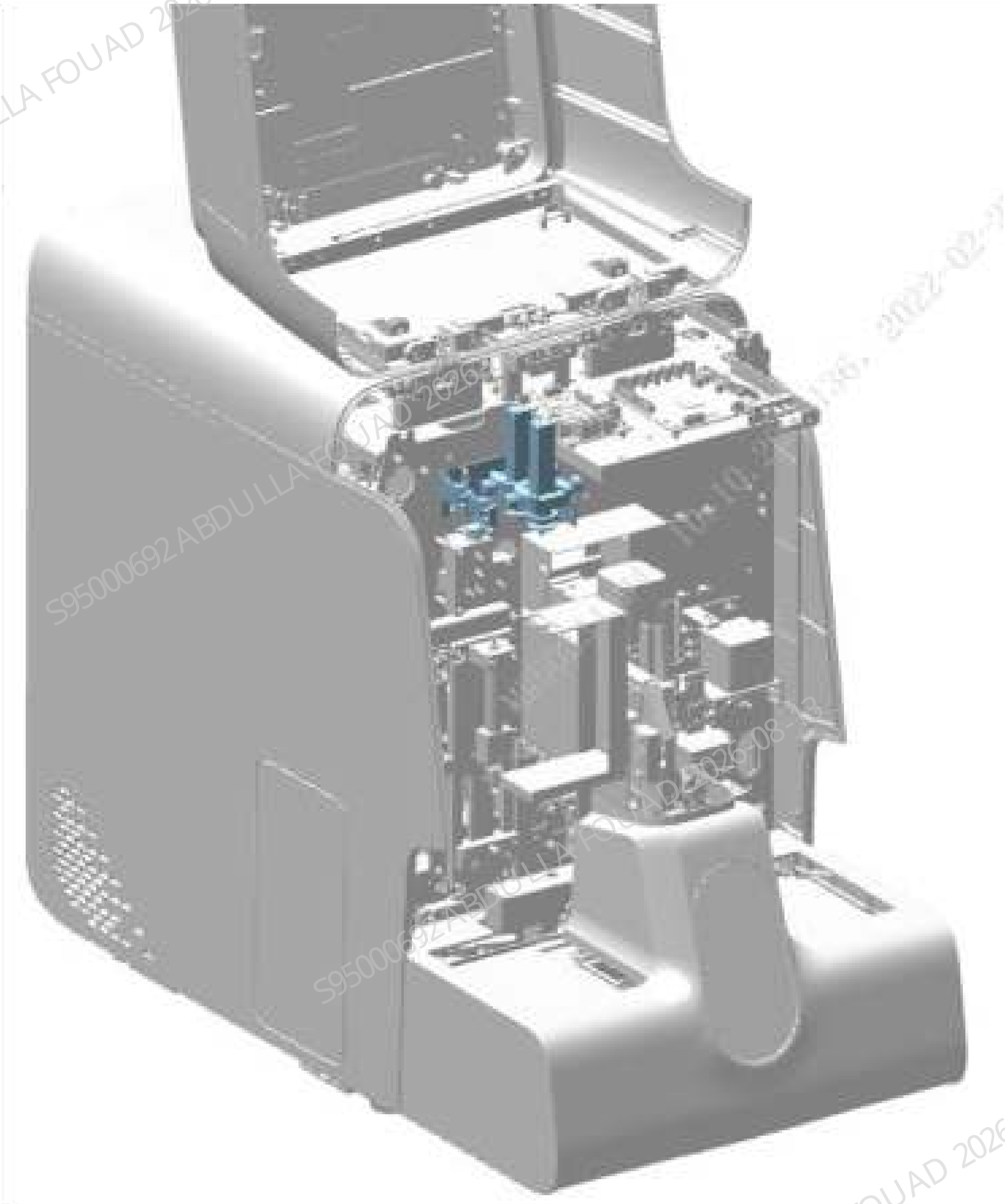
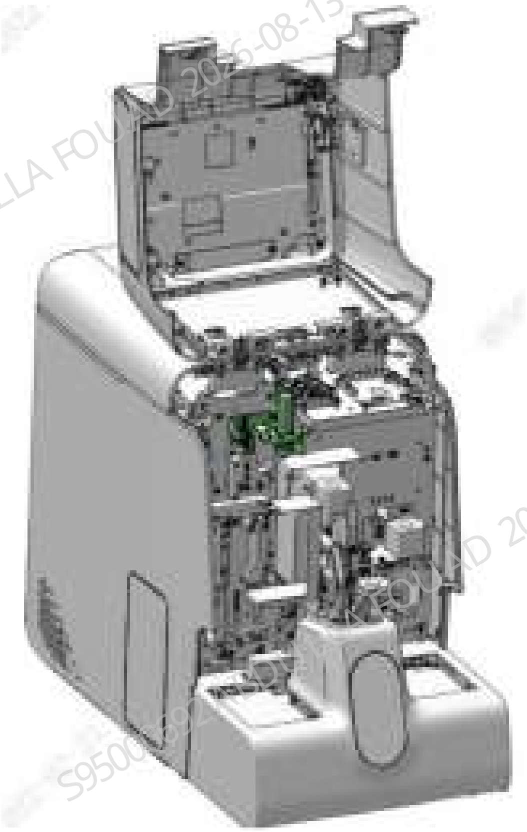


Figure8–585 Diagram B of Location of the Flyorescence Dye Pump Asm (Exclude RET)



2. Loosen the three screws fixing the flyorescence dye pump assembly (for the one with RET, see Figure A below; for the one without RET, see Figure B below), and pull out the assembly slightly.

Figure8–586 Diagram A of Screw Fixing Location of the Flyorescence Dye Pump Asm (Include RET)

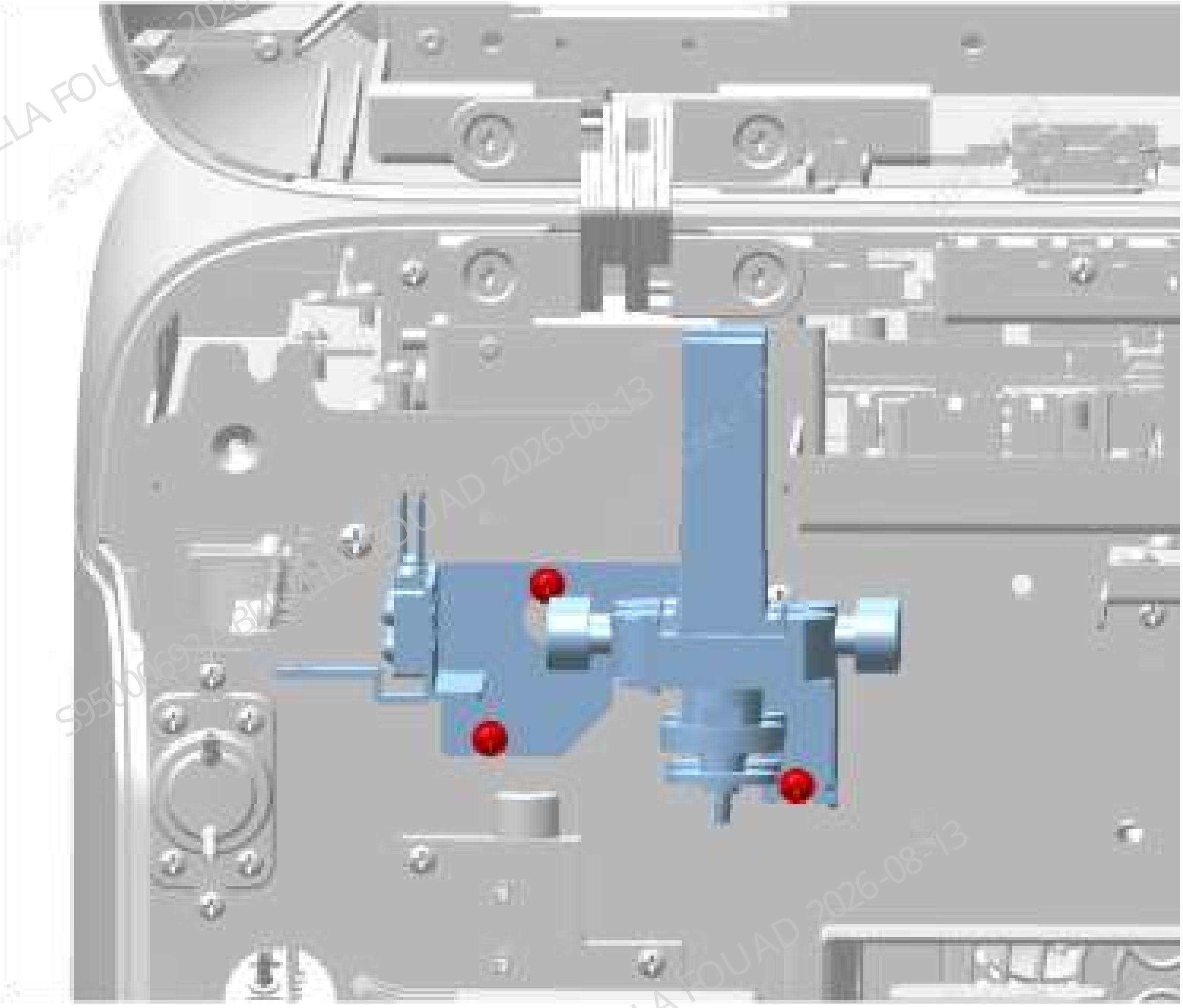
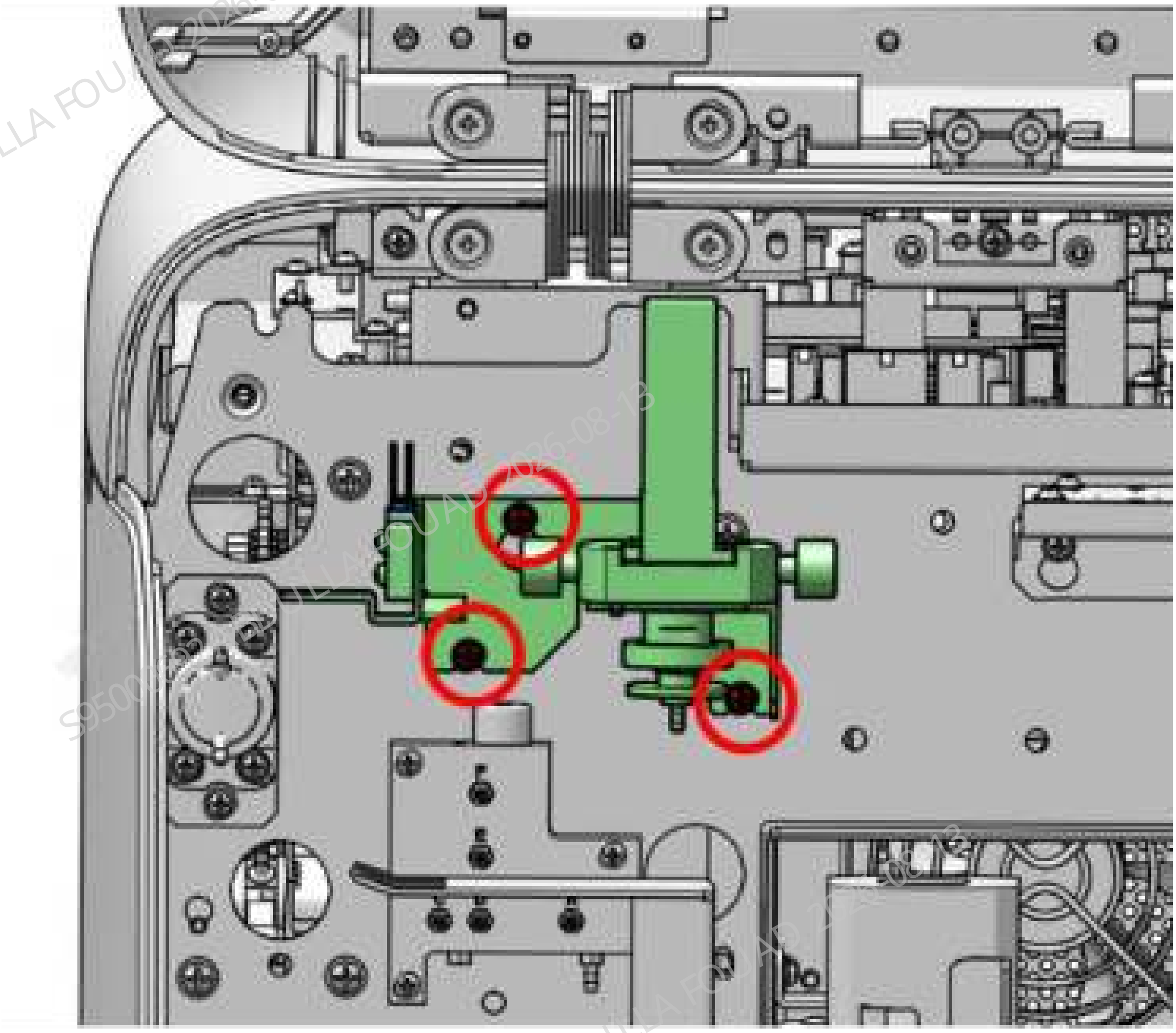


Figure8–587 Diagram A of Screw Fixing Location of the Flyorescence Dye Pump Asm (Exclude RET)



3. Loosen the terminals of the 20 μL diaphragm pump assembly and the rubber tube of the bottom connector, loosen the two set screws of the 20 μL diaphragm pump assembly, unscrew the two connectors (CNS 1/4-28), and remove the 20 μL diaphragm pump assembly, as shown in the figure below.

Figure8–588 Diagram of Removing 20 μL Metering Pump Assembly (Include RET)

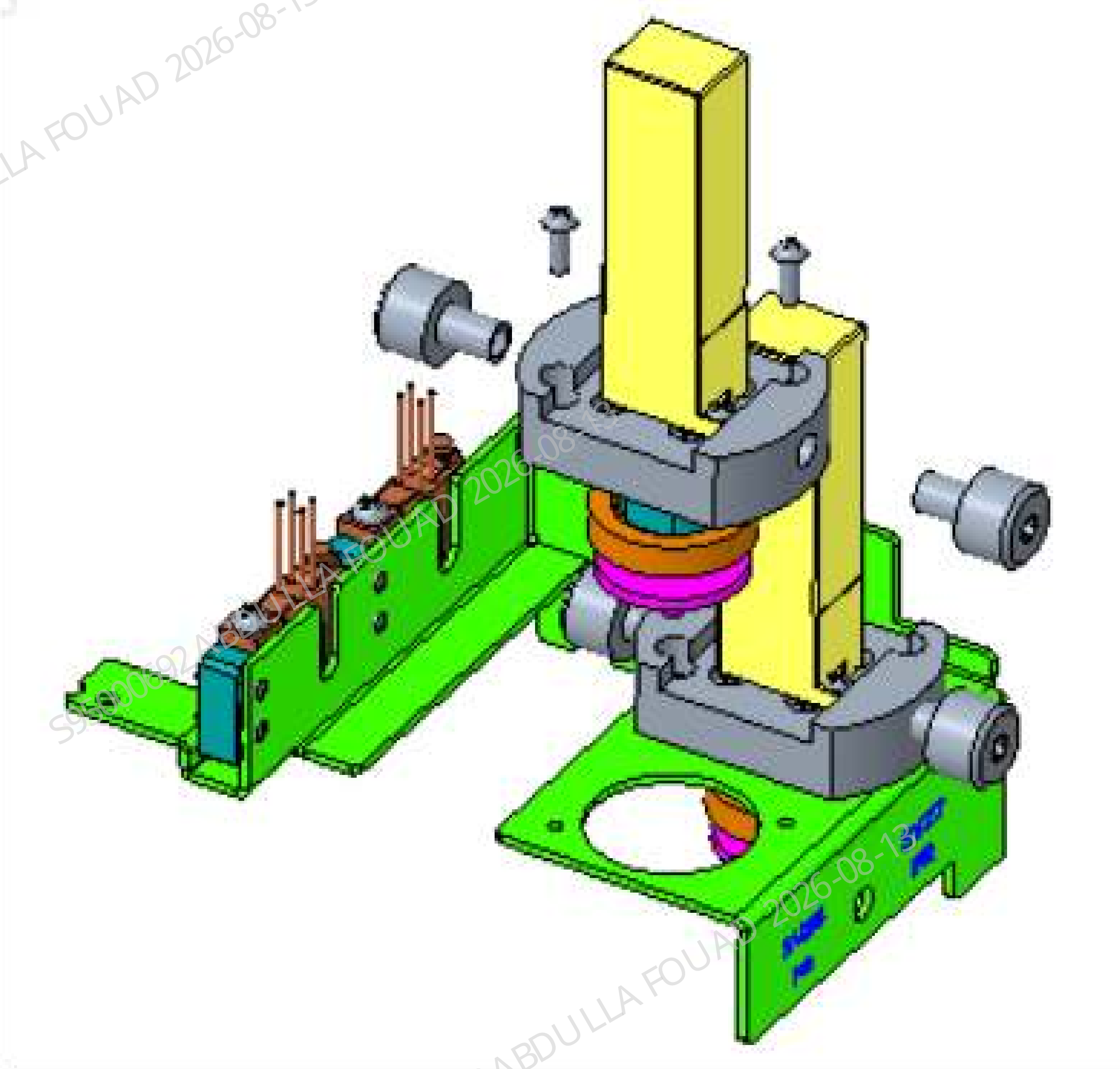
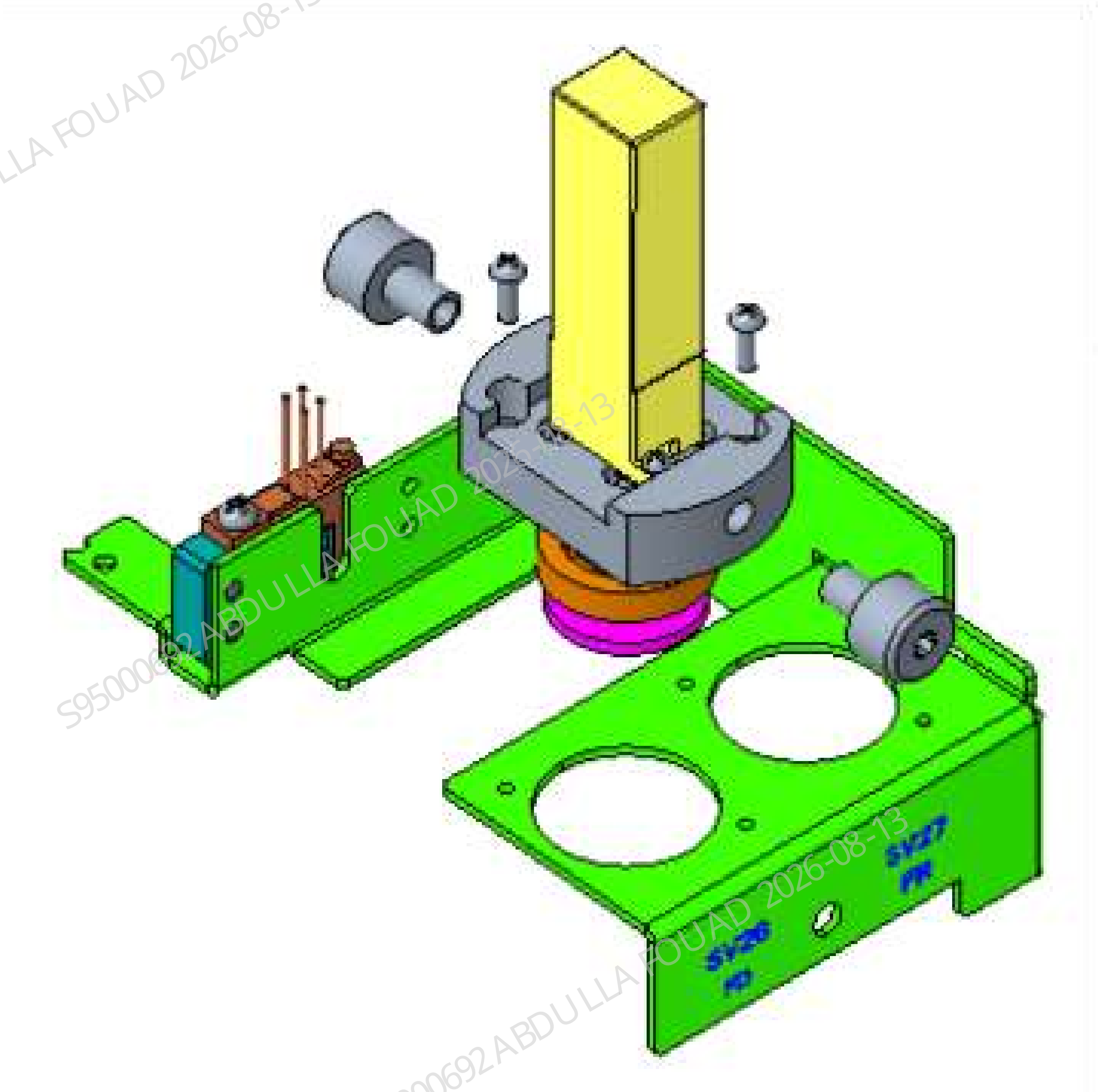


Figure8–589 Diagram of Removing 20 μL Metering Pump Assembly (Exclude RET)



Subsequent Requirements

1. Install two connectors (CNS 1/4-28) on the 20 μL diaphragm pump assembly, tighten two screws to fix the 20 μL diaphragm pump assembly to the sheet metal bracket, and connect the terminals of the 20 μL diaphragm pump assembly and the bottom connector of the rubber tube.
2. Install the flyorescence dye pump assembly (include or exclude RET) back into the analyzer and tighten the three screws.
3. Turn over the front cover to reset it.

8.66.3 Commissioning and Verification

Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

Commissioning:

1. Start the analyzer, and log in the software at the service access level.
2. Click **Reagent Setup**, click the **Setup** button, and use the barcode scanner to input the barcode of each dye. Then click **OK**.
3. Check whether the software screen reports an error of **No Dye**, if so, click the **Eliminate Error** button.

Verification:

1. Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.
2. Test controls or calibrators or random samples, and confirm that the DIFF/RET scattergram of the corresponding channel and its measured value are normal.

8.67 Dye Pump Assembly (No RET)

8.67.1 General Information

Figure8–590 Photo of Dye Pump Assembly (No RET)

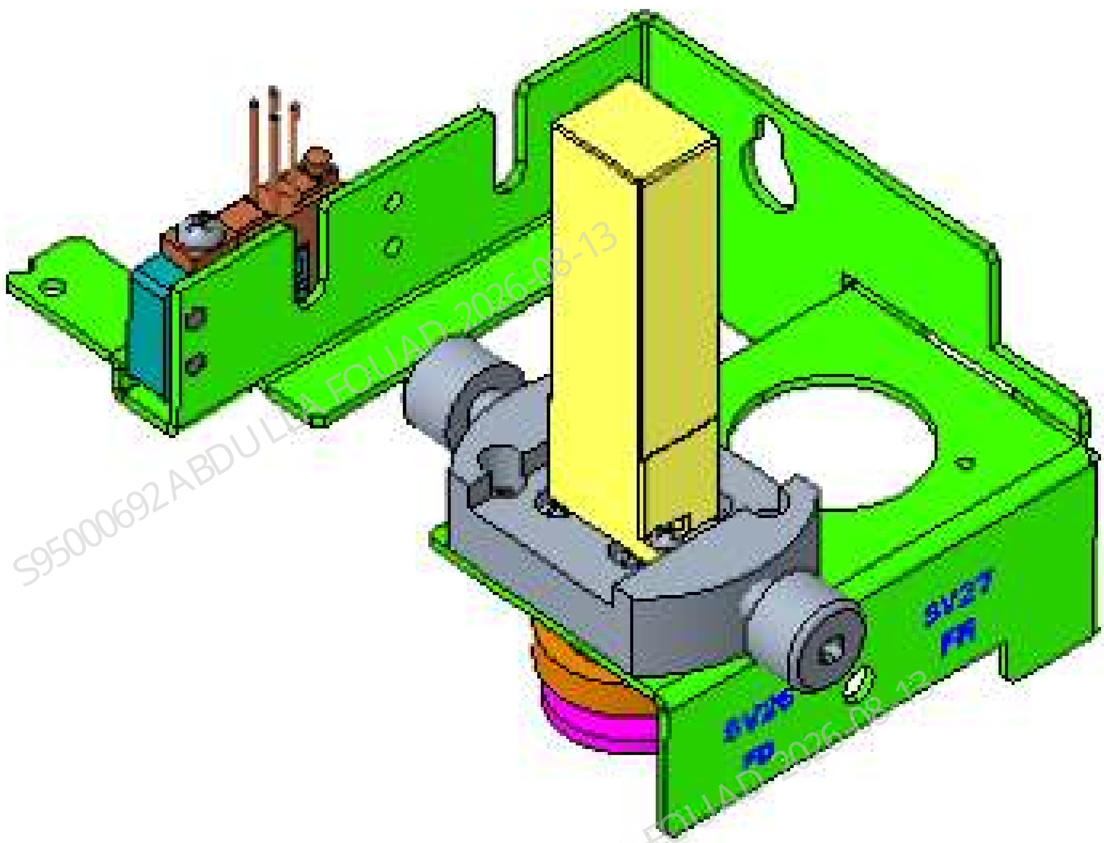


Table 8–71 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081840-00	Fluorescence Dye Pump Assembly (Without RET)	N/A	N/A

Function:
Absorbing dye reagent and detecting reagent.

8.67.2 Disassembly and Assembly

Note

1. Prevent liquid from entering the terminals during disassembly.
2. During the disassembly process, no kinks should occur. If there are kinks, the related tubes need to be replaced.
3. Be careful not to damage the connector when removing the tube.

Required tools

Cross-headed screwdriver
One pair of cutting nippers

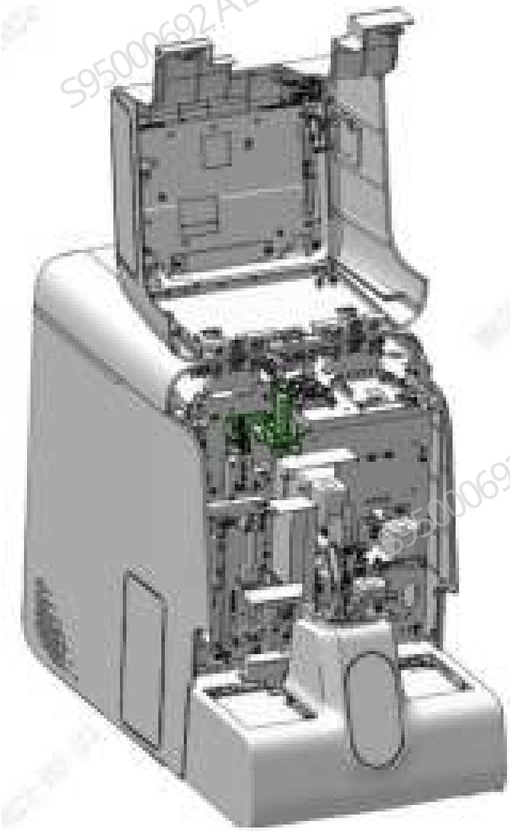
Pre-requirements

Power off the main unit, and unplug the power cord.

Procedures

1. Flip the front cover up and dock on the top cover, as shown in the following figure.

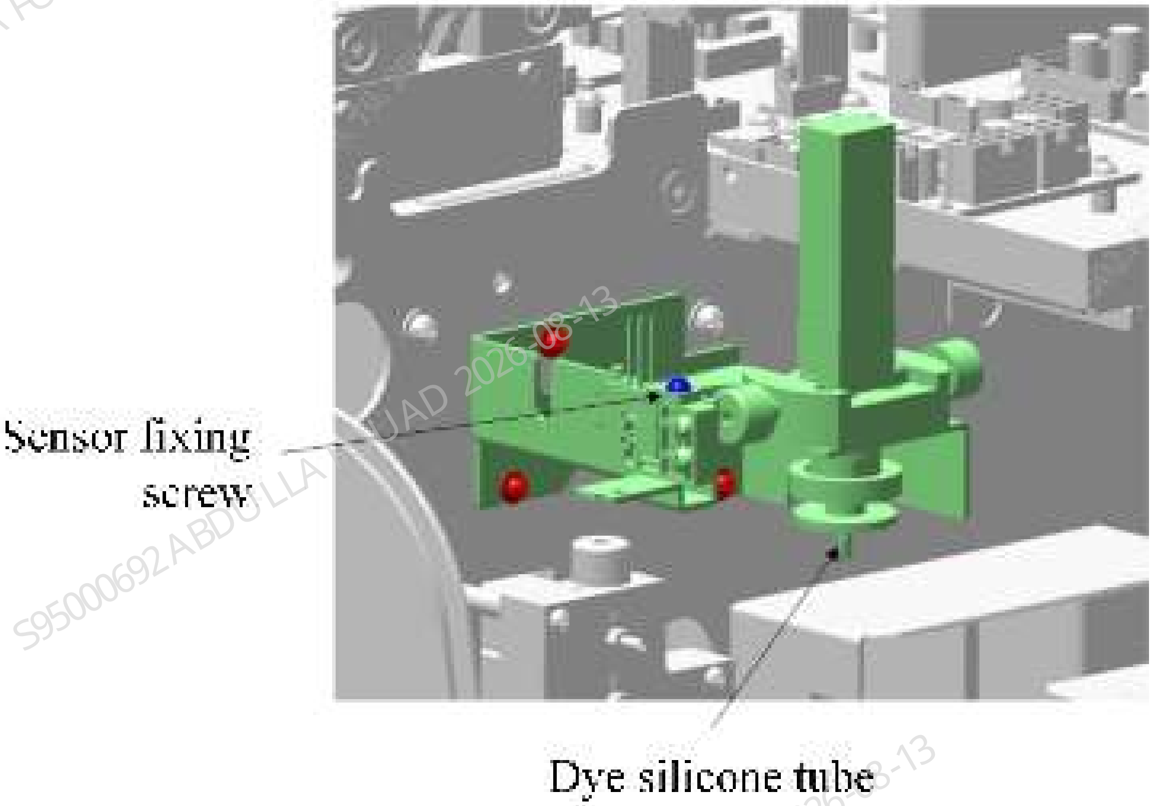
Figure8–591 Diagram of Removing the Flyorescence Dye Pump Asm (Exclude RET)



2. Loosen the wiring terminal of the sensor on the assembly, pull out the dye rubber tube connected to the reaction bath at the lower end of the metering pump, remove the M3X8 cross recessed panhead screws on the two dye liquor sensors respectively, and then take out the

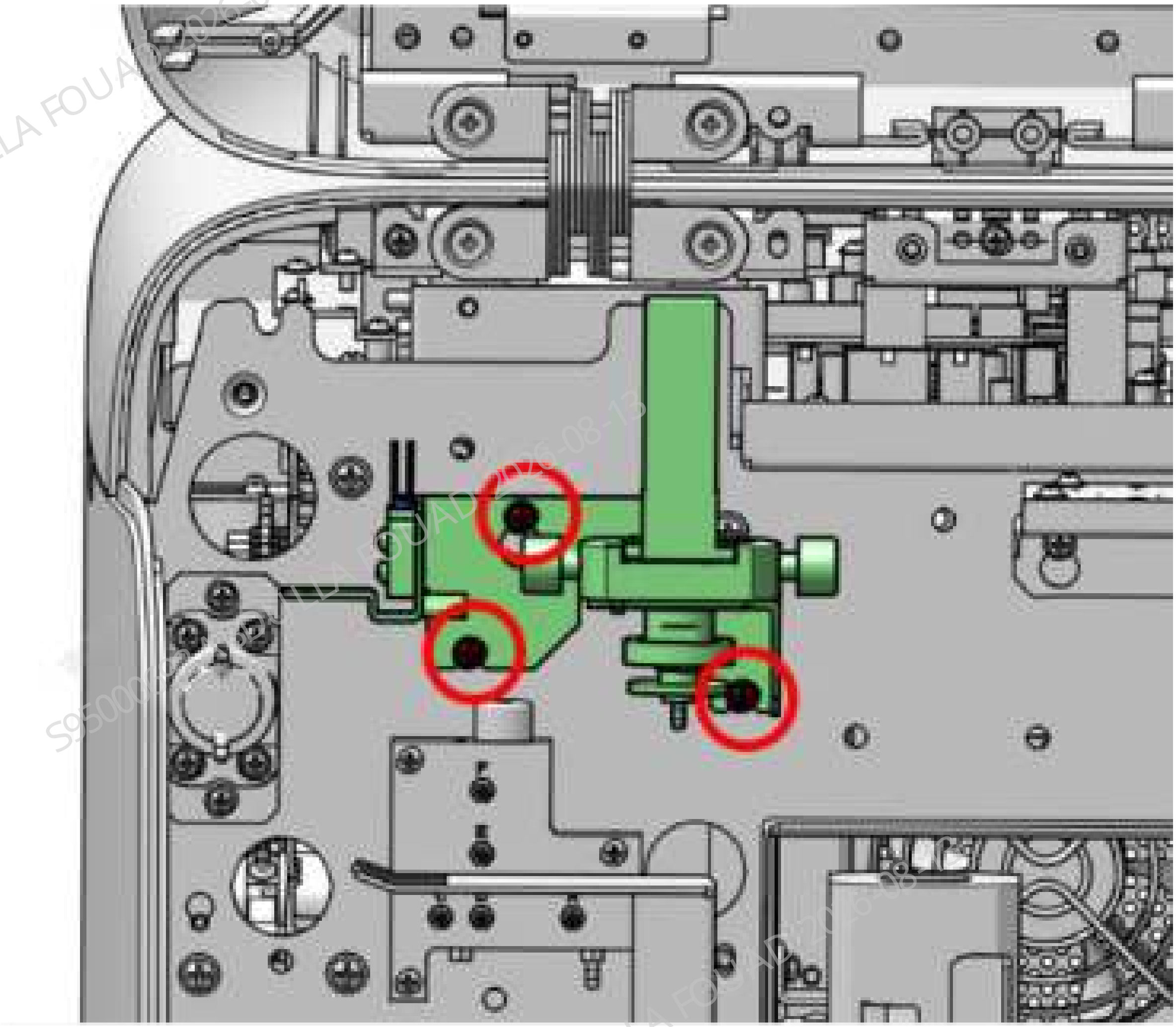
rubber tube connected to the dye reagent probe from the sensor, as shown in the following figure.

Figure8–592 Diagram of Removing the Rubber Tube of Flyorescence Dye Pump Asm (Exclude RET)



3. Loosen the three screws fixing the assembly, as shown in the figure below, and remove the assembly outward. Note not to fold the tube.

Figure8–593 Diagram of Screw Fixing Location of the Flyorescence Dye Pump Asm (Exclude RET)



4. Remove the snap connector on the dye metering pump respectively.

Subsequent Requirements

Installation procedure:

1. Use a snap connector to connect the rubber tube connected to the dye reagent probe to the metering pump (Note: Connect them correctly according to the pipe label), clip the rubber hose into the groove of the dye sensor, and then tighten the M3x8 cross recessed panhead screws locking the dye sensor.
2. Install the assembly back into the analyzer and tighten the screws to fix it.
3. Connect the dye pipeline connected to the reaction bath to the metering pump.

NOTICE

Connect them correctly according to the pipe label.

4. Reset the front cover.

8.67.3 Commissioning and Verification

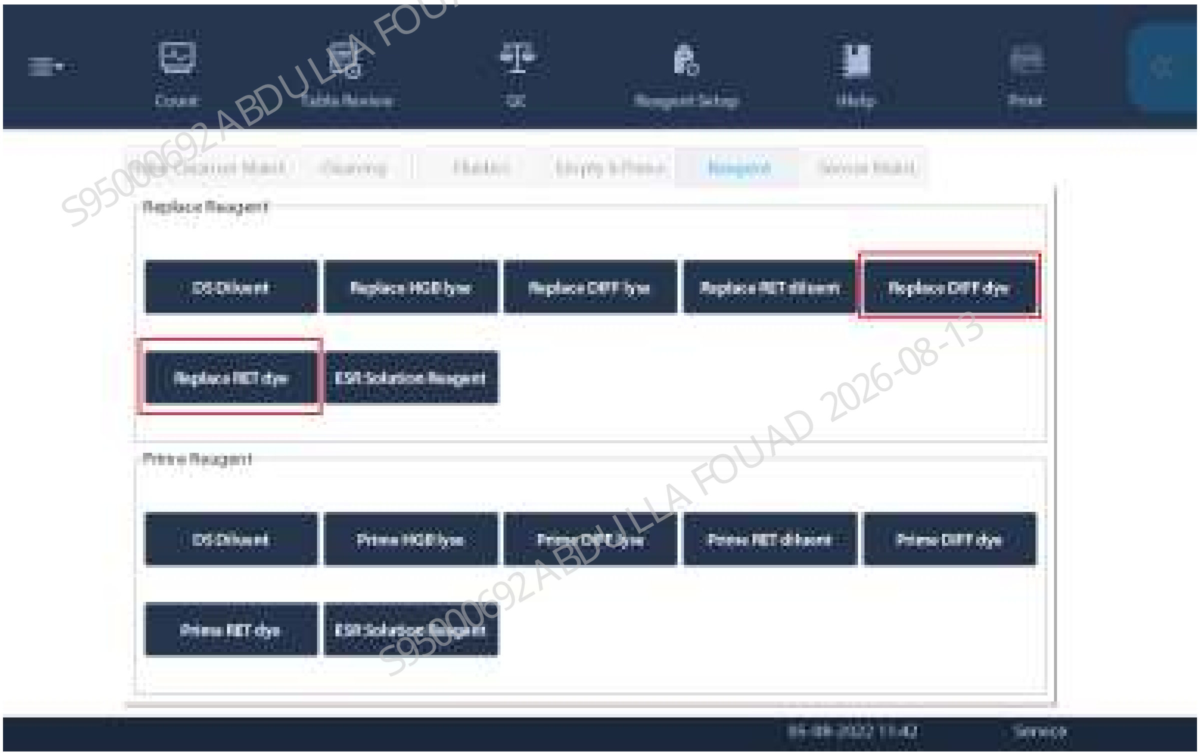
Required tools

Cross-headed screwdriver
One Allen wrench

Procedures

1. Select **Service>>Maintenance>>Reagent>>Reagent Replacement**, click **FD Dye** and **FR Dye** (depending on the configuration) respectively, and complete the reagent replacement operation (as shown in the figure below).

Figure8–594 Reagent Replacement screen



2. Select **Service>>Commissioning & Self-Test>>Sensor Commissioning>>Dye Sensors**, click **Calibrate All**, check whether the test voltage is within the reference range. If yes, click **Save All**. (As shown in the following figure.)

Figure8–595 Dye Liquor Sensor Gain Calibration Interface

8.68 Liquid detect assembly (3109)

8.68.1 General Information

Figure8–596 Photo of Liquid Detect Assembly (3109)

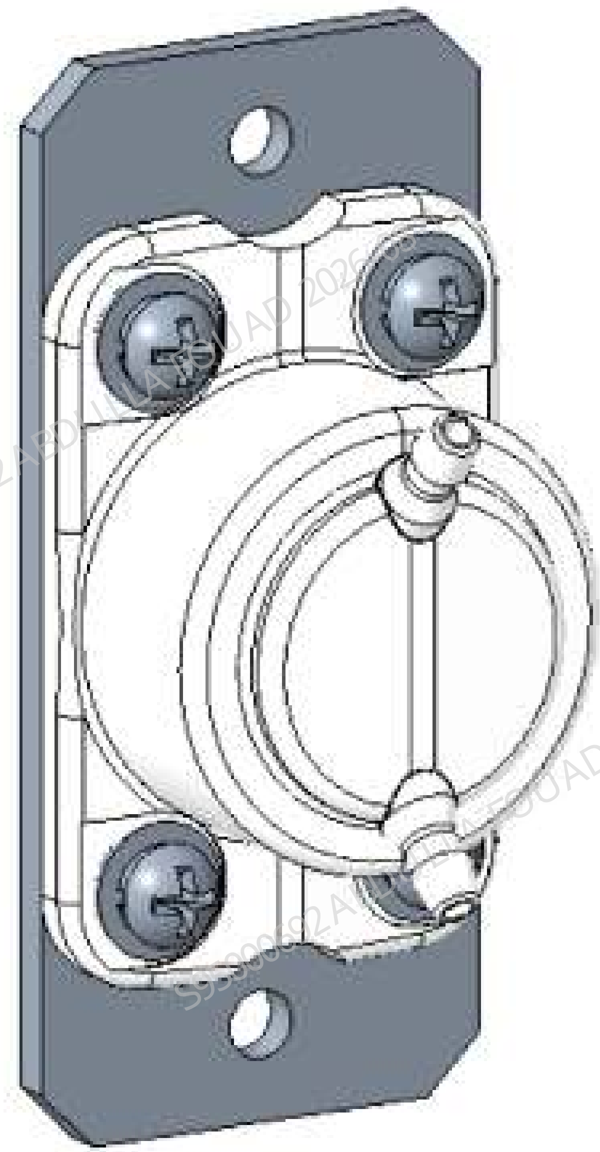


Table 8–72 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-033322-00	Fluid pressure detection assembly (3109)	N/A	N/A

Function:
Detecting the fluid pressure at the outlet of the 250 µL syringe.

8.68.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the main unit, and unplug the power cord. The sample probe will leak a small amount of liquid, so toilet paper should be laid under the sample probe in advance.

Procedures

1. Flip the front cover up and dock on the top cover, as shown in the following figure.

Figure8–597 Position of liquid detect assembly (3109)

2. Unplug the rubber tube connected to the connector of the liquid detect assembly (3109).
3. Loosen the set screws of the liquid detect assembly (3109), pull the assembly outward slightly, loosen the wiring terminals connected to the main unit, and remove the assembly.

Figure8–598 Positions of the set screws of the liquid detect assembly (3109)

Subsequent Requirements

Installation procedure:

1. Reconnect the terminals of the liquid detect assembly (3109).
2. Install the liquid detect assembly (3109) back into the analyzer, tighten two screws to fix it, and insert the corresponding tube into the tube connector of the assembly.
3. Flip down the front cover to reset it.

8.68.3 Commissioning and Verification

Required tools

Cross-headed screwdriver

Procedures

1. After the replacement is completed, start the analyzer, run the initialization program, log in at the service access level, and check whether there is a liquid pressure error reported. If not, the replacement succeeds.
2. Select **Service**>>**Maintenance**>>**Cleaning** on the main unit software and perform the **overall cleaning** action.

3. Click**Status**>>**Temperature and Pressure**>>**Liquid Pressure** in turn to check whether the liquid pressure is within the range.

Verification:

Perform a blank count test to confirm that the analyzer has no alarms and that the blank count results of various parameters are within the index range.

8.69 DS diluent cap assembly

8.69.1 General Information

N/A

8.69.2 Disassembly and Assembly

N/A

8.69.3 Commissioning and Verification

N/A

8.70 Lyse container cap assembly (red, open mold connector, 0.85m

8.70.1 General Information

N/A

8.70.2 Disassembly and Assembly

N/A

8.70.3 Commissioning and Verification

N/A

8.71 Lyse container cap assembly (green, open mold connector, 0.8

8.71.1 General Information

N/A

8.71.2 Disassembly and Assembly

N/A

8.71.3 Commissioning and Verification

N/A

8.72 Lyse container cap assembly (black, open mold connector, 0.8

8.72.1 General Information

N/A

8.72.2 Disassembly and Assembly

N/A

8.72.3 Commissioning and Verification

N/A

8.73 Waste Cap Assembly (3120)

8.73.1 General Information

N/A

8.73.2 Disassembly and Assembly

N/A

8.73.3 Commissioning and Verification

N/A

8.74 Lyse Container Cap Assembly (Purple, OPEN MOLD CONNector, 0.

8.74.1 General Information

N/A

8.74.2 Disassembly and Assembly

N/A

8.74.3 Commissioning and Verification

N/A

8.75 Fluorescent reagent bag cap assembly (FD)

8.75.1 General Information

N/A

8.75.2 Disassembly and Assembly

N/A

8.75.3 Commissioning and Verification

N/A

8.76 Fluorescent reagent bag cap assembly (FN)

8.76.1 General Information

N/A

8.76.2 Disassembly and Assembly

N/A

8.76.3 Commissioning and Verification

N/A

8.77 Cleanser Cap Assembly (Orange, Open Mold Connector, 0.85m)

8.77.1 General Information

N/A

8.77.2 Disassembly and Assembly

N/A

8.77.3 Commissioning and Verification

N/A

8.78 PCBA, Air Pressure Detection Board, Analog, 5 Channels

8.78.1 General Information

Figure8–599 Physical map of air pressure detection board

Table 8–73 General information of air pressure detection board

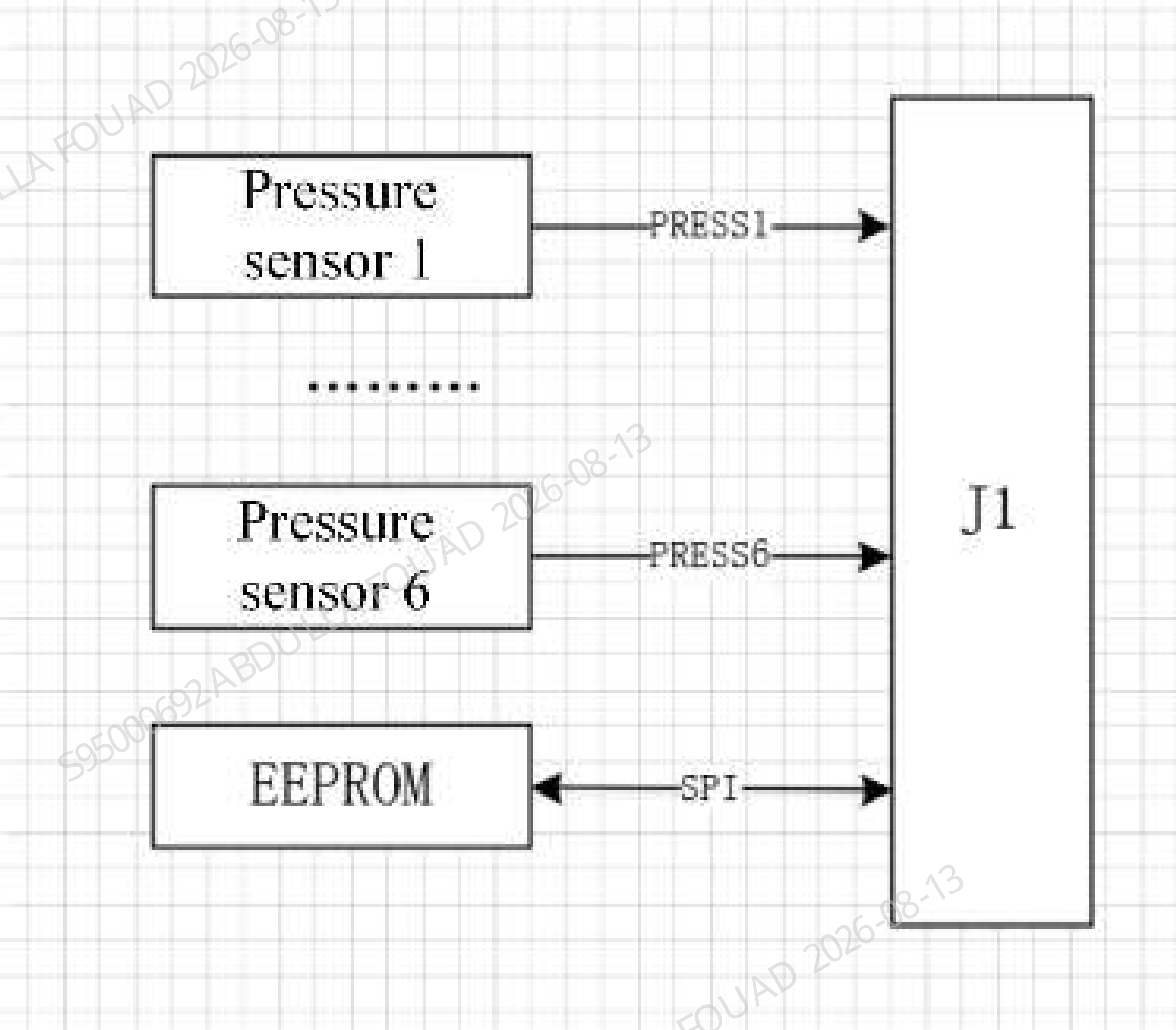
FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
051-004205-00	Air pressure detection board	No lower-level FRU	No lower-level FRU

Function:

Detect air pressure of +50a, +40a, +90Kpa, -30KPa and -40Kpa.

Functional block diagram

Figure8–600 Functional block diagram of the board



Connectors

Figure8–601 Position numbers of interfaces of air-pressure detection board

Table 8–74 Definitions of interfaces of air-pressure detection board

Position No.	Interface function and definition
J1	+3.3V/+12V/SPI communication interface/air pressure sensor signal output

Table 8–75 PIN position definitions of interfaces of air-pressure detection board PCBA

PIN No.	Interface function and definition	PIN No.	Interface function and definition
J1.1	VDD/3.135V - 3.465V	13	PRESS_4
J1.2	GND	14	AGND
J1.7	PRESS_1	15	PRESS_5
J1.9	PRESS_2	16	AGND
J1.10	AGND	17	PRESS_6
J1.11	PRESS_3	18	AGND
J1.12	AGND	20	A12V/+11.4V – 12.6V

Indicators

Figure8–602 Indicator positions of air pressure detection board

Table 8–76 Description of indicator position numbers of air pressure detection board

Area	Indicator position number	Meaning	Normal status
A	D1	A5V power indicator	Green, steady on
B	D2	VDD3.3V power supply indicator	Green, steady on

Test Points

Figure8–603 Positions of test points of air pressure detection board

Table 8–77 Description of position numbers of test points of air pressure detection board

Position numbers for test points	Interface function and definition	Position numbers for test points	Interface function and definition
TP1	PRESS_1/0.2-4.7V	TP6	PRESS_6/0.2-4.7V
TP2	PRESS_2/0.2-4.7V	TP11	A5V/4.75-5.25V
TP3	PRESS_3/0.2-4.7V	TP87	AGND
TP4	PRESS_4/0.2-4.7V	TP13	VDD/3.135V - 3.465V
TP5	PRESS_5/0.2-4.7V	TP14	GND

8.78.2 Disassembly and Assembly

Required tools

Cross-head screwdriver (M3)(1 set)
Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Open the front cover, and support and fix it, as shown in the following figure.

Figure8–604 Opening the front cover

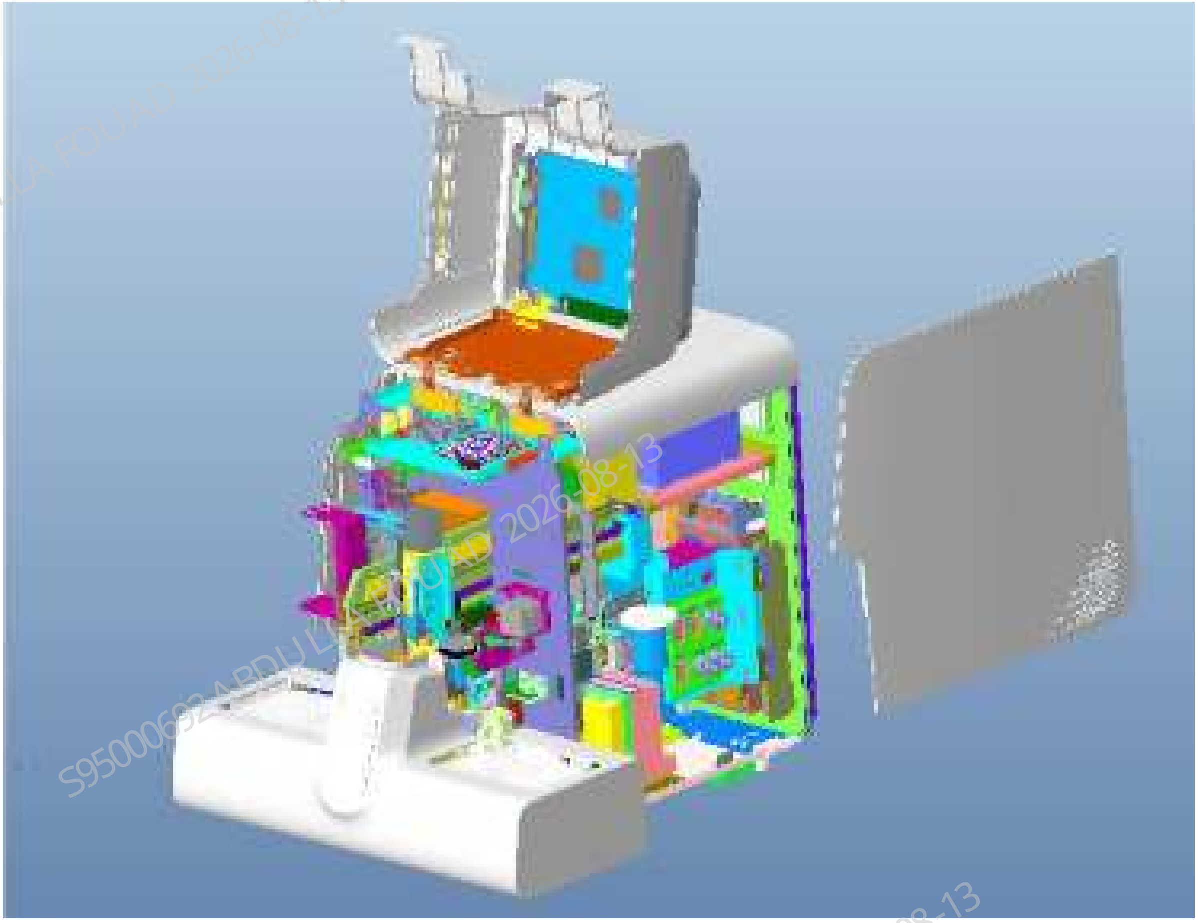
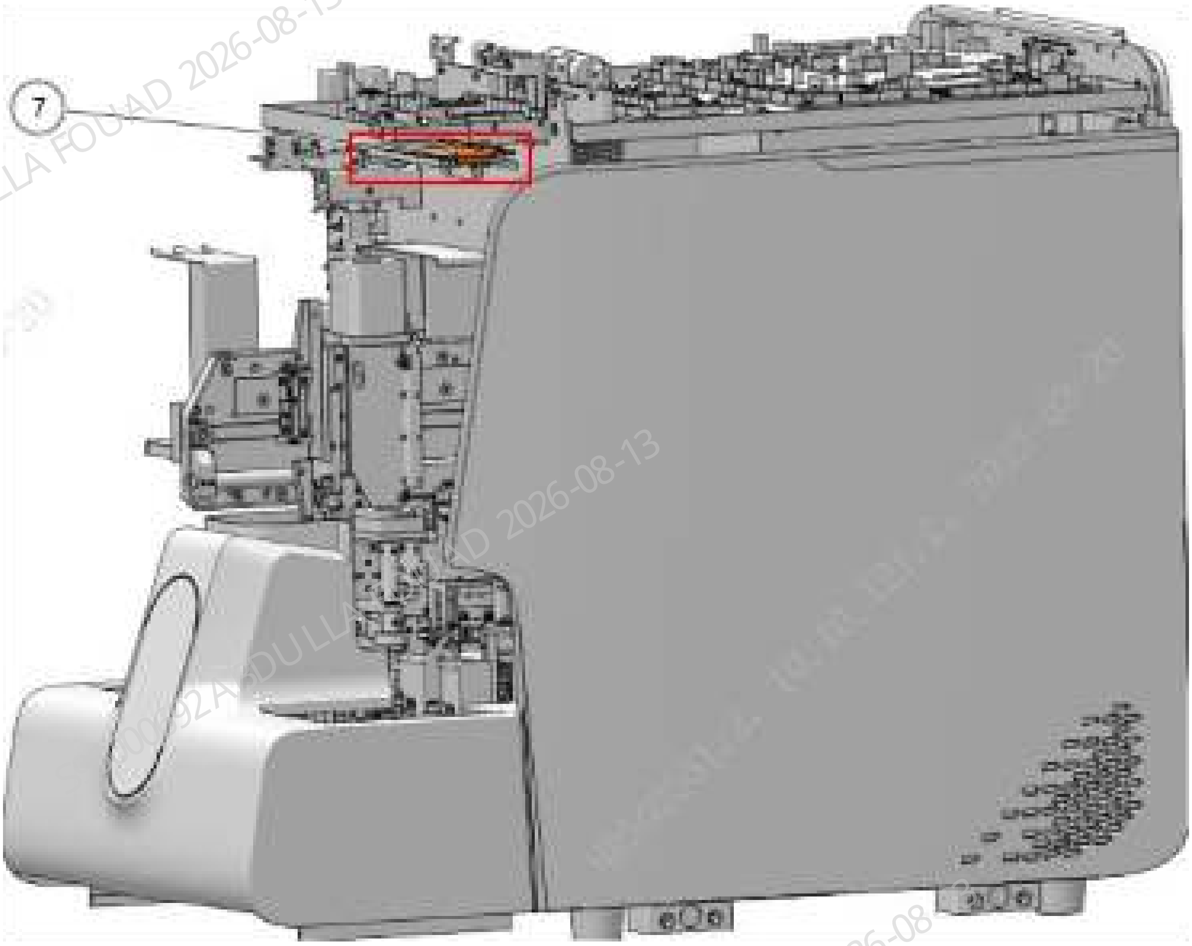


Figure8–605 Diagram of Board Position



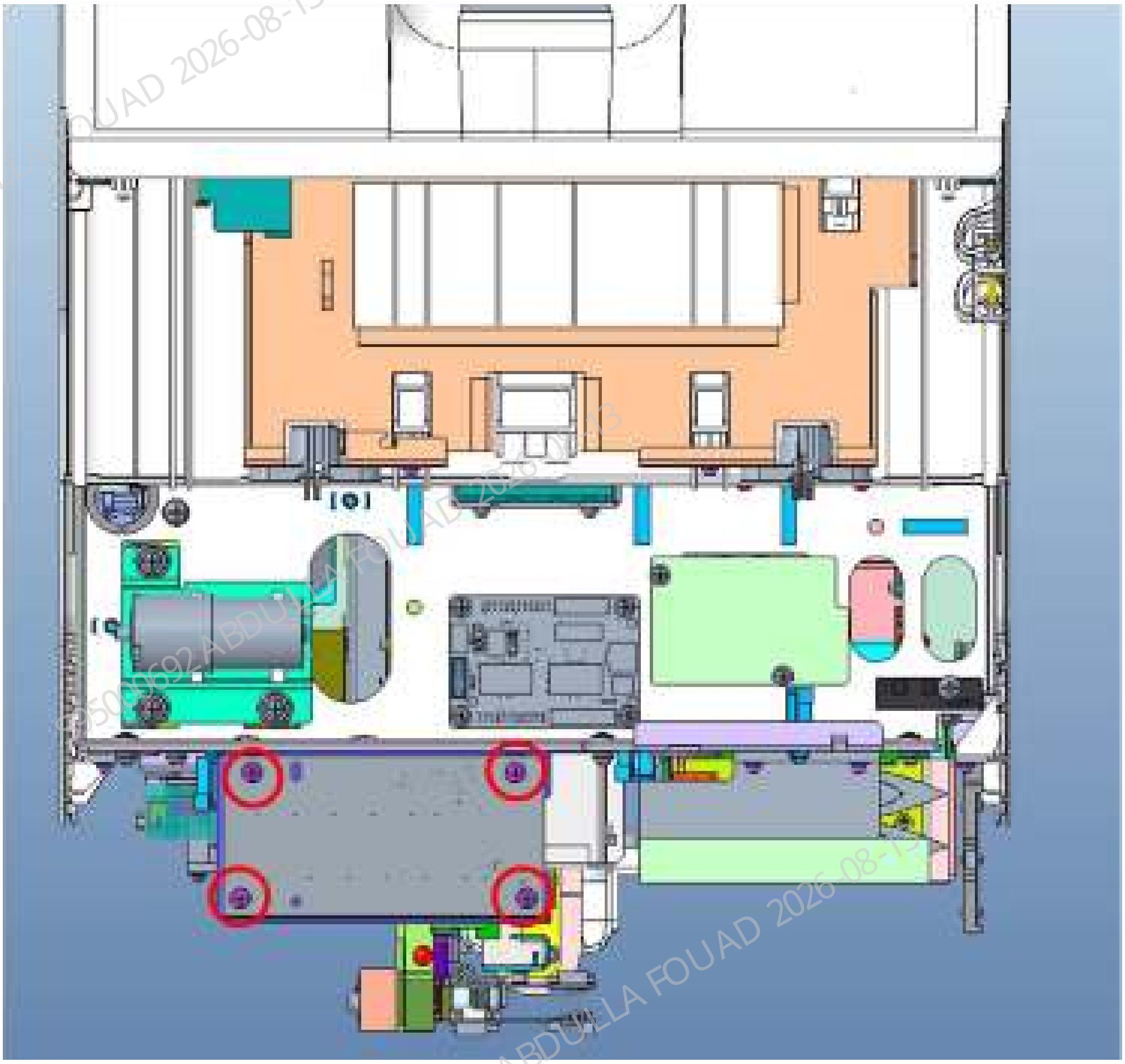
3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Disassemble as follows:

1. Confirm that the air pressure detection board is identified, pull out the wires on the J1 socket and the air tubes on all sensors.
2. Remove all set screws in turn, and take out the board. There are four screws, and their positions are shown in the figure below:

Figure8–606 Positions of screws of air pressure detection board



Install as follows:

1. Make sure the system is powered off.
2. Identify the correct orientation of the board and place the board to the installation position, confirm that all wire plugs are not pressed by the board, and confirm that the air tube is not squeezed or deformed by the board.
3. Install all set screws.
4. Plug all the wires and air tubes, and check whether the connections of tubes and wires are correct and complete according to the interface definition table of the air pressure detection board PCBA.
5. Install the cover of the air pressure detection board.

8.78.3 Commissioning and Verification

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Verification Procedure in the analyzer

1. Start the analyzer and check whether an alarm is generated during startup.
2. Take a blank tube, select the **OV-WB-CD** mode, and after startup, confirm whether the test process alarms.

8.79 PCBA, Motor Board

8.79.1 General Information

Figure8–607 Physical Map of PCBA, Motor Board

Table 8–78 General Information about PCBA, Motor Board

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
051-004344-03	PCBA, Motor board	No lower-level FRU	No lower-level FRU

Function:

Responsible for motor drive and sensor detection of electromechanical components.

Connectors

Figure8–608 Diagram of Position Number of PCBA, Motor Board Interface

Table 8–79 Definition of PCBA, Motor Board Interface

Position No.	Interface function and definition	Position No.	Interface function and definition
J1	Power and communication interface	J7	Photocoupler interface
J2	Sensor interface	J8	Photocoupler interface
J3	Photocoupler interface	J9	Motor interface
J4	Autoloader type identification interface	J10	Motor interface

Table 8–79 Definition of PCBA, Motor Board Interface(continued)

Position No.	Interface function and definition	Position No.	Interface function and definition
J5	FPGA burning and commissioning interface	J11	Motor interface
J6	MCU burning interface		

Indicators

Table 8–80 Indicator position number of the board

Indicator position number	Meaning	Normal status
D2	24V power supply indicator	Constantly on
D3	Digital 5V power supply indicator	Constantly on
D4	FPGA operating status indicator	Blinking
D5	M1 channel motor operating status indicator	Blinking
D6	M2 channel motor operating status indicator	Blinking
D7	M3 channel motor operating status indicator	Blinking
D8	M4 channel motor operating status indicator	Blinking
D9	M5 channel motor operating status indicator	Blinking
D10	M5 channel motor operating status indicator	Blinking
D11	MCU operating status indicator	Blinking

Test Points

Table 8–81 Position numbers of TP test points of the board

Position numbers for test points	Interface function and definition	Position numbers for test points	Interface function and definition
TP29	24V		
TP19/TP26	GND		
TP3/TP118	D5V		
TP5	D3V3		
TP6	D2V5		
TP7	D1V2		

8.79.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver
Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Remove the top cover: Turn up the front cover, loosen the three fixing screws of the top cover, and remove the top cover.

Figure8–609 Opening the front cover

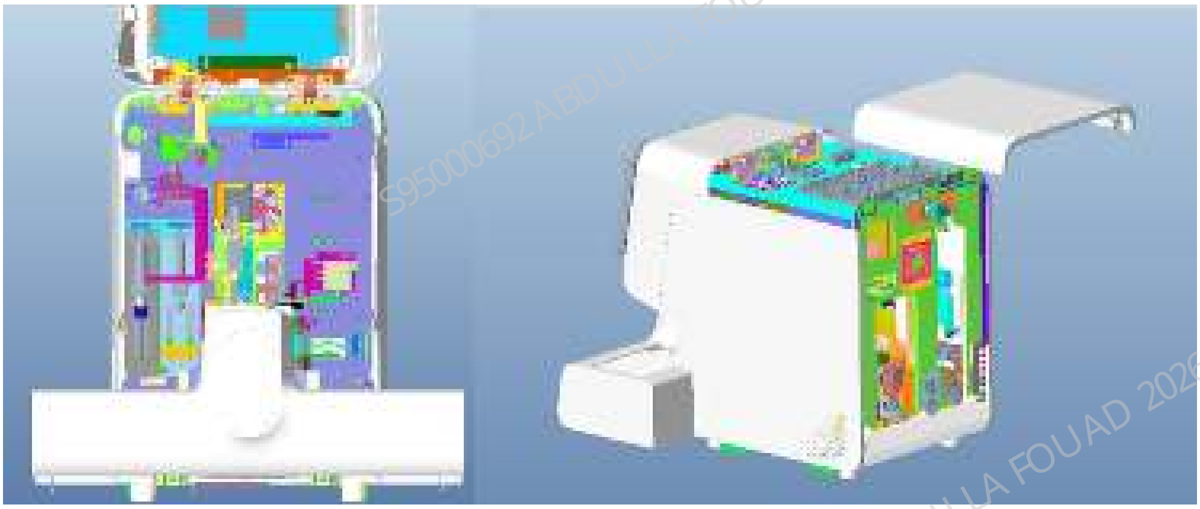
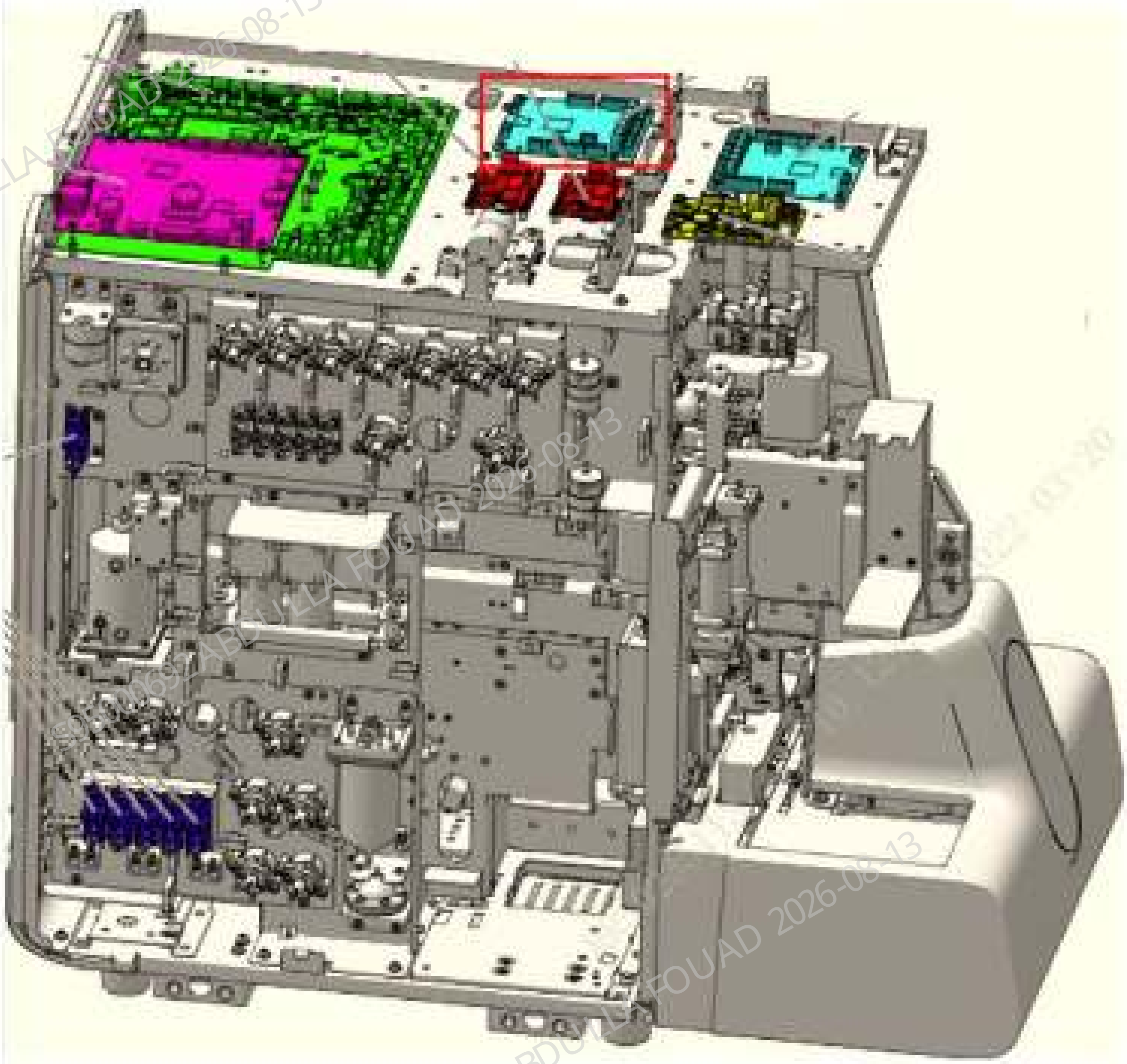


Figure8–610 Diagram of Board Position



3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

8.79.3 Commissioning and Verification

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Verification Procedure in the analyzer

1. Start the analyzer and check whether an alarm is generated during startup.
2. Log in to the system at the service access level.
3. Select **Service>>Commissioning & Self-Test>>Self-Test**, select **Mix Mechanism**, execute **Assembly Initialization**, **Venous Blood Mix Assembly Self-Test**, and **Capillary Blood Mix Assembly Self-Test**, and confirm if self-test succeeds.

8.80 PCBA, Valve Driver Board

8.80.1 General Information

Figure8–611 Physical map of valve driver board

Table 8–82 General information

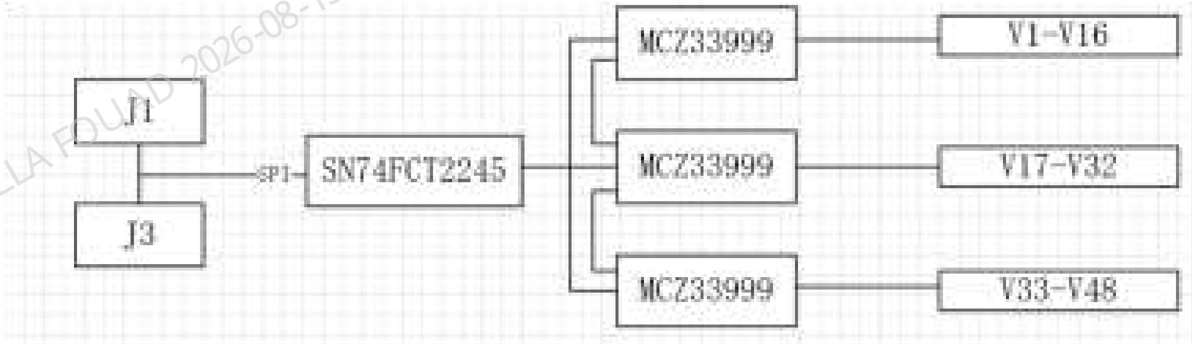
FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU
051-002805-00	Valve driver board PCBA	No lower-level FRU	No lower-level FRU

Function:

Control up to 48 valves.

Functional block diagram

Figure8–612 Functional block diagram of valve driver board



Connectors

Figure8–613 Position numbers of interfaces of the valve driver board

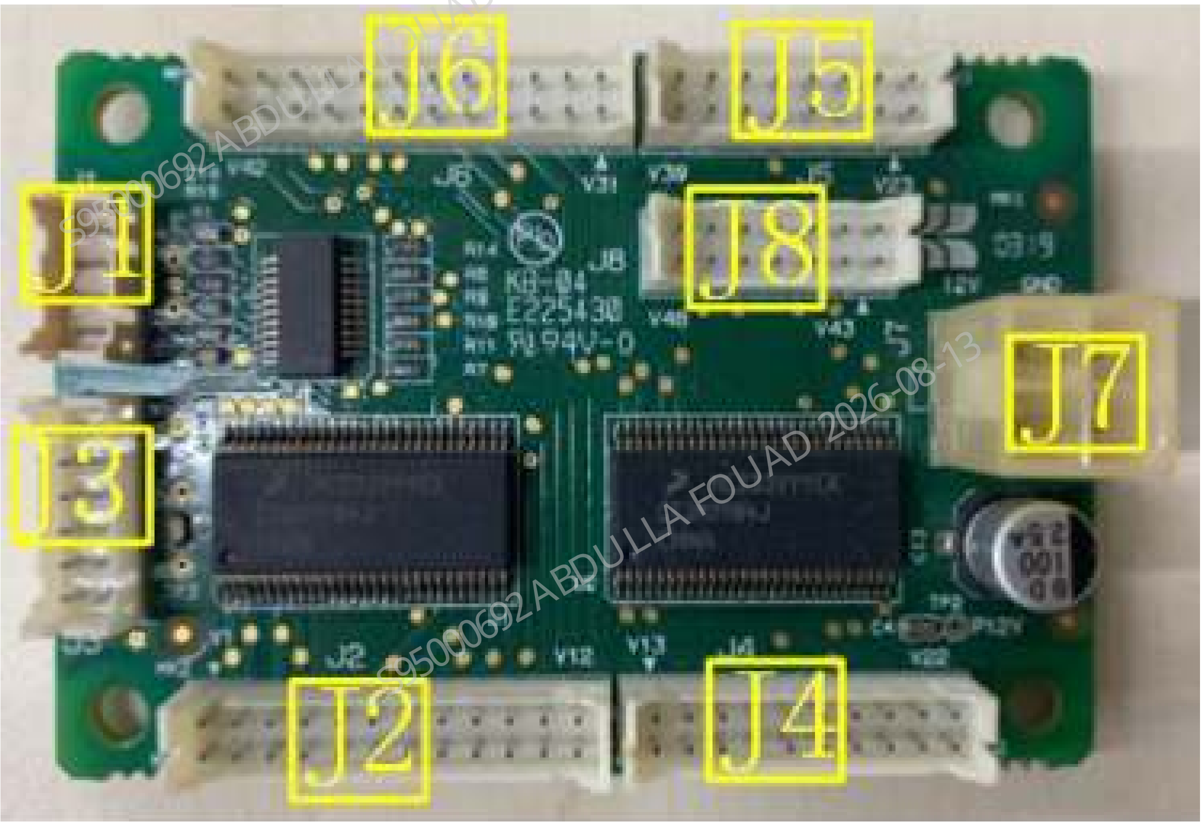


Table 8–83 Definitions of interfaces of the valve driver board

Position No.	Interface function and definition	Position No.	Interface function and definition
J1	SPI communication interface	J5	Eight valve control interfaces
J2	12 valve control interfaces	J6	12 valve control interfaces

Table 8–83 Definitions of interfaces of the valve driver board(continued)

Position No.	Interface function and definition	Position No.	Interface function and definition
J3	SPI communication interface	J7	12V power supply interface
J4	10 valve control interfaces	J8	6 valve control interfaces

Table 8–84 Definitions of interfaces of the valve driver board

PIN No.	Interface function and definition	PIN No.	Interface function and definition
J1/6	VCC/4.75-5.25V	J3/8,9,10,11,12	GND
J1/8	GND	J7/1,2	GND
J3/6	VCC/4.75-5.25V	J7/3,4	P12V/11.4-12.6V

Test Points

Figure8–614 Positions of test points of valve driver board

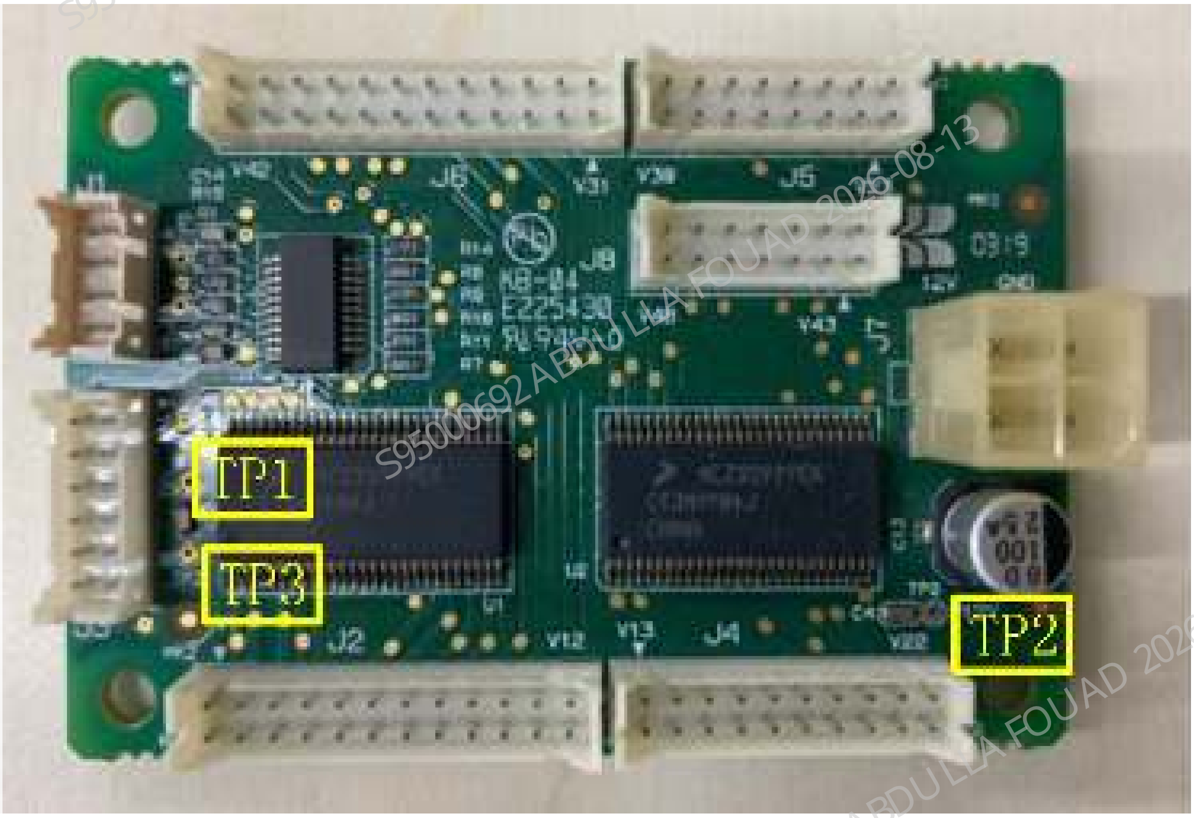


Table 8–85 Position numbers of test points of valve driver board

Position numbers for test points	Interface function and definition	Position numbers for test points	Interface function and definition
TP1	VCC/4.75-5.25V	TP3	GND
TP2	P12V/11.4-12.6V	N/A	N/A

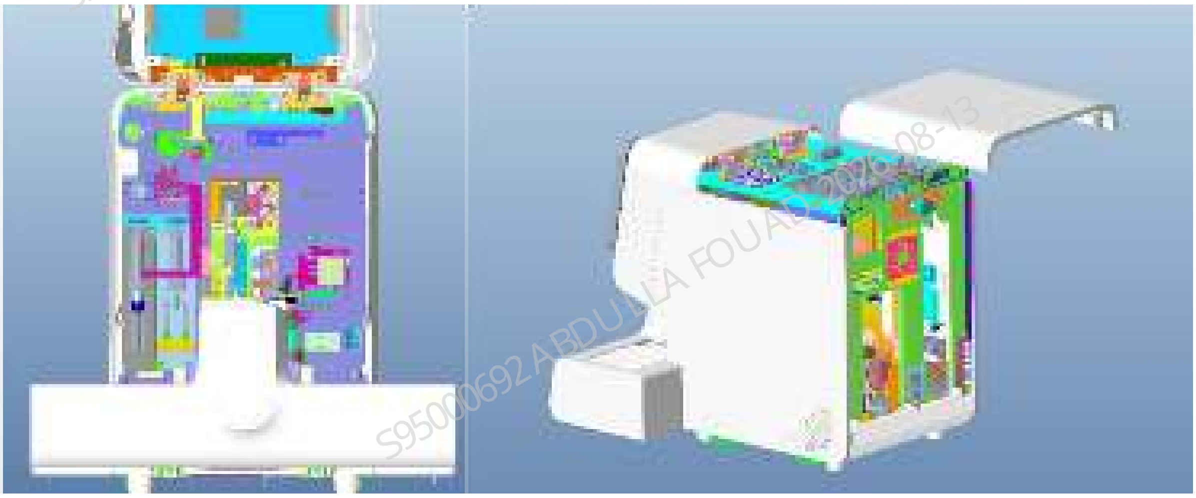
8.80.2 Disassembly and Assembly

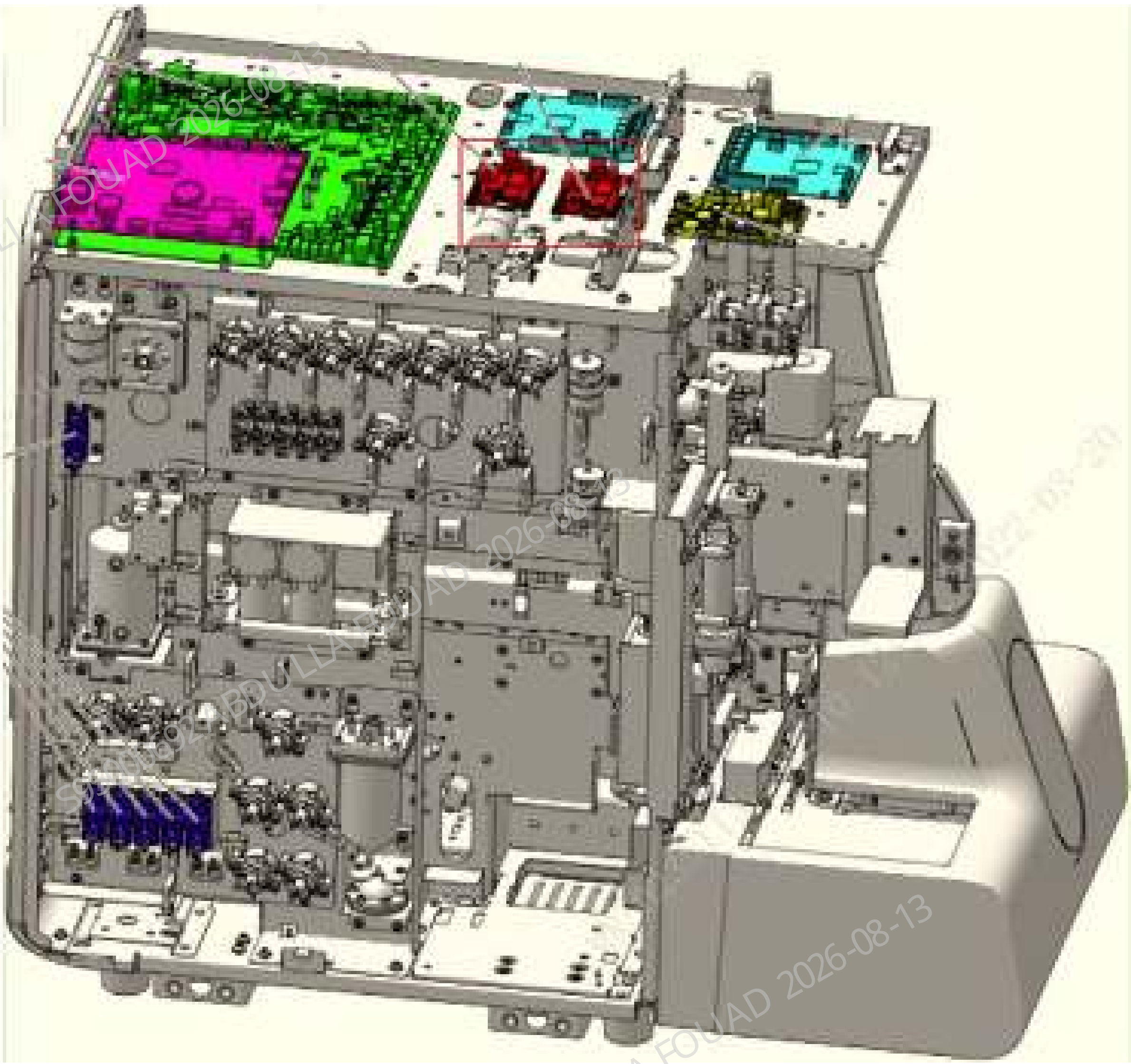
Required tools

Cross-head screwdriver (M3)(1 set)
Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Remove the top cover: Turn up the front cover, loosen the three fixing screws of the top cover, and remove the top cover.





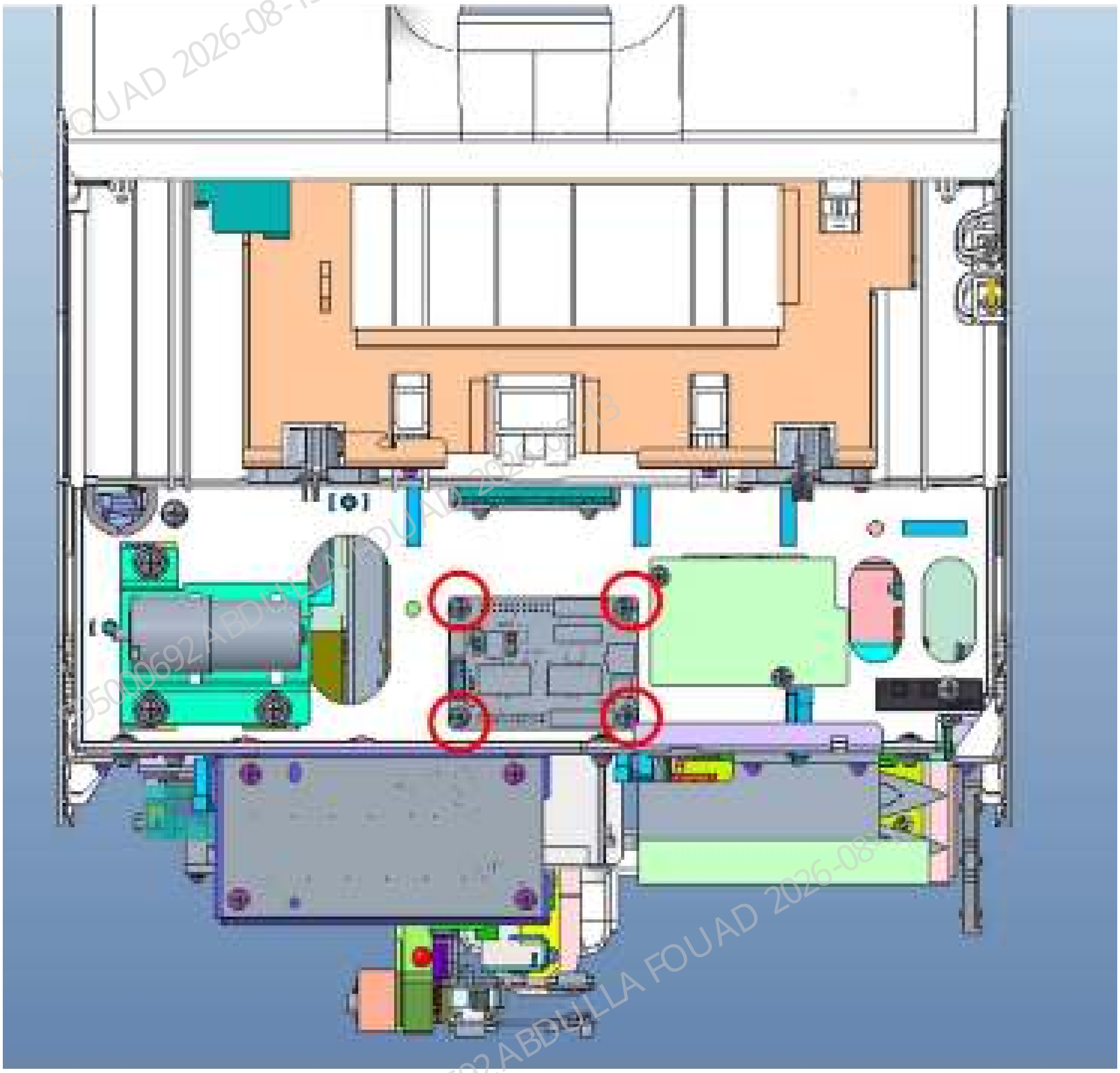
3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

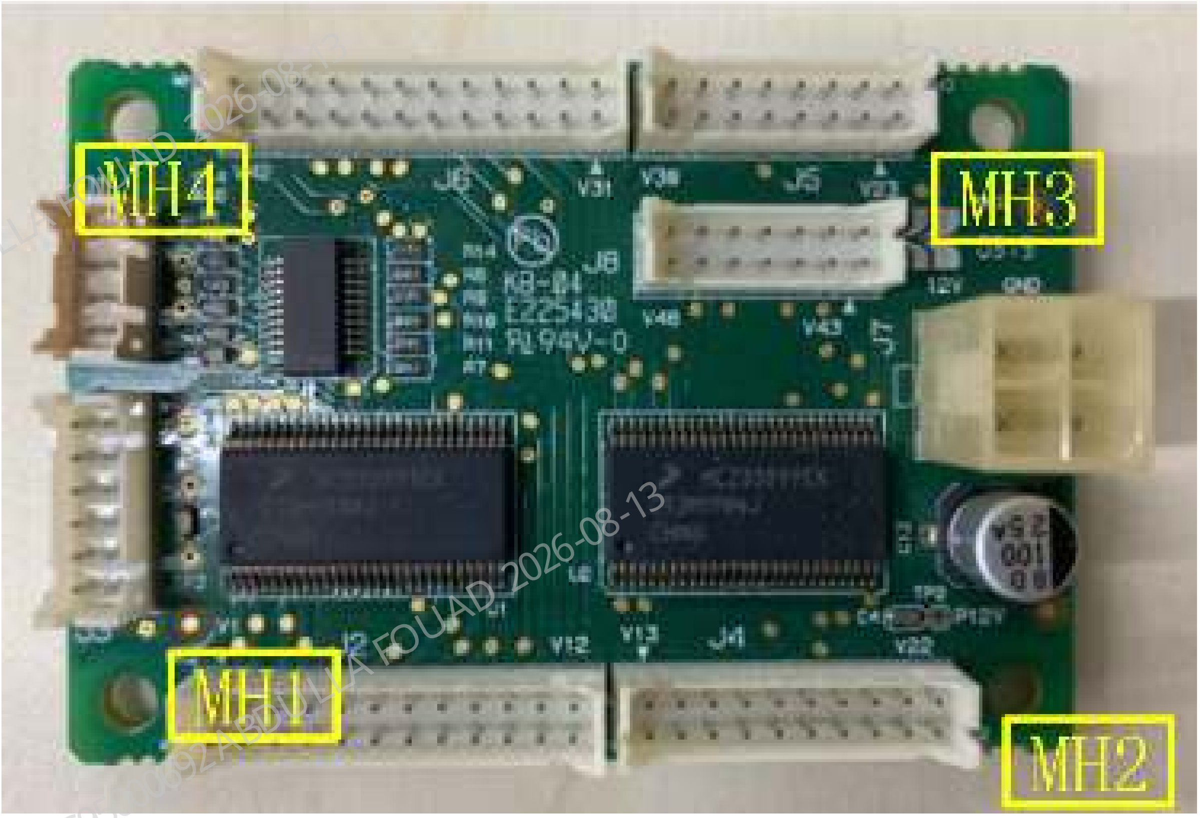
Procedures

Disassemble as follows:

1. Confirm that the valve drive board is identified, and pull out all the wires.
2. Remove all set screws in turn, and take out the board. There are four screws, and their positions are shown in the figure below.

Figure8–615 Positions of screws of the valve drive board





Specific assembly procedure:

1. Make sure the system is powered off.
2. Identify the correct direction of the board and place the board on the installation position. Make sure that all the wire plugs are not pressed by the board.
3. Install all set screws.
4. Connect all wires.

8.80.3 Commissioning and Verification

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Verification Procedure in the analyzer

1. Start the analyzer and check whether an alarm is generated during startup.
2. Log in to the system at the service access level.
3. Select**Service**>**>Commissioning & Self-Test**>**>Self-Test**, and select the valve assembly, perform self-test on all valves, and listen to the sound of valve operation to confirm that all valves can operate.

4. Take a blank tube, select the **OV-WB-CD** mode, and after startup, confirm whether the test process alarms.

8.81 Liquid Detection Board PCBA

8.81.1 General Information

Figure8–616 Level detection board (front)

Table 8–86 General information of liquid detection board

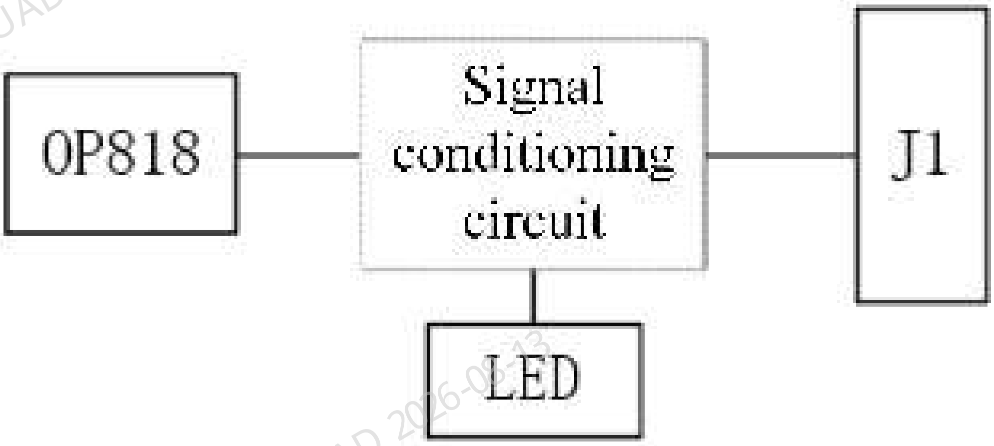
FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
051-001621-01	Liquid Detection Board PCBA	No lower-level FRU	No lower-level FRU

Function:

Detect the reagents. The liquid tube led from the reagent bottle passes through the sensor. When there is no liquid in the liquid tube, the analog quantity output by the sensor is different. After analog-to-digital conversion, the board indicates the presence or absence of reagents in the corresponding channel by outputting high and low levels to the upper computer board and turning on/off the indicator on its own board.

Functional block diagram

Figure8–617 Functional block diagram of the board



Connectors

Figure8–618 Position numbers of board interfaces

Table 8–87 Definition of board interfaces - Level detection board

Position No.	Interface function and definition
J1	Provide power supply, and level output interface for liquid level status.

Table 8–88 Definitions of board interfaces

PIN No.	Interface function and definition
J1/1	VCC
J1/2	LIQ_OUT/Liquid level status indication, high level - With liquid, low level - Without liquid.
J1/3	GND

Indicators

Indicator position and description

Figure8–619 Indicator position of the board

Table 8–89 Indicator position number of the board

Area	Indicator position number	Meaning	Normal status
A	D1	Liquid level status indicator	Off with liquid, and always on red without liquid.

Test Points

Positions and descriptions of test points

Figure8–620 Positions of TP test points of the board

Table 8–90 Position numbers of TP test points of the board

Position numbers for test points	Interface function and definition	Position numbers for test points	Interface function and definition
TP1	VCC +5V	TP4	High level 5V - Without liquid, low level 0V - With liquid.
TP2	GND: Grounding	TP5	High level 5V - With liquid, low level 0V - Without liquid.
TP3	High level - With liquid, low level - Without liquid.		

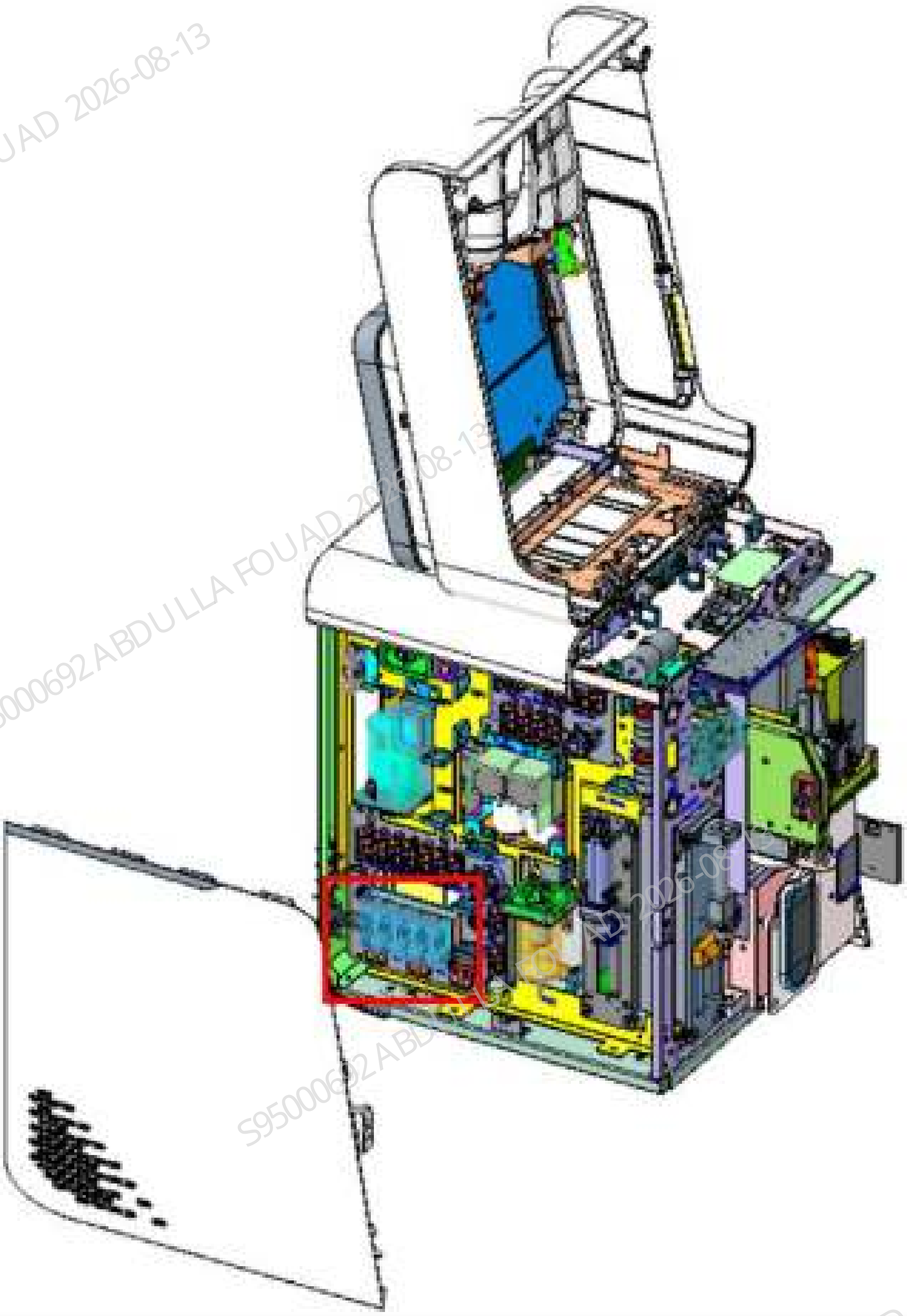
8.81.2 Disassembly and Assembly

Required tools

Cross-head screwdriver (M3)(1 set)
Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Remove the left cover, turn up the front cover, dock it on the top cover, loosen the two set screws on the left cover, and remove the left cover.



3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Disassemble as follows:

1. Confirm that the air pressure detection board is identified, pull out the wire on the (J1) socket and the liquid tube on the sensor.
2. and take out the board (in snap installation mode).

Install as follows:

1. First, insert the connecting wire of the liquid detection board into the J1 socket of the level detection board.
2. Insert the reagent detection tube into the board.
3. and install the board (snap installation method).

8.81.3 Commissioning and Verification

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Verification Procedure in the analyzer

1. Confirm that there is liquid in the reagent bottle corresponding to the liquid detection board, the analyzer is turned on, and there is no alarm.
2. If the No Reagent alarm occurs, the error can be eliminated with one key in the alarm screen.

8.82 PCBA, RFID Card Reader

8.82.1 General Information

Figure8–621 Physical Map of RFID Card Reader PCBA

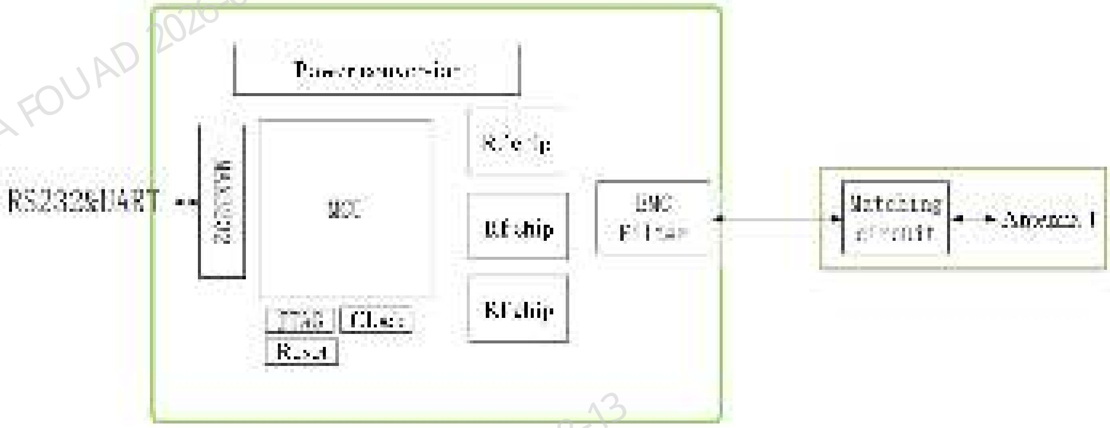
Table 8–91 General Information about RFID Card Reader PCBA

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU
051-004706-00	051-004706-00	No lower-level FRU	No lower-level FRU

Function:
Providing the RFID label reading and writing functions on closed reagent and ESR authorization card.

Functional block diagram

Figure8–622 Functional block diagram of the board



Connectors

Figure8–623 Diagram of RFID Card Reader Interface Position Number

Table 8–92 Definition of RFID Card Reader Interface

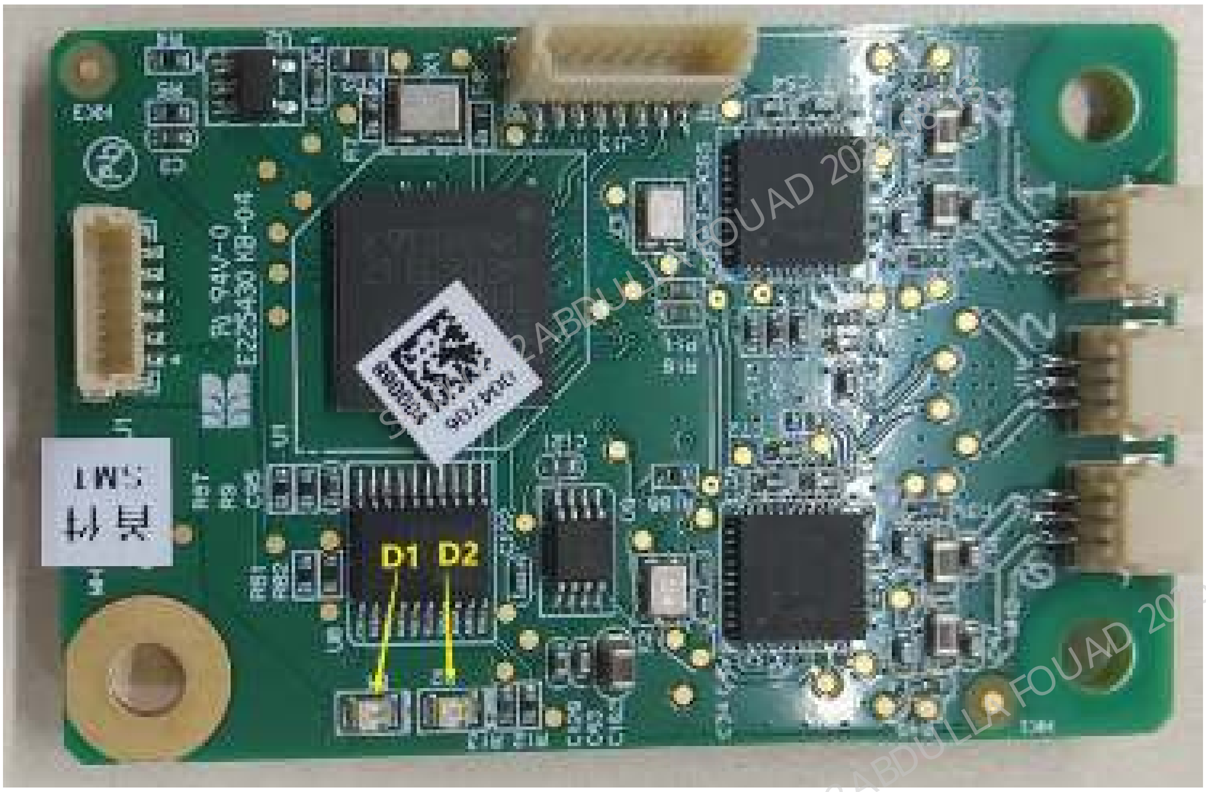
Position No.	Interface function and definition
J1	Power interface (VDD_5V, GND), communication interface (connected to the main control board)
J13	JTAG commissioning interface
J9	Antenna interface
J10	Reserved, not in use
J11	Reserved, not in use

Table 8–93 Definition of RFID Card Reader J1 Power Input Interface

Position No.	Interface function and definition
J1.1	VDD_5V,+4.75~5.25V
J1.2	GND

Indicators

Figure8–624 LED Position Diagram of RFID Card Reader Board



Indicator position number	Meaning	Normal status
D1	RFID operation indicator	Flashing when RFID reads information
D2	MCU operation indicator	Green, blinking

8.82.2 Disassembly and Assembly

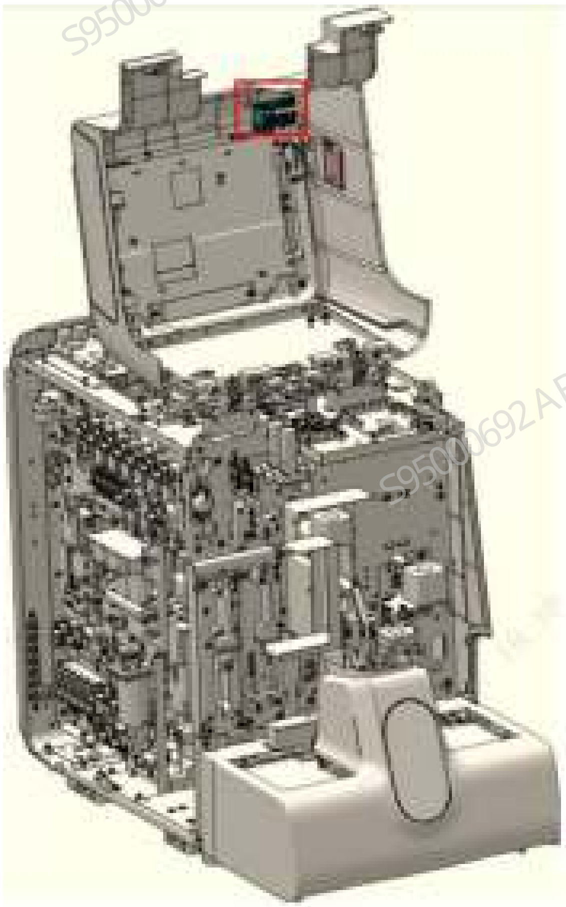
Required tools

Cross-head screwdriver (M3)(1 set)
One set of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Open the front cover, and support and fix it.

Figure8–625 Opening the front cover



3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Disassemble according to the steps below:

1. On the front cover, confirm to identify the RFID card reader board, and pull out the wires on all the sockets in turn.
2. Remove all the three fixing screws, and take out the board.

Figure8–626 Diagram of RFID Card Reader Screw Installation Positions

Install according to the steps below:

1. Make sure the system is powered off.
2. Identify the correct direction of the board and place the board on the installation position. Make sure that all the wire plugs are not pressed by the board.
3. Install all set screws.
4. Insert all the wires according to the wire markings, and recheck whether the wiring is complete according to Table 1.

Subsequent Requirements

Continue as follows after replacing the board:

1. Replace the board, and power on the analyzer.

2. If no upgrade prompt is displayed, skip this step. If an upgrade prompt is displayed, insert a USB flash drive that contains the software upgrade package and upgrade the instrument. After an upgrade success prompt is displayed, shut down and then restart the instrument.
3. Restart the analyzer. The analyzer enters the initialization process. Wait for the analyzer to be restarted.

8.82.3 Commissioning and Verification

Pre-requirements

Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Verification procedure at the analyzer level:

1. Start the analyzer normally and check whether there is an error alarm of abnormal communication of closed reagent RFIID card during startup.
2. If no upgrade prompt is displayed on the operation interface, skip this step. If an upgrade prompt is displayed, insert a USB flash drive that contains the software upgrade package and upgrade the analyzer. After an upgrade success prompt is displayed, shut down and then restart the analyzer.
3. Restart the analyzer. The analyzer enters the initialization process. Then log in at the service access level.
4. Log in to the system at the service access level.
5. Select **Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Other Parts** to enter the interface, and select RFID. Perform the following steps in turn to confirm whether the following requirements are met: Put one label in the card reading area, and click "Self-Test" to perform the operation successfully.
6. Take a blank tube, select the **OV-WB-CD** mode, and after startup, confirm whether the test process alarms.

8.83 PCBA, RFID Antenna

8.83.1 General Information

Figure8–627 Physical Map of RFID Antenna PCBA

Table 8–94 General Information about RFID Antenna PCBA

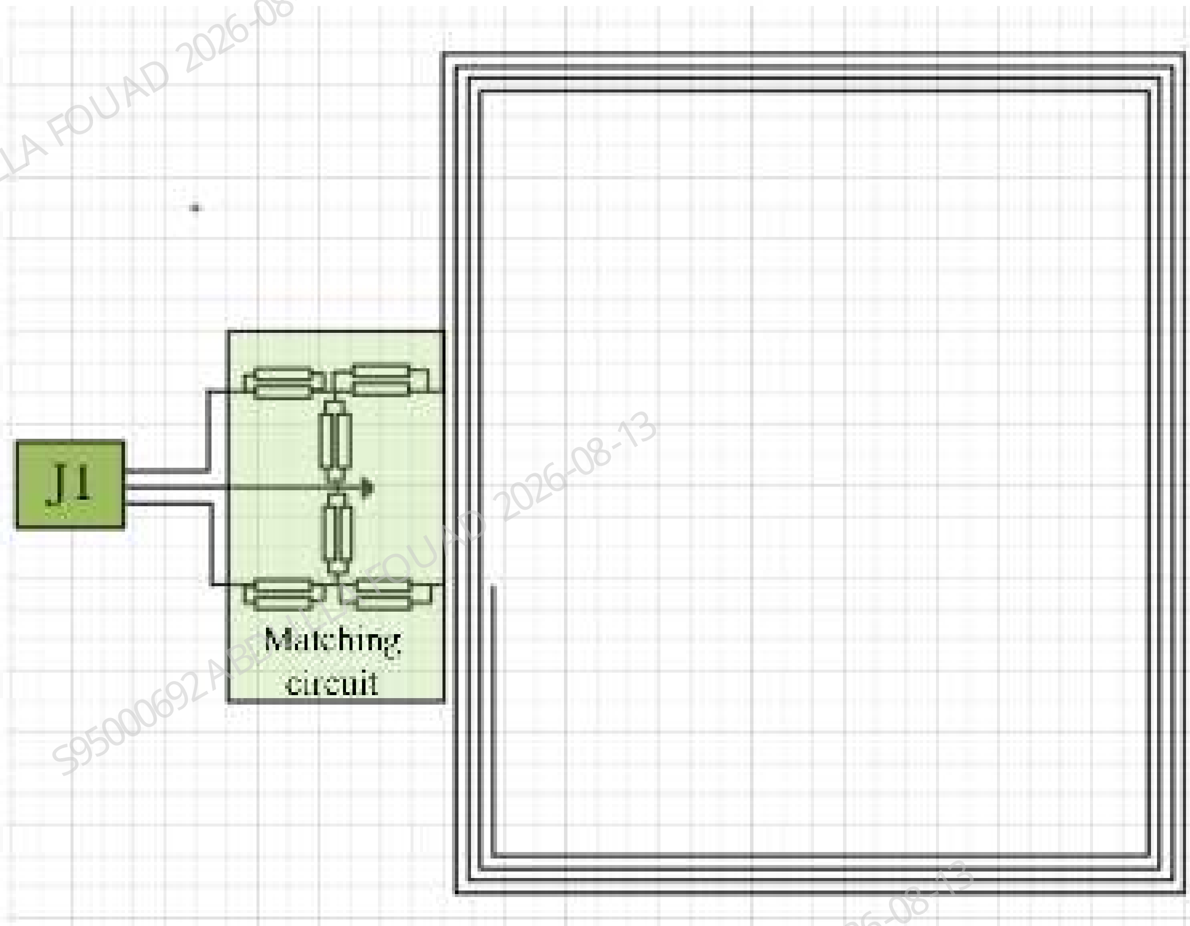
FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU
051-004707-00	051-004707-00	No lower-level FRU	No lower-level FRU

Function:

Connected to the RFID card reader to generate a RFID excitation magnetic field as antenna and implement signal interaction with the RFID card.

Functional block diagram

Figure8–628 Functional block diagram of the board



Connectors

Figure8–629 Diagram of RFID Antenna Interface Position Number

Table 8–95 Definition of RFID Antenna Interface

Position No.	Interface function and definition
J1	Antenna and card reader interface

8.83.2 Disassembly and Assembly

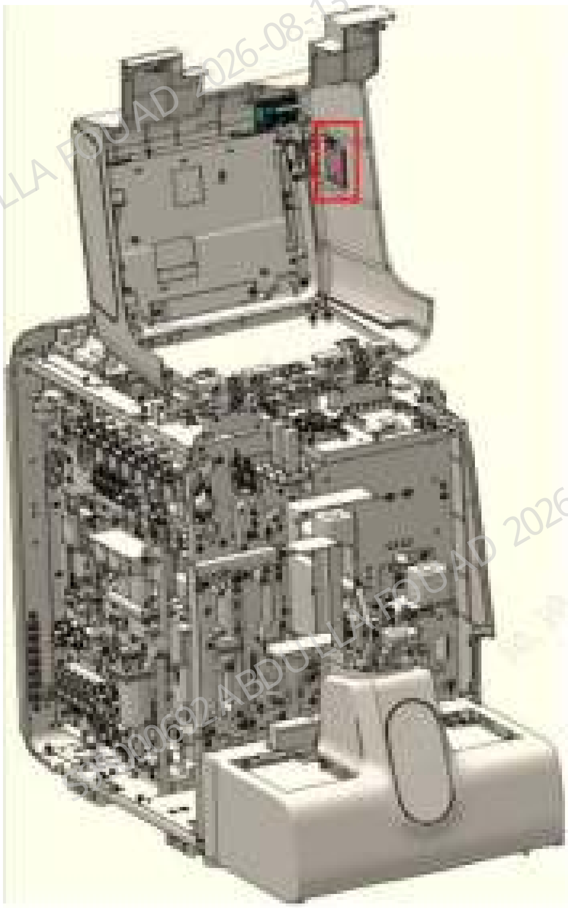
Required tools

- Cross-head screwdriver (M3)(1 set)
- Two sets of ESD gloves

Pre-requirements

1. Press the power button on the back to power off the analyzer. Make sure that the software system and the analyzer screen are off.
2. Open the front cover, and support and fix it.

Figure8–630 Opening the front cover



3. Commissioning/verification personnel should wear ESD gloves. Otherwise, they should not touch the electronic parts with fingers when handling the board. Hold the edge of the board or the blank space.

Procedures

Disassemble according to the steps below:

1. On the front cover, confirm to identify the RFID antenna, and pull out the wires on all the sockets in turn.
2. Remove all the three fixing screws, and take out the board.

Figure8–631 Diagram of RFID Antenna Screw Installation Positions

Install according to the steps below:

1. Make sure the system is powered off.
2. Identify the correct direction of the board and place the board on the installation position. Make sure that all the wire plugs are not pressed by the board.
3. Install all set screws.

8.83.3 Commissioning and Verification

Commissioning/test/verification not required

8.84 Fan assembly

8.84.1 General Information

Figure8–632 Photo of Fan Assembly

Table 8–96 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-059450-00	Fan Assembly	N/A	N/A

Function:
Dissipating heat for the main unit.

8.84.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

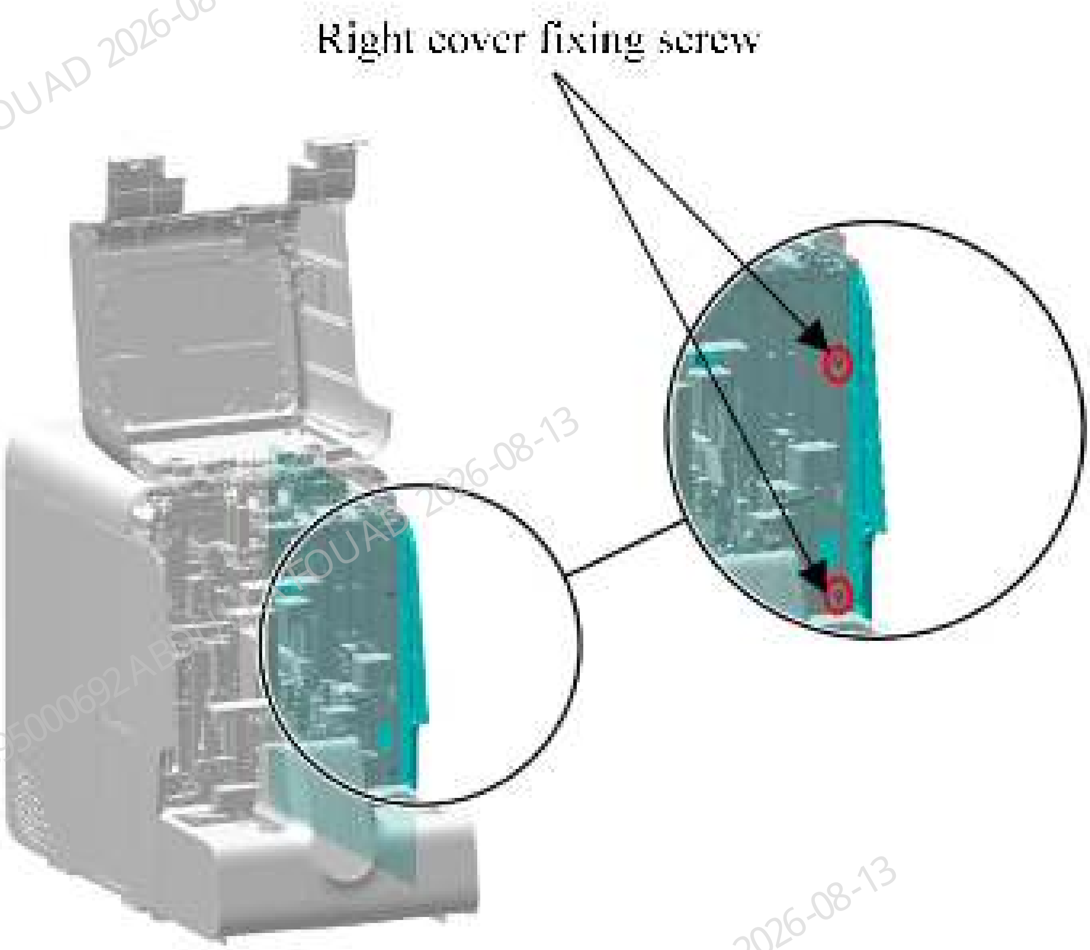
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

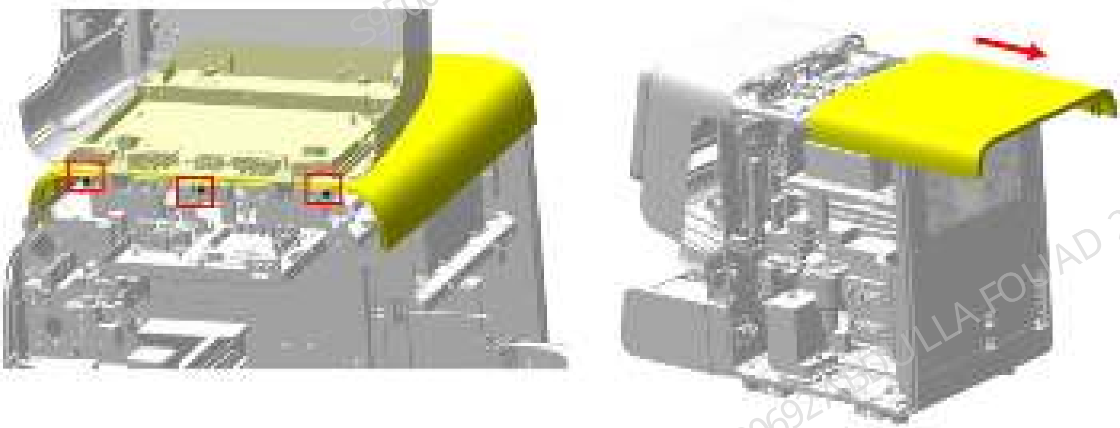
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–633 Removing the right cover



2. As shown in the figure below, remove the three screws fixing the top cover, and drag and remove the top cover in the direction of the arrow.

Figure8–634 Diagram of Removing the Top Cover



3. As shown in Figure A below, buckle off the six screw rubber plugs on the rear cover; as shown in Figure B below, remove the six screws fixing the rear cover, and remove the rear cover.

Figure8–635 Diagram A of Removing the Screw Rubber Plugs on the Rear Cover

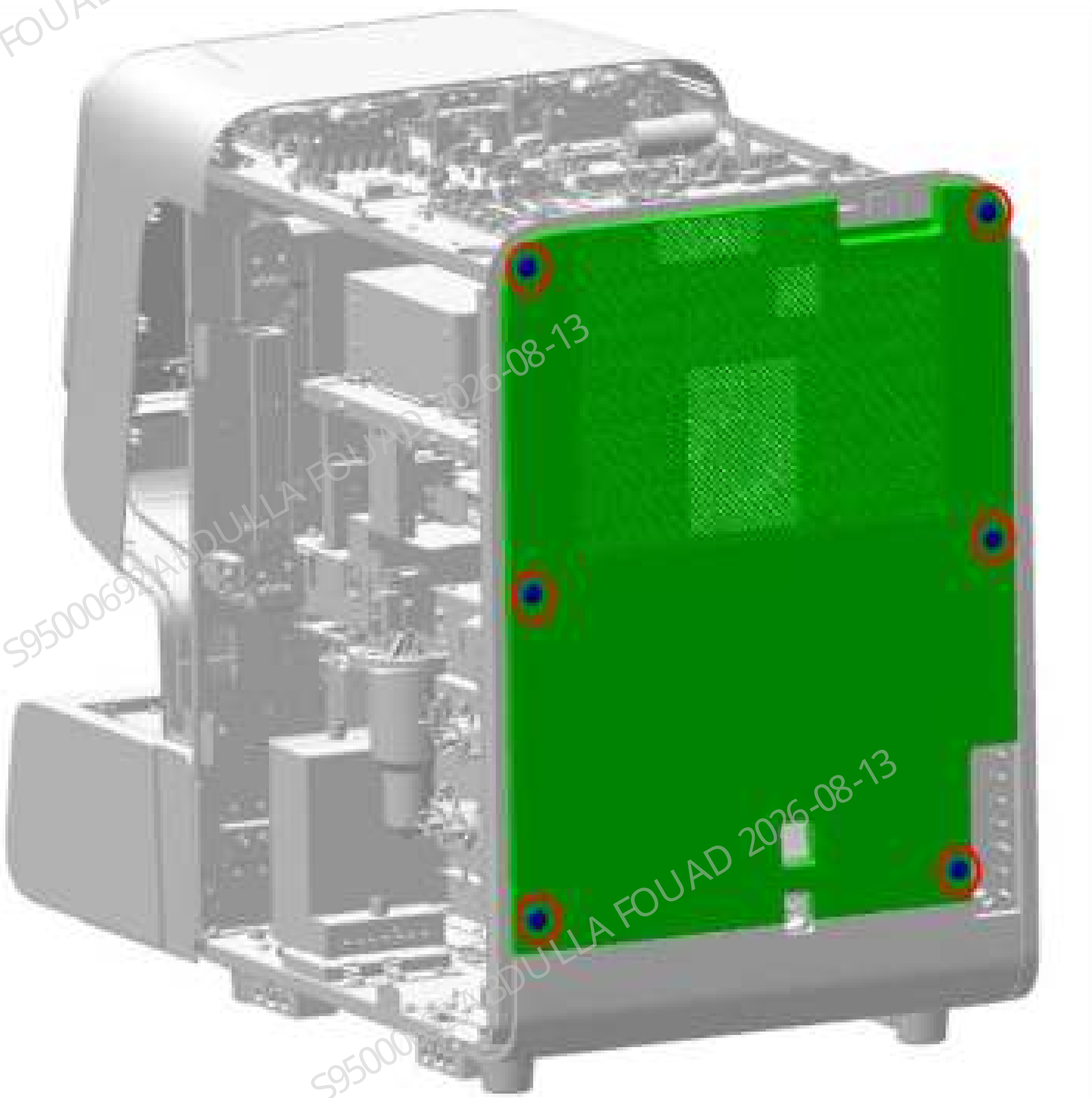
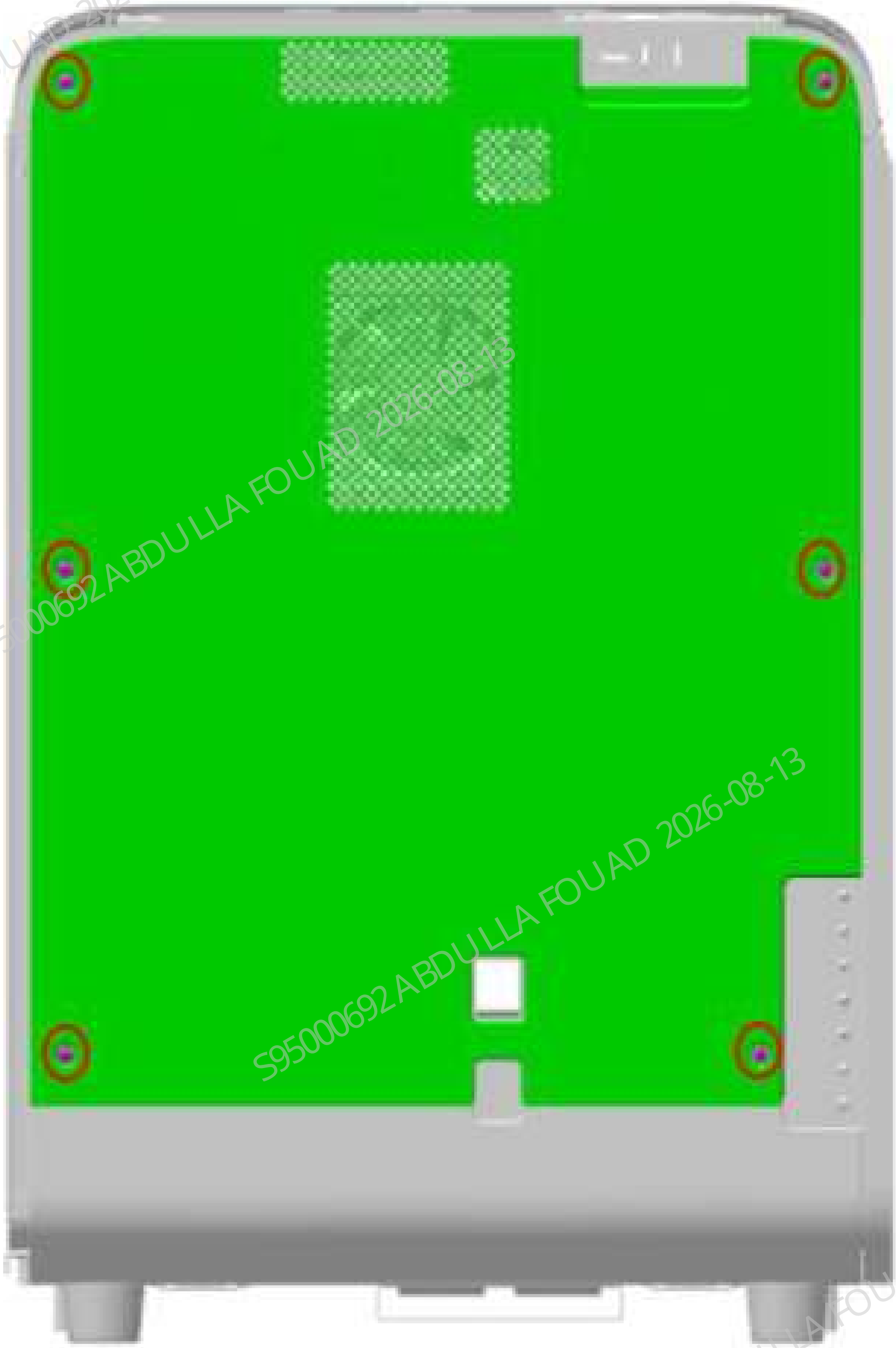
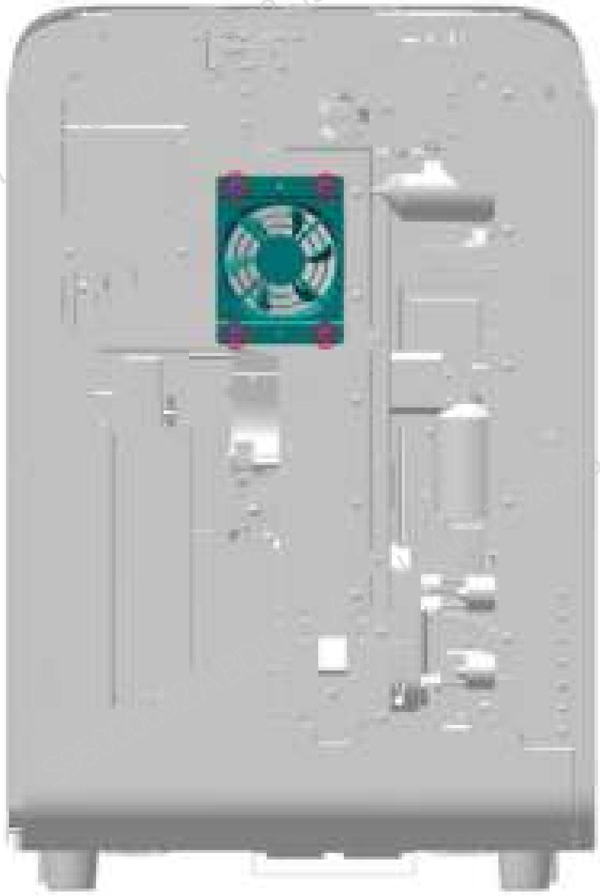


Figure8–636 Diagram B of Removing the Rear Cover Fastening Screws



4. As shown in the figure below, remove the four fastening screws of the fan assembly, remove the fan assembly, and pull out the wire plug of the fan assembly to complete the disassembly.

Figure8–637 Diagram of Removing the Fan Assembly



Subsequent Requirements

Installation procedure:

1. Insert the fan assembly wire.
2. Use four screws to lock the fan assembly on the rear panel.
3. Restore the right, top, and rear covers.
4. Insert back the screw rubber plugs on the rear cover.
5. Put down the front cover for resetting.

8.84.3 Commissioning and Verification

Procedures

Commissioning:

1. After logging in at the service access level, click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Other Parts**, and select fan, as shown in the figure below.

Figure8–638 Diagram of Fan Assembly Commissioning Interface

2. Click "Close Fan" of system fan to observe whether the fan is turned off; click "Open Fan" of system fan to observe whether the fan is turned on.
3. If the fans are all functioning normally, the commissioning and verification of the board fan assembly is completed.
4. If the fan is functioning abnormally, check whether the installation is correct and perform commissioning again.
5. If the fan function problems persists, replace the assembly and repeat the disassembly/assembly and commissioning methods.

8.85 Board Fan Assembly

8.85.1 General Information

Figure8–639 Photo of Board Fan Assembly

Table 8–97 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-078552-00	Board Fan Assembly	N/A	N/A

Function:
Dissipating heat for the main unit.

8.85.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

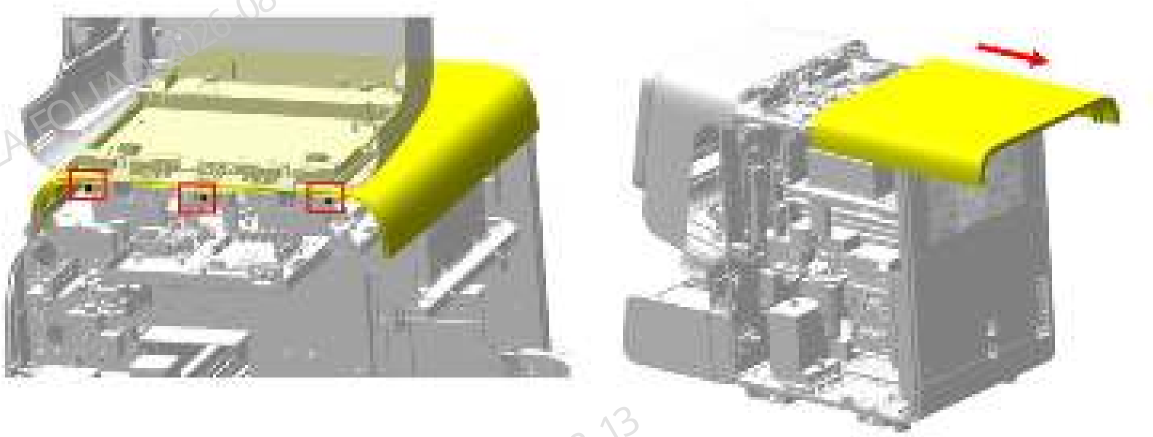
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. As shown in the figure below, open the front cover and lean it on the top cover, remove the three screws fixing the top cover, put the top cover down for resetting, and drag and remove the top cover in the direction of the arrow.

Figure8–640 Diagram of Removing the Top Cover



2. As shown in Figure A below, buckle off the six screw rubber plugs on the rear cover; as shown in Figure B below, remove the six screws fixing the rear cover, and remove the rear cover.

Figure8–641 Diagram A of Removing the Screw Rubber Plugs on the Rear Cover

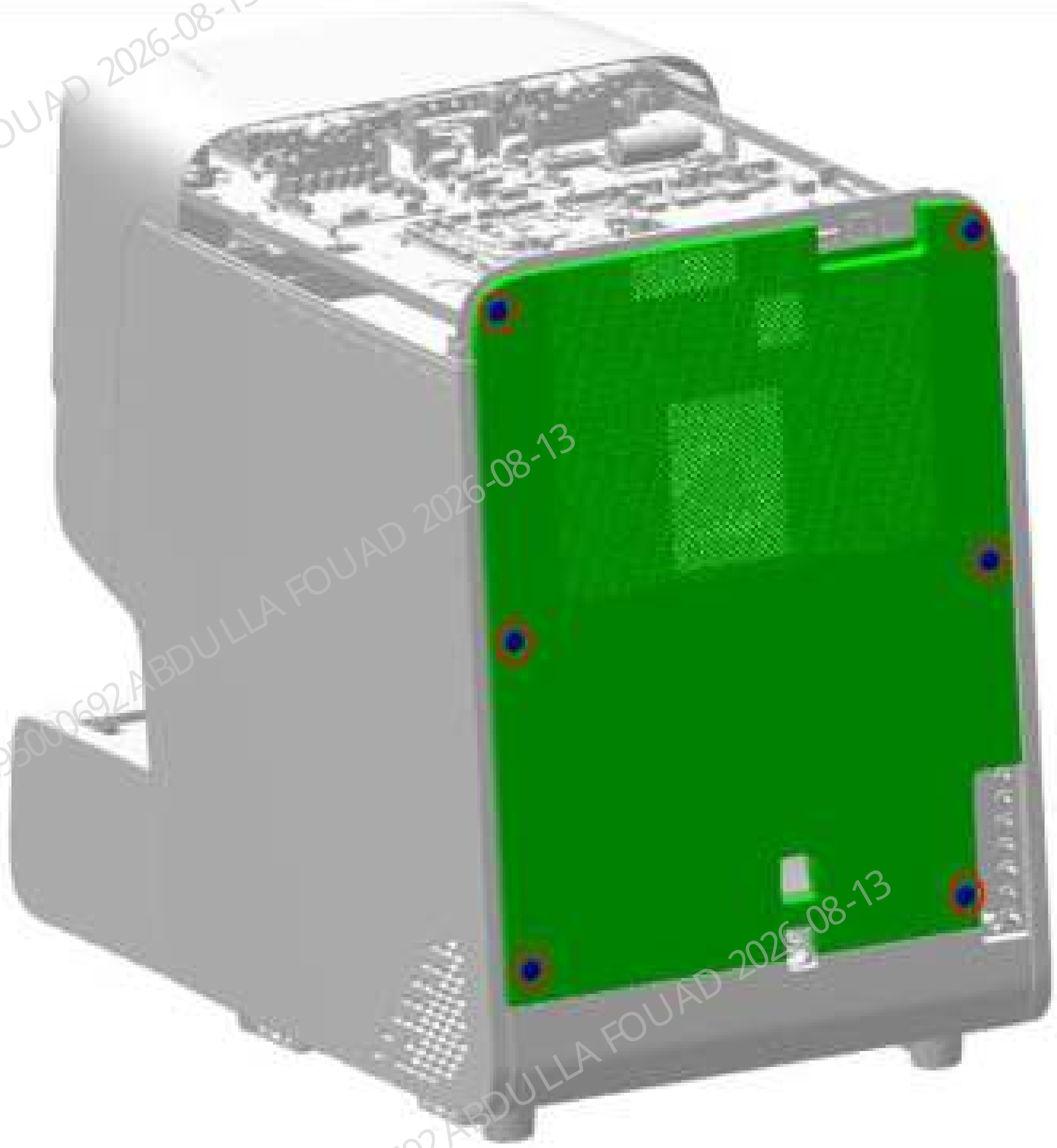
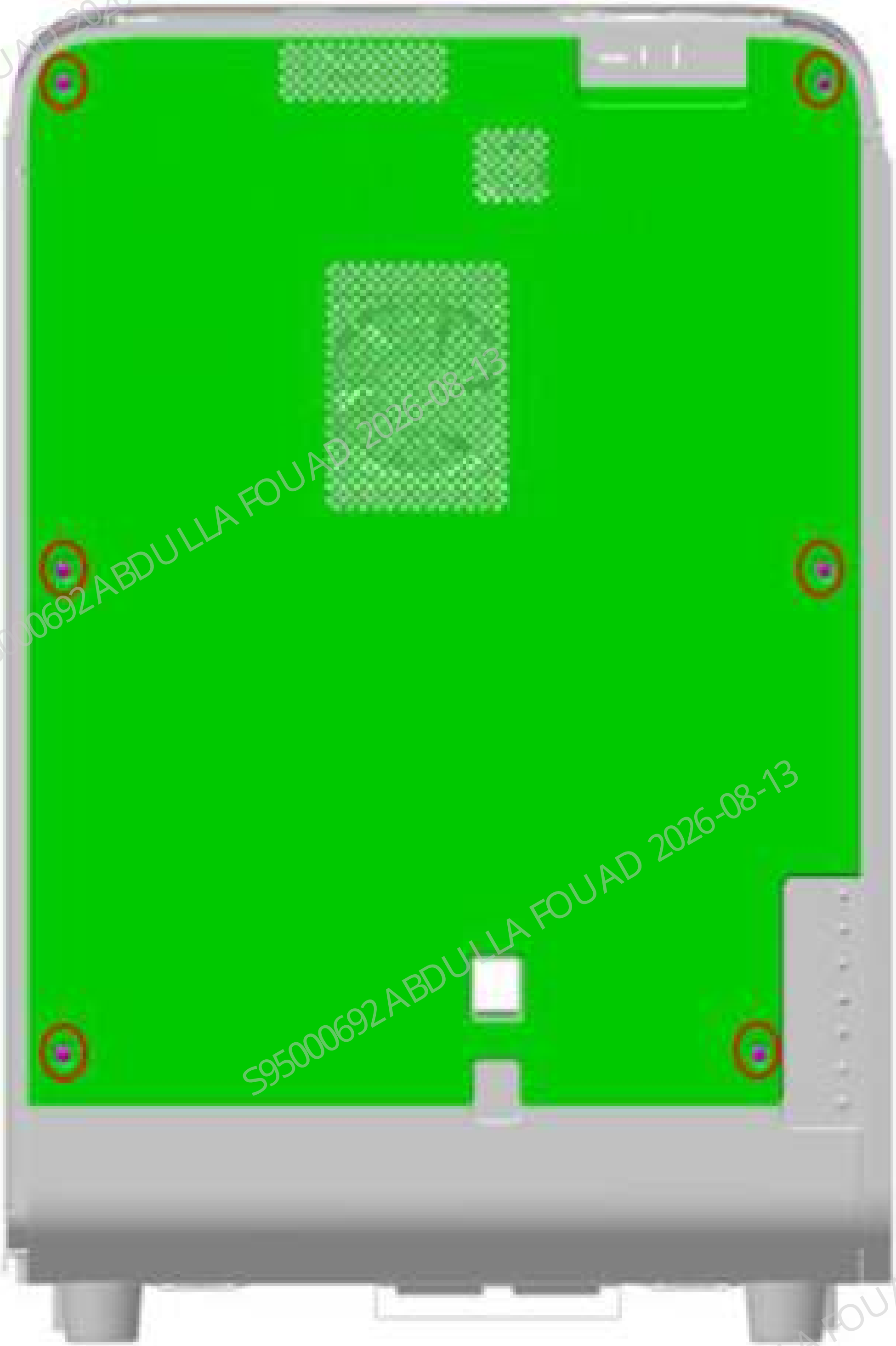
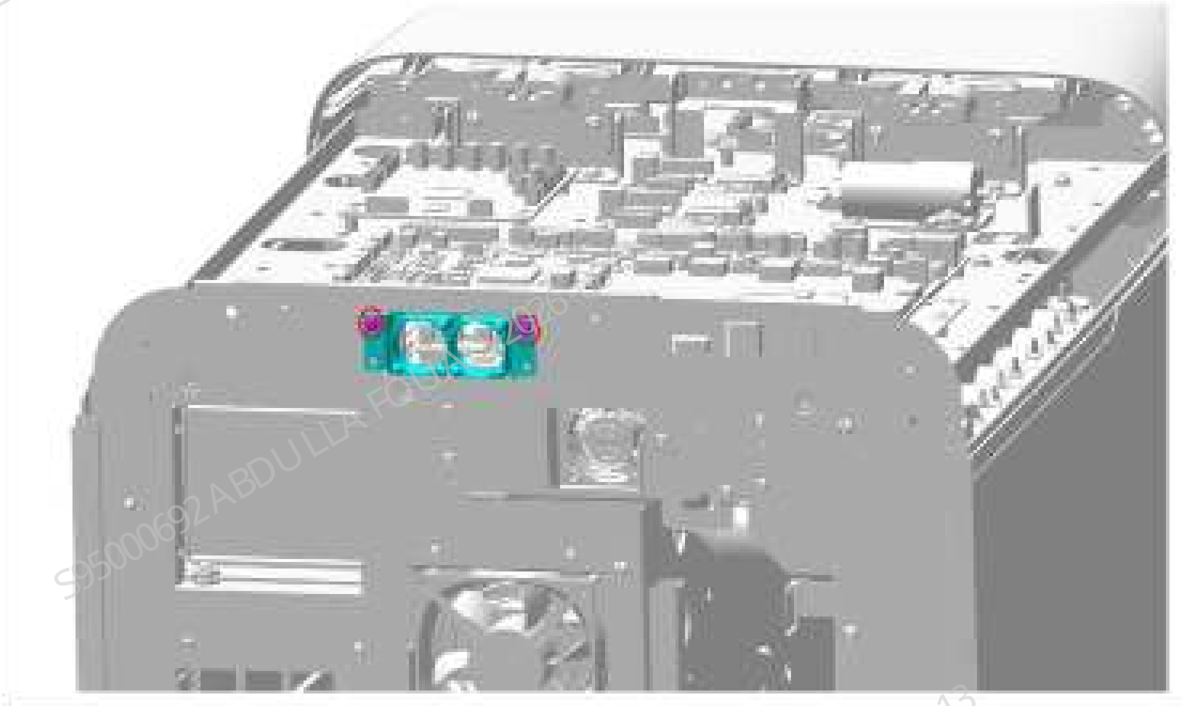


Figure8–642 Diagram B of Removing the Rear Cover Fastening Screws



3. As shown in the figure below, remove the two fastening screws of the board fan assembly, remove the board fan assembly, and pull out the wire plug of the board fan assembly to complete the disassembly.

Figure8–643 Diagram of Removing the Board Fan Assembly



Subsequent Requirements

Installation procedure:

1. Insert the wire of the board fan assembly.
2. Use two screws to lock the board fan assembly on the rear panel.
3. Restore the top and rear covers.
4. Insert back the screw rubber plugs on the rear cover.
5. Put down the front cover for resetting.

8.85.3 Commissioning and Verification

Procedures

Commissioning:

1. After logging in at the service access level, click **Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Other Parts**, and select fan, as shown in the figure below.

Figure8–644 Diagram of Board Fan Assembly Commissioning Interface

2. Click "Close Fan" of board fan 1 to observe whether the fan is turned off; click "Open Fan" of board fan 1 to observe whether the fan is turned on.
3. Click "Close Fan" of board fan 2 to observe whether the fan is turned off; click "Open Fan" of board fan 2 to observe whether the fan is turned on.
4. If the fans are all functioning normally, the commissioning and verification of the board fan assembly is completed.
5. If the fan is functioning abnormally, check whether the installation is correct and perform commissioning again.
6. If the fan function problems persists, replace the assembly and repeat the disassembly/ assembly and commissioning methods.

8.86 Sensor Temperature 5Kohm B3470K Threaded Installation

8.86.1 General Information

Figure8–645 Photo of Ambient Temperature Sensor

Table 8–98 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
024-000110-00	Ambient temperature sensor	N/A	N/A

Function:
Detecting the temperature of the surrounding environment where the analyzer is located.

8.86.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. As shown in Figure A below, open the front cover and lean it on the top cover, loosen the two screws fixing the autoloader, as shown in Figure B below, drag the autoloader to an appropriate distance, then pull out the autoloader wire, and finally completely separate the autoloader from the main unit.

Figure8–646 Diagram A of Removing the Autoloader

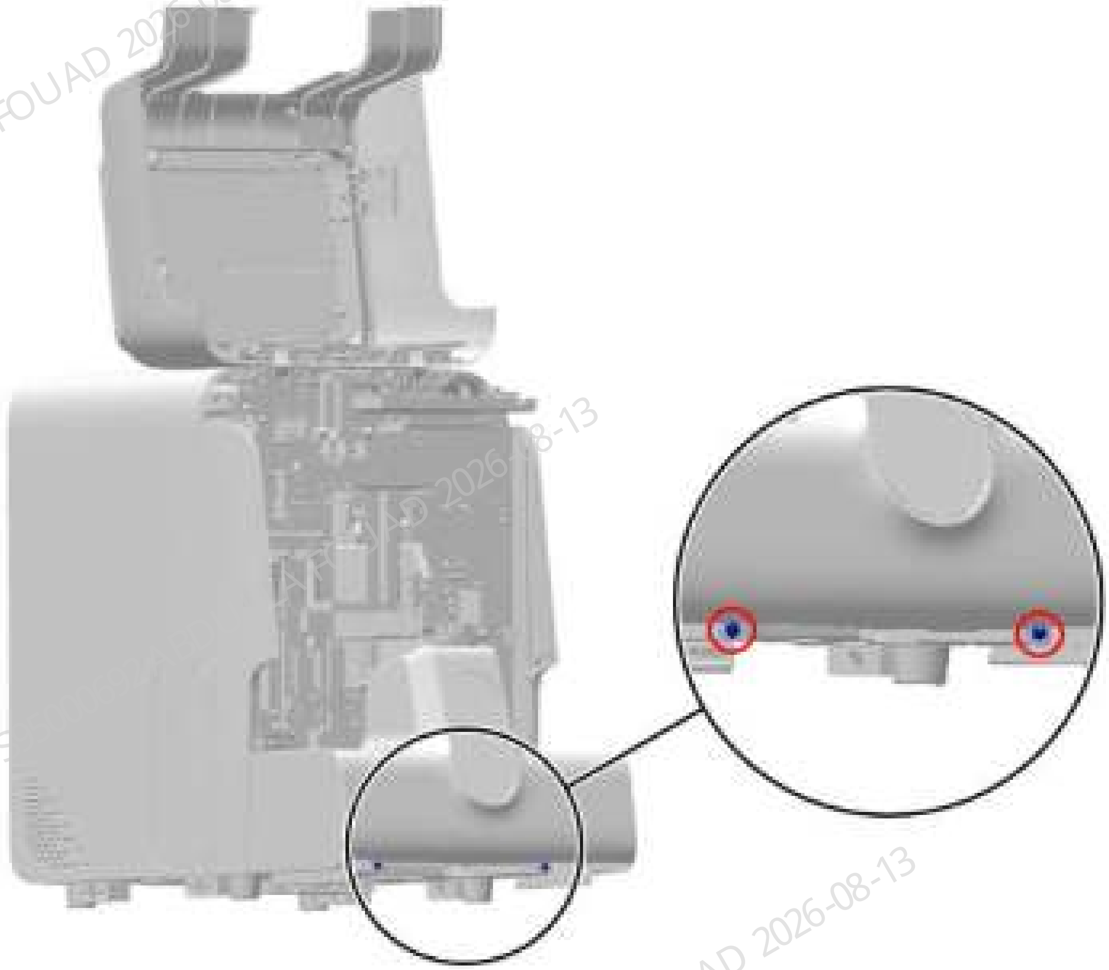
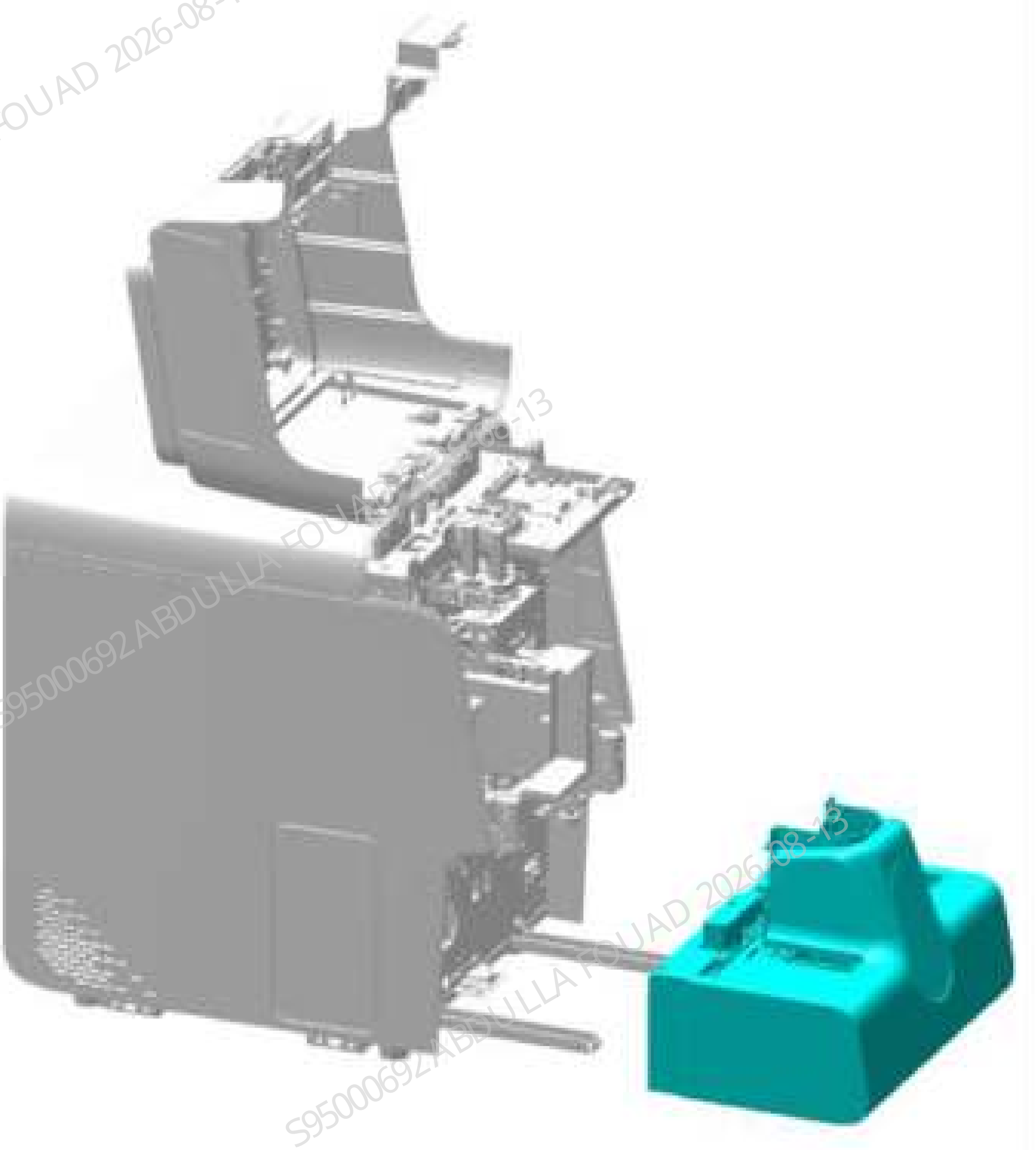
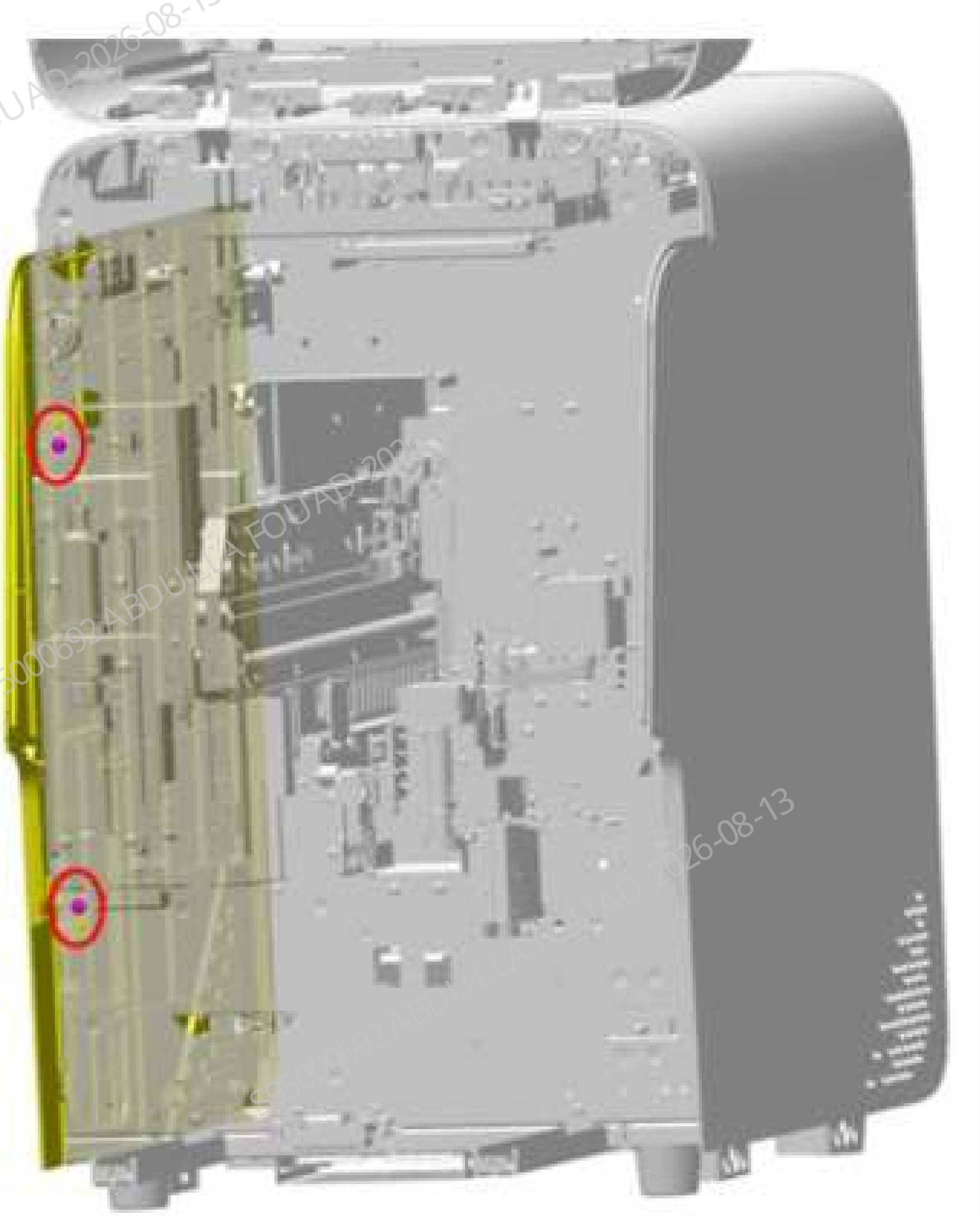


Figure8–647 Diagram B of Removing the Autoloader



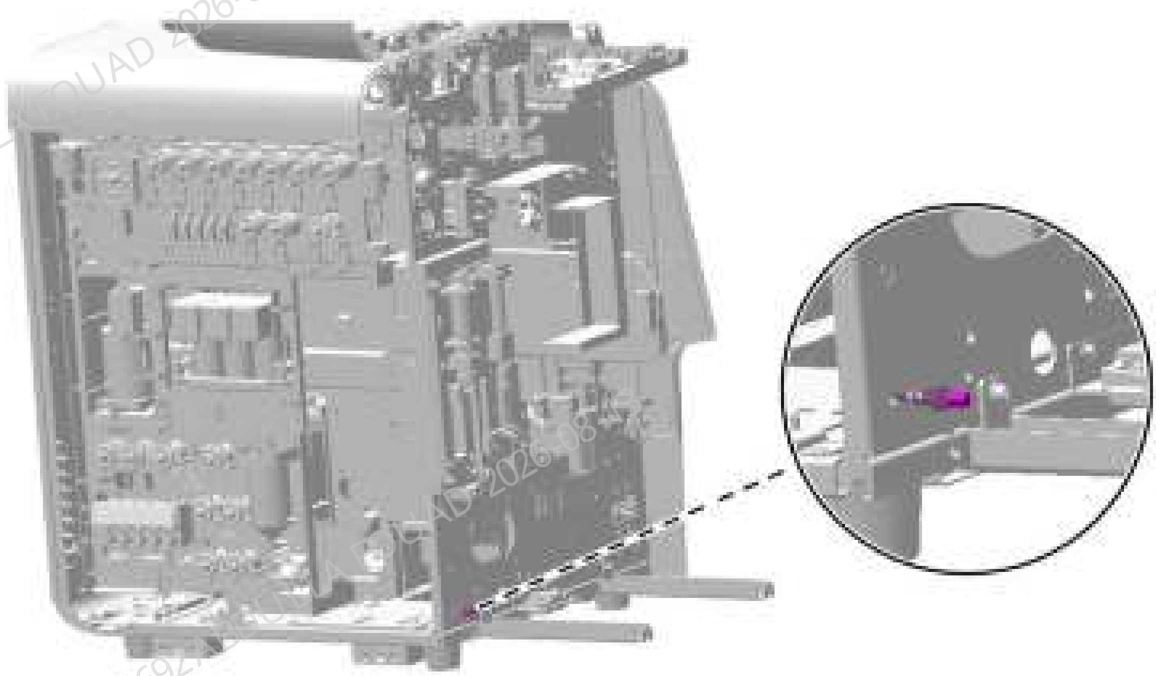
2. As shown below, loosen the two screws that fix the left cover, and remove the left cover.

Figure8–648 Diagram of Removing the Left Cover



3. As shown in the figure below, loosen and remove the nut on the head of the ambient temperature sensor.

Figure8–649 Diagram of Removing the Ambient Temperature Sensor



4. As shown in the figure below, pull out the plug of the ambient temperature sensor to remove the temperature sensor.

Figure8–650 Diagram of Removing the Ambient Temperature Sensor Wire

Subsequent Requirements

Installation procedure:

1. Insert the ambient temperature sensor plug.
2. Install the ambient temperature sensor on the main unit and lock it using nuts.
3. Restore the left cover.
4. Connect the autoloader wire and install it back on the main unit.
5. Put down the front cover for resetting.

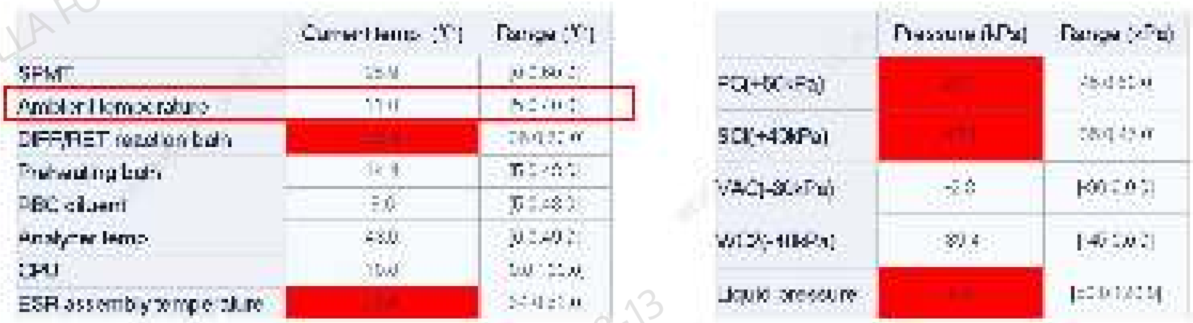
8.86.3 Commissioning and Verification

Procedures

Commissioning:

1. After logging in at the service access level, click **Status**>>**Temperature & Pressure**, as shown in the figure below.

Figure8–651 Diagram of Ambient Temperature Sensor Commissioning Interface



2. Check if the ambient temperature is within the required range.
3. If the temperature is normal, the commissioning and verification of the ambient temperature sensor is completed.
4. If the temperature is abnormal, check whether the installation is correct and perform commissioning again.
5. If the temperature is still abnormal, replace the ambient temperature sensor and repeat the disassembly/assembly and commissioning methods.

8.87 Power Assembly

8.87.1 General Information

Figure8–652 Photo of power supply assembly

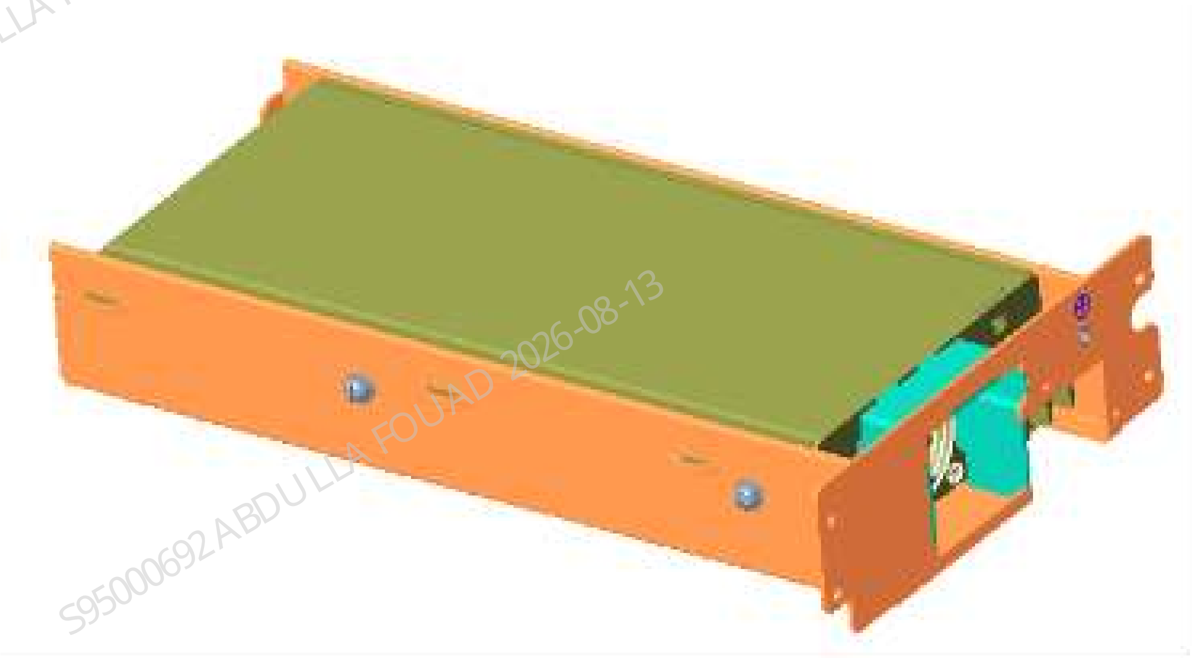


Table 8–99 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-074574-00	Power Supply Assembly	N/A	N/A

Function:
Provide power for the analyzer.

8.87.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

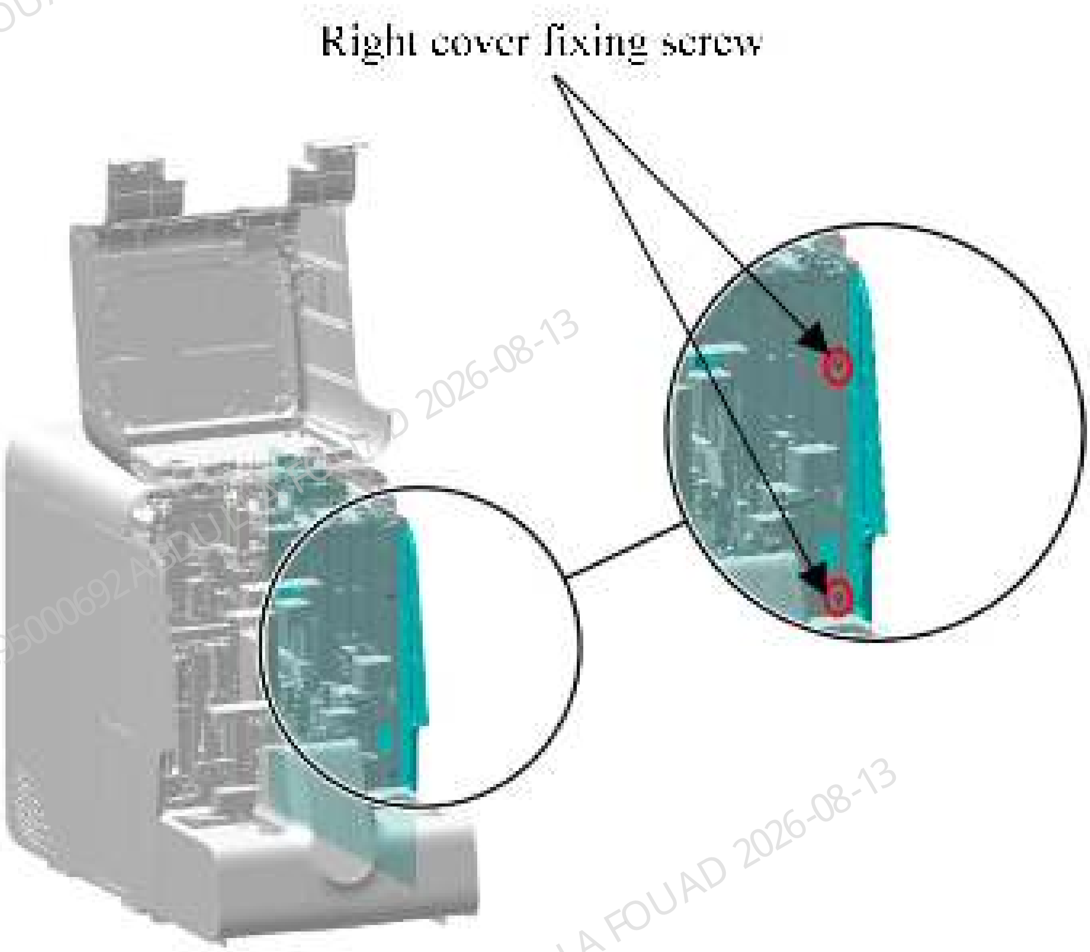
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

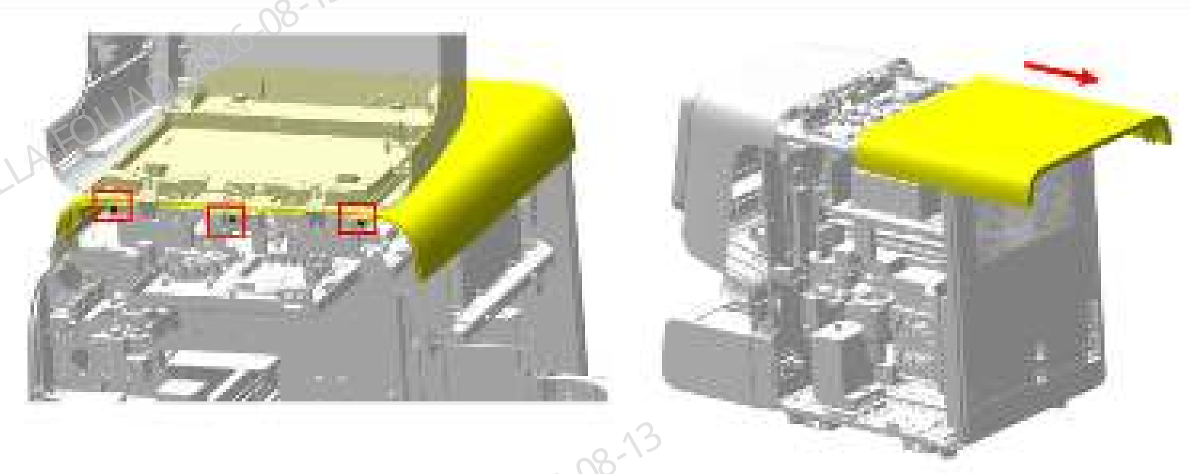
1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–653 Diagram of Removing the Right Cover



2. As shown in the figure below, remove the three screws fixing the top cover, put the top cover down for resetting, and drag and remove the top cover in the direction of the arrow.

Figure8–654 Diagram of Removing the Top Cover



3. As shown in Figure A below, buckle off the six screw rubber plugs on the rear cover; as shown in Figure B below, remove the six screws fixing the rear cover, and remove the rear cover.

Figure8–655 Diagram A of Removing the Screw Rubber Plugs on the Rear Cover

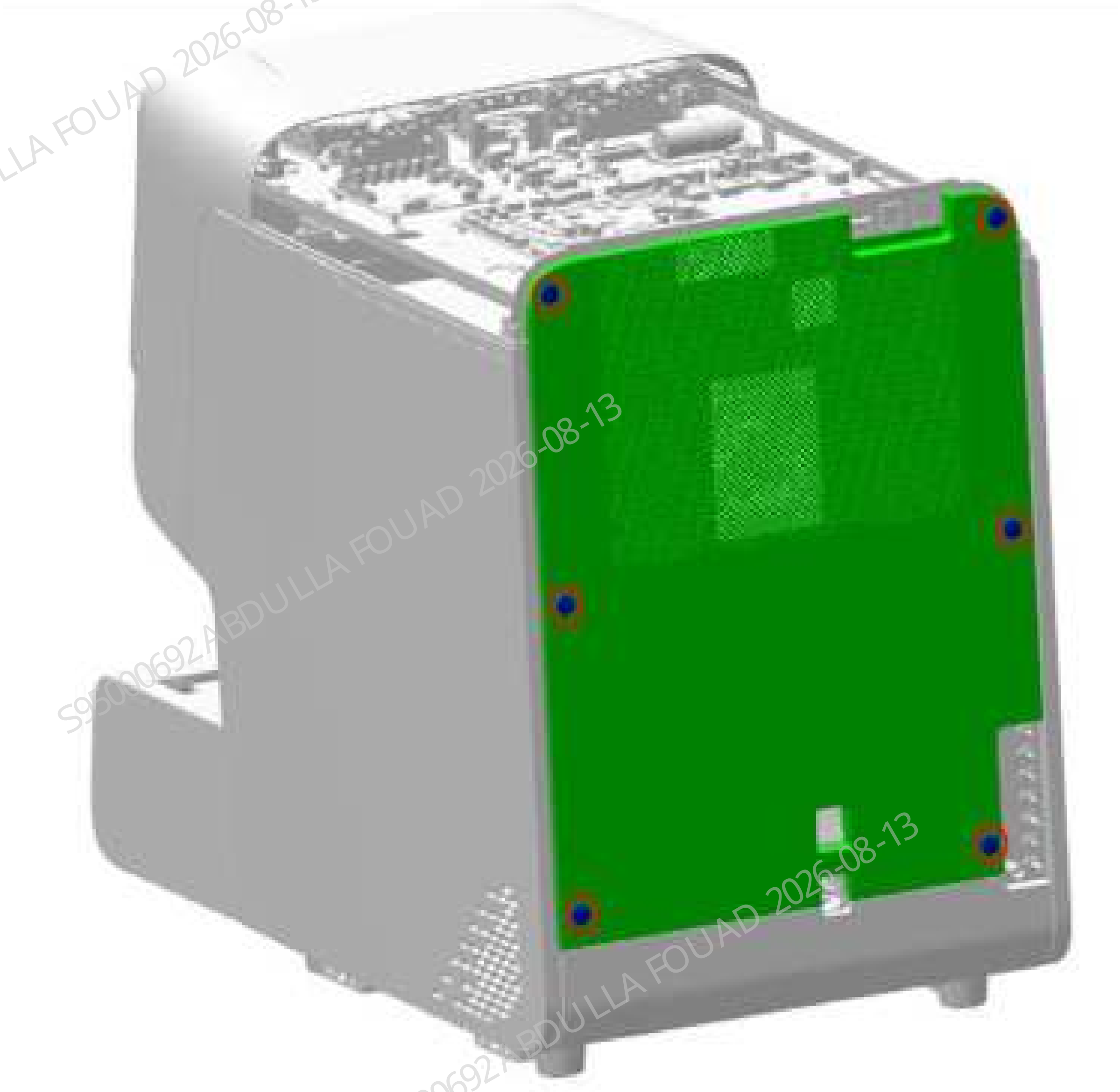
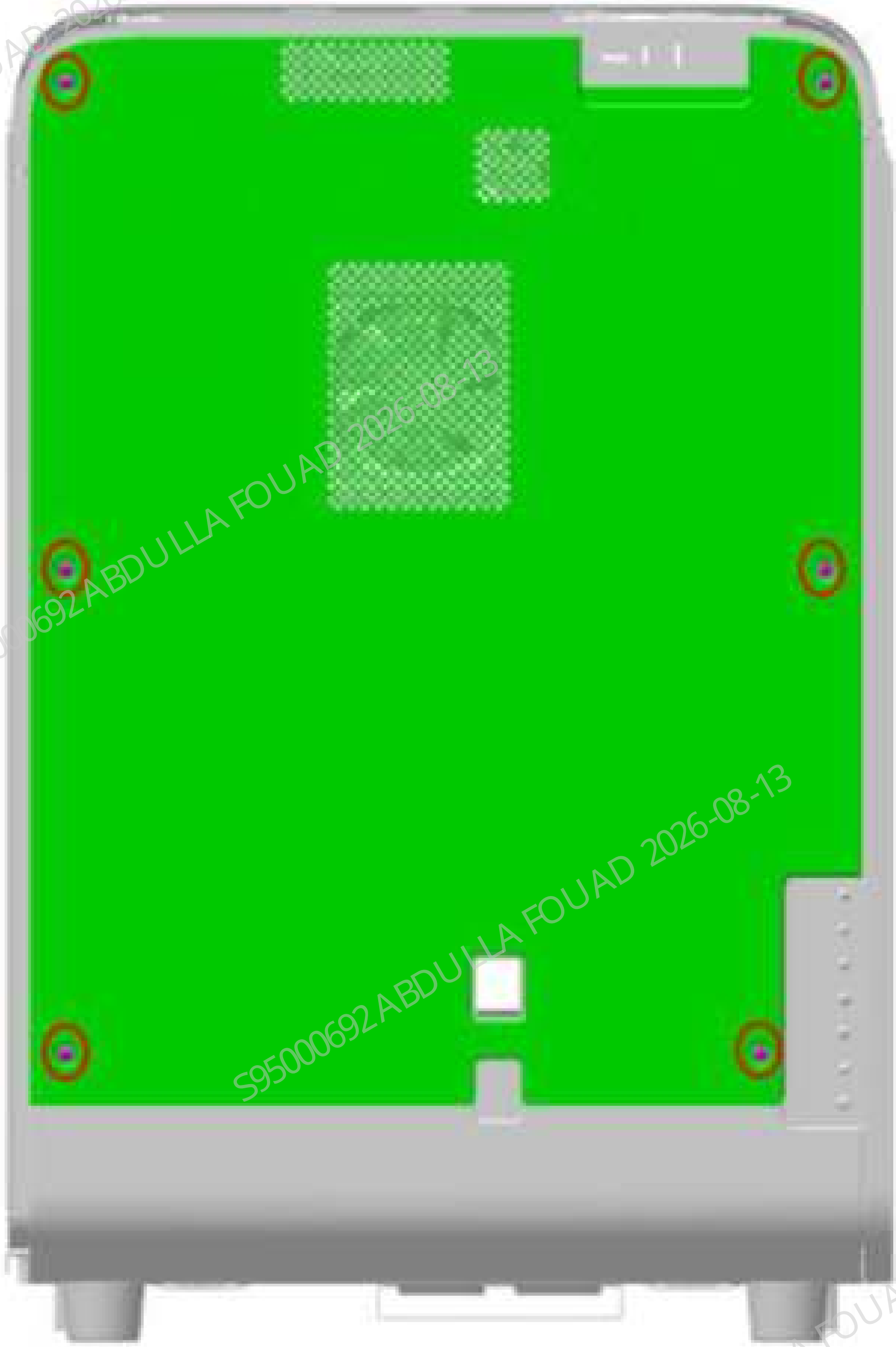


Figure8–656 Diagram of Removing the Rear Cover Fastening Screws



4. As shown in Figure A below, remove the two fastening screws of the power terminal shielding cover; as shown in Figure B below, remove the wire at the power input end from the terminal.

Figure8–657 Figure A of Removing Power Terminal Shielding Cover

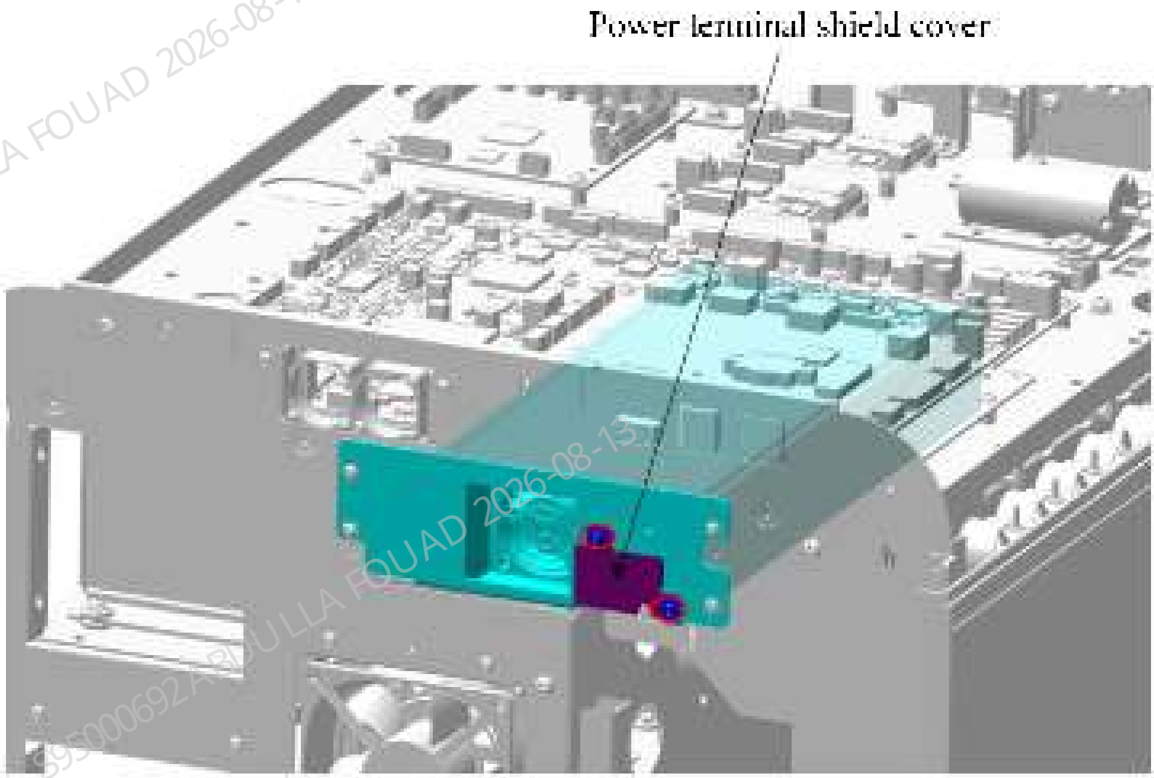
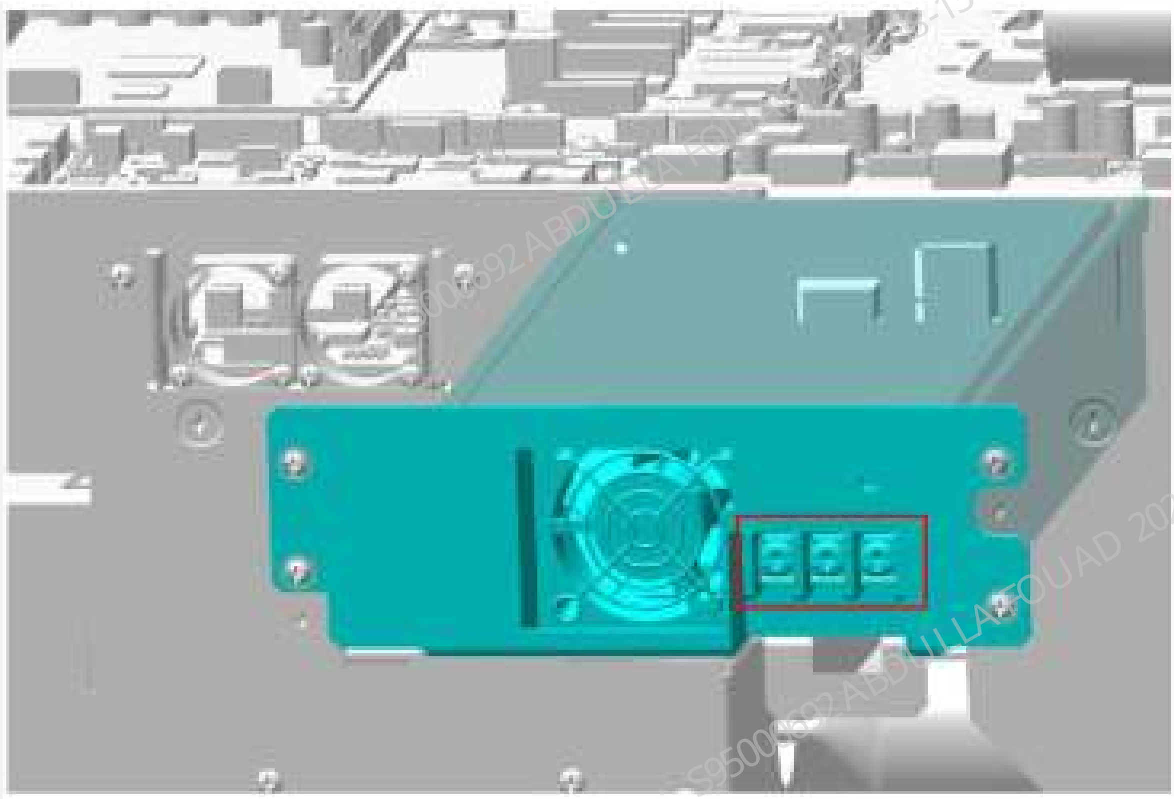
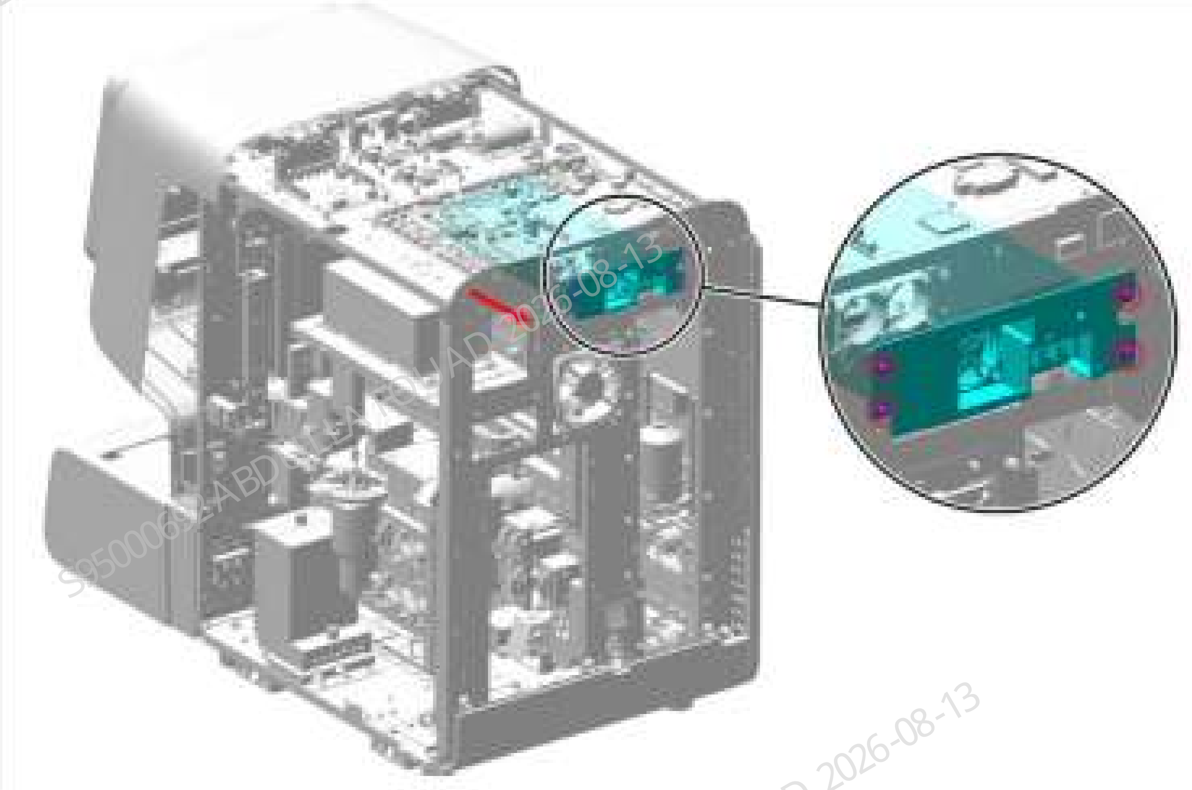


Figure8–658 Figure B of Removing Power Input End Wire



5. As shown in the figure below, remove the four fastening screws fixing the power supply assembly, pull the power supply assembly out of the main unit along the arrow direction, and remove the wire at the power output end. Then, the power supply assembly has been removed.

Figure8–659 Diagram of Removing the Power Supply Assembly



Subsequent Requirements

Installation procedure:

1. Insert the wire at the output end of the power supply assembly.
2. Push the power supply assembly into the main unit and insert the input end wire of the power supply assembly.
3. Use for screws to fix the power supply assembly onto the main unit.
4. Use two screws to fix the power terminal shielding cover on the power supply assembly.
5. Restore the top, right, and rear covers.
6. Insert back the screw rubber plugs on the rear cover.
7. Put down the front cover for resetting.

8.87.3 Commissioning and Verification

Procedures

Commissioning:

1. Start-up.
2. If the analyzer is started normally, the power supply assembly - commissioning/test and verification is completed.
3. If the analyzer cannot be started normally, return to the part of the power supply assembly - disassembly and assembly method to check whether the assembly is installed correctly. After troubleshooting, if the analyzer can be started again normally, the power supply assembly - commissioning/test and verification is completed.
4. If the assembly is installed correctly and the analyzer cannot be started normally, select another power supply assembly and carry out the disassembly and assembly method until the analyzer is started normally.
5. The power supply assembly - commissioning/test and verification is completed.

8.88 Power Switch Assembly

8.88.1 General Information

Figure8–660 Photo of Power Supply Assembly

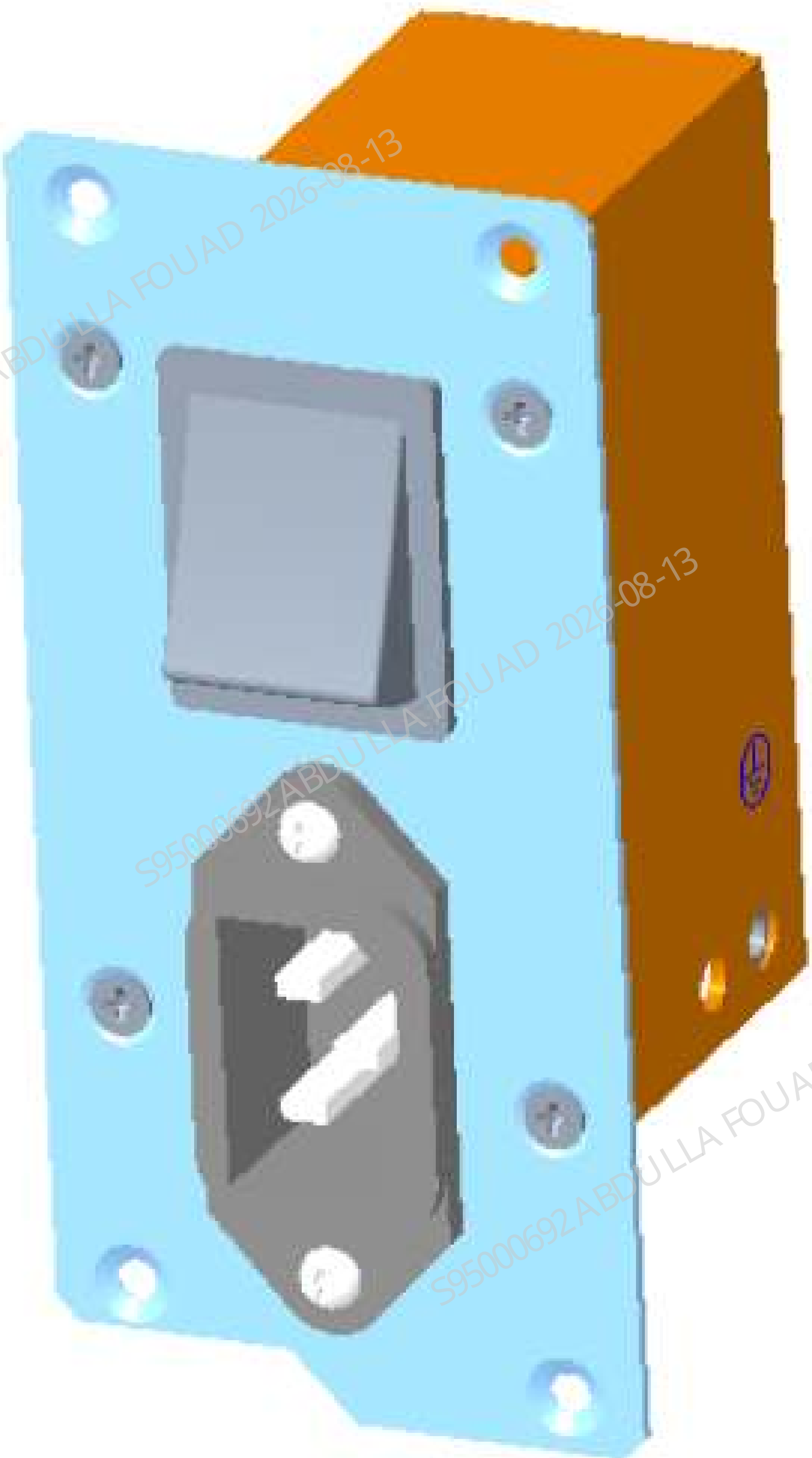


Table 8–100 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-074584-00	Power Switch Assembly	N/A	N/A

Function:
Controlling connection/disconnection of the analyzer power supply.

8.88.2 Disassembly and Assembly

Required tools

Cross-headed screwdriver

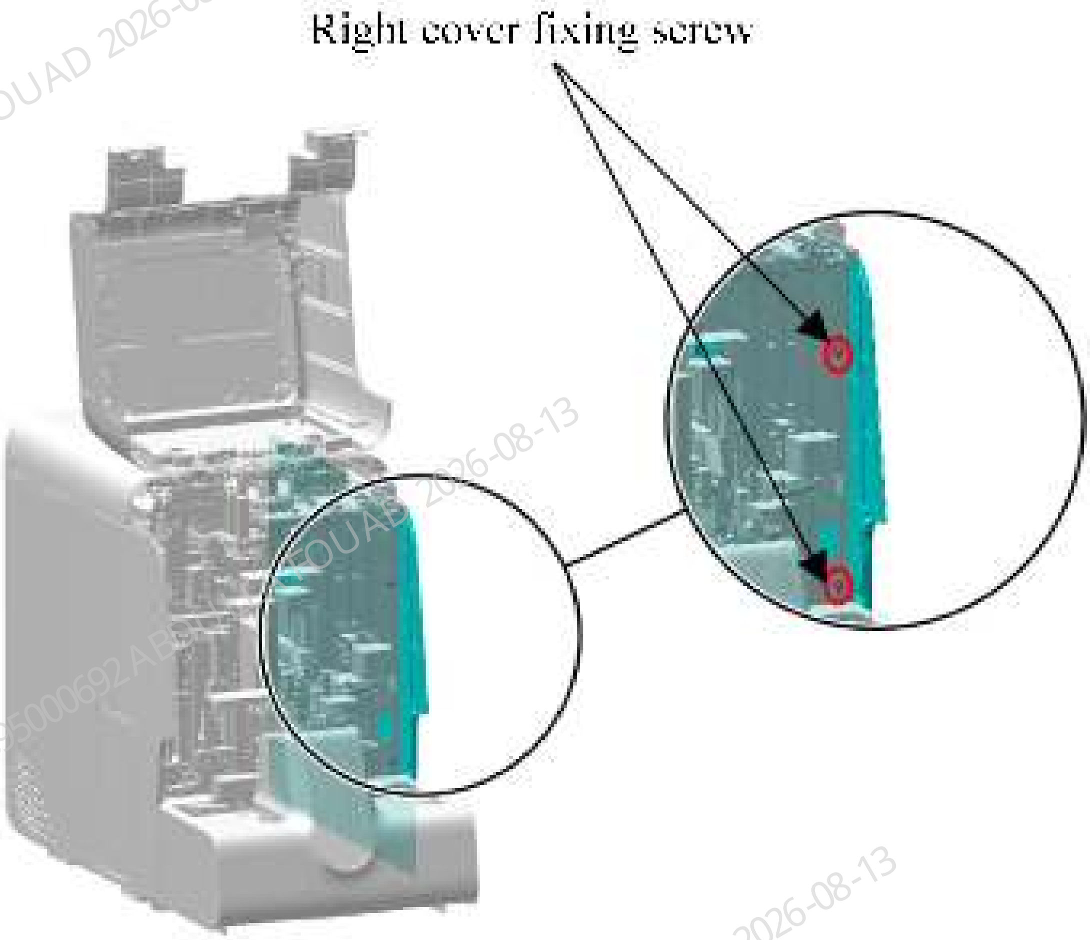
Pre-requirements

Power off the analyzer, and unplug the power cord.

Procedures

1. Open the front cover and dock it on the top cover, loosen the two screws that fix the right cover, and remove the right cover, as shown in the figure below.

Figure8–661 Diagram of Removing the Right Cover



2. As shown in Figure A below, buckle off the six screw rubber plugs on the rear cover; as shown in Figure B below, remove the six screws fixing the rear cover, and remove the rear cover.

Figure8–662 Diagram A of Removing the Screw Rubber Plugs on the Rear Cover

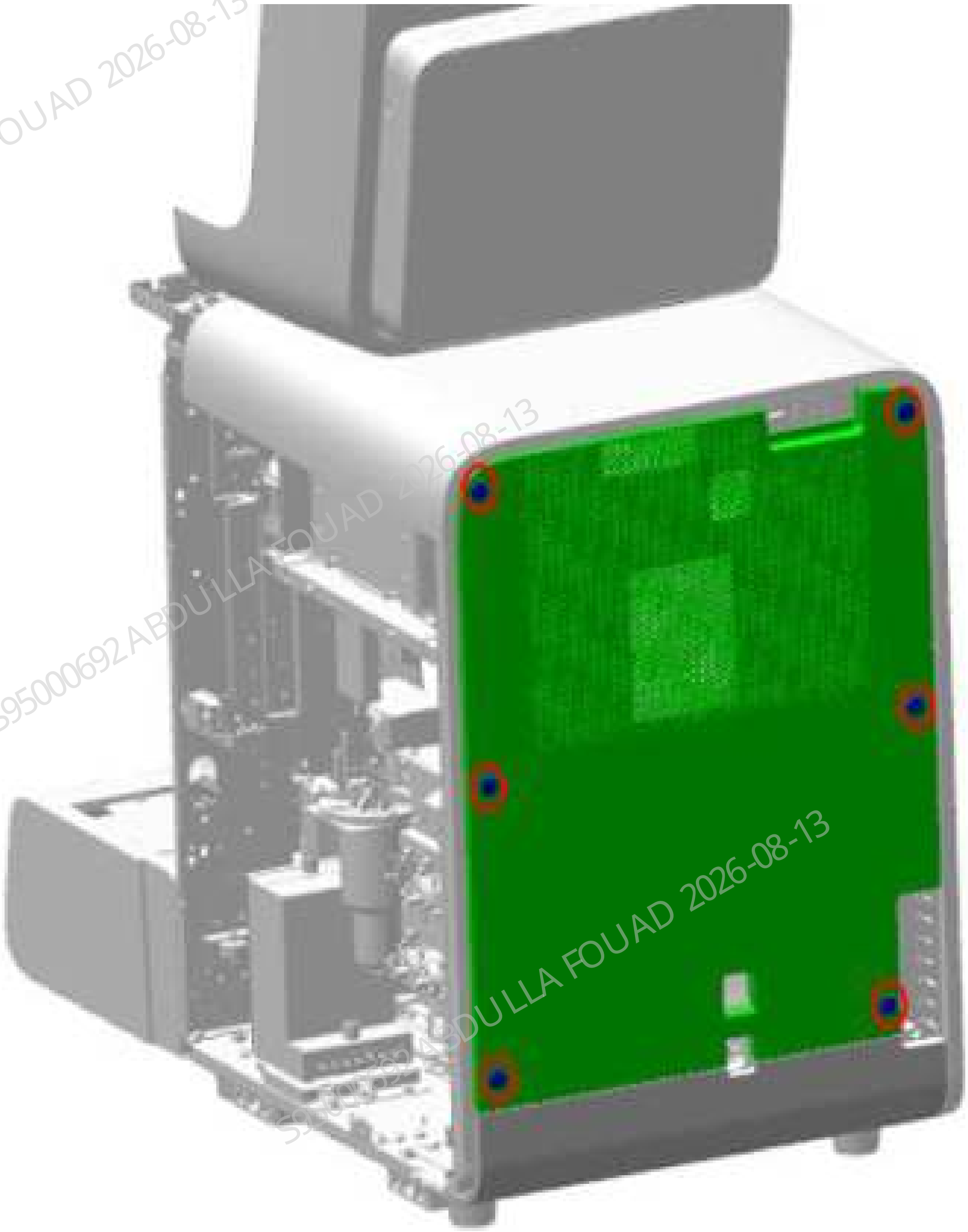
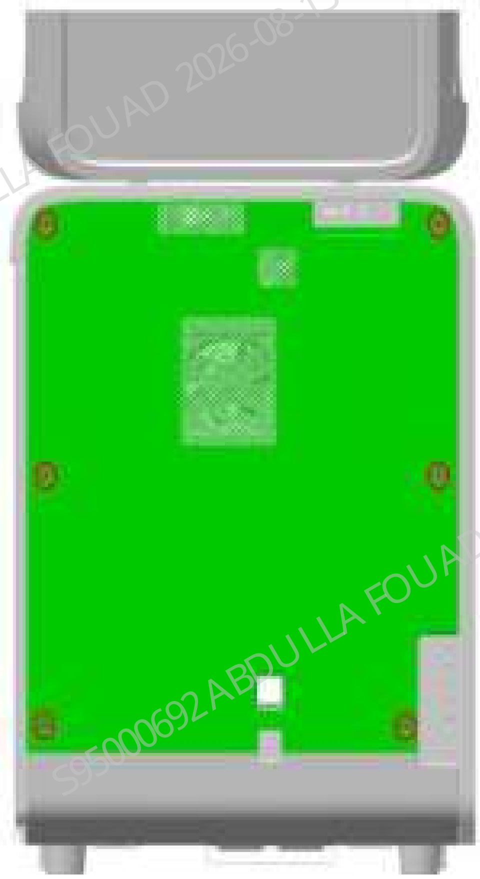
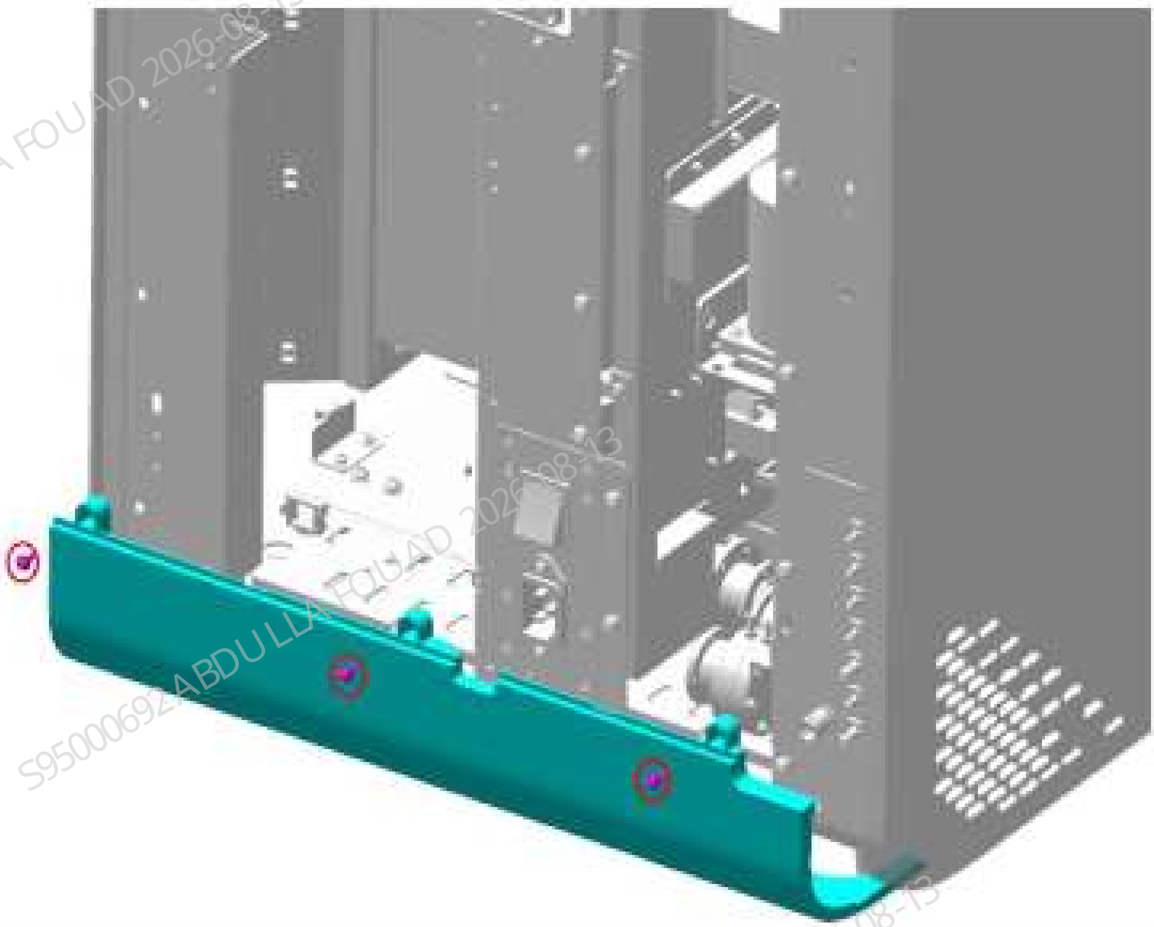


Figure8–663 Diagram B of Removing the Rear Cover Fastening Screws



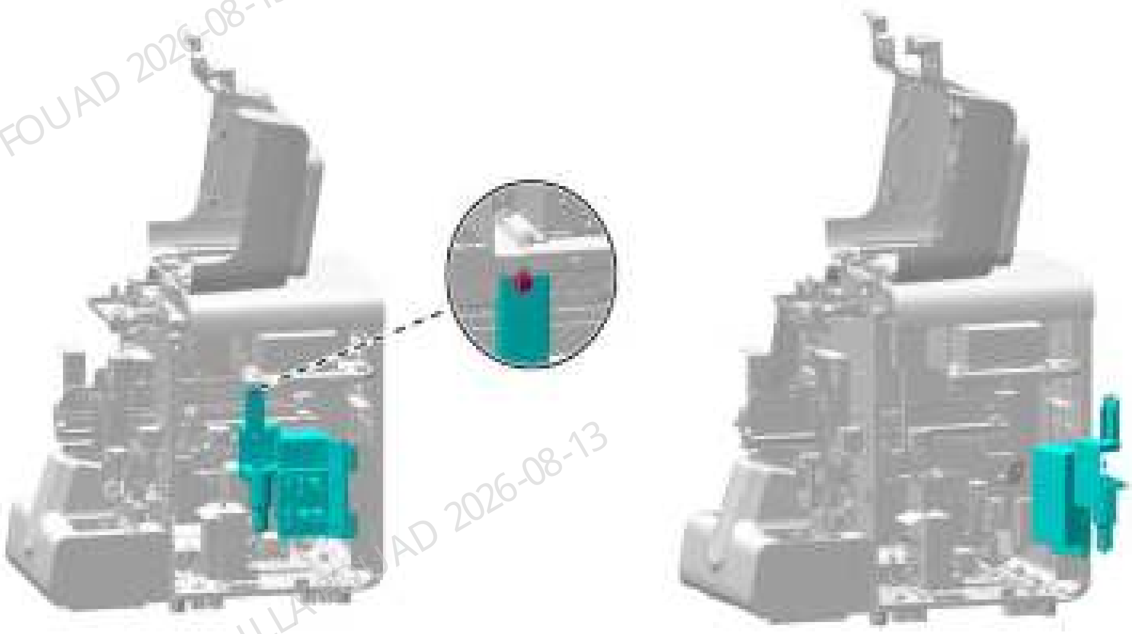
3. Remove the three screws fixing the bottom shell, and take down the bottom shell, as shown below.

Figure8–664 Removing the Bottom Shell



4. As shown below, unscrew the one screw that fixes the right bath subassembly, as shown in the figure below, lift the hook of the right bath subassembly upward, disengage it, and turn it outwards.

Figure8–665 Opening the right cover assembly



5. As shown in Figure A below, remove the two screws of the power terminal shielding cover and remove the power terminal shielding cover. As shown in Figure B below, remove the wire at the power input end.

Figure8–666 Figure A of Removing Power Terminal Shielding Cover

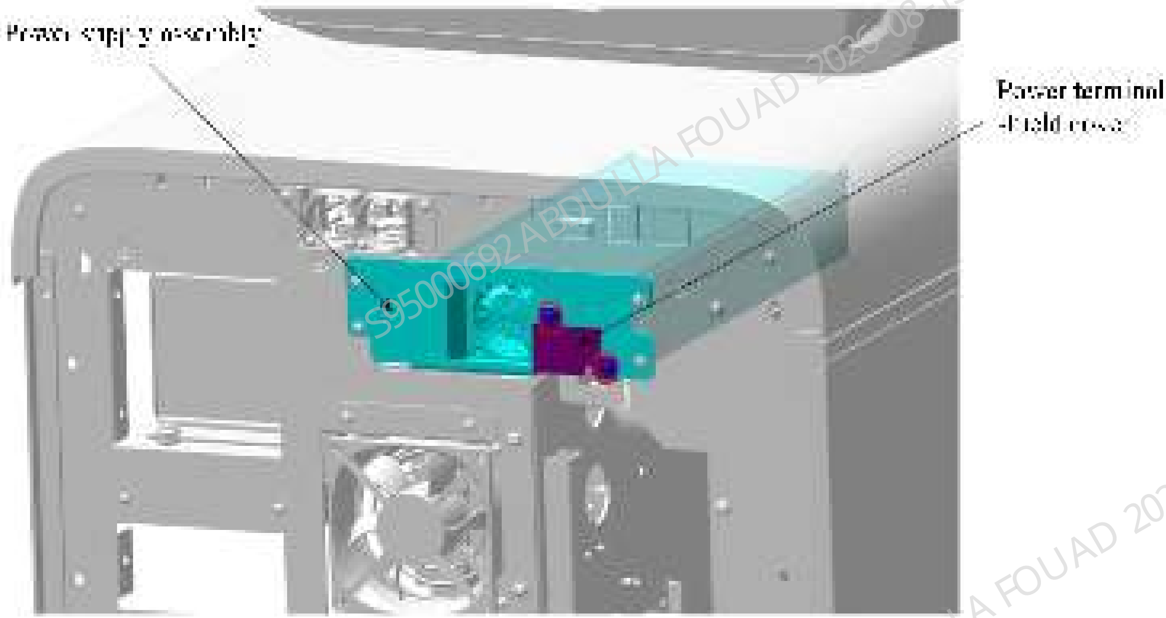
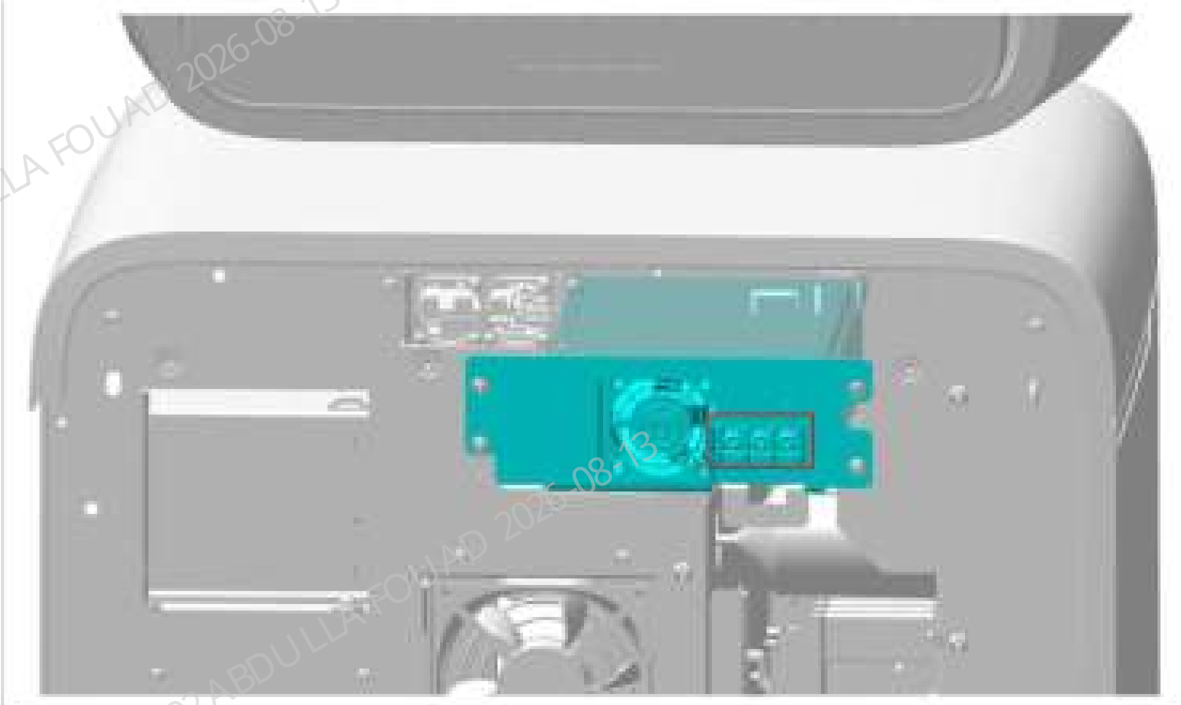


Figure8–667 Figure B of Removing Power Input End Wire



6. As shown in Figure A below, remove the screw fixing the power cord shielding cover from the right side of the main unit. As shown in Figure B below, remove the other two screws fixing the power cord shielding cover and remove the power cord shielding cover.

Figure8–668 Figure A of Removing Power Cord Shielding Cover

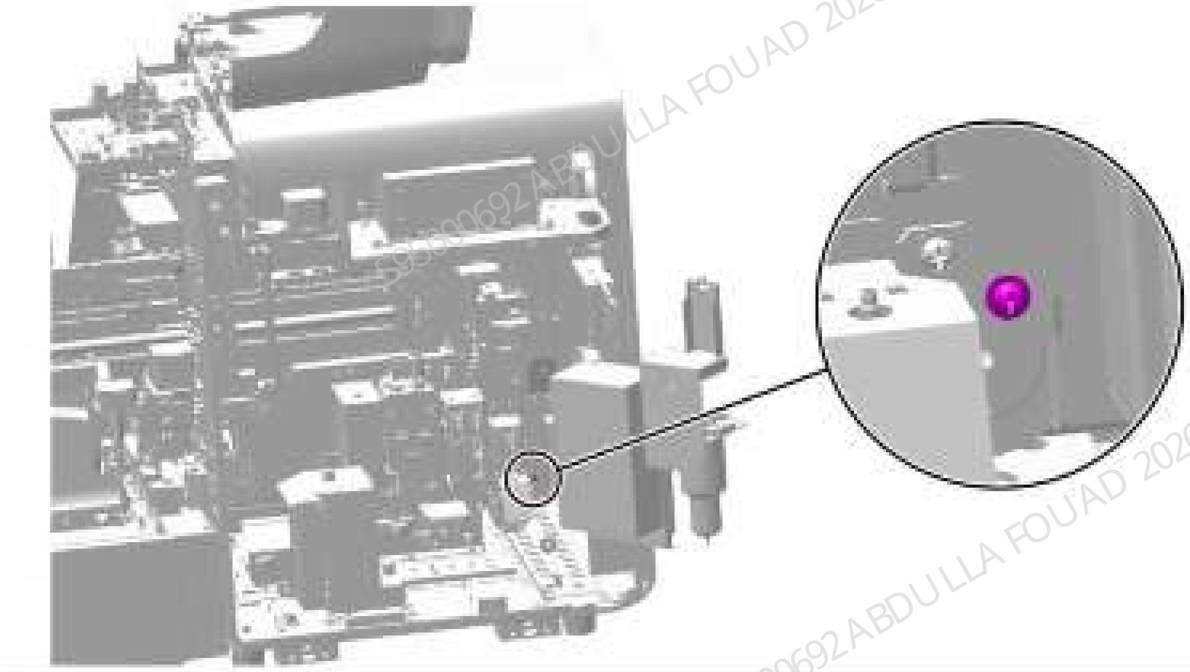
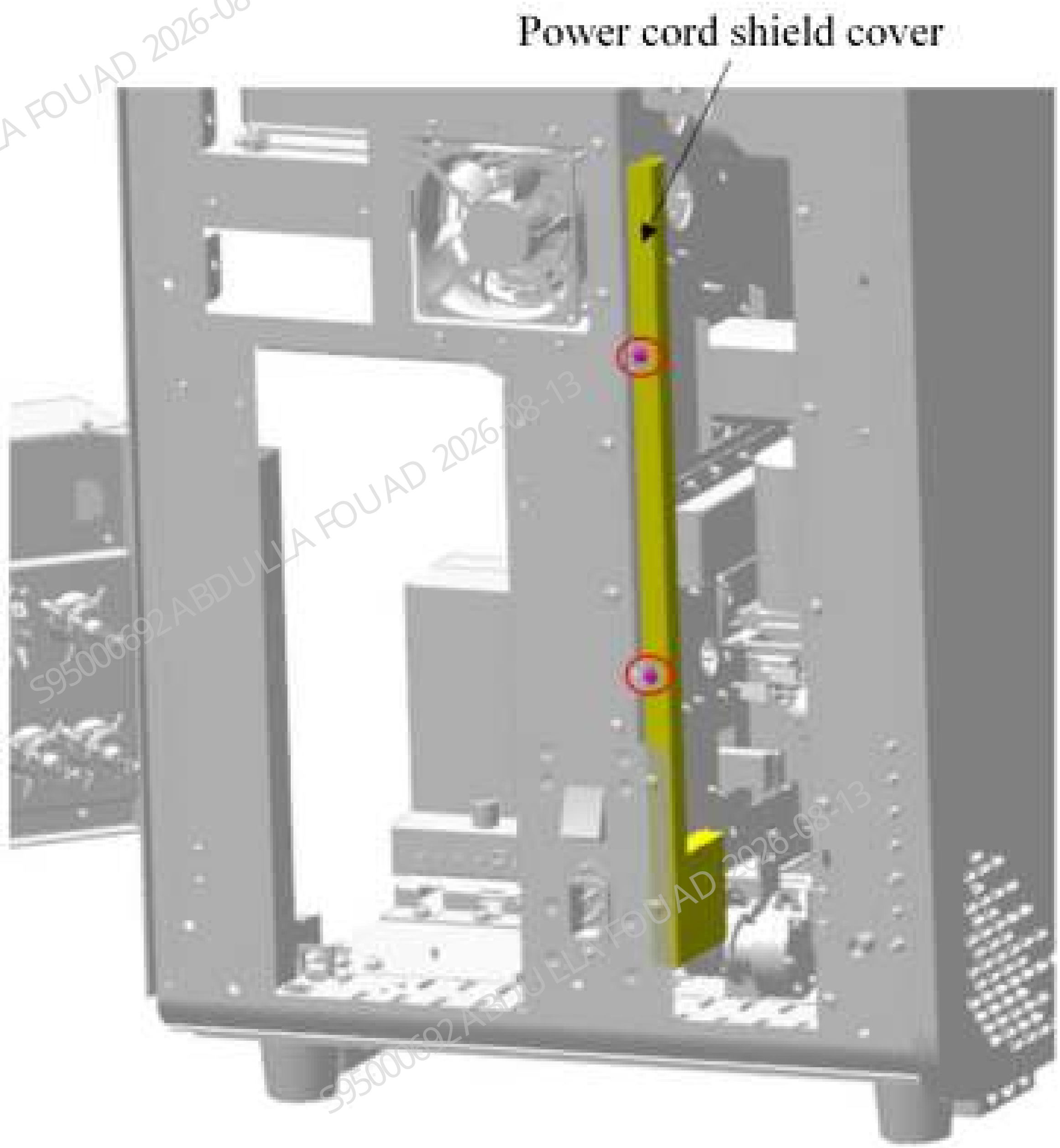
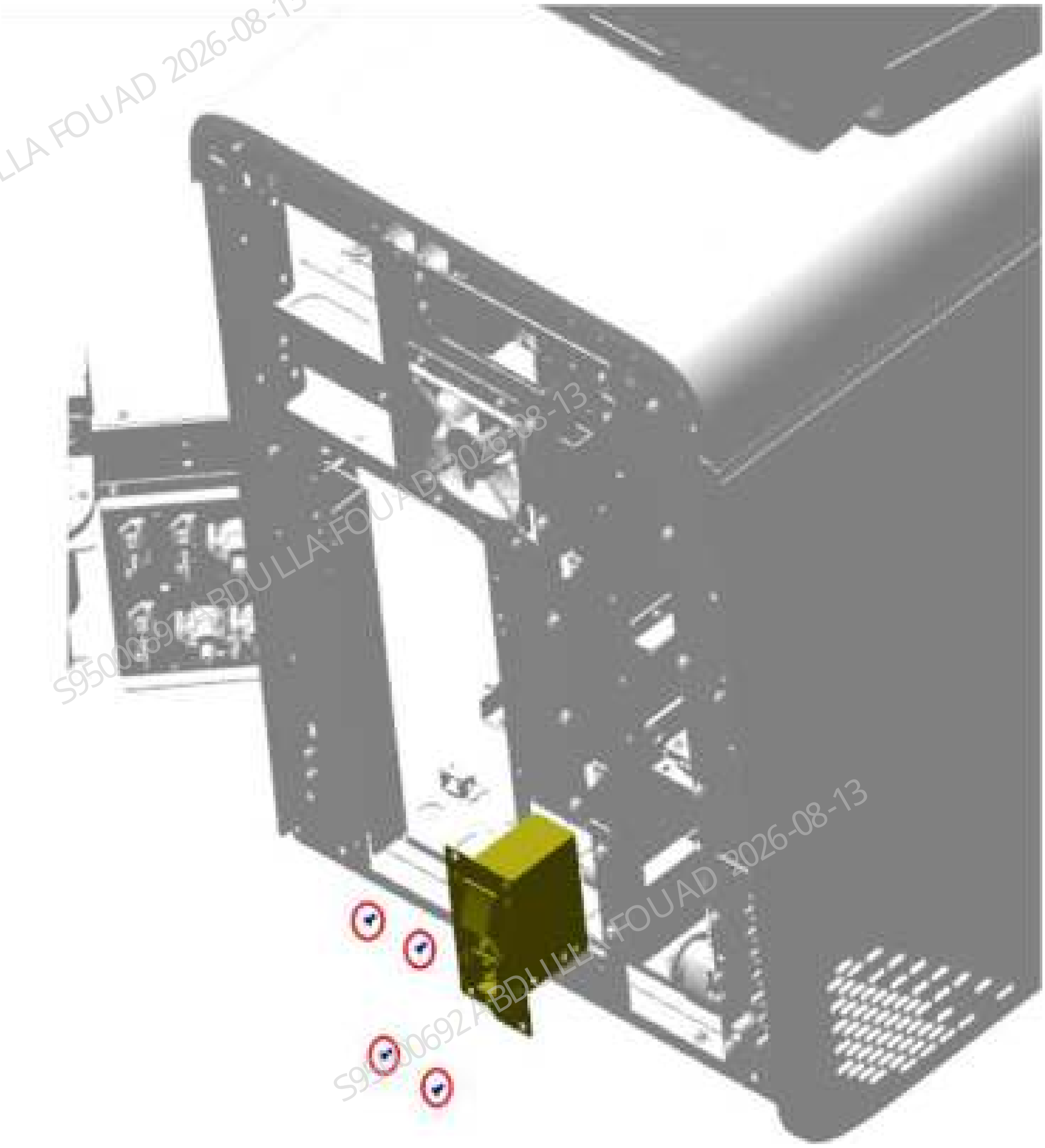


Figure8–669 Figure B of Removing Power Cord Shielding Cover



7. As shown in the figure below, remove the four countersunk head screws fixing the power switch assembly, drag the power switch assembly outward, and then remove the two power supply lines on the power switch assembly to take out the power switch assembly.

Figure8–670 Diagram of Removing the Power Switch Assembly



Subsequent Requirements

Installation procedure:

1. Insert the two power supply lines on the power switch assembly, use four countersunk head screws to fix the power switch assembly on the main unit, and then re-lay the power supply line along the original path and fix the other end of the power supply line on the power terminal.
2. Use two screws to install the power terminal shielding cover on the power switch assembly.
3. Use three screws to install the rear cover on the main unit.
4. Restore the right and rear covers.

5. Insert back the screw rubber plugs on the rear cover.
6. Put down the front cover for resetting.

8.88.3 Commissioning and Verification

Procedures

Commissioning:

1. Connect the main unit to the 220V power and start the analyzer.
2. After normal startup, the power switch assembly - commissioning/test and verification is completed.
3. If the analyzer cannot be started normally, return to the part of the power switch assembly - disassembly and assembly method to check whether the assembly is installed correctly. After troubleshooting, if the analyzer can be started again normally, the power switch assembly - commissioning/test and verification is completed.
4. If the assembly is installed correctly and the analyzer cannot be started normally, select another power switch assembly and carry out the disassembly and assembly method until the analyzer is started normally.
5. The power switch assembly - commissioning/test and verification is completed.

8.89 paired wire, mother-motor boardA, M end

8.89.1 General Information

Figure8–671

Table 8–101 paired wire, mother-motor boardA, M end

FRU Code	FRU Name	Lower-Level FRU Code	Lower-level FRU Name
115-106216-00	paired wire, mother-motor boardA, M end	XXX	XXX

8.89.2 Disassembly and Assembly

N/A

8.89.3 Commissioning and Verification

N/A

8.90 Wire, sample type detect sensor

8.90.1 General Information

N/A

8.90.2 Disassembly and Assembly

N/A

8.90.3 Commissioning and Verification

N/A

8.91 Wire, mother board to motor board B

8.91.1 General Information

N/A

8.91.2 Disassembly and Assembly

N/A

8.91.3 Commissioning and Verification

N/A

8.92 Paired wire, latex probe asm, board end

8.92.1 General Information

N/A

8.92.2 Disassembly and Assembly

N/A

8.92.3 Commissioning and Verification

N/A

8.93 Wire, mother board to motor board C

8.93.1 General Information

N/A

8.93.2 Disassembly and Assembly

N/A

8.93.3 Commissioning and Verification

N/A

8.94 Wire, mother board-scanner

8.94.1 General Information

N/A

8.94.2 Disassembly and Assembly

N/A

8.94.3 Commissioning and Verification

N/A

8.95 Wire, mother board to fan, ext.

8.95.1 General Information

N/A

8.95.2 Disassembly and Assembly

N/A

8.95.3 Commissioning and Verification

N/A

8.96 Wire, HGB sensor

8.96.1 General Information

N/A

8.96.2 Disassembly and Assembly

N/A

8.96.3 Commissioning and Verification

N/A

8.97 Wire, hydraulic pressure sensors, ext.

8.97.1 General Information

N/A

8.97.2 Disassembly and Assembly

N/A

8.97.3 Commissioning and Verification

N/A

8.98 Wire, impedance test module, ext.

8.98.1 General Information

N/A

8.98.2 Disassembly and Assembly

N/A

8.98.3 Commissioning and Verification

N/A

8.99 Wire, temperature sensors, ext.

8.99.1 General Information

N/A

8.99.2 Disassembly and Assembly

N/A

8.99.3 Commissioning and Verification

N/A

8.100 Wire, dye detection sensor, ext.

8.100.1 General Information

N/A

8.100.2 Disassembly and Assembly

N/A

8.100.3 Commissioning and Verification

N/A

8.101 Wire, M board-air press. board PWR&Com.

8.101.1 General Information

N/A

8.101.2 Disassembly and Assembly

N/A

8.101.3 Commissioning and Verification

N/A

8.102 Paired wire, M board-RFID board, M board

8.102.1 General Information

N/A

8.102.2 Disassembly and Assembly

N/A

8.102.3 Commissioning and Verification

N/A

8.103 Wire, Mother board to floats

8.103.1 General Information

N/A

8.103.2 Disassembly and Assembly

N/A

8.103.3 Commissioning and Verification

N/A

8.104 Wire, ESR module

8.104.1 General Information

N/A

8.104.2 Disassembly and Assembly

N/A

8.104.3 Commissioning and Verification

N/A

8.105 Wire mother board to heaters, ext.

8.105.1 General Information

N/A

8.105.2 Disassembly and Assembly

N/A

8.105.3 Commissioning and Verification

N/A

8.106 Paired wire, val.board2 val. grp1, board

8.106.1 General Information

N/A

8.106.2 Disassembly and Assembly

N/A

8.106.3 Commissioning and Verification

N/A

8.107 Paired wire, val.board1 val. grp3, board

8.107.1 General Information

N/A

8.107.2 Disassembly and Assembly

N/A

8.107.3 Commissioning and Verification

N/A

8.108 Wire, valve board 1 valve group 5

8.108.1 General Information

N/A

8.108.2 Disassembly and Assembly

N/A

8.108.3 Commissioning and Verification

N/A

8.109 Paired wire, TEMP brd-CRP&Itx asm, board

8.109.1 General Information

N/A

8.109.2 Disassembly and Assembly

N/A

8.109.3 Commissioning and Verification

N/A

8.110 Wire, Motherboard to jog switch

8.110.1 General Information

N/A

8.110.2 Disassembly and Assembly

N/A

8.110.3 Commissioning and Verification

N/A

8.111 Paired wire, AL board to CB motor, M end

8.111.1 General Information

N/A

8.111.2 Disassembly and Assembly

N/A

8.111.3 Commissioning and Verification

N/A

8.112 Wires, sampling asm&syringe motor sensor

8.112.1 General Information

N/A

8.112.2 Disassembly and Assembly

N/A

8.112.3 Commissioning and Verification

N/A

8.113 Wire, valve board 1 to valve board 2

8.113.1 General Information

N/A

8.113.2 Disassembly and Assembly

N/A

8.113.3 Commissioning and Verification

N/A

8.114 Wire,valve board 1 Pow & Com

8.114.1 General Information

N/A

8.114.2 Disassembly and Assembly

N/A

8.114.3 Commissioning and Verification

N/A

8.115 Paired wire, val.board 2 val. Grp 2, board

8.115.1 General Information

N/A

8.115.2 Disassembly and Assembly

N/A

8.115.3 Commissioning and Verification

N/A

8.116 Wire, valve board 1 valve group 2

8.116.1 General Information

N/A

8.116.2 Disassembly and Assembly

N/A

8.116.3 Commissioning and Verification

N/A

8.117 Wire, mother board to TEMP brd

8.117.1 General Information

N/A

8.117.2 Disassembly and Assembly

N/A

8.117.3 Commissioning and Verification

N/A

8.118 Wire, injector earthing

8.118.1 General Information

N/A

8.118.2 Disassembly and Assembly

N/A

8.118.3 Commissioning and Verification

N/A

8.119 Paired wire, mother board to CRP brd

8.119.1 General Information

N/A

8.119.2 Disassembly and Assembly

N/A

8.119.3 Commissioning and Verification

N/A

8.120 Wire,valve board 2 Power

8.120.1 General Information

N/A

8.120.2 Disassembly and Assembly

N/A

8.120.3 Commissioning and Verification

N/A

8.121 Wire, mother board to RFID board

8.121.1 General Information

N/A

8.121.2 Disassembly and Assembly

N/A

8.121.3 Commissioning and Verification

N/A

8.122 Wire, RFID reader to Antenna

8.122.1 General Information

N/A

8.122.2 Disassembly and Assembly

N/A

8.122.3 Commissioning and Verification

N/A

8.123 Sample Switch wire

8.123.1 General Information

N/A

8.123.2 Disassembly and Assembly

N/A

8.123.3 Commissioning and Verification

N/A

8.124 wire,autoloader

8.124.1 General Information

Figure8–672

Table 8–102 wire,autoloader

FRU Code	FRU Name	Lower-Level FRU Code	Lower-level FRU Name
115-106215-00	wire,autoloader	XXX	XXX

8.124.2 Disassembly and Assembly

N/A

8.124.3 Commissioning and Verification

N/A

8.125 Wire, SEN (M)

8.125.1 General Information

N/A

8.125.2 Disassembly and Assembly

N/A

8.125.3 Commissioning and Verification

N/A

8.126 Wire, motor (M)

8.126.1 General Information

N/A

8.126.2 Disassembly and Assembly

N/A

8.126.3 Commissioning and Verification

N/A

8.127 Wire, motor (S)

8.127.1 General Information

N/A

8.127.2 Disassembly and Assembly

N/A

8.127.3 Commissioning and Verification

N/A

8.128 Wire, SEN (S)

8.128.1 General Information

N/A

8.128.2 Disassembly and Assembly

N/A

8.128.3 Commissioning and Verification

N/A

8.129 Switch cable

8.129.1 General Information

N/A

8.129.2 Disassembly and Assembly

N/A

8.129.3 Commissioning and Verification

N/A

8.130 Wire, pusher (M)

8.130.1 General Information

N/A

8.130.2 Disassembly and Assembly

N/A

8.130.3 Commissioning and Verification

N/A

8.131 Wire, pusher (S)

8.131.1 General Information

N/A

8.131.2 Disassembly and Assembly

N/A

8.131.3 Commissioning and Verification

N/A

8.132 Patch wire, PB electric motor end

8.132.1 General Information

N/A

8.132.2 Disassembly and Assembly

N/A

8.132.3 Commissioning and Verification

N/A

8.133 Handheld Bar Code Scanner

8.133.1 General Information

N/A

8.133.2 Disassembly and Assembly

N/A

8.133.3 Commissioning and Verification

N/A

8.134 LCD Display 21.5 16:9

8.134.1 General Information

N/A

8.134.2 Disassembly and Assembly

N/A

8.134.3 Commissioning and Verification

N/A

8.135 PC A6-8570 4GB 500GB Dual Adapter DVD Windows 10

8.135.1 General Information

N/A

8.135.2 Disassembly and Assembly

N/A

8.135.3 Commissioning and Verification

N/A

8.136 Power Cord (GB/10A250V/2.5M)

8.136.1 General Information

N/A

8.136.2 Disassembly and Assembly

N/A

8.136.3 Commissioning and Verification

N/A

8.137 Network Cable (connecting the analyzer and PC)

8.137.1 General Information

N/A

8.137.2 Disassembly and Assembly

N/A

8.137.3 Commissioning and Verification

N/A

8.138 HP LaserJet Printer

8.138.1 General Information

N/A

8.138.2 Disassembly and Assembly

N/A

8.138.3 Commissioning and Verification

N/A

8.139 3202 tube rack assembly

8.139.1 General Information

N/A

8.139.2 Disassembly and Assembly

N/A

8.139.3 Commissioning and Verification

N/A

8.140 5-position tube rack assembly

8.140.1 General Information

N/A

8.140.2 Disassembly and Assembly

N/A

8.140.3 Commissioning and Verification

N/A

8.141 CRP CAL tube rack (CAL)

8.141.1 General Information

N/A

8.141.2 Disassembly and Assembly

N/A

8.141.3 Commissioning and Verification

N/A

8.142 5-position tube rack for CRP(CAL)

8.142.1 General Information

N/A

8.142.2 Disassembly and Assembly

N/A

8.142.3 Commissioning and Verification

N/A

8.143 CAL adapter (for high speed CRP)

8.143.1 General Information

N/A

8.143.2 Disassembly and Assembly

N/A

8.143.3 Commissioning and Verification

N/A

8.144 Factory Installer CD

8.144.1 General Information

N/A

8.144.2 Disassembly and Assembly

N/A

8.144.3 Commissioning and Verification

N/A

8.145 Paired wire, 10-position and 5-row closed-sampling autoloade

8.145.1 General Information

N/A

8.145.2 Disassembly and Assembly

N/A

8.145.3 Commissioning and Verification

N/A

8.146 SIM card-IOT card

8.146.1 General Information

N/A

8.146.2 Disassembly and Assembly

N/A

8.146.3 Commissioning and Verification

N/A

8.147 PC (with 3G installation package)

8.147.1 General Information

N/A

8.147.2 Disassembly and Assembly

N/A

8.147.3 Commissioning and Verification

N/A

8.148 4G wireless routing USB port 802.11 b/g/n

8.148.1 General Information

N/A

8.148.2 Disassembly and Assembly

N/A

8.148.3 Commissioning and Verification

N/A

8.149 Motor Step 3.6V 1.8 Degree/Step Hold Torque 0.53Nm

8.149.1 General Information

N/A

8.149.2 Disassembly and Assembly

N/A

8.149.3 Commissioning and Verification

N/A

8.150 PCBA, Motor Board

8.150.1 General Information

N/A

8.150.2 Disassembly and Assembly

N/A

8.150.3 Commissioning and Verification

N/A

8.151 Ring for proof blood spurt 2

8.151.1 General Information

N/A

8.151.2 Disassembly and Assembly

N/A

8.151.3 Commissioning and Verification

N/A

8.152 Diluent temperature detection assembly (3206)

8.152.1 General Information

N/A

8.152.2 Disassembly and Assembly

N/A

8.152.3 Commissioning and Verification

N/A

8.153 Power cord (American standard/13A110V/2.5M)

8.153.1 General Information

N/A

8.153.2 Disassembly and Assembly

N/A

8.153.3 Commissioning and Verification

N/A

8.154 Power cord (European standard/10A250V/2.5M)

8.154.1 General Information

N/A

8.154.2 Disassembly and Assembly

N/A

8.154.3 Commissioning and Verification

N/A

8.155 Brazil power cord 250V 10A 3M

8.155.1 General Information

N/A

8.155.2 Disassembly and Assembly

N/A

8.155.3 Commissioning and Verification

N/A

8.156 Australian standard transparent plug power cord, orange, 10A

8.156.1 General Information

N/A

8.156.2 Disassembly and Assembly

N/A

8.156.3 Commissioning and Verification

N/A

8.157 Sample compartment assembly

8.157.1 General Information

N/A

8.157.2 Disassembly and Assembly

N/A

8.157.3 Commissioning and Verification

N/A

8.158 Bottom shell (laser-etched)

8.158.1 General Information

Figure8–673 043-016694-00 bottom shell (laser-etched) photo

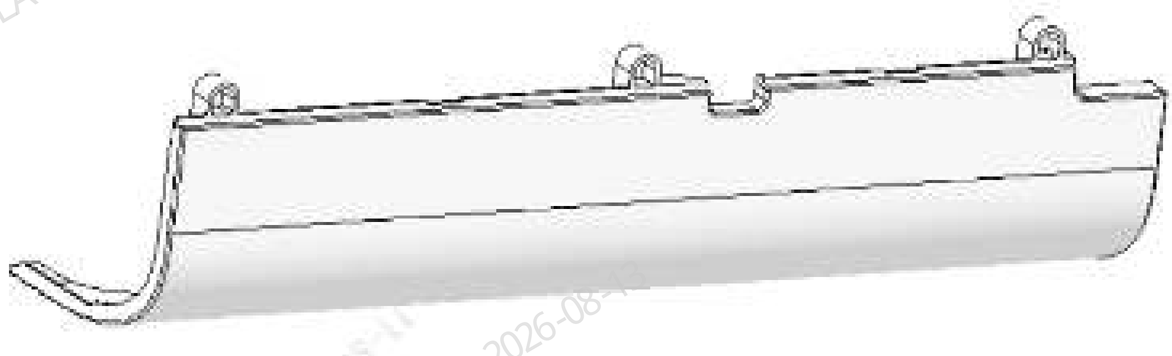


Table 8–103 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
043-016694-00	Bottom shell (laser-etched)	N/A	N/A

Functions:
Analyzer appearance.

8.158.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

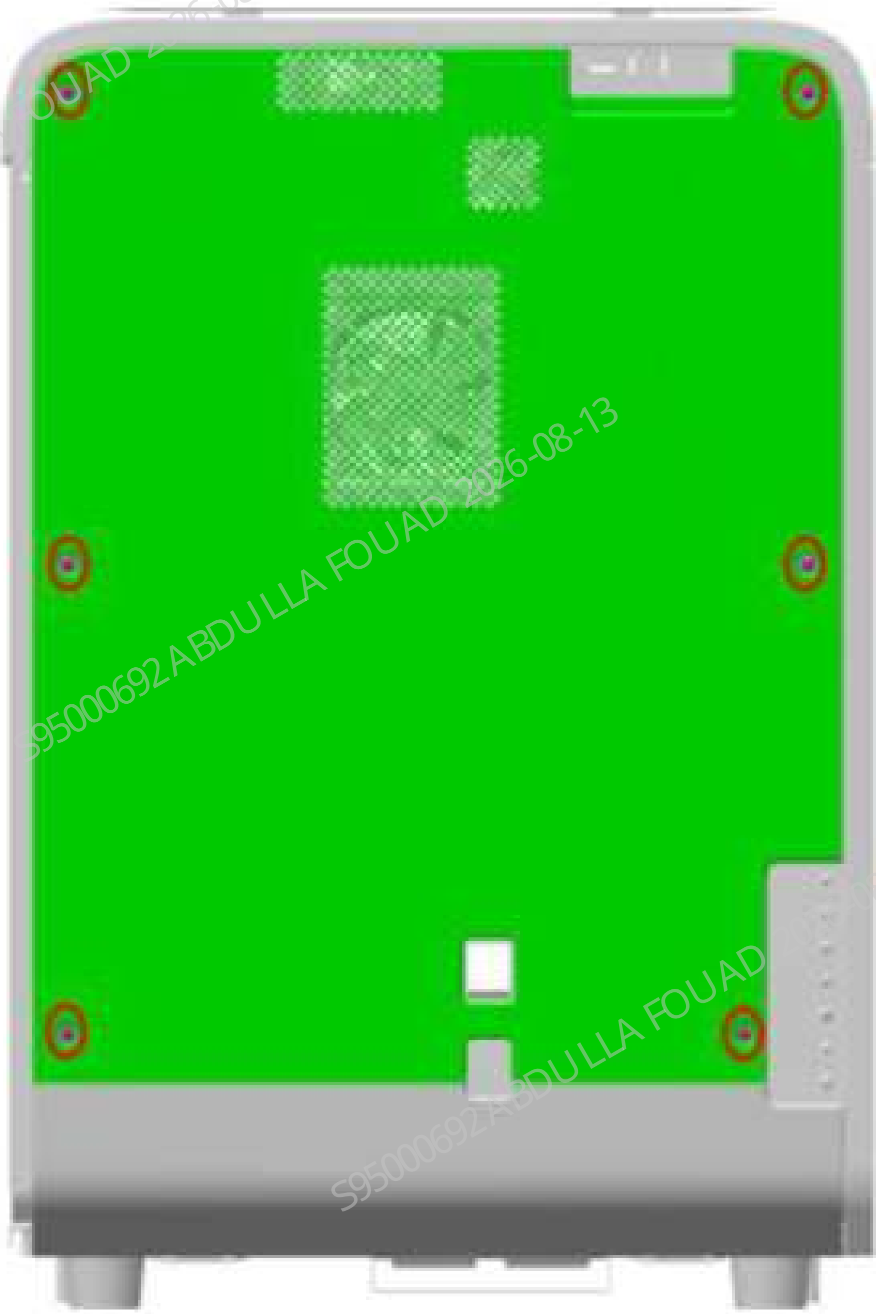
Prerequisites

N/A

Steps

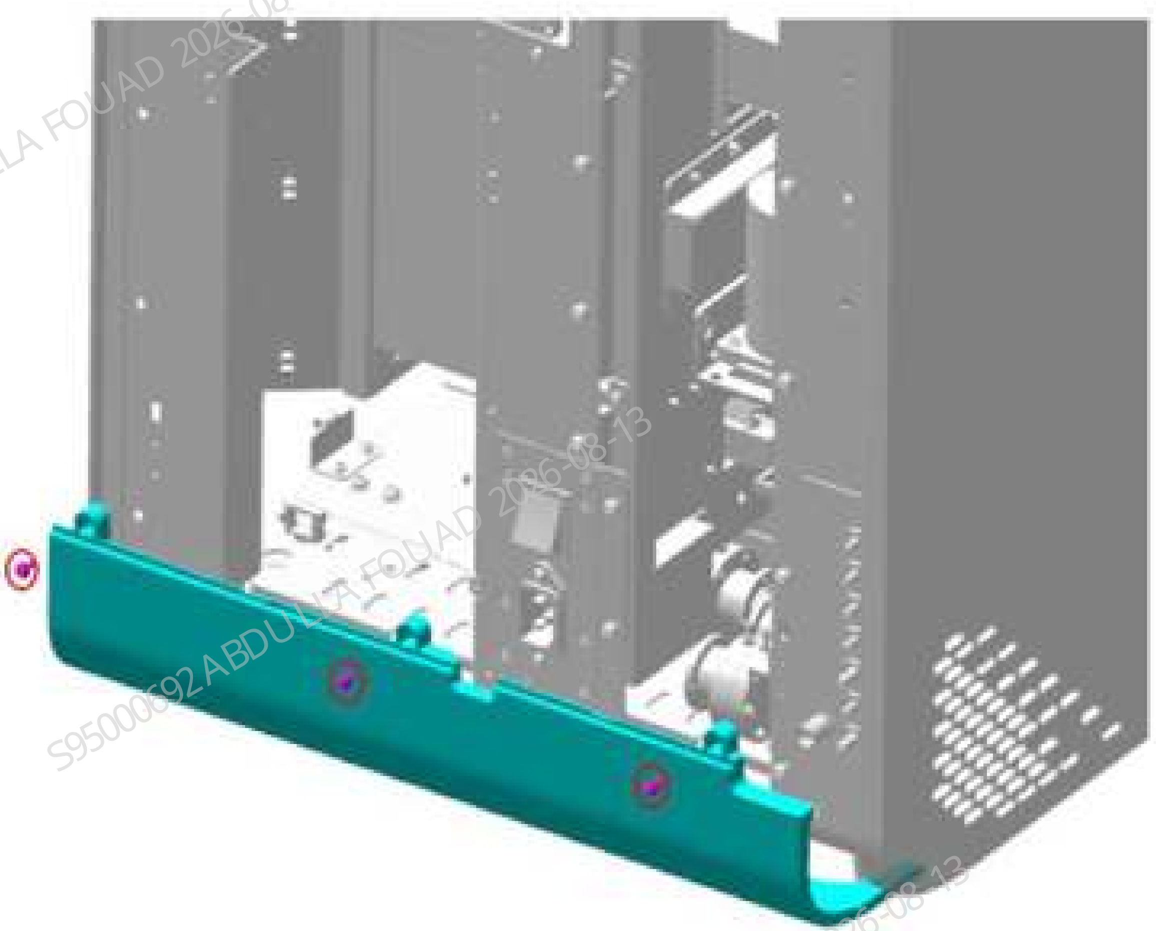
1. Loosen the six fixing screws on the rear shell, and remove the rear panel, as shown in the following figure.

Figure8–674 Step 1 of removing the rear shell



2. Loosen the three fixing screws on the rear shell, and remove the rear panel, as shown in the following figure.

Figure8–675 Step 2 of removing the rear shell



Subsequent Requirements

N/A

8.158.3 Commissioning and Verification

N/A

8.159 Left shell (5X6) (BC-760 series)

8.159.1 General Information

Figure8–676 115-098558-00 left shell (5X6) photo

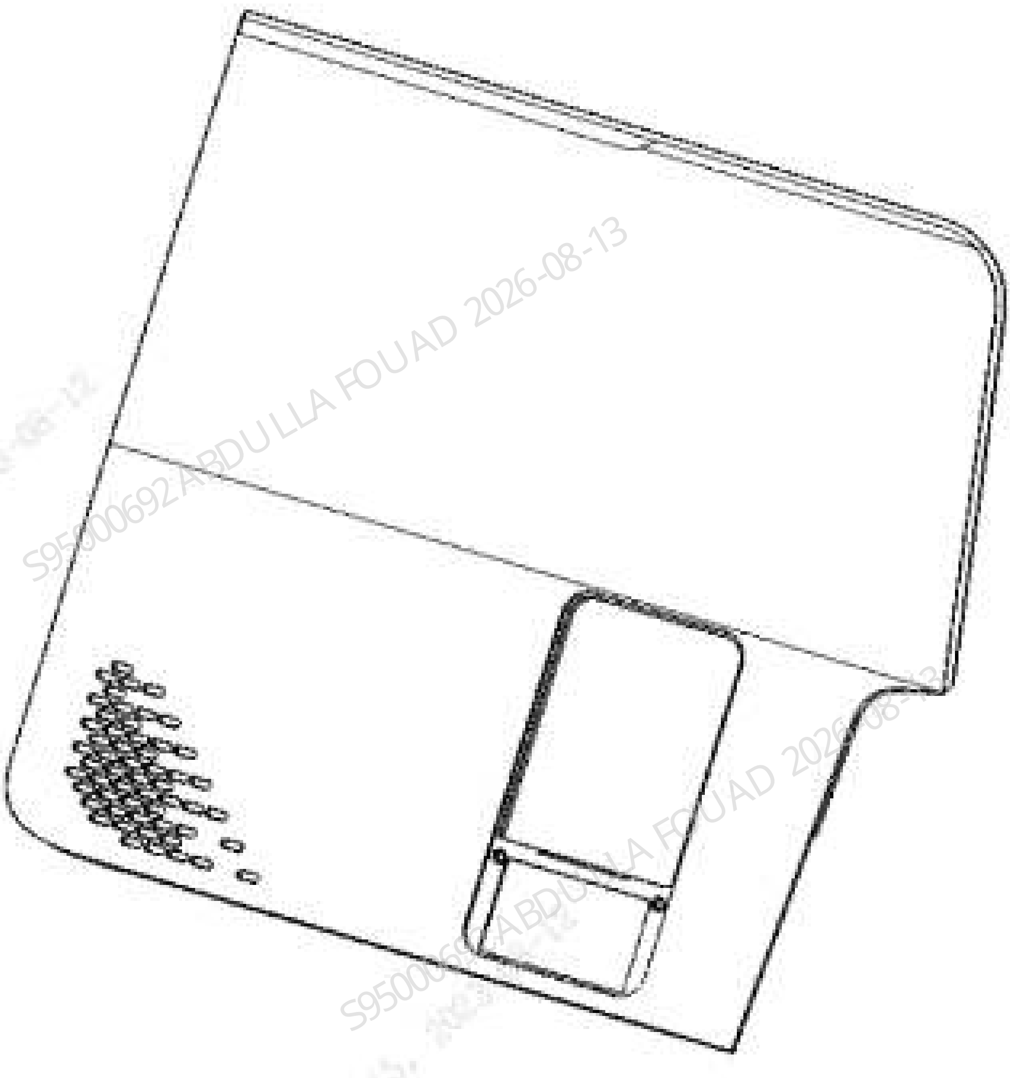


Table 8–104 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-098558-00	Left shell (5X6)	N/A	N/A

Functions:

Appearance part.

8.159.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver

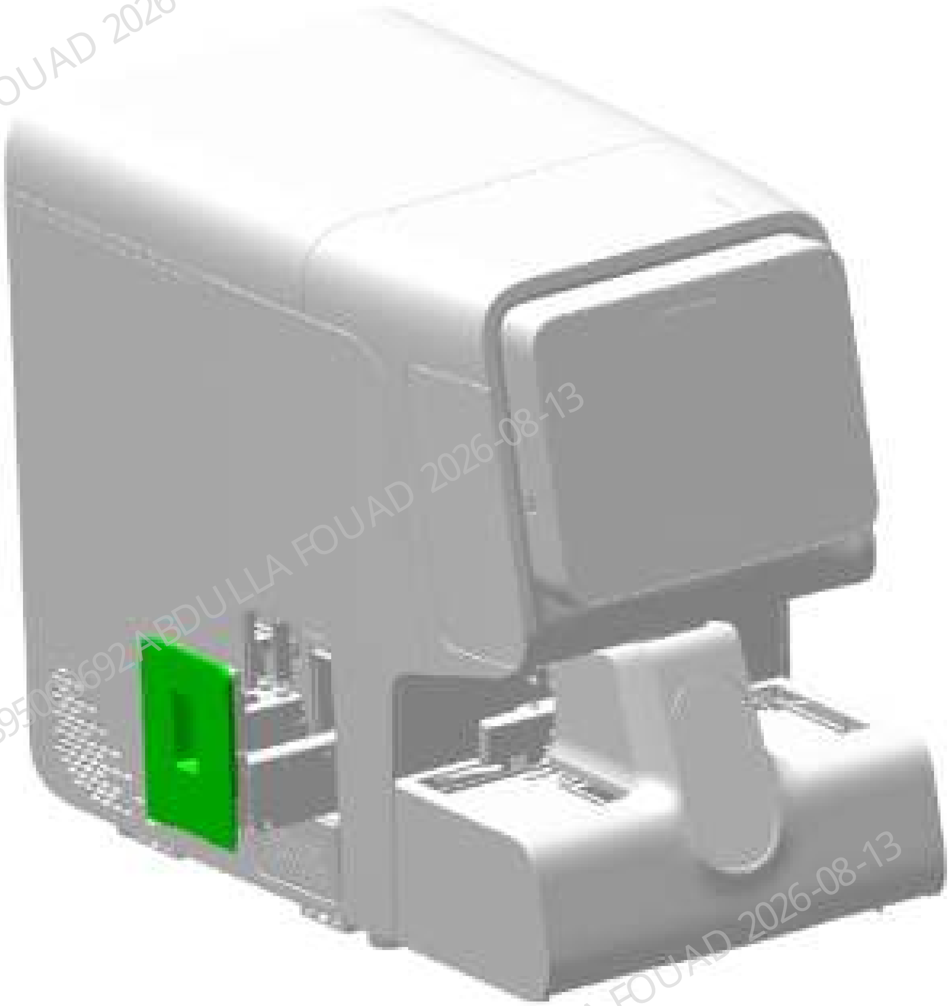
Prerequisites

Before removal, click the screen to lift the aspirate probe of the latex reagent thermostat assembly, and pull out and remove the latex reagent compartment door.

Steps

1. As shown in the figure below, lift the latex reagent probe to the upper limit position on the software screen, and pull out the latex reagent compartment door (along with the sliding mechanism of the latex reagent assembly).

Figure8–677 Latex reagent compartment door disassembly procedure 1



2. As shown in the figure below, unscrew one M3 cross screw fixing the latex reagent compartment door, and remove the latex reagent compartment door.

Figure8–678 Latex reagent compartment door disassembly procedure 2

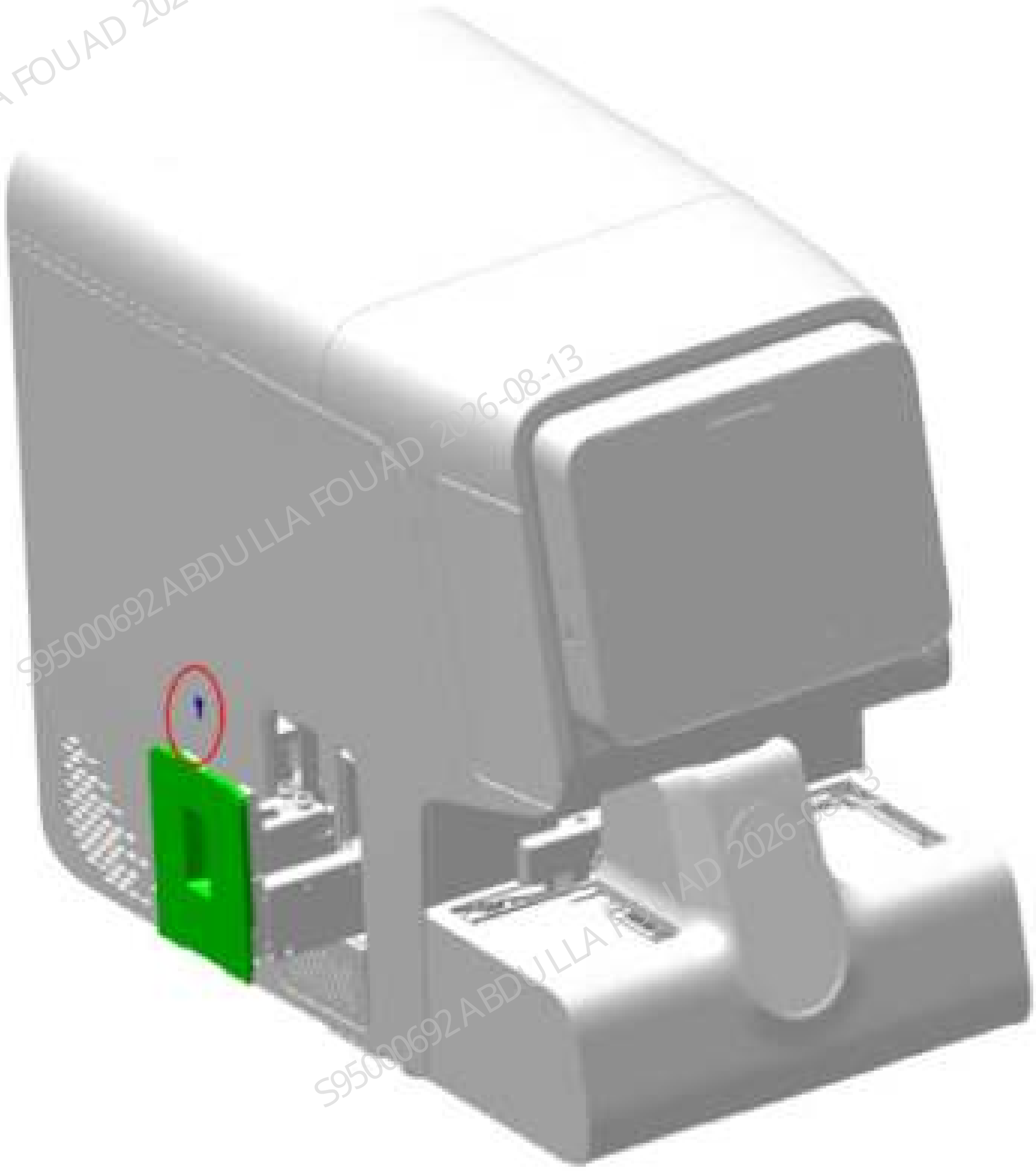


Figure8–679 Removing the left shell screw

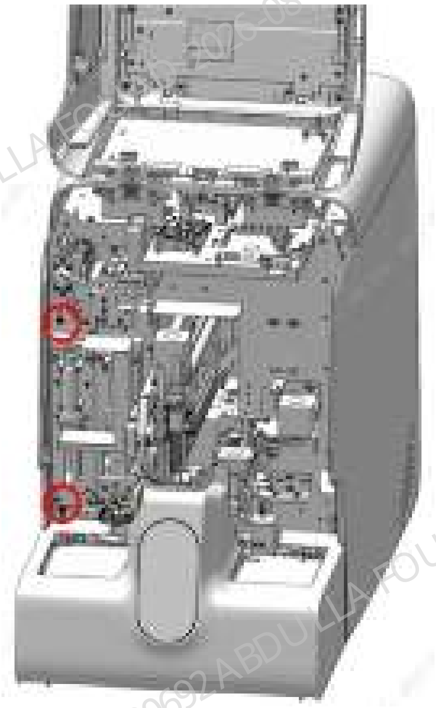


Figure8–680 Removing the left shell

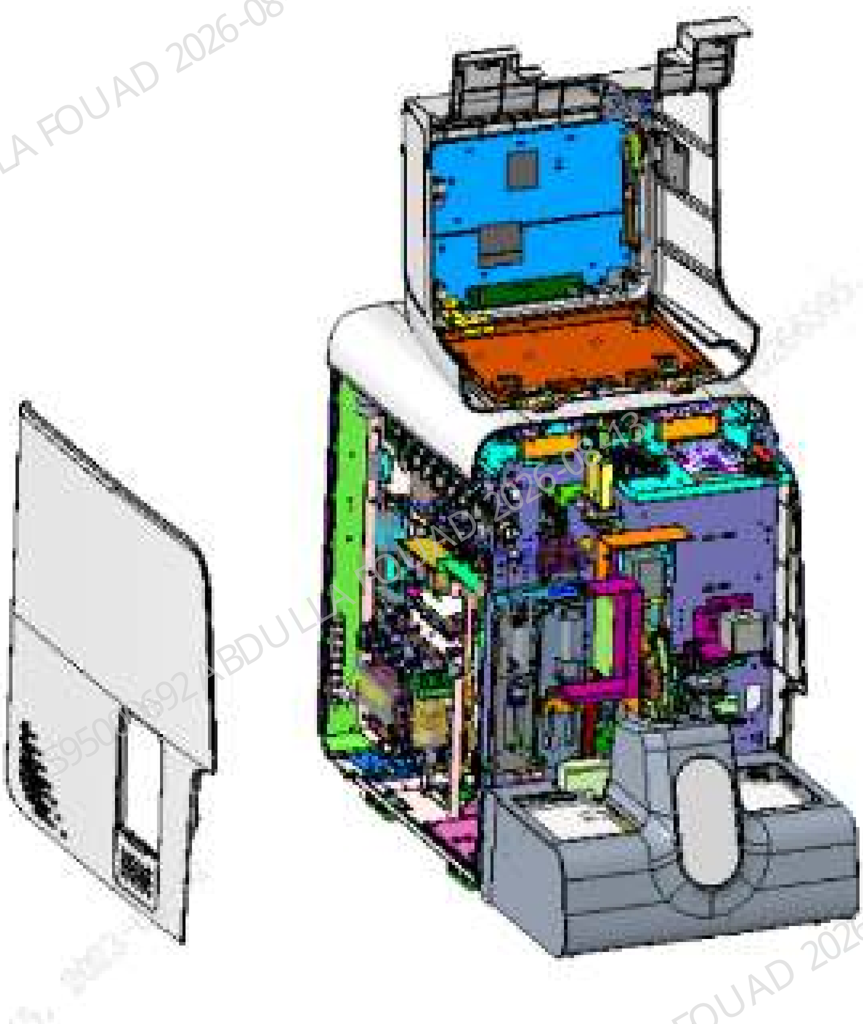
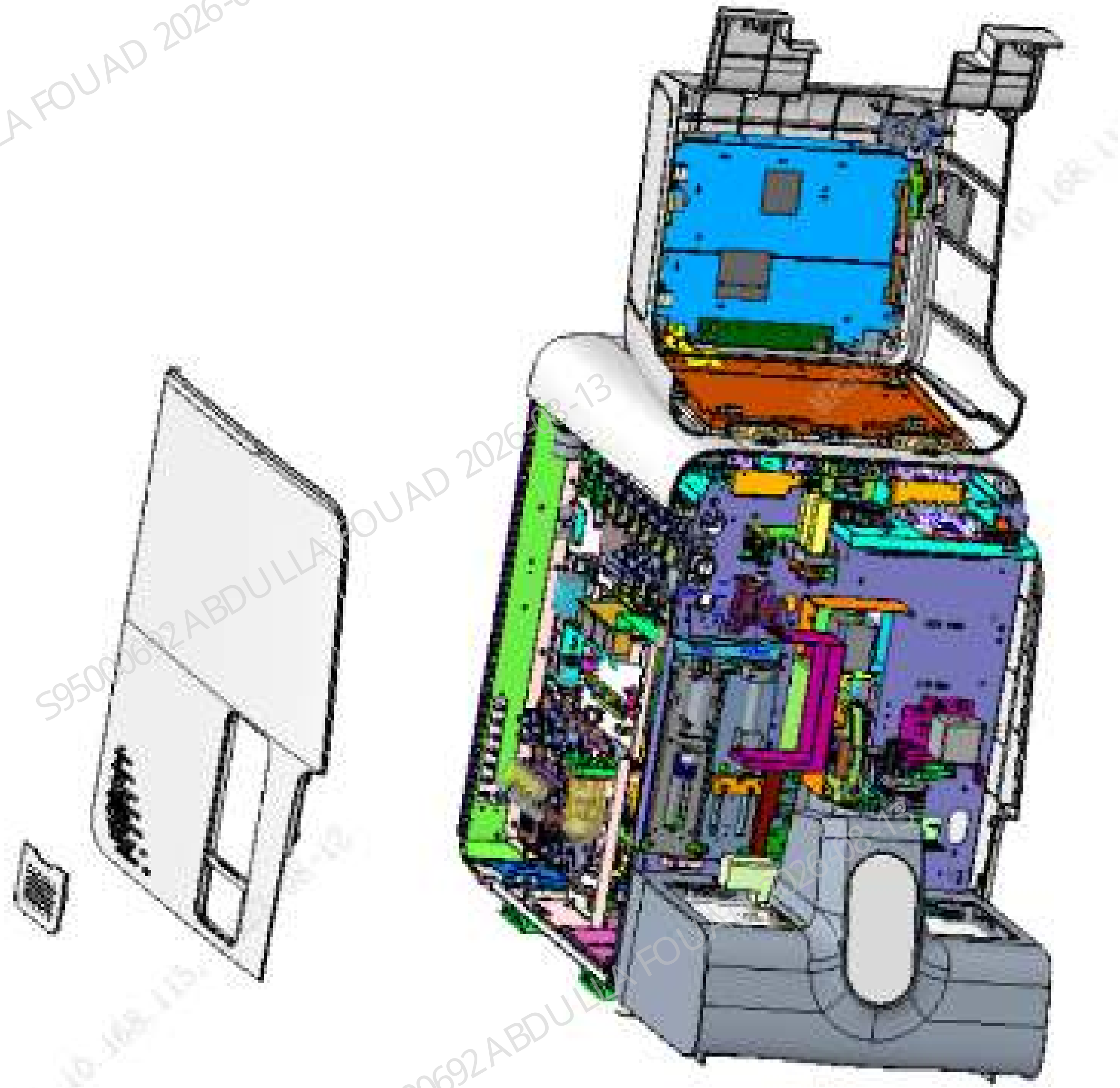


Figure8–681 Removing the small door of the filter screen



Subsequent Requirements

Install the latex reagent compartment door on the front door of cold chamber, and close it. Then, click the screen to lower the aspirate probe of the latex reagent thermostat assembly.

8.159.3 Commissioning and Verification

N/A

8.160 Metal model nameplate (BC-760)

8.160.1 General Information

Figure8–682 047-036812-00 metal model nameplate (BC-760) photo

Table 8–105 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
047-036812-00	Metal model nameplate (BC-760)	N/A	N/A

Functions:
Analyzer model label.

8.160.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

Prerequisites

N/A

Steps

Stick the metal model label at the following position, as shown in the figure below.

Figure8–683 Step 1 of sticking the metal model nameplate (BC-760)

Subsequent Requirements

N/A

8.160.3 Commissioning and Verification

N/A

8.161 Metal model nameplate (BC-780)

8.161.1 General Information

Figure8–684 047-036813-00 metal model nameplate (BC-780) photo

Table 8–106 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
047-036813-00	Metal model nameplate (BC-780)	N/A	N/A

Functions:
Analyzer model label.

8.161.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

Prerequisites

N/A

Steps

Stick the metal model label at the following position, as shown in the figure below.

Figure8–685 Step 1 of sticking the metal model nameplate (BC-780)

Subsequent Requirements

N/A

8.161.3 Commissioning and Verification

N/A

8.162 Top shell

8.162.1 General Information

Figure8–686 043-015109-00 top shell photo

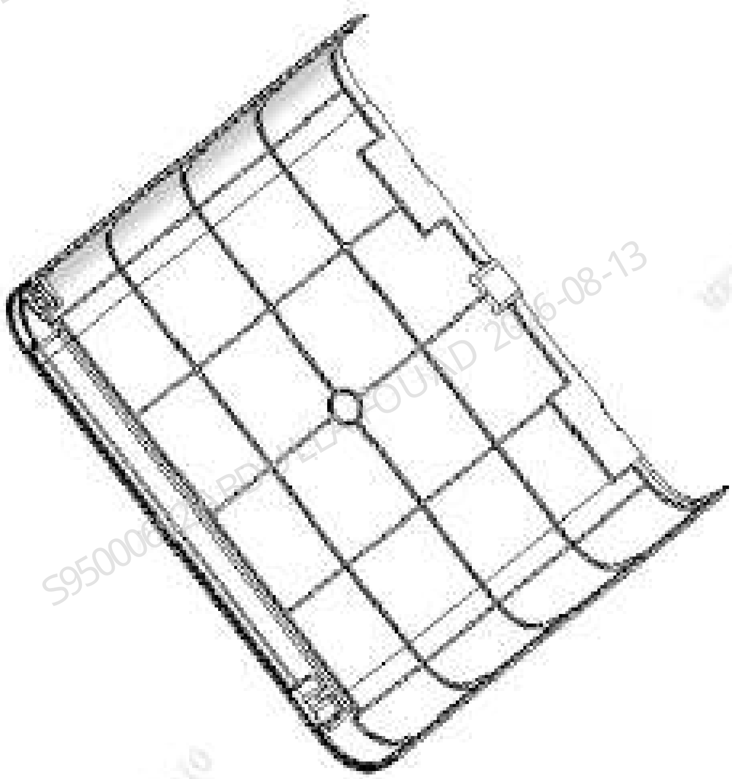


Table 8–107 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
043-015109-00	Top shell	N/A	N/A

Functions:
Analyzer appearance.

8.162.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

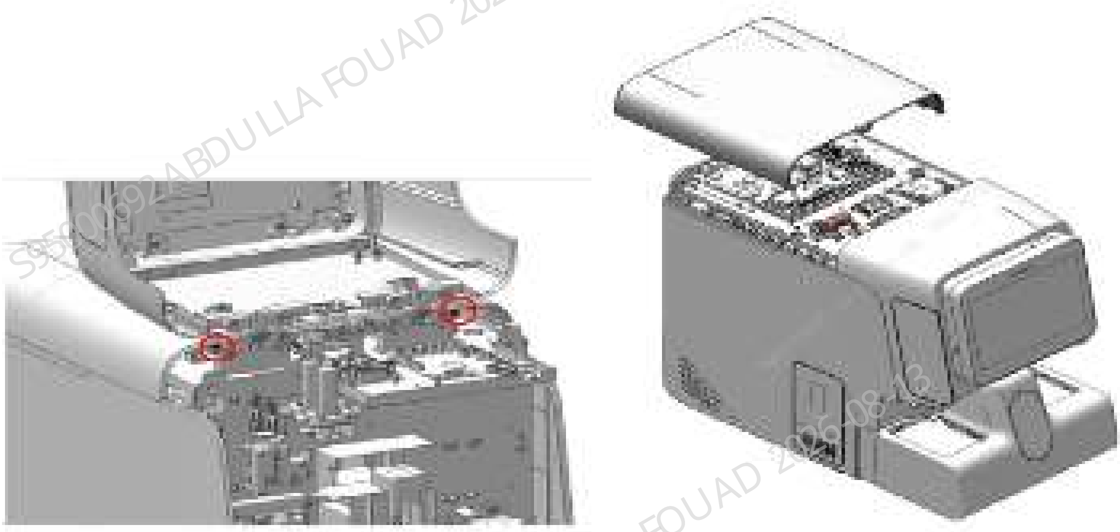
Prerequisites

N/A

Steps

Open the front shell as shown in the figure below, loosen the two fixing screws, and remove the top shell, as shown in the figure below.

Figure8–687 Step 1 of removing the top shell



Subsequent Requirements

N/A

8.162.3 Commissioning and Verification

N/A

8.163 CS rear shell (sy)

8.163.1 General Information

Figure8–688 043-015890-00 CS rear shell (sy) photo

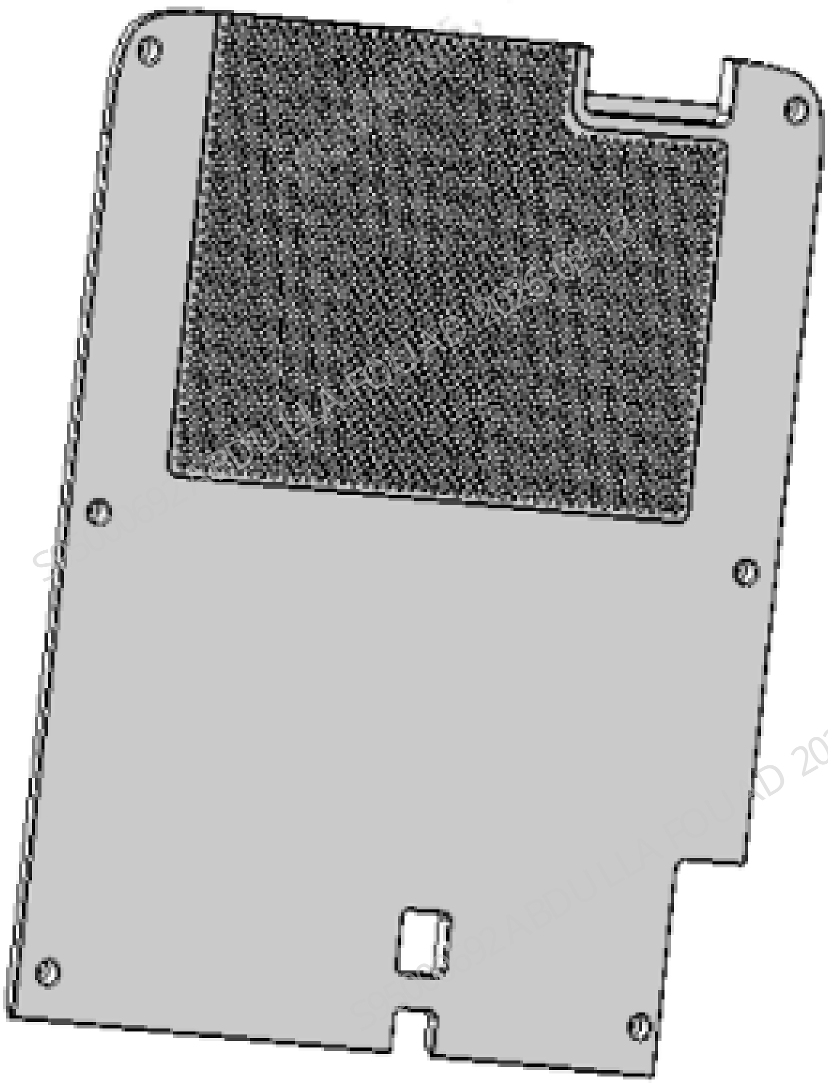


Table 8–108 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
043-015890-00	CS rear shell (sy)	N/A	N/A

Functions:

Analyzer appearance.

8.163.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

Prerequisites

N/A

Steps

Loosen the six fixing screws on the rear shell, and remove the rear panel, as shown in the following figure.

Figure8–689 Procedure of removing the rear shell

Subsequent Requirements

N/A

8.163.3 Commissioning and Verification

N/A

8.164 Right shell (5X6) (BC-760 series)

8.164.1 General Information

Figure8–690 115-098559-00 right shell (5X6) photo

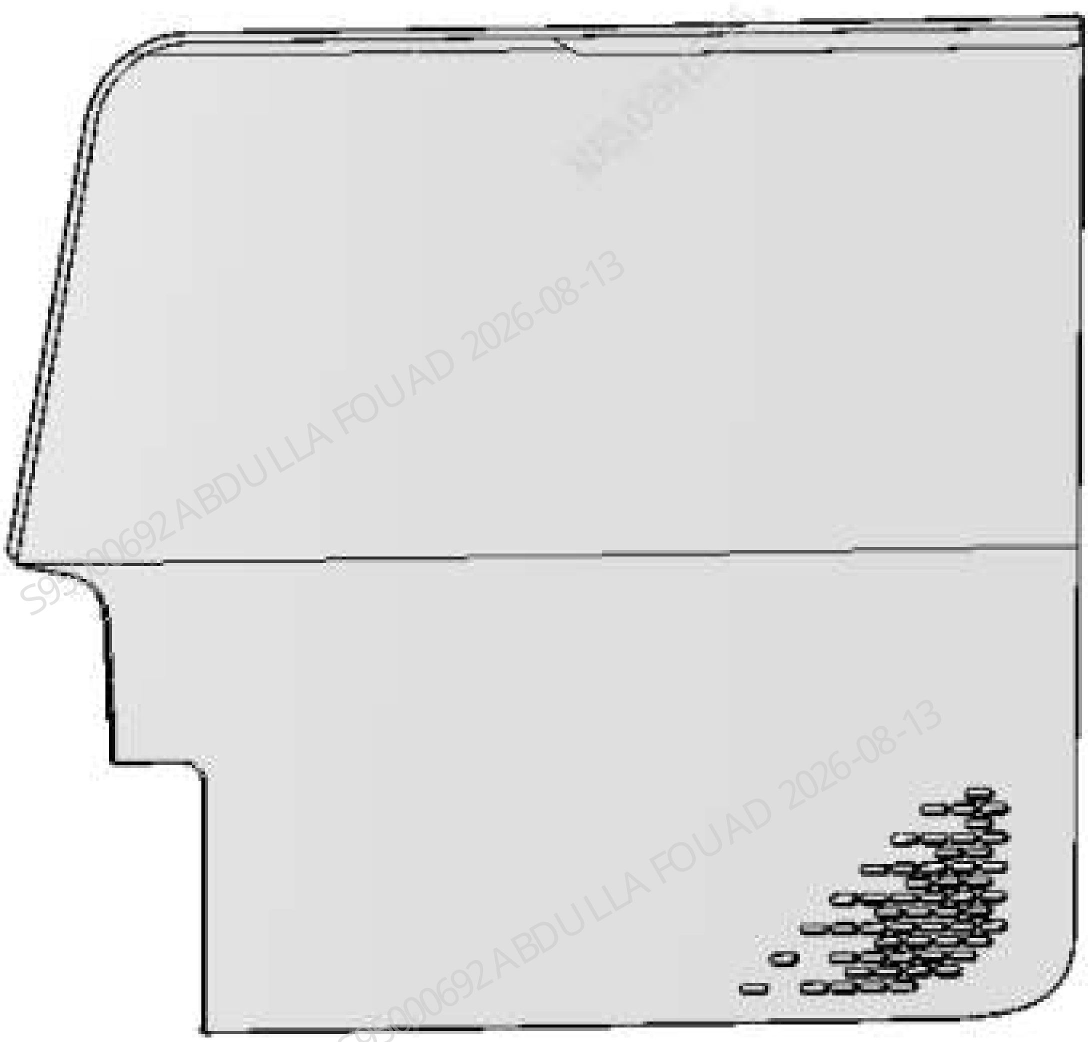


Table 8–109 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-098559-00	Right shell (5X6)	N/A	N/A

Functions:
Analyzer appearance.

8.164.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver

Prerequisites

N/A

Steps

Open the front shell and dock it on the top shell, loosen the two screws that fix the right shell, and remove the right shell, as shown in the figure below.

Figure8–691 Latex reagent compartment door disassembly procedure 1

Subsequent Requirements

N/A

8.164.3 Commissioning and Verification

N/A

8.165 Internal barcode scanner (including the rotated wheel compre

8.165.1 General Information

N/A

8.165.2 Disassembly and Assembly

N/A

8.165.3 Commissioning and Verification

N/A

8.166 Small fluorescent 10*5 autoloader upgrade package

8.166.1 General Information

Table 8–110 Small fluorescent 10*5 autoloader upgrade package

FRU Code	FRU Name	Lower-Level FRU Code	Lower-level FRU Name
115-091934-02	Small fluores cent10*5autoloaderup grade package	/	/

Functions:

1. Provides the function of touch control.
2.
3.

8.166.2 Disassembly and Assembly

N/A

8.166.3 Commissioning and Verification

N/A

8.167 3128 front shell (sy)

8.167.1 General Information

Figure8–692 043-015891-00 3128 front shell (sy) photo

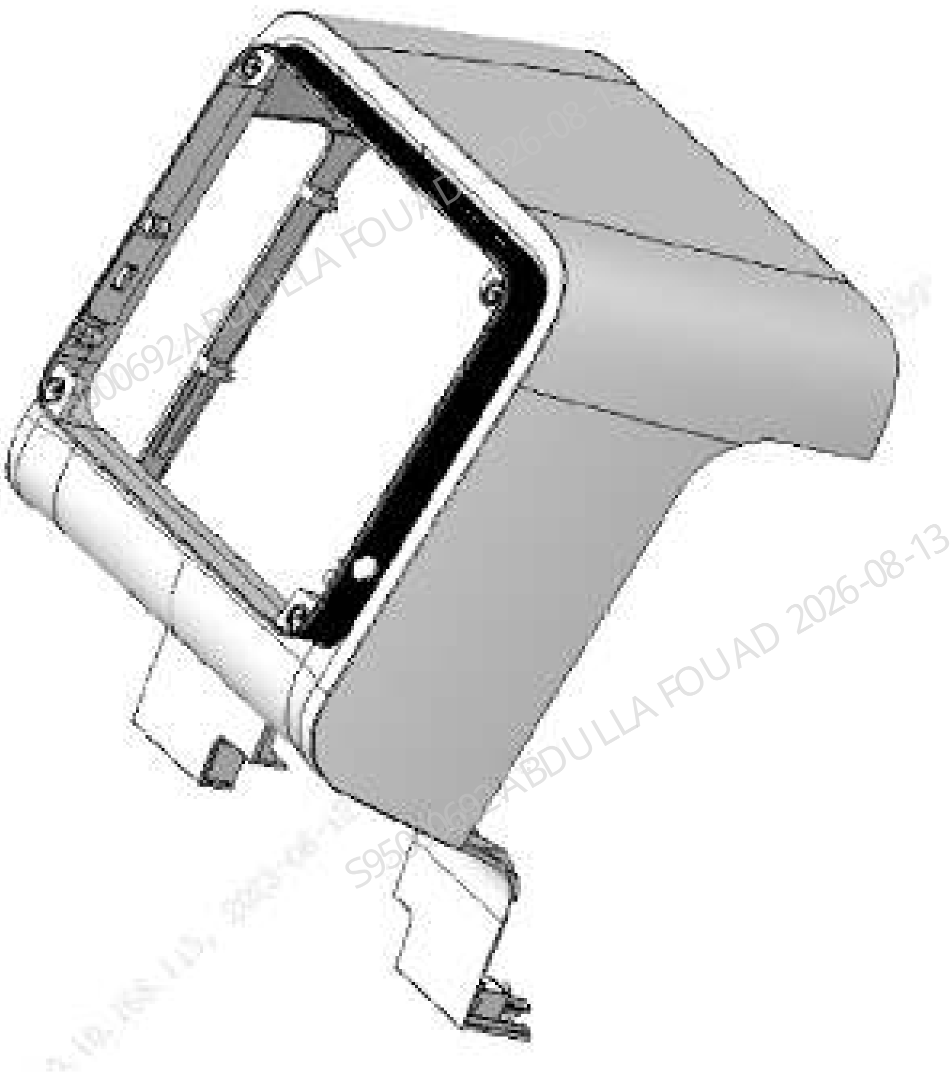


Table 8–111 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
043-015891-00	Front shell (sy)	N/A	N/A

Functions:

Analyzer appearance.

8.167.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

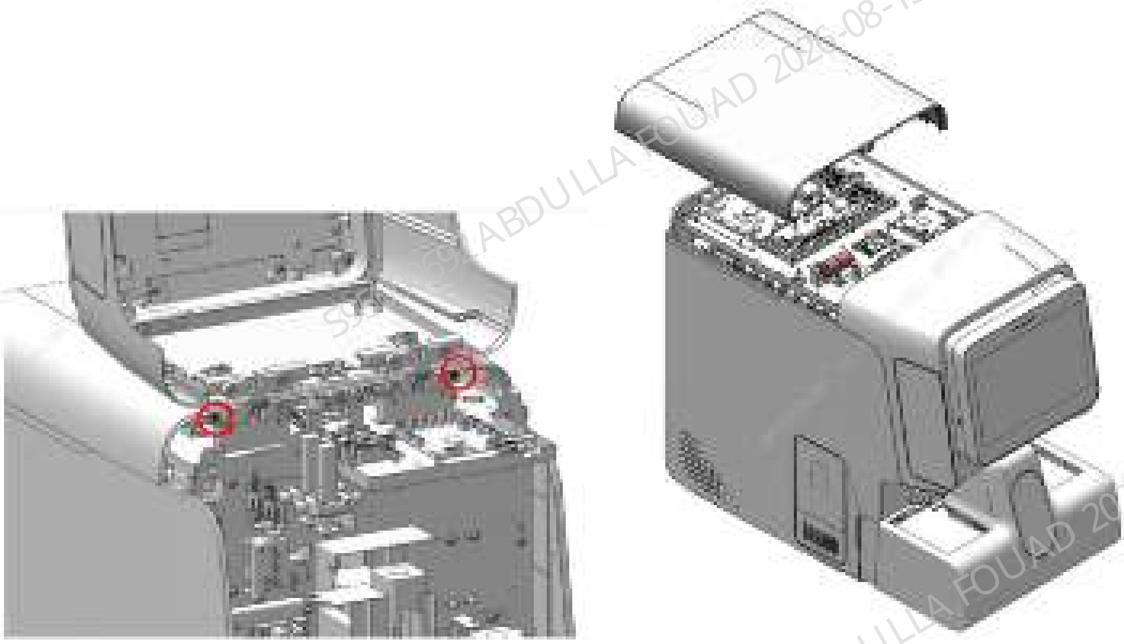
Prerequisites

N/A

Steps

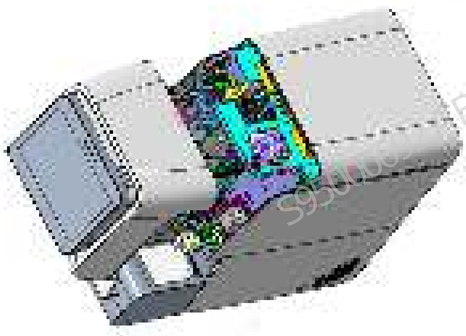
1. Open the front shell assembly as shown in the following figure, loosen the two fixing screws, and remove the top shell.

Figure8–693 Step 1 of removing the front shell (sy)



2. Remove the top plug as shown in the following figure, loosen the lower hinge, and remove the front shell assembly.

Figure8–694 Step 2 of removing the front shell (sy)



3. Loosen the two screws on the right side as shown in the following figure and remove the RFID board bracket.

Figure8–695 Step 3 of removing the front shell (sy)

4. Loosen the two fixing screws as shown in the figure below and remove the On/Off button.

Figure8–696 Step 4 of removing the front shell (sy)

5. Loosen the two fixing screws as shown in the following figure and remove the printed circuit board.

Figure8–697 Step 5 of removing the front shell (sy)

6. Loosen the two fixing screws as shown in the figure below and remove the USB assembly.

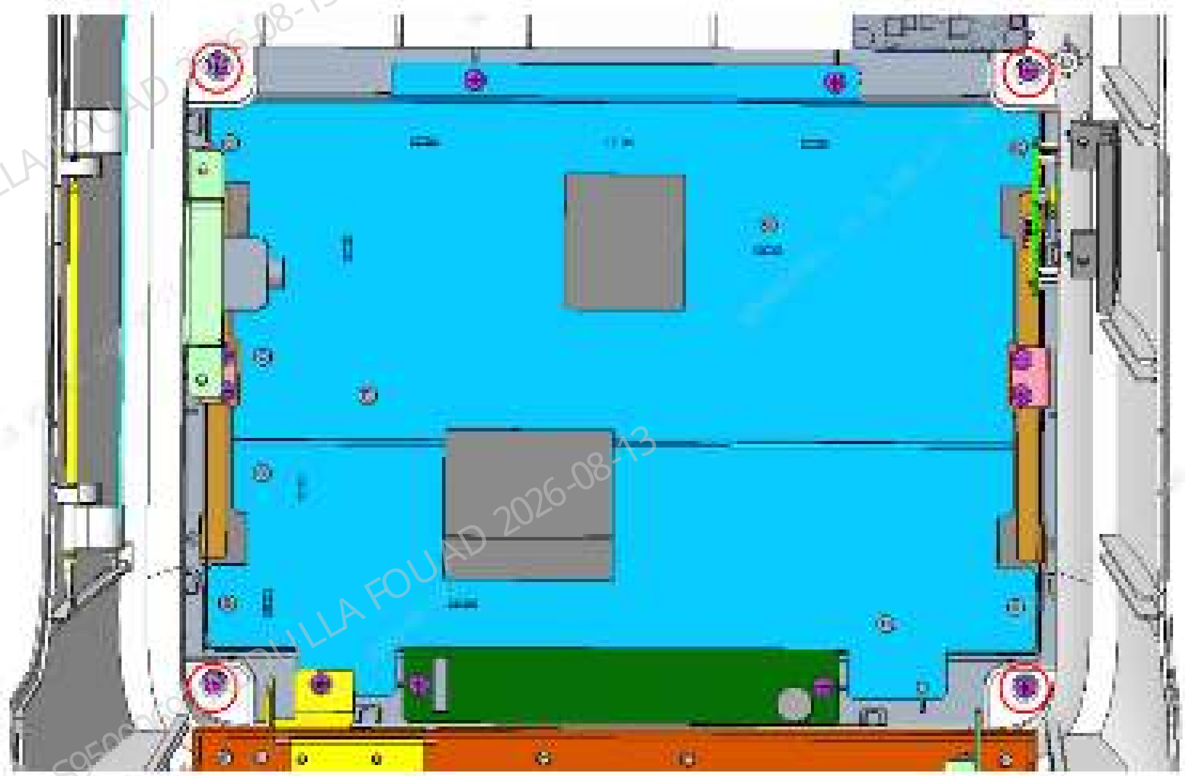
Figure8–698 Step 6 of removing the front shell (sy)

7. As shown in the figure below, loosen the left rotation shaft (circlip) and remove the compartment door.

Figure8–699 Step 7 of removing the front shell (sy)

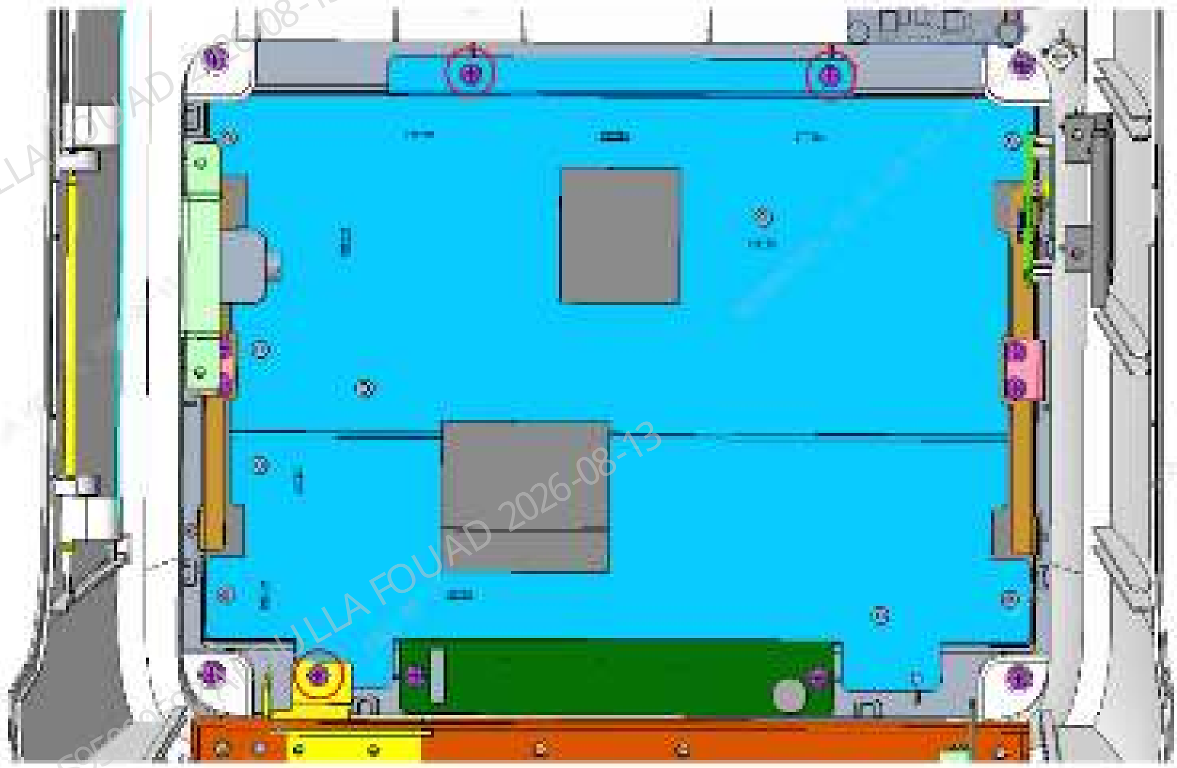
8. Loosen the four fixing screws as shown in the figure below and remove the fully-fitted screen assembly.

Figure8–700 Step 8 of removing the front shell (sy)



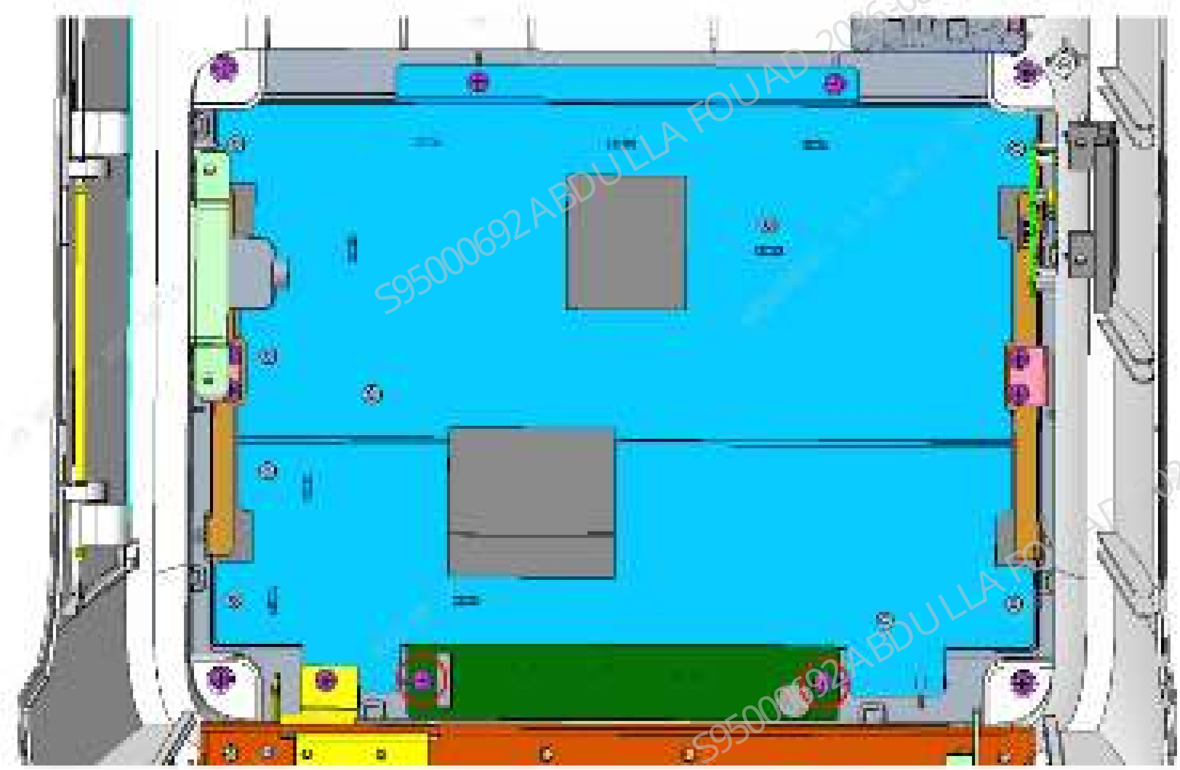
9. Loosen the three fixing screws as shown in the figure below and remove the screen cable cover.

Figure8–701 Step 9 of removing the front shell (sy)



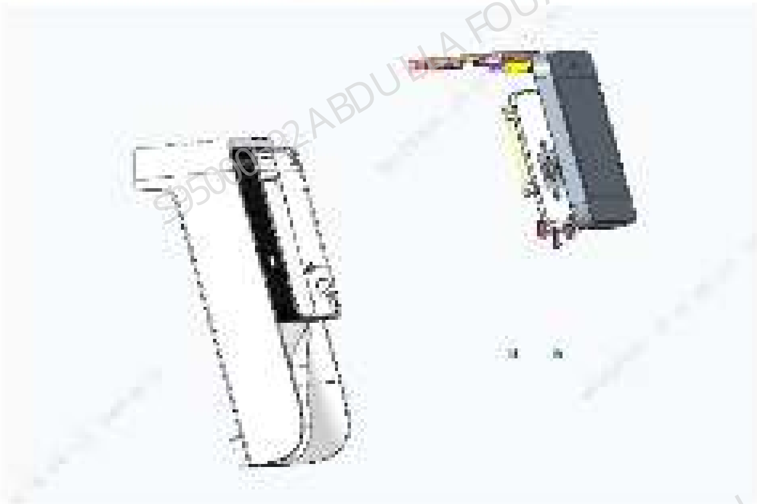
10.Loosen the two fixing screws as shown in the following figure and remove the indicator board.

Figure8–702 Step 10 of removing the front shell (sy)



11.Loosen the ten lower screws as shown in the figure below, and remove the hinge bracket of the front shell to complete the separation of the front shell (sy) from the front shell assembly.

Figure8–703 Step 11 of removing the front shell (sy)



Subsequent Requirements

N/A

8.167.3 Commissioning and Verification

N/A

8.168 Autoloader shell assembly (5X6)

8.168.1 General Information

Figure8–704 115-081771-00 autoloader shell assembly (5X6) photo

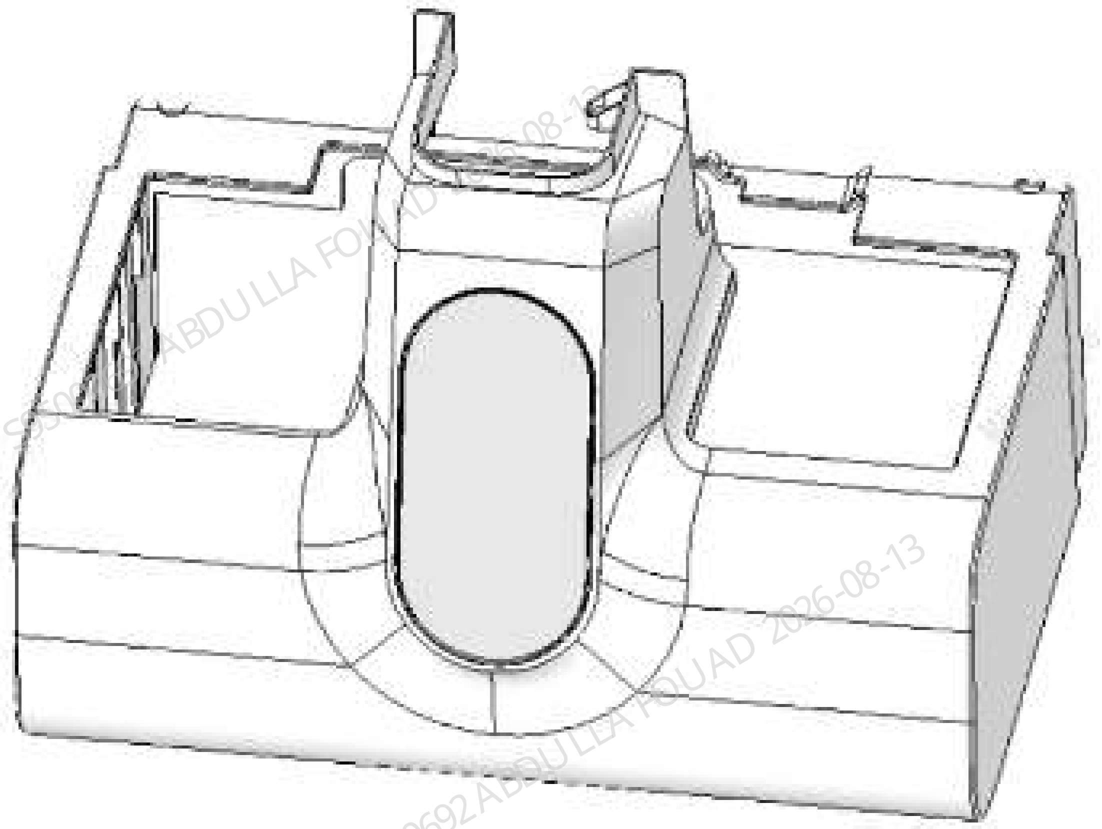


Table 8–112 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081771-00	Autoloader shell assembly (5X6)	N/A	N/A

Functions:
Appearance shell.

8.168.2 Disassembly and Assembly

Note:

N/A

Required Tools

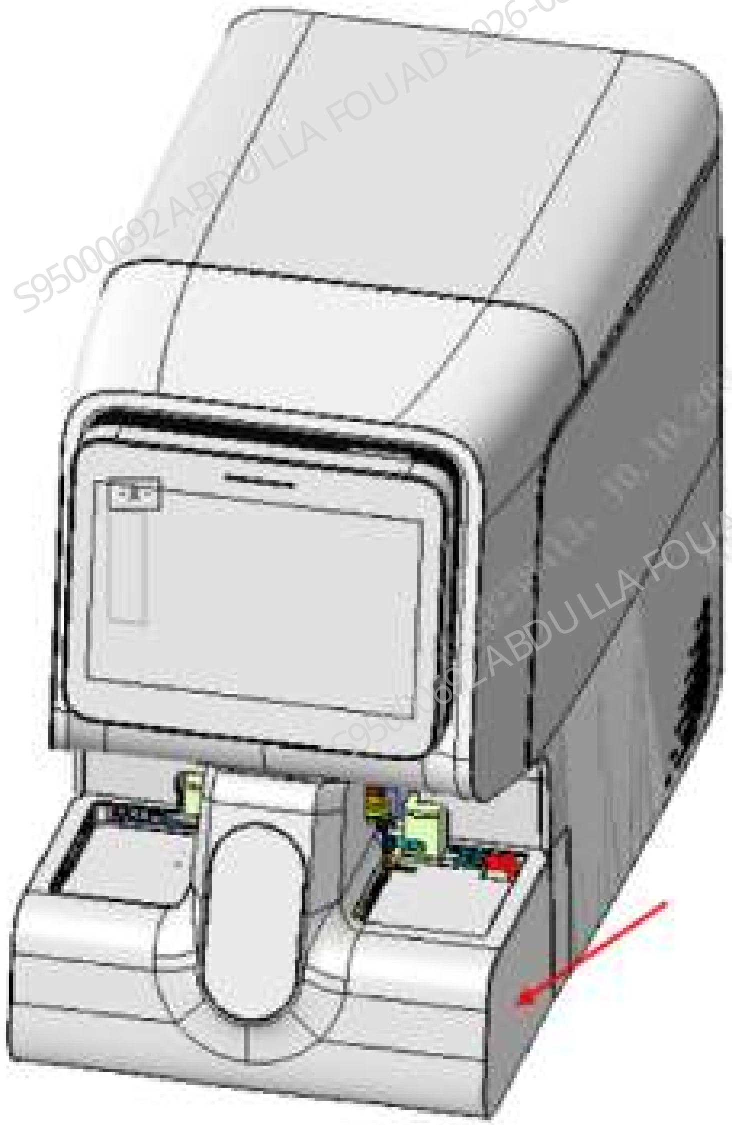
One cross-headed screwdriver

Prerequisites

Power off the analyzer, and unplug the power cord.

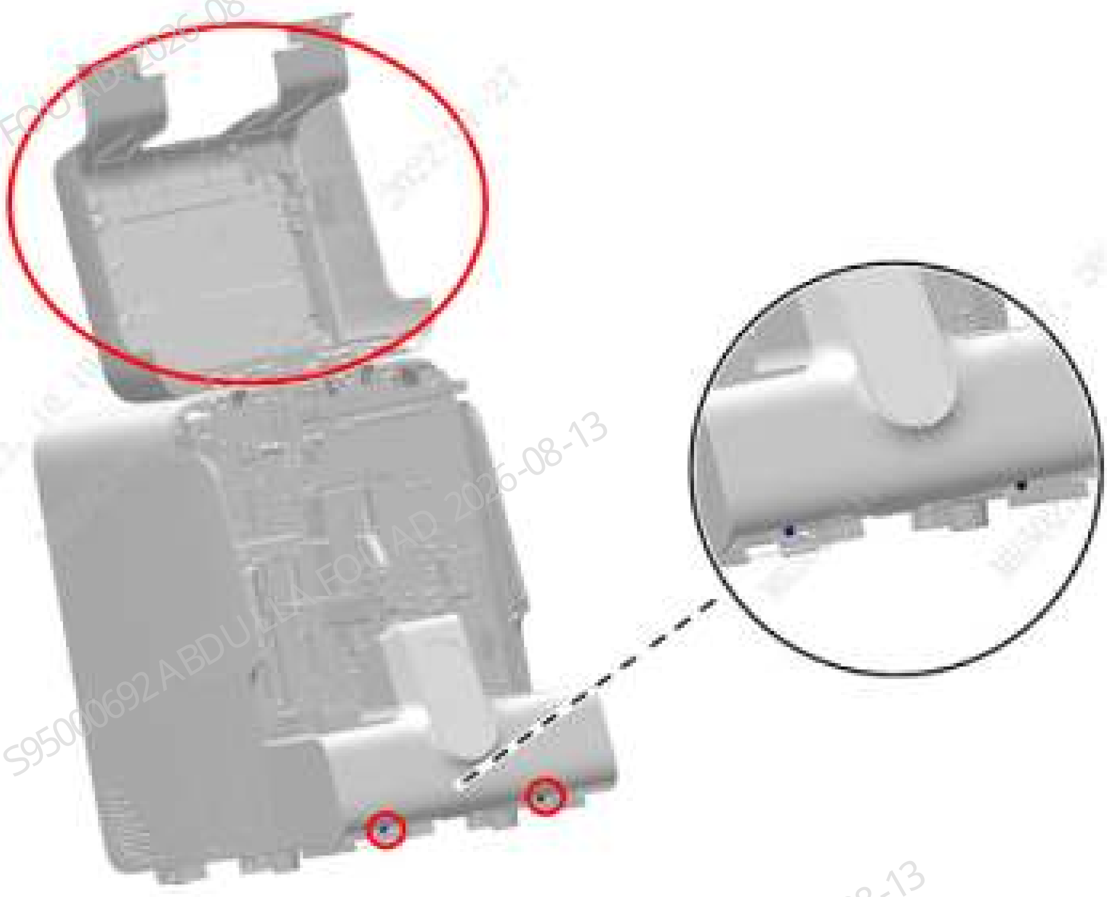
Steps

Figure8–705 Position of the Autoloader Shell Assembly (5x6) on the Overall Unit



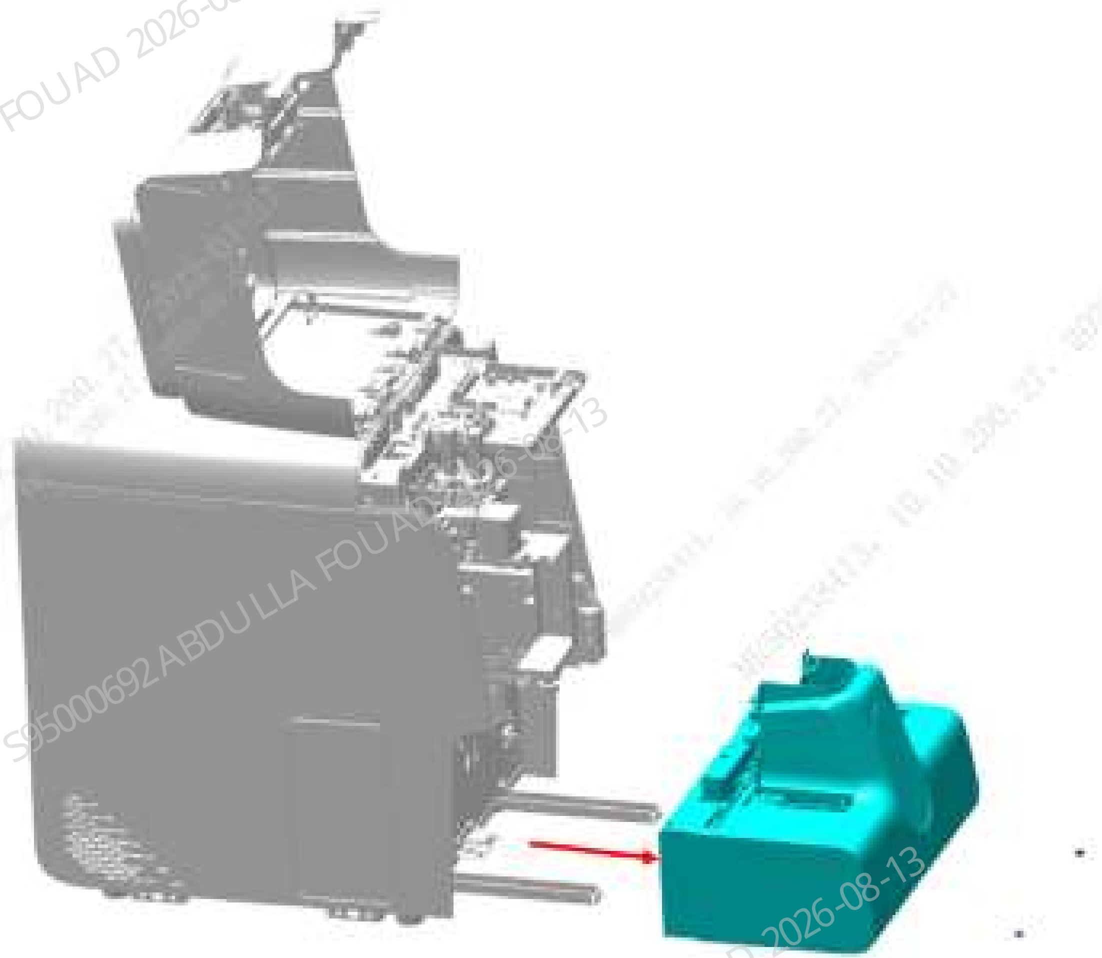
1. Open the analyzer's front cover, and put it against the top cover, as shown in the figure below.

Figure8–706 Opening the front cover



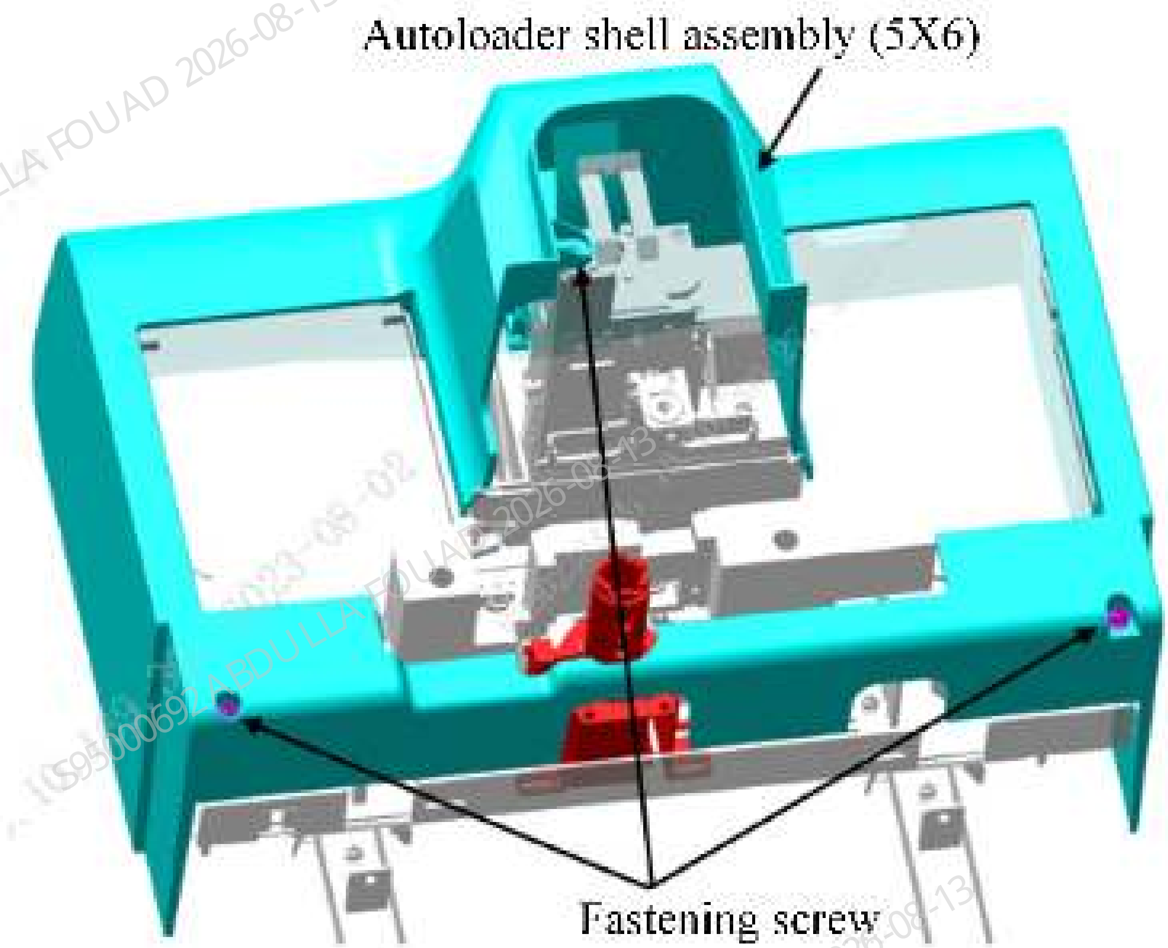
2. Unscrew the two M4x8 cross-head screws securing the autoloader on the autoloader crossbeam, as shown in the figure above.
3. Pull the autoloader to an appropriate distance in the direction of the arrow and towards the operator, and disconnect the wires connecting the analyzer and the autoloader.
4. Pull the autoloader out of the autoloader crossbeam to remove the 5X6 closed autoloader assembly, as shown in the figure below.

Figure8–707 Removing the autoloader assembly



5. Unscrew the three M4x8 cross-head screws securing the autoloader shell assembly (5x6), and remove the autoloader shell assembly (5x6), as shown in the figure below.

Figure8–708 Removing the autoloader cover assembly (10X5)



Subsequent Requirements

Installation procedure:

1. Install the autoloader cover assembly (5x6) to the autoloader.
2. Place the autoloader on the autoloader crossbeam, push it towards the analyzer's front panel for a proper distance, and connect the wires connecting the analyzer and the autoloader.
3. Push the autoloader along the autoloader crossbeam to attach it to the analyzer's front panel.
4. Secure the autoloader to the autoloader crossbeam with two M4X8 cross-head screws.
5. Flip down the front cover.

8.168.3 Commissioning and Verification

N/A

8.169 Front shell assembly (ZD)

8.169.1 General Information

Figure8–709 115-078084-00 front shell assembly (ZD) photo

Table 8–113 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-078084-00	Front shell assembly (ZD)	N/A	N/A

Functions:

Analyzer appearance.

8.169.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver
One set of Allen wrenches
Cutting nipper x 1

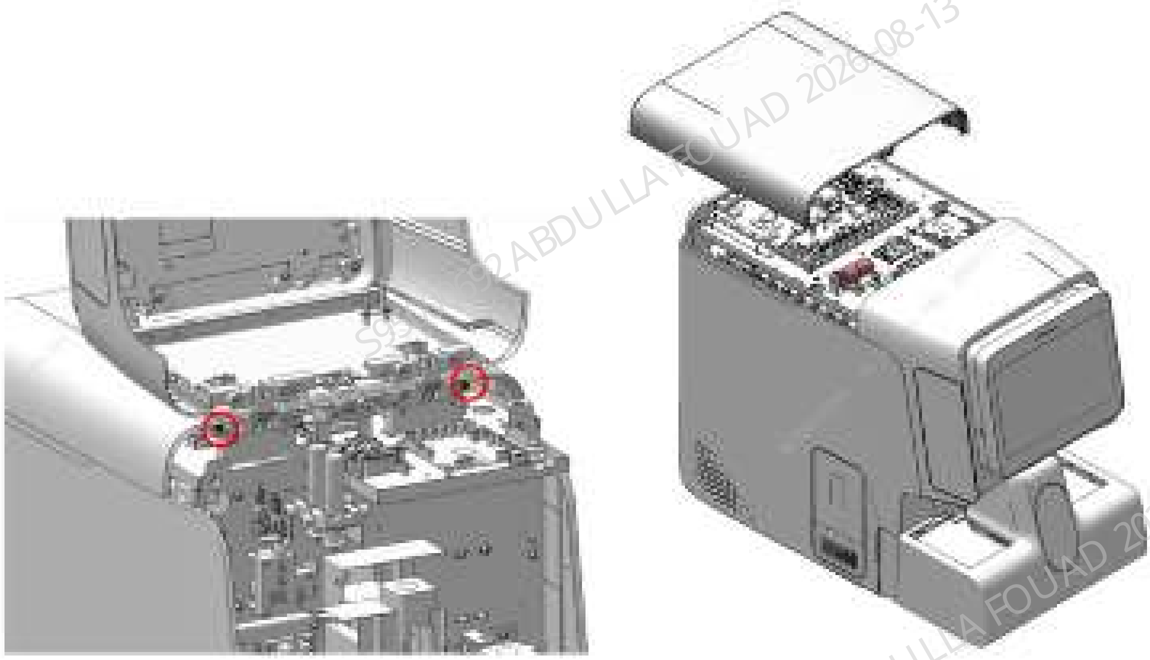
Prerequisites

N/A

Steps

1. Open the front shell, loosen the four fixing screws on the rear shell, and remove the rear panel, as shown in the following figure.

Figure8–710 Step 1 of removing the upper shell



2. Remove the top plug on the screen assembly as shown in the following figure, loosen the lower hinge, and remove the front shell assembly.

Figure8–711 Step 2 of removing the upper shell

Subsequent Requirements

N/A

8.169.3 Commissioning and Verification

N/A

8.170 Autoloader shell assembly (10X5)

8.170.1 General Information

Figure8–712 115-081770-00 autoloader shell assembly (10X5) photo

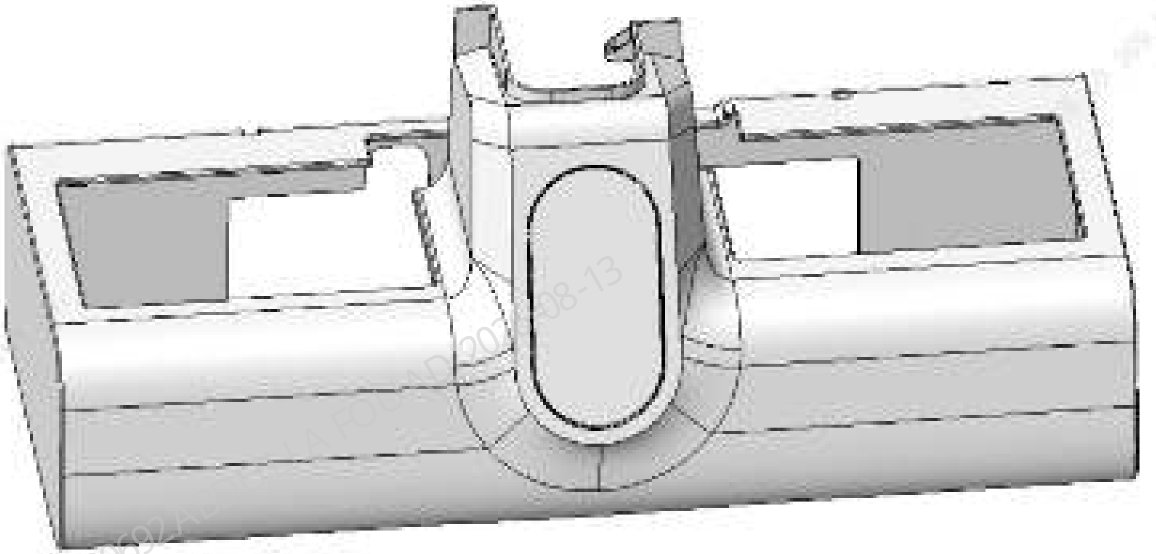


Table 8–114 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-081770-00	Autoloader shell assembly (10X5)	N/A	N/A

Functions:
Appearance shell.

8.170.2 Disassembly and Assembly

Note:

N/A

Required Tools

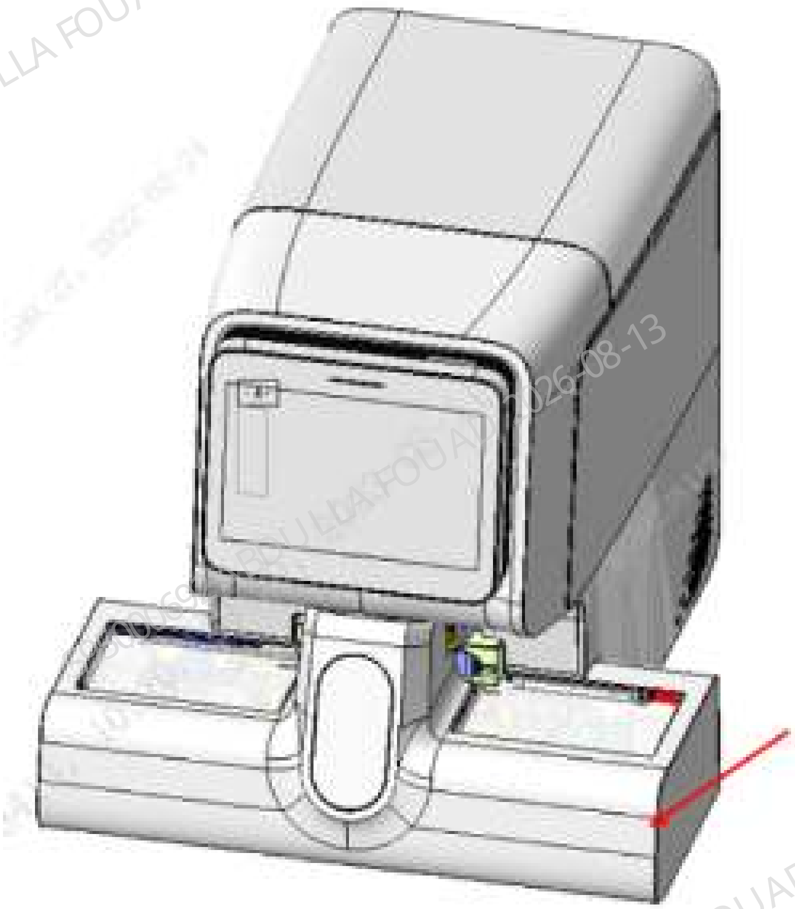
One cross-headed screwdriver

Prerequisites

Power off the analyzer, and unplug the power cord.

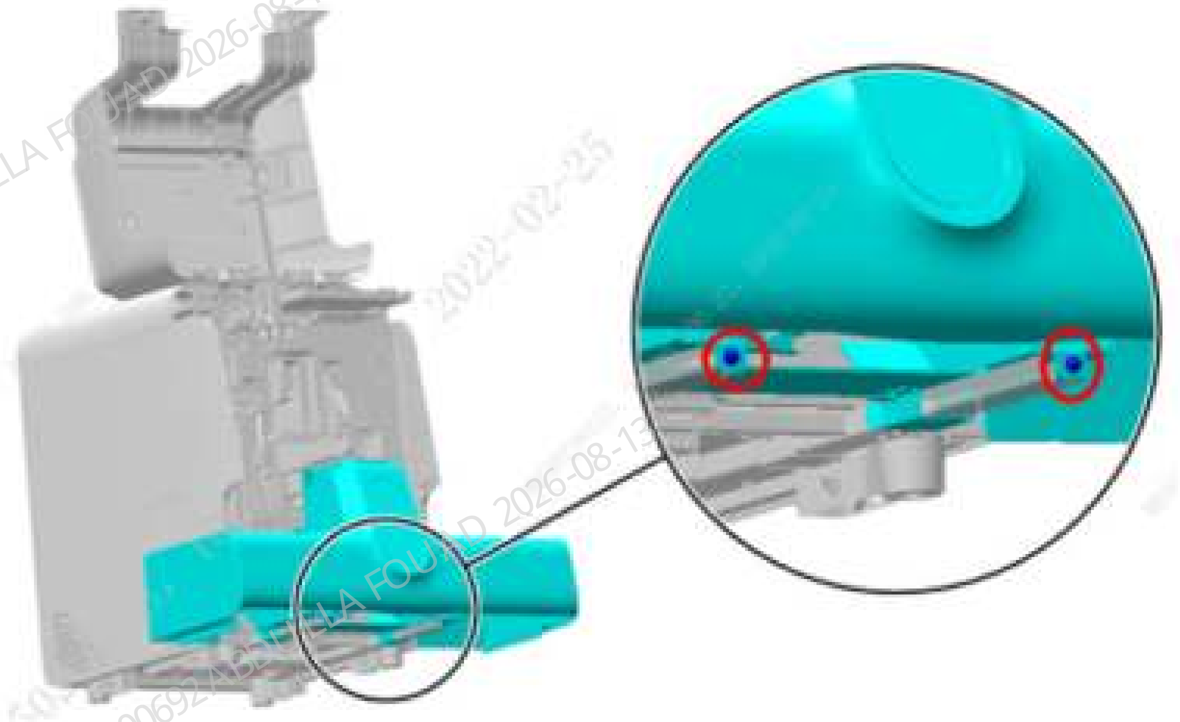
Steps

Figure8–713 Position of the Autoloader Shell Assembly (10X5) on the Overall Unit



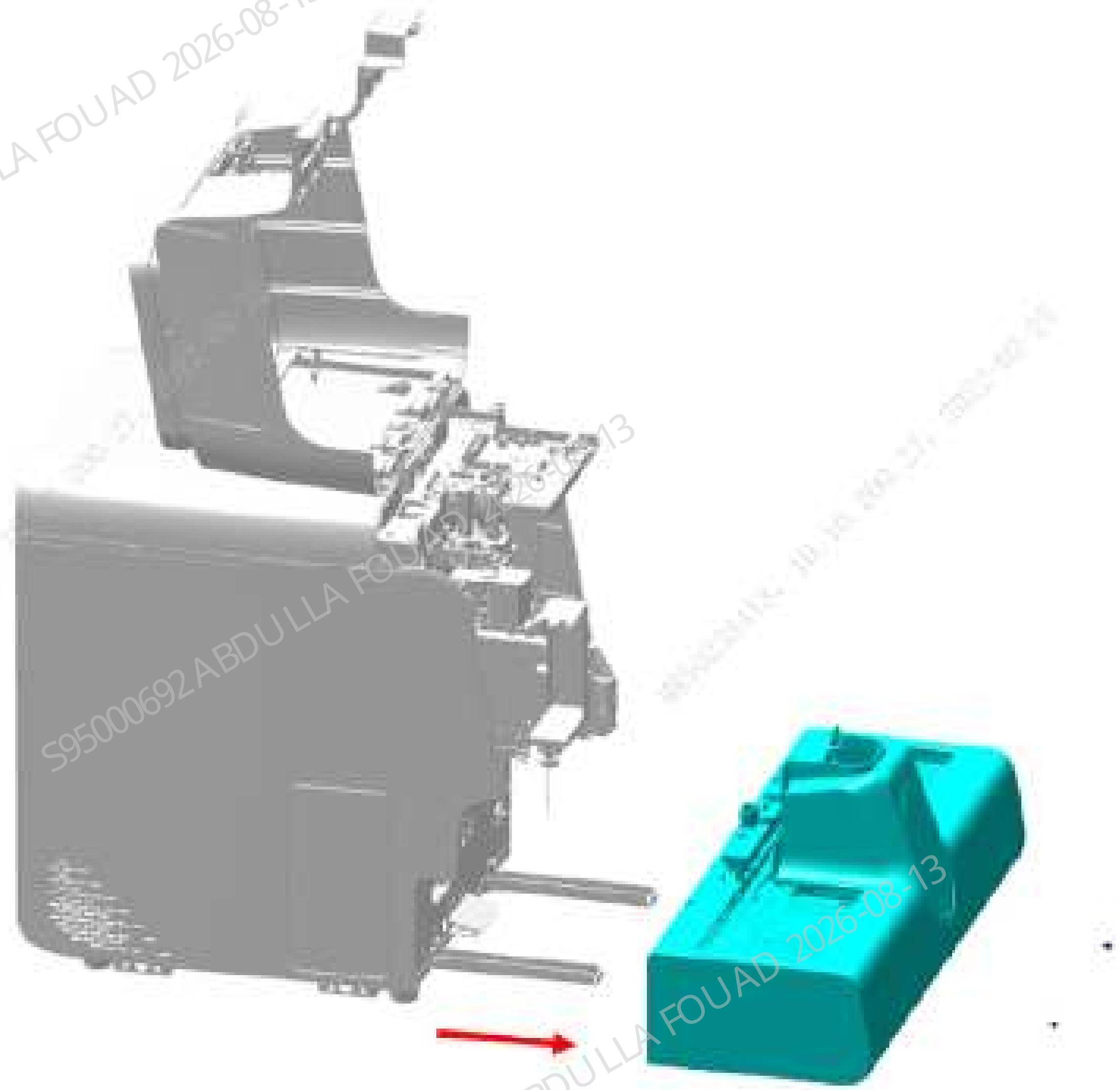
1. Open the analyzer's front cover, and put it against the top cover, as shown in the figure below.

Figure8–714 Opening the front cover



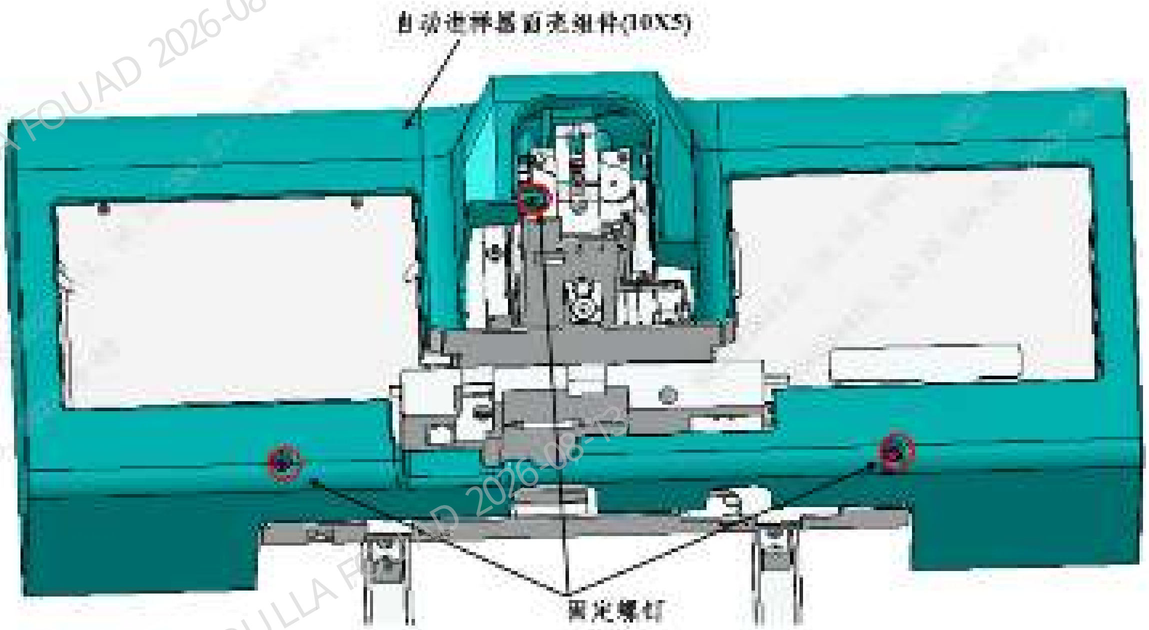
2. Unscrew the two M4x8 cross-head screws securing the autoloader on the autoloader crossbeam, as shown in the figure above.
3. Pull the autoloader to an appropriate distance in the direction of the arrow and towards the operator, and disconnect the wires connecting the analyzer and the autoloader.
4. Pull the autoloader out of the autoloader crossbeam to remove the position-10 and row-5 closed autoloader assembly, as shown in the figure below.

Figure8–715 Removing the autoloader assembly



5. Unscrew the three M4x8 cross-head screws securing the autoloader cover assembly (10X5), and remove the autoloader cover assembly (10X5), as shown in the figure below.

Figure8–716 Removing the autoloader cover assembly (10X5)



Subsequent Requirements

Installation procedure:

1. Install the autoloader cover assembly (10X5) to the autoloader.
2. Place the autoloader on the autoloader crossbeam, push it towards the analyzer's front panel for a proper distance, and connect the wires connecting the analyzer and the autoloader.
3. Push the autoloader along the autoloader crossbeam to attach it to the analyzer's front panel.
4. Secure the autoloader to the autoloader crossbeam with two M4X8 cross-head screws.
5. Flip down the front cover.

8.170.3 Commissioning and Verification

N/A

8.171 Left shell (10X5) (BC-760 series)

8.171.1 General Information

Figure8–717 115-098561-00 left shell (10x5) photo

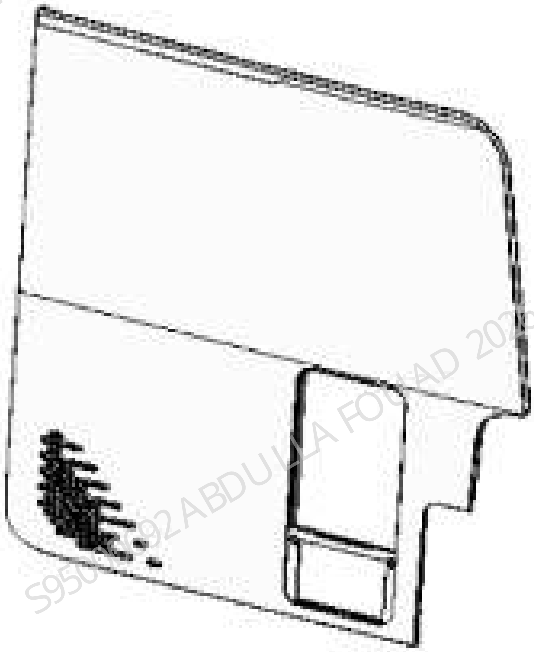


Table 8–115 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-098561-00	Left shell (10x5)	N/A	N/A

Functions:
Appearance part.

8.171.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver

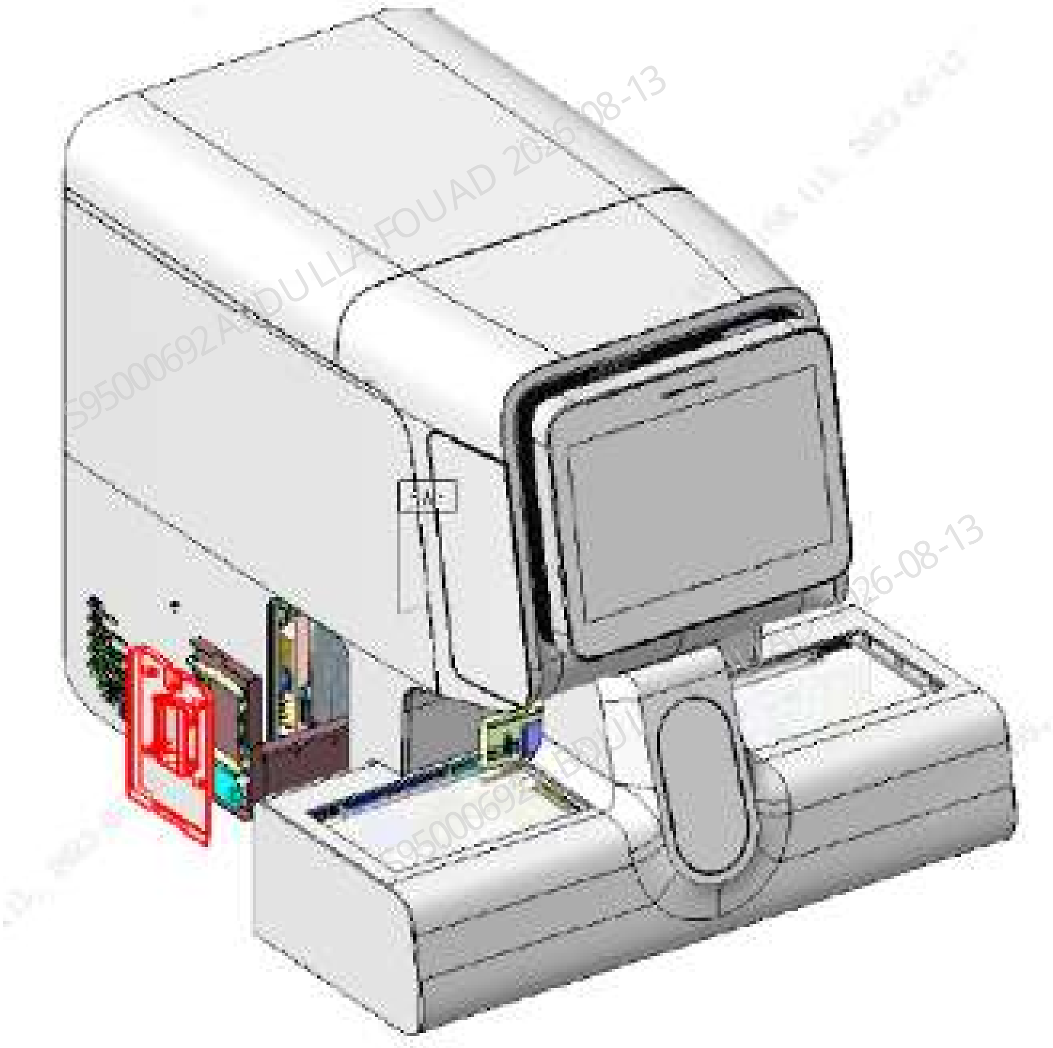
Prerequisites

Before removal, click the screen to lift the aspirate probe of the latex reagent thermostat assembly, and pull out and remove the latex reagent compartment door.

Steps

1. As shown in the figure below, lift the latex reagent probe to the upper limit position on the software screen, and pull out the latex reagent compartment door (along with the sliding mechanism of the latex reagent assembly).

Figure8–718 Latex reagent compartment door disassembly procedure 1



2. As shown in the figure below, unscrew one M3 cross screw fixing the latex reagent compartment door, and remove the latex reagent compartment door.

Figure8–719 Latex reagent compartment door disassembly procedure 1

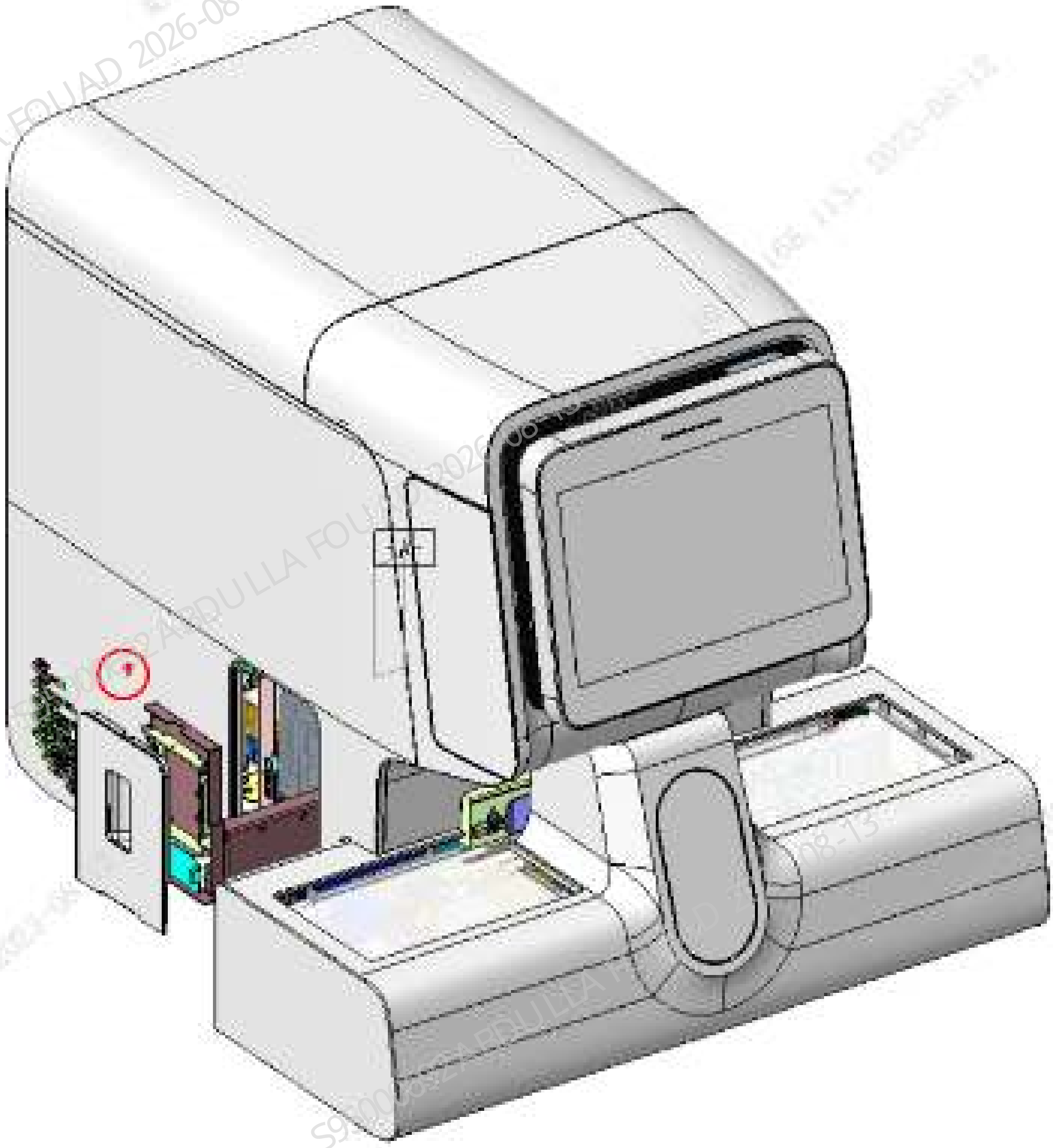


Figure8–720 Turning the front shell assembly

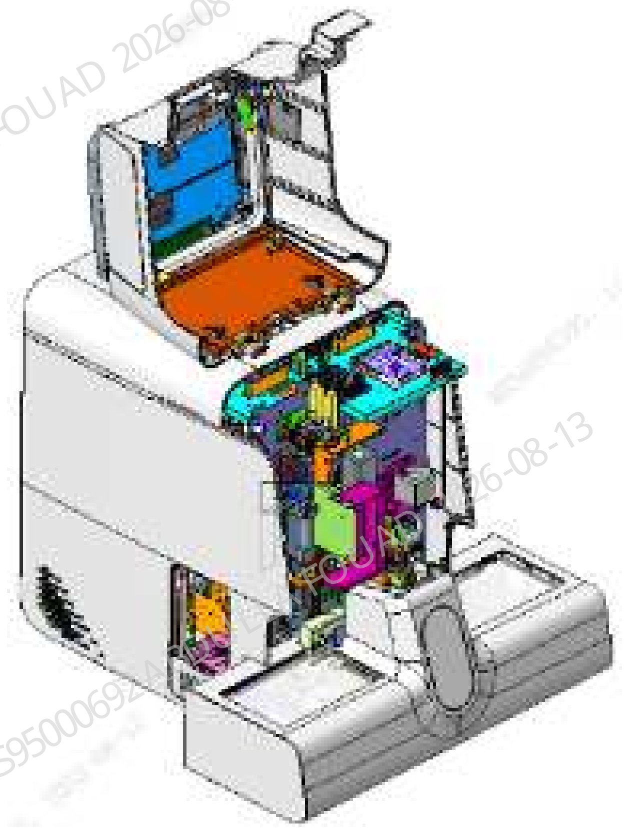


Figure8–721 Removing the left shell screw

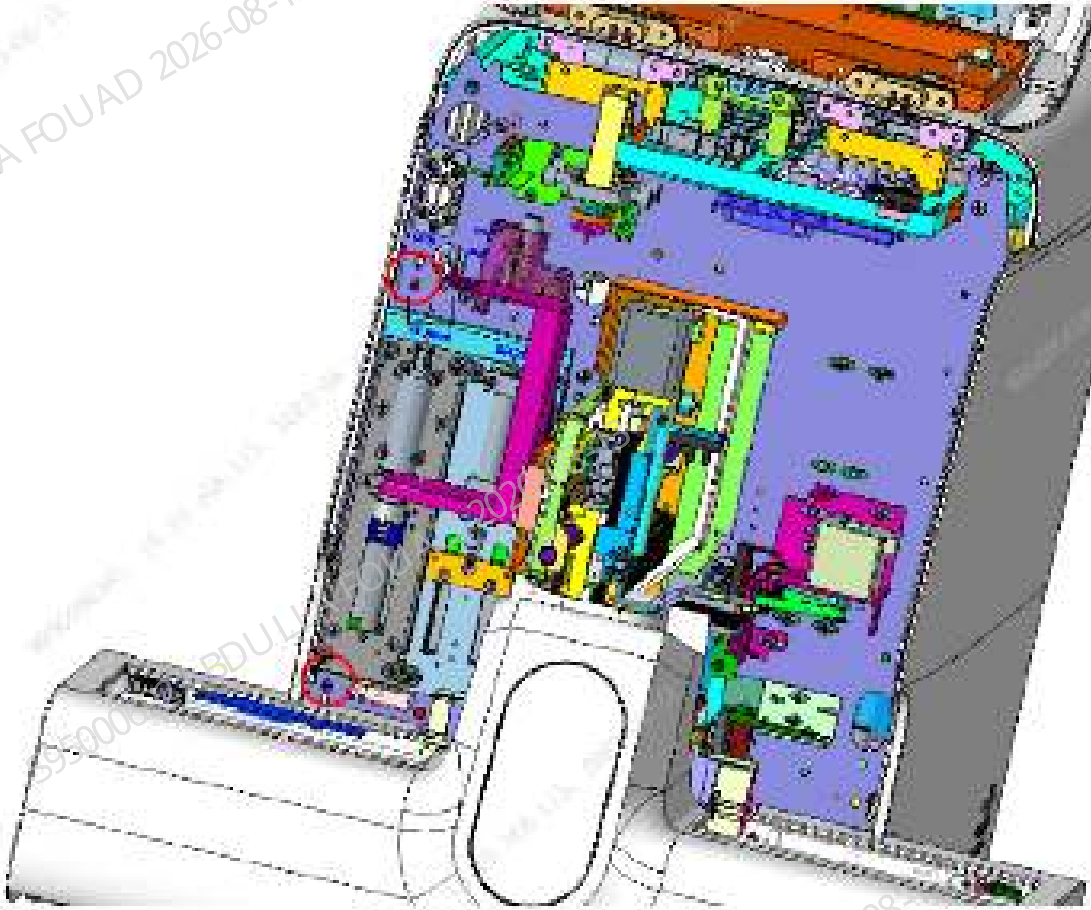


Figure8–722 Removing the left shell

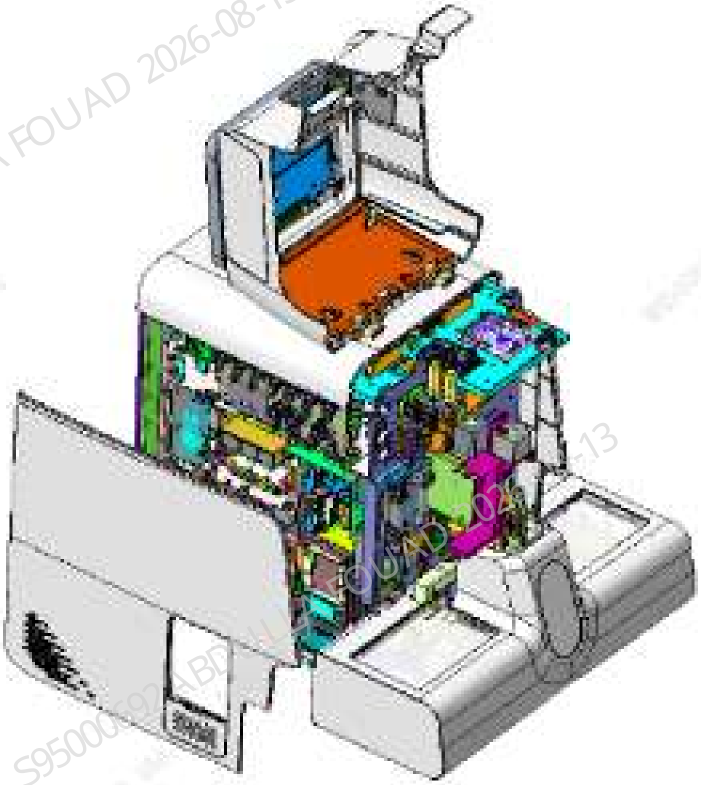
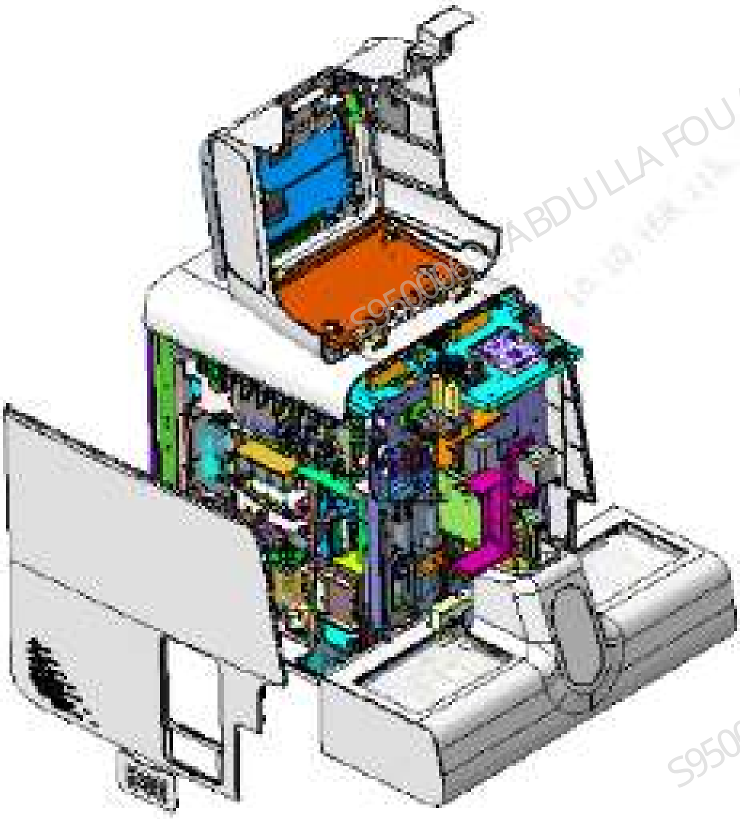


Figure8–723 Removing the small door of the filter screen



Subsequent Requirements

Install the latex reagent compartment door on the front door of cold chamber, and close it. Then, click the screen to lower the aspirate probe of the latex reagent thermostat assembly.

8.171.3 Commissioning and Verification

N/A

8.172 Right shell (10X5) (BC-760 series)

8.172.1 General Information

Figure8–724 115-098564-00 right shell (10X5) photo

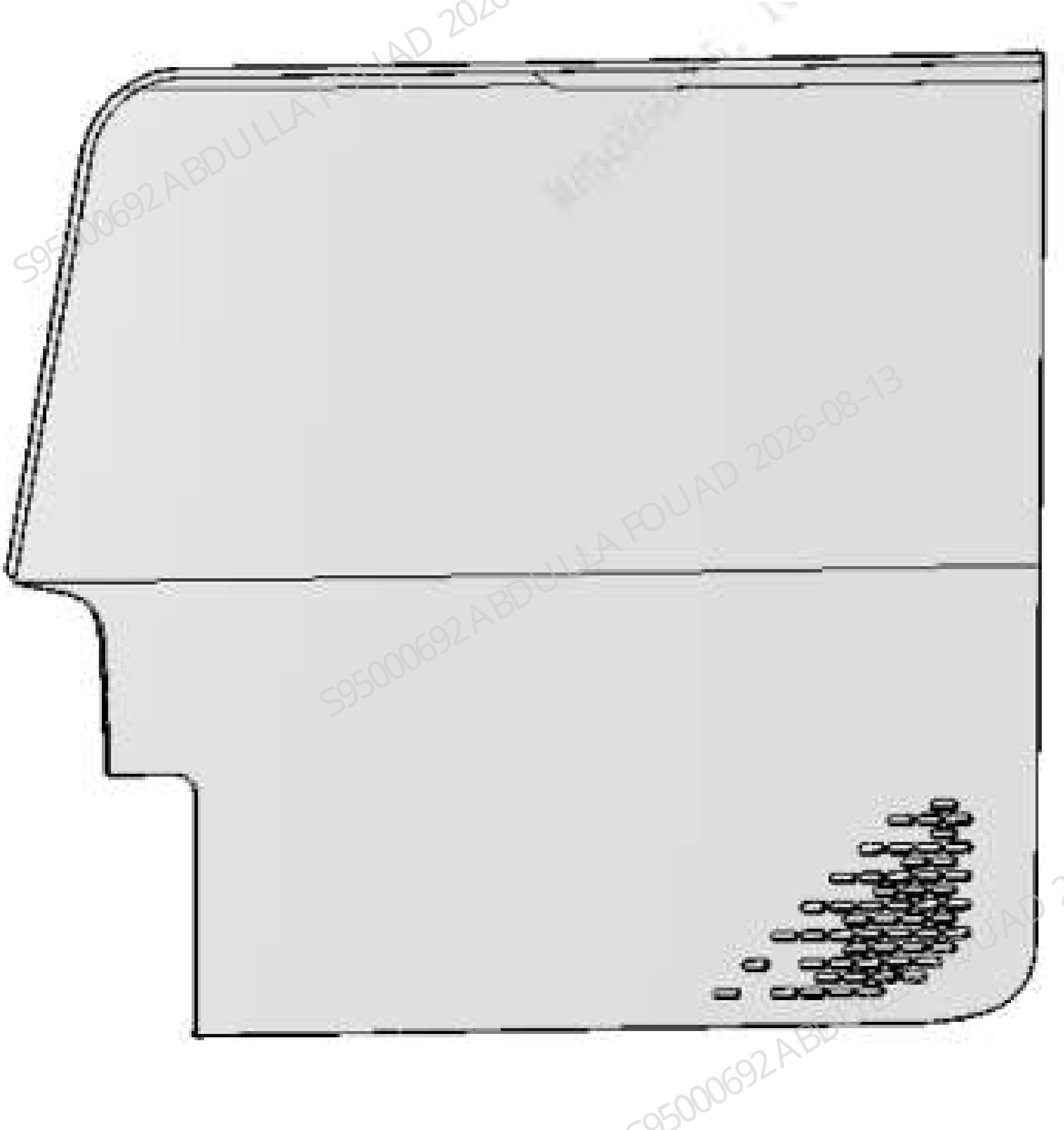


Table 8–116 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-098564-00	Right shell (10X5)	N/A	N/A

Functions:
Analyzer appearance.

8.172.2 Disassembly and Assembly

Note:

N/A

Required Tools

One cross-headed screwdriver

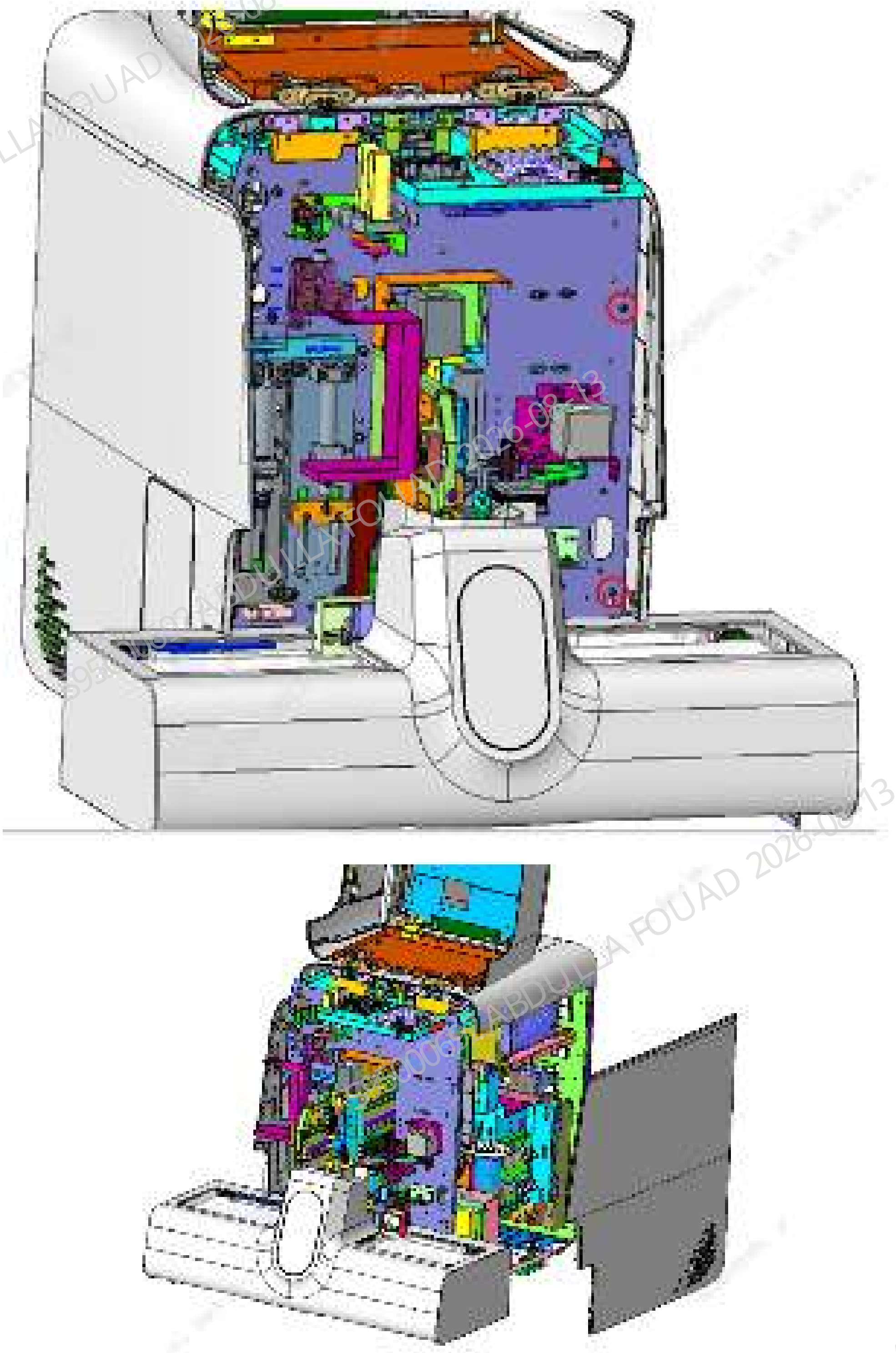
Prerequisites

N/A

Steps

Open the front shell and dock it on the top shell, loosen the two screws that fix the right shell, and remove the right shell, as shown in the figure below.

Figure8–725 Latex reagent compartment door disassembly procedure 1



Subsequent Requirements

N/A

8.172.3 Commissioning and Verification

N/A

8.173 Internal barcode scanner (excluding the rotated wheel compre

8.173.1 General Information

N/A

8.173.2 Disassembly and Assembly

N/A

8.173.3 Commissioning and Verification

N/A

8.174 Tube detection assembly

8.174.1 General Information

Figure8–726 115-004878-00 tube detection assembly photo

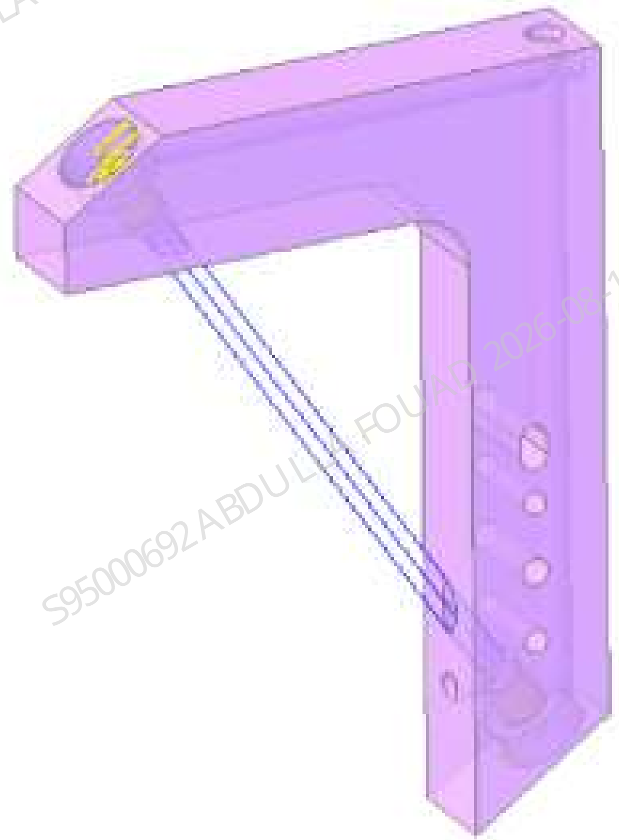


Table 8–117 Information About the Assembly

FRU Code	FRU Name	Lower-Level FRU Code	Lower-Level FRU Name
115-004878-00	Tube detection assembly	N/A	N/A

Functions:
Check if there is any information about the test tube on the test tube rack in the analysis area on the autoloader.

8.174.2 Disassembly and Assembly

Note:

N/A

Required Tools

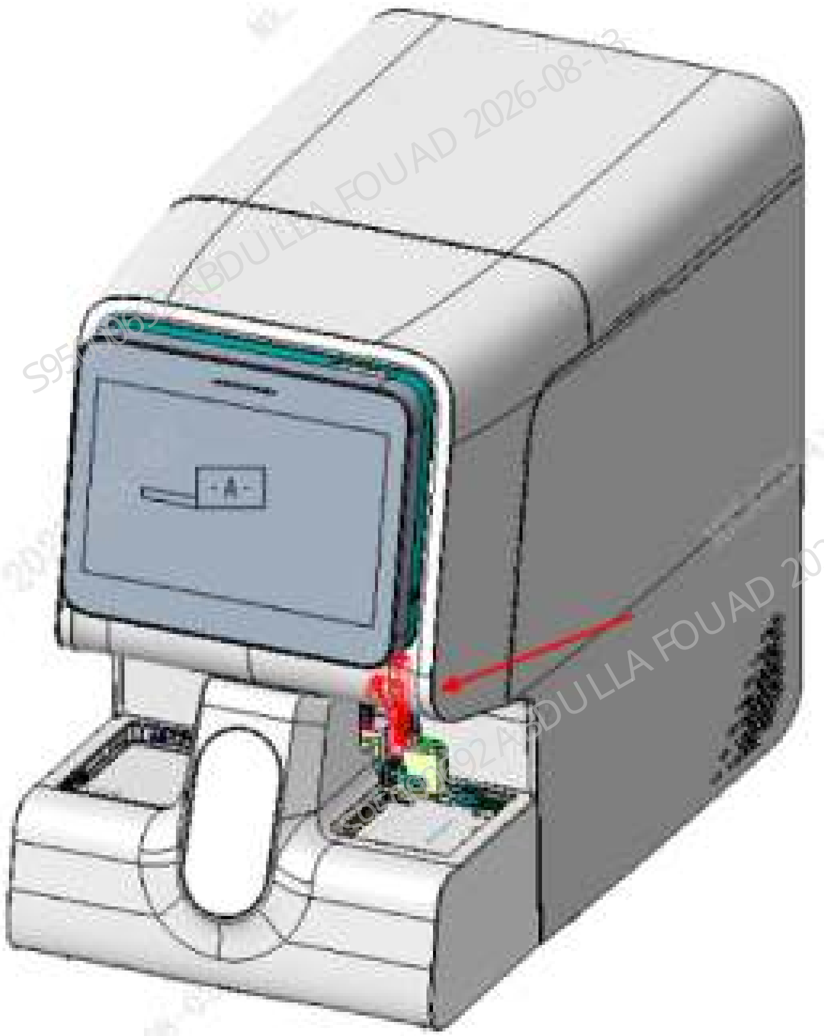
One cross-headed screwdriver

Prerequisites

Power off the analyzer, and unplug the power cord.

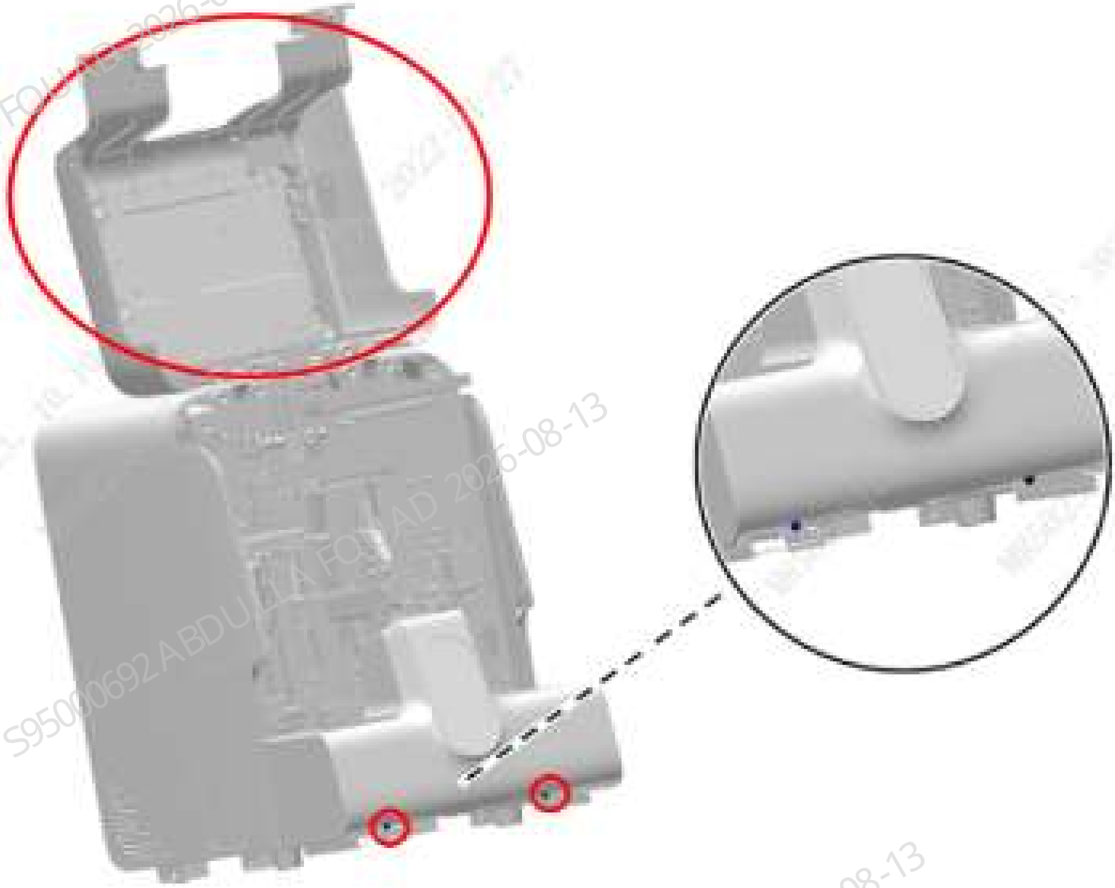
Steps

Figure8–727 Position of Tube Detection Assembly on the Overall Unit



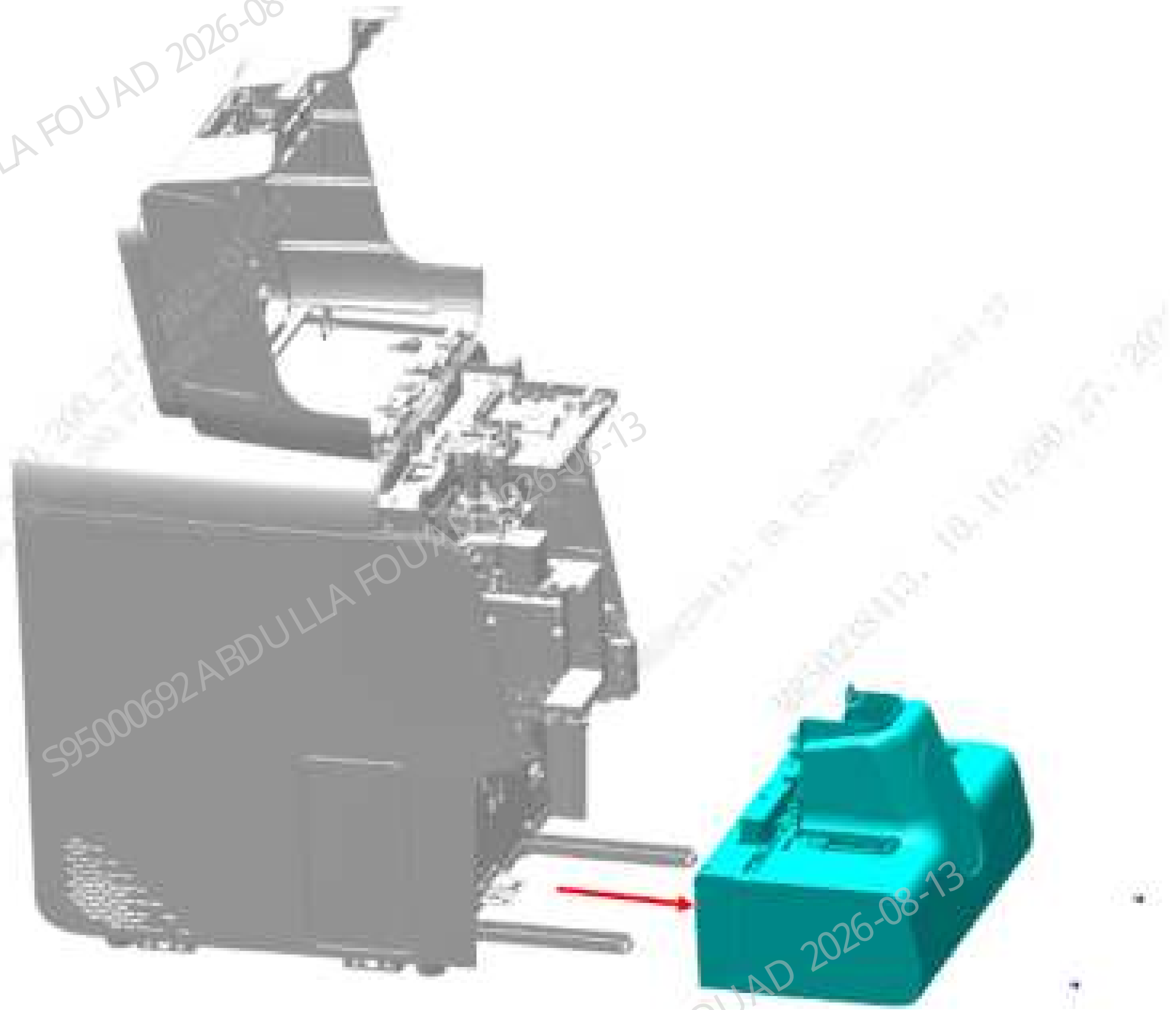
1. Open the analyzer's front cover, and put it against the top cover, as shown in the figure below.

Figure8–728 Opening the front cover



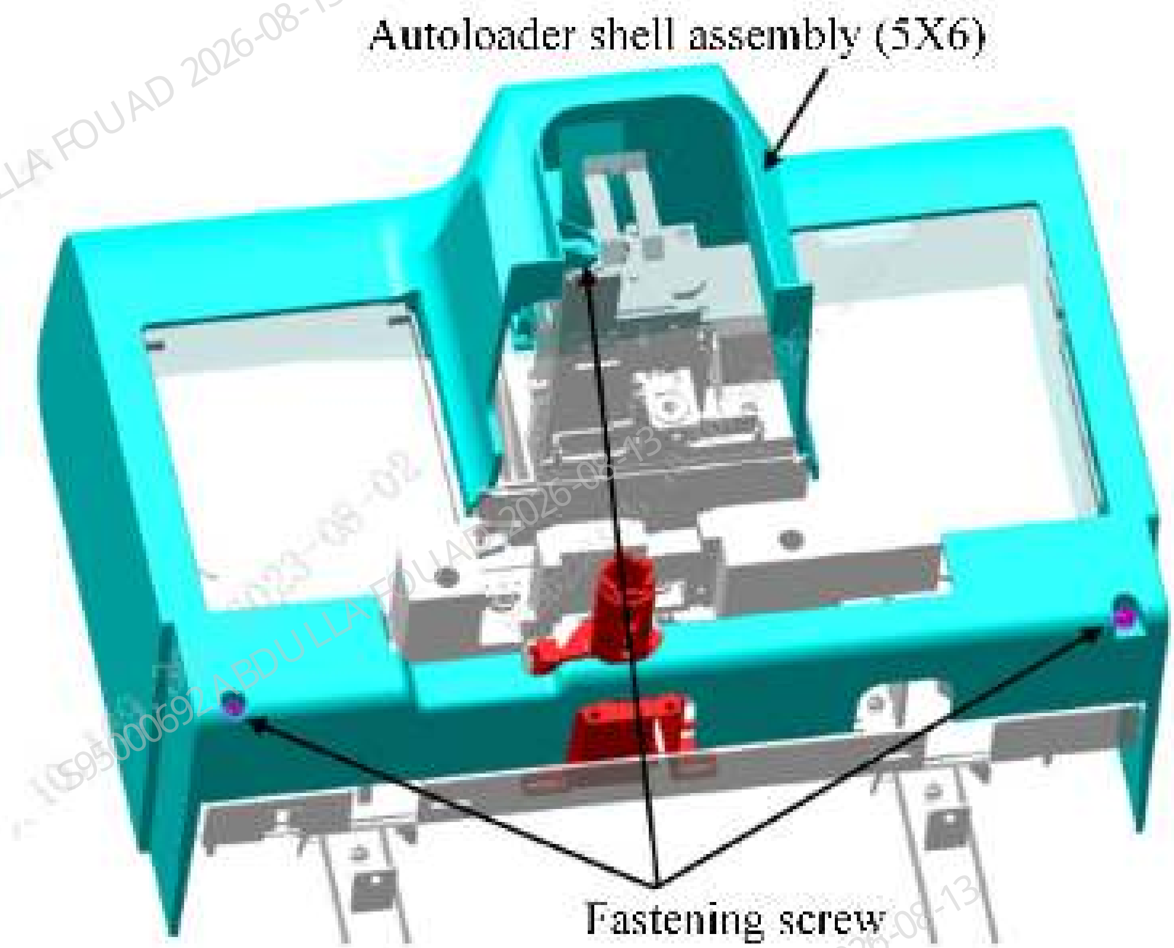
2. Unscrew the two M4x8 cross-head screws securing the autoloader on the autoloader crossbeam, as shown in the figure above.
3. Pull the autoloader to an appropriate distance in the direction of the arrow and towards the operator, and disconnect the wires connecting the analyzer and the autoloader.
4. Pull the autoloader out of the autoloader crossbeam to remove the 5X6 closed autoloader assembly, as shown in the figure below.

Figure8–729 Removing the autoloader assembly



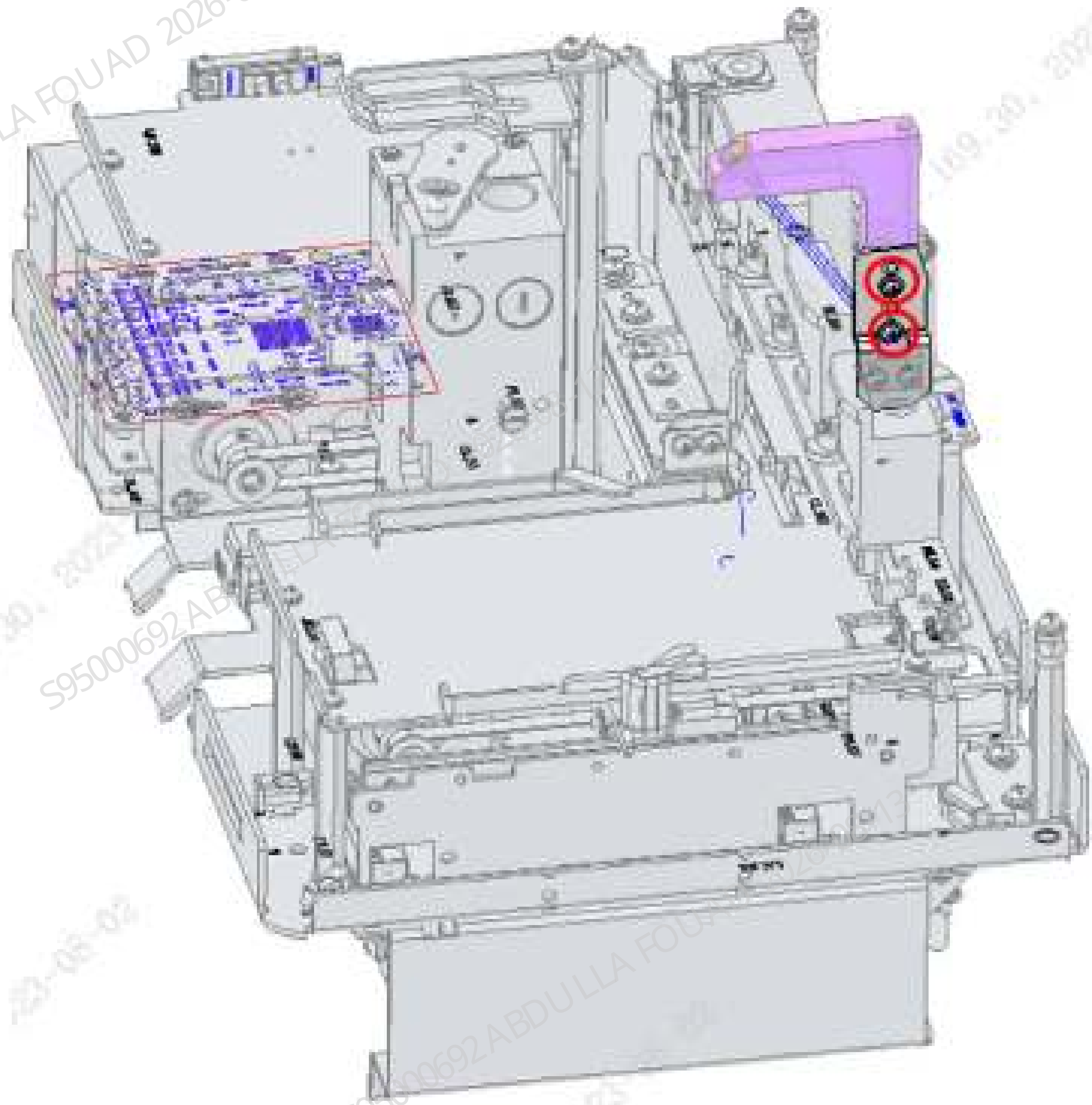
5. Unscrew the three M4x8 cross-head screws securing the autoloader shell assembly (5x6), and remove the autoloader shell assembly (5x6), as shown in the figure below.

Figure8–730 Removing the autoloader cover assembly (5x6)



6. Remove the two M3X8 screws securing the tube detection assembly, unplug the wire for the tube detection assembly, and then remove the tube detection assembly, as shown in the figure below.

Figure8–731 Diagram of Removing the Tube Detection Assembly



Subsequent Requirements

Installation procedure:

1. Fix the tube detection assembly to the main body of the autoloader using two M3x8 screws, and insert the wire for the tube detection assembly.
2. Install the autoloader cover assembly (5x6) to the autoloader.
3. Place the autoloader on the autoloader crossbeam, push it towards the analyzer's front panel for a proper distance, and connect the wires connecting the analyzer and the autoloader.
4. Push the autoloader along the autoloader crossbeam to attach it to the analyzer's front panel.
5. Secure the autoloader to the autoloader crossbeam with two M4X8 cross-head screws.
6. Flip down the front cover.

8.174.3 Commissioning and Verification

Verification contents:
Use the auto feeding function to perform the sample test and verify the function.

8.175 Left shell closed door (CDR-ZD)

8.175.1 General Information

N/A

8.175.2 Disassembly and Assembly

N/A

8.175.3 Commissioning and Verification

N/A

9 Troubleshooting

9.1 Errors with Codes

9.1.1 30200: HGB baseline abnormal

Error Code

30200

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	HGB baseline abnormal
Service engineer level	HGB baseline abnormal

Involved FRU

1. Power Supply Assembly
2. PCBA (Motherboard)

Error Mechanism

During the power-on self-test process, the current HGB baseline voltage value (the ADC detection voltage value when the LED is turned off) exceeds the normal voltage range of **100–500**.

Possible Causes

1. Power supply assembly abnormal
2. PCBA (Motherboard) error
3. Unknown Cause

Resolve Path

1. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
2. Refer to Chapter 8 Replacing, Commissioning, and Verifying Power Supply Assembly FRUs and confirm whether the power supply assembly is faulty.
3. Observe whether the four indicators D8 to D11 on the motherboard are always on green, and observe whether the electronic material is damaged or corroded on the motherboard. For error

confirmation methods, please refer to Analog Board Section of Chapter 8 FRU Replacement, Commissioning and Verification.
4. None of the above methods can solve the problem.

Solution

1. Refer to Power Supply Assembly **8.87.2** Disassembly and Assembly, **8.87.3** Commissioning and Verification.
2. To replace PCBA (motherboard), see **8.9.2** Disassembly and Assembly.
3. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.2 30201: HGB blank voltage abnormal

Error Code

30201

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	HGB blank voltage abnormal
Service engineer level	HGB blank voltage abnormal

Involved FRU

1. PCBA (Motherboard)
2. HGB assembly
3. Two-position ways LVM valve assembly (with valve board and connector)
4. Filter, 100 μm

Error Mechanism

HGB blank voltage out of range [3.2 V, 4.8V]

Possible Causes

1. HGB bath contaminated/There are bubbles in diluent in HGB bath
2. The motherboard samples the output signal of the HGB sensor to detect link abnormal: Device corrosion, loose plug, etc.
3. HGB assembly abnormal_LED abnormal
4. Diluent priming branch abnormal_SV16 abnormal
5. Waste channel abnormal_SV09 abnormal, leakage results in low reaction level and abnormal
6. Unknown Cause

Resolve Path

1. The HGB background voltage alarm occurs frequently during continuous testing.
2. Soak the HGB bath with probe cleanser or clean the HGB bath to eliminate the error.
3. Run the accurate diagnosis program for abnormal HGB background voltage, and report the diagnosis result of motherboard and its branches abnormal.
4. Run the accurate diagnosis program for abnormal HGB background voltage, and report the diagnosis result of HGB assembly abnormal_LED abnormal.
5. Run the accurate diagnosis program for abnormal HGB background voltage, and report the diagnosis result of Diluent priming branch abnormal_SV16 abnormal.
6. Run the accurate diagnosis program for abnormal HGB background voltage, and report the diagnosis result of waste channel abnormal_SV09 abnormal.
7. None of the above methods can solve the problem.

Solution

1. HGB bath probe cleanser maintenance/HGB bath cleaning.
2. To replace the motherboard and its wire J47, see PCBA, motherboard **8.9.2** Disassembly and Assembly.
3. Open the shielding cover and observe whether the LED lights up normally. If it lights up normally, the luminous intensity of the LED is abnormal; select **Menu**>**>Setup**>**>Gain Setup**, click the **Auto set HGB gain** button, and re-calibrate the gain. If the error exists after the gain calibration, replace the HGB assembly directly. See **8.41.2** Disassembly and Assembly.
4. Open the shielding cover and observe whether the LED lights up normally. If the LED is off, replace the HGB assembly. See **8.41.2** Disassembly and Assembly.
5. Check whether the tubes T86, T85, T84, T30, T28, T27, and T26 are bent or fallen off. If so, replace the tube.
6. Select **Menu**>**>Service**>**>Commissioning & Self-Test** >**>Self-Test**, and self-test the SV16 valve. If abnormal, replace the valve.
7. Check whether the tubes T88, T89, T90, and T91 are bent or fallen off. If so, replace the tube.
8. Select **Menu**>**>Service**>**>Commissioning & Self-Test** >**>Self-Test**, and self-test the SV09 valve. If abnormal, replace the valve.
9. Check whether the filter LF2 is leaking, if so, replace the filter by referring to **8.63.2** Disassembly and Assembly.
10. Collect the INF original data and logs of the error, and transfer them to R&D for processing. For specific paths, see the appendix.

9.1.3 30202-30203: HGB Waste Channels Clogged

Error Code

30202-30203

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste channels clogged
Service engineer level	HGB waste channel clogged

Involved FRU

1. Mindray two-way valve (PEIZH)
2. Filter, 100 μm
3. PCBA, Air pressure detection board, analog, 5 channels

Error Mechanism

1. During the measurement process, the CBC_W sensor detects that the detection pressure of the waste channel of the HGB bath is out of range
2. During the probe cleanser measurement process, the CBC_W sensor detects that the detection pressure of the waste channel of the HGB bath is out of range

Possible Causes

1. Clogged valve or clogged filter
2. Abnormal waste sensor and related tubes
3. Unknown Cause

Resolve Path

1. The SV09 valve is clogged or fails to open.
2. The LF2 filter is clogged.
3. The tubes T88, T89, T90, and T91 are bent or crystallized, etc.
4. When CBC_W waste sensor and its common tubes T190, T189, T187, T184, T185, T186, T183, T182, T181, T200 and T298 are abnormal, other measurement channels will report at the same time that the waste channel is clogged.
5. None of the above methods can solve the problem.

Solution

1. Check whether the tubes T88, T89, T90, T91 are bent, squeezed, crystallized, etc. If so, replace the tube.
2. Select **Menu**>>**Service**>>**Commissioning & Self-Test** >>**Self-Test**, and self-test the SV09 valve. If abnormal, replace the valve.
3. Check whether the filter LF2 is clogged, if so, replace the filter by referring to **8.63.2** Disassembly and Assembly.

4. To replace the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.
5. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.4 30000: RET Reaction Bath Temperature Control Abnormal (High)

Error Code

30000

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Reaction bath temperature high
Service engineer level	Reaction bath detection temperature high

Involved FRU

Reaction bath assembly

Error Mechanism

The detection temperature exceeds the set target by 3 °C (not indicating that the real temperature is high)

Possible Causes

1. Temperature sensor error
2. Heating rod error

Resolve Path

Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **DIFF/RET Reaction Bath** to confirm the temperature.

Solution

1. Unplug and replug the temperature sensor connector.
2. Unplug and replug the heating rod connector.
3. To replace the reaction bath assembly, see **8.36.2** Disassembly and Assembly.
(Neither heating rod nor temperature sensor can be replaced separately at the client)

9.1.5 30001: Preheating Bath Temperature Control Abnormal (High)

Error Code

30001

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	The diluent needs temperature-raising
Service engineer level	Detected preheating bath temperature extremely high

Involved FRU

Diluent heating assembly

Error Mechanism

The detected temperature is higher than the set target by 8°C

Possible Causes

1. Diluent temperature extremely low
2. Temperature sensor error
3. Heating rod error

Resolve Path

1. Check whether the diluent temperature is room temperature.
2. Click **Menu**>>**Status**>>**Temperature & Pressure**to access the temperature and pressure display screen, and then view **Preheating Bath Temperature** to confirm the **temperature inside the analyzer**.
3. If the diluent temperature can gradually approach the temperature inside the analyzer, it may be caused by the diluent temperature-raising but not the error of the analyzer. If the diluent temperature is low and does not rise, the function of the heating rod may be damaged.

Solution

1. The diluent shall be used after being placed at room temperature for 1 ~ 2 days. The diluent in cold storage cannot be used for the analyzer. The diluent can be used only when its temperature is close to room temperature.
2. Unplug and replug the temperature sensor connector.

3. Unplug and replug the heating rod connector.
4. To replace the preheating bath assembly, see **8.44.2** Disassembly and Assembly.
5. Heating function is lost.

9.1.6 30002: Internal Analyzer Detection Temperature High

Error Code

30002

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Temperature rise inside the analyzer
Service engineer level	Detected internal analyzer temperature high

Involved FRU

1. PCBA (Motherboard)
2. Board Fan Assembly

Error Mechanism

The internal temperature of the analyzer is higher than the ambient temperature by more than 6°C

Possible Causes

1. Motherboard damage
2. The board fan assembly is damaged and the fan does not rotate, resulting in incomplete scattering

Resolve Path

1. Check whether the motherboard has electronic material damage or corrosion
2. Observe whether the scattering fan of the analyzer rotates

Solution

1. To replace the motherboard FRU, see **8.9.2** Disassembly and Assembly.
2. To replace the board fan assembly FRU, see **8.85.2** Disassembly and Assembly.

9.1.7 30003: Reaction Bath Temperature Control Abnormal (Low)

Error Code

30003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Reaction bath temperature low
Service engineer level	Detected reaction bath temperature extremely low

Involved FRU

Reaction bath assembly

Error Mechanism

The detected temperature is lower than the set target by more than 3 °C

Possible Causes

1. Temperature sensor error
2. Heating rod error

Resolve Path

Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **DIFF/RET Reaction Bath** to confirm the temperature.

Solution

1. Unplug and replug the temperature sensor connector.
2. Unplug and replug the heating rod connector.
3. To replace the reaction bath assembly, see **8.36.2** Disassembly and Assembly.
(Neither heating rod nor temperature sensor can be replaced separately at the client)

9.1.8 30004: Preheating Bath Temperature Control Abnormal (Low)

Error Code

30004

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	The diluent needs temperature-raising
Service engineer level	Detected preheating bath temperature extremely low

Involved FRU

Diluent heating assembly

Error Mechanism

The detected temperature is lower than the set target by more than 8°C

Possible Causes

1. Diluent temperature extremely low
2. Temperature sensor error
3. Heating rod error

Resolve Path

1. Check whether the diluent temperature is room temperature.
2. Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **Preheating Bath Temperature** to confirm the **temperature inside the analyzer**.
3. If the diluent temperature can gradually approach the temperature inside the analyzer, it may be caused by the diluent temperature-raising but not the error of the analyzer. If the diluent temperature is low and does not rise, the function of the heating rod may be damaged.

Solution

1. The diluent shall be used after being placed at room temperature for 1 ~ 2 days. The diluent in cold storage cannot be used for the analyzer. The diluent can be used only when its temperature is close to room temperature.
2. Unplug and replug the temperature sensor connector.
3. Unplug and replug the heating rod connector.
4. To replace the preheating bath assembly, see **8.44.2** Disassembly and Assembly.
5. Heating function is lost.

9.1.9 30005: Detected Ambient Temperature Over 42°C

Error Code

30005

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Ambient temperature high
Service engineer level	The detected ambient temperature is over 42°C

Involved FRU

Sensor Temperature 5Kohm B3470K Threaded installation

Error Mechanism

The detected ambient temperature is over 42°C

Possible Causes

1. Ambient temperature does not meet the requirements
2. Heat source radiation, sunlight and poor ventilation
3. Temperature sensor error

Resolve Path

1. Confirm whether the working environment and temperature of the analyzer meet the requirements.
2. Click **Menu**>>**Status**>>**Temperature & Pressure**to access the temperature and pressure display screen, and then view **Ambient Temperature** to confirm the **temperature inside the analyzer**.

Solution

1. Ensure that the working environment of the analyzer meets the requirements (including temperature, temperature rise, direct sunlight, etc.).
2. Unplug and replug the connector of ambient temperature sensor or temperature sensor.

9.1.10 30006: Detected Ambient Temperature Below 2°C

Error Code

30006

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Ambient temperature low
Service engineer level	The detected ambient temperature is below 2°C

Involved FRU

Sensor Temperature 5Kohm B3470K Threaded installation

Error Mechanism

The detected ambient temperature is below 2°C

Possible Causes

1. Ambient temperature does not meet the requirements
2. Heat source radiation, sunlight and poor ventilation
3. Temperature sensor error

Resolve Path

1. Confirm whether the working environment and temperature of the analyzer meet the requirements.
2. Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **Ambient Temperature** to confirm the **temperature inside the analyzer**.

Solution

1. Ensure that the working environment of the analyzer meets the requirements (including temperature, temperature rise, direct sunlight, etc.).
2. Unplug and replug the connector of ambient temperature sensor or temperature sensor.

9.1.11 30007: Detected Ambient Temperature Over 37°C

Error Code

30007

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Ambient temperature high
Service engineer level	The detected ambient temperature is over 37°C

Involved FRU

Sensor Temperature 5Kohm B3470K Threaded installation

Error Mechanism

The detected ambient temperature is over 37°C

Possible Causes

1. Ambient temperature does not meet the requirements
2. Heat source radiation, sunlight and poor ventilation
3. Temperature sensor error

Resolve Path

1. Confirm whether the working environment and temperature of the analyzer meet the requirements.
2. Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **Ambient Temperature** to confirm the **temperature inside the analyzer**.

Solution

1. Ensure that the working environment of the analyzer meets the requirements (including temperature, temperature rise, direct sunlight, etc.).
2. Unplug and replug the connector of ambient temperature sensor or temperature sensor.

9.1.12 30008: Detected Ambient Temperature Below 13°C

Error Code

30008

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Ambient temperature low
Service engineer level	The detected ambient temperature is below 13°C

Involved FRU

Sensor Temperature 5Kohm B3470K Threaded installation

Error Mechanism

The detected ambient temperature is below 13°C

Possible Causes

1. Ambient temperature does not meet the requirements
2. Heat source radiation, sunlight and poor ventilation
3. Temperature sensor error

Resolve Path

1. Confirm whether the working environment and temperature of the analyzer meet the requirements.
2. Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **Ambient Temperature** to confirm the **temperature inside the analyzer**.

Solution

1. Ensure that the working environment of the analyzer meets the requirements (including temperature, temperature rise, direct sunlight, etc.).
2. Unplug and replug the connector of ambient temperature sensor or temperature sensor.

9.1.13 30009: Reaction Bath Temperature Control Damaged

Error Code

30009

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Reaction bath temperature control assembly is damaged
Service engineer level	Reaction bath temperature control assembly is damaged

Involved FRU

Reaction bath assembly

Error Mechanism

Slow heating during startup

Possible Causes

1. Temperature sensor error
2. Heating rod error

Resolve Path

Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **DIFF/RET Reaction Bath** to confirm the temperature.

Solution

1. Unplug and replug the temperature sensor connector.
2. Unplug and replug the heating rod connector.
3. To replace the reaction bath assembly, see **8.36.2** Disassembly and Assembly.

(Neither heating rod nor temperature sensor can be replaced separately at the client)

9.1.14 30010: Preheating Bath Temperature Control Abnormal

Error Code

30010

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Preheating bath temperature control abnormal
Service engineer level	Preheating bath temperature control abnormal

Involved FRU

Diluent heating assembly

Error Mechanism

Slow heating during startup

Possible Causes

1. Diluent temperature extremely low
2. Temperature sensor error
3. Heating rod error

Resolve Path

1. Check whether the diluent temperature is room temperature.
2. Click **Menu>>Status>>Temperature & Pressure**to access the temperature and pressure display screen, and then view **Preheating Bath Temperature** to confirm the **temperature inside the analyzer**.
3. If the diluent temperature can gradually approach the temperature inside the analyzer, it may be caused by the diluent temperature-raising but not the error of the analyzer. If the diluent temperature is low and does not rise, the function of the heating rod may be damaged.

Solution

1. The diluent shall be used after being placed at room temperature for 1 ~ 2 days. The diluent in cold storage cannot be used for the analyzer. The diluent can be used only when its temperature is close to room temperature.
2. Unplug and replug the temperature sensor connector.
3. Unplug and replug the heating rod connector.
4. To replace the preheating bath assembly, see **8.44.2** Disassembly and Assembly.
5. Heating function is lost.

9.1.15 30011–30012: Specific Protein Assembly Temperature Abnormal

N/A

9.1.16 30400: ESR Assembly Communication Abnormal

Error Code

30400

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR assembly communication error
Service engineer level	ESR assembly communication error

Involved FRU

1. ESR Measurement Assembly
2. Paired wire, sensor vertical motor of sampling assembly and ESR
3. Wire, ESR assembly
4. Motherboard

Error Mechanism

1. ESR assembly control serial port communication failed.
2. ESR assembly data serial port has no valid data.

Possible Causes

1. ESR measurement assembly error
2. Poor connection between the motherboard and the ESR measurement assembly
3. Motherboard error

Resolve Path

1. Replace the ESR measurement assembly and confirm whether the ESR measurement assembly is faulty.
2. Check whether the "009-012555-00, paired wire, sampling assembly sensor vertical motor and ESR" is broken, damaged, loose, poor contact, etc., and whether there is splashing, corrosion, crystallization, etc.
3. Check whether the "009-011170-00, connection wire, ESR assembly" is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
4. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, replace the motherboard, and confirm whether the motherboard is faulty.

Solution

1. To replace ESR measurement assembly, see **8.12.2** Disassembly and Assembly, and **8.12.3** Commissioning and Verification of the ESR measurement assembly.
2. Replace the connection wire between the motherboard and the ESR measurement assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.17 30401: ESR baseline abnormal

Error Code

30401

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR baseline abnormal
Service engineer level	ESR baseline abnormal

Involved FRU

1. ESR Measurement Assembly
2. PCBA, Motherboard

Error Mechanism

During the boot process, the measured ESR baseline value exceeds the reference range

Possible Causes

1. ESR measurement assembly error
2. Motherboard error

Resolve Path

1. Replace the ESR measurement assembly and confirm whether the ESR measurement assembly is faulty.
2. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, replace the motherboard, and confirm whether the motherboard is faulty.

Solution

1. To replace ESR measurement assembly, see **8.12.2** Disassembly and Assembly, and **8.12.3** Commissioning and Verification of the ESR measurement assembly.
2. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.18 30402: ESR background intensity abnormal

Error Code

30402

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR background intensity abnormal
Service engineer level	ESR background intensity abnormal

Involved FRU

1. ESR Measurement Assembly
2. T33

Error Mechanism

ESR background intensity is out of range

Possible Causes

1. Light intensity attenuation of LED of ESR measurement assembly
2. Dirty sampling tube/air bubbles
3. ESR measurement assembly error

Resolve Path

1. Select **Menu** >**>Service**>**>Commissioning & Self-Test** >**>Sensor Commissioning** to enter the **ESR LED Current** screen. Adjust the LED current setting value and adjust the ESR background intensity to 50000–60000.
2. Perform probe cleaning or probe cleanser maintenance.
3. Replace the ESR measurement assembly and confirm whether the ESR measurement assembly is faulty.

Solution

1. Carry out the maintenance according to troubleshooting.
2. To replace ESR measurement assembly, see **8.12.2** Disassembly and Assembly, and **8.12.3** Commissioning and Verification of the ESR measurement assembly.

9.1.19 30403: ESR Assembly Temperature Low

Error Code

30403

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR assembly temperature low
Service engineer level	ESR assembly temperature low

Involved FRU

ESR Measurement Assembly

Error Mechanism

The ESR assembly temperature is lower than the lower limit

Possible Causes

1. The temperature sensor connector is loose
2. ESR measurement assembly error

Resolve Path

1. Make sure that the temperature sensor connector J4 of the ESR measurement assembly is inserted in place.
2. Replace the ESR measurement assembly and confirm whether the ESR measurement assembly is faulty.

Solution

1. Unplug and replug the temperature sensor.
2. To replace ESR measurement assembly, see **8.12.2** Disassembly and Assembly, and **8.12.3** Commissioning and Verification of the ESR measurement assembly.

9.1.20 30404: ESR Assembly Temperature High

Error Code

30404

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR assembly temperature high
Service engineer level	ESR assembly temperature high

Involved FRU

ESR Measurement Assembly

Error Mechanism

The ESR assembly temperature is higher than the upper limit

Possible Causes

1. The temperature sensor connector is loose
2. ESR measurement assembly error

Resolve Path

1. Make sure that the temperature sensor connector J4 of the ESR measurement assembly is inserted in place.
2. Replace the ESR measurement assembly and confirm whether the ESR measurement assembly is faulty.

Solution

1. Unplug and replug the temperature sensor.
2. To replace ESR measurement assembly, see **8.12.2** Disassembly and Assembly, and **8.12.3** Commissioning and Verification of the ESR measurement assembly.

9.1.21 30405: ESR Assembly Version Mismatching

Error Code

30405

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR assembly version mismatching
Service engineer level	ESR assembly version mismatching

Involved FRU

ESR Measurement Assembly

Error Mechanism

The hardware version number of ESR assembly read during startup is higher than the compatible software version number

Possible Causes

The new ESR version incompatible with the host software is replaced

Resolve Path

1. Confirm ESR assembly is replaced.
2. The error is eliminated after upgrading to the latest version of host software.

Solution

Upgrade to the latest version of host software.

9.1.22 10000: No DS Diluent

Error Code

10000

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No DS diluent
Service engineer level	No DS diluent

Involved FRU

1. DS diluent container cap assembly
2. Lyse container cap assembly (red, open mold connector, 0.85m)
3. Lyse container cap assembly (black, open mold connector, 0.85m)
4. Lyse container cap assembly (green, open mold connector, 0.85m)
5. Lyse container cap assembly (orange, open mold connector, 0.85m)
6. Mindray two-way valve (PEIZH)
7. Mindray three-way valve (PEIZH)
8. 0.5mL diaphragm pump (same direction as EPDM connector)
9. Three-way LVM valve assembly 105R
10. Liquid Detection Board PCBA

Error Mechanism

The level detection board detects the presence or absence of the reagent

Possible Causes

1. No reagent
2. Poor connection of container cap assembly and its tube
3. Poor connection of valve and its tube
4. Poor connection of level detection board

Resolve Path

1. Check whether the reagent bottle still has reagent (excluding the bottom residue that cannot be aspirated).
2.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the DS reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the DS diluent container cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T1/T5/T15/T16/T17/T18/T19/T20/T21/T22/T23).
3.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the LD reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the LD lyse cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T3/T7/T10/T13/P3/P4).
4.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the DR reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the DR diluent container cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T4/T8/T11/T14/P5/P6).
5.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the LH reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the LH lyse cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T2/T6/T9/T12/T238/P1/P2).
6.

1) Check that the remaining volume in the reagent bottle is sufficient, and the ESR cleanser is replaced normally, but the error cannot be eliminated.
2) Remove the ESR cleanser bag cap assembly and check for broken connector.
3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube:T300/T301/T302/T303/P32/P33).

7.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace DS Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve (DS: SV13) is connected incorrectly, and whether the tubes T22, T23, and T205 are abnormal.
3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (DS: SV13), and check whether the valve is damaged, etc.

8.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (LD: SV02/04), check whether the valve is damaged, etc.
4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T103, T104, T13, and SV02 branch/T173, T176 and SV04 branch).

9.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (DR: SV01/03), check whether the valve is damaged, etc.
4) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T95, T96, T14, and SV01 branch/T170, T172 and SV03 branch).

10.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>>**Service**>>**Maintenance**>>**Replace LH Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (LH: SV05/10), check whether the valve is damaged, etc.

4) Check whether the diaphragm pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T85, T86, T94, T12 and SV10 branch/T177, T178 and SV05 branch).

11.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>>**Service**>>**Maintenance**>>**Replace ESR Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (ESR cleanser: SV04/40), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T303, T304, and SV40 branch/T176, T233, T173, T306 and SV04 branch).

12. Check whether the light is not off when the detection tube is full of liquid, if so, it is the level detection board.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace DS diluent container cap assembly, see **8.69.2** Disassembly and Assembly.
3. To replace lyse container cap assembly (red, open mold connector, 0.85m), see **8.70.2** Disassembly and Assembly.
4. To replace lyse container cap assembly (black, open mold connector, 0.85m), see **8.72.2** Disassembly and Assembly.
5. To replace lyse container cap assembly (green, open mold connector, 0.85m), see **8.71.2** Disassembly and Assembly.
6. To replace lyse container cap assembly (orange, open mold connector, 0.85m), see **8.74.2** Disassembly and Assembly.
7. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.

8. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
9. To replace 0.5 mL dosing pump (same direction as EPDM connector), see **8.52.2** Disassembly and Assembly.
10. To replace three-way LVM valve assembly 105R, see .
11. To replace liquid detection board PCBA, see **8.81.2** Disassembly and Assembly.
12. Replace the tube.

9.1.23 10001: No LD lyse

Error Code

10001

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No LD lyse
Service engineer level	No LD lyse

Involved FRU

1. DS diluent container cap assembly
2. Lyse container cap assembly (red, open mold connector, 0.85m)
3. Lyse container cap assembly (black, open mold connector, 0.85m)
4. Lyse container cap assembly (green, open mold connector, 0.85m)
5. Lyse container cap assembly (orange, open mold connector, 0.85m)
6. Mindray two-way valve (PEIZH)
7. Mindray three-way valve (PEIZH)
8. 0.5mL diaphragm pump (same direction as EPDM connector)
9. Three-way LVM valve assembly 105R
10. Liquid Detection Board PCBA

Error Mechanism

The level detection board detects the presence or absence of the reagent

Possible Causes

1. No reagent
2. Poor connection of container cap assembly and its tube
3. Poor connection of valve and its tube
4. Poor connection of level detection board

Resolve Path

1. Check whether the reagent bottle still has reagent (excluding the bottom residue that cannot be aspirated).
2.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the DS reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the DS diluent container cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T1/T5/T15/T16/T17/T18/T19/T20/T21/T22/T23).
3.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the LD reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the LD lyse cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T3/T7/T10/T13/P3/P4).
4.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the DR reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the DR diluent container cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T4/T8/T11/T14/P5/P6).
5.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the LH reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the LH lyse cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T2/T6/T9/T12/T238/P1/P2).
6.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the ESR cleanser is replaced normally, but the error cannot be eliminated.
   2) Remove the ESR cleanser bag cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube:T300/T301/T302/T303/P32/P33).
7.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace DS Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve (DS: SV13) is connected incorrectly, and whether the tubes T22, T23, and T205 are abnormal.
3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (DS: SV13), and check whether the valve is damaged, etc.

8.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (LD: SV02/04), check whether the valve is damaged, etc.
4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T103, T104, T13, and SV02 branch/T173, T176 and SV04 branch).

9.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (DR: SV01/03), check whether the valve is damaged, etc.
4) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T95, T96, T14, and SV01 branch/T170, T172 and SV03 branch).

10.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace LH Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.

3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (LH: SV05/10), check whether the valve is damaged, etc.

4) Check whether the diaphragm pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T85, T86, T94, T12 and SV10 branch/T177, T178 and SV05 branch).

11.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace ESR Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (ESR cleanser: SV04/40), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T303, T304, and SV40 branch/T176, T233, T173, T306 and SV04 branch).

12. Check whether the light is not off when the detection tube is full of liquid, if so, it is the level detection board.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace DS diluent container cap assembly, see **8.69.2** Disassembly and Assembly.
3. To replace lyse container cap assembly (red, open mold connector, 0.85m), see **8.70.2** Disassembly and Assembly.
4. To replace lyse container cap assembly (black, open mold connector, 0.85m), see **8.72.2** Disassembly and Assembly.
5. To replace lyse container cap assembly (green, open mold connector, 0.85m), see **8.71.2** Disassembly and Assembly.
6. To replace lyse container cap assembly (orange, open mold connector, 0.85m), see **8.74.2** Disassembly and Assembly.
7. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
8. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
9. To replace 0.5 mL dosing pump (same direction as EPDM connector), see **8.52.2** Disassembly and Assembly.
10. To replace three-way LVM valve assembly 105R, see .
11. To replace liquid detection board PCBA, see **8.81.2** Disassembly and Assembly.
12. Replace the tube.

9.1.24 10002: No DR Diluent

Error Code

10002

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No DR diluent
Service engineer level	No DR diluent

Involved FRU

1. DS diluent container cap assembly
2. Lyse container cap assembly (red, open mold connector, 0.85m)
3. Lyse container cap assembly (black, open mold connector, 0.85m)
4. Lyse container cap assembly (green, open mold connector, 0.85m)
5. Lyse container cap assembly (orange, open mold connector, 0.85m)
6. Mindray two-way valve (PEIZH)
7. Mindray three-way valve (PEIZH)
8. 0.5mL diaphragm pump (same direction as EPDM connector)
9. Three-way LVM valve assembly 105R
10. Liquid Detection Board PCBA

Error Mechanism

The level detection board detects the presence or absence of the reagent

Possible Causes

1. No reagent
2. Poor connection of container cap assembly and its tube
3. Poor connection of valve and its tube
4. Poor connection of level detection board

Resolve Path

1. Check whether the reagent bottle still has reagent (excluding the bottom residue that cannot be aspirated).
2.
    1) Check that the remaining volume in the reagent bottle is sufficient, and the DS reagent is replaced normally, but the error cannot be eliminated.

2) Remove the DS diluent container cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T1/T5/T15/T16/T17/T18/T19/T20/T21/T22/T23).

3.

1) Check that the remaining volume in the reagent bottle is sufficient, and the LD reagent is replaced normally, but the error cannot be eliminated.

2) Remove the LD lyse cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T3/T7/T10/T13/P3/P4).

4.

1) Check that the remaining volume in the reagent bottle is sufficient, and the DR reagent is replaced normally, but the error cannot be eliminated.

2) Remove the DR diluent container cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T4/T8/T11/T14/P5/P6).

5.

1) Check that the remaining volume in the reagent bottle is sufficient, and the LH reagent is replaced normally, but the error cannot be eliminated.

2) Remove the LH lyse cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T2/T6/T9/T12/T238/P1/P2).

6.

1) Check that the remaining volume in the reagent bottle is sufficient, and the ESR cleanser is replaced normally, but the error cannot be eliminated.

2) Remove the ESR cleanser bag cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube:T300/T301/T302/T303/P32/P33).

7.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace DS Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve (DS: SV13) is connected incorrectly, and whether the tubes T22, T23, and T205 are abnormal.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (DS: SV13), and check whether the valve is damaged, etc.

8.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>>**Service**>>**Maintenance**>>**Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (LD: SV02/04), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T103, T104, T13, and SV02 branch/T173, T176 and SV04 branch).

9.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>>**Service**>>**Maintenance**>>**Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (DR: SV01/03), check whether the valve is damaged, etc.

4) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T95, T96, T14, and SV01 branch/T170, T172 and SV03 branch).

10.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>>**Service**>>**Maintenance**>>**Replace LH Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (LH: SV05/10), check whether the valve is damaged, etc.

4) Check whether the diaphragm pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T85, T86, T94, T12 and SV10 branch/T177, T178 and SV05 branch).

11.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace ESR Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (ESR cleanser: SV04/40), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T303, T304, and SV40 branch/T176, T233, T173, T306 and SV04 branch).

12. Check whether the light is not off when the detection tube is full of liquid, if so, it is the level detection board.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace DS diluent container cap assembly, see **8.69.2** Disassembly and Assembly.
3. To replace lyse container cap assembly (red, open mold connector, 0.85m), see **8.70.2** Disassembly and Assembly.
4. To replace lyse container cap assembly (black, open mold connector, 0.85m), see **8.72.2** Disassembly and Assembly.
5. To replace lyse container cap assembly (green, open mold connector, 0.85m), see **8.71.2** Disassembly and Assembly.
6. To replace lyse container cap assembly (orange, open mold connector, 0.85m), see **8.74.2** Disassembly and Assembly.
7. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
8. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
9. To replace 0.5 mL dosing pump (same direction as EPDM connector), see **8.52.2** Disassembly and Assembly.
10. To replace three-way LVM valve assembly 105R, see .
11. To replace liquid detection board PCBA, see **8.81.2** Disassembly and Assembly.
12. Replace the tube.

9.1.25 10003: No LH lyse

Error Code

10003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No LH lyse
Service engineer level	No LH lyse

Involved FRU

1. DS diluent container cap assembly
2. Lyse container cap assembly (red, open mold connector, 0.85m)
3. Lyse container cap assembly (black, open mold connector, 0.85m)
4. Lyse container cap assembly (green, open mold connector, 0.85m)
5. Lyse container cap assembly (orange, open mold connector, 0.85m)
6. Mindray two-way valve (PEIZH)
7. Mindray three-way valve (PEIZH)
8. 0.5mL diaphragm pump (same direction as EPDM connector)
9. Three-way LVM valve assembly 105R
10. Liquid Detection Board PCBA

Error Mechanism

The level detection board detects the presence or absence of the reagent

Possible Causes

1. No reagent
2. Poor connection of container cap assembly and its tube
3. Poor connection of valve and its tube
4. Poor connection of level detection board

Resolve Path

1. Check whether the reagent bottle still has reagent (excluding the bottom residue that cannot be aspirated).
2.
    1) Check that the remaining volume in the reagent bottle is sufficient, and the DS reagent is replaced normally, but the error cannot be eliminated.
    2) Remove the DS diluent container cap assembly and check for broken connector.
    3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T1/T5/T15/T16/T17/T18/T19/T20/T21/T22/T23).
3.

1) Check that the remaining volume in the reagent bottle is sufficient, and the LD reagent is replaced normally, but the error cannot be eliminated.

2) Remove the LD lyse cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T3/T7/T10/T13/P3/P4).

4.

1) Check that the remaining volume in the reagent bottle is sufficient, and the DR reagent is replaced normally, but the error cannot be eliminated.

2) Remove the DR diluent container cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T4/T8/T11/T14/P5/P6).

5.

1) Check that the remaining volume in the reagent bottle is sufficient, and the LH reagent is replaced normally, but the error cannot be eliminated.

2) Remove the LH lyse cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T2/T6/T9/T12/T238/P1/P2).

6.

1) Check that the remaining volume in the reagent bottle is sufficient, and the ESR cleanser is replaced normally, but the error cannot be eliminated.

2) Remove the ESR cleanser bag cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube:T300/T301/T302/T303/P32/P33).

7.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace DS Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve (DS: SV13) is connected incorrectly, and whether the tubes T22, T23, and T205 are abnormal.

3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (DS: SV13), and check whether the valve is damaged, etc.

8.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (LD: SV02/04), check whether the valve is damaged, etc.
4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T103, T104, T13, and SV02 branch/T173, T176 and SV04 branch).

9.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (DR: SV01/03), check whether the valve is damaged, etc.
4) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T95, T96, T14, and SV01 branch/T170, T172 and SV03 branch).

10.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace LH Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.
2) Check whether the reagent valve is misconnected.
3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (LH: SV05/10), check whether the valve is damaged, etc.
4) Check whether the diaphragm pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T85, T86, T94, T12 and SV10 branch/T177, T178 and SV05 branch).

11.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace ESR Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (ESR cleanser: SV04/40), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T303, T304, and SV40 branch/T176, T233, T173, T306 and SV04 branch).

12. Check whether the light is not off when the detection tube is full of liquid, if so, it is the level detection board.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace DS diluent container cap assembly, see **8.69.2** Disassembly and Assembly.
3. To replace lyse container cap assembly (red, open mold connector, 0.85m), see **8.70.2** Disassembly and Assembly.
4. To replace lyse container cap assembly (black, open mold connector, 0.85m), see **8.72.2** Disassembly and Assembly.
5. To replace lyse container cap assembly (green, open mold connector, 0.85m), see **8.71.2** Disassembly and Assembly.
6. To replace lyse container cap assembly (orange, open mold connector, 0.85m), see **8.74.2** Disassembly and Assembly.
7. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
8. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
9. To replace 0.5 mL dosing pump (same direction as EPDM connector), see **8.52.2** Disassembly and Assembly.
10. To replace three-way LVM valve assembly 105R, see .
11. To replace liquid detection board PCBA, see **8.81.2** Disassembly and Assembly.
12. Replace the tube.

9.1.26 10004: No LS lyse

N/A

9.1.27 10005: No FD dye

Error Code

10005

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No FD dye
Service engineer level	No FD dye

Involved FRU

1. Fluorescent Reagent Container Cap Assembly (FD)
2. Fluorescent Reagent Container Cap Assembly (FR)
3. Three-way LVM body without valve nozzle (V026)
4. Three-way LVM body without valve nozzle (V027)
5. Mindray three-way valve (PEIZH)
6. 20 μL Dosing Pump Assembly
7. Optical switch COJ-MY03-04
8. Wire, dye detecting sensor, ext.

Error Mechanism

The sensor of the corresponding channel cannot detect dye liquor in the pipe or detects that bubbles flow through the pipe.

Possible Causes

1. No reagent
2. Poor fluorescent reagent bag cap assembly
3. Poor connection of valve and its tube
4. Poor dye detection sensor
5. Poor connecting wire of dye sensor
6. Conventional drive board error

Resolve Path

1. Remove the fluorescent dye bag and determine whether there is any reagent in the bag (excluding the bottom residue that cannot be aspirated).
2.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the FD reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the FD dye bag cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: P16/J20/J21/T113).
3.

1) Check that the remaining volume in the reagent bottle is sufficient, and the FR reagent is replaced normally, but the error cannot be eliminated.
2) Remove the FR dye bag cap assembly and check for broken connector.
3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: P12/J17/J18/T111).

4.
1) Make sure that there is reagent in the fluorescent dye bag and the cap assembly is normal, but the error cannot be eliminated.
2) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (FD: SV026/SV04), check whether the valve is damaged, etc.
3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T113, J21 and SV26 branch/T175, T176, T233 and SV04 branch).

5.
1) Make sure that there is reagent in the fluorescent dye bag and the cap assembly is normal, but the error cannot be eliminated.
2) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (FR: SV027/SV03), check whether the valve is damaged, etc.
3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T111, J18 and SV27 branch abnormal/T171, T172, T232 and SV03 branch abnormal).

6.
1) Make sure that there is reagent in the fluorescent dye bag and the cap assembly is normal, but the error cannot be eliminated.
2) Check whether the optical switch COJ-MY03-04 and its connecting wire are broken, damaged, loose, poor contact, etc., and whether there is splashing, corrosion, crystallization, etc.
3) Select **Menu**>**>Status**>**>Sensor Status**>**>Fluorescent Reagent Sensor**, and check whether **Fluorescent Reagent Sensor** is in the **No Reagent** status.

7. Check whether the connecting wire of dye sensor is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
8. Confirm that the wiring is normal. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the conventional driver board is faulty.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace fluorescent reagent container cap assembly (FD), see **8.75.2** Disassembly and Assembly.
3. To replace fluorescent reagent container cap assembly (FR), see **8.76.2** Disassembly and Assembly.
4. To replace three-way LVM valve body, see .
5. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
6. To replace 20ul dosing pump assembly, see **8.66.2** Disassembly and Assembly.
7. To replace optical switch COJ-MY03-04, see **8.65.2** Disassembly and Assembly.
8. Replace the tube.
9. Replace the connection wire of dye sensor.
10. Replace the common driver board.

9.1.28 10006: No FR dye

Error Code

10006

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No FR dye
Service engineer level	No FR dye

Involved FRU

1. Fluorescent Reagent Container Cap Assembly (FD)
2. Fluorescent Reagent Container Cap Assembly (FR)
3. Three-way LVM body without valve nozzle (V026)
4. Three-way LVM body without valve nozzle (V027)
5. Mindray three-way valve (PEIZH)
6. 20 μL Dosing Pump Assembly
7. Optical switch COJ-MY03-04
8. Wire, dye detecting sensor, ext.

Error Mechanism

The sensor of the corresponding channel cannot detect dye liquor in the pipe or detects that bubbles flow through the pipe.

Possible Causes

1. No reagent
2. Poor fluorescent reagent bag cap assembly
3. Poor connection of valve and its tube
4. Poor dye detection sensor
5. Poor connecting wire of dye sensor
6. Conventional drive board error

Resolve Path

1. Remove the fluorescent dye bag and determine whether there is any reagent in the bag (excluding the bottom residue that cannot be aspirated).
2.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the FD reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the FD dye bag cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: P16/J20/J21/T113).
3.
   1) Check that the remaining volume in the reagent bottle is sufficient, and the FR reagent is replaced normally, but the error cannot be eliminated.
   2) Remove the FR dye bag cap assembly and check for broken connector.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: P12/J17/J18/T111).
4.
   1) Make sure that there is reagent in the fluorescent dye bag and the cap assembly is normal, but the error cannot be eliminated.
   2) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (FD: SV026/SV04), check whether the valve is damaged, etc.
   3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T113, J21 and SV26 branch/T175, T176, T233 and SV04 branch).
5.
   1) Make sure that there is reagent in the fluorescent dye bag and the cap assembly is normal, but the error cannot be eliminated.

2) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (FR: SV027/SV03), check whether the valve is damaged, etc.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T111, J18 and SV27 branch abnormal/T171, T172, T232 and SV03 branch abnormal).

6.

1) Make sure that there is reagent in the fluorescent dye bag and the cap assembly is normal, but the error cannot be eliminated.

2) Check whether the optical switch COJ-MY03-04 and its connecting wire are broken, damaged, loose, poor contact, etc., and whether there is splashing, corrosion, crystallization, etc.

3) Select **Menu>>Status>>Sensor Status>>Fluorescent Reagent Sensor**, and check whether **Fluorescent Reagent Sensor** is in the **No Reagent** status.

7. Check whether the connecting wire of dye sensor is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
8. Confirm that the wiring is normal. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the conventional driver board is faulty.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace fluorescent reagent container cap assembly (FD), see **8.75.2** Disassembly and Assembly.
3. To replace fluorescent reagent container cap assembly (FR), see **8.76.2** Disassembly and Assembly.
4. To replace three-way LVM valve body, see .
5. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
6. To replace 20ul dosing pump assembly, see **8.66.2** Disassembly and Assembly.
7. To replace optical switch COJ-MY03-04, see **8.65.2** Disassembly and Assembly.
8. Replace the tube.
9. Replace the connection wire of dye sensor.
10. Replace the common driver board.

9.1.29 10007: No CRP Latex Reagent

N/A

9.1.30 10008: No SAA Latex Reagent

N/A

9.1.31 10009: No ESRCLN Diluent

Error Code

10009

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	No ESRCLN cleanser
Service engineer level	No ESRCLN cleanser

Involved FRU

1. DS diluent container cap assembly
2. Lyse container cap assembly (red, open mold connector, 0.85m)
3. Lyse container cap assembly (black, open mold connector, 0.85m)
4. Lyse container cap assembly (green, open mold connector, 0.85m)
5. Lyse container cap assembly (orange, open mold connector, 0.85m)
6. Mindray two-way valve (PEIZH)
7. Mindray three-way valve (PEIZH)
8. 0.5mL diaphragm pump (same direction as EPDM connector)
9. Three-way LVM valve assembly 105R
10. Liquid Detection Board PCBA

Error Mechanism

The level detection board detects the presence or absence of the reagent

Possible Causes

1. No reagent
2. Poor connection of container cap assembly and its tube
3. Poor connection of valve and its tube
4. Poor connection of level detection board

Resolve Path

1. Check whether the reagent bottle still has reagent (excluding the bottom residue that cannot be aspirated).

2.

1) Check that the remaining volume in the reagent bottle is sufficient, and the DS reagent is replaced normally, but the error cannot be eliminated.

2) Remove the DS diluent container cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T1/T5/T15/T16/T17/T18/T19/T20/T21/T22/T23).

3.

1) Check that the remaining volume in the reagent bottle is sufficient, and the LD reagent is replaced normally, but the error cannot be eliminated.

2) Remove the LD lyse cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T3/T7/T10/T13/P3/P4).

4.

1) Check that the remaining volume in the reagent bottle is sufficient, and the DR reagent is replaced normally, but the error cannot be eliminated.

2) Remove the DR diluent container cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T4/T8/T11/T14/P5/P6).

5.

1) Check that the remaining volume in the reagent bottle is sufficient, and the LH reagent is replaced normally, but the error cannot be eliminated.

2) Remove the LH lyse cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T2/T6/T9/T12/T238/P1/P2).

6.

1) Check that the remaining volume in the reagent bottle is sufficient, and the ESR cleanser is replaced normally, but the error cannot be eliminated.

2) Remove the ESR cleanser bag cap assembly and check for broken connector.

3) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube:T300/T301/T302/T303/P32/P33).

7.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace DS Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve (DS: SV13) is connected incorrectly, and whether the tubes T22, T23, and T205 are abnormal.

3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (DS: SV13), and check whether the valve is damaged, etc.

8.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (LD: SV02/04), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T103, T104, T13, and SV02 branch/T173, T176 and SV04 branch).

9.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace LD Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (DR: SV01/03), check whether the valve is damaged, etc.

4) Check whether the reagent tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T95, T96, T14, and SV01 branch/T170, T172 and SV03 branch).

10.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu**>**>Service**>**>Maintenance**>**>Replace LH Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (LH: SV05/10), check whether the valve is damaged, etc.

4) Check whether the diaphragm pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T85, T86, T94, T12 and SV10 branch/T177, T178 and SV05 branch).

11.

1) Check that the remaining volume in the reagent bottle is sufficient. Select **Menu>>Service>>Maintenance>>Replace ESR Reagent**. Try to replace the reagent, but no reagent is seen transported from the reagent container to the inside of the analyzer.

2) Check whether the reagent valve is misconnected.

3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (ESR cleanser: SV04/40), check whether the valve is damaged, etc.

4) Check whether the dosing pump tube is blocked, bent, leaking, etc.; whether the tube connectors are loose, damaged, etc.; (Tube: T303, T304, and SV40 branch/T176, T233, T173, T306 and SV04 branch).

12. Check whether the light is not off when the detection tube is full of liquid, if so, it is the level detection board.

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. To replace DS diluent container cap assembly, see **8.69.2** Disassembly and Assembly.
3. To replace lyse container cap assembly (red, open mold connector, 0.85m), see **8.70.2** Disassembly and Assembly.
4. To replace lyse container cap assembly (black, open mold connector, 0.85m), see **8.72.2** Disassembly and Assembly.
5. To replace lyse container cap assembly (green, open mold connector, 0.85m), see **8.71.2** Disassembly and Assembly.
6. To replace lyse container cap assembly (orange, open mold connector, 0.85m), see **8.74.2** Disassembly and Assembly.
7. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
8. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
9. To replace 0.5 mL dosing pump (same direction as EPDM connector), see **8.52.2** Disassembly and Assembly.
10. To replace three-way LVM valve assembly 105R, see .
11. To replace liquid detection board PCBA, see **8.81.2** Disassembly and Assembly.
12. Replace the tube.

9.1.32 10200: DS diluent expired

Error Code

10200

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DS diluent expired
Service engineer level	DS diluent expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.33 10201: LD Lyse Expired

Error Code

10201

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	LD Dye expired
Service engineer level	LD Dye expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.34 10202: DR diluent expired

Error Code

10202

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DR Dye expired
Service engineer level	DR Dye expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.35 10203: LH Lyse Expired

Error Code

10203

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	LH Dye expired
Service engineer level	LH Dye expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.36 10204: LS Lyse Expired

N/A

9.1.37 10205: FD Dye Expired

Error Code

10205

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	FD Dye expired
Service engineer level	FD Dye expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.38 10206: FR Dye Expired

Error Code

10206

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	FR Dye expired
Service engineer level	FR Dye expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.39 10207: CRP Lax Reagent Expired

N/A

9.1.40 10208: SAA Lax Reagent Expired

N/A

9.1.41 10209: ESRCLN diluent expired

Error Code

10209

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	ESR cleanser expired
Service engineer level	ESR cleanser expired

Involved FRU

N/A

Error Mechanism

The reagent validity countdown starts at user registering the reagent information to the analyzer by scanning the reagent barcode. The analyzer reports error when a reagent passes its open-vial validity period or the expiry date.

Possible Causes

1. Reagent expired
2. Software failure

Resolve Path

1. Check whether the service time of the reagent is beyond the effective date after opening the bottle or the effective date of package.
2. Check that the bottle opening time and use time are within the effective date, but the error alarm cannot be eliminated. You can check and update the latest version of the software system.

Solution

1. Tap the **Reagent** button on the maintenance screen, to load the information of an effective reagent.
2. Update the latest system version.

9.1.42 10400: Insufficient DS diluent

Error Code

10400

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Insufficient DS diluent
Service engineer level	Insufficient DS diluent

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag
2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.43 10401: LD reagent low volume

Error Code

10401

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	LD Dye low volume
Service engineer level	LD Dye low volume

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag
2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.44 10402: LH reagent low volume

Error Code

10402

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	LH Dye low volume
Service engineer level	LH Dye low volume

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag
2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.45 10403: DR reagent low volume

Error Code

10403

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DR Dye low volume
Service engineer level	DR Dye low volume

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag
2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.46 10404: FD reagent low volume

Error Code

10404

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	FD Dye low volume
Service engineer level	FD Dye low volume

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag
2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.47 10405: FR reagent low volume

Error Code

10405

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	FR Dye low volume
Service engineer level	FR Dye low volume

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag

2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.48 10406: LS reagent low volume

N/A

9.1.49 10407: Insufficient CRP Latex Reagent

N/A

9.1.50 10408: Insufficient SAA Latex Reagent

N/A

9.1.51 10409: Insufficient ESRCLN diluent

Error Code

10409

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Insufficient ESR cleanser
Service engineer level	Insufficient ESR cleanser

Involved FRU

N/A

Error Mechanism

When a reagent is newly registered, the software refreshes the reagent initial volume, and each time a sequence is run, the software reduces the corresponding consumption. When the total consumption reaches the threshold, the analyzer reports error.

Possible Causes

1. Reagent low volume
2. Software failure

Resolve Path

1. Check for insufficient reagent residue in the reagent bottle or reagent bag
2. Reagent residue is sufficient but the alarm cannot be eliminated

Solution

1. Replace with a new reagent, and tap "Remove Error". Enter the new and effective reagent barcode information in the popped up message box, and replace the reagent.
2. Update the latest system version.

9.1.52 10100: DIL Bath Not Fulfilled

Error Code

10100

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Preheating bath priming abnormal
Service engineer level	DIL cistern preheating bath priming abnormal

Involved FRU

1. WC2 Cistern
2. Mindray two-way valve (PEIZH)
3. Filter.Inline Filter43um 1/8"I.D.Tubing
4. Two-way LVM valve assembly (with valve board and connector) SV31
5. Liquid Detection Board PCBA

Error Mechanism

The level detection board at the outlet of DIL cistern detects the presence or absence of the reagent

Possible Causes

1. The waste pump (PUMP_WASTE) is abnormal
2. The waste cistern (WC2) leaks
3. The SV13 is abnormal or the filter (LF1) is clogged

4. SV31 abnormal or SV34 abnormal
5. The DIL cistern level sensor (DT5) is abnormal

Resolve Path

1.
   1) -40kpa pressure value exceeds the range and alarm is given.
   2) Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the pressure value of -40kpa channel is out of range. If the waste pump does not move, the air pump is highly likely damaged.
   3) Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether each pressure value has changed and stabilized within the normal range.
2. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
3.
   1) Check that the remaining volume of reagent in the reagent bottle is sufficient and that the container cap assembly is normal.
   2) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV13 valve, and check whether the valve is damaged or defective.
   3) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
4.
   1) Check that the remaining volume of reagent in the reagent bottle is sufficient and that the container cap assembly is normal.
   2) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV31/SV34 valve, and check whether the valve is damaged or defective.
   3) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
5. Check whether the light is not off when the detection tube is full of liquid, if so, it is the level detection board.

Solution

1. To replace WC2 cistern, see **8.43.2** Disassembly and Assembly.

2. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
3. To replace filter.Inline Filter43um 1/8"I.D.Tubing, see **8.61.2** Disassembly and Assembly.
4. To replace two-way LVM valve assembly (with valve board and connector) SV31, see .
5. To replace liquid detection board PCBA, see **8.81.2** Disassembly and Assembly.

9.1.53 10101: SCI Cistern Floater Status Abnormal

Error Code

10101

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Cistern floater abnormal
Service engineer level	SCI cistern floater status abnormal

Involved FRU

1. SCI bath assembly
2. Two-position ways LVM valve assembly (with valve board and connector)
3. Mindray two-way valve (PEIZH) SV34
4. WC2 cistern
5. Wire, motherboard to floater, ext.

Error Mechanism

At a specific node of the time sequence (a specific node refers to the node that the cistern has been emptied or filled in the time sequence), the SCI cistern floater is checked for empty or full, and if the floater status is not empty or full, an error will be reported

Possible Causes

1. The SCI cistern leaks
2. SV30 abnormal
3. SCI cistern (SCI)_Floater signal abnormal
4. SV31 abnormal or SV34 abnormal
5. WC2 cistern assembly abnormal
6. Floater connecting wire error
7. Conventional drive board error

Resolve Path

1. Select **Menu>>Status>>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
2.
   1) Check that the remaining volume of reagent in the reagent bottle is sufficient and that the container cap assembly is normal.
   2) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV030 (SCI) valve, and check whether the valve is damaged or defective.
   3) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
3. Check whether the floater connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
4.
   1) Check that the remaining volume of reagent in the reagent bottle is sufficient and that the container cap assembly is normal.
   2) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV31/SV34 valve, and check whether the valve is damaged or defective.
   3) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
5.
   1) Select **Menu>>Status>>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
   2) Check whether the floater connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
6. Check whether the floater connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
7. Confirm that the wiring is normal. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the conventional driver board is faulty.

Solution

1. To replace SCI bath assembly, see **8.46.2** Disassembly and Assembly.
2. To replace two-way LVM valve assembly (with valve board and connector), see .
3. To replace Mindray two-way valve (PEIZH) SV34, see **8.5.2** Disassembly and Assembly.
4. To replace WC2 cistern, see **8.43.2** Disassembly and Assembly.
5. To replace the connection wire between the motherboard and the extended floater, see **8.103.2** Disassembly and Assembly.

9.1.54 10102: WC2 Cistern Floater Status Abnormal

Error Code

10102

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste cistern floater abnormal
Service engineer level	WC2 cistern floater status abnormal

Involved FRU

1. SCI bath assembly
2. Two-position ways LVM valve assembly (with valve board and connector)
3. Mindray two-way valve (PEIZH) SV34
4. WC2 cistern
5. Wire, motherboard to floater, ext.

Error Mechanism

At a specific node of the time sequence (a specific node refers to the node that the waste cistern has been emptied or filled in the time sequence), check the WC2 floater status in the analyzer, and if it is not empty or full, an error will be reported

Possible Causes

1. The SCI cistern leaks
2. SV30 abnormal
3. SCI cistern (SCI)_Floater signal abnormal
4. SV31 abnormal or SV34 abnormal
5. WC2 cistern assembly abnormal
6. Floater connecting wire error
7. Conventional drive board error

Resolve Path

1. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
2.
    1) Check that the remaining volume of reagent in the reagent bottle is sufficient and that the container cap assembly is normal.
    2) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV030 (SCI) valve, and check whether the valve is damaged or defective.
    3) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
3. Check whether the floater connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
4.
    1) Check that the remaining volume of reagent in the reagent bottle is sufficient and that the container cap assembly is normal.
    2) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV31/SV34 valve, and check whether the valve is damaged or defective.
    3) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
5.
    1) Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
    2) Check whether the floater connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
6. Check whether the floater connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
7. Confirm that the wiring is normal. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the conventional driver board is faulty.

Solution

1. To replace SCI bath assembly, see **8.46.2** Disassembly and Assembly.
2. To replace two-way LVM valve assembly (with valve board and connector), see .
3. To replace Mindray two-way valve (PEIZH) SV34, see **8.5.2** Disassembly and Assembly.
4. To replace WC2 cistern, see **8.43.2** Disassembly and Assembly.
5. To replace the connection wire between the motherboard and the extended floater, see **8.103.2** Disassembly and Assembly.

9.1.55 10103: Waste Container Full

Error Code

10103

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste container full
Service engineer level	Waste container full

Involved FRU

Waste Cap Assembly (3120)

Error Mechanism

When the waste in the waste container has reached a certain level, the floater ball floats upward, and the sensor status changes and triggers error report

Possible Causes

1. Waste container collapsing
2. Poor contact of waste floater BNC connector
3. Floater connecting wire 2 error
4. Waste container cap assembly error
5. Conventional drive board error

Resolve Path

1. Visually check if the waste container collapses. When the container collapses, the analyzer will reports the error even when the waste is not half the container.
2. Check whether the BNC connector of the waste floater on the back of the analyzer is dropped, loose, damaged, etc.

3. Check whether the floater connecting wire 2. is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
4.
   1) If the waste container is not full, and the connector wiring is normal, take out the waste container cap assembly (3120) and check whether the floater and the floating rod are stuck.
   2) If the statuses of the floater and the floating rod are normal, continue to select **Menu>>Status>>Floater status** to enter the floater status display screen, and check whether the status of the waste container changes to **empty**.
5. Confirm that the wiring is normal. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the conventional driver board is faulty.

Solution

1. Install the waste container in a carton box, and use a supporting board to lock the mouth of the container and prevent the container from collapsing.
2. Tighten the BNC connector of the waste floater.
3. Replace floater cable 2.
4. To replace the waste container cap assembly, see **8.73.2** Disassembly and Assembly.
5. Replace the common driver board.

9.1.56 10107: SCI Priming Timeout

Error Code

10107

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	SCI priming timeout
Service engineer level	SCI priming timeout

Involved FRU

N/A

Error Mechanism

The syringe fails to complete a liquid pushing action within the specified time. This error is mostly found in R&D sequence action inspection

Possible Causes

This error is mostly found in R&D sequence action inspection, and often demonstrated as communication abnormalities at customer site

Resolve Path

Check whether there are other errors leading to abnormal communication before this error occurs

Solution

Remove the other errors according to the prompt

9.1.57 10108-10110: Probe Wipe Waste Channels Clogged

Error Code

10108-10110

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste channels clogged
Service engineer level	The waste channel of aspiration module wipe is clogged

Involved FRU

1. Mindray two-way valve (PEIZH)
2. Filter assembly (injection molding)
3. PCBA, Air pressure detection board, analog, 5 channels
4. Wire, motherboard to air pressure detection board, ext.

Error Mechanism

At the sequence-defined time point, the system detects that the pressure detected by the CBC_W sensor exceeds the sequence-defined range

Possible Causes

1. The SV08 is abnormal or the filter (LF6) is clogged
2. The waste pump (PUMP_VAC) is abnormal
3. The SV07 is abnormal or the filter (LF4) is clogged
4. The SV06 is abnormal or the filter (LF3) is clogged
5. The air pressure detection board/single channel air pressure detection board is defective
6. Poor contact of air pressure detection board

Resolve Path

1.

   1) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV08 valve, and check whether the valve is damaged or defective.

   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.

2. Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Fluidic Commissioning**. Click Wipe Waste Pump (P2) to perform self-test on the PUMP_VAC valve, and check whether the pump is damaged or defective.

3.

   1) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV07 valve, and check whether the valve is damaged or defective.

   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.

4.

   1) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV06 valve, and check whether the valve is damaged or defective.

   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.

5.

   1) Select **Menu>>Status>>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the pressure value of each channel is out of range. If the pressure is low, you can further check whether splashed liquid, corrosion, crystallization, etc. occurs on the air pressure detection board.

   2) For troubleshooting of air pressure detection board errors, see sections Air Pressure Detection Board/Single Air Pressure Detection Board in Chapter 8 "FRU Replacement, Commissioning and Verification".

6. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.

Solution

1. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
2. To replace the filter assembly (injection molding), see **8.62.2** Disassembly and Assembly.
3. To replace PCBA, the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.

4. To replace the wire between motherboard to air pressure detection board and extended communicating cable, see **8.101.2** Disassembly and Assembly.

9.1.58 10104-10106: Waste Channel Abnormal

Error Code

10104-10106

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste channel abnormal
Service engineer level	DIL cistern preheating bath priming abnormal

Involved FRU

1. Mindray two-way valve (PEIZH)
2. Filter assembly (injection molding)
3. PCBA, Air pressure detection board, analog, 5 channels
4. Wire, motherboard to air pressure detection board, ext.

Error Mechanism

At the sequence-defined time point, the system detects that the pressure detected by the CBC_W sensor exceeds the sequence-defined range

Possible Causes

1. The SV08 is abnormal or the filter (LF6) is clogged
2. The waste pump (PUMP_VAC) is abnormal
3. The SV07 is abnormal or the filter (LF4) is clogged
4. The SV06 is abnormal or the filter (LF3) is clogged
5. The air pressure detection board/single channel air pressure detection board is defective
6. Poor contact of air pressure detection board

Resolve Path

1.

   1) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV08 valve, and check whether the valve is damaged or defective.

   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.

2. Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Fluidic Commissioning**. Click Wipe Waste Pump (P2) to perform self-test on the PUMP_VAC valve, and check whether the pump is damaged or defective.
3.
   1) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV07 valve, and check whether the valve is damaged or defective.
   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
4.
   1) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the SV06 valve, and check whether the valve is damaged or defective.
   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
5.
   1) Select **Menu>>Status>>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the pressure value of each channel is out of range. If the pressure is low, you can further check whether splashed liquid, corrosion, crystallization, etc. occurs on the air pressure detection board.
   2) For troubleshooting of air pressure detection board errors, see sections Air Pressure Detection Board/Single Air Pressure Detection Board in Chapter 8 "FRU Replacement, Commissioning and Verification".
6. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.

Solution

1. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
2. To replace the filter assembly (injection molding), see **8.62.2** Disassembly and Assembly.
3. To replace PCBA, the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.
4. To replace the wire between motherboard to air pressure detection board and extended communicating cable, see **8.101.2** Disassembly and Assembly.

9.1.59 10111: SCI Bath Priming Failed

Error Code

10111

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	SCI bath priming failed
Service engineer level	SCI Syringe busy

Involved FRU

N/A

Error Mechanism

During the SCI priming process, the syringe receives more than 2 instructions, resulting in syringe action conflict. This error is mostly found in R&D sequence action inspection

Possible Causes

This error is mostly found in R&D sequence action inspection, and often demonstrated as communication abnormalities at customer site

Resolve Path

Check whether there are other errors leading to abnormal communication before this error occurs

Solution

Remove the other errors according to the prompt

9.1.60 30600-30602: DIFF Waste Channels Clogged

Error Code

30600-30602

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste channels clogged
Service engineer level	DIFF waste channel clogged

Involved FRU

1. Mindray two-way valve (PEIZH)
2. Filter assembly (injection molding)

3. PCBA, Air pressure detection board, analog, 5 channels
4. Wire, motherboard to air pressure detection board, ext.

Error Mechanism

At the sequence-defined time point, the system detects that the pressure detected by the CBC_W sensor exceeds the sequence-defined range

Possible Causes

1. The SV08 is abnormal or the filter (LF6) is clogged
2. The waste pump (PUMP_VAC) is abnormal
3. The SV07 is abnormal or the filter (LF4) is clogged
4. The SV06 is abnormal or the filter (LF3) is clogged
5. The air pressure detection board/single channel air pressure detection board is defective
6. Poor contact of air pressure detection board

Resolve Path

1.
   1) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV08 valve, and check whether the valve is damaged or defective.
   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
2. Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Fluidic Commissioning**. Click Wipe Waste Pump (P2) to perform self-test on the PUMP_VAC valve, and check whether the pump is damaged or defective.
3.
   1) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV07 valve, and check whether the valve is damaged or defective.
   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
4.
   1) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV06 valve, and check whether the valve is damaged or defective.
   2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
5.

1) Select **Menu>>Status>>Temperature & Pressure** to enter the **Temperature & Pressure** status display screen. Check whether the pressure value of each channel is out of range. If the pressure is low, you can further check whether splashed liquid, corrosion, crystallization, etc. occurs on the air pressure detection board.

2) For troubleshooting of air pressure detection board errors, see sections Air Pressure Detection Board/Single Air Pressure Detection Board in Chapter 8 "FRU Replacement, Commissioning and Verification".

6. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.

Solution

1. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
2. To replace the filter assembly (injection molding), see **8.62.2** Disassembly and Assembly.
3. To replace PCBA, the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.
4. To replace the wire between motherboard to air pressure detection board and extended communicating cable, see **8.101.2** Disassembly and Assembly.

9.1.61 30603-30605: RET Waste Channels Clogged

Error Code

30603-30605

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Waste channels clogged
Service engineer level	RET waste channel clogged

Involved FRU

1. Mindray two-way valve (PEIZH)
2. Filter assembly (injection molding)
3. PCBA, Air pressure detection board, analog, 5 channels
4. Wire, motherboard to air pressure detection board, ext.

Error Mechanism

At the sequence-defined time point, the system detects that the pressure detected by the CBC_W sensor exceeds the sequence-defined range

Possible Causes

1. The SV08 is abnormal or the filter (LF6) is clogged
2. The waste pump (PUMP_VAC) is abnormal
3. The SV07 is abnormal or the filter (LF4) is clogged
4. The SV06 is abnormal or the filter (LF3) is clogged
5. The air pressure detection board/single channel air pressure detection board is defective
6. Poor contact of air pressure detection board

Resolve Path

1.
    1) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV08 valve, and check whether the valve is damaged or defective.
    2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
2. Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Fluidic Commissioning**. Click Wipe Waste Pump (P2) to perform self-test on the PUMP_VAC valve, and check whether the pump is damaged or defective.
3.
    1) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV07 valve, and check whether the valve is damaged or defective.
    2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
4.
    1) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the SV06 valve, and check whether the valve is damaged or defective.
    2) No error is found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
5.
    1) Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the pressure value of each channel is out of range. If the pressure is low, you can further check whether splashed liquid, corrosion, crystallization, etc. occurs on the air pressure detection board.
    2) For troubleshooting of air pressure detection board errors, see sections Air Pressure Detection Board/Single Air Pressure Detection Board in Chapter 8 "FRU Replacement, Commissioning and Verification".

6. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.

Solution

1. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
2. To replace the filter assembly (injection molding), see **8.62.2** Disassembly and Assembly.
3. To replace PCBA, the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.
4. To replace the wire between motherboard to air pressure detection board and extended communicating cable, see **8.101.2** Disassembly and Assembly.

9.1.62 00305: After the DIL Syringe Acts, the Initial Position Sensor Is Not Blocked Unexpectedly

Error Code

00305

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DIL syringe action abnormal
Service engineer level	After the DIL syringe action, the initial position sensor is not blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. 10 mL lead screw driving syringe assembly
3. Wire, sampling asm&syringe motor sensor
4. PCBA (Motherboard)

Error Mechanism

After the aspiration/dispensation action, the initial position sensor of the syringe should be blocked, but it is actually not

Possible Causes

1. The initial position sensor of the DIL syringe fails
2. The DIL syringe drive assembly fails
3. The wires of the DIL syringe fails
4. Motherboard error

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the status of the DIL (250 μL) syringe sensor changes accordingly on the Motherboard Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service** >>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the DIL (250 μL) syringe in the fluidic commissioning screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the DIL (250 μL) syringe. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motherboard.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 10 mL lead screw driving syringe assembly, see **8.48.2** Disassembly and Assembly.
3. To replace the connection wire between the sampling assembly and the syringe motor sensor, see **8.112.2** Disassembly and Assembly.
4. To replace PCBA (motherboard), see **8.9.2** Disassembly and Assembly.

9.1.63 00306: Before the DIL Syringe Action, the Initial Position Sensor Is Blocked Unexpectedly

Error Code

00306

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DIL syringe action abnormal
Service engineer level	Before the DIL syringe action, the initial position sensor is blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. 10 mL lead screw driving syringe assembly
3. Wire, sampling asm&syringe motor sensor

4. PCBA (Motherboard)

Error Mechanism

Before the aspiration/dispensation action, the initial position sensor of the syringe should not be blocked, but it actually is

Possible Causes

1. The initial position sensor of the DIL syringe fails
2. The DIL syringe drive assembly fails
3. The wires of the DIL syringe fails
4. Motherboard error

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the status of the DIL (250 μL) syringe sensor changes accordingly on the **Motherboard Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service** >**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a self-test on the DIL (250 μL) syringe in the fluidic commissioning screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a self-test on the DIL (250 μL) syringe. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motherboard.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 10 mL lead screw driving syringe assembly, see **8.48.2** Disassembly and Assembly.
3. To replace the connection wire between the sampling assembly and the syringe motor sensor, see **8.112.2** Disassembly and Assembly.
4. To replace PCBA (motherboard), see **8.9.2** Disassembly and Assembly.

9.1.64 00307: Before the DIL Syringe Action, the Initial Position Sensor Is Not Blocked Unexpectedly

Error Code

00307

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DIL syringe action abnormal
Service engineer level	Before the DIL syringe action, the initial position sensor is not blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. 10 mL lead screw driving syringe assembly
3. Wire, sampling asm&syringe motor sensor
4. PCBA (Motherboard)

Error Mechanism

Before the aspiration/dispensation action, the initial position sensor of the syringe should be blocked, but it is actually not

Possible Causes

1. The initial position sensor of the DIL syringe fails
2. The DIL syringe drive assembly fails
3. The wires of the DIL syringe fails
4. Motherboard error

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the status of the DIL (250 µL) syringe sensor changes accordingly on the Motherboard Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service** >>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the DIL (250 µL) syringe in the fluidic commissioning screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the DIL (250 µL) syringe. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motherboard.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 10 mL lead screw driving syringe assembly, see **8.48.2** Disassembly and Assembly.
3. To replace the connection wire between the sampling assembly and the syringe motor sensor, see **8.112.2** Disassembly and Assembly.
4. To replace PCBA (motherboard), see **8.9.2** Disassembly and Assembly.

9.1.65 00308: DIL Syringe Aspirated Volume Too High

Error Code

00308

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DIL syringe action abnormal
Service engineer level	DIL syringe aspirated volume too high

Involved FRU

Main Control Board

Error Mechanism

At the current moment, the required aspirated volume + the already aspirated volume > the maximum aspirated volume

Possible Causes

The DIL syringe receives a command of abnormal aspirated volume

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.66 00309: DIL Syringe Dispensed Volume Too High

Error Code

00309

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DIL syringe action abnormal
Service engineer level	DIL syringe dispensed volume too high

Involved FRU

Main Control Board

Error Mechanism

At the current moment, the required aspirated volume > the already aspirated volume

Possible Causes

The DIL syringe receives a command of abnormal dispensed volume

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.67 00310: DIL Syringe Action Timeout

Error Code

00310

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	DIL syringe action abnormal
Service engineer level	DIL syringe action timeout

Involved FRU

Main Control Board

Error Mechanism

The time from the start of the action to the completion of the action is compared with the time transmitted. If it is greater than the time transmitted, a timeout will be reported

Possible Causes

The time from the start of the action to the completion of the action is compared with the time transmitted. If it is greater than the time transmitted, a timeout will be reported

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.68 30300: Aperture Voltage Abnormal

Error Code

30300

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Clogging
Service engineer level	Aperture voltage abnormal

Involved FRU

1. Sheath fluid impedance assembly
2. Diluent heating assembly
3. Two-position ways LVM valve assembly (10R6 with valve board, 3120)
4. SV30, SV31, SV33, Mindray two-way valve (PEIZH), SV32, SV34, and SV35
5. PCBA (Motherboard)
6. Power Supply Assembly

Error Mechanism

The aperture voltage exceeds the allowable range, triggering the clogging alarm mechanism

Possible Causes

1. Aperture clogging by blood sample/RBC bath clogging
2. Diluent conductivity abnormal
3. Diluent temperature abnormal
4. Diluent temperature detection unit assembly abnormal
5. NTC temperature sensor wire 1 defective
6. Tube error
7. Valve error
8. Sheath fluid count bath/sheath fluid impedance assembly defective
9. Motherboard error
10. Power supply assembly abnormal
11. Unknown Cause

Resolve Path

1. Aperture clogging occurs frequently in continuous testing.
2. Failure can be eliminated by performing aperture cleaning and probe cleanser maintenance/ aperture clogging removal/RBC bath cleaning.
3. Expired diluent, wrong diluent model, etc.
4. Freezing and thawing of diluent (excessive temperature difference between day and night, etc.).
5. Confirm whether the room temperature is within the working temperature range of the analyzer. If the temperature exceeds the standard, eliminate the error after the temperature is within the normal working range.
6. If the room temperature is in the normal range, click **Menu**>>**Status**>>**Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the temperatures of **Preheating Bath** and **RBC Diluent** are within the normal range. If the temperatures exceed the normal range and the temperature difference between them is small, a thermometer or hand can be used to feel whether the temperature control system works normally.

7. It is confirmed that the room temperature and **Preheating Bath** temperature are within the normal range, but **RBC Diluent** temperature exceeds the standard range and has a great difference with the temperature control system.
8. Check whether the diluent temperature detection unit assembly is loose or has damaged wires.
9. Check whether the connection wire of NTC temperature sensor is broken, damaged, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
10. Check whether diluent is present in the RBC assembly tubes and whether the tubes are aging, bent, silted, deformed, or damaged, etc. (check tubes T52, T60, T66, T64, T65, T68, and T43).
11. Check and confirm the defective valve: Check whether the tubes and valves (**Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Valve Self-test**) associated with the sheath flow impedance assembly are clogged. No error found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
12. If the above operation cannot troubleshoot the error, try to replace the sheath fluid count bath/ sheath fluid impedance assembly.
13. Observe whether the indicators on the motherboard are always on green, and observe whether the electronic material is damaged or corroded on the motherboard.
14. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
15. Refer to the power supply assembly in the Hardware System Part in the maintenance manual to confirm whether the power supply assembly has has any error.
16. None of the above methods can troubleshoot the error.

Solution

1. Aperture probe cleanser maintenance/aperture clogging removal/RBC bath cleaning
2. Replace with new DS diluent
3. For temperature control system error, please see Chapter 14 Temperature Control System.
4. Replace the diluent temperature detection unit assembly.
5. Unplug and replug the temperature sensor or replace the diluent temperature detection unit assembly
6. Replace the tubes T52, T60, T66, T64, T65, T68, and T43.
7. Replace the valves SV30, SV31, SV32, SV33, SV34, and SV35.
8. To replace the sheath fluid count bath/sheath fluid impedance assembly, see **8.38.2** Disassembly and Assembly and **8.38.3** Commissioning and Verification for sheath fluid count bath assembly, as well as **8.37.2** Disassembly and Assembly and **8.37.3** Commissioning and Verification for sheath fluid impedance assembly.
9. To replace the motherboard and confirm the error, see PCBA, motherboard **8.9.2** Disassembly and Assembly and **8.9.3** Commissioning and Verification.
10. For the treatment measures of power supply assembly, see the power supply assembly in maintenance manual **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.

11.To collect the original error data, see the error data export part in the chapter of software.

9.1.69 30301: Aperture Voltage Abnormal

Error Code

30301

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Clogging
Service engineer level	Aperture voltage abnormal

Involved FRU

1. Sheath fluid impedance assembly
2. Diluent heating assembly
3. Two-position ways LVM valve assembly (10R6 with valve board, 3120)
4. SV30, SV31, SV33, Mindray two-way valve (PEIZH), SV32, SV34, and SV35
5. PCBA (Motherboard)
6. Power Supply Assembly

Error Mechanism

The aperture voltage exceeds the allowable range by about 2000AD - 3000AD (certain different for individual analyzers), triggering the clogging alarm mechanism (This error alarm only appears when the service engineer uses the aging function. With the same mechanism, the difference can not limit the continuous measurement due to the clogging error)

Possible Causes

1. Aperture clogging by blood sample/RBC bath clogging
2. Diluent conductivity abnormal
3. Diluent temperature abnormal
4. Diluent temperature detection unit assembly abnormal
5. NTC temperature sensor wire 1 defective
6. Tube error
7. Valve error
8. Sheath fluid count bath/sheath fluid impedance assembly defective
9. Motherboard error
10. Power supply assembly abnormal
11. Unknown Cause

Resolve Path

1. Aperture clogging occurs frequently in continuous testing.
2. Failure can be eliminated by performing aperture cleaning and probe cleanser maintenance/ aperture clogging removal/RBC bath cleaning.
3. Expired diluent, wrong diluent model, etc.
4. Freezing and thawing of diluent (excessive temperature difference between day and night, etc.).
5. Confirm whether the room temperature is within the working temperature range of the analyzer. If the temperature exceeds the standard, eliminate the error after the temperature is within the normal working range.
6. If the room temperature is in the normal range, click **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the temperatures of **Preheating Bath** and **RBC Diluent** are within the normal range. If the temperatures exceed the normal range and the temperature difference between them is small, a thermometer or hand can be used to feel whether the temperature control system works normally.
7. It is confirmed that the room temperature and **Preheating Bath** temperature are within the normal range, but **RBC Diluent** temperature exceeds the standard range and has a great difference with the temperature control system.
8. Check whether the diluent temperature detection unit assembly is loose or has damaged wires.
9. Check whether the connection wire of NTC temperature sensor is broken, damaged, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
10. Check whether diluent is present in the RBC assembly tubes and whether the tubes are aging, bent, silted, deformed, or damaged, etc. (check tubes T52, T60, T66, T64, T65, T68, and T43).
11. Check and confirm the defective valve: Check whether the tubes and valves (**Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-test**>**>Valve Self-test**) associated with the sheath flow impedance assembly are clogged. No error found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
12. If the above operation cannot troubleshoot the error, try to replace the sheath fluid count bath/ sheath fluid impedance assembly.
13. Observe whether the indicators on the motherboard are always on green, and observe whether the electronic material is damaged or corroded on the motherboard.
14. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
15. Refer to the power supply assembly in the Hardware System Part in the maintenance manual to confirm whether the power supply assembly has has any error.
16. None of the above methods can troubleshoot the error.

Solution

1. Aperture probe cleanser maintenance/aperture clogging removal/RBC bath cleaning

2. Replace with new DS diluent
3. For temperature control system error, please see Chapter 14 Temperature Control System.
4. Replace the diluent temperature detection unit assembly.
5. Unplug and replug the temperature sensor or replace the diluent temperature detection unit assembly
6. Replace the tubes T52, T60, T66, T64, T65, T68, and T43.
7. Replace the valves SV30, SV31, SV32, SV33, SV34, and SV35.
8. To replace the sheath fluid count bath/sheath fluid impedance assembly, see **8.38.2** Disassembly and Assembly and **8.38.3** Commissioning and Verification for sheath fluid count bath assembly, as well as **8.37.2** Disassembly and Assembly and **8.37.3** Commissioning and Verification for sheath fluid impedance assembly.
9. To replace the motherboard and confirm the error, see PCBA, motherboard **8.9.2** Disassembly and Assembly and **8.9.3** Commissioning and Verification.
10. For the treatment measures of power supply assembly, see the power supply assembly in maintenance manual **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.
11. To collect the original error data, see the error data export part in the chapter of software.

9.1.70 30302: Sample Preparation Abnormal

Error Code

30302

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Clogging
Service engineer level	RBC sample preparation abnormal

Involved FRU

1. Two-way LVM valve assembly (SV36)
2. Two-way LVM valve assembly (SV28, SV29)
3. PCBA (Motherboard)
4. Power Supply Assembly

Error Mechanism

Abnormal particle number during RBC sample preparation

Possible Causes

1. Valve leakage or error
2. Valve leakage or lax closure
3. Tube error
4. Motherboard error
5. Power supply assembly abnormal
6. Unknown Cause

Resolve Path

1. The puncture debris of the test tube cap enters the HGB bath together with the sample probe, and is pulled to the SV36 valve position during RBC sample preparation and stuck in the SV36 valve diaphragm, resulting in leakage due to lax closure of SV36 valve.
2. Check and confirm the defective valve: Check whether the tubes and valves (**Menu** >>**Service** >>**Commissioning & Self-Test**>>**Self-test**>>**Valve Self-test**) associated with the sheath flow impedance assembly are clogged. No error found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
3. Check and confirm the defective valve: Check whether the tubes and valves (**Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Valve Self-test**) associated with the sheath flow impedance assembly are clogged. No error found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
4. Check whether the RBC assembly tubes are aging, bent, silted, crystallized, deformed, or damaged, etc. (check tubes T86, T87, T41, T42, T43, T47, T52, T48, T49, T50, T51, and T192).
5. Observe whether the indicators on the motherboard are always on green, and observe whether the electronic material is damaged or corroded on the motherboard.
6. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
7. Refer to the power supply assembly in the Hardware System Part in the maintenance manual to confirm whether the power supply assembly has has any error.
8. None of the above methods can troubleshoot the error.

Solution

1. One click to eliminate the error, or replace the valve SV36.
2. Replace the valves SV28 and SV20.
3. Replace the tubes T86, T87, T41, T42, T43, T47, T52, T48, T49, T50, T51, and T192.
4. To replace the motherboard and confirm the error, see PCBA, motherboard **8.9.2** Disassembly and Assembly and **8.9.3** Commissioning and Verification.
5. For the treatment measures of power supply assembly, see the power supply assembly in maintenance manual **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.
6. To collect the original error data, see the error data export part in the chapter of software.

9.1.71 30303: Sample Preparation Abnormal

Error Code

30303

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Clogging
Service engineer level	RBC sample preparation abnormal

Involved FRU

1. Two-way LVM valve assembly (SV36)
2. Two-way LVM valve assembly (SV28, SV29)
3. PCBA (Motherboard)
4. Power Supply Assembly

Error Mechanism

Abnormal particle number during RBC sample preparation (This error alarm only appears when the service engineer uses the aging function, and the continuous measurement is not limited by the clogging error)

Possible Causes

1. Valve leakage or error
2. Valve leakage or lax closure
3. Tube error
4. Motherboard error
5. Power supply assembly abnormal
6. Unknown Cause

Resolve Path

1. The puncture debris of the test tube cap enters the HGB bath together with the sample probe, and is pulled to the SV36 valve position during RBC sample preparation and stuck in the SV36 valve diaphragm, resulting in leakage due to lax closure of SV36 valve.
2. Check and confirm the defective valve: Check whether the tubes and valves (**Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Valve Self-test**) associated with the sheath flow impedance assembly are clogged. No error found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.

3. Check and confirm the defective valve: Check whether the tubes and valves (**Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-test**>>**Valve Self-test**) associated with the sheath flow impedance assembly are clogged. No error found during the valve self-check does not mean that the valve itself has no error. Replace the valve as appropriate to troubleshoot the error.
4. Check whether the RBC assembly tubes are aging, bent, silted, crystallized, deformed, or damaged, etc. (check tubes T86, T87, T41, T42, T43, T47, T52, T48, T49, T50, T51, and T192).
5. Observe whether the indicators on the motherboard are always on green, and observe whether the electronic material is damaged or corroded on the motherboard.
6. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
7. Refer to the power supply assembly in the Hardware System Part in the maintenance manual to confirm whether the power supply assembly has has any error.
8. None of the above methods can troubleshoot the error.

Solution

1. One click to eliminate the error, or replace the valve SV36.
2. Replace the valves SV28 and SV20.
3. Replace the tubes T86, T87, T41, T42, T43, T47, T52, T48, T49, T50, T51, and T192.
4. To replace the motherboard and confirm the error, see PCBA, motherboard **8.9.2** Disassembly and Assembly and **8.9.3** Commissioning and Verification.
5. For the treatment measures of power supply assembly, see the power supply assembly in maintenance manual **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.
6. To collect the original error data, see the error data export part in the chapter of software.

9.1.72 30304-30306: RBC Sample Preparation Cleaning Channel Blocked

N/A

9.1.73 30100: Optical PMT Voltage Abnormal

Error Code

30100

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	PMT voltage abnormal
Service engineer level	PMT voltage abnormal

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

Alarm indicating PMT voltage exceeds the threshold value

Possible Causes

1. Communication error between optical system and motherboard
2. Optical assembly error
3. Motherboard error

Resolve Path

1. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
2. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. Unplug and re-plug the wire, and replace the wire if nothing happens.
2. To replace the optical system, see **8.42.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.74 30103: Optical DIFF Channel FS Background Photovoltage Abnormal

Error Code

30103

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Flow cell contaminated
Service engineer level	Optical DIFF channel FS blank voltage abnormal

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

At the **Voltage&Current Status** screen, the optical FS background voltage exceeds the threshold value identified in the screen

Possible Causes

1. The inner hole of the flow cell is dirty
2. Communication error between optical system and motherboard
3. Optical assembly error
4. Motherboard error

Resolve Path

1. On the **Maintenance** screen, perform probe cleanser maintenance for the flow cell and observe whether the optical FS background voltage decreases after maintenance.
2. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
3. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
4. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. If the optical FS background voltage decreases after one maintenance, continue probe cleanser maintenance for the flow cell until the optical FS background voltage meets the threshold range identified in the screen.
2. Unplug and re-plug the wire, and replace the wire if nothing happens.
3. To replace the optical system, see **8.42.2** Disassembly and Assembly.
4. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.75 30104: Optical RET Channel FS Background Photovoltage Abnormal

N/A

9.1.76 30105: Optical FS Baseline Abnormal

Error Code

30105

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Optical FS baseline abnormal
Service engineer level	Optical FS baseline abnormal

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

FS baseline exceeds the threshold value

Possible Causes

1. Communication error between optical system and motherboard
2. Optical assembly error
3. Motherboard error

Resolve Path

1. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
2. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. Unplug and re-plug the wire, and replace the wire if nothing happens.
2. To replace the optical system, see **8.42.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.77 30106: Optical SS Baseline Abnormal

Error Code

30106

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Optical SS baseline abnormal
Service engineer level	Optical SS baseline abnormal

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

SS baseline exceeds the threshold value

Possible Causes

1. Communication error between optical system and motherboard
2. Optical assembly error
3. Motherboard error

Resolve Path

1. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
2. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. Unplug and re-plug the wire, and replace the wire if nothing happens.

2. To replace the optical system, see **8.42.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.78 30107: Optical FL Baseline Abnormal

Error Code

30107

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Optical FL baseline abnormal
Service engineer level	Optical FL baseline abnormal

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

FL baseline exceeds the threshold value

Possible Causes

1. Communication error between optical system and motherboard
2. Optical assembly error
3. Motherboard error

Resolve Path

1. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
2. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. Unplug and re-plug the wire, and replace the wire if nothing happens.
2. To replace the optical system, see **8.42.2** Disassembly and Assembly.

3. To replace the motherboard, see 8.9.2 Disassembly and Assembly.

9.1.79 30108: Optical System Shielding Box Open

Error Code

30108

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Optical system shielding box is open
Service engineer level	Optical system shielding box is open

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

The level signal of micro switch inside the optical system is abnormal

Possible Causes

1. The optical system cover is loose
2. Communication error between optical system and motherboard
3. Optical assembly error
4. Motherboard error

Resolve Path

1. Check whether the optical system cover is loose
2. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
3. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
4. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. If the cover is loose, re-install the cover and tighten the screws.

2. Unplug and re-plug the wire, and replace the wire if nothing happens.
3. To replace the optical system, see **8.42.2** Disassembly and Assembly.
4. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.80 30109: Optical Signal Board Communication Timeout

Error Code

30109

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Optical signal communication timeout
Service engineer level	Optical signal communication timeout

Involved FRU

1. Optical system
2. Motherboard

Error Mechanism

Relevant signal exceeds the threshold value during communication between optical system and motherboard

Possible Causes

1. Communication error between optical system and motherboard
2. Optical assembly error
3. Motherboard error

Resolve Path

1. Check whether the connection wires (009-013519-00) between the motherboard and the optical assembly are broken, damaged, loose, or in poor contact.
2. Refer to Chapter 8 Replacing, Commissioning, and Verifying Optical System FRUs and confirm whether the optical assembly is faulty.
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 Replacing, Commissioning, and Verifying Optical Motherboard FRUs, and confirm whether the motherboard is faulty.

Solution

1. Unplug and re-plug the wire, and replace the wire if nothing happens.
2. To replace the optical system, see **8.42.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.81 40001: System Time Error

Error Code

40001

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	System time error
Service engineer level	System time error

Involved FRU

M05-100R29-02

Error Mechanism

After power on, the software confirms the system time. If the time is not between [2017.6.1 ~ 2035.12.31], an alarm will be given to the user that the system time is wrong

Possible Causes

1. Time setup error
2. Main button cell is used up
3. Unknown Cause

Resolve Path

1. Whether the current time is between [2017.6.1 ~ 2035.12.31]
2. Use a multimeter to see whether the voltage of the button battery is lower than 1.8V
3. None of the above methods can solve the problem

Solution

1. Select **Setup**>>**System Setup**>>**Date/Time Setup** to correctly set the current system time.
2. Replace the button cell.
3. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.82 40003: Export KEY File

Error Code

40003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Export KEY File
Service engineer level	Export KEY File

Involved FRU

N/A

Error Mechanism

The anti-regional dumping model is enabled, but the KEY of the agent has not been imported

Possible Causes

1. KEY not imported
2. Unknown Cause

Resolve Path

1. The authorization button on the reagent setting interface is grayed out
2. None of the above methods can solve the problem

Solution

1. Import the KEY after agent anti-regional dumping: **Advanced Toolbox**>>**Commissioning Setup**>>**Anti-regional Dumping of Reagents**>>**Import KEY**.
2. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.83 20200: Power Radiator Fan Clogged

Error Code

20200

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Analyzer fan does not work
Service engineer level	Power radiator fan clogged

Involved FRU

1. Lead-out wire
2. Power Supply Assembly
3. PCBA, Motherboard

Error Mechanism

After the fan is turned on, it is detected that the power supply fan is blocked, with other error alarms at the same time

Possible Causes

1. Poor connection for fan of power supply assembly
2. Power supply assembly abnormal
3. Motherboard

Resolve Path

1. Check whether the power fan connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
3. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" and confirm whether the power supply assembly is faulty.
4. After it is confirmed that the power fan wiring and fan are normal, if the error still can not be eliminated, please refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the motherboard is faulty.

Solution

1. Restore or replace the wire for fan of power supply assembly
2. Refer to Power Supply Assembly **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.84 20002: Air Pressure Detection Board Communication Abnormal

Error Code

20002

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Air pressure detection board error
Service engineer level	Air pressure detection board communication abnormal

Involved FRU

1. Wire, motherboard to air pressure detection board, ext.
2. Air pressure detection board PCBA
3. PCBA (Motherboard)

Error Mechanism

Failed to verify the data (pressure calibration parameters) read from EEPROM (two consecutive readings are inconsistent in comparison)

Possible Causes

1. Poor contact of air pressure detection board
2. Air pressure detection board PCBA abnormal
3. Motherboard error

Resolve Path

1. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Check whether there is splashing, corrosion, crystallization, etc. on the air pressure detection board PCBA.
3. For troubleshooting of air pressure detection board PCBA errors, see Chapter 8 "FRU Replacement, Commissioning and Verification".
4. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. To restore or replace air pressure detection board of 5 channels, see **8.101.2** Disassembly and Assembly.
2. To replace the air pressure detection board PCBA, see **8.78.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.85 20003: Air Pressure Detection Board Calibration Parameters Abnormal

Error Code

20003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Air pressure detection board error
Service engineer level	Air pressure detection board calibration parameters abnormal

Involved FRU

1. Wire, motherboard to air pressure detection board, ext.
2. Air pressure detection board PCBA
3. PCBA (Motherboard)

Error Mechanism

The calibration value read from the air pressure board EEPROM (4 channels) is out of the normal range

Possible Causes

1. Poor contact of air pressure detection board
2. Air pressure detection board PCBA abnormal
3. Motherboard error

Resolve Path

1. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Check whether there is splashing, corrosion, crystallization, etc. on the air pressure detection board PCBA.
3. For troubleshooting of air pressure detection board PCBA errors, see Chapter 8 "FRU Replacement, Commissioning and Verification".

4. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. To restore or replace air pressure detection board of 5 channels, see **8.101.2** Disassembly and Assembly.
2. To replace the air pressure detection board PCBA, see **8.78.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.86 20004: Motherboard Power 12V Voltage Abnormal

Error Code

20004

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	12V voltage abnormal
Service engineer level	12V voltage abnormal

Involved FRU

1. Wire, motherboard to valve board 1 Pow & Com, ext.
2. Wire, motherboard to optical board
3. Power Supply Assembly
4. PCBA (Motherboard)

Error Mechanism

The current voltage is out of range [11.50, 12.50]V

Possible Causes

1. The power cord of valve drive board (power 12V) / optical signal board connecting wire is unreliable
2. Power supply assembly abnormal
3. Motherboard error

Resolve Path

1. Check whether the connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
3. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" and confirm whether the power supply assembly is faulty.
4. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. To restore or replace the power cord of valve drive board (power 12V) / optical signal board connecting wire, see "Wire, motherboard to valve board 1 Pow & Com, ext." **8.114.2** Disassembly and Assembly.
2. Refer to the treatment measures for Power Supply Assembly **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.87 20005: Motherboard Power 24V Voltage Abnormal

Error Code

20005

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	24V voltage abnormal
Service engineer level	24V voltage abnormal

Involved FRU

1. Power Supply Assembly
2. PCBA (Motherboard)

Error Mechanism

The current voltage is out of range [22.80, 25.20]V

Possible Causes

1. Power supply assembly abnormal
2. Motherboard error

Resolve Path

1. Make sure that the power input and output wires are in good condition, and check whether the power indicator is normal.
2. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" and confirm whether the power supply assembly is faulty.
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. Refer to the treatment measures for Power Supply Assembly **8.87.2** Disassembly and Assembly and **8.87.3** Commissioning and Verification.
2. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.88 20201: Board Radiator Fan Clogged

Error Code

20201

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Analyzer fan does not work
Service engineer level	Board radiator fan1 clogged

Involved FRU

1. Lead-out wire, board fan
2. Board Fan Assembly
3. PCBA, Motherboard

Error Mechanism

After the fan is turned on, it is detected that the board fan is blocked

Possible Causes

1. Poor connection for board fan
2. Board fan assembly error
3. Motherboard

Resolve Path

1. Check whether the board fan connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Click **Menu>>Maintenance>>Self-test Screen>>Other Components**, and select Fan, to view whether the fan can normally work.
3. After it is confirmed that the 4 channels of fan wiring and fan are normal, if the error still can not be eliminated, please refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the motherboard is faulty.

Solution

1. Restore or replace the wire for board fan.
2. To replace the fan assembly, see **8.84.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.89 20202: Board Radiator Fan 2 Clogged

Error Code

20202

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Analyzer fan does not work
Service engineer level	Board radiator fan 2 clogged

Involved FRU

1. Lead-out wire, board fan
2. Board Fan Assembly
3. PCBA, Motherboard

Error Mechanism

After the fan is turned on, it is detected that the board fan is blocked

Possible Causes

1. Poor connection for board fan
2. Board fan assembly error
3. Motherboard

Resolve Path

1. Check whether the board fan connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Click **Menu>>Maintenance>>Self-test Screen>>Other Components**, and select Fan, to view whether the fan can normally work.
3. After it is confirmed that the 4 channels of fan wiring and fan are normal, if the error still can not be eliminated, please refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the motherboard is faulty.

Solution

1. Restore or replace the wire for board fan.
2. To replace the fan assembly, see **8.84.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.90 20008: RFID Communication Abnormal, Closed Reagent Model

Error Code

20008

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	RFID board, closed reagent model
Service engineer level	RFID communication abnormal, closed reagent model

Involved FRU

1. Paired wire, motherboard to RFID board 1&2, motherboard end
2. Wire, front cover
3. PCBA, RFID Reader
4. PCBA (Motherboard)

Error Mechanism

The motherboard sends an command to RFID reader, but receives no reply within the specified time

Possible Causes

1. Poor connection between motherboard and RFID reader board
2. RFID reader board abnormal
3. Motherboard error

Resolve Path

1. Check whether the connecting wire between the motherboard and RFID reader board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Does RFID reader board have any error?
3. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. To replace the wire between motherboard and RFID reader board, see **8.102.2** Disassembly and Assembly.
2. To replace RFID reader board, see **8.82.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.91 20203: Analyzer Radiator Fan Clogged

Error Code

20203

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Analyzer fan does not work
Service engineer level	Radiator fan in the analyzer is clogged

Involved FRU

1. Lead-out wire
2. Main unit fan assembly
3. PCBA, Motherboard

Error Mechanism

The system detects that the fan at the back of main unit stops.

Possible Causes

1. Poor connection for analyser fan
2. Fan assembly error
3. Motherboard

Resolve Path

1. Check whether the analyzer fan connecting wire is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.
2. Click **Menu**>>**Maintenance**>>**Self-test Screen**>>**Other Components**, and select Fan, to view whether the fan can normally work.
3. After it is confirmed that the analyzer fan wiring and fan assembly are normal, if the error still can not be eliminated, please refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" to confirm whether the motherboard is faulty.

Solution

1. Restore or replace the wire for analyzer fan.
2. To replace the fan assembly, see **8.84.2** Disassembly and Assembly.
3. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.92 20011: Motherboard Power MCU Communication Timeout

Error Code

20011

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Motherboard power MCU communication timeout
Service engineer level	Motherboard power MCU communication timeout

Involved FRU

1. PCBA, Main Control Board, iMX8
2. PCBA (Motherboard)

Error Mechanism

The main control board sends an command to the motherboard power MCU, but receives no reply within the specified time.

Possible Causes

1. Main control board abnormal
2. Motherboard error

Resolve Path

1. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" and confirm whether the main control board is faulty.
2. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. To replace the main control board, see **8.8.2** Disassembly and Assembly.
2. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.93 20012: Motherboard Drive MCU Communication Timeout

Error Code

20012

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Motherboard drive MCU communication timeout
Service engineer level	Motherboard drive MCU communication timeout

Involved FRU

1. PCBA, Main Control Board, iMX8
2. PCBA (Motherboard)

Error Mechanism

The main control board sends an command to the motherboard drive MCU, but receives no reply within the specified time

Possible Causes

1. Main control board abnormal

2. Motherboard error

Resolve Path

1. Refer to Chapter 8 "FRU Replacement, Commissioning, and Verification" and confirm whether the main control board is faulty.
2. If the error still cannot be eliminated after the above possible causes are excluded, consider whether the motherboard is abnormal, you can refer to Chapter 8 "FRU Replacement, Commissioning and Verification", and confirm whether the motherboard is faulty.

Solution

1. To replace the main control board, see **8.8.2** Disassembly and Assembly.
2. To replace the motherboard, see **8.9.2** Disassembly and Assembly.

9.1.94 10300: M Aspiration Abnormal

Error Code

10300

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	50kPa pressure out of range
Service engineer level	50kPa pressure out of range

Involved FRU

1. Air tank assembly
2. Air pump assembly
3. Mindray three-way valve (PEIZH)
4. Mindray two-way valve (PEIZH)
5. PCBA, Air pressure detection board, analog, 5 channels
6. Wire, motherboard to air pressure detection board, ext.

Error Mechanism

At the sequence-defined time point, the system detects that the pressure of the 50kPa (PC) pressure sensor exceeds the sequence-defined range

Possible Causes

1. The air tank assembly (PC) is leaking or the air pump (GP) is abnormal or the SV20 is abnormal
2. Air tank assembly (PC) leaks
3. SV18/19 branch is abnormal
4. The air pressure detection board/single channel air pressure detection board is defective
5. Poor contact of air pressure detection board

Resolve Path

1. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
2. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the pressure value of each channel exceeds the range. If the air pump has no movement, it has a high possibility of air pump damage. Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges.
3.
    1) Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges
    2) Check if the valve is misconnected.
    3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (SV20), check whether the valve is damaged, etc.
    4) Check whether the tube is blocked, bent, disengaged, etc.; whether the tube connectors are loose, damaged, etc.; (Tubes: T148\T149\T150\T151\T152\T153).
4. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
5.
    1) Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges
    2) Check if the valve is misconnected.

3) Select **Menu>>Service>>Commissioning & Self-Test>>Self-Test>>Valve Self-Test**. Perform self-test on the relevant valve (SV18/19), check whether the valve is damaged, etc.

4) Check whether the tube is blocked or bent (Tube: T155/T165/T167/T231/J41/J42).

6. Select **Menu>>Status>>Temperature & Pressure** to enter the **Temperature & Pressure** status display screen. Check whether the pressure value of each channel is out of range. If the pressure is low, you can further check whether splashed liquid, corrosion, crystallization, etc. occurs on the air pressure detection board. For troubleshooting of air pressure detection board errors, see sections Air Pressure Detection Board/Single Air Pressure Detection Board in Chapter 8 "FRU Replacement, Commissioning and Verification".
7. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.

Solution

1. To replace the air tank assembly, see **8.45.2** Disassembly and Assembly.
2. Replace the air pump assembly.
3. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
4. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
5. To replace PCBA, the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.
6. To replace the wire between motherboard to air pressure detection board and extended communicating cable, see **8.101.2** Disassembly and Assembly.

9.1.95 10301: M Aspiration Insufficient/Sample Abnormal

Error Code

10301

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	PC pressure release abnormal
Service engineer level	PC pressure release abnormal

Involved FRU

1. Air tank assembly
2. Air pump assembly
3. Mindray three-way valve (PEIZH)
4. Mindray two-way valve (PEIZH)
5. PCBA, Air pressure detection board, analog, 5 channels

6. Wire, motherboard to air pressure detection board, ext.

Error Mechanism

At the sequence-defined time point, the system detects that the pressure of the 50kPa (PC) pressure sensor exceeds the sequence-defined range

Possible Causes

1. The air tank assembly (PC) is leaking or the air pump (GP) is abnormal or the SV20 is abnormal
2. Air tank assembly (PC) leaks
3. SV18/19 branch is abnormal
4. The air pressure detection board/single channel air pressure detection board is defective
5. Poor contact of air pressure detection board

Resolve Path

1. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.
2. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Check whether the pressure value of each channel exceeds the range. If the air pump has no movement, it has a high possibility of air pump damage. Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges.
3.
   1) Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges
   2) Check if the valve is misconnected.
   3) Select **Menu**>**>Service**>**>Commissioning & Self-Test**>**>Self-Test**>**>Valve Self-Test**. Perform self-test on the relevant valve (SV20), check whether the valve is damaged, etc.
   4) Check whether the tube is blocked, bent, disengaged, etc.; whether the tube connectors are loose, damaged, etc.; (Tubes: T148\T149\T150\T151\T152\T153).
4. Select **Menu**>**>Status**>**>Temperature & Pressure** to enter the Temperature & Pressure status display screen. If the pressure is too low, click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized

within the normal ranges. At this time, you can feel whether there is air leakage near the air tank of the analyzer.

5.
   1) Select **Menu**>>**Status**>>**Temperature & Pressure** to enter the **Temperature & Pressure** status display screen. Click **Pressure Buildup** and **Pressure Release** respectively, and observe whether the pressure values have changed and stabilized within the normal ranges
   2) Check if the valve is misconnected.
   3) Select **Menu**>>**Service**>>**Commissioning & Self-Test**>>**Self-Test**>>**Valve Self-Test**. Perform self-test on the relevant valve (SV18/19), check whether the valve is damaged, etc.
   4) Check whether the tube is blocked or bent (Tube: T155/T165/T167/T231/J41/J42).
6. Select **Menu**>>**Status**>>**Temperature & Pressure** to enter the **Temperature & Pressure** status display screen. Check whether the pressure value of each channel is out of range. If the pressure is low, you can further check whether splashed liquid, corrosion, crystallization, etc. occurs on the air pressure detection board. For troubleshooting of air pressure detection board errors, see sections Air Pressure Detection Board/Single Air Pressure Detection Board in Chapter 8 "FRU Replacement, Commissioning and Verification".
7. Check whether the connecting wire of air pressure detection board is broken, damaged, loose, or in poor contact, and whether there is splashing, corrosion, crystallization, etc.

Solution

1. To replace the air tank assembly, see **8.45.2** Disassembly and Assembly.
2. Replace the air pump assembly.
3. To replace Mindray three-way valve (PEIZH), see **8.4.2** Disassembly and Assembly.
4. To replace Mindray two-way valve (PEIZH), see **8.5.2** Disassembly and Assembly.
5. To replace PCBA, the air pressure detection board, analog, 5 channels, see **8.78.2** Disassembly and Assembly.
6. To replace the wire between motherboard to air pressure detection board and extended communicating cable, see **8.101.2** Disassembly and Assembly.

9.1.96 00400: LS Syringe Receives an Unrecognizable Action Command

N/A

9.1.97 00401: LS Syringe Receives a New Command When Action Not Completed

N/A

9.1.98 00402: LS Syringe Fails to Return to Initial Position

N/A

9.1.99 00403: LS Syringe Fails to Leave the Initial Position

N/A

9.1.100 00404: After the LS Syringe Action, the Initial Position Sensor Is Blocked Unexpectedly

N/A

9.1.101 00405: After the LS Syringe Acts, the Initial Position Sensor Is Not Blocked Unexpectedly

N/A

9.1.102 00406: Before the LS Syringe Action, the Initial Position Sensor Is Blocked Unexpectedly

N/A

9.1.103 00407: Before the LS Syringe Action, the Initial Position Sensor Is Not Blocked Unexpectedly

N/A

9.1.104 00408: LS Syringe Aspirated Volume Too High

N/A

9.1.105 00409: LS Syringe Dispensed Volume Too High

N/A

9.1.106 00410: LS Syringe Action Timeout

N/A

9.1.107 01000: The Blending Assembly Receives an Invalid Command

Error Code

01000

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending assembly receives an invalid command

Involved FRU

Main Control Board

Error Mechanism

The blending assembly receives an unrecognizable action command

Possible Causes

The blending assembly receives an invalid command

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.108 01001: Conflicting Command to Blending Assembly

Error Code

01001

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Conflicting command to blending assembly

Involved FRU

Main Control Board

Error Mechanism

The command to operate the blending assembly is received, but the previous action is not finished

Possible Causes

The command to operate the blending assembly is received, but the previous action is not finished

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.109 01002: Blending Assembly Status Error (Software)

Error Code

01002

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Blending assembly status abnormal (software)

Involved FRU

N/A

Error Mechanism

Software status abnormal

Possible Causes

Software status abnormal

Resolve Path

N/A

Solution

Upgrade the software

9.1.110 01003: Blending Motor Board Communication Timeout

Error Code

01003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Blending motor board communication timeout
Service engineer level	Blending motor board communication timeout

Involved FRU

Blending motor board

Error Mechanism

Software communication timeout

Possible Causes

Software communication timeout

Resolve Path

N/A

Solution

1. Check whether the communication line is connected normally.
2. If yes, copy the log and return to R&D.
3. Upgrade the software according to advice from R&D.

4. If the error cannot be eliminated after software upgrading, replace the blending motor board FRU and upgrade the software after replacement. See PCBA, motor board 8.79.2 Disassembly and Assembly.

9.1.111 01100: When the Blending Assembly Returns, Y-direction Home Position Sensor Is Mistakenly Triggered

Error Code

01100

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	When the blending assembly returns, Y-direction home position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of returning to the initial position in Y-direction. After Y-direction home position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Sampling assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor

status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.112 01101: The Blending Assembly Fails to Leave Y-direction Home Position

Error Code

01101

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending assembly fails to leave Y-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

In Y-direction initialization process for the blending assembly, after the blending assembly executes the command of leaving home position sensor in Y-direction, the Y-direction home position sensor should not be blocked, but it actually is.

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.113 01102: The Blending Assembly Fails to Return to Y-direction Home Position

Error Code

01102

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending assembly fails to return to Y-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

In Y-direction initialization process for the blending assembly, after the blending assembly executes the command of returning to home position sensor in Y-direction, the Y-direction home position sensor should not blocked, but it actually is not.

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.114 01103: The Blending Tube Clamp Y-direction Home Position Sensor Status Abnormal

Error Code

01103

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Y-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of moving from home position to mid position in Y-direction, the Y-direction home position sensor should not be blocked, but it actually is.

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.115 01104: The Blending Tube Clamp Y-direction Home Position Sensor Status Abnormal

Error Code

01104

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Y-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of stretching in Y-direction (before retracting in Y-direction), the Y-direction home position sensor should not be blocked, but it actually is.

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.116 01105: The Blending Tube Clamp Fails to Retract from Mid Position to Home Position in Y-direction

Error Code

01105

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to retract from mid position to home position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of finding the home position in Y direction, and the home position sensor is not triggered after a fixed number of steps

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.

2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.117 01106: The Blending Tube Clamp Y-direction Home Position Sensor Status Abnormal

Error Code

01106

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Y-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of moving from mid position to home position in Y-direction, the Y-direction home position sensor should be blocked, but it actually is not.

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor

status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.118 01107: The Blending Tube Clamp Y-direction Home Position Sensor Is Mistakenly Triggered

Error Code

01107

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The Blending tube clamp Y-direction home position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of moving from mid position to home position in Y-direction. After Y-direction home position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Sampling assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.119 01108: The Blending Tube Clamp Y-direction Home Position Sensor Status Abnormal

Error Code

01108

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Y-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of moving from end position to mid position in Y-direction, the Y-direction home position sensor should not be blocked, but it actually is.

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.

2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.120 01109: The Blending Tube Clamp Fails to Retract from Front Position to Home Position in Y-direction

Error Code

01109

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to retract from front position to home position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of finding the home position in Y direction, and the home position sensor is not triggered after a fixed number of steps

Possible Causes

1. Blending assembly Y-direction home position sensor Y_S_S4 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor

status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.121 01200: The Blending Tube Clamp Fails to Stretch out to Mid Position in Y-direction

Error Code

01200

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to stretch out to mid position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Y-direction mid position sensor with a fixed number of steps, but until the motor steps are all completed, the Y-direction mid position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending assembly Y-direction mid position sensor Y_M_S23 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.122 01201: The Blending Tube Clamp Stretches out to Mid Position in Y-direction, Sensor Mistakenly Triggered

Error Code

01201

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to stretch out to mid position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of moving from home position to mid position in Y-direction. After Y-direction mid position sensor has been triggered and the compensation steps have been completed, the Y-direction mid position sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending assembly Y-direction mid position sensor Y_M_S23 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.123 01202: The Blending Tube Clamp Fails to Retract to Mid Position in Y-direction

Error Code

01202

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to retract to mid position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of moving from front position to mid position in Y-direction, but until the motor steps are all completed, the Y-direction mid position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending assembly Y-direction mid position sensor Y_M_S23 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.124 01203: The Blending Tube Clamp Retracts to Mid Position in Y-direction, Sensor Mistakenly Triggered

Error Code

01203

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp retracts to mid position in Y-direction, sensor mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly

5. Motor board

Error Mechanism

The blending assembly executes the command of moving from front position to mid position in Y-direction. After Y-direction mid position sensor has been triggered and the compensation steps have been completed, the Y-direction mid position sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending assembly Y-direction mid position sensor Y_M_S23 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.125 01204: The Blending Tube Clamp Y-direction Mid Position Sensor Status Abnormal

Error Code

01204

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Y-direction mid position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of moving from mid position to home position or end position in Y-direction, the Y-direction mid position sensor should not be blocked, but it actually is.

Possible Causes

1. Blending assembly Y-direction mid position sensor Y_M_S23 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.

4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.126 01205: The Blending Tube Clamp Y-direction Mid Position Sensor Status Abnormal

Error Code

01205

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Y-direction mid position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of moving from home position or end position to mid position in Y-direction, the Y-direction mid position sensor should not be blocked, but it actually is.

Possible Causes

1. Blending assembly Y-direction mid position sensor Y_M_S23 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal

4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.127 01300: After the Blending Tube Clamp Moves from End Position to Home Position in Y-direction, End Position Sensor Is Blocked Unexpectedly

Error Code

01300

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Blending tube clamp Y-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending assembly executes the command of moving from front position to home position in Y-direction, the front position sensor after action completion should not be blocked, but it actually is.

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.128 01301: After the Blending Tube Clamp Stretches from Mid Position to End Position in Y-direction, End Position Sensor Is Mistakenly Triggered

Error Code

01301

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Blending tube clamp Y-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of moving to front position in Y-direction. After front position sensor has been triggered and the compensation steps have been completed, the Y-direction front position sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.129 01302: The Blending Tube Clamp Fails to Stretch out from Home Position to End Position in Y-direction

Error Code

01302

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to stretch out from home position to end position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Y-direction end position sensor with a fixed number of steps, but until the motor steps are all completed, the Y-direction end position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.130 01303: The Blending Tube Clamp Fails to Stretch out from Mid Position to End Position in Y-direction

Error Code

01303

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to stretch out from mid position to end position in Y-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Y-direction end position sensor with a fixed number of steps, but until the motor steps are all completed, the Y-direction end position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.131 01304: Before the Blending Tube Clamp Moves from Mid Position to End Position in Y-direction, End Position Sensor Is Blocked Unexpectedly

Error Code

01304

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Before the blending tube clamp moves from mid position to end position in Y-direction, end position sensor is blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Y-direction end position sensor with a fixed number of steps, but until the motor steps are all completed, the Y-direction end position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.132 01305: After the Blending Tube Clamp Stretches from Home Position to End Position Y-direction, End Position Sensor Is Mistakenly Triggered

Error Code

01305

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Tube gripper Y direction end position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly

3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the Y-direction end position sensor generates a non-blocked → blocked jump signal, the motor continues to take compensation steps. When the motor steps are all completed, it is found that the Y-direction end position sensor is not blocked

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.133 01306: After the Blending Tube Clamp Moves from End Position to Mid Position in Y-direction, End Position Sensor Is Blocked Unexpectedly

Error Code

01306

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	After the blending tube clamp moves from end position to mid position in Y-direction, end position sensor is blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the blending tube clamp executes the command of retracting from front position to mid position in Y-direction, the end position sensor after action completion should not be blocked, but it actually is.

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.134 01307: Before the Blending Tube Clamp Moves from Home Position to Mid Position in Y-direction, End Position Sensor Is Blocked Unexpectedly

Error Code

01307

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Before the blending tube clamp moves from home position to mid position in Y-direction, end position sensor is blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

Before the blending tube clamp moves from home position to mid position in Y-direction, end position sensor is blocked unexpectedly

Possible Causes

1. Blending tube clamp Y-direction end position sensor Y_C_S5 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.135 01401: The Blending Tube Clamp Z-direction Home Position Sensor Is Mistakenly Triggered

Error Code

01401

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction home position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending assembly executes the command of returning to the home position in Z-direction. After Z-direction home position sensor has been triggered and the compensation steps have been completed, the Z-direction home position sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.136 01402: The Blending Tube Clamp Fails to Leave Z-direction Home Position

Error Code

01402

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to leave Z-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

During initialization, the motor leaves the Z-direction home position with a fixed number of steps, but after the motor steps are all completed, the Z-direction home position sensor is still blocked

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.137 01403: The Blending Tube Clamp Fails to Return to Z-direction Home Position

Error Code

01403

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to leave Z-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Z-direction home position sensor with a fixed number of steps, but until the motor steps are all completed, the Z-direction home position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.138 01404: The Blending Tube Clamp Z-direction Home Position Sensor Status Abnormal

Error Code

01404

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

Before dropping, Z-direction home position sensor has been blocked

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.139 01405: The Blending Tube Clamp Fails to Drop in Z-Direction

Error Code

01405

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	tube gripper fails to drop in the Z direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Z-direction home position sensor with a fixed number of steps, but until the motor steps are all completed, the Z-direction home position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.140 01406: The Blending Tube Clamp Z-direction Home Position Sensor Is Mistakenly Triggered

Error Code

01406

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction home position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the Z-direction home position sensor generates a non-blocked → blocked jump signal, the motor continues to take compensation steps. When the motor steps are all completed, it is found that the Z-direction home position sensor is not blocked

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.141 01407: With Tube Straightening, the Blending Tube Clamp Z-direction Home Position Sensor Is Blocked Unexpectedly

Error Code

01407

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	With tube straightening, the blending tube clamp Z-direction home position sensor is blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

With tube straightening, the blending tube clamp Z-direction home position sensor is blocked unexpectedly

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.142 01408: The Blending Tube Clamp Z-direction Home Position Sensor Is Blocked Unexpectedly

Error Code

01408

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction home position sensor is blocked unexpectedly

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After rising, the Z-direction home position sensor is blocked unexpectedly

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.143 01409: The Blending Tube Clamp Fails to Drop in Z-Direction at Peripheral Blood Position

Error Code

01409

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to drop in Z-Direction at peripheral blood position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending tube clamp drops at mid position in Y-direction. After a fixed number of steps, the Z-direction home position sensor has never produced a non-blocked → blocked signal change

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.144 01410: The Blending Tube Clamp Z-direction Home Position Sensor Is Mistakenly Triggered at Peripheral Blood Position

Error Code

01410

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction home position sensor is mistakenly triggered at peripheral blood position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending tube clamp drops at the mid position in Y-direction. After Z-direction home position sensor has a non-blocked → blocked jump signal, and all compensation steps have been completed, the Z-direction home position sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.

4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.145 01411: The Blending Tube Clamp Z-direction Home Position Sensor Status Abnormal

Error Code

01411

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The Y-direction home position sensor is still blocked after rise in Z-direction.

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.146 01412: With Tube Straightening, the Blending Tube Clamp Z-direction Home Position Sensor Status Abnormal

Error Code

01412

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	With tube straightening, the blending tube clamp Z-direction home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly

3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

At micro tube position, when the blending tube clamp rises to the extent of tube straightening, the Z-direction initialization sensor shall not be blocked, but it actually is

Possible Causes

1. Blending assembly Z-direction home position sensor Z_B_S6 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.147 01500: The Blending Tube Clamp Z-direction End Position Sensor Status Abnormal

Error Code

01500

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

Before rising, Z-direction end position sensor has been blocked

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.148 01501: The Blending Tube Clamp Z-direction End Position Sensor Is Mistakenly Triggered

Error Code

01501

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The blending tube clamp rises in Z-direction. After Z-direction end position sensor has a non-blocked → blocked jump signal, and all compensation steps have been completed, the end position sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.149 01502: The Blending Tube Clamp Fails to Rise in Z-Direction

Error Code

01502

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to rise in Z-direction

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor

4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the Z-direction end position sensor with a fixed number of steps, but until the motor steps are all completed, the Z-direction end position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.150 01503: The Blending Tube Clamp Z-direction End Position Sensor Status Abnormal

Error Code

01503

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

In the initialization process, Z-direction end position sensor is still blocked after the tube clamp returns to the Z-direction home position

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.151 01504: The Blending Tube Clamp Z-direction End Position Sensor Status Abnormal

Error Code

01504

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

At micro blood position, Z-direction end position is still blocked after the movement to the lower position in Z-direction

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.152 01505: The Blending Tube Clamp Z-direction End Position Sensor Is Mistakenly Triggered at Peripheral Blood Position

Error Code

01505

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor is mistakenly triggered at peripheral blood position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly

5. Motor board

Error Mechanism

After the Z-direction end position sensor generates a non-blocked → blocked jump signal, the motor continues to take compensation steps. When the motor steps are all completed, it is found that the Z-direction end position sensor is not blocked

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.153 01506: The Blending Tube Clamp Z-direction End Position Sensor Status Abnormal

Error Code

01506

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

Before the peripheral blood blending position rises, Z-direction end position sensor has been blocked

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.154 01507: The Blending Tube Clamp Fails to Rise in Z-Direction at Peripheral Blood Position

Error Code

01507

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to rise in Z-Direction at peripheral blood position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

At peripheral blood position, the motor looks for the Z-direction end position sensor with a fixed number of steps, but until the motor steps are all completed, the Z-direction end position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.155 01508: The Blending Tube Clamp Z-direction End Position Sensor Status Abnormal

Error Code

01508

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor

4. Wires for blending assembly
5. Motor board

Error Mechanism

Before R-direction blending, the Z-direction end position sensor is not blocked

Possible Causes

1. Blending tube clamp Z-direction end position sensor Z_T_S7 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.156 01600: The Blending Tube Clamp R-direction Home Position Sensor Is Mistakenly Triggered

Error Code

01600

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp Z-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

After the R-direction home position sensor generates a non-blocked → blocked jump signal, the motor continues to take compensation steps. When the motor steps are all completed, it is found that the R-direction home position sensor is not blocked

Possible Causes

1. Blending tube clamp R-direction home position sensor R_S11 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.157 01601: The Blending Tube Clamp Fails to Leave R-direction Home Position

Error Code

01601

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to leave the R-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor leaves the R-direction home position with a fixed number of steps, but after the motor steps are all completed, the R-direction home position sensor is still blocked

Possible Causes

1. Blending tube clamp R-direction home position sensor R_S11 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.158 01602: The Blending Tube Clamp Fails to Return to R-direction Home Position

Error Code

01602

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to return to the R-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the R-direction home position sensor with a fixed number of steps, but until the motor steps are all completed, the R-direction home position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp R-direction home position sensor R_S11 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.159 01603: The Blending Tube Clamp R-direction End Position Sensor Status Abnormal

Error Code

01603

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp R-direction end position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor leaves the R-direction home position with a fixed number of steps, but after the motor steps are all completed, the R-direction home position sensor is still blocked

Possible Causes

1. Blending tube clamp R-direction home position sensor R_S11 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.160 01605: The Blending Tube Clamp Fails to Move to R-direction Home Position

Error Code

01605

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	The blending tube clamp fails to move to R-direction home position

Involved FRU

1. Photoelectric Position Sensor
2. Blending Assembly
3. Wire, rotary scanner and blending motor
4. Wires for blending assembly
5. Motor board

Error Mechanism

The motor looks for the R-direction home position sensor with a fixed number of steps, but until the motor steps are all completed, the R-direction home position sensor still does not generate a non-blocked → blocked jump signal

Possible Causes

1. Blending tube clamp R-direction home position sensor R_S11 abnormal
2. Blending assembly driving abnormal
3. Blending assembly wire abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform an initialization test on the blending assembly in the blending mechanism screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a self-test on the blending assembly. If the motor is observed no response and no running sound, then the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the blending assembly, see **8.11.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. Replace the wires for blending assembly.
5. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.161 02000: Scanning Mechanism Error

Error Code

02000

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Rotary scanning assembly action error
Service engineer level	Scanning mechanism error

Involved FRU

Main Control Board

Error Mechanism

Rotary scanning assembly software module abnormal

Possible Causes

Rotary scanning assembly software module abnormal

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.162 02001: Scanner Communication Abnormal

Error Code

02001

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Rotary scanning assembly action error
Service engineer level	Scanner communication abnormal

Involved FRU

N/A

Error Mechanism

After opening the scanner, the host computer cannot receive the communication signal

Possible Causes

Scanner error

Resolve Path

The error cannot be eliminated even after restart

Solution

Replace FRU

9.1.163 02002: Tube Fixing Mechanism Fails to Stretch

Error Code

02002

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Tube fixing mechanism fails to stretch

Involved FRU

1. Photoelectric Position Sensor
2. Rotary Compressing Assembly
3. Wire, rotary scanner and blending motor
4. Motor board

Error Mechanism

The tube fixing mechanism executes the command of stretching, and the fixing counter sensor does not generate the falling edge of the signal

Possible Causes

1. Fixing counter sensor C_S21 abnormal
2. Rotary scanning unit driving abnormal
3. Rotary scanning unit wiring abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If it is observed that the motor has no response and no running sound, the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.

4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the rotary compressing assembly, see **8.25.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.164 02003: Tube Fixing Mechanism Fails to Retract

Error Code

02003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Tube fixing mechanism fails to retract

Involved FRU

1. Photoelectric Position Sensor
2. Rotary Compressing Assembly
3. Wire, rotary scanner and blending motor
4. Motor board

Error Mechanism

The tube fixing mechanism executes the command of retracting, and the fixing counter sensor does not generate the rising edge of the signal

Possible Causes

1. Fixing counter sensor C_S21 abnormal
2. Rotary scanning unit driving abnormal
3. Rotary scanning unit wiring abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If it is observed that the motor has no response and no running sound, the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the rotary compressing assembly, see **8.25.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.165 02004: Tube Fixing Mechanism Fails to Leave Home Position

Error Code

02004

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Tube fixing mechanism failed to leave the home position

Involved FRU

1. Photoelectric Position Sensor
2. Rotary Compressing Assembly
3. Wire, rotary scanner and blending motor
4. Motor board

Error Mechanism

The tube fixing mechanism leaves the home position. After a fixed number of steps, the fixing counter sensor is blocked unexpectedly

Possible Causes

1. Fixing counter sensor C_S21 abnormal
2. Rotary scanning unit driving abnormal
3. Rotary scanning unit wiring abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If it is observed that the motor has no response and no running sound, the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the rotary compressing assembly, see **8.25.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.166 02005: Tube Fixing Mechanism Fails to Return to Home Position

Error Code

02005

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Tube fixing mechanism failed to return to home position

Involved FRU

1. Photoelectric Position Sensor
2. Rotary Compressing Assembly
3. Wire, rotary scanner and blending motor
4. Motor board

Error Mechanism

The tube fixing mechanism returns to the home position. After a fixed number of steps, the fixing counter sensor is not blocked unexpectedly

Possible Causes

1. Fixing counter sensor C_S21 abnormal
2. Rotary scanning unit driving abnormal
3. Rotary scanning unit wiring abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If it is observed that the motor has no response and no running sound, the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the rotary compressing assembly, see **8.25.2** Disassembly and Assembly.

3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.167 02006: Sensor Mistakenly Triggered When Tube Fixing Mechanism Returns to Home Position

Error Code

02006

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Sensor mistakenly triggered during tube fixing mechanism returning to home position

Involved FRU

1. Photoelectric Position Sensor
2. Rotary Compressing Assembly
3. Wire, rotary scanner and blending motor
4. Motor board

Error Mechanism

The tube fixing mechanism executes the command of returning to the home position. After the tube fixing counter sensor has been triggered and the compensation steps have been completed, the sensor is not blocked unexpectedly.

Possible Causes

1. Fixing counter sensor C_S21 abnormal
2. Rotary scanning unit driving abnormal
3. Rotary scanning unit wiring abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Blending Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If it is observed that the motor has no response and no running sound, the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the rotary compressing assembly, see **8.25.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.168 02007: Tube Fixing Mechanism Slide Rail Jammed

Error Code

02007

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Mix assembly action abnormal
Service engineer level	Tube fixing mechanism slide rail jammed

Involved FRU

1. Photoelectric Position Sensor
2. Rotary Compressing Assembly
3. Wire, rotary scanner and blending motor
4. Motor board

Error Mechanism

The remaining steps after the tube fixing mechanism triggers the sensor are not equal to the expected steps

Possible Causes

1. Fixing counter sensor C_S21 abnormal

2. Rotary scanning unit driving abnormal
3. Rotary scanning unit wiring abnormal
4. Motor board abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Blending Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If the test fails several times, replace this FRU.
3. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a test on the fixing assembly in the autoloading screen. If it is observed that the motor has no response and no running sound, the hardware environment is abnormal. Observe whether the wires of the motor and sensor are normal.
4. If the wires are confirmed normal, consider replacing the motor board.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the rotary compressing assembly, see **8.25.2** Disassembly and Assembly.
3. To replace the connection wire between the rotary scanner and the blending motor sensor, see **8.111.2** Disassembly and Assembly.
4. To replace the motor board, see **8.79.2** Disassembly and Assembly.

9.1.169 03000: Sample Type Detection Sensor Is Blocked Unexpectedly

N/A

9.1.170 03001: Tube Misplaced in Tube Rack

N/A

9.1.171 04000: Autoloading Status Error

Error Code

04000

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader action abnormal
Service engineer level	Autoloading status error (software)

Involved FRU

Main Control Board

Error Mechanism

The current state of the autoloader cannot respond to the command from host computer

Possible Causes

The current state of the autoloader cannot respond to the command from host computer

Resolve Path

N/A

Solution

1. Copy the log and return to R&D.
2. Upgrade the software.
3. If the error cannot be eliminated after software upgrading, replace the main control board FRU and upgrade the software after replacement. See PCBA, main control board and iMX8 **8.8.2** Disassembly and Assembly.

9.1.172 04001: Tube Rack Present in Feeding Channel

Error Code

04001

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Tube racks are left in the analysis area
Service engineer level	Tube racks are left in the analysis area

Involved FRU

N/A

Error Mechanism

Before the feeding mechanism executes the first feeding command, test tube racks not pushed out are left in the feeding channel

Possible Causes

Before the feeding mechanism executes the first feeding command, test tube racks not pushed out are left in the feeding channel

Resolve Path

Observe whether tube racks are left in the analysis area

Solution

Remove the tube rack

9.1.173 04002: Tube Rack Abnormally Moved

Error Code

04002

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Manually move of tube rack

Involved FRU

N/A

Error Mechanism

In the process of motor returning to the right home position after feeding from the left, the counter sensor is not blocked

Possible Causes

The loader is disturbed by the external environment

Resolve Path

During the switching between the left and right feeding positions of the lateral feeding assembly, the tube rack in the loader may be moved manually, or the counter may be subject to unexpected pressure and impact

Solution

Click "Remove Error"

9.1.174 04003: Wrong Counter Count Jump Times

Error Code

04003

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Wrong counter count jump times

Involved FRU

Counter unit

Error Mechanism

During the movement of the feeding mechanism, the jump times of the left and right counters do not meet the expectations

Possible Causes

1. Abnormal status of left and right feeding positions
2. Counter unit abnormal

Resolve Path

1. In the autoloader self-test screen, observe whether the single feed by the lateral feeding assembly can feed the tube rack in place.
2. Keep the count unit plug connected, conduct compressing test of the counter and make the counter rebound freely. Select **Status** >>**Sensor Status** in turn. Observe whether the status of the corresponding sensor changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

Solution

1. Observe and debug the left and right feeding positions of the loader.
2. Replace the counter unit

9.1.175 04004: Tube Rack Abnormally Moved

Error Code

04004

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Manually move of tube rack

Involved FRU

N/A

Error Mechanism

When the feeding mechanism executes the 1-bit feed command, the jump times of the counter do not meet the expectations

Possible Causes

The loader is disturbed by the external environment

Resolve Path

During the switching between the left and right feeding positions of the lateral feeding assembly, the tube rack in the loader may be moved manually, or the counter may be subject to unexpected pressure and impact

Solution

Click "Remove Error"

9.1.176 04100: Loading Mechanism Fails to Leave Home Position

Error Code

04100

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism fails to leave home position

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the loading mechanism executes the command of leaving the home position, the home position sensor should not be blocked, but it actually is

Possible Causes

1. Loading home position sensor LS_S1 abnormal
2. Loading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an assembly initialization test on the loading assembly in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.177 04101: Loading Mechanism Fails to Return to Home Position

Error Code

04101

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism fails to return to the home position

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the loading mechanism executes the command of returning to the home position, the home position sensor should be blocked, but it actually is not

Possible Causes

1. Loading home position sensor LS_S1 abnormal
2. Loading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an assembly initialization test on the loading assembly in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.178 04102: Loading Mechanism Jammed

Error Code

04102

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism jammed

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The remaining steps after the loading mechanism triggers the sensor are not equal to the expected steps

Possible Causes

1. The user did not place the test tube rack properly
2. Loading home position sensor LS_S1 abnormal
3. Loading assembly driving abnormal

Resolve Path

1. Observe, and restart auto-loading
2. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
3. Check whether foreign matters are present in the loader that may hinder the movement of the assembly.
4. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a loading test on the loading assembly in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. Remove Error
2. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
3. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.

4. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.179 04103: Loading Mechanism Home Position Sensor Status Abnormal

Error Code

04103

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism home position sensor status error

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the loading mechanism executes the loading action, the loading home position sensor should not be blocked, but it actually is

Possible Causes

1. Loading home position sensor LS_S1 abnormal
2. Loading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Autoloader Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.180 04104: Loading Mechanism End Position Sensor Status Abnormal

Error Code

04104

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism end position sensor status error

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

1. After the loading mechanism executes the action of returning to the home position, the end position sensor should not be blocked, but it actually is
2. After the loading mechanism executes the loading action, the end position sensor should be blocked, but it actually is not

Possible Causes

1. Loading end position sensor LE_S abnormal
2. Loading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a loading test on the loading assembly in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.181 04105: Sensor Mistakenly Triggered When Loading Mechanism Returns to Home Position

Error Code

04105

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism home position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The loading assembly executes the command of returning to the home position. After home position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Loading end position sensor LE_S abnormal
2. Loading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.182 04106: Loading End Position Sensor Is Mistakenly Triggered

Error Code

04106

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Loading mechanism end position sensor is mistakenly triggered

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The loading assembly executes the loading command. After end position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Loading end position sensor LE_S abnormal
2. Loading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, loosen the sensor screw, take out the sensor, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a loading test on the loading assembly in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.183 04107: Autoloading Sensor Abnormal

Error Code

04107

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Autoloading sensor abnormal

Involved FRU

1. PHOTOELEC emitter 900nm 400mm conn
2. PHOTOELEC detector 900nm 400mm conn

Error Mechanism

After the autoloading function is enabled, the loading mechanism executes the loading command, and the micro switch is not triggered. At this time, the autoloading sensor should not be blocked, but it actually is

Possible Causes

1. Autoloading sensor (AD/AE) abnormal
2. Poor contact

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Check whether the sensor connector is reliable.

Solution

1. To replace PHOTOELEC emitter 940nm 400mm conn, see **8.30.2** Disassembly and Assembly.
2. To replace PHOTOELEC detector 900nm 400mm conn, see **8.31.2** Disassembly and Assembly.
3. Check whether the sensor connector is reliable.

9.1.184 04200: Unloading Mechanism Fails to Leave Home Position

Error Code

04200

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Unloading mechanism fails to leave home position

Involved FRU

1. Photoelectric position sensor (10-position loader)
2. Photoelectric position sensor (5-position loader)
3. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
4. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the unloading mechanism executes the command of leaving the home position, the home position sensor should not be blocked, but it actually is

Possible Causes

1. Unloading home position sensor UL_S11 abnormal
2. Unloading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.185 04201: Unloading Mechanism Fails to Return to Home Position

Error Code

04201

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Unloading mechanism fails to return to the home position

Involved FRU

1. Photoelectric position sensor (10-position loader)
2. Photoelectric position sensor (5-position loader)
3. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
4. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The unloading assembly executes the command of returning to the home position. After home position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Unloading home position sensor UL_S11 abnormal
2. Unloading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.186 04202: Unloading Tray Is Full

Error Code

04202

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Unloading tray is full
Service engineer level	Unloading tray is full

Involved FRU

PHOTOELEC photosensor reflective 940nm

Error Mechanism

The unloading tray full sensor is blocked

Possible Causes

1. Unloading tray is full
2. Unloading tray full detection sensor UF_S13 abnormal

Resolve Path

1. Observe whether the unloading area is full, and timely remove the tube racks after measurement
2. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

Solution

1. Remove the tube racks in time.
2. To replace PHOTOELEC sensor reflective 940nm, see .

9.1.187 04203: Sensor Mistakenly Triggered When Unloading Mechanism Returns to Home Position

Error Code

04203

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sensor mistakenly triggered when unloading mechanism returns to home position

Involved FRU

1. Photoelectric position sensor (10-position loader)
2. Photoelectric position sensor (5-position loader)
3. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
4. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The unloading assembly executes the command of returning to the home position. After home position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Unloading home position sensor UL_S11 abnormal
2. Unloading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.188 04204: Unloading Mechanism Fails to Stretch

Error Code

04204

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Unloading mechanism protruding failed

Involved FRU

1. Photoelectric position sensor (10-position loader)
2. Photoelectric position sensor (5-position loader)
3. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
4. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The unloading mechanism executes the command of stretching. After a fixed number of steps, the unloading home position sensor is still blocked

Possible Causes

1. Unloading home position sensor UL_S11 abnormal
2. Unloading assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Check whether foreign matters are present in the loader that may hinder the movement of the assembly.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an unloading test on the unloading assembly in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.189 04300: Feeding Mechanism Fails to Leave the Sensor in Initialization

Error Code

04300

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Failed to break away from the sensor during feed mechanism initialization

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the feeding mechanism executes the command of leaving the home position, the right position sensor should not be blocked, but it actually is

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.190 04301: Feeding Mechanism Fails to Return to Home Position

Error Code

04301

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Failed to break away from the sensor during feed mechanism initialization

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the feeding mechanism executes the command of returning to the right home position, the right position sensor should be blocked, but it actually is not

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.191 04302: Feeding Mechanism Fails to Return to Right Position Sensor

Error Code

04302

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Feeding mechanism fails to return to right position sensor

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the feeding assembly executes the command of returning and finding the right home position, the right position sensor is still not triggered after a fixed number of steps

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.192 04303: Feeding Mechanism Fails to Leave Home Position

Error Code

04303

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Feeding mechanism fails to leave the home position

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the feeding mechanism executes the command of leaving the home position, the right position sensor is still not triggered to generate blocked→non-blocked jump after a fixed number of steps

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** **>>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the Autoloader Motor Board Sensor page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.193 04304: Sensor Mistakenly Triggered When Feeding Mechanism Returns to Home Position

Error Code

04304

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sensor mistakenly triggered when feeding mechanism returns to home position

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding assembly executes the command of returning to the home position. After the right position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal

2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.194 04305: Sensor Mistakenly Triggered When Feeding Mechanism Returns to Right Position

Error Code

04305

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sensor mistakenly triggered when feeding mechanism returns to right position

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding mechanism executes the command of right resetting. After the right position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.195 04306: Step Loss in Feeding Mechanism Left Resetting

Error Code

04306

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Step loss in feeding mechanism left resetting

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding mechanism executes the command of left resetting. After the left sensor is triggered, the remaining steps of feeding motor are not equal to the expected steps

Possible Causes

1. Left feeding home position sensor FL_S8 abnormal

2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Check whether foreign matters are present in the loader that may hinder the movement of the assembly.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.196 04307: Step Loss in Feeding Mechanism Right Resetting

Error Code

04307

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Step loss in feeding mechanism right resetting

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding mechanism executes the command of left resetting. After the left sensor is triggered, the remaining steps of feeding motor are not equal to the expected steps

Possible Causes

1. Right feeding home position sensor FR_S7 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Check whether foreign matters are present in the loader that may hinder the movement of the assembly.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.197 04308: Feeding Mechanism Left Sensor Abnormal in Initialization

Error Code

04308

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Feeding mechanism left sensor abnormal in initialization

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the feeding mechanism executes the command of initializing, the left position sensor should not be blocked at any time, but it actually is

Possible Causes

1. Left feeding home position sensor FL_S8 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a feeding test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.198 04309: Feeding Mechanism Fails to Reach Left Position Sensor

Error Code

04309

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Feeding mechanism fails to reach left position sensor

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

After the feeding mechanism executes the command of moving the pusher to the left side, the left position sensor is still not triggered after a fixed number of steps

Possible Causes

1. Left feeding home position sensor FL_S8 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Check whether foreign matters are present in the loader that may hinder the movement of the assembly.
3. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a feeding test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.199 04310: Sensor Mistakenly Triggered When Feeding Mechanism Reaches Left Position

Error Code

04310

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sensor mistakenly triggered when feeding mechanism reaches left position

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding mechanism executes the command of moving the pusher to the left side. After the left position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Left feeding home position sensor FL_S8 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a feeding test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.200 04311: Same Barcode for Adjacent Positions on the Tube Rack

Error Code

04311

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Same barcode for adjacent positions on the tube rack

Involved FRU

1. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
2. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding mechanism executes the command of feeding the tube rack. The feeding pusher does not rise, resulting in immobility of the tube rack, so the scanner scans the same barcode twice in a row

Possible Causes

Lateral feeding assembly driving abnormal

Resolve Path

If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a feeding test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
2. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.201 04312: Step Loss in Feeding Mechanism Going Back

Error Code

04312

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Step loss in feeding mechanism going back

Involved FRU

1. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
2. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

The feeding mechanism executes the command of left resetting or right resetting in the process of going back. After the left and right position sensors are triggered, the remaining steps of feeding motor are not equal to the expected steps

Possible Causes

Lateral feeding assembly driving abnormal

Resolve Path

1. Check whether foreign matters are present in the loader that may hinder the movement of the assembly.
2. Select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform a going back test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
2. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.202 04400: Counter Sensor Status Error

Error Code

04400

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Counter sensor status error

Involved FRU

1. Photoelectric Position Sensor
2. 5X6 Closed-sampling autoloader (Select according to the existing analyzer configuration)
3. 10X5 Closed-sampling autoloader (Select according to the existing analyzer configuration)

Error Mechanism

1. Before and after the movement of the feeding mechanism, the left and right counter sensors should be blocked, but they are actually in the non-blocked state
2. In the movement process of the feeding mechanism (where the feeding pusher does not rise), the left and right counter sensors should be blocked, but they are actually in the non-blocked state

Possible Causes

1. Left feeding home position sensor FL_S8 abnormal
2. Lateral feeding assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform a feeding test in the autoloader screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the 5X6 closed-sampling autoloader, see **8.28.2** Disassembly and Assembly.
3. To replace the 10X5 closed-sampling autoloader, see **8.35.2** Disassembly and Assembly.

9.1.203 04500: Sample Compartment Fails to Leave Home Position

Error Code

04500

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sample compartment fails to leave home position

Involved FRU

1. Photoelectric Position Sensor
2. Sample compartment assembly

Error Mechanism

The sample compartment executes the command of leaving the home position sensor, and the home position sensor of the closed compartment does not change from the blocked state to the non-blocked state after a fixed number of steps

Possible Causes

1. Sample compartment home position sensor CL_S5 abnormal
2. Closed sample compartment assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in Other Components screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. Replace the sample compartment assembly.

9.1.204 04501: Sample Compartment Fails to Return to Home Position

Error Code

04501

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sample compartment fails to return to home position

Involved FRU

1. Photoelectric Position Sensor
2. Sample compartment assembly

Error Mechanism

The sample compartment executes the command of returning to the home position sensor, and the home position sensor of the closed compartment does not change from the non-blocked state to the blocked state after a fixed number of steps

Possible Causes

1. Sample compartment home position sensor CL_S5 abnormal
2. Closed sample compartment assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in Other Components screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. Replace the sample compartment assembly.

9.1.205 04502: Sample Compartment Fails to Open

Error Code

04502

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sample compartment failed to open

Involved FRU

1. Photoelectric Position Sensor
2. Sample compartment assembly

Error Mechanism

The sample compartment executes the opening process. After a fixed number of steps, the sample compartment should not be blocked, but it actually is

Possible Causes

1. Sample compartment home position sensor CL_S5 abnormal
2. Closed sample compartment assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. Check whether foreign matters are present in the loader that may hinder the movement of the assembly. Note whether the bottom guide rail is rusted.
3. Select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an opening test in Other Components screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. Replace the sample compartment assembly.

9.1.206 04503: Sensor Mistakenly Triggered When Sample Compartment Returns to Home Position

Error Code

04503

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sensor mistakenly triggered when sample compartment returns to home position

Involved FRU

1. Photoelectric Position Sensor
2. Sample compartment assembly

Error Mechanism

The sample compartment executes the command of returning to the home position sensor. After sample compartment home position sensor has been triggered and the compensation steps have been completed, the sensor shall be blocked, but it is actually in the non-blocked state

Possible Causes

1. Sample compartment home position sensor CL_S5 abnormal
2. Closed sample compartment assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an opening test in Other Components screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. Replace the sample compartment assembly.

9.1.207 04504: Sample Compartment Home Position Sensor Status Abnormal

Error Code

04504

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sample compartment home position sensor status abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Sample compartment assembly

Error Mechanism

Before and after the sample compartment executes the opening or closing command, the compartment home position sensor state is inconsistent with the expected blocking state

Possible Causes

1. Sample compartment home position sensor CL_S5 abnormal
2. Closed sample compartment assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. Select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an opening test in Other Components screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. Replace the sample compartment assembly.

9.1.208 04505: Sample Compartment Status Error

Error Code

04505

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sample compartment status error

Involved FRU

1. Photoelectric Position Sensor
2. Sample compartment assembly

Error Mechanism

The <opening> or <closing> status of the sample compartment is inconsistent with the expected blocking state of sample compartment home position

Possible Causes

1. Sample compartment home position sensor CL_S5 abnormal
2. Closed sample compartment assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. Select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an opening test in Other Components screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. Replace the sample compartment assembly.

9.1.209 04600: Sensor Mistakenly Triggered When Micro-blood Mixer Returns to Home Position

Error Code

04600

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Sensor mistakenly triggered when micro-blood mixer returns to home position

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

In the initialization process, the micro-blood mixer executes the command of returning to the home position sensor. After home position sensor has been triggered and the compensation steps have been completed, the sensor shall not be blocked, but it is actually in the blocked state

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.210 04601: Micro-blood Mixer Fails to Leave Home Position

Error Code

04601

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer fails to leave home position

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

In the initialization process, the micro-blood mixer executes the command of leaving the home position sensor. The sensor is still not triggered to generate non-blocked→blocked jump after a fixed number of steps

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.211 04602: Micro-blood Mixer Fails to Return to Home Position

Error Code

04602

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer fails to return to home position

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

In the initialization process, the micro-blood mixer executes the command of returning to the home position sensor. The sensor is still not triggered to generate blocked→non-blocked jump after a fixed number of steps

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.212 04603: Micro-blood Mixer Sensor Status Abnormal

Error Code

04603

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer fails to return to home position

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

When the micro-blood mixer executes the blending command, the status of blending home position sensor does not meet the expectation

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.

2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.213 04604: Micro-blood Mixer Sensor Jump Abnormal

Error Code

04604

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer sensor jump abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

When the micro-blood mixer executes the blending command, the number of times saved by the blending counter does not meet the expectation

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.214 04605: Sensor Mistakenly Triggered When Micro-blood Mixer Returns to Home Position

Error Code

04605

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer sensor jump abnormal

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

In the blending process, the micro-blood mixer executes the command of returning to the home position sensor. After home position sensor has been triggered and the compensation steps have been completed, the sensor shall not be blocked, but it is actually in the blocked state

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >**>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>**>Commissioning & Self-Test**>**>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.215 04606: Micro-blood Mixer Fails to Leave Home Position

Error Code

04606

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer fails to leave home position

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

In the blending process, the micro-blood mixer executes the command of leaving the home position sensor. The sensor is still not triggered to generate non-blocked→blocked jump after a fixed number of steps

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status >>Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service>>Commissioning & Self-Test>>Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.216 04607: Micro-blood Mixer Fails to Return to Home Position

Error Code

04607

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader abnormal
Service engineer level	Micro-blood mixer fails to return to home position

Involved FRU

1. Photoelectric Position Sensor
2. Peripheral Blood Blending Assembly

Error Mechanism

In the blending process, the micro-blood mixer executes the command of returning to the home position sensor. The sensor is still not triggered to generate blocked→non-blocked jump after a fixed number of steps

Possible Causes

1. Micro blending home position sensor MB_S14 abnormal
2. Peripheral blood blending assembly driving abnormal

Resolve Path

1. Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the sensor status changes accordingly on the **Autoloader Motor Board Sensor** page. If the status does not change, the sensor fails.
2. If the photoelectric position sensor test is normal, select **Service**>>**Commissioning & Self-Test**>>**Self-Test** in turn, and perform an initialization test in the blending mechanism screen. If the test fails several times, replace this autoloader FRU.

Solution

1. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.
2. To replace the peripheral blood blending assembly, see .

9.1.217 06000: Latex Probe Fails to Leave Lower Position

N/A

9.1.218 06001: Latex Probe Fails to Return Lower Position

N/A

9.1.219 06002: Latex Probe Lower Sensor Is Blocked Unexpectedly

N/A

9.1.220 06003: Latex Probe Fails to Move to Upper Position

N/A

9.1.221 06005: Latex Probe Upper Sensor Is Blocked Unexpectedly

N/A

9.1.222 06007: Latex Probe Lower Sensor Is Mistakenly Triggered

N/A

9.1.223 06008: Latex Probe Upper Sensor Is Mistakenly Triggered

N/A

9.1.224 06009: Specific-Protein Motor Board Communication Timeout

N/A

9.1.225 99800: Startup Initiation Not Performed

Error Code

99800

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Startup initiation not performed
Service engineer level	Startup initiation not performed

Involved FRU

N/A

Error Mechanism

The alarm is triggered when the startup process is interrupted

Possible Causes

1. The startup process is interrupted by the error
2. Unknown Cause

Resolve Path

1. Check the alarm information during the startup
2. None of the above methods can solve the problem

Solution

1. Restart after removing the error that interrupts the startup process.
2. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.226 99801: Exiting Short Standby Status Failed

Error Code

99801

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Exiting standby status failed
Service engineer level	Exiting standby status failed

Involved FRU

N/A

Error Mechanism

This alarm will be triggered when the process of exiting shallow standby status is interrupted

Possible Causes

1. The process of exiting shallow standby status is interrupted
2. Unknown Cause

Resolve Path

1. Check the alarm information during the standby exiting
2. None of the above methods can solve the problem

Solution

1. Restart the analyzer after removing the error that interrupts the startup process.
2. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.227 99802: Exiting Long Standby Status Failed

Error Code

99802

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Exiting standby status failed
Service engineer level	Exiting standby status failed

Involved FRU

N/A

Error Mechanism

This alarm will be triggered when the process of exiting deep standby status is interrupted

Possible Causes

1. The process of exiting shallow standby status is interrupted
2. Unknown Cause

Resolve Path

1. Check the alarm information during the standby exiting
2. None of the above methods can solve the problem

Solution

1. Restart the analyzer after removing the error that interrupts the startup process.
2. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.228 99805: Background Abnormal

Error Code

99805

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Background abnormal
Service engineer level	Background abnormal

Involved FRU

N/A

Error Mechanism

The background results tested during the startup process are out of specification range

WBC ≤ 0.10 × 109 /L

RBC ≤ 0.02× 1012/L

HGB ≤ 1 g/L

PLT ≤ 3 × 109 /L

Possible Causes

1. Diluent abnormal
2. Abnormal operations such as long-term standby of the analyzer and unsuccessful shutdown
3. Unknown Cause

Resolve Path

1. Observe the open-vial date of the diluent and check whether the diluent is crystallized or contaminated.
2. View the log and check whether there is any abnormality in the last startup and last shutdown of the analyzer.

3. None of the above methods can solve the problem.

Solution

1. Replace with new diluent and perform overall cleaning.
2. Execute probe cleanser maintenance and cleaning (This is for the whole device).
3. Collect the INF original data and logs of the error, and transfer them to R&D for processing.

9.1.229 99804: Front Cover Opened

Error Code

99804

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Front cover is open
Service engineer level	Front cover is open

Involved FRU

Photoelectric Position Sensor

Error Mechanism

The cover is not closed, or the set-top sensor is invalid

Possible Causes

The cover is not closed, or the set-top sensor is invalid

Resolve Path

Keep the sensor plug connected, use an opaque object to block it, and select **Status** >>**Sensor Status** in turn. Observe whether the status of cover-closing sensor changes accordingly on the **Motherboard Sensor** page. If the status does not change, the sensor fails.

Solution

1. Confirm whether the front cover is open.
2. To replace the photoelectric position sensor, see **8.20.2** Disassembly and Assembly.

9.1.230 99806: Maintenance Failed

Error Code

99806

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Maintenance failed
Service engineer level	Maintenance failed

Involved FRU

N/A

Error Mechanism

Report in case of probe cleanser maintenance failure (pop-up shows limited or interrupted use after liquid filling) and shutdown failure

Possible Causes

Maintenance interruption due to other faults

Resolve Path

Check whether there are other faults leading to maintenance interruption before this fault occurs

Solution

Remove the faults according to the prompt

9.1.231 04005: Autoloader Board Communication Timeout

Error Code

04005

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	Autoloader board communication out of time
Service engineer level	Autoloader board communication out of time

Involved FRU

Autoloader motor board

Error Mechanism

Software communication timeout

Possible Causes

Software communication timeout

Resolve Path

N/A

Solution

1. Check whether the communication line is connected normally.
2. If yes, copy the log and return to R&D.
3. Upgrade the software according to advice from R&D.
4. If the fault cannot be eliminated after software upgrading, replace the autoloader motor board FRU and upgrade the software after replacement.

9.1.232 20204: CS Analyzer Fan Clogged

N/A

9.1.233 20205: CS Board Fan Clogged

N/A

9.1.234 07204: Horizontal Action Execution Time of Sampling Assembly Is Greater Than Expected Time of Software

Error Code

07204

Error Description

Access Level for Viewing Faults	Corresponding Error Description under Access Level
User and administrator levels	Sampling assembly action abnormal
Servicer engineer level	Sampling assembly actions in horizontal direction time out.

Involved FRU

Main control board

Severity

/

Handling Level

/

Error Mechanism

The horizontal action execution time of the sampling assembly is greater than the expected time.

Possible Causes

Sampling assembly actions in horizontal direction time out.

Resolve Path

/

Solution

1. Copy the log and return it to the R&D department.
2. Upgrade the software.
3. After the software is upgraded, if the fault cannot be eliminated, replace the main control board FRU, and then upgrade the software.

9.2 Errors without Codes

9.2.1 39900M: WBC Scattergram Abnormal

Error Code

39900M

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	WBC Scattergram Abn
Service engineer level	WBC Scattergram Abn

Involved FRU

N/A

Error Mechanism

The distribution of leukocyte particle clusters was different from that of normal samples, and the distribution of different particle clusters was overlapped.

Possible Causes

1. Scenarios with classification results: Dirty flow cell; insufficient dye addition; inappropriate gain settings; sheath fluid instability due to clogged sheath fluid related valves.
2. Results of scenarios without classification: Dirty flow cell; insufficient dye addition; inappropriate gain settings; sheath fluid instability due to clogged sheath fluid related valves; laser failure; poor hemolysis; high circuit noise.

Resolve Path

Check the basic status:
1. Background verification.
2. QC verification.
3. Replace the dye.
4. Replace the lyse.
5. Optical gain calibration.
6. Flow cell maintenance.

Troubleshooting operation: Enter the status screen to view: Temperature, pressure, laser current, optical baseline.

Solution

According to the troubleshooting operation, if the corresponding FRU is abnormal, replace it.

9.2.2 39901M: RET Analysis Abnormal

Error Code

39901M

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	RET Analy. abn.
Service engineer level	RET Analy. abn.

Involved FRU

N/A

Error Mechanism

The fluorescence center of gravity of the RBC particle cluster is significantly lower or there is a serious error in the RET measurement process

Possible Causes

Particle stability abnormal

Resolve Path

1. Check for leaks in the valves related to the sample preparation line.
2. Replace the SP syringe.

Solution

According to the troubleshooting operation, if the corresponding FRU is abnormal, replace it.

9.2.3 39902M: PLT_O Analysis Abnormal

Error Code

39902M

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	PLT_O Analy. abn.
Service engineer level	PLT_O Analy. abn.

Involved FRU

N/A

Error Mechanism

There is an error in the secondary background of PLT_O

Possible Causes

DS diluent is dirty

Resolve Path

1. Replace the DS diluent.
2. Replace the DS container cap assembly.

Solution

According to the troubleshooting operation, if the corresponding FRU is abnormal, replace it. To replace DS container cap assembly, see **8.69.2** Disassembly and Assembly.

9.2.4 39903M: DIFF Analysis Abnormal

Error Code

39903M

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	PLT_O Analy. abn.
Service engineer level	PLT_O Analy. abn.

Involved FRU

N/A

Error Mechanism

The DIFF measurement process may be seriously faulty

Possible Causes

Particle stability abnormal

Resolve Path

1. Check for leaks in the valves related to the sample preparation line.

2. Replace the SP syringe.

Solution

According to the troubleshooting operation, if the corresponding FRU is abnormal, replace it.

9.2.5 39904M: WBC Analysis Abnormal

N/A

9.2.6 39905X: WBC Not Classified

N/A

9.2.7 39906X: WBC Results with *

N/A

9.2.8 39907: RET Scattergram Abnormal

Error Code

39907M

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	RET Scattergram abnormal
Service engineer level	RET Scattergram abnormal

Involved FRU

N/A

Error Mechanism

The fluorescence center of gravity of the RBC particle cluster is significantly lower or there is a serious error in the RET measurement process

Possible Causes

1. Insufficient fluorescent signal
2. Abnormal RBC volume
3. High noise

Resolve Path

1. Measure the QC, and judge whether the addition of FR dye is abnormal through the QC scattergram. If the QC chart is abnormal, replace the FR dye or check whether the FR dye addition channel is abnormal.
2. Replace the DR diluent or check whether the DR diluent addition channel is abnormal.

Solution

According to the troubleshooting operation, if the corresponding FRU is abnormal, replace it.

9.2.9 39908X: WBC QC Results Out of Range

N/A

9.2.10 39909X: WBC Results High

N/A

9.2.11 39910X: WBC Results Low

N/A

9.2.12 39911X: RET Results High

N/A

9.2.13 39912X: RET Results Low

N/A

9.2.14 39700M: RBC Analysis Abnormal

N/A

9.2.15 39701M: PLT Analysis Abnormal

N/A

9.2.16 39702X: RBC Results with *

N/A

9.2.17 39703M: RBC Histogram Abnormal

N/A

9.2.18 39704X: PLT Results with *

N/A

9.2.19 39705M: PLT Histogram Abnormal

N/A

9.2.20 39706X: RBC QC Results Out of Range

N/A

9.2.21 39707X: PLT QC Results Out of Range

N/A

9.2.22 39708X: MCV QC Results Out of Range

N/A

9.2.23 39709X: RBC Results High

N/A

9.2.24 39710X: RBC Results Low

N/A

9.2.25 39711X: PLT Results High

N/A

9.2.26 39712X: PLT Results Low

N/A

9.2.27 39713X: MCV Results High

N/A

9.2.28 39714X: MCV Results Low

N/A

9.2.29 39500M: ESR Analysis Abnormal

N/A

9.2.30 39501X: ESR Results with *

N/A

9.2.31 39502X: ESR Calibration Failed

N/A

9.2.32 39503X: ESR QC Results Out of Range

N/A

9.2.33 39504X: ESR Results High

N/A

9.2.34 39505X: ESR Results Low

N/A

9.2.35 39600M: CRP Analysis Abnormal

N/A

9.2.36 39601M: SAA Analysis Abnormal

N/A

9.2.37 39602X: CRP Results with *

N/A

9.2.38 39603X: SAA Results with *

N/A

9.2.39 39604X: CRP Calibration Failed

N/A

9.2.40 39605X: SAA Calibration Failed

N/A

9.2.41 39606X: CRP QC Results Out of Range

N/A

9.2.42 39607X: SAA QC Results Out of Range

N/A

9.2.43 39609X: CRP Results Low

N/A

9.2.44 39610X: SAA Results High

N/A

9.2.45 39611X: SAA Results Low

N/A

9.2.46 39800M: HGB Analysis Abnormal

N/A

9.2.47 39801X: HGB Results with *

N/A

9.2.48 39802M: HGB Interference/Turbidity

N/A

9.2.49 39803X: HGB Calibration Failed

N/A

9.2.50 39804X: HGB QC Results Out of Range

N/A

9.2.51 39805X: HGB Results High

N/A

9.2.52 39806X: HGB Results Low

N/A

9.2.53 19900X: Liquid Leakage/Overflow

N/A

9.2.54 19901X: DIFF Reaction Bath Leakage/Overflow

N/A

9.2.55 19902X: RET Reaction Bath Leakage/Overflow

N/A

9.2.56 19903X: WNB Bath Leakage/Overflow

N/A

9.2.57 19904X: HGB Reaction Bath Leakage/Overflow

N/A

9.2.58 19905X: CRP/SAA Measurement Assembly Leakage/Overflow

N/A

9.2.59 19906X: SCI Bath Leakage

N/A

9.2.60 19907X: Preheating Bath Leakage

N/A

9.2.61 19908X: WC Bath Leakage

N/A

9.2.62 19909X: Optical System Leakage

N/A

9.2.63 19910X: Syringe Leakage

N/A

9.2.64 19911X: Solenoid Valve Leakage

N/A

9.2.65 19912X: Waste Pump Leakage

N/A

9.2.66 19913X: Fluid Drop Suspends on or Falls from Sampling Probe Tip

N/A

9.2.67 07099X: Abnormal Noise (Sampling Assembly)

Error Code

07099X

Error Description

View the access level of the error	Error description under this access level
Operator and administrator's level	N/A
Service engineer level	Abnormal noise of sampling assembly

Involved FRU

Sampling assembly

Error Mechanism

Abnormal noise of sampling assembly

Possible Causes

Abnormal noise of sampling assembly

Resolve Path

N/A

Solution

To replace the sampling assembly, see **8.13.2** Disassembly and Assembly.

9.2.68 99901X: Cover Damaged

N/A

9.2.69 99902X: Touch Screen Invalid

N/A

9.2.70 99903X: Peripheral Errors

N/A

9.2.71 99904X: Hand-Held Scanner

N/A

9.2.72 99905X: External Printer

N/A

9.2.73 99906X: Computer (PC)

N/A

9.2.74 99907X: Keyboard/Mouse

N/A

9.2.75 99801X: Analyzer Not Started/Power Off

N/A

9.2.76 99802X: Black Screen/Blue Screen/Blurred Screen

N/A

9.2.77 99803X: Analyzer Generating Smoke

N/A

9.2.78 99804X: Crash/Screen Frozen

N/A

9.2.79 99805X: Host Display Broken

N/A

10 Analyzer Emptying and Relocation

10.1 Overview of Relocation and Packaging

When the analyzer needs to be relocated, or not used for a long time, or needs to be transported for a long distance, refer to the contents of this chapter.

10.2 Relocation Preparations

Tools

Tool List

Type	Name	Quantity
Tools for Relocation	Scissors or cutting pliers	1
	Inner hexagon spanner (M8)	1
	Cross-headed screwdriver (M3)	1
	Cross-headed screwdriver (M4)	1
	Empty container (capacity: □ 5L, open)	1

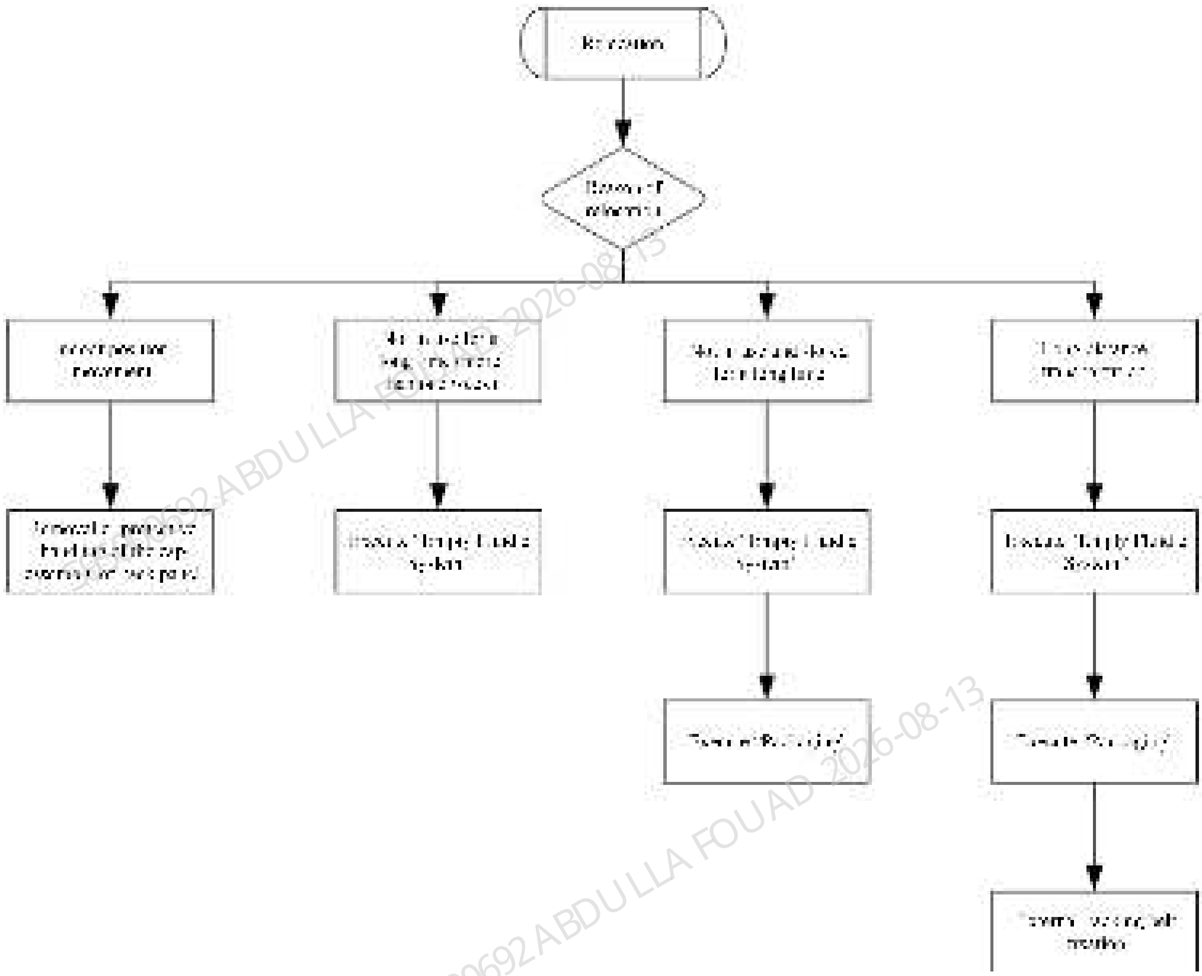
Consumables

Consumable list

Type	Name	Quantity
Frequently-used	Deionized water	5L
	Compact bag	20
	Wire tie	20
	Gloves	2
	High temperature adhesive paper	1 (roll)
	Tissues	1 (roll)

Type	Name	Quantity
	Duct tape	1 (roll)
	Binding straps	1 (roll)

10.3 Relocation Procedure



10.4 Main Unit Emptying Procedure

1. Select **Menu**>>**Service**>>**Maintenance**, select **Overall maintenance**, and perform **Packaging**, as shown below:

Figure10–1 Fluidic Packaging screen

The analyzer pops up the following prompt box, click **Yes** to confirm the execution of fluidic packaging operation.

Figure10–2 Fluidic Packaging screen

2. The analyzer will pop up the following prompt box. Take out the container cap assemblies on the rear panel except the waste container cap from the reagent container, and put the container cap assemblies in an empty clean wide-mouth container, as shown in the figure below. Take the fluorescent dye bag cap assembly on the front panel out of the reagent bag and place it in a clean compact bag or other clean location. When finished, click **OK** to proceed to the next step.

Figure10–3 Container cap assembly lifting prompt

Figure10–4 Container cap assembly removal diagram

3. After the reagents inside the analyzer are drained, the following prompt box will pop up. Pour at least 50mL of deionized water into the wide-mouth container.

Figure10–5 Fluidic packaging screen - Container cap assembly, pour deionized water

Figure10–6 Placing the container cap assembly in the deionized water

4. After the deionized water priming is completed, the analyzer will pop up the following prompt box. Take out the container cap assembly from the wide-mouth container, then pour out all the deionized water, and place the container cap assembly back into the empty wide-mouth container, as shown in the figures A and B. Place the dye bag cap assembly in a clean compact bag or other clean locations. When finished, click **OK** to proceed to the next step.

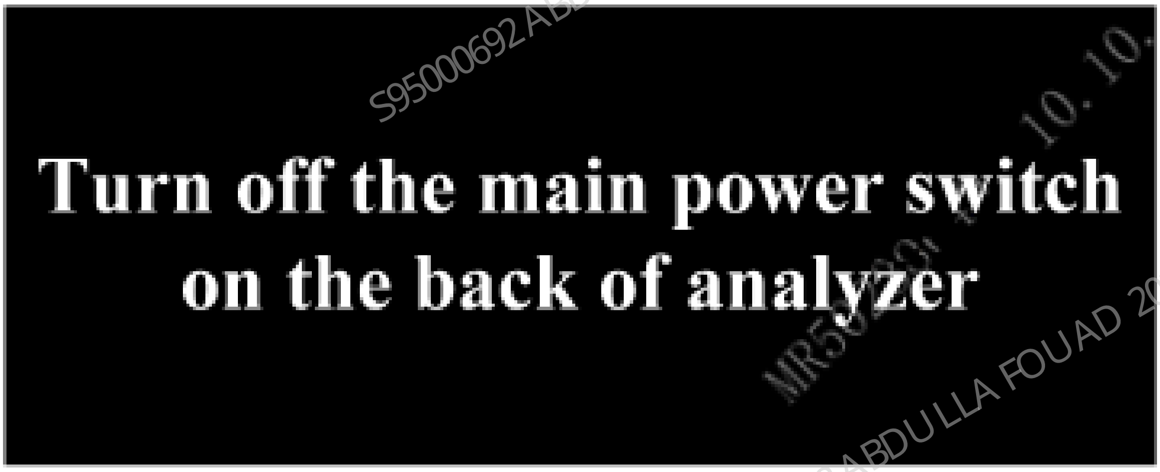
Figure10–7 Fluidic packaging screen - Container cap assembly vacant

Figure10–8 Removing the container cap assembly from the container (A)

Figure10–9 Dumping the water out of the container and putting it back into the cap assembly (B)

5. After the analyzer completes all emptying actions, the following prompt box will pop up. Power off the analyzer directly.

Figure10–10 Power off prompt



10.5 Data Clearing

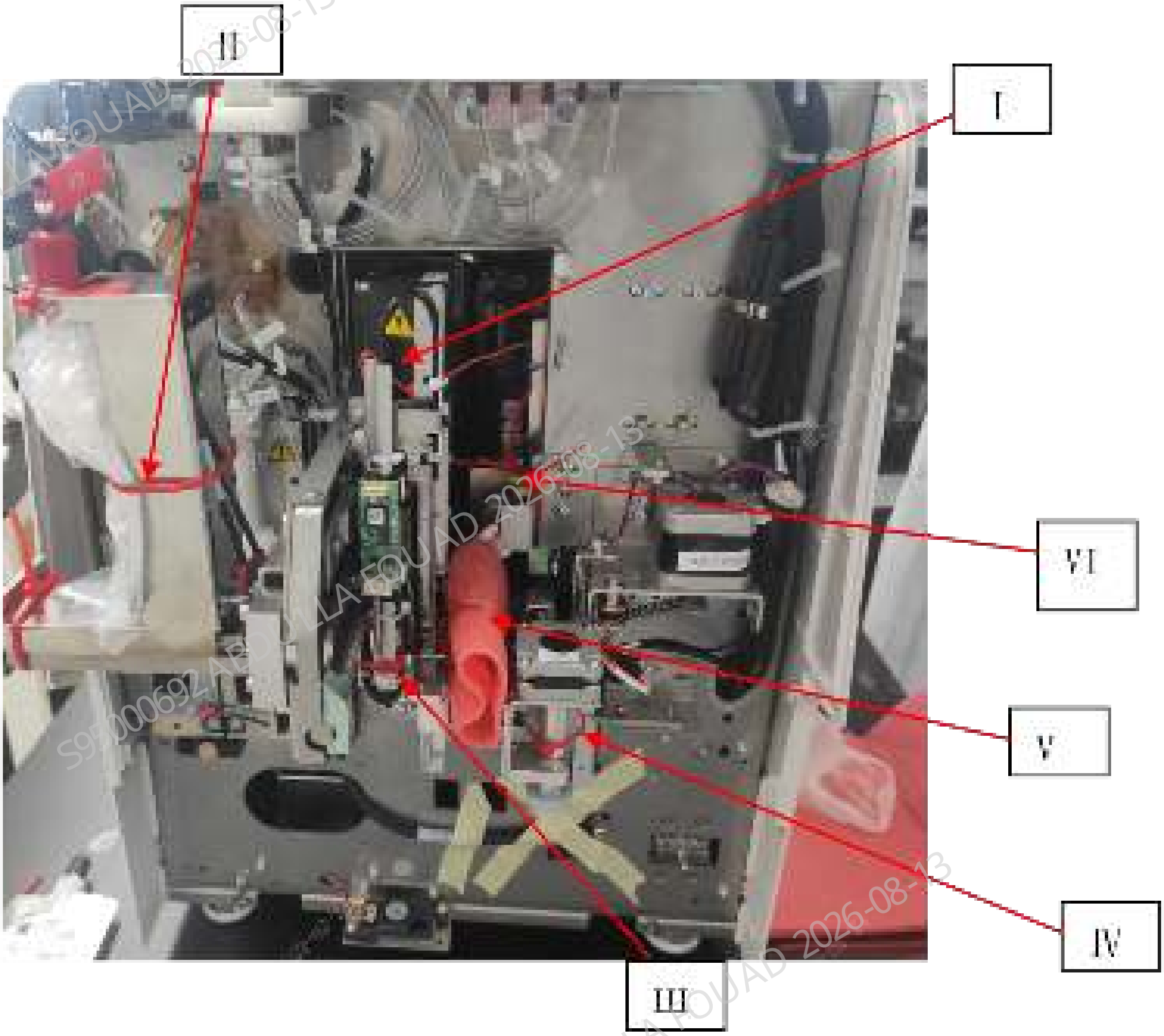
The analyzer automatically clears the data during the packaging operation, so you do not need to clear the data separately.

10.6 Overall packing

Pack the analyzer in a box and operate according to the following requirements:

1. Check the appearance of the analyzer, and remove the external power cord, container cap assembly, network cable, etc. of the analyzer.
2. Pack the back panel container cap assembly in a compact bag, and install the front panel fluorescent dye bag cap assembly onto the fluorescent reagent holder after packed in a compact bag. Place the cap assembly and related accessories in the accessory box.
3. Lift up the front cover of the analyzer, and use wire ties to fix the sampling assembly, fluorescent reagent container cap assembly, rotary compressing assembly and blending assembly as shown in the figure below.

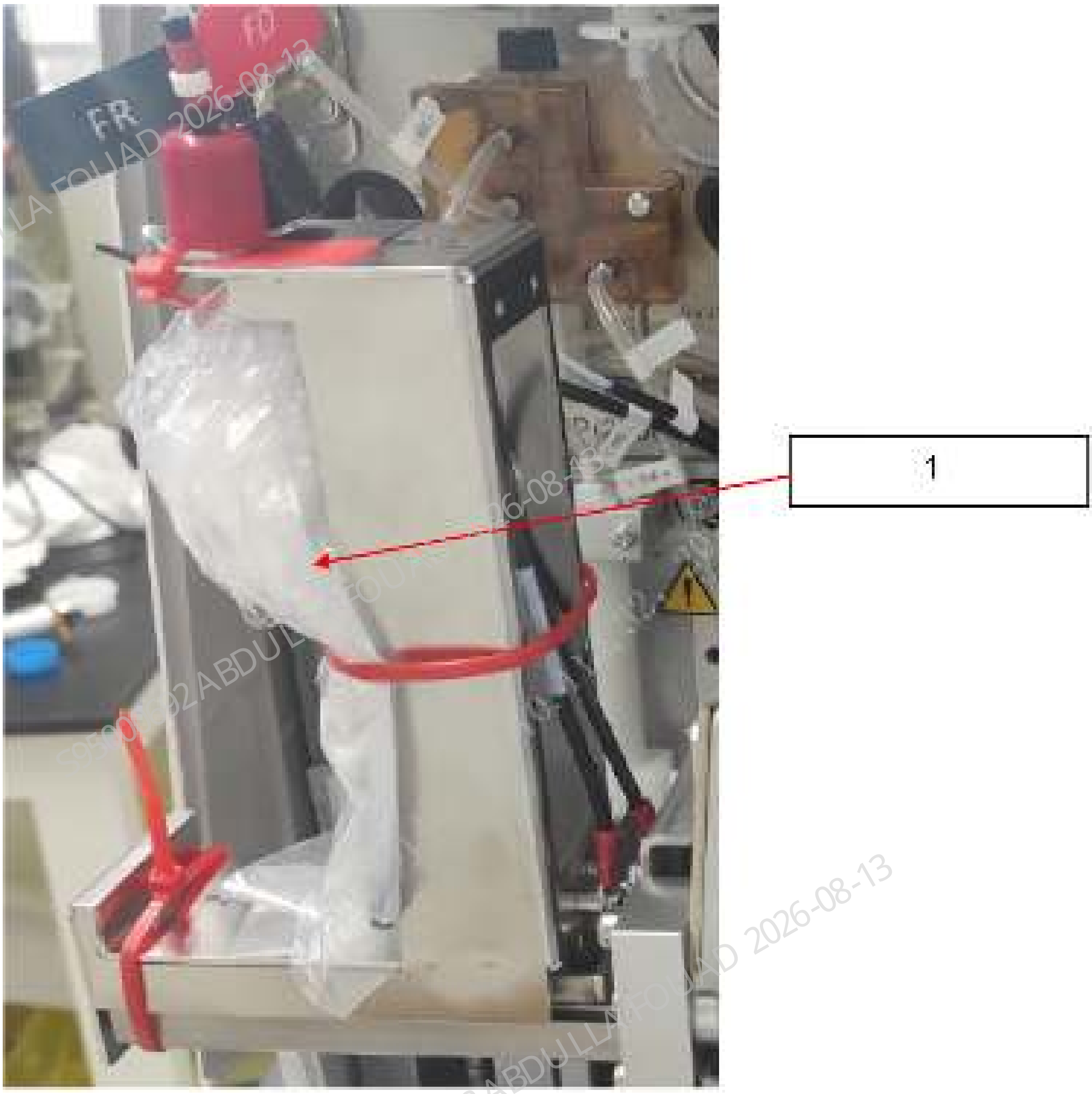
Figure10–11 Layout of wire ties



□	At sampling Z-axis assembly
□	At dye container cap assembly
III	At sampling tube rack
□	At rotary compressing assembly*
V	At tube clamp*
VI	At blending Z-axis assembly

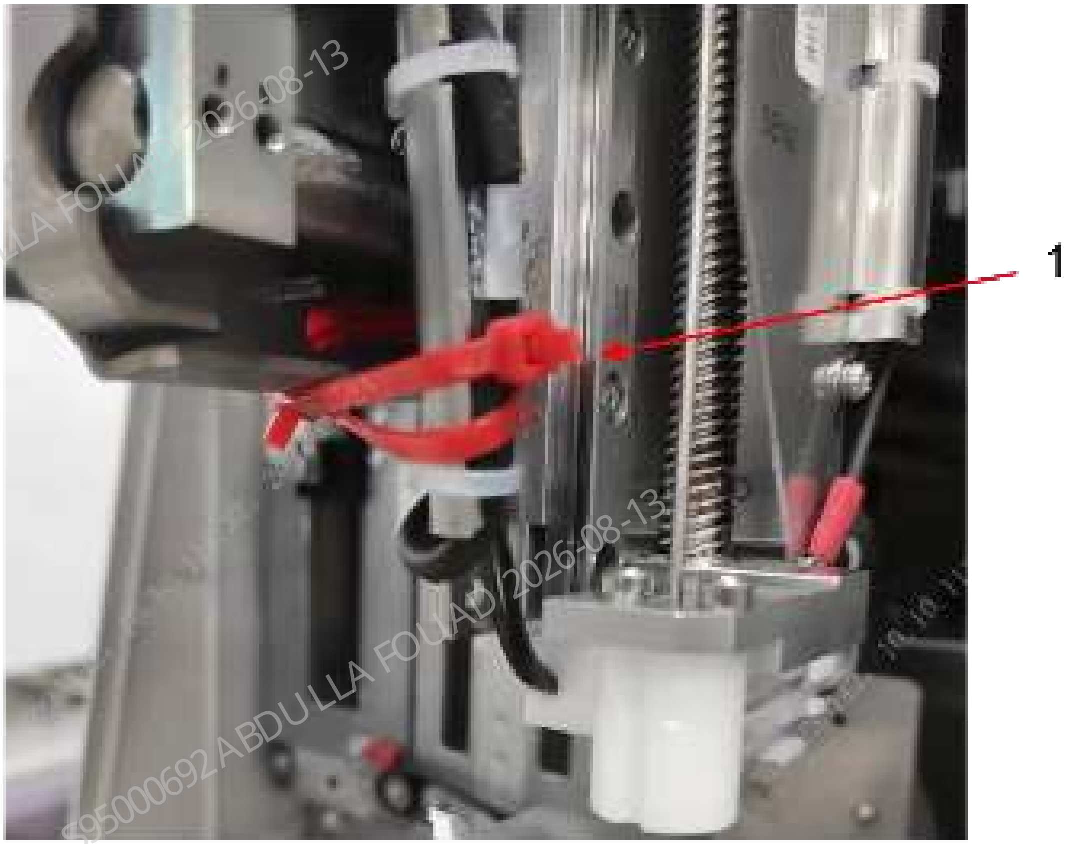
(The places with * only involve the autoloader models, not the open models)

I: Position of sampling Z-axis assembly



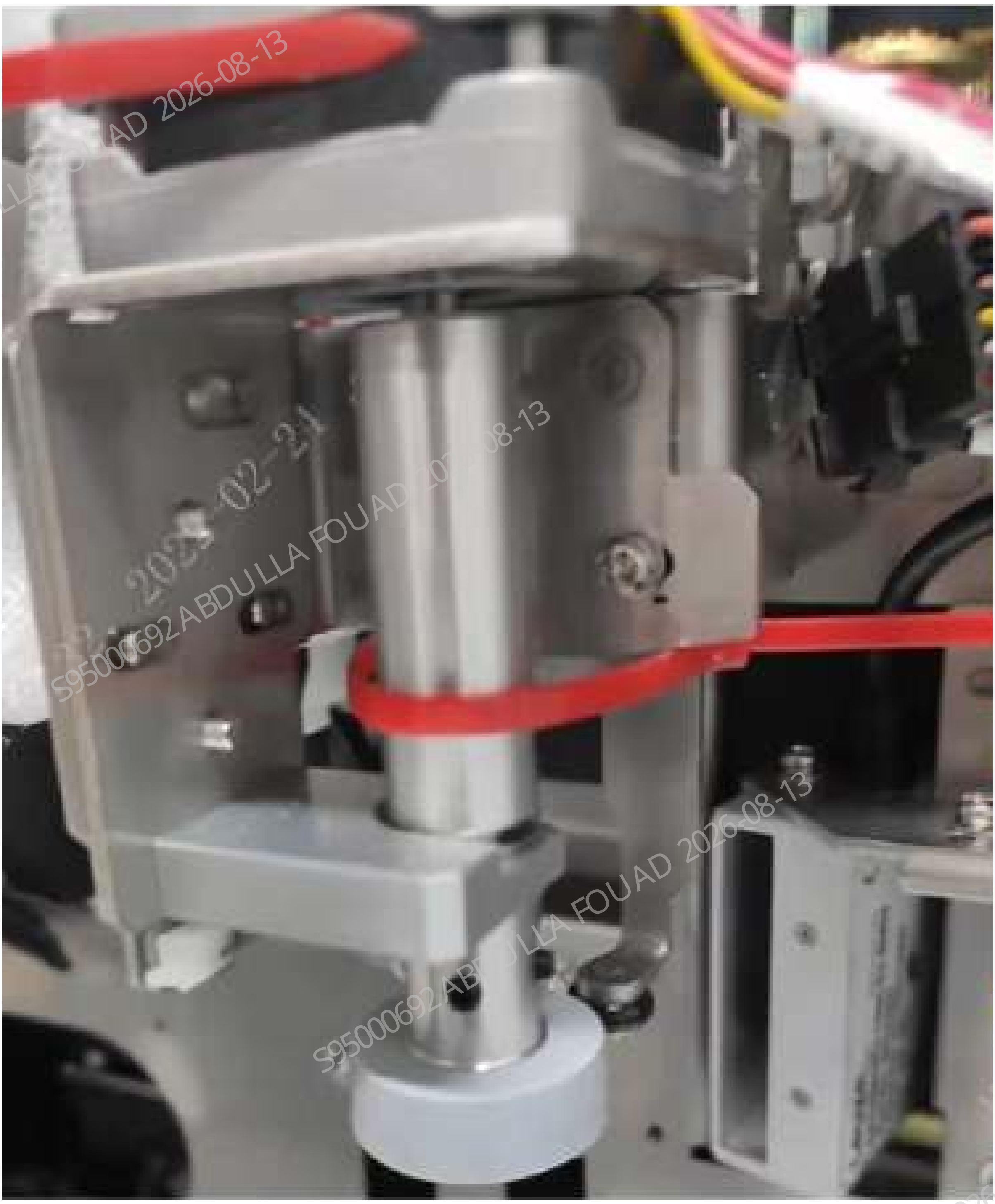
1 The compact bag covers the dye probe

□: Dye container cap assembly



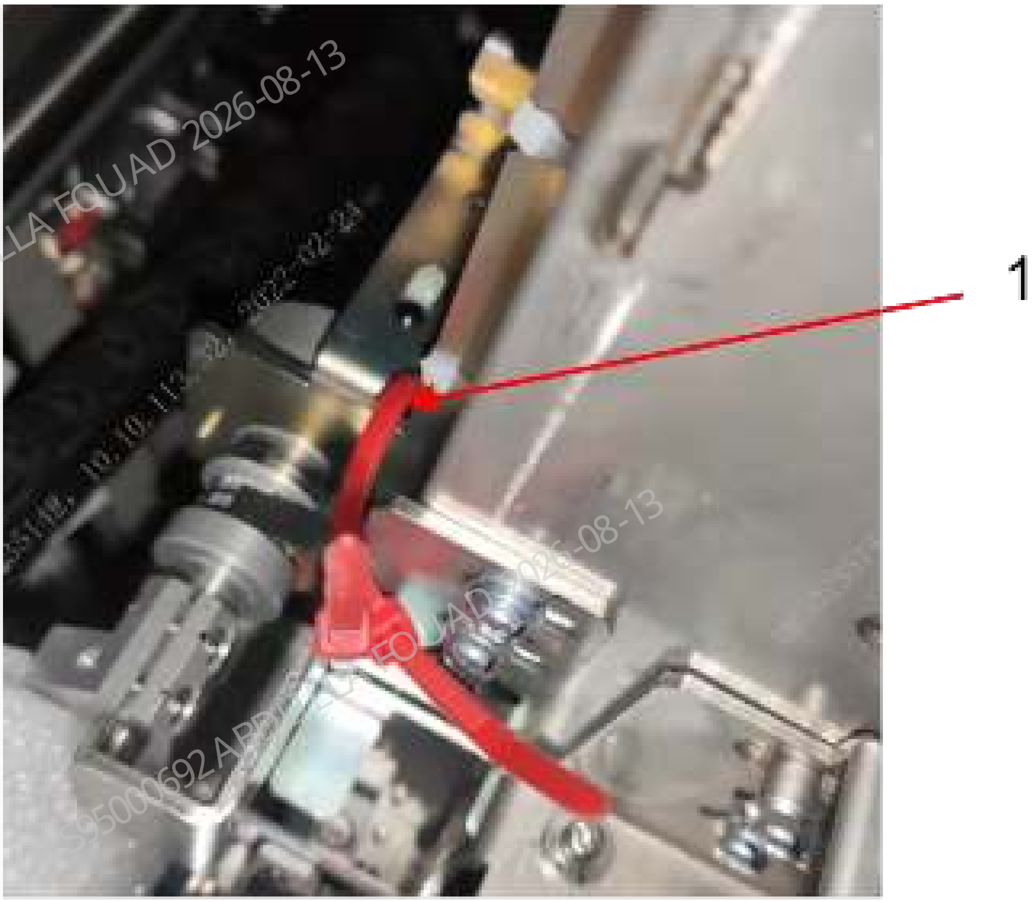
1 Two wire ties

III: Sampling tube rack



IV. Rotary compressing assembly

V. Tube clamp

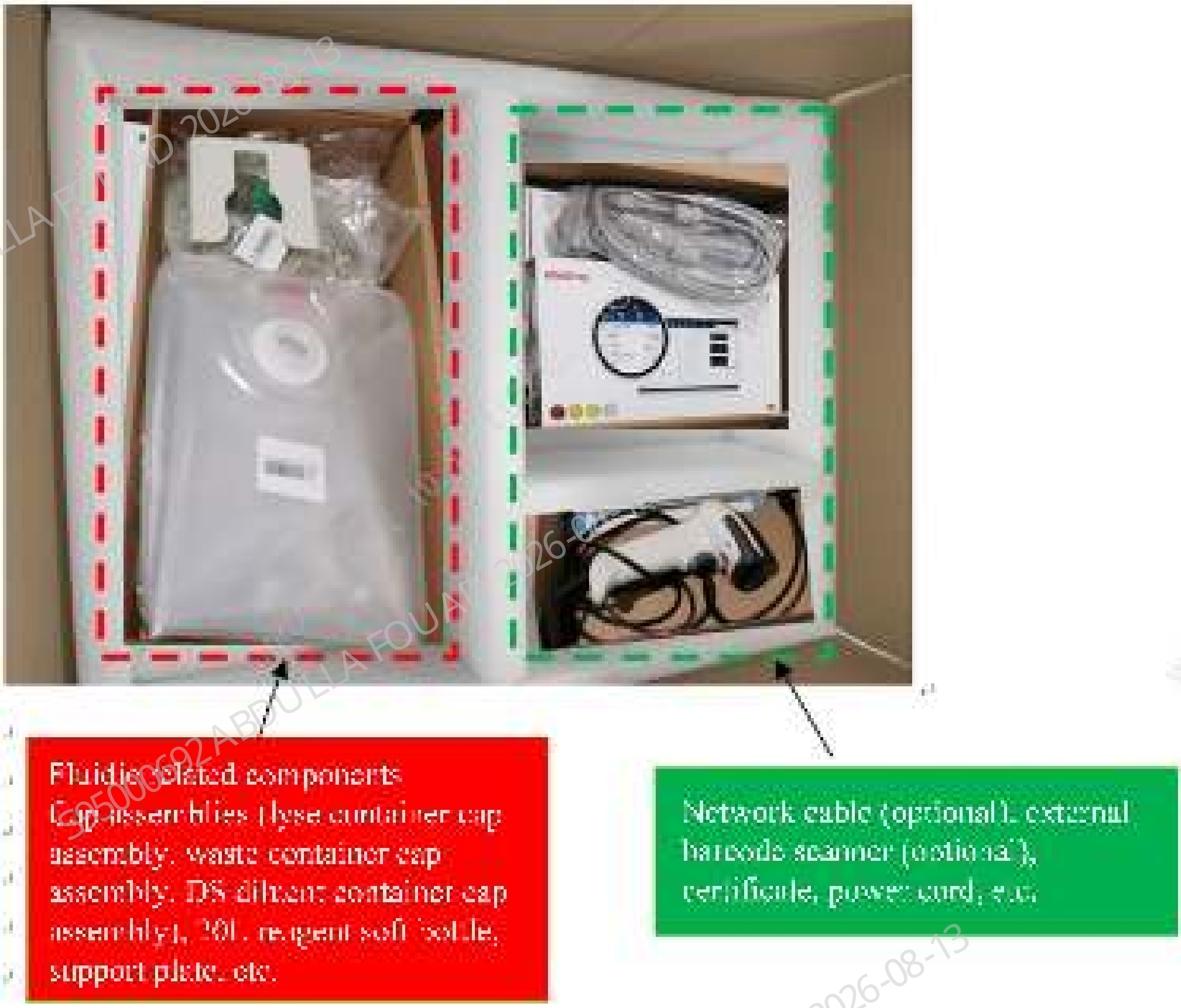


1 Two wire ties

VI. Blending Z-axis assembly

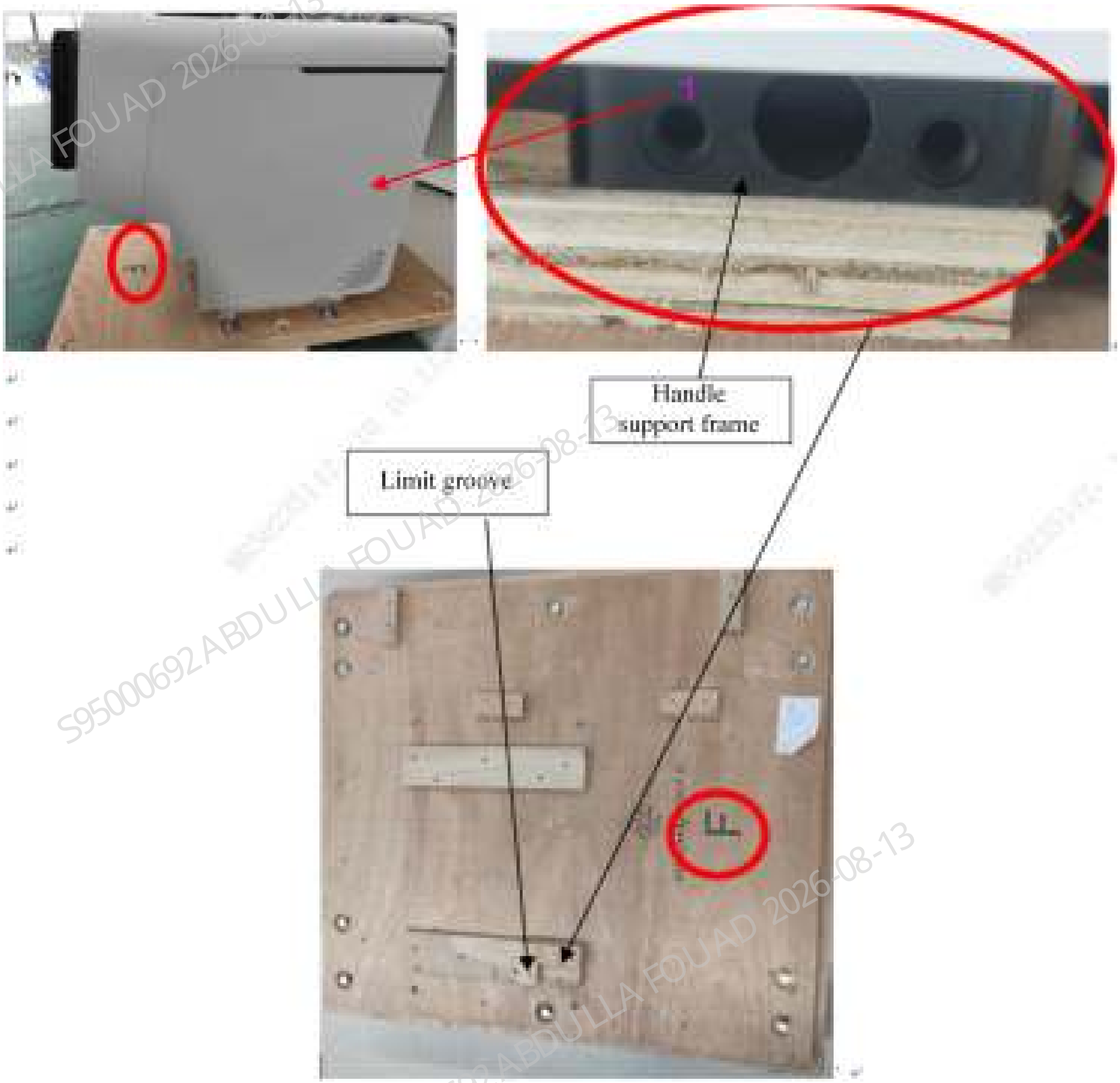
Description of packing the main unit

Seal the accessories with appropriate compact bags, then put them into the accessory box and seal them with transparent tape, as shown in the figure below.



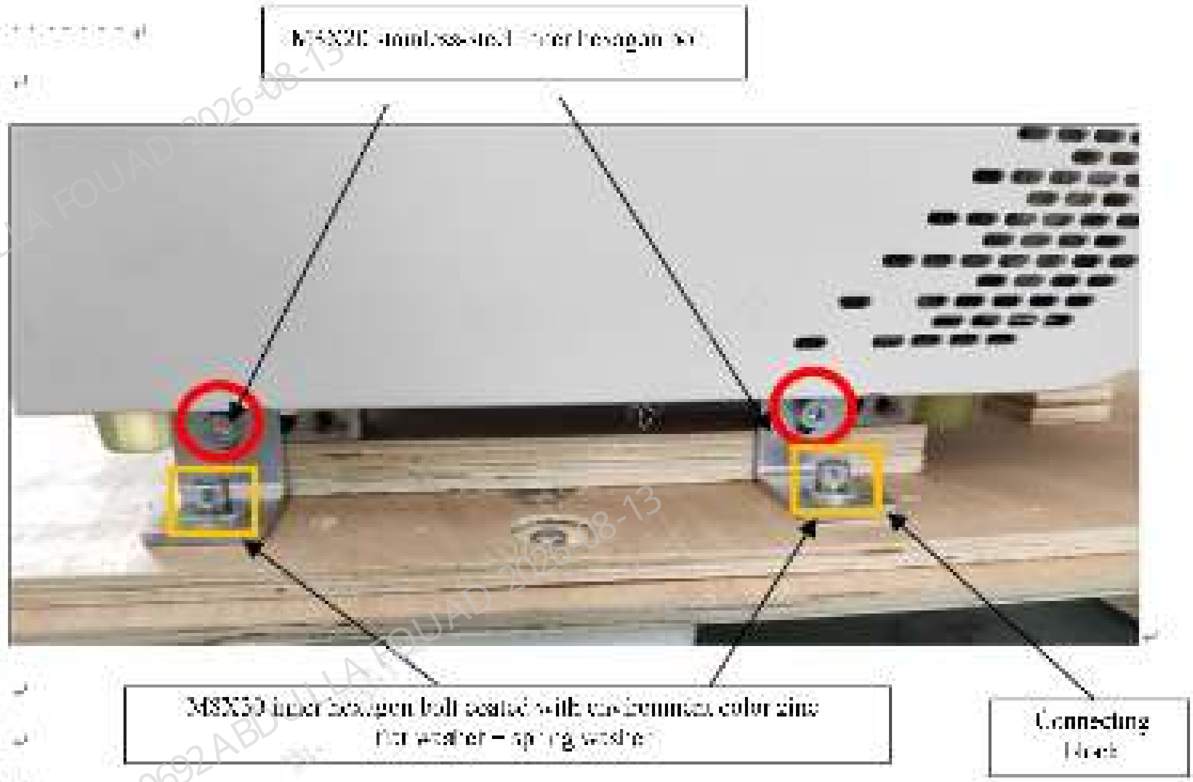
Place the main unit on the wooden base with handle, where the word **F** on the base is in front of the main unit. Stand in front of the main unit, and take the limit slot as the positioning benchmark to limit the left handle support frame, as shown in the figure below.

Figure10–12 Placing the Main Unit (Analyzer)

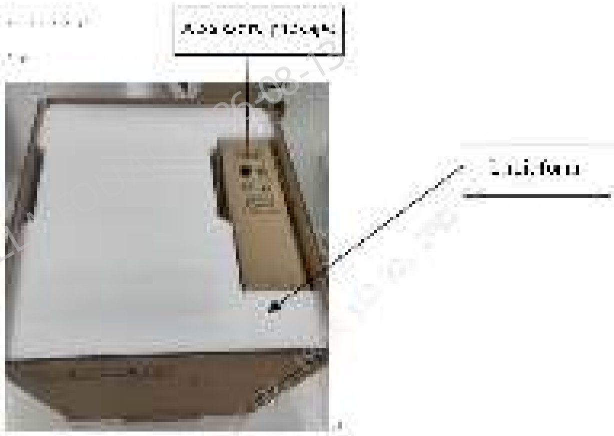


1 Place foam

Use 4 - M8X20 stainless steel hexagon socket bolts to lock the four connecting blocks on the four handle support frames respectively, use 4 - M8x30 hexagon socket bolts plated with environmental protection colored zinc, together with flat pad and elastic pad to lock the four connecting blocks on the embedded nuts of the base, and cover the main unit carton on the wooden base, as shown in the figure below.

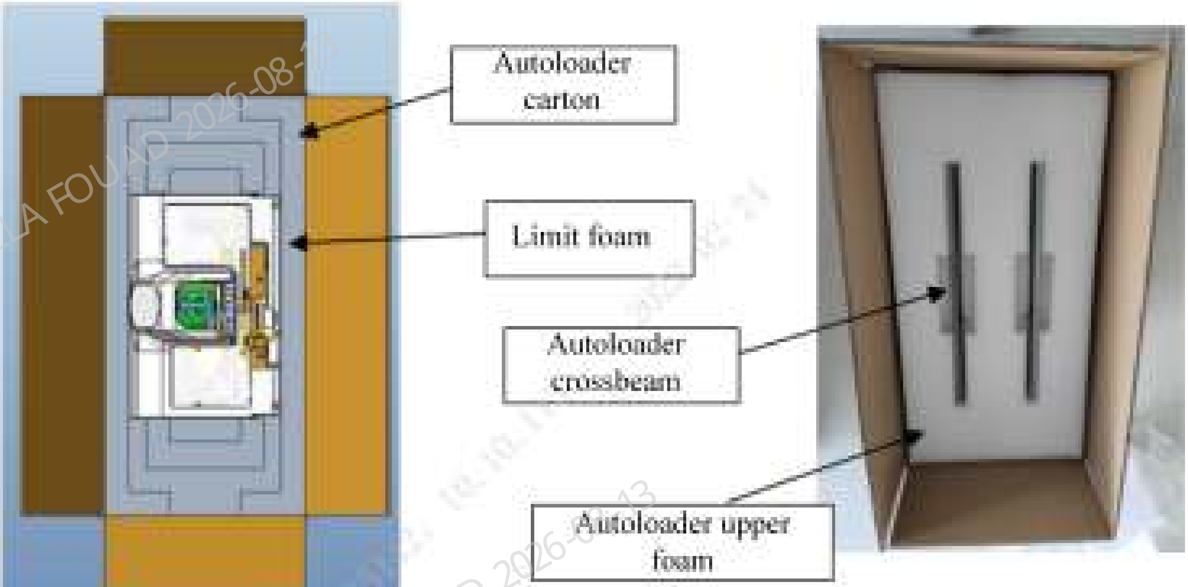


Clamp the limit foam on the top of the main unit, insert the accessory package into the position shown in the figure, install the top cover of the outer package, and fix two binding straps on the outer package, as shown in the figure below.



Description of packing the loader

Put the limit foam into the loader carton, and clamp the loader to the limit foam. Put the autoloader upper foam equipped with two loader beams into the carton and seal the carton with tape, as shown in the figure below.


Autoloader carton
Limit foam
Autoloader crossbeam
Autoloader upper foam


Tape

11 Appendix

11.1 Fluidic Chart

Figure11–1 BC-760 & BC-780(No RET ,No CRP, No SAA) Fluid Chart

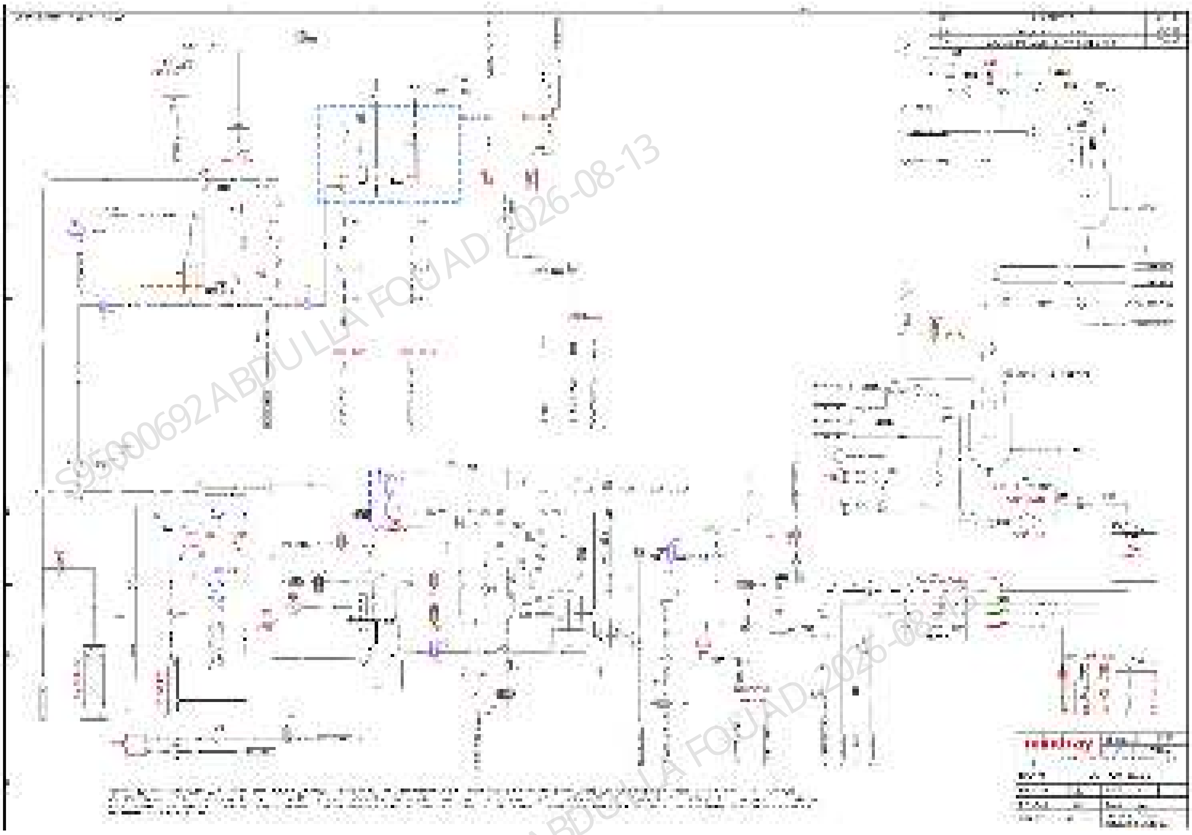
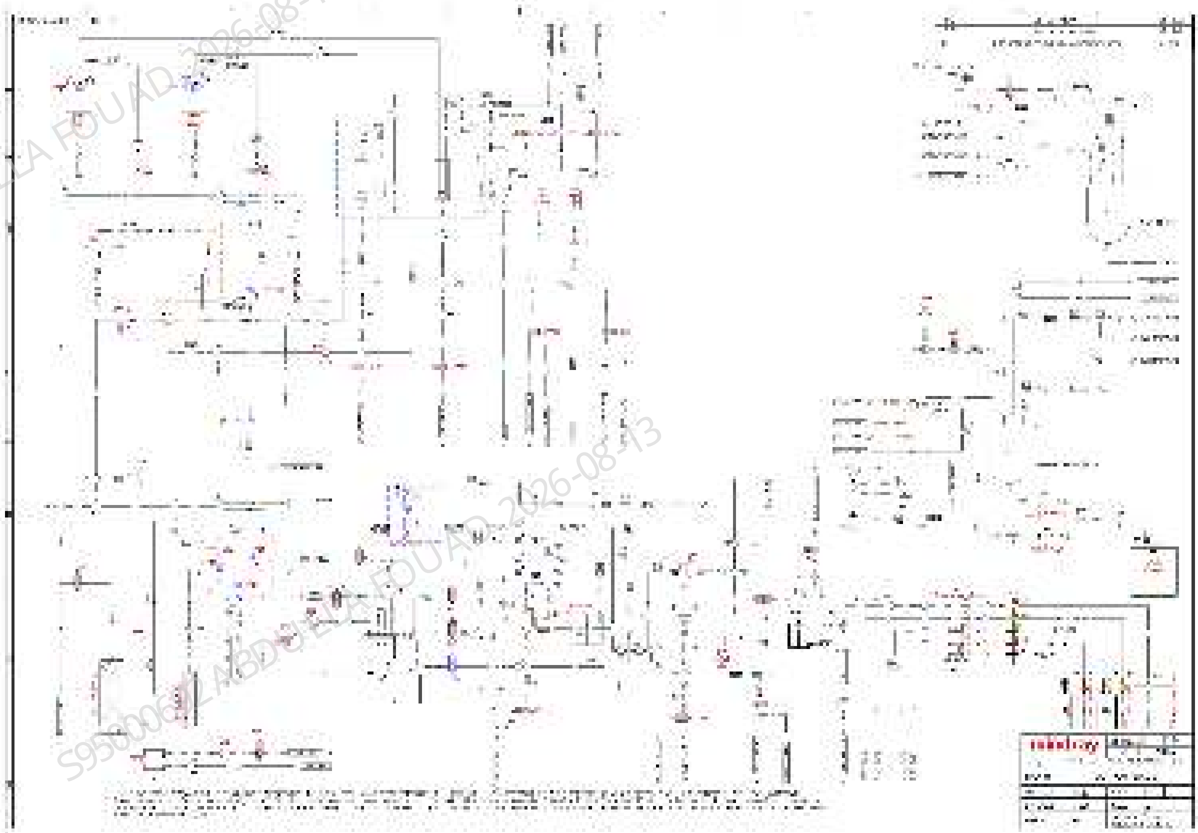


Figure11–2 BC-760 & BC-780(with RET ,No CRP and SAA) Fluid Chart



11.2 Tube and Connector Checklist

Table 11–1 Tube and Connector Checklist

Names	FRU Code
T40,T41,T102,T110,T111,T112,T113,T114, T121,T123,T124,T128,T134,T305	0040-10-32301
T86,T92,T93,T95,T96,T100,T103,T104,T108, T304	082-000055-00
T16,T22,T23,T53,T54,T55,T56,T57,T58,T59, T60,T62,T63,T64,T65,T72,T73,T215	082-000108-00
T61,T167	082-000109-00
T1	082-000314-00
T99,T107	082-000316-00
T33,T36,T38,T39,T125,T129	082-000422-00
T6,T7,T8,T9,T10,T11,T12,T13,T14,T25,T26, T27,T28,T29,T30,T31,T32,T34,T35,T44,T45, T74,T80,T81,T82,T83,T84,T85,T94,T97,T98,	082-000432-00

Table 11–1 Tube and Connector Checklist(continued)

Names	FRU Code
T105,T106,T115,T118,T120,T238,T301,T302,T303,T311	
T42,T126,T130,T132,T308	082-000710-00
T49,T66,T67,T68,T69,T75,T76,T77,T88,T91,T135,T138,T139,T142,T143,T144,T147,T150,T151,T152,T155,T158,T159,T160,T161,T162,T165,T166,T169,T170,T171,T172,T173,T175,T176,T177,T178,T183,T184,T185,T186,T187,T188,T189,T190,T191,T192,T193,T194,T195,T199,T200,T201,T202,T204,T205,T206,T208,T209,T210,T211,T212,T214,T216,T231,T232,T233,T298,T306,T307	082-003550-00
T50,T51,T89,T90,T136,T137,T140,T141,T168,T180,T181,T182,T196,T197	A21-000002—
T78,T79,T122,T148,T153,T154,T156,T157,T163,T164,T198,T203,T207,T213	M6G-020006—
T43,T71,T87,T127,T131	M6G-020008—
T2,T3,T4,T300	M6G-020034—
T17,T18,T21,T24,T149,T310	M6G-020054—
T5,T15,T19,T20,T116,T117,T119,T179	M6G-020055—
T46,T52,T70,T133	M90-100031—
T145,T146	M90-100071—

Table 11–2 Comparison table of transfer tube

Names	FRU Code
J17,J20	0030-20-13339
J8,J9,J13,J15,J37,J38,J55,J56	082-000055-00
J19,J22	082-000614-00
J18,J21	082-000664-00
J1,J2,J3,J4,J5,J6,J7,J10,J11,J12,J14,J16,J23,J24,J25,J26,J27,J28,J29,J30,J31,J32,J33,J34,J35,J36,J39,J40,J41,J42,J57,J58,J59,J60,J61,J62,J63,J64	082-000710-00

Table 11–3 Tube comparison table

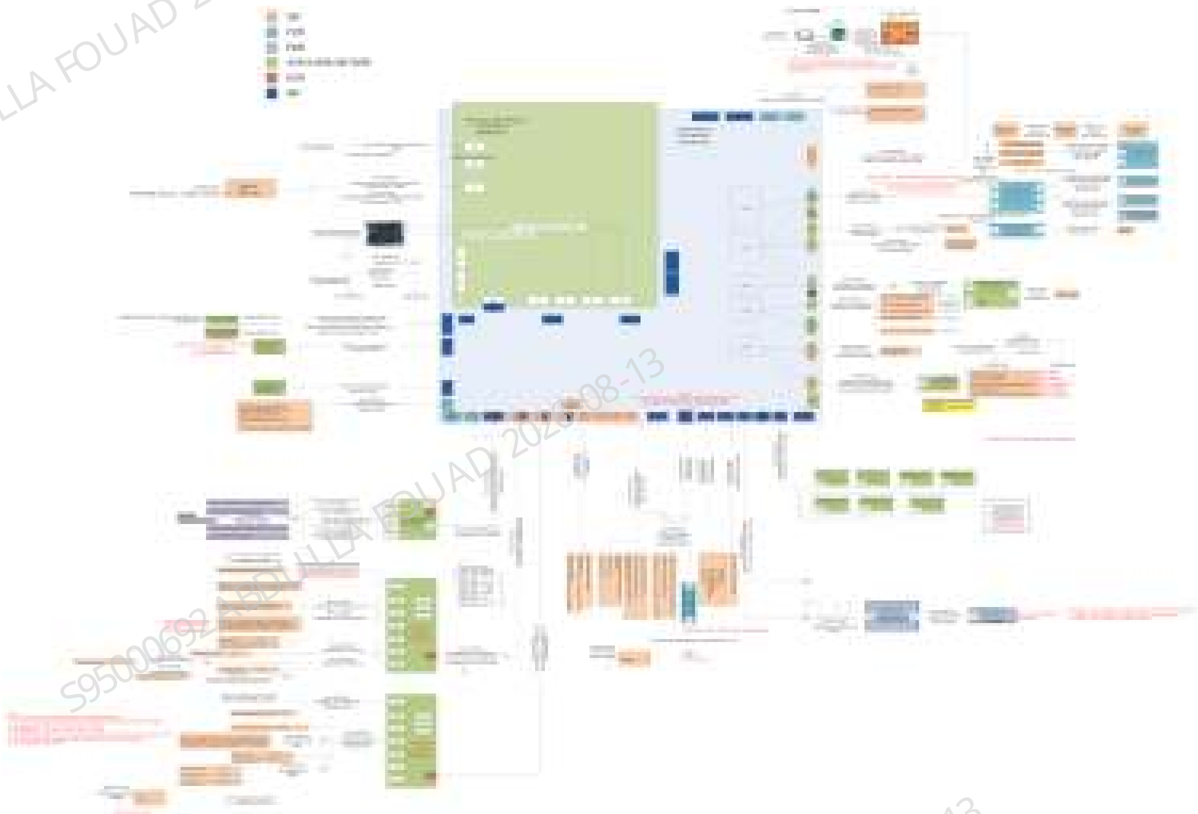
Names	FRU Code
P12,P16	082-000664-00
P7,P8,P9,P10,P11,P13,P14,P15,P17,P18, P19,P20,P21,P22,P23	082-000709-00
P1,P2,P3,P4,P5,P6,P32,P33	A21-000002—

Table 11–4 Tube connector comparison table

Names	FRU Code
C12,C28,C52,C53,C54	043-000880-00
C34,C38,C59,C60,C89	043-000892-00
C27,C146	043-012116-00
C9,C40,C42,C44,C46	082-000525-00
C8,C41,C43,C45,C47	082-000526-00
C22,C23,C31,C32,C33,C35,C36,C37,C39, C51,C75,C93,C94,C114,C150,C151	082-001105-00
C142,C144	082-001131-00
C141,C143	082-001133-00
C77	082-001280-00
C15,C16,C29,C30,C55,C56,C57,C58,C78, C87	082-002131-00
C24,C25,C65,C69,C70,C85	082-002745-00
C76	3102-20-69219
C49,C50	M90-100027—
C4,C5,C6,C7,C19,C20,C26,C72,C74,C79, C83,C86,C90,C91	M90-100028—
C10,C13,C17,C18,C21,C62,C67,C68,C71, C81,C82,C84,C88,C145	M90-100028-03
C1,C2	M90-100065—
C3,C48	M90-100066—
C61,C63,C64,C66,C80,C92,C111,C112,C113, C139	M90-100100—

11.3 Hardware Connection Diagram

Figure11–3 3128CDR Automatic Model Hardware System Connection Diagram



11.4 Cable Checklist

Table 11–5 3128CDR Automatic Model Cable Checklist

Cable material silkscreen	FRU Code	Name
C012952	009-012952-00	Paired wire, motherboard to motor board A, motherboard end
C012959	009-012959-00	Wire, sample type detection sensor
C012960	009-012960-00	Wire, motherboard to motor board B
C012970	009-012970-00	Wire, motherboard to scanner
C012981	009-012981-00	Wire, motherboard to fan, ext.
C012982	009-012982-00	Wire, HGB sensor, ext.
C012983	009-012983-00	Wire, fluid pressure sensor wire, ext.

Table 11–5 3128CDR Automatic Model Cable Checklist(continued)

Cable material silkscreen	FRU Code	Name
C012984	009-012984-00	Wire, impedance detection module, ext.
C012985	009-012985-00	Wire, temperature detection wire, ext.
C012986	009-012986-00	Wire, dye detecting sensor, ext.
C012987	009-012987-00	Wire, motherboard to air pressure detection board, ext.
C012989	009-012989-00	Paired wire, motherboard to RFID board 1&2, motherboard end
C012990	009-012990-00	Wire, motherboard to floater, ext.
C012991	009-012991-00	Wire, ESR assembly
C012993	009-012993-00	Wire, motherboard to heater, ext.
C013000	009-013000-00	Paired wire, valve board 1 valve group 3, board end
C013002	009-013002-00	Wire, valve board 1 valve group 5
C013010	009-013010-00	Wire, motherboard to jog switch (automatic)
C013020	009-013020-00	Wire, rotary scanner and blending motor
C013021	009-013021-00	Wire, sampling asm&syringe motor sensor
C013072	009-013072-00	Wire, motherboard to valve board 1 Pow & Com, ext.
C013193	009-013193-00	Wire, valve board 1 valve group 2
C013235	009-013235-00	Wire, syringe grounding
C012607	009-012607-00	Wire, front cover sheet metal ground wire
C013186	009-013186-00	Wire, front cover
C013064	009-013064-00	Wire, RFID reader to antenna

Table 11–5 3128CDR Automatic Model Cable Checklist(continued)

Cable material silkscreen	FRU Code	Name
C012969	009-012969-00	Paired wire, sensor vertical motor of sampling assembly and ESR
C012962	009-012962-00	Paired wire, motor and sensor of latex probe assembly, latex probe assembly end
C013234	009-013234-00	Lead-out wire, latex reader
C008191	009-008191-00	Micro-switch connection cable
C005387	009-005387-00	Wire, semiconductor refrigeration sheet
C013019	009-013019-00	Wire, digital sensor (S), A.5 (Design)
C013018	009-013018-00	Wire, motor (S)
C012948	009-012948-00	Wire, pusher (S)
C012953	009-012953-00	Paired wire, motherboard to motor board A, motor board end
C49040	3100-20-49040	Micro-switch connection cable
C013022	009-013022-00	Wire, digital sensor (M)
C013023	009-013023-00	Wire, motor (M)
C013024	009-013024-00	Wire, pusher (M)

11.5 LIS Communication Connection and Troubleshooting

Overview

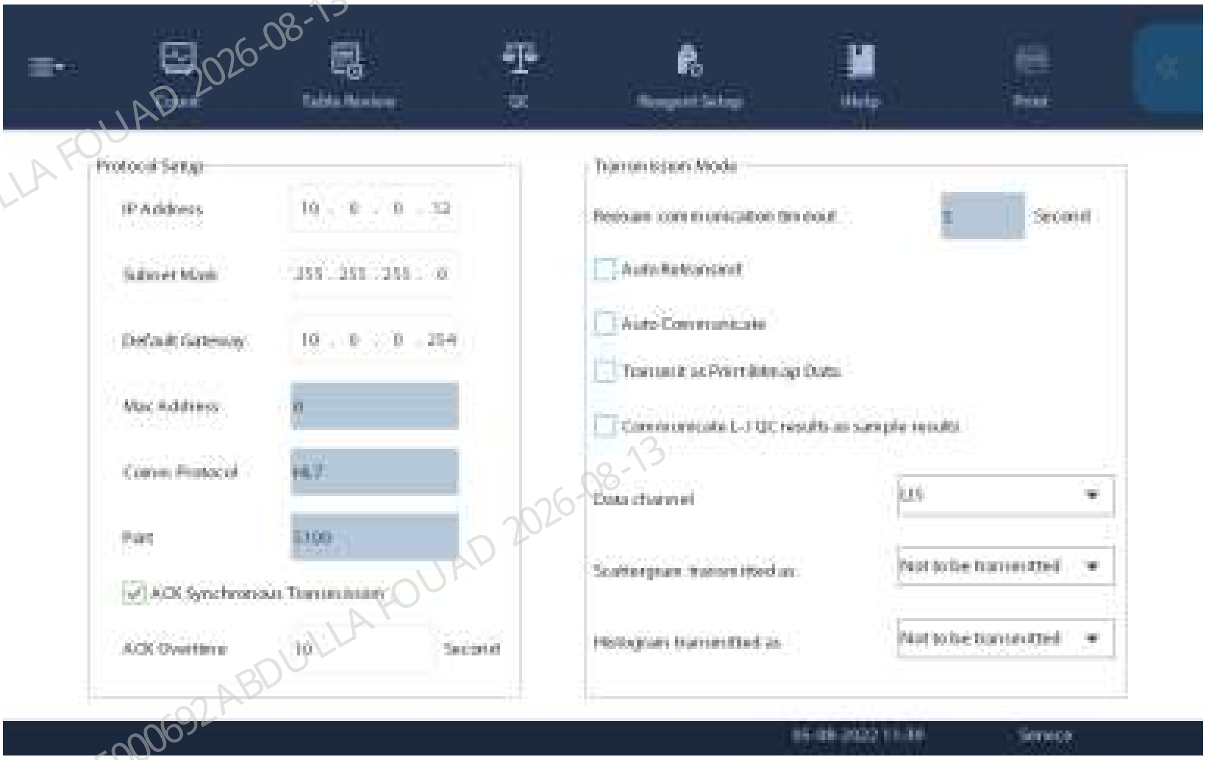
This chapter is to enable engineers to effectively cooperate with LIS engineers to complete LIS linkage, and focuses on the basic work content of LIS linkage and common LIS failures. This chapter mainly introduces the communication parameter settings of Laboratory Information System (LIS) and analyzer software, and the sample test and result transmission based on LIS connection.

LIS Connection

Analyzer Setup

Communication (Menu>Setup>System Setup>Communication)

Figure11–4 Communication: Select LIS for Data Channel



NOTICE

If LIS is connected, set the analyzer as the server, and the LIS serves as the client. The port No. is 5100.

Network Setup Verification

Physical Connection

Check whether the network wire is properly connected; and whether the physical network connection is correct.

Comm. Setup

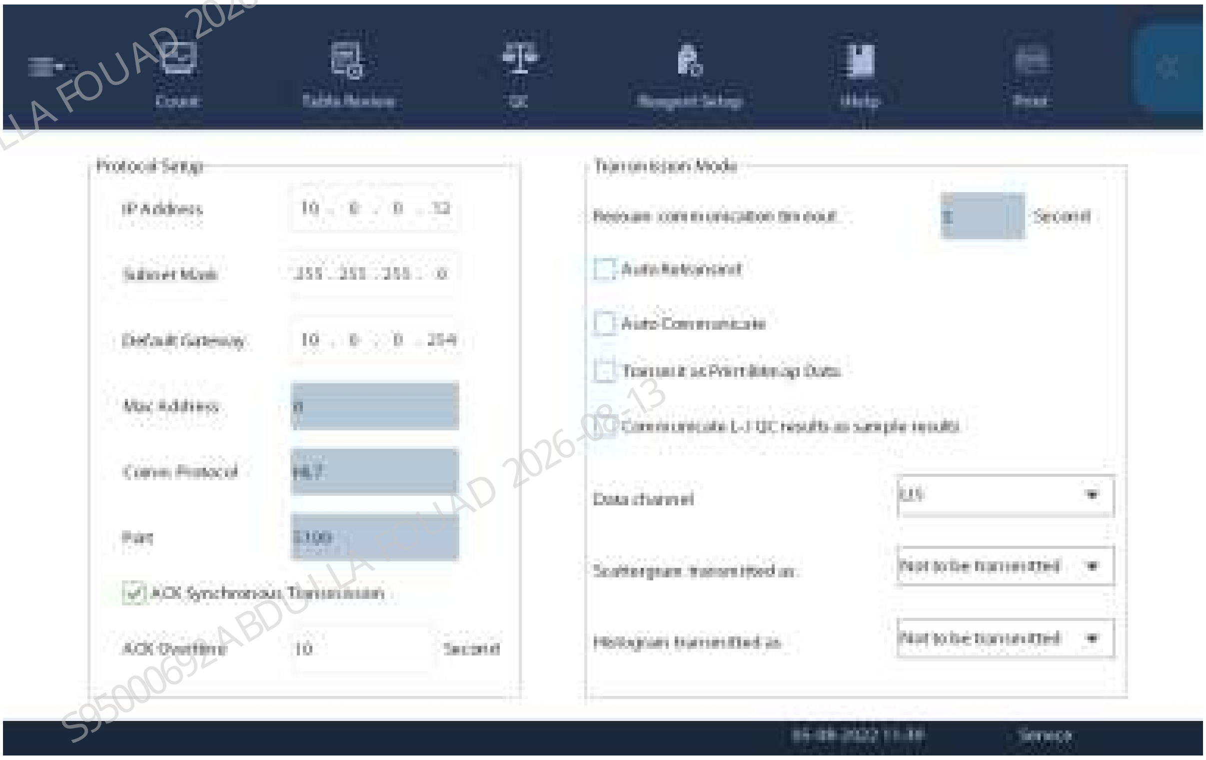
Check whether the network settings (including the communication setup for both analyzer and LIS) are correct.

If both physical connection and communication settings above are verified to be correct, enter ping command (e. g. ping 10.0.0.12, which is the IP address of the analyzer) in the console command window on the computer at the LIS side. A successful response indicates the correction hardware connection.

Communication Parameter Setup

Communication setup options (**Menu>Setup>System Setup>Communication**)

Figure11–5 Comm. Setup



NOTICE

The default IP address of analyzer is 10.0.0.12. The IP address of analyzer may be set by user as required.
A user with the permissions of administrator level is allowed to set up the following communication settings:

- **Protocol Setup**
- **Transmission Mode**

Protocol Setup

Click **Menu>Setup>System Setup>Communication** to access the **Communication** screen. Refer to the Figure **Figure11–5** for the specific setup screen.
The communication protocol options are currently unadjustable, which is protocol HL7 by default.

Transmission Mode

Click **Menu>Setup>System Setup>Communication** to access the **Communication** screen. Set up the transmission mode according to the practical needs of the laboratory.
See below for setting descriptions:

Transmission Mode	Check to enable one or more transmission functions: Auto retransmit Auto communication Auto acquisition of sample information Transmission of bitmap data printed	Only when **ACK Synchronous Transmission** is enabled, the Auto Retransmit is available.
Data channel	Select from the pull-down list: LIS labXpert	When you are using Mindray's labXpert software, select **PC Software** as the data channel.
Scattergram transmitted as /Histogram transmitted as	Select from the pull-down list: Not to be transmitted Bitmap transmission Data transmission	

Common LIS Failures

Effective Use of ACK

Communication data can be transmitted and received only with the effective communication linkage between labXpert and LIS port. With ACK synchronous transmission enabled, labXpert will add all test results to be transmitted to the communication queue. Before transmitting the sample test results, it is required to verify the effectiveness of the ACK of last test results transmitted. In case of no ACK or any ACK error after LIS receives the test results from labXpert, the smooth communication of the next sample cannot be guaranteed and the communication blocking will be caused by the increasing sample size. After receiving the test results from the analyzer, LIS must reply within a specified time with an ACK message string in a fixed format, which is different from 06 or the string ACK. ACK message string consists of MSH, MSA, message event ACK^R01 and message ID. LIS may not correctly receive and analyze the raw data of the analyzer while processing ACK error or ACK overtime.

On the contrary, with ACK synchronous transmission enabled by labXpert, LIS can correctly process ACK and acknowledge within the time threshold. To a great extent, this can cut the load for LIS to process the data received. LabXpert will transmit the test results at a rate dependent on LIS acknowledgment.

Large-scale Grade 3A hospitals are recommended to enable ACK synchronous transmission for the purpose of flow control. Before checking the box of ACK synchronous transmission, it is available to verify the ACK capability and format of LIS according to the logs generated by the analyzer and LIS communication.

It is not recommended to enable ACK synchronous transmission for Grade 2 hospitals or any hospital with a daily testing volume of less than 200 samples, since communication blocking will be caused by improper ACK processing or no ACK processing.

The fault phenomenons below are for your information:

- More than 50 test results are in the communication queue of labXpert software, and the feedback from LIS indicates an extremely slow transmission rate which takes more than ten minutes to transmit one test result.
- The feedback from LIS indicates that labXpert transmitted the same test results for four times.
- The feedback from LIS indicates that the system keeps receiving the test results with no sign of stopping.
- The quantities of results displayed by LIS system are different from those of labXpert. Many results are not displayed by LIS and required to be manually transmitted.
- The feedback from LIS indicates that the system keeps receiving the previous results (historical results).

Example of acknowledgment:

MSH|^~\&|||Mindray||20190703094835||ACK^R01|1|P|2.3.1||||0||ASCII|||MSA|AA|1||||0|

Insufficient Results Transmitted by LIS

Error phenomenon

- According to the laboratory personnel and LIS engineers, LIS has failed to receive the results of many samples. However, the test results of all samples are available on labXpert and required to be manually transmitted by laboratory personnel.
- According to the LIS engineers, the test results received by LIS are incomplete, and the results of some report parameters are not transmitted.

With a successful LIS connection between labXpert and LIS, LabXpert is powerful in data communication and data processing. If ACK synchronous transmission is enabled and LIS properly process ACK within the specified time limit,

the sample test results will be immediately transmitted to LIS. If ACK synchronous transmission is not enabled, labXpert will instantaneously transmit sample test results to LIS. However, LIS communication performance is affected to a certain extent due to the simultaneous processing by LIS (data receiving, data analysis, data processing, log backup). Any graphic data contained in the test results transmitted by labXpert will double the communication volume.

Cause of Insufficient Transmission

1. The sample test result transmission rate of labXpert does not match with the sample test result processing rate of LIS. This issue can be solved by enabling ACK synchronous transmission and realizing the proper ACK in the specified time limit.
2. LIS shall judge the integrity of sample test results, and verify whether the data contains both start character and end character.

Mode Related Issues

Advice:

For the communication between labXpert and LIS, the former serves as the client. With a successful LIS connection between labXpert and LIS, labXpert is capable of automatically monitoring the LIS connection status and automatically connecting to LIS.

With an effective LIS connection between labXpert and LIS, in the process of bi-directional LIS communication, the instrument sends the worklist request to LIS through labXpert after scanning the barcode. After receiving the worklist request, LIS shall send back the worklist information in the proper fixed format and within the specified time limit. Bi-directional LIS inquiry will fail in case of improper or no processing by LIS.

Bi-directional LIS communication varies for hematology and biochemistry. Biochemical bi-directional instrument determines the test items according to the channel number for LIS transmission. This is not applicable to hematology, however, which cannot be tested based on the parameters of WBC, RBC and PLT, etc. Hematology is tested based on the modes. For example, the test mode **CBC+DIFF** shall be used for blood routine examination.

There are actually many modes for hematology, which can be combined. LIS engineers are responsible for the logic of LIS ports, such as how to convert the test items in the barcode into the test modes supported by the instrument.

Start/End Character Related Issues

TCP/IP is a byte stream protocol, providing no message boundaries. That is, with an effective LIS connection between labXpert and LIS and the normal transmission of test results, no obvious start/end mark is available. Data integrity is judged by LIS, including the start and end characters of data in the LIS ports. Many LIS engineers do not check the existence of such characters or use a method incapable of integrity judgment.

For example, many engineers used to judge the data integrity by verifying whether carriage return character is the end character of OBX line. In consideration of no uniqueness for such judgment, it is impossible to judge the integrity of test results in this way.

The control characters include start characters, end characters and carriage return character, none of them is macroscopic. Where manual operation is required during LIS processing, the start character will be processed as 0x0B, VT and the end character as 0x1C, FS, in order to facilitate the understanding of communication protocol. For LIS port procedure, LIS engineers shall well process the control characters based on ASCII table, other than simply the + (hyphen) character string. The control characters added shall make sense.

Test Tool User Guide

The Mindray test tool mainly receives the data and simulate 2-way LIS communication, including but not limited to the following functions:

- Preliminary troubleshooting of LIS transmission problems

- Newly installed hematology assembly line simulates LIS communication test, 2-way test, and pressure test
- Help LIS engineers quickly understand the principle of 2-way LIS communication
- Help Mindray engineers quickly understand the principle of 2-way LIS communication

Tool Instructions

Double-click **Mindray.exe** and click the Server button.

The tool, as a server, and labXpert, as a client, participate in LIS communication. Because the labXpert software can monitor the LIS connection status, when labXpert detects that the LIS connection is out, it will try to recover the LIS connection, in another word, it can automatically reconnect to the LIS. Many LISs, as clients, are very unstable in the LIS communication. After the LIS is disconnected, you need to restart the computer to resume the connection.

Figure11–6 Main Interface of Mindray Tool

Figure11–7 The tool establishes connection with the labXpert software

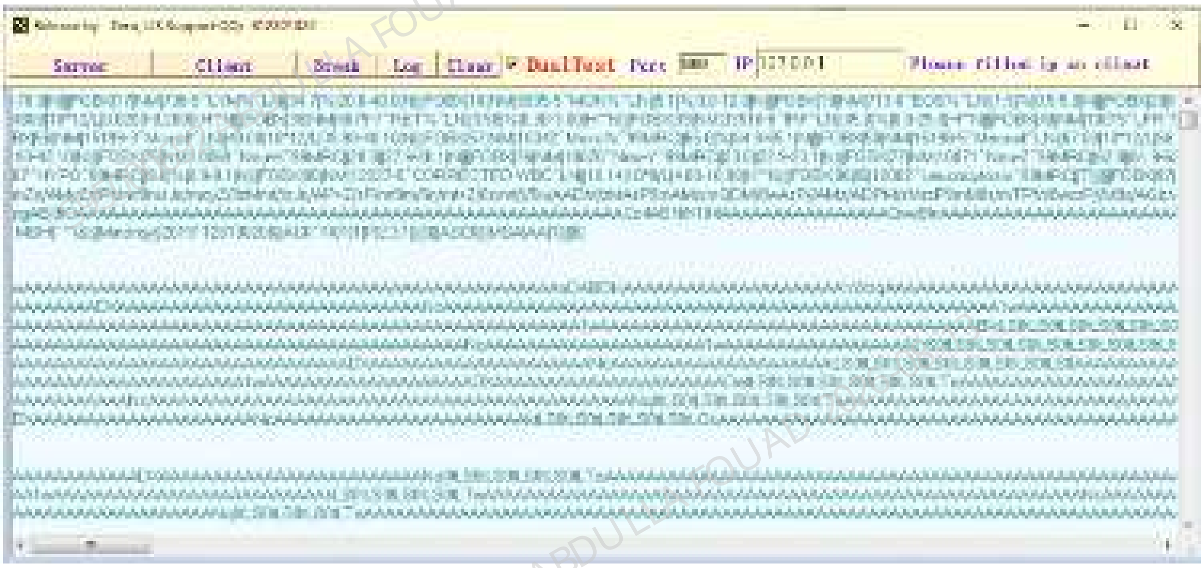
When labXpert acts as a client and LIS acts as a server in LIS communication, you need to enter the IP address and port number. The IP address to be filled can be considered as the IP address pointing to the LIS interface. LabXpert and the LIS interface or tool port need to have the same settings and the IP addresses are set correctly to successfully establish the LIS connection. After the LIS connection is established, the LIS icon in the upper right corner will change from yellow to green.

Figure11–8 LIS Disconnection Status

Figure11–9 LIS Connection Status

After labXpert establishes the LIS connection with the tool, the sample can be selected for communication. The tool can display the original data received from the analyzer in real time. You can select a single sample or sample results in batches for transmission.

Figure11–10 Communication Test



Click the **Log** button to generate a log. The log path is **LisDebug \lislog** where the tool is located, and click the **Clear** button to clear the received original data.

The tool has an auto response function, whether the ACK synchronization communication button is ticked or not, it will not affect the normal communication.

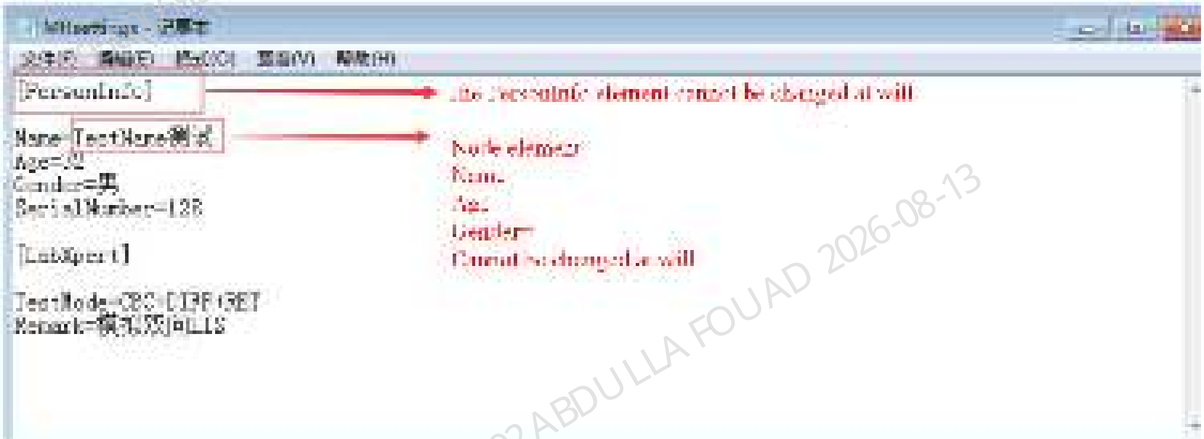
Two-way LIS Communication Settings

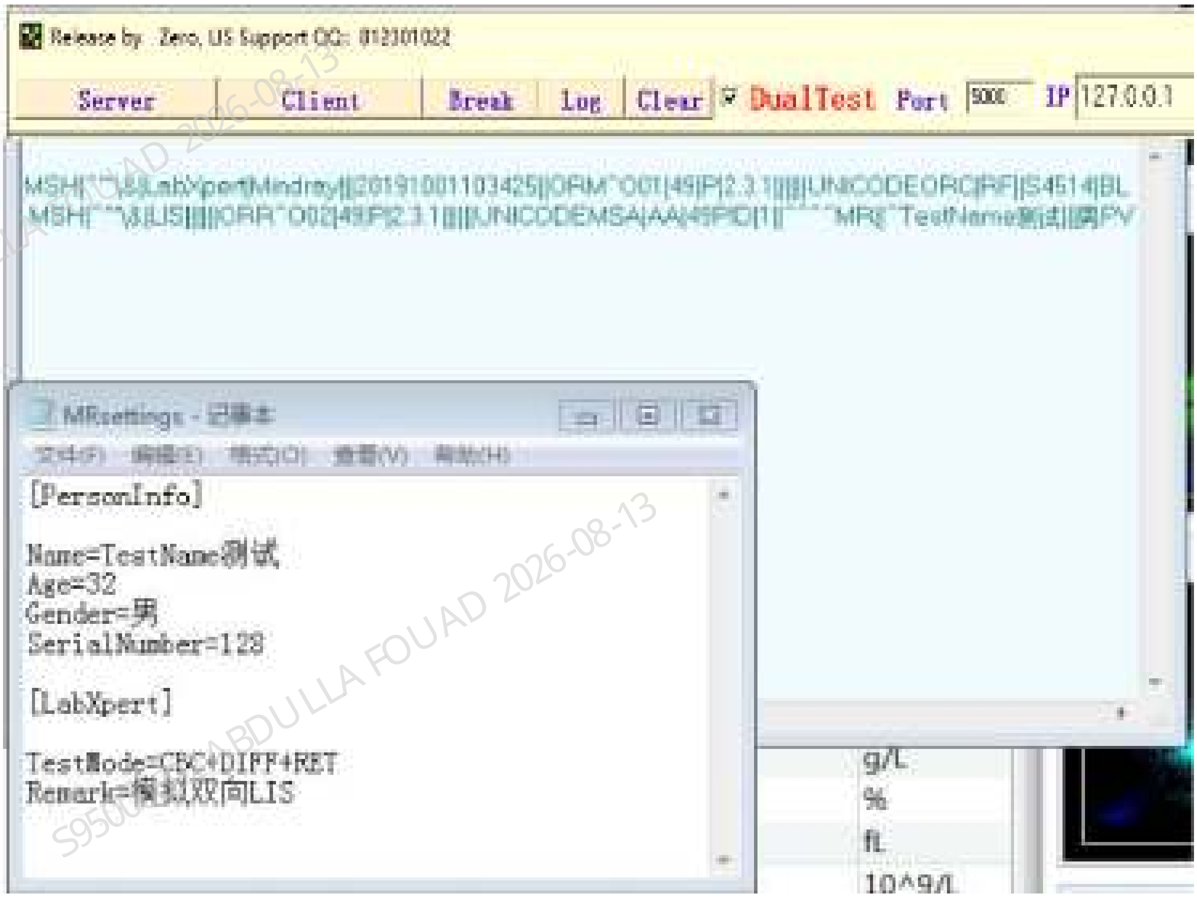
MRsettings.ini works with Mindray.exe to establish the LIS connection with labXpert, which can simulate 2-way LIS communication and instruct LIS engineers to develop 2-way LIS communication.

MRsettings. ini is a configuration file where engineers can enter sample-related clinical information. During the 2-way LIS communication process, the information entered by the engineer in the configuration file will be organized and spliced and sent to labXpert in a fixed format. If the message sent by the LIS or tool is in a correct format within the specified time, the labXpert software will receive it correctly, which will be displayed in the clinical information column of the labXpert software. The sample is also detected in the correct mode (the received mode).

Figure11–11 Configuration Files

You can edit relevant clinical information in the configuration file, but do not change the names of the node elements without authorization.





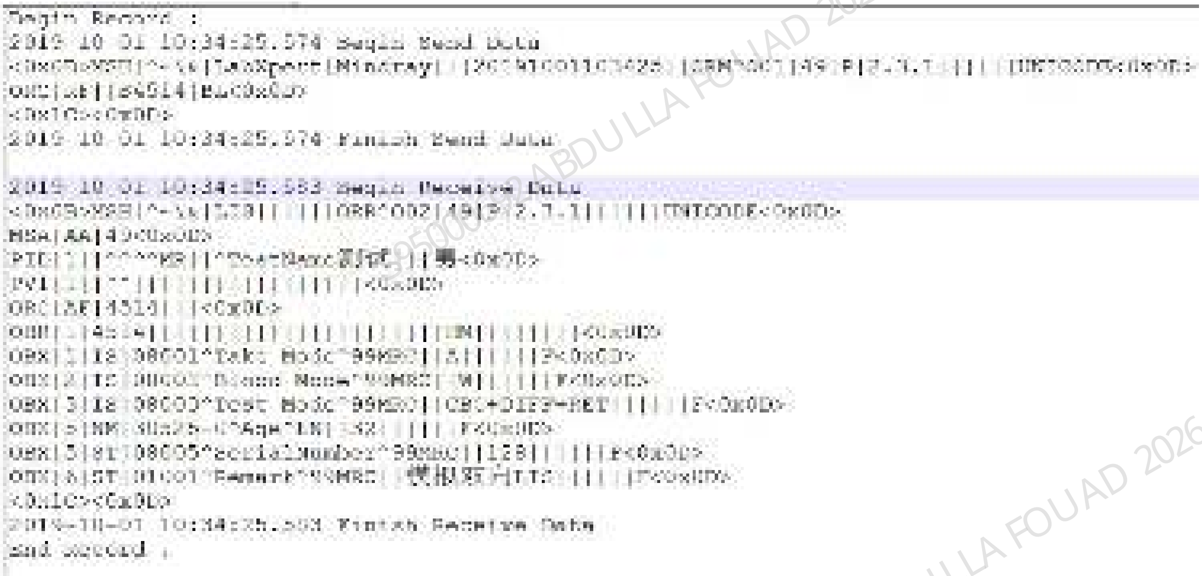
Logs Related to 2-Way LIS Communication

During 2-way LIS communication, the tool will display the interaction with labXpert information in real time.

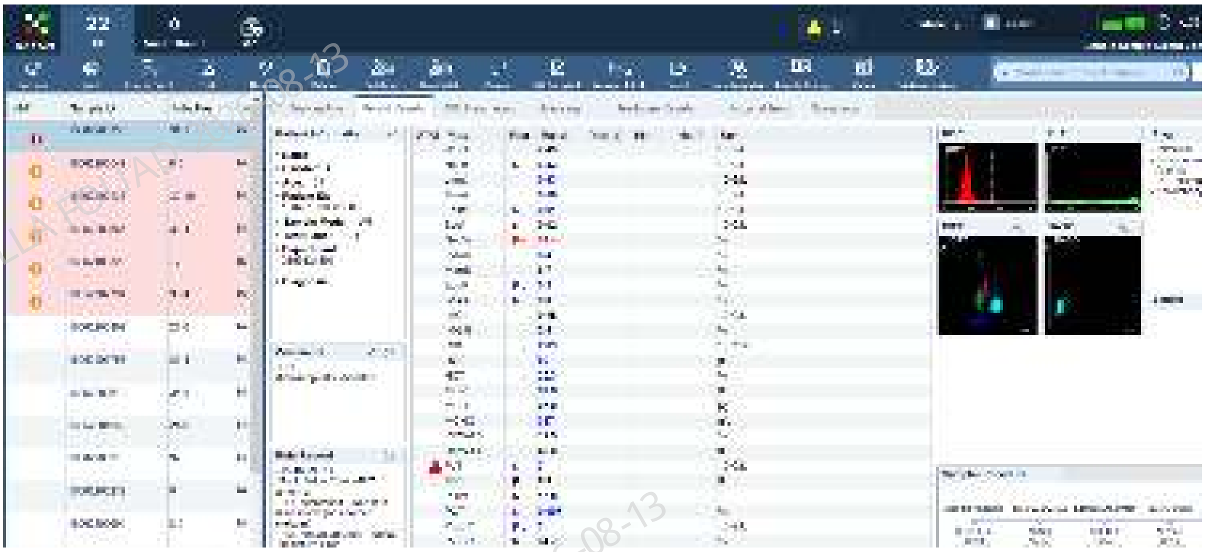
Figure11–12 Logs Generated by Real-Time Interactions

The log path of labXpert communication is D:\labXpertServerData\Log\LisLog\2019-10-01

Figure11–13 Communication Logs Generated by labXpert

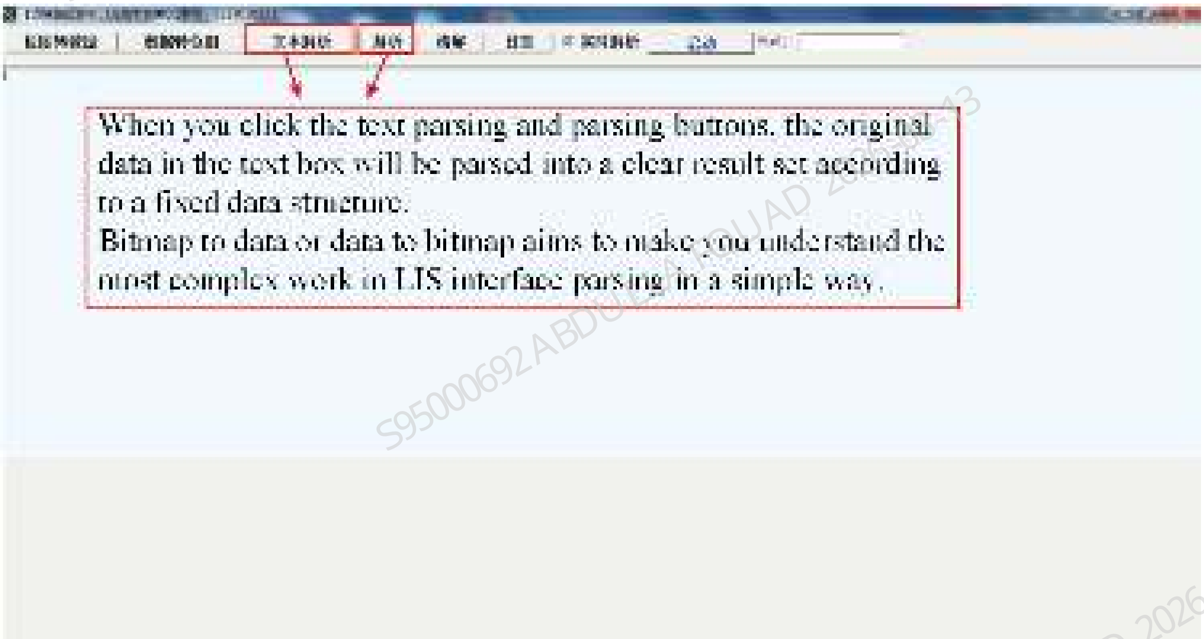


After the 2-way LIS communication is established, the patient information bar on labXpert will display relevant clinical information.

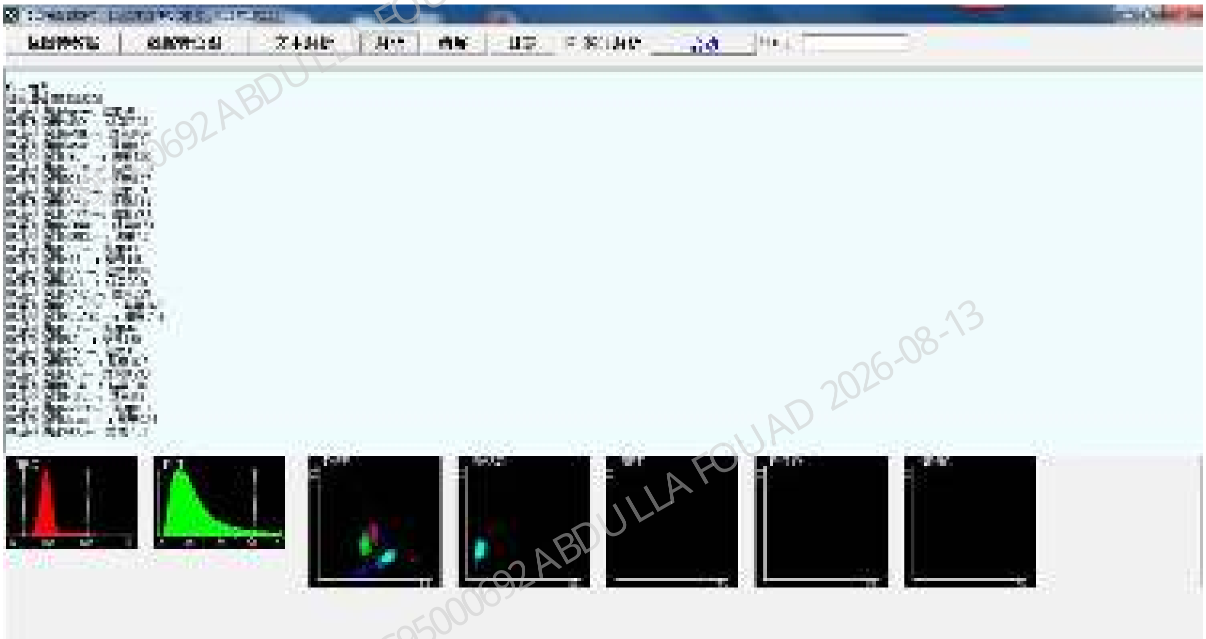


Data Analysis Tool User Guide

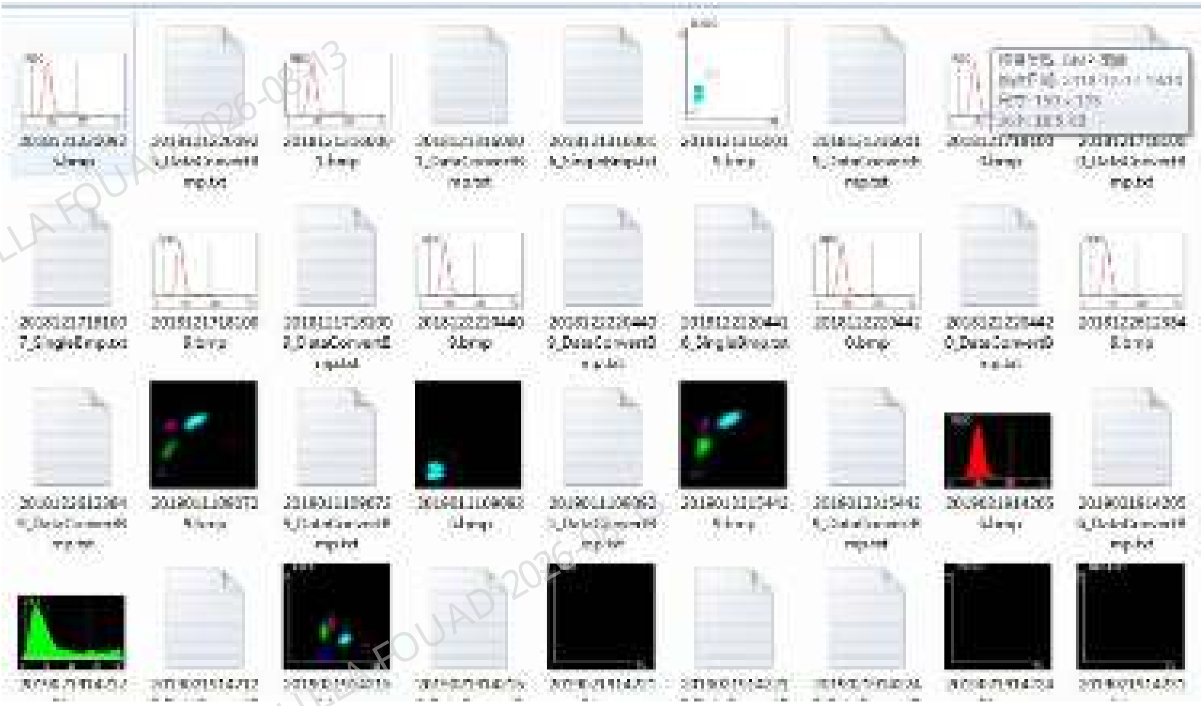
The main purpose of this guide is to let the engineers understand the main work content in the LIS interface development process, so as to more effectively support the LIS interface engineer to complete the LIS interface development work. This tool mainly introduces bitmap data processing and the original data of the analyzer.



When there is doubt about the original data of the analyzer, copy the original data to the data analysis tool. After analysis, compare the results with those results on the analyzer.



When you click Analyze, the tool will perform data analyzing and record operation logs for reference.



Communication Protocol

Overview

The analyzer can communicate with the laboratory computer via Ethernet. It sends test results on the analyzer to the laboratory computer, and receives work orders from the laboratory computer.

This communication protocol is defined on the basis of the HL7 standard. HL7 is an electronic data interchange standard in the medical field, originally defined by the United States and now adopted by many countries. This protocol is defined based on HL7 v2.3.1. For details of HL7 standards, see *HL7 Interface Standards Version 2.3.1*.

HL7 message layer protocol

Introduction to HL7 protocol

- **HL7 basic syntax**

1. Message construction rules

   Each HL7 message consists of some segments, each segment ends with characters of <CR> (0x0D).

   Each segment consists of a three-character segment name and a field with variable number; each field consists of a component and a subcomponent. Delimiters for fields, components, and subcomponents are defined in the MSH segment of each message.

   For example:

   MSH|^~\&|Mindray|||20060427194802||ORU^R01|1|P|2.3.1||||||UNICODE

   Where:

The five characters after MSH define the delimiters used to distinguish fields, components, and subcomponents. While these characters can be any non-text characters, the HL7 standard recommends the following characters:

Table 11–6 HL7 delimiter

Character	Meaning
|	Field delimiter
^	Component delimiter
&	Subcomponent delimiter
~	Repeating delimiter
\	Escape character

The first two fields of the MSH include the respective delimiters. Some of the following fields are empty because they are optional and the Mindray HL7 interfaces do not use them. For detailed field definitions and selections, see the segment definition section.
For any kind of message, the segments after the MSH segment are displayed in a fixed sequence. The following sections will describe these sequences in detail. The following syntax structures are used to indicate the sequence of segments:
The segment in [ ] is optional.
The segment in { } can be repeated once or several times.

2. String escape rules
In ST, TX, FT, CF and other types of field data, such as remarks, diagnostic information, user-defined gender and other string data, escape delimiters may appear, and the delimiters in the original string should be escaped during encoding to the escape character sequence, which is then restored when decoding. The HL7 interface uses the following escape rules:

Escape character sequence	Original character
\F\	Field delimiter
\S\	Component delimiter
\T\	Subcomponent delimiter
\R\	Repeating delimiter
\E\	Escape character
\.br\	<CR>, end of segment character

NOTICE

'\' in the escape string sequence represents the escape delimiter, and its value is defined in the MSH segment.

HL7 lower layer protocol

As an upper layer protocol, HL7 is message-based and does not provide a message ending mechanism. To determine message boundaries, we use the Minimal Lower Layer Protocol (MLLP) (*HL7 Interface Standards Version 2.3.1* also describes this protocol accordingly).

Communication layer

Messages are sent in the following format:

<SB> *ddddd* <EB><CR>

Where:

<SB> = **Start Block character (1 byte)**

ASCII <VT>, ie, <0x0B>. Do not confuse it with the characters SOH or STX in ASCII.

ddddd = **Data (variable number of bytes)**

ddddd is the valid data of the HL7 message, expressed as a string. Strings in HL7 messages for analyzer communication use UTF-8 encoding.

<EB> = **End Block character (1 byte)**

ASCII <FS>, ie <0x1C. Do not confuse it with the characters ETX or EOT in ASCII.

<CR> = **Carriage Return (1 byte)**

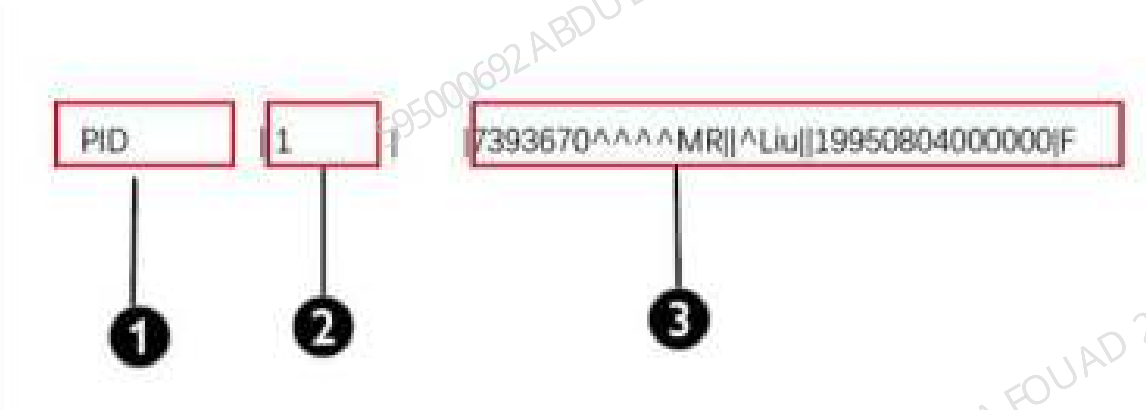
ASCII carriage return, i.e. <0x0D>.

HL7 segment definitions involved

The detailed definitions of the fields contained in each segment will be explained in the list below. A row in the table corresponds to a field in the segment, and the meanings of the columns in the table are as follows:

1. Sequence number: The HL7 segment starts with a 3-character segment name, followed by each field delimiter and the content of the field. The sequence number is the sequence or the position of the field in the HL7 segment.
   Example:

Figure11–14 Message example - Sequence number example of HL7 field



1	Message segment name
2	Field 1
3	Field 3

NOTICE

The MSH segment is slightly different. The field delimiter immediately after the segment name is considered to be the first field, and is used to describe the value of the field delimiter used in the entire message.

2. Field name: The logical meaning of the field.
3. Data type: The HL7 standard type of the field, the structure of which will be described in Table **Table 11–7**.
4. Maximum recommended length: HL7 standard recommended length. However, in the actual message transmission process, the length of the actual transmission will exceed this value, so when the message is parsed, the message field should be read according to the delimiter.
5. Description: A description of the actual value of the field.
   Example: An example of the actual value of the field.

MSH

The MSH (Message Header) segment contains the general Information of the HL7 message, including the value of the message delimiter, the type of the message, and the encoding method of the message, etc. It is the first field of each HL7 message.
Message example:
MSH|^~\&||Mindray|||20101012092538||ORU^R01|1|P|2.3.1||||||UNICODE
The field definitions used in the MSH segment are shown in Table **Table 11–7**.

Table 11–7 MSH field definition table

No.	Field name	Field name	Maximum recommend-ed length	Description	Example
1	Field Seperator	ST	1	It contains the first field delimiter after the segment name, which is used to specify the values of the field delimiters for the rest of the message.	|
2	Encoding Characters	ST	4	It contains component	^~\&

Table 11–7 MSH field definition table(continued)

No.	Field name	Field name	Maximum recommend-ed length	Description	Example
				delimiter, repeating delimiter, escape delimiter, and subcomponent delimiter.	
3	Sending application	EI	180	Sender application. If the analyzer sends a message, the value is	
4	Sending Facility	EI	180	Sender The value is **Mindray** (in Chinese or English)	Mindray
7	Date/Time Of Message	TS	26	Message creation time (in the form of YYYY[MM[DD[HH[MM[SS]]]]]), which takes the system time value.	20101012092538
9	Message Type	CM	7	Message type, in the form of **Message Type**, **>Event Type**.	ORU^R01
10	Message Control ID	ST	20	Message control ID, used to uniquely	1

Table 11–7 MSH field definition table(continued)

No.	Field name	Field name	Maximum recommend-ed length	Description	Example
				identify a message.	
11	Processing ID	PT	3	Message processing ID. Value: **P>Query information for sample, work order**. **Q>QC count result information**. In the Ack message, it is consistent with the previously received message.	P
12	Version ID	VID	60	HL7 version number, the value is **2.3.1**.	2.3.1
18	Character Set	ID	10	Character set. The value is **UNICODE**, that is, the communication message is represented by a UTF-8 encoded string.	UNICODE

MSA

The MSA (Message Acknowledgment) segment contains message acknowledgment information.
Message example:
MSA|AA|1
The field definitions used are shown in the table below.

Table 11–8 MSA field definition table

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Acknowledgment Code	ID	2	Confirmation codes: **AA** - Accept, **AE** - Error, **AR** - Reject.	AA
2	Message Control ID	ST	20	Message control ID, the same as MSH-10 in the corresponding received message.	1
6	Error Condition	CE	100	Error condition (status code), optional transmission, or including error condition description information, see Table 3 for the value.	

Table 11–9 Error Codes of MSA-6 Field

Status Code (MSA-6)	Status Text (MSA-3)	Description/Remarks
Succeeded:		AA
0	Message accepted	Succeeded
Error status code:		AE

Table 11–9 Error Codes of MSA-6 Field(continued)

Status Code (MSA-6)	Status Text (MSA-3)	Description/Remarks
100	Segment sequence error	In the message, the segments are not in the correct sequence, or a required segment is missing
101	Required field missing	Missing required field in a segment
102	Data type error	The data type of the field is wrong, for example, a number is set to a character
103	Table value not found	Table value not found, not used for now
Rejected status code:		AR
200	Unsupported message type	Unsupported message type
201	Unsupported event code	Unsupported event code
202	Unsupported processing id	Unsupported processing ID
203	Unsupported version id	Unsupported version ID
204	Unknown key identifier	Unknown key identifier, such as transmission of a non-existent patient information
205	Duplicate key identifier	Duplicated key identifier
206	Application record locked	Transactions cannot be performed at the application storage level, e.g. the database is locked
207	Application internal error	Other unknown application errors
Skip:		AS

PID

The PID (Patient Identification) segment contains the general Information of a patient.
Message example:
Example (in Chinese)
PID|1||C1^^^^MR||^Zhangsan||20101005084346|Male
Example (in other languages than Chinese)
PID|1||C1^^^^MR|| Jordan^Michael ||20101005084346|Male
The field definitions used are shown in the table below.

Table 11–10 PID field definition table

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Set ID - PID	SI	4	Sequence number, used to identify different PID segments in a message.	1
3	Patient Identifier List	CX	20	In the sample test result message, it is used as the medical record number, and is shown as **Medical Record Num ber^^^^MR**. In QC messages, it is used to indicate the QC lot number.	C1^^^^MR
5	Patient Name	XPN	48	Patient name (divided into FirstName and LastName), in the form of **LastName^ FirstName**	^Liu

Table 11–10 PID field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
7	Date/Time of Birth	TS	26	In the sample result message, it is used as the birth time, in the form of YYYY[MM[DD[HH[MM[SS]]]]]. In the QC information, it is used as the validity period of QC.	20101005084346
8	Sex	IS	1	Gender, string. Same as the string displayed on the screen.	Male.

PV1

PV1 (Patient Visit) contains the patient's medical information.
Message example:
PV1|1|Outpatient|Medicine^^BN1|||||||||||||||||MedicalInsurance
The field definitions used are shown in the table below.

Table 11–11 Definitions of PV1 Fields

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Set ID - PV1	SI	4	Sequence number, used to identify different PV1 segments in a message.	1
2	Patient Class	IS	1	Patient type, string, any content. Same as the string displayed on the screen.	Outpatient
3	Assigned Patient Location	PL	80	Patient location information, shown as **Department ^ ^ Bed No.**.	Medicine^^BN1
20	Financial Class	FC	50	Expense type, string, any content.	MedicalInsurance

OBR

The OBR (Observation Request) segment mainly contains observation report information.
Message example:
OBR|1||TestSampleID1|00001^Automated Count^99MRC||20101006084439|20101009091515|||Li|||Cold|20101007084458||||||||||HM||||||||admin
The field definitions used are shown in the table below.

Table 11–12 OBR field definition table

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Set ID - OBR	SI	4	Sequence number, used to identify different OBR segments in a message.	1
2	Placer Order Number	EI	22	In the work order query response message (ie ORR^O02 message), it is used as the sample ID.	
3	Filler Order Number +	EI	22	In the sample observation result message, it is used as the sample ID. In the QC message, it is used as the file No.	TestSampleID1
4	Universal Service ID	CE	200	Universal service ID, used to identify different count result types.	00001^Automated Count^99MRC
6	Requested Date/time	TS	26	Requested date/time. Used to indicate	20101006084439

Table 11–12 OBR field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
				sampling time.	
7	Observation Date/Time #	TS	26	Observation date/time.	20101009091515
10	Collector Identifier *	XCN	60	Sample collector. It is used here to indicate the submitter.	Li
13	Relevant Clinical Info.	ST	300	Relevant clinical information. It can be used to represent clinical diagnosis information in patient information.	Cold
14	Specimen Received Date/Time *	TS	26	Sample received date/time. It is used to indicate submission time.	20101007084458
15	Specimen Source *	CM	300	Sample source. Reserved field of the analyzer.	
22	Results Rpt/ Status Chng - Date/Time +	TS	26	Results report/Status change - date/time It is used as review time.	

Table 11–12 OBR field definition table(continued)

No.	Field name	Data type	Maximum recommended length	Description	Example
24	Diagnostic Serv Sect ID	ID	10	Diagnosis ID, the value is **HM** which means Hematology.	HM
28	Result Copies To	XCN	150	Result copies to It is used here to indicate the sample reviewer.	
32	Principal Result Interpreter +	CM	200	Principal result interpreter. In sample messages, it is used to indicate the observer. In QC count messages, it is used to indicate the operator.	admin

OBX

The OBX (Observation/Result) segment mainly contains the parameter information of each observation result. A complete sample/QC result/two-way query transmission message package contains multiple OBX segments. Depending on the content of the transmission, the OBX segment included in the message varies.

Message example:

OBX|8|NM|6690-2^WBC^LN||2.20|10*9/L|4.00-10.00|L~A|||F

The field definitions used are shown in the table below.

Table 11–13 OBX field definition table

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Set ID - OBX	SI	10	Sequence number, used to identify different OBX segments in a message.	8
2	Value Type	ID	3	Data type of the observation result, which can be **ST**, **NM**, **ED**, **IS**, etc.	NM
3	Observation Identifier	CE	590	Observation Identifier. In the form of **ID^Name^EncodeSys**, ID is the observation identifier, Name is the observation description information, and EncodeSys is the observation coding system.	6690-2^WBC^LN

Table 11–13 OBX field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
				NOTICE **ID and EncodeSys are used to uniquely identify an observation parameter, while Name is mainly used for description and cannot be used as an identifier.**	
5	Observation Value	*	65535	The observation result data can be numbers, strings, enumeration values, binary data, etc. Binary data such as histograms and scattergrams are converted	2.20

Table 11–13 OBX field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
				using Base64 encoding.	
6	Units	CE	60	Observation units. It is shown by standard units defined by HL7.	10*9/L
7	References Range	ST	60	Observation references range. The format is as follows: **Lower limit of references range>Higher limit of references range**, or **<Higher limit of references range**, or **>Lower limit of references range**.	4.00-10.00
8	Abnormal Flags	ID	5	Observation result flag. The values are as follows: **N** - Normal **A** - Abnormal **H** - Higher than the	L~A

Table 11–13 OBX field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
				upper limit of references range **L** – Lower than the lower limit of references range NOTICE **The field may have abnormal flags or high and low alarm signs at the same time. At this time, multiple flags are connected with ~, such as H~A**	
11	Observ Result Status	ID	1	Observation result status. The value is **F** - (Final Result), indicating the final result.	F

Table 11–13 OBX field definition table(continued)

No.	Field name	Data type	Maximum recommended length	Description	Example
13	User Defined Access Checks	ST	20	User defined content, here is used to store reagent expiration flags and modification flags, etc. The form is **flag 1~flag 2**. There are three flags: O – Reagent expired flag E – User edited result flag e – Calculation edited result flag	

ORC

The ORC (Common Order) segment mainly contains general information about the Order.
Message example:
ORC|RF||SampleID||IP
The definitions of its fields are shown in the table below.

Table 11–14 ORC field definition table

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Order Control	ID	2	Order control field. Value: **RF** in the ORM message, meaning **Refill the order request** **AF** in the ORR message, meaning **Order refilling confirmation**	RF
2	Placer Order Number	EI	22	Number of order placer. In the ORM message, the value is null, and in the ORR message, the value is the sample ID.	

Table 11–14 ORC field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
3	Filler OrderNum	EI	22	Number of order receiver. In the ORM message, the value is the sample ID, and in the ORR message, the value is null.	SampleID
5	Order Status	ID	2	Order status. In the work order information query communication, the value in the ORM message is fixed as **IP**, which means **Order is being processed, but the result has not yet been obtained**; the value in the ORR message is null.	IP

NTE

The NTE (Note and comment messages) segment contains note and comments messages.

Message example:

NTE|9|P|This is comment.|GR

The field definitions used are shown in the table below.

Table 11–15 ORC field definition table

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
1	Set ID – NTE	SI	4	Sequence number, used to identify different NTE segments in a message.	9
2	Source of comment	ID	8	Source of comment. There are three values, as follows: L: Ancillary (filler) department is source of comment . P: Orderer (placer) is source of comment . O: Other system is source of comment . The field is always filled with "P" for morphology analysis result.	P

Table 11–15 ORC field definition table(continued)

No.	Field name	Data type	Maximum recommend-ed length	Description	Example
3	Comment	FT	64K	Content for comments	This is comment.
5	Comment Type	CE	60	Comment type. There are eight values as follows: PI: Patient Instructions . AI: Ancillary Instructions . GI: General Instructions . 1R: Primary Reason . 2R: Secondary Reason . GR: General Reason . RE: Remark . DR: Duplicate/ Interaction Reason . The field is always filled with "GR" for morphology analysis result.	GR

Complete message example

The following two messages demonstrate the process of sample data communication.

Sample message

```
MSH|^~\&||Mindray|||20111124091140||ORU^R01|1|P|2.3.1||||||UNICODE
PID|1||^^^^MR
PV1|1
OBR|1||ste5|00001^Automated Count^99MRC|||20111101170410|||||||||||||||||HM
OBX|1|IS|08001^Take Mode^99MRC||O||||||F
OBX|2|IS|08002^Blood Mode^99MRC||W||||||F
OBX|3|IS|08003^Test Mode^99MRC||CBC+DIFF||||||F
OBX|4|IS|01002^Ref Group^99MRC||Universal||||||F
OBX|5|NM|6690-2^WBC^LN||6.58|10*9/L|4.00-10.00|N|||F
OBX|6|NM|704-7^BAS#^LN||0.02|10*9/L|0.00-0.10|N|||F
OBX|7|NM|706-2^BAS%^LN||0.4|%|0.0-1.0|N|||F
OBX|8|NM|751-8^NEU#^LN||4.81|10*9/L|2.00-7.00|N|||F
OBX|9|NM|770-8^NEU%^LN||73.2|%|50.0-70.0|H~N|||F
OBX|10|NM|711-2^EOS#^LN||0.21|10*9/L|0.02-0.50|N|||F
OBX|11|NM|713-8^EOS%^LN||3.2|%|0.5-5.0|N|||F
OBX|12|NM|731-0^LYM#^LN||1.38|10*9/L|0.80-4.00|N|||F
OBX|13|NM|736-9^LYM%^LN||21.0|%|20.0-40.0|N|||F
OBX|14|NM|742-7^MON#^LN||0.15|10*9/L|0.12-1.20|N|||F
OBX|15|NM|5905-5^MON%^LN||0.2|%|3.0-12.0|L~N|||F
OBX|16|NM|789-8^RBC^LN||3.77|10*12/L|3.50-5.50|A|||F
OBX|17|NM|718-7^HGB^LN||105|g/L|110-160|L~N|||F
OBX|18|NM|787-2^MCV^LN||84.4|fL|80.0-100.0|A|||F
OBX|19|NM|785-6^MCH^LN||27.9|pg|27.0-34.0|A|||F
OBX|20|NM|786-4^MCHC^LN||330|g/L|320-360|A|||F
OBX|21|NM|788-0^RDW-CV^LN||14.9|%|11.0-16.0|A|||F
OBX|22|NM|21000-5^RDW-SD^LN||51.7|fL|35.0-56.0|A|||F
OBX|23|NM|4544-3^HCT^LN||31.8|%|37.0-54.0|L~A|||F
OBX|24|NM|777-3^PLT^LN||228|10*9/L|100-300|N|||F
OBX|25|NM|32623-1^MPV^LN||7.3|fL|7.0-11.0|N|||F
OBX|26|NM|32207-3^PDW^LN||15.4||9.0-17.0|N|||F
OBX|27|NM|10002^PCT^99MRC||0.166|%|0.108-0.282|N|||F
OBX|28|NM|10014^PLCR^99MRC||*****|%|11.0-45.0|N|||F
OBX|29|NM|10013^PLCC^99MRC||*****|10*9/L|30-90|N|||F
OBX|35|NM|15051^RBC Histogram. Left Line^99MRC||25||||||F
OBX|36|NM|15052^RBC Histogram. Right Line^99MRC||149||||||F
OBX|37|NM|15053^RBC Histogram. Binary Meta Length^99MRC||1||||||F
OBX|38|NM|15057^RBC Histogram. Total^99MRC||52978||||||F
OBX|39|NM|15111^PLT Histogram. Left Line^99MRC||3||||||F
OBX|40|NM|15112^PLT Histogram. Right Line^99MRC||41||||||F
OBX|41|NM|15113^PLT Histogram. Binary Meta Length^99MRC||1||||||F
OBX|42|NM|15117^PLT Histogram. Total^99MRC||3010||||||F
OBX|43|NM|15203^WBC DIFF Scattergram. Meta len^99MRC||4||||||F
```

OBX|44|NM|15205^WBC DIFF Scattergram. Fsc dimension^99MRC||0||||||F
OBX|45|NM|15206^WBC DIFF Scattergram. Ssc dimension^99MRC||0||||||F
OBX|46|NM|15207^WBC DIFF Scattergram. FL dimension^99MRC||0||||||F
OBX|47|NM|15208^WBC DIFF Scattergram. FSC-LOG dimension^99MRC||0||||||F

Sample response message

When the analyzers communicate synchronously, each sample result requires a response message. The sample response message contains two segments: MSH and MSA. Pay attention to two points for the correct response message: The content of the MSH-9 field needs to be filled with ACK^R01, indicating that the type of this message is a sample response message. The value of the MSA-2 field is the same as the value of the MSH-10 field of the received count result, indicating which count result the response message corresponds to. In this example, the value of the MSA-2 field is 1.
MSH|^~\&|LIS||||20111124091140||ACK^R01|1|P|2.3.1||||||UNICODE
MSA|AA|1

QC message example

The content and format of the QC message is different from that of the sample count result message: The MSH-11 value of the QC message is Q, which means that the message type is QC data; a QC message corresponds to a QC point of the analyzer software, which may contain multiple count results, for example, an L-J QC message contains one count result, and an X-R QC message may contain two count results and an average count result.
The QC message consists of an MSH message header and multiple count results. Each count result starts with a PID and OBR segment containing sample information, followed by multiple OBX segments to carry parameter results and other information. The OBR-4 field of each count result indicates the type of count result.
Take an L-J QC message as example: MSH|^~\&||Mindray|||20111124091422||ORU^R01|1|Q|2.3.1||||||UNICODE
PID|1||1||||20111103000000
OBR|1||1|00003^LJ QCR^99MRC|||20111103150042|||||||||||||||||HM||||||||Development engineer
OBX|1|IS|05001^Qc Level^99MRC||M||||||F
OBX|2|IS|08001^Take Mode^99MRC||O||||||F
OBX|3|IS|08002^Blood Mode^99MRC||W||||||F
OBX|4|IS|08003^Test Mode^99MRC||CBC+DIFF||||||F
OBX|5|NM|6690-2^WBC^LN||3.91|10*9/L|0.10-0.50|H~N|||F
OBX|6|NM|704-7^BAS#^LN||0.02|10*9/L||N|||F
OBX|7|NM|706-2^BAS%^LN||0.5|%||N|||F
OBX|8|NM|751-8^NEU#^LN||2.13|10*9/L||N|||F
OBX|9|NM|770-8^NEU%^LN||54.3|%||N|||F
OBX|10|NM|711-2^EOS#^LN||0.06|10*9/L||N|||F
OBX|11|NM|713-8^EOS%^LN||1.6|%||N|||F

OBX|12|NM|731-0^LYM#^LN||1.54|10*9/L||N|||F
OBX|13|NM|736-9^LYM%^LN||39.4|%||N|||F
OBX|14|NM|742-7^MON#^LN||0.16|10*9/L||N|||F
OBX|15|NM|5905-5^MON%^LN||4.2|%||N|||F
OBX|16|NM|789-8^RBC^LN||4.16|10*12/L||N|||F
OBX|17|NM|718-7^HGB^LN||132|g/L|2-8|H~N|||F
OBX|18|NM|787-2^MCV^LN||85.2|fL||N|||F
OBX|19|NM|785-6^MCH^LN||31.7|pg||N|||F
OBX|20|NM|786-4^MCHC^LN||373|g/L||N|||F
OBX|21|NM|788-0^RDW-CV^LN||11.4|%||N|||F
OBX|22|NM|21000-5^RDW-SD^LN||41.7|fL||N|||F
OBX|23|NM|4544-3^HCT^LN||35.4|%||N|||F
OBX|24|NM|777-3^PLT^LN||176|10*9/L|1-3|H~N|||F
OBX|25|NM|32623-1^MPV^LN||8.6|fL||N|||F
OBX|26|NM|32207-3^PDW^LN||16.3|||N|||F
OBX|27|NM|10002^PCT^99MRC||0.151|%||N|||F
OBX|28|NM|10014^PLCR^99MRC||0.0|%||N|||F
OBX|29|NM|10013^PLCC^99MRC||0|10*9/L||N|||F

Example of QC response message

There is only one difference between the QC response message and the count result response message: The value of the MSH-11 field is Q.
The following example is a QC message ACK:
MSH|^~\&|LIS||||20111124091422||ACK^R01|1|Q|2.3.1||||||UNICODE
MSA|AA|1

Example of two-way LIS query request

The 2-way LIS query request message contains the sample ID. After the LIS receives the message, it queries the patient and sample information corresponding to the sample.
The query request message contains two segments: MSH and ORC. The MSH segment is basically the same as the sample count result segment, except that the MSH-9 message type field takes the value ORM^O01. The recipient number is filled in ORC-3, and the sample ID is filled in here. In the example, this field is filled with SampleID1.

NOTICE

When the auto sample loading count initiates a query, if the built-in barcode is scanned incorrectly, the value of the sample ID field is Invalid.

Example of query request message. MSH| ^~ \&|| Mindray||| 20081120174836|| ORM^O01| 4| P| 2.3.1||||||UNICODE

ORC|RF||SampleID1||IP

Example of 2-way LIS query request response

When LIS receives the query request message, it needs to reply with a query result response message. The first two segments of the query response message are: MSH and MSA. Fill in the field of MSH-9 message type with ORR^O02. For the writing method of MSA segment, see the Sample Message Acknowledgment Example section. If the query is successful, it will include segments PID, PV1, ORC, OBR, and OBX to describe the patient and sample information. The description method of the information is the same as that of the sample data communication message. The ORC segment in the query success message is indispensable. The value in the ORC-1 message is AF, and the ORC-2 field is filled with the query primary key, that is, the sample ID. Note that the OBR-2 field is the sample ID information, and the value must be the same as the ORC-2 field, otherwise the message is considered an error.

The following is an example of a message with a successful query result:

MSH|^~\&|LIS||||20081120174836||ORR^O02|1|P|2.3.1||||||UNICODE
MSA|AA|4
PID|1||ChartNo^^^^MR||^FName||19810506|NT
PV1|1|E|Internal Medicine^^Bn4|||||||||||||||NewCharge
ORC|AF|SampleID1|||
OBR| 1| SampleID1|||| 20060506|||| tester||| Diagnose content....| 20060504||||||| 20080821|| HM|||| Reviewer||||Observer
OBX|1|IS|08001^Take Mode^99MRC||A||||||F
OBX|2|IS|08002^Blood Mode^99MRC||W||||||F
OBX|3|IS|08003^Test Mode^99MRC||CBC||||||F
OBX|4|IS|01002^Ref Group^99MRC||XXXX||||||F
OBX|5|NM|30525-0^Age^LN||1|hr|||||F
OBX|6|ST|01001^Remark^99MRC||remark content....||||||F

The following is an example of a response message for a query failure. The MSA-2 field indicates the response result, and the value here is AR.

If the query operation is rejected, the value can also be AE, indicating an error in processing the query operation: MSH|^~\&|LIS||||20081120175238||ORR^O02|1|P|2.3.1||||||UNICODE
MSA|AR|9

Use definitions of HL7 data types

CE − Code Element

<identifier (ST)> ^ <text (ST)> ^ <name of coding system (ST)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (ST)>

CM − Composite

The format is defined by concrete fields.

CX − Extended composite ID with check digit

<ID (ST)> ^ <check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ < assigning authority (HD)> ^ <identifier type code (IS)> ^ < assigning facility (HD)>

ED – Encapsulate Data

<source application(HD)^ <type of data(ID)^ <data sub type(ID)^ <encoding(ID)^ <data(ST)

EI – Entity Identifier

<entity identifier (ST)> ^ <namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

FC – Financial Class

<financial class(IS)^ <effective date(TS)

HD – Hierarchic designator

<namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

Used only as part of EI and other data types.

FT – Formatted text

This data type is derived from the string data type by allowing the addition of embedded formatting instructions. These instructions are limited to those that are intrinsic and independent of the circumstances under which the field is being used.

IS – Coded value for user--defined tables

The value of such a field follows the formatting rules for an ST field except that it is drawn from a site-defined (or user-defined) table of legal values. There shall be an HL7 table number associated with IS data types.

ID – Coded values for HL7 tables

The value of such a field follows the formatting rules for an ST field except that it is drawn from a table of legal values. There shall be an HL7 table number associated with ID data types.

NM – Numeric

A number represented as a series of ASCII numeric characters consisting of an optional leading sign (+ or -), the digits and an optional decimal point.

PL – Person location

<point of care (IS)> ^ <room (IS)> ^ <bed (IS)> ^ <facility (HD)> ^ < location status (IS)> ^ <person location type (IS)> ^ <building (IS)> ^ <floor (IS)> ^ <location description (ST)>

PT – Processing type

<processing ID (ID)> ^ <processing mode (ID)>

SI – Sequence ID

A non-negative integer in the form of an NM field. The uses of this data type are defined in the chapters defining the segments and messages in which it appears.

ST – String

TS – Time stamp

YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][+/-ZZZZ] ^ <degree of precision>

XCN – Extended composite ID number and name

In Version 2.3, use instead of the CN data type. <ID number (ST)> ^ <family name (ST)> & <last_name_prefix (ST) ^ <given name (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (ST)> ^ <source table (IS)> ^ <assigning authority (HD)> ^ <name type code (ID)> ^ <identifier check digit (ST)> ^ <code identifying the

check digit scheme employed (ID)> ^ <identifier type code (IS)> ^ <assigning facility (HD)> ^ <name representation code (ID)>

XPN − Extended person name

In Version 2.3, replaces the PN data type. <family name (ST)> ^ <given name (ST)> & <last_ name_prefix (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <name type code (ID) > ^ <name representation code (ID) >

VID − Version identifier

<version ID (ID)> ^ <internationalization code (CE)> ^ <international version ID (CE)>

11.6 Communication Protocol